Gauge-Covariant Variational Free Energy and Renormalization

Covariant Information Geometry, Timeless Inference Histories, Local–Collective Bounds, and Exact Effective Scale Theory

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Date: September 2026

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Part I

General Gauge-Variational Foundations

Chapter 1

Scope, Claims, and Reading Map

This document develops a gauge-covariant variational theory from typed probability objects and then asks which parts survive coarse description. Its ambient geometry consists of one principal gauge bundle $\mathscr{P}_G \to \mathcal{C}$ with structure group G, and two associated law bundles $\mathcal{E}_b = \mathscr{P}_G \times_{\widehat{\rho}_b} \mathcal{B}_b$ and $\mathcal{E}_m = \mathscr{P}_G \times_{\widehat{\rho}_m} \mathcal{B}_m$. The representations ρ_b and ρ_m may be inequivalent and may act on fibers of different dimensions; the hats denote their induced actions on probability laws. Belief and model coordinates may use different local sections of $\mathscr{P}_G$. Their unique relative principal-frame field is a G-valued comparison, not a map between the two associated fibers. The declared cross-channel morphisms

$$\Phi : \mathcal{E}_b \longrightarrow \mathcal{E}_m, \qquad \widetilde{\Phi} : \mathcal{E}_m \longrightarrow \mathcal{E}_b$$

cover $\mathrm{id}_{\mathcal{C}}$; they are neither principal-bundle morphisms nor assumed inverses. Separate connections induce same-channel parallel transports Ω_γ on $\mathcal{E}_b$ and $\widetilde{\Omega}_\gamma$ on $\mathcal{E}_m$. Parallelness of either cross map is an additional hypothesis, measured otherwise by its covariant defect. The two channels may use different connections on the same $\mathscr{P}_G$; neither a common local frame nor an equal connection is assumed. Independently declared graph links are denoted Θ^b_e, Θ^m_e on oriented edge copies (abbreviated by endpoints when the copy is unique), and are identified with base parallel transport only after a curve assignment and an equality hypothesis. These distinctions are the type contract of Chapter 2. [DEFINITION]

The primitive descriptive data are finitely many section-bearing agents, their agent-indexed recognition laws and evaluated generative mechanisms, and normalized kernels for the records produced at declared interactions. Complete population laws enter only after an explicit construction or selection. They encode the dependence required by evidence and collective variational free energy, but they are not additional agents and need not be represented by any participant. [DEFINITION]

The local-first architecture is

$$\{\mathcal{A}^i, G_{\theta,i,D}, \mathbb{Q}_{i,o,X}, K_a\}_{i\in V, a\in\mathfrak{A}} \longrightarrow \begin{cases} \mathbb{P}_{\theta,V} & \text{normalized construction,} \\ \mathbb{Q}_{V,o,X} \in \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}) & \text{declared measurable selection,} \end{cases} \longrightarrow \mathcal{L}_V. \tag{1.1}$$

The constructed joint $\mathbb{P}_{\theta,V}$ and selected coupling $\mathbb{Q}_{V,o,X}$ therefore precede the local conditional and collective objectives. [DEFINITION]

Under the stated absolute-continuity and integrability hypotheses, there is one exact identity,

$$\log p_{\theta,V}(o \mid X) = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + D_{\mathrm{KL}}\left(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}\right). \tag{1.2}$$

It is invariant under independent belief- and model-coordinate pushforwards. These passive re-expressions form a product of frame-coordinate changes; they must not be confused with replacing the ambient diagonal G-gauge symmetry by a principal $G \times G$-bundle. Equality holds exactly when the recognition law is the posterior as a measure; matching coordinate marginals is insufficient. Excluding the posterior makes every pointwise gap positive, but does not by itself make the optimized infimum positive or attained. [ESTABLISHED]

The logical order is therefore

typed bundles and kernels $\longrightarrow$ normalized joint
$\longrightarrow$ local and collective ELBOs
$\longrightarrow$ covariant informational pullbacks
$\longrightarrow$ relational inference histories
$\longrightarrow$ general coarse maps $\longrightarrow$ exact effective RG
$\longrightarrow$ constructive dynamical and spatial realizations
$\longrightarrow$ Gaussian realization.

The general coarse layer distinguishes three operations that are often conflated. A normalized Markov kernel pushes laws forward, preserves the observation evidence when it acts only on latent variables, and contracts relative entropy. Energy precomposition restricts an unnormalized energy along a sample-space map and requires a separately declared coarse reference measure and finite normalizer. Parameter aggregation is a conditional closure statement inside a selected graph-exponential family. A renormalization further requires comparison or rescaling maps that type different scales; without them there is only a changing-space coarse diagram, not a fixed point problem. [ESTABLISHED]

For an arbitrary finite agent hypergraph, normalized interaction-record kernels give one collective VFE and, by disintegration, a conditional VFE for every agent or block. A unilateral local change equals the corresponding change of the collective VFE, so local evolution is a blockwise collective descent. The local objectives do not sum symmetrically because shared factors occur in every incident coordinate objective. A categorical source label places attention inside the fixed joint, and a standard-Borel observation kernel is equivalent to a message from an operational environment node. These results are proved in Chapter 7. [ESTABLISHED]

A chosen connection turns the derivative of a smooth statistical section into a vertical first jet. Pulling the vertical Fisher and Amari tensors back by that jet gives gauge-invariant, connection-relative tensors on the contextual base; the Fisher tensor may be degenerate. This tensor pullback is distinct both from restricting a parameter manifold to a submodel and from a dynamical-systems pullback attractor. The subsequent relational construction quotients regular natural-gradient solution

curves by orientation-preserving reparameterization and measures them by Fisher length. The resulting histories and durations are information-geometric objects on a declared configuration space. These distinctions are developed in Chapters 8 and 9. [ESTABLISHED]

The construction does not identify those histories or durations with a physical clock. [NOT-CLAIMED]

The exact law-level RG of Chapter 12 pushes the baseline, evidence submeasure, recognition law, posterior, and joint attention-event law through the same structural channel. Exact closure retains induced hyperedges, simultaneous root-framed holonomy and boundary data, and path memory. Cross-scale agent–meta kernels follow from posterior bridges, while differential beta functions and fixed points are typed only after an explicit rescaling returns the changing scale spaces to one common space. [ESTABLISHED]

The ordered path coordinate is neither a relational inference history nor a physical-time coordinate, and RG depth is a scale index. [NOT-CLAIMED]

The constructive extensions give explicit realizations of several earlier conditional statements. Chapter 14 constructs compatible parent sections over an actual overlap cover, handles changing active agents through joint event masses, and retains coarse connection defects. Chapter 15 gives a normalized parent-sensitive mechanism and retained path laws with an exact finite-memory information cost. These constructions use declared ambient laws, channels, and mechanisms; they do not infer a joint law from independently specified singleton objectives. [ESTABLISHED]

At one base point, Chapter 16 constructs a nonlinear cyclic family for every prescribed finite depth. Its declared VFE-descending diffusion, source jumps, and natural-gradient model learning produce successive coherence episodes. Exact normalized parents retain all source assignments and loop marks, activate from the preceding actual retained law, and continue learning from their own laws. The weakest interaction energy, continuous mobility, and source proposal rate stay fixed at one. Separate bounds control conditional relative entropy, complete model scores, continuous trajectories, and entire discrete source histories. Theorem 16.1 and Chapter A give the joined assumptions and proof. [ESTABLISHED]

This family uses supplied interaction contrasts, restricted source support, and prepared initial laws at positive temperatures that decrease with the regime parameter. The finite-depth construction does not assert that source learning creates those contrasts, that arbitrary initial laws select the same hierarchy, or that an infinite-depth or physical-time limit exists. [NOT-CLAIMED]

Chapter 17 supplies an infinite-volume high-temperature lattice realization, complete generated interactions, a local RG operator on an open interaction neighborhood, and a constructive basin bridge. Its Fisher geometry is regular only on the stated fixed-depth parameter regions. The lattice domain is a worked realization, not a restriction imposed on the ambient gauge-variational theory. [ESTABLISHED]

Chapters 18 and 19 distinguish branchwise normalization from one global geometry factor, transport the complete observation experiment and scalar normalization, and prove pressure and specific-VFE limits for retained finite geometry labels. Gauge, Fisher, and base curvature require different

geometric data. Label erasure can destroy Gibbs regularity, and a disappearing label can make KL to a limiting joint infinite while the finite-ensemble VFE density remains finite. [ESTABLISHED]

Critical universality, continuum field measures, smooth geometry across phase ties, uniform Fisher invertibility at unbounded depth, and a self-consistent curvature action remain separate construction obligations. The available results do not establish autonomous agency or a physical law. [OPEN]

The multivariate Gaussian enters only after that general structure. It realizes the smooth statistical tier in information coordinates and supplies explicit determinant gaps, Fisher charts, matrix-weighted interaction operators, aggregation formulas, Kron counterexamples, and generalized spectra. The Laplacian-plus-self-term interaction cone is a declared subfamily, not a consequence of every linear-Gaussian directed model. [HYPOTHESIS]

Within its exact domain, aggregation is closed, but closure alone does not select that cone or a nondegenerate partition. [ESTABLISHED]

Unrestricted matrix-weighted Kron closure fails, whereas the later Gaussian coarse chapter records the exact closed congruence-diagonal domain and its qualified maximality boundary. [ESTABLISHED]

The RG conclusions are deliberately narrower. Rescaling is declared rather than derived. [DEFINITION]

Bi-additive Gaussian parameters are closed under additive pushforward, and an identified homogeneous hierarchy gives an operator fixed ray, but the ray lies on the boundary of the full-space Gaussian probability domain. [ESTABLISHED]

For the Gaussian operator family just described, general infinite-hierarchy attraction and blocking-scheme independence remain open. These Gaussian obligations are distinct from the constructive lattice results below. [OPEN]

Finite numerical gates test specified diagnostics and controls; they do not prove those asymptotic claims. [NUMERICAL]

Identifying any repaired limiting object with a physical law remains the semantic claim E.1; it requires a complete probability repair, a target system, a validation criterion, and discriminating evidence. [OPEN]

Epistemic contract

Every nontrivial statement carries one of the following tags. The central open-obligation appendix collects unresolved conjectures and problems once; established claims remain with their hypotheses or proofs, and numerical claims remain bound to the numerical-provenance appendix.

A gesture is either discharged by proof, calculation, or a checked source, or converted into an open obligation with exact hypotheses. Likewise, `NOT-CLAIMED` is never used for a refuted statement. The obstruction chapter contains exact negative results; the interpretation chapter marks readings and empirical commitments without promoting them to theorems.

Table 1.1: Status taxonomy: tags used throughout.

Tag	What it promises the reader
[ESTABLISHED]	Proved here, or a standard result cited to a source that has been checked.
[DEFINITION]	A declared type, construction, or convention. Nothing is being proved and the text says so.
[HYPOTHESIS]	A restriction the development adopts by choice. What it excludes, and where it is used, are stated.
[CONJECTURE]	Believed and precisely stated, but not proved. Stated sharply enough to be attacked.
[NUMERICAL]	Supported by computation only, with its measurement, seed, and control reported. Computation is not proof.
[OPEN]	Unsettled. What would settle it, and what obstructs it, are named.
[NOT-CLAIMED]	A statement the development deliberately declines to make. Declining is not refuting, and the text says which it is doing.

A physicist's fast path

A first reading may take the object and frame-covariance contract from Chapters 1 and 2, then read only the normalized-joint and ELBO identities in Chapters 3 to 5. The next pass should read the local–collective potential theorem and interaction-label attention in Chapter 7, then the covariant pullback and relational-history constructions in Chapters 8 and 9. The next pass should take the three-operation distinction, Markov coarse theorem, exact effective hypergraph/path-space closure, and typed scale diagram in Chapters 10 to 12. For the constructive continuation, read the cover and memory results in Chapters 14 and 15, then the finite-depth pointwise dynamics in Chapter 16 and the lattice theorem in Chapter 17. The geometry normalization and pressure chapters, Chapters 18 and 19, explain which geometric factors enter the VFE and survive blocking. Chapter 20 gives the separate strict-phase, fixed-tie, and reciprocal-volume Fisher results with their source and regularity hypotheses. The Gaussian spine then consists of the interaction cone, recognition costs and Fisher geometry in Chapters 22 to 24, followed by Gaussian aggregation, congruence-diagonal Kron closure, holonomy, fixed-ray qualifications, generalized spectra, and numerical gates in Chapters 25 and 26. Finally, Chapters 27 and 28 separate exact no-go results and repairs from empirical and interpretive commitments. Disintegration proofs, convex-duality details, long rational witnesses, ledger rows, and protocol metadata can be recovered through their semantic cross-references on a second reading.

Chapter 2

The Geometric Setting

The ambient geometry has one principal gauge bundle over the contextual base and two associated statistical bundles, one for belief laws and one for model laws. The two associated actions of the same group may be inequivalent and may act on fibers of different dimensions. Belief and model frames may nevertheless be chosen separately as local sections of the common principal bundle, and the two channels may carry different connections. Cross-channel maps are additional associated-bundle morphisms; neither a relative frame nor a shared structure group supplies such a map by itself. A product-gauge theory with two independent principal bundles is recorded only as the optional extension in Section 2.12.

2.1 The contextual base

Definition 2.1 (Contextual base). Let $\mathcal{C}$ be a finite-dimensional smooth, second countable, Hausdorff manifold whose points c are called *contexts*. [DEFINITION]

This declaration fixes a type and proves nothing. The base is neither the population index set $V = \{1, \ldots, N\}$ nor its interaction graph, neither latent sample space, and not physical spacetime. Its smooth structure supports curves and differential forms. A connection on a bundle over $\mathcal{C}$ defines transport in that bundle along curves in $\mathcal{C}$; it does not, unless separately specified as a connection on $T\mathcal{C}$, define parallel transport of tangent vectors of the base. The finite design $D = \{c_a\}_{a=1}^{M}$ used later is a finite subset of $\mathcal{C}$, not a continuum limit or a discretization theorem.

Throughout this development, $\mathcal{C}$ is fixed and timeless. A curve in $\mathcal{C}$ orders contexts and does not by itself represent an agent's evolution. Belief or model change at a fixed context is a curve inside one associated-bundle fiber, while simultaneous change over all contexts is a curve in a declared space of sections. A trajectory whose projection moves in $\mathcal{C}$ is a different construction and requires the horizontal–vertical distinction made by a chosen connection. These are typing declarations rather than physical conclusions. [DEFINITION]

2.2 One principal bundle and two associated statistical bundles

Definition 2.2 (Common principal gauge bundle). Let

$$\pi : \mathscr{P}_G \longrightarrow \mathcal{C} \tag{2.1}$$

be a smooth principal right G-bundle for a Lie group G. For each agent i, choose an open support $\mathcal{C}_i \subseteq \mathcal{C}$ and two local principal-frame sections

$$u_i^b : \mathcal{C}_i \longrightarrow \mathscr{P}_G, \qquad u_i^m : \mathcal{C}_i \longrightarrow \mathscr{P}_G. \tag{2.2}$$

[DEFINITION]

The superscripts distinguish two choices of local section, not two principal bundles. They may be unequal. Because each principal fiber is a right G-torsor, there is a unique smooth map

$$h_i : \mathcal{C}_i \longrightarrow G, \qquad u_i^m = u_i^b \cdot h_i, \tag{2.3}$$

called the *relative principal-frame field*. Its existence uses only the common principal bundle and is independent of all associated representations. [ESTABLISHED]

Let $(\mathsf{K}, \mathscr{K})$ and $(\mathsf{M}, \mathscr{M})$ be the belief and model sample fibers in local coordinates, and declare normalized-law fibers

$$\mathcal{B}_b \subseteq \mathcal{P}(\mathsf{K}), \qquad \mathcal{B}_m \subseteq \mathcal{P}(\mathsf{M}). \tag{2.4}$$

The two fibers contain normalized probability laws and are not kernel fibers. A generative transition belongs to a separately declared collection of normalized Markov kernels between specified measurable spaces. No density chart, common dominating measure, affine structure, or Fisher metric follows from (2.4).

The model-law value records uncertainty over a declared space of model presentations. It becomes generative only when paired with the measurable evaluator introduced in Definition 3.5; the evaluator maps a presentation to a normalized conditional mechanism. Thus an agent may have both beliefs and generative models without identifying a probability law over models with the model kernel itself. No vector addition or unrestricted scalar multiplication is part of either law fiber; the smooth, Fisher, and Gaussian structures used later belong to explicitly restricted regular tiers. [DEFINITION]

Suppose the same group acts measurably on the two sample fibers through representations

$$\rho_b : G \longrightarrow \mathrm{Aut}(\mathsf{K}), \qquad \rho_m : G \longrightarrow \mathrm{Aut}(\mathsf{M}). \tag{2.5}$$

The representations need not be equivalent, faithful, or of equal dimension. Denote their induced actions on laws by

$$\widehat{\rho}_b(g)q = (\rho_b(g))_{\#}q, \qquad \widehat{\rho}_m(g)s = (\rho_m(g))_{\#}s. \tag{2.6}$$

[DEFINITION]

Thus $(\widehat{\rho}_b(g)q)(A) = q(\rho_b(g)^{-1}A)$, and similarly in the model channel. The hats distinguish actions on laws from actions on samples. Multiplying a matrix into a density is not the pushforward in (2.6). [ESTABLISHED]

Hypothesis 2.3 (Selected smooth tier). When differential statements are made, $\mathcal{B}_b$ and $\mathcal{B}_m$ are finite-dimensional smooth parametrized-measure models, each is closed under its G-action, and

$$(g, q) \longmapsto (\rho_b(g))_{\#}q, \qquad (g, s) \longmapsto (\rho_m(g))_{\#}s \tag{2.7}$$

are smooth. [HYPOTHESIS]

A finite-dimensional statistical interpretation additionally requires differentiability in quadratic mean, square-integrable scores, and the integrability required by each tensor. The Fisher tensor is a Riemannian metric only when it is nondegenerate. Stratified and infinite-dimensional fibers require their own categories and are not included silently.

Definition 2.4 (Belief and model associated bundles). On invariant law fibers define

$$\mathcal{E}_b = \mathscr{P}_G \times_{\widehat{\rho}_b} \mathcal{B}_b, \qquad \mathcal{E}_m = \mathscr{P}_G \times_{\widehat{\rho}_m} \mathcal{B}_m, \tag{2.8}$$

using the quotient convention

$$(u \cdot g, \beta) \sim (u, g \cdot \beta). \tag{2.9}$$

[DEFINITION]

At the general tier these are associated measurable bundles. Under Hypothesis 2.3 they are smooth associated bundles. Smoothness of $\mathscr{P}_G$ alone does not upgrade an arbitrary law fiber.

The field h_i compares the two principal sections, even when ρ_b and ρ_m are inequivalent. It then produces two different represented operators, $\widehat{\rho}_b(h_i)$ on $\mathcal{B}_b$ and $\widehat{\rho}_m(h_i)$ on $\mathcal{B}_m$. It does not canonically produce an operator from $\mathcal{B}_b$ to $\mathcal{B}_m$. If "frame" instead means an arbitrary basis of an associated vector fiber that is not induced from a section of this same $\mathscr{P}_G$, the existence of a G-valued relative field is not guaranteed. Moreover, when a representation is nonfaithful, represented frame data alone determine at most a coset modulo its kernel; the principal sections, rather than their represented images, are what make h_i unique.

2.3 Frame choices, gauge symmetry, and relative frames

Independent changes of the two local trivializations may be written

$$u_i^{b\prime} = u_i^b \cdot a_i, \qquad u_i^{m\prime} = u_i^m \cdot b_i, \qquad a_i, b_i : \mathcal{C}_i \to G. \tag{2.10}$$

They change associated coordinates by $\beta_b' = \widehat{\rho}_b(a_i)^{-1}\beta_b$ and $\beta_m' = \widehat{\rho}_m(b_i)^{-1}\beta_m$. The unique relative field becomes

$$h_i' = a_i^{-1} h_i b_i. \tag{2.11}$$

[ESTABLISHED]

These two passive coordinate choices must not be confused with a product gauge symmetry. The principal gauge group of the ambient theory is the single group of G-equivariant automorphisms

of $\mathscr{P}_G$ over $\mathrm{id}_{\mathcal{C}}$, and each such automorphism acts simultaneously on both associated bundles. For such an automorphism F, define its local functions by $F(u_i^b) = u_i^b \cdot k_i^b$ and $F(u_i^m) = u_i^m \cdot k_i^m$. Equivariance and (2.3) imply

$$k_i^m = h_i^{-1} k_i^b h_i. \tag{2.12}$$

Thus the two local descriptions of one gauge transformation are related, not independent. By contrast, (2.10) simply permits two atlases to be replaced independently. Applying one principal gauge automorphism to both sections leaves their relative principal element unchanged, since equivariance gives $F(u_i^m) = F(u_i^b \cdot h_i) = F(u_i^b) \cdot h_i$.

2.4 Cross-associated-bundle morphisms

Definition 2.5 (Cross morphisms). At the general measurable tier, declare fiber-preserving measurable maps

$$\Phi : \mathcal{E}_b \longrightarrow \mathcal{E}_m, \qquad \widetilde{\Phi} : \mathcal{E}_m \longrightarrow \mathcal{E}_b, \qquad \varpi_m \circ \Phi = \varpi_b, \quad \varpi_b \circ \widetilde{\Phi} = \varpi_m, \tag{2.13}$$

where ϖ_b and ϖ_m are the associated-bundle projections. Both maps cover $\mathrm{id}_{\mathcal{C}}$. [DEFINITION]

These maps are extra structure. They are not principal-bundle morphisms, gauge automorphisms, same-channel transports, or assumed inverses. Under Hypothesis 2.3, differential statements additionally assume that they are smooth. In the frames u_i^b, u_i^m, let ϕ_i and $\widetilde{\phi}_i$ denote their local fiber representatives. The rechoices in (2.10) give

$$\phi_i' = \widehat{\rho}_m(b_i)^{-1} \circ \phi_i \circ \widehat{\rho}_b(a_i), \tag{2.14}$$

$$\widetilde{\phi}_i' = \widehat{\rho}_b(a_i)^{-1} \circ \widetilde{\phi}_i \circ \widehat{\rho}_m(b_i). \tag{2.15}$$

[ESTABLISHED]

Proof. Write $\beta_m = \phi_i(\beta_b)$ and substitute $\beta_b = \widehat{\rho}_b(a_i)\beta_b'$ and $\beta_m' = \widehat{\rho}_m(b_i)^{-1}\beta_m$. The second law follows by interchanging the channels. □

Proposition 2.6 (Equivariant fiber maps induce cross morphisms). Suppose maps $\phi : \mathcal{B}_b \to \mathcal{B}_m$ and $\widetilde{\phi} : \mathcal{B}_m \to \mathcal{B}_b$ satisfy, for every $g \in G$,

$$\phi \circ \widehat{\rho}_b(g) = \widehat{\rho}_m(g) \circ \phi, \qquad \widetilde{\phi} \circ \widehat{\rho}_m(g) = \widehat{\rho}_b(g) \circ \widetilde{\phi}. \tag{2.16}$$

Then

$$[p, \beta_b] \longmapsto [p, \phi(\beta_b)], \qquad [p, \beta_m] \longmapsto [p, \widetilde{\phi}(\beta_m)] \tag{2.17}$$

are well-defined cross morphisms. [ESTABLISHED]

Proof. The quotient representatives $(p \cdot g, \beta)$ and $(p, g \cdot \beta)$ have the same image under (2.17) exactly when the corresponding equality in (2.16) holds. □

Equivariance is sufficient for the representative-independent construction (2.17). A general bundle morphism is instead characterized by local maps satisfying the overlap laws

$$\begin{aligned} \phi_i \circ \widehat{\rho}_b(T^b_{ij}) &= \widehat{\rho}_m(T^m_{ij}) \circ \phi_j, \\ \widetilde{\phi}_i \circ \widehat{\rho}_m(T^m_{ij}) &= \widehat{\rho}_b(T^b_{ij}) \circ \widetilde{\phi}_j. \end{aligned} \tag{2.18}$$

In any selected regularity category admitting the two mapping objects and the composition actions displayed below, their local representatives first require conversion to one principal-frame atlas. In the belief atlas define

$$f_i := \widehat{\rho}_m(h_i) \circ \phi_i, \qquad \widetilde{f}_i := \widetilde{\phi}_i \circ \widehat{\rho}_m(h_i^{-1}). \tag{2.19}$$

Then f_i is the local representative of a section of the associated map bundle with action $f \mapsto \widehat{\rho}_m(g) \circ f \circ \widehat{\rho}_b(g)^{-1}$, and $\widetilde{f}_i$ obeys the analogous law with the two channels interchanged. Indeed, (2.18) and $T^m_{ij} = h_i^{-1} T^b_{ij} h_j$ give $f_i \widehat{\rho}_b(T^b_{ij}) = \widehat{\rho}_m(T^b_{ij}) f_j$ and $\widetilde{f}_i \widehat{\rho}_m(T^b_{ij}) = \widehat{\rho}_b(T^b_{ij}) \widetilde{f}_j$. Thus the compatible family is a map-bundle section only after the displayed conversion, and h_i does not replace the cross-fiber map. For example, if Φ is induced by the first map in (2.17), then $\phi_i = \widehat{\rho}_m(h_i^{-1}) \circ \phi$ and hence $f_i = \phi$. [ESTABLISHED]

2.5 Chosen connections, parallel transport, and covariant defects

Definition 2.7 (Two connections on the common principal bundle). Under Hypothesis 2.3, choose

$$\omega_b, \omega_m \in \Omega^1(\mathscr{P}_G, \mathfrak{g}) \tag{2.20}$$

as principal connections on $\mathscr{P}_G$. They induce horizontal distributions H^b, H^m on the two associated bundles and, for every piecewise-smooth $\gamma : [0,1] \to \mathcal{C}$, parallel transports

$$\Omega_\gamma : (\mathcal{E}_b)_{\gamma(0)} \longrightarrow (\mathcal{E}_b)_{\gamma(1)}, \qquad \widetilde{\Omega}_\gamma : (\mathcal{E}_m)_{\gamma(0)} \longrightarrow (\mathcal{E}_m)_{\gamma(1)}. \tag{2.21}$$

[DEFINITION]

Explicitly, ω_x selects the unique horizontal lift of γ through a chosen point of $(\mathscr{P}_G)_{\gamma(0)}$; carrying the associated-law coordinate along that lift defines the corresponding map in (2.21). This is transport *over* the base curve, not transport of vectors in $T\mathcal{C}$. A tangent-vector transport would require a separately chosen affine connection on $T\mathcal{C}$, unless $\mathscr{P}_G$ and its representation have explicitly been identified with the frame bundle of $\mathcal{C}$.

The two connections need not agree. Their difference $\omega_m - \omega_b$ is horizontal and Ad-equivariant, hence descends to an $\mathrm{Ad}(\mathscr{P}_G)$-valued one-form on $\mathcal{C}$. In an associated vector-bundle realization, the induced horizontal distributions also define covariant derivatives ∇^b, ∇^m. No linear covariant derivative is assumed on a merely measurable law fiber.

Let $A_i^b = (u_i^b)^*\omega_b$ and $A_i^m = (u_i^m)^*\omega_m$. Under (2.10),

$$A_i^{b\prime} = \mathrm{Ad}_{a_i^{-1}} A_i^b + a_i^{-1} da_i, \tag{2.22}$$

$$A_i^{m\prime} = \mathrm{Ad}_{b_i^{-1}} A_i^m + b_i^{-1} db_i. \tag{2.23}$$

If endpoint coordinates are changed by a_0, a_1 in the belief frame and b_0, b_1 in the model frame, the represented transports obey

$$\Omega'_\gamma = \widehat{\rho}_b(a_1)^{-1} \circ \Omega_\gamma \circ \widehat{\rho}_b(a_0), \tag{2.24}$$

$$\widetilde{\Omega}'_\gamma = \widehat{\rho}_m(b_1)^{-1} \circ \widetilde{\Omega}_\gamma \circ \widehat{\rho}_m(b_0). \tag{2.25}$$

[ESTABLISHED]

These are the standard connection and endpoint laws for the quotient convention (2.9) [50, 62]. They also show why a same-point frame comparison and transport along a nonconstant curve are different objects.

Definition 2.8 (Covariant defects). Let $e \in \mathcal{E}_b$ and $f \in \mathcal{E}_m$ lie over $c \in \mathcal{C}$, and let $X \in T_c\mathcal{C}$. For smooth associated fibers, define the vertical-valued tensors

$$(\mathcal{D}\Phi)_e(X) = T_e\Phi\big(H_e^b X\big) - H^m_{\Phi(e)} X, \tag{2.26}$$

$$(\mathcal{D}\widetilde{\Phi})_f(X) = T_f\widetilde{\Phi}\big(H_f^m X\big) - H^b_{\widetilde{\Phi}(f)} X. \tag{2.27}$$

[DEFINITION]

Each difference is vertical because its two terms project to the same X. The maps are parallel exactly when these defects vanish, equivalently when every base curve satisfies

$$\widetilde{\Omega}_\gamma \circ \Phi_{\gamma(0)} = \Phi_{\gamma(1)} \circ \Omega_\gamma, \tag{2.28}$$

$$\Omega_\gamma \circ \widetilde{\Phi}_{\gamma(0)} = \widetilde{\Phi}_{\gamma(1)} \circ \widetilde{\Omega}_\gamma. \tag{2.29}$$

[ESTABLISHED]

Proof. Vanishing defect says that the image of every horizontal lift is horizontal. Integration along γ gives the two commutative squares, and differentiation at the initial point gives the converse. □

Only in a fiberwise-linear associated-vector-bundle specialization do the defects reduce to

$$D\Phi = \nabla^m \circ \Phi - \Phi \circ \nabla^b, \qquad D\widetilde{\Phi} = \nabla^b \circ \widetilde{\Phi} - \widetilde{\Phi} \circ \nabla^m. \tag{2.30}$$

Write V_b, V_m for the two representation spaces in this linear specialization. The old-to-new coordinate maps induced by (2.10) are, by definition,

$$R_i^b := \rho_b(a_i)^{-1} : V_b \longrightarrow V_b, \qquad R_i^m := \rho_m(b_i)^{-1} : V_m \longrightarrow V_m, \tag{2.31}$$

so that $v'_b = R^b_i v_b$ and $v'_m = R^m_i v_m$. These are invertible coordinate maps, not cross-channel morphisms or coarse restriction operators. In these coordinates the defect matrices obey

$$(D\Phi)'_i = R^m_i (D\Phi)_i (R^b_i)^{-1}, \qquad (D\widetilde{\Phi})'_i = R^b_i (D\widetilde{\Phi})_i (R^m_i)^{-1}. \tag{2.32}$$

[ESTABLISHED]

The matrix laws are not asserted for nonlinear law manifolds. The connections in (2.20) are chosen data; no curvature or transport is inferred from the agent frames. [NOT-CLAIMED]

2.6 Agents and finite-moment pushforward

Definition 2.9 (Agent). Agent i is the section-bearing object

$$\mathcal{A}^i = \big(\mathcal{C}_i; q_i, q_i^m, u_i^b, u_i^m\big), \tag{2.33}$$

where

$$q_i \in \Gamma(\mathcal{C}_i, \mathcal{E}_b|_{\mathcal{C}_i}), \qquad q_i^m \in \Gamma(\mathcal{C}_i, \mathcal{E}_m|_{\mathcal{C}_i}). \tag{2.34}$$

[DEFINITION]

The sections are smooth only in the selected smooth tier. A sample

$$Y(c) = \big((k_1(c), m_1(c)), \ldots, (k_N(c), m_N(c))\big) \tag{2.35}$$

is a random element of a product of sample fibers, not a tuple of agents. The continuum sections above and the finite recognition law introduced later are separately declared objects; their compatibility is a hypothesis in the probability chapter.

A population sample and a population law are compatibility and evaluation objects, not agents. The compatibility hypothesis between continuum section values and finite recognition blocks remains separate from the definitions of both objects. [DEFINITION]

Proposition 2.10 (Finite-second-moment congruence). Let a belief law q on $\mathbb{R}^K$ have finite mean μ and covariance Σ, and let $R = \rho_b(g)$ be invertible and linear for $g \in G$. Then

$$\mu' = R\mu, \qquad \Sigma' = R\Sigma R^\top. \tag{2.36}$$

If $\Sigma \succ 0$ and $J = \Sigma^{-1}$, the pushed law has precision

$$J' = (\Sigma')^{-1} = R^{-\top} J R^{-1}. \tag{2.37}$$

The model channel has the same statement in its own dimension. [ESTABLISHED]

Proof. For $X \sim q$, linearity gives $\mathbb{E}[RX] = R\mu$ and $\mathbb{E}[(RX - R\mu)(RX - R\mu)^\top] = R\Sigma R^\top$. Inverting the latter identity gives (2.37). □

2.7 Pointwise frame comparisons

Fix $c \in \mathcal{C}_i \cap \mathcal{C}_j$. Freeness and transitivity give unique elements of the same group G such that

$$u_j^b(c) = u_i^b(c) \cdot T_{ij}^b(c), \qquad u_j^m(c) = u_i^m(c) \cdot T_{ij}^m(c). \tag{2.38}$$

They act through $\rho_b(T_{ij}^b)$ and $\rho_m(T_{ij}^m)$ in the two associated fibers. Under (2.10),

$$T_{ij}^{b\prime} = a_i^{-1} T_{ij}^b a_j, \qquad T_{ij}^{m\prime} = b_i^{-1} T_{ij}^m b_j. \tag{2.39}$$

Moreover, the relative fields force

$$T_{ij}^m = h_i^{-1} T_{ij}^b h_j. \tag{2.40}$$

[ESTABLISHED]

Proof. The transformation law follows by substitution and uniqueness in the principal fiber. The two expressions $u_j^m = u_j^b h_j = u_i^b T_{ij}^b h_j$ and $u_j^m = u_i^m T_{ij}^m = u_i^b h_i T_{ij}^m$ give (2.40). □

The comparisons are same-point changes of coordinates. They involve no curve and no connection. The two group elements are related by (2.40), while their represented actions can be inequivalent.

2.8 The Čech picture and bundle topology

Let $\mathcal{C}_0 = \bigcup_i \mathcal{C}_i$, and let $\mathcal{N}$ be the nerve of this open cover:

$$\{i_0, \dots, i_r\} \in \mathcal{N} \iff \mathcal{C}_{i_0 \cdots i_r} = \bigcap_{a=0}^{r} \mathcal{C}_{i_a} \neq \varnothing. \tag{2.41}$$

For either frame atlas $x \in \{b, m\}$, the maps $T_{ij}^x : \mathcal{C}_{ij} \to G$ obey

$$T_{ii}^x = e_G, \qquad T_{ji}^x = (T_{ij}^x)^{-1}, \qquad T_{ij}^x T_{jl}^x = T_{il}^x. \tag{2.42}$$

[ESTABLISHED]

Proof. On a triple overlap, composing the unique change from l to j with the unique change from j to i gives the unique change from l to i. □

Let $\mathcal{G}$ be the sheaf of smooth G-valued maps. There is one principal bundle and hence one nonabelian Čech class:

$$[T^b] = [T^m] = [\mathscr{P}_G|_{\mathcal{C}_0}] \in \check{H}^1(\{\mathcal{C}_i\}; \mathcal{G}). \tag{2.43}$$

Indeed, (2.40) is exactly the coboundary equivalence between the two cocycles. The class is distinguished exactly when $\mathscr{P}_G|_{\mathcal{C}_0}$ is trivial, equivalently when, for either atlas, there are smooth $U_i^x : \mathcal{C}_i \to G$ such that

$$T_{ij}^x = U_i^x (U_j^x)^{-1}. \tag{2.44}$$

[ESTABLISHED]

Proof. If (2.44) holds, the rechoice $u_i^{x\prime} = u_i^x U_i^x$ makes all transitions identity, so the sections glue to a global section. A global section trivializes a principal bundle. Conversely, a global trivialization supplies the U_i^x. This is the standard global-section characterization of a trivial principal bundle [42, 50, 80]. □

The single principal class does not make the two associated bundles isomorphic. Inequivalent representations can send the same principal cocycle to nonisomorphic associated bundles and to different induced characteristic data.

Hypothesis 2.11 (Common principal trivialization). Whenever global principal-frame coordinates are used, declare $\mathscr{P}_G|_{\mathcal{C}_0}$ trivial and choose one reference section σ_0 such that

$$u_i^b = \sigma_0 \cdot (U_i^b)^{-1}, \qquad u_i^m = \sigma_0 \cdot (U_i^m)^{-1}, \qquad h_i = U_i^b (U_i^m)^{-1}. \tag{2.45}$$

[HYPOTHESIS]

This is one topological hypothesis, not two channelwise hypotheses. The general theory retains local frames and their common Čech class. A global section of an associated bundle does not in general trivialize $\mathscr{P}_G$: if an associated fiber contains a G-fixed point β_0, then $c \mapsto [p, \beta_0]$ is well defined for any $\mathscr{P}_G$. [ESTABLISHED]

2.9 Channel-specific graph links and trivializing subfamilies

Definition 2.12 (Typed graph links). Let $\mathfrak{I} = (V, E, {}^-)$ be a finite interaction multigraph declared independently of $\mathcal{C}$. Its oriented edge copies have a fixed-point-free reversal involution, and parallel copies are allowed. For $e : j \to i$, declare

$$\Theta_e^b, \Theta_e^m \in G, \qquad \Theta_{\bar{e}}^x = (\Theta_e^x)^{-1}. \tag{2.46}$$

Under independently changed channel coordinates at the vertices,

$$\Theta_e^{b\prime} = (a_i^b)^{-1} \Theta_e^b a_j^b, \qquad \Theta_e^{m\prime} = (a_i^m)^{-1} \Theta_e^m a_j^m. \tag{2.47}$$

Here the $a_i^x \in G$ are passive vertex-coordinate choices. They are induced by smooth frame rechoices only after vertices are anchored at contexts $c_i \in \mathcal{C}_i$, in which case $a_i^b = a_i(c_i)$ and $a_i^m = b_i(c_i)$. Allowing two coordinate descriptions of two graph-link data sets does not enlarge the principal gauge group to $G \times G$. In the old-to-new sample-coordinate convention of Section 4.6, the contextual elements denoted there by h_i^x satisfy $h_i^x = (a_i^x)^{-1}$. The superscripted h_i^x is a discrete contextual coordinate re-expression and is unrelated to the relative principal-frame field h_i of (2.3). [DEFINITION]

The links are channel-specific data. Neither is determined by the other, by h_i, by Φ or $\widetilde{\Phi}$, by pointwise frame comparison, or by a principal connection. For a closed edge-labeled walk $\gamma = (e_0, \dots, e_{r-1})$,

where $e_a : i_{a+1} \to i_a$ and $i_r = i_0$, define

$$H^x(\gamma) = \prod_{a=0}^{r-1} \Theta^x_{e_a}. \tag{2.48}$$

Its conjugacy class and the condition $H^x(\gamma) = e_G$ are invariant under the corresponding coordinate changes. [ESTABLISHED]

Proposition 2.13 (Trivializing criterion). Fix a channel x, a subfamily $F \subseteq V$, and a G-valued comparison assignment Ξ^x_e on its internal oriented edge copies. The following are equivalent:

$$\begin{aligned} &\exists V^x_i \in G \ (i \in F) \quad \Xi^x_e = V^x_i (V^x_j)^{-1} \quad \text{for every internal } e : j \to i, \\ &\prod_{a=0}^{r-1} \Xi^x_{e_a} = e_G \quad \text{for every closed internal walk.} \end{aligned} \tag{2.49}$$

[ESTABLISHED]

Proof. The first condition telescopes around every closed walk. Conversely, choose a base vertex in each connected component and define V^x_i by an ordered path product. Two choices differ by a closed walk and hence agree. Appending $e : j \to i$ gives the required edge factorization. □

A subfamily is *jointly trivializing* when the criterion holds separately for both link data sets. For T^x evaluated at one common context the criterion is automatic by the Čech cocycle law. Its nontrivial graph content concerns the independently declared Θ^x. Pointwise graph triviality is weaker than triviality of $\mathscr{P}_G$, which requires smooth overlap data on the entire cover.

Hypothesis 2.14 (Graph links realized by base transport). Choose contexts $c_i \in \mathcal{C}_i$ and, for every $e : j \to i$, a piecewise-smooth curve γ_e from c_j to c_i. In the chosen endpoint frames impose

$$\widehat{\rho}_b(\Theta^b_e) = \Omega_{\gamma_e}, \qquad \widehat{\rho}_m(\Theta^m_e) = \widetilde{\Omega}_{\gamma_e}. \tag{2.50}$$

[HYPOTHESIS]

Without the curve assignment and these equalities, graph holonomy and base-connection holonomy are unrelated. Under the hypothesis, represented graph-loop products equal the corresponding parallel transports around concatenated base loops. This is a connection-holonomy statement, not a claim of nontrivial principal-bundle topology.

2.10 Flat graph links and residuals

Hypothesis 2.15 (Pointwise flat-link specialization). At one common context, declare separately for the two link data sets that

$$\Theta_e^b = T_{ij}^b = U_i^b (U_j^b)^{-1}, \qquad \Theta_e^m = T_{ij}^m = U_i^m (U_j^m)^{-1}, \qquad e : j \to i. \tag{2.51}$$

[HYPOTHESIS]

This excludes represented graph holonomy in either channel. It does not assert that either principal connection is flat or that the two channel frames coincide. Let μ_i be a belief-channel vector coordinate, set $z_i = \rho_b(U_i^b)^{-1}\mu_i$, and suppress ρ_b only in the defining representation. Then

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(U_i^b)(z_i - z_j). \tag{2.52}$$

For an independent graph link,

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(U_i^b)(z_i - H_e^b z_j), \qquad H_e^b = \rho_b(U_i^b)^{-1}\rho_b(\Theta_e^b)\rho_b(U_j^b). \tag{2.53}$$

More generally, if $\Theta_e^b = V_i^b (V_j^b)^{-1}$ on a belief-trivializing subfamily and $z_i = \rho_b(V_i^b)^{-1}\mu_i$, then

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(V_i^b)(z_i - z_j). \tag{2.54}$$

[ESTABLISHED]

These identities follow by substitution. Belief-link trivialization says nothing about model links, either cross morphism, or equality of the belief and model frames.

2.11 Common-frame and common-connection specializations

The one-principal-bundle construction is already the ambient theory. A *common-frame* specialization makes the stronger declaration

$$u_i^b = u_i^m = u_i, \qquad h_i = e_G, \qquad T_{ij}^b = T_{ij}^m = T_{ij}. \tag{2.55}$$

Even then the represented frame changes generally differ: choose a common old-to-new sample-coordinate re-expression $g_i : \mathcal{C}_i \to G$, so that the corresponding principal-section rechoice is $u_i' = u_i \cdot g_i^{-1}$. Then

$$R_i^b = \rho_b(g_i), \qquad R_i^m = \rho_m(g_i). \tag{2.56}$$

Thus R_i^b and R_i^m have the same principal source g_i but act within different representation spaces; neither is a map between the belief and model fibers. An independent *equal-connection* specialization declares $\omega_b = \omega_m = \omega$. The underlying principal holonomy is then common, while its actions on belief and model laws remain $\widehat{\rho}_b$ and $\widehat{\rho}_m$. A further common-link specialization

declares $\Theta_e^b = \Theta_e^m = \Theta_e \in G$. None of these three stronger hypotheses follows from the others. [HYPOTHESIS]

The cross maps remain extra equivariant structure governed by (2.16); they are not forced to be inverses or parallel. Thus inequivalent representations do not obstruct a common principal frame. In a linear associated-vector-bundle specialization they may obstruct a nonzero sample-fiber intertwiner. Law-level equivariance is weaker: an equivariant Markov kernel may induce an equivariant map on laws without supplying a linear map between the sample fibers. [ESTABLISHED]

2.12 Optional product-gauge extension

If the application genuinely requires independently nonisomorphic principal bundles or two independent gauge symmetries, replace the ambient $\mathscr{P}_G$ by

$$\pi_b : P_b \to \mathcal{C}, \qquad \pi_m : P_m \to \mathcal{C}, \qquad Q = P_b \times_{\mathcal{C}} P_m, \tag{2.57}$$

where Q is a principal $G_b \times G_m$-bundle. The belief and model associated bundles are then formed using the two projections. [DEFINITION]

There is no canonical relative field valued in one common group between frames in P_b and P_m without additional comparison data. When $G_b = G_m = G$, sufficient principal-level data are a chosen principal G-bundle isomorphism $B : P_b \to P_m$ over $\mathrm{id}_{\mathcal{C}}$ satisfying $B(p \cdot g) = B(p) \cdot g$. For possibly different groups, the corresponding declared data are a homomorphism $\kappa : G_b \to G_m$ and a bundle map $B : P_b \to P_m$ over $\mathrm{id}_{\mathcal{C}}$ satisfying $B(p \cdot g_b) = B(p) \cdot \kappa(g_b)$. The latter data alone do not create a canonical relative field valued in one common group: no such common group or inverse identification has been declared. [ESTABLISHED]

In a category with the required mapping object, a belief-to-model morphism is a section of

$$Q \times_{G_b \times G_m} \mathrm{Map}(\mathcal{B}_b, \mathcal{B}_m), \qquad (g_b, g_m) \cdot f = \widehat{\rho}_m(g_m) \circ f \circ \widehat{\rho}_b(g_b)^{-1}. \tag{2.58}$$

[DEFINITION]

A constant fiber map descends only if it is fixed by every independent pair. Given the κ-equivariant map B above, a fiber map $\phi : \mathcal{B}_b \to \mathcal{B}_m$ satisfying $\phi \circ \widehat{\rho}_b(g_b) = \widehat{\rho}_m(\kappa(g_b)) \circ \phi$ induces a compatible cross morphism. These principal and fiber data are sufficient, but a general mapping-bundle section need not arise that way. Thus the product bundle supplies neither a relative field nor either cross morphism automatically. [ESTABLISHED]

This extension carries additional topology and an independent product gauge freedom. It should therefore be invoked only when those degrees of freedom are intended, not merely because belief and model use different associated representations of one group.

Chapter 3

Probability Semantics

Chapter 2 fixed a geometry. This chapter fixes the measurable typing on which every probabilistic statement rests, and it fixes it before any density is written. The order is a substantive commitment: probability measures and normalized kernels are declared first, densities appear afterward as Radon-Nikodym derivatives against explicitly declared reference measures, and logarithms of density ratios appear only under an absolute-continuity convention stated in advance. That convention looks like bookkeeping and is not. It is what makes the exact evidence bound of Chapter 5 an identity rather than an inequality with hidden exceptions, and it is what later forbids degenerate recognition laws whose support is a null set of the posterior. A development that introduces densities first and normalizes later can write those degenerate objects down without noticing, and the resulting statements are false rather than merely imprecise.

Throughout, $\mathcal{P}(\mathsf{S}, \mathscr{S})$ denotes the set of probability measures on a measurable space $(\mathsf{S}, \mathscr{S})$; the calligraphic symbol with a measurable-space argument is unrelated to the total space $\mathscr{P}_G$ of the principal bundle in Chapter 2.

3.1 The finite design and its evaluation

Definition 3.1 (Finite design and incidence). A *design* is a finite set of distinct contexts

$$D = \{c_a\}_{a=1}^{M} \subseteq \mathcal{C}, \tag{3.1}$$

indexed by $a \in \{1, \dots, M\}$. The *agent–design incidence set* is $\mathfrak{I}_D = \{(i, a) : c_a \in \mathcal{C}_i\}$, where $\mathcal{C}_i$ is the support of agent i from Definition 2.2. Define

$$D_i := \{a : (i, a) \in \mathfrak{I}_D\}, \qquad V_a := \{i : (i, a) \in \mathfrak{I}_D\}.$$

Design points with no participating agent are omitted, so $V_a \neq \varnothing$ for every a. For the reference presentation $D \subseteq \bigcap_{i \in V} \mathcal{C}_i$, so $\mathfrak{I}_D = V \times \{1, \dots, M\}$ is rectangular, $D_i = \{1, \dots, M\}$, and $V_a = V$. The symbol $\mathfrak{A}$ is reserved for the finite interaction-record index set introduced below. [DEFINITION]

Nothing is proved by this. The definitions below use $\mathfrak{I}_D$ throughout to make one point explicit: a sparse design is handled by indexing over declared incidences, never by writing a product over all (i, a) and then deleting the coordinates that happen to be undefined. Deleting coordinates from a

product silently changes the sample space, and a bound computed on one sample space is not a bound on another.

Definition 3.2 (Sample spaces and product sigma-algebras). At each admissible (i,a) declare measurable state, model, and observation fibers $(\mathsf{K}_{i,a},\mathscr{K}_{i,a})$, $(\mathsf{M}_{i,a},\mathscr{M}_{i,a})$, and $(\mathsf{O}_{i,a},\mathscr{O}_{i,a})$, and write $\mathsf{Y}_{i,a} := \mathsf{K}_{i,a}\times\mathsf{M}_{i,a}$. Define the agent block

$$\mathsf{Y}_{i,D} := \prod_{a\in D_i}(\mathsf{K}_{i,a}\times\mathsf{M}_{i,a}), \qquad \mathscr{Y}_{i,D} := \bigotimes_{a\in D_i}(\mathscr{K}_{i,a}\otimes\mathscr{M}_{i,a}).$$

The finite latent and observation spaces are

$$\mathsf{Y}_D = \prod_{i\in V}\mathsf{Y}_{i,D} \cong \prod_{a=1}^{M}\prod_{i\in V_a}\big(\mathsf{K}_{i,a}\times\mathsf{M}_{i,a}\big), \qquad \mathscr{Y}_D = \bigotimes_{i\in V}\mathscr{Y}_{i,D}, \tag{3.2}$$

$$\mathsf{O}_D = \prod_{a=1}^{M}\prod_{i\in V_a}\mathsf{O}_{i,a}, \qquad \mathscr{O}_D = \bigotimes_{a=1}^{M}\bigotimes_{i\in V_a}\mathscr{O}_{i,a}. \tag{3.3}$$

The stacked latent is written $Y\in\mathsf{Y}_D$ and the realized observation $o\in\mathsf{O}_D$. In the Gaussian realization $\mathsf{K}_{i,a}=\mathbb{R}^K$ and $\mathsf{M}_{i,a}=\mathbb{R}^{d_m}$ with their Borel sigma-algebras. [DEFINITION]

Hypothesis 3.3 (Measurable context-indexed section response). For every agent i, and for the measurable configuration space fixed in Definition 3.4, declare measurable spaces $(\mathfrak{S}_i^b,\mathscr{S}_i^b)$ and $(\mathfrak{S}_i^m,\mathscr{S}_i^m)$ whose underlying sets satisfy

$$\mathfrak{S}_i^b \subseteq \Gamma(\mathcal{C}_i,\mathcal{E}_b|_{\mathcal{C}_i}), \qquad \mathfrak{S}_i^m \subseteq \Gamma(\mathcal{C}_i,\mathcal{E}_m|_{\mathcal{C}_i}).$$

Each law fiber carries its declared sigma-algebra. For $x\in\{b,m\}$ and $c\in\mathcal{C}_i$, the evaluation map

$$\mathrm{eval}_{i,c}^x : \mathfrak{S}_i^x \longrightarrow (\mathcal{E}_x)_c, \qquad s\longmapsto s(c),$$

is measurable. The admitted response family is the measurable map

$$\mathrm{Resp}_i : (\mathsf{O}_D\times\mathsf{X},\mathscr{O}_D\otimes\mathscr{X}) \longrightarrow (\mathfrak{S}_i^b\times\mathfrak{S}_i^m,\mathscr{S}_i^b\otimes\mathscr{S}_i^m), \qquad (o,X)\longmapsto (q_i^{b;o,X},q_i^{m;o,X}). \tag{3.4}$$

At every $(i,a)\in\mathfrak{I}_D$, the fixed measurable fiber identifications send these evaluated values into $\mathcal{P}(\mathsf{K}_{i,a},\mathscr{K}_{i,a})$ and $\mathcal{P}(\mathsf{M}_{i,a},\mathscr{M}_{i,a})$, respectively. Thus their design-point evaluations are measurable functions of (o,X). For fixed (o,X), these sections occupy the generic section slots q_i and q_i^m of (2.34); the superscripted response family is not a bare alias for those variables, because its admitted dependencies and measurability are additional declared data. [HYPOTHESIS] Evaluation and section evaluation have different codomains, and the difference is the one Chapter 2 insisted on. For a sample-field realization $y=((k_i,m_i))_{i\in V}$ defined on the admitted incidences, the evaluation map

$$\mathrm{Ev}_D(y) = \big((k_i(c_a),m_i(c_a))_{i\in V_a}\big)_{a=1}^{M} \in \mathsf{Y}_D \tag{3.5}$$

returns a point of the latent space. Evaluating a distribution-valued section returns something else entirely: $q_i^{b;o,X}(c_a)$ is a probability measure on $\mathsf{K}_{i,a}$ and $q_i^{m;o,X}(c_a)$ is a probability measure on $\mathsf{M}_{i,a}$, and neither is an element of Y_D. The finite random element Y takes values in Y_D; the section values are laws. Section 3.7 is where the two are related, and it relates them by hypothesis.

Missing or aggregated observations are not represented by removing latent coordinates from (3.2). They require a separately declared observation map or kernel into a coarser observation space, and the declaration changes O_D rather than Y_D.

3.2 Structural data and measurable typing

Definition 3.4 (Structural configuration and kernel signatures). Fix a measurable configuration space $(\mathsf{X}, \mathscr{X})$. Every $X \in \mathsf{X}$ uses the same design D, the same agent index set V, the same incidence set $\mathfrak{I}_D$, and the same measurable identifications of the fibers in (3.2) and (3.3); therefore $(\mathsf{Y}_D, \mathscr{Y}_D)$ and $(\mathsf{O}_D, \mathscr{O}_D)$ do not vary with X. A configuration may carry the finite interaction complex $\mathfrak{I}$ of Definition 2.12 with its edge set, the frames U_i or the links L_{ij}, and the parameters of the observation and transition mechanisms.

For every $i \in V$, let $(\mathsf{I}_i, \mathscr{I}_i)$ and $(\mathsf{Y}_{i,D}, \mathscr{Y}_{i,D})$ be standard Borel, and declare a measurable access map and a normalized local recognition kernel

$$\mathrm{Acc}_i : \mathsf{O}_D \times \mathsf{X} \longrightarrow \mathsf{I}_i, \qquad \mathbb{Q}_i : \mathsf{I}_i \rightsquigarrow \mathsf{Y}_{i,D}, \qquad \mathbb{Q}_{i,o,X} := \mathbb{Q}_i(\mathrm{Acc}_i(o, X), \cdot). \tag{3.6}$$

The identity access map on $\mathsf{O}_D \times \mathsf{X}$ is the unrestricted interface. Any decentralized information claim requires a stricter declared Acc_i.

Let $\mathfrak{A}$ be a finite set of interaction records. Each $a \in \mathfrak{A}$ has a scope $\partial a \subseteq V$, a standard-Borel record space $(\mathsf{O}_a, \mathscr{O}_a)$, and a normalized kernel

$$K_a : (\mathsf{X} \times \mathsf{Y}_{\partial a,D}) \rightsquigarrow \mathsf{O}_a, \qquad K_a(X, y_{\partial a}, \mathsf{O}_a) = 1. \tag{3.7}$$

The observation space is identified measurably with $\prod_{a\in\mathfrak{A}} \mathsf{O}_a$. A singleton-scope agent observation kernel is one admissible K_a after its record ownership is fixed. Each record coordinate occurs once.

The complete laws have the signatures

$$\mathbb{P}_{\theta,V} : (\mathsf{X}, \mathscr{X}) \longrightarrow \mathcal{P}\big(\mathsf{Y}_D \times \mathsf{O}_D, \mathscr{Y}_D \otimes \mathscr{O}_D\big), \qquad X \longmapsto \mathbb{P}_{\theta,V}(\cdot \mid X), \tag{3.8}$$

$$\mathbb{Q}_V : (\mathsf{O}_D \times \mathsf{X}, \mathscr{O}_D \otimes \mathscr{X}) \longrightarrow \mathcal{P}\big(\mathsf{Y}_D, \mathscr{Y}_D\big), \qquad (o, X) \longmapsto \mathbb{Q}_{V,o,X}. \tag{3.9}$$

Equation (3.8) denotes the normalized population law constructed in Chapter 4; it is not primitive data. Equation (3.9) denotes the selected coupling constructed in Section 3.6. Both displayed maps are measurable in the kernel sense. The agent's generative-model structure is the pair (q_i^m, ev_i), where q_i^m is a law-valued section over model presentations and $\mathrm{ev}_i := (\mathrm{ev}_{i,a})_{a\in D_i}$ is the typed finite

family of normalized evaluated mechanisms defined below. The model law, presentation, evaluated mechanism, record kernel, and population law are different types. [DEFINITION]

Signature (3.8) is the typing on which several later results depend, and it is worth reading for what it omits. The arguments of the generative kernel are θ and X and nothing else. A recognition law, a recognition parameter, and a posterior are not among them, so no factor of the generative model may read one. This is a definitional requirement of the fixed-joint construction rather than a modeling preference: the joint must be a single measure fixed before an observation is seen, and a factor that reads a posterior makes the joint a function of the object it is supposed to be conditioned into. Nothing is proved here beyond the type; the consequences are drawn in Chapter 4, where the joint is built, and again in Chapter 27, where a construction that violates the typing is examined and found inadmissible.

Later chapters may make X random, but only within this fixed measurable space, and at a different probability level. Doing so introduces a configuration prior in $\mathcal{P}(\mathsf{X}, \mathscr{X})$ and a configuration recognition kernel from $(\mathsf{O}_D, \mathscr{O}_D)$ into $\mathcal{P}(\mathsf{X}, \mathscr{X})$; neither changes D, $\mathfrak{I}_D$, or the identified fibers, and neither is denoted by any symbol reserved above for a state-level object.

Definition 3.5 (Model presentations and evaluated kernels). For every admissible incidence $(i,a) \in \mathfrak{I}_D$, fix structural data X, standard-Borel source and target spaces $(\mathsf{S}^X_{i,a}, \mathscr{S}^X_{i,a})$ and $(\mathsf{T}^X_{i,a}, \mathscr{T}^X_{i,a})$ containing all source and output coordinates required by the declared generative factorization, and the standard-Borel model-presentation space $(\mathsf{M}_{i,a}, \mathscr{M}_{i,a})$. A model sample, or presentation coordinate, is $m_i \in \mathsf{M}_{i,a}$, while the model-law section has incidence value $q_i^m(c_a) \in \mathcal{P}(\mathsf{M}_{i,a}, \mathscr{M}_{i,a})$. Declare

$$\mathrm{ev}_{i,a} : \mathsf{M}_{i,a} \longrightarrow \mathrm{Kern}(\mathsf{S}^X_{i,a}, \mathsf{T}^X_{i,a}), \qquad m_i \longmapsto K^{X,a}_{i,m_i}, \tag{3.10}$$

such that $(m_i, u) \mapsto K^{X,a}_{i,m_i}(u, B)$ is measurable for every $B \in \mathscr{T}^X_{i,a}$ and $K^{X,a}_{i,m_i}(u, \mathsf{T}^X_{i,a}) = 1$ for every (m_i, u). The finite family is jointly measurable in the tagged sense: for every collection $(B_{i,a})_{(i,a)\in\mathfrak{I}_D}$ with $B_{i,a} \in \mathscr{T}^X_{i,a}$, the map

$$((i,a), m_i, u) \longmapsto K^{X,a}_{i,m_i}(u, B_{i,a})$$

is measurable on the finite measurable disjoint union of the spaces $\mathsf{M}_{i,a} \times \mathsf{S}^X_{i,a}$. The bundled notation

$$\mathrm{ev}_i := (\mathrm{ev}_{i,a})_{a\in D_i}$$

therefore denotes a typed finite family, not one map into an undeclared common kernel space. Each source space retains every observation, structural-parent, or other parent coordinate required by its declared factor; (3.10) does not silently reduce any factor to a state-to-observation kernel. The law $q_i^m(c_a)$, the sampled coordinate m_i, and its evaluated kernel $K^{X,a}_{i,m_i}$ are three different types. In the definite-presentation case $q_i^m(c_a) = \delta_{m_i}$. Neither that law, a recognition parameter, nor a posterior is an argument of the evaluated generative kernel. [DEFINITION]

Injectivity of any $\mathrm{ev}_{i,a}$ is not assumed: distinct presentations in one incidence fiber may evaluate to the same normalized kernel. One may declare $m \sim m'$ within that fiber when their evaluated

kernels agree, but the standard-Borel, Hausdorff, or smooth regularity of the presentation quotient is not supplied by measurability of (3.10). A quotient-level model fiber requires a separate regularity theorem; the default probability construction retains presentations. [NOT-CLAIMED]

3.3 Reference measures

Definition 3.6 (Declared reference measures). Each continuous Euclidean coordinate carries its Borel sigma-algebra and Lebesgue base measure. Each discrete coordinate carries its power-set sigma-algebra and counting base measure. Each mixed real coordinate carries the measure declared in Corollary 3.8. Any other fiber carries an explicitly declared sigma-finite base measure. Writing $\nu^k_{i,a}$, $\nu^m_{i,a}$, and $\nu^o_{i,a}$ for the component measures, the finite product reference measures are

$$\nu^Y_D = \bigotimes_{a=1}^{M} \bigotimes_{i \in V_a} (\nu^k_{i,a} \otimes \nu^m_{i,a}), \qquad \nu^O_D = \bigotimes_{a=1}^{M} \bigotimes_{i \in V_a} \nu^o_{i,a}. \tag{3.11}$$

[DEFINITION]

Sigma-finiteness of every component is required and is not decoration. It is the hypothesis of the Radon-Nikodym theorem, without which the densities of Section 3.4 need not exist, and it is the hypothesis of the Fubini-Tonelli theorem, without which the iterated integrals used to marginalize Y out of the joint need not agree with the product integral. A finite product of sigma-finite measures is sigma-finite, so (3.11) inherits the property from its factors; this is the reason the construction is stated on a finite design and is one of the reasons the continuum question of Section 3.8 stays open.

Mixed coordinates are the case where a declaration is genuinely needed rather than conventional. A reference measure used by a parameterized family must be fixed for the whole family; choosing a new atomic reference after seeing the parameter would not define one density model.

Theorem 3.7 (Common domination for a mixed family). Let $(\mu_t)_{t \in T}$ be any family of Borel probability measures on $\mathbb{R}$ with no singular-continuous part, written

$$\mu_t = f_t \lambda + \sum_{a \in A_t} p_t(a) \delta_a, \qquad A_t := \{a \in \mathbb{R} : \mu_t(\{a\}) > 0\}.$$

There exists a single sigma-finite Borel measure dominating every μ_t if and only if $A_* := \bigcup_{t \in T} A_t$ is countable. When $A_* = \{a_1, a_2, \ldots\}$, one such parameter-independent measure is

$$\nu_{\text{mix}} := \lambda + \sum_{n \geq 1} 2^{-n} \delta_{a_n}. \tag{3.12}$$

In particular, the moving-atom mixed family $\mu_t = \frac{1}{2}\mathcal{N}(0,1) + \frac{1}{2}\delta_t$, $t \in \mathbb{R}$, has no common sigma-finite dominating measure. [ESTABLISHED]

Proof. If A_* is countable, (3.12) is sigma-finite and a ν_{mix}-null set is both Lebesgue null and contains no point of A_*. Hence every continuous and atomic term of every μ_t vanishes on it, so $\mu_t \ll \nu_{\mathrm{mix}}$. Conversely, suppose a sigma-finite ξ dominates every μ_t. For each $a \in A_*$ some t has $\mu_t(\{a\}) > 0$, whence $\xi(\{a\}) > 0$. A sigma-finite measure has at most countably many positive-mass point atoms: on each finite-measure member of a countable cover, the points of mass at least $1/n$ are finite, and the countable union over the cover and n exhausts them. Thus A_* is countable. The displayed moving-atom family has $A_* = \mathbb{R}$, proving the last assertion. □

Corollary 3.8 (Dominating measure for one mixed coordinate). Let μ be a Borel probability measure on $\mathbb{R}$ whose Lebesgue decomposition has no singular continuous part, so that $\mu = \mu_{\mathrm{ac}} + \mu_{\mathrm{d}}$ with $\mu_{\mathrm{ac}} \ll \lambda$ and μ_{d} carried by a countable set A. Put

$$\nu = \lambda + \sum_{x \in A} \delta_x. \tag{3.13}$$

Then ν is sigma-finite and $\mu \ll \nu$. [ESTABLISHED]

Proof. Apply Theorem 3.7 to the singleton family $\{\mu\}$. The measure displayed here and the weighted atomic reference in (3.12) have the same null sets, so either dominates μ. □

The hypothesis excluding a singular-continuous part is a restriction on this particular Lebesgue-plus-atoms formula, not an intrinsic nondominability claim. A family carrying Cantor-type or other singular-continuous components can be dominated when a common sigma-finite singular reference is available, but that reference must be declared as part of the family before any density is written. Equation (3.12) supplies no such component.

3.4 Kernels first, densities second

Definition 3.9 (Normalized kernels; densities as derivatives). The objects (3.8) and (3.9) are probability kernels, hence normalized by type. When $\mathbb{P}_{\theta,V}(\cdot \mid X) \ll \nu_D^Y \otimes \nu_D^O$, its density is the Radon-Nikodym derivative $p_{\theta,V}(o, y \mid X)$ determined by

$$\mathbb{P}_{\theta,V}(dy, do \mid X) = p_{\theta,V}(o, y \mid X)\nu_D^Y(dy)\nu_D^O(do), \tag{3.14}$$

and the evidence is the normalized marginal

$$p_{\theta,V}(o \mid X) = \int_{\mathsf{Y}_D} p_{\theta,V}(o, y \mid X)\nu_D^Y(dy). \tag{3.15}$$

When $\mathbb{Q}_{V,o,X} \ll \nu_D^Y$, its density is written $q_{V,o,X}(y)$. Densities are determined only up to a null set of the declared reference measure, and any two versions agreeing outside such a set denote the same density. [DEFINITION]

The order in Definition 3.9 is the content. A normalized kernel is declared, and only then is a density extracted; the reverse order, in which a nonnegative function is written down and a normalizer is invoked afterward, is a different construction whose normalizer must be shown finite before anything it appears in denotes a probability. This document uses the first order everywhere, and where a later chapter writes an unnormalized energy it says so and exhibits the normalizer.

Regular-conditional convention. Whenever a later result conditions on a realized observation, assume in addition that $(\mathsf{O}_D, \mathscr{O}_D)$ and $(\mathsf{Y}_D, \mathscr{Y}_D)$ are standard Borel spaces. This holds for the finite products of Euclidean and countable discrete fibers used in the worked construction; it is an additional hypothesis for a general measurable fiber. For fixed (θ, X), write

$$\mathbb{P}^{O}_{\theta,V,X}(A) := \mathbb{P}_{\theta,V}(\mathsf{Y}_D \times A \mid X), \qquad A \in \mathscr{O}_D, \tag{3.16}$$

for the observation marginal. The standard-Borel hypothesis supplies a regular conditional probability

$$\mathbf{\Pi}_{\theta,V,o,X}(B) = \mathbb{P}_{\theta,V}(B \mid o, X), \qquad B \in \mathscr{Y}_D, \tag{3.17}$$

unique only for $\mathbb{P}^{O}_{\theta,V,X}$-almost every o.

When the domination in (3.14) holds, Tonelli's theorem makes the function in (3.15) a density of $\mathbb{P}^{O}_{\theta,V,X}$ relative to ν^{O}_{D}. After altering it on a ν^{O}_{D}-null set if necessary, there is a measurable set $\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ with

$$\mathbb{P}^{O}_{\theta,V,X}\big(\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}\big) = 1, \qquad 0 < p_{\theta,V}(o \mid X) < \infty, \tag{3.18}$$

such that, for every $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ and $B \in \mathscr{Y}_D$,

$$\mathbf{\Pi}_{\theta,V,o,X}(B) = \frac{\displaystyle\int_B p_{\theta,V}(o, y \mid X)\nu^{Y}_{D}(dy)}{p_{\theta,V}(o \mid X)}. \tag{3.19}$$

The displayed version is obtained directly from the density on the positive-finite evidence set and completed arbitrarily off that set. Fubini's theorem shows why the qualification is unavoidable: changing a joint density on a $(\nu^{Y}_{D} \otimes \nu^{O}_{D})$-null set can change an entire slice at an exceptional observation, while it changes (3.19) only on an observation-marginal-null set. An exact witness is available on $\mathbb{R}^2$. Let ϕ be the standard normal density and let ψ be any different probability density. The functions

$$p(o, y) = \phi(o)\phi(y), \qquad \widetilde{p}(o, y) = \begin{cases} \phi(o)\phi(y), & o \neq 0, \\ \phi(0)\psi(y), & o = 0 \end{cases}$$

are versions of the same joint density because they differ only on the Lebesgue-null slice $\mathbb{R} \times \{0\}$. Both displayed evidence representatives equal $\phi(0)$ at $o = 0$, but their density-ratio conditionals there are $\phi(y)dy$ and $\psi(y)dy$. The joint measure therefore determines neither pointwise posterior at that exceptional observation. Accordingly, every later fixed-observation posterior, ELBO, or free-energy statement is asserted for $\mathbb{P}^{O}_{\theta,V,X}$-almost every o, represented by $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$. An everywhere pointwise statement instead requires a particular jointly measurable density, evidence

representative, and regular-conditional version to be declared as part of the model data; its exceptional-point values then belong to that declaration and are not determined by the joint measure alone. [HYPOTHESIS]

The same order is what makes the group action of Chapter 2 well typed. The action (2.6) is defined on measures. A frame change that does not preserve the reference measure changes the density and the base-measure Jacobian together, and only their combination is invariant, so a rule that multiplies a matrix into a density and stops is not the represented action unless the Jacobian is identically one. [ESTABLISHED]

Theorem 3.10 (Jointly measurable Radon–Nikodym version for a dominated kernel). Let $(\mathsf{E}, \mathscr{E})$ be a measurable space, let $(\mathsf{F}, \mathscr{F})$ be standard Borel, and let K and L be probability kernels from E to F satisfying $K(x,\cdot) \ll L(x,\cdot)$ for every x. Then there is a finite-valued $\mathscr{E} \otimes \mathscr{F}$-measurable $r : \mathsf{E} \times \mathsf{F} \to [0,\infty)$ such that

$$K(x,B) = \int_B r(x,y) L(x,dy) \quad \text{for every } x \in \mathsf{E}, B \in \mathscr{F}.$$

The theorem supplies one simultaneous version; independently selecting an RN derivative for each x does not establish joint measurability. [ESTABLISHED]

Proof. Standard Borelness gives finite refining measurable partitions $(\mathcal{P}_n)_{n\geq 1}$ whose generated sigma-algebras increase to $\mathscr{F}$. With $0/0 := 0$, set

$$r_n(x,y) = \sum_{A \in \mathcal{P}_n} \frac{K(x,A)}{L(x,A)} \mathbf{1}_A(y), \qquad r(x,y) = \begin{cases} \limsup_{n\to\infty} r_n(x,y), & \limsup_n r_n(x,y) < \infty, \\ 0, & \limsup_n r_n(x,y) = +\infty. \end{cases} \tag{3.20}$$

Every r_n and therefore the finite-valued r is jointly measurable. For each fixed x, if $h_x = dK(x,\cdot)/dL(x,\cdot)$, then $r_n(x,\cdot)$ is the conditional expectation of h_x on the finite sigma-algebra generated by $\mathcal{P}_n$. Martingale convergence gives $r_n(x,\cdot) \to h_x$ at $L(x,\cdot)$-almost every point. Because h_x is finite $L(x,\cdot)$-almost everywhere, the second branch of (3.20) changes only an $L(x,\cdot)$-null set, and $r(x,\cdot)$ is the required derivative for every x. □

Corollary 3.11 (Joint densities against one reference). Let ν be a fixed sigma-finite measure on a standard Borel $(\mathsf{F}, \mathscr{F})$, and let K be a probability kernel from $(\mathsf{E}, \mathscr{E})$ to F with $K(x,\cdot) \ll \nu$ for every x. Then K has a finite jointly measurable density $k(x,y)$ against ν. Consequently every finite or countable declared collection of kernels dominated by the same fixed ν may be represented by jointly measurable densities against that common reference. In particular this applies to the generative and recognition kernels in (3.8)–(3.9) when they are dominated by the declared product references. [ESTABLISHED]

Proof. Sigma-finiteness supplies measurable w with $0 < w < \infty$ ν-almost everywhere and $\int w d\nu = 1$. The constant probability kernel $L(x,dy) = w(y)\nu(dy)$ is equivalent to ν. Apply

Theorem 3.10 to (K, L) to obtain finite jointly measurable $r = dK/dL$ and set $k(x, y) = r(x, y)w(y)$, completing w arbitrarily on its ν-null exceptional set. Applying this construction to each member proves the collection statement. □

Proposition 3.12 (Integration against a fixed sigma-finite measure). If $f : \mathsf{E} \times \mathsf{F} \to [0, \infty]$ is $\mathscr{E} \otimes \mathscr{F}$-measurable and ν is a fixed sigma-finite measure on $(\mathsf{F}, \mathscr{F})$, then

$$x \longmapsto \int_{\mathsf{F}} f(x, y)\nu(dy)$$

is $\mathscr{E}$-measurable. For an extended signed integrand the same conclusion holds whenever at least one of the positive- and negative-part integrals is finite; no expression of the form $+\infty - \infty$ is defined. [ESTABLISHED]

Proof. Choose a measurable disjoint partition $\mathsf{F} = \bigsqcup_{m\geq 1} F_m$ with $\nu(F_m) < \infty$. For fixed m, let

$$\mathscr{D}_m = \{C \in \mathscr{E} \otimes \mathscr{F} : x \longmapsto \nu(C_x \cap F_m) \text{ is } \mathscr{E}\text{-measurable}\}.$$

This class contains every measurable rectangle. It is a Dynkin system: $\nu((C^c)_x \cap F_m) = \nu(F_m) - \nu(C_x \cap F_m)$, where finiteness of $\nu(F_m)$ makes the complement step legitimate, and section measures of a countable disjoint union are the corresponding pointwise sums. The π–λ theorem therefore gives $\mathscr{D}_m = \mathscr{E} \otimes \mathscr{F}$. Hence, for every product-measurable C,

$$x \longmapsto \nu(C_x) = \sum_{m\geq 1} \nu(C_x \cap F_m)$$

is measurable. This proves the assertion for indicators; linearity proves it for nonnegative simple functions, and monotone convergence proves it for every nonnegative measurable f. Apply that result to f^+ and f^- for the signed statement. On the measurable domain where they are not both infinite, their extended-real difference is measurable and no infinity cancellation is used. □

Corollary 3.13 (Measurability of kernel relative entropy). Let κ and η be probability kernels from $(\mathsf{E}, \mathscr{E})$ to a standard Borel space $(\mathsf{F}, \mathscr{F})$. Then $x \mapsto D_{\mathrm{KL}}(\kappa(x, \cdot)\|\eta(x, \cdot))$ is $\mathscr{E}$-measurable as an $[0, \infty]$-valued map. More explicitly, for finite refining partitions $(\mathcal{P}_n)$ generating $\mathscr{F}$,

$$D_{\mathrm{KL}}(\kappa(x, \cdot)\|\eta(x, \cdot)) = \sup_{n\geq 1} \sum_{A\in\mathcal{P}_n} \kappa(x, A) \log \frac{\kappa(x, A)}{\eta(x, A)}, \tag{3.21}$$

with $0 \log(0/b) = 0$ and $a \log(a/0) = +\infty$ for $a > 0$. [ESTABLISHED]

Proof. Put $L = (\kappa + \eta)/2$. Applying Theorem 3.10 to (κ, L) and (η, L) gives finite jointly measurable densities $a = d\kappa/dL$ and $b = d\eta/dL$. Fix x and abbreviate $L_x = L(x, \cdot)$. If $\mathscr{G}_n = \sigma(\mathcal{P}_n)$, then the cellwise ratios in (3.21) are versions of

$$a_n = \mathbb{E}_{L_x}[a(x, \cdot) \mid \mathscr{G}_n], \qquad b_n = \mathbb{E}_{L_x}[b(x, \cdot) \mid \mathscr{G}_n].$$

Define the nonnegative lower-semicontinuous convex perspective

$$\Phi(u, v) = u \log(u/v) - u + v,$$

with $\Phi(0, v) = v$, $\Phi(u, 0) = +\infty$ for $u > 0$, and $\Phi(0, 0) = 0$. Because both kernels have unit mass, the nth partition expression $D_n(x)$ equals $\int \Phi(a_n, b_n) dL_x$. Conditional Jensen under refinement makes $D_n(x)$ nondecreasing, while the log-sum inequality gives

$$D_n(x) \le D_{\mathrm{KL}}(\kappa(x, \cdot) \| \eta(x, \cdot)).$$

Martingale convergence gives $(a_n, b_n) \to (a, b)$ L_x-almost everywhere. Fatou's lemma and the preceding upper bound yield

$$D_{\mathrm{KL}}(\kappa(x, \cdot) \| \eta(x, \cdot)) = \int \Phi(a, b) dL_x \le \liminf_n D_n(x) \le \limsup_n D_n(x) \le D_{\mathrm{KL}}(\kappa(x, \cdot) \| \eta(x, \cdot)).$$

Thus $D_n(x) \uparrow D_{\mathrm{KL}}(\kappa(x, \cdot) \| \eta(x, \cdot))$, proving (3.21) without a cofinality assertion. Each D_n is measurable in x, so its countable supremum is measurable. □

Proposition 3.14 (Kernel integration preserves measurability). Let κ be a probability kernel from $(\mathsf{E}, \mathscr{E})$ to $(\mathsf{F}, \mathscr{F})$ and let $f : \mathsf{E} \times \mathsf{F} \to [0, \infty]$ be $\mathscr{E} \otimes \mathscr{F}$-measurable. Then

$$x \longmapsto \int_{\mathsf{F}} f(x, y) \kappa(x, dy) \tag{3.22}$$

is $\mathscr{E}$-measurable. Consequently $X \mapsto p_{\theta,V}(o \mid X)$ and the $\mathbb{Q}_{V,o,X}$-expectations appearing in the bound of Chapter 5 are measurable in (o, X) whenever their integrands are jointly measurable. [ESTABLISHED]

Proof. Let $\mathcal{D}$ be the collection of $C \in \mathscr{E} \otimes \mathscr{F}$ for which $x \mapsto \kappa(x, C_x)$ is $\mathscr{E}$-measurable, where $C_x = \{y : (x, y) \in C\}$. Measurable rectangles $A \times B$ lie in $\mathcal{D}$, since $\kappa(x, (A \times B)_x) = \mathbf{1}_A(x) \kappa(x, B)$ and $x \mapsto \kappa(x, B)$ is measurable by the definition of a kernel; the rectangles form a π-system generating $\mathscr{E} \otimes \mathscr{F}$. The collection $\mathcal{D}$ contains $\mathsf{E} \times \mathsf{F}$, is closed under proper differences because $\kappa(x, \cdot)$ is a finite measure, and is closed under increasing limits by continuity from below, hence is a λ-system; Dynkin's lemma gives $\mathcal{D} = \mathscr{E} \otimes \mathscr{F}$. The claim therefore holds for indicators, hence for nonnegative simple functions by linearity, hence for all nonnegative measurable f by monotone approximation and monotone convergence. The full development of kernels and their integrals is standard [46]. □

This proposition is stated because the measurability of the evidence and of the bound in their arguments is used implicitly whenever one writes an expectation over configurations, and an implicit use of an unproved measurability claim is a gap. It is discharged here rather than assumed.

3.5 The absolute-continuity convention

Definition 3.15 (Standing convention on logarithmic density ratios). Wherever this document writes a logarithmic density ratio, or an expectation of one, the measure appearing in the numerator is absolutely continuous with respect to the measure appearing in the denominator. Relative entropy is understood in the measure-theoretic sense,

$$D_{\mathrm{KL}}(Q\|P) = \begin{cases} \displaystyle\int \log\Big(\frac{dQ}{dP}(y)\Big) Q(dy), & Q \ll P, \\ +\infty, & \text{otherwise}, \end{cases} \tag{3.23}$$

following the measure-theoretic formulations of Kullback and Leibler [51] and Csiszár [25]. [DEFINITION]

The convention is not a restriction on what may be written; the second branch of (3.23) assigns a value in every case. It is a statement about which of those values is finite, and it is load-bearing precisely because the infinite branch is not an edge case to be ignored but the mechanism by which whole classes of candidate recognition laws are excluded. The next statement makes the criterion checkable in terms of densities, which is the form in which it is applied later.

Proposition 3.16 (Density criterion for absolute continuity). Let Q and P be measures on a common measurable space with $Q \ll \nu$ and $P \ll \nu$ for a sigma-finite ν, with densities q and p. Write $Z = \{p = 0\}$. Then

$$Q \ll P \iff Q(Z) = 0 \iff q = 0 \quad \nu\text{-almost everywhere on } Z. \tag{3.24}$$

[ESTABLISHED]

Proof. First, $P(Z) = \int_Z p(x)\nu(dx) = 0$. If $Q \ll P$ then $Q(Z) = 0$ follows at once. Conversely suppose $Q(Z) = 0$ and let A satisfy $P(A) = 0$, that is $\int_A p(x)\nu(dx) = 0$ with $p \geq 0$; then $\nu(A \cap \{p > 0\}) = 0$, so $Q(A \cap \{p > 0\}) = \int_{A\cap\{p>0\}} q(x)\nu(dx) = 0$, while $Q(A \cap Z) \leq Q(Z) = 0$. Adding gives $Q(A) = 0$. The second equivalence is immediate from $Q(Z) = \int_Z q(x)\nu(dx)$ and $q \geq 0$. □

The consequence used later has two cases that must not be conflated. Proposition 3.16 applies when both measures are dominated by the same ν. If a candidate Q is singular relative to that reference measure, its failure of absolute continuity must instead be checked directly from sets. Namely, if there is a measurable S with

$$Q(S) = 1, \qquad P(S) = 0, \tag{3.25}$$

then $Q \not\ll P$ by the definition of absolute continuity, and (3.23) gives $D_{\mathrm{KL}}(Q\|P) = +\infty$. For example, let P be a nondegenerate Lebesgue-dominated law on $\mathbb{R}^d$ and let Q be supported on a proper affine subspace S. The subspace is Lebesgue null, hence $P(S) = 0$, whereas $Q(S) = 1$;

the direct argument (3.25), not Proposition 3.16, proves the claim. This distinction matters because a subspace-supported Q has no Lebesgue density to which the density criterion could be applied. The specific instance, in which an identification of latent coordinates across a cluster is shown to be a generative construction rather than a restriction on the recognition law, is developed in Part II.

Hypothesis 3.17 (Regular positive-finite observation). For fixed (θ, X), the selected observation satisfies $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ under the regular-conditional convention above, and

$$0 < p_{\theta,V}(o \mid X) < \infty. \tag{3.26}$$

[HYPOTHESIS]

This hypothesis selects a pointwise posterior version and a finite real log evidence. It imposes no support or entropy condition on a candidate recognition law; those conditions are needed only for the classical split representation below.

Hypothesis 3.18 (Classical split-ELBO domain). In addition to Hypothesis 3.17, the recognition law satisfies

$$\mathbb{Q}_{V,o,X} \ll \mathbf{\Pi}_{\theta,V,o,X}$$

and

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\Big[\big|\log p_{\theta,V}(o, Y \mid X)\big| + \big|\log q_{V,o,X}(Y)\big|\Big] < \infty. \tag{3.27}$$

[HYPOTHESIS]

These are choices, and each excludes something identifiable. Membership in $\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ makes the posterior the selected regular-conditional version and makes the statement invariant under changes of density version outside a marginal-full set. The lower bound in (3.26) excludes observations the selected representative deems impossible, for which the log evidence is $-\infty$. The separate absolute-continuity requirement in Hypothesis 3.18 excludes exactly the degenerate laws just described. Condition (3.27) excludes recognition laws whose tails make the two classical expectations individually infinite even when their formal difference suggests cancellation. Chapter 5 first defines an extended functional for every probability law and invokes Hypothesis 3.18 only when splitting it into separate expected-log-joint and entropy terms.

3.6 Recognition marginals, couplings, and selection

Definition 3.19 (Local recognition laws and their coupling class). Let $\mathrm{pr}_i : \mathsf{Y}_D \to \mathsf{Y}_{i,D}$ be the agent-block projection. For each admitted (o, X) define

$$\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}) := \left\{ R \in \mathcal{P}\left(\prod_{i \in V} \mathsf{Y}_{i,D}\right) : (\mathrm{pr}_i)_{\#} R = \mathbb{Q}_{i,o,X} \text{for every } i \right\}. \tag{3.28}$$

With $\mathrm{pr}^k_{i,a}$ and $\mathrm{pr}^m_{i,a}$ the within-block coordinate projections, define

$$q^{b;o,X}_{i,a} = (\mathrm{pr}^k_{i,a})_{\#}\mathbb{Q}_{i,o,X}, \qquad q^{m;o,X}_{i,a} = (\mathrm{pr}^m_{i,a})_{\#}\mathbb{Q}_{i,o,X}. \tag{3.29}$$

These coordinate laws and the agent-block laws are local data. A member of (3.28) is additional population-level dependence data. [DEFINITION]

Proposition 3.20 (Existence and convexity of the coupling class). For every admitted (o, X), the class $\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$ is nonempty and convex. Moreover,

$$(o, X) \longmapsto \bigotimes_{i\in V} \mathbb{Q}_{i,o,X} \tag{3.30}$$

is a measurable product-coupling kernel. [ESTABLISHED]

Proof. Finite kernel induction constructs the product in (3.30). On measurable rectangles its value is the product of the measurable functions $\mathbb{Q}_{i,o,X}(B_i)$; a monotone-class argument extends measurability to the product sigma-algebra. Its ith pushforward is $\mathbb{Q}_{i,o,X}$ because all other factors have unit mass, so it belongs to the coupling class. If R_0, R_1 are couplings and $t \in [0, 1]$, then marginalization is affine:

$$(\mathrm{pr}_i)_{\#}\big((1-t)R_0 + tR_1\big) = (1-t)\mathbb{Q}_{i,o,X} + t\mathbb{Q}_{i,o,X} = \mathbb{Q}_{i,o,X}.$$

Thus every convex combination is also a coupling. □

Hypothesis 3.21 (Measurable population-coupling selection). Equip probability-law spaces with their evaluation sigma-algebras. A selection rule S is declared such that

$$\mathbb{Q}_{V,o,X} = \mathsf{S}_{o,X}\big((\mathbb{Q}_{i,o,X})_{i\in V}\big) \in \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}), \tag{3.31}$$

and $(o, X) \mapsto \mathbb{Q}_{V,o,X}(B)$ is measurable for every $B \in \mathscr{Y}_D$. [HYPOTHESIS]

A copula, a common latent variable, ordered recognition conditionals, a graphical family, or the product restriction may supply the selector. None is supplied by the marginal laws alone.

The selector in (3.31) is applied once. For any later declared blockwise bimeasurable frame re-expression R, with R_i its restriction to $\mathsf{Y}_{i,D}$, the valid transformed recognition joint is the whole-law pushforward $R_{\#}\mathbb{Q}_{V,o,X}$. The identity

$$\mathsf{S}_{o,X'}\big(((R_i)_{\#}\mathbb{Q}_{i,o,X})_{i\in V}\big) = R_{\#}\mathsf{S}_{o,X}\big((\mathbb{Q}_{i,o,X})_{i\in V}\big)$$

would be a separate selector-equivariance hypothesis. It is neither assumed here nor claimed by the later gauge-invariance corollary.

Hypothesis 3.22 (Constrained coupling feasibility). For every admitted (o, X), any imposed restricted family $\mathrm{Cpl}_{\mathcal{R}}(\mathbb{Q}_{\bullet,o,X})$ is nonempty. [HYPOTHESIS]

Hypothesis 3.23 (Constrained measurable selection). There is a selector from the feasible restricted fibers for which $(o, X) \mapsto \mathsf{S}_{o,X}((\mathbb{Q}_{i,o,X})_{i\in V})(B)$ is measurable for every $B \in \mathscr{Y}_D$. [HYPOTHESIS]

These hypotheses are independent obligations. The unconstrained product coupling proves neither feasibility of the restricted intersection nor measurability of a selector from it.

Proposition 3.24 (Coordinate marginals do not determine a sufficiently rich joint). For a finite family of standard-Borel block spaces, the unconstrained class $\mathrm{Cpl}((\mu_i)_{i\in V})$ is a singleton if and only if at most one marginal μ_i is non-Dirac. Hence two non-Dirac local recognition laws imply that the local marginals do not determine $\mathbb{Q}_{V,o,X}$. [ESTABLISHED]

Proof. Suppose first that two marginals, denoted μ and ν, are non-Dirac. On a standard-Borel space a non-Dirac probability measure admits a measurable set of probability strictly between zero and one. Indeed, if every member of a countable separating generator had probability zero or one, the intersection of the generator members or complements having probability one would itself have probability one and contain at most one point, forcing a Dirac law. Choose such sets A, B and, for $0 < |\varepsilon| < 1$, define

$$dR_\varepsilon(x, y) = [1 + \varepsilon(1_A(x) - \mu(A))(1_B(y) - \nu(B))]\, d\mu(x)d\nu(y). \tag{3.32}$$

Each centered indicator lies in $[-1, 1]$, so the bracket is at least $1 - |\varepsilon| > 0$. Its integral is one because both centered indicators have mean zero. Integrating in y removes the perturbation and leaves μ; integrating in x leaves ν. Finally,

$$R_\varepsilon(A \times B) = \mu(A)\nu(B) + \varepsilon\mu(A)(1 - \mu(A))\nu(B)(1 - \nu(B)),$$

so the coupling varies with ε. Tensoring R_ε with all remaining marginals gives distinct couplings of the full finite family.

Conversely, suppose at most one marginal is non-Dirac. Every coupling assigns mass one to each fixed Dirac coordinate. Because there are finitely many such coordinates, it assigns mass one to their intersection, on which only the possible non-Dirac coordinate can vary. Its law there is fixed by that remaining marginal. Thus the product of the fixed Dirac masses and the one possible non-Dirac marginal is the unique coupling. □

As an optional Gaussian illustration, $\mathcal{N}(0, I_2)$ and $\mathcal{N}(0, \Sigma_r)$ with $\Sigma_r = \left(\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}\right)$ and $0 < |r| < 1$ are distinct couplings of two standard normal marginals. The perturbation proof above is the general argument and does not depend on Gaussian structure.

3.7 Continuum sections versus finite evaluation

Chapter 2 gave each agent continuum sections $q_i^{b;o,X}$ and $q_i^{m;o,X}$ over its support. This chapter has given the finite recognition law $\mathbb{Q}_{V,o,X}$ and its marginals. There are now two ways to obtain each state or model-coordinate law at (i,a): sample the corresponding continuum section at c_a, or marginalize the finite law. Write the first as

$$\mathrm{Samp}_{D,i}\big(q_i^{b;o,X}\big) = \big(q_i^{b;o,X}(c_a)\big)_{a\in D_i}, \qquad \mathrm{Samp}_{D,i}\big(q_i^{m;o,X}\big) = \big(q_i^{m;o,X}(c_a)\big)_{a\in D_i}. \tag{3.33}$$

The two constructions have different domains. The sampling map acts on the declared response spaces $\mathfrak{S}_i^b$ and $\mathfrak{S}_i^m$, which are typed subspaces of the two section spaces in (3.4); the maps of Definition 3.19 act on $\mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D)$. Neither is defined in terms of the other, so their outputs coincide only if that is declared. The following statement shows that the declaration is substantive in one direction and empty in the other, so that calling it a consequence would be wrong in both.

Proposition 3.25 (Compatibility determines neither side). Suppose the compatibility relation

$$q_i^{b;o,X}(c_a) = q_{i,a}^{b;o,X}, \qquad q_i^{m;o,X}(c_a) = q_{i,a}^{m;o,X} \qquad \text{for every } (i,a)\in\mathfrak{I}_D \tag{3.34}$$

holds. Then (i), when at least two local agent-block laws are non-Dirac, (3.34) does not determine $\mathbb{Q}_{V,o,X}$; and (ii), if some support $\mathcal{C}_i$ contains a point outside the finite design D and the selected section class admits a nonzero smooth deformation supported away from D, the relation does not determine the sections $q_i^{b;o,X}$ and $q_i^{m;o,X}$. The positive-dimensional Gaussian tier supplies such deformations. If every support is zero-dimensional and exhausted by its design points, the finite values can instead determine a section pointwise. [ESTABLISHED]

Proof. (i) By Proposition 3.24 there are distinct laws in $\mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D)$ with the same coordinate marginals, and (3.34) constrains $\mathbb{Q}_{V,o,X}$ only through those marginals; both such laws satisfy it against the same sections. (ii) Fix $c^\star \in \mathcal{C}_i \setminus D$. In the positive-dimensional Gaussian realization, choose a neighborhood of $c^\star$ disjoint from the finite set D and a smooth bump function $\chi : \mathcal{C}_i \to [0,1]$ supported there with $\chi(c^\star) = 1$. The section obtained from $q_i^{b;o,X}$ by replacing $\mu_i(c)$ with $\mu_i(c)+\chi(c)v$ for any fixed $v \neq 0$ is a smooth section of the same bundle, differs from $q_i^{b;o,X}$ at $c^\star$, and agrees with it on all of D; it therefore satisfies (3.34) against the same marginals. By contrast, if $\mathcal{C}_i = D \cap \mathcal{C}_i$ as a zero-dimensional support, equality at those points is pointwise equality of the section, so the nonuniqueness conclusion has no such witness and can fail. □

Hypothesis 3.26 (Sampling and marginalization are compatible). Where a result requires the continuum sections and the finite recognition law to describe the same coordinate laws, relation (3.34) is adopted. [HYPOTHESIS]

This is a choice and is named as one. It excludes configurations in which an agent's continuum section and the finite law disagree at a design point, which is a genuine possibility because the

two objects are declared independently. It is invoked only where a statement about the finite law is transferred to the geometric layer, or the reverse; a result stated entirely about the finite construction does not need it. Proposition 3.25 records that, under its richness hypothesis, adopting compatibility does not reconstruct the joint from its marginals and, when the support extends beyond the design and the selected fiber is locally deformable, does not reconstruct a continuum section from finite samples. Singleton finite fibers and zero-dimensional supports exhausted by the design are the explicit exceptions.

3.8 What the finite construction does not supply

Open Problem 3.27 (Continuum probabilistic theory). The finite construction supplies probability spaces and kernels associated with the declared design D. It supplies no probability measure on a space of sections over $\mathcal{C}$, no unique extension of the finite marginals to such a space, and no continuum limit. Whether one exists for this construction is not settled here. [OPEN]

What would close it can be listed exactly, and listing it is the point of declaring the gap rather than gesturing at it. One would need a refining family of designs; a compatible, that is projective, system of finite laws indexed by those designs, so that a Kolmogorov-type extension theorem applies [46]; a declared topology and sigma-algebra on the candidate space of sections, since the extension theorem produces a measure on a product space and further work is needed to place it on a space of smooth sections; tightness or compactness sufficient to control the limit; convergence of the functionals and observables that the finite theory computes; control of the gauge choices of Chapter 2 and of the reference measures of Section 3.3 under refinement, since neither is canonical; and a proof that any proposed continuum interaction energy is normalizable.

The last item is where the obstruction is concrete rather than merely laborious. The reference measures (3.11) are finite products, and sigma-finiteness was inherited from finitely many sigma-finite factors; the same construction does not extend to infinitely many non-probability factors, so a continuum theory cannot obtain its reference measure by the route used here and would have to declare a measure on the section space directly. Until that is done, a continuum energy has no density to be the exponent of, and the objects that the finite theory calls densities have no continuum counterparts. Every evidence-bound claim in this document is accordingly a finite-design claim, and none of them is asserted in a continuum limit.

Chapter 4

The Fixed Normalized Generative Law

This chapter fixes one normalized joint before inference begins. The latent channels inherit the two representations of the common principal gauge group from Chapter 2. Their local coordinates may be re-expressed independently, so the covariance formulas carry a passive product of frame changes without introducing a second principal bundle. The linear-Gaussian realization is deferred to Chapter 21.

4.1 Parameters, structure, and kernel types

Fix the design $D = \{c_a\}_{a=1}^{M}$, the agent set $V = \{1, \dots, N\}$, and the measurable spaces and sigma-finite reference measures of Chapter 3. Structural data $X \in \mathsf{X}$ contain a finite directed acyclic graph $\Gamma = (V, E)$, design incidence, and the geometric data. Writing $\mathrm{pa}(i)$ for the parents of i, the incidence is parent closed: if $(i, a) \in \mathfrak{I}_D$ and $j \in \mathrm{pa}(i)$, then $(j, a) \in \mathfrak{I}_D$. Thus the induced graph Γ_a on V_a contains every parent of each participating vertex and is acyclic. Numerical parameters that a learning coordinate may move are denoted θ. This split is declared rather than derived; all typing requirements below apply to both symbols. [DEFINITION]

Write $\mathcal{R} = \{r : \mathrm{pa}(r) = \varnothing\}$. At design point c_a, for each $i \in V_a$, declare normalized Markov kernels

$$P^{m}_{\theta,i}(dm_i \mid m_{\mathrm{pa}(i)}, X), \qquad P^{k}_{\theta,i}(dk_i \mid k_{\mathrm{pa}(i)}, m_i, X), \qquad P^{o}_{\theta,i}(do_i \mid k_i, m_i, X). \tag{4.1}$$

At a root, the first factor is a proper model law and the second is a normalized bridge from the same-agent model coordinate to the belief coordinate. The observation component is retained as a normalized singleton-scope record mechanism; in the population construction below it is used only through the owned record kernel K_r to which it is assigned. [DEFINITION]

For each admissible incidence (i, a), specialize the source and target spaces of Definition 3.5 by

$$(\mathsf{S}^{X}_{i,a}, \mathscr{S}^{X}_{i,a}) := \left(\prod_{j \in \mathrm{pa}(i)} \mathsf{K}_{j,a}, \bigotimes_{j \in \mathrm{pa}(i)} \mathscr{K}_{j,a} \right), \qquad (\mathsf{T}^{X}_{i,a}, \mathscr{T}^{X}_{i,a}) := (\mathsf{K}_{i,a}, \mathscr{K}_{i,a}),$$

while retaining the already declared presentation space $(\mathsf{M}_{i,a}, \mathscr{M}_{i,a})$. The empty parent product is a singleton with its trivial sigma-algebra. For fixed (θ, X), the incidence evaluator used by the state

factor is

$$\mathrm{ev}_{i,a}(m_i) = K^{X,a}_{i,m_i}, \qquad K^{X,a}_{i,m_i}(k_{\mathrm{pa}(i)}, B) := P^{k}_{\theta,i}(B \mid k_{\mathrm{pa}(i)}, m_i, X), \tag{4.2}$$

for every $m_i \in \mathsf{M}_{i,a}$, every complete parent-state tuple in the displayed source, and $B \in \mathscr{K}_{i,a}$. Thus m_i is the sampled presentation supplied to the evaluator, while every parent-state coordinate remains in the source of its normalized kernel. Together with the separate model factor, which retains every parent-model coordinate, this specialization reads the complete source block $y_{\mathrm{pa}(i),a}$. [DEFINITION]

For $i \in V_a$, write $\mathsf{Y}_{\mathrm{pa}(i),a} = \prod_{j\in\mathrm{pa}(i)} \mathsf{Y}_{j,a}$. The ordered composite of the model and state kernels is the normalized agent mechanism

$$\begin{aligned} G_{\theta,i,a} &: (\mathsf{X} \times \mathsf{Y}_{\mathrm{pa}(i),a}) \rightsquigarrow \mathsf{Y}_{i,a}, \\ G_{\theta,i,a}\big(d(k_i, m_i) \mid y_{\mathrm{pa}(i),a}, X\big) &:= P^{m}_{\theta,i}(dm_i \mid m_{\mathrm{pa}(i)}, X) K^{X,a}_{i,m_i}(k_{\mathrm{pa}(i)}, dk_i). \end{aligned} \tag{4.3}$$

Its conditioning coordinates precede its target block, and integration in k_i followed by m_i gives unit mass. The model factor retains the complete parent-model tuple and the evaluated state factor retains the complete parent-state tuple, so their composition reads the full $y_{\mathrm{pa}(i),a}$ source. Under the design-point conditional-independence hypothesis, the finite-design agent mechanism is

$$G_{\theta,i,D} := \bigotimes_{a\in D_i} G_{\theta,i,a}. \tag{4.4}$$

The two factors inside each $G_{\theta,i,a}$ are composed in their displayed dependency order. The tensor product in $G_{\theta,i,D}$ is the product kernel across independent design points. [DEFINITION]

The channels are asymmetric by declaration. At a root incidence (i, a), the parent-state source in (4.2) is the singleton with element $*$. For $s \in \mathcal{P}(\mathsf{M}_{i,a}, \mathscr{M}_{i,a})$, the evaluated kernel induces the incidence-level map on laws

$$\widetilde{\phi}_{i,a}(s)(A) = \int_{\mathsf{M}_{i,a}} K^{X,a}_{i,m}(*, A) s(dm) = \int_{\mathsf{M}_{i,a}} P^{k}_{\theta,i}(A \mid m, X) s(dm), \qquad A \in \mathscr{K}_{i,a}. \tag{4.5}$$

Only this pushforward-on-laws map can represent the design-point value of the local associated-bundle morphism $\widetilde{\Phi} : (\mathcal{E}_m)_{c_a} \to (\mathcal{E}_b)_{c_a}$, and only if its contextual and gauge compatibility is declared. At a nonroot, the evaluated kernel is indexed by the complete parent belief tuple; it does not define a map $\mathcal{E}_m \to \mathcal{E}_b$ unless that parent is fixed or the domain is enlarged. Otherwise it is a separately declared kernel-valued bridge. In every case the generative dependence runs from model samples to belief samples. The geometric existence of $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ neither inserts a reverse factor into (4.1) nor forces cyclic generative dependence. A model using both directions would need a separately normalized joint and cannot infer its factorization from the two associated-bundle morphisms.

4.2 Iterated composition and latent densities

For each a, let $N_a := |V_a|$ and choose a topological ordering $v_{a,1}, \dots, v_{a,N_a}$ of Γ_a. The one-design-point latent law is the successive kernel composition

$$\mathbb{P}^Y_{\theta,V,a}(dy_a \mid X) := \prod_{\ell=1}^{\overrightarrow{N_a}} G_{\theta,v_{a,\ell},a}\big(dy_{v_{a,\ell},a} \mid y_{\mathrm{pa}(v_{a,\ell}),a}, X\big). \tag{4.6}$$

The arrow denotes ordered application of kernels, not multiplication of measures. For bounded measurable f, its meaning is

$$\int f(y_a)\mathbb{P}^Y_{\theta,V,a}(dy_a \mid X) = \int G_{\theta,v_{a,1},a}(dy_{v_{a,1},a} \mid X) \cdots \int G_{\theta,v_{a,N_a},a}\big(dy_{v_{a,N_a},a} \mid y_{\mathrm{pa}(v_{a,N_a}),a}, X\big) f(y_a). \tag{4.7}$$

Every conditioning coordinate has been introduced before its kernel is applied.

Construction 4.1 (Finite directed latent law). Equation (4.7) defines a normalized measurable probability kernel $X \mapsto \mathbb{P}^Y_{\theta,V,a}(\cdot \mid X)$. [ESTABLISHED]

Proof. Integrating a bounded measurable function against a probability kernel preserves boundedness and measurability in the remaining variables. Finite induction along the topological order therefore defines a measure and a measurable kernel in X; applying the induction to $f \equiv 1$ gives total mass one. This is the finite kernel construction underlying directed graphical models [49, 54, 87]. □

Distinct design points are conditionally independent given X:

$$\mathbb{P}^Y_{\theta,V}(dy \mid X) = \bigotimes_{a=1}^{M} \mathbb{P}^Y_{\theta,V,a}(dy_a \mid X). \tag{4.8}$$

Equivalently, each agent contributes the bundled mechanism $G_{\theta,i,D}$ of (4.4). This modeling hypothesis excludes residual cross-design dependence and is used in the finite-design normalization and factor count. [HYPOTHESIS]

When the component kernels have densities relative to the declared component reference measures,

$$g_{\theta,i,a}(k_i, m_i \mid y_{\mathrm{pa}(i),a}, X) = p^m_{\theta,i}(m_i \mid m_{\mathrm{pa}(i)}, X) p^k_{\theta,i}(k_i \mid k_{\mathrm{pa}(i)}, m_i, X), \qquad p^Y_{\theta,V,a}(y_a \mid X) = \prod_{i \in V_a} g_{\theta,i,a}. \tag{4.9}$$

Consequently

$$p^Y_{\theta,V}(y \mid X) = \prod_{a=1}^{M} p^Y_{\theta,V,a}(y_a \mid X), \qquad \mathbb{P}^Y_{\theta,V}(dy \mid X) = p^Y_{\theta,V}(y \mid X)\nu^Y_D(dy). \tag{4.10}$$

The measure $\mathbb{P}^{Y}_{\theta,V}$ is authoritative; $p^{Y}_{\theta,V}$ denotes only its Radon–Nikodym density.

4.3 The typing prohibition

Requirement 4.2 (Typing prohibition). *Each factor in (4.1) is a function only of θ, X, and its displayed parent and same-agent latent coordinates. No generative kernel may take as an argument the recognition law $\mathbb{Q}_{V,o,X}$, one of its local marginals or parameters, or the posterior $\mathbf{\Pi}_{\theta,V,o,X}$.* [DEFINITION]

This requirement defines the fixed-joint construction and proves nothing. A transition built from a live peer recognition marginal is a map from recognition laws to kernels, not a factor of a joint fixed before inference.

Proposition 4.3 (No distinguished target). Let $\mathcal{Q}$ be a nonempty family of recognition laws. If a factor in (4.6) is replaced by a nonconstant Q-indexed family, producing a normalized joint $P_{\theta,Q}$ for each $Q \in \mathcal{Q}$, assume at the displayed observation that $0 < p_{\theta,Q}(o \mid X) < \infty$, that $Q \ll P_{\theta,Q}(\cdot \mid o, X)$, and that the log-density terms defining the ELBO and KL divergence are integrable. Then for each fixed Q

$$\log p_{\theta,Q}(o \mid X) = \mathcal{L}(Q; X) + D_{\mathrm{KL}}\left(Q \| P_{\theta,Q}(\cdot \mid o, X)\right). \tag{4.11}$$

The family itself selects no joint as "the model." Writing $e(Q) = \log p_{\theta,Q}(o \mid X)$, two members satisfy

$$\begin{aligned}\mathcal{L}(Q'; X) - \mathcal{L}(Q; X) = e(Q') - e(Q) \\ - \left[D_{\mathrm{KL}}\big(Q' \| P_{\theta,Q'}(\cdot \mid o, X)\big) - D_{\mathrm{KL}}\big(Q \| P_{\theta,Q}(\cdot \mid o, X)\big)\right].\end{aligned} \tag{4.12}$$

Thus a divergence decrease certifies improvement relative to one fixed reference value exactly when the two members lie on one level set of e. [ESTABLISHED]

Proof. Apply the exact ELBO identity to each fixed pair $(Q, P_{\theta,Q})$ and subtract the two instances. No member is distinguished because no selection rule was declared. □

Proposition 4.4 (The stronger consequences fail). Nonconstant dependence of $P_{\theta,Q}$ on Q does not imply that e varies, that no ascent can improve a fixed quantity, or that no single member's evidence bounds the family of bounds. [ESTABLISHED]

Proof. Take one binary latent y, one binary observation o, and $Q_\beta = \mathrm{Ber}(\beta)$ for $\beta \in (0, 1)$. Let $g(\beta) = \beta/2 + 1/4$ and define

$$P_{\theta,Q_\beta}(o, y \mid X) = \tfrac{1}{2}\,\mathrm{Ber}\big(g(\beta)\big)(y). \tag{4.13}$$

The joint is normalized and varies injectively with β, but $e(Q_\beta) = \log(1/2)$ for every β. The posterior is $\mathrm{Ber}(g(\beta))$, so the gap at $\beta = 9/10, 7/10, 1/2$ is respectively

$$\tfrac{9}{10}\log\tfrac{9}{7} + \tfrac{1}{10}\log\tfrac{1}{3} = 0.1163218\ldots, \qquad \tfrac{7}{10}\log\tfrac{7}{6} + \tfrac{3}{10}\log\tfrac{3}{4} = 0.0216009\ldots, \qquad 0. \tag{4.14}$$

The bound therefore increases to the fixed evidence, and the member at $\beta = 1/2$ has evidence equal to the supremum of all these bounds. □

These are closed-form relative entropies, not numerical experiments. The development declines the three stronger statements as false in this witness, while retaining the absence of a distinguished member. [NOT-CLAIMED]

The typing prohibition does not ban population coupling through latents inside a fixed joint; it bans a generative factor from reading the posterior that the joint is supposed to determine. [DEFINITION]

4.4 Normalized record extension

Proposition 4.5 (Normalization by latent composition and record attachment). Under the declared conditional independence of record coordinates given (X, y), the full population law is the kernel

$$\mathbb{P}_{\theta,V}(dy, do \mid X) = \mathbb{P}^{Y}_{\theta,V}(dy \mid X) \prod_{a \in \mathfrak{A}} K_a(X, y_{\partial a,D}, do_a). \tag{4.15}$$

It is normalized for every admissible (θ, X):

$$\mathbb{P}_{\theta,V}(\mathsf{Y}_D \times \mathsf{O}_D \mid X) = 1. \tag{4.16}$$

If the record kernels have jointly measurable densities $K_a(X, y_{\partial a,D}, do_a) = \ell_a(o_a \mid X, y_{\partial a,D})\nu_a(do_a)$ and $\nu^{O}_{D} = \bigotimes_{a \in \mathfrak{A}} \nu_a$, then

$$p_{\theta,V}(o, y \mid X) = p^{Y}_{\theta,V}(y \mid X) \prod_{a \in \mathfrak{A}} \ell_a(o_a \mid X, y_{\partial a,D}) \tag{4.17}$$

is its normalized density. [ESTABLISHED]

Proof. Measurability follows by finite kernel composition. For normalization, apply Tonelli's theorem and integrate each record coordinate first. Every K_a has unit mass in o_a, so their finite product disappears and leaves $\mathbb{P}^{Y}_{\theta,V}(dy \mid X)$. The latter has unit mass by the finite topological-order construction and the finite design product. The same calculation with the displayed Radon–Nikodym derivatives proves the density statement. □

Cycles in the finite record hypergraph do not obstruct (4.15): record coordinates are conditionally attached outputs of one already constructed latent law. By contrast, reciprocal latent-state conditionals require a declared order or schedule, a common-cause construction, or a globally normalized undirected model. Local normalization of those reciprocal conditionals alone does not supply a joint.

4.5 The normalized undirected alternative

An undirected cyclic model is valid when its global normalizer exists. For nonnegative node and edge potentials, define

$$\Psi_X(o,y) = \prod_{i\in V} \psi_i(o_i,k_i,m_i;X) \prod_{\{i,j\}\in E} \psi_{ij}(k_i,m_i,k_j,m_j;X), \tag{4.18}$$

$$Z_X = \int \Psi_X(o,y)\nu_D^O(do)\nu_D^Y(dy), \qquad 0 < Z_X < \infty, \tag{4.19}$$

$$p^{\mathrm{Gibbs}}_{\theta,V}(o,y \mid X) = Z_X^{-1}\Psi_X(o,y). \tag{4.20}$$

Finiteness and positivity of Z_X are model-existence conditions. [DEFINITION]

Proposition 4.6 (Local normalization is insufficient). Normalized node potentials and an everywhere positive, finite, continuous edge potential need not give finite Z_X. [ESTABLISHED]

Proof. On $\mathbb{R}^2$ take

$$\psi_i(y_i) = (2\pi)^{-1/2}e^{-y_i^2/2}, \qquad \psi_{12}(y_1,y_2) = e^{cy_1y_2}, \qquad c \geq 1. \tag{4.21}$$

The quadratic precision is

$$\Lambda_{\mathrm{ex}} = \begin{pmatrix} 1 & -c \\ -c & 1 \end{pmatrix}, \qquad \operatorname{spec}\Lambda_{\mathrm{ex}} = \{1-c, 1+c\}. \tag{4.22}$$

In orthonormal sum-and-difference coordinates the integral factorizes, and the factor in the $1-c$ direction diverges. □

The document chooses the directed construction because its normalization does not depend on the parameters, not because normalized cyclic Gaussian Markov random fields are invalid.

4.6 Covariance under independent channel-frame re-expression

At each admissible incidence $(i,a) \in \mathfrak{I}_D$, independently re-express the belief and model frame coordinates by $g^b_{a,i} = g^b_i(c_a) \in G$ and $g^m_{a,i} = g^m_i(c_a) \in G$. Define the old-to-new represented sample-coordinate maps

$$R^b_{a,i} := \rho_b(g^b_{a,i}) : \mathsf{K} \longrightarrow \mathsf{K}, \qquad R^m_{a,i} := \rho_m(g^m_{a,i}) : \mathsf{M} \longrightarrow \mathsf{M}. \tag{4.23}$$

With the passive-section convention of (2.10), this sample-coordinate transformation corresponds to the section rechoice $u_i^{x\prime} = u_i^x \cdot (g_i^x)^{-1}$, whose value at c_a uses $(g^x_{a,i})^{-1}$. Thus $R^x_{a,i}$ acts within the channel's sample fiber, while the group element $(g_i^x)^{-1} \in G$, not the represented operator $(R^x_{a,i})^{-1}$, multiplies the principal frame on the right. This distinction remains necessary when the representation is not faithful. Within a formula for one design point, suppress the index a on the

participating vertices and write g_i^x, R_i^x. They act on latent coordinates as $k_i' = R_i^b k_i$ and $m_i' = R_i^m m_i$, while observations are gauge inert. [DEFINITION]

To permit all such independent coordinate changes, set $D_{ij} := D_i \cap D_j$ for each edge (i,j). The finite-design structural data carry graph-link copies $\Theta^x_{a,ij}$ exactly for $a \in D_{ij}$. At fixed $a \in D_{ij}$ write Θ^x_{ij}; their discrete vertex frame changes are

$$h^x_{a,i} := g^x_{a,i} = g^x_i(c_a), \qquad a^x_{a,i} := (h^x_{a,i})^{-1}, \tag{4.24}$$

in the notation of (2.47). The contextual coordinate element $h^x_{a,i}$, abbreviated h^x_i at one design point, is unrelated to the relative principal-frame field h_i defined by (2.3). The suppressed one-design-point law is

$$\Theta^{b\prime}_{ij} = h^b_i \Theta^b_{ij} (h^b_j)^{-1}, \qquad \Theta^{m\prime}_{ij} = h^m_i \Theta^m_{ij} (h^m_j)^{-1}. \tag{4.25}$$

Consequently the represented links obey

$$\rho_x(\Theta^{x\prime}_{ij}) = R^x_i \rho_x(\Theta^x_{ij})(R^x_j)^{-1}, \qquad x \in \{b, m\}, \tag{4.26}$$

and the inverse-orientation rule is preserved. Equations (2.47) and (4.25) are therefore the same passive law written with mutually inverse rechoice conventions. [ESTABLISHED]

If instead one structural link is constrained to be identical at every point of D_{ij}, an admissible frame rechoice must preserve that constraint: for every $a, a' \in D_{ij}$ and every constrained edge (i,j),

$$h^x_{a,i} \Theta^x_{ij} (h^x_{a,j})^{-1} = h^x_{a',i} \Theta^x_{ij} (h^x_{a',j})^{-1}. \tag{4.27}$$

[DEFINITION]

At one fixed shared-link datum, define the admissible family to be the solution set of (4.27). Define the group that acts on the entire constrained submodel to be the setwise stabilizer of the shared-link constraint set. A larger transformation law requires an explicitly declared parameter compensation. [DEFINITION]

Proposition 4.7 (Shared-link admissibility need not be closed under composition). For $G = \mathrm{GL}(2,\mathbb{R})$, two design points at which both endpoints participate, one constrained edge, and $\Theta_{ij} = I$, the solution set of (4.27) is not closed under pointwise multiplication. [ESTABLISHED]

Proof. Let $S = \mathrm{diag}(2,1)$ and $C = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. The rechoice $h_{1,i} = h_{1,j} = I$, $h_{2,i} = h_{2,j} = S$ sends the shared link to I at both design points. The rechoice $\widetilde{h}_{a,i} = C$, $\widetilde{h}_{a,j} = I$ sends it to C at both points. For their pointwise product $h\widetilde{h}$, the two transformed links are C and $SCS^{-1} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, respectively, so the product violates (4.27). □

Context-independent rechoices form a sufficient subgroup, but the first rechoice in the proof shows that they are not necessary. [ESTABLISHED]

At the unconstrained finite-design level, declaring the coordinate re-expressions independently gives the passive product $\prod_{(i,a)\in\mathfrak{I}_D}(G \times G)$. If they must arise as values of smooth maps $g^x_i : \mathcal{C}_i \to G$,

the realized group is the image of the corresponding incidence-restricted evaluation map; it equals the displayed product whenever that evaluation map is surjective. For example, this holds when each relevant $\mathcal{C}_i$ is smooth and positive-dimensional, the points $\{c_a : a \in D_i\}$ are distinct, and G is connected, by smooth interpolation in disjoint coordinate neighborhoods. This product records two independently chosen trivializations of associated bundles induced from the same $\mathscr{P}_G$; it is not a second principal structure group. The active principal gauge group is $\mathrm{Aut}_G(\mathscr{P}_G)$ over $\mathrm{id}_{\mathcal{C}}$; one automorphism acts on both associated bundles, with local functions related by (2.12). Its image in the passive product is contained in the pointwise h_i-twisted diagonal and is contained in the ordinary diagonal in the common-frame specialization $h_i = e_G$. Equality with the corresponding pointwise diagonal holds only when the relevant active-gauge evaluation map is surjective; disconnected structure groups or global bundle constraints can obstruct that surjectivity. It is not the principal right G-action itself. A context-independent frame re-expression is the subgroup for which $g^x_{a,i}$ is constant as a ranges over D_i. [DEFINITION]

Hypothesis 4.8 (Kernel covariance). The transformed parameters (θ', X') are declared so that every nonroot factor obeys

$$P^m_{\theta',i}\left(\cdot \mid R^m_{\mathrm{pa}(i)} m_{\mathrm{pa}(i)}, X'\right) = (R^m_i)_{\#} P^m_{\theta,i}\left(\cdot \mid m_{\mathrm{pa}(i)}, X\right), \tag{4.28}$$

$$P^k_{\theta',i}\left(\cdot \mid R^b_{\mathrm{pa}(i)} k_{\mathrm{pa}(i)}, R^m_i m_i, X'\right) = (R^b_i)_{\#} P^k_{\theta,i}\left(\cdot \mid k_{\mathrm{pa}(i)}, m_i, X\right), \tag{4.29}$$

$$P^o_{\theta',i}\left(\cdot \mid R^b_i k_i, R^m_i m_i, X'\right) = P^o_{\theta,i}\left(\cdot \mid k_i, m_i, X\right), \tag{4.30}$$

with the analogous root identities. Every interaction-record kernel also obeys

$$K_a\left(X', R_{\partial a,D} y_{\partial a,D}, \cdot\right) = K_a\left(X, y_{\partial a,D}, \cdot\right), \qquad a \in \mathfrak{A}, \tag{4.31}$$

where $R_{\partial a,D}$ is the product restriction of R to the incident agent blocks. Record coordinates are gauge inert. [HYPOTHESIS]

The second identity is the covariance requirement for the model-to-belief sample kernel. At each root incidence it implies that the induced law map (4.5) obeys the design-point form of (2.15); at a nonroot it remains a parent-indexed kernel law unless an enlarged cross-bundle domain is declared. Neither route uses Φ.

Proposition 4.9 (Joint measure covariance and evidence invariance). Let $T(y, o) = (Ry, o)$, where $R = \bigoplus_{(i,a)\in\mathfrak{I}_D} (R^b_{a,i} \oplus R^m_{a,i})$. Under Hypothesis 4.8,

$$\mathbb{P}_{\theta',V}(\cdot \mid X') = T_{\#}\mathbb{P}_{\theta,V}(\cdot \mid X), \tag{4.32}$$

as probability measures, without a density or reference-measure hypothesis. Because T is the identity on observations, the observation marginals agree. Relative to the common observation

reference measure this gives

$$p_{\theta',V}(o \mid X') = p_{\theta,V}(o \mid X) \quad \text{for } \nu_D^O\text{-almost every } o. \tag{4.33}$$

If latent densities exist and $R_\# \nu_D^Y \ll \nu_D^Y$, define

$$j_R(y') := \frac{d(R_\# \nu_D^Y)}{d\nu_D^Y}(y'). \tag{4.34}$$

Then the density statement is

$$p_{\theta',V}(o, y' \mid X') = p_{\theta,V}(o, R^{-1}y' \mid X) j_R(y') \quad \text{for } (\nu_D^Y \otimes \nu_D^O)\text{-almost every } (y', o). \tag{4.35}$$

On the further differentiable equal-dimensional Euclidean tier, when every represented block is an invertible linear map and ν_D^Y is product Lebesgue measure, $j_R = |\det R|^{-1}$ and hence

$$p_{\theta',V}(o, y' \mid X') = |\det R|^{-1} p_{\theta,V}(o, R^{-1}y' \mid X). \tag{4.36}$$

Equal dimensionality here is between the source and target coordinates within each channel; it does not impose $K = d_m$. The determinant formula is not asserted for mixed or atomic reference measures. [ESTABLISHED]

Proof. Apply (4.28) and (4.29) along the topological ordering of each Γ_a. Finite kernel induction gives

$$\mathbb{P}^Y_{\theta',V,a}(\cdot \mid X') = R_{a\#} \mathbb{P}^Y_{\theta,V,a}(\cdot \mid X),$$

where R_a is the product restriction of R to the block at design point c_a, and the product over design points in (4.8) therefore gives $\mathbb{P}^Y_{\theta',V}(\cdot \mid X') = R_\# \mathbb{P}^Y_{\theta,V}(\cdot \mid X)$. Now let F be a bounded measurable function on the joint latent–observation space. Applying (4.15), then the latent pushforward, gives

$$\begin{aligned}
\int F d\mathbb{P}_{\theta',V} &= \int \mathbb{P}^Y_{\theta,V}(dy \mid X) \int F(Ry, o) \prod_{r \in \mathfrak{A}} K_r \left(X', R_{\partial r,D} y_{\partial r,D}, do_r\right) \\
&= \int \mathbb{P}^Y_{\theta,V}(dy \mid X) \int F(Ry, o) \prod_{r \in \mathfrak{A}} K_r \left(X, y_{\partial r,D}, do_r\right) \\
&= \int F \circ T d\mathbb{P}_{\theta,V},
\end{aligned}$$

where the second equality applies (4.31) to every record factor. This proves (4.32) without using the legacy singleton observation kernel $P^o_{\theta,i}$. Projection to the observation coordinate now shows that the observation measures agree, and their densities satisfy (4.33). The joint measure identity and the Radon–Nikodym representations give

$$\int F(y', o) p_{\theta',V}(o, y' \mid X') \nu_D^Y(dy') \nu_D^O(do) = \int F(Ry, o) p_{\theta,V}(o, y \mid X) \nu_D^Y(dy) \nu_D^O(do).$$

Rewriting the right side against $R_{\#}\nu_D^Y$ and applying (4.34) gives (4.35). On the stated Lebesgue tier, linear change of variables gives $d(R_{\#}\nu_D^Y)/d\nu_D^Y = |\det R|^{-1}$, proving (4.36). □

Evidence invariance is a redundancy constraint, not an empirical prediction: the action changes latent coordinates while leaving the observation law fixed. The linear-Gaussian chapter verifies Hypothesis 4.8 parameter by parameter and then displays the stronger common-frame reduction.

The passive coordinate action on the full parameter space can have nontrivial stabilizers, so its orbit space need not be a manifold. Subtracting the dimension of the coordinate group from a parameter count is unjustified without freeness and properness. Determining generic stabilizers and a regular quotient remains open; nothing later depends on that quotient. [OPEN]

Chapter 5

The Exact Single Evidence Bound

The previous chapter fixed one normalized joint. This chapter introduces one recognition law for the entire population and derives the exact measure-level identity that relates the two. Its primary form is an extended-real lower bound defined for every probability law; the familiar difference of expected log densities is a finite-domain representation of that functional. Conditional and local bounds used later are disintegrations or restrictions of this same construction, not competing definitions of evidence.

The chapter is organized around three things that are easy to get wrong. The recognition law is a joint law and is not recoverable from its coordinate marginals, so a calculation that substitutes marginals into an entropy is computing something else and the difference has a name and a sign. The identity holds under stated hypotheses, one of which, absolute continuity with respect to the posterior, is a genuine restriction and is exactly what later forbids degenerate recognition laws. And the exact coordinates of the alternating scheme are two specific objects; an optimizer direction is a proposal that must pass an acceptance test before it is entitled to the conclusions that attach to a coordinate.

There is also a measure-theoretic qualification that applies to every fixed-observation formula in this chapter. For fixed (θ, X), let $\mathbb{P}^{O}_{\theta,V,X}$, $\mathbf{\Pi}_{\theta,V,o,X}$, and the marginal-full set $\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ be the observation marginal, selected regular conditional probability, and regular-observation set of Chapter 3. Unless an everywhere-defined density and conditional version has been declared as part of the model data, every posterior and ELBO statement below is a $\mathbb{P}^{O}_{\theta,V,X}$-almost-everywhere statement, represented by selecting $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$. Changing a density on a null slice can change exceptional point values but cannot change these almost-everywhere statements.

5.1 The correlated population recognition kernel

Definition 5.1 (Correlated population recognition kernel). Fix the finite design D, the observed value $o \in \mathsf{O}_D$, and the structural value $X \in \mathsf{X}$ of Chapter 3. The population recognition kernel is the

declared measurable selection

$$\mathbb{Q}_V : (\mathsf{O}_D \times \mathsf{X}, \mathscr{O}_D \otimes \mathscr{X}) \longrightarrow \mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D), \qquad (o, X) \longmapsto \mathbb{Q}_{V,o,X} = \mathsf{S}_{o,X}\big((\mathbb{Q}_{i,o,X})_{i\in V}\big) \in \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}). \tag{5.1}$$

Thus $\mathbb{Q}_{V,o,X}(\mathsf{Y}_D) = 1$ and, for every $B \in \mathscr{Y}_D$, the map $(o, X) \mapsto \mathbb{Q}_{V,o,X}(B)$ is $\mathscr{O}_D \otimes \mathscr{X}$-measurable. The product coupling $\bigotimes_{i\in V} \mathbb{Q}_{i,o,X}$ is one admissible restriction and is not the default identity. If the selection is required to lie in a constrained family $\mathrm{Cpl}_{\mathcal{R}}(\mathbb{Q}_{\bullet,o,X})$, nonemptiness of every constrained fiber and measurability of a selector from those fibers are separate hypotheses; the unconstrained product coupling proves neither. [DEFINITION]

The law is permitted to correlate agents with one another, the state channel with the model channel, and one design point with another. No factorization is imposed here; the restrictions that impose one are the subject of Chapter 23, and they are restrictions on this object rather than replacements for it. [DEFINITION]

Write the belief- and model-coordinate marginals as the corresponding pushforwards defined in Definition 3.19. By Proposition 3.24, these marginals do not reconstruct $\mathbb{Q}_{V,o,X}$ whenever the latent space contains at least two nondegenerate real coordinates.

The proposition is a statement about a single pair of coordinates, and the general situation is worse in a specific way. The set of joint laws with a prescribed family of marginals is the Fréchet class of those marginals; it is convex, it contains the product law, and in continuous Euclidean coordinates it is parameterized by the dependence structure, the copula, which the marginals leave entirely free [65]. When second moments exist, every cross-agent, cross-channel, and cross-design covariance block is unspecified by the marginals. The consequence for this chapter is not stylistic. Marginal recognition laws cannot be substituted for the population law in an exact entropy or evidence calculation, and the following makes the error exact rather than merely warning against it.

Theorem 5.2 (Extended total-correlation chain identity). Let $Y = (Y_b)_{b=1}^{B}$ take values in a finite product of standard Borel spaces, let Q be any joint probability law with marginals Q_b, let $(\rho_b)_{b=1}^{B}$ be arbitrary probability laws, and put $R = \bigotimes_b \rho_b$. With

$$\mathrm{TC}(Q) := D_{\mathrm{KL}}\left(Q \,\middle\|\, \bigotimes_b Q_b\right) \in [0, +\infty],$$

the identity

$$D_{\mathrm{KL}}\left(Q \,\middle\|\, \bigotimes_b \rho_b\right) = \mathrm{TC}(Q) + \sum_b D_{\mathrm{KL}}(Q_b \| \rho_b) \tag{5.2}$$

holds in $[0, +\infty]$. It includes both singular branches and never subtracts infinities. [ESTABLISHED]

Proof. For each block choose finite refining partitions $\mathcal{P}_{b,n}$ that generate its sigma-algebra, and let π_n be the resulting finite product quantization. On the finite product alphabet, direct summation

gives

$$D_{\mathrm{KL}}(\pi_{n\#}Q\|\bigotimes_b \pi_{b,n\#}\rho_b) = D_{\mathrm{KL}}(\pi_{n\#}Q\|\bigotimes_b \pi_{b,n\#}Q_b) + \sum_b D_{\mathrm{KL}}(\pi_{b,n\#}Q_b\|\pi_{b,n\#}\rho_b).$$

All displayed terms are nonnegative extended numbers, so this finite identity uses no signed-log cancellation. The refining-partition characterization of relative entropy, (3.21) applied to constant kernels, makes the left side increase to $D_{\mathrm{KL}}(Q\|\bigotimes_b \rho_b)$, the first term on the right increase to $\mathrm{TC}(Q)$, and each of the finitely many remaining terms increase to $D_{\mathrm{KL}}(Q_b\|\rho_b)$. Passing to the monotone limits proves (5.2) in $[0,+\infty]$.

For an explicit branch audit, if $Q \not\ll R$ then the left side is $+\infty$. If some $Q_b \not\ll \rho_b$, the corresponding marginal term is $+\infty$. Otherwise $\bigotimes_b Q_b \ll R$; transitivity shows that $Q \ll \bigotimes_b Q_b$ would contradict $Q \not\ll R$, so $\mathrm{TC}(Q) = +\infty$. Thus the right side is also $+\infty$ in every singular branch. □

The infinite branch is realized by a continuous diagonal, not merely by a defective density calculation. If $U \sim \mathrm{Unif}[0,1]$ and $Q = \mathrm{law}(U,U)$, then both marginals are uniform but Q is singular with respect to $Q_1 \otimes Q_2$, so $\mathrm{TC}(Q) = +\infty$. Taking $\rho_b = Q_b$ makes (5.2) read $+\infty = +\infty + 0 + 0$ in the extended nonnegative reals.

Proposition 5.3 (Finite-density total correlation and the two signs). Partition the coordinates of Y into blocks indexed by b, let $\mathbb{Q}_{V,o,X}$ have density $q_{V,o,X}$ with respect to ν_D^Y and block marginal densities q_b, and suppose the joint and all block differential entropies are finite. Let

$$\mathrm{TC}(\mathbb{Q}_{V,o,X}) = D_{\mathrm{KL}}\left(\mathbb{Q}_{V,o,X}\middle\|\bigotimes_b \mathbb{Q}_{V,o,X,b}\right) \geq 0 \tag{5.3}$$

be the total correlation of the blocks. Then

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log q_{V,o,X}(Y)\right] = \sum_b \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log q_b(Y_b)\right] + \mathrm{TC}(\mathbb{Q}_{V,o,X}), \tag{5.4}$$

with equality of the two log-density expectations exactly when the blocks are independent under $\mathbb{Q}_{V,o,X}$. For the following free-energy and pseudo-ELBO identities, assume in addition

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left|\log p_{\theta,V}(o,Y \mid X)\right| < \infty,$$

so the common expected negative-log-joint term and both finite entropy terms define finite real quantities. Define the substituted free energy and its negative pseudo-ELBO by

$$\widetilde{\mathcal{F}} := \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log p_{\theta,V}(o,Y \mid X) + \sum_b \log q_b(Y_b)\right], \tag{5.5}$$

$$\widetilde{\mathcal{L}} := -\widetilde{\mathcal{F}}. \tag{5.6}$$

Then

$$\widetilde{\mathcal{F}} = \mathcal{F}_V - \mathrm{TC}(\mathbb{Q}_{V,o,X}), \qquad \widetilde{\mathcal{L}} = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + \mathrm{TC}(\mathbb{Q}_{V,o,X}). \tag{5.7}$$

Thus it is the *substituted free energy* that is smaller by total correlation, while its negative *pseudo-ELBO* is larger by total correlation and need not be a lower bound on the log evidence. In the Gaussian case with joint covariance C and diagonal blocks C_{bb},

$$\mathrm{TC}(\mathbb{Q}_{V,o,X}) = \tfrac{1}{2}\left(\sum_b \log\det C_{bb} - \log\det C\right). \tag{5.8}$$

[ESTABLISHED]

Proof. This is the finite-density corollary of Theorem 5.2 with $\rho_b = \mathbb{Q}_{V,o,X,b}$, after the stated entropy hypotheses permit the logarithms to be split. Expanding gives $\mathrm{TC}(\mathbb{Q}_{V,o,X}) = \mathbb{E}_{\mathbb{Q}_{V,o,X}}[\log q_{V,o,X}] - \sum_b \mathbb{E}_{\mathbb{Q}_{V,o,X}}[\log q_b]$, which is (5.4). Subtracting (5.5) from $\mathcal{F}_V = \mathbb{E}_{\mathbb{Q}_{V,o,X}}[-\log p_{\theta,V} + \log q_{V,o,X}]$ gives $\mathrm{TC}(\mathbb{Q}_{V,o,X})$, and negation gives the second identity in (5.7). For (5.8), substitute the Gaussian differential entropy $\frac{1}{2}\log\det(2\pi e C)$ for the joint and for each block; the dimensional constants cancel because the block dimensions sum to d. □

The sign in Proposition 5.3 matters and is the reason the point is made here rather than in a remark. Substituting marginal entropies produces a smaller free-energy functional and, after negation, a larger pseudo-ELBO that is not a bound at all. A scheme that reports the latter as an evidence lower bound is reporting a quantity that can exceed the log evidence, and by an amount that grows with exactly the correlation the recognition law was introduced to represent.

5.2 Hypotheses

Hypothesis 5.4 (Evidence-domain hypotheses). The extended identity of Section 5.3 uses (H1)–(H2); its classical split into two expected log densities additionally uses (H3)–(H4). The selection is subject to the standing qualification $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$; equivalently, the measure-level theorem holds for $\mathbb{P}^{O}_{\theta,V,X}$-almost every observation. Together (H1)–(H4) are the consolidated classical evidence domain and reproduce Hypotheses 3.17 and 3.18. [HYPOTHESIS]

(H1) Types and conditional version. The constructed joint kernel $\mathbb{P}_{\theta,V}(\cdot \mid X)$ of Chapter 4 and the selected recognition kernel $\mathbb{Q}_{V,o,X}$ of Definition 5.1 are normalized and measurable as declared. The generative density $p_{\theta,V}(o, y \mid X)$ exists relative to the declared reference, and $\mathbf{\Pi}_{\theta,V,o,X}$ denotes the selected regular conditional probability represented by (3.19) on $\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$. No density of $\mathbb{Q}_{V,o,X}$ is required at the extended tier. An everywhere pointwise version may replace the almost-everywhere convention only when that version is declared as model data.

(H2) Positive finite evidence. $0 < p_{\theta,V}(o \mid X) < \infty$. This excludes observed values that the model assigns zero density, at which the conditional posterior is undefined and there is nothing to approximate. It is used to divide by the evidence in (5.19).

(H3) Absolute continuity and recognition density for the classical split. $\mathbb{Q}_{V,o,X} \ll \mathbf{\Pi}_{\theta,V,o,X}$. Since the selected posterior slice is dominated by ν_D^Y, this implies $\mathbb{Q}_{V,o,X} \ll \nu_D^Y$; write its RN density as $q_{V,o,X}(y)$. For parameterized recognition kernels, Theorem 3.10 and corollary 3.11 supplies a jointly measurable version under their stated hypotheses. This is a genuine restriction on the

finite classical representation, but not on the extended functional, which assigns value $-\infty$ to the singular branch. Proposition 5.5 below makes the exclusion precise.

(H4) Log-integrability.

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[|\log p_{\theta,V}(o, Y \mid X)| + |\log q_{V,o,X}(Y)|\right] < \infty. \tag{5.9}$$

This is not implied by (H1) through (H3), and a witness settles the point rather than leaving it asserted: take $d = 1$ with a standard Gaussian conditional posterior and a standard Cauchy recognition law. Both have strictly positive Lebesgue densities, so (H1) through (H3) hold; but $\log p_{\theta,V}(o, y \mid X)$ differs from $-y^2/2$ by a constant, and the Cauchy law has no second moment, so (5.9) fails. Relative entropy and the extended ELBO remain well defined without (H4). What fails is only the split into two separately finite expectations: the formal difference can be an undefined $\infty - \infty$.

Hypothesis (H3) deserves its own statement, because it is the one with consequences later.

Proposition 5.5 (Subspace support violates absolute continuity). Suppose the conditional posterior $\mathbf{\Pi}_{\theta,V,o,X}$ is dominated by Lebesgue measure on $\mathbb{R}^d$, as it is in the linear-Gaussian realization of Chapter 4 with the positive definite covariances (21.9), and suppose $\mathbb{Q}_{V,o,X}$ is supported on a proper affine subspace $\mathcal{S} \subsetneq \mathbb{R}^d$, meaning $\mathbb{Q}_{V,o,X}(\mathcal{S}) = 1$ with $\dim \mathcal{S} < d$. Then (H3) fails and

$$D_{\mathrm{KL}}\left(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}\right) = +\infty. \tag{5.10}$$

[ESTABLISHED]

Proof. A proper affine subspace of $\mathbb{R}^d$ is Lebesgue null, so $\mathbf{\Pi}_{\theta,V,o,X}(\mathcal{S}) = 0$ while $\mathbb{Q}_{V,o,X}(\mathcal{S}) = 1$. Absolute continuity fails on the set $\mathcal{S}$. By the definition of relative entropy for measures that are not absolutely continuous, the divergence is $+\infty$ [25, 51]. □

This is the exact mechanism that forbids subspace-supported recognition laws, and its scope should be stated exactly. It does not say that constraining the recognition law is illegitimate; it says that a constraint which collapses the support onto a lower-dimensional set puts the law outside the family for which the bound is finite, so the resulting cost is not a large finite number to be traded against something else but an infinite one. Chapter 23 takes this up in the case that matters, where a coarse-graining is expressed by tying groups of coordinates to a common value: the tied law is supported on the range of an aggregation matrix, hence exactly of the form excluded here, and the admissible finite operation constrains the recognition mean instead, with a cost that the same chapter computes exactly. The forward reference is deliberate; the present chapter supplies the reason, not the application.

5.3 The exact evidence identity

Definition 5.6 (Slice measure and extended ELBO). Assume (H1)–(H2), fix $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$, and abbreviate the finite latent slice measure, its mass, and its normalization by

$$M_{\theta,V,o,X}(B) := \int_B p_{\theta,V}(o, y \mid X)\nu_D^Y(dy) = p_{\theta,V}(o \mid X)\mathbf{\Pi}_{\theta,V,o,X}(B), \qquad p_{\theta,V}(o \mid X) = M_{\theta,V,o,X}(\mathsf{Y}_D) \in (0, \infty). \tag{5.11}$$

For the selected population recognition law, define

$$\mathcal{L}_V^{\mathrm{ext}}(\mathbb{Q}_{V,o,X}; X, o) := \log p_{\theta,V}(o \mid X) - \mathrm{KL}\left(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}\right) \in [-\infty, \log p_{\theta,V}(o \mid X)] \tag{5.12}$$

and

$$\mathcal{F}_V^{\mathrm{ext}}[\mathbb{Q}_{V,o,X}; X, o] := -\mathcal{L}_V^{\mathrm{ext}}(\mathbb{Q}_{V,o,X}; X, o) = -\log p_{\theta,V}(o \mid X) + D_{\mathrm{KL}}\big(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}\big). \tag{5.13}$$

The same formulas define $\mathcal{L}_V^{\mathrm{ext}}(Q; X, o)$ and $\mathcal{F}_V^{\mathrm{ext}}[Q; X, o]$ for an arbitrary theorem-local probability law Q on $(\mathsf{Y}_D, \mathscr{Y}_D)$. These are well-defined extended-real quantities because $\log p_{\theta,V}(o \mid X)$ is finite and relative entropy lies in $[0, +\infty]$. [DEFINITION]

Theorem 5.7 (Extended ELBO gap identity). For every probability law Q on the latent space,

$$\log p_{\theta,V}(o \mid X) - \mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = \mathcal{F}_V^{\mathrm{ext}}[Q; X, o] + \log p_{\theta,V}(o \mid X) = D_{\mathrm{KL}}\big(Q \| \mathbf{\Pi}_{\theta,V,o,X}\big). \tag{5.14}$$

Consequently $\mathcal{L}_V^{\mathrm{ext}} \le \log p_{\theta,V}(o \mid X)$, with equality exactly at $Q = \mathbf{\Pi}_{\theta,V,o,X}$ as full probability measures. No unconditional formula of the form $\log p_{\theta,V} = \mathcal{L}_V^{\mathrm{ext}} + D_{\mathrm{KL}}$ is asserted, because its right side is undefined when $\mathcal{L}_V^{\mathrm{ext}} = -\infty$ and $D_{\mathrm{KL}} = +\infty$. [ESTABLISHED]

Proof. Both equalities are direct substitutions from (5.12)–(5.13); they only add or subtract the finite number $\log p_{\theta,V}(o \mid X)$. Nonnegativity and the zero condition for relative entropy give the bound and equality case. □

Proposition 5.8 (Relative-log representation). If $Q \ll M_{\theta,V,o,X}$ and $r = dQ/dM_{\theta,V,o,X}$, then

$$\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = -\int_{\mathsf{Y}_D} \log r(y) Q(dy), \qquad \int_{\{r<1\}} -\log r dQ = \int_{\{r<1\}} -r \log r dM_{\theta,V,o,X} \le \frac{p_{\theta,V}(o \mid X)}{e}. \tag{5.15}$$

Thus the integral is well defined in $(-\infty, +\infty]$ and its negative lies in $[-\infty, +\infty)$. If $Q \not\ll M_{\theta,V,o,X}$, then $\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = -\infty$. [ESTABLISHED]

Proof. Because $\mathbf{\Pi}_{\theta,V,o,X} = M_{\theta,V,o,X}/p_{\theta,V}(o \mid X)$, the RN chain rule gives $dQ/d\mathbf{\Pi}_{\theta,V,o,X} = p_{\theta,V}(o \mid X)r$ when $Q \ll M_{\theta,V,o,X}$. The negative part of $\log r$ is Q-integrable by (5.15), since $-u \log u \le 1/e$ on $0 \le u \le 1$ and $M_{\theta,V,o,X}(\mathsf{Y}_D) = p_{\theta,V}(o \mid X)$. Hence

$$D_{\mathrm{KL}}(Q \| \mathbf{\Pi}_{\theta,V,o,X}) = \int \log\big(p_{\theta,V}(o \mid X)r\big) dQ = \log p_{\theta,V}(o \mid X) + \int \log r dQ$$

is a legitimate extended identity, and substitution into (5.12) proves the first branch. Since $M_{\theta,V,o,X}$ and $\mathbf{\Pi}_{\theta,V,o,X}$ have the same null sets, $Q \not\ll M_{\theta,V,o,X}$ implies $D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X}) = +\infty$, proving the second. □

Under the full classical domain Hypothesis 5.4, the relative-log functional has the separately integrable density representation

$$\mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) = \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log p_{\theta,V}(o, Y \mid X) - \log q_{V,o,X}(Y)\right]. \tag{5.16}$$

The notation keeps the fixed observation o explicit; the parameter subscript is restored on $\mathcal{L}_{V,\vartheta}$ whenever two parameter values are compared. The variational free energy is the opposite sign,

$$\mathcal{F}_V[\mathbb{Q}_{V,o,X}; X, o] = -\mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o). \tag{5.17}$$

In this domain $\mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) = \mathcal{L}_V^{\mathrm{ext}}(\mathbb{Q}_{V,o,X}; X, o)$ and $\mathcal{F}_V = \mathcal{F}_V^{\mathrm{ext}}$; outside it only the extended definitions are available.

Theorem 5.9 (Classical finite exact evidence identity). Under Hypothesis 5.4, for $\mathbb{P}^O_{\theta,V,X}$-almost every observation o (equivalently, for the selected $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$),

$$\log p_{\theta,V}(o \mid X) = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + D_{\mathrm{KL}}\left(\mathbb{Q}_{V,o,X}\|\mathbf{\Pi}_{\theta,V,o,X}\right). \tag{5.18}$$

[ESTABLISHED]

Proof. For $B \in \mathscr{Y}_D$, the regular-observation construction (3.19), together with (H1) and (H2), gives the dominated conditional posterior

$$\mathbf{\Pi}_{\theta,V,o,X}(B) = \frac{1}{p_{\theta,V}(o \mid X)} \int_B p_{\theta,V}(o, y \mid X)\nu_D^Y(dy), \qquad \frac{d\mathbf{\Pi}_{\theta,V,o,X}}{d\nu_D^Y}(y) = \frac{p_{\theta,V}(o, y \mid X)}{p_{\theta,V}(o \mid X)}, \tag{5.19}$$

which is normalized because the denominator is the evidence. Writing $q_{V,o,X}(y) = d\mathbb{Q}_{V,o,X}/d\nu_D^Y$ and using (H3) with the Radon–Nikodym chain rule gives, $\mathbb{Q}_{V,o,X}$-almost surely,

$$\frac{d\mathbb{Q}_{V,o,X}}{d\mathbf{\Pi}_{\theta,V,o,X}}(y) = \frac{q_{V,o,X}(y)p_{\theta,V}(o \mid X)}{p_{\theta,V}(o, y \mid X)}. \tag{5.20}$$

Hypothesis (H4) permits integration of its logarithm term by term, so

$$\begin{aligned} D_{\mathrm{KL}}\left(\mathbb{Q}_{V,o,X}\|\mathbf{\Pi}_{\theta,V,o,X}\right) &= \int_{\mathsf{Y}_D} \mathbb{Q}_{V,o,X}(dy) \log \frac{d\mathbb{Q}_{V,o,X}}{d\mathbf{\Pi}_{\theta,V,o,X}}(y) \\ &= \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log q_{V,o,X}(Y) - \log p_{\theta,V}(o, Y \mid X)\right] + \log p_{\theta,V}(o \mid X) \\ &= -\mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + \log p_{\theta,V}(o \mid X). \end{aligned} \tag{5.21}$$

Rearranging gives (5.18). □

The almost-everywhere qualifier is part of the theorem, not a regularity detail suppressed by notation. If $\widetilde{p}_{\theta,V}$ is another joint-density version, Fubini's theorem gives equality of the two slices for ν_D^Y-almost every y outside a ν_D^O-null set of observations, and the two evidence and posterior versions therefore agree outside a $\mathbb{P}^O_{\theta,V,X}$-null set. At an exceptional observation, the values of (5.19) and (5.16) can be changed without changing the joint measure. The theorem consequently makes no version-independent claim there.

Every corollary and coordinate statement below inherits this regular-observation qualification. For a comparison of finitely many parameter values, the observation must lie in the intersection of their marginal-full regular sets. An optimization over an uncountable parameter space requires either a common full-measure set proved for that family or jointly measurable pointwise density and conditional versions declared as part of the parameterized model; separate almost-everywhere versions for each parameter do not by themselves define a pointwise objective on the whole parameter space.

Corollary 5.10 (Lower bound and exact equality condition). Under Hypothesis 5.4, $\mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) \leq \log p_{\theta,V}(o \mid X)$, and equality holds if and only if

$$\mathbb{Q}_{V,o,X} = \mathbf{\Pi}_{\theta,V,o,X} \qquad \text{as probability measures on } (\mathsf{Y}_D, \mathscr{Y}_D). \tag{5.22}$$

Equality of all coordinate marginals is not sufficient. [ESTABLISHED]

Proof. The inequality is nonnegativity of relative entropy. For the equality case, strict convexity of $u \mapsto u \log u$ gives that the divergence vanishes exactly when the Radon–Nikodym derivative (5.20) equals one $\mathbf{\Pi}_{\theta,V,o,X}$-almost surely, which is equality of the two full measures [25, 51]. Insufficiency of marginal agreement is Proposition 3.24, whose Gaussian witnesses have the same coordinate marginals but are distinct measures. □

5.4 The general restriction principle

Fix (θ, X), select $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ with $0 < p_{\theta,V}(o \mid X) < \infty$, and write $P^\star = \mathbf{\Pi}_{\theta,V,o,X}$. For every probability law Q on the latent space,

$$\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = \log p_{\theta,V}(o \mid X) - D_{\mathrm{KL}}(Q \| P^\star). \tag{5.23}$$

Proposition 5.11 (Restriction principle). Let $\mathcal{Q}' \subseteq \mathcal{Q}$ be arbitrary sets of probability laws. Then

$$\sup_{Q \in \mathcal{Q}'} \mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = \log p_{\theta,V}(o \mid X) - \inf_{Q \in \mathcal{Q}'} D_{\mathrm{KL}}(Q \| P^\star), \tag{5.24}$$

and set inclusion gives

$$\sup_{Q \in \mathcal{Q}'} \mathcal{L}_V^{\mathrm{ext}}(Q; X, o) \leq \sup_{Q \in \mathcal{Q}} \mathcal{L}_V^{\mathrm{ext}}(Q; X, o).$$

If both divergence infima below are finite, their optimized loss is the finite identity

$$\sup_{\mathcal{Q}} \mathcal{L}_V^{\mathrm{ext}} - \sup_{\mathcal{Q}'} \mathcal{L}_V^{\mathrm{ext}} = \inf_{\mathcal{Q}'} D_{\mathrm{KL}}(Q\|P^\star) - \inf_{\mathcal{Q}} D_{\mathrm{KL}}(Q\|P^\star) \geq 0. \tag{5.25}$$

The identities do not require attainment of the suprema or infima. [ESTABLISHED]

Proof. Equation (5.23) holds pointwise in Q. Taking suprema converts a finite constant minus a nonnegative extended divergence into that constant minus its infimum, including the value $-\infty$ when the infimum is $+\infty$. Set inclusion proves monotonicity. Only when both infima are finite is subtraction of the two supremum identities legitimate, and it gives (5.25). □

Constancy of the evidence is load bearing: the generative kernel is fixed once (θ, X) is fixed and does not read Q. If a proposed model changes with the recognition law, Proposition 4.3 and Proposition 4.4 show that no common evidence comparison follows. Thus every later closed-form "restriction cost" inherits its meaning from the fixed-target hypothesis.

Infinite-cost restrictions. Equation (5.25) is written only on its finite domain. If each $Q \in \mathcal{Q}'$ has a measurable set S_Q with

$$Q(S_Q) = 1, \qquad P^\star(S_Q) = 0, \tag{5.26}$$

then $Q \not\ll P^\star$ and $D_{\mathrm{KL}}(Q\|P^\star) = +\infty$. A proper closed support subset alone is insufficient: the uniform law on $[0, \frac{1}{2}]$ is absolutely continuous with respect to the uniform law on $[0, 1]$ and has divergence $\log 2$. The null-set condition, not proper closedness, is the obstruction. [ESTABLISHED]

Orientation, separation, and attainment. The infimum is a reverse information projection because Q is the first argument of the divergence. Excluding $P^\star$ does not ensure a positive or attained optimum: the family $\{\mathcal{N}(\mu, 1) : \mu > 0\}$ excludes $\mathcal{N}(0, 1)$, but its gaps $\mu^2/2$ have unattained infimum zero. A positive optimized gap requires $\inf_{Q\in\mathcal{Q}} D_{\mathrm{KL}}(Q\|P^\star) > 0$. Nor does uniqueness follow from convexity of a product family, because mixtures of product laws need not be products. The Gaussian restrictions of Chapter 23 instead prove convexity and coercivity in the moment parameters they optimize. [ESTABLISHED]

Corollary 5.12 (Independent channel-frame invariance). Let

$$T : \mathsf{Y}_D \times \mathsf{O}_D \longrightarrow \mathsf{Y}_D \times \mathsf{O}_D, \qquad T(y, o) = (Ry, o),$$

be an independent belief–model frame re-expression as in Chapter 4, where $R = \bigoplus_{(i,a)\in\mathfrak{I}_D} (R^b_{a,i} \oplus R^m_{a,i})$ is the assembled represented bimeasurable bijection of Y_D. Only admitted incidence blocks occur in this direct sum; no frame map is quantified off $\mathfrak{I}_D$. Transport the entire slice by

$$M'_{\theta',V,o,X'} = R_\# M_{\theta,V,o,X}, \qquad \mathbf{\Pi}_{\theta',V,o,X'} = R_\# \mathbf{\Pi}_{\theta,V,o,X}, \qquad \mathbb{Q}'_{V,o,X'} = R_\# \mathbb{Q}_{V,o,X}.$$

Here $\mathbb{Q}_{V,o,X}$ is selected before the frame change and then transported as a complete joint law. The primed recognition symbol in this corollary is defined by that pushforward, not by rerunning the

selector on transformed marginals. Then $M'_{\theta',V,o,X'}(\mathsf{Y}_D) = M_{\theta,V,o,X}(\mathsf{Y}_D) = p_{\theta,V}(o \mid X)$ and

$$D_{\mathrm{KL}}\left(\mathbb{Q}'_{V,o,X'} \middle\| \mathbf{\Pi}_{\theta',V,o,X'}\right) = D_{\mathrm{KL}}\left(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}\right), \qquad \mathcal{L}'^{\mathrm{ext}}_V(\mathbb{Q}'_{V,o,X'}; X', o) = \mathcal{L}^{\mathrm{ext}}_V(\mathbb{Q}_{V,o,X}; X, o). \tag{5.27}$$

This includes the singular branch. Whenever both sides satisfy the classical split hypothesis, the same equality holds for $\mathcal{L}'_V$ and $\mathcal{L}_V$. [ESTABLISHED]

Proof. By the joint pushforward identity of Chapter 4 and the fact that T acts as the identity on the observation coordinate, the conditional posterior transforms as $\mathbf{\Pi}_{\theta',V,o,X'} = R_{\#}\mathbf{\Pi}_{\theta,V,o,X}$. Relative entropy is invariant under a common bimeasurable bijection. In the absolutely continuous branch, the RN derivative of the pushforwards is the original derivative composed with R^{-1}, and change of variables preserves its integral. In the singular branch, a posterior-null set carrying positive $\mathbb{Q}_{V,o,X}$-mass maps to a pushed-posterior-null set carrying the same positive pushed-recognition mass, so both divergences equal $+\infty$. The slice mass is invariant under pushforward, and substitution in (5.12) proves the second identity without subtracting infinities. The classical claim follows from its equality with the extended functional on the split domain. □

Corollary 5.12 states that the bound is unchanged under common transport of the fixed joint slice, posterior, and already selected recognition coupling. It does not assert that rerunning a coordinate-dependent selector on transformed local marginals produces the pushed coupling. That identification requires a separate selector-equivariance hypothesis, which is not assumed here. The invariants used later are accordingly stated in generalized-eigenvalue form rather than in terms of absolute eigenvalues.

5.5 The factorwise decomposition

Hypothesis 5.13 (Factorwise log-integrability). Write $d \in \{1, \dots, M\}$ for the design index, retain $a \in \mathfrak{A}$ for the interaction-record index, and let $\mathcal{R} \subseteq V$ be the root set. To display the factor expectations separately, assume in this section that the absolute logarithm of every component density in the finite directed factorization and every attached record density is $\mathbb{Q}_{V,o,X}$-integrable. [HYPOTHESIS]

This is strictly stronger than (H4), which controls only the complete log joint, and it excludes cancellation between individually undefined extended-real summands; a decomposition displayed without it can be a formal rearrangement of divergent quantities. [ESTABLISHED]

Proposition 5.14 (Exact factorwise decomposition). Under Hypotheses 5.4 and 5.13, substituting the normalized latent density (4.10) and the record attachment (4.17) into (5.17) yields exactly

one decomposition,

$$\begin{aligned}\mathcal{F}_V[\mathbb{Q}_{V,o,X}; X, o] = &\sum_{d=1}^{M}\sum_{r\in\mathcal{R}\cap V_d}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log p^{m}_{\theta,r,d}(m_{r,d}\mid X)\right]\\ &+\sum_{d=1}^{M}\sum_{r\in\mathcal{R}\cap V_d}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log p^{k}_{\theta,r,d}(k_{r,d}\mid m_{r,d}, X)\right]\\ &+\sum_{d=1}^{M}\sum_{i\in V_d\setminus\mathcal{R}}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log p^{m}_{\theta,i,d}\left(m_{i,d}\mid m_{\mathrm{pa}(i),d}, X\right)\right]\\ &+\sum_{d=1}^{M}\sum_{i\in V_d\setminus\mathcal{R}}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log p^{k}_{\theta,i,d}\left(k_{i,d}\mid k_{\mathrm{pa}(i),d}, m_{i,d}, X\right)\right]\\ &+\sum_{a\in\mathfrak{A}}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log \ell_a(o_a\mid X, Y_{\partial a,D})\right]\\ &+\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log q_{V,o,X}(Y)\right].\end{aligned} \tag{5.28}$$

The generative sectors contain $2|\mathfrak{I}_D|+|\mathfrak{A}|$ factors in total, equal to $2MN+|\mathfrak{A}|$ on the reference rectangular design. The final line is one full-joint recognition log-density, equivalently the negative joint differential entropy relative to ν_D^Y when that entropy is finite. It is not a sum of marginal entropies. [ESTABLISHED]

Proof. Write $h_{\theta,d}$ for the one-design-point latent log density obtained from (4.9). Equations (4.10) and (4.17) give

$$\log p_{\theta,V}(o, y\mid X) = \sum_{d=1}^{M} h_{\theta,d}(y_d; X) + \sum_{a\in\mathfrak{A}}\log \ell_a(o_a\mid X, y_{\partial a,D}).$$

Taking $\mathbb{Q}_{V,o,X}$-expectations of the negative complete log density and adding the one joint term $\mathbb{E}_{\mathbb{Q}_{V,o,X}}[\log q_{V,o,X}]$ reproduces (5.28) term by term. Denoting that right side by $\mathcal{F}_V^{\mathrm{fact}}$, the residual is

$$\begin{aligned}\mathcal{F}_V[\mathbb{Q}_{V,o,X}; X, o] - \mathcal{F}_V^{\mathrm{fact}} = &\,\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log q_{V,o,X}(Y) - \log p_{\theta,V}(o, Y\mid X)\right]\\ &-\left\{\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log q_{V,o,X}(Y)\right] - \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\sum_{d=1}^{M} h_{\theta,d}(Y_d; X) + \sum_{a\in\mathfrak{A}}\log \ell_a(o_a\mid X, Y_{\partial a,D})\right]\right\} = 0,\end{aligned} \tag{5.29}$$

where the same joint recognition term appears once on each side and cancels as a whole, and no step replaces it by marginal entropies. Every incidence contributes one model and one state factor, and every owned record contributes one normalized record factor. Root state bridges replace nonroot state transitions at roots; they are not additional factors. □

Two readings of (5.28) must be resisted. The first is the substitution warned against in Proposition 5.3: the last line is a single number determined by the joint law, and replacing it by a sum of marginal entropies inflates the bound by the total correlation. The second is that the decomposition

licenses independent optimization of its terms. Every expectation is taken under the same joint $\mathbb{Q}_{V,o,X}$, so the terms are coupled through their common measure; the decomposition is an exact rewriting of one functional, not a decoupling of it into separately minimizable pieces.

5.6 Exact coordinates and accepted proposals

All coordinate statements below refer to the same normalized family $\mathbb{P}_{\theta,V}$ of Chapter 4. Because Y_D is a finite product of Euclidean and discrete spaces it is a standard Borel space, so regular conditional distributions of the posterior exist and the relative entropy chain rule is available [49].

Proposition 5.15 (Exact E-coordinate in the extended domain). Let $\mathcal{B}$ be a block of latent coordinates with complement $-\mathcal{B}$, put

$$P := \mathbf{\Pi}_{\theta,V,o,X}, \qquad P_- := (\mathrm{pr}_{-\mathcal{B}})_{\#}\mathbf{\Pi}_{\theta,V,o,X}, \qquad R := (\mathrm{pr}_{-\mathcal{B}})_{\#}\mathbb{Q}_{V,o,X},$$

with R an arbitrary probability law. Choose a Markov-kernel version $C(dY_{\mathcal{B}} \mid Y_{-\mathcal{B}})$ that agrees P_--almost everywhere with a regular conditional law of P, completing it arbitrarily off that full set. Then among *all* probability laws with complement marginal R, the extended-real divergence from P has minimum $D_{\mathrm{KL}}(R\|P_-) \in [0, +\infty]$, attained by

$$\mathbb{Q}^{\mathrm{new}}_{V,o,X}\left(dY_{\mathcal{B}}, dY_{-\mathcal{B}}\right) = C\left(dY_{\mathcal{B}} \mid Y_{-\mathcal{B}}\right) R\left(dY_{-\mathcal{B}}\right). \tag{5.30}$$

It therefore maximizes $\mathcal{L}^{\mathrm{ext}}_V$ on that marginal class. The minimizer is unique as a joint measure when $D_{\mathrm{KL}}(R\|P_-) < \infty$; when this divergence is $+\infty$, every law in the class has divergence $+\infty$ and extended ELBO $-\infty$, so uniqueness is not asserted. If, in addition, the minimum is finite and the new law satisfies the log-integrability condition

$$\mathbb{E}_{\mathbb{Q}^{\mathrm{new}}_{V,o,X}}\left[\left|\log p_{\theta,V}(o, Y \mid X)\right| + \left|\log q^{\mathrm{new}}_{V,o,X}(Y)\right|\right] < \infty, \tag{5.31}$$

then it satisfies (H3)–(H4) and maximizes the classical $\mathcal{L}_V$ among the laws with complement marginal R that satisfy Hypothesis 5.4. Finite complement relative entropy by itself does not imply (5.31). [ESTABLISHED]

Proof. Projection onto $Y_{-\mathcal{B}}$ and the extended data-processing inequality give

$$D_{\mathrm{KL}}(Q\|P) \geq D_{\mathrm{KL}}(R\|P_-)$$

for every theorem-local law Q in the marginal class. If $D_{\mathrm{KL}}(R\|P_-) < \infty$, then $R \ll P_-$, the completion of C is irrelevant R-almost everywhere, and the relative-entropy chain rule on a standard Borel product gives

$$D_{\mathrm{KL}}\left(Q\|P\right) = D_{\mathrm{KL}}\left(R\|P_-\right) + \mathbb{E}_R\left[D_{\mathrm{KL}}\left(Q_{\mathcal{B}|-\mathcal{B}}\middle\|P(\cdot \mid Y_{-\mathcal{B}})\right)\right]. \tag{5.32}$$

The second term is an integral of nonnegative extended-real quantities and is zero exactly when the two conditional laws agree for R-almost every $Y_{-\mathcal{B}}$. For (5.30) it vanishes, giving the minimum and uniqueness. Moreover, if $r = dR/dP_-$, direct disintegration gives

$$\frac{d\mathbb{Q}^{\mathrm{new}}_{V,o,X}}{dP} = r(Y_{-\mathcal{B}}) \quad P\text{-almost everywhere},$$

so $\mathbb{Q}^{\mathrm{new}}_{V,o,X} \ll P$ and its relative entropy is exactly $D_{\mathrm{KL}}(R\|P_-)$. If instead $D_{\mathrm{KL}}(R\|P_-) = +\infty$, data processing makes every candidate divergence $+\infty$, including the constructed law, so the extended optimum is attained but is nonunique. In both cases (5.12) converts divergence minimization into extended-ELBO maximization. Under the separate finite and log-integrability assumptions, Theorem 5.9 gives the classical statement. □

The classical extra obligation cannot be recovered from complement absolute continuity. First take P_- to be standard Gaussian and R standard Cauchy on $\mathbb{R}$. Then $R \ll P_-$, but $\log(dR/dP_-)(x)$ contains $x^2/2$ up to logarithmic and constant terms, and the Cauchy second moment is infinite; hence the extended coordinate is well defined but has value $-\infty$. Second, finite relative entropy still does not imply (H4). On the countable space $\{2, 3, \ldots\}$, let

$$P_-(\{n\}) = R(\{n\}) = \frac{c}{n(\log n)^2},$$

where c normalizes the series. Then $D_{\mathrm{KL}}(R\|P_-) = 0$, but for all sufficiently large n, $|\log P_-(\{n\})| \geq \frac{1}{2}\log n$, and therefore

$$\sum_{n\geq 2} P_-(\{n\})\,|\log P_-(\{n\})| \geq \frac{c}{2}\sum_{n \text{ large}} \frac{1}{n\log n} = +\infty.$$

With a trivial observation and conditional block this is precisely a $\mathbb{Q}^{\mathrm{new}}_{V,o,X} = P$ of zero relative entropy for which the separate log expectations in (H4) are not integrable. Equation (5.31) is therefore an independent hypothesis, not a consequence of the KL chain rule.

The exact full E-step is the special case $\mathcal{B}$ equal to all coordinates, giving $\mathbb{Q}^{\mathrm{new}}_{V,o,X} = \mathbf{\Pi}_{\theta,V,o,X}$ and a zero extended gap by Theorem 5.7; (H4) is needed only for its classical split. A product-family coordinate is a different object entirely: it optimizes within a restricted family, is not given by (5.30), and leaves a gap that Chapter 23 quantifies.

Proposition 5.16 (Exact M-coordinate). Hold $\mathbb{Q}_{V,o,X}$ fixed. Suppose the parameterized family supplies the common regular-observation set or the jointly measurable pointwise versions required above, and suppose Hypothesis 5.4 holds for $\mathbb{Q}_{V,o,X}$ at every candidate parameter being compared. Specifically,

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[|\log q_{V,o,X}(Y)|\right] < \infty, \qquad \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[|\log p_{\vartheta,V}(o, Y \mid X)|\right] < \infty$$

for every such ϑ. If a maximizer exists in the declared parameter space, then any

$$\theta^{\mathrm{new}} \in \operatorname*{argmax}_{\vartheta} \mathbb{E}_{\mathbb{Q}_{V,o,X}} \left[\log p_{\vartheta,V}(o, Y \mid X)\right] \tag{5.33}$$

also maximizes $\mathcal{L}_{V,\vartheta}(\mathbb{Q}_{V,o,X}; X, o)$ over the same space, because $\mathbb{E}_{\mathbb{Q}_{V,o,X}}[\log q_{V,o,X}(Y)]$ does not depend on ϑ. [ESTABLISHED]

The finiteness assumption is load-bearing. On $\{2, 3, \ldots\}$, let

$$q_n = \frac{c}{n(\log n)^2}, \qquad h_n = \mathbf{1}_{\{n=2\}} - q_2, \qquad r_n = 1 + \varepsilon h_n,$$

where c normalizes q and $0 < \varepsilon < 1/q_2$. Then r is bounded, strictly positive, nonconstant, and $\sum_n q_n r_n = 1$. With a trivial observation, set $p_0 = q$ and $p_1 = qr$. The entropy of q is infinite, so

$$\mathbb{E}_q[\log p_0] = \mathbb{E}_q[\log p_1] = -\infty.$$

An extended-real argmax of the expected complete log-likelihood therefore treats both candidates as maximizers. The separately integrated ELBO of (5.16) is undefined because (H4) fails, while its canonical relative-log extension distinguishes them:

$$\mathbb{E}_q\left[\log \frac{p_0}{q}\right] = 0, \qquad \mathbb{E}_q\left[\log \frac{p_1}{q}\right] = \mathbb{E}_q[\log r] = -D_{\mathrm{KL}}(q \| p_1) < 0.$$

Thus cancellation of a parameter-independent infinite entropy does not justify the M-coordinate claim.

The existence proviso in (5.33) is a hypothesis and not a formality. [HYPOTHESIS]

The covariance parameters of Chapter 4 range over an open cone, and the expected complete log-likelihood is unbounded above along sequences that drive a conditional covariance to the boundary whenever the corresponding expected residual second moment is singular. In that situation the supremum is not attained and (5.33) defines nothing. Under a nondegenerate $\mathbb{Q}_{V,o,X}$ the expected residual second moments are positive definite and the maximizer exists, so the hypothesis is mild in the Gaussian realization; it is stated because it is a hypothesis, not because it usually fails. [ESTABLISHED]

Definition 5.17 (Generalized M-coordinate acceptance). A generalized M-coordinate need not attain the maximum, but it must pass an acceptance condition:

$$\mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{new}}}(\mathbb{Q}_{V,o,X}; X, o) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{old}}}(\mathbb{Q}_{V,o,X}; X, o). \tag{5.34}$$

The condition is a declaration about what counts as a step; it is checkable for arbitrary $\mathbb{Q}_{V,o,X}$ whenever each candidate has a selected positive-finite slice as in (H1)–(H2). No recognition density or (H4) is required. On the classical split domain it reduces to the same inequality for (5.16);

in particular, the expected-complete-log-likelihood update of Proposition 5.16 is a finite-domain sufficient condition. [DEFINITION]

Proposition 5.18 (Evidence monotonicity). Suppose (H1)–(H2) hold for the fixed $\mathbb{Q}_{V,o,X}$ at both θ^{old} and θ^{new}; thus o lies in both regular-observation sets, or common pointwise density, slice, and conditional versions have been declared as model data. If the E-coordinate is exact at the old parameter, so that $\mathbb{Q}_{V,o,X} = \mathbf{\Pi}_{\theta^{\mathrm{old}},V,o,X}$, and the M-coordinate satisfies (5.34), then

$$\log p_{\theta^{\mathrm{new}},V}(o \mid X) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{new}}}(\mathbb{Q}_{V,o,X}; X, o) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{old}}}(\mathbb{Q}_{V,o,X}; X, o) = \log p_{\theta^{\mathrm{old}},V}(o \mid X). \tag{5.35}$$

The first inequality is Theorem 5.7 at θ^{new}, the second is (5.34), and the final equality is the zero-gap case at θ^{old}. This proof never writes the undefined singular-branch sum $\mathcal{L}^{\mathrm{ext}}_V + D_{\mathrm{KL}}$. Under (H3)–(H4) at both candidates it reduces to the classical conclusion [27, 64]. [ESTABLISHED]

A natural-gradient direction and an ordinary optimizer direction are proposals, not accepted coordinates. The distinction is not pedantic, and the following states exactly what is and is not guaranteed.

Proposition 5.19 (Finite natural-gradient steps need not be monotone). Let $\mathsf{G} \succ 0$ be a preconditioner, write $\nabla\mathcal{L}_V$ for the gradient of $\vartheta \mapsto \mathcal{L}_{V,\vartheta}(\mathbb{Q}_{V,o,X}; X, o)$, and consider the step $\vartheta^+ = \vartheta + \alpha\mathsf{G}^{-1}\nabla\mathcal{L}_V$. Define $\varphi(\alpha) = \mathcal{L}_{V,\vartheta^+}(\mathbb{Q}_{V,o,X}; X, o)$. At a nonstationary point the direction is an ascent direction, since

$$\varphi'(0) = \nabla\mathcal{L}_V^{\top}\mathsf{G}^{-1}\nabla\mathcal{L}_V > 0. \tag{5.36}$$

This is an infinitesimal statement and does not extend to finite α. Suppose $\mathcal{L}_{V,\xi} = \mathcal{L}^{\star}_V - \frac{1}{2}(\xi - \theta^{\star})^{\top}H(\xi - \theta^{\star})$ with $H \succ 0$ throughout a neighborhood $\mathcal{U}$, let $d_{\max}$ be the largest generalized eigenvalue of the pair (H, G), and let v be a corresponding generalized eigenvector normalized so that $v^{\top}\mathsf{G}v = 1$. Then at any $\vartheta = \theta^{\star} + cv \in \mathcal{U}$ with $c \neq 0$, for every step whose endpoint ϑ^+ also lies in $\mathcal{U}$,

$$\mathcal{L}_{V,\vartheta^+} - \mathcal{L}_{V,\vartheta} = \tfrac{1}{2}d_{\max}c^2\left[1 - (1 - \alpha d_{\max})^2\right] < 0 \qquad \text{whenever} \qquad \alpha > \frac{2}{d_{\max}}. \tag{5.37}$$

[ESTABLISHED]

Proof. Equation (5.36) is the chain rule with G^{-1} positive definite. For the second part write $e = \vartheta - \theta^{\star}$, so $\nabla\mathcal{L}_V = -He$ and $e^+ = (I - \alpha\mathsf{G}^{-1}H)e$. With $Hv = d_{\max}\mathsf{G}v$, $v^{\top}\mathsf{G}v = 1$, and $e = cv$ one gets $e^+ = (1 - \alpha d_{\max})cv$ and $\mathcal{L}_{V,\vartheta} = \mathcal{L}^{\star}_V - \frac{1}{2}d_{\max}c^2$. The endpoint hypothesis permits the same quadratic formula to be evaluated at ϑ^+. Substituting gives (5.37), and the bracket is negative exactly when $|1 - \alpha d_{\max}| > 1$, that is when $\alpha d_{\max} > 2$. □

The content of Proposition 5.19 is that monotonicity at finite step size is an additional requirement, supplied by a line search satisfying a sufficient-increase condition, by a trust region with an acceptance test, or by any other rule that verifies (5.34) on the same functional. Absent such a test the proposal is not entitled to (5.35), and the failure is not exotic: it occurs for a quadratic objective at any step length exceeding twice the reciprocal of the largest preconditioned curvature. Nor is

the difficulty removed by improving the preconditioner, since (5.37) is stated in the preconditioned spectrum. Separately, optimizing a different objective, for instance a population energy that is not (5.16), supplies no guarantee for the fixed-joint evidence at all: two functionals with different stationary points can move in opposite directions under the same update, and only a bound with matched stationarity would transfer the conclusion.

5.7 What this chapter does not settle

Three limits of the development are recorded so that later chapters do not inherit more than was proved. Existence of a maximizer of (5.16) over a restricted recognition family is not established here and is not implied by Theorem 5.9; the theorem is an identity at a fixed pair of measures and says nothing about the attainability of any optimum. Tightness of the bound for any particular family is likewise not claimed, and the gap for the restrictions this document uses is computed in Chapter 23 rather than bounded here.

The third limit is a genuine open problem.

Open Problem 5.20 (Convergence of the alternating scheme). Whether the alternating scheme built from block E-coordinates (5.30) and accepted M-coordinates (5.34) converges to a stationary point of the common extended ELBO under the present hypotheses is not settled. [OPEN]

Block E-coordinates at fixed target and accepted parameter proposals make the compared objective values nondecreasing when their acceptance tests are evaluated on that same functional. They do not by themselves make the evidence nondecreasing. The evidence chain (5.35) additionally requires a full exact E phase, $\mathbb{Q}_{V,o,X} = \mathbf{\Pi}_{\theta^{\mathrm{old}},V,o,X}$, before each accepted M phase. Under that stronger schedule the evidence values are nondecreasing and converge when bounded above; convergence of the parameter sequence is a separate question requiring regularity conditions of the type identified for the classical algorithm [90], and global optimality does not follow even when they hold. [ESTABLISHED]

Closing the problem in the present setting requires either verifying those conditions for the declared parameter space, which is an open cone so that compactness of the relevant level sets must be checked rather than assumed, or exhibiting a counterexample within the declared family. [OPEN]

Nothing in Chapter 10 or Chapter 11 depends on the answer, because those chapters analyze the structure of the objective and its transformation under aggregation rather than the trajectory of any optimizer. [ESTABLISHED]

Chapter 6

Belief Fibers and Exponential Families

The objects carried by the belief and model associated bundles of Chapter 2 are normalized laws, while generative fibers carry kernels. Coordinates, smoothness, domination, and exponential-family structure are additional choices. This chapter states that hierarchy before selecting the finite-dimensional exponential subclass used by the general coarse theory.

6.1 Law fibers before coordinates

Definition 6.1 (Law and kernel fibers). For a standard Borel space $(\mathsf{Z}, \mathscr{Z})$, let

$$\mathcal{B}(\mathsf{Z}) \subseteq \mathcal{P}(\mathsf{Z}) \tag{6.1}$$

be a declared collection of normalized probability laws. For standard Borel spaces X, Y, let

$$\mathcal{K}(\mathsf{X}, \mathsf{Y}) \subseteq \mathrm{Markov}(\mathsf{X}, \mathsf{Y}) \tag{6.2}$$

be a declared collection of normalized Markov kernels. A law is not a kernel, and neither declaration supplies a common density, affine chart, manifold, Fisher metric, or convex cone. [DEFINITION]

The belief and model bundles may use different law fibers. Cross-bundle maps $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ and $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$ act between those associated bundles and are not iterations on either fiber. To prevent collision with that reserved notation, a generic endomorphism of a parameter set will be written Ψ.

Definition 6.2 (Partial realization). A parametrized model consists of a parameter space Θ, a domain $\Theta_{\mathrm{law}} \subseteq \Theta$, and a map

$$\mathfrak{r} : \Theta_{\mathrm{law}} \longrightarrow \mathcal{B}(\mathsf{Z}), \qquad \vartheta \longmapsto Q_\vartheta. \tag{6.3}$$

Outside Θ_{law}, a parameter may still denote an algebraic operator, but it does not denote a law through $\mathfrak{r}$. [DEFINITION]

A bimeasurable gauge action $g : \mathsf{Z} \to \mathsf{Z}$ acts on laws by

$$Q \longmapsto g_{\#}Q, \tag{6.4}$$

and on kernels by

$$(g_{\mathrm{out}} \cdot K \cdot g_{\mathrm{in}}^{-1})(x, B) = K(g_{\mathrm{in}}^{-1}x, g_{\mathrm{out}}^{-1}B). \tag{6.5}$$

These formulas type the action. [DEFINITION]

Closure of a declared fiber under the action is a separate hypothesis. [HYPOTHESIS]

6.2 Dominated and smooth tiers

Domination is not automatic. An uncountable family of Dirac laws on an uncountable standard Borel space admits no common sigma-finite dominating measure: any such measure would have to assign positive mass to uncountably many disjoint atoms, which a sigma-finite measure cannot do. [ESTABLISHED]

Hence density charts cannot be inferred from the general law fiber. [ESTABLISHED]

The differential tier used below is a finite-dimensional smooth parametrized-measure model. Differentiability in quadratic mean, square-integrable scores, and the interchange of differentiation and integration are hypotheses. The Fisher form is a Riemannian metric only after nondegeneracy is proved. Parametrized-measure models supply an intrinsic framework for this tier [8]. [HYPOTHESIS]

Stratified and infinite-dimensional tiers require their own tangent and integration theories; no universal global Fisher metric or natural chart is claimed. [NOT-CLAIMED]

6.3 Regular finite-dimensional exponential families

Definition 6.3 (Canonical family). Let ν be a nonzero sigma-finite measure on $(\mathsf{Z}, \mathscr{Z})$, let $T : \mathsf{Z} \to \mathsf{T}$ be measurable into a finite-dimensional inner-product space, and define

$$\mathsf{A}(\theta) = \log \int_{\mathsf{Z}} e^{\langle \theta, T(z) \rangle} \nu(dz), \qquad \mathsf{N} = \{\theta : \mathsf{A}(\theta) < \infty\}. \tag{6.6}$$

For $\theta \in \mathsf{N}$,

$$q_\theta(z) = \exp\{\langle \theta, T(z) \rangle - \mathsf{A}(\theta)\}, \qquad Q_\theta(dz) = q_\theta(z)\nu(dz). \tag{6.7}$$

Nothing is asserted for $\theta \notin \mathsf{N}$. [DEFINITION]

Proposition 6.4 (Finite-domain interface to ELBO coordinates). The map $(\theta, z) \mapsto q_\theta(z)$ in (6.7) is jointly measurable on $\mathsf{N} \times \mathsf{Z}$, and every law, density ratio, Fisher tensor, or ELBO coordinate built from this canonical representation is restricted to parameters in N. For a comparison of several canonical representations, either the comparison is made intrinsically at the measure level or the representations must have one declared common sigma-finite base measure; its density domain is the intersection of their natural domains. An extended-ELBO update additionally requires a positive-finite selected evidence slice at every compared parameter, while the classical expected-log-density

representation separately requires Hypothesis 3.18. An algebraic operator that leaves N therefore ceases to define a probability or ELBO coordinate even if it remains an element of the operator cone. [ESTABLISHED]

Proof. Lower semicontinuity makes A Borel measurable, and measurability of T makes $(\theta, z) \mapsto \langle \theta, T(z) \rangle$ jointly measurable; exponentiation proves the first claim. Normalization is finite exactly on N by (6.6). The remaining statements are direct domain checks against Definitions 5.6 and 6.2 and hypothesis 3.18: a density comparison needs a fixed reference, the extended functional needs a normalized posterior slice with finite positive mass, and its classical split needs the additional support and log-integrability hypotheses. □

Proposition 6.5 (Convexity and lower semicontinuity). A is convex and lower semicontinuous as an extended-real function, and N is convex. [ESTABLISHED]

Proof. For $0 < t < 1$, Hölder's inequality gives

$$\int e^{\langle t\theta+(1-t)\vartheta, T\rangle} d\nu \leq \left(\int e^{\langle \theta, T\rangle} d\nu \right)^{t} \left(\int e^{\langle \vartheta, T\rangle} d\nu \right)^{1-t}, \tag{6.8}$$

and taking logarithms proves convexity and convexity of the effective domain. Fatou's lemma applied to a convergent parameter sequence proves lower semicontinuity. □

Proposition 6.6 (Cumulants and Fisher information). At every interior parameter where local exponential domination permits differentiation under the integral,

$$\nabla \mathsf{A}(\theta) = \mathbb{E}_\theta[T], \qquad \nabla^2 \mathsf{A}(\theta) = \mathrm{Cov}_\theta(T). \tag{6.9}$$

The Hessian is the Fisher information in the natural chart. [ESTABLISHED]

Proof. Differentiate the normalizer once and twice. The score is $T - \mathbb{E}_\theta T$, so its second moment is the covariance in (6.9). Local exponential domination follows by enclosing the parameter in a compact subset of the interior and applying the elementary bound

$$|x|^r \leq C_{r,\varepsilon} \left(e^{\varepsilon x} + e^{-\varepsilon x} \right) \tag{6.10}$$

to each coordinate of T. □

Hypothesis 6.7 (Regularity and minimality). The family is regular, so N is open, and minimal, so no nonzero $a \in \mathsf{T}$ makes $\langle a, T \rangle$ constant ν-almost everywhere. Regularity excludes natural-domain boundary points; minimality excludes affine redundancy. [HYPOTHESIS]

Proposition 6.8 (Minimality and nondegenerate information). Under the preceding hypothesis, $\nabla^2 \mathsf{A}(\theta)$ is positive definite, A is strictly convex, and $\theta \mapsto Q_\theta$ is injective. Failure of minimality makes the Hessian singular. [ESTABLISHED]

Proof. For every a,

$$a^\top \nabla^2 \mathsf{A}(\theta) a = \mathrm{Var}_{Q_\theta}(\langle a, T \rangle).$$

The variance vanishes exactly when the displayed statistic is almost surely constant. Since $q_\theta > 0$, Q_θ and ν have the same null sets, so minimality gives positive definiteness. Strict convexity follows. Equality of two densities makes $\langle \theta - \vartheta, T \rangle$ constant almost everywhere, and minimality then gives $\theta = \vartheta$. □

6.4 Operator and probability layers

Definition 6.9 (Operator layer). Let $\mathcal{C} \subseteq \mathsf{T}$ be a separately declared closed convex cone preserved by the algebraic maps under study. Its probability layer is $\mathcal{C} \cap \mathsf{N}$. The cone is chosen for algebraic closure, not normalization, and no such cone is asserted for an arbitrary law fiber. [DEFINITION]

Proposition 6.10 (A natural-domain boundary carries no law). Under regularity, if $\theta^\star \in \partial \mathsf{N}$, then

$$\int e^{\langle \theta^\star, T(z) \rangle} \nu(dz) = +\infty. \tag{6.11}$$

For $\theta_\tau = (1-\tau)\theta_0 + \tau\theta^\star$, with $\theta_0 \in \mathsf{N}$ and $w = \theta^\star - \theta_0$,

$$\mathsf{A}(\theta_\tau) \to +\infty, \qquad \langle w, \nabla \mathsf{A}(\theta_\tau) \rangle \to +\infty, \qquad \int_0^1 \mathrm{Var}_{Q_{\theta_\tau}}(\langle w, T \rangle) d\tau = +\infty. \tag{6.12}$$

[ESTABLISHED]

Proof. Regularity puts $\theta^\star$ outside N, proving (6.11). Convexity puts every θ_τ, $\tau < 1$, in N, while lower semicontinuity forces the first limit. For $g(\tau) = \mathsf{A}(\theta_\tau)$, convexity makes g' nondecreasing. If g' were bounded above, then g would remain bounded, a contradiction. The identities

$$g' = \langle w, \nabla \mathsf{A} \rangle, \qquad g'' = \mathrm{Var}_{Q_{\theta_\tau}}(\langle w, T \rangle)$$

and integration of g'' prove the remaining assertions. □

Thus the declared representation supplies no finite log partition, mean parameter, Fisher metric, relative entropy from the nonexistent endpoint law, or instance of the exact ELBO at $\theta^\star$. [ESTABLISHED]

This does not rule out a quotient, pin, new reference measure, or boundary stratum; each is a different declared object. [NOT-CLAIMED]

Proposition 6.11 (An operator fixed point need not be a law). Let $\Psi : \mathcal{C} \to \mathcal{C}$ be continuous. Every iterate may lie in $\mathcal{C} \cap \mathsf{N}$ while the unique fixed point lies on $\partial\mathsf{N}$. For $\mathsf{Z} = \mathbb{R}$, $T(z) = -z^2/2$, and Lebesgue ν,

$$\mathsf{A}(\theta) = -\tfrac{1}{2}\log\theta + \tfrac{1}{2}\log 2\pi, \qquad \mathsf{N} = (0,\infty), \qquad \mathcal{C} = [0,\infty), \tag{6.13}$$

the map $\Psi(\theta) = \theta/2$ has this property. [ESTABLISHED]

Proof. Starting at one gives $\theta_\ell = 2^{-\ell} > 0$, while $\theta_\ell \to 0$, the unique fixed point, and $\mathsf{A}(\theta_\ell) \to +\infty$. The constant kernel at zero has infinite Lebesgue integral. □

6.5 Declared repairs

Definition 6.12 (Pinning). Choose $\theta_{\mathrm{m}} \in \mathsf{N}$ and replace $\mathcal{C}$ by $\theta_{\mathrm{m}} + \mathcal{C}$; the pin does not flow. [DEFINITION]

Proposition 6.13 (Pinning under a recession condition). If $\theta_{\mathrm{m}} \in \operatorname{int}\mathsf{N}$ and every $\eta \in \mathcal{C}$ is a recession direction of N, then $\theta_{\mathrm{m}} + \mathcal{C} \subseteq \operatorname{int}\mathsf{N}$. [ESTABLISHED]

Proof. Translate an open ball around θ_{m} by η. The recession hypothesis keeps the translated ball inside N, so its center is interior. □

The pin changes the family, may change the fixed set, and introduces a parameter whose removal need not commute with the long-scale limit. Whether the mass sector is relevant, marginal, or irrelevant is an open spectral question for the declared scale map. [OPEN]

Definition 6.14 (Quotienting). Suppose $\mathsf{Z} = \mathbb{R}^n$, ν is Lebesgue measure, and the energy is invariant along a nonzero linear subspace $\mathcal{V}$. Declare the new sample space $\mathcal{W} = \mathcal{V}^\perp$, its Lebesgue measure, and the restricted statistic. [DEFINITION]

Proposition 6.15 (Quotient normalization is not full-space normalization). Writing $z = v + w$ in the orthogonal splitting,

$$\int_{\mathbb{R}^n} e^{\langle\theta, T(z)\rangle} dz = \lambda_{\mathcal{V}}(\mathcal{V}) \int_{\mathcal{W}} e^{\langle\theta, T(w)\rangle} dw = +\infty. \tag{6.14}$$

The quotient law is normalized exactly when the second factor is finite. [ESTABLISHED]

Proof. The change of variables has unit Jacobian and the integrand is independent of v. Tonelli's theorem gives (6.14); the first factor is infinite. □

Definition 6.16 (Retaining a mass sector). Enlarge the state by a declared coercive parameter that remains in the natural domain under the scale map. This is neither pinning nor quotienting: the mass is a running coordinate and its scaling law must be stated. [DEFINITION]

A Moore–Penrose inverse and pseudodeterminant inserted into an interior full-space density formula do not repair normalization. They compute the finite quotient expression, whose dimension and reference measure differ, while the original integral remains infinite by (6.14). [ESTABLISHED]

6.6 Relative entropy and affine information projection

For a regular minimal family, let $\tau_\theta = \nabla \mathsf{A}(\theta)$ and let A^* be the convex conjugate. Direct expectation of the log-density ratio gives

$$D_{\mathrm{KL}}(Q_\vartheta \| Q_\theta) = \mathsf{A}(\theta) - \mathsf{A}(\vartheta) - \langle \nabla \mathsf{A}(\vartheta), \theta - \vartheta \rangle = D_{\mathsf{A}}(\theta, \vartheta), \tag{6.15}$$

and Legendre duality gives

$$D_{\mathrm{KL}}(Q_\vartheta \| Q_\theta) = D_{\mathsf{A}^*}(\tau_\vartheta, \tau_\theta). \tag{6.16}$$

This is the exact exponential-family divergence; in general it is not a finite-displacement Fisher quadratic [2, 20, 25]. [ESTABLISHED]

Proposition 6.17 (Affine mean constraints attain a unique projection). Assume the extended A is Legendre, as in a regular minimal canonical family, and let $\mathsf{M} = \operatorname{int} \operatorname{dom} \mathsf{A}^*$. If C is a closed affine subspace with $C \cap \mathsf{M} \neq \varnothing$, then for every $\tau_\theta \in \mathsf{M}$ there is a unique $\widehat{\tau} \in C \cap \mathsf{M}$ minimizing

$$\tau \longmapsto D_{\mathsf{A}^*}(\tau, \tau_\theta). \tag{6.17}$$

The corresponding natural parameter is uniquely $\widehat{\vartheta} = \nabla \mathsf{A}^*(\widehat{\tau})$. [ESTABLISHED]

Proof. The finite-dimensional Legendre Bregman projection theorem gives existence [10], uniqueness, and interiority on every nonempty closed convex feasible set meeting the interior domain. Strict convexity gives uniqueness, and essential smoothness excludes boundary attainment. For $C = \{\tau : B\tau = b\}$, the solution equivalently satisfies

$$B\widehat{\tau} = b, \qquad \nabla \mathsf{A}^*(\widehat{\tau}) - \theta + B^\top \lambda = 0, \tag{6.18}$$

which also proves the stated natural-parameter formula. □

Feasibility, closedness, and convexity are load bearing. For a Bernoulli mean domain $(0, 1)$, $C = \{0\}$ is infeasible; an open feasible interval may have a finite unattained infimum; and a closed nonconvex two-point set can have two minimizers. Gauge compatibility and commutation of this projection with a coarse parameter map remain open and require an intertwining theorem; existence alone does not provide them. [OPEN]

6.7 Boundary of this chapter

This chapter supplies the general law-fiber hierarchy, the exponential-family interior, its operator boundary, and the Bregman projection theorem. It does not contain graph aggregation, Gaussian determinants, Schur complements, or matrix pencils. Graph-exponential energy closure is developed in Chapter 10; the multivariate-Gaussian realization and its quadratic boundary rates appear later. An operator RG additionally requires a preserved parameter category, scale comparisons, rescaling, and topology. A law-level RG additionally requires normalized laws or kernels, compatible reference measures, and finite normalizers at every scale. A Fisher or ELBO reading additionally requires the hypotheses of the corresponding earlier chapters. [HYPOTHESIS]

Chapter 7

Local and Collective Free Energy

The bundle geometry of Chapter 2 is the inherited foundation of this chapter. The probabilistic construction below is additional structure. It is not inferred from the existence of the principal bundle, its group, or its two associated statistical fibers. The construction answers the local–global question by putting every interaction record in one fixed normalized joint law and then deriving each local objective by disintegration of that law. In this sense local and collective free energies are compatible, but they are not the same accounting object: the graph-local objectives below are coordinate potentials, and summing their incident-factor formulas overcounts shared records.

7.1 A normalized interaction-record model

Hypothesis 7.1 (Agent state and interaction kernels). Use the standard-Borel context, agent, and finite-design spaces of Definition 3.4, let V be the finite agent set, and let $\mathfrak{A}$ be the finite interaction-record index set. Agent i has the standard-Borel block space $(\mathsf{Y}_{i,D}, \mathscr{Y}_{i,D})$. Its state may contain a belief-law point in $(\mathcal{E}_b)_{c_i}$, a model-law point in $(\mathcal{E}_m)_{c_i}$, and declared auxiliary latent variables. Write $(\mathsf{Y}_D, \mathscr{Y}_D) = \prod_{i\in V}(\mathsf{Y}_{i,D}, \mathscr{Y}_{i,D})$. The normalized baseline is the latent population kernel constructed in Construction 4.1,

$$\mathbb{P}^{Y}_{\theta,V} : (\mathsf{X}, \mathscr{X}) \rightsquigarrow (\mathsf{Y}_D, \mathscr{Y}_D). \tag{7.1}$$

Each record a has a finite scope $\partial a \subseteq V$, a standard Borel record space $(\mathsf{O}_a, \mathscr{O}_a)$, and a normalized Markov kernel

$$K_a : (\mathsf{X} \times \mathsf{Y}_{\partial a,D}, \mathscr{X} \otimes \mathscr{Y}_{\partial a,D}) \rightsquigarrow (\mathsf{O}_a, \mathscr{O}_a). \tag{7.2}$$

of the abstract type (3.7). For the density presentation used in this chapter, one fixed probability or sigma-finite measure ν_a dominates the whole family $\{K_a(X, y_{\partial a,D}, \cdot) : (X, y_{\partial a,D}) \in \mathsf{X} \times \mathsf{Y}_{\partial a,D}\}$, and a finite jointly measurable version

$$\ell_a : \mathsf{X} \times \mathsf{Y}_{\partial a,D} \times \mathsf{O}_a \longrightarrow [0, +\infty), \qquad K_a(X, y_{\partial a,D}, do_a) = \ell_a(o_a \mid X, y_{\partial a,D})\nu_a(do_a) \tag{7.3}$$

is fixed once and for all. Thus $\int \ell_a(o_a \mid X, y_{\partial a,D})\nu_a(do_a) = 1$ for every $(X, y_{\partial a,D})$. The common ν_a and the displayed joint measurability are family-level hypotheses, not recordwise choices. The kernels and $\mathbb{P}^{Y}_{\theta,V}$ are fixed by the generative parameters and structural data; none reads a recognition law, a variational parameter, or a live posterior. Their local coordinate presentations are

required to transform by the common passive pushforward induced by a change of bundle frames, so the underlying measures and record probabilities are frame independent. [HYPOTHESIS]

This hypothesis permits correlated baseline states. A graph-local Markov blanket requires the additional condition that the regular conditionals of $\mathbb{P}^Y_{\theta,V}$ have the corresponding locality. Bundle covariance is imposed by requiring each K_a to be invariant under simultaneous changes of the incident agent frames and the factor frame. One concrete realization gives factor a its own principal frame and incidence-leg transports $U^x_{a\leftarrow i}$ in each channel $x \in \{b, m\}$; the likelihood is then a measurable function of the dressed variables $\rho_x(U^x_{a\leftarrow i})y^x_i$. The two representations remain distinct, and a belief–model comparison still requires the declared cross morphisms of Section 2.4. [DEFINITION]

Conditional independence of the records given the agent configuration defines the same joint law constructed by the normalized extension Proposition 4.5,

$$\mathbb{P}_{\theta,V}(dy, do \mid X) = \mathbb{P}^Y_{\theta,V}(dy \mid X) \prod_{a\in\mathfrak{A}} K_a(X, y_{\partial a,D}, do_a). \tag{7.4}$$

Proposition 7.2 (Normalization on arbitrary interaction cycles). The law in (7.4) is normalized for every finite hypergraph, including cyclic ones. [ESTABLISHED]

Proof. Integrate the record coordinates in any order. Each normalized kernel integrates to one, leaving the normalized latent law $\mathbb{P}^Y_{\theta,V}(\cdot \mid X)$. The factor scopes are permitted to overlap because the record variables are conditionally independent children of one latent configuration. This argument would not validate an arbitrary cycle of separately prescribed agent conditionals, which need not be compatible with any joint law. □

For the remainder of this chapter fix one $X \in \mathsf{X}$. We suppress that fixed argument consistently: $\mathbb{P}^Y_{\theta,V}(dy) := \mathbb{P}^Y_{\theta,V}(dy \mid X)$, $K_a(y_{\partial a}, do_a) := K_a(X, y_{\partial a}, do_a)$, and $\ell_a(o_a \mid y_{\partial a}) := \ell_a(o_a \mid X, y_{\partial a})$. Every descendant symbol $L_o, H_o, p_{\theta,V}(o \mid X), \mathbf{\Pi}_{\theta,V,o,X}$ and its block analogue is relative to this same fixed context unless an X argument is restored explicitly.

With $\nu = \bigotimes_{a\in\mathfrak{A}} \nu_a$, the complete record density is the finite jointly measurable function

$$\ell(o \mid y) = \prod_{a\in\mathfrak{A}} \ell_a(o_a \mid y_{\partial a}), \qquad \mathbb{P}_{\theta,V}(dy, do \mid X) = \mathbb{P}^Y_{\theta,V}(dy)\ell(o \mid y)\nu(do). \tag{7.5}$$

For a fixed record o, define its likelihood before taking logarithms:

$$L_o(y) := \ell(o \mid y) = \prod_{a\in\mathfrak{A}} \ell_a(o_a \mid y_{\partial a}) \in [0, +\infty). \tag{7.6}$$

Its extended interaction energy is

$$H_o(y) := -\log L_o(y) = \sum_{a\in\mathfrak{A}} -\log \ell_a(o_a \mid y_{\partial a}), \qquad -\log 0 := +\infty. \tag{7.7}$$

Because each factor density is finite, H_o never equals $-\infty$, although individual negative log densities and their finite sum can be negative. A binary success record with $K_a(1 \mid y) = \exp[-E_a(y)]$ is a valid special case only when $E_a \in [0, +\infty]$. General observations are represented by the normalized kernels, not by treating an arbitrary exponential as a probability. [ESTABLISHED]

7.2 One collective VFE

Fix a regular record o for which

$$0 < p_{\theta,V}(o \mid X) = \int_{\mathsf{Y}_D} L_o(y) \mathbb{P}^Y_{\theta,V}(dy) < +\infty, \tag{7.8}$$

and let

$$\mathbf{\Pi}_{\theta,V,o,X}(dy) = p_{\theta,V}(o \mid X)^{-1} L_o(y) \mathbb{P}^Y_{\theta,V}(dy) \tag{7.9}$$

be the conditional posterior. The regular-observation qualifications of Chapter 5 apply.

Theorem 7.3 (Collective interaction VFE). For every probability measure Q on Y_D, define the collective extended ELBO and VFE by

$$\boxed{\mathcal{F}^{\mathrm{ext}}_o(Q) := -\mathcal{L}^{\mathrm{ext}}_o(Q) := -\log p_{\theta,V}(o \mid X) + D_{\mathrm{KL}}(Q \| \mathbf{\Pi}_{\theta,V,o,X}).} \tag{7.10}$$

This takes values in $\mathbb{R} \cup \{+\infty\}$ and is the specialization of Definition 5.6 and theorem 5.7. Equivalently, $\mathcal{L}^{\mathrm{ext}}_o(Q) \le \log p_{\theta,V}(o \mid X)$ for every Q, with equality exactly when $Q = \mathbf{\Pi}_{\theta,V,o,X}$ as full measures. In particular, $Q = \mathbb{Q}_{V,o,X}$ is the selected population recognition instance. No domination or mean-field assumption on Q is needed. [ESTABLISHED]

Proof. Apply the extended gap theorem Theorem 5.7 with reference $\mathbb{P}^Y_{\theta,V}$ and likelihood L_o. Nonnegativity and the equality condition are the corresponding properties of relative entropy. □

If $Q \ll \mathbb{P}^Y_{\theta,V}$ and the separate complexity and energy terms are defined without an indeterminate $+\infty - \infty$, the same quantity has the familiar energy presentation

$$\mathcal{F}^{\mathrm{ext}}_o(Q) = D_{\mathrm{KL}}(Q \| \mathbb{P}^Y_{\theta,V}) + \mathbb{E}_Q H_o. \tag{7.11}$$

This is a derived representation, not the definition at singular Q. [ESTABLISHED]

There is one term in H_o for each actual interaction record. An undirected record shared by agents i and j occurs once. Writing both E_{ij} and E_{ji} in the global law asserts two directed records, not two descriptions of one record. [ESTABLISHED]

7.3 The conditional VFE on an agent block

Let $B \subseteq V$ be nonempty and write $b = y_{B^c}$. The same construction therefore treats both the coordinates of a singleton agent $B = \{i\}$ and an arbitrary nonempty aggregate block B without changing probability models. Choosing a block does not construct a meta-agent. That term is reserved for a separately constructed coarse state, observation interface, evaluator, recognition law, and update rule. Define

$$\mathfrak{A}_B = \{a \in \mathfrak{A} : \partial a \cap B \neq \varnothing\}, \qquad g_{B,o}(y_B; b) = \prod_{a \in \mathfrak{A}_B} \ell_a(o_a \mid y_{\partial a}), \tag{7.12}$$

and the likelihood of records whose scopes miss B,

$$L_{\bar{B},o}(b) = \prod_{a:\partial a \cap B = \varnothing} \ell_a(o_a \mid b_{\partial a}), \qquad L_o(y_B, b) = g_{B,o}(y_B; b) L_{\bar{B},o}(b). \tag{7.13}$$

The corresponding extended energies retain the useful additive notation

$$H_{B,o}(y_B; b) = -\log g_{B,o}(y_B; b) = \sum_{a \in \mathfrak{A}_B} -\log \ell_a(o_a \mid y_{\partial a}) \tag{7.14}$$

and

$$H_{\bar{B},o}(b) = -\log L_{\bar{B},o}(b), \qquad H_o(y) = H_{B,o}(y_B; b) + H_{\bar{B},o}(b), \tag{7.15}$$

as an identity in $\mathbb{R} \cup \{+\infty\}$.

Fix once and for all a measurable regular-conditional version $b \mapsto \mathbb{P}^Y_{\theta,V,B}(\cdot \mid b)$ in the disintegration

$$\mathbb{P}^Y_{\theta,V}(dy_B, db) = \mathbb{P}^Y_{\theta,V,B^c}(db)\mathbb{P}^Y_{\theta,V,B}(dy_B \mid b). \tag{7.16}$$

Every pointwise formula in b below is relative to this declared version. Two baseline-conditional versions agree $\mathbb{P}^Y_{\theta,V,B^c}$-almost everywhere and hence, by (7.19), $\mathbf{\Pi}_{\theta,V,o,X,B^c}$-almost everywhere. No pointwise version invariance is claimed beyond these almost-sure statements: their values and all descendant formulas at exceptional b are not determined by the joint law. Two different conditional likelihoods are needed. The incident-factor evidence is

$$Z_B(b) = \int g_{B,o}(y_B; b)\mathbb{P}^Y_{\theta,V,B}(dy_B \mid b), \tag{7.17}$$

whereas the full likelihood of the outside state is defined directly by

$$w_B(b) = \int L_o(y_B, b)\mathbb{P}^Y_{\theta,V,B}(dy_B \mid b). \tag{7.18}$$

The direct definition is valid even where the formal product $L_{\bar{B},o}(b)Z_B(b)$ would have the indeterminate form $0 \cdot \infty$. Tonelli's theorem gives

$$p_{\theta,V}(o \mid X) = \int w_B(b)\mathbb{P}^Y_{\theta,V,B^c}(db), \qquad \mathbf{\Pi}_{\theta,V,o,X,B^c}(db) = p_{\theta,V}(o \mid X)^{-1} w_B(b)\mathbb{P}^Y_{\theta,V,B^c}(db). \quad (7.19)$$

Let $\mathsf{R}_{B,o}$ be the measurable set on which

$$0 < w_B(b) < +\infty, \quad 0 < L_{\bar{B},o}(b) < +\infty, \quad 0 < Z_B(b) < +\infty, \quad w_B(b) = L_{\bar{B},o}(b)Z_B(b). \quad (7.20)$$

Then $\mathbf{\Pi}_{\theta,V,o,X,B^c}(\mathsf{R}_{B,o}) = 1$. Indeed, finite collective evidence makes $w_B < +\infty$ outside a $\mathbb{P}^Y_{\theta,V,B^c}$-null set, and the posterior outside is supported on $w_B > 0$. On that support the finite-factor hypothesis makes $L_{\bar{B},o}$ finite; positivity of w_B forces $L_{\bar{B},o} > 0$, after which ordinary factorization under the conditional integral gives $Z_B = w_B/L_{\bar{B},o} \in (0, +\infty)$. The regular set need not be $\mathbb{P}^Y_{\theta,V,B^c}$-full. [ESTABLISHED]

For $b \in \mathsf{R}_{B,o}$ relative to the fixed baseline-conditional version, the posterior conditional is

$$\mathbf{\Pi}_{\theta,V,o,X,B}(dy_B \mid b) = Z_B(b)^{-1} g_{B,o}(y_B; b)\mathbb{P}^Y_{\theta,V,B}(dy_B \mid b), \quad (7.21)$$

and, after choosing any measurable probability-kernel extension off $\mathsf{R}_{B,o}$ (for example the fixed baseline conditional there), it is a version for which

$$\mathbf{\Pi}_{\theta,V,o,X}(dy_B, db) = \mathbf{\Pi}_{\theta,V,o,X,B^c}(db)\mathbf{\Pi}_{\theta,V,o,X,B}(dy_B \mid b). \quad (7.22)$$

The posterior joint determines this kernel only $\mathbf{\Pi}_{\theta,V,o,X,B^c}$-almost everywhere. In particular, neither the conditional nor the associated pointwise local VFE at an outside-null b is an invariant of the joint law. All almost-sure local/global statements below use $Q_{B^c} \ll \mathbf{\Pi}_{\theta,V,o,X,B^c}$ and are therefore independent of these choices. [ESTABLISHED]

Theorem 7.4 (Local multi-agent ELBO). For $b \in \mathsf{R}_{B,o}$ and every probability measure $r_B(\cdot \mid b)$, define

$$\boxed{\mathcal{F}^{\mathrm{ext}}_{B,o}(r_B; b) := -\mathcal{L}^{\mathrm{ext}}_{B,o}(r_B; b) := -\log Z_B(b) + D_{\mathrm{KL}}\big(r_B(\cdot \mid b) \| \mathbf{\Pi}_{\theta,V,o,X,B}(\cdot \mid b)\big).} \quad (7.23)$$

Thus $\mathcal{L}^{\mathrm{ext}}_{B,o}(r_B; b) \le \log Z_B(b)$, with equality exactly at $r_B = \mathbf{\Pi}_{\theta,V,o,X,B}$. The singleton $B = \{i\}$ is one agent's local VFE; an arbitrary nonempty B is the corresponding block-coordinate aggregate VFE. No domination or mean-field condition on r_B is needed. [ESTABLISHED]

Proof. Apply Theorem 5.7 conditionally with reference $\mathbb{P}^Y_{\theta,V,B}(\cdot \mid b)$ and likelihood $g_{B,o}(\cdot; b)$. The equality condition follows from vanishing relative entropy. □

On the dominated energy tier, when its two terms are separately defined without an indeterminate infinity, the local functional is also

$$\mathcal{F}^{\mathrm{ext}}_{B,o}(r_B;b) = D_{\mathrm{KL}}\big(r_B(\cdot \mid b)\|\mathbb{P}^{Y}_{\theta,V,B}(\cdot \mid b)\big) + \mathbb{E}_{r_B(\cdot|b)}H_{B,o}(Y_B;b). \tag{7.24}$$

The exact relative-entropy chain rule is most safely written against the posterior. If $Q(dy_B, db) = Q_{B^c}(db)r_B(dy_B \mid b)$ and $Q_{B^c} \ll \mathbf{\Pi}_{\theta,V,o,X,B^c}$, then, in $[0,+\infty]$,

$$D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X}) = D_{\mathrm{KL}}(Q_{B^c}\|\mathbf{\Pi}_{\theta,V,o,X,B^c}) + \mathbb{E}_{Q_{B^c}} D_{\mathrm{KL}}\big(r_B(\cdot \mid Y_{B^c})\|\mathbf{\Pi}_{\theta,V,o,X,B}(\cdot \mid Y_{B^c})\big). \tag{7.25}$$

The baseline chain rule, whenever its regular conditional versions are used, likewise reads

$$D_{\mathrm{KL}}(Q\|\mathbb{P}^{Y}_{\theta,V}) = D_{\mathrm{KL}}(Q_{B^c}\|\mathbb{P}^{Y}_{\theta,V,B^c}) + \mathbb{E}_{Q_{B^c}} D_{\mathrm{KL}}\big(r_B(\cdot \mid Y_{B^c})\|\mathbb{P}^{Y}_{\theta,V,B}(\cdot \mid Y_{B^c})\big). \tag{7.26}$$

Theorem 7.5 (Exact local–global potential identity). Let $Q = Q_{B^c}r_B$ and $Q' = Q_{B^c}r'_B$ have the same outside marginal with $Q_{B^c} \ll \mathbf{\Pi}_{\theta,V,o,X,B^c}$. This fixed outside marginal condition is load bearing. Suppose $D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X}) < +\infty$ and $D_{\mathrm{KL}}(Q'\|\mathbf{\Pi}_{\theta,V,o,X}) < +\infty$; the chain rule then makes both conditional-KL fields integrable, so the following subtraction is finite. Then

$$\mathcal{F}^{\mathrm{ext}}_{o}(Q') - \mathcal{F}^{\mathrm{ext}}_{o}(Q) = \mathbb{E}_{Q_{B^c}}\left[\mathcal{F}^{\mathrm{ext}}_{B,o}(r'_B;Y_{B^c}) - \mathcal{F}^{\mathrm{ext}}_{B,o}(r_B;Y_{B^c})\right]. \tag{7.27}$$

Thus every exact local conditional update is a coordinate update of the same collective VFE. [ESTABLISHED]

Proof. Apply (7.25) to both laws. Their outside KL terms agree. The common finite constant $-\log p_{\theta,V}(o \mid X)$ cancels, as does the pointwise term $-\log Z_B(Y_{B^c})$ inside the two local functionals. The finite-difference hypotheses permit the remaining subtraction under the integral. □

When all baseline KL terms and positive and negative energy parts are integrable, the same result admits the familiar finite decomposition

$$\begin{aligned}\mathcal{F}^{\mathrm{ext}}_{o}(Q) = {} & D_{\mathrm{KL}}(Q_{B^c}\|\mathbb{P}^{Y}_{\theta,V,B^c}) \\ & + \mathbb{E}_{Q_{B^c}}\left[H_{\bar{B},o}(Y_{B^c}) + \mathcal{F}^{\mathrm{ext}}_{B,o}(r_B;Y_{B^c})\right].\end{aligned} \tag{7.28}$$

This theorem licenses sequential exact coordinate minimization, damped updates accepted by the collective objective, and continuous block-gradient dynamics. Independently replacing all correlated full conditionals need not define any joint recognition law, so such a parallel prescription is not licensed. [ESTABLISHED]

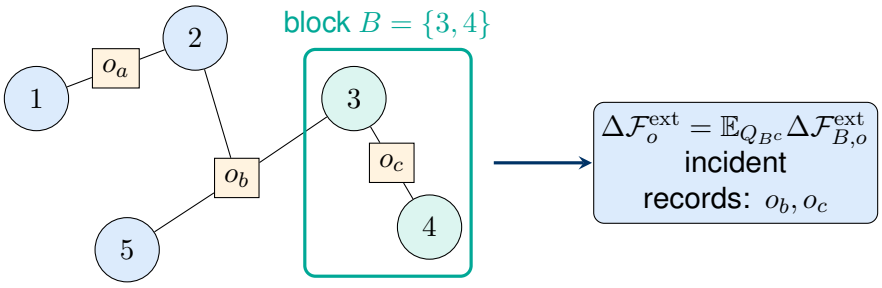


Figure 7.1: A block-local VFE contains every factor incident to the block. The factor o_b crosses the boundary and therefore enters the block conditional; o_a is outside and cancels. The equality is an exact-potential identity, not a sum of overlapping local objectives.

7.4 Additive accounting is a different construction

When $\mathbb{P}_{\theta,V}^{Y} = \bigotimes_i \rho_i$, define the total correlation of a theorem-local joint recognition law Q by

$$\mathrm{TC}(Q) = D_{\mathrm{KL}}\left(Q \middle\| \bigotimes_i Q_i\right). \tag{7.29}$$

Without assuming mean-field factorization, the extended chain rule gives

$$D_{\mathrm{KL}}(Q\|\mathbb{P}_{\theta,V}^{Y}) = \mathrm{TC}(Q) + \sum_i D_{\mathrm{KL}}(Q_i\|\rho_i) \quad \text{in } [0, +\infty]. \tag{7.30}$$

Consequently, whenever the energy presentation (7.11) is defined without an indeterminate infinity,

$$\mathcal{F}_o^{\mathrm{ext}}(Q) = \mathrm{TC}(Q) + \sum_i D_{\mathrm{KL}}(Q_i\|\rho_i) + \mathbb{E}_Q\left[\sum_{a\in\mathfrak{A}} E_{a,o}\right], \qquad E_{a,o} = -\log \ell_a(o_a \mid Y_{\partial a}). \tag{7.31}$$

[ESTABLISHED]

Equation (7.31) is an additive ledger: each interaction appears once inside the pointwise extended factor sum. It can be rewritten as $\sum_a \mathbb{E}_Q E_{a,o}$ when every $E_{a,o} \in L^1(Q)$, or under another condition excluding opposing infinities. Moreover, $\mathrm{TC}(Q) = 0$ exactly when $Q = \bigotimes_i Q_i$; mean field is thus a special case of the ledger, not a premise of it.

For the singleton conditional objectives actually defined above, assume the declared interaction factors have nonempty scopes. Their incident energies then obey the exact pointwise identity

$$H_{\{i\},o}(y_i; y_{-i}) = \sum_{a: i\in\partial a} E_{a,o}(y_{\partial a}), \qquad \sum_{i\in V} H_{\{i\},o}(y_i; y_{-i}) = \sum_{a\in\mathfrak{A}} |\partial a| E_{a,o}(y_{\partial a}). \tag{7.32}$$

An empty-scope record, if separately permitted, is instead a label-independent global constant, and the right-hand sum must then be restricted to $|\partial a| > 0$. Because every $E_{a,o}$ lies in $\mathbb{R} \cup \{+\infty\}$ and both index sets are finite, the rearrangement is valid as an extended pointwise identity. Thus the specific local VFEs of Theorem 7.4 have overlapping incident-energy components and are not

themselves the single-count ledger (7.31). This is a counting identity, not a universal no-go for arbitrary decompositions.

Ordered chain rules, a separate residual account, or nonlocal counting-number shares—including compensating negative shares assigned away from an incident coordinate—can restore an additive sum. Such a construction depends on extra bookkeeping and is not the conditional local VFE of Theorem 7.4. [ESTABLISHED]

7.5 Attention as a latent interaction label

Attention enters the fixed joint by a latent source label, not by allowing the generative law to inspect recognition beliefs. Let $\mathcal{I}_{\mathrm{att}} \subseteq V$ be the finite set of receiving agents. For each $i \in \mathcal{I}_{\mathrm{att}}$, let J_i augment the baseline independently as

$$\mathbb{P}^{Y,\mathrm{aug}}_{\theta,V}(dy, dj) = \mathbb{P}^{Y}_{\theta,V}(dy) \prod_{i \in \mathcal{I}_{\mathrm{att}}} \pi_i(dj_i). \tag{7.33}$$

Suppose $\pi_{ij} > 0$ on the finite declared source set and that every $D_{ij} : \mathsf{Y}_D \to \mathbb{R}$ is finite and measurable. At the selected regular interaction record, suppose its density has source-dependent energy

$$\ell_i(o_i \mid y, J_i = j) = c_i(o_i, y) \exp[-D_{ij}(y)/\tau_i], \qquad \tau_i > 0, \tag{7.34}$$

where c_i is independent of j and the normalized observation kernel is part of the model data. The energy D_{ij} is a fixed gauge-invariant function of latent sample-level states and declared transports. It does not contain a live recognition distribution. Equation (7.34) is an identity on the selected observation slice; normalization of the kernel and its values away from that slice remain part of the declared generative model. More strongly, at the complete selected record the full augmented likelihood is required to factor as

$$L^{\mathrm{aug}}_o(y, j) = L^{Y}_o(y) \prod_{i \in \mathcal{I}_{\mathrm{att}}} c_i(o_i, y) \exp[-D_{ij_i}(y)/\tau_i], \tag{7.35}$$

where $L^Y_o : \mathsf{Y}_D \to [0, +\infty)$ is a finite measurable function independent of every source label and no other record or factor in the generative law reads any J_i. Equation (7.35) is a label-exclusivity hypothesis on the selected likelihood slice; normalization of the augmented observation kernel over all possible records remains declared model data. Without exclusivity, an additional J_i-dependent likelihood factor would enter Bayes' formula and generally change the displayed softmax below. [HYPOTHESIS]

The constant-row recognition calculation below additionally restricts the augmented recognition law to the full factorization

$$Q(dy, dj) = Q_Y(dy) \prod_{i \in \mathcal{I}_{\mathrm{att}}} \beta^Q_i(dj_i), \qquad \beta^Q_i(dj_i) \text{ independent of } y. \tag{7.36}$$

This assumption imposes both conditional independence among the source labels and independence of every label row from the latent state. It is not implied by the generative law. For the finite-valued formulas below, also assume $\mathbb{E}_{Q_Y}|D_{ij}| < +\infty$ on every active source and $\mathbb{E}_{Q_Y}|\log c_i(o_i, Y)| < +\infty$. [HYPOTHESIS]

Proposition 7.6 (Exact posterior and recognition attention). Under the generative label-exclusivity hypothesis (7.35), conditioning on y and the complete interaction record gives, on the positive-likelihood set $L_o^Y(y)\prod_{k\in\mathcal{I}_{\mathrm{att}}} c_k(o_k, y) > 0$, the generative posterior row

$$\beta_{ij}^P(y) = \frac{\pi_{ij}\exp[-D_{ij}(y)/\tau_i]}{\sum_k \pi_{ik}\exp[-D_{ik}(y)/\tau_i]}. \tag{7.37}$$

and the same display may be chosen as its version off that set. Under the additional recognition restriction (7.36), the exact categorical contribution of row i to the collective VFE is

$$-\mathbb{E}_{Q_Y}\log c_i(o_i, Y) + \mathcal{F}_i^{\mathrm{att}}(\beta_i^Q), \tag{7.38}$$

where the first term is independent of β_i^Q and

$$\mathcal{F}_i^{\mathrm{att}}(\beta_i^Q) = D_{\mathrm{KL}}(\beta_i^Q\|\pi_i) + \frac{1}{\tau_i}\sum_j \beta_{ij}^Q \mathbb{E}_{Q_Y} D_{ij}. \tag{7.39}$$

Its unique interior minimizer on the positive-prior source simplex is

$$\beta_{ij}^{Q\star} = \frac{\pi_{ij}\exp[-\mathbb{E}_{Q_Y} D_{ij}/\tau_i]}{\sum_k \pi_{ik}\exp[-\mathbb{E}_{Q_Y} D_{ik}/\tau_i]}. \tag{7.40}$$

[ESTABLISHED]

Proof. The product label prior and the exclusive likelihood (7.35) make the conditional posterior over J factor by rows; every factor not containing j_i cancels. Bayes' formula then gives (7.37). With $Z_i^{\mathrm{att}}(y) = \sum_k \pi_{ik}\exp[-D_{ik}(y)/\tau_i]$, expanding the categorical relative entropy to that posterior gives

$$D_{\mathrm{KL}}(\beta_i^Q\|\beta_i^P) = D_{\mathrm{KL}}(\beta_i^Q\|\pi_i) + \tau_i^{-1}\sum_j \beta_{ij}^Q D_{ij} + \log Z_i^{\mathrm{att}}(y).$$

Subtracting $\log Z_i^{\mathrm{att}}(y)$ and the source-independent log likelihood $\log c_i(o_i, y)$, then taking the recognition expectation, proves (7.38) and (7.39). Lagrange multiplication under $\sum_j \beta_{ij}^Q = 1$ proves (7.40). □

The two softmax rows answer different variational questions. $\beta_i^P(y)$ is the generative posterior at a fixed latent state, whereas $\beta_i^{Q\star}$ is the optimum after forcing one recognition row to be constant in y. In general $\beta_i^{Q\star} \neq \mathbb{E}_{Q_Y}[\beta_i^P(Y)]$; the two formulas agree pointwise when all differences $D_{ij}(Y) - D_{ik}(Y)$ are Q_Y-almost surely constant. [ESTABLISHED]

For completeness, still under (7.35), let an arbitrary recognition law disintegrate as $Q(dy, dj) = Q_Y(dy)Q_{J|Y}(dj \mid y)$. With $Q_{J_i|Y}$ denoting its conditional row marginals and the label-free term $-\mathbb{E}_{Q_Y} \log L_o^Y(Y)$ accounted for in the state sector, the exact attention-label contribution is

$$\begin{aligned} &- \sum_{i \in \mathcal{I}_{\mathrm{att}}} \mathbb{E}_{Q_Y} \log c_i(o_i, Y) + \mathbb{E}_{Q_Y} \operatorname{TC}(Q_{J|Y}) \\ &+ \sum_{i \in \mathcal{I}_{\mathrm{att}}} \mathbb{E}_{Q_Y} \left[D_{\mathrm{KL}}(Q_{J_i|Y} \| \pi_i) + \frac{1}{\tau_i} \mathbb{E}_{Q_{J_i|Y}} D_{iJ_i}(Y) \right], \end{aligned} \tag{7.41}$$

where $\operatorname{TC}(Q_{J|Y=y}) = D_{\mathrm{KL}}\left(Q_{J|Y=y} \middle\| \prod_{i \in \mathcal{I}_{\mathrm{att}}} Q_{J_i|Y=y}\right)$. Thus merely requiring every row marginal to be independent of y does not justify (7.39): a correlated conditional law retains the nonnegative conditional-total-correlation term. If conditional dependence is unrestricted, minimizing over all $Q_{J|Y=y}$ gives the product of the pointwise rows (7.37). Under a constrained coupled family, the coupling term or constraint enters the coordinate optimum, so a row softmax is not asserted. [ESTABLISHED]

The familiar row functional $\sum_j \beta_{ij} \mathbb{E}_{Q_Y} D_{ij} + \tau_i D_{\mathrm{KL}}(\beta_i \| \pi_i)$ is $\tau_i \mathcal{F}_i^{\mathrm{att}}$. It has the same row minimizer, but for $\tau_i \neq 1$ it is not an independently weighted sector of one standard global ELBO unless the entire objective is scaled or a different generalized free energy is declared. [ESTABLISHED]

Holding the latent recognition law Q_Y, and hence every $\mathbb{E}_{Q_Y} D_{ij}$, fixed during this row-coordinate flow, the Fisher natural gradient of (7.39) on the open simplex is the replicator equation

$$\dot{\beta}_{ij}^Q = -\gamma_i \beta_{ij}^Q \left(c_{ij} - \sum_k \beta_{ik}^Q c_{ik} \right), \qquad c_{ij} = \log \frac{\beta_{ij}^Q}{\pi_{ij}} + \frac{\mathbb{E}_{Q_Y} D_{ij}}{\tau_i}. \tag{7.42}$$

for $\gamma_i \geq 0$. Its partial contribution to the row-free-energy rate is

$$\left. \frac{d}{dt} \mathcal{F}_i^{\mathrm{att}} \right|_{\beta_i} = -\gamma_i \operatorname{Var}_{\beta_i^Q}(c_{ij}) \leq 0. \tag{7.43}$$

If Q_Y evolves simultaneously, its additional chain-rule term must be included. [ESTABLISHED]

Chapter 16 supplies a different dynamical realization inside a fixed normalized joint: source assignments jump at positive rates while the full correlated recognition law and a model parameter evolve. The inherited parent rates average the microscopic rates against the optimizing hidden conditional; they are not obtained by inserting a moving latent law into (7.42). Its complete joint entropy and own-law score estimates account for those coupled changes. [ESTABLISHED]

7.6 Local evolution is collective descent

Let a regular product recognition family Q_η on an open parameter domain have coordinates $\eta = (\eta_i)_i$ and block-diagonal Fisher metric $G(\eta) = \bigoplus_i G_i(\eta_i)$, including categorical blocks when present. Assume $F(\eta) := \mathcal{F}_o^{\mathrm{ext}}(Q_\eta) \in \mathbb{R}$ is C^1, every $G_i(\eta_i)$ is positive definite, every $\gamma_i \geq 0$, and the trajectory remains in this domain. By Theorem 7.5, the gradient of the outside-averaged local VFE

is the corresponding block gradient of F. Hence

$$\dot{\eta}_i = -\gamma_i G_i^{-1} \nabla_{\eta_i} F \tag{7.44}$$

obeys

$$\frac{d}{dt} F = -\sum_i \gamma_i (\nabla_{\eta_i} F)^\top G_i^{-1} (\nabla_{\eta_i} F) \leq 0. \tag{7.45}$$

Thus an agent following its exact local VFE follows a block of the collective VFE flow. [ESTABLISHED]

Block orthogonality is load bearing. With a nondiagonal Fisher metric, the global natural gradient mixes agents and independent local inversions are a different dynamics. Hard support boundaries and finite steps also require a separate projected flow, line search, damping, or acceptance test. [ESTABLISHED]

The variable t in (7.44) parameterizes the declared block flow. Passing from a regular solution curve to its oriented unparameterized orbit, and then measuring that orbit by Fisher length, are the separate constructions of Definition 9.4 and theorem 9.54. A common positive scalar mobility changes only the parameterization, whereas unequal block rates γ_i generally change the orbit. [ESTABLISHED]

No physical-time interpretation of t or of the resulting Fisher duration is asserted. [NOT-CLAIMED]

7.7 Are observations only agent–agent interactions?

Definition 7.7 (Operational environment node). An operational environment node is a standard Borel state space with a normalized state law and a measurable message kernel. This definition does not assert autonomy, self-modeling, or biological agency. [DEFINITION]

Theorem 7.8 (Observation–interaction kernel equivalence). Every normalized standard-Borel observation kernel $K(do \mid y)$ can be realized as a message from an operational environment node: there are $U \sim \mathrm{Unif}[0,1]$ and a measurable map F such that

$$O = F(Y, U), \qquad \Pr(F(y, U) \in A) = K(A \mid y). \tag{7.46}$$

Conversely, marginalizing any environment state and message policy produces a normalized observation kernel. [ESTABLISHED]

Proof. The forward statement is the randomization lemma for kernels on standard Borel spaces [46]. For the converse, if the environment has law $\rho(du)$ and policy $M(y, u, A)$, then $K(A \mid y) = \int M(y, u, A)\rho(du)$ is measurable in y, countably additive in A, and normalized. □

The theorem gives an exact operationally agent-only presentation when every datum is represented as a clamped state or message of a boundary node and all exogenous noise is included among those nodes. It does not eliminate the conditioning role. If O denotes the complete record map,

the ELBO is conditioned on the retained sigma-algebra $\mathscr{O} = \sigma(O)$. Replacing the environment by boundary nodes preserves the same random variable O and hence the same $\mathscr{O}$; it does not delete it. In particular,

$$I(Y;O) = \int D_{\mathrm{KL}}\big(P(dY \mid o) \| P^{Y}(dY)\big) P^{O}(do), \tag{7.47}$$

with values in $[0, +\infty]$. This quantity measures how much the retained interaction record changes the latent law on average. [ESTABLISHED]

More generally, retaining only a statistic $S = T(O)$ gives the posterior $P(dY \mid S)$, not $P(dY \mid O)$. The deterministic-statistic chain rule

$$I(Y;O) = I(Y;S) + I(Y;O \mid S) \tag{7.48}$$

shows that this replacement is posterior-sufficient exactly when $I(Y;O \mid S) = 0$ (up to the usual almost-sure choice of conditional versions). Complete deletion makes S constant and leaves P^{Y}. A finite witness is $Y \sim \mathrm{Bernoulli}(1/2)$ with the deterministic message $O = Y$: then $I(Y;O) = \log 2$, $P(dY \mid O = o) = \delta_o$, while deletion returns the nondegenerate prior. Thus an agent-only redescription cannot remove the conditioning sigma-algebra unless it also accepts the corresponding loss of information. [ESTABLISHED]

Nor does the operational equivalence prove that a random seed is an autonomous agent. If physical agency requires persistent state, a Markov blanket, action, or its own local VFE, those are additional hypotheses. An Y-independent kernel is a boundary message with zero mutual information and therefore a null interaction in the informational sense. [ESTABLISHED]

The resulting answer is two-tiered. The probability theory can be written entirely as interactions among agent and environment nodes, while the ELBO still conditions on the sigma-algebra generated by those interaction records. Whether every environment node deserves the stronger ontological name “agent” is not determined by the principal bundle or by variational inference. [NOT-CLAIMED]

Chapter 8

Connection-Relative Informational Pullback Geometry

The preceding chapters separate the principal bundle, the two associated law bundles, and the normalized probabilistic model. This chapter adds a smooth information-geometric layer without identifying those objects. A chosen connection converts the derivative of an agent section into a vertical first jet. The Fisher metric and the Amari–Chentsov tensor can then be pulled back to the contextual base. The resulting tensors are invariant under passive changes of frame, but they depend on the selected connection and may be degenerate. These qualifications are part of the construction rather than defects to be removed by notation.

The same construction also supplies the differential interface to coarse-graining. A coarse morphism whose scale-connection defect form takes values in the isotropy subalgebra of the coarse section value intertwines the covariant first jets, while a parameter-independent Markov channel contributes a positive-semidefinite vertical Fisher defect. That isotropy condition is the only meaning this chapter attaches to compatibility of the two connections, and it is stated as (8.56) rather than left to the reader. The vertical defect composes by an exact unconditional cocycle identity, whereas its pullback to the base composes only up to an exact residual that this chapter displays. Deterministic energy precomposition and Galerkin aggregation do not satisfy the Markov hypothesis merely because they are called coarse maps.

8.1 Statistical tensors on the associated fibers

Hypothesis 8.1 (Regular channel models). For each channel $x \in \{b, m\}$, let $\mathcal{B}_x$ be a finite-dimensional smooth parametrized-measure model satisfying the selected smooth-tier hypothesis of Hypothesis 2.3. Assume differentiability in quadratic mean, a positive-definite Fisher form, third-power integrability of every score direction used below, smooth dependence of the resulting second- and third-order tensors, and domination sufficient to differentiate the divergence expressions below through third order. Assume also that the represented action $\widehat{\rho}_x(g)$ is induced by a parameter-independent bimeasurable change of sample coordinates and preserves $\mathcal{B}_x$. These assumptions exclude singular statistical strata, changing-support models without the stated differentiability, and arbitrary actions on parameter coordinates. [HYPOTHESIS]

At $p \in \mathcal{B}_x$, write ℓ_u for the score in direction $u \in T_p\mathcal{B}_x$. Define

$$g^F_{x,p}(u,v) = \mathbb{E}_p[\ell_u \ell_v], \tag{8.1}$$

$$\mathcal{T}_{x,p}(u,v,w) = \mathbb{E}_p[\ell_u \ell_v \ell_w]. \tag{8.2}$$

The first is the Fisher metric and the second is the symmetric Amari–Chentsov tensor. Their existence here is a statement about the selected parametrized-measure models, not about arbitrary law fibers. [DEFINITION]

Proposition 8.2 (Descent of the statistical tensors). Under Hypothesis 8.1, the represented G-action preserves g^F_x and $\mathcal{T}_x$. Consequently they descend to smooth vertical tensors, denoted by the same symbols, on $\mathcal{E}_x = \mathscr{P}_G \times_{\widehat{\rho}_x} \mathcal{B}_x$. [ESTABLISHED]

Proof. Let r_g be the bimeasurable sample-coordinate change representing g. Pushforward defines the unitary map $L^2(p) \to L^2((r_g)_{\#}p)$ given by $f \mapsto f \circ r_g^{-1}$. Because r_g is independent of the statistical parameter, differentiability in quadratic mean identifies the score of the pushed-forward tangent direction with $\ell_u \circ r_g^{-1}$. The pushforward integration identity preserves the second and third score moments in (8.1) and (8.2). An invariant tensor on the fiber descends through the associated-bundle quotient. Explicitly, let $\iota_u : \mathcal{B}_x \to \mathcal{E}_x$ send β to the class $[u, \beta]$ of a principal frame u. The quotient convention (2.9) gives $\iota_{ug} = \iota_u \circ \widehat{\rho}_x(g)$, so a fiber tensor τ has frame-independent transport $\tau^u := (\iota_u^{-1})^*\tau$ exactly when $\widehat{\rho}_x(g)^*\tau = \tau$ for every g. Smoothness of the descended tensor follows by evaluating this construction on local smooth sections of $\mathscr{P}_G$. □

The bimeasurable sample action in this proposition is load bearing. Closure of a set of parameter values under an arbitrary diffeomorphism of its parameter chart would not prove statistical isometry and would not define the same descended tensors. [ESTABLISHED]

8.2 Covariant first jets and pullback tensors

Fix one channel x temporarily and write $E = \mathcal{E}_x$, $\mathcal{B} = \mathcal{B}_x$, $\varpi = \varpi_x$, $g^F = g^F_x$, $\mathcal{T} = \mathcal{T}_x$, and $\omega = \omega_x$. The connection induces a horizontal subbundle $H^\omega E \subset TE$ and a vertical projection $\mathrm{ver}^\omega : TE \to VE$. For a smooth section $s : \mathcal{U} \to E$ over an open set $\mathcal{U} \subseteq \mathcal{C}$, define

$$D^\omega s := \mathrm{ver}^\omega \circ Ts : T\mathcal{U} \longrightarrow s^*VE. \tag{8.3}$$

The connection-split first jet is the pair $j^1_\omega s = (s, D^\omega s)$. For nonlinear statistical fibers, $D^\omega s$ is a vertical-valued one-form along s; it is not a linear covariant derivative on a vector space of sections. [DEFINITION]

Definition 8.3 (Covariant informational pullbacks). The connection-relative Fisher and Amari pullbacks of s are

$$h_s^\omega(X,Y) = g^F_{s(c)}(D^\omega s(X), D^\omega s(Y)), \tag{8.4}$$

$$c_s^\omega(X,Y,Z) = \mathcal{T}_{s(c)}(D^\omega s(X), D^\omega s(Y), D^\omega s(Z)), \tag{8.5}$$

for $X, Y, Z \in T_c\mathcal{C}$. Thus $h_s^\omega \in \Gamma(\mathrm{Sym}^2 T^*\mathcal{U})$ and $c_s^\omega \in \Gamma(\mathrm{Sym}^3 T^*\mathcal{U})$. The first is called a semimetric until its nondegeneracy is proved. [DEFINITION]

Theorem 8.4 (Passive gauge covariance of the covariant pullbacks). Under a passive local change of principal frame, with the connection and section coordinates transformed by their associated laws, both h_s^ω and c_s^ω are unchanged as tensors on $\mathcal{U}$. [ESTABLISHED]

Proof. In a local frame, the vertical jet transforms by the tangent representation,

$$D^{\omega'} s'(X) = T\widehat{\rho}_x(g^{-1}) D^\omega s(X), \tag{8.6}$$

where $g : \mathcal{U} \to G$ is the frame change and the inverse follows from the quotient convention (2.9). The connection law is (2.22) or (2.23), according to the channel. By Proposition 8.2, the tangent representation is an isometry of g^F and preserves $\mathcal{T}$. Substitution in (8.4) and (8.5) cancels the frame change in every argument. □

The word invariance in Theorem 8.4 means passive covariance: two coordinate descriptions of the same pair (ω, s) give the same tensor. It does not say that two different connections give the same pullback, and it does not say that the tensors are fixed by the active gauge group at a fixed connection. For the active statement, take the trivial principal $\mathbb{R}$-bundle over $\mathbb{R}$ with the normal location fiber, the zero connection, and the zero section, so that $h = 0$; the bundle automorphism $F(u(x)g) = u(x)(x+g)$ carries that section to $s'(x) = \mathcal{N}(x,1)$ and hence h to dx^2. [ESTABLISHED]

8.3 Exact dependence on the chosen connection

Let ω' be a second principal connection on $\mathscr{P}_G$. Their difference is horizontal and Ad-equivariant; denote its descended form by

$$a \in \Omega^1(\mathcal{C}, \mathrm{Ad}(\mathscr{P}_G)). \tag{8.7}$$

For the associated action, let $\vartheta_e : \mathrm{Ad}(\mathscr{P}_G)_{\varpi(e)} \to V_eE$ denote the fundamental vertical map. Along s, set

$$R^s_a(X) = \vartheta_{s(c)}(a_c(X)). \tag{8.8}$$

The sign convention is fixed frame-free by (8.8): in a local trivialization the horizontal lift of X at (c,β) is $(X, -\zeta_{A(X)}\beta)$, where ζ_ξ is the fundamental vertical field of $\xi \in \mathfrak{g}$ for the represented action, so replacing A by $A + a$ adds $\vartheta(a(X))$ to the vertical jet. For a general law fiber $\mathcal{B} \subseteq \mathcal{P}(\mathsf{K})$ the fiber is not a vector space and the linear expression $\widehat{\rho}_{x*}(A)$ is undefined; it is available only in a declared

fiberwise-linear associated-vector-bundle realization, where it agrees with $\zeta_{A(X)}$. The convention is compatible with (2.9). [DEFINITION]

Proposition 8.5 (Connection-change formulas). The two vertical jets and their Fisher pullbacks obey

$$D^{\omega'}s = D^{\omega}s + R^s_a, \tag{8.9}$$

$$h^{\omega'}_s(X,Y) = h^{\omega}_s(X,Y) + g^F(R^s_aX, D^{\omega}sY) + g^F(D^{\omega}sX, R^s_aY) + g^F(R^s_aX, R^s_aY). \tag{8.10}$$

The Amari pullback satisfies the exact polarized identity

$$c^{\omega'}_s(X,Y,Z) = \mathcal{T}((D^{\omega}s + R^s_a)X, (D^{\omega}s + R^s_a)Y, (D^{\omega}s + R^s_a)Z). \tag{8.11}$$

It therefore contains the original term, three terms with one occurrence of R^s_a, three terms with two occurrences, and one term with three occurrences. [ESTABLISHED]

Proof. In one local frame, replacing the horizontal lift $(X, -\zeta_{A(X)}\beta)$ by $(X, -\zeta_{A(X)+a(X)}\beta)$ shifts the vertical part of Ts by $\vartheta_{s(c)}(a(X)) = R^s_a(X)$, which proves (8.9). Bilinearity of g^F gives (8.10), and trilinearity of $\mathcal{T}$ gives (8.11). The formulas are independent of the selected local frame because a is an $\mathrm{Ad}(\mathscr{P}_G)$-valued one-form and every displayed contraction is gauge invariant. □

The dependence can occur even in the simplest regular family. Let $\mathcal{C} = \mathbb{R}$, let $\mathscr{P}_G$ be the trivial principal translation bundle, and let $\mathcal{B} = \{\mathcal{N}(\mu, 1) : \mu \in \mathbb{R}\}$. For the constant section $\mu = 0$, the zero connection gives $h = 0$, whereas the local connection $A' = a_0 dx$ gives

$$h^{A'} = a_0^2 dx^2. \tag{8.12}$$

This example refutes a connection-independent interpretation of h^{ω}_s; it does not refute the gauge invariance of a fixed connection–section pair. [ESTABLISHED]

8.4 Two channel geometries and their types

Restoring the channel labels gives the ordered pair

$$(h^{\omega_b}_{i,b}, h^{\omega_m}_{i,m}) \in \Gamma(\mathrm{Sym}^2 T^*\mathcal{U}) \times \Gamma(\mathrm{Sym}^2 T^*\mathcal{U}), \tag{8.13}$$

and an analogous pair of cubic tensors. This pair is determined by the two selected statistical models, sections, and connections. A single scalar geometry is not part of those data. [ESTABLISHED]

Hypothesis 8.6 (Weighted product geometry). If one joint geometry is required, choose constants $w_b, w_m > 0$ and define

$$h^{\mathrm{prod}} = w_b h^{\omega_b}_{i,b} + w_m h^{\omega_m}_{i,m}. \tag{8.14}$$

The weights fix the relative normalization of the two statistical channels. Unit weights are a specialization rather than a consequence of the common principal bundle. [HYPOTHESIS]

Whenever weighted products at two different scales are compared, two further declarations are consumed and neither follows from any fiberwise contraction theorem. Writing μ and $\bar{\mu}$ for the finite positive base measures used to integrate the two tensors and $w_x, \bar{w}_x$ for the two sets of channel weights, the comparison hypotheses are

$$\textbf{(X1)} \quad f_{\#}\mu = \bar{\mu}, \qquad \textbf{(X2)} \quad \bar{w}_x \circ f \leq w_x \quad \text{for } x \in \{b, m\}. \tag{8.15}$$

Under a channelwise comparison the two-channel difference is

$$h^{\mathrm{prod}} - f^{*}\bar{h}^{\mathrm{prod}} = \sum_{x \in \{b,m\}} \left[w_x \left(h_x - f^{*}\bar{h}_x \right) + \left(w_x - \bar{w}_x \circ f \right) f^{*}\bar{h}_x \right], \tag{8.16}$$

so independently declared coarse weights with $(\bar{w}_x \circ f) > w_x$ break positivity even when every channel has zero horizontal defect and a genuine Markov fiber map. Failure of (X1) reverses the integrated comparison just as directly: take $\mathcal{C} = \bar{\mathcal{C}}$ the disjoint union of two copies of $\mathbb{R}$ with $f = \mathrm{id}$, let h vanish on the first copy and equal dx^2 on the second, and let μ and $\bar{\mu}$ be unit point masses on the first and the second copy respectively; the fine integrated tensor is zero and the coarse one is one, with vanishing fiberwise defect throughout. [ESTABLISHED]

Proposition 8.7 (Radical of the weighted product). Under Hypotheses 8.1 and 8.6,

$$\operatorname{rad} h^{\mathrm{prod}} = \ker D^{\omega_b} q_i \cap \ker D^{\omega_m} q_i^m. \tag{8.17}$$

Hence the two degenerate channel tensors can jointly be nondegenerate exactly when their null distributions have zero intersection. [ESTABLISHED]

Proof. For every X,

$$h^{\mathrm{prod}}(X, X) = w_b g_b^F (D^{\omega_b} q_i X, D^{\omega_b} q_i X) + w_m g_m^F (D^{\omega_m} q_i^m X, D^{\omega_m} q_i^m X).$$

Both summands are nonnegative and both weights are positive. Their sum vanishes exactly when both vertical jets vanish. Polarization then identifies the radical. □

The common principal group acts separately through $\widehat{\rho}_b$ and $\widehat{\rho}_m$. It supplies no bilinear cross tensor on $V\mathcal{E}_b \times V\mathcal{E}_m$. Such a cross term requires a separately declared invariant bilinear form, and a comparison of channel points requires Φ or $\widetilde{\Phi}$ from Section 2.4. Neither the relative principal frame nor equal channel dimensions supplies either object. [ESTABLISHED]

8.5 Rank, radicals, and quotient geometry

Theorem 8.8 (Constant-rank quotient theorem). Let $s : \mathcal{U} \to E$ satisfy Hypothesis 8.1. Pointwise,

$$\operatorname{rad} h_s^\omega = \ker D^\omega s, \qquad \operatorname{rank} h_s^\omega = \operatorname{rank} D^\omega s. \tag{8.18}$$

If $D^\omega s$ has constant rank on $\mathcal{U}$, then $K_s^\omega = \ker D^\omega s$ is a smooth subbundle and

$$Q_s^\omega = T\mathcal{U}/K_s^\omega, \tag{8.19}$$

has a positive-definite metric and a symmetric cubic tensor defined by

$$\bar{h}_s^\omega([X],[Y]) = h_s^\omega(X,Y), \tag{8.20}$$

$$\bar{c}_s^\omega([X],[Y],[Z]) = c_s^\omega(X,Y,Z). \tag{8.21}$$

The induced map $Q_s^\omega \to \operatorname{im} D^\omega s$ is an isometric vector-bundle isomorphism. [ESTABLISHED]

Proof. If $D^\omega sX = 0$, then (8.4) gives $h_s^\omega(X,Y) = 0$ for all Y. Conversely, if X lies in the radical, then

$$0 = h_s^\omega(X,X) = g^F(D^\omega sX, D^\omega sX),$$

and positive definiteness of g^F forces $D^\omega sX = 0$. This proves (8.18). The applicable constant-rank statement is the one for morphisms of smooth vector bundles over a fixed base, not the constant-rank theorem for smooth maps of manifolds: $D^\omega s$ is a vector-bundle morphism $T\mathcal{U} \to s^*VE$ over id, and a vector-bundle morphism of locally constant rank has smooth kernel and image subbundles by local adaptation of frames. Replacing an argument of (8.20) or (8.21) by a representative differing by an element of K_s^ω leaves its vertical jet unchanged. The quotient tensors are therefore well defined. The factored map induced by $D^\omega s$ is injective onto its image and preserves the displayed tensors; the quotient metric is positive definite. □

The quotient in Theorem 8.8 is a vector bundle. It is not automatically the tangent bundle of a quotient manifold. For that interpretation, require a smooth Hausdorff quotient manifold $\overline{\mathcal{U}}$ and a surjective submersion $q_K : \mathcal{U} \to \overline{\mathcal{U}}$ whose fibers are exactly the null leaves and whose tangent kernel is $\ker Tq_K = K_s^\omega$. Consequently, the null distribution must be involutive,

$$D^\omega s([X,Y]) = 0 \quad \text{for all } X, Y \in \Gamma(K_s^\omega), \tag{8.22}$$

but involutivity alone supplies neither the quotient manifold nor the surjective-submersion quotient map. [ESTABLISHED]

Separately, the tensors must be basic for the null foliation. Their contraction with a null vector already vanishes by construction, but invariance along null leaves is an additional requirement:

$$\mathcal{L}_Z h_s^\omega = 0, \qquad \mathcal{L}_Z c_s^\omega = 0 \quad \text{for every } Z \in \Gamma(K_s^\omega). \tag{8.23}$$

Without this condition, even a smooth Hausdorff leaf-space quotient need not carry tensors whose pullbacks are h^ω_s and c^ω_s. [ESTABLISHED]

Basicness of the quadratic tensor is checkable directly on the jet, but the bundle-valued one-form $D^\omega s$ has no ordinary Lie derivative. Choose a g^F-compatible auxiliary connection ∇ on s^*VE and, for a s^*VE-valued one-form α, write

$$(\mathcal{L}^\nabla_Z \alpha)(X) = \nabla_Z(\alpha(X)) - \alpha([Z, X]). \tag{8.24}$$

Then for $Z \in \Gamma(K^\omega_s)$ and arbitrary X, Y, the Leibniz rule for the ordinary Lie derivative of h^ω_s gives

$$(\mathcal{L}_Z h^\omega_s)(X, Y) = g^F\left((\mathcal{L}^\nabla_Z D^\omega s)(X), D^\omega sY\right) + g^F\left(D^\omega sX, (\mathcal{L}^\nabla_Z D^\omega s)(Y)\right). \tag{8.25}$$

Metric compatibility supplies this identity directly, while $\iota_Z h^\omega_s = 0$ is already automatic from $Z \in \ker D^\omega s$. If $\nabla' = \nabla + B$ is another g^F-compatible connection, then B_Z is g^F-skew-adjoint, so the changes in the two displayed covariant summands cancel. Thus the displayed equality with the ordinary, canonical $\mathcal{L}_Z h^\omega_s$ is independent of the selected auxiliary metric connection, even though neither covariant summand is separately canonical. [ESTABLISHED]

Proposition 8.9 (A constant-rank nonintegrable radical). There is a regular one-dimensional statistical fiber, a smooth connection, and a section over $\mathbb{R}^3$ for which h^ω_s has constant rank one while $\ker D^\omega s$ is not involutive. [ESTABLISHED]

Proof. Use the normal location family and the trivial principal translation bundle over $\mathbb{R}^3$ with coordinates (x, y, z). In the constant section $\mu = 0$, choose the local connection form

$$\alpha = \mathrm{d}z - x\mathrm{d}y. \tag{8.26}$$

Then $D^\omega s = \alpha$ and

$$h^\omega_s = (\mathrm{d}z - x\mathrm{d}y)^2. \tag{8.27}$$

The rank is one everywhere, but

$$\alpha \wedge \mathrm{d}\alpha = -\mathrm{d}z \wedge \mathrm{d}x \wedge \mathrm{d}y \neq 0. \tag{8.28}$$

The Frobenius criterion therefore shows that $\ker \alpha$ is a contact distribution rather than a foliation. □

Constant rank is also necessary for the smooth quotient-bundle conclusion. For the same normal location family on $\mathcal{C} = \mathbb{R}$, with the zero connection and $s(x) = \mathcal{N}(x^2, 1)$,

$$h_s = 4x^2 dx^2, \tag{8.29}$$

has rank one away from the origin and rank zero at the origin. The pointwise quotients do not assemble into a vector bundle across that point. [ESTABLISHED]

8.6 Divergence jets and transported local comparison

Let $\mathscr{D} : \mathcal{B} \times \mathcal{B} \to \mathbb{R}$ be a C^3 contrast in a neighborhood of the diagonal: $\mathscr{D}(p, q) \geq 0$ and $\mathscr{D}(p, q) = 0$ exactly on the diagonal in that neighborhood. For tangent vectors below, assume additionally that $\mathscr{D}$ is invariant under the simultaneous represented action, $\mathscr{D}(\widehat{\rho}(g)p, \widehat{\rho}(g)q) = \mathscr{D}(p, q)$, so that it defines an unambiguous comparison of two points in one associated-bundle fiber. Without this invariance, the following jets are only local-frame objects. Assume also that the mixed second form defined below is positive definite on the selected regular tier. A contrast such as $\mathscr{D}(\mu, \nu) = (\mu - \nu)^4$ is positive off the diagonal but has a zero diagonal Hessian; it yields degenerate jets rather than a divergence- induced dualistic geometry. [HYPOTHESIS]

For tangent vectors $u, v, w \in T_p\mathcal{B}$, let $(u, 0)$ and $(0, u)$ denote their lifts to the first and second factors. In one common local chart, extend these vectors as constant coordinate vector fields and define the mixed diagonal jets

$$g_{\mathscr{D}}(u, v) = -[(u, 0)(0, v)\mathscr{D}]_{(p,p)}, \tag{8.30}$$

$$\Gamma_{\mathscr{D}}(u, v, w) = -[(u, 0)(v, 0)(0, w)\mathscr{D}]_{(p,p)}, \tag{8.31}$$

$$\Gamma^*_{\mathscr{D}}(u, v, w) = -[(0, u)(0, v)(w, 0)\mathscr{D}]_{(p,p)}, \tag{8.32}$$

$$\mathcal{T}_{\mathscr{D}} = \Gamma_{\mathscr{D}} - \Gamma^*_{\mathscr{D}}. \tag{8.33}$$

The individual third-order arrays are connection coefficients in a chart; their difference is a cubic tensor. The sign convention in (8.33) is chosen so that forward KL on an exponential family gives the score-third-moment tensor (8.2). [DEFINITION]

Proposition 8.10 (KL divergence jets). On a regular statistical model satisfying the integrability hypotheses of Hypothesis 8.1, the mixed second jet of $D_{\mathrm{KL}}(p\|q)$ is the Fisher metric, and the difference of the two mixed third jets is the Amari–Chentsov tensor. [ESTABLISHED]

Proof. Use a local parameter θ for the first argument and η for the second. Write $\ell_i = \partial_i \log p_\theta$ and let primed derivatives act on η. Normalization gives $\mathbb{E}_\theta[\ell_i] = 0$, while $\partial_i p_\theta = p_\theta \ell_i$ and $\partial_i\partial_j p_\theta = p_\theta(\ell_i\ell_j + \partial_i\ell_j)$. Direct differentiation at $\eta = \theta$ therefore gives

$$-\partial_i\partial'_j D_{\mathrm{KL}}(p_\theta\|p_\eta) = \mathbb{E}_\theta[\ell_i\ell_j], \tag{8.34}$$

$$-\partial_i\partial_j\partial'_k D_{\mathrm{KL}}(p_\theta\|p_\eta) = \mathbb{E}_\theta[(\ell_i\ell_j + \partial_i\ell_j)\ell_k], \tag{8.35}$$

$$-\partial'_i\partial'_j\partial_k D_{\mathrm{KL}}(p_\theta\|p_\eta) = \mathbb{E}_\theta[\ell_k\partial_i\ell_j]. \tag{8.36}$$

The score Hessian is symmetric in the chosen smooth chart, so subtracting the last line from the second cancels its score-derivative term and leaves $\mathbb{E}_\theta[\ell_i\ell_j\ell_k]$. Polarization gives the asserted formulas for arbitrary u, v, w. Differentiation under the integral is licensed by Hypothesis 8.1; these are the standard divergence-induced dualistic structures of information geometry [3, 7]. □

For a smooth base curve γ with $\gamma(0) = c$, let $\Omega^{\omega}_{0\to\epsilon}$ be the channel transport along $\gamma|_{[0,\epsilon]}$ and bring the section value back to the initial fiber by

$$\widehat{s}_{\gamma}(\epsilon) = (\Omega^{\omega}_{0\to\epsilon})^{-1} s(\gamma(\epsilon)). \tag{8.37}$$

Then

$$\frac{d}{d\epsilon}\widehat{s}_{\gamma}(\epsilon)\Big|_{\epsilon=0} = D^{\omega}s(\dot{\gamma}(0)). \tag{8.38}$$

[ESTABLISHED]

Corollary 8.11 (Quadratic transported divergence). Let $g_{\mathscr{D}}$ be the metric induced by (8.30) and set $h^{\omega}_{s,\mathscr{D}} = (D^{\omega}s)^{*}g_{\mathscr{D}}$. Then

$$\mathscr{D}(s(c), \widehat{s}_{\gamma}(\epsilon)) = \frac{\epsilon^2}{2} h^{\omega}_{s,\mathscr{D}}(\dot{\gamma}(0), \dot{\gamma}(0)) + O(\epsilon^3). \tag{8.39}$$

If $g_{\mathscr{D}} = g^{F}$, in particular for the regular KL contrast of Proposition 8.10, then $h^{\omega}_{s,\mathscr{D}} = h^{\omega}_{s}$. [ESTABLISHED]

Proof. A contrast and its first derivative vanish on the diagonal. Its quadratic Taylor term in the second argument is the induced metric. Substituting (8.38) gives (8.39). □

The coefficient of ϵ^3 in a one-sided expansion such as (8.39) is not determined by $c^{\omega}_{s,\mathscr{D}} := (D^{\omega}s)^{*}\mathcal{T}_{\mathscr{D}}$ alone. It also contains the covariant acceleration of $\widehat{s}_{\gamma}$ and the connection coefficient selected by the one-sided divergence jet. The divergence-induced cubic is recovered from the difference of the two mixed third jets in (8.33), not from an arbitrary one-sided cubic Taylor coefficient. For regular KL, it is the Amari–Chentsov tensor and $c^{\omega}_{s,\mathscr{D}} = c^{\omega}_{s}$. [ESTABLISHED]

8.7 Vertical information length

For a smooth curve $\Gamma : J \to E$ over $\gamma = \varpi \circ \Gamma$, define its connection-relative vertical velocity by

$$v^{\omega}(\Gamma) = \mathrm{ver}^{\omega}(\dot{\Gamma}) = \dot{\Gamma} - H^{\omega}_{\Gamma}\dot{\gamma}, \tag{8.40}$$

and its vertical Fisher length by

$$L^{\omega}(\Gamma) = \int_{J} \sqrt{g^{F}(v^{\omega}(\Gamma), v^{\omega}(\Gamma))}\, d\lambda. \tag{8.41}$$

Here λ is an arbitrary curve parameter rather than a declared physical time. [DEFINITION]

If $\omega' = \omega + a$, define along the curve

$$R^{\Gamma}_{a}(\dot{\gamma}) = \vartheta_{\Gamma(\lambda)}\left(a_{\gamma(\lambda)}(\dot{\gamma}(\lambda))\right). \tag{8.42}$$

With the convention of (8.9), the exact curve identity is $v^{\omega'}(\Gamma) = v^{\omega}(\Gamma) + R^{\Gamma}_{a}(\dot{\gamma})$. [ESTABLISHED]

Proposition 8.12 (Vertical, horizontal, and section curves). The length L^ω is invariant under orientation-preserving C^1 reparameterization with positive derivative. If Γ is vertical, then $L^\omega(\Gamma)$ is independent of the connection. If Γ is ω-horizontal, then $L^\omega(\Gamma) = 0$. For a section curve $\Gamma = s \circ \gamma$,

$$L^\omega(s \circ \gamma) = \int_J \sqrt{h^\omega_s(\dot\gamma, \dot\gamma)}\, d\lambda. \tag{8.43}$$

For a mixed curve, changing ω to ω' changes the vertical velocity by $R^\Gamma_a(\dot\gamma)$ in the convention of (8.9); its length is therefore generally connection dependent. [ESTABLISHED]

Proof. Under a reparameterization $\lambda = \phi(r)$, vertical velocity is multiplied by $\phi'(r)$, and positive homogeneity of the square root converts that factor into the Jacobian in the one-dimensional change-of-variables formula. For a vertical curve, $\dot\gamma = 0$, so the horizontal correction vanishes for every connection. Horizontality is the equation $v^\omega(\Gamma) = 0$. If $\Gamma = s \circ \gamma$, then $v^\omega(\Gamma) = D^\omega s(\dot\gamma)$, and (8.43) follows from (8.4). The connection-change statement is (8.9) along the curve. □

The functional (8.41) measures only vertical statistical change. It assigns zero length to horizontal displacement and does not define a metric on TE without a separately chosen base metric. It also selects neither an inference orbit nor an orientation. Those require a separately declared configuration manifold together with an objective differential and a mobility or metric sharp map that defines its vector field. [NOT-CLAIMED]

A base curve γ orders contexts, and $s \circ \gamma$ evaluates one fixed section along that ordering; neither object is an agent inference history. Updating at one fixed c is instead a vertical curve in the relevant statistical fiber, while evolution of a complete contextual agent is a curve in a separately declared section configuration space. [DEFINITION]

The finite-dimensional length above applies pointwise or on a declared finite-design product. It does not construct a continuum section-space length or a probability law on a continuum space of sections, because no base measure, channel weighting, or gauge-quotient rule has been declared here. [NOT-CLAIMED]

8.8 Coarse-morphism naturality

Consider smooth associated bundles $E \to \mathcal{C}$ and $\bar E \to \bar{\mathcal{C}}$ with connections ω and $\bar\omega$. Let $f : \mathcal{C} \to \bar{\mathcal{C}}$ and let $\Psi : E \to \bar E$ be a smooth bundle morphism over f. Suppose sections s and $\bar s$ are related as follows. Before that relation is imposed, $\Psi \circ s$ is only a section of the pullback bundle $f^*\bar E$ along f; a fine section does not by itself define a section on $\bar{\mathcal{C}}$. [DEFINITION]

We adopt the descent hypothesis that there exists a smooth coarse section $\bar s$ satisfying

$$\Psi \circ s = \bar s \circ f. \tag{8.44}$$

For a surjective submersion f, this excludes fine sections whose Ψ-images vary along a fiber of f. [HYPOTHESIS]

Write $T^V\Psi : VE \to V\bar{E}$ for the vertical differential. Following the cross-channel defect of (2.26), define

$$(\mathcal{D}\Psi)_e(X) = A_\Psi(e;X) := \mathrm{ver}^{\bar{\omega}}\left(T_e\Psi(H^\omega_e X)\right) = T_e\Psi(H^\omega_e X) - H^{\bar{\omega}}_{\Psi(e)}(T_c f X). \tag{8.45}$$

The sign convention is fixed and load bearing: the transported fine horizontal lift comes first and the coarse horizontal lift is subtracted. The difference is vertical because $T\bar{\varpi}\circ T\Psi(H^\omega_e X) = T_c f X = T\bar{\varpi}\circ H^{\bar{\omega}}(T_c f X)$, and the second displayed form follows because the $\bar{\omega}$-horizontal part of $T\Psi(H^\omega_e X)$ is exactly $H^{\bar{\omega}}_{\Psi(e)}(T_c f X)$. Its type is $A_\Psi \in \Gamma(E; \varpi^* T^*\mathcal{C} \otimes \Psi^* V\bar{E})$, and along a section $A_\Psi(s;\cdot) \in \Gamma(\mathcal{U}; T^*\mathcal{U} \otimes f^* \bar{s}^* V\bar{E})$. The abbreviation A_Ψ is used below wherever the vertical values must be paired with $\bar{g}^F$. [DEFINITION]

Theorem 8.13 (Sharp section descent). Let f be a surjective smooth submersion, let Ψ be a smooth bundle morphism over f, and let $Q \in \Gamma(\mathcal{C}, E)$. Consider the three conditions: (P1) there exists $\bar{Q} \in \Gamma(\bar{\mathcal{C}}, \bar{E})$ with $\Psi \circ Q = \bar{Q} \circ f$; (P2) $\Psi \circ Q$ is constant on each fiber of f; and

$$\textbf{(P3)} \qquad T^V\Psi\left(D^\omega Q(X)\right) + A_\Psi\left(Q(c);X\right) = 0 \quad \text{for every } c \text{ and every } X \in \ker T_c f. \tag{8.46}$$

Then (P1) and (P2) are equivalent, both imply (P3), and (P3) implies (P2) when the fibers of f are connected. Under (P2) the descended $\bar{Q}$ is unique, is automatically smooth, and is automatically a section. [ESTABLISHED]

Proof. That (P1) implies (P2) is immediate. Conversely, a surjective smooth submersion is a smooth quotient map, so a smooth map constant on its fibers descends uniquely through it and the descended factor is smooth; this is where smoothness of $\bar{Q}$ becomes a theorem rather than an obligation. Surjectivity gives uniqueness, and $\bar{\varpi}\bar{Q}f = \bar{\varpi}\Psi Q = f\varpi Q = f$ with f surjective gives $\bar{\varpi}\bar{Q} = \mathrm{id}$. For (P2) implying (P3), take $X \in \ker T_c f$, split $T_c Q X = H^\omega_{Q(c)} X + D^\omega Q(X)$, apply $T\Psi$, and use $T_c f X = 0$ to annihilate the coarse horizontal lift; what remains is $T(\Psi Q)(X) = A_\Psi(Q(c);X) + T^V\Psi(D^\omega Q(X))$, which fiber constancy forces to vanish. The same computation shows that (P3) makes $T(\Psi Q)$ annihilate $\ker Tf$, and a smooth map with vanishing derivative along a connected embedded submanifold is constant on it. □

The submersion hypothesis is load bearing rather than decorative. With $\mathcal{C} = \bar{\mathcal{C}} = \mathbb{R}$, $f(x) = x^3$ a smooth bijection that fails to be a submersion at the origin, the trivial bundle with fiber $\{\mathcal{N}(\mu, 1)\}$, Ψ the identity on fibers, and $Q(x) = \mathcal{N}(x, 1)$, the unique descent $\bar{Q}(y) = \mathcal{N}(y^{1/3}, 1)$ is continuous but not differentiable at $y = 0$. [ESTABLISHED]

Proposition 8.14 (A pointwise bundle morphism is not a map on configurations). Suppose f is a surjective submersion, $\ker T_{c_0} f \neq 0$ at some c_0, there are $e_0 \in E_{c_0}$ and $w \in V_{e_0}E$ with $T^V\Psi(w) \neq 0$, and a global smooth section Q_0 satisfies $Q_0(c_0) = e_0$. Then the projectable set $\Gamma_{\mathrm{proj}}(\Psi)$ of sections satisfying (P2) is a proper subset of $\Gamma(\mathcal{C}, E)$. [ESTABLISHED]

Proof. Choose a chart in which $\ker Tf$ is spanned by $\partial_1, \ldots, \partial_k$ with $k \geq 1$, and use the declared global section Q_0 through e_0. If Q_0 already fails (P2) there is nothing to prove. Otherwise let W

be a smooth vertical field in that local trivialization of E with $W(Q_0(c_0)) = w$ and local flow Φ^W, take $\chi \in C_c^\infty$ supported inside this trivializing chart with $\chi(c_0) = 0$ and $\partial_1\chi(c_0) \neq 0$, and define the global smooth section Q_ϵ by $Q_\epsilon(c) = \Phi^W_{\epsilon\chi(c)}(Q_0(c))$ on the support chart and $Q_\epsilon = Q_0$ off its support. Since $\chi(c_0) = 0$, both $A_\Psi(Q_\epsilon(c_0); \partial_1)$ and $T^V\Psi$ at that point are independent of ϵ, so differentiating the left-hand side of (8.46) at $\epsilon = 0$ gives $\partial_1\chi(c_0)T^V\Psi(w) \neq 0$ and (P3) fails for small nonzero ϵ. □

A concrete witness is the total collapse $\mathcal{C} = S^1$, $\bar{\mathcal{C}} = \{*\}$, which is a surjective submersion, with trivial bundles carrying the unit-variance normal location fiber, Ψ the identity on fibers, and $Q(x) = \mathcal{N}(\sin x, 1)$. Here $A_\Psi = 0$ and $\ker Tf = TS^1$, while $T^V\Psi(D^\omega Q\partial_x) = \cos x\partial_\mu$ is not identically zero, so no coarse section exists. In the linear realization with configuration space $L^2(S^1)$ the descendable set is exactly the constants: a closed subspace of infinite codimension and empty interior. This statement is scoped to a collapsing f; if f is a diffeomorphism every section descends, so no unscoped claim that a bundle morphism induces a configuration map nowhere is available. [ESTABLISHED]

Theorem 8.15 (Covariant first-jet chain rule). Under (8.44),

$$D^{\bar\omega}\bar s(Tf X) = T^V\Psi(D^\omega sX) + (\mathcal{D}\Psi)_{s(c)}(X). \tag{8.47}$$

Hence the square of covariant first jets commutes exactly when $\mathcal{D}\Psi$ vanishes along s. In that case, every vertical covariant tensor $\bar\tau$ on $\bar E$ satisfies

$$f^*[(D^{\bar\omega}\bar s)^*\bar\tau] = (D^\omega s)^*[(T^V\Psi)^*\bar\tau]. \tag{8.48}$$

[ESTABLISHED]

Proof. Differentiate (8.44) and decompose $TsX = H^\omega_s X + D^\omega sX$. Applying $T\Psi$ to the horizontal term gives the target horizontal lift plus $(\mathcal{D}\Psi)_{s(c)}(X)$ by (8.45); applying it to the vertical term gives $T^V\Psi(D^\omega sX)$. The vertical part of the other side is $D^{\bar\omega}\bar s(Tf X)$, which proves (8.47). When the defect vanishes, applying $\bar\tau$ to the intertwined vertical jets proves (8.48). □

The base comparison of the two Fisher pullbacks can now be written exactly, with no compatibility hypothesis of any kind. Along s write $u_X = D^\omega sX$ for the fine vertical jet, $L = T^V\Psi$ for the vertical differential, $a_X = A_\Psi(s(c); X)$ for the horizontal defect, and $\bar u_X = D^{\bar\omega}\bar s(T_c f X) = Lu_X + a_X$ for the coarse vertical jet. With the vertical Fisher defect Δ^Ψ_F of (8.60) below, set

$$\delta_\Psi(X, Y) = \Delta^\Psi_F(u_X, u_Y), \tag{8.49}$$

$$\mathcal{X}_\Psi(X, Y) = \bar g^F(Lu_X, a_Y) + \bar g^F(a_X, Lu_Y), \tag{8.50}$$

$$\mathcal{Q}_\Psi(X, Y) = \bar g^F(a_X, a_Y). \tag{8.51}$$

Theorem 8.16 (Exact signed base Fisher comparison). Under (8.44) alone,

$$h^\omega_s - f^*\bar h^{\bar\omega}_{\bar s} = \delta_\Psi - \mathcal{X}_\Psi - \mathcal{Q}_\Psi \tag{8.52}$$

as symmetric two-tensors on the fine domain. Here $\mathcal{X}_\Psi$ is symmetric and sign indefinite while $\mathcal{Q}_\Psi \succeq 0$, so both retained tensors enter with a minus sign and neither may be summarized as one vertical mismatch term. [ESTABLISHED]

Proof. By (8.47), $f^*\bar{h}(X,Y) = \bar{g}^F(Lu_X + a_X, Lu_Y + a_Y) = \bar{g}^F(Lu_X, Lu_Y) + \mathcal{X}_\Psi(X,Y) + \mathcal{Q}_\Psi(X,Y)$, and by (8.60), $\bar{g}^F(Lu_X, Lu_Y) = g^F(u_X, u_Y) - \Delta_F^\Psi(u_X, u_Y)$. Subtracting from $h_s^\omega(X,Y) = g^F(u_X, u_Y)$ gives (8.52). □

Theorem 8.17 (Exact signed positivity criterion). Assume in addition the Markov hypotheses of Theorem 8.21, so that $\delta_\Psi \succeq 0$. Then at a fixed c the following are equivalent: the base comparison $h_s^\omega - f^*\bar{h}^{\bar{\omega}}_{\bar{s}}$ is positive semidefinite; the coarse jet is no longer than the fine jet, meaning $\|D^{\bar{\omega}}\bar{s}(T_c f X)\|_{\bar{g}^F} \le \|D^\omega s X\|_{g^F}$ for every X; and

$$2\bar{g}^F\left(T^V \Psi D^\omega s X, A_\Psi(s(c); X)\right) + \|A_\Psi(s(c); X)\|^2_{\bar{g}^F} \le \delta_\Psi(X,X) \quad \text{for every } X. \tag{8.53}$$

Consequently vanishing of the horizontal defect is *sufficient* for base positivity and is *not necessary*: whenever the cross term $\bar{g}^F(Lu_X, a_X)$ is negative and large enough in modulus, (8.53) holds strictly with $a_X \neq 0$. The triangle-inequality margin $\|a_X\|_{\bar{g}^F} \le \sqrt{h(X,X)} - \sqrt{h(X,X) - \delta_\Psi(X,X)}$ is likewise sufficient and not necessary. [ESTABLISHED]

Proof. The first two conditions are (8.52) read on the diagonal, using $h(X,X) = \|u_X\|^2_{g^F}$ and $f^*\bar{h}(X,X) = \|Lu_X + a_X\|^2_{\bar{g}^F}$. Expanding that square and substituting $\|Lu_X\|^2_{\bar{g}^F} = \|u_X\|^2_{g^F} - \delta_\Psi(X,X)$ gives (8.53). □

Both failure directions are realized on the normal location fiber, and the arithmetic is exact. Take $\mathcal{C} = \bar{\mathcal{C}} = \mathbb{R}$, f the identity, $\mathcal{B} = \{\mathcal{N}(\mu, 1)\}$ with $g^F = 1$, the sample kernel $N(x, \cdot) = \mathcal{N}(x, 1)$ so that $\bar{\mathcal{B}} = \{\mathcal{N}(\mu, 2)\}$ with $\bar{g}^F = \frac{1}{2}$ and $\Delta_F^\Psi = \frac{1}{2}$, the section $s(x) = \mathcal{N}(x, 1)$, zero source connection, and a target connection contributing $a_X = b\partial_\mu$. Then $h_s^\omega - f^*\bar{h}^{\bar{\omega}}_{\bar{s}} = 1 - \frac{1}{2}(1+b)^2$, which equals $\frac{79}{200} > 0$ at $b = \frac{1}{10}$ and $\frac{23}{25} > 0$ at $b = -\frac{3}{5}$, both with nonzero horizontal defect, and equals $-\frac{1}{8} < 0$ at $b = \frac{1}{2}$. At $b = -\frac{3}{5}$ the margin above is violated, since $\|a_X\|_{\bar{g}^F} = \frac{3}{5\sqrt{2}}$ exceeds $1 - \frac{1}{\sqrt{2}}$, so the margin is genuinely only sufficient. Strict negativity survives even at zero information loss: with the identity kernel, so $\Delta_F^\Psi = 0$, constant related sections, source horizontal field ∂_x, and target horizontal field $\partial_x + a\partial_\mu$ with $a \neq 0$, one gets $u = 0$, $a_X = -a\partial_\mu$, and $h_s^\omega - f^*\bar{h}^{\bar{\omega}}_{\bar{s}} = -a^2 dx^2 \prec 0$. [ESTABLISHED]

The horizontal defect of an induced morphism is not an arbitrary vertical field. Suppose Ψ is induced by an equivariant principal scale map $\mathcal{P} : \mathscr{P} \to \bar{\mathscr{P}}$ over $f : \mathcal{C} \to \bar{\mathcal{C}}$ with Lie-group homomorphism $\kappa : G \to \bar{G}$ and law-fiber map q, in the sense of (11.15). Define the *scale-connection defect form*

$$\mathfrak{A}_\mathcal{P} = \mathcal{P}^*\bar{\omega} - \mathrm{d}\kappa \circ \omega \in \Omega^1(\mathscr{P}, \bar{\mathfrak{g}}). \tag{8.54}$$

Theorem 8.18 (Isotropy criterion for a vanishing horizontal defect). The form $\mathfrak{A}_{\mathcal{P}}$ is horizontal and $\mathrm{Ad}\circ\kappa$-equivariant, hence descends to an element of $\Omega^1(\mathcal{C}; f^*\,\mathrm{Ad}\,\bar{\mathscr{P}})$, and

$$A_\Psi(e;X) = \vartheta_{\Psi(e)}\left(\mathfrak{A}_{\mathcal{P}}(X)\right), \tag{8.55}$$

so the horizontal defect is a fundamental vertical field determined by a $\bar{\mathfrak{g}}$-valued one-form on the base rather than by the fiber point. Consequently, for related sections satisfying $\Psi\circ s = \bar{s}\circ f$, the horizontal defect vanishes along the fine section, meaning $A_\Psi(s(c);X)=0$ for every $c\in\mathcal{C}$ and $X\in T_c\mathcal{C}$, if and only if

$$\mathfrak{A}_{\mathcal{P}}(X) \in \bar{\mathfrak{g}}_{\bar{s}(f(c))} \quad \text{for every } c \text{ and every } X, \tag{8.56}$$

the isotropy subalgebra of the coarse section value under $\widehat{\bar{\rho}}$. The principal-level identity $\mathfrak{A}_{\mathcal{P}}=0$ is sufficient, and is necessary only when the represented action is infinitesimally effective at $\bar{s}(f(c))$. Moreover $\mathcal{Q}_\Psi = \mathfrak{a}^*\mathfrak{k}_{\bar{s}(f(c))}$ with $\mathfrak{k}_{\bar{\beta}}(\xi,\eta) = \bar{g}^F(\bar{\zeta}_\xi\bar{\beta},\bar{\zeta}_\eta\bar{\beta}) \succeq 0$, whose radical is exactly $\bar{\mathfrak{g}}_{\bar{\beta}}$. [ESTABLISHED]

Proof. Evaluating $\mathcal{P}^*\bar{\omega}$ on the fundamental field of ξ gives $\bar{\omega}(\bar{\zeta}_{\mathrm{d}\kappa\xi}) = \mathrm{d}\kappa\xi$, so the difference annihilates vertical vectors; equivariance follows from $R_g^*\mathcal{P}^*\bar{\omega} = \mathrm{Ad}_{\kappa(g)^{-1}}\,\mathcal{P}^*\bar{\omega}$ together with $\mathrm{d}\kappa\circ\mathrm{Ad}_{g^{-1}} = \mathrm{Ad}_{\kappa(g)^{-1}}\circ\mathrm{d}\kappa$. In local trivializations write $\Psi(c,\beta) = (f(c),\psi_c\beta)$ with $\psi_c = \widehat{\bar{\rho}}(\varsigma(c))\circ q$. Using $H^\omega_{(c,\beta)}X = (X, -\zeta_{A(X)}\beta)$ and the differentiated intertwining $Tq\circ\zeta_\xi = \bar{\zeta}_{\mathrm{d}\kappa(\xi)}\circ q$, obtained from (11.14) at $g=\exp(t\xi)$, gives $A_\Psi(X) = \bar{\zeta}_{\mathfrak{a}(X)}(\psi_c\beta)$ with $\mathfrak{a} = \mathrm{Ad}_\varsigma(u^*\mathfrak{A}_{\mathcal{P}})$, which is (8.55). A fundamental field vanishes at $\bar{\beta}$ exactly when its generator lies in $\bar{\mathfrak{g}}_{\bar{\beta}}$, and conjugation by Ad_ς moves both the generator and the isotropy algebra together, which gives (8.56). The last statement is (8.55) substituted into (8.51). □

Equation (8.56) is the definition of connection compatibility used throughout this manuscript. No weaker or informal reading is intended, and no result below relies on a compatibility notion that has not been reduced to it. [DEFINITION]

Theorem 8.19 (Ordered composition of horizontal defects). For composable morphisms $E_0 \xrightarrow{\Psi_{01}} E_1 \xrightarrow{\Psi_{12}} E_2$ over f_{01}, f_{12}, with $\Psi_{02} = \Psi_{12}\circ\Psi_{01}$ and $f_{02} = f_{12}\circ f_{01}$,

$$A_{\Psi_{02}}(e;X) = T^V\Psi_{12}\big|_{\Psi_{01}(e)}\left(A_{\Psi_{01}}(e;X)\right) + A_{\Psi_{12}}\left(\Psi_{01}(e);T_cf_{01}X\right). \tag{8.57}$$

At the connection level this reads $\mathfrak{A}_{\mathcal{P}_{02}} = \mathcal{P}_{01}^*\mathfrak{A}_{\mathcal{P}_{12}} + \mathrm{d}\kappa_{12}\circ\mathfrak{A}_{\mathcal{P}_{01}}$. [ESTABLISHED]

Proof. Expand $T\Psi_{01}(H^{\omega_0}_e X) = H^{\omega_1}_{\Psi_{01}e}(Tf_{01}X) + A_{\Psi_{01}}(e;X)$ and apply $T\Psi_{12}$. The horizontal summand contributes $H^{\omega_2}_{\Psi_{02}e}(Tf_{02}X) + A_{\Psi_{12}}(\Psi_{01}e;Tf_{01}X)$ by (8.45) at the second stage, and the vertical summand contributes $T^V\Psi_{12}(A_{\Psi_{01}}(e;X))$ because a bundle morphism preserves verticality. Subtracting $H^{\omega_2}(Tf_{02}X)$ gives (8.57); the connection-level form follows by $T^V\Psi_{12}\circ\vartheta = \vartheta\circ\mathrm{d}\kappa_{12}$. □

The order in (8.57) is not cosmetic. The earlier defect takes values in $V\bar{E}_1$ and the later one in VE_2, so writing $A_{\Psi_{02}} = A_{\Psi_{01}} + A_{\Psi_{12}}$ is a type error before any question of correctness arises: the earlier defect must be pushed by the *later* vertical differential, and the later defect must be evaluated at

the image point on the *pushed* base vector. Correspondingly, $A_{\Psi_{02}}(e;X)=0$ holds if and only if

$$T^V\Psi_{12}\big|_{\Psi_{01}(e)}A_{\Psi_{01}}(e;X)=-A_{\Psi_{12}}\left(\Psi_{01}(e);T_cf_{01}X\right). \tag{8.58}$$

Factorwise vanishing of $A_{\Psi_{01}}$ on $T\mathcal{C}_0$ and of $A_{\Psi_{12}}$ on $Tf_{01}(T\mathcal{C}_0)$ at the image points is sufficient, but it is not necessary because the two typed terms can cancel after the later vertical differential. On the abelian three-level instance $f_{01}(x)=2x$, $f_{12}(y)=3y$ with local connection forms 0, a_1dy, and a_2dz and identity fiber maps, the composition law reads $2a_1+2(3a_2-a_1)=6a_2$ identically in (a_1,a_2); thus $a_2=0$ and $a_1\neq 0$ give a vanishing composite defect while both factor defects are nonzero and cancel. Dropping the base pushforward gives a_1+3a_2, and dropping the vertical differential fails likewise. [ESTABLISHED]

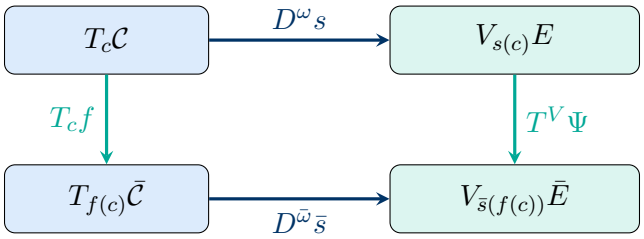


Figure 8.1: Naturality of the connection-split first jet. The square commutes exactly under the isotropy criterion (8.56). When the horizontal defect is nonzero, (8.47) records the missing vertical term and (8.52) records the two retained base tensors $-\mathcal{X}_\Psi$ and $-\mathcal{Q}_\Psi$ rather than suppressing them.

Corollary 8.20 (Null directions under a parallel coarse morphism). If $\mathcal{D}\Psi=0$ along s, then

$$Tf(\ker D^\omega s)\subseteq\ker D^{\bar\omega}\bar s. \tag{8.59}$$

Thus a coarse morphism can create additional null directions; first-jet naturality does not imply rank preservation. [ESTABLISHED]

Proof. Set $D^\omega sX=0$ in (8.47) and use $\mathcal{D}\Psi=0$. □

8.9 Fisher defects and their composition law

Suppose now that the fiber map underlying Ψ is the pushforward of a parameter-independent Markov kernel N between the two regular statistical experiments, normalized in the strong sense that $N(x,\bar{\mathsf{K}})=1$ for every x rather than for almost every x, and that it is equivariant under the represented gauge actions. Three further hypotheses are consumed here and none of them follows from affinity of the induced law map on the cone of measures. First, *family closure*: the induced law map $N_\star$ must satisfy $N_\star(\mathcal{B})\subseteq\bar{\mathcal{B}}$, so that $q:=N_\star|_{\mathcal{B}}$ is defined at all. Second, *smoothness of* q between the two declared parametrized-measure models, together with a jointly measurable

and parameter-smooth version selection for $p \mapsto T^V_p\Psi$. Third, Hypothesis 8.1 is instantiated at the coarse scale as well as the fine one; its clause that the represented action is induced by a parameter-independent bimeasurable sample re-coordinatization preserving the model is exactly the $\widehat{\bar{\rho}}(\bar{G})$-invariance of $\bar{g}^F$, and that invariance is what cancels the context-dependent factor $\widehat{\bar{\rho}}(\varsigma(c))$ in the local representative $\psi_c = \widehat{\bar{\rho}}(\varsigma(c)) \circ q$ of Ψ. Only in a frame pair of the form $(u(c), \mathcal{P}(u(c)))$ is $T^V\Psi$ literally Tq; in an arbitrary frame pair the identification requires that cancellation. Finally, for the DQM transfer used below, assume a common σ-finite dominating measure μ for the selected fine family, with a fixed density version p_θ in the parameter neighborhood under discussion. This common-domination tier is sufficient for the argument below; no broader nondominated transfer theorem is asserted here. Jointly measurable, parameter-smooth conditional-score versions needed for pointwise bundle statements remain separate hypotheses. [HYPOTHESIS]

The coarse Fisher metric $\bar{g}^F$ can then be pulled back by $T^V\Psi$. Define the vertical Fisher defect by

$$\Delta^\Psi_F = g^F - (T^V\Psi)^* \bar{g}^F. \tag{8.60}$$

[DEFINITION]

Theorem 8.21 (Fisher defect pullback and contraction). Assume the preceding Markov hypotheses and $\mathcal{D}\Psi = 0$ along s. Then

$$\Delta^\Psi_F \succeq 0, \tag{8.61}$$

$$h^\omega_s - f^* h^{\bar{\omega}}_{\bar{s}} = (D^\omega s)^* \Delta^\Psi_F \succeq 0. \tag{8.62}$$

Equality on a base tangent vector X holds exactly when the fine score in the direction $D^\omega sX$ is measurable with respect to the coarse output. [ESTABLISHED]

Proof. Work in a frame pair $(u(c), \mathcal{P}(u(c)))$, where $T^V\Psi = Tq$. The step that carries an arbitrary quadratic-mean path through the channel is not a consequence of parameter independence by itself. Let $J_\theta(dx, dy) = P_\theta(dx)N(x, dy)$ and $\nu(dx, dy) = \mu(dx)N(x, dy)$. Then $\mathrm{d}J_\theta/\mathrm{d}\nu = p_\theta(x)$, so the fine DQM remainder has exactly the same squared $L^2(\nu)$ norm in the joint experiment. Apply Pollard's preservation theorem for measurable statistics [75] to the deterministic statistic $(x, y) \mapsto y$: the pushed family $P_\theta N$ is DQM with directional score $\bar{\ell}_w(y) = \mathbb{E}_{J_\theta}[\ell_w(X) \mid Y = y]$. Squared Hellinger contraction is compatible corroboration, but it alone does not establish this score identification. Given the joint-experiment reduction and Pollard's theorem, the law of total variance gives

$$\Delta^\Psi_F(u, u) = \mathbb{E}\,\mathrm{Var}(\ell_u \mid Y) \geq 0, \tag{8.63}$$

with equality exactly when ℓ_u is Y-measurable. Under the vanishing horizontal defect, (8.48) applied to $\bar{g}^F$ gives

$$f^* h^{\bar{\omega}}_{\bar{s}} = (D^\omega s)^* (T^V\Psi)^* \bar{g}^F.$$

Subtracting this identity from (8.4) proves (8.62). □

The guard $\mathcal{D}\Psi = 0$ in Theorem 8.21 may not be dropped silently, and what survives when it is dropped is stated exactly rather than gestured at. Without the guard the vertical conclusion (8.61) is unchanged, while the base conclusion (8.62) is replaced by the exact signed identity (8.52) and its criterion (8.53); positivity then holds on exactly the directions satisfying that criterion, and fails on the others. [ESTABLISHED]

Corollary 8.22 (Coarse or meta-agent perceived geometry). The coarse section $\bar{s}$ carries its own connection-relative semimetric

$$\bar{h}^{\bar{\omega}}_{\bar{s}} = (D^{\bar{\omega}}\bar{s})^* \bar{g}^F. \tag{8.64}$$

Under the exact section relation (8.44) and the compatibility condition $\mathcal{D}\Psi = 0$ along s,

$$f^* \bar{h}^{\bar{\omega}}_{\bar{s}} = (D^{\omega} s)^* (T^V \Psi)^* \bar{g}^F. \tag{8.65}$$

If the fiber map is also a genuine parameter-independent Markov channel, then the fine-minus-coarse perceived-geometry defect is

$$h^{\omega}_{s} - f^* \bar{h}^{\bar{\omega}}_{\bar{s}} = (D^{\omega} s)^* \Delta^{\Psi}_F \succeq 0. \tag{8.66}$$

[ESTABLISHED]

Proof. The first equation is the coarse-channel definition. The second is (8.48) with $\bar{\tau} = \bar{g}^F$, and the third is (8.62). □

The inequality in (8.62) is contravariant: the coarse tensor is pulled back to the fine tangent space. No canonical pushforward of a Riemannian metric is being used. If the coarse operation is a restriction along an embedding, an energy precomposition, a Galerkin map, or a parameter-dependent kernel, the Markov proof does not apply. [ESTABLISHED]

A perceived-geometry beta is not defined by subtracting tensors that live on different level bases. If the reference-space declarations of Chapter 11 include base diffeomorphisms $i_\ell : \mathcal{C}_\ell \to \mathcal{C}_\star$ and a block factor $b_\ell > 1$, then one may first define

$$\widetilde{h}_\ell = (i_\ell^{-1})^* h_\ell, \qquad \mathfrak{B}_{h,\ell} = \frac{\widetilde{h}_{\ell+1} - \widetilde{h}_\ell}{\log b_\ell} \quad \text{on } \mathcal{C}_\star. \tag{8.67}$$

This is the exact discrete perceived-geometry beta in that scheme. On a differentiable reference-space curve $s \mapsto \widetilde{h}_s$, its continuous counterpart is $\beta_h(s) = \partial_s \widetilde{h}_s$. Both definitions are conditional on the declared identifications and scale normalization. [DEFINITION]

Without those identifications, $h_{\ell+1} - h_\ell$ is ill typed rather than a beta function. [ESTABLISHED]

Theorem 8.23 (Vertical Fisher defect cocycle). Let $E_0 \xrightarrow{\Psi_{01}} E_1 \xrightarrow{\Psi_{12}} E_2$ be smooth bundle morphisms over base maps $f_{01} : \mathcal{C}_0 \to \mathcal{C}_1$ and $f_{12} : \mathcal{C}_1 \to \mathcal{C}_2$, carrying principal connections

$\omega_0, \omega_1, \omega_2$ and vertical Fisher metrics g_0^F, g_1^F, g_2^F. Whenever all three defects are defined,

$$\Delta_F^{\Psi_{12}\circ\Psi_{01}} = \Delta_F^{\Psi_{01}} + (T^V\Psi_{01})^*\Delta_F^{\Psi_{12}}. \tag{8.68}$$

This identity is unconditional: it uses neither the connections nor any section. If both arrows are parameter-independent Markov coarse maps, the two terms on the right are positive semidefinite. [ESTABLISHED]

Proof. Contravariance of tensor pullback gives

$$\begin{aligned}\Delta_F^{\Psi_{12}\circ\Psi_{01}} &= g_0^F - (T^V\Psi_{01})^*(T^V\Psi_{12})^* g_2^F \\ &= g_0^F - (T^V\Psi_{01})^* g_1^F + (T^V\Psi_{01})^*\left[g_1^F - (T^V\Psi_{12})^* g_2^F\right],\end{aligned}$$

which is (8.68). Positivity follows from (8.61) for each Markov arrow. □

Pulling (8.68) back to the base does *not* give an additive identity in general, and the exact residual is available in closed form. Fix sections s_0, s_1, s_2 related at each stage, write $\delta_{jk} = (D^{\omega_j}s_j)^*\Delta_F^{\Psi_{jk}}$ for the three base defects, and along s_0 write $v_X = T^V\Psi_{01}(D^{\omega_0}s_0X)$, $A_X = A_{\Psi_{01}}(s_0; X)$, and $\bar{u}_X = D^{\omega_1}s_1(Tf_{01}X) = v_X + A_X$.

Theorem 8.24 (Base Fisher defect cocycle: exact residual and sharp criterion). The residual $\mathcal{N} := \delta_{02} - \delta_{01} - f_{01}^*\delta_{12}$ has the three equivalent exact forms

$$\mathcal{N}(X,Y) = \Delta_F^{\Psi_{12}}(v_X, v_Y) - \Delta_F^{\Psi_{12}}(\bar{u}_X, \bar{u}_Y), \tag{8.69}$$

$$= -\left[\Delta_F^{\Psi_{12}}(\bar{u}_X, A_Y) + \Delta_F^{\Psi_{12}}(A_X, \bar{u}_Y)\right] + \Delta_F^{\Psi_{12}}(A_X, A_Y), \tag{8.70}$$

$$= -\left[\Delta_F^{\Psi_{12}}(v_X, A_Y) + \Delta_F^{\Psi_{12}}(A_X, v_Y)\right] - \Delta_F^{\Psi_{12}}(A_X, A_Y). \tag{8.71}$$

The sign of the quadratic term is coupled to the choice of argument in the cross terms: the coarse jet $\bar{u}$ carries $+\Delta_F^{\Psi_{12}}(A, A)$ and the pushed fine jet v carries $-\Delta_F^{\Psi_{12}}(A, A)$. Mixing the two conventions inflates the residual by exactly $2\Delta_F^{\Psi_{12}}(A_X, A_Y)$, which is two copies of the second-arrow vertical Fisher defect evaluated on the first-arrow horizontal defect. Consequently the sharp base cocycle $\delta_{02} = \delta_{01} + f_{01}^*\delta_{12}$ holds *if and only if*

$$\Delta_F^{\Psi_{12}}(v_X, v_X) = \Delta_F^{\Psi_{12}}(\bar{u}_X, \bar{u}_X) \quad \text{for every } X, \tag{8.72}$$

equivalently $\Delta_F^{\Psi_{12}}(A_X, 2v_X + A_X) = 0$ for every X. If the second arrow is a Markov coarse map, so that $\Delta_F^{\Psi_{12}} \succeq 0$, this is equivalently equality of the two $\Delta_F^{\Psi_{12}}$-seminorms. [ESTABLISHED]

Proof. Apply $(D^{\omega_0}s_0)^*$ to (8.68) to get $\delta_{02} = \delta_{01} + \Delta_F^{\Psi_{12}}(v_X, v_Y)$, and note that by definition $f_{01}^*\delta_{12}(X,Y) = \Delta_F^{\Psi_{12}}(\bar{u}_X, \bar{u}_Y)$. Subtracting gives (8.69). Substituting $\bar{u} = v + A$ in the second term gives (8.71), and substituting $v = \bar{u} - A$ in the first gives (8.70). Since $\mathcal{N}$ is symmetric, vanishing on the diagonal is vanishing, and (8.69) on the diagonal is exactly (8.72). □

Three conditions are sufficient for (8.72), in decreasing strength: the stage-one horizontal defect vanishes, which by Theorem 8.18 is the isotropy condition (8.56) at the first arrow; the second arrow is Fisher lossless, $\Delta_F^{\Psi_{12}} = 0$; or, more generally, $A_X \in \operatorname{rad} \Delta_F^{\Psi_{12}}$ for every X, so that the stage-one anomaly is invisible to the stage-two information loss. The third suffices because $\mathcal{N}$ is linear in the product $\Delta_F^{\Psi_{12}} A$. None of the three is necessary, since (8.72) is an equality of quadratic forms and admits accidental solutions: on the three-level normal location instance with fibers of variance $1, 2, 3$, sections $x \mapsto x$, $y \mapsto y/2$, $z \mapsto z/6$, base maps $f_{01}(x) = 2x$ and $f_{12}(y) = 3y$, and local connection forms 0, $a_1 dy$, $a_2 dz$, the residual is $\mathcal{N} = -\frac{2}{3} a_1(a_1 + 1)$, which vanishes at $a_1 = -1$ where the stage-one horizontal defect equals $-2 \neq 0$. On the same data the convention-mixed expression evaluates to $\frac{2}{3} a_1(a_1 - 1)$, exceeding $\mathcal{N}$ by $\frac{4}{3} a_1^2 = 2\Delta_F^{\Psi_{12}}(A, A)$, and at $a_1 = \frac{1}{10}$ the two numbers are $-\frac{11}{150}$ and $-\frac{3}{50}$. [ESTABLISHED]

Two distinct hypotheses are involved here and conflating them is the reading this chapter forbids. The sharp base cocycle needs only the stage-one condition (8.72). Reading the middle term δ_{12} as the stage-two information loss $h_1 - f_{12}^* h_2$ requires in addition that the stage-two horizontal defect vanish, by Theorem 8.16 applied at the second arrow. Only when both hold does (8.68) become an additive decomposition of information loss on the base. [ESTABLISHED]

The Amari defect

$$\Delta_{\mathcal{T}}^{\Psi} = \mathcal{T} - (T^V \Psi)^* \bar{\mathcal{T}}, \tag{8.73}$$

obeys the same algebraic cocycle identity. It has no analogous positive-semidefinite order because a nonzero cubic changes sign under $u \mapsto -u$. Exact Amari naturality therefore requires a $\mathcal{T}$-preserving statistical morphism; Markov coarse-graining alone does not supply a monotonicity statement for the cubic tensor. [ESTABLISHED]

8.9.1 Intrinsic classification of connection defects

The positivity and cocycle criteria admit complete finite-dimensional parameterizations that retain the action constraint (8.55). Fix the sections, principal scale map, fiber morphism, and source connection. At one base point put $V = T_c\mathcal{C}$, $E_0 = V_{s(c)}E$, and $E_1 = V_{\bar{s}(f(c))}\bar{E}$, with positive Fisher inner products g_0 and g_1. Write $U = D^\omega s$, $L = T^V \Psi$, and $v = LU$ at this point. Let $\mathfrak{a}_c = (f^* \operatorname{Ad} \bar{\mathscr{P}})_c$ and $\mathsf{J} : \mathfrak{a}_c \to E_1$ be the actual fundamental-action map at $\bar{s}(f(c))$. A descended principal connection defect $\eta : V \to \mathfrak{a}_c$ produces the anomaly $A = \mathsf{J}\eta$, whose image lies in the orbit tangent $O = \operatorname{im} \mathsf{J}$. The two problems are

$$(v + \mathsf{J}\eta)^* g_1 (v + \mathsf{J}\eta) \preceq U^* g_0 U, \tag{8.74}$$

$$(v + \mathsf{J}\eta)^* \Delta (v + \mathsf{J}\eta) = v^* \Delta v, \tag{8.75}$$

where Δ in the second problem is the second-arrow vertical Fisher defect. These are separate predicates; imposing both means intersecting their solution sets for the same η. Varying the source connection would also change U and is a different problem. [DEFINITION]

Theorem 8.25 (Complete positivity parameterization). Let P_O be the g_1-orthogonal projection onto O and set

$$v_\perp = (I - P_O)v, \qquad w = P_O v, \qquad H = U^* g_0 U - v_\perp^* g_1 v_\perp. \tag{8.76}$$

There is a pointwise solution of (8.74) if and only if $H \succeq 0$. In that case let $q_H : V \to V_H = V/\operatorname{rad} H$ be the quotient, with its inner product induced by H. All solutions are exactly

$$\mathsf{J}\eta = Dq_H - w, \qquad D : (V_H, H) \longrightarrow (O, g_1) \text{ a linear contraction.} \tag{8.77}$$

Every displayed right side has an adjoint lift, and all lift ambiguity is $\operatorname{Hom}(V, \ker \mathsf{J})$. Zero-dimensional spaces and noninjective contractions are included. This is an established algebraic classification; principal realization is addressed separately below. [ESTABLISHED]

Proof. The coarse jet is $v_\perp + C$, where $C = w + \mathsf{J}\eta$ takes values in O. Orthogonality reduces the inequality to $C^* g_1 C \preceq H$. This forces $H \succeq 0$ and $Cx = 0$ whenever $H(x, x) = 0$. For a positive-semidefinite form the latter diagonal null set is its radical: positivity of $H(x + ty, x + ty)$ for both signs of small t forces $H(x, y) = 0$. Thus $D([x]) = Cx$ is well defined, and the inequality says exactly that D is a contraction. Conversely every such D satisfies the inequality. Since $Dq_H - w$ takes values in O, a linear right inverse of J on its image gives a lift; any two lifts differ by a map into its kernel. If $H \succeq 0$, $D = 0$ proves existence. □

For a Markov first arrow the vertical Fisher contraction gives $\delta = U^*(g_0 - L^* g_1 L)U \succeq 0$, and

$$H = \delta + w^* g_1 w \succeq 0. \tag{8.78}$$

The algebraic feasible set is therefore nonempty and contains the zero anomaly. It is convex: for feasible C_0, C_1 and $0 \le t \le 1$, the difference between $(1-t)\|C_0 x\|^2 + t\|C_1 x\|^2$ and $\|(1-t)C_0 x + tC_1 x\|^2$ is $t(1-t)\|(C_0 - C_1)x\|^2 \ge 0$. Adjoint lifting preserves this convexity. These facts do not establish descent to a target connection. [ESTABLISHED]

Theorem 8.26 (Complete positive-semidefinite cocycle parameterization). Suppose $\Delta \succeq 0$. Let $\pi_\Delta : E_1 \to E_\Delta = E_1/\operatorname{rad}\Delta$ be the metric quotient and define

$$F = \pi_\Delta \mathsf{J}, \qquad O_\Delta = \operatorname{im} F, \qquad b = \pi_\Delta v, \qquad W = P_\Delta b, \tag{8.79}$$

where P_Δ is orthogonal projection onto O_Δ. Every solution of (8.75) is given by

$$F\eta = (T - \iota)W, \qquad T : \operatorname{im} W \longrightarrow O_\Delta \text{ a linear isometric embedding,} \tag{8.80}$$

and every such choice is a solution. Here ι is inclusion. The lift ambiguity is $\operatorname{Hom}(V, \ker F)$, which includes both action isotropy and action directions invisible to Δ. This establishes the complete pointwise classification of the accidental solutions in the positive-semidefinite case. [ESTABLISHED]

Proof. The component $(I - P_\Delta)b$ is unchanged by $F\eta$ and is orthogonal to O_Δ. Removing its identical contribution from both sides reduces the cocycle to

$$(W + F\eta)^*(W + F\eta) = W^*W. \tag{8.81}$$

Here the stars denote the quotient inner products. If $Wx = 0$, equality of squared norms forces $(W + F\eta)x = 0$. Therefore $T(Wx) = (W + F\eta)x$ is well defined and linear. Polarization proves that it is isometric. Conversely an isometric T gives (8.81). Every right side of (8.80) lies in the image of F, so it lifts, with exactly the stated kernel ambiguity. □

The radical and isotropy sufficient conditions correspond to $T = \iota$. The choice $T = -\iota$ gives $F\eta = -2W$, exhibiting the other solutions without requiring a vanishing anomaly or a lossless second arrow. When $W = 0$ the equation forces $F\eta = 0$, whereas $\Delta = 0$ makes every η a solution. Extending T to an orthogonal map on all of O_Δ is possible pointwise by completing orthonormal bases, but is not needed by the classification. [ESTABLISHED]

Theorem 8.27 (Complete signed cocycle parameterization). Let Δ be any symmetric form on E_1. Put $N = \mathrm{rad}(\Delta|_O)$, choose a complement $O = S \oplus N$, and write Δ_S for the nondegenerate restriction to S, with positive and negative indices (p, q). Define $W : V \to S$ and $\ell_x \in N^*$ by

$$\Delta_S(Wx, s) = \Delta(vx, s) \quad (s \in S), \qquad \ell_x(n) = \Delta(vx, n) \quad (n \in N). \tag{8.82}$$

For a map $B : V \to N$ set

$$G_B(x, y) = \Delta_S(Wx, Wy) - \ell_x(By) - \ell_y(Bx). \tag{8.83}$$

Choose precisely those B for which the positive and negative indices (a, b) of G_B obey $a \le p$ and $b \le q$. With $K_B = \mathrm{rad}\, G_B$, choose a complement $V = D_B \oplus K_B$, an isometric embedding $I_B : (D_B, G_B|_{D_B}) \to (S, \Delta_S)$, and a linear map $Z : K_B \to I_B(D_B)^{\perp_{\Delta_S}}$ with totally isotropic image, meaning $\Delta_S(Zx, Zy) = 0$ for all $x, y \in K_B$. Define

$$C(d + k) = I_B d + Zk, \qquad \mathsf{J}\eta = C - W + B \quad (d \in D_B,\ k \in K_B). \tag{8.84}$$

These are all solutions of (8.75), with arbitrary lifts through J and ambiguity $\mathrm{Hom}(V, \ker \mathsf{J})$. Zero spaces are allowed. The map Z may vanish regardless of $\dim K_B$; its rank is at most $\min(p - a, q - b)$. This is an established finite-dimensional classification without a Markov hypothesis. [ESTABLISHED]

Proof. If $s \in S$ is orthogonal to S, it is also orthogonal to N, since N is orthogonal to all of O. Thus $s \in N \cap S = 0$, proving nondegeneracy of Δ_S and uniqueness of W. Decompose $\mathsf{J}\eta = A_S + B$ and put $C = W + A_S$. The pairings within N and between S and N vanish. Expanding the cocycle, and replacing each pairing of v with S by the pairing of W with S, gives exactly

$$C^*\Delta_S C = G_B. \tag{8.85}$$

The terms involving ℓ remain because the radical of $\Delta|_O$ need not be orthogonal to v.

The restriction of G_B to D_B is nondegenerate. Hence any Gram factor C restricts to an injective isometry I_B on D_B, so its image has signature (a, b) and the inertia inequalities are necessary. Its restriction to K_B is orthogonal to that image and is totally isotropic. Conversely these properties give the correct block on D_B and zero cross terms and block on K_B, proving (8.85). The inertia inequalities suffice for I_B: diagonalize the two nondegenerate forms using arbitrary positive inner products, rescale their eigenvectors to squared norms $+1$ or -1, and map like-signed basis vectors to each other. The finite spectral step follows by maximizing a symmetric quadratic form on the unit sphere to obtain an eigenvector and then inducting on its invariant orthogonal complement. Taking $Z = 0$ gives existence. Finally, in a space of signature (r, s) each projection of a totally isotropic subspace onto the positive or negative summand is injective; a vector with either projection zero and zero norm must vanish. Its dimension is therefore at most $\min(r, s)$. Apply this to the complement of $I_B(D_B)$ and then lift through J. □

The restricted radical cannot be replaced by the ambient radical. For $\Delta = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $O = \operatorname{span}(e_1)$, and $v(X) = Xe_2$, one has $N = O$ and $S = 0$. A candidate $B(X) = bXe_1$ gives $G_B(X, X) = -2bX^2$, so the cocycle forces $b = 0$. If $v = 0$, instead, every such B is allowed. This computation also shows why deleting the ℓ terms changes the answer. [ESTABLISHED]

Proposition 8.28 (Smooth solutions on regular strata). Let the preceding data be smooth finite-rank bundles and bundle maps on a paracompact smooth base region. For positivity assume constant ranks of J and H, with $H \succeq 0$. For the positive-semidefinite cocycle assume constant ranks of Δ, F, and W. Then the smooth solutions are exactly the smooth contractions or isometric bundle embeddings in eqs. (8.77) and (8.80), with smooth lifts and arbitrary smooth kernel-valued maps. For the signed case assume constant ranks of J and $\Delta|_O$, choose a smooth complement S, and restrict to smooth B for which G_B has constant rank and signature. The smooth solutions are exactly the smooth I_B, Z and lifts of (8.84). Inertia bounds give local smooth I_B but do not alone give a global isometric embedding into the specified S bundle. Global embeddings and agreement of the resulting C on overlaps are explicit parts of the parameterization. [ESTABLISHED]

Proof. A nonzero rank-size minor of a constant-rank smooth map stays invertible locally. Its selected columns give a smooth image frame, and solving with that minor gives smooth preimages and kernel frames. The kernels, images, and quotients above are consequently smooth bundles. Choose positive bundle metrics by patching local metrics with a partition of unity. Orthogonal projections are smooth because their local matrices invert positive Gram matrices. Restriction to the orthogonal complement of a kernel gives a smooth isomorphism onto the image, hence a smooth right inverse for J or F. These constructions also prove the converse: the maps $D([x]) = (w + \mathsf{J}\eta)x$ and $T(Wx) = (W + F\eta)x$ induced by a smooth solution are smooth. In the signed case I_B and Z are restrictions of the smooth C to the chosen smooth splitting. Local frames for a nondegenerate symmetric form with fixed signature are obtained by retaining a basis with nonzero successive squared norms near a point, orthogonalizing, and dividing by the square roots of their absolute norms. Mapping like-signed vectors gives local smooth embeddings whenever the inertia bounds

hold. This construction makes no claim that independent local isometries glue: their assembled maps must agree on overlaps. □

The classified sets are intrinsic. Passive frame changes carry U, L, J, v, their kernels and images, and the Fisher or signed forms by their tensor transformation laws. Contraction, isometry, inertia, and total isotropy are preserved. Auxiliary metrics and splittings only change the parameters describing the same solution set; no positive invariant norm on a noncompact adjoint representation is assumed. The regular-stratum result makes no automatic extension assertion across a rank change. [ESTABLISHED]

Theorem 8.29 (Exact realization by a target connection). Fix the source connection ω, the principal scale map $\mathcal{P}$, and a target reference connection $\bar{\omega}_0$. Let η_0 be its descended defect $\mathcal{P}^*\bar{\omega}_0 - \mathrm{d}\kappa \circ \omega$. An adjoint-valued one-form η is realized by a target principal connection if and only if

$$\eta = \eta_0 + f^*\beta, \qquad \beta \in \Omega^1(\bar{\mathcal{C}}; \operatorname{Ad} \bar{\mathscr{P}}). \tag{8.86}$$

For a surjective smooth submersion f, this holds exactly when $\zeta = \eta - \eta_0$ annihilates $\ker Tf$ and satisfies

$$\zeta_c(X) = \zeta_{c'}(X') \quad \text{whenever} \quad f(c) = f(c'), \quad T_c f X = T_{c'} f X'. \tag{8.87}$$

Both values belong to the same target adjoint fiber. All components of each fiber are included. The descended β is unique and smooth. Thus the principal-realizable solutions are exactly the algebraic parameterizations, with all lift freedom retained, intersected with (8.86). [ESTABLISHED]

Proof. The difference of two principal connections is horizontal and equivariant, hence is the tensorial form represented by a unique β on the target base. Conversely adding such a form preserves the connection's value on fundamental vectors and its equivariance. Pullback by $\mathcal{P}$ therefore gives exactly (8.86). A pullback annihilates $\ker Tf$ and has the stated equality on each fiber. Conversely define $\beta_y(Y) = \zeta_c(X)$ using any c over y and any X with $T_c f X = Y$. The conditions make this definition independent of both choices and linear in Y. In local submersion coordinates a smooth local section σ exists, and $\beta = \sigma^*\zeta$ proves smoothness. Surjectivity gives uniqueness. □

When f is a diffeomorphism every smooth adjoint lift is realizable. For general f, failure of one chosen lift to descend does not exclude a different lift corrected by a kernel-valued one-form. Realization requires at least one lift to satisfy (8.86). No flatness or curvature equation is imposed on the target connection in this statement. [ESTABLISHED]

Action rank can obstruct a smooth lift. Consider trivial $\mathrm{SO}(2)$ principal bundles over the identity map of $\mathbb{R}$. The fine and coarse fibers are normal-location families on $\mathbb{R}^2$ with covariances I and $2I$; the normalized, rotation-equivariant channel adds an independent $\mathcal{N}(0, I)$ vector. Choose zero source connection and related mean sections $\mu(x) = (x^2, 0)$. On the unit base vector,

$$U = (2x, 0), \qquad L = I, \qquad g_0 = I, \qquad g_1 = \tfrac{1}{2}I, \qquad \mathrm{J}_x(t) = (0, x^2 t). \tag{8.88}$$

The smooth anomaly $A_x = (0, x)$ lies in $\operatorname{im} \mathsf{J}_x$ at every point and obeys $\|U + A_x\|_{g_1}^2 = \frac{5}{2}x^2 \le 4x^2 = \|U\|_{g_0}^2$. Its unique adjoint lift at $x \neq 0$ is $\eta_x = 1/x$. Thus no continuous connection defect realizes it across zero. The obstruction is action rank, although both vertical Fisher metrics remain positive. [ESTABLISHED]

A smooth defect need not have an ambient isometry extension. Put $a(x) = e^{-1/x^2}$ for $x \neq 0$ and $a(0) = 0$, and $\mu(x) = \int_0^x a(t)dt$. Use scalar unit-variance normal-location families for the first two levels, their identity first arrow, translation group $\mathbb{R}$, identity base maps, and zero source connection. An additive unit-normal noise second arrow gives $\Delta = \frac{1}{2}$. Writing one-form coefficients relative to dx gives $v = a(x)$ and $\mathsf{J} = 1$. Define the target connection defect by

$$\eta_x = \begin{cases} -2a(x), & x > 0, \\ 0, & x \le 0. \end{cases} \tag{8.89}$$

It is smooth: every one-sided derivative at zero is a polynomial in $1/x$ times e^{-1/x^2} and tends to zero. Its coarse jet is $-a$ for $x > 0$ and a for $x < 0$, so both the cocycle and first-arrow positivity hold with equality. The isometry in (8.80) is uniquely -1 on the positive half-line and $+1$ on the negative half-line. It has no continuous extension as a field in $\operatorname{End}(\mathbb{R})$, although the realized defect is smooth. [ESTABLISHED]

This does not assert discontinuity on the actual rank-varying image. If continuous W, Y obey $Y^*Y = W^*W$, their induced isometry $T(Wx) = Yx$ is continuous on the image total space. In local bundle coordinates, if $(c_n, z_n) \to (c_0, z_0)$ and $W_{c_0}u_0 = z_0$, then

$$\|T_{c_n}z_n - T_{c_0}z_0\| \le C_0\|z_n - W_{c_n}u_0\| + \|Y_{c_n}u_0 - Y_{c_0}u_0\| \longrightarrow 0. \tag{8.90}$$

Here $C_0 < \infty$ follows from uniform local equivalence of the continuous positive metrics with the coordinate norms. The preceding example therefore concerns extension to a fixed ambient domain, not the continuity of this induced map. [ESTABLISHED]

Smooth algebraic solutions can fail connection descent. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(x, y) = x$, use trivial translation principal bundles and the principal map $(x, y, g) \mapsto (x, g)$, and use identical scalar unit-variance normal-location fibers with identity first fiber map. Both mean sections are zero. A source connection $-dy$ and target reference connection zero give $v = -dy$ and $\eta_0 = dy$. The smooth candidate $\eta = 2dy$ has coarse algebraic jet dy and satisfies positivity with equality. Adding a unit-normal noise second arrow gives $\Delta = \frac{1}{2}$ and also satisfies the cocycle. But $\eta - \eta_0 = dy$ does not annihilate ∂_y, so no target connection realizes it. The translation action is injective and leaves no alternative adjoint lift. [ESTABLISHED]

Disconnected fibers require the full equality (8.87). Let $\mathcal{C} = \mathbb{R} \times \{+, -\}$, $f(x, \epsilon) = x$, and use the translation-equivariant normal-location arrows with variances $1 \to 2 \to 3$, zero source and target reference connections, and related mean sections $\mu = x$. Then $v = dx$, $\mathsf{J} = 1$, and $\Delta = \frac{1}{6}$. Choose $\eta = 0$ on the positive copy and $\eta = -2dx$ on the negative copy. All relevant algebraic ranks are

constant, and both branches satisfy $(v+\eta)^2 = v^2$ and $\frac{1}{2}(v+\eta)^2 \le v^2$. Here $\ker Tf = 0$, but no single target form pulls back to the two different values. This proves that vertical annihilation and smoothness cannot replace agreement across fiber components. All four examples use proper normal laws and normalized channels; a one-point observation of likelihood one may be appended with evidence one. [ESTABLISHED]

Signed isotropic solutions occur outside the Markov domain. For unit-covariance normal-location fibers on $\mathbb{R}^2$, a smooth second mean map with derivative $L_2 = \mathrm{diag}(0, \sqrt{2})$ has $\Delta = I - L_2^\top L_2 = \mathrm{diag}(1, -1)$. Use full translation action, a constant related mean section, zero source connection, and identity first arrow. Then $v = 0$, $\mathsf{J} = I$, and the constant target connection inducing $A(X) = (X, X)$ realizes a nonzero cocycle solution $A^*\Delta A = 0$. It belongs to the totally isotropic Z term, although the ambient radical is zero. All normal laws are proper, and the second morphism is equivariant for the translation homomorphism L_2. Its Fisher increase excludes its interpretation as a parameter-independent Markov arrow under Theorem 8.21. [ESTABLISHED]

The classifications settle the finite-dimensional algebra and, on the stated regular strata, its smooth parameterization. They retain the separate requirement identified after Theorem 8.24: interpreting a cocycle as additive base information loss still requires stage-two connection compatibility. No automatic global solution with prescribed boundary data across changing ranks, infinite-dimensional closed-range result, or compatible connection for every scale map is asserted by these theorems. [NOT-CLAIMED]

8.10 Boundary of the construction

The chapter has constructed vertical tensors and their functorial defects. A scalar gauged sigma energy would additionally require a base cometric, a base density, channel weights, boundary conditions, and a decision about whether the connection is fixed or dynamical. None is selected by h_s^ω or c_s^ω. [NOT-CLAIMED]

Vertical Fisher length on a fixed regular curve does not itself select an orbit, identify physical time, or compare independently evolved fine and coarse paths. Such comparisons require an oriented semiconjugacy in addition to (8.62). The semiconjugacy criterion, sufficient natural-gradient conditions, and duration relation are proved in Chapter 9; the configuration tier of Definition 9.6 supplies a concrete example. [ESTABLISHED]

For further declared renormalization maps, a general construction of the required dynamical compatibility remains open. Proposition 9.72 gives one sufficient route: exhibit functional compatibility on an open region containing the orbit and prove horizontal conformality of the coarse configuration map. The pointwise tensor classification above does not supply these functional or dynamical hypotheses, and this sufficient route is not asserted necessary. [OPEN]

The scale index carried by (8.68) and by Theorem 8.24 is renormalization depth. It is not an inference-orbit coordinate and not a duration, and no result of this chapter converts it into either. [NOT-CLAIMED]

The general coarse-graining chapter uses the typed distinction exposed here. Its normalized Markov channels inherit the Fisher contraction theorem. Its energy precompositions and operator restrictions require their own probability and closure arguments and do not inherit that theorem by terminology. [ESTABLISHED]

Chapter 9

Relational Inference Histories and Fisher Information Clocks

The contextual base $\mathcal{C}$ is fixed data in the present theory. It organizes the domains on which agents are represented, but it carries no distinguished temporal parameter. Inference therefore has to be described as change in a statistical fiber or as change of a section, rather than as motion of the base. A curve in $\mathcal{C}$ orders contexts and can be used to probe a section; it is not, by that fact, an agent history. [HYPOTHESIS]

This chapter supplies two structures that the preceding pullback construction deliberately left open. A declared VFE gradient ray selects an oriented unparameterized inference orbit, and the Fisher line element assigns length to that already selected orbit. The resulting information clock is agent-relative and invariant under orientation-preserving reparameterization. It is neither a physical time coordinate nor a scalar that automatically exists on the whole configuration space. [DEFINITION]

9.1 Fixed-base kinematics

For agent i, restrict the belief and model bundles to its contextual domain $\mathcal{C}_i$ and form

$$\mathcal{E}_i = \mathcal{E}_b|_{\mathcal{C}_i} \times_{\mathcal{C}_i} \mathcal{E}_m|_{\mathcal{C}_i}, \qquad \varpi_i : \mathcal{E}_i \longrightarrow \mathcal{C}_i. \tag{9.1}$$

The pair $\boldsymbol{\omega}_i = (\omega_b, \omega_m)$ supplies a product horizontal distribution and a vertical projection $\mathrm{ver}^{\boldsymbol{\omega}_i}$. Its vertical Fisher form is not fixed until positive channel weights or a joint statistical metric are declared. The common bundle projection supplies neither choice. [DEFINITION]

Definition 9.1 (Vertical, horizontal, and mixed curves). Let J be a connected interval and let $\Gamma : J \to \mathcal{E}_i$ be C^1, with $\gamma = \varpi_i \circ \Gamma$. Relative to $\boldsymbol{\omega}_i$, its velocity has the unique splitting

$$\dot{\Gamma} = H^{\boldsymbol{\omega}_i}_{\Gamma}\dot{\gamma} + v_{\boldsymbol{\omega}_i}(\Gamma), \qquad v_{\boldsymbol{\omega}_i}(\Gamma) = \mathrm{ver}^{\boldsymbol{\omega}_i}(\dot{\Gamma}). \tag{9.2}$$

The primary labels are pointwise. At a parameter value λ the velocity is *stationary* when $\dot{\Gamma}(\lambda) = 0$; *vertical* when $\dot{\gamma}(\lambda) = 0$ and $\dot{\Gamma}(\lambda) \neq 0$; *$\boldsymbol{\omega}_i$-horizontal* when $v_{\boldsymbol{\omega}_i}(\Gamma)(\lambda) = 0$ and $\dot{\gamma}(\lambda) \neq 0$; and *mixed* when both $\dot{\gamma}(\lambda)$ and $v_{\boldsymbol{\omega}_i}(\Gamma)(\lambda)$ are nonzero. These four cases partition every parameter value. The interval-level labels are then defined by quantification: the curve is vertical, horizontal, or mixed

on J when it has that pointwise type at every $\lambda \in J$. The interval-level labels are therefore *not* exhaustive, since a curve may change pointwise type along J. Every use of an interval label below is on a subinterval of one fixed pointwise type, so no downstream statement is affected. Verticality is intrinsic to the bundle projection; horizontality and the mixed decomposition depend on the selected connection. [DEFINITION]

Two further curve types occur in this chapter and neither is a curve in $\mathcal{E}_i$. A base curve $\gamma \subset \mathcal{C}_i$ carries no verticality or horizontality predicate at all, since those predicates are statements about a total-space velocity. A curve of configurations, defined in Section 9.2, lives in a section space; verticality statements about it are statements about its adjoint evaluation at a fixed context, by (9.4), and not about the configuration curve itself. Applying a verticality predicate directly to a base curve or to a configuration curve is a type error. [DEFINITION]

The definition agrees with the standard connection splitting of a bundle tangent space [50]. Since $T\varpi_i \circ H^{\boldsymbol{\omega}_i}$ is the identity and $T\varpi_i \circ \mathrm{ver}^{\boldsymbol{\omega}_i} = 0$, applying $T\varpi_i$ to (9.2) recovers $\dot{\gamma}$. Thus a vertical curve stays in one fiber, whereas a horizontal lift is a connection-selected comparison of points in different fibers. Neither operation moves the fixed base itself. [ESTABLISHED]

Proposition 9.2 (Horizontality is connection relative). The same nonvertical total-space curve can be mixed for one connection and horizontal for another. [ESTABLISHED]

Proof. Take the trivial line bundle $\mathbb{R}_x \times \mathbb{R}_y \to \mathbb{R}_x$ and the curve $\Gamma(r) = (r, r)$. For the horizontal distribution $H_0 = \mathrm{span}\{\partial_x\}$, its vertical velocity is ∂_y, so the curve is mixed. For $H_1 = \mathrm{span}\{\partial_x + \partial_y\}$, the same velocity is horizontal. The base trace $x = r$ is unchanged. □

9.2 Curves of sections and pointwise vertical change

Hypothesis 9.3 (Regular section configuration). Let $\mathfrak{S}_i$ be a declared regular space of pairs of smooth sections $Q = (q, s)$ of $\mathcal{E}_i \to \mathcal{C}_i$. Whenever a tangent-space calculation is used, assume either a finite-dimensional configuration submanifold or a specified smooth locally convex manifold structure for which evaluation is differentiable. No such structure follows merely from writing down all smooth sections. [HYPOTHESIS]

For a C^1 representative $Q_i : I \to \mathfrak{S}_i$, define its adjoint family by

$$\widehat{\Sigma}_i : I \times \mathcal{C}_i \longrightarrow \mathcal{E}_i, \qquad \widehat{\Sigma}_i(\lambda, c) = Q_i(\lambda)(c), \qquad \varpi_i \circ \widehat{\Sigma}_i(\lambda, c) = c. \tag{9.3}$$

Differentiation in the history parameter at fixed c gives

$$T\varpi_i \left(\partial_\lambda \widehat{\Sigma}_i(\lambda, c) \right) = 0, \qquad \partial_\lambda \widehat{\Sigma}_i(\lambda, c) \in V_{Q_i(\lambda)(c)} \mathcal{E}_i. \tag{9.4}$$

Thus a curve of full sections produces a vertical statistical update at every fixed context, even though its state is an entire section. [ESTABLISHED]

At each history position the instantaneous sections also induce a pair of connection-relative perceived geometries on the same fixed base:

$$\mathbf{h}_i(\lambda) = \left(h^{\omega_b}_{q_i(\lambda)}, h^{\omega_m}_{q_i^m(\lambda)}\right). \tag{9.5}$$

Consequently a section history determines an ordered family of base semimetrics without making the base itself evolve. Under $\widetilde{Q}_i = Q_i \circ \phi$, this family changes only by $\widetilde{\mathbf{h}}_i = \mathbf{h}_i \circ \phi$. The connections are held fixed in this statement; allowing them to change would add a separately declared connection history and the correction terms of Proposition 8.5. [ESTABLISHED]

Definition 9.4 (Oriented unparameterized history). Two regular representatives $Q_i : I \to \mathfrak{S}_i$ and $\widetilde{Q}_i : \widetilde{I} \to \mathfrak{S}_i$ represent the same oriented history when

$$\widetilde{Q}_i = Q_i \circ \phi, \qquad \phi : \widetilde{I} \longrightarrow I \text{ is an orientation-preserving } C^1 \text{ diffeomorphism.} \tag{9.6}$$

The resulting abstract oriented curve is denoted by $\mathscr{H}_i$. Its adjoint evaluation has the required type

$$\Sigma_i : \mathscr{H}_i \times \mathcal{C}_i \longrightarrow \mathcal{E}_b \times_{\mathcal{C}} \mathcal{E}_m, \qquad \varpi_i \circ \Sigma_i(r, c) = c. \tag{9.7}$$

If a representative is an embedding, $\mathscr{H}_i$ may be identified with its oriented image. With self-intersections, $\mathscr{H}_i$ retains traversal order and multiplicity and is not merely the set-theoretic image. [DEFINITION]

The distinction between a base probe and a section history is exposed by a diagonal evaluation. Let $\gamma : I \to \mathcal{C}_i$ and set $\Gamma(\lambda) = Q_i(\lambda)(\gamma(\lambda))$. Write

$$D^{\boldsymbol{\omega}_i} Q_i(\lambda) = (D^{\omega_b} q_i(\lambda), D^{\omega_m} q_i^m(\lambda)). \tag{9.8}$$

The chain rule and the connection splitting give

$$v_{\boldsymbol{\omega}_i}(\Gamma) = \partial_\lambda \widehat{\Sigma}_i(\lambda, \gamma(\lambda)) + D^{\boldsymbol{\omega}_i} Q_i(\lambda) \dot{\gamma}(\lambda). \tag{9.9}$$

The first term is history change at a fixed context. The second is the vertical response of one instantaneous section to a changing contextual probe. If γ is constant, only the pointwise history velocity remains; if Q_i is constant, only the section jet remains. When γ is constant, the diagonal curve is vertical exactly when the remaining vertical velocity is nonzero and is stationary when it vanishes. On an open subinterval where $\dot{\gamma} \neq 0$, the diagonal curve is mixed exactly when the sum in (9.9) is nonzero, and it is horizontal when that sum vanishes. [ESTABLISHED]

A smooth section is injective because $\varpi_i \circ Q = \mathrm{id}_{\mathcal{C}_i}$, and it is an embedding under the usual smooth-bundle hypotheses. A total-space curve can therefore be written as $Q \circ \gamma$ exactly when it lies in the section image, in which case $\gamma = \varpi_i \circ \Gamma$ is unique. For a general statistical parameterization that is not a section, such descent is only local under a local embedding hypothesis and is globally unique only under global injectivity. In particular, a nonconstant curve confined to one statistical fiber cannot be the pushforward of a nonconstant base curve through a section. [ESTABLISHED]

9.3 VFE selection of an oriented orbit

Hypothesis 9.5 (Regular metric inference domain). Let $\mathcal{Q}_i \subseteq \mathfrak{S}_i$ be a regular configuration manifold equipped with a positive-definite Fisher Riemannian metric G_i^F. In an infinite-dimensional realization, assume it is a strong metric so that the musical map is invertible on the covectors used below. Let $\mathcal{F}_i : \mathcal{Q}_i \to \mathbb{R}$ be C^2. A quotient-metric domain may replace $\mathcal{Q}_i$ only after the quotient has been constructed. [HYPOTHESIS]

A declared hypothesis with no exhibited model is vacuous, so the following construction exhibits one. It assumes no configuration manifold and no strong metric; it builds both from the bundle data already declared.

Definition 9.6 (An exhibited finite-dimensional configuration tier). Declare at scale ℓ a compact smooth base $\mathcal{C}_\ell$ with a finite positive Borel measure μ_ℓ; a trivial principal bundle $\mathscr{P}_\ell = \mathcal{C}_\ell \times G$ with $G = (\mathbb{R}^K, +)$ acting on the fiber by translation of the mean and ω the flat connection; the fiber $\mathcal{B} = \{\mathcal{N}(m, \Sigma_0) : m \in \mathbb{R}^K\}$ with fixed $\Sigma_0 \succ 0$, whose fiber Fisher form is the constant $g^F = \Sigma_0^{-1}$; a measurable weight $w : \mathcal{C}_\ell \to (0, \infty)$ bounded above and below; smooth fields $\phi_1, \dots, \phi_N : \mathcal{C}_\ell \to \mathbb{R}^K$ that are linearly independent in $L^2(w\mu_\ell; \Sigma_0^{-1})$ and whose span contains every constant field $c \mapsto g$ for $g \in \mathbb{R}^K$; and

$$\mathcal{Q}_\ell = \left\{ s_\xi : \xi \in \mathbb{R}^N \right\}, \qquad s_\xi(c) = \mathcal{N}\left(\sum_a \xi_a \phi_a(c), \Sigma_0 \right), \tag{9.10}$$

Let $\mathsf{L}_\ell : \mathbb{R}^K \to \mathbb{R}^N$ be the unique coefficient map satisfying $\sum_a (\mathsf{L}_\ell g)_a \phi_a(c) = g$ for every c. Constant mean translations preserve the selected span and induce the affine coefficient action

$$\Theta_g^\ell(\xi) = \xi + \mathsf{L}_\ell g. \tag{9.11}$$

Thus the gauge action, its coefficient action, and its orbit directions are declared on $\mathcal{Q}_\ell$, rather than inferred from an arbitrary basis. Its configuration metric is

$$\mathsf{G}_\ell(V, V) = \int_{\mathcal{C}_\ell} w(c) g^F \left(\partial_V s_\xi(c), \partial_V s_\xi(c) \right) \mathrm{d}\mu_\ell(c), \tag{9.12}$$

declared as a *weighted product of marginal fiber metrics* and not as the pullback of a joint-law Fisher metric. [DEFINITION]

Theorem 9.7 (The exhibited tier satisfies the regular metric hypothesis). Under Definition 9.6: the configuration space $\mathcal{Q}_\ell$ is a nonempty finite-dimensional smooth manifold diffeomorphic to $\mathbb{R}^N$, with $\xi = 0$ the constant configuration $c \mapsto \mathcal{N}(0, \Sigma_0)$; the metric (9.12) is the constant Gram form

$$\mathsf{G}_\ell = \Phi, \qquad \Phi_{ab} = \int_{\mathcal{C}_\ell} \phi_a(c)^\top \Sigma_0^{-1} \phi_b(c) w(c) \mathrm{d}\mu_\ell(c), \tag{9.13}$$

independent of ξ, smooth, finite, and positive definite; any Riemannian metric on a finite-dimensional manifold is automatically strong; every C^1 functional has a unique gradient $\Phi^{-1}\nabla\mathcal{F}$, and for $\mathcal{F} \in C^2$

the field $X_\ell = -\Phi^{-1}\nabla\mathcal{F}$ is locally Lipschitz with locally unique integral curves; and the gauge orbit tangent generated by constant translations of the mean is $\mathsf{L}_\ell(\mathbb{R}^K)$, a linear subspace of $\mathbb{R}^N$, hence closed and complemented, so that the infimum in (9.101) is attained and equals the Φ-orthogonal projection. [ESTABLISHED]

Proof. The bundle is trivial and s_ξ is smooth jointly in (c,ξ). If $s_\xi = s_\eta$, then $\sum_a(\xi_a-\eta_a)\phi_a = 0$ in $L^2(w\mu_\ell;\Sigma_0^{-1})$, so the declared independence gives $\xi=\eta$; hence the selected section family is diffeomorphic to its coefficient space $\mathbb{R}^N$. Since $\partial_V s_\xi(c) = \sum_a v_a\phi_a(c)$ does not depend on ξ, the integrand of (9.12) is the quadratic form $v^\top\Phi v$ with Φ as displayed; finiteness is compactness of the base together with boundedness of w. The form Φ is positive definite: $v^\top\Phi v = 0$ forces $\sum_a v_a\phi_a = 0$ in $L^2(w\mu_\ell;\Sigma_0^{-1})$, and the fields are declared independent. Strongness in finite dimensions is the statement that every inner product on a finite-dimensional space induces the norm topology and that its musical map is a linear isomorphism. The gradient statement is invertibility of Φ. Finally, the span condition defines L_ℓ and gives $s_{\Theta_g^\ell(\xi)} = g\cdot s_\xi$; its finite-dimensional image is closed and complemented. □

Nondegeneracy is a rank condition that can be checked before any dynamics is written. With an M-atom design $\mu_\ell = \sum_{a=1}^M \rho_a\delta_{c_a}$ the Gram matrix has rank at most MK, so whenever $N > MK$ the metric is degenerate and, by Proposition 9.52, the natural-gradient equation has no solution or many. A worked instance is $\mathcal{C}_\ell = S^1$ with normalized arclength, $K=1$, $\Sigma_0 = 1$, $w\equiv 1$, and $\{\phi_a\} = \{1,\cos\theta,\sin\theta\}$, for which $\Phi = \mathrm{diag}(1,\frac12,\frac12)$ has determinant $\frac14$ and eigenvalues $\{1,\frac12\}$, so $\mathcal{Q}_\ell \cong \mathbb{R}^3$ carries a positive definite constant Gram metric. [ESTABLISHED]

For the integrated normal-mean metric just displayed, two choices of parameter topology exhibit the strong-metric and weak-metric boundary. On $\mathcal{Q}^{L^2} = L^2(\mu;\mathbb{R}^K)$ with $\mathsf{G}^{L^2}(V,W) = \int V^\top\Sigma_0^{-1}W w\mathrm{d}\mu$, two-sided bounds $0 < w_- \le w \le w_+ < \infty$ and $\lambda_- I \preceq \Sigma_0^{-1} \preceq \lambda_+ I$ give $w_-\lambda_-\|V\|_{L^2}^2 \le \mathsf{G}^{L^2}(V,V) \le w_+\lambda_+\|V\|_{L^2}^2$, so the musical map is a topological isomorphism and the metric is strong; the price is that $\mathcal{F}$ must be C^1 on L^2, which excludes gradient-energy objectives. Either bound failing alone destroys the conclusion. On a Sobolev tier H^s with the same integrated metric the topology of G^{L^2} is strictly coarser, the musical map is injective and bounded but not surjective, and the metric is weak. The failure is realized: on $\mathcal{C} = S^1$ with $\mathcal{Q} = H^1(S^1)$ and $\mathcal{F}(Q) = \frac12\int|Q'|^2$, the configuration $Q = \sum_{k\ge1} k^{-2}\sin k\theta$ has $\|Q\|_{H^1}^2$ finite while a gradient would be $-Q'' = \sum_k \sin k\theta$ with infinite L^2 norm, so no natural-gradient vector field exists at Q. The missing ingredient is Riesz representability of $\mathrm{d}\mathcal{F}_Q$, ensured for instance by $|\mathrm{d}\mathcal{F}_Q[V]| \le C_Q\|V\|_{L^2}$ on a dense domain. No claim of an exhaustive classification of section-space metrics follows: these examples contrast a strong L^2 tier with a Sobolev tier requiring actual representability of the derivative in its tangent space. The strong Hilbert theorem below applies whenever its own metric and action hypotheses are established. [ESTABLISHED]

Hypothesis 9.8 (Standing configuration tier). Wherever this manuscript asserts a natural-gradient vector field or a Fisher duration on a configuration manifold, the manifold and its metric are those of Definition 9.6, or those of the L^2 tier above with its two-sided bounds and a C^1-on-L^2 objective. This is a restriction adopted by choice. It excludes the space of all smooth sections

with no declared manifold topology, every weak-metric realization in which the musical map is not surjective, and every configuration metric labeled as an exact joint-law Fisher metric without the block-orthogonality or independence condition of (9.19). It is used at Sections 9.3, 9.7 and 9.11 to 9.13 and at every cross-scale duration comparison. Its model class is nonempty by Theorem 9.7, so nothing downstream is vacuous. Whether the recognition families a particular application declares can be brought to one of these two tiers is a separate modeling obligation and is not settled here. [HYPOTHESIS]

Hypothesis 9.9 (Exact multi-agent VFE lift). Calling the configuration objective an exact local or collective VFE requires more than its pair of marginal sections. Fix a block B, an outside marginal Q_{B^c}, and a declared manifold $\mathfrak{R}_B$ of admissible conditional recognition kernels. There must be a declared smooth marginalization or configuration-extraction map

$$\pi_i^{\mathrm{conf}} : \mathfrak{R}_B \longrightarrow \mathcal{Q}_i \tag{9.14}$$

on the admissible family. Supply a smooth right-inverse lift

$$\iota_i : \mathcal{Q}_i \longrightarrow \mathfrak{R}_B, \qquad Q_i \longmapsto r_B^{Q_i}(\cdot \mid b), \qquad \pi_i^{\mathrm{conf}} \circ \iota_i = \mathrm{id}_{\mathcal{Q}_i}, \tag{9.15}$$

so the lifted conditional law actually reproduces the displayed agent configuration. Equivalently, the full recognition law $Q_{B^c}(db) r_B^{Q_i}(dy_B \mid b)$ must have the declared marginals or section statistics encoded by Q_i. Without this commuting condition, ι_i is merely a parameterized family of laws, not a lift of the agent configuration.

For every Q_i in the inference domain, impose the support and finiteness hypotheses of Theorems 7.4 and 7.5: $Q_{B^c} \ll \Pi_{o,B^c}$, $Z_B \in (0, +\infty)$ Q_{B^c}-almost everywhere, $r_B^{Q_i}(\cdot \mid b) \ll P_{0,B}(\cdot \mid b)$, and integrability of the KL terms and positive and negative energy parts, so the displayed local, collective, and difference functionals are finite. Assume also that the lifted family is C^2 and satisfies the following chartwise domination condition. For every chart $\kappa : V \to \mathbb{R}^d$ of $\mathcal{Q}_i$ and every multi-index α with $|\alpha| \le 2$, require a jointly measurable version in (ξ, b) of $\partial_\xi^\alpha \mathcal{F}_{B,o}(r_B^{\kappa^{-1}(\xi)}; b)$. For every compact $K \subset \kappa(V)$, there is a $G_K \in L^1(Q_{B^c})$ such that, for every multi-index α with $|\alpha| \le 2$,

$$\sup_{\xi \in K} \left| \partial_\xi^\alpha \mathcal{F}_{B,o}\left(r_B^{\kappa^{-1}(\xi)}; b \right) \right| \le G_K(b) \quad \text{for } Q_{B^c}\text{-almost every } b. \tag{9.16}$$

The case $|\alpha| = 0$ controls the objective integrand, and the cases $|\alpha| = 1, 2$ justify two differentiations under the Q_{B^c} expectation. Then define the outside-averaged objective

$$\overline{\mathcal{F}}_{B,o}(Q_i) = \mathbb{E}_{Q_{B^c}} \left[\mathcal{F}_{B,o}(r_B^{Q_i}; Y_{B^c}) \right]. \tag{9.17}$$

The lift is responsible for all correlation or copula data not determined by the displayed marginal sections. A fixed product or fixed-copula family is a special hypothesis that can supply it; the pair (q, s) alone cannot. Alternatively, a population configuration may be lifted directly to a full recognition law and the collective VFE pulled back along that lift. If G_i^F is called the exact recognition Fisher

metric, also require

$$\mathsf{G}_i^F = \iota_i^* G^F_{\mathfrak{R}_B} \tag{9.18}$$

on a nondegenerate tier, or use its justified metric quotient. A weighted product of marginal-fiber metrics equals this joint pullback only under a separately proved block-orthogonality or fixed-dependence hypothesis. [HYPOTHESIS]

The criterion behind that last sentence is exact and worth stating, because it also shows that no order relation is available in either direction. Let L be the joint score in a configuration direction and let $L_b = \Pi_b L$ and $L_m = \Pi_m L$ be its orthogonal projections onto the centered $\sigma(Y_b)$- and $\sigma(Y_m)$-measurable subspaces, marginalization being a deterministic parameter-independent Markov kernel. Then

$$\|L\|^2 - \big(\|L_b\|^2 + \|L_m\|^2\big) = \|L - L_b - L_m\|^2 - 2\langle L_b, L_m\rangle, \tag{9.19}$$

whose first term is nonnegative and whose second is signed. Equality at unit weights therefore holds exactly when $\|L - L_b - L_m\|^2 = 2\langle L_b, L_m\rangle$; independence of the two channels is the clean sufficient structural condition, and under independence with independently excitable nondegenerate marginals the weights are forced to $w_b = w_m = 1$. A fixed non-independence copula does not suffice, since the derivative of its log density along the configuration parameter is generally nonzero. No Loewner ordering holds either way: with jointly Gaussian precision $\Lambda = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ the difference $\Lambda - (1-\rho^2)I_2 = \left(\begin{smallmatrix} \rho^2 & \rho \\ \rho & \rho^2 \end{smallmatrix}\right)$ has determinant $\rho^2(\rho^2 - 1) < 0$, taking the value $2\rho(1+\rho)$ along $(1,1)$ and $2\rho(\rho-1)$ along $(1,-1)$. Finally, distinct smooth right inverses of the same extraction map π_i^{conf} can, but need not, induce different pullback metrics. For the two-parameter normal-location family with Fisher metric $d\theta^2 + d\eta^2$ and extraction $\pi(\theta,\eta) = \theta$, the distinct lifts $\iota_0(x) = (x,0)$ and $\iota_c(x) = (x,c)$, for $c \neq 0$, both induce dx^2, whereas $\iota_\Delta(x) = (x,x)$ induces $2dx^2$. Thus the configuration Fisher metric is not in general determined by the displayed configuration alone; a selected lift must be supplied unless a separately proved canonicity or lift-independence theorem removes that choice. [ESTABLISHED]

By Theorem 7.5, $\overline{\mathcal{F}}_{B,o}$ differs from the collective VFE restricted along $Q_i \mapsto Q_{B^c} r_B^{Q_i}$ only by a Q_i-independent outside term. Their differentials and natural-gradient rays therefore agree when the same block metric is used. A single realized conditional objective $\mathcal{F}_{B,o}(r_B; b)$ need not have that gradient, and independently moving outside marginals do not define the same fixed objective. [ESTABLISHED]

9.4 A finite correlated right-inverse witness

Definition 9.10 (Three-agent parity lift). Let

$$\mathcal{I}_6 = \{1,2,3\} \times \{b,m\}, \qquad x = (x_j)_{j\in\mathcal{I}_6} \in \{0,1\}^{\mathcal{I}_6}, \qquad \theta = (\theta_j)_{j\in\mathcal{I}_6} \in (0,1)^{\mathcal{I}_6}. \tag{9.20}$$

The channel label in a typed agent permutation is preserved. Define

$$p_0(x;\theta) = \prod_{j\in\mathcal{I}_6} \theta_j^{x_j}(1-\theta_j)^{1-x_j}, \qquad D(\theta) = \prod_{j\in\mathcal{I}_6} \theta_j(1-\theta_j), \tag{9.21}$$

$$\chi(x) = (-1)^{\sum_{j\in\mathcal{I}_6} x_j}, \qquad p_\kappa(x;\theta) = p_0(x;\theta) + \kappa\chi(x)D(\theta), \qquad |\kappa| < 1. \tag{9.22}$$

The map $\iota_\kappa : (0,1)^6 \to \Delta_{63}^\circ$ sends θ to $p_\kappa(\cdot;\theta)$. In this definition κ is fixed relational or nomological data, not an inferred coordinate. [DEFINITION]

Theorem 9.11 (Positive smooth lift with the declared even relabeling symmetry). For every fixed $|\kappa| < 1$, the law in (9.22) is strictly positive and normalized. Every proper marginal is the corresponding product-Bernoulli marginal of p_0. Consequently the map ι_κ is a smooth right inverse of the map that extracts the six one-bit marginals, and it is an immersion.

Let $H_{\mathrm{pair}} = (\mathbb{Z}_2)^3 \rtimes S_3$ act by simultaneous complement of the two channel bits of any selected agent and by typed agent permutations $(a,r) \mapsto (\sigma(a),r)$. Apply the same complements and permutations to the coordinates of θ. Then

$$\iota_\kappa(g\theta) = g_\# \iota_\kappa(\theta) \qquad \text{for every } g \in H_{\mathrm{pair}}. \tag{9.23}$$

For an odd single-channel complement F_j, by contrast,

$$(F_j)_\# p_\kappa(\cdot;\theta) = p_{-\kappa}(\cdot;F_j\theta). \tag{9.24}$$

Thus odd independent complements are equivariant only if κ is promoted to a pseudoscalar that changes sign, or if $\kappa = 0$. [ESTABLISHED]

Proof. For each atom,

$$\frac{p_0(x;\theta)}{D(\theta)} = \prod_{j:x_j=1} \frac{1}{1-\theta_j} \times \prod_{j:x_j=0} \frac{1}{\theta_j} > 1. \tag{9.25}$$

Hence $p_\kappa \geq p_0 - |\kappa|D > (1-|\kappa|)D > 0$. Normalization follows from $\sum_x \chi(x) = \prod_{j\in\mathcal{I}_6}(1-1) = 0$. If a marginal sums out at least one coordinate, its parity term contains the same zero factor, so the entire perturbation cancels and the remaining law is the product marginal. Writing π for extraction of the one-bit success probabilities gives $\pi \circ \iota_\kappa = \mathrm{id}_{(0,1)^6}$. Polynomial dependence gives smoothness, and differentiation of the right-inverse identity gives $D\pi_{\iota_\kappa(\theta)} D\iota_{\kappa,\theta} = I_6$, so $D\iota_{\kappa,\theta}$ is injective.

A paired complement changes two bits, so it preserves χ; it also preserves D, while the product law transforms by ordinary Bernoulli relabeling. A typed agent permutation preserves all three expressions in (9.21)–(9.22). This proves (9.23). A single-channel complement changes the sign of χ, proving (9.24). □

For a fixed interior tuple θ, the Frechet fiber of positive laws with those six one-bit marginals has dimension $63 - 6 = 57$; varying κ traces only its one-dimensional parity line. The theorem therefore exhibits one correlated lift direction and does not classify or parameterize the other fifty-six

dependence directions. For $\kappa \neq 0$, it also gives an exact same-marginal control against any claim that pairwise independence forces the full joint to be a product. [ESTABLISHED]

Corollary 9.12 (Agent-block local–global VFE identity). Fix an agent block $B_a = \{(a,b),(a,m)\}$, fix the four outside parameters $\theta_{B_a^c}$, and let

$$r_{a,\theta_{B_a}}(x_{B_a} \mid x_{B_a^c}) = \frac{p_\kappa(x;\theta)}{p_{0,B_a^c}(x_{B_a^c};\theta_{B_a^c})}. \tag{9.26}$$

For any two interior block parameters θ_{B_a} and θ'_{B_a}, and any fixed strictly positive posterior target Π_o on the Boolean cube, all relative entropies are finite and

$$\begin{aligned}&\mathcal{F}_o^{\mathrm{ext}}\left(p_\kappa(\cdot;\theta'_{B_a},\theta_{B_a^c})\right) - \mathcal{F}_o^{\mathrm{ext}}\left(p_\kappa(\cdot;\theta_{B_a},\theta_{B_a^c})\right)\\ &\quad = \mathbb{E}_{p_{0,B_a^c}}\left[\mathcal{F}_{B_a,o}^{\mathrm{ext}}\left(r_{a,\theta'_{B_a}};X_{B_a^c}\right) - \mathcal{F}_{B_a,o}^{\mathrm{ext}}\left(r_{a,\theta_{B_a}};X_{B_a^c}\right)\right].\end{aligned} \tag{9.27}$$

Thus the outside-averaged agent-block conditional VFE is exactly the restriction of one global VFE up to a block-parameter-independent constant. For every block tangent $V \in T_{\theta_{B_a}}(0,1)^2$, differentiation gives

$$d\mathcal{F}_o^{\mathrm{ext}}[V] = \mathbb{E}_{p_{0,B_a^c}}\left[d\mathcal{F}_{B_a,o}^{\mathrm{ext}}[V]\right], \tag{9.28}$$

where both differentials are evaluated at the same lifted conditional law and the outside expectation is fixed. [ESTABLISHED]

Proof. The proper-marginal statement in Theorem 9.11 makes the outside marginal exactly p_{0,B_a^c}, independent of θ_{B_a}, and strict positivity makes (9.26) a positive conditional law. The finite state space and positive target make every KL term finite. Equation (9.27) is then Theorem 7.5 with this fixed outside marginal. The summands are smooth on the finite positive family, so differentiating that finite identity proves (9.28). □

Proposition 9.13 (Exact joint-versus-weighted-marginal score residual). Let P_u be a regular dominated joint family on $(Y_1,\dots,Y_m)$, with square-integrable joint score $L = \nabla_u \log p_u(Y)$. Let $L_i = \mathbb{E}[L \mid Y_i]$ be the embedded score of its i-th marginal and let w_i be fixed scalar weights. Define

$$G_J = \mathbb{E}[LL^\top], \qquad G_i = \mathbb{E}[L_iL_i^\top], \qquad G_w = \sum_i w_iG_i, \tag{9.29}$$

$$R = L - \sum_i L_i, \qquad K = \mathbb{E}[RR^\top], \qquad C = \sum_{i\neq j}\mathbb{E}[L_iL_j^\top]. \tag{9.30}$$

The ordered sum makes C symmetric, but it can be indefinite. The exact identity is

$$G_J - G_w = K - C + \sum_i (1-w_i)G_i. \tag{9.31}$$

On a declared tangent subspace T, equality $G_J = G_w$ holds exactly when the bilinear restriction

$$\left(K - C + \sum_i (1 - w_i) G_i \right) \Big|_{T \times T} = 0. \tag{9.32}$$

[ESTABLISHED]

Proof. The marginal-score identity gives $\mathbb{E}[L L_i^\top] = \mathbb{E}[L_i L_i^\top] = G_i$, and likewise after transpose. Expanding $K = \mathbb{E}[(L - \sum_i L_i)(L - \sum_i L_i)^\top]$ therefore gives $K = G_J - \sum_i G_i + C$. Rearrangement proves (9.31). □

Corollary 9.14 (Parity-family Fisher residual). For the family in Definition 9.10, every pair of bits is independent, so $C = 0$. With unit marginal weights,

$$G_J - \sum_{j \in \mathcal{I}_6} G_j = K \succ 0 \qquad \text{when } \kappa \neq 0. \tag{9.33}$$

At $\theta_j = 1/2$ for all j, set $c = \kappa/64$. Then

$$G_J = \frac{4}{1 - c^2} I_6, \qquad G_J - \sum_{j \in \mathcal{I}_6} G_j = \frac{4c^2}{1 - c^2} I_6. \tag{9.34}$$

In particular, $\kappa = 1/2$ gives the residual

$$G_J - \sum_{j \in \mathcal{I}_6} G_j = \frac{4}{16383} I_6. \tag{9.35}$$

[ESTABLISHED]

Proof. All two-bit marginals are product by Theorem 9.11; centered marginal scores from distinct bits therefore have zero cross expectation. If s_j is the one-bit Bernoulli score and L the joint score, direct differentiation gives

$$R_j = L_j^{\text{joint}} - s_j = \frac{\kappa \chi(x) D(\theta)}{p_\kappa(x; \theta)} \times \left(\frac{1 - x_j}{\theta_j} - \frac{x_j}{1 - \theta_j} \right). \tag{9.36}$$

If $v^\top R$ vanishes on every atom, positivity and $\kappa D \neq 0$ imply that the sum of the six bracketed one-bit functions, weighted by v_j, vanishes on the Boolean cube. Comparing two atoms that differ only in coordinate j forces $v_j = 0$. Thus no nonzero v lies in the null space of $K = \mathbb{E}[R R^\top]$, proving positive definiteness.

At the center, $D = 1/4096$ and

$$p_\kappa(x; \tfrac{1}{2}\mathbf{1}) = \frac{1}{64} \left(1 + c \chi(x) \right). \tag{9.37}$$

The derivative of D vanishes there, each product-marginal score has squared magnitude 4, and averaging $(1+c\chi)^{-1} = (1-c\chi)/(1-c^2)$ over the Boolean cube gives (9.34). Substitution of $c = (1/2)/64 = 1/128$ gives (9.35). □

At the parity center, the six unit-coordinate marginal tensors are $G_j = 4e_j e_j^\top$. Therefore equality $G_J = G_w$ on the full six-dimensional tangent space holds exactly when

$$w_j = \frac{1}{1-c^2} \qquad \text{for every } j \in \mathcal{I}_6. \tag{9.38}$$

For unit weights this holds exactly when $c = 0$, equivalently $\kappa = 0$. [ESTABLISHED]

The positive-semidefinite ordering in (9.33) is specific to this parity family. The older two-bit signed Fisher witness at the symmetric point has joint pullback $63^{-1}\left(\begin{smallmatrix} 256 & -32 \\ -32 & 256 \end{smallmatrix}\right)$ and product-marginal metric $4I_2$; their difference is $63^{-1}\left(\begin{smallmatrix} 4 & -32 \\ -32 & 4 \end{smallmatrix}\right)$, with eigenvalues $36/63$ and $-28/63$. Hence the parity-family ordering does not support a general Loewner-order claim. [ESTABLISHED]

Definition 9.15 (Declared three-agent parity hyperedge). At the symmetric point of (9.37), let the latent prior be $U(x) = 1/64$ and supply the binary record kernel

$$K_{\mathrm{hyp}}(1 \mid x) = \frac{1}{2}\,(1 + c\chi(x))\,, \qquad K_{\mathrm{hyp}}(0 \mid x) = \frac{1}{2}\,(1 - c\chi(x))\,. \tag{9.39}$$

This is a normalized three-agent hyperedge record by declaration. [DEFINITION]

Proposition 9.16 (Exact posterior and tight VFE for the declared hyperedge). The evidence of record $O = 1$ is $P(O=1) = 1/2$, and its posterior is exactly $p_\kappa(\cdot; \frac{1}{2}\mathbf{1})$. For

$$\mathcal{F}_1(Q) = D_{\mathrm{KL}}(Q\|U) - \mathbb{E}_Q \log K_{\mathrm{hyp}}(1 \mid X), \tag{9.40}$$

the posterior attains the tight value $\mathcal{F}_1 = \log 2$. The product lift of its six one-bit marginals is U, and its exact excess is

$$\mathcal{F}_1(U) - \log 2 = -\frac{1}{2}\log(1-c^2). \tag{9.41}$$

[ESTABLISHED]

Proof. Exactly half of the atoms have each parity, so averaging (9.39) under U gives evidence $1/2$. Bayes' rule then gives posterior $64^{-1}(1+c\chi)$, which is (9.37). The standard VFE gap identity gives the tight posterior value $-\log P(O=1) = \log 2$. Every posterior one-bit marginal is uniform, so its product lift is U, and

$$-\mathbb{E}_U \log K_{\mathrm{hyp}}(1 \mid X) = \log 2 - \frac{1}{2}\log(1-c^2), \tag{9.42}$$

because the two parities have equal U-mass. This proves (9.41). □

The hyperedge kernel in (9.39) is supplied generative data. No derivation from pairwise locality, conditional independence, autonomous agency, or a canonical agent inventory is claimed. No within-pair belief–model swap is used as a universal typed symmetry. [NOT-CLAIMED]

The finite group action above supplies no principal $\mathrm{GL}(K)$ action, Lie algebra, connection, parallel transport, or holonomy. The promoted-family rank calculation in Corollary 9.50 identifies κ as a full-joint direction invisible to singleton retention, but rank alone does not make that direction an agent or intervention variable. [NOT-CLAIMED]

Open Problem 9.17 (Scope beyond the finite parity witness). Theorem 9.11 closes one finite categorical right-inverse instance. Arbitrary-family existence on a separately declared restricted morphism class, coarse-preserving nonproduct selection, $\mathrm{GL}(K)$-equivariance, agency, a continuum or section-space lift, physical geometry, physical time, units, and renormalization remain open. The absolute single-valued selector on the admitted split class is not open: it is refuted by Theorem 9.21. [OPEN]

9.5 Canonical dependence selection: exact finite boundary

The following finite results are bound to the released certificate with target digest `8112f0083a554a4d f3b5de3875174d0b0cbfeee36a7bc2ea20e78c8cf9da6b39`. They distinguish an absolute selector, a reference-relative selector, and a retained Fisher quotient. None supplies autonomous agency, continuum dynamics, or physical units.

Definition 9.18 (Finite marginal selection problem). For a finite nonempty typed list $X = (X_i)_{i=1}^n$, set

$$\mathcal{J}_X = \Delta\left(\prod_i X_i\right), \qquad \mathcal{M}_X = \prod_i \Delta(X_i), \qquad m_X : \mathcal{J}_X \longrightarrow \mathcal{M}_X, \tag{9.43}$$

where m_X extracts the singleton marginals. A law-valued section is a map $S_X : \mathcal{M}_X \to \mathcal{J}_X$ with $m_X S_X = \mathrm{id}$. A fixed-arity local morphism is an independently tensored finite Markov kernel $K = \bigotimes_i K_i$. An admitted arity-changing kernel $K : \prod_i X_i \rightsquigarrow \prod_j Y_j$ is marginal compatible when a specified $\bar{K} : \mathcal{M}_X \to \mathcal{M}_Y$ satisfies

$$m_Y K_\# = \bar{K} m_X. \tag{9.44}$$

Naturality on a declared morphism class means $K_\# S_X = S_Y \bar{K}$ for every morphism in that class. This definition does not admit all finite stochastic kernels silently. [DEFINITION]

Theorem 9.19 (Tensor-local naturality uniquely selects product law). If $(S_X)_X$ is natural under every fixed-arity coordinatewise finite Markov kernel with independently tensored randomness, then

$$S_X(\mu_1, \ldots, \mu_n) = \bigotimes_{i=1}^n \mu_i. \tag{9.45}$$

This conclusion is throughout the declared local morphism class, not all of $\mathrm{FinStoch}$. [ESTABLISHED]

Proof. For each i, let the preparation kernel from a singleton typed space send its unique point to μ_i. Their tensor product sends the unique law on the singleton product to $\bigotimes_i \mu_i$, while the induced marginal map sends the unique marginal tuple to $(\mu_i)_i$. Naturality forces (9.45). □

Corollary 9.20 (Maximal product-preserving category). Among wide marginal-compatible categories on the frozen typed objects that contain every fixed-arity local kernel, a natural section exists exactly when every added kernel preserves product laws. When it exists it is uniquely the product section, so the product-preserving marginal-compatible kernels form the maximal such morphism class. [ESTABLISHED]

Proof. The preceding theorem forces the product section. Its naturality for an added kernel is precisely the assertion that the kernel sends each product law to the product of its pushed singleton marginals. □

Theorem 9.21 (Admitted correlated refinements refute an absolute selector). For a typed source bit and typed target bits (Y, Z), explicitly admit both split kernels

$$R_\rho(y, z \mid x) = \mathbf{1}_{\{y=x\}} \begin{cases} (1+\rho)/2, & z = x, \\ (1-\rho)/2, & z \neq x, \end{cases} \qquad \rho \in \{1/3, 1/2\}. \tag{9.46}$$

Both are normalized and marginal compatible; admitting them is a declaration. No single-valued marginal section can be natural under both. This no-go uses single-valuedness and the admission of both displayed splits, but it uses no local-naturality premise. [ESTABLISHED]

Proof. Push the fair source law through the two splits. Both descended target data are the same fair/fair marginal pair, while the joint atom multisets are

$$\{1/3, 1/3, 1/6, 1/6\} \quad \text{and} \quad \{3/8, 3/8, 1/8, 1/8\}. \tag{9.47}$$

Naturality would require one value of the target section at that marginal pair to equal both laws. Their atom multisets are unequal even after any sample relabeling, a contradiction. □

Corollary 9.22 (No faithful marginal recovery of every joint law). Marginalization is noninjective even on strictly positive binary laws. Consequently no marginal section can also satisfy $S_X(m_X Q) = Q$ for every compatible positive full joint law. [ESTABLISHED]

Proof. The two positive split-output laws in (9.47) have the same singleton marginals and are distinct. A function of their common marginal datum cannot return both. □

Proposition 9.23 (Singleton marginals do not determine full VFE). There is no function of only the recognition singleton tuple, posterior singleton tuple, and positive evidence z that

equals $-\log z + D_{\mathrm{KL}}(Q\|P)$ for every compatible full recognition law Q and positive posterior law P. [ESTABLISHED]

Proof. On two fair bits take P to be the uniform product law, first take $Q_0 = P$, and then take the positive $\rho = 1/2$ split-output law Q_1. Both recognition inputs have the same singleton marginals and the posterior and evidence inputs are unchanged. Nevertheless $D_{\mathrm{KL}}(Q_0\|P) = 0$ while $D_{\mathrm{KL}}(Q_1\|P) > 0$ by strictness of finite KL. □

Proposition 9.24 (Singleton families do not determine full Fisher tensor). For $\theta \in (0,1)^6$, write

$$P_\theta(x) = \prod_{i=1}^{6} \theta_i^{x_i}(1-\theta_i)^{1-x_i}, \qquad Q_{\theta,\kappa}(x) = P_\theta(x)\big[1 + \kappa\chi(x)P_\theta(\bar{x})\big], \tag{9.48}$$

where $\chi(x) = (-1)^{\sum_i x_i}$ and $|\kappa| < 1$. The fixed families $\kappa = 0$ and $\kappa = 1/2$ have identical singleton-family maps, but at $\theta_i = 1/2$ their full-joint Fisher tensors are

$$4I_6 \quad \text{and} \quad \frac{65536}{16383} I_6. \tag{9.49}$$

Thus no rule of only the complete singleton-family map recovers every full-joint Fisher tensor. [ESTABLISHED]

Proof. Since $P_\theta(x)P_\theta(\bar{x}) = \prod_i \theta_i(1-\theta_i)$, the bracket in (9.48) is positive. Summing over any missing bit cancels the parity term, so every proper marginal, and hence every singleton-family map, is independent of κ. Direct finite score summation at the symmetric point gives (9.49). □

The finite intervention result below is bound to the released certificate with target digest `efe715cba81c2000403811015e6cecb599f4023133543b5cb0ae77288bccc874`. It closes one universal target inside the declared finite category; it does not make that category ontologically canonical. [ESTABLISHED]

Definition 9.25 (Finite typed-intervention presentations). Let $\mathsf{FinTIP}^{\mathrm{iso}}_{R,O}$ be the groupoid whose objects are tuples

$$P = \big(V, \tau, (X_v)_{v\in V}, D, (\kappa_v)_{v\in V}, r, o\big). \tag{9.50}$$

Here V is finite and nonempty, every state space X_v is finite and nonempty, $\tau : V \to \mathcal{T}$ is the declared role/type map, $D = (V, E_D)$ is a finite DAG, the distinguished nodes r, o have the retained-state and record roles under τ, and each $\kappa_v : X_{\mathrm{pa}(v)} \rightsquigarrow X_v$ is a normalized finite Markov kernel. A hard intervention replaces exactly the selected local kernels by point masses. An isomorphism is a type-preserving node bijection with state-space bijections that fixes the retained roles, carries the DAG and every local kernel to the target presentation, and intertwines all hard interventions and their induced laws. Let $\mathsf{FinObs}^{\mathrm{iso}}_{R,O}$ contain the complete retained objects (X_R, X_O, P_{RO}) with typed boundary relabelings. Passive marginalization defines $U_{\mathrm{pass}} : \mathsf{FinTIP}^{\mathrm{iso}}_{R,O} \to \mathsf{FinObs}^{\mathrm{iso}}_{R,O}$. Complete joint tables are used even when intervened cells vanish, so no conditioning on a null event is introduced. [DEFINITION]

Definition 9.26 (Right-override intervention monoid). For P, let A_P be the set of all partial sections $i : \operatorname{dom} i \to \coprod_{v\in V} X_v$ with $i(v) \in X_v$. Sequential hard interventions compose by total right override:

$$(i \star j)(v) = \begin{cases} j(v), & v \in \operatorname{dom} j, \\ i(v), & v \in \operatorname{dom} i \setminus \operatorname{dom} j, \\ \text{undefined}, & v \notin \operatorname{dom} i \cup \operatorname{dom} j. \end{cases} \tag{9.51}$$

The empty section is the identity, and associativity holds pointwise because both parenthesizations return the last assignment that names v. Writing P^i for the intervened presentation, one has $(P^i)^j = P^{i\star j}$. Its retained response is

$$\Phi_P(i) = \big(p_{P^i}\big)_{RO}. \tag{9.52}$$

Thus $(A_P, \star, \varnothing)$ is a finite monoid acting by hard-kernel replacement. [ESTABLISHED]

Definition 9.27 (Two-sided operational reduction). Contexts are behaviorally equivalent when their complete response signatures agree in every left and right context:

$$i \equiv_P j \quad \Longleftrightarrow \quad \Phi_P(u \star i \star v) = \Phi_P(u \star j \star v) \quad \text{for all } u, v \in A_P. \tag{9.53}$$

This formula defines the two-sided behavioral relation. [DEFINITION]

Proposition 9.28 (Behavioral quotient is well-defined). The relation $\equiv_P$ is an equivalence relation. Associativity of $\star$ also gives $a \star i \star b \equiv_P a \star j \star b$, so it is a two-sided monoid congruence. Hence $\bar{A}_P = A_P/\equiv_P$ is a monoid and $\bar{\Phi}_P([i]) = \Phi_P(i)$ is well-defined. Define

$$\operatorname{Red}(P) = \big(X_R, X_O, \bar{A}_P, [\varnothing], \bar{\Phi}_P\big). \tag{9.54}$$

The groupoid $\mathsf{FinRIE}^{\mathrm{iso}}_{R,O}$ has these reduced experiments as objects; its arrows are identity-preserving monoid isomorphisms together with typed boundary relabelings that intertwine every retained response. [ESTABLISHED]

Proof. Reflexivity, symmetry, and transitivity follow pointwise from equality of the complete signature indexed by (u, v). If $i \equiv_P j$, then for all $u, v, a, b \in A_P$, associativity gives

$$\Phi_P\big(u \star (a \star i \star b) \star v\big) = \Phi_P\big((u \star a) \star i \star (b \star v)\big) = \Phi_P\big((u \star a) \star j \star (b \star v)\big) = \Phi_P\big(u \star (a \star j \star b) \star v\big).$$

Thus the relation is a two-sided congruence and quotient multiplication is representative independent. Taking $u = v = \varnothing$ in (9.53) proves that $\bar{\Phi}_P$ is well-defined. □

Proposition 9.29 (Universal property of the contextual operational quotient). Let A be any monoid with identity 1, let Y be a response set, and let $\Phi : A \to Y$. Define

$$a \sim_\Phi b \quad \Longleftrightarrow \quad \Phi(uav) = \Phi(ubv) \quad \text{for every } u, v \in A. \tag{9.55}$$

Then $\sim_\Phi$ is the largest two-sided monoid congruence contained in the kernel relation of Φ. Write $\mathrm{Syn}(\Phi) = A/\sim_\Phi$, with quotient map π and response $\bar{\Phi}([a]) = \Phi(a)$. If $q : A \to B$ is a surjective unital monoid homomorphism and $\Phi = \psi q$, there is one unique surjective unital homomorphism

$$h : B \longrightarrow \mathrm{Syn}(\Phi), \qquad \pi = hq, \qquad \bar{\Phi} h = \psi. \tag{9.56}$$

Thus the contextual quotient is terminal when arrows point from finer response-compatible quotients to coarser ones, and is contextually fully abstract relative to the fixed data (A, Φ, Y). If A is finite, it has minimum protocol-class cardinality; equality of cardinalities makes h an isomorphism uniquely over A. The cardinality conclusion is finite-only and does not minimize a latent state space, DAG, kernel family, parameterization, or computational realization. After forgetting the map from A, a bare response monoid need not be rigid. [ESTABLISHED]

Proof. Equation (9.55) is an equivalence relation, and associativity shows that $a \sim_\Phi b$ implies $xay \sim_\Phi xby$. Taking $u = v = 1$ places it inside the kernel of Φ. Conversely, if $\approx$ is a two-sided congruence contained in that kernel and $a \approx b$, then $uav \approx ubv$ for all u, v, so $a \sim_\Phi b$. Hence it is the largest such congruence. The kernel congruence of q is therefore contained in $\sim_\Phi$, and $h(q(a)) = [a]$ is well-defined, unital, multiplicative, surjective, and the only map satisfying $hq = \pi$. The response identity follows from $\Phi = \psi q$. When A is finite, q makes B finite and the surjection h gives $|B| \geq |\mathrm{Syn}(\Phi)|$; equality makes h bijective. Uniqueness over A follows from surjectivity of q. □

Attribution. This proposition is not new. Relation (9.55) is the classical *syntactic congruence*, $\mathrm{Syn}(\Phi)$ is the *syntactic monoid*, and the terminality and finite-minimality statements are the standard universal property of that construction, due to Myhill [61] and Nerode [66] and developed by Schützenberger [78]; textbook treatments are Eilenberg [29, Ch. B.II] and Pin [73, Ch. 2]. The proof above is included because it is short and because the quantifiers matter downstream, not because the result is original here. What is specific to this setting is only the reading of A as a protocol monoid and of Φ as a retained operational response, together with the explicit fence that the cardinality conclusion is finite-only and minimizes no latent realization.

Theorem 9.30 (Compact operational quotient). Suppose that A is a compact metrizable monoid with jointly continuous multiplication, Y is metrizable Hausdorff, and $\Phi : A \to Y$ is continuous. For any countable dense $D \subseteq A$ containing 1, the signature

$$S_D(a) = \big(\Phi(uav)\big)_{(u,v)\in D\times D} \tag{9.57}$$

has kernel $\sim_\Phi$. Its compact metrizable image is homeomorphic to $\mathrm{Syn}(\Phi)$, whose descended multiplication and response are continuous. Only the coding depends on D; multiplication on the image is induced by $S_D(ab)$, not coordinatewise multiplication. If a topological response-compatible triple has compact Hausdorff monoid B, continuous q and ψ, and quotient map $q : A \to B$, then the factor h in (9.56) is the unique continuous surjection with those identities. [ESTABLISHED]

Proof. Contextual equivalence implies equal signatures. Conversely, approximate arbitrary $u, v \in A$ by sequences in D; continuity of multiplication and Φ promotes equality on $D \times D$ to equality in all contexts. The signature therefore factors to a continuous bijection from the compact quotient onto the Hausdorff image $S_D(A)$, hence a homeomorphism. The image is compact metrizable. The product quotient map $\pi \times \pi$ is quotient because it is a continuous surjection from a compact space to a Hausdorff space, so $\bar{m}(\pi \times \pi) = \pi m$ proves continuity of quotient multiplication; $\bar{\Phi}\pi = \Phi$ proves continuity of the response. Finally, $hq = \pi$ and the quotient property of q prove continuity of the algebraically unique factor. □

Proposition 9.31 (Functorial operational reduction). Typed transport of nodes and states carries contexts, right override, and complete retained responses compatibly. It therefore descends through $\equiv_P$, and the assignments

$$\mathrm{Red} : \mathsf{FinTIP}^{\mathrm{iso}}_{R,O} \longrightarrow \mathsf{FinRIE}^{\mathrm{iso}}_{R,O}, \qquad \overline{U}_{\mathrm{pass}} : \mathsf{FinRIE}^{\mathrm{iso}}_{R,O} \longrightarrow \mathsf{FinObs}^{\mathrm{iso}}_{R,O}, \tag{9.58}$$

are functors, where $\overline{U}_{\mathrm{pass}}$ retains the response of the identity class. [ESTABLISHED]

Proof. A raw isomorphism induces a bijective homomorphism of context monoids and pushes each retained response through its boundary relabeling. Surjectivity of that context map transports the two universal quantifiers in (9.53); applying the inverse gives the converse. The induced quotient map preserves multiplication, identity, and responses, and respects identity arrows and composition. The identity class is preserved, which makes $\overline{U}_{\mathrm{pass}}$ well-typed. □

Proposition 9.32 (Independent-null operational collapse). Let S be the split chain on $\{R, E, O\}$, and let S^N add an independent isolated binary node N. Forgetting the N-coordinate induces an identity- and response-preserving quotient-monoid isomorphism $\mathrm{Red}(S^N) \cong \mathrm{Red}(S)$. Consequently, the empty context, $\mathrm{do}(N = 0)$, and $\mathrm{do}(N = 1)$ lie in one behavioral class. [ESTABLISHED]

Proof. Let $\pi : A_{S^N} \to A_S$ forget N. It is a surjective monoid homomorphism. Independence and normalization give $\Phi_{S^N}(i) = \Phi_S(\pi i)$ for every context; equivalently, the independent N-factor sums to one whether unassigned or replaced by either point mass. Therefore $i \equiv_{S^N} j$ exactly when $\pi i \equiv_S \pi j$, and π descends to the claimed isomorphism. The exact finite control checks both values of N against all $3^3 = 27$ partial contexts on $\{R, E, O\}$. □

Proposition 9.33 (Conditional intervention-recovery no-go). Let $U : \mathcal{E} \to \mathcal{B}$ be a declared forgetful functor. If two admitted enriched objects E_0, E_1 over one observational datum satisfy $U(E_0) = U(E_1)$ but are not type-preservingly intervention-isomorphic, then there is no reconstruction $R : \mathcal{B} \to \mathcal{E}$ with

$$RU(E) \cong E \quad \text{for every compatible } E. \tag{9.59}$$

A conventional right-inverse choosing one representative per observational fiber is not excluded. [ESTABLISHED]

Proof. Equation (9.59) would make both E_0 and E_1 isomorphic to $R(U(E_0)) = R(U(E_1))$, hence isomorphic to each other, contradicting the hypothesis. □

Theorem 9.34 (Finite typed-intervention nonidentifiability). In the declared binary-symmetric-channel subcategory, equality of the complete passive retained law does not determine the reduced protocol-indexed retained response experiment. Write

$$C_e(y \mid x) = \begin{cases} 1-e, & y = x, \\ e, & y \neq x, \end{cases} \tag{9.60}$$

and let $L(a, b)$ denote the chain $R \to E \to O$ with $R \sim \mathrm{Bernoulli}(1/2)$, kernel C_a on $R \to E$, and kernel C_b on $E \to O$. Set

$$L_1 = L(a = 1/4, b = 1/3), \qquad L_2 = L(a = 1/3, b = 1/4). \tag{9.61}$$

They have the same node roles, binary state cardinalities, edges, and mediator intervention target, and both have passive crossover $5/12$ and retained law. In every response vector below, coordinates are ordered lexicographically as $(R, O) = (0, 0), (0, 1), (1, 0), (1, 1)$:

$$\Phi_{L_1}(\varnothing) = \Phi_{L_2}(\varnothing) = (7/24, 5/24, 5/24, 7/24). \tag{9.62}$$

Each reduced protocol monoid has fifteen behavioral classes, yet the L_1 response

$$\Phi_{L_1}\big(\mathrm{do}(E = 0)\big) = (1/3, 1/6, 1/3, 1/6), \tag{9.63}$$

is absent from the complete L_2 reduced response image under all four independent typed binary boundary relabelings of R and O. Therefore $\mathrm{Red}(L_1)$ and $\mathrm{Red}(L_2)$ are nonisomorphic. [ESTABLISHED]

Proof. BSC composition gives $a + b - 2ab = 5/12$ for both parameter pairs. Direct two-sided reduction gives fifteen classes in each model. The full-support laws in the complete L_2 response image are exactly

$$(7/24, 5/24, 5/24, 7/24), \qquad (3/8, 1/8, 3/8, 1/8), \qquad (1/8, 3/8, 1/8, 3/8); \tag{9.64}$$

every other L_2 response has at least one zero entry. The four boundary flips send the displayed L_1 response only to itself or to $(1/6, 1/3, 1/6, 1/3)$. Both are strictly positive and neither equals one of the three displayed L_2 laws, so neither occurs in the complete L_2 response image. A reduced isomorphism would have to map the $\mathrm{do}(E = 0)$ class to a class with exactly such a relabeled response, a contradiction. The mediator-output total-variation contrasts, $1/3$ for L_1 and $1/2$ for L_2, are diagnostics only: an arbitrary protocol-class bijection need not preserve the named mediator pair. The complete response-image mismatch is the proof invariant. □

Corollary 9.35 (No universal two-sided finite recovery). The functor $\overline{U}_{\mathrm{pass}}$ is not essentially injective on the declared BSC subcategory. Consequently there is no functor $\mathcal{R} : \mathsf{FinObs}^{\mathrm{iso}}_{R,O} \to \mathsf{FinRIE}^{\mathrm{iso}}_{R,O}$ with $\mathcal{R}\overline{U}_{\mathrm{pass}} \cong \mathrm{id}_{\mathsf{FinRIE}^{\mathrm{iso}}_{R,O}}$ on that subcategory. This does not exclude the conventional representative-selecting condition $\overline{U}_{\mathrm{pass}}\mathcal{R} \cong \mathrm{id}_{\mathsf{FinObs}^{\mathrm{iso}}_{R,O}}$. [ESTABLISHED]

Proof. Apply Proposition 9.33 to the equal passive images and nonisomorphic reduced experiments in Theorem 9.34. □

The operational extensions below are bound to the released certificate with target digest `af08539e8868b09e5165943d91c488c6e06a00ac7a00b1d408ae22ddca6ee7e1`. They retain one ordered external boundary, target/type coloring where retained, and one protocol-independent response intertwiner. They do not make those choices canonical. [ESTABLISHED]

Definition 9.36 (Normalized soft-intervention monoid). For a finite DAG on finite state spaces, let J_v be a compact subset of the normalized stochastic-kernel polytope at node v, with its Euclidean, equivalently evaluation-subspace, topology. Adjoin an isolated symbol $\perp_v$ selecting the baseline kernel and set

$$\mathcal{A}_J = \prod_v \big(\{\perp_v\} \sqcup J_v\big), \qquad (a \star b)_v = \begin{cases} b_v, & b_v \neq \perp_v, \\ a_v, & b_v = \perp_v. \end{cases} \tag{9.65}$$

This is a compact metrizable topological monoid. Every replacement is a normalized kernel, and every coordinate of the retained response is a finite sum of finite products of continuous kernel evaluations. Hence its response is continuous and Theorem 9.30 supplies a compact operational quotient. [ESTABLISHED]

Theorem 9.37 (Marked soft BSC nonidentifiability). Retain the mediator-replacement face, the ordered roles R as input/parameter and O as output/observation, and one global typed response intertwiner for every protocol. For $0 < \epsilon < 1/2$, replace the mediator kernel in $L(a, b)$ by

$$K_t(E = 0 \mid R = r) = t_r, \qquad t \in [\epsilon, 1-\epsilon]^2, \tag{9.66}$$

and write $Q_b(t)$ for the retained (R, O)-law. Then

$$\mathrm{TV}\big(Q_b(t), Q_b(t')\big) = \frac{|1-2b|}{2}\big(|t_0 - t_0'| + |t_1 - t_1'|\big), \tag{9.67}$$

so the marked-face diameters for $L_1 = L(1/4, 1/3)$ and $L_2 = L(1/3, 1/4)$ are respectively

$$\frac{1-2\epsilon}{3} \qquad \text{and} \qquad \frac{1-2\epsilon}{2}. \tag{9.68}$$

The passive retained laws agree, but the marked soft experiments are nonisomorphic in this category. The obstruction is not confined to hard endpoints: for every $\epsilon < s_- < s_+ < 1-\epsilon$, the two parent-independent interior preparations have separation $|1-2b|(s_+ - s_-)$. Target erasure, exchange of R with O, and time reversal are not admitted. [ESTABLISHED]

Proof. Conditionally on $R = r$, one has $Q_b(t)(O = 0 \mid R = r) = b + (1 - 2b)t_r$. Since R is uniform, summing the absolute changes in both output atoms of each row gives (9.67). Its diameter gives (9.68). A typed boundary bijection preserves total variation and an admitted protocol map preserves the marked mediator face, so unequal diameters forbid an isomorphism. Substituting the two constant interior preparations gives the final formula. □

Definition 9.38 (Affine independently randomized interventions). For a finite hard reduced protocol monoid S, put $\mathrm{Rand}(S) = \Delta(S)$. Independent sequential selectors compose by convolution and the unobserved response is affine:

$$(p * q)(z) = \sum_{xy=z} p(x)q(y), \qquad \Phi_{\mathrm{R}}(p) = \sum_{x \in S} p(x)\phi(x). \tag{9.69}$$

The unit is the point mass at the hard identity. This semantics assumes independent selector draws; a correlated or shared-noise selector requires a joint selector object. [DEFINITION]

Theorem 9.39 (Randomized hard-intervention nonidentifiability). For each released fifteen-class BSC monoid, the deterministic complete two-sided contextual vectors are linearly independent. An exact fifteen-coordinate minor, with δ denoting passive crossover, has

$$\det M(b, \delta) = \frac{(2b - 1)^6 (2\delta - 1)^3}{32}, \tag{9.70}$$

which is nonzero for both models at $\delta = 5/12$. Therefore randomized contextual equivalence is equality. Any admitted response-compatible affine unital convolution-monoid isomorphism maps simplex vertices to vertices and restricts to a response-compatible hard-monoid isomorphism, already ruled out by Theorem 9.34. Equal passive retained laws therefore do not identify the independently randomized response experiment. Convexification does invalidate the old unmatched-response invariant: the response in (9.63) is the mixture with weights $5/6$ and $1/6$ of the two mediator-point responses of L_2. [ESTABLISHED]

Proof. Equality in every deterministic left and right context gives $\sum_x (p(x) - q(x))c_x = 0$, where c_x is the complete contextual vector. The nonzero determinant in (9.70) forces $p = q$. An affine bijection of finite simplices preserves their extreme points; convolution and unit preservation then make its vertex permutation a unital hard-monoid isomorphism. The single global response intertwiner makes that restriction response-compatible, giving the contradiction. □

Proposition 9.40 (Standard-Borel intervention semantics). Let a finite DAG have standard-Borel node spaces and standard-Borel replacement palettes. Assume that every baseline and replacement mechanism is a declared normalized pointwise kernel and that, for every measurable output set C, its evaluation $(\theta, x_{\mathrm{pa}(v)}) \mapsto K_v(\theta, x_{\mathrm{pa}(v)}; C)$ is jointly measurable. Finite topological-order recursion then produces a normalized joint law for every protocol and a Borel retained response when the retained-law space carries its evaluation sigma-algebra. The construction supplies an algebraic contextual quotient but does not by itself establish a standard-Borel quotient;

that requires an exhibited smooth classifier or stronger topological hypotheses. No conditional on a passive-law null set is inferred by this construction. [ESTABLISHED]

Proof. Iterate the declared kernels in topological order. The standard monotone class argument for parameterized kernel integration makes the integral of every bounded measurable cylinder function measurable in the protocol. Finite induction gives a normalized joint kernel, and retained-set evaluations remain measurable. These evaluations generate the declared law sigma-algebra. Standard Borelness alone does not make the uncountable contextual equivalence relation smooth. □

Corollary 9.41 (Compact-Feller operational quotient). If the node spaces are compact Polish, the palettes are compact metrizable, each $\perp_v$ is isolated, and all baseline and replacement kernels are jointly Feller in state and palette parameter, then the finite-coordinate right-override monoid in (9.65) has weakly continuous retained response. Its contextual quotient is compact metrizable with continuous multiplication and response. Finite-coordinate does not mean finite cardinality. [ESTABLISHED]

Proof. Successive reverse-topological integration sends every continuous test function to a continuous function of the remaining state and palette parameters. Thus protocol evaluation of every continuous retained test function is continuous, which is weak continuity. Apply Theorem 9.30. □

Theorem 9.42 (Circle heat-intervention nonidentifiability). Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, let m be normalized Haar law, and let $H_\tau e_n = e^{-n^2\tau} e_n$ be its heat kernels. For $0 < s < t$, the ordered chains

$$P_1 = m(dR)H_s(R, dE)H_t(E, dO), \qquad P_2 = m(dR)H_t(R, dE)H_s(E, dO) \tag{9.71}$$

have the same passive retained law $m(dR)H_{s+t}(R, dO)$. Give the marked mediator the palette $\mathcal{P}(\mathbb{T})$ in the weak topology and constant-parent replacements $K_\nu(dE \mid R = r) = \nu(dE)$. The palette is compact metrizable, and the family $(K_\nu)_{\nu\in\mathcal{P}(\mathbb{T})}$ is jointly Feller. Nevertheless,

$$H_t = H_s H_{t-s}, \qquad \text{but no Markov kernel } L \text{ satisfies } H_s = H_t L, \tag{9.72}$$

and the soft output sets obey

$$\{\nu H_t : \nu \in \mathcal{P}(\mathbb{T})\} \subsetneq \{\nu H_s : \nu \in \mathcal{P}(\mathbb{T})\}. \tag{9.73}$$

For every fixed x_0 and $0 < \rho < t - s$, the positive smooth law $\nu_\rho = H_\rho(x_0, \cdot)$ supplies a strictness witness. Hence passive retained-law equality does not identify this compact-Feller mediator experiment under maps that preserve the marked mediator, circle heat geometry, ordered R-to-O roles, compatible protocol map, and one global response intertwiner. Boundary exchange and time reversal are not admitted. [ESTABLISHED]

Proof. The heat-semigroup law proves passive retained-law equality and the forward garbling in (9.72). If $H_s = H_t L$, put $g = Le_1$. Markov positivity gives $|g| \le 1$, while the first Fourier coefficient of $H_t g = e^{-s} e_1$ would satisfy $|\widehat{g}(1)| = e^{t-s} > 1$, a contradiction. The forward garbling gives the inclusion in (9.73). If $H_{\rho+s}(x_0, \cdot) = \nu H_t$, comparison of first Fourier magnitudes gives $|\widehat{\nu}(1)| = e^{t-s-\rho} > 1$, again impossible. Since $\rho > 0$, the witness has a positive smooth heat density. □

Open Problem 9.43 (Category-independent typed-intervention recovery). The hard theorem and its extensions establish protocol-class minimality relative to fixed operational data, finite normalized marked-soft and independently randomized no-go results, declared standard-Borel response semantics, a compact-Feller positive tier, and the frozen circle heat-kernel no-go. They do not choose the protocol category, targets, or ordered boundary canonically; prove a target-erasing soft theorem; cover correlated, shared-noise, or adaptive selectors; identify point interventions from almost-sure observational conditionals; supply a noncompact quotient theorem; compare every latent dilation; or construct a minimal raw DAG/generative realization. They give no fixed-observation ELBO or VFE equality, no criterion for autonomous agency, and no continuum base-manifold, gauge/RG, or physical bridge. Those category-independent and interpretive recovery questions remain open. [OPEN]

Theorem 9.44 (Finite reference-relative I-projection). Let A be finite, let $p \in \Delta(A)$, let $T : A \to \mathbb{R}^d$, and prescribe $m \in \mathbb{R}^d$. A finite-KL feasible law with $\mathbb{E}_q T = m$ exists, and the minimizer of $D_{\mathrm{KL}}(q\|p)$ exists, exactly when

$$m \in \operatorname{conv} T(\operatorname{supp} p). \tag{9.74}$$

The minimizing law $q_{p,m}$ is unique relative to (p, T, m). Its support is the inverse image in $\operatorname{supp} p$ of the minimal face containing m, and every finite-KL feasible q satisfies

$$D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\|q_{p,m}) + D_{\mathrm{KL}}(q_{p,m}\|p). \tag{9.75}$$

After removing affine redundancies, the exponential multiplier is unique and the selected law is analytic on each fixed relative-interior face stratum. Equivariance requires coherent transport of the reference and the complete constraint diagram. [ESTABLISHED]

Proof. Condition (9.74) is exactly finite convex feasibility on $\operatorname{supp} p$. Restriction to the minimal face gives a compact feasible simplex; lower semicontinuity gives attainment and strict convexity gives uniqueness of the law. The finite exponential family calculation on that face gives (9.75) and the stated stratum regularity. □

Theorem 9.45 (Deterministic reference completion). Let $f : A \to B$ be deterministic, let $p \in \Delta(A)$, and let $r \ll f_{\#}p$. The unique KL-minimizing completion with $f_{\#}q = r$ is

$$L_f^p(r)(x) = \begin{cases} r(f(x))p(x)/(f_{\#}p)(f(x)), & (f_{\#}p)(f(x)) > 0, \\ 0, & (f_{\#}p)(f(x)) = 0. \end{cases} \tag{9.76}$$

For another deterministic map $g : B \to C$ and $s \ll (g \circ f)_{\#}p$, completion composes strictly when the intermediate stage uses the pushed reference:

$$L_f^p\Big(L_g^{f_{\#}p}(s)\Big) = L_{g\circ f}^p(s). \tag{9.77}$$

[ESTABLISHED]

Proof. The finite conditional KL chain rule separates the divergence into the fixed coarse divergence plus a nonnegative conditional term, which vanishes only when the conditional law within each positive-reference fiber equals that of p. This gives (9.76); substitution proves (9.77). □

Corollary 9.46 (Qualified retained VFE selection and descent). Under the complete retained-joint and retained-conditioning equivalence, positive evidence, common feasible retained constraints, and the support condition in Theorem 9.44, the unique retained optimizer and optimized VFE value descend. Deterministic posterior completion minimizes the full auxiliary VFE, but the presentation-specific auxiliary law need not descend. The retained envelope differential descends only for a positive C^1 parameterized family with a common locally fixed feasible set and support stratum and a unique C^1 optimizer. [ESTABLISHED]

Theorem 9.47 (Retained Fisher radical and quotient). For a smooth retained-law map $\rho : \Theta \to N$, a positive-semidefinite target tensor g, and $h = \rho^* g$,

$$\operatorname{rad} h_\theta = (d\rho_\theta)^{-1}(\operatorname{rad} g_{\rho(\theta)}). \tag{9.78}$$

Hence $\operatorname{rad} h_\theta = \ker d\rho_\theta$ exactly when $\operatorname{im} d\rho_\theta \cap \operatorname{rad} g_{\rho(\theta)} = \{0\}$. On a stratum where the radical has locally constant rank, $T\Theta/\operatorname{rad} h$ is a smooth positive-definite quotient bundle. A global quotient manifold additionally requires a simple Hausdorff null-leaf quotient. Under the preceding transversality equality $\operatorname{rad} h = \ker d\rho$ and a regular connected-leaf quotient, the specific pullback h is basic because ρ factors through that quotient; basicness remains extra for arbitrary tensors and declared block projectors. [ESTABLISHED]

Proof. For every $v \in T_\theta\Theta$, positivity and Cauchy–Schwarz for the semidefinite form give $h_\theta(v, \cdot) = 0$ exactly when $d\rho_\theta v \in \operatorname{rad} g$, proving (9.78). The remaining statements are the constant-rank quotient construction and the separate regular-leaf-space hypothesis. □

Proposition 9.48 (Declared-block quotient boundary). Assume the transversality case $K = \ker d\rho = \operatorname{rad} h$ on a constant-rank stratum and declare a smooth splitting $T\Theta = \bigoplus_a B_a$. The

following are equivalent: the quotient images of the B_a form a direct sum, their retained images form a direct sum, $K = \bigoplus_a (K \cap B_a)$, and every quotient tangent has a unique block decomposition. Additive Fisher energy further requires pairwise Fisher orthogonality. With $D_a = B_a + K$, local block-distribution projectability is

$$[\Gamma(K), \Gamma(D_a)] \subseteq \Gamma(D_a), \tag{9.79}$$

and invariance across disconnected components of a retained fiber is an additional global requirement. Smooth pointwise directness and orthogonality alone do not imply descent. [ESTABLISHED]

Proposition 9.49 (Law and Fisher geometry do not canonically agentize blocks). At the uniform seven-outcome law, no rule depending only on the unlabeled law and its Fisher metric and natural under every outcome relabeling can select an unordered decomposition of the six-dimensional tangent space into three two-dimensional blocks. Declared typing or other symmetry-breaking data can supply blocks, but the law and Fisher tensor alone do not. [ESTABLISHED]

Proof. Such a rule would induce a homomorphism $S_7 \to S_3$. Its restriction to the perfect group A_7 is trivial. The sum-zero six-dimensional A_7-representation is irreducible: an equivariant endomorphism extends by zero on the invariant $\mathbb{R}\mathbf{1}$ line, and the resulting permutation commutant calculation makes it scalar. No invariant rank-two summand can therefore exist. □

Corollary 9.50 (Promoted parity has one hidden retained direction). For the seven-parameter family $(\theta, \kappa) \mapsto Q_{\theta,\kappa}$ in (9.48), the full-joint derivative has rank seven everywhere on $(0,1)^6 \times (-1,1)$. The singleton-retained map has rank six and

$$\ker d(mQ) = \operatorname{span}\{\partial_\kappa\}. \tag{9.80}$$

Thus κ is identifiable from the full law and invisible to singleton retention; this does not agentize it. [ESTABLISHED]

Proof. The singleton means are exactly θ_i, so a zero full-law derivative first has zero θ-components. The remaining derivative is a scalar multiple of $\partial_\kappa Q = \chi \prod_i \theta_i(1-\theta_i)$, which is nonzero in the open domain; hence the full derivative is injective. The same singleton identity gives rank six and (9.80). □

Local Markov naturality therefore selects the product law uniquely, while under the declared full naturality and canonicity requirements, selecting nonproduct dependence requires either restricting the admitted morphism class or adding declared relational, constraint, or reference data. A positive reference supplies a unique relative selector only under the feasibility and support hypotheses above, and the retained Fisher quotient identifies only directions visible at the declared retained scope. The absolute split-selector target depends only on Theorem 9.21; none of the positive replacements is a premise. Arbitrary-family coarse-preserving nonproduct selection, autonomous agency, continuum and renormalization limits, physical geometry, units, and time remain open. The finite hard BSC recovery target is refuted by Theorem 9.34. Relative to fixed target coloring where

retained, ordered boundary roles, and admitted morphisms, its operational quotient is universal and minimal as stated in Proposition 9.29; marked soft and independently randomized extensions remain nonidentifiable by Theorems 9.37 and 9.39, and Theorem 9.42 supplies a compact-Feller continuous witness. Category-independent intervention semantics remains open as stated in Open Problem 9.43.

9.6 Natural-gradient selection of an oriented orbit

The Fisher natural gradient is characterized by

$$\mathsf{G}_i^F\left(\operatorname{grad}^F \mathcal{F}_i, Z\right) = d\mathcal{F}_i[Z] \quad \text{for every } Z \in T\mathcal{Q}_i. \tag{9.81}$$

At a noncritical configuration Q, define the positive descent ray

$$\mathcal{R}^-_{\mathcal{F}_i}(Q) = \left\{-a \operatorname{grad}^F \mathcal{F}_i(Q) : a > 0\right\}. \tag{9.82}$$

A natural-gradient history for the declared objective is an oriented unparameterized history whose regular representatives satisfy

$$\dot{Q}_i(\lambda) = -a_i(\lambda) \operatorname{grad}^F \mathcal{F}_i(Q_i(\lambda)), \qquad a_i(\lambda) > 0. \tag{9.83}$$

The objective alone does not select this ray: the metric, or an explicitly declared mobility replacing it, is load-bearing structure. When $\mathcal{F}_i$ is the pullback specified in Hypothesis 9.9, this is an exact VFE history; otherwise it is a history for a generic declared objective and no exact-VFE identification is asserted. [DEFINITION]

Proposition 9.51 (Scalar mobility preserves the oriented orbit). On a noncritical domain, multiplication of the natural-gradient vector field by any smooth positive scalar mobility changes only its parameterization. A positive-definite anisotropic mobility can change the unparameterized orbit. [ESTABLISHED]

Proof. If $X = -\operatorname{grad}^F \mathcal{F}_i$ and $X_a = aX$ with $a > 0$, then the integral curves have the same tangent line and orientation. Along an X-integral curve, the change of parameter defined by a positive integral of a^{-1} converts one differential equation into the other. For the second claim, use the Euclidean plane with

$$\mathcal{F}(x, y) = \frac{x^2 + y^2}{2}, \qquad M = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}. \tag{9.84}$$

Ordinary gradient descent from (x_0, y_0) satisfies $y = (y_0/x_0)x$, whereas $(\dot{x}, \dot{y})^\top = -M\nabla\mathcal{F}$ satisfies

$$y = y_0 \left(\frac{x}{x_0}\right)^2. \tag{9.85}$$

For generic nonzero initial data these are different images. □

A VFE natural-gradient history is not generally a Fisher geodesic. On the same Euclidean Fisher plane, choose

$$\mathcal{F}(x,y) = \frac{x^2 + 2y^2}{2}. \tag{9.86}$$

Its natural-gradient flow obeys $\dot{x} = -x$ and $\dot{y} = -2y$, hence follows the curved trajectory in (9.85). Euclidean geodesics are straight lines, so gradient selection and geodesic selection are distinct even in this elementary Fisher geometry. [ESTABLISHED]

Proposition 9.52 (A semidefinite tensor cannot be silently inverted). If the proposed Fisher tensor on configuration space is semidefinite, the natural-gradient equation can fail to have a solution or can have nonunique solutions. [ESTABLISHED]

Proof. On $\mathbb{R}^2$ let $\mathsf{G} = dx^2$. For $\mathcal{F} = y$, the equation $\mathsf{G}(\operatorname{grad}\mathcal{F}, \cdot) = dy$ has no solution because the left side annihilates ∂_y. For $\mathcal{F} = x$, every vector $\partial_x + b\partial_y$ solves the equation. A justified metric quotient or a separately declared regularization is therefore required before an inverse or pseudoinverse is used. □

9.7 Fisher duration on a selected history

Definition 9.53 (Fisher length and information clock). For a C^1 representative $Q : I \to \mathcal{Q}_i$, define its Fisher speed, length, and cumulative clock from a chosen origin λ_0 by

$$\nu_F(\lambda) = \sqrt{\mathsf{G}_i^F(\dot{Q}(\lambda), \dot{Q}(\lambda))}, \tag{9.87}$$

$$L_F(Q; [a,b]) = \int_a^b \nu_F(\lambda) d\lambda, \tag{9.88}$$

$$\tau_{Q,\lambda_0}(\lambda) = \int_{\lambda_0}^{\lambda} \nu_F(u) du. \tag{9.89}$$

The origin is relational data attached to the history. The auxiliary parameter λ is not part of the resulting arc length. [DEFINITION]

Theorem 9.54 (Reparameterization invariance and unit-speed form). Let $\widetilde{Q} = Q \circ \phi$, where $\phi : \widetilde{I} \to I$ is an orientation-preserving C^1 diffeomorphism. Then corresponding subarcs have the same Fisher length and their cumulative clocks obey

$$\tau_{\widetilde{Q},\widetilde{\lambda}_0}(\widetilde{\lambda}) = \tau_{Q,\lambda_0}(\phi(\widetilde{\lambda})), \qquad \lambda_0 = \phi(\widetilde{\lambda}_0). \tag{9.90}$$

If $\nu_F > 0$, the cumulative clock is strictly increasing and gives the unit-speed representative of the oriented history. If Q satisfies (9.83), then on its noncritical part

$$\frac{d\tau}{d\lambda} = a_i \left\| \operatorname{grad}^F \mathcal{F}_i \right\|_F , \tag{9.91}$$

$$\frac{dQ}{d\tau} = -\frac{\operatorname{grad}^F \mathcal{F}_i}{\left\| \operatorname{grad}^F \mathcal{F}_i \right\|_F}, \tag{9.92}$$

$$\frac{d\mathcal{F}_i}{d\tau} = -\left\| \operatorname{grad}^F \mathcal{F}_i \right\|_F . \tag{9.93}$$

[ESTABLISHED]

Proof. The chain rule gives $\dot{\widetilde{Q}} = (\dot{Q} \circ \phi)\phi'$ and hence $\widetilde{\nu}_F = (\nu_F \circ \phi)\phi'$ because $\phi' > 0$. The substitution $u = \phi(\widetilde{\lambda})$ proves both length invariance and (9.90). Positive speed makes τ locally and globally invertible on the interval, and division by $d\tau/d\lambda$ gives unit speed. Substitution of (9.83) into (9.87) gives (9.91). The remaining identities follow from (9.81). □

This is the standard Riemannian arc-length construction applied to the selected statistical configuration manifold [3, 70]. Fisher geometry supplies duration; the VFE descent ray supplies orientation. A null segment in a semimetric or a critical stationary segment has $\nu_F = 0$, so the cumulative clock stalls and cannot be used as a coordinate there. [ESTABLISHED]

Proposition 9.55 (Endpoints do not determine an update duration). Let A and B lie in one path component of a regular Fisher manifold. Those endpoints alone determine neither an update curve nor its realized duration. For every actual update Q from A to B,

$$d_F(A, B) = \inf_{\substack{\Gamma(a)=A \\ \Gamma(b)=B}} L_F(\Gamma; [a, b]) \le L_F(Q; [a, b]), \tag{9.94}$$

and the inequality can be strict. [ESTABLISHED]

Proof. The first equality is the definition of Riemannian distance as the infimum of curve lengths. Since the realized update Q is one admissible curve, the inequality follows. In the Euclidean Fisher geometry of a two-parameter normal location family, a curved path between two points is longer than the straight geodesic joining them, which proves possible strictness. □

The information clock is also distinct from the amount of auxiliary algorithmic parameter used to approach a limit. In the unit-variance normal location family, the path $\mu(t) = e^{-t}$ has Fisher length

$$\int_0^\infty |\dot{\mu}(t)| dt = \int_0^\infty e^{-t} dt = 1, \tag{9.95}$$

although t ranges over an unbounded interval. No physical-time interpretation follows from the length construction. [NOT-CLAIMED]

9.8 Pointwise, design, continuum, and quotient clocks

At one fixed $c \in \mathcal{C}_i$, choose positive belief and model weights $w_b(c)$ and $w_m(c)$. The pointwise squared speed of a section history is

$$\nu_{i,c}^2 = w_b(c) g_b^F(\partial_\lambda q_i(c), \partial_\lambda q_i(c)) + w_m(c) g_m^F(\partial_\lambda q_i^m(c), \partial_\lambda q_i^m(c)). \tag{9.96}$$

Integration of $\nu_{i,c}$ defines a pointwise clock. This uses only vertical evaluation velocities from (9.4); it does not integrate over the base. [DEFINITION]

For a finite design $D = \{c_a\}_{a=1}^M$, declare positive design weights ρ_a and define

$$\nu_{i,D}^2 = \sum_{a=1}^{M} \rho_a \left[w_b(c_a) g_b^F(\partial_\lambda q_i(c_a), \partial_\lambda q_i(c_a)) + w_m(c_a) g_m^F(\partial_\lambda q_i^m(c_a), \partial_\lambda q_i^m(c_a)) \right]. \tag{9.97}$$

This is the product Fisher speed for the declared design and weights. It is not automatically a positive-definite metric on the full section space: any nonzero section variation vanishing at all c_a lies in its radical. On a smooth positive-dimensional base, a bump variation supported away from D is an explicit witness whenever the vertical fiber has a nonzero local direction there. [ESTABLISHED]

A continuum clock requires a finite positive base measure μ_i, measurable positive channel weights, square-integrable evaluation velocities, and a declared treatment of gauge redundancy. Under those hypotheses its proposed squared speed is

$$\nu_{i,\mu}^2 = \int_{\mathcal{C}_i} \left[w_b(c) g_b^F(\partial_\lambda q_i(c), \partial_\lambda q_i(c)) + w_m(c) g_m^F(\partial_\lambda q_i^m(c), \partial_\lambda q_i^m(c)) \right] d\mu_i(c). \tag{9.98}$$

The contextual base, the common principal bundle, and the VFE functional do not by themselves select μ_i, the relative channel weights, or a strong manifold topology. Without those declarations, (9.98) is not an available clock and only the pointwise or finite-design constructions are defined. [ESTABLISHED]

The single-integral form in (9.98) requires that the actual configuration Fisher bilinear form equal the displayed local integral. Contextual dependence of a law alone does not settle this identity; it must be derived from that law's score covariance. If independent section directions with disjoint contextual supports have nonzero Fisher pairing, no local single-integral metric on those directions represents that pairing. A general correlated law may contribute a nonlocal operator, and even an ordinary double-integral kernel representation requires its own regularity. Such contributions must be retained in the exact metric; omitting them defines a declared local approximation. [HYPOTHESIS]

9.8.1 Strong Hilbert quotients and attained duration

The finite-dimensional sufficiency statement extends to a precise Hilbert category. Closed infinitesimal orbit ranges for smooth proper Banach-Lie actions, and smooth Hilbert quotients for free proper actions, are established results of Jotz and Neeb [45, Theorem 2.1 and Corollary 2.5]. The proof

below establishes the isometric strong-metric case directly and constructs the orthogonal projection that the duration formula requires. [ESTABLISHED]

Theorem 9.56 (Strong Hilbert quotient duration). Let $\mathcal{M}$ be a nonempty Hausdorff C^∞ real Hilbert manifold with a C^∞ strong Riemannian metric g. Here strong means that the metric norm gives the declared tangent topology and, in each Hilbert chart, its positive metric operator is boundedly invertible and varies smoothly in operator norm. Let a Hausdorff C^∞ real Banach-Lie group $\mathcal{G}$, with Lie algebra $\mathfrak{g}$, act jointly smoothly, freely, and isometrically on $\mathcal{M}$. Assume that the action map

$$\mathfrak{a} : \mathcal{G} \times \mathcal{M} \longrightarrow \mathcal{M} \times \mathcal{M}, \qquad \mathfrak{a}(a, Q) = (a \cdot Q, Q), \tag{9.99}$$

is closed with compact fibers. Then the infinitesimal action

$$A_Q \xi = \left.\frac{d}{dt}\right|_{t=0} \exp(t\xi) \cdot Q, \tag{9.100}$$

is injective with closed range V_Q. The spaces V_Q and $H_Q = V_Q^{\perp_{\mathrm{g}}}$ are smooth complementary subbundles. The orbit space $\mathcal{M}/\mathcal{G}$ is a Hausdorff smooth Hilbert manifold, componentwise, and its projection π is a smooth principal bundle and a Riemannian submersion for a strong quotient metric $\overline{\mathrm{g}}$. For every $Q \in \mathcal{M}$ and $v \in T_Q\mathcal{M}$,

$$\|d\pi_Q v\|^2_{\overline{\mathrm{g}}} = \min_{\xi \in \mathfrak{g}} \mathrm{g}_Q(v - A_Q\xi, v - A_Q\xi). \tag{9.101}$$

The minimizing ξ is unique and smooth in (Q, v). Consequently the quotient speed and its integrated duration are independent of a C^1 history-dependent choice of representative. No metric completeness or finite-dimensionality of $\mathcal{G}$ is assumed. [ESTABLISHED]

Proof. First, closedness with compact fibers implies compact inverse images of compact sets. Indeed, an open cover of such an inverse image has a finite subcover on each fiber. If O is the union of the finitely many sets covering one fiber, closedness of $\mathfrak{a}$ gives a neighborhood of the target point whose entire inverse image lies in O. A finite collection of these neighborhoods covers the compact target set and supplies the required finite subcover. Compact-preimage properness gives the same hypothesis here: fibers are compact, and a convergent sequence of images of a closed set lifts into the compact inverse image of that sequence and its limit. First countability supplies a convergent subsequence of lifts, so the image is sequentially closed and hence closed.

The length distance of a strong metric induces the manifold topology on each connected component. To verify the needed local fact, choose a Hilbert chart at Q. Strongness and operator-norm continuity give constants $0 < c < C < \infty$ and a coordinate ball on which

$$c\|w\|^2 \le \mathrm{g}_y(w, w) \le C\|w\|^2. \tag{9.102}$$

Straight segments give the upper distance bound. A path from the chart center that exits a smaller ball of radius r has length at least $\sqrt{c}r$; thus sufficiently small length distance forces the endpoint into every prescribed neighborhood. This argument uses no minimizing geodesic.

For $c_\xi(t) = \exp(t\xi) \cdot Q$, the group law and isometry give

$$\dot{c}_\xi(t) = T_Q \Phi_{\exp(t\xi)} A_Q \xi, \qquad \|\dot{c}_\xi(t)\|_{\mathfrak{g}} = \|A_Q \xi\|_{\mathfrak{g}_Q}, \qquad d_{\mathfrak{g}}(Q, \exp(\xi) \cdot Q) \le \|A_Q \xi\|_{\mathfrak{g}_Q}, \tag{9.103}$$

where $\Phi_a(Q) = a \cdot Q$. If $A_Q \xi = 0$, the curve is constant. Freeness gives $\exp(t\xi) = e$ for every t, hence $\xi = 0$.

Suppose that unit vectors ξ_n in the original Banach norm satisfied $\|A_Q \xi_n\|_{\mathfrak{g}_Q} \to 0$. Choose a fixed $\delta > 0$ whose closed Lie-algebra ball lies in an exponential chart, and set $a_n = \exp(\delta \xi_n)$. Equation (9.103) and the strong-metric topology imply $a_n \cdot Q \to Q$. Properness over the compact convergent sequence $\{(a_n \cdot Q, Q)\} \cup \{(Q, Q)\}$ gives a subsequence $(a_n, Q) \to (a, Q)$, where $a \cdot Q = Q$. Freeness gives $a = e$. The continuous local logarithm would then give $\delta \xi_n \to 0$, a contradiction. Therefore, for some $c_Q > 0$,

$$\|A_Q \xi\|_{\mathfrak{g}_Q} \ge c_Q \|\xi\|_{\mathfrak{g}} \quad \text{for every } \xi \in \mathfrak{g}. \tag{9.104}$$

This is vacuous if $\mathfrak{g} = \{0\}$. Otherwise it also shows closedness: a convergent sequence $A_Q \xi_n$ makes ξ_n Cauchy, and completeness puts its limit in the range.

Fix Q_0. The inner product $\langle \xi, \eta \rangle_{\mathfrak{g},H} = \mathrm{g}_{Q_0}(A_{Q_0}\xi, A_{Q_0}\eta)$ has a complete norm equivalent to the original Banach norm by (9.104). Thus $\mathfrak{g}$ is Hilbertizable. Only with this fixed equivalent Hilbert norm do we take adjoints. In a tangent chart write $\mathrm{g}_y(v, w) = \langle v, M_y w \rangle$, and define

$$B_y = A_y^* M_y A_y, \qquad P_y = A_y B_y^{-1} A_y^* M_y, \qquad \xi_*(y, v) = B_y^{-1} A_y^* M_y v. \tag{9.105}$$

The action makes A_y smooth in operator norm. Near each fixed point, (9.104) and metric continuity give $B_y \ge \kappa I$ for one $\kappa > 0$. Such a self-adjoint operator has closed range, since $\|B_y \xi\| \ge \kappa \|\xi\|$, and dense range, since its orthogonal complement is $\ker B_y = \{0\}$. It is therefore invertible. The local Neumann series proves smoothness of inversion.

Direct multiplication gives $P_y^2 = P_y$, $\operatorname{ran} P_y = V_y$, and $A_y^* M_y (v - P_y v) = 0$. Thus P_y is the metric-orthogonal vertical projection. Its ranges and kernels are smooth subbundles: near a fixed point x, the smooth operator

$$S_y = P_y P_x + (I - P_y)(I - P_x), \qquad P_y S_y = S_y P_x, \tag{9.106}$$

equals I at x and remains invertible nearby, identifying the fixed splitting at x with that at y. Orthogonality now gives

$$\|v - A_y \xi\|_{\mathrm{g}_y}^2 = \|(I - P_y)v\|_{\mathrm{g}_y}^2 + \|A_y(\xi - \xi_*(y, v))\|_{\mathrm{g}_y}^2. \tag{9.107}$$

This proves attainment and uniqueness, before any quotient chart is used.

We next construct those charts. Properness and freeness imply that for each identity neighborhood $W \subseteq \mathcal{G}$, there is a neighborhood N of Q such that

$$aN \cap N \neq \varnothing \quad \Longrightarrow \quad a \in W. \tag{9.108}$$

If not, choose $y_n, z_n \to Q$ and $a_n \notin W$ with $a_n y_n = z_n$. Properness applied to the compact set of pairs (z_n, y_n) and its limit gives a convergent subsequence of (a_n, y_n), whose limiting group element fixes Q, hence is e. This contradicts $a_n \notin W$.

Choose a local submanifold S_0 through Q, obtained in a chart from the closed complemented space H_Q. For $F : \mathcal{G} \times S_0 \to \mathcal{M}$, $F(a, s) = a \cdot s$, its derivative at (e, Q) is $(\xi, h) \mapsto A_Q \xi + h$, a bounded isomorphism by the orthogonal splitting and (9.104). The Banach inverse function theorem states that a smooth map with a boundedly invertible derivative is a diffeomorphism between neighborhoods of that point and its image. It therefore gives a product $W \times S_1$ on which F is a diffeomorphism onto an open set. Shrink to $S \subseteq S_1 \cap N$, using (9.108) for this W. If $as = t$ for $s, t \in S$, then $a \in W$; injectivity of the local product gives $a = e$ and $s = t$. Consequently

$$\mathcal{G} \times S \longrightarrow \mathcal{G} \cdot S, \qquad (a, s) \longmapsto a \cdot s, \tag{9.109}$$

is bijective. Its target is open, being the union of translates of $F(W \times S)$, and it is a local diffeomorphism at every point by translation of the local product. Thus it is a diffeomorphism. These are principal trivializations, with smooth transition maps given by the slice components of their inverses.

The orbit relation is closed by (9.99), and π is open because saturations of open sets are open. A product neighborhood disjoint from the orbit relation separates two distinct orbits in the quotient, proving Hausdorffness. There is also a compatible quotient metric for topology: truncate the strong length distance at 1, and give points in different connected components distance 1, to obtain an invariant metric d_0 inducing the topology of $\mathcal{M}$. Then

$$d_B([Q], [R]) = \inf_{a \in \mathcal{G}} d_0(Q, a \cdot R) \tag{9.110}$$

is positive for distinct orbits because every orbit is closed. Symmetry follows by inversion, and the triangle inequality follows from isometry and multiplication in $\mathcal{G}$. Its balls give the quotient topology: their inverse images are open, and an open saturated neighborhood of Q contains a d_0-ball at Q, whose quotient ball remains in that neighborhood. Thus standard definitions requiring paracompact Hilbert manifolds are satisfied, since metric spaces are paracompact. This topological metric is not asserted to be the untruncated Riemannian quotient distance.

The action transports V_Q by $T\Phi_a A_Q \xi = A_{aQ} \operatorname{Ad}_a \xi$; isometry transports H_Q as well. Define

$$\overline{\mathrm{g}}_{[Q]}(d\pi_Q v, d\pi_Q w) = \mathrm{g}_Q((I - P_Q)v, (I - P_Q)w). \tag{9.111}$$

Adding vertical vectors to the lifts leaves this expression unchanged, and equivariance of the orthogonal projection proves independence of Q within its orbit. In each slice it is smooth. The map

$(I - P_s)|_{T_sS} : T_sS \to H_s$ is a bounded isomorphism, since the local product gives $T_s\mathcal{M} = V_s \oplus T_sS$. Hence the pulled-back inner product is strong. Equations (9.107) and (9.111) prove (9.101).

Finally, for $\widetilde{Q}(t) = a(t) \cdot Q(t)$,

$$\dot{\widetilde{Q}}(t) = T\Phi_{a(t)}\dot{Q}(t) + A_{\widetilde{Q}(t)}\eta(t), \tag{9.112}$$

where $\eta(t)$ is the right logarithmic derivative of $a(t)$. The second term is vertical and the first preserves horizontal norm. This proves representative independence of speed and therefore duration. The unique coefficient $\xi_*(Q, v)$ also defines a smooth principal connection: it sends $A_Q\xi$ to ξ and transforms by Ad_a, as follows from uniqueness and equivariance of P. A global gauge section is unnecessary. □

For the configuration notation of this chapter, apply Theorem 9.56 with $(\mathcal{M}, \mathsf{g}) = (\mathcal{Q}_i, \mathsf{G}_i^F)$ only after its Hilbert topology, strong metric, Banach-Lie gauge group, and smooth proper isometric action have been established. Passive re-trivialization changes coordinates; it adds no physical segment to a history. The theorem supplies no singular quotient when the action is not free or proper. [ESTABLISHED]

9.8.2 A proper action with a weak Fisher quotient of zero speed

Topological closedness and complementedness do not suffice for a weak metric. The following counterexample establishes failure even when the smooth quotient and a global topological splitting already exist. [ESTABLISHED]

Proposition 9.57 (Proper weak-Fisher quotient obstruction). Let $\mathcal{C} = \mathbb{R}/\mathbb{Z}$ carry normalized arclength dc, and set

$$\mathcal{M} = H^1(\mathcal{C}; \mathbb{R}), \qquad \|f\|_{H^1}^2 = \int_{\mathcal{C}} (|f|^2 + |f'|^2)dc, \qquad \mathcal{K} = \{h \in \mathcal{M} : h(0) = 0\}. \tag{9.113}$$

Use the continuous absolutely continuous periodic representative of each Sobolev class. The additive Hilbert Lie group $\mathcal{K}$, with the inherited H^1 topology, acts by $h \cdot f = f + h$. Equip $\mathcal{M}$ with

$$\mathsf{g}_f(v, w) = \int_{\mathcal{C}} v(c)w(c)dc. \tag{9.114}$$

The action is smooth, free, proper, and isometric. Its smooth quotient is $\mathbb{R}$, via $f \mapsto f(0)$, and $\mathcal{K}$ is closed and complemented in H^1. Nevertheless

$$\mathcal{K}^{\perp_{\mathsf{g}}} = \{0\}, \qquad \inf_{k \in \mathcal{K}} \|v - k\|_{\mathsf{g}}^2 = 0 \quad \text{for every } v \in H^1. \tag{9.115}$$

The infimum is attained exactly when $v(0) = 0$. The form g is the exact Fisher metric of the identifiable normalized joint observation family

$$P_f(dc, dx) = \varphi(x - f(c))dcdx, \qquad \varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}, \tag{9.116}$$

which is differentiable in quadratic mean in the H^1 parameter topology. Thus every nonzero quotient tangent has zero induced speed and no minimizing lift, despite the smooth proper quotient. [ESTABLISHED]

Proof. Absolute continuity and Cauchy–Schwarz give

$$\|f\|_\infty \le \|f\|_2 + \|f'\|_2 \le \sqrt{2}\|f\|_{H^1}. \tag{9.117}$$

For the first inequality, bound $|f(c)|$ by $|f(t)| + \int_{\mathcal{C}} |f'|$ and average over t. The trace $E(f) = f(0)$ is bounded, and

$$f \longmapsto (f - f(0)\mathbf{1}, f(0)) \quad \text{identifies} \quad H^1 \cong \mathcal{K} \oplus \mathbb{R}, \tag{9.118}$$

by bounded inverse maps. Completeness in this norm can be checked directly: an H^1-Cauchy sequence converges uniformly by (9.117), its derivatives converge in L^2, and passage to the integral identity $f(c) - f(0) = \int_0^c f'(t)dt$ supplies an absolutely continuous periodic limit with that derivative. This also verifies the Hilbert structure used for the group and manifold.

The action is translation in the $\mathcal{K}$ factor of (9.118). It is smooth and free, and its quotient is the displayed line. The action map $(h, f) \mapsto (f + h, f)$ is a homeomorphism onto the closed relation $\{(u, f) : E(u - f) = 0\}$, with inverse $(u, f) \mapsto (u - f, f)$. It is therefore closed with singleton or empty fibers; inverse images of compact sets are compact as well. This proves properness in the convention of (9.99). Every translation has identity derivative and preserves g.

Choose $\beta(t) = \exp(1 - (1 - t^2)^{-1})$ for $|t| < 1$, and $\beta(t) = 0$ otherwise. For $n \ge 3$, let $\eta_n(c) = \beta(nc)$ in the signed circle coordinate $[-1/2, 1/2]$, extended periodically. These are smooth, $\eta_n(0) = 1$, and with $a_\beta = \int_{-1}^1 \beta^2 > 0$ and $b_\beta = \int_{-1}^1 (\beta')^2 > 0$,

$$\|\eta_n\|_2^2 = \frac{a_\beta}{n}, \qquad \|\eta_n'\|_2^2 = n b_\beta, \qquad k_n = v - v(0)\eta_n \in \mathcal{K}. \tag{9.119}$$

Thus $\|v - k_n\|_2 = |v(0)|\sqrt{a_\beta/n} \to 0$. If v is orthogonal to $\mathcal{K}$, then $0 = \lim_n \mathrm{g}(v, k_n) = \|v\|_2^2$, so $v = 0$. The same approximation proves the zero infimum in (9.115). Attainment would imply $v - k = 0$, because the L^2 form is positive definite on H^1; this is equivalent to $v(0) = 0$.

The metric is weak in the declared topology. Its musical map is injective, but cannot represent the continuous trace covector: if $E = \mathrm{g}(v, \cdot)$, then

$$1 = E(\eta_n) = \int_{\mathcal{C}} v\eta_n dc, \qquad \left| \int_{\mathcal{C}} v\eta_n dc \right| \le \|v\|_2 \sqrt{a_\beta/n} \longrightarrow 0, \tag{9.120}$$

a contradiction. The derivative norm in (9.119) diverges, so none of this asserts H^1 convergence of the bumps or contradicts properness in that topology.

It remains to verify that (9.114) is an actual statistical tensor. Equation (9.116) normalizes by integrating each conditional Gaussian and then the unit circle measure. Equality of two joint laws gives equality of their conditional Gaussian means almost everywhere in c; continuity gives $f = \widetilde{f}$ everywhere, proving identifiability. Write $\psi_\mu(x) = \sqrt{\varphi(x - \mu)}$ and $S(f)(c, x) = \psi_{f(c)}(x)$.

The translation-invariant constant $C_2 = \|\partial_\mu^2 \psi_\mu\|_{L^2(dx)}$ is finite, since the derivative is a quadratic polynomial times a Gaussian square root. Taylor's formula in $L^2(dx)$, followed by integration over c, gives

$$\begin{aligned} DS(f)[v](c,x) &= \tfrac{1}{2} v(c)(x - f(c))S(f)(c,x), \\ \|S(f+h) - S(f) - DS(f)[h]\|_2 &\le \tfrac{1}{2} C_2 \|h\|_4^2 \le \tfrac{1}{2} C_2 \|h\|_\infty \|h\|_2 = O(\|h\|_{H^1}^2). \end{aligned} \tag{9.121}$$

This proves differentiability in quadratic mean with square-integrable directional score $v(c)(x - f(c))$. The conditional mean and variance of $X - f(C)$ are zero and one, respectively, so

$$\mathbb{E}_f[\dot{\ell}_f[v]] = 0, \qquad \mathbb{E}_f[\dot{\ell}_f[v]\dot{\ell}_f[w]] = \int_{\mathcal{C}} v(c)w(c)dc. \tag{9.122}$$

This is the Fisher metric of the observed pair (C, X), with C retained. For each fixed $h \in \mathcal{K}$, the bimeasurable observation change $T_h(c,x) = (c, x + h(c))$ pushes P_f to P_{f+h}. Orbit equivalence is therefore implemented by a declared coordinate action; it does not assert equality of these laws in fixed recorded coordinates. □

Proposition 9.58 (The trace quotient is not a fixed observation channel). No parameter-independent Markov kernel $K : \mathcal{C} \times \mathbb{R} \rightsquigarrow \mathbb{R}$ satisfies

$$P_f K = \mathcal{N}(f(0), 1) \quad \text{for every } f \in H^1(\mathcal{C}; \mathbb{R}). \tag{9.123}$$

Thus attaching this normalized output family to the smooth parameter quotient does not give a statistical coarse channel or its quotient Fisher metric. [ESTABLISHED]

Proof. Use $\mathrm{TV}(P,Q) = \sup_B |P(B) - Q(B)|$, half the L^1 distance for densities. Integrating the derivative of the normal density gives

$$\int_{\mathbb{R}} |\varphi(x-t) - \varphi(x)|dx \le |t| \int_{\mathbb{R}} |\varphi'(x)|dx = |t|\sqrt{2/\pi}, \qquad \mathrm{TV}(P_f, P_0) \le \frac{\|f\|_1}{\sqrt{2\pi}}. \tag{9.124}$$

Fix $a > 0$ and take $f_n = a\eta_n$. Its trace is a, whereas $\|f_n\|_1 = an^{-1}\int_{-1}^1 \beta \to 0$. Every Markov kernel contracts total variation: for an output event B, the function $r(z) = K(z,B)$ lies in $[0,1]$, and writing $r(z) = \int_0^1 \mathbf{1}_{\{r(z)>t\}}dt$ bounds its expectation difference by $\mathrm{TV}(P,Q)$. Taking the supremum over B proves the contraction. If (9.123) held, it would therefore imply $\mathrm{TV}(\mathcal{N}(a,1), \mathcal{N}(0,1)) \to 0$. But the fixed event $(a/2, \infty)$ has probability difference

$$\mathcal{N}(a,1)((a/2,\infty)) - \mathcal{N}(0,1)((a/2,\infty)) = \int_{-a/2}^{a/2} \varphi(x)dx > 0, \tag{9.125}$$

a contradiction. A continuous conditional version at $C = 0$ is allowed, but selecting it at this zero-probability context is not the action of a fixed kernel on one joint observation. □

The exact distinction is metric orthogonality. In any inner-product space, a lift $v - k_*$ minimizes squared norm over a subspace V exactly when $v - k_* \in V^\perp$: differentiating along every element of V proves necessity, and Pythagoras proves sufficiency. Hence all classes have minimizers exactly when the tangent space is $V \oplus V^\perp$. A topological complement alone is insufficient, as Proposition 9.57 proves. A smooth quotient metric additionally needs a smooth compatible horizontal structure. [ESTABLISHED]

For arbitrary recognition families and completed section spaces, admission to the strong Banach-Lie/Hilbert category remains a separate obligation. A weak metric, a Fréchet group or manifold, a singular action, or a changing support is not covered by Theorem 9.56. A proposed extension must supply its manifold calculus, actual statistical metric, smooth quotient, and metric-orthogonal splitting. The theorem and counterexample do not decide whether an arbitrary specified family admits such additional structure. [OPEN]

9.9 The obstruction to a global unit-speed clock

On a connected noncritical open set $U \subseteq \mathcal{Q}_i$, set

$$N = \left\|\operatorname{grad}^F \mathcal{F}_i\right\|_F, \qquad U_F = -\frac{\operatorname{grad}^F \mathcal{F}_i}{N}, \qquad \alpha_F = U_F^\flat = -\frac{d\mathcal{F}_i}{N}. \tag{9.126}$$

The form α_F evaluates to one on the unit descent field U_F and annihilates the Fisher-orthogonal complement of that field. [DEFINITION]

Theorem 9.59 (Exact criterion for a global orthogonal clock). There is a smooth scalar $T : U \to \mathbb{R}$ satisfying

$$dT = \alpha_F, \tag{9.127}$$

if and only if α_F is exact. Equivalently, it must be closed and have vanishing period around every piecewise C^1 closed curve in U:

$$d\alpha_F = 0, \qquad \oint_\gamma \alpha_F = 0 \quad \text{for every closed } \gamma \subset U. \tag{9.128}$$

On a simply connected U, closedness suffices. Whenever T exists, every unit-speed VFE history in U obeys $dT/d\tau = 1$, and the level sets of T are Fisher-orthogonal to the descent field. [ESTABLISHED]

Proof. If (9.127) holds, exact forms are closed and have zero periods. Conversely, fix $Q_0 \in U$ and define $T(Q)$ as the integral of α_F from Q_0 to Q. Closedness and vanishing periods make this integral path independent, and the standard line-integral argument gives $dT = \alpha_F$. Since $\alpha_F(U_F) = \mathsf{G}_i^F(U_F, U_F) = 1$, the asserted unit rate follows. The kernel of $dT = U_F^\flat$ is the orthogonal complement of U_F. □

The local differential obstruction is explicit:

$$d\alpha_F = \frac{1}{N^2} dN \wedge d\mathcal{F}_i. \tag{9.129}$$

Thus the norm of the gradient must be locally constant along each connected regular level set of $\mathcal{F}_i$ for the normalized form to be closed. Even after local closedness is established, the period condition in (9.128) is the remaining global obligation. [ESTABLISHED]

For a concrete local obstruction, take the Euclidean plane with $\mathcal{F}(x, y) = xy$ on the first quadrant. Then $N = (x^2 + y^2)^{1/2}$ and

$$d\alpha_F = \frac{x^2 - y^2}{(x^2 + y^2)^{3/2}} dx \wedge dy, \tag{9.130}$$

which is not identically zero on any open neighborhood. Every individual regular orbit still has its cumulative arc-length clock, but those orbitwise origins do not assemble into the orthogonal scalar (9.127). At critical points $N = 0$ and α_F is undefined, so extension across critical strata is a separate problem. The objective $\mathcal{F}_i$ itself may remain a Lyapunov scalar, but it is not generally a unit Fisher clock. [ESTABLISHED]

9.10 Operational record clocks and exact Markov loss

Let $\{P_\theta\}$ be a regular fine statistical experiment and let $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ be a normalized, parameter-independent Markov kernel. Along a C^1 fine history $\theta = \theta(\lambda)$, write ℓ_λ^X for its directional score and define the record history by

$$P_\lambda^Y = P_\lambda^X K. \tag{9.131}$$

This equation identifies the record family as the image of the same fine path; it does not define a separately optimized record trajectory. [HYPOTHESIS]

Theorem 9.60 (Same-path record and meta-agent clock contraction). Under the preceding regularity and Markov hypotheses, the record score and the exact Fisher-speed defect are

$$\ell_\lambda^Y(Y) = \mathbb{E}_\lambda[\ell_\lambda^X(X) \mid Y], \tag{9.132}$$

$$(\nu_F^X)^2 - (\nu_F^Y)^2 = \mathbb{E}_\lambda \operatorname{Var}_\lambda(\ell_\lambda^X(X) \mid Y) \geq 0. \tag{9.133}$$

Consequently, on every common parameter interval,

$$L_F(P^Y) \leq L_F(P^X), \qquad \tau_Y(\lambda_1) - \tau_Y(\lambda_0) \leq \tau_X(\lambda_1) - \tau_X(\lambda_0). \tag{9.134}$$

Equality holds exactly when the fine directional score is measurable with respect to the record for almost every path position. The same result applies to a meta-agent only when its statistical law is the normalized parameter-independent Markov image of this same fine history. [ESTABLISHED]

Proof. Attach K to the fine experiment. Because K has no θ dependence, the joint score is $\ell_\lambda^X(X)$. Projection onto the record sigma algebra gives (9.132). The law of total variance gives (9.133), including its equality condition. Taking nonnegative square roots and integrating proves (9.134). This is the pathwise form of the Fisher score-projection theorem [8]. □

Parameter independence is load bearing. If a dominated channel has density $k_\lambda(y \mid x)$, its output score is instead

$$\ell_\lambda^Y(Y) = \mathbb{E}_\lambda \left[\ell_\lambda^X(X) + \partial_\lambda \log k_\lambda(Y \mid X) \,\middle|\, Y \right]. \tag{9.135}$$

The additional term can create Fisher information. For example, a parameter-independent input combined with a channel that emits a Bernoulli variable of success probability $(1 + e^{-\lambda})^{-1}$ has zero fine Fisher speed and positive output Fisher speed. A fitted coarse approximation, an energy precomposition, a Galerkin restriction, or a mismatch between generative and recognition parameters is likewise outside Theorem 9.60. [ESTABLISHED]

One apparent counterexample to Theorem 9.60 must be retired rather than answered. The independent b-fold replication $\{\mathcal{N}(x\mathbf{1}_b, I_{\mathbb{R}^b})\}$ has Fisher information b against the input family's value 1, which invites the reading that some coarse arrow increased information. There is no such arrow: if a parameter-independent normalized Markov kernel K satisfied $\mathcal{N}(x,1)K = \mathcal{N}(x\mathbf{1}_b, I_{\mathbb{R}^b})$ for every x, then (9.133) would force $b \le 1$, so no such K exists for $b \ge 2$. What the pair witnesses is a change of reference tangent space, in which the replication lift $\mathscr{I}_b$ of Proposition 12.10 has norm $\sqrt{b}$; the composite $\mathscr{L}_b = U_b \mathscr{I}_b$ therefore obeys $\|\mathscr{L}_b\| \le \sqrt{b}$ rather than $\|\mathscr{L}_b\| \le 1$, and no contradiction arises. The replication pair may not be cited as a Markov-contraction counterexample. [ESTABLISHED]

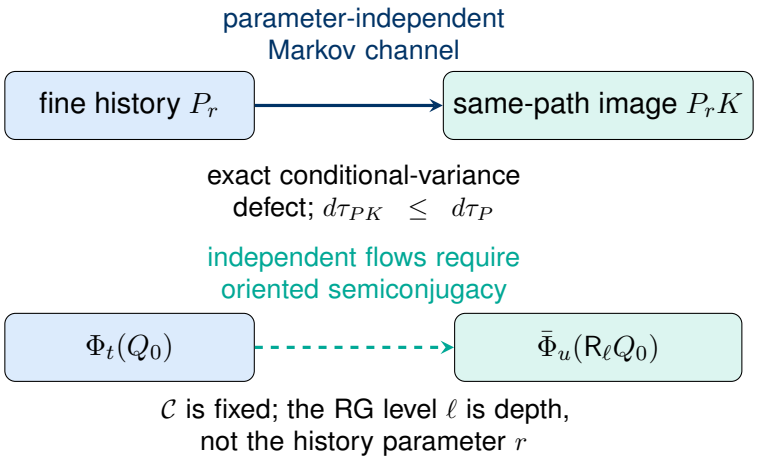


Figure 9.1: The upper arrow is the same-path comparison covered by Fisher contraction. The lower arrow compares independently generated flows and exists only after the additional semiconjugacy condition is proved.

9.11 The coarse configuration map

The coarse configuration map is separately declared data with its own symbol. Write

$$\mathsf{R}_\ell : \mathcal{Q}_\ell \longrightarrow \mathcal{Q}_{\ell+1}, \tag{9.136}$$

smooth, and never identified with the sample kernel K_ℓ, the induced law map, the associated-bundle morphism of Chapter 8, the nonlinear action map $\mathcal{R}^H$, the block measure-pair map $\mathcal{R}_b$, the derivative D_ℓ, the associated scale map $C_{\ell,s}$, the reference-space endomorphism $\widehat{\mathcal{R}}_\ell$, the conjugated retained projection $\widehat{R}_\ell$, the root-vertex set $\mathcal{R}$, or the descent ray $\mathcal{R}^-_{\mathcal{F}_i}$. The notation appendix carries the corresponding row. [DEFINITION]

A pointwise bundle morphism induces a map on configurations only on the projectable set of Theorem 8.13. Under the reachability hypotheses of Proposition 8.14, including a global smooth section through the witnessing fiber point, that set is a proper subset whenever the base map collapses a direction on which the vertical differential does not vanish; in the linear tier it can have empty interior. [ESTABLISHED]

No result here establishes that the projectable set is a smooth submanifold carrying a smooth induced arrow; that would require an additional transversality or elliptic-regularity hypothesis. The generic separately declared map R_ℓ is therefore neither constructed nor identified through section descent. [NOT-CLAIMED]

The averaging construction is the standard separately declared instance, and it carries an exact defect rather than an unconditional contraction. Let $f : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$ be measurable between standard Borel spaces, let μ be finite positive with $\bar{\mu} = f_\# \mu$, and let $\{\kappa_{\bar{c}}\}$ disintegrate μ over f. Suppose the coarse fiber $\bar{\mathcal{B}}$ is convex in a locally convex space, the structure group acts by restrictions of linear maps, and Ψ acts fiberwise as the restriction of a continuous affine map with linear part L, with the integrand Bochner integrable and the barycenter inside the fiber. Define

$$(\mathsf{R}_\ell s)(\bar{c}) = \int_{f^{-1}(\bar{c})} \Psi(s(c))\, \kappa_{\bar{c}}(\mathrm{d}c). \tag{9.137}$$

[DEFINITION]

For this averaging instance, if $\Psi \circ s$ is f-basic with unique descended section $\bar{s}$, then, for $\bar{\mu}$-almost every $\bar{c}$, the integrand equals $\bar{s}(\bar{c})$ for $\kappa_{\bar{c}}$-almost every point of the fiber. Normalization of the disintegration therefore gives $(\mathsf{R}_\ell s)(\bar{c}) = \bar{s}(\bar{c})$ for $\bar{\mu}$-almost every $\bar{c}$. [ESTABLISHED]

Hypothesis 9.61 (Chart joint convexity). In the declared affine chart of the coarse fiber, the map $(\bar{\beta}, v) \mapsto \bar{g}^F_{\bar{\beta}}(v, v)$ is jointly convex on $\bar{\mathcal{B}} \times \bar{W}$. The special case in which $\bar{g}^F$ is constant in the chart is written (JC-const). This hypothesis is a restriction on the chart, not a property of Fisher metrics: it holds in the location sector with fixed fiber covariance and in the law chart, and it fails in the covariance sector of the Gaussian moment chart. [HYPOTHESIS]

Theorem 9.62 (Exact averaging defect). Assume the preceding structure, Hypothesis 9.61, the fiberwise Fisher contraction $\Delta_F^{\Psi} = g^F - L^* \bar{g}^F L \succeq 0$, the weight condition $\bar{w} \circ f \leq w$ of (8.15), and finiteness of every integral. Then $\mathsf{R}_\ell s$ is a section, R_ℓ is gauge equivariant because the transition maps act linearly and commute with barycenters, and for every $Z \in T_s \mathcal{Q}_\ell$ the defect

$$\Delta_{\mathrm{avg}}(Z) = \int_{\mathcal{C}_\ell} w g^F_{s(c)}(Z, Z) \mathrm{d}\mu - \int_{\mathcal{C}_{\ell+1}} \bar{w} \bar{g}^F_{(\mathsf{R}_\ell s)(\bar{c})} \left(T\mathsf{R}_\ell Z, T\mathsf{R}_\ell Z\right) \mathrm{d}\bar{\mu} \tag{9.138}$$

is nonnegative. Under (JC-const) it is exactly the sum of three separately nonnegative terms,

$$\Delta_{\mathrm{avg}}(Z) = \underbrace{\int_{\mathcal{C}_\ell} w \Delta_F^{\Psi}(Z, Z) \mathrm{d}\mu}_{\text{channel loss}} + \underbrace{\int_{\mathcal{C}_\ell} (w - \bar{w} \circ f) \left(L^* \bar{g}^F\right)(Z, Z) \mathrm{d}\mu}_{\text{weight gap}} + \underbrace{\int_{\mathcal{C}_{\ell+1}} \bar{w} \, \mathrm{Var}^{\bar{g}^F}_{\kappa_{\bar{c}}}(LZ) \mathrm{d}\bar{\mu}}_{\text{context gap}}, \tag{9.139}$$

where $\mathrm{Var}^{\bar{g}}_{\kappa}(V) = \int \bar{g}(V, V) \mathrm{d}\kappa - \bar{g}(\int V \mathrm{d}\kappa, \int V \mathrm{d}\kappa)$. [ESTABLISHED]

Proof. Affineness gives $T\mathsf{R}_\ell Z(\bar{c}) = \int LZ(c) \kappa_{\bar{c}}(\mathrm{d}c)$, so the pair $(\mathsf{R}_\ell s(\bar{c}), T\mathsf{R}_\ell Z(\bar{c}))$ is the $\kappa_{\bar{c}}$-barycenter of $(\Psi s(c), LZ(c))$. Joint convexity and Jensen's inequality give $\bar{g}^F_{\mathsf{R}_\ell s(\bar{c})}(T\mathsf{R}_\ell Z, T\mathsf{R}_\ell Z) \leq \int \bar{g}^F_{\Psi s(c)}(LZ, LZ) \mathrm{d}\kappa_{\bar{c}}$. Multiplying by $\bar{w}$, integrating against $\bar{\mu}$, and using the disintegration converts the right-hand side into $\int (\bar{w} \circ f)(L^* \bar{g}^F)(Z, Z) \mathrm{d}\mu$; the weight condition and $\Delta_F^{\Psi} \succeq 0$ then give nonnegativity. When $\bar{g}^F$ is constant in the chart the Jensen step is an equality minus the variance term, and expanding $g^F = \Delta_F^{\Psi} + L^* \bar{g}^F$ separates the channel and weight contributions, which gives (9.139). □

Hypothesis 9.61 is not decorative, and the statement that averaging over the base loses information is false without it. Take two contexts with uniform κ, unit weights, Ψ the identity so that $\Delta_F^{\Psi} \equiv 0$, and the centered normal fiber $\{\mathcal{N}(0, \Sigma)\}$ in the moment chart, where $g^F(A, A) = A^2/(2\Sigma^2)$. At the configuration $(\Sigma_1, \Sigma_2) = (1, \delta)$ with tangent $Z = (1, 0)$ the fine integrated metric is $\frac{1}{4}$ and the chart-barycenter image metric is $1/(2(1+\delta)^2)$, so their ratio is $2/(1+\delta)^2$, which equals $\frac{20000}{10201} = 1.9605921\ldots$ at $\delta = 10^{-2}$ and tends to the block size 2 as $\delta \to 0^+$. The defect is therefore strictly negative, tending to $-\frac{1}{4}$, and is negative exactly for $\delta < \sqrt{2} - 1$. Every other hypothesis of Theorem 9.62 holds on this datum; only joint convexity fails, and the reason is exact: the Hessian of $(\Sigma, A) \mapsto A^2/(2\Sigma^2)$ has determinant $-A^2\Sigma^{-6} < 0$ for $A \neq 0$, whereas the law-chart integrand $\dot{p}^2/p$ has Hessian determinant zero and positive trace and is jointly convex. Accordingly no contraction statement may be attached to a generic averaging or variational coarse map. [ESTABLISHED]

Proposition 9.63 (Smoothness and gauge descent at the exhibited tier). Extend Definition 9.6 to scale $\ell + 1$ with base $\mathcal{C}_{\ell+1}$, measure $\bar{\mu}$, weight $\bar{w}$, coarse fiber $\{\mathcal{N}(m, \bar{\Sigma}_0)\}$ with $\bar{\Sigma}_0 \succeq \Sigma_0$, and a coarse basis $\psi_1, \ldots, \psi_M$ linearly independent in $L^2(\bar{w}\bar{\mu}; \bar{\Sigma}_0^{-1})$ whose span contains every constant field. Let $\mathsf{L}_{\ell+1} : \mathbb{R}^K \to \mathbb{R}^M$ represent those constants in the coarse basis as in (9.11). Suppose the fiberwise average of each fine basis field lies in the coarse span, $\int_{f^{-1}(\bar{c})} \phi_a \mathrm{d}\kappa_{\bar{c}} = \sum_b \mathsf{E}_{ba} \psi_b(\bar{c})$ for a constant matrix $\mathsf{E} \in \mathbb{R}^{M \times N}$ which intertwines the coefficient actions,

$$\mathsf{E}\mathsf{L}_\ell = \mathsf{L}_{\ell+1}. \tag{9.140}$$

Then the averaging map (9.137) reads $\mathsf{R}_\ell(\xi) = \mathsf{E}\xi$ in coordinates. It is linear, hence C^∞, with tangent map E everywhere; it is gauge equivariant for the affine coefficient actions because $\mathsf{E}\Theta_g^\ell = \Theta_g^{\ell+1}\mathsf{E}$; and its metric compatibility is exactly the matrix inequality

$$\mathsf{E}^\top \bar{\Phi}\mathsf{E} \preceq \Phi, \qquad \bar{\Phi}_{bc} = \int \psi_b^\top \bar{\Sigma}_0^{-1}\psi_c \bar{w}\mathrm{d}\bar{\mu}, \tag{9.141}$$

which is $\mathsf{R}_\ell^*\mathsf{G}_{\ell+1} \preceq \mathsf{G}_\ell$ and is implied by Theorem 9.62. [ESTABLISHED]

Proof. Averaging a section of the declared family produces the mean field $\bar{c} \mapsto \sum_a \xi_a \int \phi_a \mathrm{d}\kappa_{\bar{c}} = \sum_b (\mathsf{E}\xi)_b \psi_b(\bar{c})$, which is the coarse configuration $s_{\mathsf{E}\xi}$; smoothness and the tangent map are those of a linear map. For equivariance, the constant-field hypotheses give $\Theta_g^\ell(\xi) = \xi + \mathsf{L}_\ell g$ and $\Theta_g^{\ell+1}(\eta) = \eta + \mathsf{L}_{\ell+1} g$, while (9.140) gives $\mathsf{E}\Theta_g^\ell(\xi) = \Theta_g^{\ell+1}(\mathsf{E}\xi)$. Equivalently, $\int (m+g)\mathrm{d}\kappa_{\bar{c}} = \int m\mathrm{d}\kappa_{\bar{c}} + g$ because $\kappa_{\bar{c}}$ is a probability measure and both resulting fields remain in their declared spans. Compatibility is Theorem 9.62 evaluated on the finite-dimensional tangent, and it takes the displayed matrix form because both metrics are constant Gram forms. □

Metric compatibility is therefore carried as an explicit hypothesis discharged by Theorem 9.62, and it is not a corollary of any fiberwise contraction theorem. [HYPOTHESIS]

9.12 Independently recomputed flows

Let X_ℓ and $X_{\ell+1}$ be the declared fine and coarse natural-gradient vector fields on regular configuration manifolds $\mathcal{Q}_\ell$ and $\mathcal{Q}_{\ell+1}$, with local flows Φ_t and $\bar{\Phi}_u$. Pointwise contraction of $T\mathsf{R}_\ell$ does not state that R_ℓ maps an X_ℓ-orbit to an $X_{\ell+1}$-orbit. [ESTABLISHED]

Definition 9.64 (Oriented semiconjugacy). For an open $U \subseteq \mathcal{Q}_\ell$, the pair $(\mathsf{R}_\ell, a_\ell)$ is an oriented semiconjugacy of X_ℓ onto $X_{\ell+1}$ over U when $a_\ell : U \to (0,\infty)$ is continuous and

$$T_Q\mathsf{R}_\ell X_\ell(Q) = a_\ell(Q) X_{\ell+1}(\mathsf{R}_\ell Q) \quad \text{for every } Q \in U. \tag{9.142}$$

[DEFINITION]

Lemma 9.65 (The factor is determined and regular off the coarse critical set). If $X_{\ell+1}(\mathsf{R}_\ell Q) \neq 0$, then $a_\ell(Q)$ is unique and

$$a_\ell(Q) = \frac{\mathsf{G}_{\ell+1}\left(T_Q\mathsf{R}_\ell X_\ell(Q), X_{\ell+1}(\mathsf{R}_\ell Q)\right)}{\|X_{\ell+1}(\mathsf{R}_\ell Q)\|^2_{\mathsf{G}_{\ell+1}}}, \tag{9.143}$$

so, for any integer $k \geq 0$, a_ℓ is C^k wherever the maps $Q \mapsto T_Q\mathsf{R}_\ell X_\ell(Q)$, $Q \mapsto X_{\ell+1}(\mathsf{R}_\ell Q)$, and the pulled-back metric coefficients are C^k. Sufficient primitive hypotheses are $\mathsf{R}_\ell \in C^{k+1}$ and X_ℓ, $X_{\ell+1}$, and $\mathsf{G}_{\ell+1}$ of class C^k, with the displayed compositions defined. If instead $X_{\ell+1}(\mathsf{R}_\ell Q) = 0$, then (9.142) forces $T_Q\mathsf{R}_\ell X_\ell(Q) = 0$ and imposes no algebraic pointwise value on $a_\ell(Q)$. The definition

still requires a positive continuous function on U: limits from the noncritical set may determine the value at Q, or may obstruct any continuous positive extension. [ESTABLISHED]

Proof. Pair (9.142) with $X_{\ell+1}(\mathsf{R}_\ell Q)$ and divide; strong nondegeneracy of $\mathsf{G}_{\ell+1}$ makes the denominator positive. □

Continuity of a_ℓ is therefore a consequence on the noncritical part rather than an extra assumption, and the phrase noncritical must name the field it governs: the load-bearing reading is $X_{\ell+1} \neq 0$ on $\mathsf{R}_\ell(U)$. [DEFINITION]

Theorem 9.66 (Oriented semiconjugacy and maximal intervals). Assume (9.142) on an open U, with $X_{\ell+1}$ locally Lipschitz. Fix $Q \in U$, let J_Q be the connected component of 0 in the set of t for which $\Phi_s(Q) \in U$ for every s between 0 and t, and set $\sigma_Q(t) = \int_0^t a_\ell(\Phi_s Q)\mathrm{d}s$. Then σ_Q is a strictly increasing C^1 diffeomorphism of J_Q onto an open interval $\Sigma_Q \ni 0$, and

$$\Sigma_Q \subseteq \bar{J}^{\max}_{\mathsf{R}_\ell Q}, \qquad \mathsf{R}_\ell\left(\Phi_t(Q)\right) = \bar{\Phi}_{\sigma_Q(t)}\left(\mathsf{R}_\ell Q\right) \quad \text{for every } t \in J_Q, \tag{9.144}$$

where $\bar{J}^{\max}$ is the maximal interval of the coarse flow through $\mathsf{R}_\ell Q$. More explicitly, write $J_Q = (\alpha_Q, \beta_Q)$ and $\bar{J}^{\max}_{\mathsf{R}_\ell Q} = (\bar{\alpha}_Q, \bar{\beta}_Q)$, with endpoints in the extended real line. Then

$$\Sigma_Q = \left(\lim_{t \downarrow \alpha_Q} \sigma_Q(t), \lim_{t \uparrow \beta_Q} \sigma_Q(t) \right), \tag{9.145}$$

and the full maximal coarse orbit is traversed if and only if the two limits equal $\bar{\alpha}_Q$ and $\bar{\beta}_Q$, respectively. Conversely, differentiating an oriented flow semiconjugacy of this form at $t = 0$ yields (9.142). [ESTABLISHED]

Proof. Put $c(t) = \mathsf{R}_\ell(\Phi_t Q)$, so that $\dot{c} = a_\ell(\Phi_t Q) X_{\ell+1}(c)$ by (9.142). The integrand of σ_Q is continuous and strictly positive, so σ_Q is a C^1 diffeomorphism onto an open interval. Let $\theta = \sigma_Q^{-1}$ and $d = c \circ \theta$; then $d'(u) = X_{\ell+1}(d(u))$ on Σ_Q with $d(0) = \mathsf{R}_\ell Q$, so d is an integral curve of $X_{\ell+1}$, and by local Lipschitz continuity the maximal integral curve through $\mathsf{R}_\ell Q$ contains it. That gives both the domain inclusion and the flow identity. The order matters: the reparameterized curve is constructed first and the coarse flow is then shown to extend over Σ_Q, rather than the coarse flow being assumed defined at $\sigma_Q(t)$. The converse is the chain rule at $t = 0$. □

When the semiconjugacy holds along the entire complete fine orbit ($J_Q = \mathbb{R}$) and the coarse maximal interval is also $\mathbb{R}$, the endpoint criterion is equivalently

$$\int_0^{+\infty} a_\ell(\Phi_s Q)\mathrm{d}s = +\infty, \qquad \int_{-\infty}^{0} a_\ell(\Phi_s Q)\mathrm{d}s = +\infty. \tag{9.146}$$

A positive lower bound for a_ℓ is sufficient, but it is not necessary: for $X_\ell = \partial_x$, $X_{\ell+1} = \partial_y$, and $\mathsf{R}_\ell = \operatorname{arsinh}$ on $\mathbb{R}$, one has $a_\ell(x) = (1 + x^2)^{-1/2}$ and $\inf a_\ell = 0$, while $\sigma_0(t) = \operatorname{arsinh}(t)$ maps $\mathbb{R}$ onto $\mathbb{R}$. Nor does positivity alone guarantee full traversal: on $\mathbb{R}$ with $X_\ell = \partial_x$, $X_{\ell+1} = \partial_y$, and $\mathsf{R}_\ell = \arctan$, the factor $a_\ell = (1 + x^2)^{-1}$ is positive while $\Sigma_0 = (-\pi/2, \pi/2)$ is a proper subinterval

of $\mathbb{R}$ even though both flows are complete. Positivity of the factor is also the entire orientation content: on $\mathbb{R}$ with $\mathcal{F}_\ell = x^2/2$, $\mathcal{F}_{\ell+1} = -y^2/2$, and $\mathsf{R}_\ell(x) = -x$, the orbit sets agree while $a_\ell \equiv -1$ and coarse descent becomes ascent, so requiring only $a_\ell \neq 0$ loses orientation. [ESTABLISHED]

Proposition 9.67 (Noncollapse is a separate hypothesis). Condition (9.142) alone permits total collapse: with $\mathsf{R}_\ell \equiv Q^*$ constant and $X_{\ell+1} \equiv 0$, both sides vanish identically, so it holds for every positive a_ℓ while every nonconstant fine orbit maps to a point and every coarse Fisher duration is zero. A nonconstant shared oriented history therefore additionally requires $T_Q \mathsf{R}_\ell X_\ell(Q) \neq 0$ on every nontrivial subarc, equivalently $X_{\ell+1}(\mathsf{R}_\ell Q) \neq 0$ there given $a_\ell > 0$, together with the maximal-interval condition of Theorem 9.66, namely the endpoint equalities in (9.145); for complete real-line flows these are the two divergence conditions in (9.146). Neither is implied by (9.142). [ESTABLISHED]

Theorem 9.68 (The semiconjugacy parameter rate cancels from geometric length). Under (9.142), write $\nu_\ell(r) = \|X_\ell(Q^{(\ell)}(r))\|_{\mathsf{G}_\ell}$ and $\nu^{\mathrm{img}}_{\ell+1}(r) = \|\frac{\mathrm{d}}{\mathrm{d}r}\mathsf{R}_\ell Q^{(\ell)}(r)\|_{\mathsf{G}_{\ell+1}}$. Then

$$\nu^{\mathrm{img}}_{\ell+1}(r) = a_\ell\left(Q^{(\ell)}(r)\right)\left\|X_{\ell+1}\left(\mathsf{R}_\ell Q^{(\ell)}(r)\right)\right\|_{\mathsf{G}_{\ell+1}}, \tag{9.147}$$

and for $r_0 < r_1$ in J_Q the accumulated coarse duration is

$$\tau^{(\ell+1)}(r_1) - \tau^{(\ell+1)}(r_0) = \int_{\sigma_Q(r_0)}^{\sigma_Q(r_1)} \left\|X_{\ell+1}\left(\bar{\Phi}_u(\mathsf{R}_\ell Q)\right)\right\|_{\mathsf{G}_{\ell+1}} \mathrm{d}u, \tag{9.148}$$

which is the intrinsic $\mathsf{G}_{\ell+1}$-arc length of the coarse orbit arc and does not depend on a_ℓ. [ESTABLISHED]

Proof. Equation (9.147) is (9.142) together with positive homogeneity of the norm and $a_\ell > 0$. For (9.148), substitute (9.147) and change variables $u = \sigma_Q(s)$, whose Jacobian is $\mathrm{d}u = a_\ell(\Phi_s Q)\mathrm{d}s$; the factor a_ℓ in the integrand cancels against that Jacobian exactly. □

The factor $a_\ell = \mathrm{d}\sigma_Q/\mathrm{d}r$ is the coarse-flow-parameter rate relative to the chosen fine orbit parameter. It is neither physical time nor a Fisher-duration rate, and it cancels from a final geometric arc length. For an unoriented relation with $a_\ell < 0$, ordinary arc length still uses $|a_\ell|$ and is unchanged by the orientation reversal; positivity is the extra condition selecting an oriented semiconjugacy. [ESTABLISHED]

Theorem 9.69 (Exact duration comparison criterion). Under (9.142), along a fixed orbit and from a common origin, the coarse and fine durations agree identically if and only if $\|T\mathsf{R}_\ell X_\ell\|_{\mathsf{G}_{\ell+1}} = \|X_\ell\|_{\mathsf{G}_\ell}$ almost everywhere along the orbit; they agree up to a constant factor $\kappa > 0$ if and only if the same holds with that factor; and the coarse duration has nonincreasing increments relative to the fine duration on every subinterval if and only if

$$\|T\mathsf{R}_\ell X_\ell\|_{\mathsf{G}_{\ell+1}} \leq \|X_\ell\|_{\mathsf{G}_\ell} \quad \text{pointwise along the orbit.} \tag{9.149}$$

[ESTABLISHED]

Proof. Both durations are integrals of continuous nonnegative densities from a common origin. Two such integrals agree for every upper limit exactly when their densities agree almost everywhere, by Lebesgue differentiation, and the scaled and ordered versions follow in the same way. □

The necessary and sufficient condition is thus the *scalar* inequality (9.149) evaluated on the single direction X_ℓ. The tensorial Loewner condition $\mathsf{R}_\ell^* \mathsf{G}_{\ell+1} \preceq \mathsf{G}_\ell$ along the curve is sufficient and strictly stronger; it is what one declares when the comparison must hold in every direction, and it is what Theorem 9.62 and proposition 9.63 deliver. Neither follows from any fiberwise Fisher contraction theorem, and two independent mechanisms show why. [ESTABLISHED]

The first mechanism is that the two configuration metrics are separately declared data. On $\mathcal{Q}_\ell = \mathcal{Q}_{\ell+1} = \mathbb{R}$ with $X_\ell = X_{\ell+1} = \partial_x$ and R_ℓ the identity, (9.142) holds with $a_\ell \equiv 1$, there is no collapse, and the map is a diffeomorphism; declaring $\mathsf{G}_\ell = dx^2$ and $\mathsf{G}_{\ell+1} = 4dx^2$ makes $\tau^{(\ell+1)} = 2\tau^{(\ell)}$. The first realization is one context with an identity fiber map, so that $\Delta_F^\Psi = 0$ exactly, and channel weights 1 and 4; this is the zero-information-loss metric-declaration witness. Separately, fine fiber $\{\mathcal{N}(\mu, 1)\}$ against coarse fiber $\{\mathcal{N}(\mu, \frac{1}{4})\}$ with the identity parameter map gives the non-Markov Fisher-increase witness $\Delta_F^\Psi = 1 - 4 = -3$, while the duration still doubles. The second mechanism is that a contraction theorem compares the image of the fine orbit, not an independently recomputed coarse orbit. On $\mathbb{R}^2$ with R_ℓ and Ψ both the identity, so that contraction is an equality and image durations agree, the objectives $\mathcal{F}_\ell = \frac{1}{2}(x^2 + y^2)$ and $\mathcal{F}_{\ell+1} = \frac{1}{2}(x^2 + 4y^2)$ give the fine history (e^{-t}, e^{-t}) of total duration $\sqrt{2}$ and the independently recomputed coarse history (e^{-t}, e^{-4t}), which is not a straight segment and has strictly greater length $\int_0^1 \sqrt{1 + 16u^6}\mathrm{d}u > \sqrt{2}$; and (9.142) itself fails there, since $(-x, -y) = a(-x, -4y)$ forces $a = 1$ and then $y = 4y$. The two mechanisms are logically independent and neither implies the other, which is the exact content of the statement that either condition without the other is insufficient. [ESTABLISHED]

Theorem 9.70 (When fiberwise contraction does lift). Assume the two scales share the base $\mathcal{C}_i$, the same finite base measure μ_i, and the same positive channel weights w_x; assume R_ℓ acts pointwise as $(\mathsf{R}_\ell Q)(c) = \Psi_c(Q(c))$ with each Ψ_c the pushforward of a normalized parameter-independent Markov kernel with vertical differential $T^V\Psi_c$; assume both configuration metrics are the corresponding weighted integrals of the fiber Fisher metrics; and assume every integral is finite. Then for every $Z \in T_Q\mathcal{Q}_\ell$,

$$\|Z\|^2_{\mathsf{G}_\ell} - \|T\mathsf{R}_\ell Z\|^2_{\mathsf{G}_{\ell+1}} = \int_{\mathcal{C}_i} \sum_x w_x \Delta_F^{\Psi_c}(Z_x(c), Z_x(c))\mathrm{d}\mu_i \geq 0, \tag{9.150}$$

so $\nu_{\ell+1}^{\mathrm{img}} \leq \nu_\ell$ pointwise and the coarse duration has nonincreasing increments, with equality on a subinterval exactly when the fine score in the direction $Z_x(c)$ is measurable with respect to the coarse output for μ_i-almost every c and every channel. [ESTABLISHED]

Proof. Differentiate the pointwise action, insert into the coarse metric, and subtract the fine one; the integrand is $\Delta_F^{\Psi_c}$, nonnegative by Theorem 8.21. Monotonicity of the integral and of the square

root gives the speed inequality, and integrating in the orbit parameter gives the duration statement. □

Theorem 9.70 is the case of a shared base; Theorem 9.62 is an alternative sufficient extension that admits a collapsing base map, which is what a renormalization step performs. It additionally assumes affine-fiber, convexity, and barycenter hypotheses, so no general logical-strength comparison with the shared-base result is asserted. Its second hypothesis excludes this manuscript's own Galerkin aggregation, because by Proposition 24.5 that coarse operator is a restriction rather than a Markov pushforward, the two differing by a positive semidefinite Schur term with the restriction larger in the Loewner order. Its first hypothesis is exactly what the metric-declaration witness above violates. [ESTABLISHED]

9.13 Sufficient conditions for natural-gradient semiconjugacy

Throughout this section $X_\ell = -\operatorname{grad}^{\mathsf{G}_\ell}\mathcal{F}_\ell$ and $X_{\ell+1} = -\operatorname{grad}^{\mathsf{G}_{\ell+1}}\mathcal{F}_{\ell+1}$ with both objectives C^2 and both metrics strong. Nothing in this section is automatic; each conclusion consumes a stated hypothesis.

Equality of objectives does not intertwine natural gradients. The differential $\mathrm{d}\mathcal{F}$ is metric free while the gradient is not, so with R_ℓ the identity on $\mathbb{R}^2$, one objective $\frac{1}{2}(x^2+2y^2)$, and metrics $\operatorname{diag}(1,1)$ and $\operatorname{diag}(1,\kappa)$ with $\kappa \neq 1$, condition (9.142) would require $(x,2y) = a(x,2y/\kappa)$, forcing $a=1$ and then $\kappa = 1$; it therefore fails on the dense open set where $xy \neq 0$. [ESTABLISHED]

Hypothesis 9.71 (Functional compatibility). There is a C^1 function χ_ℓ with $\chi_\ell' > 0$ and an open $U \supseteq \mathscr{O}$ containing the fine orbit on which $\mathcal{F}_\ell = \chi_\ell \circ \mathcal{F}_{\ell+1} \circ \mathsf{R}_\ell$. This supplies orientation and noncollapse; it does not by itself supply (9.142). [HYPOTHESIS]

Proposition 9.72 (Metric sufficient conditions for semiconjugacy). Let R_ℓ be a surjective submersion with closed orbit-tangent splitting, let $\mathcal{H} = (\ker T\mathsf{R}_\ell)^{\perp_{\mathsf{G}_\ell}}$, and assume Hypothesis 9.71. If R_ℓ is a Riemannian submersion, meaning that $T\mathsf{R}_\ell|_{\mathcal{H}}$ is a linear isometry onto $T\mathcal{Q}_{\ell+1}$, then (9.142) holds with $a_\ell(Q) = \chi_\ell'(\mathcal{F}_{\ell+1}(\mathsf{R}_\ell Q)) > 0$. More generally, if R_ℓ is horizontally conformal with dilation $\varphi_\ell > 0$, then (9.142) holds with

$$a_\ell(Q) = \chi_\ell'\left(\mathcal{F}_{\ell+1}(\mathsf{R}_\ell Q)\right)\varphi_\ell(Q)^2 > 0, \qquad Q \in U. \tag{9.151}$$

[ESTABLISHED]

Proof. Under Hypothesis 9.71 the gradient $u = \operatorname{grad}^{\mathsf{G}_\ell}\mathcal{F}_\ell$ lies in $\mathcal{H}$. For $Z \in \mathcal{H}$, horizontal conformality gives $\mathsf{G}_{\ell+1}(T\mathsf{R}_\ell u, T\mathsf{R}_\ell Z) = \varphi_\ell^2 \mathsf{G}_\ell(u,Z) = \varphi_\ell^2 \mathrm{d}\mathcal{F}_\ell[Z] = \varphi_\ell^2\chi_\ell'\mathrm{d}\mathcal{F}_{\ell+1}[T\mathsf{R}_\ell Z] = \varphi_\ell^2\chi_\ell'\mathsf{G}_{\ell+1}(w, T\mathsf{R}_\ell Z)$ with $w = \operatorname{grad}^{\mathsf{G}_{\ell+1}}\mathcal{F}_{\ell+1}\circ\mathsf{R}_\ell$. Since $T\mathsf{R}_\ell(\mathcal{H}) = T\mathcal{Q}_{\ell+1}$, nondegeneracy gives $T\mathsf{R}_\ell u = \varphi_\ell^2\chi_\ell' w$; negating both sides gives the claim. The Riemannian-submersion case is $\varphi_\ell \equiv 1$. □

A sanity check and a nonempty model are both available. On Euclidean $\mathbb{R}$ with $\mathsf{R}_\ell(x) = \lambda x$ one has $\varphi_\ell = |\lambda|$ and $\mathcal{F}'_\ell(x) = \lambda \mathcal{F}'_{\ell+1}(\lambda x)$, so $T\mathsf{R}_\ell \operatorname{grad} \mathcal{F}_\ell = \lambda^2 \mathcal{F}'_{\ell+1}(\lambda x)$, matching $a_\ell = \lambda^2$. On the exhibited tier of Definition 9.6 with $\mathcal{C}_\ell = S^1$, basis $\{1, \cos\theta, \sin\theta\}$, fine fiber $\{\mathcal{N}(m,1)\}$, coarse fiber $\{\mathcal{N}(m,2)\}$ realized by the sample kernel $N(x,\cdot) = \mathcal{N}(x,1)$, and $\mathcal{F}(\xi) = \frac{1}{2}|\xi|^2$, one has $\mathsf{G}_\ell = \Phi = \operatorname{diag}(1, \frac{1}{2}, \frac{1}{2})$, $\mathsf{G}_{\ell+1} = \frac{1}{2}\Phi$, R_ℓ the identity, hence $\varphi_\ell^2 = \frac{1}{2}$ and $\chi_\ell = \mathrm{id}$; the proposition predicts $a_\ell = \frac{1}{2}$, which is what the direct computation $X_\ell = -\Phi^{-1}\xi$ and $X_{\ell+1} = -2\Phi^{-1}\xi$ returns, with speed ratio $\nu^{\mathrm{img}}_{\ell+1}/\nu_\ell = 1/\sqrt{2}$ and both flows complete. [ESTABLISHED]

Where this manuscript's own coarse maps stand can now be stated exactly rather than gestured at. Under the hypotheses of Theorem 9.70, horizontal conformality with dilation φ_ℓ is equivalent to the integrated bilinear identity

$$\int_{\mathcal{C}_i} \sum_x w_x \Delta_F^{\Psi_c}\big(Z_x(c), W_x(c)\big)\, \mathrm{d}\mu_i(c) = (1-\varphi_\ell^2)\mathsf{G}_\ell(Z,W) \quad \text{for every } Z, W \in \mathcal{H}. \tag{9.152}$$

This is necessary and sufficient at the declared configuration tier. It does not imply the corresponding pointwise identity for a finite basis span: take a base which is the disjoint union of two equal-weight contexts, the shared constant-mean tangent $(1,1)$ in fine fibers $\mathcal{N}(\mu,1)$, and at the two contexts add independent Gaussian noises of variances 3 and $1/3$. The resulting Markov pushforwards have pulled-back location Fisher forms $1/4$ and $3/4$, respectively. Their integrated ratio is $\varphi_\ell^2 = 1/2$, but their pointwise ratios differ, so they are not pointwise conformal with that common dilation. The pointwise conclusion is available only after declaring a decomposable, localizable tangent tier (closed under compactly supported localization, with a full-support base measure), which permits the identity to be tested on every local pair. For a coarse objective defined by exact contraction, $\mathcal{F}_{\ell+1}[Q'] = \inf\{\mathcal{F}_\ell[q] : \mathsf{R}_\ell q = Q'\}$, one has $\mathcal{F}_{\ell+1}(\mathsf{R}_\ell q) \le \mathcal{F}_\ell(q)$ with equality exactly on the attaining set, so Hypothesis 9.71 holds with $\chi_\ell = \mathrm{id}$ there and nowhere else, and its differential consequence needs the orbit to lie in the interior of that set. A structured sufficient route to semiconjugacy for the manuscript's own renormalization histories is to exhibit χ_ℓ and an open U containing the orbit on which Hypothesis 9.71 holds, to prove for the exact-contraction objective that the orbit lies in the interior of the attaining set, and then to verify horizontal conformality of the declared R_ℓ. That route is sufficient only: it is not asserted necessary or minimal, and it does not establish (9.142) for the manuscript's own maps. It is stated in objects the manuscript already declares and is satisfiable in the exhibited tier. [OPEN]

9.14 RG depth is not inference time

A hierarchy of coarse objects may be indexed by

$$Q^{(\ell+1)} = \mathsf{R}_\ell(Q^{(\ell)}), \qquad \ell = 0, 1, 2, \ldots. \tag{9.153}$$

The integer ℓ records RG depth. It is not a parameter on an inference history, and the sequence in (9.153) is not a temporal trajectory. A history can instead be present at every level, written $Q^{(\ell)}(r)$, with r labeling positions on its oriented orbit and $\tau^{(\ell)}(r)$ its accumulated Fisher duration at that level.

Only when

$$Q^{(\ell+1)}(r) = \mathsf{R}_\ell(Q^{(\ell)}(r)), \tag{9.154}$$

and R_ℓ is realized by a parameter-independent Markov pushforward in the sense of Theorem 9.70 does the same-path clock contraction compare adjacent levels. Attaching that Markov predicate to a configuration arrow is a substantive hypothesis with content, not a typing remark, and it can fail: no parameter-independent Markov kernel realizes the joint mean-and-variance barycenter of two normal laws, although the centered variance barycenter and the fixed-common-variance location barycenter are both realizable. Independently recomputed levelwise histories additionally require Theorem 9.66 together with the noncollapse conditions of Proposition 9.67. [ESTABLISHED]

Proposition 9.73 (Typed non-identification of RG depth, orbit parameter, and duration). Scale depth ℓ, an orbit parameter r, and Fisher duration $\tau^{(\ell)}$ are distinct typed quantities: ℓ is a discrete scale index, r is a chosen oriented parameter on a selected orbit, and τ is metric-relative arc length from a chosen origin. After a metric, origin, orientation, and parameterized path have been chosen, τ can be a function of r. It is nevertheless not canonically determined by an unparameterized orbit or by scale depth: changing the metric rescales it, and changing the origin or orientation changes its parameterized representation. Thus no physical-time or canonical-clock identification follows from these objects. Strict monotonicity of $\tau^{(\ell)}$ does not make it a regular coordinate: with $h_s = 4x^2dx^2$ one gets $\tau(r) = r^2$, whose inverse is not differentiable at the origin, while a zero-speed subinterval destroys strict monotonicity outright. [ESTABLISHED]

Proof. The non-identification follows from the definitions: only the extra metric, origin, orientation, and parameterized-path data produces a duration function. Metric relativity is homogeneity of the norm in (9.87); the regularity statement is the displayed integral $\int_0^R 2r\mathrm{d}r = R^2$ together with nondifferentiability of the square root at zero. □

Finally, $\tau^{(\ell)}$ is not a function on the contextual base and is not physical time. A regional clock potential requires the exactness and vanishing period conditions of Theorem 9.59, and those can fail locally, as (9.130) shows. No operational identification of $\tau^{(\ell)}$ with a clock reading is made here, and none follows from anything proved above. [NOT-CLAIMED]

Cross-scale covariant first jets, their horizontal-defect term, the unconditional vertical Fisher-defect cocycle, and the exact residual of its base pullback belong to Chapter 8. They compare typed geometry across a scale map; they do not turn the scale index into time. The construction in this chapter therefore yields oriented, parameter-free inference histories and agent-relative information duration while leaving any identification with physical time, any Lorentzian signature, any causal structure, and any canonical choice of connection outside the established theory. [NOT-CLAIMED]

Part II

General Coarse Maps and Renormalization

Chapter 10

General Coarse-Graining

Coarse-graining is not one operation. A normalized Markov channel pushes probability laws forward and obeys data processing. Identification of sample variables precomposes an unnormalized energy and requires a new reference measure and normalizer. A Galerkin map restricts an operator and becomes a propagator only after a reference form and signal maps are supplied. This chapter separates these types before any Gaussian matrix formula is used.

10.1 Markov coarse maps

Definition 10.1 (Coarse channel). For standard Borel spaces X, Y, a coarse channel is a parameter-independent Markov kernel

$$K : \mathsf{X} \rightsquigarrow \mathsf{Y}.$$

It sends $P \in \mathcal{P}(\mathsf{X})$ to

$$(PK)(B) = \int_{\mathsf{X}} K(x, B) P(dx). \tag{10.1}$$

A measurable statistic $c : \mathsf{X} \to \mathsf{Y}$ is the deterministic kernel $K_c(x, B) = \mathbf{1}_B(c(x))$. [DEFINITION]

Proposition 10.2 (Markov kernels form a category). If $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ and $L : \mathsf{Y} \rightsquigarrow \mathsf{Z}$, then

$$(KL)(x, C) = \int_{\mathsf{Y}} L(y, C) K(x, dy) \tag{10.2}$$

is a Markov kernel, composition is associative, the deterministic identity is an identity, and

$$P(KL) = (PK)L. \tag{10.3}$$

[ESTABLISHED]

Proof. Kernel integration gives measurability, and monotone convergence gives countable additivity. Tonelli's theorem gives, for measurable C,

$$[P(KL)](C) = \int_{\mathsf{X}} \int_{\mathsf{Y}} L(y, C) K(x, dy) P(dx) = \int_{\mathsf{Y}} L(y, C)(PK)(dy),$$

which proves (10.3); applying the identity to a third kernel proves associativity. □

A statistical experiment $\mathcal{E} = (\mathsf{X}, \{P_\theta\}_{\theta\in\Theta})$ is therefore sent functorially to $\mathcal{E}K = (\mathsf{Y}, \{P_\theta K\}_{\theta\in\Theta})$. Parameter independence is load bearing: a family K_θ is a parameter-dependent model transformation, not one channel between experiments. [DEFINITION]

10.2 KL contraction, equality, and recovery

Theorem 10.3 (Extended KL data processing). For arbitrary probability laws $P, Q \in \mathcal{P}(\mathsf{X})$ and every Markov kernel $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$,

$$D_{\mathrm{KL}}(PK\|QK) \le D_{\mathrm{KL}}(P\|Q) \qquad \text{in } [0, +\infty]. \tag{10.4}$$

No absolute-continuity or finiteness hypothesis is required. [ESTABLISHED]

Proof. If $P \not\ll Q$, the right side is $+\infty$ and the inequality is immediate. If $P \ll Q$, put $r = dP/dQ$ and form $\mathbb{Q}(dx, dy) = Q(dx)K(x, dy)$. The calculation against bounded test functions gives $d(PK)/d(QK) = \bar{r}$, where $\bar{r}(Y) = \mathbb{E}_{\mathbb{Q}}[r(X) \mid Y]$. For the nonnegative convex generator $\phi_0(t) = t\log t - t + 1$, conditional Jensen and monotone approximation give

$$D_{\mathrm{KL}}(PK\|QK) = \mathbb{E}_{\mathbb{Q}}\phi_0(\bar{r}(Y)) \le \mathbb{E}_{\mathbb{Q}}\phi_0(r(X)) = D_{\mathrm{KL}}(P\|Q),$$

including the case in which the last expectation is $+\infty$. No signed integral is rearranged. □

Theorem 10.4 (Data processing with its exact equality condition). Let $P, Q \in \mathcal{P}(\mathsf{X})$, $P \ll Q$, and $D_{\mathrm{KL}}(P\|Q) < \infty$. For a Markov kernel $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$, form

$$\mathbb{Q}(dx, dy) = Q(dx)K(x, dy), \qquad r = \frac{dP}{dQ}, \qquad \bar{r}(y) = \mathbb{E}_{\mathbb{Q}}[r(X) \mid Y = y].$$

Then

$$\frac{d(PK)}{d(QK)} = \bar{r}, \qquad D_{\mathrm{KL}}(PK\|QK) \le D_{\mathrm{KL}}(P\|Q), \tag{10.5}$$

and equality holds exactly when

$$r(X) = \bar{r}(Y) \quad \mathbb{Q}\text{-almost surely.} \tag{10.6}$$

[ESTABLISHED]

Proof. For bounded measurable f,

$$\int f\, d(PK) = \mathbb{E}_{\mathbb{Q}}[f(Y)r(X)] = \mathbb{E}_{\mathbb{Q}}[f(Y)\bar{r}(Y)],$$

which proves the Radon–Nikodym formula. Conditional Jensen for the strictly convex function $\phi(t) = t \log t$ gives

$$\mathbb{E}_{\mathbb{Q}}\phi(\bar{r}(Y)) \leq \mathbb{E}_{\mathbb{Q}}\phi(r(X)).$$

Finite divergence and strict convexity make equality equivalent to (10.6). This is the information-loss theorem of Csiszár [25], Kullback and Leibler [51] in kernel form. □

Corollary 10.5 (Pairwise Bayes recovery). Let $R_Q(y, dx)$ be a regular conditional law of X given $Y = y$ under $\mathbb{Q}$. Then $QKR_Q = Q$, and under (10.6), $PKR_Q = P$. Conversely, any one kernel R recovering both P and Q forces equality in (10.5). [ESTABLISHED]

Proof. Disintegration gives the first identity. Under equality,

$$(PKR_Q)(A) = \int \bar{r}(y) R_Q(y, A)(QK)(dy) = \iint \mathbf{1}_A(x) r(x) \mathbb{Q}(dx, dy) = P(A).$$

The converse follows by applying data processing through K and then R. □

Corollary 10.6 (Infinite equality carries no recovery conclusion). The finiteness hypothesis of Theorem 10.4 cannot be removed from its equality characterization. Equality $D_{\mathrm{KL}}(PK\|QK) = D_{\mathrm{KL}}(P\|Q) = +\infty$ alone need not imply a common reverse kernel recovering P and Q. [ESTABLISHED]

Proof. Let X $= \{a, b, c\}$, Y $= \{u, v\}$, and let the deterministic channel send a to u and both b, c to v. Set

$$P = \tfrac{1}{2}\delta_a + \tfrac{1}{2}\delta_b, \qquad Q = \tfrac{1}{2}\delta_b + \tfrac{1}{2}\delta_c.$$

Both fine and coarse divergences $D_{\mathrm{KL}}(P\|Q)$ and $D_{\mathrm{KL}}(PK\|QK)$ are $+\infty$, because $P(a) > 0 = Q(a)$ and $(PK)(u) > 0 = (QK)(u)$. If one reverse kernel R recovered both, $QK = \delta_v$ would force $R(v, \cdot) = Q$. But then $(PK)R = \frac{1}{2}R(u, \cdot) + \frac{1}{2}Q$ assigns at least $1/4$ to c, contradicting $P(c) = 0$. □

This recovery is pairwise. It may depend on the chosen reference Q. For a dominated experiment, one common dominating probability P_0 and simultaneous finite equalities

$$D_{\mathrm{KL}}(P_\theta K\|P_0 K) = D_{\mathrm{KL}}(P_\theta\|P_0) \quad (\theta \in \Theta)$$

make the single Bayes kernel R_{P_0} recover every P_θ. Without those simultaneous hypotheses, experiment-level sufficiency means directly that one parameter-independent R satisfies $P_\theta KR = P_\theta$ for all θ. Pairwise equality does not prove that stronger statement. [ESTABLISHED]

10.3 Fisher contraction is local

Theorem 10.7 (Score projection and Fisher loss). Let $\{P_\theta\}_{\theta \in \Theta \subseteq \mathbb{R}^d}$ be differentiable in quadratic mean at θ_0, with score $\ell_{\theta_0} \in L_0^2(P_{\theta_0}; \mathbb{R}^d)$, and let K be a parameter-independent Markov kernel normalized for every input. Under the joint law $J_{\theta_0}(dx, dy) = P_{\theta_0}(dx)K(x, dy)$, the coarse family is

differentiable in quadratic mean with score

$$\bar{\ell}_{\theta_0}(Y) = \mathbb{E}_{J_{\theta_0}}[\ell_{\theta_0}(X) \mid \sigma(Y)] \quad (P_{\theta_0}K)\text{-almost surely,} \tag{10.7}$$

Its Fisher information obeys

$$I_{\mathsf{Y}}(\theta_0) \preceq I_{\mathsf{X}}(\theta_0), \qquad I_{\mathsf{X}} - I_{\mathsf{Y}} = \mathbb{E}\,\mathrm{Cov}(\ell_{\theta_0}(X) \mid Y) \succeq 0. \tag{10.8}$$

Equality holds exactly when the score is Y-measurable. [ESTABLISHED]

Proof. Write Pollard's Le Cam DQM decomposition at θ_0 as $P_{\theta_0+h} = p_h P_{\theta_0} + P_h^{\perp}$, where $P_h^{\perp} \perp P_{\theta_0}$, $P_h^{\perp}(\mathsf{X}) = o(\|h\|^2)$, and

$$\int \left(\sqrt{p_h} - 1 - \tfrac{1}{2}h^{\top}\ell_{\theta_0}\right)^2 dP_{\theta_0} = o(\|h\|^2).$$

Choose a P_{θ_0}-null set A_h supporting $P_h^{\perp}$. The lifted joint family has the decomposition

$$J_{\theta_0+h} = p_h(x)J_{\theta_0} + J_h^{\perp}, \qquad J_h^{\perp} := P_h^{\perp}K.$$

The singular term is supported on $A_h \times \mathsf{Y}$, which is J_{θ_0}-null because the kernel is normalized, and its mass is $J_h^{\perp}(\mathsf{X} \times \mathsf{Y}) = P_h^{\perp}(\mathsf{X}) = o(\|h\|^2)$. The squared square-root remainder of the absolutely continuous term is unchanged by the lift, since integration against J_{θ_0} integrates the kernel to one. Thus the joint experiment is DQM with score $\ell_{\theta_0}(X)$, without a common dominating measure. Applying Pollard's preservation theorem for measurable statistics [75] to the deterministic statistic $(x, y) \mapsto y$ gives DQM of $P_\theta K$ and the displayed conditional score as an $L^2(P_{\theta_0}K; \mathbb{R}^d)$ equivalence class. Although the projected singular term can acquire an absolutely continuous part, its total mass remains $o(\|h\|^2)$. Taking second moments and using total covariance proves (10.8). Conditional expectation is an orthogonal projection, so equality holds precisely when the fine score is Y-measurable; in a direction v, equality holds precisely when $v^{\top}\ell_{\theta_0}(X)$ is Y-measurable. □

When the fine and coarse experiments form smooth statistical law bundles, the Markov channel is realized by a bundle morphism satisfying the exact isotropy criterion (8.56), and the fine and coarse sections satisfy the related-section condition (8.44), Theorem 8.21 pulls this same score-variance defect back to a positive-semidefinite defect between contextual-base tensors. Along one declared inference path, the corresponding record-level Fisher-clock contraction is Theorem 9.60. [ESTABLISHED]

Neither statement proves that a natural-gradient flow recomputed after coarse-graining traces the image of the fine orbit; that stronger comparison requires the oriented semiconjugacy conditions of Theorem 9.66. [NOT-CLAIMED]

Fisher equality at one parameter is local score sufficiency, not global recovery. Let $X = (A, B)$ with independent Bernoulli coordinates,

$$\Pr_\theta(A = 1) = \tfrac{1}{2} + \tfrac{\theta}{4}, \qquad \Pr_\theta(B = 1) = \tfrac{1}{2} + \tfrac{\theta^2}{4},$$

and let K discard B. At $\theta = 0$, the B-score vanishes, so the fine and coarse Fisher informations agree and are nonzero. Yet $\Pr_\theta(B = 1 \mid A)$ depends on θ, so no parameter-independent reverse kernel can recover the whole experiment. [ESTABLISHED]

10.4 The fixed-evidence ELBO consequence

Theorem 10.8 (Evidence-preserving coarse channel). Let O, X, and Y be standard Borel. Fix one normalized joint $P(do, dx)$ and fix a measurable regular-conditional kernel $o \mapsto P_o$ of X given O under P. Let $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ be a Markov channel satisfying $K(x, \mathsf{Y}) = 1$ for every $x \in \mathsf{X}$, and suppose that it does not read the arbitrary recognition law Q_o. Define

$$\bar{P}(do, dy) = \int K(x, dy)P(do, dx), \qquad \bar{P}_o := P_oK \text{ for every } o, \qquad \bar{Q}_o := Q_oK. \tag{10.9}$$

The declared kernel $o \mapsto \bar{P}_o$ is a regular conditional law of Y given O under $\bar{P}$, including its selected exceptional-point values. Fix a selected regular observation o and use the same declared evidence density representative $p(o)$ for the fine and coarse slices. Then $\bar{P}^O = P^O$ and the extended ELBO of Definition 5.6 satisfies

$$\bar{\mathcal{L}}^{\mathrm{ext}}(Q_oK; o) = \log p(o) - D_{\mathrm{KL}}(Q_oK \| P_oK) \geq \log p(o) - D_{\mathrm{KL}}(Q_o \| P_o) = \mathcal{L}^{\mathrm{ext}}(Q_o; o). \tag{10.10}$$

[ESTABLISHED]

Proof. Normalization of K preserves the observation marginal. Kernel integration makes $o \mapsto (P_oK)(B)$ measurable for every measurable $B \subseteq \mathsf{Y}$. Moreover, for measurable $A \subseteq \mathsf{O}$ and $B \subseteq \mathsf{Y}$,

$$\int_A (P_oK)(B)P^O(do) = \int_A \int_{\mathsf{X}} K(x, B)P_o(dx)P^O(do) = \bar{P}(A \times B).$$

A π–λ extension proves the regular-conditional identity. Thus the pointwise declaration $\bar{P}_o := P_oK$ selects a compatible version rather than applying an almost-sure identity at an arbitrarily fixed exceptional observation. Since the same evidence representative is used and $\bar{P}^O = P^O$, Equation (10.10) follows pointwise for the selected versions from Theorem 10.3; when $Q_o \not\ll P_o$ its rightmost value is $-\infty$, and the inequality remains valid without any undefined sum. □

Pointwise parent specialization. Fix a nonempty finite child block I, a parent label A, structural data X with $X_A = \chi_A(X)$ outside the random channel, and the unchanged observation space O. Let $\mathbb{P}_I(Do, DY \mid X)$ have a selected measurable posterior-version kernel $o \mapsto \mathbf{\Pi}_{I,o,X}$, and let $C_A : \mathsf{Y}_I \rightsquigarrow \mathsf{Z}_A$ be normalized and independent of the realized observation, recognition law, and posterior. Specializing Theorem 10.8 gives

$$\mathbb{P}_A(Do, Dz \mid X) = \int_{\mathsf{Y}_I} C_A(Y, Dz)\mathbb{P}_I(Do, DY \mid X), \qquad \mathbf{\Pi}_{A,o,X} = \mathbf{\Pi}_{I,o,X}C_A, \tag{10.11}$$

for every o in the selected version family. Indeed, for bounded measurable f on O and g on Z_A,

$$\int f(o)g(z)\mathbb{P}_A(do, dz \mid X) = \int f(o)\left[\int g(z)(\mathbf{\Pi}_{I,o,X}C_A)(dz)\right]\mathbb{P}_I^O(do \mid X), \tag{10.12}$$

which is the defining test-function identity for the displayed selected parent posterior version. This is a typed specialization of the preceding theorem, not a second posterior-pushforward theorem. The associated KL/VFE loss is the common-channel chain rule of Theorem 12.1, specialized in Theorem 12.2. [ESTABLISHED]

Thus different latent inventories are not by themselves a prohibition. ELBOs are comparable under an evidence-preserving common Markov pushforward applied to the posterior and recognition law. Absent that relation, separately declared models may have different evidences, and an ELBO difference mixes evidence with variational tightness. [ESTABLISHED]

The increase in (10.10) reflects discarded information; it is not model improvement or EM evidence ascent. [NOT-CLAIMED]

10.5 Pushforward, energy precomposition, and identification

Let $\iota : \mathsf{Z} \to \mathsf{X}$ be a measurable injection. The formula

$$\bar{E}(z) = E(\iota z), \qquad \bar{P}(dz) = \frac{e^{-\bar{E}(z)}}{\bar{Z}}\bar{\nu}(dz) \tag{10.13}$$

defines a new model only after $\bar{\nu}$ and a finite $\bar{Z}$ are declared. It is not a channel from X to Z, and its pushforward can be supported on a fine-measure-null image. Therefore KL data processing, evidence preservation, and (10.10) do not follow. [DEFINITION]

Exact marginalization is a special Markov pushforward. Hard identification is energy precomposition. Restricting a recognition family is neither: it keeps the fixed joint and optimizes the same ELBO over a subset. A proof about any one of these operations cannot be transferred to another by calling all three coarse-graining. [ESTABLISHED]

10.6 Graph-exponential energy closure

Definition 10.9 (Graph-exponential energy). Let V be finite, Z_0 standard Borel with sigma-finite ν_0, and let $t : \mathsf{Z}_0 \to \mathsf{T}_0$ and $u : \mathsf{Z}_0^2 \to \mathsf{T}_1$ be measurable, with u symmetric. Set

$$\mathcal{E}_\theta(z) = \sum_{i\in V}\langle \alpha_i, t(z_i)\rangle + \sum_{\{i,j\}}\langle \beta_{ij}, u(z_i, z_j)\rangle. \tag{10.14}$$

This is an unnormalized energy until its integral relative to $\nu_0^{\otimes V}$ is finite. [DEFINITION]

For a partition $\mathcal{P}$, define

$$\iota_{\mathcal{P}} : \mathsf{Z}_0^{\mathcal{P}} \to \mathsf{Z}_0^V, \qquad (\iota_{\mathcal{P}}\bar{z})_i = \bar{z}_{I(i)}.$$

Assume the diagonal-affinity hypothesis

$$u(z,z) = \Upsilon t(z) + u_0 \tag{10.15}$$

for a linear $\Upsilon : \mathsf{T}_0 \to \mathsf{T}_1$. [HYPOTHESIS]

This excludes pair statistics whose diagonal restriction cannot be represented by the declared node statistic. [ESTABLISHED]

Theorem 10.10 (Graph-exponential trace closure). Under (10.15),

$$\mathcal{E}_\theta(\iota_{\mathcal{P}}\bar{z}) = \sum_I \langle \alpha_I, t(\bar{z}_I)\rangle + \sum_{I<J} \langle \beta_{IJ}, u(\bar{z}_I, \bar{z}_J)\rangle + c_{\mathcal{P}}(\theta), \tag{10.16}$$

where

$$\alpha_I = \sum_{i\in I} \alpha_i + \Upsilon^* \sum_{\{i,j\}\subseteq I} \beta_{ij}, \qquad \beta_{IJ} = \sum_{\substack{i\in I\\ j\in J}} \beta_{ij}, \qquad c_{\mathcal{P}}(\theta) = \left\langle \sum_I \sum_{\{i,j\}\subseteq I} \beta_{ij}, u_0 \right\rangle. \tag{10.17}$$

The induced parameter map is linear. A normalized coarse law exists if and only if

$$\bar{Z}_{\mathcal{P}}(\theta) = \int e^{\mathcal{E}_\theta(\iota_{\mathcal{P}}\bar{z})} \nu_0^{\otimes \mathcal{P}}(d\bar{z}) < \infty. \tag{10.18}$$

[ESTABLISHED]

Proof. Substitute the identified variables into (10.14). Cut edges sum into β_{IJ}. Internal edges use (10.15); the linear part joins α_I and the constant joins $c_{\mathcal{P}}$. This gives (10.16)–(10.17). The proof is algebraic and does not prove (10.18). □

If $u(z,z)$ is constant, one may choose $\Upsilon = 0$, so internal edges are annihilated and node parameters add. Translation invariance implies this constant-diagonal property, but the converse is false because diagonal data do not determine u off the diagonal. Uniqueness of the coordinate choice $\Upsilon = 0$ requires minimal effective node statistics or quotienting null parameter directions. [ESTABLISHED]

This theorem states conditional energy closure, not normalized-law closure. Its reference measure, normalizer, and any gauge action must be propagated separately. A compatible normalized family must satisfy

$$P_\theta K = \bar{P}_{\rho(\theta)}$$

for a declared channel K and parameter map ρ; energy precomposition alone does not prove that equation. [HYPOTHESIS]

10.7 Galerkin operators and message passing

Let $P : V_c \to V_f$ be injective, let $R \succ 0$ be a fine reference form, and let L be a fine bilinear form. Define

$$R_c = P^\top RP, \qquad C = R_c^{-1} P^\top R, \qquad H = R^{-1} L, \qquad H_c = CHP = R_c^{-1} P^\top LP. \tag{10.19}$$

Then $CP = I$. The form $P^\top LP$ is the Galerkin coarse form, but H_c is the typed coarse propagator. [DEFINITION]

Proposition 10.11 (Exact message-passing residual). The reduction and propagation obey

$$CH - H_c C = CH(I - PC). \tag{10.20}$$

Hence exact coarse message passing holds on all fine signals exactly when the right side vanishes. If H is self-adjoint in the R-inner product, $\operatorname{range} P$ being H-invariant is a sufficient and, for orthogonal projection PC, equivalent condition. [ESTABLISHED]

Proof. Substitute $H_c = CHP$ and use $CP = I$:

$$CH - H_c C = CH - CHPC = CH(I - PC).$$

For an R-self-adjoint operator, invariance of a closed subspace is equivalent to reduction by its R-orthogonal projector. □

Thus exact quadratic-form restriction, generalized spectral approximation, and message-passing preservation are different statements. The first follows from Galerkin algebra; the other two require additional invariant-subspace or error hypotheses. A neural pooling layer also needs a signal map and a partition rule. No GNN or oversmoothing theorem follows from energy closure alone. [NOT-CLAIMED]

10.8 Gauge and permutation equivariance

Proposition 10.12 (Equivariant Markov channels). Let a group G act bimeasurably on X, Y, and assume

$$K(gx, B) = K(x, g^{-1}B). \tag{10.21}$$

Then

$$(g_\# P)K = g_\#(PK), \tag{10.22}$$

and composites of equivariant kernels are equivariant. [ESTABLISHED]

Proof. For measurable B,

$$[(g_{\#}P)K](B) = \int K(gx, B)P(dx) = \int K(x, g^{-1}B)P(dx) = [g_{\#}(PK)](B).$$

Substitution in (10.2) proves closure under composition. □

For independently chosen belief and model frame coordinates, the passive coordinate group is $G \times G$, although the underlying principal bundle has structure group G. A joint coarse channel must intertwine the two represented coordinate actions. Under a genuine principal gauge transformation, those actions are the ρ_b- and ρ_m-images of the same element of G. KL equality does not automatically make its Bayes recovery equivariant; an equivariant conditional version is an additional hypothesis or theorem, especially for noncompact groups. [OPEN]

A data-dependent hard partition selector $S(X)$ must also be natural under node relabeling. If P permutes fine nodes and Q permutes coarse labels,

$$S(P \cdot X) = PS(X)Q^{\top} \tag{10.23}$$

is the required law. Gauge-carrying inputs must be reduced to invariant assignment scores, while represented messages transform equivariantly. These conditions do not select a unique partition; they only type an admissible selector. Constructing a selector satisfying them and the later holonomy conditions remains open. [OPEN]

10.9 Holonomy-conditioned marginal-law modes

Fix a channel $x \in \{b, m\}$, a finite connected cluster I, and one declared internal path groupoid. The paths may be edge-labeled graph paths for typed graph links or base curves for the selected channel connection; the two cases are not identified. Let each represented transport $T^x_\gamma : \mathsf{Z}^x_i \to \mathsf{Z}^x_j$, for $\gamma : i \to j$, be a bimeasurable bijection satisfying composition and inversion. Assume that the parent-law family $\mathscr{M}^x_i \subseteq \mathcal{P}(\mathsf{Z}^x_i)$ is equivariant: $(T^x_\gamma)_{\#}\mathscr{M}^x_i = \mathscr{M}^x_j$. Under a passive section rechoice $u^{x\prime}_i = u^x_i a^x_i$, write $R^x_i = \rho_x(a^x_i)^{-1}$ for the old-to-new sample-coordinate map; equivalently, the coordinate element is $g^x_i = (a^x_i)^{-1}$ and $R^x_i = \rho_x(g^x_i)$. Also assume that $\mathscr{M}^{x\prime}_i = (R^x_i)_{\#}\mathscr{M}^x_i$. [HYPOTHESIS]

Bimeasurability excludes lossy transports, and family equivariance excludes parent classes tied to one root or frame; both hypotheses are used in the invariance proof below. [ESTABLISHED]

Choose a root $r \in I$, paths $\gamma_i : i \to r$, marginal laws $q^x_i \in \mathcal{P}(\mathsf{Z}^x_i)$, and weights $w_i > 0$ with $\sum_{i\in I} w_i = 1$. Define

$$P^x_i = (T^x_{\gamma_i})_{\#}q^x_i, \qquad \mathfrak{H}^x_I(r) = \{T^x_\lambda : \lambda : r \to r\}, \tag{10.24}$$

$$\mathscr{Q}^x_{I,\mathrm{fix}}(r) = \{Q \in \mathscr{M}^x_r : (H)_{\#}Q = Q \text{ for every } H \in \mathfrak{H}^x_I(r)\}, \tag{10.25}$$

and the extended-real forward-KL score

$$\mathfrak{D}_I^x = \inf_{Q \in \mathscr{Q}_{I,\mathrm{fix}}^x(r)} \sum_{i \in I} w_i D_{\mathrm{KL}}(P_i^x \| Q), \qquad \inf \varnothing := +\infty. \tag{10.26}$$

Any minimizer, when one exists, is a holonomy-conditioned marginal-law mode. These are definitions, not existence assertions. [DEFINITION]

Theorem 10.13 (Invariant holonomy-conditioned marginal-law mode). The value in (10.26) is independent of the chosen paths, root, and passive gauge coordinates. If the infimum is attained, then

$$\mathfrak{D}_I^x = 0 \quad \Longleftrightarrow \quad \text{there is } Q_\star \in \mathscr{Q}_{I,\mathrm{fix}}^x(r) \text{ such that } (T_{\gamma_i}^x)_\# q_i^x = Q_\star \text{ for every } i \in I. \tag{10.27}$$

The right side is precisely a holonomy-stabilized parallel marginal-law section. If $\mathscr{Q}_{I,\mathrm{fix}}^x(r) = \varnothing$, then $\mathfrak{D}_I^x = +\infty$. If $\mathfrak{D}_I^x = 0$ but the infimum is not attained, the conclusion is only forward-KL approximability by admissible fixed parent laws; no member $Q_\star$ of that family has been produced. [ESTABLISHED]

Proof. For a bimeasurable bijection A, if $P \not\ll Q$, then $A_\# P \not\ll A_\# Q$, so both divergences below are infinite. If $P \ll Q$, then

$$\frac{d(A_\# P)}{d(A_\# Q)} = \frac{dP}{dQ} \circ A^{-1} \quad A_\# Q\text{-almost surely}, \tag{10.28}$$

and integration gives

$$D_{\mathrm{KL}}(A_\# P \| A_\# Q) = D_{\mathrm{KL}}(P \| Q), \tag{10.29}$$

as an identity in $[0, +\infty]$. If γ_i' is another path from i to r, then $H_i = T_{\gamma_i'}^x \circ (T_{\gamma_i}^x)^{-1}$ belongs to $\mathfrak{H}_I^x(r)$. Hence, for every fixed parent Q,

$$D_{\mathrm{KL}}((H_i)_\# P_i^x \| Q) = D_{\mathrm{KL}}((H_i)_\# P_i^x \| (H_i)_\# Q) = D_{\mathrm{KL}}(P_i^x \| Q). \tag{10.30}$$

Thus every objective value, and therefore its infimum, is path independent.

For a new root r' and a path $\delta : r \to r'$, transport conjugates $\mathfrak{H}_I^x(r)$ onto $\mathfrak{H}_I^x(r')$. Equivariance of $\mathscr{M}^x$ therefore makes $Q \mapsto (T_\delta^x)_\# Q$ a bijection between the two fixed parent families, while the paths $\delta \circ \gamma_i$ send every P_i^x by that same bijection. Equation (10.29) preserves every term. Path independence then covers arbitrary paths to the new root.

Under the same passive section rechoices, use the old-to-new represented maps $R_i^x = \rho_x(a_i^x)^{-1}$ declared above. Then

$$T_{\gamma_i}^{x\prime} = R_r^x \circ T_{\gamma_i}^x \circ (R_i^x)^{-1}, \qquad q_i^{x\prime} = (R_i^x)_\# q_i^x. \tag{10.31}$$

Consequently all transported marginals are pushed forward by R_r^x, the holonomy group is conjugated by the same map, and parent-family equivariance carries the fixed family bijectively. A final application of (10.29) proves gauge invariance.

Relative entropy is nonnegative and vanishes exactly when its two probability measures agree. If a minimizer $Q_\star$ exists, positivity of every w_i makes its weighted sum zero exactly when $P_i^x = Q_\star$ for all i, which proves (10.27). Holonomy-fixity makes the reconstructed laws $((T_{\gamma_i}^x)^{-1})_\# Q_\star$ independent of path, so they form the stated parallel section. The empty-family assertion is the convention in (10.26). Finally, if the zero infimum is not attained, its definition supplies, for every $n \geq 1$, some $Q_n \in \mathscr{Q}_{I,\mathrm{fix}}^x(r)$ with $\sum_i w_i D_{\mathrm{KL}}(P_i^x \| Q_n) < 1/n$, and supplies no minimizer. □

The score matches marginal laws after transport. The cluster and its weights are inputs, so it does not select a partition. It supplies neither a Markov kernel nor a normalized coarse channel, and it does not recover the dependence structure of a joint law. These stronger conclusions are not claimed. [NOT-CLAIMED]

For a statistical experiment $\{P_\theta\}_{\theta\in\Theta}$, exact common recovery instead requires parameter-independent Markov kernels C and R satisfying

$$P_\theta C R = P_\theta \qquad \text{for every } \theta \in \Theta. \tag{10.32}$$

Here C, R are local kernel notation, not the Galerkin reduction and reference form in (10.19). This is the common-recovery condition, not a consequence of (10.26). [DEFINITION]

Under the richness hypothesis of Proposition 3.24, distinct joint laws can have identical coordinate marginals. Identical transported marginals therefore do not imply (10.32). [ESTABLISHED]

Corollary 10.14 (Parallel cross morphisms preserve fixed marginal-law sections). Let $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ be a cross-associated-bundle morphism on law fibers that is parallel for the chosen path system. Writing $\widehat{T}_\gamma^x = (T_\gamma^x)_\#$, this means

$$\widehat{T}_\gamma^m \circ \Phi_i = \Phi_j \circ \widehat{T}_\gamma^b \qquad (\gamma : i \to j). \tag{10.33}$$

For base-connection paths this is ordinary parallelness. For independently declared graph paths, (10.33) is an additional link-intertwining hypothesis and does not follow from smooth parallelness. Assume also $\Phi_r(\mathscr{M}_r^b) \subseteq \mathscr{M}_r^m$. [HYPOTHESIS]

Then

$$\Phi_r\big(\mathscr{Q}_{I,\mathrm{fix}}^b(r)\big) \subseteq \mathscr{Q}_{I,\mathrm{fix}}^m(r). \tag{10.34}$$

Moreover, applying Φ_i fiberwise to a holonomy-stabilized parallel belief-law section gives such a model-law section. If $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$ is likewise parallel for the chosen path system and $\widetilde{\Phi}_r(\mathscr{M}_r^m) \subseteq \mathscr{M}_r^b$, then

$$\widetilde{\Phi}_r\big(\mathscr{Q}_{I,\mathrm{fix}}^m(r)\big) \subseteq \mathscr{Q}_{I,\mathrm{fix}}^b(r), \tag{10.35}$$

and the analogous model-to-belief statement holds. [ESTABLISHED]

Proof. If $Q \in \mathscr{Q}^b_{I,\mathrm{fix}}(r)$ and $\lambda : r \to r$, then

$$\widehat{T}^m_\lambda(\Phi_r(Q)) = \Phi_r(\widehat{T}^b_\lambda(Q)) = \Phi_r(Q). \tag{10.36}$$

Parent-family compatibility proves (10.34). If the belief marginals transport to Q, the same commutative square shows that their Φ_i-images transport to $\Phi_r(Q)$. Interchanging b, m and replacing Φ by $\widetilde{\Phi}$ proves (10.35). □

This corollary does not identify $\mathcal{E}_b$ with $\mathcal{E}_m$, does not require equal fiber dimensions or equivalent representations ρ_b, ρ_m. No KL-preservation hypothesis is imposed, so no equality or ordering between $\mathfrak{D}^b_I$ and $\mathfrak{D}^m_I$ is claimed. Existence, parallelness, and parent-family compatibility of each cross morphism are additional structure. [NOT-CLAIMED]

10.10 Projective systems of laws

Theorem 10.15 (Cylinder-level projective law). Let $\mathcal{I}$ be an index set and each coordinate space Z_i standard Borel. For every finite $D \subset \mathcal{I}$, let P_D be a probability law on $\mathsf{Z}_D = \prod_{i\in D} \mathsf{Z}_i$. If

$$(\pi_{E,D})_\# P_E = P_D \qquad (D \subset E), \tag{10.37}$$

then there is a unique law on the product cylinder sigma-algebra with finite-dimensional marginals P_D. Compatible finite-dimensional gauge equivariance passes to the limit. [ESTABLISHED]

Proof. This is the Kolmogorov extension theorem for standard Borel coordinate spaces [46]. The transformed and original laws have identical finite-dimensional marginals under compatible equivariance, so uniqueness gives the last assertion. □

Consistency alone does not put the limit on continuous or smooth sections, define a continuum reference density, justify an ELBO limit, or produce an RG flow. Those conclusions require support regularity, compatible kernels and rescalings, convergence or uniform integrability of the functionals, and a declared topology. [OPEN]

Physicist summary. A general coarse-graining is a noisy detector. Detectors compose, and they cannot increase KL distinguishability or Fisher information. Equality in KL means that a reverse detector recovers the particular laws; equality in Fisher means only that the infinitesimal score survived. Restricting an energy to block-constant fields is a different construction: it is a new model until its measure and normalizer are supplied. Galerkin restriction is exact for the quadratic form, but propagation needs a mass form and has the explicit residual (10.20). Holonomy determines the admissible parallel marginal-law modes. When the infimum is attained, transported forward KL tests whether the actual marginals occupy one; without attainment, a zero infimum establishes only approximability. Neither construction selects a partition or proves joint-law recovery. These distinctions are the general spine that the later multivariate-Gaussian chapters realize.

Chapter 11

General Renormalization

Coarse arrows alone form a scale diagram. They become a renormalization dynamics only after comparison and rescaling data identify what is being compared across levels. In the present theory the state includes one principal gauge bundle, two associated statistical bundles, their possibly distinct connections, and both cross-bundle morphisms. A scalar normalization of one operator cannot stand for that full state.

11.1 The scale category and comparison data

Definition 11.1 (Scale diagram). Let S be a category whose objects are resolutions ℓ. Choose a target category $\mathscr{K}$: sets and maps for a deterministic state, standard Borel spaces and Markov kernels for a stochastic state, an explicitly declared category of topological vector spaces and continuous linear maps for an operator state, or an explicitly declared product of such categories for a mixed state. In the analytic tier used below, the operator objects are Banach spaces and the morphisms are bounded linear maps. A state functor $F : \mathsf{S} \to \mathscr{K}$ assigns an object $\mathfrak{X}_\ell = F(\ell)$ and morphisms

$$C_{\ell k} \in \mathrm{Hom}_{\mathscr{K}}(\mathfrak{X}_\ell, \mathfrak{X}_k), \qquad C_{kr} \circ C_{\ell k} = C_{\ell r}. \tag{11.1}$$

Here $\circ$ denotes abstract composition in $\mathscr{K}$, with componentwise composition in a product category. Separately, for the stochastic component only, use right-acting kernel juxtaposition

$$K_{\ell r} := K_{\ell k} K_{kr}, \qquad (\mu K_{\ell k}) K_{kr} = \mu K_{\ell r}, \tag{11.2}$$

this is the kernel representing the abstract composite $K_{kr} \circ K_{\ell k}$. One symbol is never asked to be simultaneously a map, a kernel, and an operator. [DEFINITION]

Comparison requires more data. Either choose levelwise re-embeddings $\jmath_\ell : \mathfrak{X}_{\ell+1} \to \mathfrak{X}_\ell$, or identify each level with a reference object by declared isomorphisms $I_\ell : \mathfrak{X}_\ell \xrightarrow{\sim} \mathfrak{X}_\star$ in $\mathscr{K}$. Field, base, unit, and reference-measure rescalings are part of these maps. If an externally parameterized dynamical model is separately supplied, its parameter rescaling also belongs here; the scale diagram itself introduces no time variable, and RG depth is not the intrinsic inference duration of Chapter 9. Only the isomorphism option supplies the inverse used in the reference-space transformations

$$\widehat{\mathcal{R}}_\ell = I_{\ell+1} C_{\ell,\ell+1} I_\ell^{-1} : \mathfrak{X}_\star \longrightarrow \mathfrak{X}_\star. \tag{11.3}$$

[DEFINITION]

A positive scalar ζ_ℓ can be one component of this comparison; it cannot by itself identify changing fields, measures, or bundle objects. [ESTABLISHED]

If probability laws are present, every scale declares its sample space and reference measure. A law-level arrow is a normalized Markov kernel as in Chapter 10. An energy-level arrow additionally owes the coarse reference measure, its Radon–Nikodym or Jacobian rule, and a finite normalizer. Algebraic closure does not supply those data. [HYPOTHESIS]

11.2 Exact measure-pair arrows

Definition 11.2 (Scale-indexed measure pair). For each scale ℓ, let $(\mathsf{X}_\ell, \mathscr{X}_\ell)$ be a standard-Borel space, let ρ_ℓ be a probability measure, and let $m_\ell \ll \rho_\ell$ be a finite positive measure. Write

$$m_\ell(dx) = \exp[-H_\ell(x)]\rho_\ell(dx), \qquad 0 < M_\ell := m_\ell(\mathsf{X}_\ell) < \infty, \qquad \pi_\ell = \frac{m_\ell}{M_\ell}. \tag{11.4}$$

The extended-value convention is $\exp(-\infty) = 0$; thus the action is defined only ρ_ℓ-almost everywhere and no finite action is silently assumed. A scale arrow is a normalized, parameter-independent Markov kernel

$$K_\ell : \mathsf{X}_\ell \rightsquigarrow \mathsf{X}_{\ell+1}, \qquad \rho_{\ell+1} = \rho_\ell K_\ell, \qquad m_{\ell+1} = m_\ell K_\ell. \tag{11.5}$$

The downstream action is the Radon–Nikodym representative

$$H_{\ell+1} = -\log \frac{dm_{\ell+1}}{d\rho_{\ell+1}} \quad \rho_{\ell+1}\text{-almost everywhere.} \tag{11.6}$$

This is a definition of an exact law-level RG arrow. The requirement that the spaces be standard Borel is used later to choose regular conditional versions; it does not identify distinct scale spaces or produce an autonomous operator. [DEFINITION]

Proposition 11.3 (Exact composition and normalized law). For two composable normalized kernels K_ℓ and $K_{\ell+1}$, the measure-pair arrows compose exactly:

$$(\rho_\ell, m_\ell)(K_\ell K_{\ell+1}) = \big((\rho_\ell K_\ell)K_{\ell+1}, (m_\ell K_\ell)K_{\ell+1}\big). \tag{11.7}$$

Moreover $m_{\ell+1} \ll \rho_{\ell+1}$, $m_{\ell+1}(\mathsf{X}_{\ell+1}) = M_\ell$, and

$$\pi_{\ell+1} = \pi_\ell K_\ell. \tag{11.8}$$

Thus the normalized law and the unnormalized evidence mass are distinct components of the same exact construction. [ESTABLISHED]

Proof. Kernel composition and Tonelli's theorem give (11.7). If $(\rho_\ell K_\ell)(A) = 0$, then $K_\ell(x, A) = 0$ for ρ_ℓ-almost every x, hence also for m_ℓ-almost every x, which proves absolute continuity at the next scale. Normalization of K_ℓ preserves the mass of m_ℓ; division by that same mass gives (11.8). □

The subsequent finite-network realization takes each X_ℓ to be a finite product of agent coordinate spaces and supplies the action calculus, contraction theorems, and interaction content that this general arrow does not itself supply. The exact nonlinear interaction-coordinate map, its tilted derivative, and its retained-space residual are defined only after the additional product-equivalence premises of Section 12.3; they are not linear conditional-expectation maps away from the unperturbed action. In particular, a scale sequence is a cocycle until the comparison data of Section 11.1 identify its varying spaces. [ESTABLISHED]

11.3 The common-principal geometric state

Definition 11.4 (Geometric RG state). At level ℓ, declare a smooth base $\mathcal{C}_\ell$, a Lie group G_ℓ, and one principal right G_ℓ-bundle $\varpi_\ell : \mathscr{P}_\ell \to \mathcal{C}_\ell$. Let $\mathcal{B}_{\ell,b}$ and $\mathcal{B}_{\ell,m}$ be statistical fibers with possibly different left actions $\widehat{\rho}_{\ell,b}$ and $\widehat{\rho}_{\ell,m}$ of the same group. The belief and model bundles are

$$\mathcal{E}_{\ell,b} = \mathscr{P}_\ell \times_{\widehat{\rho}_{\ell,b}} \mathcal{B}_{\ell,b}, \qquad \mathcal{E}_{\ell,m} = \mathscr{P}_\ell \times_{\widehat{\rho}_{\ell,m}} \mathcal{B}_{\ell,m}. \tag{11.9}$$

Choose two principal connections $\omega_{\ell,b}, \omega_{\ell,m} \in \Omega^1(\mathscr{P}_\ell, \mathfrak{g}_\ell)$. They induce horizontal distributions H^b_ℓ, H^m_ℓ and parallel transports $\Omega_{\ell,\gamma}$ and $\widetilde{\Omega}_{\ell,\gamma}$ on the two associated bundles. Thus the general geometric state contains

$$\mathfrak{X}_\ell = \left(\mathcal{C}_\ell;\ G_\ell, \mathscr{P}_\ell;\ \mathcal{E}_{\ell,b}, H^b_\ell;\ \mathcal{E}_{\ell,m}, H^m_\ell;\ \Phi_\ell, \widetilde{\Phi}_\ell;\ \ldots\right). \tag{11.10}$$

Here

$$\Phi_\ell : \mathcal{E}_{\ell,b} \to \mathcal{E}_{\ell,m}, \qquad \widetilde{\Phi}_\ell : \mathcal{E}_{\ell,m} \to \mathcal{E}_{\ell,b}$$

are separately declared associated-bundle morphisms over the identity. They need not be invertible or mutual inverses. A fixed fiber map $f^{bm}_\ell : \mathcal{B}_{\ell,b} \to \mathcal{B}_{\ell,m}$ induces such a morphism only when it intertwines the actions,

$$f^{bm}_\ell \circ \widehat{\rho}_{\ell,b}(g) = \widehat{\rho}_{\ell,m}(g) \circ f^{bm}_\ell \quad \text{for every } g \in G_\ell, \tag{11.11}$$

and analogously in the reverse direction. Hence a common principal bundle does not manufacture either cross map. The ellipsis in (11.10) denotes law, kernel, or operator components declared by a realization. [DEFINITION]

When levelwise agent or meta-agent sections and regular vertical Fisher tensors are also declared, the geometric state may include their connection-relative perceived semimetrics. These components are not supplied by the associated bundles alone. If a scale arrow also supplies a bundle morphism, related fine and coarse sections, and a vanishing horizontal defect, their exact cross-scale naturality

is Corollary 8.22. Its positive-semidefinite information-loss defect further requires that the fiber map be a normalized parameter-independent Markov channel. The discrete or continuous beta is defined only after the reference-base identifications in (8.67). Thus a meta-agent has its own pullback geometry, but no tensor on two different bases is subtracted before transport to the common reference space. [ESTABLISHED]

Local belief and model frames may nevertheless be chosen separately. On a common trivializing neighborhood, two sections $u^b_{\ell,i}, u^m_{\ell,i} : U_i \to \mathscr{P}_\ell$ determine a unique smooth relative principal-frame field

$$u^m_{\ell,i} = u^b_{\ell,i} \cdot h_{\ell,i}, \qquad h_{\ell,i} : U_i \to G_\ell. \tag{11.12}$$

Existence and uniqueness follow because each principal fiber is a free and transitive G_ℓ-torsor; the statement is independent of the representations. If the sections are re-chosen as $u^b_{\ell,i}a^b_{\ell,i}$ and $u^m_{\ell,i}a^m_{\ell,i}$, then $h_{\ell,i} \mapsto (a^b_{\ell,i})^{-1}h_{\ell,i}a^m_{\ell,i}$. When the representations differ, $\widehat{\rho}_{\ell,b}(h_{\ell,i})$ and $\widehat{\rho}_{\ell,m}(h_{\ell,i})$ act on different fibers; there is no matrix quotient of the associated frames and no cross-fiber identification without extra intertwining data such as (11.11). If only associated-frame data are given and either action is nonfaithful, the principal element $h_{\ell,i}$ need not be reconstructible without chosen lifts to $\mathscr{P}_\ell$. [ESTABLISHED]

Scale change must also be typed. Let $c_\ell : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$ be smooth, let $\kappa_\ell : G_\ell \to G_{\ell+1}$ be a Lie-group homomorphism, and let $\mathcal{P}_\ell : \mathscr{P}_\ell \to \mathscr{P}_{\ell+1}$ cover c_ℓ and obey

$$\mathcal{P}_\ell(p \cdot g) = \mathcal{P}_\ell(p) \cdot \kappa_\ell(g). \tag{11.13}$$

Existence of such a map is a genuine topological condition on the declared data, not a normalization: an equivariant $\mathcal{P}_\ell$ over c_ℓ exists if and only if the extended bundle $\mathscr{P}_\ell \times_{\kappa_\ell} G_{\ell+1}$ is isomorphic to the pullback $c_\ell^*\mathscr{P}_{\ell+1}$ as a principal $G_{\ell+1}$-bundle over $\mathcal{C}_\ell$. It can fail, for instance for the Hopf bundle over S^2 against the trivial bundle with c_ℓ the identity. This is a scope note on a declared datum rather than a defect of any theorem below, and every statement that uses $\mathcal{P}_\ell$ inherits it. [ESTABLISHED]

For $s \in \{b, m\}$, given a declared κ_ℓ-equivariant $\mathcal{P}_\ell$, a fiber map $q_{\ell,s} : \mathcal{B}_{\ell,s} \to \mathcal{B}_{\ell+1,s}$ descends to a bundle map if and only if

$$q_{\ell,s} \circ \widehat{\rho}_{\ell,s}(g) = \widehat{\rho}_{\ell+1,s}(\kappa_\ell(g)) \circ q_{\ell,s}. \tag{11.14}$$

It then induces

$$C_{\ell,s} : \mathcal{E}_{\ell,s} \to \mathcal{E}_{\ell+1,s}, \qquad C_{\ell,s}[p, z] = [\mathcal{P}_\ell(p), q_{\ell,s}(z)], \tag{11.15}$$

covering c_ℓ. The biconditional is a two-line computation on representatives. Two representatives of one point of $\mathcal{E}_{\ell,s}$ are (p, z) and $(pg, \widehat{\rho}_{\ell,s}(g)^{-1}z)$; their images under the displayed formula are $[\mathcal{P}_\ell(p), q_{\ell,s}(z)]$ and $[\mathcal{P}_\ell(p)\kappa_\ell(g), q_{\ell,s}(\widehat{\rho}_{\ell,s}(g)^{-1}z)]$, and the second equals $[\mathcal{P}_\ell(p), \widehat{\rho}_{\ell+1,s}(\kappa_\ell(g))q_{\ell,s}(\widehat{\rho}_{\ell,s}(g)^{-1}z)]$. These agree for every (p, z, g) exactly when (11.14) holds, which proves both directions. One degenerate case must be recorded rather than absorbed: if $\mathcal{B}_{\ell+1,s}$ reduces to a single $G_{\ell+1}$-fixed point, then (11.14) is vacuously true and is therefore not necessary in any informative sense. Two further hypotheses are consumed whenever the induced morphism is used at the information-geometric tier, and neither follows from affinity of the law pushforward on the cone of measures: family closure

of the underlying kernel, meaning that the pushed laws remain in the declared coarse model, and smoothness of $q_{\ell,s}$ between the two declared parametrized-measure models. [ESTABLISHED]

More general associated-bundle maps may be declared directly, but they owe the same transition compatibility. For a genuine multi-step scale functor, the maps c, κ, $\mathcal{P}$, and $q_{\ell,s}$ also obey their respective composition laws. Explicitly, for composable steps the required identities are $c_{\ell+1} \circ c_\ell$, $\kappa_{\ell+1} \circ \kappa_\ell$, $\mathcal{P}_{\ell+1} \circ \mathcal{P}_\ell$, $q_{\ell+1,s} \circ q_{\ell,s}$, and consequently $C_{\ell+1,s} \circ C_{\ell,s}$, with the rightmost factor acting first at every level; at the law-fiber level the composite is the pushforward of the composed kernel by the Chapman–Kolmogorov identity. Identity arrows have identity components, so these laws make the family a functor on the thin category of the finite scale set. Adjacent maps without these laws form only a sequence of coarse steps. [ESTABLISHED]

If local comparison functions a_ℓ^s satisfy

$$\mathcal{P}_\ell(u_\ell^s(x)) = u_{\ell+1}^s(c_\ell x) \cdot a_\ell^s(x),$$

then equivariance of $\mathcal{P}_\ell$ forces

$$a_\ell^b \kappa_\ell(h_\ell) = (h_{\ell+1} \circ c_\ell) a_\ell^m. \tag{11.16}$$

Thus the two represented frame channels remain compatible even when their fiber dimensions or actions differ. [ESTABLISHED]

For general nonlinear associated fibers, the cross-scale morphism defects are the ordered pairs of maps

$$\mathbf{D}_\ell^{\Phi} = (C_{\ell,m} \circ \Phi_\ell, \Phi_{\ell+1} \circ C_{\ell,b}), \qquad \mathbf{D}_\ell^{\widetilde{\Phi}} = \left(C_{\ell,b} \circ \widetilde{\Phi}_\ell, \widetilde{\Phi}_{\ell+1} \circ C_{\ell,m}\right). \tag{11.17}$$

The defect vanishes when the two components of the corresponding pair are equal. For a base curve $\gamma : x \to y$, the cross-scale connection defects are likewise the pairs

$$\begin{aligned} \mathbf{D}_{\ell,\gamma}^{b} &= (C_{\ell,b,y} \circ \Omega_{\ell,\gamma}, \Omega_{\ell+1,c_\ell\gamma} \circ C_{\ell,b,x}), \\ \mathbf{D}_{\ell,\gamma}^{m} &= \left(C_{\ell,m,y} \circ \widetilde{\Omega}_{\ell,\gamma}, \widetilde{\Omega}_{\ell+1,c_\ell\gamma} \circ C_{\ell,m,x}\right). \end{aligned} \tag{11.18}$$

Independently, within-scale nonparallelness is measured by

$$\begin{aligned} \mathbf{A}_{\ell,\gamma}^{\Phi} &= \left(\widetilde{\Omega}_{\ell,\gamma} \circ \Phi_{\ell,x}, \Phi_{\ell,y} \circ \Omega_{\ell,\gamma}\right), \\ \mathbf{A}_{\ell,\gamma}^{\widetilde{\Phi}} &= \left(\Omega_{\ell,\gamma} \circ \widetilde{\Phi}_{\ell,x}, \widetilde{\Phi}_{\ell,y} \circ \widetilde{\Omega}_{\ell,\gamma}\right). \end{aligned} \tag{11.19}$$

[DEFINITION]

Subtraction of the two components is reserved for a declared vector or affine realization. [DEFINITION]

A sufficient principal-level condition for $\mathbf{D}_{\ell,\gamma}^s$ to vanish is

$$\mathcal{P}_\ell^* \omega_{\ell+1,s} = \mathrm{d}\kappa_\ell \circ \omega_{\ell,s}, \qquad s \in \{b, m\}. \tag{11.20}$$

If a representation is not faithful, vanishing of its represented transport defect need not imply this principal-level identity. [ESTABLISHED]

The infinitesimal object measuring failure of (11.20) is the scale-connection defect form $\mathfrak{A}_{\mathcal{P}_\ell} = \mathcal{P}_\ell^* \omega_{\ell+1,s} - \mathrm{d}\kappa_\ell \circ \omega_{\ell,s}$ of (8.54), and by Theorem 8.18 the represented horizontal defect vanishes along a coarse section exactly under the isotropy condition (8.56), which is strictly weaker than (11.20) whenever the represented action fails to be infinitesimally effective there. Across composable steps these defects obey an *ordered* law, not a sum:

$$\mathfrak{A}_{\mathcal{P}_{02}} = \mathcal{P}_{01}^* \mathfrak{A}_{\mathcal{P}_{12}} + \mathrm{d}\kappa_{12} \circ \mathfrak{A}_{\mathcal{P}_{01}}, \tag{11.21}$$

whose represented form is (8.57). Writing the composite defect as an unweighted sum of the two stage defects is a type error before it is an error of fact: the two take values in different vertical bundles, the earlier one must be pushed forward by the later vertical differential, and the later one must be evaluated at the image point on the pushed base tangent. [ESTABLISHED]

Proposition 11.5 (Covariance of the defect diagrams). Under admissible local-frame changes in the belief and model associated bundles, both components of every pair in (11.17)–(11.19) acquire the same endpoint coordinate changes. Consequently the equality locus of each pair is gauge invariant. In a fiberwise-linear vector-bundle realization, subtracting the two components gives the familiar Hom-bundle defect with the same left–right law. [ESTABLISHED]

Proof. Let $\mathscr{G}_{\ell,b,x}$ and $\mathscr{G}_{\ell,m,x}$ denote the fiber coordinate changes induced by the selected belief and model frame re-trivializations at x, represented through $\widehat{\rho}_{\ell,b}$ and $\widehat{\rho}_{\ell,m}$. For example,

$$\Phi_{\ell,x} \mapsto \mathscr{G}_{\ell,m,x} \circ \Phi_{\ell,x} \circ (\mathscr{G}_{\ell,b,x})^{-1}, \quad \Omega_{\ell,\gamma} \mapsto \mathscr{G}_{\ell,b,y} \circ \Omega_{\ell,\gamma} \circ (\mathscr{G}_{\ell,b,x})^{-1},$$

and a cross-scale map obeys

$$C_{\ell,b,x} \mapsto \mathscr{G}_{\ell+1,b,c_\ell(x)} \circ C_{\ell,b,x} \circ (\mathscr{G}_{\ell,b,x})^{-1},$$

with analogous model and endpoint laws. Substitution shows that both components of $\mathbf{A}^{\Phi}_{\ell,\gamma}$ acquire the same model coordinate change at y and inverse belief coordinate change at x. The equality of the two components is therefore independent of the chosen frames. The other pairs are identical calculations. In a linear realization, subtraction factors out the common endpoint actions. □

The defect data may be retained as running fields; the theory does not require them to vanish. A numerical size requires a declared metric or divergence on each target fiber. The compositions $\widetilde{\Phi}_\ell \Phi_\ell$ and $\Phi_\ell \widetilde{\Phi}_\ell$ are independent running endomorphisms and are not identities unless imposed. Smooth parallel transport remains distinct from separately declared graph links $\Theta^b_{ij}, \Theta^m_{ij}$. [ESTABLISHED]

The default common-principal construction does not impose a common local frame or a common connection. The stronger common-frame hypothesis is $u^b_{\ell,i} = u^m_{\ell,i}$, equivalently $h_{\ell,i} = e$, on the chosen patches. Independently, the equal-connection hypothesis is $\omega_{\ell,b} = \omega_{\ell,m}$. Even then, different representations generally produce different associated transports on different fibers: they represent

the same principal holonomy but are not literally the same map. If a further associated-bundle isomorphism $\Xi_\ell : \mathcal{E}_{\ell,b} \xrightarrow{\sim} \mathcal{E}_{\ell,m}$ is declared, its existence is additional global data; equal fiber type is necessary but not sufficient in general. Its parallel and RG-preservation conditions are

$$\widetilde{\Omega}_{\ell,\gamma}\Xi_{\ell,x} = \Xi_{\ell,y}\Omega_{\ell,\gamma}, \qquad C_{\ell,m}\Xi_\ell = \Xi_{\ell+1}C_{\ell,b}.$$

The optional Ξ_ℓ is unavailable when the associated bundles are nonisomorphic, as can occur when ranks or fiber types differ. No such isomorphism is needed to define Φ_ℓ and $\widetilde{\Phi}_\ell$. [HYPOTHESIS]

Optional product-gauge extension. If an application truly requires independent gauge groups, independent principal-bundle topology, or independently acting gauge automorphisms, one may instead introduce

$$P_{\ell,b} \to \mathcal{C}_\ell, \qquad P_{\ell,m} \to \mathcal{C}_\ell, \qquad Q_\ell = P_{\ell,b} \times_{\mathcal{C}_\ell} P_{\ell,m},$$

where Q_ℓ is principal for $G_{\ell,b} \times G_{\ell,m}$. Each associated bundle then uses its own factor. This extension carries extra topological and gauge data and is not part of the minimal theory. In a category admitting the required mapping object, its belief-to-model maps are sections of

$$Q_\ell \times_{G_{\ell,b}\times G_{\ell,m}} \mathrm{Map}(\mathcal{B}_{\ell,b}, \mathcal{B}_{\ell,m}), \qquad (g_b, g_m) \cdot f = \widehat{\rho}_{\ell,m}(g_m) \circ f \circ \widehat{\rho}_{\ell,b}(g_b)^{-1}.$$

A constant fiber map descends only if it is fixed by this action, a generally restrictive condition for independently varying (g_b, g_m). Cross maps therefore still have to be supplied, and a diagonal reduction to one group requires additional group and bundle-identification data. [DEFINITION]

11.4 Autonomous maps and cocycles

Definition 11.6 (Dynamical type). The sequence $\widehat{\mathcal{R}}_\ell$ in (11.3) is: autonomous when it is one map $\widehat{\mathcal{R}}$; periodic when $\widehat{\mathcal{R}}_{\ell+p} = \widehat{\mathcal{R}}_\ell$; stationary random when it is a measurable stationary cocycle; and general otherwise. [DEFINITION]

The correct invariant object depends on this type. An autonomous fixed object satisfies $\widehat{\mathcal{R}}x = x$. A periodic scheme has a fixed object of its monodromy or an attracting p-cycle. A stationary random scheme may have a random invariant section. A general cocycle may have pullback or forward attracting sections. The term *fixed ray* additionally requires a positive cone and projectivization. [DEFINITION]

Proposition 11.7 (Uniform contraction need not give one fixed ray). Uniform Hilbert-projective contraction of a nonautonomous positive cocycle can erase initial conditions without selecting one scale-independent ray. [ESTABLISHED]

Proof. On $\mathbb{R}^2_{>0}$, let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Both are strictly positive and have a uniform projective contraction coefficient below one. Their Perron rays differ. Under the alternating cocycle, even iterates converge to the Perron ray $[1, \sqrt{2}]$ of

$$BA = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix},$$

while odd iterates converge to the distinct ray $A[1, \sqrt{2}]$. The attractor is period two, not one fixed ray. □

For maps satisfying $\tau(\widehat{\mathcal{R}}_\ell) \le q < 1$, Birkhoff contraction gives

$$d_H(\widehat{\mathcal{R}}_{\ell-1} \cdots \widehat{\mathcal{R}}_0 x, \widehat{\mathcal{R}}_{\ell-1} \cdots \widehat{\mathcal{R}}_0 y) \le q^\ell d_H(x, y), \tag{11.22}$$

where defined [16, 68]. This proves forgetting of initial rays, not time independence of the attracting section.

11.4.1 Ordered derivative cocycles and generalized cross-scale modes

A scale sequence by itself supplies no eigenvalue problem. It supplies an ordered family of maps between different spaces, and the invariant content of that family is a composition law together with compatible mode sections. This subsection fixes those types once, so that the interaction realization of Chapter 12 can instantiate them without inventing an ambient common space.

Definition 11.8 (Scale-indexed evolution and its derivative cocycle). Let $(X_\ell, \|\cdot\|_\ell)_{\ell \ge 0}$ be Banach spaces and let $T_\ell : \mathcal{U}_\ell \to X_{\ell+1}$ be continuously Fréchet differentiable on an open set $\mathcal{U}_\ell \subseteq X_\ell$. The ordered nonlinear evolution is

$$\Phi_{n \leftarrow \ell} = T_{n-1} \circ \cdots \circ T_\ell, \qquad \Phi_{\ell \leftarrow \ell} = \mathrm{id}_{X_\ell}, \tag{11.23}$$

defined at those points whose successive forward images remain in the successive domains. Along an exact orbit $x_{\ell+1} = T_\ell(x_\ell)$ contained in those domains, write

$$D_\ell = DT_\ell(x_\ell) \in \mathcal{L}(X_\ell, X_{\ell+1}), \qquad D_{n \leftarrow \ell} = D_{n-1} \cdots D_\ell, \qquad D_{\ell \leftarrow \ell} = I_{X_\ell}. \tag{11.24}$$

The rightmost factor acts first. Nothing is proved here; the display fixes domains, codomains, and order. [DEFINITION]

Proposition 11.9 (Ordered composition of the evolution and its derivative). For $\ell \le n \le r$ and an exact orbit as above,

$$\Phi_{r \leftarrow n} \circ \Phi_{n \leftarrow \ell} = \Phi_{r \leftarrow \ell}, \qquad D_{n \leftarrow \ell} = D\Phi_{n \leftarrow \ell}(x_\ell), \qquad D_{r \leftarrow n} D_{n \leftarrow \ell} = D_{r \leftarrow \ell}. \tag{11.25}$$

[ESTABLISHED]

Proof. The first identity is associativity of composition of maps. The second is proved by induction on $n-\ell$: it is the definition when $n=\ell$, and if it holds at n then the Fréchet chain rule applied to $\Phi_{n+1\leftarrow\ell}=T_n\circ\Phi_{n\leftarrow\ell}$ at x_ℓ, using $\Phi_{n\leftarrow\ell}(x_\ell)=x_n$, gives $D\Phi_{n+1\leftarrow\ell}(x_\ell)=DT_n(x_n)D_{n\leftarrow\ell}=D_nD_{n\leftarrow\ell}$. The third identity is the associativity of the ordered operator product, which is also the second identity applied to the first. □

The composition law in (11.25) is a two-parameter law, not a one-parameter semigroup. It reduces to a discrete semigroup only after declared isomorphisms $J_\ell : X_\star \to X_\ell$ make the identified maps $\widehat{T}_\ell = J_{\ell+1}^{-1}T_\ell J_\ell$ independent of ℓ; a p-periodic identified sequence produces a monodromy rather than a one-step autonomous map, in the sense of Definition 11.6. Writing a general segment as a power of one unnamed operator erases the base scale and, when the factors do not commute, also reverses the order: for $A=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $B=\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$ one has $BA=\left(\begin{smallmatrix}1&1\\1&2\end{smallmatrix}\right)$ and $AB=\left(\begin{smallmatrix}2&1\\1&1\end{smallmatrix}\right)$, which differ. [ESTABLISHED]

For a mixed state in the sense of Definition 11.1, the composition law is owed in every declared component category. Composition of one linear derivative tier proves nothing about the base, principal, law-fiber, associated-bundle, action, or configuration components. [HYPOTHESIS]

Definition 11.10 (Compatible cross-scale mode line). A rank-one mode line of index a along the orbit is a family of nonzero vectors $v_{\ell,a}\in X_\ell$ together with scalars $\lambda_{\ell,a}\in\mathbb{C}$ satisfying

$$D_\ell v_{\ell,a}=\lambda_{\ell,a}v_{\ell+1,a}. \tag{11.26}$$

This is a compatibility equation between adjacent fibers. The expression $D_\ell v=\lambda v$ is not used: its two sides inhabit $X_{\ell+1}$ and X_ℓ, which are different spaces unless an identification has been declared. For a mode subbundle of finite rank d_a, choose injective frames $V_{\ell,a}:\mathbb{C}^{d_a}\to X_\ell$ and matrices $A_{\ell,a}$ with

$$D_\ell V_{\ell,a}=V_{\ell+1,a}A_{\ell,a}. \tag{11.27}$$

[DEFINITION]

Proposition 11.11 (Ordered product law for mode coefficients). Under (11.26),

$$D_{n\leftarrow\ell}v_{\ell,a}=\lambda_{n\leftarrow\ell,a}v_{n,a},\qquad \lambda_{n\leftarrow\ell,a}=\prod_{k=\ell}^{n-1}\lambda_{k,a}, \tag{11.28}$$

with the empty product equal to one. Under (11.27) the corresponding finite-step matrix is the ordered product $A_{n-1,a}\cdots A_{\ell,a}$. If one factor vanishes, the propagated mode is annihilated from that step onward. The complex phase or real sign of each factor is retained separately from any growth exponent. [ESTABLISHED]

Proof. Induction on $n-\ell$ using (11.25) and (11.26): the case $n=\ell$ is the empty product, and $D_{n+1\leftarrow\ell}v_{\ell,a}=D_nD_{n\leftarrow\ell}v_{\ell,a}=\lambda_{n\leftarrow\ell,a}D_nv_{n,a}=\lambda_{n\leftarrow\ell,a}\lambda_{n,a}v_{n+1,a}$. The frame case is the same induction with matrix factors, whose order is fixed by (11.27). □

An ill-typed eigenvalue equation is not repaired by notation. With $X_0 = \mathbb{R}$, $X_1 = \mathbb{R}^2$, and $D_0 x = (x, 0)$, the expression $D_0 v = \lambda v$ equates an element of $\mathbb{R}^2$ with one of $\mathbb{R}$; the compatible line $v_0 = 1$, $v_1 = (1, 0)$, $\lambda_0 = 1$ is well typed and carries the same information. [ESTABLISHED]

Definition 11.12 (Cumulative log block scale and growth rates). Let $b_k > 1$ be the declared adjacent block ratios and set

$$\Delta s_k = \log b_k, \qquad s_{n\leftarrow\ell} = \sum_{k=\ell}^{n-1} \Delta s_k, \tag{11.29}$$

and assume $s_{n\leftarrow\ell} \to \infty$. For $u \in X_\ell \setminus \{0\}$ with $D_{n\leftarrow\ell} u \neq 0$, the finite-time growth rate relative to the declared norms is

$$\chi_{n\leftarrow\ell}(u) = \frac{\log\|D_{n\leftarrow\ell} u\|_n - \log\|u\|_\ell}{s_{n\leftarrow\ell}}, \tag{11.30}$$

with the convention that the rate is $-\infty$ once the image vanishes. Its upper and lower asymptotic rates are the corresponding limit superior and limit inferior; a Lyapunov exponent is asserted only when the two agree. For a mode line, the scalar finite-time exponent is

$$\upsilon_{n\leftarrow\ell,a} = \frac{\log|\lambda_{n\leftarrow\ell,a}|}{s_{n\leftarrow\ell}} = \frac{\sum_{k=\ell}^{n-1} \log|\lambda_{k,a}|}{s_{n\leftarrow\ell}}, \qquad \log 0 = -\infty. \tag{11.31}$$

A mode is called relevant, marginal, or irrelevant according as the declared asymptotic version of (11.31) is positive, zero, or negative. That trichotomy is a classification relative to the declared norms, the declared extensive normalization of the mode sections, the declared block ratios, and the declared comparison maps; it is not an intrinsic property of the scale sequence. [DEFINITION]

Proposition 11.13 (Scalar and norm growth agree for tempered mode sections). For a mode line and every $n > \ell$ with $\lambda_{n\leftarrow\ell,a} \neq 0$,

$$\chi_{n\leftarrow\ell}(v_{\ell,a}) = \upsilon_{n\leftarrow\ell,a} + \frac{\log\|v_{n,a}\|_n - \log\|v_{\ell,a}\|_\ell}{s_{n\leftarrow\ell}}. \tag{11.32}$$

The two rates therefore have the same asymptotic upper and lower values whenever the mode sections are normalized to unit norm, or more generally whenever $\log\|v_{n,a}\|_n$ is sublinear in $s_{n\leftarrow\ell}$. Under a scalar regauging $v'_{\ell,a} = c_{\ell,a} v_{\ell,a}$ with $c_{\ell,a} \neq 0$,

$$\lambda'_{\ell,a} = \lambda_{\ell,a} \frac{c_{\ell,a}}{c_{\ell+1,a}}, \qquad \lambda'_{n\leftarrow\ell,a} = \lambda_{n\leftarrow\ell,a} \frac{c_{\ell,a}}{c_{n,a}}, \tag{11.33}$$

so the scalar exponent is unchanged exactly when $|\log|c_{n,a}||/s_{n\leftarrow\ell} \to 0$. [ESTABLISHED]

Proof. Take norms in (11.28) and substitute into (11.30); the displayed remainder is exactly the endpoint norm difference divided by $s_{n\leftarrow\ell}$. Substituting $v'_{\ell,a} = c_{\ell,a} v_{\ell,a}$ into (11.26) and comparing coefficients gives the first equality in (11.33), and the telescoping product gives the second. Dividing $\log|\lambda'_{n\leftarrow\ell,a}|$ by $s_{n\leftarrow\ell}$ isolates the stated tempering condition. □

Theorem 11.14 (Exponent invariance under tempered comparison trivializations). Let $J_k : X_\star \to X_k$ be bounded linear isomorphisms with bounded inverses, and set

$$\widehat{D}_k = J_{k+1}^{-1} D_k J_k, \qquad \widehat{D}_{n\leftarrow\ell} = J_n^{-1} D_{n\leftarrow\ell} J_\ell. \tag{11.34}$$

If the complete trivializations and their inverses are tempered relative to the cumulative log block scale,

$$\frac{\log^+\|J_n\| + \log^+\|J_n^{-1}\|}{s_{n\leftarrow\ell}} \longrightarrow 0, \tag{11.35}$$

then for every $u \in X_\ell \setminus \{0\}$ the asymptotic upper and lower rates of u in the native norms coincide with those of $J_\ell^{-1}u$ computed for $\widehat{D}_{n\leftarrow\ell}$ in the reference norm. Together with the scalar tempering condition of Proposition 11.13, this preserves both the scalar mode exponent and the norm growth rate. [ESTABLISHED]

Proof. Write $x_n = D_{n\leftarrow\ell}u$ and $\widehat{x}_n = J_n^{-1}x_n = \widehat{D}_{n\leftarrow\ell}J_\ell^{-1}u$, which is the second identity in (11.34). Bounded invertibility gives

$$\frac{\|x_n\|_n}{\|J_n\|} \le \|\widehat{x}_n\|_\star \le \|J_n^{-1}\|\|x_n\|_n. \tag{11.36}$$

Taking logarithms changes the numerator of (11.30) by at most $\log^+\|J_n\| + \log^+\|J_n^{-1}\|$, and the fixed initial scale contributes the constant $|\log\|J_\ell\|| + |\log\|J_\ell^{-1}\||$. Dividing by $s_{n\leftarrow\ell}$ and using (11.35) and $s_{n\leftarrow\ell} \to \infty$ makes the difference of the two finite-time rates tend to zero, so their limits superior and inferior agree. The scalar statement is (11.33). □

Proposition 11.15 (Superexponential distortion changes the apparent exponent). Tempering the scalar mode representatives alone is not sufficient. Let $X_k = \mathbb{R}$ with the native absolute value, let every D_k be the identity, let $b_k = e$, and let every scalar gauge factor be one, so that the native exponent is zero and the scalar normalization is tempered. Take $J_k u = e^{k^2}u$. Then $\widehat{D}_k = e^{-2k-1}I$ and $\widehat{D}_{n\leftarrow 0} = e^{-n^2}I$, so the apparent reference-space rate is $-n^2/n = -n \to -\infty$. The complete comparison trivialization and its inverse, not only the mode normalization, must satisfy (11.35). [ESTABLISHED]

Proof. $\widehat{D}_k = J_{k+1}^{-1}J_k = e^{k^2-(k+1)^2} = e^{-2k-1}$, and the telescoping product over $k = 0, \dots, n-1$ is e^{-n^2}. Here $s_{n\leftarrow 0} = n$, so (11.30) gives $-n$, while the native rate is zero. Condition (11.35) fails because $\log\|J_n\| = n^2$. □

The definitions above are deliberately weaker than an existence statement. A multiplicative ergodic theorem would supply a measurable splitting into finitely or countably many exponent subspaces, but it requires data this scale diagram does not contain: a measure-preserving base flow, measurability of the cocycle in the base variable, integrability of $\log^+\|D\|$ and, for the two-sided splitting, of $\log^+\|D^{-1}\|$, and either finite dimension or a compactness or quasi-compactness hypothesis in the infinite-dimensional case [4]. A deterministic scale sequence supplies none of these by itself. This document therefore defines the growth rates above and does not assert an Oseledets splitting for the RG cocycle. [NOT-CLAIMED]

11.5 Scheme dependence and invariant candidates

Let $g(t)$ be running coordinates. Under a smooth reparameterization $g' = f(g)$ and scale change $t' = ct$,

$$\beta'(g') = \frac{1}{c} Df(g)\beta(g). \tag{11.37}$$

[ESTABLISHED]

Hence raw couplings, beta-function components, and a chosen blocking statistic are scheme dependent. They remain valid diagnostics when their scheme is stated, but are not universal observables. [ESTABLISHED]

For a differentiable homogeneous positive endomorphism at a fixed ray, let μ_0 be the radial eigenvalue and μ_a a transverse eigenvalue. The projective ratios and block-normalized dimensions are

$$\rho_a = \frac{\mu_a}{\mu_0}, \qquad \upsilon_a = \frac{\log|\rho_a|}{\log b}, \tag{11.38}$$

for declared block factor b. [DEFINITION]

They are invariant under smooth coordinate conjugacy at the ray once gauge, normalization, and other redundant directions have been quotiented and the same scale normalization is used. [ESTABLISHED]

Candidate cross-scheme quantities include conjugacy classes, invariant manifolds, projective eigenvalue ratios, dimensionless amplitude ratios, and matched spectral or heat exponents. Their existence, convergence, and completeness must be proved in the chosen realization. Universality is a statement about basins and invariant observables, not algebraic closure alone. [HYPOTHESIS]

11.6 Two-index thermodynamic and RG limits

A finite-volume RG family has two indices:

$$X_{n,\ell}, \qquad 0 \le \ell \le L(n), \tag{11.39}$$

where n controls system size and ℓ RG depth. Fixed-depth, maximal-depth, and diagonal limits are different:

$$\lim_{\ell\to\infty}\lim_{n\to\infty} X_{n,\ell}, \qquad \lim_{n\to\infty} X_{n,L(n)}, \qquad \lim_{n\to\infty} X_{n,\ell(n)}. \tag{11.40}$$

[DEFINITION]

Proposition 11.16 (Fixed-depth and maximal-depth limits need not agree). Fix an integer $b \ge 2$. For $n \ge 0$, put $m = b^n$ and let Path_m be the unweighted path on $\{1,\dots,m\}$, with combinatorial Laplacian L_m. At depth $0 \le \ell \le n$, form the consecutive blocks

$$B_j^{(\ell)} = \{(j-1)b^\ell + 1, \dots, jb^\ell\}, \qquad 1 \le j \le b^{n-\ell}. \tag{11.41}$$

Join two blocks exactly when a path edge crosses between them. The resulting simple quotient is exactly $\mathrm{Path}_{b^{n-\ell}}$. For any m divisible by b, let the unnormalized contiguous block-indicator prolongation be

$$S_{m,b} \in \mathbb{R}^{m\times(m/b)}, \qquad (S_{m,b})_{ij} = \mathbf{1}_{\{(j-1)b+1,\dots,jb\}}(i). \tag{11.42}$$

It realizes the exact operator quotient

$$S_{m,b}^{\mathsf{T}} L_m S_{m,b} = L_{m/b}. \tag{11.43}$$

Set $S_{m,1} = I_m$. At depth $1 \le \ell \le n$, define

$$S_{m,b^\ell} := S_{m,b} S_{m/b,b} \cdots S_{m/b^{\ell-1},b}, \tag{11.44}$$

this is the indicator of the blocks in (11.41) and obeys

$$S_{m,b^\ell}^{\mathsf{T}} L_m S_{m,b^\ell} = L_{m/b^\ell}. \tag{11.45}$$

The identities use unnormalized indicators. With orthonormalized one-step indicators $\widetilde{S}_{m,b} = b^{-1/2} S_{m,b}$, the left side of (11.43) is instead $b^{-1} L_{m/b}$, and the depth-ℓ factor is $b^{-\ell}$.

For every $m \ge 1$, the eigenvalues of L_m, including multiplicity, are

$$\lambda_{m,k} = 2 - 2\cos\left(\frac{\pi k}{m}\right), \qquad 0 \le k \le m-1. \tag{11.46}$$

Let

$$\mu_m = \frac{1}{m}\sum_{k=0}^{m-1} \delta_{\lambda_{m,k}}, \qquad N_m(\lambda) = \mu_m((-\infty,\lambda]), \qquad U_m(\lambda) = \frac{1}{m}\#\{1 \le k \le m-1 : \lambda_{m,k} \le \lambda\}. \tag{11.47}$$

These are right-continuous full and zero-mode-excluded counts, and $U_m = N_m - m^{-1}\mathbf{1}_{[0,\infty)}$. More explicitly, for $0 \le \lambda \le 4$, set $\theta(\lambda) = \arccos(1-\lambda/2)$; then

$$N_m(\lambda) = \frac{1 + \min\{m-1, \lfloor m\theta(\lambda)/\pi \rfloor\}}{m}, \qquad U_m(\lambda) = \frac{\min\{m-1, \lfloor m\theta(\lambda)/\pi \rfloor\}}{m}. \tag{11.48}$$

Both counts vanish for $\lambda < 0$; for $\lambda > 4$, they equal 1 and $1 - m^{-1}$, respectively. Specifically, for $m \ge 2$, $N_m(\lambda) = m^{-1}$ on $[0, \lambda_{m,1})$, while U_m is zero there and jumps to m^{-1} at the included threshold $\lambda_{m,1}$; at the included upper endpoint $\lambda_{m,m-1}$, the counts are 1 and $1 - m^{-1}$.

For every fixed ℓ, $b^{n-\ell} \to \infty$ as $n \to \infty$, and both counts converge pointwise to the right-continuous function

$$N_\infty(\lambda) = U_\infty(\lambda) = \begin{cases} 0, & \lambda < 0, \\ \dfrac{1}{\pi}\arccos\left(1 - \dfrac{\lambda}{2}\right), & 0 \le \lambda \le 4, \\ 1, & \lambda > 4. \end{cases} \tag{11.49}$$

Thus the zero-mode-excluded lower-edge count obeys

$$U_\infty(\lambda) = \frac{1}{\pi}\arccos\left(1-\frac{\lambda}{2}\right) \sim \frac{1}{\pi}\sqrt{\lambda} \qquad (\lambda \downarrow 0). \tag{11.50}$$

At maximal depth $\ell = n$, however, the exact quotient is Path_1 for every n, so $\mu_1 = \delta_0$, $N_1 = \mathbf{1}_{[0,\infty)}$, and $U_1 \equiv 0$. Hence the fixed-depth thermodynamic limit and the maximal-depth Path_1 limit are different. [ESTABLISHED]

Proof. Each block in (11.41) is connected, and exactly one original edge joins block j to block $j+1$, with no other interblock edge; this proves the exact quotient statement. More explicitly, for every $z \in \mathbb{R}^{m/b}$,

$$z^{\mathsf{T}} S_{m,b}^{\mathsf{T}} L_m S_{m,b} z = \sum_{i=1}^{m-1} \big((S_{m,b}z)_{i+1} - (S_{m,b}z)_i\big)^2 = \sum_{j=1}^{m/b-1} (z_{j+1}-z_j)^2 = z^{\mathsf{T}} L_{m/b} z.$$

Equality of the symmetric quadratic forms proves (11.43); iteration proves (11.45), and scalar normalization gives the stated factors. The discrete cosine eigenvectors of the path Laplacian give (11.46). Monotonicity of $k \mapsto \lambda_{m,k}$ then gives (11.48), including its threshold and upper endpoint. For every bounded continuous f,

$$\int f \mathrm{d}\mu_m = \frac{1}{m}\sum_{k=0}^{m-1} f\left(2-2\cos\frac{\pi k}{m}\right) \longrightarrow \frac{1}{\pi}\int_0^\pi f(2-2\cos\theta)\mathrm{d}\theta.$$

The limiting distribution has the continuous cumulative function in (11.49); removing the one zero atom changes a finite count by m^{-1} and hence has the same limit. Finally, $\arccos(1-\lambda/2) \sim \sqrt{\lambda}$, while $L_1 = 0$. □

Therefore a thermodynamic RG claim must specify a triangular family, projective consistency at fixed depth, uniform control of the coarse error, and a diagonal scaling $\ell(n)$. Operator, law, connection, and cross-morphism convergence are separate obligations. Proving an exchange of limits or a nontrivial diagonal limit remains open. [OPEN]

Put $U(\lambda) = N(\lambda) - N(0)$. Suppose U is nondecreasing and right-continuous, $U(0+) = 0$, $Z_+(t_0) < \infty$ for some $t_0 > 0$, and, for $c > 0$, $\alpha > 0$, and a function L slowly varying at infinity,

$$U(\lambda) \sim c\lambda^\alpha L(1/\lambda) \qquad (\lambda \downarrow 0).$$

Define the zero-mode-excluded heat moments

$$Z_+(t) = M_0(t) := \int_{(0,\infty)} e^{-t\lambda}\, dU(\lambda), \qquad M_k(t) := \int_{(0,\infty)} \lambda^k e^{-t\lambda}\, dU(\lambda).$$

The Abelian/Karamata direction of the regular-variation theorem [15, §1.7.1] gives

$$M_k(t) \sim c\alpha\Gamma(\alpha+k)t^{-(\alpha+k)}L(t) \qquad (t \to \infty;\ k = 0,1,2).$$

With $\pi_t(d\lambda) = e^{-t\lambda} dU(\lambda)/Z_+(t)$ and $S_+(t) = \log Z_+(t) + tM_1(t)/Z_+(t)$, the spectral susceptibility obeys

$$-\frac{dS_+}{d\log t} = t^2 \left(\frac{M_2}{Z_+} - \left(\frac{M_1}{Z_+} \right)^2 \right) \longrightarrow \alpha. \tag{11.51}$$

This is the Abelian forward direction; the converse Tauberian direction is not asserted here. [ESTABLISHED]

Dividing entropy by $\log |V_n|$ divides the susceptibility by the same factor and generally destroys a nonzero thermodynamic plateau. Zero-mode atoms and normalization must therefore be declared before interpreting a plateau as spectral dimension. [ESTABLISHED]

11.7 Projective laws and refinement

Compatible finite-dimensional laws obey the projective theorem of Section 10.10. An energy recursion alone does not provide those laws. A probabilistic RG needs normalized kernels $K_{\ell k}$ with

$$\mu_k = \mu_\ell K_{\ell k}, \qquad K_{\ell k} K_{kr} = K_{\ell r},$$

and corresponding rules for the two bundle sectors and their reference measures. Smooth-section support, continuum ELBO convergence, and connection convergence remain additional open obligations. [OPEN]

If the coarse map Q is noninjective, no deterministic two-sided inverse exists. A refinement kernel R can at most be a stochastic right section,

$$Q_\# R(c, \cdot) = \delta_c, \tag{11.52}$$

which redraws a compatible microstate. It cannot recover every discarded realized microstate. [ESTABLISHED]

Infinite divisibility may construct such refinements in distribution, but does not turn a many-to-one coarse-graining into a group inverse. [ESTABLISHED]

11.8 Typed comparison with Bayesian renormalization

Berman, Klinger, and Stapleton build on the dynamical Bayesian-inference flow of Berman et al. [12], using a centered-KL posterior flow on model-parameter space. In the regular late-data regime they approximate the posterior by a Gaussian with covariance $T^{-1} I^{-1}$. Set $\tau = 1/T$, and consider an inverse problem with a differentiable forward map G. The delta method gives the leading local predictive approximation

$$G(\theta) \mathrel{\dot\sim} \mathcal{N}\left(G(\mu_\tau), \quad \tau J_G(\mu_\tau) I(\mu_\tau)^{-1} J_G(\mu_\tau)^\top \right).$$

It is exact when G is affine, but a nonlinear Gaussian pushforward is not in general Gaussian. Accordingly, the pushed inverse Fisher cometric

$$D_\tau = J_G(\mu_\tau) I(\mu_\tau)^{-1} J_G(\mu_\tau)^\top \tag{11.53}$$

is the local diffusion tensor used in their ERG-form Fokker–Planck equation; the corresponding small-τ transition covariance is τD_τ [13]. Their continuation of this tensor beyond the late-data, small-τ regime is a renormalization construction, not an exact nonlinear pushforward theorem or a Bayesian asymptotic theorem. [ESTABLISHED]

This is an exact, carefully scoped literature statement. It does not identify an arbitrary recognition Fisher tensor with the BKS generative-model Fisher metric, transfer their stiff/sloppy eigenspaces to another operator sector, or prove that a graph aggregation runs in the opposite direction. [NOT-CLAIMED]

In the small-τ approximation, covariance broadens like τI^{-1} if I is held fixed, whereas a later Gaussian sufficient-statistic construction may add precision. That is a typed local contrast. A theorem of opposite flow directions would require a common state space, an explicit scale map, and a transported monotone including

$$\frac{d}{d\tau}\left[\tau I(\mu_\tau)^{-1}\right] = I^{-1} - \tau I^{-1} \frac{dI}{d\tau} I^{-1}.$$

No such bridge is asserted. [OPEN]

Theorem 10.13 gives a separate admissibility statement: represented holonomy restricts the marginal-law modes available inside an already proposed block. The BKS Fisher scale instead orders model-parameter directions by statistical distinguishability; it neither chooses node partitions nor tests holonomy. [ESTABLISHED]

Constructing a nondegenerate partition selector compatible with these separate structures remains open. [OPEN]

Mehta and Schwab give an exact variational-RG/deep-network correspondence under their RBM kernel and trace condition [60]. It is a historical exact precedent inside that construction, not a theorem that deterministic energy precomposition is a neural layer. Likewise, scalar Laplacian RG provides a heat-spectrum scale diagnostic and a heuristic real-space blocker; lifting its entrywise blocker covariantly to the two represented frame channels requires a new invariant statistic. These are constraints on applications, not definitions of the present general RG. [ESTABLISHED]

Physicist summary. Blocking removes detail; renormalization adds the maps that say how the blocked theory is compared with the original units. Here that comparison must carry one principal bundle, its two differently represented statistical bundles, two connection choices that may coincide, and two separately supplied cross morphisms. Separate belief and model frames are local views of the same principal geometry; their relative element lies in G_ℓ, even when its two representations act on unlike spaces. Failures to commute with blocking are covariant defect fields. Once all scales

are identified, the dynamics may be autonomous, periodic, random, or a general cocycle, and each case has a different invariant object. System size and RG depth then form a two-index limit. Fixed points, universality, and continuum laws are meaningful only after these types and limits have been declared.

Chapter 12

Exact Renormalization of Agent Networks

This chapter constructs an effective variational theory for the interaction model of Chapter 7. Its primary objects are normalized measures and Markov kernels. Actions, couplings, and differential beta functions appear only after a reference and a scale identification have been declared. Exact closure is obtained by retaining induced hyperedges, marked attention events, complete holonomy data, and path-space memory. A pairwise, memoryless model with one scalar attention row and one averaged group link is therefore a truncation, not the general effective theory.

12.1 Law-level coarse-graining and the VFE identity

Let $P(do, dy)$ be the fixed normalized joint law on standard-Borel spaces, let o be regular, let $\Pi_o(dy)$ be its posterior, and let $Q_o \ll \Pi_o$ be a recognition law. A coarse channel between standard-Borel latent spaces is a fixed Markov kernel

$$C : \mathsf{Y} \rightsquigarrow \mathsf{Z} \tag{12.1}$$

that does not read Q_o and does not alter the observation coordinate. Define

$$P^c(do, dz) = \int_{\mathsf{Y}} C(y, dz) P(do, dy), \qquad \Pi_o^c = \Pi_o C, \qquad Q_o^c = Q_o C. \tag{12.2}$$

Theorem 12.1 (Exact coarse VFE and discarded conditional information). Let the fine and coarse VFE expressions be defined as extended-real sums, with the common finite evidence term separated from their KL gaps, and let

$$\widehat{Q}_o(dy, dz) = Q_o(dy) C(y, dz), \qquad \widehat{\Pi}_o(dy, dz) = \Pi_o(dy) C(y, dz). \tag{12.3}$$

Then P^c has the same observation evidence as P, and the extended-real additive identity is

$$\mathcal{F}_P(Q_o) = \mathcal{F}_{P^c}(Q_o^c) + \int_{\mathsf{Z}} D_{\mathrm{KL}}\big(\widehat{Q}_o(dy \mid z) \| \widehat{\Pi}_o(dy \mid z)\big) Q_o^c(dz). \tag{12.4}$$

In particular, the fine VFE is at least the coarse VFE. If the fine KL is finite, their ordinary real-valued difference equals the displayed conditional KL. The corresponding coarse ELBO rises only because a conditional inference gap has been discarded; the evidence has not increased. [ESTABLISHED]

Proof. The observation marginal is unchanged because $C(y, \mathsf{Z}) = 1$. Attaching the same channel to both measures preserves the fine relative entropy, $D_{\mathrm{KL}}(Q_o \| \Pi_o) = D_{\mathrm{KL}}(\widehat{Q}_o \| \widehat{\Pi}_o)$. Disintegration on z and the relative-entropy chain rule split the latter into $D_{\mathrm{KL}}(Q_o^c \| \Pi_o^c)$ plus the integral in (12.4). Substitute the exact evidence identities at the two scales. □

The theorem does not cover different generative and recognition channels, a fitted rather than pushed-forward P^c, or a simultaneous coarse-graining of the observation. Such operations generally change the comparison target or the evidence event. An invertible or evidence-sufficient observation channel is an exception, but its preserved event and posterior must be proved explicitly. [ESTABLISHED]

Theorem 12.2 (Full pointwise probabilistic datum for a candidate parent). Fix a nonempty finite child block I, a parent label A, the common neighborhood

$$U_A = \bigcap_{i \in I} \mathcal{C}_i, \qquad c_* \in U_A, \tag{12.5}$$

and structural data $X = X(c_*)$ with $X_A = \chi_A(X)$. Every random space and law in this theorem is evaluated over the single fiber above c_*; no family over U_A is being asserted. Let $\mathsf{O}, \mathsf{Y}_{I,c_*}, \mathsf{B}_{A,c_*}, \mathsf{M}_{A,c_*}, \mathbf{\Xi}_{A,c_*}$, and H_{A,c_*} be nonempty standard-Borel spaces, and put

$$\mathsf{Z}_{A,c_*} = \mathsf{B}_{A,c_*} \times \mathsf{M}_{A,c_*} \times \mathbf{\Xi}_{A,c_*} \times \mathsf{H}_{A,c_*}. \tag{12.6}$$

Keep X and X_A outside the random channel. Supply the fixed normalized generative joint

$$\mathbb{P}_I(Do, DY \mid X),$$

and its selected measurable posterior-version kernel $o \mapsto \mathbf{\Pi}_{I,o,X}$. At an admitted o with a finite positive evidence representative, supply a normalized correlated recognition law $\mathbb{Q}_{I,o,X} \ll \mathbf{\Pi}_{I,o,X}$. Declare one normalized measurable vertical Markov-kernel channel over the identity on c_*:

$$C_{A,c_*} : \mathsf{Y}_{I,c_*} \rightsquigarrow \mathsf{Z}_{A,c_*}. \tag{12.7}$$

Equivalently, its source and target are the evaluated fibers over c_* and its base projection is id_{c_*}. It is fixed independently of recognition, posterior, recognition parameters, and realized observation. It acts only on the fine random variables and leaves the observation coordinate in O unchanged.

Define

$$\mathbb{P}_A(Do, Dz \mid X) = \int_{\mathsf{Y}_{I,c_*}} C_{A,c_*}(Y, Dz)\mathbb{P}_I(Do, DY \mid X), \tag{12.8}$$

$$\mathbf{\Pi}_{A,o,X} = \mathbf{\Pi}_{I,o,X}C_{A,c_*}, \tag{12.9}$$

$$\mathbb{Q}_{A,o,X} = \mathbb{Q}_{I,o,X}C_{A,c_*} \quad \text{(same normalized Markov-kernel channel).} \tag{12.10}$$

Then all three parent laws are normalized, the parent observation marginal equals the fine one, $o \mapsto \mathbf{\Pi}_{A,o,X}$ is the selected parent posterior-version kernel, and $\mathbb{Q}_{A,o,X} \ll \mathbf{\Pi}_{A,o,X}$.

Declare a jointly measurable normalized parent evaluator

$$\mathrm{ev}_{A,c_*} : \mathsf{M}_{A,c_*} \longrightarrow \mathrm{Kern}(\mathbf{\Xi}_{A,c_*}, \mathsf{B}_{A,c_*} \times \mathsf{O} \times \mathsf{H}_{A,c_*}), \qquad m_A \longmapsto K^{X_A}_{A,m_A}. \tag{12.11}$$

It is compatible with the parent generative law either when it is the family induced by a selected standard-Borel disintegration, or, for a family declared independently of $\mathbb{P}_A$, under the separate almost-sure compatibility hypothesis

$$\mathbb{P}_A(db_A, do, dh_A \mid \xi_A, m_A, X) = K^{X_A}_{A,m_A}(\xi_A; db_A, do, dh_A), \tag{12.12}$$

for the parent (M_{A,c_*}, Ξ_{A,c_*}) marginal almost surely. Disintegration proves existence of the induced tier; it does not validate an arbitrary predeclared evaluator, select its null extension canonically, make evaluation injective, or regularize a quotient of presentations.

With $\mathbb{P}^Z_A(D \mid X) = \mathbb{P}_A(\mathsf{O} \times D \mid X)$ and locally fixed (o, X), the parent recognition, prior, and posterior marginals are the coordinate pushforwards

$$q^b_A = (\mathrm{pr}_b)_\# \mathbb{Q}_{A,o,X}, \qquad q^m_A = (\mathrm{pr}_m)_\# \mathbb{Q}_{A,o,X}, \tag{12.13}$$

$$p^b_A = (\mathrm{pr}_b)_\# \mathbb{P}^Z_A(\cdot \mid X), \qquad p^m_A = (\mathrm{pr}_m)_\# \mathbb{P}^Z_A(\cdot \mid X), \tag{12.14}$$

$$\mathbf{\Pi}^b_{A,o,X} = (\mathrm{pr}_b)_\# \mathbf{\Pi}_{A,o,X}, \qquad \mathbf{\Pi}^m_{A,o,X} = (\mathrm{pr}_m)_\# \mathbf{\Pi}_{A,o,X}. \tag{12.15}$$

Under the richness hypothesis of Proposition 3.24 — that the relevant parent coordinate space carry at least two nondegenerate coordinates — no displayed marginal pair reconstructs the corresponding full law. The hypothesis is necessary and not decorative: if Z_{A,c_*} degenerates to a single point, its unique law is trivially determined by its marginals, and the released finite witnesses exhibit nonreconstruction only in the nondegenerate case.

Apply the same normalized Markov-kernel channel C_{A,c_*} to the fine recognition and posterior laws and disintegrate both lifts in Y conditional on z. The additive identity in $[0, +\infty]$ is

$$D_{\mathrm{KL}}(\mathbb{Q}_{I,o,X} \| \mathbf{\Pi}_{I,o,X}) = D_{\mathrm{KL}}(\mathbb{Q}_{A,o,X} \| \mathbf{\Pi}_{A,o,X}) + \Delta_A(o, X), \tag{12.16}$$

where

$$\Delta_A(o, X) = \int_{\mathsf{Z}_{A,c_*}} D_{\mathrm{KL}}\left(\widehat{\mathbb{Q}}_{I,o,X}(dY \mid z) \middle\| \widehat{\mathbf{\Pi}}_{I,o,X}(dY \mid z)\right) \mathbb{Q}_{A,o,X}(dz) \geq 0. \tag{12.17}$$

Adding the same finite real $-\log p_X(o)$ to both KL terms gives the extended-real identity $\mathcal{F}_I = \mathcal{F}_A + \Delta_A$; when finite, a VFE may be negative. Without a finite-fine-KL premise, $\Delta_A = 0$ exactly when the two discarded conditional laws in (12.17) agree $\mathbb{Q}_{A,o,X}$-almost surely. Finite fine KL is required for the ordinary subtraction $\mathcal{F}_I - \mathcal{F}_A = \Delta_A$ and for the stated two-way pairwise common-recovery equivalence. Family-wide common recovery requires simultaneous hypotheses for every family member. A bare equality $+\infty = +\infty$ supplies neither the zero-defect criterion nor a recovery consequence. [ESTABLISHED]

Proof. Normalization and the observation marginal. Since $C_{A,c_*}(Y, \mathsf{Z}_{A,c_*}) = 1$ for every $Y \in \mathsf{Y}_{I,c_*}$, integrating (12.8) against the constant function gives $\mathbb{P}_A(\mathsf{O} \times \mathsf{Z}_{A,c_*} \mid X) = 1$, and the same computation applied to $\mathbf{\Pi}_{I,o,X}$ and $\mathbb{Q}_{I,o,X}$ normalizes the other two laws in (12.10). Testing (12.8) with a bounded measurable function of o alone gives $\mathbb{P}_A^{\mathsf{O}} = \mathbb{P}_I^{\mathsf{O}}$, because C_{A,c_*} leaves the observation coordinate unchanged.

Posterior version. Let f be bounded measurable on Z_{A,c_*} and g bounded measurable on O, and put $h(Y) = \int f(z) C_{A,c_*}(Y, dz)$, which is bounded and measurable because C_{A,c_*} is a kernel. Then

$$\int g\Big(\int f d\mathbf{\Pi}_{A,o,X}\Big)\mathbb{P}_A^{\mathsf{O}}(do) = \int g\Big(\int h d\mathbf{\Pi}_{I,o,X}\Big)\mathbb{P}_I^{\mathsf{O}}(do) = \int g(o)h(Y)\mathbb{P}_I(do, dY \mid X) = \int g(o)f(z)\mathbb{P}_A(do, dz \mid X),$$

using the preserved observation marginal, then the defining property of the selected fine version, then (12.8). Hence $o \mapsto \mathbf{\Pi}_{A,o,X}$ is a selected parent posterior version, including at its declared exceptional points. The step uses that C_{A,c_*} acts on the conditioned variable only; a channel that also coarsened O would not descend.

Absolute continuity. If $\mathbf{\Pi}_{A,o,X}(N) = 0$ then $\int C_{A,c_*}(Y, N)\mathbf{\Pi}_{I,o,X}(dY) = 0$, so $C_{A,c_*}(\cdot, N) = 0$ holds $\mathbf{\Pi}_{I,o,X}$-almost surely, hence $\mathbb{Q}_{I,o,X}$-almost surely by $\mathbb{Q}_{I,o,X} \ll \mathbf{\Pi}_{I,o,X}$, and therefore $\mathbb{Q}_{A,o,X}(N) = \int C_{A,c_*}(Y, N)\mathbb{Q}_{I,o,X}(dY) = 0$.

Evaluator. On standard-Borel spaces a selected disintegration of $\mathbb{P}_A(\cdot \mid X)$ over (M_{A,c_*}, Ξ_{A,c_*}) exists and is jointly measurable, which supplies the induced tier of (12.11) together with (12.12) by construction. For a family declared independently of $\mathbb{P}_A$, (12.12) is the stated separate hypothesis; disintegration alone does not supply it.

Marginals. Equations (12.13) and the accompanying prior and posterior displays are coordinate pushforwards by definition, obtained by testing each parent law with functions of a single coordinate.

Chain identity and defect. Form the joint lifts of (12.3) with $C = C_{A,c_*}$, $Q_o = \mathbb{Q}_{I,o,X}$ and $\Pi_o = \mathbf{\Pi}_{I,o,X}$. Applying Theorem 12.1 verbatim yields (12.17) and the extended-real additive identity, every term lying in $[0, +\infty]$ with no infinite quantity subtracted. Adding the common finite real $-\log p_X(o)$ to both KL terms gives $\mathcal{F}_I = \mathcal{F}_A + \Delta_A$ in the extended reals; the sum of a finite real and a nonnegative extended real is no longer nonnegative, which is why a finite VFE may be negative.

Zero defect. Δ_A is the integral of a nonnegative measurable integrand against $\mathbb{Q}_{A,o,X}$, so it vanishes exactly when that integrand vanishes $\mathbb{Q}_{A,o,X}$-almost surely; the integrand is a relative entropy, which vanishes exactly when its two arguments agree. No finiteness hypothesis enters here.

Finite tier. If the fine KL is finite then both right-hand terms are finite, so the subtraction $\mathcal{F}_I - \mathcal{F}_A = \Delta_A$ is an ordinary difference of reals. On that tier the selected posterior reverse kernel recovers both declared laws under $\Delta_A = 0$; conversely, one normalized kernel R recovering both laws sandwiches the fine and coarse KL values by data processing through C_{A,c_*} and then through R, forcing equality and hence $\Delta_A = 0$. This is the equality case of data processing, i.e. sufficiency of C_{A,c_*} for the pair, and it is pairwise: a single kernel serving an entire family requires the hypothesis for every member simultaneously. If both KL values are $+\infty$ the equality is vacuous and yields neither conclusion. □

This theorem constructs a full normalized, generally lossy probabilistic datum only at the fixed base point c_*. The vertical coarse channel and child membership are declared inputs, not canonical selections. It does not construct a neighborhood family over U_A, so it establishes neither neighborhood gluing nor connection compatibility. It also supplies no base coarse map f_ℓ, no section-level RG, no parent local section, geometric meta-agent, autonomous dynamics, agency, nonequilibrium persistence, continuum limit, physical time, unique latent DAG, unique microscopic physics, or ontology. Those are separate obligations. [NOT-CLAIMED]

12.2 The exact effective likelihood and action

Fix one adjacent scale arrow and suppress its index temporarily. Its fine space is a finite product of standard-Borel agent spaces, so it is itself standard Borel. Let $\rho = P_0$ be a probability reference and let

$$m_o(dy) = L_o(y)\rho(dy), \qquad L_o(y) = \exp[-H_o(y)], \qquad 0 < M_o := m_o(\mathsf{Y}) < \infty, \qquad \pi(dy) = M_o^{-1} m_o(dy) \tag{12.18}$$

be the unnormalized evidence submeasure, its normalized law, and its strictly positive finite mass. The action may be extended valued; the displayed density convention is $e^{-\infty} = 0$ and all equalities involving actions are therefore almost-everywhere statements. Push both measures through the normalized, recognition-independent channel C:

$$\rho^c = \rho C, \qquad m_o^c = m_o C, \qquad \pi^c = \pi C = M_o^{-1} m_o^c. \tag{12.19}$$

Absolute continuity $m_o^c \ll \rho^c$ follows from $m_o \ll \rho$, and the channel preserves the evidence mass. Define

$$L_o^c(z) = \frac{dm_o^c}{d\rho^c}(z), \qquad H_o^c(z) = -\log L_o^c(z). \tag{12.20}$$

Theorem 12.3 (Conditional-partition formula and composition). Let $R_\rho(z, dy)$ be a reverse conditional of Y given $Z = z$ under $\rho(dy)C(y, dz)$. Then

$$\exp[-H_o^c(z)] = \int \exp[-H_o(y)] R_\rho(z, dy) \tag{12.21}$$

ρ^c-almost everywhere, and

$$Z(o) = \int \exp[-H_o^c(z)]\rho^c(dz). \tag{12.22}$$

For a second coarse channel D, the direct and staged effective likelihoods agree almost everywhere when their conditional versions are chosen from the same joint tower. [ESTABLISHED]

Proof. For bounded measurable f,

$$\int f(z)m_o^c(dz) = \iint f(z)L_o(y)C(y,dz)\rho(dy) = \int f(z)\mathbb{E}_\rho[L_o(Y) \mid Z = z]\rho^c(dz).$$

This identifies the Radon–Nikodym derivative and proves (12.21). Setting $f = 1$ proves (12.22). The tower property proves composition. □

For binary interaction successes, $0 \le L_o \le 1$ implies $0 \le L_o^c \le 1$, so one effective binary interaction record realizes the entire coarse action. For general record densities, L_o^c remains a valid density relative to the pushed probability reference, although it need not be bounded by one. An infinite Lebesgue reference cannot be pushed through an arbitrary projection without checking sigma-finiteness; this is why the probability pair (ρ, m_o) is primary. [ESTABLISHED]

12.2.1 Local bounded action chart and its derivatives

The unnormalized pair tracks the evidence mass, while the local action calculus is based at its normalized law. Because the fine and coarse spaces are standard Borel, fix a jointly measurable reverse kernel $\Pi(z, dy)$ for the normalized joint law:

$$\pi(dy)C(y,dz) = \pi^c(dz)\Pi(z,dy). \tag{12.23}$$

Values of Π on a π^c-null set are not used. Write $\mathfrak{B} = L^\infty(\pi;\mathbb{R})$ and choose $0 < \epsilon < \log 2$. The local chart is the open ball

$$\mathcal{U}_\epsilon = \{\varphi \in \mathfrak{B} : \|\varphi\|_\infty < \epsilon\}. \tag{12.24}$$

For every $\varphi \in \mathfrak{B}$, a perturbation acts on the complete measure pair by $m_o^\varphi = e^{-\varphi}m_o$. Its exact coarse action increment is

$$Q(\varphi)(z) = -\log \frac{d\big((e^{-\varphi}m_o)C\big)}{dm_oC}(z) = -\log \int e^{-\varphi(y)}\Pi(z,dy). \tag{12.25}$$

The last expression is finite and belongs to $L^\infty(\pi^c;\mathbb{R})$: it lies between $-\|\varphi\|_\infty$ and $\|\varphi\|_\infty$. Thus the Radon–Nikodym formula defines $Q : \mathfrak{B} \to L^\infty(\pi^c;\mathbb{R})$ on the entire bounded space. The real-analytic theorem below is asserted on $\mathcal{U}_\epsilon$, not on a general L^p neighborhood. Where both actions are finite, it equals the literal difference $\mathcal{R}^H[H_o + \varphi;\rho] - \mathcal{R}^H[H_o;\rho]$. For an extended-valued base action, the Radon–Nikodym ratio in (12.25) is the definition and no expression of the form $+\infty - (+\infty)$ is used. [DEFINITION]

Theorem 12.4 (Exact bounded action calculus). The map

$$Q : \mathcal{U}_\epsilon \longrightarrow L^\infty(\pi^c; \mathbb{R}) \tag{12.26}$$

is real analytic. If

$$(U\varphi)(z) = \int \varphi(y)\Pi(z, dy), \tag{12.27}$$

then its first two Fréchet derivatives at the origin are the bounded typed maps

$$DQ(0)[\varphi] = U\varphi, \tag{12.28}$$

$$D^2Q(0)[\varphi, \psi] = -\operatorname{Cov}_{\Pi(z,\cdot)}(\varphi, \psi). \tag{12.29}$$

More generally, with

$$\Pi^\varphi(z, dy) = \frac{e^{-\varphi(y)}\Pi(z, dy)}{\int e^{-\varphi(u)}\Pi(z, du)}, \tag{12.30}$$

the derivatives at $\varphi \in \mathcal{U}_\epsilon$ are

$$DQ(\varphi)[h](z) = \int h(y)\Pi^\varphi(z, dy), \tag{12.31}$$

$$D^2Q(\varphi)[h, k](z) = -\operatorname{Cov}_{\Pi^\varphi(z,\cdot)}(h, k). \tag{12.32}$$

All covariance expressions are in $L^\infty(\pi^c; \mathbb{R})$. [ESTABLISHED]

Proof. For $\|\varphi\|_\infty < \epsilon$, the exponential power series converges in $L^\infty(\pi)$, and positivity plus unitality of U give

$$\|U(e^{-\varphi}) - 1\|_\infty \le e^\epsilon - 1 < 1. \tag{12.33}$$

The Banach-algebra logarithm series therefore converges in $L^\infty(\pi^c)$, proving analyticity of $Q = -\log U(e^{-\varphi})$. Differentiating the convergent series, or using the ordinary Fréchet chain rule for the exponential, bounded linear operator U, and logarithm, gives (12.31). Differentiating that normalized integral in direction k gives

$$\left.\frac{d}{dt}\right|_{t=0} \int h d\Pi^{\varphi+tk} = -\int hk d\Pi^\varphi + \left(\int h d\Pi^\varphi\right)\left(\int k d\Pi^\varphi\right), \tag{12.34}$$

which is (12.32). Setting $\varphi = 0$ proves (12.28) and (12.29). The uniform lower bound $U(e^{-\varphi}) \ge e^{-\epsilon}$ and boundedness of h, k make every displayed derivative a bounded map into the stated target. □

Proposition 12.5 (Bounded recentering gives analyticity at every bounded action). For every $\varphi \in \mathfrak{B}$, define the positive unital operator

$$U^\varphi h := \frac{U(e^{-\varphi}h)}{U(e^{-\varphi})}. \tag{12.35}$$

The denominator lies between $e^{-\|\varphi\|_\infty}$ and $e^{\|\varphi\|_\infty}$, so this is a bounded $L^\infty(\pi) \to L^\infty(\pi^c)$ operator. For every bounded increment h,

$$Q(\varphi + h) - Q(\varphi) = -\log U^\varphi(e^{-h}). \tag{12.36}$$

Consequently, for each $0 < \epsilon < \log 2$, the left side is real analytic as a function of h on $\|h\|_\infty < \epsilon$, and

$$DQ(\varphi)[h] = U^\varphi h. \tag{12.37}$$

Thus Q is locally Fréchet analytic at every bounded action, even though (12.24) was the one origin-centered chart used in Theorem 12.4. [ESTABLISHED]

Proof. The bounds follow from positivity and unitality of U. Substitution in (12.25) gives

$$-\log U(e^{-\varphi-h}) + \log U(e^{-\varphi}) = -\log \frac{U(e^{-\varphi}e^{-h})}{U(e^{-\varphi})},$$

which is (12.36). Positivity and unitality of U^φ imply $\|U^\varphi(e^{-h}) - 1\|_\infty \le e^\epsilon - 1 < 1$. The same Banach-algebra exponential and logarithm series used in Theorem 12.4 therefore converge on this increment ball. Their first derivative at $h = 0$ is $U^\varphi h$. □

For $\varphi, \psi \in \mathfrak{B}$, the nonlinear map is order preserving and additively homogeneous. Indeed, $\varphi \le \psi + c$ implies $e^{-\varphi} \ge e^{-c}e^{-\psi}$, and positivity of U gives $Q(\varphi) \le Q(\psi) + c$. Applying this implication in both directions with $c = \|\varphi - \psi\|_\infty$ proves the exact bounded-action estimate

$$\|Q(\varphi) - Q(\psi)\|_\infty \le \|\varphi - \psi\|_\infty. \tag{12.38}$$

Under isometric cross-scale identifications, this prevents relevance from arising in the bounded measure-pair action sector. It does not constrain an extensive interaction lift or a separately proved weighted action space. [ESTABLISHED]

The full action space remembers the perturbed mass $m_o^\varphi(\mathrm{Y})$. If an application deliberately discards that scalar, it instead uses the projective action space

$$\overline{\mathfrak{B}} = \mathfrak{B}/\mathbb{R}1. \tag{12.39}$$

Since $Q(\varphi + c) = Q(\varphi) + c$ and $U1 = 1$, both Q and U descend to the quotient. The quotient is not used to erase an evidence constant in the normalized measure-pair theory. [ESTABLISHED]

For comparison, normalize the perturbed fine law itself:

$$\widehat{\pi}^\varphi = \frac{e^{-\varphi}\pi}{\pi(e^{-\varphi})}, \qquad \widehat{Q}(\varphi) := -\log \frac{d(\widehat{\pi}^\varphi C)}{d\pi^c} = Q(\varphi) + \log \pi(e^{-\varphi}). \tag{12.40}$$

The two action increments therefore agree only modulo a constant. Their derivatives make that distinction explicit:

$$D\widehat{Q}(0)[h] = Uh - \pi(h), \tag{12.41}$$

$$D^2\widehat{Q}(0)[h,k] = -\operatorname{Cov}_{\Pi(z,\cdot)}(h,k) + \operatorname{Cov}_{\pi}(h,k). \tag{12.42}$$

The score of the normalized path $t \mapsto \widehat{\pi}^{th}$ is $-h + \pi(h)$, and its coarse score is $-Uh + \pi(h)$. Consequently the defect in (12.44), applied to the centered score, is exactly the Fisher-information loss under the fixed Markov channel. This score statement remains an L^2 tangent result; it does not enlarge the nonlinear bounded action chart. [ESTABLISHED]

At a boundary where an action representative is infinite, or outside the bounded action space, neither a literal action difference nor a differential beta functional is supplied by this calculus. Within the bounded space, Proposition 12.5 supplies local analyticity at every center. At an extended-valued boundary the primary object remains the measure pair and its Radon–Nikodym action; an action beta requires the separately declared finite-valued vector space and subtraction hypotheses of Section 12.9. [ESTABLISHED]

12.2.2 Canonical contraction and the bounded Dobrushin certificate

Theorem 12.6 (Conditional-expectation contraction and exact defect). For every $1 \le p \le \infty$, the operator in (12.27) extends uniquely to a contraction

$$U : L^p(\pi;\mathbb{R}) \longrightarrow L^p(\pi^c;\mathbb{R}), \qquad \|U\varphi\|_{L^p(\pi^c)} \le \|\varphi\|_{L^p(\pi)}. \tag{12.43}$$

For every $1 \le p \le \infty$, it preserves the mean and hence maps centered tangent spaces to centered tangent spaces. At $p = 2$, the loss is exactly

$$\|\varphi\|^2_{L^2(\pi)} - \|U\varphi\|^2_{L^2(\pi^c)} = \int \operatorname{Var}_{\Pi(z,\cdot)}(\varphi)\pi^c(dz) \ge 0. \tag{12.44}$$

Equality holds exactly when there is a measurable g on the coarse space such that $\varphi(y) = g(z)$ for $\pi(dy)C(y,dz)$-almost every (y,z). For a deterministic coarse map $z = c(y)$, this is precisely $\varphi = g \circ c$ almost everywhere under π. [ESTABLISHED]

Proof. Conditional Jensen gives $|U\varphi|^p \le U|\varphi|^p$ for finite p; integrating the disintegration (12.23) proves (12.43). The $p = \infty$ case follows from the pointwise bound by the essential supremum. The same disintegration gives $\int U\varphi d\pi^c = \int \varphi d\pi$. Finally, the conditional variance identity

$$\int \varphi^2 d\pi = \int (U\varphi)^2 d\pi^c + \int \operatorname{Var}_{\Pi(z,\cdot)}(\varphi)\pi^c(dz) \tag{12.45}$$

proves (12.44). A nonnegative conditional variance integrates to zero exactly when it is zero π^c-almost everywhere, which is equivalent to the stated joint measurability condition. □

On $L^\infty(\pi)/\mathbb{R}1$, use

$$\|[\varphi]\|_{\mathrm{osc}} = \inf_{c\in\mathbb{R}} \|\varphi - c\|_\infty = \frac{1}{2}\left(\operatorname{esssup}\varphi - \operatorname{essinf}\varphi\right). \tag{12.46}$$

For the fixed reverse-kernel version, define its Dobrushin coefficient by

$$\delta = \operatorname*{esssup}_{(z,z')\sim(\pi^c)^{\otimes 2}} \sup_{A\in\mathscr{Y}} \left|\Pi(z,A) - \Pi(z',A)\right|. \tag{12.47}$$

Proposition 12.7 (Dobrushin contraction is a sufficient cocycle certificate). The quotient map satisfies

$$\|\overline{U}\|_{\mathrm{osc}\to\mathrm{osc}} \le \delta. \tag{12.48}$$

For a nonautonomous sequence of such arrows with block factors $b_k > 1$, set

$$B_{n\leftarrow\ell} = \prod_{k=\ell}^{n-1} b_k, \qquad \overline{U}_{n\leftarrow\ell} = \overline{U}_{n-1}\cdots\overline{U}_\ell. \tag{12.49}$$

If $B_{n\leftarrow\ell}\to\infty$ and

$$\limsup_{n\to\infty} \frac{\sum_{k=\ell}^{n-1}\log\delta_k}{\log B_{n\leftarrow\ell}} < 0, \qquad \log 0 = -\infty, \tag{12.50}$$

then every bounded projective perturbation has strictly negative upper growth rate in this oscillation norm. With a fixed isometric cross-scale identification, fixed $b > 1$, and $\delta < 1$, this gives the autonomous bound $\log\delta/\log b < 0$. The condition is sufficient only: failure of (12.50) proves neither marginality nor relevance. [ESTABLISHED]

Proof. For probability measures μ,ν, $\left|\int f d\mu - \int f d\nu\right|$ is at most $\operatorname{osc}(f)\sup_A|\mu(A)-\nu(A)|$. Apply this to the two reverse laws in (12.47), divide by two, and take the essential supremum to obtain (12.48). Iteration gives

$$\|\overline{U}_{n\leftarrow\ell}\|_{\mathrm{osc}\to\mathrm{osc}} \le \prod_{k=\ell}^{n-1}\delta_k. \tag{12.51}$$

Taking logarithms with the stated zero convention proves the claimed upper growth bound. Two controls show why neither one-step behavior nor failure of the displayed certificate can be promoted to a necessary condition. For the first, take reverse matrices

$$R_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1/2 & 1/2 \end{pmatrix}, \qquad R_1 = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{12.52}$$

with compatible laws $\pi_0 = (1/2,1/2)$, $\pi_1 = (1/3,1/3,1/3)$, and $\pi_2 = (2/3,1/3)$. Both one-step coefficients are one, but the two rows of R_1R_0 are both $(1/2,1/2)$, so the composed coefficient is zero. Thus individual strict Dobrushin contraction is not necessary for complete two-step forgetting. For the opposite control, take $\pi_k = (1/2,1/2)$ at every scale and let J have both rows $(1/2,1/2)$.

Each matrix below is then a reverse kernel compatible with these laws:

$$R_k = (1 - a_k)I + a_k J, \qquad a_k = 2^{-k-2}. \tag{12.53}$$

Then $\delta_k = 1 - a_k < 1$ at every step, but $\prod_k \delta_k > 0$. With fixed $b_k = b > 1$, the normalized accumulated logarithmic rate can therefore be zero. One-step strict contraction is not sufficient for the nonautonomous conclusion. The identity channel has $\delta = 1$ and leaves every bounded mode marginal, while a two-cycle has an eigenvalue -1; these controls also prevent a phase-blind conclusion. □

12.2.3 Score tangents, the extensive block lift, and an inhabited spectrum

Estimate (12.38) shows that the bounded measure-pair action sector is nonexpansive, so under isometric cross-scale identifications no bounded action mode in that sector can grow. A growing mode must therefore come from somewhere else, and this subsection identifies the first of the two places where it does: the score tangent carries an *extensive* replication factor that the one-arrow contraction theorem cannot see. The second place is the extensive interaction assembly of Section 12.3.

Definition 12.8 (Quadratic-mean differentiability and score). Let $(\mathsf{W}, \mathscr{W}, \mu)$ be a probability space and write

$$L_0^2(\mu) = \left\{ h \in L^2(\mu) : \int h d\mu = 0 \right\}. \tag{12.54}$$

A family $(\mu_t)_{|t|<t_0}$ of probability laws with $\mu_t \ll \mu$ and $\mu_0 = \mu$ is differentiable in quadratic mean at $t = 0$ with score $h \in L^2(\mu)$ when

$$\left\| \sqrt{\frac{d\mu_t}{d\mu}} - 1 - \frac{t}{2} h \right\|_{L^2(\mu)} = o(|t|), \qquad t \to 0, \tag{12.55}$$

where both one-sided limits are required. The Fisher information of the path at $t = 0$ is $\|h\|^2_{L^2(\mu)}$. Nothing is proved by this declaration. [DEFINITION]

Centering is forced rather than assumed. Writing $p_t = d\mu_t/d\mu$ and using $\int p_t d\mu = 1$ gives the exact identity $\int (\sqrt{p_t} - 1) d\mu = -\frac{1}{2} \int (\sqrt{p_t} - 1)^2 d\mu$. Inserting (12.55) makes the left side $\frac{t}{2} \int h d\mu + o(|t|)$ and the right side $-\frac{t^2}{8} \|h\|_2^2 + o(t^2)$, so $\int h d\mu = 0$. The quadratic-mean tangent set is therefore contained in $L_0^2(\mu)$. [ESTABLISHED]

Lemma 12.9 (Every centered square-integrable score is realized two-sided). Let $h \in L_0^2(\mu)$ and put $a = \|h\|_2^2/4$. For every real t, define

$$p_t = \frac{(1 + th/2)^2}{1 + at^2}, \qquad \mu_t = p_t \mu. \tag{12.56}$$

Then every μ_t is a probability law, $\mu_0 = \mu$, and the path is differentiable in quadratic mean at $t = 0$ with score h, for $t \to 0^+$ and for $t \to 0^-$. No exponential moment, and no boundedness of h, is used. [ESTABLISHED]

Proof. Centering gives $\int (1 + th/2)^2 d\mu = 1 + t\mu(h) + \frac{t^2}{4}\mu(h^2) = 1 + at^2$, so $p_t \geq 0$ integrates to one. Put $s_t = 1 + th/2$ and $N_t = 1 + at^2 \geq 1$, so that $\sqrt{p_t} = |s_t|/\sqrt{N_t}$ and

$$\sqrt{p_t} - s_t = \frac{|s_t| - s_t}{\sqrt{N_t}} + s_t \left(\frac{1}{\sqrt{N_t}} - 1 \right). \tag{12.57}$$

On $\{s_t < 0\}$ one has $th/2 < -1$, hence $|h| > 2/|t|$ and $|s_t| = |th/2| - 1 \leq |th|/2$, so $|s_t| - s_t = 2|s_t| \leq |th|$ there and the difference vanishes elsewhere. Since $N_t \geq 1$,

$$\frac{1}{t^2} \left\| \frac{|s_t| - s_t}{\sqrt{N_t}} \right\|_2^2 \leq \int h^2 \mathbf{1}_{\{|h| > 2/|t|\}} d\mu \longrightarrow 0 \tag{12.58}$$

by dominated convergence, because h^2 is integrable. For the second term, $\|s_t\|_2 = \sqrt{N_t}$, so its norm is exactly $|1 - \sqrt{N_t}| = O(t^2)$. Adding the two bounds and using $s_t = 1 + \frac{t}{2}h$ gives (12.55). □

Proposition 12.10 (Independent block replication of a differentiable path). Fix an integer $b \geq 1$. Let (μ_t) be any path satisfying (12.55) with score h, and form the product path $\mu_t^{\otimes b}$ at $\mu^{\otimes b}$. Then the product path is differentiable in quadratic mean at $t = 0$ with score

$$(\mathscr{I}_b h)(x_1, \ldots, x_b) = \sum_{i=1}^{b} h(x_i), \tag{12.59}$$

and the replication map is a bounded linear injection

$$\mathscr{I}_b : L_0^2(\mu) \longrightarrow L_0^2(\mu^{\otimes b}), \qquad \|\mathscr{I}_b h\|^2_{L^2(\mu^{\otimes b})} = b\|h\|^2_{L^2(\mu)}, \tag{12.60}$$

so $\|\mathscr{I}_b\| = \sqrt{b}$ and $b^{-1/2}\mathscr{I}_b$ is an isometry onto its range. The symbol $\mathscr{I}_b$ is the score replication lift; it is not the Hoeffding assembly E_ℓ of (12.87). [ESTABLISHED]

Proof. Write $r_t = \sqrt{d\mu_t/d\mu} = 1 + \frac{t}{2}h + t\varepsilon_t$ with $\|\varepsilon_t\|_{L^2(\mu)} \to 0$, which is (12.55). The product amplitude is $\prod_{i=1}^b r_t(x_i) = \prod_{i=1}^b (1 + u_i)$ with $u_i(x) = \frac{t}{2}h(x_i) + t\varepsilon_t(x_i)$. Expanding the finite product,

$$\prod_{i=1}^{b}(1 + u_i) = 1 + \sum_{i=1}^{b} u_i + \sum_{|S| \geq 2} \prod_{i \in S} u_i. \tag{12.61}$$

Because the coordinates are independent and each factor depends on one coordinate, $L^2(\mu^{\otimes b})$ norms of products factor, so $\|\prod_{i \in S} u_i\|_2 = \prod_{i \in S} \|u_i\|_2 = O(|t|^{|S|})$, and every term with $|S| \geq 2$ is $O(t^2)$. The linear part is $\frac{t}{2}\mathscr{I}_b h + t\sum_i \varepsilon_t(x_i)$, whose remainder has norm at most $b|t|\|\varepsilon_t\|_2 = o(|t|)$ by the triangle inequality. This proves quadratic-mean differentiability with score $\mathscr{I}_b h$. Independence and centering give $\mathbb{E}(\mathscr{I}_b h)^2 = \sum_i \mathbb{E}h(X_i)^2 + \sum_{i \neq j} \mathbb{E}h(X_i)\mathbb{E}h(X_j) = b\|h\|_2^2$, which is (12.60). □

The replication is extensive. Replacing (12.59) by the arithmetic mean without simultaneously rescaling the path parameter would change the Fisher information from $b\|h\|_2^2$ to $\|h\|_2^2/b$, which is a different theory. [ESTABLISHED]

Theorem 12.11 (Pushed score and the exact extensive Fisher defect). Let $C_b : \mathsf{W}^b \rightsquigarrow \mathsf{Z}$ be a normalized, parameter-independent Markov kernel between standard-Borel spaces, put $\mu^{(b)} = \mu^{\otimes b}$, $J_b(dx, dz) = \mu^{(b)}(dx)C_b(x, dz)$, and $\mu^{(b),c} = \mu^{(b)}C_b$. Let $U_b : L^2(\mu^{(b)}) \to L^2(\mu^{(b),c})$ be the reverse conditional expectation of Theorem 12.6 for this arrow. Then the pushed path $\mu_t^{\otimes b}C_b$ is differentiable in quadratic mean at $\mu^{(b),c}$ with score

$$\mathscr{L}_b h := U_b \mathscr{I}_b h = \mathbb{E}_{J_b}\left[\sum_{i=1}^{b} h(X_i)\middle| Z\right], \tag{12.62}$$

so that $\mathscr{L}_b : L_0^2(\mu) \to L_0^2(\mu^{(b),c})$ is linear with $\|\mathscr{L}_b\| \le \sqrt{b}$. Moreover the exact information budget is

$$b\|h\|^2_{L^2(\mu)} - \|\mathscr{L}_b h\|^2_{L^2(\mu^{(b),c})} = \int \mathrm{Var}_{J_b}\left(\mathscr{I}_b h | Z = z\right)\mu^{(b),c}(dz) \ge 0, \tag{12.63}$$

with equality exactly when $\mathscr{I}_b h$ agrees J_b-almost surely with a measurable function of Z. Equivalently $b^{-1/2}\mathscr{L}_b$ is a contraction of L_0^2 norms. [ESTABLISHED]

Proof. Suppose first that h is bounded and use the path (12.56). For $|t|$ small, $s_t \ge 1-|t|\|h\|_\infty/2 > 0$, so $p_t = s_t^2/N_t = 1 + th + O(t^2)$ uniformly, and the product density is $\prod_i p_t(x_i) = 1 + t\mathscr{I}_b h + O(t^2)$ uniformly. Conditional expectation of a uniformly $O(t^2)$ remainder is uniformly $O(t^2)$, and

$$\frac{d(\mu_t^{\otimes b}C_b)}{d\mu^{(b),c}}(z) = \mathbb{E}_{J_b}\left[\frac{d\mu_t^{\otimes b}}{d\mu^{(b)}}(X)\middle| Z = z\right] = 1 + t\mathscr{L}_b h(z) + O(t^2) \tag{12.64}$$

uniformly, exactly as in (12.21). Since $\sqrt{1+w} = 1 + w/2 + O(w^2)$ uniformly for $|w| \le 1/2$, the square root of (12.64) is $1 + \frac{t}{2}\mathscr{L}_b h + O(t^2)$ uniformly, hence in $L^2(\mu^{(b),c})$. This is (12.55) for the pushed path.

For general $h \in L_0^2(\mu)$, choose bounded centered $h_n \to h$ in $L^2(\mu)$ and let $\mu_t^{(n)}$ be their paths (12.56). Squared Hellinger distance is an f-divergence with the convex $f(u) = (\sqrt{u}-1)^2$, so conditional Jensen applied to the disintegration of the pushed joint law gives the data-processing bound

$$\left\|\sqrt{\frac{d(\mu_t^{\otimes b}C_b)}{d\mu^{(b),c}}} - \sqrt{\frac{d((\mu_t^{(n)})^{\otimes b}C_b)}{d\mu^{(b),c}}}\right\|_2 \le \left\|\sqrt{\frac{d\mu_t^{\otimes b}}{d\mu^{(b)}}} - \sqrt{\frac{d(\mu_t^{(n)})^{\otimes b}}{d\mu^{(b)}}}\right\|_2. \tag{12.65}$$

Both input paths are differentiable in quadratic mean with scores $\mathscr{I}_b h$ and $\mathscr{I}_b h_n$, so the right side divided by $|t|$ has limit superior $\frac{1}{2}\|\mathscr{I}_b(h-h_n)\|_2 = \frac{\sqrt{b}}{2}\|h-h_n\|_2$. Adding and subtracting the

bounded-case expansion, and using $\|\mathscr{L}_b(h-h_n)\|_2 \le \sqrt{b}\|h-h_n\|_2$,

$$\limsup_{t\to 0}\frac{1}{|t|}\left\|\sqrt{\frac{d(\mu_t^{\otimes b}C_b)}{d\mu^{(b),c}}}-1-\frac{t}{2}\mathscr{L}_b h\right\|_2 \le \sqrt{b}\|h-h_n\|_2. \tag{12.66}$$

The left side does not depend on n, so letting $n\to\infty$ makes it zero, which is the general case.

Conditional Jensen gives $\|U_b\|\le 1$ and preservation of the mean, so $\mathscr{L}_b$ has the stated type and $\|\mathscr{L}_b\|\le\|\mathscr{I}_b\|=\sqrt{b}$. Finally (12.45) applied to $\mathscr{I}_b h$, together with (12.60), is exactly (12.63), and a nonnegative conditional variance integrates to zero precisely when it vanishes almost everywhere. □

Two boundaries of (12.63) deserve a sentence. A channel that maps every block to one point sends every score to zero, so a fixed channel can annihilate the entire tangent. In the opposite direction, Fisher equality at one parameter is not recovery of the experiment: let A and B be independent with $\Pr_\theta(A=1)=\frac{1}{2}+\frac{\theta}{4}$ and $\Pr_\theta(B=1)=\frac{1}{2}+\frac{\theta^2}{4}$, and let the channel retain only A. At $\theta=0$ the B-score vanishes, so the full score is A-measurable and equality holds, yet the conditional law of B varies with θ and no parameter-independent reverse kernel reconstructs the experiment. [ESTABLISHED]

Proposition 12.12 (Bounded action classes embed isometrically in the Fisher tangent). For $\varphi\in\mathfrak{B}=L^\infty(\pi;\mathbb{R})$, the normalized exponential path $\widehat{\pi}^{t\varphi}$ of (12.40) is differentiable in quadratic mean with score

$$\mathscr{S}_\pi[\varphi]=-\varphi+\pi(\varphi), \tag{12.67}$$

which depends only on the class $[\varphi]\in\overline{\mathfrak{B}}$. Equip the action quotient with the Fisher norm

$$\|[\varphi]\|_F=\inf_{c\in\mathbb{R}}\|\varphi-c\|_{L^2(\pi)}=\|\varphi-\pi(\varphi)\|_{L^2(\pi)}. \tag{12.68}$$

Then $\mathscr{S}_\pi:(\overline{\mathfrak{B}},\|\cdot\|_F)\to L^2_0(\pi)$ is a linear isometry whose range is the bounded centered subspace; that range is dense but generally proper, so the Fisher-norm completion of the bounded action quotient is canonically $L^2_0(\pi)$. The two quotient norms are ordered by $\|[\varphi]\|_F\le\|[\varphi]\|_{\mathrm{osc}}$. Moreover the centering square commutes,

$$\mathscr{S}_{\pi^c}\big(\overline{U}[\varphi]\big)=U\mathscr{S}_\pi[\varphi], \tag{12.69}$$

and for the replicated block action $\Phi_\varphi(x_1,\dots,x_b)=\sum_i\varphi(x_i)$ one has $\mathscr{S}_{\mu^{(b)}}[\Phi_\varphi]=\mathscr{I}_b\mathscr{S}_\mu[\varphi]$ and, after the block arrow, $\mathscr{L}_b\mathscr{S}_\mu[\varphi]$. [ESTABLISHED]

Proof. Boundedness gives the uniform expansions $e^{-t\varphi}=1-t\varphi+O(t^2)$ and $\pi(e^{-t\varphi})=1-t\pi(\varphi)+O(t^2)$, so the density ratio of $\widehat{\pi}^{t\varphi}$ is $1-t[\varphi-\pi(\varphi)]+O(t^2)$ uniformly and its square root is $1+\frac{t}{2}\mathscr{S}_\pi[\varphi]+O(t^2)$, which is (12.55). Adding a constant to φ leaves (12.67) unchanged. The L^2 minimizer over constants is the mean, giving (12.68) and hence the isometry; $\inf_c\|\varphi-c\|_2\le\inf_c\|\varphi-c\|_\infty$ gives the norm ordering. Bounded functions are dense in $L^2(\pi)$ and centering is continuous, so the range is dense; it is proper whenever $L^2_0(\pi)$ contains an unbounded element, as

for $\pi = N(0,1)$ and $h(x) = x$. Identity (12.69) is mean preservation of U together with linearity, and the replication statement is (12.59). □

The two norms on $\overline{\mathfrak{B}}$ do different work and must not be exchanged. The oscillation norm of (12.46) controls the nonlinear map Q; the Fisher norm controls the quadratic-mean tangent. Completing in the Fisher norm introduces unbounded scores on which the nonlinear bounded action map of Theorem 12.4 is not defined. A spectral statement proved on L^2_0 is therefore not a statement about the nonlinear bounded action chart, and conversely. [ESTABLISHED]

Two further separations belong here, and each is stated with the witness that makes it sharp. First, every class in the Fisher completion is realized as a two-sided quadratic-mean score by the *quadratic* path of Lemma 12.9, not by the exponential-action path $\widehat{\pi}^{t\varphi}$, which is defined only where its normalizer is finite. The witness that separates the two must therefore be a direction whose exponential normalizer diverges on *both* sides. The correct witness is an odd Hermite polynomial of degree at least three: for $\varphi = \mathrm{He}_3(x) = x^3 - 3x \in L^2_0(\gamma)$, the log integrand $-t\mathrm{He}_3(x) - x^2/2$ has an exact divergent tail on either side. For $t > 0$, substitute $x = -r$ to obtain

$$-t\mathrm{He}_3(-r) - \frac{r^2}{2} = tr^3 - \frac{r^2}{2} - 3tr \longrightarrow +\infty \qquad (r \to +\infty),$$

for $t < 0$, substitute $x = r$ to obtain

$$-t\mathrm{He}_3(r) - \frac{r^2}{2} = |t|r^3 - \frac{r^2}{2} - 3|t|r \longrightarrow +\infty.$$

The leading cubic is positive in both cases, so the exponential integrand is eventually at least one on an infinite tail and $N_3(t) = +\infty$ for *every* $t \neq 0$, while He_3 is a perfectly good Fisher tangent direction with $\mathbb{E}_\gamma[\mathrm{He}_3] = 0$ and $\mathbb{E}_\gamma[\mathrm{He}_3^2] = 6$. The degree-two direction $\varphi = -x^2$ is *not* such a witness and must not be cited as one: its normalizer is $\pi(e^{tx^2}) = (1-2t)^{-1/2}$, which is finite on the two-sided interval $(-\frac{1}{2}, \frac{1}{2})$, taking the values $\sqrt{6}/3$ at $t = -\frac{1}{4}$ and $\sqrt{2}$ at $t = \frac{1}{4}$. This is the classification already recorded in Proposition 24.7. [ESTABLISHED]

Second, the extensive replication lift is not a channel and no data-processing statement applies to it in the coarse-to-fine direction. The operator $\mathscr{I}_b$ of Proposition 12.10 maps $L^2_0(\pi)$ into $L^2_0(\pi^{\otimes b})$ with $\|\mathscr{I}_b\| = \sqrt{b} > 1$; it is a lift between two different reference tangent spaces, not a pushforward along any arrow. No parameter-independent normalized Markov kernel realizes b-fold replication for $b \geq 2$: if K satisfied $\mathcal{N}(x,1)K = \mathcal{N}(x\mathbf{1}_b, I_{\mathbb{R}^b})$ for every x, then Theorem 12.11 would force the output Fisher information b to satisfy $b \leq 1$. Consequently the replication pair may not be cited as a Markov-contraction counterexample or as evidence that Fisher information increased along a coarse-graining arrow, because there is no such arrow; the factor $\sqrt{b}$ belongs to the reference identifications I_ℓ of Definition 12.28 and to a declared configuration-metric normalization, and to nothing else. [ESTABLISHED]

Theorem 12.13 (Gaussian block spectrum in the normalized Hermite basis). Let $\gamma = N(0,1)$, let $b \geq 2$ be an integer, let $X_1, \ldots, X_b$ be independent with law γ, and let the block channel be the

deterministic normalized sum

$$Z = b^{-1/2} \sum_{i=1}^{b} X_i, \tag{12.70}$$

whose law is again γ. It is that normalization, and not a general theorem, that makes source and target the same space and hence makes

$$\mathscr{L}_b = U_b \mathscr{I}_b : L_0^2(\gamma) \longrightarrow L_0^2(\gamma) \tag{12.71}$$

an endomorphism. Let He_k be the probabilists' Hermite polynomials and $e_k = \mathrm{He}_k/\sqrt{k!}$. Then $\{e_k\}_{k\geq 1}$ is a complete orthonormal basis of $L_0^2(\gamma)$ and

$$\mathscr{L}_b e_k = b^{1-k/2} e_k, \qquad k \geq 1. \tag{12.72}$$

Consequently $\mathscr{L}_b$ is self-adjoint, positive, and diagonal in that basis; it is Hilbert–Schmidt with $\sum_{k\geq 1} b^{2-k} = b^2/(b-1)$, hence compact; its norm and spectral radius are $\sqrt{b}$; and

$$\sigma(\mathscr{L}_b) = \left\{ b^{1-k/2} : k \geq 1 \right\} \cup \{0\}. \tag{12.73}$$

Every displayed nonzero value is a simple eigenvalue. The value 0 is *not* an eigenvalue: it is the unique accumulation point of the spectrum and belongs to the continuous spectrum. This is a linear tangent-space statement about the derivative at the Gaussian law; it is not a nonlinear convergence theorem. [ESTABLISHED]

Proof. Completeness of the normalized Hermite system in $L^2(\gamma)$ is standard, and deleting $e_0 = 1$ gives a complete orthonormal basis of the centered subspace. For each i, the pair (X_i, Z) is jointly centered Gaussian with unit variances and $\mathrm{Cov}(X_i, Z) = b^{-1/2}$, so $X_i \mid Z = z \sim N(z/\sqrt{b}, 1 - 1/b)$. Taking the conditional expectation of the generating function $\exp(tx - t^2/2)$ and completing the square,

$$\mathbb{E}\left[e^{tX_i - t^2/2} \middle| Z = z \right] = \exp\left(\frac{tz}{\sqrt{b}} + \frac{t^2}{2}\left(1 - \frac{1}{b}\right) - \frac{t^2}{2} \right) = \exp\left(sz - \frac{s^2}{2} \right), \quad s = \frac{t}{\sqrt{b}}. \tag{12.74}$$

The generating identity $\exp(sz - s^2/2) = \sum_{k\geq 0} \mathrm{He}_k(z) s^k/k!$ converges in $L^2(\gamma)$ locally uniformly in s, and conditional expectation is L^2 continuous, so the two power series in t may be compared coefficient by coefficient. This gives the Mehler regression identity $\mathbb{E}[\mathrm{He}_k(X_i) \mid Z] = b^{-k/2} \mathrm{He}_k(Z)$. Summing over the b replicated coordinates and dividing by $\sqrt{k!}$ proves (12.72); the argument is uniform in b and is not an induction from the binary case.

Since $\mathscr{L}_b$ is bounded and agrees with the displayed diagonal action on a complete orthonormal basis, it is that diagonal operator, hence self-adjoint and positive with eigenvalues $b^{1-k/2} \downarrow 0$. Square summability of the eigenvalues is the stated geometric series, so $\mathscr{L}_b$ is Hilbert–Schmidt and compact, and its norm is the largest eigenvalue $b^{1/2}$. The spectrum of a diagonal operator is the closure of its eigenvalue set, which is (12.73), and the eigenvalues are pairwise distinct so each is simple. Every eigenvalue is strictly positive, so $\ker \mathscr{L}_b = \{0\}$ and 0 is not an eigenvalue.

The range contains every finite Hermite expansion and is therefore dense, but it is proper: the vector $y = \sum_{k\geq 1} b^{1-k/2} e_k$ lies in $L^2_0(\gamma)$ because the coefficients are square summable, while any preimage would have every Hermite coefficient equal to one, which is not square summable. An injective operator with dense, nonclosed range has 0 in its continuous spectrum. □

Definition 12.14 (Relevance conventions for the Gaussian block spectrum). Fix the replication sum (12.59), the block statistic (12.70), the same Fisher norm $\|\cdot\|_{L^2(\gamma)}$ at every scale, and the cumulative log block scale $\Delta s = \log b$ of (11.29). Then the exponent of the degree-k mode is $y_k = \log_b b^{1-k/2} = 1 - k/2$. Degree one is relevant with $y_1 = \frac{1}{2}$, degree two is marginal with $y_2 = 0$, and every degree $k \geq 3$ is irrelevant with $y_k = 1 - k/2 < 0$. The constant mode $k = 0$ would have eigenvalue b on the full space $L^2(\gamma)$ but is absent from the normalized probability tangent, where scores are centered. [DEFINITION]

This is the promised inhabited relevance spectrum, and it locates precisely where the growth comes from. The one-arrow operator U_b is a contraction by Theorem 12.6, and the bounded action sector is nonexpansive by (12.38). The eigenvalue $b^{1-k/2}$ exceeds one for $k = 1$ only because the replication factor $\|\mathscr{I}_b\| = \sqrt{b}$ of (12.60) is extensive while the norm is held fixed across scales. Relevance in this theory is therefore a joint property of the operator, the extensive normalization, the declared norms, and the block scale, in exactly the sense of Definition 11.12. Removing any one of the four removes the conclusion. [ESTABLISHED]

Proposition 12.15 (Convention dependence and forbidden extensions). The spectrum (12.73) is specific to the scalar independent Gaussian realization and its declared normalization. Four exact boundaries hold. First, if the coordinates are jointly centered Gaussian with unit variances and common pairwise correlation $\rho > -1/(b-1)$, and the statistic is normalized to unit variance as $Z = (\sum_i X_i)/\sqrt{b[1+(b-1)\rho]}$, then $\mathrm{Corr}(X_i, Z) = \alpha = \sqrt{[1+(b-1)\rho]/b}$ and

$$\mathscr{L}_{b,\rho} e_k = b\alpha^k e_k = b^{1-k/2}\,[1+(b-1)\rho]^{k/2}\, e_k, \tag{12.75}$$

which differs from (12.72) for every $\rho \neq 0$; keeping the uncorrelated normalization instead destroys the property that the output law is γ, so the operator is then not an endomorphism at all. Second, for independent $N(0, I_d)$ coordinates and componentwise normalized sums the tensor Hermite functions satisfy $\mathscr{L}_b e_\alpha = b^{1-|\alpha|/2} e_\alpha$, so degree k has multiplicity $\binom{d+k-1}{k}$ rather than one. Third, a common orthogonal action commutes with the componentwise normalized sum and preserves each degree space, whereas a general $\mathrm{GL}(d)$ transformation changes the reference covariance and therefore requires an explicitly transported family of laws and tangent norms; no gauge link, holonomy, or curvature datum enters the scalar theorem. Fourth, (12.59) is the score of an independent product perturbation and is a restricted direction inside $L^2_0(\gamma^{\otimes b})$; the theorem says nothing about interaction scores or higher Hoeffding sectors, which are the subject of Section 12.3. [ESTABLISHED]

Proof. For the first statement, (X_i, Z) is again jointly centered Gaussian with unit variances and correlation α, so $X_i \mid Z = z \sim N(\alpha z, 1-\alpha^2)$; the computation (12.74) with $b^{-1/2}$ replaced by α gives

$\mathbb{E}[\mathrm{He}_k(X_i) \mid Z] = \alpha^k \mathrm{He}_k(Z)$, and summing over i gives (12.75). Substituting $\alpha^2 = [1 + (b-1)\rho]/b$ gives the second form. For the second statement, tensor Hermite functions factor over coordinates and each factor obeys the scalar identity with the same $b^{-1/2}$, so the eigenvalue depends only on the total degree $|\alpha|$, and the number of multi-indices of total degree k in d variables is $\binom{d+k-1}{k}$. For the third, an orthogonal map preserves $N(0, I_d)$ and commutes with the componentwise sum, hence preserves each finite-dimensional degree space, while a general invertible linear map sends $N(0, I_d)$ to $N(0, QQ^\top)$, which is a different reference law. The fourth statement is the definition of $\mathscr{I}_b$. □

The linearization also does not by itself establish nonlinear convergence. The spectrum controls the derivative at the Gaussian law, while a basin of attraction, a remainder estimate, and the behavior of the marginal direction are separate obligations recorded in Chapter E. [OPEN]

The same qualification is made in the primary source: Jona-Lasinio [44] obtains the binary case of (12.72) and states that the nonlinear terms required to complete a limit theorem are not pursued there. [ESTABLISHED]

Source scope. Section 2 of Jona-Lasinio [44] treats the binary normalized sum, linearizes the induced map on laws around a centered unit-variance Gaussian under normalization, centering, and variance constraints, and states the Hermite eigenvalues $2^{1-k/2}$; Section 5 identifies the linearized renormalization at a self-similar Gaussian fixed point with a conditional expectation and states the corresponding generalized Hermite eigenvalue equation; and Section 7 gives the two-tangent-space form of that eigenvalue equation together with its multiplicative composition law across scales. Those four items are what the citation supports. The arbitrary-integer statement (12.72), the quadratic-mean realization of Lemma 12.9, the extensive Fisher budget (12.63), the exact spectrum (12.73) including the status of 0, the correlated boundary (12.75), and the typed cocycle of Section 11.4.1 are proved here and are not attributed to that source. [ESTABLISHED]

12.2.4 Essential spectrum and topology dependence

Theorem 12.16 (Qualified positive-unital essential-spectrum theorem). Let X be a complex Banach lattice with quasi-interior unit $\mathbf{1} \in X_+$, let $U \in \mathcal{L}(X)$ be positive and unital, and write $q : \mathcal{L}(X) \to \mathcal{L}(X)/\mathcal{K}(X)$ for the Calkin quotient by the compact-operator ideal. Use the convention

$$r_{\mathrm{ess}}(U) = r\big(q(U)\big). \tag{12.76}$$

Assume that whenever $r(U) > r_{\mathrm{ess}}(U)$, the value $r(U)$ is a pole of the resolvent and U^* has a nonzero positive eigenfunctional λ satisfying $U^*\lambda = r(U)\lambda$. Then

$$r(U) > 1 \quad \Longrightarrow \quad r(U) = r_{\mathrm{ess}}(U). \tag{12.77}$$

This does not assert quasi-compactness or a Perron theorem under weaker hypotheses. [ESTABLISHED]

Proof. The quotient map is contractive, so $r_{\mathrm{ess}}(U) \le r(U)$. Suppose that $r(U) > 1$ and that the inequality is strict. The stated hypothesis gives $\lambda \ge 0$, $\lambda \ne 0$, and $U^*\lambda = r(U)\lambda$. Quasi-interiority implies $\lambda(\mathbf{1}) > 0$: otherwise positivity makes λ zero on the principal ideal generated by $\mathbf{1}$, hence on its dense closure. But unitality yields

$$r(U)\lambda(\mathbf{1}) = (U^*\lambda)(\mathbf{1}) = \lambda(U\mathbf{1}) = \lambda(\mathbf{1}), \tag{12.78}$$

a contradiction. Therefore strict inequality is impossible, and the two radii agree. □

Proposition 12.17 (Circle witness for norm-dependent radius). Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ carry its geodesic metric and Haar probability λ, let $D(z) = 2z \bmod 1$, and set

$$K(y, A) = \frac{1}{2} \sum_{z \in D^{-1}\{y\}} \mathbf{1}_A(z) \qquad (A \subseteq \mathbb{T} \text{ Borel}). \tag{12.79}$$

The kernel preserves Haar measure and its reverse is $\Pi(z, dy) = \delta_{D(z)}(dy)$, so the conditional-expectation operator is $Uf = f \circ D$. Its spectral radius is one on $L^\infty(\lambda)$. For every $0 < \alpha \le 1$, its spectral radius on complex periodic $C^\alpha(\mathbb{T})$ equipped with the sup-plus-Hölder norm is 2^α. At $\alpha = 1$, this means the Lipschitz space $C^{0,1}$, not C^1. The Hölder space with this standard norm is not the Banach lattice of Theorem 12.16; the witness is a separate topology comparison. Hence the same exact kernel has distinct spectral conclusions in two declared topologies. [ESTABLISHED]

Proof. For bounded f, Haar preservation gives $\|f \circ D^n\|_\infty = \|f\|_\infty$, so $\|U^n\| = 1$ and the first radius is one. With $[f]_\alpha$ the Hölder seminorm,

$$[f \circ D^n]_\alpha \le 2^{\alpha n}[f]_\alpha, \qquad \|U^n f\|_\infty \le \|f\|_\infty, \tag{12.80}$$

so the second radius is at most 2^α. For the fixed Fourier mode $f(t) = e^{2\pi i t}$, evaluate $f \circ D^n$ at 0 and $2^{-(n+1)}$. Their values differ by two while their circle distance is $2^{-(n+1)}$, whence

$$[f \circ D^n]_\alpha \ge 2^{1+\alpha(n+1)}. \tag{12.81}$$

After division by the fixed nonzero C^α norm of f, this gives the matching lower spectral-radius bound. □

12.3 Exact finite-network interaction coordinates

For the graph-level treatment, fix one base point c_*. Unless a neighborhood family is explicitly introduced, every graph variable in this and the following constructions is evaluated over that single fixed fiber. The index ℓ denotes RG depth.

The following construction is an action-coordinate construction, not a claim that an arbitrary coarse channel preserves a product law. At every scale ℓ, let V_ℓ be an arbitrary finite set, let $\mathsf{X}_{\ell i}$ be standard

Borel for $i \in V_\ell$, and put

$$\mathsf{X}_\ell = \prod_{i \in V_\ell} \mathsf{X}_{\ell i}, \qquad \nu_\ell = \bigotimes_{i \in V_\ell} \nu_{\ell i}, \qquad \pi_\ell \sim \nu_\ell. \tag{12.82}$$

Here each $\nu_{\ell i}$ is a probability law and $\sim$ denotes mutual absolute continuity. Thus $L^\infty(\pi_\ell)$ and $L^\infty(\nu_\ell)$ have the same equivalence classes and the same essential-supremum norm, without requiring bounded Radon–Nikodym ratios. The product structure of ν_ℓ, rather than equivalence alone, makes the coordinate integrals below representative-independent by Fubini. For a scale arrow, the target premise $\pi_{\ell+1} = \pi_\ell K_\ell \sim \nu_{\ell+1}$ is separately required; it does not follow from $\pi_\ell \sim \nu_\ell$. The evidence mass M_ℓ of Definition 11.2 is retained independently of these normalized references. [DEFINITION]

For $V_\ell = \varnothing$, the product is the one-point probability space, $\mathcal{G}_\ell = \{0\}$, and all assembly, extraction, and interaction maps below are the unique zero maps. The nontrivial estimates that divide by $3^{|V_\ell|} - 1$ are stated only when $V_\ell \neq \varnothing$.

Proposition 12.18 (Product equivalence is an admitted, not an automatic, scale premise). Let $\mathsf{X}_0 = \{0,1\}$ with its uniform law $\pi_0 = \nu_0$, and let K be the deterministic diagonal-cloning channel

$$K(x, \{(x,x)\}) = 1 \quad \text{from } \{0,1\} \text{ to } \{0,1\}^2. \tag{12.83}$$

Then $\pi_1 = \pi_0 K$ is supported on $\{(0,0),(1,1)\}$. There is no product probability $\nu_1 = \nu_{11} \otimes \nu_{12}$ on $\{0,1\}^2$ equivalent to π_1. In particular the uniform product reference gives positive mass to all four points and is not equivalent. No target Hoeffding decomposition relative to a product reference is admitted at this target. [ESTABLISHED]

Proof. Equivalence would make both diagonal atoms positive under ν_1. Thus each marginal would charge both 0 and 1, forcing both off-diagonal atoms to have positive product mass. But π_1 assigns them mass zero, contradicting equivalence. The uniform product is the stated special case. □

For $A \subseteq V_\ell$, let $C_{\ell,A}$ be conditional integration against the product complement, lifted back to X_ℓ:

$$(C_{\ell,A} f)(x_A) = \int f(x_A, y_{A^c}) \nu_{\ell, A^c}(dy_{A^c}), \qquad C_{\ell,\varnothing} f = \nu_\ell(f). \tag{12.84}$$

The operators commute and obey $C_{\ell,A} C_{\ell,B} = C_{\ell,A\cap B}$. Their Boolean-lattice Mobius projectors are

$$P_{\ell,A} = \sum_{B \subseteq A} (-1)^{|A|-|B|} C_{\ell,B} = \prod_{i \in A} (I - C_{\ell, V_\ell \setminus \{i\}}) C_{\ell,A}, \qquad A \subseteq V_\ell. \tag{12.85}$$

The second expression only records the commuting-product identity; the first is the definition. Let

$$\mathcal{H}_{\ell,A} = P_{\ell,A} L^\infty(\nu_\ell), \qquad \mathcal{G}_\ell = \bigoplus_{\varnothing \neq A \subseteq V_\ell}^{\ell^1} \mathcal{H}_{\ell,A}, \qquad \|g\|_{\mathcal{G}_\ell} = \sum_{\varnothing \neq A \subseteq V_\ell} \|g_A\|_\infty. \tag{12.86}$$

Thus g_A depends only on x_A and has zero product-reference mean in every active coordinate. This includes every nonempty hyperedge; it is finite at each ℓ, with no bound imposed on $|V_\ell|$. Define the action quotient and its assembly/extraction maps by

$$\overline{\mathfrak{B}}_\ell = L^\infty(\nu_\ell)/\mathbb{R}1, \qquad \phi_g := \sum_{\varnothing \neq A \subseteq V_\ell} g_A, \qquad E_\ell g = [\phi_g], \qquad \mathsf{H}_\ell[f] = (P_{\ell,A} f)_{\varnothing \neq A \subseteq V_\ell}. \tag{12.87}$$

Theorem 12.19 (Hoeffding assembly is an exact finite-network action isomorphism). For every finite V_ℓ, E_ℓ and H_ℓ in (12.87) are bounded inverse linear isomorphisms,

$$\mathsf{H}_\ell E_\ell = I_{\mathcal{G}_\ell}, \qquad E_\ell \mathsf{H}_\ell = I_{\overline{\mathfrak{B}}_\ell}, \qquad \|E_\ell\| \le 1, \qquad \|\mathsf{H}_\ell\| \le 3^{|V_\ell|} - 1. \tag{12.88}$$

The latter bound is valid for every finite network but is not dimension-free. Its exponential dependence is sharp in the worst case: for biased Rademacher product coordinates and $f(x) = \prod_i x_i$, the sum of component norms approaches the bound at the corresponding product scale as the bias tends to one. [ESTABLISHED]

Proof. The commuting conditional expectations give $P_{\ell,A} P_{\ell,B} = \mathbf{1}_{A=B} P_{\ell,A}$ and $\sum_{A \subseteq V_\ell} P_{\ell,A} = I$. Hence nonempty components recover g, while their sum is $f - C_{\ell,\varnothing} f$, which is the class $[f]$. The triangle inequality gives $\|E_\ell\| \le 1$. Since $P_{\ell,A}$ is an alternating sum of $2^{|A|}$ contractive coordinate expectations, its norm is at most $2^{|A|}$. Summing over nonempty A gives $\sum_{A \neq \varnothing} 2^{|A|} = 3^{|V_\ell|} - 1$. For independent $\{-1, 1\}$-coordinates with $\Pr(X_i = 1) = p \ge 1/2$, set $n = |V_\ell|$ and $f = \prod_i X_i$. Direct expansion gives $\sum_{\varnothing \neq A \subseteq V_\ell} \|P_A f\|_\infty = (4p-1)^n - (2p-1)^n$, which tends to $3^n - 1$ as $p \uparrow 1$. □

Gauge covariance and its necessary realization hypothesis. Let a gauge transformation at scale ℓ be realized by a Borel coordinate permutation σ and coordinate Borel isomorphisms whose product map ϑ_ℓ sends ν_ℓ to $\widehat{\nu}_\ell$. Write $\Theta_\ell f = f \circ \vartheta_\ell^{-1}$ and let $\overline{\Theta}_\ell$ be its action on the quotient by constants. Fubini then gives

$$\Theta_\ell C^\nu_{\ell,A} = C^{\widehat{\nu}}_{\ell,\sigma A} \Theta_\ell, \qquad \Theta_\ell P^\nu_{\ell,A} = P^{\widehat{\nu}}_{\ell,\sigma A} \Theta_\ell, \qquad \overline{\Theta}_\ell E_\ell = \widehat{E}_\ell \Theta^{\mathcal{G}}_\ell, \qquad \Theta^{\mathcal{G}}_\ell \mathsf{H}_\ell = \widehat{\mathsf{H}}_\ell \overline{\Theta}_\ell. \tag{12.89}$$

Fubini proves the first identity and the rest follow by finite Mobius sums. Consequently coordinate expectations and interaction components are gauge-covariant under this componentwise/permutation realization. Product measure preservation alone is insufficient: a product-Haar-preserving shear of $\mathbb{T}^2$, for example $(x_1, x_2) \mapsto (x_1, x_1 + x_2)$, sends a singleton function of x_2 to one depending on both coordinates and mixes a singleton component into a pair component. A grading-intertwining hypothesis is therefore necessary for a claim that hyperedge degree is gauge covariant; componentwise/permutation realization is the declared sufficient hypothesis here, not the only logically possible one. If only almost-everywhere reverse-kernel equivariance is supplied, the valid statement is instead covariance of the L^∞ equivalence classes; it must not be promoted to pointwise covariance of a selected reverse-kernel version. [ESTABLISHED]

Proposition 12.20 (Radon–Nikodym covariance of the nonlinear interaction step). At adjacent scales, suppose $\vartheta_\ell : \mathsf{X}_\ell \to \widehat{\mathsf{X}}_\ell$ and $\vartheta_{\ell+1} : \mathsf{X}_{\ell+1} \to \widehat{\mathsf{X}}_{\ell+1}$ have the componentwise/permutation Borel realizations above. Transform the complete measure-pair data by

$$\widehat{\rho}_\ell = (\vartheta_\ell)_\# \rho_\ell, \qquad \widehat{m}_\ell = (\vartheta_\ell)_\# m_\ell, \qquad \widehat{\pi}_\ell = (\vartheta_\ell)_\# \pi_\ell, \tag{12.90}$$

and suppose the transformed normalized kernel obeys

$$\widehat{K}_\ell(\vartheta_\ell x, \vartheta_{\ell+1} B) = K_\ell(x, B) \tag{12.91}$$

for every Borel B. Then, as Radon–Nikodym equivalence classes,

$$\widehat{Q}_\ell(\Theta_\ell \varphi) = \Theta_{\ell+1} Q_\ell(\varphi), \tag{12.92}$$

$$\widehat{U}_\ell^{\Theta_\ell \varphi} \Theta_\ell h = \Theta_{\ell+1} U_\ell^{\varphi} h. \tag{12.93}$$

Consequently

$$\widehat{T}_\ell^{\mathcal{G}} \Theta_\ell^{\mathcal{G}} = \Theta_{\ell+1}^{\mathcal{G}} T_\ell^{\mathcal{G}}. \tag{12.94}$$

If the retained projections also intertwine, $\widehat{R}_k \Theta_k^{\mathcal{G}} = \Theta_k^{\mathcal{G}} R_k$ for $k = \ell, \ell+1$, then both residuals are covariant:

$$\widehat{r}_{\ell+1}^{\mathcal{G}}(\Theta_\ell^{\mathcal{G}} g) = \Theta_{\ell+1}^{\mathcal{G}} r_{\ell+1}^{\mathcal{G}}(g), \tag{12.95}$$

$$\widehat{\overline{r}}_{\ell+1}^{Q}(\Theta_\ell^{\mathcal{G}} g) = \overline{\Theta}_{\ell+1} \overline{r}_{\ell+1}^{Q}(g). \tag{12.96}$$

[ESTABLISHED]

Proof. Kernel covariance gives

$$\widehat{m}_\ell \widehat{K}_\ell = (\vartheta_{\ell+1})_\# (m_\ell K_\ell), \qquad (e^{-\Theta_\ell \varphi} \widehat{m}_\ell) \widehat{K}_\ell = (\vartheta_{\ell+1})_\# ((e^{-\varphi} m_\ell) K_\ell).$$

The Radon–Nikodym derivative of two measures pushed through the same Borel isomorphism is the transported original derivative. Applying $-\log$ proves (12.92) without selecting a pointwise reverse conditional. Differentiate this identity at an arbitrary bounded φ, using Proposition 12.5, to obtain (12.93). Compose the first identity with the assembly and extraction intertwinings in (12.89) to prove (12.94). Finally insert the retained-projection intertwining into $I - R_{\ell+1}$ and into the exact residual identity (12.104); this proves the two residual formulas. □

12.3.1 Exact nonlinear interaction RG and retained projections

Let Q_ℓ be the RN-first bounded action map of Section 12.2.1; additive homogeneity makes its quotient map $\overline{Q}_\ell : \overline{\mathfrak{B}}_\ell \to \overline{\mathfrak{B}}_{\ell+1}$ well-defined on the full bounded domain. The exact nonlinear finite-network interaction step is

$$T_\ell^{\mathcal{G}} = \mathsf{H}_{\ell+1} \overline{Q}_\ell E_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}, \qquad g_{\ell+1}^{\mathrm{ex}} = T_\ell^{\mathcal{G}}(g_\ell). \tag{12.97}$$

It is exact as a change of the bounded action class, conditional on the product-equivalence premises at both scales; it does not assert that a pairwise or finite retained ansatz is closed. The displayed sum ϕ_g is the canonical zero-ν_ℓ-mean representative of $E_\ell g$, so the separately tracked perturbed evidence mass is

$$M_\ell(g) = M_\ell \pi_\ell\big(e^{-\phi_g}\big), \qquad M_{\ell+1}(g) = M_\ell(g), \tag{12.98}$$

where the second equality is normalization of K_ℓ, not an identification of the action quotient with a normalized probability path. [ESTABLISHED]

At g, define the tilted conditional-expectation operator on bounded classes by

$$U_\ell^{\phi_g} h = \frac{U_\ell(e^{-\phi_g} h)}{U_\ell(e^{-\phi_g})}. \tag{12.99}$$

It is unchanged when a constant is added to ϕ_g, and therefore descends to the action quotient. The chain rule and Proposition 12.5 give the correctly typed derivative

$$DT_\ell^{\mathcal{G}}(g) = \mathsf{H}_{\ell+1}\overline{U_\ell^{\phi_g}} E_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}. \tag{12.100}$$

Only at $g = 0$ may $U_\ell^{\phi_g}$ be replaced by the fixed reverse conditional expectation U_ℓ. Along an exact orbit these derivatives form a typed cocycle, rather than an ordinary eigenvalue equation across unequal network spaces. [ESTABLISHED]

Definition 12.21 (Interaction modes along an exact orbit). Instantiate Definition 11.8 with $X_\ell = \mathcal{G}_\ell$, $T_\ell = T_\ell^{\mathcal{G}}$, and an exact interaction orbit $g_{\ell+1} = T_\ell^{\mathcal{G}}(g_\ell)$, so that

$$D_\ell = DT_\ell^{\mathcal{G}}(g_\ell) = \mathsf{H}_{\ell+1}\overline{U_\ell^{\phi_{g_\ell}}} E_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}. \tag{12.101}$$

A generalized interaction mode of index a is a compatible line $D_\ell v_{\ell,a} = \lambda_{\ell,a} v_{\ell+1,a}$ in the sense of Definition 11.10, and its finite-step coefficient is the ordered product (11.28) taken with respect to the cumulative log block scale (11.29) of the same blocking sequence. No eigenvalue equation is written inside one unnamed interaction space. [DEFINITION]

Corollary 12.22 (Temperedness of interaction comparison trivializations). Suppose the declared comparison isomorphisms $J_\ell : \mathcal{G}_\star \to \mathcal{G}_\ell$ obey the growth bound

$$\log^+\|J_n\| + \log^+\|J_n^{-1}\| \le c\,(1 + |V_n|) \tag{12.102}$$

for a constant c independent of n. This bound holds for every trivialization assembled from the maps of (12.88) together with a bounded gauge realization, because $\|E_\ell\| \le 1$ and $\|\mathsf{H}_\ell\| \le 3^{|V_\ell|} - 1$. Then the tempering hypothesis (11.35) of Theorem 11.14 holds whenever $|V_n|/s_{n\leftarrow\ell} \to 0$. In particular it holds for every blocking sequence whose vertex counts do not increase and whose block ratios satisfy $b_k \ge b > 1$. [ESTABLISHED]

Proof. Insert (12.102) into the left side of (11.35) and divide by $s_{n\leftarrow\ell}$; the constant term vanishes because $s_{n\leftarrow\ell}\to\infty$. If the vertex counts do not increase then $|V_n|\le|V_\ell|$ while $s_{n\leftarrow\ell}\ge(n-\ell)\log b\to\infty$. The stated operator bounds are (12.88). □

This corollary is the point at which the sharp finite-size extraction bound of Theorem 12.19 becomes a hypothesis of the mode theory rather than a curiosity: the bound is exponential in $|V_\ell|$, so a scheme in which the network grows fast enough with depth can violate (11.35) and change the reported exponents, exactly as in Proposition 11.15. [ESTABLISHED]

Let $R_\ell:\mathcal{G}_\ell\to\mathcal{G}_\ell$ be a declared bounded idempotent retained projection which intertwines the componentwise/permutation gauge action in (12.89). Define the exact and retained coordinate outputs and the coordinate/action-quotient residuals:

$$g^{\mathrm{ret}}_{\ell+1}=R_{\ell+1}T^{\mathcal{G}}_\ell(g_\ell),\qquad r^{\mathcal{G}}_{\ell+1}(g_\ell)=(I-R_{\ell+1})T^{\mathcal{G}}_\ell(g_\ell), \tag{12.103}$$

$$\overline{r}^{Q}_{\ell+1}(g_\ell)=\overline{Q}_\ell E_\ell g_\ell-E_{\ell+1}R_{\ell+1}\mathsf{H}_{\ell+1}\overline{Q}_\ell E_\ell g_\ell\quad=E_{\ell+1}r^{\mathcal{G}}_{\ell+1}(g_\ell). \tag{12.104}$$

The second line is the exact projected-residual identity in the action quotient. Since $E_{\ell+1}$ is injective, $\overline{r}^{Q}_{\ell+1}(g)=0$ if and only if $r^{\mathcal{G}}_{\ell+1}(g)=0$. Consequently

$$T^{\mathcal{G}}_\ell(\operatorname{Ran}R_\ell)\subseteq\operatorname{Ran}R_{\ell+1}\quad\Longleftrightarrow\quad r^{\mathcal{G}}_{\ell+1}(g)=0\quad\text{for every } g\in\operatorname{Ran}R_\ell. \tag{12.105}$$

Thus a retained update is exact precisely under this exact-image-invariance condition; otherwise it is a projection scheme with the displayed residual. Moreover extraction and assembly give the exact coordinate identity and the two-sided finite-network norm control, for $V_{\ell+1}\neq\varnothing$,

$$\mathsf{H}_{\ell+1}\overline{r}^{Q}_{\ell+1}=r^{\mathcal{G}}_{\ell+1},\qquad\frac{\|r^{\mathcal{G}}_{\ell+1}\|_{\mathcal{G}_{\ell+1}}}{3^{|V_{\ell+1}|}-1}\le\|\overline{r}^{Q}_{\ell+1}\|_{\overline{\mathfrak{B}}_{\ell+1}}\le\|r^{\mathcal{G}}_{\ell+1}\|_{\mathcal{G}_{\ell+1}}. \tag{12.106}$$

Under the full transformed-measure, kernel, and retained-projection intertwining hypotheses of Proposition 12.20, both residuals are gauge covariant. [ESTABLISHED]

12.4 Exact closure generates hyperedges

Suppose the complete nonnegative density of the law, including the baseline, is factorized against a product or block-product reference. Marginalizing an internal variable multiplies every incident factor, including baseline factors, and integrates their product. The result is one factor on the union of the remaining incident scopes. Iterating this operation represents the normalized law by generally unnormalized factors on the elimination cliques; the inherited total normalizer, not each factor, normalizes their product. Tonelli's theorem makes the resulting joint law independent of elimination order even though the displayed factorization is scheme dependent. For an arbitrary correlated baseline without such a factorization, retain the baseline as one global factor and use the law-level pushforward instead. [ESTABLISHED]

For a finite meta-agent set $\mathcal{P}$ and a finite-valued effective action, choose an anchor z° and define

$$\Phi_A^c(z_A) = \sum_{B\subseteq A} (-1)^{|A|-|B|} H_o^c(z_B, z^\circ_{B^c}), \qquad A \subseteq \mathcal{P}. \tag{12.107}$$

Mobius inversion gives

$$H_o^c(z) = \sum_{A\subseteq\mathcal{P}} \Phi_A^c(z_A). \tag{12.108}$$

Thus the full hypergraph family is exactly closed. If H_o^c is extended valued, retaining it as one top-order factor gives exact closure without forming undefined alternating sums. [ESTABLISHED]

The anchor is a covariant background, not a coordinatewise constant under frame changes. Consequently each displayed factor obeys $\Phi_A^c(gz_A; gz^\circ) = \Phi_A^c(z_A; z^\circ)$ when the action is gauge invariant. If an anchor is held fixed only in one coordinate chart, gauge covariance is asserted only for the reconstructed sum (12.108). [ESTABLISHED]

Pairwise closure is false in general. Eliminating the center s_0 of an Ising star produces

$$\sum_{s_0=\pm1} \exp\left[s_0\left(h_0 + \sum_{r=1}^{3} J_r s_r\right)\right] = 2\cosh\left(h_0 + \sum_{r=1}^{3} J_r s_r\right), \tag{12.109}$$

whose negative log has cubic coefficient $2\,\mathrm{sech}^2(h_0)\tanh(h_0)J_1J_2J_3 + O(J^5)$ when $h_0J_1J_2J_3 \neq 0$; it is therefore nonzero for all sufficiently small nonzero J_r with $h_0 \neq 0$. [ESTABLISHED]

12.5 Fine–meta bridge kernels

The posterior bridge

$$B_o(dy, dz) = \Pi_o(dy)C(y, dz) \tag{12.110}$$

defines the full reverse kernel $R_o(z, dy) = B_o(dy \mid z)$. For fine agent j and meta-agent I, take the (Y_j, Z_I) marginal and disintegrate it in both directions:

$$\Pi_{o,j}(dy_j)K_{I\leftarrow j}(y_j, dz_I) = \Pi_{o,I}^c(dz_I)K_{j\leftarrow I}(z_I, dy_j). \tag{12.111}$$

These normalized kernels are the exact agent-to-meta and meta-to-agent pair-marginal association, or bridge, kernels. They are not thereby causal generative interactions. For bounded observables f and g, they obey

$$\int f(y_j)(K_{I\leftarrow j}g)(y_j)\Pi_{o,j}(dy_j) = \int (K_{j\leftarrow I}f)(z_I)g(z_I)\Pi_{o,I}^c(dz_I). \tag{12.112}$$

[ESTABLISHED]

Equation (12.111) does not say that Z_I alone reconstructs the full fine posterior. The exact full reverse generally reads all meta variables. A product baseline and blockwise coarse channel factorize the baseline reverse, but a likelihood coupling blocks can destroy that factorization in the

posterior. Posterior-local refinement additionally requires the conditional-independence property $Y_I \perp Z_{-I} \mid (Z_I, o)$, or another separately proved sufficient condition. [ESTABLISHED]

For a deterministic statistic $z = c(y)$, the observable pullback and posterior conditional expectation are

$$(\mathsf{P}f)(y) = f(c(y)), \qquad (\mathsf{C}g)(z) = \mathbb{E}_{\Pi_o}[g(Y) \mid c(Y) = z]. \tag{12.113}$$

They satisfy $\mathsf{CP} = I$ in $L^p(\Pi_o^c)$, equivalently Π_o^c-almost everywhere, and compose by the tower property along a fixed nested filtration. Values of a conditional kernel at unused coarse points are immaterial. For a randomized channel, the corresponding operators remain adjoint but $\mathsf{CP} = I$ is not automatic. [ESTABLISHED]

12.6 Gauge-covariant cross-scale operators

For an arbitrary block partition $\mathcal{P}$, define the component-indexed meta-agent set

$$\widehat{\mathcal{P}} = \{\widehat{I} = (I, c) : I \in \mathcal{P}, c \in \pi_0(\Gamma_I)\}. \tag{12.114}$$

Choose a rooted spanning tree inside each connected component $\widehat{I}$ and require successive trees to be nested. From this point through the rooted operator formulas, I, J denote elements of $\widehat{\mathcal{P}}$ and r_I denotes their single component root. The state of an original disconnected block is the product over its component meta-agents; collapsing that product to one root, row, or feature is an additional declared coarse channel. For channel $x \in \{b, m\}$, let $\tau^x_{I\leftarrow i}$ be the ordered product of the channel-specific graph links from i to the root r_I. Under the passive section rechoices $a_i^b = a_i$ and $a_i^m = b_i$ of Equation (2.10), it transforms as

$$\tau^{x\prime}_{I\leftarrow i} = (a^x_{r_I})^{-1}\tau^x_{I\leftarrow i}a^x_i. \tag{12.115}$$

These are two passive coordinate rechoices in one principal G-bundle, not an active $G \times G$ gauge symmetry. For one active automorphism, the two local functions instead obey $k_i^m = h_i^{-1}k_i^b h_i$.

Every non-tree edge is retained through the channel-typed based holonomy map

$$H^x_I : \pi_1(\Gamma_I, r_I) \longrightarrow G, \qquad H^{x\prime}_I(\gamma) = (a^x_{r_I})^{-1}H^x_I(\gamma)a^x_{r_I}. \tag{12.116}$$

For each marked boundary edge $e : j \to i$ from block J to block I, retain

$$V^x_e = \tau^x_{I\leftarrow i}\Theta^x_e(\tau^x_{J\leftarrow j})^{-1}, \qquad V^{x\prime}_e = (a^x_{r_I})^{-1}V^x_e a^x_{r_J}. \tag{12.117}$$

The exact datum is the simultaneous root-framed equivariant state $(\bar{z}^x_I, H^x_I, \{V^x_e\})$, or its one simultaneous root-gauge orbit; separately quotienting each holonomy loses its orientation relative to the root features and boundary legs. For noncompact G, a naive conjugacy quotient need not be standard Borel, so the measure tier retains the raw root-framed G-space unless a proper-action or standard-Borel quotient theorem is supplied. Different trees change the presentation by Nielsen

transformations; nested trees with the compatibility below give strict composition, while arbitrary choices give a root-gauge-equivariant isomorphism of presentations. [ESTABLISHED]

Full-law holonomy alternatives. The full pointwise datum of Theorem 12.2 has two declared alternatives, which need not be logically exclusive at different retained-coordinate or quotient levels. For a holonomy-blind parent, supply bimeasurable typed groupoid actions on the complete fine random space, the parent space Z_{A,c_*}, the unchanged observation space, and structural data; require the fine generative and recognition laws to be covariant; select a posterior-version family compatible with those actions; require

$$C_{A,c_*}(g \cdot Y, g \cdot D) = C_{A,c_*}(Y, D), \tag{12.118}$$

for every admitted arrow and measurable parent event; and require evaluator covariance for the typed actions on (m_A, ξ_A) and (B_A, O, H_A). Under those hypotheses, substitution in (12.8) and (12.10) makes the complete parent generative, posterior, and recognition laws covariant. Same-slice invariance at a fixed (o, X) follows only for isotropy arrows that fix (o, X) and the selected posterior versions. Separate invariance of q_A^b and q_A^m is only a projection consequence and does not replace these full-law hypotheses.

On the retention alternative, the raw root-framed state above places component roots, the complete based holonomy representations, and the dressed boundary marks in H_A and pushes their joint law forward without a holonomy-blind claim. This alternative preserves path information; it does not select I, A, a partition, or a membership kernel. [ESTABLISHED]

Scope of the word "holonomy" in the blindness alternative. The name is inherited from Theorem 10.13, where the acting object is the genuine based-loop transport group $\mathfrak{h}_I^x(r) = \{T_\lambda : \lambda \text{ a loop at } r\}$ and the statement is about parallel marginal-law sections. The full-law alternative above generalizes that statement by widening the acting object to an *arbitrary typed groupoid of bimeasurable isomorphisms*, and no connection form, horizontal distribution, lift, loop, or curvature enters its hypotheses or its proof: the argument is substitution of an equivariant channel into (12.8) and (12.10). What is proved is therefore *covariance of the full parent laws under a declared groupoid action*, with same-slice invariance on the isotropy subgroup of (o, X); it is not a statement that a holonomy has been computed or quotiented away. Loop transports are one admissible instance of that groupoid, which is why the inherited name is retained, but a reader should not infer connection-theoretic content from it. The retention alternative, by contrast, does carry literal holonomy data: the based holonomy representations are stored in H_A as raw records. [ESTABLISHED]

On a separately declared linear feature fiber V_x, let $R_x : G \to \mathrm{GL}(V_x)$ be a linear representation. Choose $w_{Ii} \geq 0$ with $\sum_{i \in I} w_{Ii} = 1$ and set

$$(\mathsf{C}_x z)_I = \sum_{i \in I} w_{Ii} R_x(\tau^x_{I \leftarrow i}) z_i, \qquad (\mathsf{P}_x \bar{z})_i = R_x(\tau^x_{I \leftarrow i})^{-1} \bar{z}_I. \tag{12.119}$$

Then $\mathsf{C}_x\mathsf{P}_x = I$. With $R_{x,f} = \bigoplus_i R_x(a_i^x)$ and $R_{x,c} = \bigoplus_I R_x(a_{r_I}^x)$, passive covariance is

$$\mathsf{C}'_x = R_{x,c}^{-1}\mathsf{C}_x R_{x,f}, \qquad \mathsf{P}'_x = R_{x,f}^{-1}\mathsf{P}_x R_{x,c}. \tag{12.120}$$

At nested levels $0 \to 1 \to 2$, let $\sigma_{A\leftarrow I}^{x,12}$ be the ordered interroot link from the root of I to the root of A. Strict composition additionally requires

$$\tau_{A\leftarrow i}^{x,02} = \sigma_{A\leftarrow I}^{x,12}\tau_{I\leftarrow i}^{x,01}, \qquad w_{Ai}^{02} = w_{AI}^{12}w_{Ii}^{01}. \tag{12.121}$$

It then follows that $\mathsf{C}_x^{02} = \mathsf{C}_x^{12}\mathsf{C}_x^{01}$ and $\mathsf{P}_x^{02} = \mathsf{P}_x^{01}\mathsf{P}_x^{12}$; nested forests alone would not imply either equality. If A_x is a fine message operator with $A'_x = R_{x,f}^{-1}A_x R_{x,f}$, the fine-to-meta, meta-to-fine, and meta-to-meta message operators are

$$A_{I\leftarrow j}^x = (\mathsf{C}_x A_x)_{Ij}, \qquad A_{j\leftarrow I}^x = (A_x\mathsf{P}_x)_{jI}, \qquad A_{I\leftarrow J}^x = (\mathsf{C}_x A_x \mathsf{P}_x)_{IJ}. \tag{12.122}$$

They transform between the corresponding endpoint frames; in particular,

$$(\mathsf{C}_x A_x \mathsf{P}_x)' = R_{x,c}^{-1}(\mathsf{C}_x A_x \mathsf{P}_x)R_{x,c}.$$

[ESTABLISHED]

These message formulas are not an energy marginalization theorem. A hard-tied quadratic energy uses $\mathsf{P}_x^* H_x \mathsf{P}_x$; an exact Gaussian marginal uses a Schur complement; a general law uses Theorem 12.3. The identity $\mathsf{C}_x\mathsf{P}_x = I$ alone does not make $\mathsf{C}_x H_x \mathsf{P}_x$ positive. [ESTABLISHED]

If $\Phi_f : V_{b,f} \to V_{m,f}$ is a separately declared linear cross-channel morphism on this tier, usually $\bigoplus_i \Phi_i$, with $\Phi'_f = R_{m,f}^{-1}\Phi_f R_{b,f}$, its compressed map is

$$\Phi^c = \mathsf{C}_m \Phi_f \mathsf{P}_b, \qquad \Phi^{c\prime} = R_{m,c}^{-1}\Phi^c R_{b,c}. \tag{12.123}$$

The common principal bundle does not create Φ, and the two representations must never be identified merely because both use G. [ESTABLISHED]

12.7 Exact attention between meta-agents

A row-stochastic matrix β_{ij} is a conditional law of a source given a receiver. It cannot be coarse-grained without the receiver law. Let $\alpha_i(y)$ be a normalized receiver occupancy and define the joint marked edge-event law

$$\eta_{ij}(y) = \alpha_i(y)\beta_{ij}(y), \qquad \sum_{i,j}\eta_{ij}(y) = 1. \tag{12.124}$$

The event probabilities are gauge-invariant scalars: $\eta'_{ij} = \eta_{ij}$. For a posterior bridge and node partition, set

$$\eta^c_{IJ}(z) = \mathbb{E}_{B_o}\left[\sum_{i\in I}\sum_{j\in J}\eta_{ij}(Y)\middle| Z = z\right], \quad \alpha^c_I(z) = \sum_J \eta^c_{IJ}(z), \quad \beta^c_{IJ}(z) = \frac{\eta^c_{IJ}(z)}{\alpha^c_I(z)}. \tag{12.125}$$

The last ratio is used on $\{\alpha^c_I > 0\}$; a row on a zero-occupancy receiver is immaterial. Pushing η and then disintegrating is normalized and associative under nested partitions by the tower property. The next scale must push the joint law η^c, not β^c alone. Consequently η^c, α^c, and β^c are gauge invariant. [ESTABLISHED]

In an augmented receiver–source-label version of Section 7.5, the fixed recognition-independent label baseline is

$$P_0^{\mathrm{att}}(dy, di, dj) = P_0^Y(dy)a_i\pi_{ij}, \qquad \sum_i a_i = 1, \quad \sum_j \pi_{ij} = 1. \tag{12.126}$$

Fix the latent state y and define the row partition function $Z_i(y) = \sum_j \pi_{ij}\exp[-D_{ij}(y)/\tau]$. At the selected record, the conditional prior inside a block with $a_I = \sum_{k\in I} a_k > 0$ is $a_{i|I} = a_i/a_I$. Define the conditional block evidence weight $W_I(y) = \sum_{k\in I} a_{k|I}c_k(o_k, y)Z_k(y)$. On $\{W_I(y) > 0\}$, its posterior is evidence weighted:

$$\alpha'_i(y) = \frac{a_{i|I}c_i(o_i, y)Z_i(y)}{W_I(y)}, \qquad \beta^c_{IJ}(y) = \sum_{i\in I}\alpha'_i(y)\sum_{j\in J}\beta_{ij}. \tag{12.127}$$

The formula holds posterior-almost everywhere. On a zero-evidence block the receiver row is an arbitrary conditional version and is removed from the effective support. The equality $\alpha'_i = a_{i|I}$ holds exactly when $c_i(o_i, y)Z_i(y)$ is constant on the conditional support in I. This guarantees equality with the pre-observation coarse mixture, although equality can also occur accidentally after source coarsening—for example, when all receivers induce the same block-source row. If all receiver labels are retained, the exact meta-label is their vector or multiset; replacing it by one categorical row is a further coarse channel whose conditional KL appears in (12.4). [ESTABLISHED]

For one categorical row and a source partition $j \mapsto J$, the exact coarse prior and energy are

$$\pi^c_J = \sum_{j\in J}\pi_j, \qquad E^c_J = -\tau\log\left[\frac{1}{\pi^c_J}\sum_{j\in J}\pi_j\exp[-E_j/\tau]\right]. \tag{12.128}$$

for every block with $\pi^c_J > 0$; a zero-prior block has zero posterior mass and is removed from the effective support. This log-sum-exp rule associates for nested partitions of the same row. It does not merge independently normalized receiver rows without (12.127). [ESTABLISHED]

The transported meta-message in channel x is represented exactly by marked parallel edges. Let the fine mark $U^x_{i\leftarrow j}$ obey the passive endpoint law and define its dressed version $V^x_{ij} = \tau^x_{I\leftarrow i}U^x_{i\leftarrow j}(\tau^x_{J\leftarrow j})^{-1}$. On the linear feature tier, define the following moment only under the conditional

Bochner-integrability condition

$$\mathbb{E}_{B_o}\left[\sum_{i\in I,j\in J}\eta_{ij}(Y)\|R_x(V_{ij}^x)\|\middle|Z\right]<\infty \quad \text{almost everywhere.} \tag{12.129}$$

Then

$$\begin{aligned}
\mathsf{T}_{IJ}^x(z)&=\mathbb{E}_{B_o}\left[\sum_{i\in I,j\in J}\eta_{ij}(Y)R_x(V_{ij}^x)\middle|Z=z\right],\\
\overline{\mathsf{U}}_{IJ}^x(z)&=\frac{\mathsf{T}_{IJ}^x(z)}{\eta_{IJ}^c(z)}\quad \text{on } \{\eta_{IJ}^c>0\},\\
\mathsf{A}_{IJ}^{x,\mathrm{att}}(z)&=\frac{\mathsf{T}_{IJ}^x(z)}{\alpha_I^c(z)}\quad \text{on } \{\alpha_I^c>0\}.
\end{aligned} \tag{12.130}$$

On $\{\eta_{IJ}^c>0\}$, $\mathsf{A}_{IJ}^{x,\mathrm{att}}=\beta_{IJ}^c\overline{\mathsf{U}}_{IJ}^x$. On $\{\alpha_I^c>0,\eta_{IJ}^c=0\}$, the attention-weighted block is zero and $\overline{\mathsf{U}}_{IJ}^x$ is left undefined as a null-event conditional. For passive root-frame rechoices with no node relabeling ($\sigma=\mathrm{id}$), corresponding transported conditional versions, and the transformed coarse Borel state $z'=\vartheta_{\ell+1}(z)$, covariance holds Π_o^c-almost everywhere in z (equivalently, under the transformed coarse law, almost everywhere in z'): $\mathsf{T}_{IJ}^{x\prime}(z')=R_x(a_{r_I}^x)^{-1}\mathsf{T}_{IJ}^x(z)R_x(a_{r_J}^x)$, so these operators transform from the J root frame to the I root frame. The symbol $R_{x,c}$ remains the direct-sum coordinate operator on the separately declared linear feature fiber in (12.120); it is not an action on the Borel state z. The conditional mean need not belong to $R_x(G)$: the average of rotations by $\pi/2$ and $-\pi/2$ is the zero matrix. Moreover, this Hom operator is exact only for its stated first operator moment on a common source input. If marks and source features are correlated, their means do not determine $\mathbb{E}[Uz]$; for example, equal-probability marks/features (I,v) and $(-I,-v)$ have zero separate means but mean product v. Full closure therefore retains the marked operator–feature kernel or all marked edges, with (12.130) as a derived moment, rather than a fictitious averaged group element. [ESTABLISHED]

At a fixed outside context, define $\beta_I^Q(J\mid z_{-I})=Q(J\mid z_{-I})$ and $K_{I\leftarrow J}^Q(dz_I\mid z_{-I})=Q(dz_I\mid J,z_{-I})$, and define the corresponding generative conditionals under $P(\cdot\mid z_{-I},o)$. When the recognition categorical and conditional kernels are absolutely continuous with respect to their generative counterparts, with the extended-value convention, the conditional KL chain rule is

$$\begin{aligned}
&D_{\mathrm{KL}}\big(Q(J,Z_I\mid z_{-I})\|P(J,Z_I\mid z_{-I},o)\big)\\
&\quad=D_{\mathrm{KL}}\big(\beta_I^Q(\cdot\mid z_{-I})\|\beta_I^P(\cdot\mid z_{-I},o)\big)\\
&\qquad+\sum_J\beta_I^Q(J\mid z_{-I})D_{\mathrm{KL}}\big(K_{I\leftarrow J}^Q(\cdot\mid z_{-I})\|K_{I\leftarrow J}^P(\cdot\mid z_{-I},o)\big).
\end{aligned} \tag{12.131}$$

Thus meta-attention and meta-agent interaction kernels are two typed parts of one conditional ELBO, not interchangeable names for the same object. [ESTABLISHED]

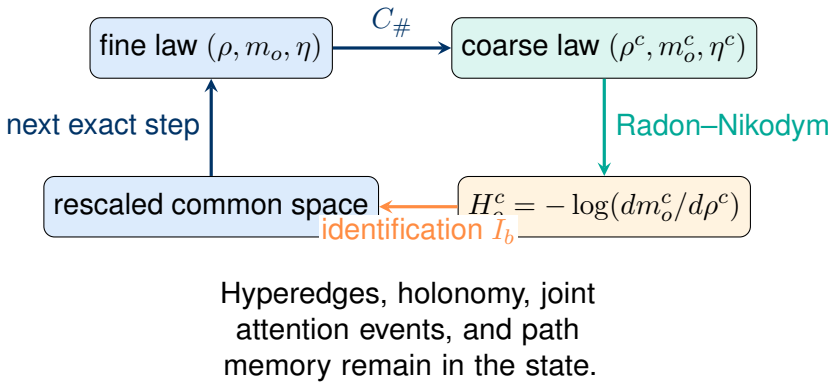


Figure 12.1: The exact RG acts first on normalized measures. An effective action and a beta function are introduced only after a pushed reference and a declared identification return the changing scale spaces to one comparison space.

12.8 Ordered-path closure and memory

For a finite ordered path interval, apply the pointwise coarse channel to the full joint path law. Its order parameter is auxiliary: it is neither a coordinate on the fixed base $\mathcal{C}$ nor the intrinsic information clock of Chapter 9. Theorem 12.1 then applies on path space without change. The resulting coarse path law is normalized and gauge covariant when the required spatial transports and, for a separately declared dynamical model, parameter-direction transports are included. It need not be first-order Markov. [ESTABLISHED]

Theorem 12.23 (Standard-Borel strong lumpability). Let $(\mathsf{Y}, \mathscr{Y})$ and $(\mathsf{Z}, \mathscr{Z})$ be standard Borel, let $c : \mathsf{Y} \to \mathsf{Z}$ be Borel and surjective, and let T be a Markov kernel on Y. Consider

$$T(y, c^{-1}B) = T(y', c^{-1}B) \quad \text{whenever } c(y) = c(y'), \qquad \text{for every } B \in \mathscr{Z}. \tag{12.132}$$

If a Markov kernel T^c on Z satisfies

$$c_\#(\mu T) = (c_\#\mu)T^c \qquad \text{for every probability } \mu \text{ on Y}, \tag{12.133}$$

then (12.132) holds and T^c is unique. Conversely, if (12.132) holds and c admits a Borel right inverse $\varsigma : \mathsf{Z} \to \mathsf{Y}$ with $c \circ \varsigma = \mathrm{id}$, then

$$T^c(z, B) = T\big(\varsigma(z), c^{-1}B\big) \tag{12.134}$$

is a Markov kernel satisfying (12.133). A Borel right inverse exists in the two cases used in this chapter: a countable coarse space, and a product projection $\mathsf{Y} = \mathsf{Z} \times \mathsf{Y}'$ with Y' nonempty, for which one fixes any point of Y'. Without a measurable selection, (12.134) still defines the coarse transition on the range of c and the selection must be declared separately. [ESTABLISHED]

Proof. Assume (12.133) and take $\mu = \delta_y$. Then $c_\#(\delta_y T)(B) = T(y, c^{-1}B)$ while $((c_\#\delta_y)T^c)(B) = T^c(c(y), B)$, so

$$T(y, c^{-1}B) = T^c(c(y), B) \qquad \text{for every } y \in \mathsf{Y},\ B \in \mathscr{Z}, \tag{12.135}$$

whose right side depends on y only through $c(y)$; this is (12.132). Surjectivity of c and (12.135) determine T^c at every z, so it is unique.

Conversely assume (12.132) and a Borel ς. For fixed z, the set map $B \mapsto T(\varsigma(z), c^{-1}B)$ is a probability measure, because c^{-1} commutes with countable disjoint unions and $c^{-1}\mathsf{Z} = \mathsf{Y}$. For fixed B, the map $z \mapsto T(\varsigma(z), c^{-1}B)$ is the composition of the Borel map ς with the measurable map $y \mapsto T(y, c^{-1}B)$, hence measurable. So (12.134) is a Markov kernel. Since $c(\varsigma(c(y))) = c(y)$, hypothesis (12.132) gives $T^c(c(y), B) = T(y, c^{-1}B)$ for every y. Therefore, for any probability μ,

$$c_\#(\mu T)(B) = \int T(y, c^{-1}B)\mu(dy) = \int T^c(c(y), B)\mu(dy) = \int T^c(z, B)(c_\#\mu)(dz),$$

which is (12.133). □

Strong lumpability is exactly the every-initial-law condition, and it is not necessary for a single initial law. Take $\mathsf{Y} = \{1, 2, 3\}$ with $c(1) = c(2) = a$ and $c(3) = \beta$, and let T send 1 to 3, send 2 to 1 or 2 with equal probability, and send 3 to 1 or 3 with equal probability. Then $T(1, \{1, 2\}) = 0$ while $T(2, \{1, 2\}) = 1$, so (12.132) fails. Started at δ_3, however, the chain never leaves $\{1, 3\}$, on which c is injective, so the coarse process is a Markov chain with transitions $a \mapsto \beta$ surely and $\beta \mapsto a$ or β with equal probability. Weak lumpability at a selected initial law is therefore a strictly weaker statement, and it is the only one available when (12.132) fails. [ESTABLISHED]

The finite-state form of the biconditional is classical and is the lumpability criterion of Kemeny and Snell [48, Ch. 6]; that source is cited only at its finite-state scope, and the standard-Borel statement above is proved here. [ESTABLISHED]

Open Problem 12.24 (Dynamic closure of the pointwise parent). The static construction of Theorem 12.2 supplies no fine evolution and no parent evolution. In a separately declared differentiable special case with fine field V_t, moving deterministic coarse map c_t, and candidate parent field $\overline{V}_t$, the exact semiconjugacy defect is

$$\delta_t = \partial_t c_t + Dc_t V_t - \overline{V}_t \circ c_t. \tag{12.136}$$

For a Markov evolution the corresponding exact condition is semigroup intertwining; a generator identity is an alternative criterion only on a declared invariant domain or core, with generation and uniqueness hypotheses sufficient to lift it to the semigroups. Proving either condition for the released static datum, together with any autonomy, moving-membership, path-law, stochastic, delayed, memory, or nonequilibrium claim, remains open. The parameter t is not thereby physical time. [OPEN]

For an explicit finite family, Theorem 16.1 gives a weaker dynamical conclusion with quantitative content: declared microscopic kinetics yield successive cyclic formation, exact marked-law parents

and actual-law activation, with separate vanishing conditional, score, continuous-trajectory and assignment-history errors. The approximation holds under the stated finite-depth preparation and scale hierarchy; it does not establish the exact every-initial-law intertwining required above. [ESTABLISHED]

Theorem 12.25 (Exact discrete projection-memory recurrence). Let X and W be vector spaces, let $T : \mathsf{X} \to \mathsf{X}$, $\mathsf{C} : \mathsf{X} \to \mathsf{W}$, and $\mathsf{P} : \mathsf{W} \to \mathsf{X}$ be linear with $\mathsf{CP} = I_{\mathsf{W}}$, and set

$$\Pi^{\mathrm{res}} = \mathsf{PC}, \qquad \mathsf{Q} = I - \Pi^{\mathrm{res}}, \tag{12.137}$$

so that Π^{res} is idempotent. For the fine recursion $x_{n+1} = Tx_n$ write $w_n = \mathsf{C}x_n$, so that $\Pi^{\mathrm{res}}x_n = \mathsf{P}w_n$. Then for every $n \geq 0$

$$\mathsf{Q}x_n = (\mathsf{Q}T\mathsf{Q})^n\mathsf{Q}x_0 + \sum_{k=0}^{n-1}(\mathsf{Q}T\mathsf{Q})^{n-1-k}\mathsf{Q}T\mathsf{P}w_k, \tag{12.138}$$

and the resolved coordinate obeys the exact closed recurrence

$$w_{n+1} = \mathsf{C}T\mathsf{P}w_n + \sum_{k=0}^{n-1}\mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^{n-1-k}\mathsf{Q}T\mathsf{P}w_k + \mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^n\mathsf{Q}x_0. \tag{12.139}$$

The operators

$$\mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^m\mathsf{Q}T\mathsf{P}, \qquad m \geq 0, \tag{12.140}$$

are the memory kernels and the last term of (12.139) is the unresolved initial-state term. Equation (12.139) is exact; no closure or Markov assumption is used. [ESTABLISHED]

Proof. Apply Π^{res} and Q to $x_{n+1} = Tx_n$ and use $x_n = \Pi^{\mathrm{res}}x_n + \mathsf{Q}x_n = \mathsf{P}w_n + \mathsf{Q}x_n$ together with $\mathsf{QQ}x_n = \mathsf{Q}x_n$. This gives the coupled system

$$w_{n+1} = \mathsf{C}T\mathsf{P}w_n + \mathsf{C}T\mathsf{QQ}x_n, \qquad \mathsf{Q}x_{n+1} = \mathsf{Q}T\mathsf{QQ}x_n + \mathsf{Q}T\mathsf{P}w_n,$$

where both lines used idempotence of Q to write $\mathsf{Q}x_n = \mathsf{Q}(\mathsf{Q}x_n)$. Solving the second line by induction on n gives (12.138): it is an identity at $n = 0$, and applying $\mathsf{Q}T\mathsf{Q}$ to the case n and adding $\mathsf{Q}T\mathsf{P}w_n$ reproduces the case $n + 1$ with the index shifted. Substituting (12.138) into the first line gives (12.139). □

Corollary 12.26 (Autonomy of the resolved recursion). Let $\mathcal{X}_0$ be an admitted initial class. The resolved recursion is autonomous on $\mathcal{X}_0$, meaning $w_{n+1} = \mathsf{C}T\mathsf{P}w_n$ for every $n \geq 0$ and every $x_0 \in \mathcal{X}_0$, if and only if the total correction in (12.139) vanishes for every such n and x_0. Two sufficient conditions are that all memory kernels (12.140) vanish and that

$$\mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^n\mathsf{Q}x_0 = 0, \qquad n \geq 0, \tag{12.141}$$

throughout $\mathcal{X}_0$. On the resolved class $\mathsf{Q}x_0 = 0$ the family (12.141) is vacuous, and there $\mathsf{Q}T\mathsf{P} = 0$ is sufficient but not necessary: on $\mathsf{X} = \mathbb{R}^2$ with $\mathsf{W} = \mathbb{R}$, $\mathsf{C}(x, y) = x$, $\mathsf{P}(w) = (w, 0)$, and $T(x, y) = (x, x)$, one has $\mathsf{Q}T\mathsf{P}w = (0, w) \neq 0$ for $w \neq 0$ while $\mathsf{C}T\mathsf{Q} = 0$, so every memory kernel vanishes and the resolved recursion $w_{n+1} = w_n$ is exactly autonomous. [ESTABLISHED]

Proof. The biconditional is (12.139) read as an identity. The two displayed families are exactly the two correction sums, so their vanishing is sufficient. For the witness, $\mathsf{Q}(x, y) = (0, y)$ and $T(0, y) = (0, 0)$ give $\mathsf{C}T\mathsf{Q} = 0$, while $T(w, 0) = (w, w)$ gives $\mathsf{Q}T\mathsf{P}w = (0, w)$. □

Recurrence (12.139) is proved here for the discrete linear observable tier. The projection-operator method of eliminating unresolved coordinates in favor of a memory kernel and an initial-state term originates with Nakajima [63] and Zwanzig [93], who work in continuous time on ensemble densities; those sources are cited as the historical antecedent of the method and not as proofs of (12.139). [ESTABLISHED]

12.9 The RG transformation and beta functions

Let $b > 1$ label one blocking ratio. A genuine RG step consists of a coarse channel C_b and a declared rescaling/identification kernel I_b that returns the target to a common measurable state space. Define the typed rescaled kernel $K_b = C_b I_b$. Compatibility means, after the declared canonical identifications,

$$K_{b_1 b_2} = K_{b_1} K_{b_2}. \tag{12.142}$$

On the primary measure pair, define

$$\mathcal{R}_b(\rho, m_o) = \big((\rho C_b) I_b, (m_o C_b) I_b\big) = (\rho K_b, m_o K_b). \tag{12.143}$$

Write $\mathcal{R}_b^{\rho}\rho = (\rho C_b) I_b = \rho K_b$ for its reference component. Equation (12.142) makes $\mathcal{R}_{b_1 b_2} = \mathcal{R}_{b_2}\mathcal{R}_{b_1}$; otherwise the sequence is a typed cocycle rather than an autonomous semigroup. [ESTABLISHED]

The dichotomy is no longer hypothetical: the compatibility law was instantiated and tested, and it failed. On 2026-08-17 the finite categorical laboratory declared the identification kernels for the nested-cycle family (the Wilson-line coarse connection composed from retained dressed boundary generators, the cyclic graph identification, and the identity state identification), verified gauge covariance of the composite step to 10^{-12} and exact holonomy conservation, and then tested (12.142) against a tolerance of 10^{-10} fixed before measurement. Blocking six agents by the full ratio against blocking in declared stages disagrees by 0.204 in sup norm on the final couplings, with a provably lossless intermediate projection, so the defect is the Bayes blocking kernel composition itself: the declared downward-kernel family is not closed under composition across levels, for the same generic reason that lumped Markov chains are not Markov. The defect is order one, between 0.12 and 0.19, at every accessible depth and factorization, including two-then-three against three-then-two on a homogeneous six-cycle where both orderings are legitimate consecutive blockings. For the exhibited construction the second

branch of the dichotomy is therefore the operative one: the scale maps of this chapter form a typed cocycle, and every statement below that assumes (12.142) is a hypothesis that the first exhibited rescaling construction fails. Implementation, pre-registered design, and measurements: `src/multiagent_elbo/finite/rescaling.py`, `coupling_readback.py`, `cocycle_flow.py`, and `docs/superpowers/specs/2026-08-17-rescaling-map-design.md`. [NUMERICAL]

On a dominated tier, the action map retains its reference argument:

$$\mathcal{R}_b^H[H;\rho] := -\log \frac{d\big((e^{-H}\rho)K_b\big)}{d(\rho K_b)}. \tag{12.144}$$

On a declared vector space of finite-valued actions that is closed under this map for the declared reference trajectory and on which subtraction exists, the exact discrete step beta functional is

$$\mathfrak{B}_b^H[H;\rho] = \frac{\mathcal{R}_b^H[H;\rho] - H}{\log b}. \tag{12.145}$$

If the evidence mass is deliberately forgotten and finite unnormalized actions are quotiented by additive constants, the same formula is interpreted in that quotient. For extended-valued actions, including points at which the numerator would be $+\infty-(+\infty)$, only the measure-pair RG is primary. A beta function on a truncated coupling vector exists only after invariance of that finite family is proved; otherwise omitted hyperedges contribute an explicit residual. [ESTABLISHED]

Proposition 12.27 (Change of reference is an inhomogeneous law for the action beta). Fix one arrow with rescaled kernel K_b and reverse conditional $R_\rho(z, dy)$ as in Theorem 12.3. Let H be an action with $m = e^{-H}\rho$ finite and positive, let Δ be bounded and measurable with $\rho(e^{-\Delta}) = 1$, and set

$$\rho' = e^{-\Delta}\rho, \qquad H' = H - \Delta, \qquad \text{so that} \qquad e^{-H'}\rho' = e^{-H}\rho = m. \tag{12.146}$$

Then ρ' is again a probability reference for the same measure pair, and $(\rho' K_b)$-almost everywhere

$$\mathcal{R}_b^H[H';\rho'] = \mathcal{R}_b^H[H;\rho] - \mathcal{R}_b^H[\Delta;\rho]. \tag{12.147}$$

On a declared finite-valued action vector space containing H, Δ, and their images, the discrete beta functional of (12.145) therefore obeys the inhomogeneous law

$$\mathfrak{B}_b^H[H';\rho'] = \mathfrak{B}_b^H[H;\rho] - \mathfrak{B}_b^H[\Delta;\rho]. \tag{12.148}$$

The added term is not a linear reparameterization of the beta functional, and it vanishes exactly when the reference-change increment is itself a fixed action, $\mathcal{R}_b^H[\Delta;\rho] = \Delta$. A beta functional is therefore a joint statement about the action and the declared reference trajectory. [ESTABLISHED]

Proof. Since $\rho(e^{-\Delta}) = 1$ and Δ is bounded, ρ' is a probability measure equivalent to ρ, and (12.146) holds by construction. Both ρK_b and $\rho' K_b$ dominate mK_b, and the chain rule for Radon–

Nikodym derivatives gives

$$\frac{d(mK_b)}{d(\rho' K_b)} = \frac{d(mK_b)}{d(\rho K_b)} \cdot \left[\frac{d(\rho' K_b)}{d(\rho K_b)}\right]^{-1}$$

almost everywhere. By Theorem 12.3 applied to the pair $(\rho, e^{-\Delta}\rho)$, the bracketed factor equals $\int e^{-\Delta(y)} R_\rho(z, dy) = \exp(-\mathcal{R}_b^H[\Delta; \rho])$, which is strictly positive and finite because Δ is bounded. Taking $-\log$ of the display and using (12.144) twice gives (12.147). Subtracting $H' = H - \Delta$, dividing by $\log b$, and regrouping gives (12.148). The vanishing criterion is $\mathcal{R}_b^H[\Delta; \rho] - \Delta = 0$. □

Definition 12.28 (Typed discrete interaction beta). Let $\mathcal{G}_\star$ be a declared reference interaction space and let

$$J_\ell : \mathcal{G}_\star \xrightarrow{\sim} \mathcal{G}_\ell \tag{12.149}$$

be bounded isomorphisms with bounded inverses. This is the inverse orientation of the reference identifications I_ℓ of (11.3), so $J_\ell = I_\ell^{-1}$ whenever both are declared. The pulled exact interaction map and the exact discrete interaction beta are

$$\widehat{T}_\ell^{\mathcal{G}} = J_{\ell+1}^{-1} T_\ell^{\mathcal{G}} J_\ell : \mathcal{G}_\star \longrightarrow \mathcal{G}_\star, \qquad \beta_\ell^{\mathrm{ex}}(g) = \frac{\widehat{T}_\ell^{\mathcal{G}}(g) - g}{\Delta s_\ell}, \qquad \Delta s_\ell = \log b_\ell. \tag{12.150}$$

Both terms of the numerator lie in $\mathcal{G}_\star$; the subtraction $T_\ell^{\mathcal{G}}(g) - g$ is not formed, because its two terms inhabit $\mathcal{G}_{\ell+1}$ and $\mathcal{G}_\ell$. The symbol $\widehat{T}_\ell^{\mathcal{G}}$ is not the retained projection R_ℓ of (12.103), and no beta functional is written before (12.149) is declared. If the coupling object is a manifold rather than a Banach space, the difference must be replaced by a declared chart difference, logarithm, or retraction into one tangent space. [DEFINITION]

Proposition 12.29 (Exact, retained, and residual interaction beta). With $\widehat{R}_\ell = J_\ell^{-1} R_\ell J_\ell$ for the bounded idempotent retained projections of (12.103), define for $g \in \operatorname{Ran} \widehat{R}_\ell$

$$\beta_\ell^{\mathrm{ret}}(g) = \frac{\widehat{R}_{\ell+1} \widehat{T}_\ell^{\mathcal{G}}(g) - g}{\Delta s_\ell}, \qquad \delta\beta_\ell(g) = \beta_\ell^{\mathrm{ex}}(g) - \beta_\ell^{\mathrm{ret}}(g). \tag{12.151}$$

Then the omitted part of the flow is exactly the transported coordinate residual divided by the log block step,

$$\delta\beta_\ell(g) = \frac{(I - \widehat{R}_{\ell+1}) \widehat{T}_\ell^{\mathcal{G}}(g)}{\Delta s_\ell} = \frac{J_{\ell+1}^{-1} r_{\ell+1}^{\mathcal{G}}(J_\ell g)}{\Delta s_\ell}, \tag{12.152}$$

and its action-quotient partner is

$$\delta\overline{\beta}_{\ell+1}^{Q}(g) = \frac{E_{\ell+1} r_{\ell+1}^{\mathcal{G}}(J_\ell g)}{\Delta s_\ell}, \qquad \mathsf{H}_{\ell+1} \delta\overline{\beta}_{\ell+1}^{Q}(g) = \frac{r_{\ell+1}^{\mathcal{G}}(J_\ell g)}{\Delta s_\ell}. \tag{12.153}$$

Consequently $\delta\beta_\ell$ vanishes on the whole retained sector if and only if the exact image of the retained sector is retained,

$$\delta\beta_\ell\big|_{\operatorname{Ran} \widehat{R}_\ell} = 0 \quad \Longleftrightarrow \quad T_\ell^{\mathcal{G}}(\operatorname{Ran} R_\ell) \subseteq \operatorname{Ran} R_{\ell+1}, \tag{12.154}$$

which is (12.105). Boundedness and idempotence of R_ℓ do not imply this invariance, so a retained beta is called exact only after (12.154) is proved, and otherwise carries $\delta\beta_\ell$ with the two-sided norm control of (12.106). [ESTABLISHED]

Proof. Subtracting the two displays in (12.151) cancels g and leaves $(I - \widehat{R}_{\ell+1})\widehat{T}^{\mathcal{G}}_\ell(g)/\Delta s_\ell$. Substituting $\widehat{R}_{\ell+1} = J^{-1}_{\ell+1} R_{\ell+1} J_{\ell+1}$ and $\widehat{T}^{\mathcal{G}}_\ell = J^{-1}_{\ell+1} T^{\mathcal{G}}_\ell J_\ell$ gives $J^{-1}_{\ell+1}(I - R_{\ell+1})T^{\mathcal{G}}_\ell(J_\ell g)/\Delta s_\ell$, which is (12.152) by the definition of $r^{\mathcal{G}}_{\ell+1}$. Applying $E_{\ell+1}$ and $\mathsf{H}_{\ell+1}$ and using the inverse identities of Theorem 12.19 together with (12.104) gives (12.153). Since $J^{-1}_{\ell+1}$ is injective and $\Delta s_\ell > 0$, $\delta\beta_\ell(g) = 0$ is equivalent to $r^{\mathcal{G}}_{\ell+1}(J_\ell g) = 0$, and J_ℓ maps $\operatorname{Ran}\widehat{R}_\ell$ onto $\operatorname{Ran} R_\ell$; quantifying over that sector and using (12.105) gives (12.154). □

A projected fixed point is not an exact fixed point. On $\mathcal{G}_\star = \mathbb{R}^2$ with $b = e$, take $R(x, y) = (x, 0)$ and $T(x, y) = (x, x)$. For every retained input $g = (x, 0)$ one has $\beta^{\mathrm{ret}}(g) = (0, 0)$ while $\beta^{\mathrm{ex}}(g) = (0, x)$, so the retained flow reports a line of fixed points although the exact step generates an omitted component whenever $x \neq 0$. [ESTABLISHED]

The interaction beta is also scheme dependent, and its scheme dependence is larger than the coupling reparameterization of (11.37). If $J'_\ell = J_\ell S_\ell$ for bounded isomorphisms S_ℓ of $\mathcal{G}_\star$, then $\widehat{T}'_\ell = S^{-1}_{\ell+1}\widehat{T}_\ell S_\ell$ and

$$\beta'_\ell(g) = \frac{S^{-1}_{\ell+1}\widehat{T}^{\mathcal{G}}_\ell(S_\ell g) - g}{\Delta s_\ell}, \tag{12.155}$$

which reduces to the usual pushforward relation only for a scale-independent $S_\ell \equiv S$. Even the identity native step acquires a nonzero beta in a moving comparison frame: on $\mathcal{G}_\ell = \mathbb{R}$ with $T_\ell = \mathrm{id}$, $b_\ell = e$, and $J_\ell u = a_\ell u$, formula (12.150) gives $\beta_\ell(g) = (a_\ell/a_{\ell+1} - 1)g$, which is zero for $a_\ell \equiv 1$ and nonzero otherwise. The comparison scheme is therefore part of the statement of any reported beta function. [ESTABLISHED]

Definition 12.30 (Smooth scale bundle, scale connection, and continuous beta). A continuous beta function is a construction on a separately declared tier and is not obtained from a discrete sequence. Declare an oriented one-dimensional C^1 scale manifold S with coordinate s, a C^1 Banach vector bundle $\pi_S : \mathscr{G} \to S$ whose fibers are the coupling spaces, and a scale connection, that is a covariant derivative ∇^{scale} on sections of $\mathscr{G}$ satisfying the usual linearity and Leibniz rules. For a C^1 running coupling section $g(s) \in \mathscr{G}_s$, the continuous beta is

$$\beta^{\mathrm{scale}}(s) = \nabla^{\mathrm{scale}}_{\partial_s} g(s) \in \mathscr{G}_s. \tag{12.156}$$

In a local trivialization $I_s : \mathscr{G}_s \to \mathcal{G}_\star$ with connection form $A_s \in \mathcal{L}(\mathcal{G}_\star)$ and $\widetilde{g}(s) = I_s g(s)$,

$$I_s\beta^{\mathrm{scale}}(s) = \partial_s\widetilde{g}(s) + A_s\widetilde{g}(s), \qquad A'_s = S_s A_s S_s^{-1} - (\partial_s S_s)S_s^{-1} \text{ under } I'_s = S_s I_s. \tag{12.157}$$

A raw coordinate derivative does not obey that transformation law. On the trivial line bundle with $g(s) = 1$ parallel in a frame with $A_s = 0$, the frame change $S_s = e^{s^2}$ gives raw derivative $2se^{s^2} \neq 0$ while $A'_s = -2s$ and the covariant derivative $\partial_s(e^{s^2}) - 2se^{s^2}$ is zero. The scale connection is a

different object from the two contextual principal connections ω_b, ω_m of Definition 11.4: the former has base S and compares coupling fibers at different resolutions, the latter have base $\mathcal{C}_\ell$ at one fixed resolution and compare associated statistical fibers along directions in $T\mathcal{C}_\ell$. It is also distinct from the inference-orbit parameter and the Fisher duration of Chapter 9. Nothing is proved by this declaration. [DEFINITION]

One realization of that tier is the following density calculation. Suppose that scale is a genuine continuous semigroup and that, relative to one common dominating measure, the positive densities $r_t, p_t \in \mathrm{Dom}(\mathcal{A}^*)$ are differentiable in a topology that permits pointwise ratios and the following dual pairings, with $p_t = r_t \exp[-H_t]$, $\dot{r}_t = \mathcal{A}^* r_t$, and $\dot{p}_t = \mathcal{A}^* p_t$. Then

$$\boxed{\dot{H}_t = -\frac{\mathcal{A}^*(r_t \exp[-H_t])}{r_t \exp[-H_t]} + \frac{\mathcal{A}^* r_t}{r_t}} \tag{12.158}$$

almost everywhere wherever the displayed ratios and their dual pairings are defined. If the reference is stationary, the second term vanishes. [ESTABLISHED]

Proof. Differentiate $H_t = -\log(p_t/r_t)$ and substitute the two forward equations. □

On a declared Banach action space with a Schauder basis $\{\psi_A\}$ and continuous coordinate functionals $\{\psi_A^*\}$, or a Hilbert space with a Riesz basis, suppose the expansion $H_t = \sum_A g_A(t)\psi_A$ converges and differentiation and coefficient extraction can be interchanged. The component beta functions are then the coefficients of the right side of (12.158):

$$\dot{g}_A = \left\langle \psi_A^*, -\frac{\mathcal{A}^*(r_t e^{-H_t})}{r_t e^{-H_t}} + \frac{\mathcal{A}^* r_t}{r_t} \right\rangle . \tag{12.159}$$

This is exact for the stated complete convergent basis. A truncated basis defines a projection and leaves an explicit residual. [ESTABLISHED]

Proposition 12.31 (Discrete endpoints do not determine a continuous beta). Let a scale evolution family on the tier of Definition 12.30 consist of maps

$$\mathsf{V}(s,t) : \mathscr{G}_t \longrightarrow \mathscr{G}_s, \qquad \mathsf{V}(r,s) \circ \mathsf{V}(s,t) = \mathsf{V}(r,t), \qquad \mathsf{V}(t,t) = I_{\mathscr{G}_t}, \tag{12.160}$$

which is the continuous analogue of (11.25) and is again two-parameter. Restriction to integer endpoints does not determine the family, hence does not determine (12.156). On $S = \mathbb{R}$ with the trivial line bundle and zero connection form, the two families

$$\mathsf{V}^{(0)}(s,t) = 1, \qquad \mathsf{V}^{(\epsilon)}(s,t) = \exp\big\{\epsilon[\sin(2\pi s) - \sin(2\pi t)]\big\} \tag{12.161}$$

both satisfy (12.160) and agree at every pair of integer endpoints, while their generators are 0 and $2\pi\epsilon\cos(2\pi s)$. The one-parameter semigroup form, in which $\mathsf{V}(s,t)$ depends only on $s-t$, additionally requires a stationary trivialization, a stationary connection, an autonomous vertical field, and a global existence domain; it is not a consequence of (12.160). [ESTABLISHED]

Proof. Both displayed families satisfy the composition and identity laws by inspection, because each is of the form $f(s)/f(t)$ with f positive. At integer s and t the sine terms vanish, so $\mathsf{V}^{(\epsilon)}(s,t) = 1 = \mathsf{V}^{(0)}(s,t)$. Differentiating $\mathsf{V}^{(\epsilon)}(s,t)$ in s at $s=t$ gives $2\pi\epsilon\cos(2\pi t)$, which is not identically zero for $\epsilon \neq 0$. □

The smooth scale base, the bundle, the interpolation, and the connection are therefore additional declared input. A discrete blocking sequence supplies (12.150) and nothing more. [ESTABLISHED]

Attention flows primarily through the joint event law. Along a differentiable path on $\{\alpha_I > 0\}$, since $\beta_{IJ} = \eta_{IJ}/\alpha_I$,

$$\dot\beta_{IJ} = \frac{\dot\eta_{IJ} - \beta_{IJ}\dot\alpha_I}{\alpha_I}. \tag{12.162}$$

For a differentiable single softmax row on a fixed strictly positive prior support, with $\tau > 0$ and $\beta_J = \pi_J \exp[-E_J/\tau]/Z$, this becomes the replicator beta function

$$\dot\beta_J = \beta_J\left(u_J - \sum_K \beta_K u_K\right), \qquad u_J = \partial_t \log \pi_J - \frac{\dot E_J}{\tau} + \frac{E_J\dot\tau}{\tau^2}. \tag{12.163}$$

At a support change or on $\{\alpha_I = 0\}$, the joint law η is the well-defined flow variable; neither the quotient row nor its logarithmic parametrization is used. [ESTABLISHED]

12.10 Fixed points, flows, and scaling operators

Theorem 12.32 (Exhaustive fixed-point equations). A normalized fixed theory is exactly a rescaled invariant pair

$$\mathcal{R}_b(\rho_*, m_*) = (\rho_*, m_*) \quad \text{for every declared } b. \tag{12.164}$$

On the dominated normalized tier this is equivalent to the conjunction

$$\mathcal{R}_b^{\rho}\rho_* = \rho_*, \qquad \mathcal{R}_b^{H}[H_*;\rho_*] = H_*, \tag{12.165}$$

almost everywhere with respect to the fixed reference. If the evidence mass is discarded and actions are considered only projectively, the weaker fixed-ray equation is

$$\mathcal{R}_b^{H}[H_*;\rho_*] = H_* + c_{\mathrm{ray}}(b). \tag{12.166}$$

It represents the normalized pair in (12.164) only when the reference is also invariant and normalization fixes $c_{\mathrm{ray}}(b) = 0$. A fixed attention law is a fixed rescaled event measure η_*. A fixed conditional row alone does not make the attention law fixed: η is fixed exactly when its receiver marginal α is fixed and its disintegrated row β is fixed α-almost everywhere. [ESTABLISHED]

Proof. The first equation is the definition of invariance on the common rescaled measure space. Invariance of its first component and taking the Radon–Nikodym derivative of its second component

give the two conditions in (12.165). Quotienting by the action's additive gauge gives (12.166), but restoring the tracked evidence mass removes that freedom. Disintegration of a fixed η gives the final statement. □

At a fixed pair, additionally require H_* to lie in the declared finite-valued action Banach space. Set

$$m_*(dy) = e^{-H_*(y)}\rho_*(dy), \qquad M_* = m_*(\mathsf{Y}), \qquad \pi_* = M_*^{-1}m_*, \qquad \pi_*^c = \pi_* K_b, \tag{12.167}$$

and fix the standard-Borel reverse-kernel version Π_* defined by

$$\pi_*(dy)K_b(y, dz) = \pi_*^c(dz)\Pi_*(z, dy). \tag{12.168}$$

Also require

$$0 < L_*^c(z) = \mathbb{E}_{\rho_*}[e^{-H_*} \mid Z = z] < \infty \quad \rho_*^c\text{-almost everywhere.} \tag{12.169}$$

For bounded perturbations $\varphi \in L^\infty(\pi_*)$, the first two variations of one exact conditional-partition step are, π_*^c-almost everywhere,

$$D_H\mathcal{R}_b^H\big|_{(H_*,\rho_*)}[\varphi](z) = \mathbb{E}_{\Pi_*(dy|z)}\varphi(Y), \qquad D_H^2\mathcal{R}_b^H\big|_{(H_*,\rho_*)}[\varphi,\varphi](z) = -\operatorname{Var}_{\Pi_*(dy|z)}\varphi(Y), \tag{12.170}$$

for the composite kernel $K_b = C_b I_b$, which already includes the declared rescaling. The bounded-chart derivation of these two variations is Theorem 12.4; the displayed conditional-expectation and covariance formulas are exact on that stated domain. [ESTABLISHED]

Definition 12.33 (Spectral and cocycle relevance conventions). An ordinary spectrum is used only after a fixed point, an autonomous blocking ratio $b > 1$, and declared bounded isometric identifications turn the linearization into one bounded endomorphism T of one complex Banach action space. An isolated eigenoperator $T\psi_a = \lambda_a\psi_a$ then has the defined exponent $\upsilon_a = \log|\lambda_a|/\log b$; it is called relevant, marginal, or irrelevant according as this number is positive, zero, or negative, while its sign or complex phase is retained. Generalized eigenspaces are used only in finite dimensions or for isolated point spectrum with the required spectral projections.

Without those identifications, the derivative is a typed cocycle between different action spaces. A generalized mode and its Lyapunov growth rate must then be stated with the comparison maps and the chosen norms, rather than written as an eigenvalue equation in one unnamed space. These are classification definitions; no autonomous spectrum, mode, or exponent is asserted for a general nonautonomous sequence. [DEFINITION]

On the declared finite family the autonomous branch of these conventions is not available, because its blocking-ratio hypothesis is the compatibility law that the measurements following (12.142) refute. What the cocycle branch licenses was measured instead. For each declared ratio, the composite of the blocking with the self-similar re-tiling identification is a self-map of the reduced homogeneous coupling space, and it is a local contraction: spectral radius 0.78 at the declared seed and 0.83 at its fixed point for ratio two, with every eigenvalue inside the unit circle. Iterating converges to a fixed structure whose pairwise block is zero at machine precision. That factorized subspace is

invariant by construction, since the block-local Bayes kernel sends factorized theories to factorized theories, and the measurements show it is attracting, with pair-sector eigenvalues near 0.17; in the level-local sense of the cocycle conventions above, interaction between parents is irrelevant at every declared ratio. The ratio-two and ratio-three fixed structures differ by 0.81 in relative sup norm, so the fixed structures themselves are typed by the ratio. The cause is architectural rather than dynamical: the declared family joins adjacent blocks through single boundary agents, its instances are quasi-one-dimensional, and a one-dimensional finite-range system with strictly positive weights has a Perron–Frobenius transfer matrix and no interacting fixed structure. These are numerical results on declared finite instances, level-local by construction; they license no universality claim and no relevance exponent. [NUMERICAL]

Every statement in the preceding paragraph describes the *passive* map, in which the coarse pairwise coupling is inherited through blocking with attention frozen as declared data, and later measurements on 2026-08-17/18 bounded that reading sharply. Thickening the coupling boundary alone does not rescue the passive channel: the one-step pair retention rises sublinearly with the number of cut couplings per block boundary (0.156, 0.441, 0.564 at one, three, and six) and saturates below one. Enlarging the parent alphabet with the smallest sector label, the block's belief-channel Z_3 charge root-framed through the spanning-tree transport, raises retention (0.209 against 0.156 at one cut coupling, 0.568 against 0.441 at three, with the constant-sector control reproducing the nine-state values exactly) while remaining below one; an earlier null (0.144) used a family-referenced charge that shifts under the sample-shift gauge and was retired by the 2026-08-18 audit. What does restore interaction is the theory's own regeneration rule: recomputing attention at each level from the flow-averaged transported divergence over the conserved connection and folding the alignment energy into the coarse action. Under that declared rule, whose coarse action satisfies the same gauge-covariance identity to 10^{-11}, the per-ratio composites converge to *interacting* fixed structures — pairwise sup 0.579 at ratio two and 0.625 at ratio three, each a quarter above its own one-step injection — so horizontal coupling in this theory is maintained by active re-binding rather than by inheritance, and the passive triviality is a statement about the frozen-attention channel only. The typing survives regeneration: the composition defect remains order one across factorizations (0.17–0.27) and the interacting fixed structures remain ratio-typed (0.64 relative sup), so the cocycle reading of this chapter stands in both channels. [NUMERICAL]

Definition 12.34 (Typed invariant and fixed objects). Write $Y_\ell^{(\tau)}$ for the object of one declared tier τ and $F_\ell^{(\tau)} : Y_\ell^{(\tau)} \to Y_{\ell+1}^{(\tau)}$ for its declared step. The general invariant object of a nonautonomous scale diagram is an *invariant section*, namely a sequence with

$$y_{\ell+1} = F_\ell^{(\tau)}(y_\ell), \tag{12.171}$$

which needs no identification. After declared isomorphisms $J_\ell : Y_\star^{(\tau)} \to Y_\ell^{(\tau)}$, define the typed reference-space step

$$\widehat{F}_\ell^{(\tau)} = J_{\ell+1}^{-1} \circ F_\ell^{(\tau)} \circ J_\ell : Y_\star^{(\tau)} \longrightarrow Y_\star^{(\tau)}. \tag{12.172}$$

A *reference fixed object* is a $y_\star \in Y_\star^{(\tau)}$ with

$$\widehat{F}_\ell^{(\tau)}(y_\star) = y_\star \qquad \text{for every } \ell, \tag{12.173}$$

which in an autonomous scheme is the ordinary equation $\widehat{F}^{(\tau)}(y_\star) = y_\star$. If the identified maps are p-periodic, the base-ℓ reference monodromy is the endomorphism

$$\widehat{\mathcal{M}}_\ell^{(\tau)} = \widehat{F}_{\ell+p-1}^{(\tau)} \circ \cdots \circ \widehat{F}_\ell^{(\tau)} : Y_\star^{(\tau)} \longrightarrow Y_\star^{(\tau)}, \tag{12.174}$$

and a *reference monodromy fixed object* satisfies $\widehat{\mathcal{M}}_\ell^{(\tau)}(y_{\star,0}) = y_{\star,0}$. Its reference p-cycle is

$$y_{\star,j+1} = \widehat{F}_{\ell+j}^{(\tau)}(y_{\star,j}) \quad (0 \le j < p), \qquad y_{\star,p} = y_{\star,0},$$

entirely in $Y_\star^{(\tau)}$; its points need not be fixed by any one-step reference map. Without a common object the expression $F_\ell^{(\tau)}(y) = y$ is ill typed, and no subtraction across tiers or scales is formed. A fixed ray additionally requires a declared cone and projectivization. [DEFINITION]

The tiers of this chapter are the following, and each carries its own step and its own invariance equation. The law tier carries the normalized law π_ℓ, the tracked measure pair (ρ_ℓ, m_ℓ) of Definition 11.2, and the marked attention event law η_ℓ of (12.124); its fixed objects are characterized by Theorem 12.32. The action tier carries H_ℓ relative to ρ_ℓ and separately the additive class $[H_\ell]$; the two are related by (12.165) and (12.166), and by Proposition 12.27 the action tier is meaningful only together with its reference trajectory. The full interaction tier carries $g_\ell \in \mathcal{G}_\ell$ with step $T_\ell^{\mathcal{G}}$, and its reference fixed objects use $\widehat{T}_\ell^{\mathcal{G}}$ of (12.150). The retained interaction tier carries $\operatorname{Ran} R_\ell$, and a retained fixed object is exact only under (12.154). The bundle tier carries the geometric state (11.10), and the configuration tier carries the section manifolds of Chapter 9 with their own smooth comparison maps; neither is the law, action, or interaction map. [DEFINITION]

Proposition 12.35 (Fixedness does not transfer between tiers). None of the following implications holds, and each is refuted by a finite witness. A fixed normalized law does not give a fixed reference-dependent action: on a two-point set hold $\mu = (1/2, 1/2)$ fixed while the reference alternates between $\rho_0 = (1/4, 3/4)$ and $\rho_1 = (3/4, 1/4)$, so that the actions $H_i = -\log(d\mu/d\rho_i)$ are interchanged. A fixed action class does not give a fixed tracked pair: H and $H + c$ are the same class, while $e^{-H}\rho$ and $e^{-c}e^{-H}\rho$ have evidence masses differing by the factor e^{-c}, which is why (12.166) is strictly weaker than (12.164) until mass preservation forces $c_{\mathrm{ray}}(b) = 0$. A fixed retained interaction does not give a fixed exact interaction: the witness $R(x, y) = (x, 0)$ and $T(x, y) = (x, x)$ of Proposition 12.29 has zero retained beta at every $(x, 0)$ and nonzero exact residual for $x \neq 0$. A fixed law does not give a fixed configuration: take the configuration manifold S^1 with antipodal step $q \mapsto -q$ and a constant configuration-to-law extraction, so that the law is fixed while the configuration step has no fixed point. A fixed conditional attention row does not give a fixed event law: holding β fixed while the receiver occupancy α_ℓ alternates makes $\eta_\ell = \alpha_\ell\beta$ alternate, which is the disintegration clause of Theorem 12.32. Finally a monodromy object is not a one-step fixed

object: the period-two maps $F_0(x) = x + 1$ and $F_1(x) = x - 1$ on $\mathbb{R}$ have monodromy $F_1 \circ F_0 = \mathrm{id}$, which fixes every point, while neither one-step map has a fixed point. [ESTABLISHED]

Proof. Each clause is the displayed computation. For the first, $d\mu/d\rho_0 = (2, 2/3)$ and $d\mu/d\rho_1 = (2/3, 2)$ are distinct. For the second, $(e^{-c}e^{-H}\rho)(\mathsf{Y}) = e^{-c}m(\mathsf{Y})$. The third is (12.152) evaluated at $g = (x, 0)$, which gives $(0, x)$. For the fourth, a constant extraction sends both q and $-q$ to the same law, while $q \mapsto -q$ has no fixed point on S^1. The fifth is $\eta_\ell = \alpha_\ell\beta$ with α_ℓ nonconstant in ℓ. For the sixth, $F_1(F_0(x)) = x$ while $F_0(x) = x$ and $F_1(x) = x$ have no solutions. □

Two prohibitions follow. First, relevance, marginality, and irrelevance are properties of a mode relative to declared norms, a declared extensive normalization, the declared block ratios, and the declared comparison maps, in the sense of Definition 11.12 and Definition 12.14; Proposition 11.15 and Corollary 12.22 show that changing the comparison data alone can change the reported exponent. Second, none of the objects above is a physical law. A fixed measure pair, a fixed action class, a fixed interaction coordinate, a monodromy object, and a fixed configuration are mathematical invariants of a declared scale diagram; identifying any of them with an empirical regularity requires the separate apparatus listed at Open Problem E.1, which this document does not supply. [NOT-CLAIMED]

The equations characterize all fixed points without claiming that every model class admits a closed-form enumeration. The following exact sectors illustrate them. The identity channel fixes every law. A terminal channel followed by its one-point identification has the constant infrared fixed theory. For independent additive vector states and the block statistic $b^{-1/\alpha}\sum_{i=1}^{b} Y_i$, every strictly α-stable baseline with constant likelihood $m_o = Z(o)\rho$ gives a fixed measure pair; the centered Gaussian is the $\alpha = 2$ case. At one fixed integer b this list is not exhaustive because b-semistable laws can also be fixed. Exhaustiveness by strict stability requires invariance over all declared block sizes (or a continuum of scales) with the usual centering and regularity hypotheses. [ESTABLISHED]

Flat graph connection data form an invariant sector under rooted contraction, but become a fixed connection only after a self-similar graph identification has been declared. Uniform attention is fixed-form only when receiver occupancies, block sizes, source labels, and self-edge conventions are also self-similar. One-hot rows are invariant boundary sectors of (12.163). These qualifications prevent an invariant form from being mislabeled as a complete fixed theory. [ESTABLISHED]

Along every exact coarse path, (12.4) makes the recognition gap nonincreasing and the ELBO nondecreasing at fixed evidence. This monotone flow is information loss under resolution, not a proof of approach to a nontrivial critical fixed point. Attraction requires spectral control of (12.170) on the declared common space. [ESTABLISHED]

12.11 Closure theorem and infinite extensions

Theorem 12.36 (Complete finite law-level gauge-VFE theory). For every finite standard-Borel agent hypergraph, suppose the model supplies the common principal bundle and its two associated statistical bundles; normalized gauge-covariant baseline and interaction-record ker-

nels with positive finite evidence; channel-specific incidence and boundary links U^x, Θ^x obeying $U^{x\prime}_{a\leftarrow i} = (a^x_a)^{-1} U^x_{a\leftarrow i} a^x_i$ and the separate belief and model representations; normalized recognition laws satisfying the stated absolute-continuity and log-integrability conditions; structural coarse channels that are recognition independent and gauge equivariant; globally gauge-equivariant jointly measurable versions of every disintegration used for the displayed bridge kernels; pushed probability references; component-rooted forests together with their simultaneous raw root-framed holonomy maps, root features, and dressed boundary generators; joint marked attention-event laws; every induced effective hyperedge; full path laws or their exact memory kernels; and rescaling kernels satisfying (12.142). Then the global and local ELBOs, agent–meta bridge kernels, meta-agent interaction kernels, meta-attention, effective likelihoods, measure-pair scale composition, and measure-pair fixed equations defined in this and the preceding chapter are simultaneously normalized, gauge covariant, and exact. [ESTABLISHED]

Proof. Normalization and the collective/local identities are Proposition 7.2 and theorems 7.3 and 7.5. The common-pushforward chain rule is Theorem 12.1; the conditional-partition and tower identities are Theorem 12.3. Mobius inversion or the retained top-order factor closes the hypergraph. Regular disintegration gives an almost-everywhere version family of bridge kernels, and the supplied equivariant-version hypothesis selects the displayed globally covariant versions. Rooted dressing gives the separate gauge-covariant channel maps while the full holonomy representation preserves the information lost by a tree presentation. Pushforward and disintegration of η give exact attention. Path pushforward supplies dynamic closure, and the explicit common-space kernel identity types and composes the measure-pair RG. Each operation is closed in the stated collection, so their finite composition is closed. □

Two classes of supplied hypotheses remain open construction obligations. First, standard-Borel disintegration supplies almost-everywhere measurable conditional versions, but it does not by itself supply versions that are jointly measurable and globally gauge equivariant. Consistent with Section 10.8, every such equivariant conditional version used by the theorem is therefore an additional hypothesis; no general construction of those versions, especially for noncompact gauge groups, is exhibited here. Second, the first proposed rescaling family did not satisfy (12.142): that declared construction fails the compatibility law at order one on its pre-registered criterion, as recorded after (12.142). The theorem remains valid as a conditional implication, but an application must separately construct and verify both the required equivariant conditional versions and a compatible rescaling-kernel family, or restate the affected composed clauses in typed-cocycle form. The finite constructions in this and the preceding chapter are not claimed to discharge either obligation for arbitrary standard-Borel and noncompact families. The explicit compatible cover and path-blocking constructions of Chapters 14 and 17 are separately admitted realizations; they do not repair the failed family or select a universal rescaling kernel. [OPEN]

Corollary 12.37 (Exact analytic action, beta, and scaling tier). In addition to the hypotheses of Theorem 12.36, suppose the pushed measures admit the declared common dominators. On the linear cross-scale tier, additionally supply the vector features and linear representations R_x,

multiplicative weights and inter-root transports in (12.121), the extra fine cross morphism Φ_f, and the full marked operator–feature kernels. These hypotheses make the fine–meta, meta–fine, meta–meta, cross-channel, and attention operators exact and compositional on their stated domains; the Hom moments also require (12.129). For the discrete beta require a declared reference trajectory and a finite-valued action vector space closed under the reference-dependent action map; a change of that reference trajectory then obeys the inhomogeneous law (12.148). For the discrete interaction beta require in addition the comparison isomorphisms (12.149), and report the residual (12.152) unless (12.154) is proved. For the continuous beta require the separately declared smooth scale bundle and scale connection of Definition 12.30, together with the generator-domain, differentiability, positivity, and dual-pairing hypotheses stated before (12.158). For component beta functions require the stated Schauder or Riesz basis and interchange conditions. For action fixed points retain invariance of the reference component. For linearization require (12.169) and use the bounded perturbations in $L^\infty(\pi_*)$ stated before (12.170). Then the effective actions, discrete and continuous beta functionals, component flows, action fixed-point equations, and linearized scaling operators are exact on their declared analytic domains. [ESTABLISHED]

Proof. The Radon–Nikodym formula gives the action. Ordinary subtraction on the finite action space gives the discrete beta; differentiation in the declared generator domain gives the continuous beta. Valid coefficient extraction gives the component flow. Invariance of both the reference and likelihood ratio is equivalent to pair invariance, and dominated differentiation under the conditional partition gives the first two variations. □

The theorem is uniform in the number and arrangement of agents; it is not a verification on one fixed graph size. For a countable agent or time index, suppose instead that the normalized finite-cylinder joints form a projectively consistent family on standard Borel coordinate spaces. Kolmogorov extension then supplies the full normalized joint [46], and the stated identities hold for every regular finite-cylinder observation and finite block. Projective consistency alone does not make one prescribed infinite record positive probability: an infinite sequence of independent positive-cost binary successes can still be null. Conditioning on an infinite record therefore requires a chosen regular conditional at an almost-everywhere record or a proved DLR, Doob-transform, or finite-volume posterior limit. An infinite-law KL pushforward additionally requires compatible $Q \ll \Pi$ and an infinite coarse channel. The required DLR and limiting hypotheses must still be proved for each family. Chapters 17 and 19 supply such proofs for the stated high-temperature lattice, retained-label, and fixed-word domains. Their scoped conclusions do not establish arbitrary volume/depth exchanges or continuum geometry; those generalizations remain in Chapter E. [ESTABLISHED]

Chapter 13

The Full Multiscale Graph VFE

The exact variational free energy was already defined in Chapter 5, extended to one normalized interacting population in Chapter 7, and transported through one declared coarse channel in Chapter 12. This chapter does not replace those results. It asks what additional probabilistic structure is required to put an adaptive two-channel graph, emergent partitions, and parent-to-child influence inside one multiscale VFE.

The answer has two levels. The exact answer is always the VFE of one normalized generative law and one normalized recognition law. A sum of self, edge, partition, and cross-scale terms is exact only when it is obtained by factorizing those two laws. Without such a factorization the same sum can be a useful composite descent potential, but it is not automatically an ELBO and does not inherit an evidence interpretation. [ESTABLISHED]

13.1 The full answer before decomposition

Fix once and for all one base point $c_* \in \mathcal{C}$ and write $X_* := X(c_*)$ for its structural data. Every graph variable, belief or model presentation, transport, membership kernel, and retained mark in this chapter is evaluated in the fiber over c_*. We suppress that fixed argument. The symbol ℓ denotes RG depth, while f_ℓ is reserved for an actual base coarse map; no such map is present in this fixed-base construction. Fix an admitted observation $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,X_*}$ and a finite depth L. Let W contain every random object retained by the multiscale model: all belief and model coordinates at depths $0, \dots, L$, edge and connection variables, partition or membership variables, retained holonomy marks, and auxiliary latents. Declare one normalized generative kernel $\mathbb{P}_\theta(do, dW \mid X_*)$ before recognition and one normalized recognition kernel $\mathbb{Q}_\phi(dW \mid o, X_*)$. With selected posterior $\mathbf{\Pi}_{\theta,o,X_*}$ and evidence density $p_\theta(o \mid X_*) \in (0,\infty)$, the full multiscale VFE is

$$\boxed{\mathcal{F}^{\mathrm{ext}}_{\mathrm{tower}}[\mathbb{Q}_\phi; X_*, o] = -\log p_\theta(o \mid X_*) + D_{\mathrm{KL}}\left(\mathbb{Q}_\phi(\cdot \mid o, X_*) \| \mathbf{\Pi}_{\theta,o,X_*}\right).} \tag{13.1}$$

Equivalently, on the finite-density domain,

$$\mathcal{F}_{\mathrm{tower}} = \mathbb{E}_{\mathbb{Q}_\phi}\left[\log \frac{q_\phi(W \mid o, X_*)}{p_\theta(o, W \mid X_*)}\right]. \tag{13.2}$$

Equations (13.1) and (13.2) are the complete VFE. Every later term is an accounting decomposition of this object or is labeled as a proposed composite potential. [ESTABLISHED]

13.2 A normalized hierarchical graph model

At depth ℓ, let V_ℓ be the vertex set and let $Z_\ell = (Z_{i,\ell})_{i\in V_\ell}$ contain the full probabilistic coordinates of that depth, including belief and model presentations rather than only their marginals. Put

$$\eta_\ell = (\eta^{b,\ell}, \eta^{m,\ell}), \qquad G_\ell = (\eta_\ell, \Omega^{b,\ell}, \Omega^{m,\ell}, \mathsf{h}_\ell), \tag{13.3}$$

where h_ℓ contains the retained root, based-holonomy, and dressed boundary marks defined below. The occupancies and normalized attention rows are obtained from η_ℓ by marginalization and disintegration, so they are not substituted for the edge-event laws in the recursive state. Let R_ℓ be a possibly stochastic membership assignment from V_ℓ to $V_{\ell+1}$. The fixed structural configuration X_* and observation O stay outside the coarse channels.

For adaptive structural coarsening, set $\mathsf{Y}_\ell = (Z_\ell, G_\ell)$ and declare one normalized recognition-independent Markov channel

$$C^{\mathrm{struct},c_*}_{\ell+1\leftarrow\ell} : \mathsf{Y}_\ell \rightsquigarrow \mathsf{R}_\ell \times \mathsf{Y}_{\ell+1}. \tag{13.4}$$

Its output includes $(R_\ell, \eta_{\ell+1}, \mathsf{h}_{\ell+1})$. This channel supplies the conditional law of $(R_\ell, \mathsf{Y}_{\ell+1})$ given Y_ℓ. Define the augmented transition state

$$\widehat{\mathsf{Y}}_{\ell+1\leftarrow\ell} := (\mathsf{Y}_\ell, R_\ell, \mathsf{Y}_{\ell+1}). \tag{13.5}$$

Apply the same channel separately to the fine generative marginal, its selected posterior, and recognition law to obtain the transition joint laws

$$\begin{aligned}
\widehat{\mathbb{P}}^{\mathrm{struct}}_{\ell+1\leftarrow\ell}(dy_\ell, dr_\ell, dy_{\ell+1}) &:= \mathbb{P}_\ell(dy_\ell) C^{\mathrm{struct},c_*}_{\ell+1\leftarrow\ell}(y_\ell; dr_\ell, dy_{\ell+1}),\\
\widehat{\mathbf{\Pi}}^{\mathrm{struct}}_{\ell+1\leftarrow\ell}(dy_\ell, dr_\ell, dy_{\ell+1}) &:= \mathbf{\Pi}_\ell(dy_\ell) C^{\mathrm{struct},c_*}_{\ell+1\leftarrow\ell}(y_\ell; dr_\ell, dy_{\ell+1}),\\
\widehat{\mathbb{Q}}^{\mathrm{struct}}_{\ell+1\leftarrow\ell}(dy_\ell, dr_\ell, dy_{\ell+1}) &:= \mathbb{Q}_\ell(dy_\ell) C^{\mathrm{struct},c_*}_{\ell+1\leftarrow\ell}(y_\ell; dr_\ell, dy_{\ell+1}).
\end{aligned} \tag{13.6}$$

These joint laws genuinely retain $(R_\ell, \eta_\ell, \mathsf{h}_\ell)$; their next-depth marginals additionally contain $(\eta_{\ell+1}, \mathsf{h}_{\ell+1})$. The induced marginals of R_ℓ may differ because the three input laws differ. The structural channel does not. If the channel is represented by an auxiliary seed, the same measurable map and seed law define all three pushforwards, and each realized seed gives the same structural map in all three applications; a law-dependent realized blocking rule is not admitted. The observation coordinate remains unchanged. [HYPOTHESIS]

A normalized top-down generative factorization is

$$\begin{aligned}
\mathbb{P}_\theta(do, dW \mid X_*) = L_\theta(do \mid Z_0, X_*) P_L(dZ_L, dG_L \mid X_*)\\
\times \prod_{\ell=0}^{L-1} P^\ell_R(dR_\ell \mid Z_{\ell+1}, G_{\ell+1}, X_*) P^\ell_G(dG_\ell \mid Z_{\ell+1}, R_\ell, X_*)\\
\times \prod_{\ell=0}^{L-1} K^\ell_\downarrow(dZ_\ell \mid Z_{\ell+1}, R_\ell, G_\ell, X_*).
\end{aligned} \tag{13.7}$$

Every factor is a normalized measurable kernel. In particular, $K_{\downarrow}^{\ell}$ is the mathematical location of downward influence: a parent belief or model changes the conditional prior of its children. The bottom-up coarse map belongs to recognition or to an induced pushforward; it is not inserted as a second incompatible generative arrow at the same time slice. [HYPOTHESIS]

For an exact chain-rule decomposition, disintegrate recognition along the same ordering:

$$\begin{aligned}\mathbb{Q}_\phi(dW \mid o, X_*) = Q_L(dZ_L, dG_L \mid o, X_*) \\ \times \prod_{\ell=0}^{L-1} Q_R^\ell(dR_\ell \mid Z_{\ell+1}, G_{\ell+1}, o, X_*) Q_G^\ell(dG_\ell \mid Z_{\ell+1}, R_\ell, o, X_*) \\ \times \prod_{\ell=0}^{L-1} Q_\ell(dZ_\ell \mid Z_{\ell+1}, R_\ell, G_\ell, o, X_*).\end{aligned} \tag{13.8}$$

This is a disintegration of a correlated law, not a mean-field assumption. A bottom-up recognition network may parameterize the same joint; the written ordering is only an accounting choice. [HYPOTHESIS]

When the hierarchy is adaptive, the displayed P_R^ℓ, P_G^ℓ and Q_R^ℓ, Q_G^ℓ are selected disintegrations of the respective joint laws induced by (13.6). They are not independently chosen structural channels.

Theorem 13.1 (Exact hierarchical decomposition). Assume the normalized factorizations above, common domination, and the integrability needed for the finite log-density form. Then

$$\begin{aligned}\mathcal{F}_{\text{tower}} = & -\mathbb{E}_{\mathbb{Q}_\phi} \log L_\theta(o \mid Z_0, X_*) + D_{\mathrm{KL}}(Q_L \| P_L) \\ & + \sum_{\ell=0}^{L-1} \mathbb{E}_{\mathbb{Q}_\phi} D_{\mathrm{KL}}\left(Q_R^\ell(\cdot \mid Z_{\ell+1}, G_{\ell+1}, o, X_*) \middle\| P_R^\ell(\cdot \mid Z_{\ell+1}, G_{\ell+1}, X_*)\right) \\ & + \sum_{\ell=0}^{L-1} \mathbb{E}_{\mathbb{Q}_\phi} D_{\mathrm{KL}}\left(Q_G^\ell(\cdot \mid Z_{\ell+1}, R_\ell, o, X_*) \middle\| P_G^\ell(\cdot \mid Z_{\ell+1}, R_\ell, X_*)\right) \\ & + \sum_{\ell=0}^{L-1} \mathbb{E}_{\mathbb{Q}_\phi} D_{\mathrm{KL}}\left(Q_\ell(\cdot \mid Z_{\ell+1}, R_\ell, G_\ell, o, X_*) \middle\| K_{\downarrow}^\ell(\cdot \mid Z_{\ell+1}, R_\ell, G_\ell, X_*)\right).\end{aligned} \tag{13.9}$$

Each conditional divergence is averaged over its displayed parent variables under $\mathbb{Q}_\phi$. These are the observation, top-prior, partition, graph, and parent–child contributions. [ESTABLISHED]

Proof. Substitute the two factorizations into the Radon–Nikodym ratio in (13.2) and apply the conditional relative-entropy chain rule. The extended-real version must use the measure-level chain rule without subtracting infinities. □

A deterministic same-time parent $Z_{\ell+1} = C_\ell(Z_\ell)$ and an independently sampled parent that generates Z_ℓ are not two free descriptions of one joint. One may use the hierarchy above, a deterministic pushforward, or a normalized undirected compatibility factor. Reciprocal same-time influence requires the last option and its partition function. Delayed Wheelerian feedback instead

lets $Z_{\ell+1}(t)$ parameterize the child kernel at $t+\Delta t$; that is a dynamical extension, not a static ELBO identity. [NOT-CLAIMED]

13.3 The scale-local two-channel graph energy

For arbitrary directed, generally non-flat transports, define

$$D_{ij}^{b,\ell} := D_{\mathrm{KL}}\left(q_{i,\ell}^{b}\middle\|(\Omega_{ij}^{b,\ell})_{\#}q_{j,\ell}^{b}\right), \qquad D_{ij}^{m,\ell} := D_{\mathrm{KL}}\left(q_{i,\ell}^{m}\middle\|(\Omega_{ij}^{m,\ell})_{\#}q_{j,\ell}^{m}\right). \tag{13.10}$$

Here and below the suppressed base argument is c_*. The edge-event laws in G_ℓ determine their receiver occupancies and normalized source rows by

$$\alpha_i^{x,\ell} := \sum_j \eta_{ij}^{x,\ell}, \qquad \beta_{ij}^{\ell} := \frac{\eta_{ij}^{b,\ell}}{\alpha_i^{b,\ell}}, \qquad \gamma_{ij}^{\ell} := \frac{\eta_{ij}^{m,\ell}}{\alpha_i^{m,\ell}}, \tag{13.11}$$

for $x \in \{b,m\}$ and positive occupancy. On a zero-occupancy row, the joint law identifies no conditional attention. An arbitrary normalized measurable version may be declared for formal disintegration, but it does not supply a measured row or a smooth extension. The cover construction in Chapter 14 retains the joint event mass and keeps that conditional domain explicit. The correct energy-unit row free energies are

$$\begin{aligned}\Phi_{i,\ell}^{b} &= \sum_j \beta_{ij}^{\ell} D_{ij}^{b,\ell} + \tau_{i,\ell}^{b} D_{\mathrm{KL}}(\beta_i^{\ell}\|\pi_i^{b,\ell}),\\ \Phi_{i,\ell}^{m} &= \sum_j \gamma_{ij}^{\ell} D_{ij}^{m,\ell} + \tau_{i,\ell}^{m} D_{\mathrm{KL}}(\gamma_i^{\ell}\|\pi_i^{m,\ell}).\end{aligned} \tag{13.12}$$

Their conditional minimizers are

$$\beta_{ij}^{\ell,*} = \frac{\pi_{ij}^{b,\ell} e^{-D_{ij}^{b,\ell}/\tau_{i,\ell}^{b}}}{\sum_k \pi_{ik}^{b,\ell} e^{-D_{ik}^{b,\ell}/\tau_{i,\ell}^{b}}}, \qquad \gamma_{ij}^{\ell,*} = \frac{\pi_{ij}^{m,\ell} e^{-D_{ij}^{m,\ell}/\tau_{i,\ell}^{m}}}{\sum_k \pi_{ik}^{m,\ell} e^{-D_{ik}^{m,\ell}/\tau_{i,\ell}^{m}}}. \tag{13.13}$$

Thus β_{ij} and γ_{ij} coevolve with the transported divergences. Omitting the row relative entropy changes the objective. [ESTABLISHED]

The quantity $1/\beta_{ij}$ is a useful monotone interaction length only in a fixed row. It is directed, depends on all competing sources through row normalization, diverges at zero, and need not obey a triangle inequality. At the Gibbs optimum the source-relative surprisal

$$\lambda_{ij}^{b,\ell} := -\tau_{i,\ell}^{b} \log \frac{\beta_{ij}^{\ell,*}}{\pi_{ij}^{b,\ell}} = D_{ij}^{b,\ell} + \tau_{i,\ell}^{b} \log Z_i^{b,\ell} \tag{13.14}$$

is the transported mismatch plus a row-dependent constant. Raw conductance, a symmetrized resistance, or diffusion time is required for a genuine graph metric. [ESTABLISHED]

If a scale-local prior is a normalized Gibbs law containing the self terms and $\sum_{ij}\eta_{ij}^{b,\ell}D_{ij}^{b,\ell}+\sum_{ij}\eta_{ij}^{m,\ell}D_{ij}^{m,\ell}$, those terms and the global normalizer belong to the exact VFE. If they are merely appended to an already fixed state-level VFE, define instead

$$\begin{aligned}\mathcal{G}_{\text{composite}} = \mathcal{F}_{\text{population}} &+ \sum_{\ell}\sum_{i}\left(\alpha_i^{b,\ell}\Phi_{i,\ell}^{b}+\alpha_i^{m,\ell}\Phi_{i,\ell}^{m}\right)\\ &+\sum_{\ell}\mathcal{F}_{\text{partition}}^{\ell}+\sum_{\ell}\mathcal{F}_{\text{cross}}^{\ell}+\sum_{\ell}\mathcal{R}_{\text{connection}}^{\ell}.\end{aligned}\tag{13.15}$$

This is a composite multiscale free-energy potential, not automatically an evidence bound. [DEFINITION]

13.4 Non-flat edges and what the parent retains

The edge transports are not assumed flat. Use the rooted construction of Section 12.6: in every connected block I, choose a root r_I and a rooted spanning tree, with nested choices across admitted depths. For $x\in\{b,m\}$, let $\tau_{I\leftarrow i}^{x,\ell}$ be the ordered tree transport from i to r_I. Under endpoint frame changes $a_i^b=a_i$ and $a_i^m=b_i$,

$$\tau_{I\leftarrow i}^{x,\ell\prime}=(a_{r_I}^x)^{-1}\tau_{I\leftarrow i}^{x,\ell}a_i^x.\tag{13.16}$$

The non-tree edges define the based holonomy representation

$$H_I^{x,\ell}:\pi_1(\Gamma_I,r_I)\longrightarrow G,\qquad H_I^{x,\ell\prime}(\gamma)=(a_{r_I}^x)^{-1}H_I^{x,\ell}(\gamma)a_{r_I}^x.\tag{13.17}$$

Nonidentity based holonomy is allowed. A holonomy-blind parent law $Q_I^{x,\ell}$ is path independent only on the stabilized sector

$$\big(H_I^{x,\ell}(\gamma)\big)_{\#}Q_I^{x,\ell}=Q_I^{x,\ell}\quad\text{for every retained based loop }\gamma.\tag{13.18}$$

This is not a flatness condition and does not select I.

For a marked boundary edge $e:j\to i$ from block J to block I, write $\Theta_e^{x,\ell}$ for its microscopic channel-x transport. Its endpoint gauge law and its root-framed dressed boundary mark are

$$\begin{aligned}&\Theta_e^{x,\ell\prime}=(a_i^x)^{-1}\Theta_e^{x,\ell}a_j^x,\\ &V_e^{x,\ell}=\tau_{I\leftarrow i}^{x,\ell}\Theta_e^{x,\ell}\big(\tau_{J\leftarrow j}^{x,\ell}\big)^{-1},\qquad V_e^{x,\ell\prime}=(a_{r_I}^x)^{-1}V_e^{x,\ell}a_{r_J}^x.\end{aligned}\tag{13.19}$$

Thus h_ℓ retains the simultaneous root-framed datum $(r_I,H_I^{x,\ell},\{V_e^{x,\ell}\})$ and its joint law. On a non-flat graph, the family of dressed boundary marks need not collapse to one group element. An exact coarse datum retains their conditional distribution; a reduced mean must carry a declared residual. Penalizing that residual to zero is an extra modeling preference, not a consequence of gauge covariance. [ESTABLISHED]

13.5 Partition variables and graph renormalization

The weights β and γ do not themselves create a parent random variable. A hierarchy intended to emerge under the same variational principle must put R_ℓ inside $\mathbb{P}$ and $\mathbb{Q}$. A common finite-temperature form is

$$\mathcal{F}^{\ell}_{\text{partition}} = \mathbb{E}_{Q^{\ell}_R}[E^{\ell}_{\text{block}}(R_\ell; Z_\ell, G_\ell)] + \tau^{\ell}_R D_{\text{KL}}(Q^{\ell}_R \| \Pi^{\ell}_R), \tag{13.20}$$

provided the normalized Gibbs factor represented by this expression is part of the generative law. The block energy may trade internal transported mismatch against boundary retention, evaluator closure, and residual cost. No canonical selector follows from the global VFE alone. [HYPOTHESIS]

Coarse attention is obtained by pushing the joint edge-event law, not by averaging normalized rows. Genuine overlapping assignment requires a normalized membership Markov kernel

$$m_\ell(I \mid i, R_\ell) \geq 0, \qquad \sum_{I \in V_{\ell+1}} m_\ell(I \mid i, R_\ell) = 1. \tag{13.21}$$

Fix the complete input $y \in \mathsf{Y}_\ell$ and a realized structural assignment $r \in \mathsf{R}_\ell$. The input event mass $\eta^{x,\ell}_{ij}(y)$ is then fixed. When the two endpoint assignments are conditionally independent at this input and assignment, the conditional coarse directed event law is

$$\eta^{x,\ell+1}_{IJ}(y, r) = \sum_{i,j \in V_\ell} \eta^{x,\ell}_{ij}(y) m_\ell(I \mid i, r) m_\ell(J \mid j, r), \qquad x \in \{b, m\}, \tag{13.22}$$

where the assignment kernels have the declared dependence on r. When endpoint assignments are correlated, replace the product by a normalized joint endpoint-assignment kernel

$$M_\ell(I, J \mid i, j, R_\ell) \geq 0, \qquad \sum_{I,J} M_\ell(I, J \mid i, j, R_\ell) = 1, \tag{13.23}$$

and use

$$\eta^{x,\ell+1}_{IJ}(y, r) = \sum_{i,j \in V_\ell} \eta^{x,\ell}_{ij}(y) M_\ell(I, J \mid i, j, r). \tag{13.24}$$

Let $\nu_\ell(dr \mid y)$ be the assignment marginal of the common structural channel (13.4), and let μ be the generative, selected-posterior, or recognition input law. The unconditional event mass is the average of the entire conditional expression:

$$\overline{\eta}^{x,\ell+1}_{IJ}[\mu] = \sum_{i,j \in V_\ell} \int \eta^{x,\ell}_{ij}(y) M_\ell(I, J \mid i, j, r) \nu_\ell(dr \mid y) \mu(dy). \tag{13.25}$$

Summing over I, J gives one by normalization of M_ℓ, η_ℓ, ν_ℓ, and μ. Conditional endpoint independence permits the product in (13.22); it does not permit factoring the expectation of a random event mass and a random assignment. The same formula therefore applies to all three input laws without an independence assumption between their event and membership marks.

The resulting random $\eta_{\ell+1}(y,r)$ is part of $G_{\ell+1}$, and its occupancies and rows are recovered by (13.11). Its unconditional mean alone does not replace that retained joint datum. A hard random partition is the deterministic conditional special case $m_\ell(I \mid i, R_\ell) = \mathbf{1}\{R_\ell(i) = I\}$. Literal unit-weight membership in several parents has total mass greater than one; it is finite-measure replication, not a probability pushforward. Arbitrary overlapping indicator sums therefore have no established probability status.

Nested assignments must satisfy Chapman–Kolmogorov composition. In the single-endpoint case this means

$$m_{\ell+2\leftarrow\ell}(A \mid i, R_\ell, R_{\ell+1}) = \sum_I m_{\ell+1}(A \mid I, R_{\ell+1}) m_\ell(I \mid i, R_\ell), \tag{13.26}$$

with the analogous normalized composition

$$M_{\ell+2\leftarrow\ell}(A, B \mid i, j, R_\ell, R_{\ell+1}) = \sum_{I,J} M_{\ell+1}(A, B \mid I, J, R_{\ell+1}) M_\ell(I, J \mid i, j, R_\ell). \tag{13.27}$$

when correlated endpoint kernels are used. The complete augmented structural channels, including retained marks, must likewise obey

$$C^{\mathrm{struct},c_*}_{\ell+2\leftarrow\ell} = C^{\mathrm{struct},c_*}_{\ell+2\leftarrow\ell+1} \circ C^{\mathrm{struct},c_*}_{\ell+1\leftarrow\ell}, \tag{13.28}$$

where the composition integrates all intermediate assignments, edge-event laws, and marks. Retained root, holonomy, and boundary marks must be included in the target of each composed channel rather than silently discarded. [ESTABLISHED]

13.6 How a parent influences its children

The normalized kernel $K^\ell_\downarrow$ supplies the cleanest static answer. For a block I with children $i \in I$, a factorization such as

$$K^\ell_\downarrow(dZ_\ell \mid Z_{\ell+1}, R_\ell, G_\ell, X_*) = \prod_{I \in V_{\ell+1}} K^\ell_I\left(dZ^\ell_I \mid Z_{I,\ell+1}, R_\ell, G_\ell, X_*\right) \tag{13.29}$$

makes the parent belief and model presentations arguments of the children's conditional generative prior. The corresponding term is

$$\mathcal{F}^\ell_{\mathrm{cross}} = \mathbb{E}_{\mathbb{Q}_\phi} D_{\mathrm{KL}}\left(Q_\ell(dZ_\ell \mid Z_{\ell+1}, R_\ell, G_\ell, o, X_*) \middle\| K^\ell_\downarrow(dZ_\ell \mid Z_{\ell+1}, R_\ell, G_\ell, X_*)\right). \tag{13.30}$$

If $K^\ell_\downarrow$ depends nontrivially on the parent coordinates and the parameterization transmits that dependence to the objective, variation with respect to a parent coordinate changes the parent update while variation with respect to a child coordinate changes the child update. Under those hypotheses, optimization is bidirectionally coupled even though the normalized generative arrow is top-down. A constant parent-independent downward kernel decouples this cross-scale term. This

is the precise static sense in which a meta-agent can influence its sub-agents without allowing a generative factor to read the posterior. [ESTABLISHED]

Calling the optimization parameter physical time requires more. Delayed feedback may use the inferred parent at time t to modify $K^{\ell}_{\downarrow}$ at $t+\Delta t$. A same-time reciprocal factor may instead be placed in a normalized Gibbs joint. Either construction needs work or flux accounting before it supports Wheelerian nonequilibrium. Static VFE descent alone proves neither sustained nonequilibrium nor downward causation in physical time. [OPEN]

13.7 Bayesian and network renormalization

Bayesian renormalization uses Fisher distinguishability on model-parameter space to define an information scale and separate stiff from sloppy directions [13]. It can rank which directions of a parent model presentation remain empirically distinguishable at a declared observational precision. It does not by itself select graph blocks, renormalize non-flat transports, or prove closure of the two-channel interaction law. Its parameter-space orientation must not be identified silently with sample-space graph diffusion.

Network renormalization addresses the absence of a lattice, momentum, and geometric length in a heterogeneous graph [33]. Laplacian RG offers a diffusion scale and spectral diagnostics [86]; the multiscale model gives a geometry-free closure family based on additive hidden variables [34]. For the present theory these are candidate partition and closure mechanisms. They do not automatically handle directed row-normalized β, γ, two gauge representations, non-flat holonomy, or simultaneous renormalization of beliefs, models, topology, and dynamics. The network-RG review identifies the last problem as open.

The proposed synthesis is hybrid: use edge-event laws or raw conductances to construct a gauge-invariant multiplex graph operator; use spectral or Bayesian information diagnostics to propose scales and relevant directions; infer R_ℓ through a normalized partition factor; push the full probabilistic law and edge-event measures; retain non-flat holonomy or prove stabilization; and measure failure of the parent family by an explicit closure residual. [HYPOTHESIS]

13.8 The remaining hierarchy question

Starting from the fixed $c_* \in \mathcal{C}$ with a finite directed multiplex datum $G_0 = (\eta_0^b, \eta_0^m, \Omega_0^b, \Omega_0^m, \mathsf{h}_0)$, construct a normalized partition law and parent–child kernels such that descent of (13.1) produces persistent nested blocks, the pushforward rules (13.22)– (13.28) compose across depths, the parent family has a controlled closure residual, and non-flat holonomy is retained or stabilized without imposing flatness. Then determine whether the coupled variational dynamics has a separated slow manifold on which depth-$\ell+1$ variables feed back consistently into depth-ℓ kernels.

Chapter 15 supplies a declared parent-sensitive transition, its normalized stationary target, and retained history laws with an exact finite-memory cost. Chapter 14 supplies compatible geometric

parents. These are established constructions at their stated hypotheses, not consequences of arbitrary descent. [ESTABLISHED]

Chapter 16 additionally realizes successive cyclic coalescence and actual-law parent learning for each prescribed finite depth in a prepared nonlinear triangle-chain family. Its exact marked parents retain all loops and source assignments, while conditional entropy, scores, continuous trajectories and assignment histories obey separate approximation bounds. The weakest energy, continuous mobility, and source proposal rate remain fixed; the stronger interaction contrasts and source support are supplied assumptions. [ESTABLISHED]

No current result proves that arbitrary descent from arbitrary β, γ produces such a hierarchy. The exact VFE tells us how a proposed hierarchy must be normalized and accounted for; it does not supply the partition selector, persistence theorem, or dynamical cross-scale closure in arbitrary families. Persistent nested lineage, a nondegenerate intrinsic selector, and a separated slow manifold remain the stronger obligations. [OPEN]

Part III

Constructive Geometric, Dynamical, and Spatial Realizations

Chapter 14

Neighborhood Parents, Availability, and Connection Descent

The pointwise construction extends over a neighborhood when the local laws, coarse channels, and geometric presentations satisfy explicit compatibility conditions. The following construction supplies those conditions through a common ambient law. It also distinguishes a parent requiring a fixed set of children from a parent whose available children vary with context. These are declared extensions of the pointwise datum, with different domains. [DEFINITION]

14.1 A cover without a common intersection

For this example take $\mathcal{C} = U = (-3/2, 3/2)^2$, with $c = (x, y)$, and four agents with labels $0, 1, 2, 3$. This local indexing replaces the generic population indexing only in this example. Define

$$\begin{aligned} I_0 &= (-8/5, -1/10), \quad I_1 = (-8/5, 4/5), \\ I_2 &= (-4/5, 8/5), \qquad I_3 = (1/10, 8/5), \\ U_i &= U \cap (I_i \times \mathbb{R}). \end{aligned} \tag{14.1}$$

These open sets and the contextual base are declared data; the construction places no probability measure on the base. [DEFINITION]

Theorem 14.1 (The overlap pattern). The sets U_i cover U. Their nonempty pairwise intersections have indices $01, 02, 12, 13, 23$, their nonempty triple intersections have indices $012, 123$, and their fourfold intersection is empty. In particular,

$$U_{012} = U \cap \{-4/5 < x < -1/10\}, \qquad U_{123} = U \cap \{1/10 < x < 4/5\}. \tag{14.2}$$

This is an established cover with genuine triple compatibility conditions. [ESTABLISHED]

Proof. An intersection of the intervals is nonempty exactly when its largest lower endpoint is smaller than its smallest upper endpoint. This gives all the listed intersections. The intervals I_1, I_2 alone cover the first coordinate of U, proving the cover assertion. Finally $U_0 \cap U_3 = \varnothing$, which also excludes the fourfold intersection. □

Let a_i, r_i be the center and half-width of I_i. The availability weights are the prescribed functions

$$w_i(c) = \begin{cases} \exp(1 - 1/t_i(c)), & t_i(c) > 0, \\ 0, & t_i(c) \le 0, \end{cases} \qquad t_i(c) = 1 - \left(\frac{x - a_i}{r_i}\right)^2. \tag{14.3}$$

These weights encode availability, not posterior source probabilities. [DEFINITION]

Theorem 14.2 (Smooth availability with an explicit null event). Each w_i is smooth, lies in $[0, 1]$, and is positive exactly on U_i. Add a null endpoint $\perp$ and define the endpoint channel

$$T_c(i, j) = w_i(c)\mathbf{1}_{j=i} + (1 - w_i(c))\mathbf{1}_{j=\perp}, \qquad T_c(\perp, j) = \mathbf{1}_{j=\perp}. \tag{14.4}$$

For any fixed finite number of directed records, conditionally independent applications of this channel to their endpoints give a smooth normalized channel on the complete record state. Consequently smooth ambient event probabilities give smooth visible joint event probabilities, including at agent entry and exit. This is an established property of the declared availability mechanism. [ESTABLISHED]

Proof. Within its positive domain, every derivative of $\exp(-1/t)$ is that exponential times a polynomial in $1/t$; all such products tend to zero as $t \downarrow 0$. Extension by zero is therefore smooth. Since $0 < t_i \le 1$ inside the interval, the stated range follows. Each row of T_c sums to one. Products of the endpoint rows remain normalized, and their entries and finite sums against smooth ambient probabilities are smooth. Applying the channel to a joint law preserves any dependence not removed by the channel; it does not replace correlated records by a product of their marginals. □

For each channel $x \in \{b, m\}$, let $\eta^x_{ij}(c)$ be the marginal law of one visible directed record, including $\perp$, and set

$$\alpha^x_i(c) = \sum_j \eta^x_{ij}(c), \qquad b^x_{ij}(c) = \frac{\eta^x_{ij}(c)}{\alpha^x_i(c)} \quad \text{only when } \alpha^x_i(c) > 0. \tag{14.5}$$

Here b^b and b^m occupy the respective recognition attention slots β and γ; a generative posterior row is obtained from its own event law by the same operation. The joint event law, occupancy, and conditional row remain different objects. [DEFINITION]

If every real ambient receiver has positive mass, then $\alpha^x_i > 0$ exactly on U_i, because its visible occupancy equals its ambient receiver mass times w_i. On this domain the conditional row is smooth. At zero occupancy the event law imposes no conditional row: all its entries in that receiver row are zero, regardless of any proposed conditional completion. Thus smoothness of η^x, α^x supplies no identifiable attention at zero occupancy. No null outcome is removed or renormalized away in this established calculation. [ESTABLISHED]

14.2 Compatible laws and the two parent domains

For the constructive specialization, adopt one smooth normalized ambient joint $\mathbb{P}_c(dY, do, de \mid X)$, a smooth selected recognition $\mathbb{Q}_{c,o,X}(dY, de)$, and a reference law on the same finite product of agent fibers and record states. The fixed structural datum is X. Assume positive finite evidence at the admitted observation and selected posterior versions that are pushed with these same laws. The continuous variables may be correlated Gaussian coordinates with positive definite ambient covariance, while records have finite laws. An individual agent's section is exposed only on its U_i. These ambient laws are additional coupling data, not a conclusion inferred from independently specified local marginals. [HYPOTHESIS]

For agent subsets $S \subset T$, let $r_{S\leftarrow T}$ project continuous coordinates to S and send every record endpoint outside S to $\perp$. Null endpoints remain null. Restriction of the base variable to a smaller open set is a separate operation from this projection. [DEFINITION]

Theorem 14.3 (Compatibility supplied by an ambient extension). Every ambient probability law in the preceding hypothesis induces compatible local laws. On nested subsets and open context domains,

$$r_{R\leftarrow S} r_{S\leftarrow T} = r_{R\leftarrow T}, \qquad (\mathbb{Q}_T r_{S\leftarrow T}) r_{R\leftarrow S} = \mathbb{Q}_T r_{R\leftarrow T}. \tag{14.6}$$

The same equality holds for the generative joint, reference, and selected posterior families. Smooth local presentations that obey their transition maps on pairwise overlaps obey the corresponding triple-overlap cocycles. This is an established sufficient construction of compatibility. [ESTABLISHED]

Proof. Coordinate projections compose. Sending a removed endpoint to $\perp$ and retaining $\perp$ also composes. Pushforward through these deterministic channels therefore gives the displayed identity. Restricting the same formula in c twice is its restriction to the smaller domain. If G_a, G_b, G_d are three local frame matrices for one associated fiber, the coordinate transformations are $G_b G_a^{-1}$, and $(G_d G_b^{-1})(G_b G_a^{-1}) = G_d G_a^{-1}$. Pushing all coordinates and full covariances by those same transformations proves the law cocycle. These are presentations of one law, not equality of different agents' beliefs. □

A general extension theorem for arbitrary local laws is not asserted here. Its unresolved obligation is to characterize which full local probability objects, supplied without the ambient extension, admit one joint extension compatible with every restriction and selected conditional version. The present proof starts with that extension and therefore cannot establish the general implication. [OPEN]

For a fixed block I_A, the fixed-child domain and the availability domain are, respectively,

$$U_A^{\cap} = \bigcap_{i\in I_A} U_i, \qquad V_A^{\cup} = \bigcup_{i\in I_A} U_i, \qquad s_A(c) = \sum_{i\in I_A} w_i(c). \tag{14.7}$$

The original U_A of the pointwise construction means $U_A^{\cap}$. On $V_A^{\cup}$, declare the different availability parent by coefficients $\lambda_{Ai} = w_i/s_A$, using all the locally available joint child data supplied by the ambient extension. [DEFINITION]

For the explicit geometric example use the standard two-dimensional representation of $\mathrm{SO}(2)$ separately for belief and model, and let G_{f}, G_A be the resulting orthogonal frame matrices for the active fine and parent coordinates. Write

$$L_A = (\lambda_{Ai} I_4)_{i \in I_A}, \qquad F_A = G_A L_A G_{\mathrm{f}}^{-1}. \tag{14.8}$$

This is a deterministic normalized channel on the continuous coordinates. Its event channel sends a child endpoint to its block label and keeps $\perp$. This declared example assumes equal standard representations; an inequivalent pair of representations requires its own intertwiners. [HYPOTHESIS]

Theorem 14.4 (Availability parents glue and compose). The declared availability parent is smooth on $V_A^{\cup}$ and agrees on all chart overlaps. For disjoint blocks, the active rows of their combined F are independent, so a positive definite fine Gaussian covariance pushes to a positive definite covariance on the active parent fiber. If a second stage weights the active blocks by $s_A/\sum_B s_B$, its direct fine coefficient is $w_i/\sum_j w_j$. Thus the direct and staged complete law channels agree. This is an established extension with changing child availability, rather than continuation of a fixed-child object by necessity. [ESTABLISHED]

Proof. On $V_A^{\cup}$, $s_A > 0$, so every coefficient is smooth. At a child boundary its vanishing bump extends smoothly as long as another child keeps the denominator positive. The formula is the same in every context chart, with the frame cocycle already proved. Each active block row has nonzero coefficients on its own disjoint child block, which gives full row rank. For a nonzero parent vector v, $v^\top F \Sigma F^\top v = (F^\top v)^\top \Sigma (F^\top v) > 0$. At two stages the coefficient product is

$$\frac{s_A}{\sum_B s_B} \frac{w_i}{s_A} = \frac{w_i}{\sum_j w_j} \quad (i \in I_A, s_A > 0). \tag{14.9}$$

An absent block contributes zero directly; no conditional coefficient for that absent block is needed. Frame matrices cancel at intermediate fibers. Endpoint maps compose with the same null convention, and equality of complete channels gives equality of all their pushed laws. For blocks 01 and 23, the final availability parent exists on all U, whereas a fixed four-child parent has empty domain by theorem 14.1. □

The full fine Gaussian family is smooth on each constant-active-set region. Across changing dimension, the construction does not supply one fixed-rank Gaussian statistical manifold. Nor does it extend a parent through a point where all its children are unavailable. These additional smooth-extension claims are declined because their rank and domain premises fail in the displayed construction. [NOT-CLAIMED]

Theorem 14.5 (Evidence and the discarded conditional law). At each fixed admitted context let C be any of the normalized, recognition-independent state channels above, keeping the observation unchanged. It preserves evidence and pushes the selected posterior version to a posterior version of the pushed joint. For finite fine relative entropy,

$$\mathcal{F}_{\mathrm{f}} = \mathcal{F}_{\mathrm{c}} + \int D_{\mathrm{KL}}\big(\mathbb{Q}(dY \mid z) \| \mathbf{\Pi}(dY \mid z, o, X)\big)\mathbb{Q}_{\mathrm{c}}(dz). \tag{14.10}$$

For a stochastic channel, Y denotes the original state in its joint input-output extension. This established identity compares coarse levels at fixed context and observation. [ESTABLISHED]

Proof. Integrating $C(Y, dz)$ over z gives one, leaving the original observation marginal. Integrating the selected original posterior against C gives a conditional version of this pushed joint by its defining integral identity. The common-channel input-output laws have the original relative entropy because their conditional output kernel is identical. Factor their density ratio into the retained marginal ratio and the conditional input ratio, and integrate. This gives the KL chain rule; adding the same finite $-\log p(o \mid X, c)$ proves the displayed VFE identity. Exceptional observation versions are transported by the chosen version family, not inferred from an almost-sure identity at an arbitrary observation. □

The channel depends on context, so theorem 14.5 does not imply Fisher contraction for derivatives in c. Such derivatives also differentiate the channel. Nor does a comparison across coarse levels give an inference-time decrease or increased evidence. These implications are declined because they compare different derivatives or different objects. [NOT-CLAIMED]

14.3 Nonflat connections and metric-dependent projections

On the trivial common principal $\mathrm{SO}(2)$ bundle over U, choose two connections independently, with local reference potentials

$$A^x_* = \frac{\kappa_x}{2}(x dy - y dx)J, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad x \in \{b, m\}. \tag{14.11}$$

In this equation only, the superscript x labels a channel; the scalar x inside the one-form is the first base coordinate. The convention is $\nabla = d + A$, so the potential in a frame G is $A = GA_*G^{-1} - dGG^{-1}$. The connections are supplied geometric data, not quantities determined by attention. [DEFINITION]

Theorem 14.6 (A nonflat parent retains its connection defect). The reference curvatures are $\mathcal{R}^x_* = \kappa_x J dx \wedge dy$. On a constant-rank parent domain, with the same channel connection repeated in its child fibers,

$$\mathcal{D}F = dF + A_{\mathrm{c}}F - FA_{\mathrm{f}} = G_{\mathrm{c}}(dL)G_{\mathrm{f}}^{-1}. \tag{14.12}$$

For composable maps their defects satisfy

$$\mathcal{D}(F_2F_1) = (\mathcal{D}F_2)F_1 + F_2(\mathcal{D}F_1). \tag{14.13}$$

These established equations permit nonzero curvature and do not make variable-weight coarse maps parallel. [ESTABLISHED]

Proof. The abelian potential has $A_* \wedge A_* = 0$, and differentiating its coefficient gives $\kappa_x J dx \wedge dy$. Differentiate $F = G_{\mathrm{c}} L G_{\mathrm{f}}^{-1}$ and substitute the displayed frame potentials. Frame derivatives cancel. The repeated reference connection commutes through scalar block weights, leaving dL. The product rule gives the second equation by adding and subtracting $F_2A_1F_1$. Thus gauge covariance is compatible with a nonzero first-order defect. □

Theorem 14.7 (An operational record sensitive to a declared base holonomy). Fix a probe agent at $c = (7/20, 1/5)$, and the closed rectangular path with successive vertices $(7/20, 1/5)$, $(13/20, 1/5)$, $(13/20, 1/2)$, $(7/20, 1/2)$, and c. Choose an agent whose domain contains this rectangle, a selector P_{pr} onto its separately typed belief and model coordinates, and fixed positive definite noise R_{pr}. Let a finite label a specify prescribed connection parameters $(\kappa_{b,a}, \kappa_{m,a})$, with positive prior masses π_a. Keep one original normalized joint $p_0(Y, e, o_0)$, independent of a. The additional record and joint are declared by

$$\begin{aligned} U_a &= G(c)\exp\left[-\mathcal{A}\,\mathrm{diag}(\kappa_{b,a}J, \kappa_{m,a}J)\right] G(c)^{-1}, \qquad \mathcal{A} = 9/100, \\ H_a &= (U_a - I_4)P_{\mathrm{pr}}, \\ p(a, Y, e, o_0, r) &= \pi_a p_0(Y, e, o_0) N(r; H_aY, R_{\mathrm{pr}}). \end{aligned} \tag{14.14}$$

This is a normalized statistical experiment whose record can distinguish base holonomies while the matter law, loop, selector, and noise are held fixed. It is an established operational witness for this declared tuple. [ESTABLISHED]

Proof. The rectangle lies inside U_2; its convexity and the strip's convexity keep every segment in that agent domain. Integrating the reference one-form along a segment gives $\kappa \det(c_j, c_{j+1})/2$. Summing segments gives $\kappa\mathcal{A}$, and the equation $\dot{v} = -A(\dot{\gamma})v$ gives the displayed exponential. Endpoint frames conjugate it at the basepoint. Each new Gaussian factor integrates to one in r, then p_0 and π normalize the full joint.

If the original matter posterior is $N(\mu_0, \Sigma_0)$, with Σ_0 positive definite, Gaussian integration gives

$$\ell_a(r) = N(r; H_a\mu_0, R_{\mathrm{pr}} + H_a\Sigma_0H_a^\top), \qquad Z(r, o_0) = Z_0(o_0)\sum_a \pi_a\ell_a(r), \qquad \omega_a = \frac{\pi_a\ell_a(r)}{\sum_d \pi_d\ell_d(r)}. \tag{14.15}$$

For example the supplied pairs $(0, 0)$ and $(7/10, -2/5)$ give $H_0 = 0$ and $H_1 \neq 0$, since neither nonzero angle is a multiple of 2π. Therefore $H_1\Sigma_0H_1^\top \neq 0$, so the two predictive Gaussian laws

have different covariance matrices. Their record laws differ with all the listed non-holonomy data fixed. No particular observed value or unique parameter recovery is required for this distinction.

Under a passive change of fine and probe coordinates by the represented orthogonal frame maps Q, V, send P_{pr} to $VP_{\mathrm{pr}}Q^{-1}$, U_a to VU_aV^{-1}, r to Vr, and R_{pr} to $VR_{\mathrm{pr}}V^{\top}$. Then $H'_a = VH_aQ^{-1}$, so the entire observation kernel is pushed coherently and its evidence is invariant. A zero-area loop instead gives $H_a = 0$ for every label and no label reweighting; its record still has an ordinary common evidence factor. □

The witness declares both a connection and a curve; it does not identify an arbitrary graph-link product with base transport. Distinct connection labels can also have the same holonomy modulo a full rotation, and a chosen matter law can leave further symmetries unidentifiable. A universal connection recovery or graph-to-base identification is therefore not claimed. [NOT-CLAIMED]

For a separate constant-rank construction, let E carry a declared Euclidean fiber metric and a compatible connection $d + A$, with $A^{\top} = -A$ in orthonormal frames. Let $F : E \to E_{\mathrm{c}}$ be a smooth surjective vector-bundle map of fixed rank. Define

$$g = (FF^{\top})^{-1}, \qquad R = F^{\top}g, \qquad P = RF, \qquad N = I - P, \qquad B = F(dR + AR) = (FA - dF)R. \tag{14.16}$$

Here P is a fiberwise orthogonal projector, not a probability law; g is the chosen metric's quotient metric. This is a declared minimum-norm splitting and its projected connection. [DEFINITION]

Theorem 14.8 (Projected connection versus exact connection descent). The connection $d + B$ preserves g. If $S = N(dR + AR)$ and $\mathcal{E} = dF + BF - FA$, then

$$dg = B^{\top}g + gB, \qquad \mathcal{E} = (dF - FA)N = g^{-1}S^{\top}. \tag{14.17}$$

Moreover F is parallel exactly when $\ker F$ is preserved by the fine connection. Its curvature satisfies

$$\mathcal{R}_B = F\mathcal{R}_A R + g^{-1}S^{\top} \wedge S, \qquad \mathcal{R}_A = dA + A \wedge A. \tag{14.18}$$

These identities hold for the declared metric and splitting of fixed rank. [ESTABLISHED]

Proof. Direct multiplication gives $FR = I$, $P = P^{\top} = P^2$, and $R^{\top}R = g$. A fine representative of y is uniquely $Ry + v$, with $v \in \ker F$, and its squared norm is $y^{\top}gy + \|v\|^2$; this proves the quotient-metric description. Multiplying B by g gives $gB = R^{\top}(dR + AR)$. Adding its transpose cancels the skew A terms and gives dg. Differentiating $FR = I$ gives the formula for $\mathcal{E}$. With $D_AP = dP + [A, P]$, differentiating $P^2 = P$ shows that D_AP has only off-diagonal blocks, and metric compatibility makes its coefficients symmetric. Hence $(D_AP)R = S$ and $\mathcal{E} = F(D_AP)N = g^{-1}S^{\top}$. Its vanishing is equivalent to preservation of the retained subbundle and, by differentiating orthogonal inner products, of its orthogonal complement $\ker F$.

On a retained section $u = Pu$, the projected connection is $P\nabla$. Expanding its square yields $P\mathcal{R}_Au + P(D_AP) \wedge (D_AP)u$: the term with $P(D_AP)P$ vanishes, and $N\nabla u = (D_AP)u$. Transporting

through R and using the preceding adjoint identities gives eq. (14.18). The second term is an endomorphism-valued two-form, not a nonnegative scalar energy. □

The projected connection can be curved even with a flat source. For

$$A = 0, \qquad F(x,y) = \begin{pmatrix} 1 & 0 & -x \\ 0 & 1 & -y \end{pmatrix}, \qquad \mathcal{R}_B(0) = -J dx \wedge dy \neq 0, \tag{14.19}$$

this is an established two-dimensional quotient example. [ESTABLISHED]

Proof. At the origin $g = I_2$ and the only nonzero row of S is $(-dx, -dy)$. Its exterior square is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) dx \wedge dy$, which gives the formula by theorem 14.8. All bundles in this example are trivial, so the curvature does not establish nontrivial topology. □

For the scalar-weight parent $L = (\lambda_i I_d)_i$, with a common internal connection A_*, the same construction instead gives

$$\sigma = \sum_i \lambda_i^2, \qquad g = \sigma^{-1} I_d, \qquad B = A_* - \tfrac{1}{2} d \log \sigma I_d, \qquad \mathcal{R}_B = \mathcal{R}_{A_*}. \tag{14.20}$$

This is an established cancellation in this specialization. [ESTABLISHED]

Proof. Insert $R = \mathrm{col}_i(\lambda_i I_d)/\sigma$ in eq. (14.16). The additional scalar one-form is exact, its square is zero, and its mixed wedge terms with A_* cancel. The quotient-orthonormal coordinates multiply y by $\sigma^{-1/2}$ and remove this scalar term from B. The original variable-weight defect need not vanish. □

Theorem 14.9 (Composition requires propagation of the quotient metric). For smooth surjections $F : E_0 \to E_1$ and $G : E_1 \to E_2$, start with metric I on E_0, give E_1 its induced metric $g_1 = (FF^\top)^{-1}$, and use that metric in the second projection. Then the quotient metrics and projected connections agree for staged and direct projection, whether or not either map is parallel. This is an established composition theorem for the specified splittings. [ESTABLISHED]

Proof. The second lift is $R_G = g_1^{-1} G^\top (G g_1^{-1} G^\top)^{-1}$. Multiplication gives

$$R_F R_G = (GF)^\top ((GF)(GF)^\top)^{-1} = R_{GF}. \tag{14.21}$$

The final quotient metric has the same denominator. The product rule and $F R_F = I$ now give $G(d + B_1) R_G = GF(d + A_0)(R_F R_G)$, which is the direct connection. Resetting the intermediate metric is a different construction: with $F = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right)$, $G = (1, x)$, and $A_0 = 0$, reset metric I gives $-x dx/(1 + x^2)$, whereas the direct connection is $-x dx/(4 + x^2)$. This verifies why the metric must be retained. □

The quotient construction does not supply a canonical effective-support quotient for arbitrary semidefinite edge weights, nor an intertwiner into an arbitrarily prescribed associated representation.

Those extensions require constant-rank support data, compatible represented actions, and an admitted quotient or bundle map. The displayed theorem closes the chosen metric construction only. [OPEN]

14.4 Actual base descent with inherited nonzero curvature

Let $B = U \times (-1, 1)$, with extra coordinate z, and let $f : B \to U$ be $f(c, z) = c$. Choose a smooth passive frame $h(z) = \exp(\gamma z J)$, and let H repeat this representation in each separately typed associated coordinate. Declare the principal and associated maps

$$\Pi((c, z), g) = (c, h(z)^{-1} g), \qquad \Psi((c, z), v) = (c, H(z)^{-1} v). \tag{14.22}$$

They cover the noninjective base map f. This example removes an explicitly redundant base coordinate; it does not average contexts. [DEFINITION]

Theorem 14.10 (Compatible section, law, and connection descent). Given smooth target sections, laws, and connections on their stated target domains, define their source presentations by H, and define

$$A_{\mathrm{s}} = H f^* A_{\mathrm{t}} H^{-1} - dH H^{-1}, \qquad F_{\mathrm{s}} = H_{\mathrm{c}} f^* F_{\mathrm{t}} H_{\mathrm{f}}^{-1}. \tag{14.23}$$

Then sections, full laws, evaluators, and channels descend through Ψ, provided observations and structural data are also pulled back. Both channel connections are compatible with descent, their curvature is $H f^* \mathcal{R}_{\mathrm{t}} H^{-1}$, and the vertical coarse defect is the conjugate pullback of the target defect. This is an established nonidentity base descent that permits nonflat target connections. [ESTABLISHED]

Proof. The principal map is equivariant since $h^{-1}(ga) = (h^{-1}g)a$. Substituting $h^{-1}g$ in the target principal connection form $g^{-1}A_{\mathrm{t}}g + g^{-1}dg$ gives eq. (14.23). Differentiating $\Psi = H^{-1}$ then gives $d\Psi + f^* A_{\mathrm{t}} \Psi - \Psi A_{\mathrm{s}} = 0$. The curvature law follows by expanding $dA_{\mathrm{s}} + A_{\mathrm{s}} \wedge A_{\mathrm{s}}$; the derivatives of H cancel. Expanding the covariant derivative of F_{s} similarly leaves the conjugate pullback of $\mathcal{D}F_{\mathrm{t}}$.

For a target kernel C_c, define the source kernel by undoing its input frame, applying C_c, and restoring its output frame. Evaluating a target measurable set gives

$$\Psi_{\mathrm{c}\#}(\mathbb{Q}_{\mathrm{s}} C_{\mathrm{s}}) = (\Psi_{\mathrm{f}\#} \mathbb{Q}_{\mathrm{s}}) C_{\mathrm{t}}. \tag{14.24}$$

The same change of variables holds for the full generative joint, reference, and selected posterior versions. Keeping observation coordinates unchanged therefore preserves evidence; common invertible presentation maps preserve relative entropy by cancellation of density Jacobians. The declared normalized evaluator transforms by its separately typed input-output square. Finally restrictions to U_i pull back to restrictions to $U_i \times (-1, 1)$, so all cover identities persist. □

For the reference potentials in eq. (14.11), $A_{\mathrm{s}}^x = f^* A_*^x - \gamma J dz$; its mixed curvature is zero and its inherited $dx \wedge dy$ curvature can be nonzero. A nonzero coefficient in the forgotten direction is

therefore compatible with exact descent. This is an established consequence of the prescribed passive frame, not a new force in that direction. [ESTABLISHED]

Theorem 14.11 (The descent criterion and its obstructions). Let $f : B \to \overline{B}$ be a given smooth surjective submersion and $\Psi : E \to \overline{E}$ a given smooth bundle map over f. A smooth section s has a unique descended section exactly when Ψs is constant on every complete fiber of f. For a given smooth surjective principal submersion Π, a connection form descends exactly when its pullbacks by the two projections of $P \times_{\overline{P}} P$ agree. These are established criteria for given maps, not existence of a canonical quotient. [ESTABLISHED]

Proof. Necessity follows by evaluating a descended object at two lifts of the same base point or tangent vector. Conversely, a submersion has local smooth sections: in local coordinates augment its full-rank map by complementary source coordinates and apply the inverse function theorem. Composing Ψs with such a local section defines a smooth target section; fiber constancy makes the definitions independent of the local section and hence consistent on overlaps. Surjectivity gives uniqueness. For forms, choose a local section j of Π and set $\overline{\omega} = j^*\omega$. The fiber-product equality applied to $(v, djd\Pi(v))$ proves $\omega = \Pi^*\overline{\omega}$, and also independence of j. Equivariance and reproduction of principal vertical generators descend by surjectivity and the equivariant group-fiber isomorphisms.

The criteria have concrete failures. The source law $N(H(z)(1 + z/4, 0)^\top, I_2)$ descends under Ψ to a law whose mean varies with z; its KL to $N(0, I_2)$ is $(1 + z/4)^2/2$, so no passive reframing makes the pair constant on the forgotten fiber. Separately, adding $\lambda z J dx$, $\lambda \neq 0$, to A_{s} adds $\lambda J dz \wedge dx$ to its curvature. Its contraction with ∂_z is nonzero, whereas every conjugate pullback curvature annihilates that vector. This excludes compatible connection descent through this f. □

Choosing a probability measure on a varying base fiber and averaging its laws would define a new mixture operation. It requires a base prior or reference, a joint base-fiber model, and the induced observation weighting. Neither fiber constancy nor a canonical such measure follows from the cover. General informative base aggregation, rank-changing quotient geometry, and a probability law on smooth random sections remain open beyond the explicit constructions above. [OPEN]

Chapter 15

Parent Inference and Controlled Memory

A normalized parent law does not by itself specify parent-sensitive inference or a closed dynamics on a smaller observation. This chapter supplies a finite inference mechanism, identifies the preparation under which its full parent is Markov, and retains the exact hidden history when a further projection has memory. Finite-memory approximation then has its own normalized path law and an exact information loss. The transition index is an auxiliary inference index; it is not physical time. [DEFINITION]

15.1 A fixed normalized target and a parent-sensitive sweep

Let I be a nonempty finite incidence set with projections $z : \mathsf{I} \to \mathsf{Z}$ and $h : \mathsf{I} \to \mathsf{H}$, where h retains the complete decorated parent. Adopt positive generative weights g_i, normalized record conditionals $\eta_i(e)$, and normalized observation likelihoods $\ell_i(o)$, all independent of recognition. At an observation for which all retained $\ell_i(o) > 0$, define

$$Z_{\mathrm{gen}} = \sum_i g_i, \qquad Z_o = \sum_i g_i \ell_i(o), \qquad p_i = \frac{g_i \ell_i(o)}{Z_o}. \tag{15.1}$$

The complete generative law is $g_i \eta_i(e) \ell_i(o)/Z_{\mathrm{gen}}$. The local symbol p denotes this finite posterior, not a new agent. Positivity and the specified incidence support are adopted hypotheses for the finite results below. [HYPOTHESIS]

For recognition $q_i \eta_i(e)$, direct substitution in the exact ELBO gives the established identity

$$\mathcal{F}(q; o) = D_{\mathrm{KL}}(q \| p) - \log(Z_o/Z_{\mathrm{gen}}). \tag{15.2}$$

A different conditional record recognition retains its averaged conditional KL. Thus matching the displayed conditionals is a specialization of the complete joint-law objective, not removal of its dependence terms. [ESTABLISHED]

At fixed z, declare an undirected graph of allowed parent changes and a symmetric proposal $a_{ij} = a_{ji} \geq 0$ whose off-diagonal row sum is at most $1/2$. Its Metropolis kernel, the parent projection,

and its posterior conditional lift are

$$\begin{aligned} M_{ij} &= a_{ij}\min(1, p_j/p_i) \quad (i \neq j), & M_{ii} &= 1 - \sum_{j \neq i} M_{ij}, \\ C_{iH} &= \mathbf{1}_{h(i)=H}, & J_{Hi} &= \frac{p_i \mathbf{1}_{h(i)=H}}{p_H(H)}, \qquad p_H = pC, \\ D &= CJ, & T &= MD. \end{aligned} \tag{15.3}$$

Only nonempty parent fibers are retained. The step D refreshes the fine conditional law at the current complete parent. It is nontrivially parent-sensitive precisely when those fine conditional laws differ. The proposal and update order are declared mechanism choices. [DEFINITION]

The complete marked version of a row kernel K is $\widetilde{K}((i,e),(j,e')) = K_{ij}\eta_j(e')$. Summing the terminal record proves normalization, and summing the initial record proves that $p_i\eta_i(e)$ is invariant whenever $pK = p$. The matched family $q_i\eta_i(e)$ is preserved. These are established lift identities; unmatched conditional recognition retains its own information terms. [ESTABLISHED]

One finite assignment graph uses interval partitions of four ordered children and one binary root mark per block. A move flips one existing mark, merges adjacent blocks retaining the left mark, or reverses such a merge while choosing the exposed right mark. There are $\sum_{r=1}^{4}\binom{3}{r-1}2^r = 54$ assignments and seven neighbors per assignment: for r blocks the counts are r, $r-1$, and $2(4-r)$. Merge-to-one and split-to-target paths show connectivity. Consequently proposal $1/14$ per edge satisfies the preceding conditions. This is an established finite realization of the move graph, not an intrinsic choice of partition or parent. [ESTABLISHED]

Theorem 15.1 (Stationarity, exact descent, and finite-component convergence). The kernels M, D, T are normalized and preserve p. Let a component mean a connected component of the union of their positive off-diagonal edges. For every initial recognition q, T preserves component masses and decreases eq. (15.2), strictly unless $q(i) = q(\mathsf{B})p(i \mid \mathsf{B})$ on each component B. On each fixed component it converges geometrically to that conditional target. These are established finite-instance results; they give no uniform rate as state counts grow or positive weights tend to zero. [ESTABLISHED]

Proof. For an edge, $p_iM_{ij} = a_{ij}\min(p_i, p_j) = p_jM_{ji}$, so M is reversible. Also $JC = I$ and $pD = p$, while $p_iD_{ij} = p_ip_j/p_H(h(i))$ on a common parent fiber, so D is reversible. Normalization follows from their definitions, and stationarity survives composition. For any stationary finite kernel K, put $q' = qK$ and $K^*_{ji} = p_iK_{ij}/p_j$. The two input-output joint laws q_iK_{ij} and p_iK_{ij} have KL $D_{\mathrm{KL}}(q\|p)$. Factoring by their output marginal proves

$$D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(qK\|p) = \sum_{j:q'_j>0} q'_j D_{\mathrm{KL}}\big(q(i \mid j)\|K^*_{j\cdot}\big). \tag{15.4}$$

Every summand is nonnegative. With positive diagonal, equality requires the ratio q_i/p_i to be constant across every positive edge: the output conditional equality identifies that ratio at i and j,

and the holding edge identifies it with the output ratio. Equality for M therefore makes q stationary under M; equality for the subsequent D additionally makes the ratio constant on parent fibers. This is exactly constancy on the stated union components.

The positive holding edges of both factors ensure that every M or D edge is a positive T edge. On a component with n states, a shortest path has length at most $n-1$; positive holding extends it to one common length $m \geq \max(1, n-1)$. Hence every entry of T^m on that component is positive. Writing its normalized target as π, let $\varepsilon = \min_{ij}(T^m)_{ij}/\pi_j > 0$. Then $T^m = \varepsilon \mathbf{1}\pi + (1-\varepsilon)R$ with a stationary stochastic R, unless $\varepsilon = 1$, when convergence is exact. Convexity and data processing give $D_{\mathrm{KL}}(qT^m \| \pi) \leq (1-\varepsilon) D_{\mathrm{KL}}(q\|\pi)$, and the same mixture contracts total variation. Iteration proves convergence. The original full-model VFE on this component includes $-\log p(\mathrm{B}) - \log(Z_o/Z_{\mathrm{gen}})$; neither constant is deleted by the conditional dynamics. □

Theorem 15.2 (The prepared complete parent is Markov). Define $L = JMC$. Then L is reversible under p_H, and

$$JT = LJ, \qquad (\nu J)T^n = \nu L^n J. \tag{15.5}$$

For a start $q_0 = \nu J$, the complete parent history has transition L. For an arbitrary start, this statement applies after the first completed D stage. It is an established prepared-history statement, not universal closure of every parent projection at the initial instant. [ESTABLISHED]

Proof. Use $T = MCJ$ and $JC = I$ to obtain $JT = JMCJ = LJ$, then induct. Moreover $p_H(H)L_{HH'} = \sum_{i:h(i)=H, j:h(j)=H'} p_i M_{ij}$, which is symmetric by reversibility of M. The final conditional refresh makes the fine law given the entire preceding parent history equal to $J_{H\cdot}$, depending only on its last parent. The next parent row is consequently $J_{H\cdot}MC = L_{H\cdot}$. Induction over histories proves the Markov assertion, including its preparation condition. □

15.2 What persistence and projection actually imply

For a finite irreducible component with stationary law π, choose a nonempty proper event A, such as a specified retained block event. Let K_A be the restriction of its transition to A. For an initial law μ supported in A, the event of remaining inside for n further transitions has probability

$$s_n = \mu K_A^n \mathbf{1}, \qquad \mathbb{E}_\mu \tau_{A^c} = \sum_{n \geq 0} s_n = \mu (I - K_A)^{-1} \mathbf{1}. \tag{15.6}$$

These are established formulas for unbroken survival with lost mass retained as lost mass. [ESTABLISHED]

Proof. Multiplication by the restricted matrix sums exactly the paths staying in A. For a positive integer-valued exit time, summing its tail gives its mean by summing nonnegative indicator functions. Every state in A has a finite positive exit path. Finiteness supplies a common number of steps and

a positive lower exit probability, so the killed tail is geometric. The convergent matrix series is the inverse displayed above. □

Theorem 15.3 (Stationary episode means and the clock control). Let $J_A = \sum_{i\notin A, j\in A} \pi_i K_{ij}$. Under the stationary entrance preparation, the inside and outside episode means are

$$\mathbb{E}\tau_{A^c} = \frac{\pi(A)}{J_A}, \qquad \mathbb{E}\tau_A = \frac{\pi(A^c)}{J_A}. \tag{15.7}$$

For whole-clock waiting $K_\lambda = (1-\lambda)I + \lambda K$, $0 < \lambda \leq 1$, every finite fixed-preparation hitting mean is multiplied by $1/\lambda$. These established identities separate stationary occupancy, crossing frequency, and survival shape. [ESTABLISHED]

Proof. The unnormalized entrance row on A is $r = \pi_{A^c} K_{A^c,A}$. Stationarity gives $\pi_A = \pi_A K_A + r$, so $r(I - K_A)^{-1} = \pi_A$. Divide by $J_A = r\mathbf{1}$ and use eq. (15.6). Stationarity equates entrance and exit flux, giving the outside formula as well. Finally $I - (K_\lambda)_A = \lambda(I - K_A)$, so its inverse rescales every finite mean by $1/\lambda$. Equal stationary occupancy and flux therefore force equal stationary episode means but need not give equal survival distributions. □

The finite mechanism does not establish universal persistent hierarchy, autonomous agency, or a preferred clock. Establishing a persistence advantage requires a stated event, preparation, and comparator after accounting for occupancy and waiting. Higher-level lineage and geometry-changing dynamics remain separate open construction problems. [OPEN]

Theorem 15.4 (The exact criterion for a universal projected chain). For a finite hidden kernel K and deterministic observation $f : \mathsf{S} \to \mathsf{A}$, one observed transition kernel works for every initial law exactly when

$$\sum_{j:f(j)=b} K_{ij} = \sum_{j:f(j)=b} K_{i'j} \quad \text{whenever } f(i) = f(i'), \quad b \in \mathsf{A}. \tag{15.8}$$

This condition is necessary and sufficient for the complete supported rows. [ESTABLISHED]

Proof. Point-mass starts at i, i' prove necessity. Under the condition, every hidden posterior given an observed history ending at a is supported on $f^{-1}(a)$, and all rows there give the same next-observation law. Averaging therefore gives that law independently of earlier observations. □

Even a stationary observation can have memory. On the deterministic three-state cycle with uniform stationary law and labels $(0, 0, 1)$, conditioning on observed histories $(0, 0)$ and $(1, 0)$ gives next symbols 1 and 0, respectively. Both histories are positive and have the same final symbol. This is an established failure of first-order Markov closure under that fixed stationary preparation. [ESTABLISHED]

15.3 The complete hidden history and its conditional state

For the rest of this chapter let Z_n be a finite hidden state with initial law ν and transition K, and let $Y_n = f(Z_n)$ be its finite observation. These are local history symbols, not the structural datum X or the original sensor observation o. Let E_a be the diagonal indicator of $f^{-1}(a)$. The declared joint finite path law and its observed marginal are

$$\nu(z_0) \prod_{t=0}^{N-2} K_{z_t z_{t+1}} \prod_{t=0}^{N-1} \mathbf{1}_{f(z_t)=y_t}, \qquad \mathbb{P}_N(y) = \nu E_{y_0} K E_{y_1} \cdots K E_{y_{N-1}} \mathbf{1}. \tag{15.9}$$

This auxiliary model is fixed before its path recognition is varied. [DEFINITION]

Theorem 15.5 (Exact filtering without a projected Markov assumption). For a positive word $w = (y_0, \ldots, y_n)$, put

$$a_w = \nu E_{y_0} K E_{y_1} \cdots K E_{y_n}, \qquad \widehat{\nu}_w = \frac{a_w}{a_w \mathbf{1}}, \qquad \mathbb{P}(Y_{n+1} = b \mid w) = \widehat{\nu}_w K E_b \mathbf{1}. \tag{15.10}$$

These formulas define the exact terminal posterior and next prediction. Every zero-mass word has zero messages for all extensions and undefined conditional readouts. The terminal posterior determines every future suffix, but its possible values need not form a finite set. The finite formulas and support statements are established. [ESTABLISHED]

Proof. Summing the hidden path law while fixing its final state gives a_w. Successive summation of E_b to I, followed by $K\mathbf{1} = \mathbf{1}$, proves normalization and prefix consistency. Division by positive word mass gives the posterior; a further matrix product gives any future suffix. Nonnegative zero-mass messages are identically zero and stay zero on multiplication. A posterior is a probability vector on the hidden set, so its sufficiency does not assert finiteness of its attainable subset. □

An auxiliary observed-word evidence and path VFE may be formed from eq. (15.9). If ν was chosen using a recognition law of the original model, that choice does not turn this path experiment into the original recognition-independent sensor model. The two VFE identities have different evidences and must remain separately typed. This identification is deliberately declined. [NOT-CLAIMED]

15.4 Normalized finite-memory projection

First take any fixed finite law $\mathbb{P}_N$ on A^N, without stationarity. For an integer $k \geq 0$ and position t, let $c_t = y_{\max(0,t-k):t-1}$. On positive context mass define

$$R_t^{(k)}(a \mid c) = \mathbb{P}_N(Y_t = a \mid c_t = c), \qquad \widehat{\mathbb{P}}_N^{(k)}(y) = \prod_{t=0}^{N-1} R_t^{(k)}(y_t \mid c_t). \tag{15.11}$$

A zero prefix has zero extensions, so the product is defined recursively without choosing a row on a null context. The hat marks a path approximation, not a recognition law in the original statistical experiment. [DEFINITION]

Theorem 15.6 (Forward-KL optimality and its exact information defect). The product in eq. (15.11) is a normalized law, has the same retained context-plus-symbol marginals as $\mathbb{P}_N$, and is positive on every $\mathbb{P}_N$-positive word. Among all normalized causal laws with time-dependent k-context rows it uniquely minimizes the forward path KL. Its minimum is

$$D_N^{(k)} := D_{\mathrm{KL}}(\mathbb{P}_N \| \widehat{\mathbb{P}}_N^{(k)}) = \sum_{t=0}^{N-1} I_{\mathbb{P}_N}\big(Y_t; Y_{0:t-k-1} \mid Y_{\max(0,t-k):t-1}\big), \tag{15.12}$$

where an empty discarded history contributes zero. These are established finite-law statements, including uniqueness as a path law rather than of irrelevant conditional completions. [ESTABLISHED]

Proof. At $t = 0$ the empty context has mass one. Suppose the retained context has its $\mathbb{P}_N$ marginal. Multiplying by its supported row gives the correct context-plus-symbol marginal, and marginalizing its oldest symbol gives the next retained context marginal. This induction proves that no positive constructed prefix enters a null context, and successive row sums prove normalization. Every positive source word uses positive factors. For a competing law with rows S_t, expansion of the finite logarithm products and regrouping by contexts gives

$$D_{\mathrm{KL}}(\mathbb{P}_N \| S) - D_N^{(k)} = \sum_{t,c:\mathbb{P}_N(c_t=c)>0} \mathbb{P}_N(c_t = c) D_{\mathrm{KL}}(R_t^{(k)}(\cdot \mid c) \| S_t(\cdot \mid c)). \tag{15.13}$$

If the competing law excludes a positive source word, its KL is infinite and the inequality is immediate; otherwise all sums are finite. Equality forces equality of every occupied row. Recursive reachability then forces the same path law. Finally subtract the exact source chain-rule logarithm from the projected product; its expected terms are the difference between entropy conditioned on the retained context and entropy conditioned on the full past. This is precisely the conditional mutual information in eq. (15.12). □

Increasing k decreases $D_N^{(k)}$, and $k \geq N-1$ gives exact recovery, by conditioning and the displayed information identity. The projection is idempotent because it retains its own defining context-plus-symbol marginals. Restriction to a shorter prefix commutes with the construction, clipping the order only at its available history. These are established finite consequences, not determination of a finite memory order for an infinite process. [ESTABLISHED]

Forward error controls bounded path observables: for finite probability laws P, Q, $2\,\mathrm{TV}(P,Q)^2 \leq D_{\mathrm{KL}}(P\|Q)$. To see this, group the KL on $\{P \geq Q\}$ and its complement; the remaining within-group divergences are nonnegative. The resulting binary KL has second derivative $1/[p(1-p)] \geq 4$, value zero, and first derivative zero at $p = q$, so it is at least $2(p-q)^2$. Boundary cases follow by continuity or infinite KL. This establishes the stated control, but does not control reverse KL or every recognition-weighted VFE difference. [ESTABLISHED]

For example, the stationary cycle labeled $(0,0,1)$ has $\mathbb{P}_3(000) = 0$ but $\widehat{\mathbb{P}}_3^{(1)}(000) = 1/6$, so the reverse KL is infinite. More generally, for separately varied ρ supported on a positive source support, the auxiliary VFE change is

$$D_{\mathrm{KL}}(\rho\|\widehat{\mathbb{P}}_N^{(k)}) - D_{\mathrm{KL}}(\rho\|\mathbb{P}_N) = \mathbb{E}_\rho \log \frac{\mathbb{P}_N}{\widehat{\mathbb{P}}_N^{(k)}}. \tag{15.14}$$

Subtraction of the two finite log ratios proves this established identity. Its worst-case bound over such ρ uses the extrema of the supported log ratio, rather than its $\mathbb{P}_N$-average. [ESTABLISHED]

15.5 Stationary entropy, duration, and effective brackets

Now assume $\nu = \pi$, with $\pi K = \pi$, and restrict the hidden set to its positive stationary support. That support is closed: stationarity and nonnegativity force $K_{ij} = 0$ if $\pi_i > 0 = \pi_j$. Stationary initialization is an additional hypothesis, excluding arbitrary initial preparations from the rate identities below. [HYPOTHESIS]

Set $H_N = H(Y_0, \dots, Y_{N-1})$, $H_0 = 0$, and

$$h_j = H(Y_j \mid Y_0, \dots, Y_{j-1}), \qquad h_\infty = \lim_{j\to\infty} h_j, \qquad r_k = h_k - h_\infty. \tag{15.15}$$

For $j = 0$ the conditional entropy means $H(Y_0)$. All entropies use natural logarithms. The existence of the limit is proved next rather than added to these definitions. [DEFINITION]

Theorem 15.7 (Stationary projection and the finite hidden information budget). The stationary projection is normalized, stationary, and prefix-consistent; it agrees with the source on every window of length at most $k+1$. The sequence h_j decreases to h_∞, and

$$\begin{aligned} D_N^{(k)} &= \sum_{t=k+1}^{N-1} (h_k - h_t), \quad \lim_{N\to\infty} \frac{D_N^{(k)}}{N} = r_k, \\ \sum_{j=0}^{\infty} r_j &\le H(Z_0) =: C_Z, \qquad\qquad r_k \le \frac{C_Z}{k+1}, \\ 0 \le r_k - \frac{D_N^{(k)}}{N} &\le \frac{C_Z}{N}, \qquad\qquad D_N^{(k)} \le (N-k-1)_+ r_k. \end{aligned} \tag{15.16}$$

An empty sum is zero. These are established bounds for any finite stationary hidden model, without irreducibility or a forgetting assumption. [ESTABLISHED]

Proof. For $k \ge 1$, the positive k-context chain shifts c to its last k symbols after appending a, with row $\mathbb{P}_{k+1}(ca)/\mathbb{P}_k(c)$. Its incoming mass at a context d is the corresponding marginal of $\mathbb{P}_{k+1}$, equal to $\mathbb{P}_k(d)$ by stationarity. This proves stationarity and exact $(k+1)$-windows; shorter prefixes follow by marginalization. For $k = 0$, the law is the stationary independent product of the one-symbol marginal.

Conditioning and stationarity give $0 \le h_{j+1} \le h_j$, so the limit exists. The entropy chain rule gives $H_N = \sum_{j<N} h_j$, whose average tends to h_∞. For $N > k$, the projected cross-entropy is $H_k + (N-k)h_k$, giving the exact KL sum and its rate; shorter horizons are exact. To bound total excess, take two adjacent finite observed blocks of lengths b, n. Their past and future are conditionally independent given the intervening hidden state, so their mutual information is at most C_Z. Expanding it by stationary entropies gives $H_n - \sum_{j=0}^{n-1} h_{b+j} \le C_Z$. With n fixed let b increase, obtaining $\sum_{j<n} r_j \le C_Z$. Increasing n proves the nonnegative series bound, and monotonicity gives $(k+1)r_k \le C_Z$. Finally, for $N > k$, $Nr_k - D_N^{(k)} = kr_k + \sum_{t=k}^{N-1} r_t \le C_Z$, while for $N \le k$, $Nr_k \le C_Z$. Bounding each nonzero KL summand by r_k gives the last inequality. □

Theorem 15.8 (Finite hidden-boundary brackets converge to the entropy rate). For $m \ge 1$, define

$$g_m = H(Y_m \mid Z_0, Y_1, \ldots, Y_{m-1}), \qquad d_m = h_{m-1} - g_m. \tag{15.17}$$

The future symbol Y_1 follows one transition from Z_0. Then

$$g_m \uparrow h_\infty, \qquad h_{m-1} \downarrow h_\infty, \qquad 0 \le d_m \le C_Z/m, \qquad \max(0, h_k - h_{m-1}) \le r_k \le h_k - g_m. \tag{15.18}$$

These are established finite-sum bounds and convergent brackets, including for reducible stationary hidden models. [ESTABLISHED]

Proof. Adding Z_1 to the conditioning and using the Markov factorization and $Y_1 = f(Z_1)$ gives $g_{m+1} \ge H(Y_{m+1} \mid Z_1, Y_2, \ldots, Y_m) = g_m$. Stationarity yields $d_m = I(Z_0; Y_m \mid Y_1, \ldots, Y_{m-1}) \ge 0$. The already proved monotonicity of h, together with that of g, makes d_m nonincreasing. The mutual-information chain rule gives $\sum_{m=1}^{M} d_m = I(Z_0; Y_1, \ldots, Y_M) \le C_Z$, whence $d_m \le C_Z/m \to 0$. Therefore both bracket endpoints have the same limit. Subtracting their inequalities from h_k proves the rate bracket. Every entropy here is a finite sum of normalized path probabilities computed from K; no infinite-past conditional is needed. □

For rational finite models, the brackets give a finite procedure for any prescribed positive entropy-rate accuracy. Choose $m \ge 2C_+/\varepsilon$, where $C_+ \ge C_Z$, and enclose each endpoint to width at most $\varepsilon/4$. The enclosing bracket then has width at most $C_Z/m + \varepsilon/2 \le \varepsilon$. Such logarithm enclosures exist from the convergent series $\log u = 2\sum_{j\ge 0} s^{2j+1}/(2j+1)$, $s = (u-1)/(u+1)$, after reducing a positive rational to $2^a u$, $1 \le u < 2$; the positive tail is bounded by a geometric series. Only finitely many positive ratios occur at a fixed depth. This establishes finite stopping for accuracy, but not efficient enumeration or finite resolution of an exact equality to a requested threshold. [ESTABLISHED]

15.6 A constructive logarithmic memory schedule under flag hypotheses

The following stronger rate uses additional support information. Put $T^a_{ij} = K_{ij}\mathbf{1}_{f(j)=a}$. Assume that for every hidden target j, a word w_j of one common positive length b is supplied such that, with $B^w = T^{w_1}\cdots T^{w_b}$,

$$S_j = \{i : (B^{w_j}\mathbf{1})_i > 0\} \neq \varnothing, \qquad B^{w_j}_{ij} > 0 \quad (i \in S_j). \tag{15.19}$$

Thus every origin capable of a selected word can follow that word to its selected endpoint. One merging pair is insufficient; the condition covers all generating origins. Positive π, fixed K, and these common flag words are the adopted hypotheses. [HYPOTHESIS]

Let $A = |f(\mathsf{S})|$, $\lambda = \min\{K_{ij} : K_{ij} > 0\}$, and define positive constants

$$\begin{aligned} q &= \lambda^b, \qquad \theta = q\min_j \pi(S_j), \\ a &= \theta^2/2, \qquad c = \theta q/2, \\ \gamma &= \min(a, c/2), \qquad C_R = 2/b + 2(A-1). \end{aligned} \tag{15.20}$$

These are declared conservative constants; they are not fitted decay rates. [DEFINITION]

Theorem 15.9 (Explicit exponential excess and logarithmic path memory). Under the flag hypotheses, for $k \geq b-1$,

$$r_k \leq C_R e^{-\gamma t(k)}, \qquad t(k) = \left\lfloor \frac{k+1}{b} \right\rfloor - 1. \tag{15.21}$$

Let s_N be the least nonnegative integer with $C_R 2^{-s_N} \leq 2/N^3$, and put

$$k_N = \min\left(N-1, b\left(\lceil s_N/\gamma \rceil + 1\right) - 1\right). \tag{15.22}$$

For this fixed flagged model, $k_N = O(\log N)$ and

$$D_N^{(k_N)} \leq 2/N^2, \qquad \mathrm{TV}(\mathbb{P}_N, \widehat{\mathbb{P}}_N^{(k_N)}) \leq 1/N. \tag{15.23}$$

This is an established sufficient schedule; its constants may be arbitrarily poor across different finite models. [ESTABLISHED]

Proof. Group b transitions into a block, whose edge-emitted alphabet has size $L \leq A^b$. For any supported suffix v, choose a hidden state $j(v)$ maximizing its suffix likelihood $P_j(v)$. Stationarity and the flag condition give the conditional probability of its preceding flag as

$$\frac{\sum_{i,j}\pi_i B^{w_{j(v)}}_{ij} P_j(v)}{\sum_j \pi_j P_j(v)} \geq \sum_i \pi_i B^{w_{j(v)}}_{i,j(v)} \geq q\pi(S_{j(v)}) \geq \theta. \tag{15.24}$$

The positive maximum bounds the denominator and cancels from the restricted numerator. Reveal a stationary word of t blocks from right to left, and count these selected flags. Their conditional success probabilities are at least θ. For indicators I_s, repeated conditional expectation bounds $\mathbb{E}\exp(-u\sum_s I_s)$ by $(1-\theta+\theta e^{-u})^t$. The centered Bernoulli log transform has value and derivative zero at zero and second derivative at most $1/4$, so it is at most $u^2/8$. Exponential Markov inequality with $u=2\theta$ proves that fewer than $\theta t/2$ flags has probability at most e^{-at}.

Condition a hidden path on one complete supported word. At a selected flag, with subsequent suffix v, its bridge transition is $B^w_{ij}P_j(v)/\sum_\ell B^w_{i\ell}P_\ell(v)$. Every reachable row assigns at least q to $j(v)$: bound the suffix likelihood denominator by its maximum and use a positive length-b path, whose probability is at least λ^b. Couple two such bridges with one shared Bernoulli q coin at each counted flag and identical transitions after meeting. Their terminal disagreement probability on a good word is at most $(1-q)^{\theta t/2}\le e^{-ct}$; if $q=1$, meeting is certain at a counted flag, without a residual row. The next-block channel contracts this total-variation bound, as do mixtures over initial hidden states.

For laws at total variation d on an L-symbol alphabet, a maximal coupling and its disagreement indicator give $|H(u)-H(v)|\le d\log L+2\sqrt{d}$. Indeed conditional entropy is at most binary disagreement entropy plus $d\log L$, and $-d\log d-(1-d)\log(1-d)\le 2\sqrt{d}$ on $[0,1]$. Let β_t be conditional block entropy after t blocks minus the block entropy rate. Conditioning additionally on the initial hidden boundary gives a lower bound on that rate: condition a longer stationary past on that hidden state, use the Markov factorization to remove the remote past, and then let its length increase. Thus β_t is bounded by the conditional information between that boundary and the next block. On good words the entropy-continuity bound applies to each origin-conditioned prediction and its mixture; on bad words this information is at most $\log L$. Consequently, also at $t=0$ by the trivial entropy bound,

$$\beta_t\le 2e^{-ct/2}+\log L(e^{-ct}+e^{-at}). \tag{15.25}$$

The entropy chain rule identifies the block rate as bh_∞, and $\beta_t=\sum_{j=bt}^{b(t+1)-1} r_j\ge br_{b(t+1)-1}$. For the displayed $t(k)$, the last index is at most k, so monotonicity, $\log L\le b\log A\le b(A-1)$, and the definition of γ prove eq. (15.21).

If the cap $N-1$ is active, the path law is exact. Otherwise $\gamma t(k_N)\ge s_N$ and $e^{-s_N}\le 2^{-s_N}$, so $r_{k_N}\le 2/N^3$. Apply $D_N^{(k)}\le Nr_k$ and the proved finite Pinsker inequality. Finally successive halving finds $s_N=O(\log N)$; multiplication by the fixed constants $b,1/\gamma$ and the cap preserve this order. This proves the entire finite schedule, not merely a limiting exponent. □

The theorem does not imply uniform contraction on every conditioned word. For a positive word, any preparation-specific posterior is a mixture of the normalized nonzero rows of its masked hidden matrix; the supremum distance between preparations is exactly those rows' diameter, attained at point-mass starts. Two disjoint killed supports with positive holding entries retain disjoint posteriors along their common constant-symbol words. This is an established obstruction to arbitrary-preparation forgetting when that support configuration exists, compatible with the stationary good-word theorem. [ESTABLISHED]

A direct representation of the projected law needs only its requested context row,

$$R_t^{(k)}(a \mid c) = \frac{\mathbb{P}_{\ell+1}(ca)}{\mathbb{P}_\ell(c)}, \qquad \ell = \min(k, t), \quad \mathbb{P}_\ell(c) > 0. \tag{15.26}$$

The exact hidden filter computes these finite word masses, starting from π with an initial observation mask and transitions between later symbols. The empty-context row is the stationary observation marginal. The reachability proof of theorem 15.6 proves that multiplying only requested rows defines the same normalized family; no complete alphabet-power table is mathematically required. Null direct queries remain undefined. This is an established representation identity, with no stronger rate or minimum-order assertion. [ESTABLISHED]

Coherent hidden and observed bijections carry all matrices, contexts, flag words, and probability laws together, preserving these results by reindexing finite sums. Interpreting such a bijection as a gauge change additionally requires the separately supplied full observation and parent squares. Neither this invariance nor memory approximation inserts gauge curvature, Fisher curvature, or base curvature into the original fixed generative VFE. These conclusions and exclusions are established consequences of the typed probability construction. [ESTABLISHED]

A useful minimum-memory criterion, uniform constants near a critical limit, geometry-changing parent transitions, unrestricted nested lineage, and a physical interpretation of the inference clock remain open. The logarithmic schedule resolves one fixed finite flagged class; it does not settle those additional quantifiers. [OPEN]

Chapter 16 gives a complementary finite-depth construction with actual successive parent activation. Every activation pushes the complete current law, including all source assignments and loop marks; subsequent parent learning uses that parent's own evolving law. Conditional entropy, continuous-trajectory displacement, and complete assignment-history disagreement receive separate estimates. This construction neither replaces the retained hidden-history law above nor turns the finite-memory projection into an exact Markov closure for arbitrary preparations. [ESTABLISHED]

Chapter 16

Finite-Depth Adaptive Formation at One Base Point

The normalized parent construction and the memory results of Chapters 12 and 15 distinguish an exact probability reduction from an autonomous reduced dynamics. This chapter constructs a finite family in which successive reductions can actually be used during inference. The family has independent moving interaction links, retained network loops, a nonlinear observation of source-dependent belief mismatch, and positive-rate changes of the actual source assignments. Every new parent begins with the preceding actual retained law and subsequently learns from its own law. The quantitative result is Theorem 16.1; its estimates are proved in Chapter A. [ESTABLISHED]

The construction adopts one fixed base point c_*, a finite triangle-chain graph, supplied interaction-energy contrasts, a proper correlated reference, a compatible observation, and calibration-assisted learning of one scalar model parameter. It is a realization inside the principal/associated-bundle framework, with no base transport. The continuous mobility, categorical proposal rule, and model metric are declared inference kinetics. A scalar VFE alone does not choose them. [HYPOTHESIS]

16.1 The pointwise carrier and its restoring flag

Fix an integer $L \geq 2$, set $b = 2^{L-1}$, and arrange b disjoint triangles in a chain. A bridge joins each consecutive pair of triangles. Each triangle is cyclically oriented; every other edge has one fixed orientation. The graph has $n = 3b$ vertices and $4b - 1$ edges. Bridge number j, between triangles j and $j + 1$, has level $\ell(j) = 2 + v_2(j)$, where $v_2(j)$ is the exponent of two dividing j. Triangle edges have level one. The numbers $J_1 \geq \cdots \geq J_L > 0$ introduced below are energy strengths, not categorical source probabilities. [DEFINITION]

Use the translation action of $(\mathbb{R}, +)$ on two scalar law fibers at c_*, and fix $0 < c < 1$. A configuration and its law presentations are

$$\begin{aligned} x = (u, v, \zeta) \in \mathbb{R}^{2n+4b-1} = \mathbb{R}^{10b-1}, & \\ q_i^b(\cdot \mid x) = \mathcal{N}(u_i, 1), \qquad & q_i^m(\cdot \mid x) = \mathcal{N}(v_i, c), \\ \Theta_e(k) = k + \zeta_e, \qquad & d_i(x) = u_i - v_i, \\ r_e(x) = u_i - u_j - \zeta_e \quad (e : j \to i). & \end{aligned} \tag{16.1}$$

Here x is a random configuration of law presentations; it is not the fixed structural datum X of the general construction. The same local translation acts on both fibers in this specialization. The independent coordinates ζ_e are graph links, with $\zeta_{\bar{e}} = -\zeta_e$. No inter-agent model-alignment record is included. This choice leaves the local model evaluator and model learning present. [DEFINITION]

Under a passive vertex-frame translation a_i, the means become $u_i + a_i, v_i + a_i$ and an edge coordinate becomes $\zeta_e + a_i - a_j$. Thus d_i and r_e are invariant. Reference centers and observation records transform with the same passive change; holding those centers fixed while moving the configuration would instead change the model. The proper reference used below is not an invariant Haar probability on a noncompact group. These are the coordinate and reference conventions of this specialization. [DEFINITION]

For triangle q, let $\mathfrak{h}_q = \sum_{e\in q} \zeta_e$ be its oriented translation holonomy and put $h_q = \mathfrak{h}_q/\sqrt{3}$. The normalized loop vector is $h = (h_q)_{q=1}^{b}$. With

$$Q_\triangle = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \end{pmatrix}, \qquad |r_q|^2 = |Q_\triangle r_q|^2 + h_q^2, \tag{16.2}$$

the second identity follows from $\sum_{e\in q} r_e = -\mathfrak{h}_q$. In particular the loop energy in this decomposition belongs to the original sum of edge mismatches. It is not an additional flattening term. [ESTABLISHED]

Let B_1 stack the n rows d_i and the $2b$ internal contrast rows $Q_\triangle r_q$. For $s \geq 2$, let B_s stack the bridge residuals of level s, and write B for their stack. The link-coordinate columns in a linear dependence among these rows force all contrast and bridge coefficients to vanish. The model-mean columns then force the remaining coefficients to vanish. Hence $\operatorname{rank} B = 6b - 1$. Every normalized loop covector is orthogonal to every row of B. If D is the oriented graph incidence matrix, define

$$\begin{aligned} &\mathcal{K} = \{(t, t, Dt) : t \in \mathbb{R}^n\}, && \mathcal{H} = \operatorname{span}\{\text{normalized loop covectors}\}, \\ &\mathcal{N} = \operatorname{range} B^\top, && \mathbb{R}^{10b-1} = \mathcal{N} \oplus \mathcal{H} \oplus \mathcal{K}, \\ &H = B^\top \operatorname{diag}(J_s I) B + J_1 P_{\mathcal{H}}. \end{aligned} \tag{16.3}$$

The direct sum is orthogonal. The matrix H has kernel $\mathcal{K}$, and $x^\top H x = J_1 \sum_i d_i(x)^2 + \sum_e J_{\ell(e)} r_e(x)^2$. These are established finite-dimensional identities. [ESTABLISHED]

Define the nested retained spaces and projections by

$$\begin{aligned} &W_s = \ker(B_1, \ldots, B_s), \quad \mathcal{N}_s = W_s \cap \mathcal{N}, \quad M_s = \mathcal{N}_s \oplus \mathcal{H}, \\ &P_s : \mathbb{R}^{10b-1} \longrightarrow W_s, \quad P_0 = I, \quad P_s P_p = P_s \quad (p < s). \end{aligned} \tag{16.4}$$

Each P_s is the Euclidean orthogonal projection. All loop coordinates and all of $\mathcal{K}$ survive. The final space is $W_L = \mathcal{H} \oplus \mathcal{K}$. Write $k_p = \dim M_p$ and $d_p = \dim W_p^\perp$; the latter is the cumulative eliminated dimension. The elementary row count gives

$$\dim W_s = 4b - 1 + \frac{b}{2^{s-1}} \quad (1 \leq s \leq L), \qquad \dim W_L = 4b. \tag{16.5}$$

These dimensions count retained continuous coordinates rather than agents. [ESTABLISHED]

Choose orthonormal columns Q_s spanning $\mathcal{N}_{s-1} \cap \mathcal{N}_s^{\perp}$, and write $x_{\mathcal{N}} = \sum_s \mathsf{Q}_s w_s$. The matrix $\mathsf{B} = B[\mathsf{Q}_1 \cdots \mathsf{Q}_L]$, including its inverse, is block lower triangular. Conjugation by $\mathrm{diag}(\sqrt{J_s} I)$ cannot enlarge a permitted lower block by a factor greater than one. Therefore

$$\|\mathsf{B}^{-1}\|_F^{-2} \sum_s J_s |w_s|^2 \le x_{\mathcal{N}}^{\top} H x_{\mathcal{N}} \le \|\mathsf{B}\|_F^2 \sum_s J_s |w_s|^2. \tag{16.6}$$

Minimizing this quadratic form over an initial group of normal layers gives the same bounds for its Schur complement. These explicit finite matrix constants underlie the restoring estimates in the proof appendix. [ESTABLISHED]

16.2 One normalized experiment with a nonlinear source record

At each vertex the model sample is $m_i \sim \mathcal{N}(v_i, c)$, and the evaluator sends m_i to $\mathcal{N}(m_i, 1-c)$ on the belief sample k_i. Among couplings with the prescribed marginals $\mathcal{N}(v_i, c)$ and $\mathcal{N}(u_i, 1)$, define

$$m_i = v_i + \xi_i, \qquad k_i = u_i + \xi_i + \varepsilon_i, \qquad \xi_i \sim \mathcal{N}(0, c), \quad \varepsilon_i \sim \mathcal{N}(0, 1-c), \tag{16.7}$$

where the two Gaussian noises are independent. This is the chosen conditional recognition coupling at a fixed configuration. [DEFINITION]

Its log-density ratio to the physical model/evaluator coupling is $\{d_i(k_i - m_i) - d_i^2/2\}/(1-c)$. Its expectation is fixed by the two prescribed marginals. Thus, for any other admissible coupling ν_i, subtracting this ratio gives

$$D_{\mathrm{KL}}(\nu_i \| \mathcal{N}(v_i, c)\,\mathrm{ev}_i) = D_{\mathrm{KL}}(\nu_i \| \nu_i^*) + \frac{d_i^2}{2(1-c)}. \tag{16.8}$$

Here ν_i^* is the law in eq. (16.7). This proves its optimality and the total restriction cost $d^{\mathrm{ev}}(x) = x^{\top} R x/2$, where $R = b_d^{\top} b_d/(1-c)$ and $b_d x = u - v$. [ESTABLISHED]

Let ϑ be the learned scalar parameter. The physical configuration prior and its Lebesgue normalizer are

$$\begin{aligned} A_0(\vartheta) &= \vartheta H + P_{\mathcal{K}} - TR, \\ \mathbb{P}_{\vartheta}^0(dx) &= Z_0(\vartheta)^{-1} \exp\left[-\frac{x^{\top} A_0(\vartheta) x}{2T}\right] dx, \\ Z_0(\vartheta) &= (2\pi T)^{(10b-1)/2} \det A_0(\vartheta)^{-1/2}. \end{aligned} \tag{16.9}$$

Assume $T > 0$, $1 \le \vartheta \le 21/10$, and $T \le J_1(1-c)/2$. Then $TR \preceq H/2$, so $A_0(\vartheta) \succeq (\vartheta/2)H + P_{\mathcal{K}} \succ 0$. The prior is a normalized correlated Gaussian. Its projector $P_{\mathcal{K}}$ is generally dense; no incident-edge-only force evaluation is claimed. [ESTABLISHED]

Fix $0 < \kappa \le 1$ and a nonzero $z_\star \in \mathcal{K}$ whose node entries are not all equal. Declare the observation kernels

$$O \mid x \sim \mathcal{N}_{\mathcal{K}}(P_{\mathcal{K}} x, (T/\kappa) I_{\mathcal{K}}), \qquad Y_{\mathrm{cal}} \mid \vartheta \sim \mathcal{N}(\vartheta, T), \tag{16.10}$$

and retain the records $O = z_\star$, $Y_{\mathrm{cal}} = 2$. The first Gaussian is a density on the n-dimensional carrier $\mathcal{K}$, not a singular full-space density. Its compatibility with the coherent root and the separate calibration record are modeling assumptions. [HYPOTHESIS]

Each receiver i has the two candidate sources $\mathcal{N}_i^{\triangle}$ inside its own triangle. The complete assignment is $S = (S_i)_i \in \mathcal{J} = \prod_i \mathcal{N}_i^{\triangle}$, with $|\mathcal{J}| = 2^n$. Every bridge retains its original interaction energy but is excluded from these source palettes. Choose one distinguished receiver i_0 and label its two sources $+$ and $-$. The independent source priors are

$$\begin{aligned} \pi_{i_0,+}(\vartheta) &= \mathrm{sigmoid}(4(\vartheta - 3/2)), \quad \pi_{i_0,-}(\vartheta) = 1 - \pi_{i_0,+}(\vartheta), \\ \pi_{i,j}(\vartheta) &= \tfrac{1}{2} \quad (i \neq i_0), \qquad\qquad \pi_S(\vartheta) = \prod_i \pi_{i,S_i}(\vartheta). \end{aligned} \tag{16.11}$$

This internal-neighbor support and the supplied prior parameterization are adopted restrictions. [HYPOTHESIS]

Set $B_0 = 10^{-4}$, $\delta = B_0/n$, and $f(r) = 1 - e^{-r^2/2}$. At each receiver add a binary observation E_i, conditionally independent given (x, S), with

$$\begin{aligned} \Pr(E_i = 1 \mid x, S_i = j) &= q_0 \exp\left[-\delta f\left(\sqrt{J_1/T}\, r_{ij}(x)\right)\right], \qquad q_0 = \tfrac{1}{2}, \\ \phi_S(x) &= \delta \sum_i f\left(\sqrt{J_1/T}\, r_{i,S_i}(x)\right), \qquad 0 \le \phi_S \le B_0. \end{aligned} \tag{16.12}$$

The complementary probability defines $E_i = 0$, and the observed record is $E_i = 1$ for every i. The standardized square satisfies $J_1 r_{ij}^2/(2T) = (J_1/T) D_{\mathrm{KL}}(\mathcal{N}(u_i, 1) \| \mathcal{N}(u_j + \zeta_{ij}, 1))$. The bounded function f specifies an additional normalized observation of that mismatch; it is not identified with the mismatch itself. The coefficient δ is fixed at every prescribed depth and is not taken to zero in the theorem. [DEFINITION]

Multiplying the normalized prior, the two observation kernels, the source priors, the Bernoulli kernels, and the model/evaluator kernels defines one complete generative joint:

$$\begin{aligned} \mathbb{P}_\vartheta = \mathbb{P}_\vartheta^0(dx) L_O(dO \mid x) L_{\mathrm{cal}}(dY_{\mathrm{cal}} \mid \vartheta) \\ \times \prod_i \left[\pi_i(dS_i \mid \vartheta) L_i(dE_i \mid x, S_i) \mathcal{N}(dm_i; v_i, c) \mathcal{N}(dk_i; m_i, 1 - c)\right]. \end{aligned} \tag{16.13}$$

Iterated integration of its conditional factors gives total mass one. None of its kernels takes a recognition law as an input. The selected observation has a strictly positive density. This establishes the normalized experiment used throughout the chapter. [ESTABLISHED]

16.3 Physical posterior, restricted optimizer, and actual recognition

Write $\mathsf{o} = (z_\star, 2, \mathbf{1})$. The chapter-local actual outer law has a smooth positive density $q(x, S)$ relative to Lebesgue measure times counting measure. Its conditional evaluator coupling is $\prod_i \nu_i^*$ from eq. (16.7). The outer density is otherwise unrestricted: assignments and configurations may be correlated. It is not required to use the source posterior as its conditional assignment law. This is the recognition family on which the restricted VFE is defined. [DEFINITION]

Set

$$\begin{aligned} \widehat{K}_\vartheta &= \vartheta H + (1+\kappa)P_{\mathcal{K}}, & G_\vartheta(x) &= \tfrac{1}{2}x^\top \widehat{K}_\vartheta x - \kappa z_\star^\top x, \\ G_\vartheta^{\mathrm{phys}}(x) &= G_\vartheta(x) - \tfrac{T}{2}x^\top R x, & U_{\vartheta,S}(x) &= G_\vartheta(x) + T\phi_S(x), \\ Z_{\mathrm{opt}}(\vartheta) &= \sum_S \pi_S(\vartheta)\int e^{-G_\vartheta(x)/T-\phi_S(x)}dx, & Z_{\mathrm{phys}}(\vartheta) &= \sum_S \pi_S(\vartheta)\int e^{-G_\vartheta^{\mathrm{phys}}(x)/T-\phi_S(x)}dx. \end{aligned} \tag{16.14}$$

Gaussian domination and $0 \le \phi_S \le B_0$ make these integrals finite and positive. The local symbol π_ϑ denotes the normalized restricted optimizer $\pi_S(\vartheta)e^{-U_{\vartheta,S}(x)/T}/Z_{\mathrm{opt}}(\vartheta)$. The physical posterior $\mathbf{\Pi}_{\vartheta,\mathsf{o}}$ on (x, S) instead uses $G_\vartheta^{\mathrm{phys}}$ and Z_{phys}. Both have full support, and their conditional assignment row at fixed x is

$$b_S(x,\vartheta) = \frac{\pi_S(\vartheta)e^{-\phi_S(x)}}{\sum_{S'}\pi_{S'}(\vartheta)e^{-\phi_{S'}(x)}}. \tag{16.15}$$

It is the product of the receiver posterior rows. Equality of these conditional source rows does not identify the two full laws or their hidden-coordinate conditionals. [ESTABLISHED]

For explicit scalar accounting, define

$$\begin{aligned} c_O &= (\kappa/(2\pi T))^{n/2}, & c_{\mathrm{cal}} &= (2\pi T)^{-1/2}, \\ B_T(\vartheta) &= T\log Z_0(\vartheta) - T\log(c_O c_{\mathrm{cal}} q_0^n) + \tfrac{1}{2}\kappa|z_\star|^2 + \tfrac{1}{2}(\vartheta-2)^2, & & \\ C_T(\vartheta) &= B_T(\vartheta) - T\log Z_{\mathrm{opt}}(\vartheta), & p_\vartheta(\mathsf{o}) &= e^{-B_T(\vartheta)/T}Z_{\mathrm{phys}}(\vartheta). \end{aligned} \tag{16.16}$$

Expansion of the complete joint and eq. (16.8) gives

$$\begin{aligned} F_T(q,\vartheta) &= T\sum_S \int q(x,S)\log q(x,S)dx \\ &\quad + \mathbb{E}_q[G_\vartheta(x) + T\phi_S(x) - T\log\pi_S(\vartheta)] + B_T(\vartheta) \\ &= TD_{\mathrm{KL}}(q\|\pi_\vartheta) + C_T(\vartheta). \end{aligned} \tag{16.17}$$

The scalar is the scaled negative ELBO of the declared evaluator restriction. Its difference from the unconstrained physical optimum is $C_T + T\log p_\vartheta(\mathsf{o}) = T\log(Z_{\mathrm{phys}}/Z_{\mathrm{opt}}) \ge 0$. The ratio, including its parameter derivative, is retained. These are exact identities, not Gaussian approximations to the nonlinear law. [ESTABLISHED]

For clarity, the actual conditional assignment density is $\rho_q(S \mid x) = q(x,S)/\sum_{S'} q(x,S')$. Its receiver row is $\beta_{ij}^Q(x) = \sum_{S:S_i=j}\rho_q(S \mid x)$; the model posterior row $\beta_{ij}^P(x,\vartheta)$ is obtained from eq. (16.15).

Receiver occupancy may be fixed at $1/n$ by a readout that selects a receiver and reports its current source. The resulting event law is the pushforward of the complete marked input law. It is not a product of independently averaged occupancies, source rows, and memberships. These are the event and recognition conventions of this family. [DEFINITION]

16.4 Exact marked parents and their inherited inference

The parent carrier at level p is $W_p \times \mathcal{J}$. For $p < s$, define the normalized deterministic Markov kernel

$$C_{s\leftarrow p}((y,S),\cdot) = \delta_{(P_s y,S)}, \qquad C_s = C_{s\leftarrow 0}. \tag{16.18}$$

The matrices P_s and the probability kernels $C_{s\leftarrow p}$ are different typed objects. Their composition follows from $P_s P_p = P_s$; no assignment is erased. The same channel, extended by the identity on observations, pushes the generative joint, physical posterior, restricted optimizer, and actual recognition. This is a declared pointwise coarse construction. [DEFINITION]

Let $\pi_\vartheta^p = (C_p)_\# \pi_\vartheta$, and similarly denote the physical and actual pushes by $\mathbf{\Pi}_{\vartheta,\mathrm{o}}^p$ and $(C_p)_\# q$. Full evaluator and boundary-mark kernels are obtained by disintegrating the same physical joint. A raw mark which is not a retained coordinate is retained through its complete joint conditional kernel, including its correlations with the other marks. Its value is not replaced by evaluation at $P_p x$. The physical reverse conditional, the optimizer reverse conditional, and the actual recognition conditional have their own roles. Finite Gaussian domination supplies positive smooth versions of the integrals in this realization. These exact probability constructions retain all original loop data and all source assignments. [ESTABLISHED]

For an original edge e, its inherited mismatch observable under either the physical posterior or the restricted optimizer is

$$\mathcal{D}_e^{p,\mu}(y,S) = \mathbb{E}_{\mu_\vartheta}\left[\tfrac{1}{2} r_e(X)^2 \mid P_p X = y, S\right], \qquad \mu_\vartheta \in \{\mathbf{\Pi}_{\vartheta,\mathrm{o}}, \pi_\vartheta\}. \tag{16.19}$$

This defines two conditional observables, not one common reverse kernel. Coherence of a retained description refers to their root-mean-square original residuals $\sqrt{2\mathcal{D}_e^{p,\mu}}$; the local d_i observables are defined by the same conditional construction. At $p = 0$ these reduce to the original configuration residuals. This definition does not identify a single residual cost with the KL of two full parent mixtures. [DEFINITION]

The root factor of π_ϑ^p is independent of the assignment and of ϑ. On M_p, let H_p be the Schur complement of $H|_{\mathcal{K}^\perp}$, and let $\gamma_{p,\vartheta}(dz \mid y)$ be the cumulative hidden Gaussian conditional of the quadratic optimizer. Its mean is independent of ϑ; its covariance is T/ϑ times a fixed inverse

precision. Integrating at fixed S gives the exact sheet potential

$$\begin{aligned} V_{p,S}(y,\vartheta) &= \tfrac{1}{2}\vartheta y^\top H_p y + C_p^{\mathrm{G}}(\vartheta) + T\psi_{p,S}(y,\vartheta), \\ \psi_{p,S}(y,\vartheta) &= -\log \mathbb{E}_{\gamma_{p,\vartheta}(\cdot|y)} e^{-\phi_S}, \qquad 0 \le \psi_{p,S} \le B_0, \\ \frac{d}{d\vartheta} C_p^{\mathrm{G}}(\vartheta) &= \frac{T d_p}{2\vartheta}. \end{aligned} \tag{16.20}$$

Here $C_0^{\mathrm{G}} = 0$, $\psi_{0,S} = \phi_S$; the unchanged root potential is appended on W_p. The Gaussian integration scalar C_p^{G} is distinct from the common objective offset C_T. The potential is an exact conditional integral, with every generated interaction retained. [ESTABLISHED]

For a density q^p on the marked parent carrier define

$$F_T^p(q^p,\vartheta) = T D_{\mathrm{KL}}(q^p \| \pi_\vartheta^p) + C_T(\vartheta). \tag{16.21}$$

The conditional KL chain rule gives, at one common parameter and one common time,

$$\begin{aligned} \Delta_{p:s}(q^p,\vartheta) &= \mathbb{E}_{(C_{s\leftarrow p})_\# q^p} D_{\mathrm{KL}}\left(q^p(dz \mid y,S) \middle\| \pi_\vartheta^p(dz \mid y,S)\right), \\ F_T^p(q^p,\vartheta) - F_T^s((C_{s\leftarrow p})_\# q^p,\vartheta) &= T\Delta_{p:s}(q^p,\vartheta). \end{aligned} \tag{16.22}$$

In the displayed conditional, z consists only of the discarded continuous normal coordinates; loops and S remain in the retained state. Nested conditional losses telescope at that same law and time. Costs evaluated at different activation times are not summed as one telescoping objective. [ESTABLISHED]

Declare unit continuous mobility on every microscopic coordinate and a rate-one Poisson proposal clock. At a proposal, redraw the whole assignment from $b(x,\vartheta)$ without changing any continuous coordinate. A proposal may return the current assignment. The microscopic generator is

$$\mathcal{L}_\vartheta^0 f(x,S) = T\Delta_x f - \nabla_x U_{\vartheta,S} \cdot \nabla_x f + \sum_{S'} b_{S'}(x,\vartheta)[f(x,S') - f(x,S)]. \tag{16.23}$$

The diffusion covariance is $2TIdt$. This is a continuous-time inference prescription on global Euclidean space times a finite set, with integrable decay at infinity and no reflecting or chart boundary. No discretization is being defined. [DEFINITION]

The inherited parent proposal row and generator are

$$\begin{aligned} R_{SS'}^p(y,\vartheta) &= \mathbb{E}_{\pi_\vartheta}[b_{S'}(X,\vartheta) \mid P_p X = y, S], \\ \mathcal{L}_\vartheta^p f(y,S) &= T\Delta_y f - \nabla_y V_{p,S} \cdot \nabla_y f + \sum_{S'} R_{SS'}^p(y,\vartheta)[f(y,S') - f(y,S)]. \end{aligned} \tag{16.24}$$

The unchanged root drift is included when y denotes all of W_p. Self-proposals are retained, so every row sums to one and every actual exit rate is at most one. The conservative jump diagonal is $R_{SS}^p - 1$. These conditional expectations specify the chosen autonomous parent kinetics; they do not assert exact temporal projection of the fine process. [DEFINITION]

Multiplying the microscopic row by $\pi_\vartheta(x,S)$ gives a quantity symmetric in S,S'. Conditional integration therefore proves detailed balance of R^p with π^p_ϑ. Furthermore

$$\begin{aligned} R^p_{SS'}(y,\vartheta) &\geq e^{-B_0}\pi_{S'}(\vartheta) > 0, \\ \mathbb{E}_{\pi^p_\vartheta}[R^p_{SS'}(Y,\vartheta) \mid P_sY = z, S] &= R^s_{SS'}(z,\vartheta). \end{aligned} \tag{16.25}$$

The second identity is the conditional-expectation tower. Thus both normalized laws and optimizer-conditioned rates compose exactly at a fixed parameter, while comparisons between their independently evolved histories still require estimates. [ESTABLISHED]

16.5 Complete model scores and dissipation

Differentiate the complete normalized experiment before conditioning on the observed record. The prior, calibration, and categorical scores have mean zero and vanishing cross terms: the categorical draws and calibration are independent of the configuration at fixed ϑ, and the remaining conditional kernels have no explicit ϑ-dependence. The complete generative Fisher metric is therefore

$$M(\vartheta) = \frac{1}{T} + \frac{1}{2}\operatorname{Tr}[(A_0(\vartheta)^{-1}H)^2] + 16\pi_{i_0,+}(\vartheta)[1-\pi_{i_0,+}(\vartheta)]. \tag{16.26}$$

The quadratic-prior term follows by the variance of its centered Gaussian quadratic score. This establishes the metric of the original full experiment; it is not the Fisher metric of a fitted coarse observation model. [ESTABLISHED]

Declare $m(\vartheta) = TM(\vartheta)$ and the model metric $m(\vartheta)/\epsilon$ for the scaled objective, with $\epsilon > 0$. At every level the actual coupled system is

$$\partial_t q^p_t = (\mathcal{L}^p_{\vartheta_p(t)})^* q^p_t, \qquad \dot\vartheta_p(t) = -\epsilon m(\vartheta_p(t))^{-1} g_p(q^p_t, \vartheta_p(t)), \qquad g_p = \partial_\vartheta F^p_T\big|_{q^p}. \tag{16.27}$$

The density is the complete actual law of that level. Its parameter path is deterministic because this feedback takes an exact-law expectation, rather than a sampled score. Parents inherit this metric; neither a new rate multiplier nor a marginal-model Fisher metric is inserted at activation. [DEFINITION]

Expanding eq. (16.21) by eq. (16.20) gives the full covector

$$\begin{aligned} g_p(q^p,\vartheta) = \vartheta - 2 + \tfrac{1}{2}\mathbb{E}_{q^p} y^\top H_p y + \frac{Td_p}{2\vartheta} + T\mathbb{E}_{q^p}\partial_\vartheta \psi_{p,S}(y,\vartheta) \\ - \frac{T}{2}\operatorname{Tr}(A_0(\vartheta)^{-1}H) - T\mathbb{E}_{q^p}\partial_\vartheta \log \pi_S(\vartheta). \end{aligned} \tag{16.28}$$

The root has no parameter score. The derivative is partial at fixed actual law, so time derivatives of actual assignment masses are not extra terms. Prior normalization, hidden determinant, source prior, and nonlinear conditional normalization are all included. At $p=0$, $d_0=0$ and $\partial_\vartheta\phi_S = 0$; this gives the microscopic score directly. [ESTABLISHED]

For positive smooth densities with finite entropy and moments, write $u_S = q_S^p/\pi_\vartheta^p(\cdot,S)$ and $c_{SS'}^p(y) = \pi_\vartheta^p(y,S)R_{SS'}^p(y,\vartheta)$. Integration by parts and the symmetry of this conductance yield

$$\begin{aligned}\frac{d}{dt}F_T^p = &-T^2\sum_S\int q_S^p\left|\nabla\log\frac{q_S^p}{\pi_{\vartheta_p}^p(\cdot,S)}\right|^2 dy\\ &-\frac{T}{2}\int\sum_{S,S'}c_{SS'}^p(u_S-u_{S'})(\log u_S-\log u_{S'})dy\\ &-\epsilon m(\vartheta_p)^{-1}g_p(q^p,\vartheta_p)^2\le 0.\end{aligned}\tag{16.29}$$

To obtain the middle line, interchange S, S' in one copy of the jump contribution and average. The resulting integrand is nonnegative. At the microscopic level the same term is $-T\int q_X[D_{\mathrm{KL}}(\rho_q\|b)+D_{\mathrm{KL}}(b\|\rho_q)]dx$. The confining moment bounds and cutoff passage in Lemmas A.2 and A.6 justify these identities for the admitted actual solutions. Thus the declared coupled inference decreases its own complete VFE. This does not assert that a discarded-conditional KL decreases separately. [ESTABLISHED]

16.6 Preparation and successive actual-law activation

The preparation uses a finite nonzero residual margin. Include each internal edge covector on $\mathcal{N}$, and for each bridge e include $r_eP_s|_{\mathcal{N}}$ for $s<\ell(e)$, in a list of m_0 linear functionals. Each is nonzero by residual-row independence. In a fixed orthonormal basis of $\mathcal{N}$, evaluate these functionals on $(1,t,\dots,t^{\dim\mathcal{N}-1})$. There are at most $m_0(\dim\mathcal{N}-1)$ forbidden integers, so at least one of the integers from zero through that bound is a simultaneous nonroot. Normalize the resulting vector to obtain $v_\star\in\mathcal{N}$ and a margin

$$m_L=\min_{e,\ s<\ell(e)}|r_e(P_sv_\star)|>0.\tag{16.30}$$

Internal edges enter this minimum only at $s=0$. A fixed sufficiently small neighborhood of $v_\star$ preserves a margin at least $m_L/2$. This is an explicit construction of preparation data, not a time-dependent merger instruction. [ESTABLISHED]

For $0<\epsilon<1$, fix the parameter family

$$Q_L=0,\qquad Q_s=2Q_{s+1}+20,\qquad J_s=\epsilon^{-Q_s},\qquad A^2=\epsilon^{3Q_1+20},\qquad T=A^2\epsilon^6,\qquad H_*=\frac{2}{\epsilon}.\tag{16.31}$$

The Q_s are scalar exponents; the orthonormal flag matrices are Q_s. The weakest energy is exactly $J_L=1$. There is no common acceleration of every energy, no vanishing source rate, and no decrease of the standardized nonlinearity δ. [DEFINITION]

Start $\vartheta(0)=1$. The root is independent of the non-root coordinates and assignments, with law $\mathcal{N}_{\mathcal{K}}(\kappa z_\star/(1+\kappa),T(1+\kappa)^{-1}I_{\mathcal{K}})$. On every assignment sheet the reference non-root law has common mean Av_0, zero loop mean, and covariance $T(H|_{\mathcal{K}^\perp})^{-1}$, where v_0 lies in the strict-margin neighborhood just constructed. Choose positive assignment masses near the product prior at

$\vartheta = 1$, so $|\lambda_{i_0,+}(0) - \mathrm{sigmoid}(-2)| < 10^{-2}$. Allow smooth within-sheet transports of this reference with coupling

$$|\widehat{K}_1^{1/2}(X_0 - X_0^{\mathrm{G}})| < \epsilon\sqrt{T} \quad \text{almost surely}, \tag{16.32}$$

and bounded log-density ratio, retaining the independent root factor. Small smooth compactly supported diffeomorphisms in standardized non-root coordinates supply such transports. The resulting neighborhood of positive smooth correlated initial laws is understood in this weighted transport and bounded-log-density topology. Its size may depend on ϵ and finite L. These are preparation hypotheses; the later actual law is not constrained to remain Gaussian or to factor across assignment sheets. [HYPOTHESIS]

Choose one finite entrance constant k_L as in the theorem below, and set

$$t_s = \frac{k_L \log(1/\epsilon)}{J_s}, \qquad I_s = [t_s, 2t_s], \qquad \tau_0 = 0, \quad \tau_s = 2t_s. \tag{16.33}$$

At τ_s, push the complete actual level-$(s-1)$ law through $C_{s\leftarrow s-1}$, copy its actual parameter, and thereafter evolve the new law by eq. (16.27). An independently continued source process retains its own law and parameter after that time. These deterministic times choose when to use a reduced description; they do not change the microscopic graph, energy hierarchy, or inference rule. [DEFINITION]

16.7 The finite-depth formation theorem

Theorem 16.1 (Adaptive formation with weakest energy one). For every prescribed finite $L \geq 2$, fix the graph, evaluator, source support, priors, observations, and microscopic kinetics above. There exist positive finite constants, depending on these fixed data and L, and $\epsilon_L > 0$, such that for every $0 < \epsilon < \epsilon_L$ the regime eq. (16.31) and the declared initial-law neighborhood admit all L successive actual-law activations. Choose

$$\begin{aligned} k_L > k_{\mathrm{geom},L}, \quad \tfrac{1}{2}c_{\mathrm{early},L}k_L > Q_1 + 20, \\ c_{\mathrm{joint},L}k_L > Q_1 + 8, \qquad c_{\mathrm{fast},L}k_L > Q_1/2 + 12, \end{aligned} \tag{16.34}$$

where the positive restoring, joint-entropy, and fast-mixing constants are those constructed in Lemmas A.6, A.10 and A.11. The finite geometric entrance constant is constructed from eqs. (16.6) and (16.30) in the coherence lemma. The following conclusions are established for the resulting actual processes. [ESTABLISHED]

The coupled diffusion, finite-state proposals, and exact-law parameter equation have a unique nonexplosive solution through H_*. Every activated parent retains positive smooth densities and satisfies

$$\begin{aligned} 1 \leq \vartheta_p(t) \leq 21/10, \quad |\dot{\vartheta}_p(t)| \leq C_L\epsilon, \\ |\vartheta_p(t) - (2 - e^{-\epsilon t})| \leq C_L r_\epsilon, \qquad r_\epsilon = A^2 J_1 + T[1 + \log(1/\epsilon)]^2. \end{aligned} \tag{16.35}$$

Here the time in the comparison is absolute time, including after activation. The reference $2 - e^{-\epsilon t}$ is an estimate deduced from the complete own-law equation, not a prescribed update. [ESTABLISHED]

Choose a fixed residual tolerance $\eta_L A$ smaller than the strict prepared separation margin. Except on an event of probability at most

$$C_L(1 + H_*) \exp(-c_L/\epsilon^6), \tag{16.36}$$

the microscopic process and its sequential parent descriptions have the successive coherent components

$$3 \cdot 2^{L-1} \longrightarrow 2^{L-1} \longrightarrow 2^{L-2} \longrightarrow \cdots \longrightarrow 1. \tag{16.37}$$

On I_s, edges already assigned to levels at most s, and the within-node model discrepancies, satisfy the coherence tolerance; future bridges remain separated by a fixed positive multiple of A. The retained description uses eq. (16.19) and its local model-discrepancy counterpart. The final coherent state persists through the finite horizon on this event. All loop coordinates remain random and retained; no identity-holonomy constraint is imposed. [ESTABLISHED]

On each active interval I_s, let $\Delta_{s-1:s}$ be the actual discarded conditional loss of eq. (16.22). At the same current law and parameter,

$$\begin{aligned} \Delta_{s-1:s} &\le \begin{cases} C_L \epsilon^{14}, & s < L, \\ C_L \epsilon^2, & s = L, \end{cases} \\ |g_{s-1}(q, \vartheta) - g_s((C_{s \leftarrow s-1})_{\#} q, \vartheta)| &\le \begin{cases} C_L T \epsilon^7, & s < L, \\ C_L T \epsilon, & s = L. \end{cases} \end{aligned} \tag{16.38}$$

Multiplication of the first bound by T gives the discarded restricted VFE. These are conditional-entropy and complete-score estimates, independently derived from the actual evolving laws. [ESTABLISHED]

Let $\lambda^p_{i_0,+}(t)$ be the actual distinguished receiver source mass in a fine, parent, or sequential process. For each actual activation time from which that process is continued,

$$\lambda^p_{i_0,+}(H_*) - \lambda^p_{i_0,+}(\tau_s) > \tfrac{1}{2}. \tag{16.39}$$

Thus the actual binary recognition-source row changes by total variation greater than one half. Moreover, between $\vartheta = 1$ and the final learned parameter, both the first hidden physical conditional and the first hidden optimizing conditional change by KL greater than $1/200$, at every fixed retained input and assignment. These compare their own normalized kernels, not one kernel substituted for another. [ESTABLISHED]

For any source process and its inherited parent begun from a common actual marked activation sample at time τ, there exists a coupling preserving that sample for which

$$\begin{aligned} &\Pr\{S_t \neq \overline{S}_t \text{ for some } t \in [\tau, H_*]\} \leq C_L \epsilon^6, \\ &\Pr\left\{ \sup_{\tau \leq t \leq H_*} |P_s X_t - \overline{X}_t| > C_L A \epsilon^2 \right\} \leq C_L (1 + H_*) e^{-c_L/\epsilon^2}. \end{aligned} \tag{16.40}$$

The first comparison is of the complete assignment path, with every receiver retained. The second is a uniform continuous-coordinate displacement bound under a compatible coupling. A finite number of successive replacements changes only the constants. No total variation convergence of continuous path laws or exact Markov lumpability is asserted. [ESTABLISHED]

Proof. The exact physical normalization, restricted scalar, and marked composition have been calculated in eqs. (16.13), (16.16), (16.22) and (16.25). Lemma A.1 proves the uniform differentiated parent bounds, including their hidden determinant and source terms. Lemma A.2 constructs the coupled solution, uses stopped moments to keep the parameter in its positive interval, and propagates the admitted moments through actual activations. Lemma A.11 proves the strict formation margins from the weighted flag. Lemma A.3 gives eq. (16.35) for separately learning parents. These arguments use neither conditional equilibration nor history averaging.

For early interfaces, Lemma A.6 keeps the finite positive jump contribution in the conditional balance. The contrast identity $J_{s+1}^2 / J_s = \epsilon^{20}$ and the entrance condition in eq. (16.34) give the first early bound in eq. (16.38). At the terminal interface the same lemma instead uses joint diffusion-plus-jump entropy contraction after logarithmic time. Conditional KL is bounded by that full entropy through the chain rule. The complete normalized hidden score then bounds the score difference by $C_L T(\sqrt{\Delta} + \Delta)$, giving the second line of eq. (16.38).

Lemma A.12 bounds the actual source mass at every activation, including the final logarithmic activation, and integrates its positive-rate equation on the last ten units of the horizon to prove eq. (16.39). The same parameter continuation supplies the hidden-kernel comparison proved there. Lemma A.8 integrates later triangular residuals exactly, leaving a common source-relevant fast optimizer marginal at every level. Lemma A.10 then compares conditional proposal-prefix kernels under the actual evolving environment. Its error is $C_L \epsilon^6$, with the common actual activation information retained. Finally Lemma A.13 supplies the separate continuous bound on a coupling compatible with those histories. Each lemma uses the same preparation, parameter regime, and finite choice eq. (16.34); the final window is of order $\log(1/\epsilon)$ and lies before $H_* = 2/\epsilon$ for small ϵ. This proves the conjunction. □

16.8 A two-triangle instance and the remaining scope

For $L = 2$, take triangles $(1, 2, 3)$ and $(4, 5, 6)$, with a single bridge $(3, 4)$. The retained carriers are

$$\mathbb{R}^{19} \times \mathcal{J} \longrightarrow \mathbb{R}^9 \times \mathcal{J} \longrightarrow \mathbb{R}^8 \times \mathcal{J}, \qquad |\mathcal{J}| = 64, \tag{16.41}$$

and the coherent agent counts are $6 \to 2 \to 1$. Both loops survive both maps. The exact regime becomes

$$J_1 = \epsilon^{-20}, \qquad J_2 = 1, \qquad A = \epsilon^{40}, \qquad T = \epsilon^{86}. \tag{16.42}$$

The first window has order $\epsilon^{20}\log(1/\epsilon)$, and the second has order $\log(1/\epsilon)$. The source and model changes occur over the longer horizon $2/\epsilon$. These are exact substitutions into the finite-L theorem, not numerical evidence for it. [ESTABLISHED]

The source-dependent term is nonquadratic in the declared configuration coordinates. For a direction with selected internal residual slope $\lambda \neq 0$, $f(\lambda t) = \lambda^2 t^2/2 - \lambda^4 t^4/8 + O(t^6)$. Every assignment selects an internal edge with nonzero first-normal projection, so its hidden log density has a nonzero fourth derivative in some first-normal direction. Exact marginalization retains its full conditional integral and the negative covariance term in its Hessian. Thus neither the physical nor optimizing first hidden conditional is silently replaced by a Gaussian conditional. This is an established coordinate-specific nonlinearity statement. [ESTABLISHED]

The theorem keeps the supplied energy hierarchy and its internal source support. It does not prove that source learning creates those contrasts. Allowing sources on different bridge levels would require a new fast-marginal and evolving-environment estimate. Uniform control as $L \to \infty$, a nondegenerate intrinsic partition selector, permanent membership under arbitrary later forcing, and autonomous agency remain separate open construction problems. The finite constants and admissible preparation neighborhoods need not be uniform in depth. [OPEN]

The parameter t in the generators is the declared inference time. This chapter declines to identify it with physical time or with Fisher length, and it does not claim computational savings or a depth-independent resource bound. Exact common-channel probability composition, approximate continuous trajectories, and complete assignment-history comparison retain the separate meanings used in Chapters 9 and 12. [NOT-CLAIMED]

Chapter 17

A constructive spatial renormalization theory

The general scale diagram admits a concrete infinite-volume realization in a positive lattice gauge–matter family. This chapter constructs its complete coarse laws, their generated interactions, a differentiable RG map on an open interaction ball, and a finite procedure with quantified errors. The family is a declared realization of the general probability theory. Its limiting law and its region of attraction do not identify a continuum theory or a physical law. [DEFINITION]

17.1 The lattice law and its geometric channels

Definition 17.1 (Dressed square-lattice source). Let $E = \mathbb{Z}^2 \times \{1, 2\}$, with $e = (x, a)$ joining x to $x + e_a$, and retain the four distinct edge copies of every unit plaquette. The matter sign is $\phi_x \in \{-1, 1\}$, the raw link is $u_e \in \{-1, 1\}$, and the dressed sign is $z_e = \phi_x u_e \phi_{x+e_a}$. In this chapter only, $b = \tanh h$ and $t = \tanh j$ denote the field and plaquette strengths; t is not time and neither parameter is the contextual point c. For a finite free edge set Λ and fixed exterior dressed signs η, define

$$W_\Lambda^\eta(z) = \prod_{e\in\Lambda}(1 + bz_e) \prod_{p:p\cap\Lambda\neq\varnothing} (1 + t\chi_p(z\eta)),$$
$$Z_\Lambda^\eta = 2^{-|\Lambda|} \sum_z W_\Lambda^\eta(z), \qquad \gamma_\Lambda^\eta(z) = \frac{2^{-|\Lambda|} W_\Lambda^\eta(z)}{Z_\Lambda^\eta}. \tag{17.1}$$

where $\chi_S(z) = \prod_{e\in S} z_e$. All touching plaquettes, including crossing plaquettes, occur. The reference is normalized counting. The finite algebra allows $0 \le b, t < 1$; subsequent theorems impose smaller domains explicitly. Periodic versions retain all sectors and require side at least three, with distinct edge and plaquette copies. These are definitions, not consequences of locality alone. [DEFINITION]

Theorem 17.2 (An actual infinite-volume source). For $0 < b < 1$ and $0 < t < 1/6$, the specification eq. (17.1) has a unique dressed Gibbs law $\mu_{b,t}$, without a translation-invariance premise. It is translation invariant. Every fixed finite cylinder is the boundary-independent limit of expanding boxes and locally embedded full tori. If $D(e, \Lambda^c)$ is co-plaquette graph distance, then, for

$A \subset \Lambda$,

$$\mathrm{TV}((\gamma^{\eta}_{\Lambda})_A, (\gamma^{\zeta}_{\Lambda})_A) \le \min\left\{1, \sum_{e\in A} (6t)^{D(e,\Lambda^c)}\right\}. \tag{17.2}$$

Independent uniform matter signs and $u_e = \phi_x z_e \phi_{x+e_a}$ give the unique Gibbs lift for the declared raw gauge–matter factors. The proof in section B.1 constructs a countably additive law and establishes its DLR equations separately from uniqueness. [ESTABLISHED]

Definition 17.3 (Straight-path blocking). For an integer $B \ge 1$, define the normalized deterministic channel C_B by retaining $\phi'_x = \phi_{Bx}$ and multiplying the raw links along $\mathcal{P}_B(x,a) = \{(Bx + se_a, a) : 0 \le s < B\}$. Its dressed output is

$$z'_{(x,a)} = \prod_{e\in\mathcal{P}_B(x,a)} z_e, \qquad \mu_B = (C_B)_*\mu_{b,t}. \tag{17.3}$$

Every output lattice is identified with $\mathbb{Z}^2$, and its cell alphabet is $\mathsf{E} = \{-1,1\}^2$. This declared identification makes fixed-law statements meaningful. On a torus of side L, require $B \mid L$ and $L/B \ge 3$. [DEFINITION]

Proposition 17.4 (Exact geometry and reference square). Distinct retained paths have disjoint edge supports, although their endpoints may coincide. The channel is gauge equivariant, pushes normalized counting to normalized counting, and satisfies $C_a C_B = C_{aB}$, including $C_2C_3 = C_3C_2 = C_6$. For any positive finite density f, its complete coarse density is $\mathbb{E}_\nu[f \mid C_B]$, where ν is the fine counting reference. A coarse plaquette holonomy equals the product of all fine plaquette holonomies in its square; a retained winding Wilson loop retains the corresponding fine winding loop. These statements retain the complete law and its normalization. [ESTABLISHED]

Proof. Along a path every internal matter sign occurs twice and cancels. Distinct axes distinguish edge copies; along each grid line consecutive intervals of length B partition its edges. Each prescribed parity has exactly half the assignments on its path, so disjoint parities and the retained uniform frames have the asserted reference. Concatenating intervals gives indices $Bs + r$, once each from zero to $aB - 1$, proving composition. Interior edges occur twice in the plaquette product and cancel. Conditional summation over counting fibers proves the density formula and its composition without any change of reference. □

Proposition 17.5 (Large blocks converge as full cylinder laws). Put

$$\rho = \tanh(h + 2j) = \frac{b(1+t^2) + 2t}{1 + t^2 + 2bt} < 1. \tag{17.4}$$

For any fixed $0 \le b, t < 1$, any Gibbs law of the displayed specification, and any M distinct retained B-paths, their parities satisfy

$$\mathrm{TV}(\mu_{B,M}, \nu_M) \le \min\left\{1, \frac{1}{2}\sum_{s=1}^{M}\binom{M}{s}\rho^{\lceil sB/7\rceil}\right\}. \tag{17.5}$$

The same bound is uniform over finite volumes and dressed boundaries when all counted path edges are free. In the source uniqueness domain, fixed blocking commutes with the thermodynamic cylinder limit. Both iterated limits and the joint box or torus limit $B_n \to \infty$, $L_n/B_n \to \infty$, under the stated embedding and divisibility conditions, yield the fair product law on each fixed coarse cylinder. The raw limit has independent uniform matter and links, and negative plaquette holonomy has probability $1/2$. [ESTABLISHED]

Proof. The single-edge conditional field has absolute value at most $h + 2j$. A greedy independent subset of an sB-edge support has at least $\lceil sB/7\rceil$ members, because co-plaquette degree is at most six. Conditional on its complement the selected signs are independent and each conditional mean has magnitude at most ρ. This bounds every nonempty Walsh moment. Fourier inversion and the triangle inequality give eq. (17.5). A fixed channel pulls a cylinder back to finitely many fine signs, so eq. (17.2) gives the fixed-block passage. The independent volume-uniform moment bound gives the increasing-block and stated joint passages directly; it does not amplify a fixed-support boundary estimate to growing support. Independent uniform frames preserve the total-variation bound, and a nonempty plaquette character is fair at the product limit. □

17.2 Every-exterior coarse laws and all generated interactions

Theorem 17.6 (A common high-temperature domain at every depth). Fix $0 < b < 1$, $0 < t < 1/12$, and choose $\theta > 0$ with $q = 12te^{\theta} < 1$. Put $C = 16q/(1-q)$ and $m = (1-\rho)^2/4$. For every integer $B \ge 1$, μ_B has a unique positive, translation-covariant, continuous specification $\gamma_A^{(B)}$ defined at every coarse exterior. It is the unique Gibbs law of that specification. For exterior patterns agreeing in the radius R neighborhood of finite cell set A,

$$\mathrm{TV}\big(\gamma_A^{(B)}(\cdot \mid y), \gamma_A^{(B)}(\cdot \mid y')\big) \le \min\{1, C|A|B^2(R+1)e^{-\theta(R+1)}\}. \tag{17.6}$$

Each one-cell atom lies between $(1-\rho^B)^2/4$ and $(1+\rho^B)^2/4$, hence is at least m. The same modulus compares a whole-block finite conditional calculation against its matching infinite kernel. These statements include infinite exterior prescriptions of probability zero, constructed by normalized finite limits. [ESTABLISHED]

The proof in section B.2 uses fine Hamming transport and column incidence. It gives $12t < 1$ independently of B; a block total-variation row estimate proportional to Bt would not give this domain. [ESTABLISHED]

Definition 17.7 (Canonical vacuum interaction). Fix the reference cell $* = (1,1)$ and a translation-invariant lexicographic order. At x, let $\eta^{x,r}$ use σ_y for past cells $y < x$ within distance r, and $*$ elsewhere. Define

$$e^{(B)}_{x,r}(\sigma) = -\log \frac{\gamma^{(B)}_x(\sigma_x \mid \eta^{x,r})}{\gamma^{(B)}_x(* \mid \eta^{x,r})}, \qquad A_{x,r} = \{x\} \cup \{y < x : |y - x|_\infty \le r\}. \tag{17.7}$$

Set $\Phi^{(B)}_{\{x\}} = e^{(B)}_{x,0}$, $\Phi^{(B)}_{A_{x,r}} = e^{(B)}_{x,r} - e^{(B)}_{x,r-1}$ for $r \ge 1$, and all other terms to zero. This convention fixes an interaction representative; it does not set the finite partition function to one. [DEFINITION]

Theorem 17.8 (Complete interaction and norm attraction). Under theorem 17.6, define

$$\begin{aligned} A_0 &= \frac{4}{1-\rho^2}, \qquad D_0 = \frac{2C}{m}, \qquad \lambda_v = \theta/8, \qquad z_v = e^{-\theta/4}, \\ \mathcal{E}_B &= A_0\rho^B + \sqrt{2A_0D_0}B\rho^{B/2}\frac{z_v(9+14z_v+z_v^2)}{(1-z_v)^4}. \end{aligned} \tag{17.8}$$

Then the entire canonical interaction generates exactly $\gamma^{(B)}$, and

$$\|\Phi^{(B)}\|_{\lambda_v} := \sum_{A \ni 0} e^{\lambda_v \operatorname{diam} A}\|\Phi^{(B)}_A\|_\infty \le \mathcal{E}_B \longrightarrow 0, \qquad \sum_{A \ni 0} \operatorname{diam}(A)\|\Phi^{(B)}_A\|_\infty \le \mathcal{E}_B/\lambda_v. \tag{17.9}$$

No generated shell is removed. Canonical reconstruction after C_a defines an exact integer-block action on this orbit, with $\mathcal{R}_a\Phi^{(B)} = \Phi^{(aB)}$, composition, and fixed point zero. The proof and explicit spatial tail are in section B.3. [ESTABLISHED]

The potential norm is a cell-interaction norm. The stronger character norm introduced next is a different topology; its basin theorem requires a separate argument. [DEFINITION]

17.3 An RG map on an open interaction ball

Definition 17.9 (Consolidated character coordinates). Let $J = (J_S)$ be a translation-invariant real coefficient family over nonempty finite edge supports, with exponential density convention $\exp(\sum_S J_S\chi_S)$. Its finite conditional energies retain all terms touching the free set. For prescribed $\kappa, \lambda > 0$, use the Banach norm

$$\|J\|_{\kappa,\lambda} = \sup_e \sum_{S \ni e} |S| e^{\kappa|S| + \lambda \operatorname{diam} S}|J_S|. \tag{17.10}$$

Diameters use fine-edge basepoints and Chebyshev distance. The empty character is a separate scalar. It is absent from these coordinates, while every finite law retains its own normalizer. [DEFINITION]

Theorem 17.10 (Same-norm differentiability, remainder, and composition). For $a \in \{2,3,6\}$, put

$$\begin{aligned} \delta_a &= \kappa(1-1/a), \qquad K_a = \frac{8ae^{\lambda(1-1/a)}}{\delta_a}, \\ \ell_a &= a^{-1}e^{-\kappa(a-1)}, \qquad M_a = 11K_a/a, \\ \varepsilon &= \min_a \left\{ \frac{1}{48}, \frac{1}{2K_a}, \frac{1-\ell_a}{2M_a} \right\} > 0. \end{aligned} \tag{17.11}$$

On the open ε-ball, the actual path push of the unique Gibbs law has one unique scalar-free representative $\mathcal{R}_a J$ in the same ball. Its specification is continuous at every exterior. The map is Fréchet C^1, and, writing $L_a = D\mathcal{R}_a(0)$,

$$\begin{aligned} (L_aJ)_T &= J_{\mathrm{lift}_a(T)}, \qquad \|L_a\| \le \ell_a < 1, \\ \|\mathcal{R}_aJ - L_aJ\| &\le M_a\|J\|^2, \qquad \|D\mathcal{R}_a(J) - L_a\| \le 2M_a\|J\|. \end{aligned} \tag{17.12}$$

It contracts radially by at most $(1+\ell_a)/2$, and $\mathcal{R}_2\mathcal{R}_3 = \mathcal{R}_3\mathcal{R}_2 = \mathcal{R}_6$. There is no area multiplier in the coefficient map. The complete coefficient construction and law passage are proved in section B.4. [ESTABLISHED]

Proposition 17.11 (The two interaction norms are not interchangeable). Arbitrarily small vacuum-vanishing interactions in the diameter norm can have infinite norm eq. (17.10). Thus theorem 17.8 alone does not imply entry into the ball of theorem 17.10. [ESTABLISHED]

Proof. On an $R \times R$ cell square put a translated term $-a_R(z_{x_R,1} - 1)\prod_{x \neq x_R} z_{x,1}$, anchored at its maximal cell, with $a_R = \eta e^{-2\lambda R}/R^4$, $R \ge 2$. Its diameter-norm sum is at most $2\eta\sum_{R\ge2} e^{-\lambda(R+1)}/R^2$. Its full-square character has coefficient a_R, distinct from every removed-corner character, so cannot cancel. Its contribution at a horizontal edge to eq. (17.10) is $\eta e^{\kappa R^2 - \lambda R - \lambda}$, which does not tend to zero. The ordinary anchored absolute coefficient sums still converge. The failure is precisely the stronger norm requirement. □

Theorem 17.12 (Complete finite-amplitude approximation). For a finite rational translation-orbit library in the strict ball, every complete Taylor polynomial $\widehat{J}_N$ of $\mathcal{R}_aJ$ is a finite rational library retaining all generated supports through order N. With $x = K_a\|J\| < 1$, its entire error satisfies

$$\|\mathcal{R}_aJ - \widehat{J}_N\| \le \frac{\|J\|}{a}\frac{x^N[(N+1) - Nx]}{(1-x)^2}. \tag{17.13}$$

Every prescribed positive rational tolerance is attainable with sufficiently large finite arithmetic and enumeration resources. For a finite word, choose one certified radius $r < \varepsilon$; then $L_a(r) = \ell_a + 2M_ar < 1$ is a Lipschitz bound. If every approximate input remains in that ball, errors propagate as $e_{k+1} \le L_{a_k}(r)e_k + \tau_k$. Incomplete enumeration gives no complete-polynomial conclusion. The proof in section B.5 counts all translations, repeated source occurrences, and omitted orders. [ESTABLISHED]

Theorem 17.13 (Constructive entry of the spatial source into any prescribed norm). Fix a compact rational box $0 < b_- \le b \le b_+ < 1$, $0 < t_- \le t \le t_+ < 1/12$, and prescribed positive rational κ, λ. There is a computable B_0 such that, for every integer $B \ge B_0$, the actual law μ_B has a unique translation-invariant scalar-free character representative $J^{(B)}$ in the open ball of theorem 17.10. An explicit bound decays exponentially in B, uniformly on that box. For rational source parameters, a finite rational library approximates $J^{(B)}$ to any prescribed positive error in the full original $\|\cdot\|_{\kappa,\lambda}$ norm, with sufficiently large finite resources. The construction and its constants are given in section B.6. [ESTABLISHED]

Once this entry holds, $\mathcal{R}_a J^{(B)} = J^{(aB)}$. For an oracle input error δ and a finite-word approximation error e, with both exact trajectories in the same certified ball, the total error is $e + \delta \prod_k L_{a_k}(r)$. This approximates the actual path-pushed law's interaction. It does not compute a discarded partition scalar or an ELBO from scalar-free coefficients. [ESTABLISHED]

17.4 Regular statistical geometry at fixed depths

Theorem 17.14 (Actual Fisher density and regular finite hierarchies). In natural parameters $g = (h, j)$, fix a compact rectangle in $h > 0$, $0 < j < \operatorname{atanh}(1/12)$, and an integer B. The complete coarse finite marginals and every constrained local row are actual C^3 functions. The complete canonical interaction is C^3 in one exponential diameter norm; its derivatives through order three tend to zero with increasing B, uniformly on each such compact rectangle. Let I_{B,Λ_L} be the Fisher matrix of the actual L-square marginal, and $I_{\mathrm{cell},B}(g \mid \mathrm{past})$ the predictive row Fisher matrix with its complete past fixed and its future integrated. Then

$$\mathcal{I}_B(g) = \mathbb{E}_{\mu_B} I_{\mathrm{cell},B}(g \mid \mathrm{past}) = \lim_{L\to\infty} L^{-2} I_{B,\Lambda_L}(g) \quad \text{in } C^2 \text{ on compact parameter sets.} \tag{17.14}$$

For each fixed depth, and for every prescribed finite list of depths, there is a nonempty open region with positive j on which these densities are positive-definite metrics. Their Levi–Civita connections are C^1, their curvatures are continuous, and the connections and curvatures of the area-normalized finite-marginal metrics converge to them on compact regular subcharts. The complete response, moving-expectation, and positive-floor arguments appear in section B.7. [ESTABLISHED]

The predictive row in eq. (17.14) integrates its future. The canonical interaction row in eq. (17.7) fixes its future to $*$. Replacing the former with the latter changes the Fisher-density construction. [DEFINITION]

Proposition 17.15 (Area-qualified information contraction). For integer blocking by a,

$$\mathcal{I}_{aB} \preceq a^2 \mathcal{I}_B, \qquad \frac{\mathcal{I}_{aB}}{(aB)^2} \preceq \frac{\mathcal{I}_B}{B^2}. \tag{17.15}$$

[ESTABLISHED]

Proof. For a fixed parameter-independent finite channel, the output score is the conditional mean of the input score. Its Fisher defect is the expected conditional covariance, hence positive semidefinite. A coarse L-square is the deterministic image of the fine $(aL+1)$-square marginal, which contains every required path. Apply the finite channel inequality and divide by L^2. The fixed-depth marginal limit in eq. (17.14), together with $(aL+1)^2/L^2 \to a^2$, gives the factor a^2. Dividing by the original fine-cell area gives the second inequality. □

Open Problem 17.16 (Limits beyond the admitted high-temperature construction). The construction does not supply an interacting critical fixed point, blocking-independent critical exponents, a continuum gauge-field measure, or a continuous-scale generator. Closing those obligations requires new parameter domains and corresponding tightness, interaction, and scaling estimates. The finite-depth Fisher floors may vanish as depth increases; an all-depth inverse or curvature limit requires estimates not supplied here. Gauge holonomy, parameter Fisher curvature, and base curvature remain distinct geometric objects. A curvature term in VFE requires its declared normalized generative model and its scalar normalizer. [OPEN]

Chapter 18

Normalized Geometry and the Variational Objective

Geometry can be an external input or a random variable of the generative model. This chapter declares a finite geometry catalog and derives the resulting VFE before taking any volume limit. A catalog label specifies a branch; it is neither an additional agent nor a replacement for the common principal bundle. The letter $\gamma \in \mathfrak{G}$ denotes a catalog label only in this chapter and the next, not a curve in the contextual base. [DEFINITION]

18.1 Two normalized geometry models

Definition 18.1 (Finite geometry ensemble). Let $\mathfrak{G}$ be finite and nonempty. For each γ, declare a finite nonempty carrier X_γ, a strictly positive probability reference ν_γ, a positive raw factor r_γ, and a positive weight a_γ. Let $\ell_\gamma(o \mid x)$ be a normalized, possibly vanishing, likelihood on a common finite observation alphabet. All are fixed generative data, independent of recognition. On the disjoint union of the branch carriers put

$$Z_\gamma = \sum_x \nu_\gamma(x) r_\gamma(x), \qquad p_\gamma(x) = \frac{\nu_\gamma(x) r_\gamma(x)}{Z_\gamma}, \qquad w_\gamma = \frac{a_\gamma}{\sum_\delta a_\delta}. \tag{18.1}$$

The hierarchical and global-factor joints are respectively

$$\mathbb{P}_{\mathrm{h}}(\gamma, x, o) = w_\gamma p_\gamma(x) \ell_\gamma(o \mid x), \tag{18.2}$$

$$\mathbb{P}_{\mathrm{f}}(\gamma, x, o) = \frac{a_\gamma \nu_\gamma(x) r_\gamma(x) \ell_\gamma(o \mid x)}{Z_{\mathrm{all}}}, \qquad Z_{\mathrm{all}} = \sum_\delta a_\delta Z_\delta. \tag{18.3}$$

These are two declared models. [DEFINITION]

Proposition 18.2 (Branch normalization changes geometry weights). Both joints are normalized. Their conditional branch laws coincide, but the factor-model geometry weight is $a_\gamma Z_\gamma / Z_{\mathrm{all}}$. For the same positive a_γ, the joints coincide exactly when all Z_γ coincide. Replacing r_γ by $d_\gamma r_\gamma$ leaves the hierarchical joint unchanged and reweights the factor ensemble by d_γ; one common $d_\gamma = d$ cancels in both. These statements are established by the following calculation. [ESTABLISHED]

Proof. Sum either joint first over o, then over x, then over γ. Equality of the joint implies equality of its label marginal, hence $Z_\gamma = Z_{\mathrm{all}}/\sum_\delta a_\delta$ for every γ; the converse follows by substitution. A branch rescaling changes Z_γ to $d_\gamma Z_\gamma$ and cancels in p_γ. Thus compensating the factor weight by a_γ/d_γ preserves the joint. In particular the choice a_γ/Z_γ converts the factor model to the hierarchy, but is a change of generative data. □

For a positive catalog reference m_γ write $a_\gamma = m_\gamma e^{-A_\gamma}$ and $r_\gamma = e^{-H_\gamma}$. Fix positive evidence $E_o = \sum_{\gamma,x} \mathbb{P}(\gamma, x, o)$ and a recognition law $\mathbb{Q}(\gamma, x) = q_\gamma Q_\gamma(x)$, defining Q_γ only when $q_\gamma > 0$. Direct expansion gives

$$\begin{aligned}\mathcal{F}_{\mathrm{h},o}(\mathbb{Q}) &= D_{\mathrm{KL}}(q\|m) + \mathbb{E}_q A_\gamma + \log \sum_\delta a_\delta \\ &\quad + \sum_\gamma q_\gamma \left[D_{\mathrm{KL}}(Q_\gamma\|\nu_\gamma) + \mathbb{E}_{Q_\gamma} H_\gamma + \log Z_\gamma - \mathbb{E}_{Q_\gamma} \log \ell_\gamma(o \mid X) \right], \qquad (18.4)\\ \mathcal{F}_{\mathrm{f},o}(\mathbb{Q}) &= D_{\mathrm{KL}}(q\|m) + \mathbb{E}_q A_\gamma + \log Z_{\mathrm{all}} \\ &\quad + \sum_\gamma q_\gamma \left[D_{\mathrm{KL}}(Q_\gamma\|\nu_\gamma) + \mathbb{E}_{Q_\gamma} H_\gamma - \mathbb{E}_{Q_\gamma} \log \ell_\gamma(o \mid X) \right]. \qquad (18.5)\end{aligned}$$

Here $\mathcal{F}_o = D_{\mathrm{KL}}(\mathbb{Q}\|\mathbf{\Pi}_o) - \log E_o$. The convention $0 \log 0 = 0$ is used; a recognition-positive atom with zero selected likelihood gives $+\infty$. These established identities contain one full recognition entropy, decomposed by its conditional chain. There is no second correlation-entropy term to append. [ESTABLISHED]

18.2 Transport of reference, scalar, and experiment

Theorem 18.3 (Complete finite transport). For each branch let $C_\gamma(x, y)$ be a fixed normalized channel retaining γ. Suppress that label within this proof and define, on reachable y,

$$\nu'(y) = \sum_x \nu(x) C(x, y), \qquad r'(y) = \frac{\sum_x \nu(x) r(x) C(x, y)}{\nu'(y)}, \qquad p'(y) = \frac{\nu'(y) r'(y)}{Z}. \qquad (18.6)$$

Then $\sum_y \nu'(y) r'(y) = Z$. The actual induced likelihood is

$$B_p(x \mid y) = \frac{p(x) C(x, y)}{p'(y)}, \qquad \ell'(o \mid y) = \sum_x B_p(x \mid y) \ell(o \mid x). \qquad (18.7)$$

It preserves the observation evidence and pushes the selected posterior by the same channel. Reference, prior, posterior, recognition, and the full observation joint compose along finite channel words. This establishes a transport of the complete experiment. [ESTABLISHED]

Proof. The reference row sums to one. Strict positivity of the source reference and raw factor implies $\nu'(y) > 0$ exactly on the prior's reachable support. Multiplication and summation in eq. (18.6) prove the partition identity. The prior reverse row B_p is normalized and $p'(y)\ell'(o \mid y) = \sum_x p(x) C(x, y) \ell(o \mid x)$. Summing over y proves evidence preservation; division by positive E_o

proves the posterior assertion. For a following channel D, interchange the finite sums over x, y to obtain the composite CD. Every assertion then follows for the complete joint and its coordinate marginals. The reference reverse row $\nu(x)C(x,y)/\nu'(y)$ is generally different from B_p and cannot replace it in the likelihood formula. On a null target, the raw quotient and reverse row are undefined; that column contributes zero, even if the next channel has a nonzero row there. □

Proposition 18.4 (Anchor scalar cocycle). Choose a reachable reference anchor and write $r = s\widehat{r}$, with $\widehat{r}(x_*) = 1$. Denote the raw push relative to νC by $C_\nu\widehat{r}$, and choose a reachable target anchor y_*. Then

$$\zeta_C = (C_\nu\widehat{r})(y_*), \qquad \widehat{r}' = \frac{C_\nu\widehat{r}}{\zeta_C}, \qquad s' = s\zeta_C, \tag{18.8}$$

and, using the same final reference and anchor,

$$\zeta_{CD}(\nu,\widehat{r}) = \zeta_C(\nu,\widehat{r})\zeta_D(\nu C,\widehat{r}'). \tag{18.9}$$

The factor-model weight must absorb s' if only $\widehat{r}'$ is used as the branch factor. The scalar cancels within a hierarchical conditional branch. These are established representative identities, not changes of principal gauge frame. [ESTABLISHED]

Proof. The numerator of the two-stage raw push is $\sum_{x,y}\nu(x)\widehat{r}(x)C(x,y)D(y,z)$ and its denominator is $(\nu CD)(z)$. Linearity allows the first anchor scalar to be pulled outside the second push. Evaluation at the final anchor gives the cocycle, and the product $s'\widehat{r}'$ recovers the complete raw factor. Substitution into eqs. (18.2) and (18.3) gives the two weight rules. Identity has scalar one only for an unchanged anchored representative. □

Proposition 18.5 (VFE loss on posterior fibers). If $\mathbb{Q} \ll \mathbf{\Pi}_o$, let $\mathbb{Q}' = \mathbb{Q}C$ and $\mathbf{\Pi}'_o = \mathbf{\Pi}_o C$. Then

$$\mathcal{F}_o(\mathbb{Q}) = \mathcal{F}'_o(\mathbb{Q}') + \sum_{\gamma,y}\mathbb{Q}'(\gamma,y)D_{\mathrm{KL}}\big(\mathbb{Q}(X \mid \gamma,y)\|\mathbf{\Pi}_o(X \mid \gamma,y)\big). \tag{18.10}$$

The loss is nonnegative and vanishes exactly when the two reverse fibers agree on every recognition-positive target. Finite-word losses add. [ESTABLISHED]

Proof. Augment both laws by the same channel. Their joint likelihood ratio is

$$\frac{\mathbb{Q}(\gamma,x)}{\mathbf{\Pi}_o(\gamma,x)}. \tag{18.11}$$

Factor this ratio into the target ratio and conditional-fiber ratio, take logs, and sum. All terms are finite on the stated finite dominated support. Evidence preservation yields eq. (18.10); applying it successively proves the telescope. □

If fine recognition is not dominated but its coarse image is, the fine VFE and a posterior-fiber loss are infinite while the coarse VFE is finite. If coarse recognition itself charges a posterior-null target, its posterior fiber is undefined and the difference of infinite VFEs is not a numerical loss. An arbitrary

null-fiber version does not identify that loss. Zero total evidence is a separate undefined-posterior case. These support boundaries follow by inspecting a positive channel incidence from each violating fine atom. [ESTABLISHED]

18.3 Scores and the three geometric structures

Proposition 18.6 (Normalized derivatives and information loss). Let the finite inputs be actual C^3 families on a fixed support over an open parameter set. For unnormalized weights u_i, $Z = \sum_i u_i > 0$, and $p_i = u_i/Z$, all ordinary derivatives through order three satisfy

$$\partial^\alpha p_i = \frac{1}{Z}\left[\partial^\alpha u_i - \sum_{0<\beta\le\alpha} \binom{\alpha}{\beta} (\partial^\beta Z)(\partial^{\alpha-\beta} p_i)\right]. \tag{18.12}$$

For a parameter-independent channel the score $s = \partial \log p$ becomes $s'(Y) = \mathbb{E}[s(X) \mid Y]$, and

$$I(p) - I(pC) = \mathbb{E}_{pC} \operatorname{Cov}_p(s \mid Y) \succeq 0. \tag{18.13}$$

These identities retain reference, raw-factor, label-weight, anchor, and likelihood derivatives. [ESTABLISHED]

Proof. Differentiate $Zp_i = u_i$ and isolate the term $Z\partial^\alpha p_i$. Increasing total derivative order gives the recurrence. Differentiating the finite push and dividing by its positive mass gives the conditional score. Write $s = (s - s') + s'$; conditional centering kills both mixed outer products. Differentiating total mass proves score centering. An observation posterior has score $s - \partial \log E_o$, where $\partial \log E_o = \mathbb{E}_{\mathbf{\Pi}_o} s$. □

A parameter-dependent channel has an additional transition-score term. A one-state source followed by a Bernoulli(θ) channel has zero input Fisher information and output information $1/[\theta(1-\theta)]$. Thus the unchanged contraction assertion is false for that channel class. [ESTABLISHED]

For a positive C^3 mass family the Fisher tensor $g_{ij} = \sum_x (\partial_i p_x)(\partial_j p_x)/p_x$ is C^2. On a neighborhood where it is positive definite it has a C^1 Levi-Civita connection and continuous curvature. The mass Jacobian has full parameter rank there: its positive weighted Gram matrix is g. Choosing a left inverse at a point and shrinking a convex neighborhood makes the derivative of the resulting coordinate map differ from the identity by less than $1/2$; integration along segments proves local injectivity. This establishes regular local statistical geometry. It supplies no inverse on a singular stratum or uniform lower eigenvalue bound over unbounded RG depth. [ESTABLISHED]

Definition 18.7 (Admitted curvature factor). Gauge curvature is the curvature F_ω of a declared principal connection. Base curvature is that of a declared nondegenerate metric on $\mathcal{C}$. Fisher curvature is that of a declared regular normalized statistical family. A curvature action requires the relevant connection, pairing, metric contraction, measure, coupling, and a finite normalizer. It enters as a specified H_γ, A_γ, or likelihood factor. These are distinct generative choices. [DEFINITION]

For a unitary representation of dimension n, the pairing $-\operatorname{Re}\operatorname{Tr}(XY)/n$ is positive on represented anti-Hermitian matrices, and the loop action $\operatorname{Re}\operatorname{Tr}(2I-U-U^*)/(2n)$ is nonnegative because it equals $\operatorname{Tr}((I-U)^*(I-U))/(2n)$. With a Riemannian base, a curvature-square action can therefore be declared as

$$\mathscr{S}_{\mathrm{g}} = -\frac{\kappa_{\mathrm{g}}}{4n}\int_{\mathcal{C}} \operatorname{Re}\operatorname{Tr}(F_{\mu\nu}F^{\mu\nu})\,d\operatorname{vol}_g, \qquad \kappa_{\mathrm{g}} \geq 0. \tag{18.14}$$

The integrand's nonnegativity is established in an orthonormal base frame by summing squared matrix norms. Normalizability of a probability law on connections is an additional hypothesis, not a consequence of this sign. [ESTABLISHED]

No positive norm on all of $\mathfrak{gl}(n,\mathbb{R})$, $n \geq 2$, is invariant under every conjugation: conjugating E_{12} by $\operatorname{diag}(s,1,\dots,1)$ multiplies it by s, contradicting positive norm invariance for $s \neq 1$. A separately specified SPD fiber metric H instead gives $\operatorname{Tr}(H^{-1}F^{\top}HF) \geq 0$, invariant under the simultaneous changes $F \mapsto gFg^{-1}$ and $H \mapsto g^{-\top}Hg^{-1}$ by cyclicity of trace. The additional metric is part of the model. [ESTABLISHED]

If links are random with normalized reference m and prior $e^{-\mathscr{S}_{\mathrm{g}}}m/Z_{\mathrm{g}}$, recognition $R(d\Theta)Q_\Theta(dy)$ gives

$$\mathcal{F} = \mathbb{E}_R\mathcal{F}_{\mathrm{matter}}(Q_\Theta; o, \Theta) + \mathbb{E}_R\mathscr{S}_{\mathrm{g}} + D_{\mathrm{KL}}(R\|m) + \log Z_{\mathrm{g}}. \tag{18.15}$$

This follows by expanding the normalized joint density. Dropping the field entropy or partition changes that VFE. For an externally fixed link, a state-independent action cancels in the normalized conditional model. [ESTABLISHED]

A Fisher-curvature action computed from the same unknown law that it reweights poses a self-consistency problem. Existence, uniqueness, normalizability, and regularity of its solutions remain to be proved. No such action, continuum random-field limit, or physical identification is asserted by the finite construction. [OPEN]

18.4 A global geometry label cannot always be erased

Theorem 18.8 (Product-branch erasure can destroy Gibbs regularity). On a countably infinite lattice choose $0 < p < q < 1$ and $0 < w < 1$. Let μ_p, μ_q be independent Bernoulli product laws. The mixture $\mu = w\mu_p + (1-w)\mu_q$ has no continuous singleton conditional version in the product topology. In particular it has no absolutely summable finite-alphabet Gibbs interaction. Both branches can lie in an arbitrarily small positive neighborhood of the fair product law. This obstruction is established for erasure of one shared global label. [ESTABLISHED]

Proof. Enumerate the exterior sites and let T_r be the event that their empirical mean tends to r. Independence gives $\mathbb{E}(\sum_{j=1}^{n}(X_j - r))^4 \leq n^2/4$. Thus the probability of a mean deviation exceeding ϵ is at most $1/(4n^2\epsilon^4)$. Summing this bound over $n \geq N$ and then taking $N \to \infty$ proves $\mu_r(T_r) = 1$. Removing finitely many sites does not alter that limit. The exterior therefore identifies the branch almost surely. Independence within a branch shows that a version of the singleton conditional is p on T_p and q on T_q.

Every nonempty finite exterior cylinder has positive probability under both branches, hence positive mixture mass intersecting each of T_p, T_q. Any other conditional version agrees off a mixture-null set, so each cylinder still contains both values. Continuity at any exterior would require a cylinder on which the oscillation is less than $q - p$, a contradiction. An absolutely summable interaction gives a uniformly convergent singleton energy sum of cylinder functions; its normalized exponential is continuous and would be such a version. Finally the one-site fields $\frac{1}{2}\log[r/(1-r)]$ tend to zero as $r \to 1/2$. Distinct p, q sufficiently close to $1/2$ make both branch norms arbitrarily small without changing the argument. □

The full labeled joint remains normalized, and its finite erasure remains an ordinary stochastic push with the posterior-fiber loss of eq. (18.10). What fails is the inference that an erased mixture belongs to the same infinite Gibbs class as each branch. [ESTABLISHED]

Chapter 19

Geometry Pressure and Specific VFE under Renormalization

This chapter retains the finite catalog $\mathfrak{G}$ of Chapter 18 and places every branch on the same square cell lattice. It constructs pressure from finite volumes and distinguishes convergence of a VFE density from convergence of its labeled target law. The spatial hypotheses are those of Chapter 17; no unrestricted mixture is presumed to be one Gibbs branch. [HYPOTHESIS]

19.1 Carrier, boundary convention, and pressure

Definition 19.1 (Native cell model). A cell $x \in \mathbb{Z}^2$ contains the outgoing signs $z_{x,1}, z_{x,2}$. For $\Lambda_L = \{0, \dots, L-1\}^2$ use normalized counting reference 4^{-L^2}, all $2L^2$ outgoing signs, and the $(L-1)^2$ fully internal plaquettes. A branch has raw factor

$$r_{\gamma,L}(z) = \prod_{e \in \Lambda_L \times \{1,2\}} (1 + b_\gamma z_e) \prod_{\square \subset \Lambda_L} (1 + t_\gamma z_{\partial\square}), \qquad 0 < b_\gamma < 1, \quad 0 < t_\gamma < 1/12. \tag{19.1}$$

Here $b_\gamma = \tanh h_\gamma$ and $t_\gamma = \tanh j_\gamma$ are branch-local parameters; t_γ is not a process coordinate. Their finite catalog lies in a fixed compact subdomain. An all-touching fixed-exterior model includes every plaquette meeting the free cells; it is a different finite boundary convention. Normalized independent gauge frames contribute total reference mass one. These are declared model data. [DEFINITION]

For local convergence, center the coarse square at $-\lfloor L/2 \rfloor$ in both coordinates and the fine BL square at $-B\lfloor L/2 \rfloor$. Every fixed cylinder then has diverging boundary distance. The all-exterior estimate of Theorem 17.6 makes free, fixed-boundary, and actual-marginal approximations converge to the same unique branch law on these cylinders. Integrate its uniform estimate over exterior mixtures to include free and marginal boundaries. A fixed corner of an uncentered positive-quadrant box does not exhaust $\mathbb{Z}^2$ and supplies no such claim. This is an established consequence of the spatial comparison. [ESTABLISHED]

Theorem 19.2 (Constructive pressure and surface comparison). Let Ψ be a translation-invariant cell interaction with

$$K(\Psi)=\sum_{A\ni 0}\|\Psi_A\|_\infty<\infty,\qquad M(\Psi)=\sum_{A\ni 0}\operatorname{diam}(A)\|\Psi_A\|_\infty<\infty, \tag{19.2}$$

using cell Chebyshev diameter. Write Z_L for its free partition relative to normalized counting and μ_{Λ_L} for the marginal of a Gibbs law with that interaction. Then

$$\mathfrak{p}(\Psi)=\lim_{L\to\infty}L^{-2}\log Z_L,\qquad \left|\mathfrak{p}(\Psi)-n^{-2}\log Z_n\right|\le\frac{4M(\Psi)}{n}, \tag{19.3}$$

$$\left|\log\frac{\mu_{\Lambda_L}(z)}{p_L^{\mathrm{free}}(z)}\right|\le 8M(\Psi)L. \tag{19.4}$$

The same mass-ratio bound holds for any fixed exterior. These conclusions are established without an infinite partition function. [ESTABLISHED]

Proof. Put $T(d)=\sum_{A\ni 0,\operatorname{diam}(A)\ge d}\|\Psi_A\|_\infty$. A support crossing the square and containing x has diameter at least $\operatorname{dist}(x,\Lambda_L^c)$. At most $4L$ cells occur at each positive boundary distance. Summing nonnegative bounds gives crossing energy at most $4L\sum_{d\ge1}T(d)=4LM$. Adding the exterior therefore changes the free Hamiltonian by at most $4LM$, its log partition by the same amount, and its normalized log mass by at most $8LM$. Integration over an exterior law preserves the multiplicative bounds.

Tile a large square by n-squares and a leftover region D. Interactions touching D have total norm at most $K|D|$. Crossings of the small squares cost at most $4Mn$ per tile. Assign independent normalized counting to D and compare partition sums exponentially. For fixed n, the leftover fraction tends to zero with the large square. Both limiting upper and lower pressures lie within $4M/n$ of $n^{-2}\log Z_n$. Their difference is at most $8M/n$ for every n, so they agree. The finite pressure bound and the surface comparison follow. □

The parameter-derivative extension is a separate open certification obligation. The proposed native C^4 pressure and actual fixed-word C^2 Fisher statements in Theorems C.6 and C.11 include local-response and boundary arguments for the explicitly admitted native source families. Their written arguments await evidence adjudication; the scalar surface estimate above does not discharge that obligation. [OPEN]

For the native interaction, the singleton energy is $-\log[(1+bz_1)(1+bz_2)]$ and each plaquette contributes $-\log(1+tz_{\partial\square})$. Its three incident cell supports give

$$K\le\frac{2b}{1-b}+\frac{3t}{1-t},\qquad M\le\frac{3t}{1-t}. \tag{19.5}$$

Indeed $-\log(1-u)\le u/(1-u)$ follows by integrating $(1-u)^{-1}$; $\log(1+u)\le-\log(1-u)$ for $0\le u<1$. Thus the bounds, and native pressure, are explicit. [ESTABLISHED]

Proposition 19.3 (Finite rational approximation of pressure). For rational native parameters and a positive rational tolerance, eqs. (19.3) and (19.5) give a terminating finite approximation procedure with sufficient finite resources. [ESTABLISHED]

Proof. Choose n so that $4M/n$ is below half the tolerance. Enumerate the 4^{n^2} rational positive raw factors to obtain Z_n exactly. For a positive rational x, find an integer e with $u = x/2^e \in [1, 2)$ and set $v = (u-1)/(u+1) \in [0, 1/3)$. The integrated finite geometric series bounds its logarithm by

$$0 \le \log u - 2\sum_{j=0}^{N-1} \frac{v^{2j+1}}{2j+1} \le \frac{2v^{2N+1}}{(2N+1)(1-v^2)}. \tag{19.6}$$

Use $v = 1/3$ for $\log 2$, and $\log x = e\log 2 + \log u$, reversing interval endpoints in the first term if $e < 0$. Increasing N attains the remaining scaled tolerance. Each inventory and arithmetic operation is finite; there is no claim of a small state count or a uniform fixed resource cap. □

19.2 Pressure of the complete coarse interaction

Fix an integer block size B before taking volume to infinity. Let Ψ_B be the canonical vacuum interaction for the full pushed law μ_B, supplied by Theorem 17.8, with vacuum $* = (+,+)$. For compact upper bounds b_+, t_+, enlarged to include the observation-shifted field below, set

$$q_0 = \frac{1+12t_+}{2}, \qquad \tau = \frac{24t_+}{1+12t_+}, \qquad C_0 = \frac{16q_0}{1-q_0}, \tag{19.7}$$

$$\rho = \frac{b_+(1+t_+^2)+2t_+}{1+t_+^2+2b_+t_+}, \qquad m_0 = \frac{(1-\rho)^2}{4}, \qquad D_B = \frac{2C_0B^2}{m_0}. \tag{19.8}$$

Both τ and ρ lie in $(0,1)$. The spatial comparison gives target variation at most $C_0|A|B^2(R+1)\tau^{R+1}$ when boundary changes are outside radius R. Revealing the two disjoint target paths successively gives each cell probability at least m_0: conditional on other fine signs, a path contains independent signs with field magnitude at most $\operatorname{atanh} b_+ + 2\operatorname{atanh} t_+$, and its parity mean has magnitude at most ρ^B. These established bounds use a common positive-coupling domain independent of B. [ESTABLISHED]

The canonical lexicographic vacuum construction writes its singleton term as a conditional log-odds and each radius-r term as the difference of successive such odds. Logarithmic Lipschitz continuity on $[m_0, 1]$ gives

$$\|\Psi_{B,(x,0)}\|_\infty \le \frac{4\rho^B}{1-\rho^2}, \qquad \|\Psi_{B,(x,r)}\|_\infty \le D_B r\tau^r \quad (r \ge 1). \tag{19.9}$$

At most $(2r+1)^2$ translated supports of radius r contain a fixed cell, and each has diameter at most $2r$. Consequently an explicit first moment bound is

$$M(\Psi_B) \le \overline{M}_B := 2D_B \sum_{r\ge 1} r^2(2r+1)^2\tau^r < \infty. \tag{19.10}$$

The analogous omitted interaction norm is bounded by $D_B \sum_{r>N} r(2r+1)^2\tau^r$. These are rationally computable series: apply $(\tau d/d\tau)^j$ to $1/(1-\tau)$ and expand $(N+1+r)^j$ to evaluate every polynomial tail. This proves the required finite first moment at each fixed depth, rather than assuming it from a finite interaction truncation. [ESTABLISHED]

Theorem 19.4 (Vacuum-cylinder pressure construction). Let $q_{n,R}$ be the probability that all $2n^2$ retained path parities in an n-square are positive, evaluated under a finite native all-touching law with a radius-R whole-block buffer and fixed exterior. All other parities are unconstrained. Then

$$\left|\mathfrak{p}(\Psi_B) + \log 4 + n^{-2}\log q_{n,R}\right| \le \frac{12\overline{M}_B}{n} + \frac{C_0 B^2(R+1)\tau^{R+1}}{m_0^{n^2}}. \tag{19.11}$$

For fixed B this constructs pressure to every positive tolerance by finite rational summation and directed logarithms. [ESTABLISHED]

Proof. The free vacuum atom of Ψ_B is $4^{-n^2}/Z_{B,n}$ because every vacuum interaction vanishes there. Its actual marginal differs in log mass by at most $8\overline{M}_B n$. Add the $4\overline{M}_B/n$ pressure error after dividing by n^2 to obtain the first term. Sequential conditional positivity gives both actual and finite vacuum probabilities at least $m_0^{n^2}$. The spatial comparison bounds their difference by $C_0 n^2 B^2(R+1)\tau^{R+1}$. Apply the mean-value bound for $\log$ and divide by n^2. First choose n, then R, then logarithmic precision. Every choice terminates. Include all free signs, unused signs, and touching plaquettes in the finite sum; a product approximation would change it. □

The separate product case $t = 0$ has native pressure zero and vacuum pressure $-2\log(1+b^B)$. These differ by a real normalization scalar; no positive-t contraction argument is needed for this control. [ESTABLISHED]

19.3 Geometry weights, selection, and ties

Hypothesis 19.5 (Controlled label growth). For the global ensemble suppose $\log a_{\gamma,L} = L^2\alpha_\gamma + u_{\gamma,L}$, with $|u_{\gamma,L}| \le U_\gamma(L) = o(L^2)$ and a supplied modulus for this sublinear bound. Use each branch's matching free partition and law, or its matching fixed-exterior pair. These hypotheses exclude arbitrary uncontrolled volume-dependent weights. [HYPOTHESIS]

For the proposed Fisher-density extension, choose the stronger parameter-regularity hypotheses in Equation (C.66): actual C^3 effective log-mass representatives with norms $O(V)$ and C^3 convergence after division by $V = L^2$. That choice excludes uncontrolled derivatives of subextensive terms and

is used in the strict-phase and crossover arguments. It is not part of the scalar-growth hypothesis above. [HYPOTHESIS]

Theorem 19.6 (Strict selection and the exact tie condition). Write $s_\gamma = \alpha_\gamma + \mathfrak{p}_\gamma$ and $W_{\gamma,L} = a_{\gamma,L}Z_{\gamma,L}/\sum_\delta a_{\delta,L}Z_{\delta,L}$. For free boundaries let $E_\gamma(L) = U_\gamma(L) + 4\overline{M}_\gamma L$; for fixed exteriors replace 4 by 8. If γ_* uniquely maximizes s_γ, then

$$1 - W_{\gamma_*,L} \le \sum_{\delta\neq\gamma_*} \exp\left[-L^2(s_{\gamma_*} - s_\delta) + E_\delta(L) + E_{\gamma_*}(L)\right] \longrightarrow 0. \tag{19.12}$$

Let S instead be the full maximal-score set, and put $v_{\gamma,L} = u_{\gamma,L} + \log Z_{\gamma,L} - L^2\mathfrak{p}_\gamma$. The labeled local law converges exactly when the finite vector

$$\left(\frac{e^{v_{\gamma,L}}}{\sum_{\delta\in S} e^{v_{\delta,L}}}\right)_{\gamma\in S} \tag{19.13}$$

converges. Pressure equality alone does not determine it. These conclusions are established under Hypothesis 19.5. [ESTABLISHED]

Proof. The pressure error bounds give $|\log(a_{\gamma,L}Z_{\gamma,L}) - L^2 s_\gamma| \le E_\gamma(L)$. Divide each competing weight by that of γ_* and bound the common denominator below by its largest term. This proves eq. (19.12). For a tied maximal set its complement has a positive gap, if nonempty, and hence vanishing mass. Within S the relative weights are exactly eq. (19.13). Branch cylinder limits exist by the centered spatial comparison. Finite summation proves sufficiency; testing the cylinder consisting only of a label proves necessity. □

With certified pressure intervals, a strict lower separation identifies a winner and the displayed growth moduli give a finite leakage threshold. Overlapping intervals establish neither equality nor a tie. Effective tied weights require a separate convergence modulus for eq. (19.13). For two identical positive-coupling branches, take $a_{1,L} = v_L/Z_L$, $a_{2,L} = 1/Z_L$. Choosing v_L to alternate between 1 and 2 gives identical bulk scores and label weights alternating between $1/2$ and $2/3$. Choosing $v_L = e^{-L}$ instead removes one label with zero bulk score gap. These established examples preserve the positive-coupling domain. [ESTABLISHED]

A fixed hierarchy $w_\gamma\mu_\gamma$ is a finite sum of probability laws and requires no infinite partition. At finite volume the factor weights $a_{\gamma,L} = w_\gamma/Z_{\gamma,L}$ reproduce it exactly. Compensation using only $e^{-L^2\mathfrak{p}_\gamma}$ leaves the surface term $\log Z_{\gamma,L} - L^2\mathfrak{p}_\gamma$. In the explicitly permitted boundary-scalar class, multiplying one of two identical raw branches by e^L leaves both pressures unchanged but makes this residual select that label. Thus a bulk counterterm does not generally restore prescribed tied weights. This is a proved scalar example, not a claim about the sign of a bare native boundary correction. [ESTABLISHED]

19.4 Observation, entropy, and specific VFE

Proposition 19.7 (Positive sensor and posterior branch). For $0 < \eta_\gamma < 1$ declare an independent edge sensor $\ell_{\eta_\gamma}(o \mid z) = (1 + \eta_\gamma o z)/2$. On the finite all-positive observation define

$$A_\gamma = \frac{1 + b_\gamma \eta_\gamma}{2}, \qquad b'_\gamma = \frac{b_\gamma + \eta_\gamma}{1 + b_\gamma \eta_\gamma}, \qquad Z^{\mathrm{obs}}_{\gamma,L} = A_\gamma^{2L^2} Z_L(b'_\gamma, t_\gamma). \tag{19.14}$$

The selected free or fixed-boundary posterior is the matching shifted branch. Reweighting actual infinite-law marginals has the same centered local posterior limit. Its evidence density is

$$e_\gamma = 2 \log A_\gamma + \mathfrak{p}(b'_\gamma, t_\gamma) - \mathfrak{p}(b_\gamma, t_\gamma). \tag{19.15}$$

This is an established finite-observation limit, not conditioning on the null infinite all-positive event. [ESTABLISHED]

Proof. The identity $(1 + bz)(1 + \eta z)/2 = A(1 + b'z)$ follows from $z^2 = 1$. Multiply it over the observed free edges; plaquette factors are unchanged. The shifted compact field bound remains below one and the coupling t remains below $1/12$. For an actual marginal, condition the finite reweighted full-space law on the window's exterior. Its conditional is exactly the shifted all-touching specification. Apply the uniform interior comparison and integrate over that reweighted exterior to prove local convergence. A mere surface log-ratio bound would not prove this step.

For densities, the prior actual/free log-ratio is bounded by $8ML$. Multiplication by the same positive likelihood bounds the evidence log-ratio by $8ML$ and the posterior log-ratio by $16ML$. The shifted free/actual comparison adds $8M'L$. These bounds vanish after division by L^2, giving eq. (19.15). The infinite all-positive event is null because finite likelihoods are bounded by $[(1 + \eta_+)/2]^{2L^2} \to 0$. □

For the proposed thermodynamic Fisher application of this sensor, choose C^4 original and shifted source maps whose images remain in compact positive native high-temperature boxes. Use the actual posterior representatives in Equations (C.68) and (C.69); the hierarchical representative retains its original physical partition divisor. These are the additional input choices for the proposed derivative bridge, not consequences of positivity of an arbitrary likelihood. [HYPOTHESIS]

For fixed positive hierarchical label weights, total evidence density is $\max_\gamma e_\gamma$ and posterior label scores are e_γ. For global weights satisfying Hypothesis 19.5, total evidence density and posterior scores are respectively

$$\max_\gamma(\alpha_\gamma + \mathfrak{p}^{\mathrm{obs}}_\gamma) - \max_\gamma(\alpha_\gamma + \mathfrak{p}_\gamma), \qquad \alpha_\gamma + \mathfrak{p}^{\mathrm{obs}}_\gamma, \tag{19.16}$$

where $\mathfrak{p}^{\mathrm{obs}}_\gamma = \mathfrak{p}_\gamma + e_\gamma$. The largest exponent and the logarithm of a finite exponential sum differ by at most $\log |\mathfrak{G}|$. This proves these rates; posterior ties still require their own subextensive ratios. [ESTABLISHED]

Theorem 19.8 (Stationary specific relative entropy). For a stationary law Q on the four-state cell lattice, define its square entropy $H(Q_{\Lambda_L})$ with $0 \log 0 = 0$. For a Gibbs law μ with interaction satisfying eq. (19.2),

$$s(Q) = \lim_{L\to\infty} L^{-2} H(Q_{\Lambda_L}) = \inf_{n\geq 1} n^{-2} H(Q_{\Lambda_n}), \tag{19.17}$$

$$e_\Psi(Q) = \sum_{A \ni 0} \frac{\mathbb{E}_Q \Psi_A}{|A|}, \tag{19.18}$$

$$h(Q\|\mu) := \lim_{L\to\infty} L^{-2} D_{\mathrm{KL}}(Q_{\Lambda_L}\|\mu_{\Lambda_L}) = \log 4 - s(Q) + e_\Psi(Q) + \mathfrak{p}(\Psi). \tag{19.19}$$

These established limits need neither ergodicity nor positive recognition atoms. They concern specific relative entropy, not ordinary KL between infinite laws. [ESTABLISHED]

Proof. Tile a large square by n-squares. Entropy subadditivity and stationarity bound its entropy by the number of tiles times $H(Q_{\Lambda_n})$ plus $|D| \log 4$. Thus the limiting upper density is no greater than every n-square density, whereas their infimum is no greater than the limiting lower density. For energy allocate each interaction equally among its sites. Absolute summability and stationarity give eq. (19.18); omitted boundary allocations are bounded by $4ML$. The finite free KL equals $-H(Q_{\Lambda_L}) + L^2 \log 4 + \mathbb{E}_Q H_L^{\mathrm{free}} + \log Z_L$. Changing the comparison to the actual marginal costs at most $8ML$. Divide by L^2 and apply Theorem 19.2. □

Stationarity alone gives no uniform finite-query entropy certificate. A uniformly random K-square four-symbol array, repeated periodically, is stationary, has zero entropy density, and has the fair independent marginal on every square of side at most K. A procedure terminating after finitely many cylinder queries on the fair infinite field receives identical answers from such an array when K exceeds every queried coordinate span. Their entropy densities differ by $\log 4$. Hence no such procedure can uniformly certify error below $\log 4/2$ for all stationary laws. A quantitative VFE approximation needs additional recognition entropy or expectation bounds. [ESTABLISHED]

Theorem 19.9 (Retained-label specific VFE). Fix recognition $\mathbb{Q} = q_\gamma Q_\gamma$ with each Q_γ stationary. Let $h_\gamma = h(Q_\gamma\|\mu_\gamma)$ and $\ell_\gamma^Q = \mathbb{E}_{Q_\gamma} \sum_{a=1}^2 \log \ell_{\eta_\gamma}(+ \mid z_{0,a})$. Writing $s_* = \max_\gamma s_\gamma$, the finite-ensemble VFE densities are

$$f_{\mathrm{h}} = \sum_\gamma q_\gamma (h_\gamma - \ell_\gamma^Q), \qquad f_{\mathrm{f}} = \sum_\gamma q_\gamma (h_\gamma - \ell_\gamma^Q + s_* - s_\gamma). \tag{19.20}$$

They exist under the stated branch and label-growth hypotheses. [ESTABLISHED]

Proof. Disintegrate finite KL over the label. For the hierarchy, $D_{\mathrm{KL}}(q\|w)/L^2 \to 0$, and branch KL limits follow from Theorem 19.8. Stationarity evaluates the likelihood sum per cell exactly. For global factors, $-L^{-2} \log W_{\gamma,L} \to s_* - s_\gamma$ by the finite exponential bounds. The finite recognition label entropy again vanishes per cell. Summing finitely many branch terms proves the formulas. Matching fixed-exterior or explicitly compared actual-marginal versions have the same density by the surface bounds. □

If $W_{\gamma,L} \to 0$ while $q_\gamma > 0$, the limiting labeled joint gives zero mass to a recognition-positive label, so KL to that limiting joint is infinite even on a finite state window. Formula eq. (19.20) can remain finite; the e^{-L} tied example even has zero label-rate penalty. Density convergence therefore does not identify VFE against a limiting joint and does not require that the label weights converge. Erasing the label instead remains subject to Theorem 18.8. [ESTABLISHED]

19.5 Scalar composition and the induced VFE density

Theorem 19.10 (Normalization density along fixed blocking words). Let Ψ and $\mathsf{R}_a\Psi$ be the complete canonical vacuum representatives of a law and its exact a-path push, each with finite first moment. For the finite raw push and its matching vacuum anchor scalar $\zeta_{a,L}$,

$$\sigma_a(\Psi) := \lim_{L\to\infty} L^{-2} \log \zeta_{a,L} = a^2\mathfrak{p}(\Psi) - \mathfrak{p}(\mathsf{R}_a\Psi), \tag{19.21}$$

$$\sigma_{ad}(\Psi) = d^2\sigma_a(\Psi) + \sigma_d(\mathsf{R}_a\Psi). \tag{19.22}$$

These are established for a finite word fixed before the volume limit, with coherent references, anchors, and centered embeddings. [ESTABLISHED]

Proof. Disjoint parity paths push normalized counting to normalized counting. The exact finite identity is $Z_{\mathrm{fine},aL} = \zeta_{a,L}\widehat{Z}_L$. The fine free/actual comparison costs $8MaL$ in log mass; pushing preserves the multiplicative bound. The coarse actual/free comparison adds $8M'L$. Thus the pushed free law and canonical coarse free law differ by at most $8(aM + M')L$ in log mass. Both anchored factors equal one at the vacuum, so evaluating that bound there compares $\log \widehat{Z}_L$ and the canonical $\log Z'_L$ by the same amount. Divide the partition identity by L^2 and use their pressure limits. Substitution and cancellation prove eq. (19.22); its d^2 counts the intermediate cells per final cell. An unchanged canonical identity has $\sigma_1 = 0$. □

The initial native factor has vacuum logarithm per cell $\chi = 2\log(1+b) + \log(1+t)$. Dividing by its vacuum value gives a finite-range vacuum representative. Its free partition and that of the canonical vacuum representative differ only by a surface term: compare both to their common actual marginal and evaluate at the vacuum. Consequently

$$\mathfrak{p}(\Psi_1) = \mathfrak{p}_{\mathrm{native}} - \chi, \qquad \widetilde{\sigma}_B = B^2\mathfrak{p}_{\mathrm{native}} - \mathfrak{p}(\Psi_B), \qquad \alpha'_B = B^2\alpha + \widetilde{\sigma}_B. \tag{19.23}$$

It follows that $\alpha'_B + \mathfrak{p}(\Psi_B) = B^2(\alpha + \mathfrak{p}_{\mathrm{native}})$. This establishes preservation of geometry scores with their scalar and area factor. Equality of infinite laws alone would not specify a raw normalizer or a representative conversion. [ESTABLISHED]

Theorem 19.11 (Specific VFE transport of the complete experiment). Retain the label and the complete fine observation alphabet through a fixed word of product block size B. Let $Q_{\gamma,B}$ be the

stationary path push of Q_γ, and let $h^{\mathrm{post}}_{\gamma,B}$ denote its specific KL to the pushed shifted branch. Then

$$B^2 f_{\mathrm{fine}} = f_{\mathrm{coarse,induced}} + k_{\mathrm{fiber}}, \qquad k_{\mathrm{fiber}} = \sum_\gamma q_\gamma \left[B^2 h^{\mathrm{post}}_{\gamma,1} - h^{\mathrm{post}}_{\gamma,B} \right] \geq 0. \tag{19.24}$$

The identity is established for each of the two ensembles above, using limits of finite quantities rather than subtraction of infinite-law KLs. [ESTABLISHED]

Proof. On a finite window the induced likelihood is $\ell'_\gamma(o \mid y) = E_{\gamma,L}(o) p^{\mathrm{post,push}}_{\gamma,L}(y) / p^{\mathrm{prior,push}}_{\gamma,L}(y)$ on its positive support, by Theorem 18.3. Apply Proposition 18.5. The coarse recognition marginal is exactly the push of its BL-square fine marginal. The actual-posterior comparison in Proposition 19.7 is $O(BL)$ uniformly in state and is preserved by the push. After division by L^2 it vanishes, so Theorem 19.8 gives both posterior state-KL limits. For the hierarchy the coarse VFE density is $\sum_\gamma q_\gamma (h^{\mathrm{post}}_{\gamma,B} - B^2 e_\gamma)$; for global factors it is $\sum_\gamma q_\gamma [h^{\mathrm{post}}_{\gamma,B} + B^2(s_* - \alpha_\gamma - \mathfrak{p}^{\mathrm{obs}}_\gamma)]$. Subtract the separately finite density limits from B^2 times their fine formulas. Evidence and geometry rates cancel, leaving the displayed fiber term. Every finite fiber loss is nonnegative, hence so is its limit. □

Choosing a new local sensor on coarse edges changes the experiment and need not preserve this identity. Likewise, a prescribed order-one geometry action A_γ has zero bulk rate but can affect tied weights; an extensive factor $e^{-L^2 A_\gamma}$ shifts α_γ by $-A_\gamma$. A state-dependent curvature action changes the branch specification and requires its own admission proof. These established normalization distinctions prevent an unaccounted geometric penalty from being appended to VFE. [ESTABLISHED]

The proposed thermodynamic Fisher construction in Chapter 20 distinguishes strict-phase C^2 densities, fixed-tie tensors normalized by V^2, and a separate reciprocal-volume statistical experiment with actual center-ratio hypotheses. Its certification remains open pending adjudication of the written native, label, and interface proofs. None of those proposed statements asserts a nonsingular extensive metric across every tie. [OPEN]

Uniform invertibility at unbounded depth, unrestricted simultaneous volume/depth limits for the full labeled experiment, self-consistent curvature actions, critical universality, and continuum field measures remain open. Each needs its own regularity, limiting, or existence construction. Identification of a physical clock or physical law also requires a separate operational bridge. [OPEN]

Chapter 20

Thermodynamic Fisher Geometry of Retained Ensembles

The results in this chapter hold on the stated finite supports, compact positive source boxes, and fixed blocking words. Strict phases, pressure ties, and reciprocal-volume experiments retain their separate hypotheses; none supplies a regular metric across arbitrary ties or at all depths.

The finite geometry label is retained as an observed component of the statistical experiment. Its score contributes information even when the conditional branch laws coincide. This chapter fixes the complete finite experiment and its exact information accounting before considering a volume limit. The accompanying proof appendix distinguishes actual finite free laws, exact fixed-word pushes, and marginals of an infinite law. It then treats strict phases, pressure ties, and their different parameter scales without identifying these regimes. [ESTABLISHED]

20.1 Normalized finite experiments and retained information

The finite geometry ensembles of Definition 18.1 are the starting point. This section keeps the geometry label in the sample space, allows every declared generative factor to vary with a statistical parameter, and distinguishes the complete observation joint from its latent marginal and a selected posterior. The structural descriptive data remain outside the stochastic channel. No volume limit or metric inverse is part of these finite definitions. [DEFINITION]

Definition 20.1 (Smooth finite branch data). Let $\Theta \subset \mathbb{R}^d$ be open, with $1 \le d < \infty$, and let $\mathfrak{G}$ be a finite nonempty geometry catalog. Each label has a finite nonempty declared carrier X_γ. On a fixed nonempty subset X_γ^+, let $\nu_{\gamma,\theta}$ be a positive probability reference and let the physical raw factor be $r_{\gamma,\theta} = s_{\gamma,\theta}\widehat{r}_{\gamma,\theta} > 0$. The scalar s_γ is part of the physical factor. Retained carrier atoms outside X_γ^+ have zero reference mass; raw-density quotients there are undefined and are never evaluated. Define

$$Z_\gamma = \sum_{x \in \mathsf{X}_\gamma^+} \nu_\gamma(x) r_\gamma(x), \qquad p_\gamma(x) = \frac{\nu_\gamma(x) r_\gamma(x)}{Z_\gamma} \quad (x \in \mathsf{X}_\gamma^+), \tag{20.1}$$

and set $p_\gamma(x) = 0$ on the remaining declared atoms. All parameter arguments are suppressed only after their dependence has been declared. A native free square is the special case of Definition 19.1:

$V = L^2$, all $2V$ signs, exactly $(L-1)^2$ internal plaquettes, and reference 4^{-V} on its complete carrier. Its source notation is $b_\gamma = \tanh h_\gamma$ and $t_\gamma = \tanh j_\gamma$. [DEFINITION]

Hypothesis 20.2 (Finite differentiable support stratum). The reference masses, positive raw factors, scalars, positive geometry weights $a_{\gamma,\theta}$, and likelihood entries on active reference atoms are actual C^3 functions on Θ. Every zero reference or likelihood entry is identically zero throughout this stratum; every other likelihood entry is strictly positive there. Each likelihood row $\ell_{\gamma,\theta}(o \mid x)$ sums to one on one fixed finite observation alphabet O. These are generative inputs independent of recognition. The fixed-support restriction excludes a family that creates or removes an atom at the parameter under consideration. It is used when differentiating scores and conditional laws; it is not an assumption of positive Fisher rank. [HYPOTHESIS]

Definition 20.3 (Two normalization modes and their complete joints). Write $A_0 = \sum_\delta a_\delta$ and $Z_{\mathrm{all}} = \sum_\delta a_\delta Z_\delta$. The hierarchical and global-factor label weights are respectively

$$W^{\mathrm{h}}_\gamma = \frac{a_\gamma}{A_0}, \qquad W^{\mathrm{f}}_\gamma = \frac{a_\gamma Z_\gamma}{Z_{\mathrm{all}}}. \tag{20.2}$$

A separately prescribed positive normalized hierarchy w_γ instead uses $W^{\mathrm{h}}_\gamma = w_\gamma$ directly. For either chosen mode, the latent and observation laws are

$$\mathbb{P}^{\mathrm{lat}}(\gamma, x) = W_\gamma p_\gamma(x), \qquad \mathbb{P}(\gamma, x, o) = W_\gamma p_\gamma(x) \ell_\gamma(o \mid x). \tag{20.3}$$

The displayed joint product is evaluated only on active reference atoms; on all remaining declared atoms its mass is defined to be zero without evaluating a raw factor or likelihood. The same convention applies to selected posterior masses below. The sample space is the disjoint union of the labeled branch carriers; identical branch distributions do not erase their labels. These formulas use the two modes of eqs. (18.2) and (18.3), with an explicit physical scalar and structural-zero convention. [DEFINITION]

Proposition 20.4 (Normalization and physical scalar dependence). Each law in eq. (20.3) is normalized. Multiplying r_γ by a positive branch scalar d_γ leaves p_γ unchanged. It leaves the hierarchy unchanged and changes the global-factor label masses by the corresponding relative factors d_γ, unless the geometry weights are compensated. One common scalar cancels from both modes. [ESTABLISHED]

Proof. Every Z_γ is finite and positive because its reference is a probability law and its factor is positive on a finite nonempty support. Summing the joint over o, then x, then γ gives one. Under the rescaling, Z_γ becomes $d_\gamma Z_\gamma$, so the same factor cancels between the numerator and denominator of p_γ. Substitution in eq. (20.2) gives the assertions about labels. Replacing a_γ by a_γ / d_γ compensates the global-factor change. Thus omitting s_γ from a physical partition is not justified by its cancellation within a conditional branch. □

Definition 20.5 (Effective label log masses and complete scores). For the chosen latent law use an effective log-mass representative λ_γ with

$$W_\gamma = \frac{e^{\lambda_\gamma}}{\sum_\delta e^{\lambda_\delta}}, \qquad \lambda_\gamma^{\mathrm{f}} = \log a_\gamma + \log Z_\gamma, \qquad \lambda_\gamma^{\mathrm{h}} = \log a_\gamma. \tag{20.4}$$

For a prescribed hierarchy use $\lambda_\gamma = \log w_\gamma$. On positive atoms set $\xi_\gamma = \nabla\lambda_\gamma$, $\bar{\xi} = \sum_\gamma W_\gamma \xi_\gamma$, and $T_\gamma(x) = \nabla \log p_\gamma(x)$. The Fisher tensor of a finite fixed-support law P_θ is

$$I(P_\theta) = \sum_{z:P_\theta(z)>0} P_\theta(z) S_\theta(z) S_\theta(z)^\top, \qquad S_\theta(z) = \nabla_\theta \log P_\theta(z). \tag{20.5}$$

A structural zero has no score and contributes no summand. The tensor is positive semidefinite. A Riemannian metric additionally requires positive definiteness. [DEFINITION]

Proposition 20.6 (Complete labeled Fisher decomposition). The latent score and Fisher tensor satisfy

$$S(\gamma, x) = \xi_\gamma - \bar{\xi} + T_\gamma(x), \tag{20.6}$$

$$I(\mathbb{P}^{\mathrm{lat}}) = \mathrm{Cov}_W(\xi_\gamma) + \sum_\gamma W_\gamma I(p_\gamma). \tag{20.7}$$

For every ordinary multi-index α with $|\alpha| \le 2$, the within-branch contribution differentiates as

$$\partial^\alpha \left(\sum_\gamma W_\gamma I(p_\gamma) \right) = \sum_\gamma \sum_{\beta \le \alpha} \binom{\alpha}{\beta} (\partial^\beta W_\gamma)(\partial^{\alpha-\beta} I(p_\gamma)). \tag{20.8}$$

A common parameter-dependent additive scalar in every λ_γ leaves these identities and their derivatives unchanged. [ESTABLISHED]

Proof. Differentiating the finite softmax gives $\nabla \log W_\gamma = \xi_\gamma - \bar{\xi}$. Differentiating $\sum_x p_\gamma(x) = 1$ on its fixed support gives $\sum_x p_\gamma(x) T_\gamma(x) = 0$. Expanding the outer product of eq. (20.6) and summing within each branch therefore removes both cross terms and yields eq. (20.7). The conditional score includes all varying factors:

$$T_\gamma(x) = \nabla \log \nu_\gamma(x) + \nabla \log s_\gamma + \nabla \log \widehat{r}_\gamma(x) - \nabla \log Z_\gamma. \tag{20.9}$$

Every positive denominator is locally bounded away from zero because the support is finite. The C^3 mass family consequently has a C^2 Fisher tensor, and the ordinary product rule gives eq. (20.8). Keeping only W_γ times a derivative of $I(p_\gamma)$ would omit terms. A common shift cancels from the normalized weights and adds the same vector to each ξ_γ, leaving its covariance unchanged. These are identities of parameter-dependent functions, so the same cancellation holds after differentiation. Finally, $v^\top I(P) v = \sum_z P(z)(v^\top S(z))^2 \ge 0$ proves semidefiniteness without an inverse or rank assumption. □

Definition 20.7 (Selected finite posterior). For a fixed observation o, define

$$Z_\gamma^o = \sum_{x \in \mathsf{X}_\gamma^+} \nu_\gamma(x) r_\gamma(x) \ell_\gamma(o \mid x), \qquad e_\gamma(o) = \frac{Z_\gamma^o}{Z_\gamma}, \qquad E_o = \sum_\gamma W_\gamma e_\gamma(o). \tag{20.10}$$

On a region where $E_o > 0$, the posterior is

$$\mathbf{\Pi}_o(\gamma, x) = \frac{W_\gamma p_\gamma(x) \ell_\gamma(o \mid x)}{E_o}, \qquad W_\gamma^o = \frac{W_\gamma e_\gamma(o)}{E_o}. \tag{20.11}$$

When $Z_\gamma^o > 0$, its conditional branch is $p_\gamma^o(x) = \nu_\gamma(x) r_\gamma(x) \ell_\gamma(o \mid x) / Z_\gamma^o$. When $Z_\gamma^o = 0$, that label remains on the declared carrier with zero posterior mass, but p_γ^o and its conditional Fisher tensor are undefined. Zero total evidence leaves the entire selected posterior undefined. [DEFINITION]

Proposition 20.8 (Posterior normalization and complete generative score). On occupied posterior labels, the unnormalized positive label masses are

$$e^{\lambda_\gamma^{o,\mathrm{f}}} = a_\gamma Z_\gamma^o, \qquad e^{\lambda_\gamma^{o,\mathrm{h}}} = a_\gamma \frac{Z_\gamma^o}{Z_\gamma}. \tag{20.12}$$

For a prescribed hierarchy replace a_γ in the second expression by w_γ. On the positive posterior support the complete score is

$$S^o(\gamma, x) = S^{\mathrm{lat}}(\gamma, x) + \nabla \log \ell_\gamma(o \mid x) - \nabla \log E_o. \tag{20.13}$$

Its labeled decomposition is eq. (20.7), applied to the occupied posterior labels and their conditional posterior laws. An empty label contributes zero weighted information and no assigned conditional score. [ESTABLISHED]

Proof. Summing eq. (20.11) over all labeled atoms gives one. Substitution of the two prior weights gives eq. (20.12); the common prior normalizer cancels, but the hierarchical branch divisor Z_γ remains. Taking the logarithm of the positive posterior mass and differentiating gives eq. (20.13). Within an occupied branch,

$$\nabla \log p_\gamma^o(x) = \nabla \log \nu_\gamma(x) + \nabla \log r_\gamma(x) + \nabla \log \ell_\gamma(o \mid x) - \nabla \log Z_\gamma^o. \tag{20.14}$$

The fixed zero pattern of the likelihood makes each unoccupied posterior fiber identically zero on the stratum. Thus its mass derivatives vanish; there is no division by zero. On occupied fibers the normalization proof of Proposition 20.6 applies verbatim to the actual conditional posterior, including the varying likelihood and evidence terms. □

For the native all-positive edge sensor of Proposition 19.7, write $A_\gamma^{\mathrm{sens}} = (1 + b_\gamma \eta_\gamma)/2$ and $b'_\gamma = (b_\gamma + \eta_\gamma)/(1 + b_\gamma \eta_\gamma)$. The finite identity and its branch normalizers are

$$(1 + bz)\frac{1 + \eta z}{2} = A^{\mathrm{sens}}(1 + b' z), \qquad Z_{\gamma,L}^o = (A_\gamma^{\mathrm{sens}})^{2V} Z_L(b'_\gamma, t_\gamma). \tag{20.15}$$

Here the scalar-free native raw factor is used; any additional physical scalar multiplies both its original and observed partitions. Substituting eq. (20.15) into eq. (20.12) retains $Z_L(b_\gamma, t_\gamma)$ in the hierarchical denominator. [ESTABLISHED]

Proof. Expand the first product using $z^2 = 1$, then multiply over the actual $2V$ free edge signs. Plaquette factors and their finite boundary inventory are unchanged, giving the second identity after summing against normalized counting. For $|b| < 1$ and $|\eta| < 1$, the scalar is positive and $1-(b')^2 = (1-b^2)(1-\eta^2)/(1+b\eta)^2 > 0$. Parameter-dependent sensor strengths remain inside these expressions when derivatives are taken. This finite calculation neither invokes an infinite all-positive event nor supplies a thermodynamic derivative theorem for an arbitrary positive likelihood. □

Definition 20.9 (Fixed stochastic channel retaining geometry). For each label choose a finite nonempty target carrier Y_γ and a nonnegative row-stochastic matrix $C_\gamma(x, y)$, independent of θ and recognition. Its lifted channel is zero between different geometry labels. In the right-acting convention of Definition 10.1, a labeled law P pushes to

$$P^C(\gamma, y) = \sum_x P(\gamma, x) C_\gamma(x, y). \tag{20.16}$$

The chosen source law can be the latent prior or a positive selected posterior. For the complete random observation joint, retain o as well and apply this channel only to x. Conditioning in the following score formula then uses (γ, y, o). [DEFINITION]

Proposition 20.10 (Stochastic score projection and retained-label loss). For an admitted source law P, its coarse score on positive targets and its Fisher loss satisfy

$$S^C(\gamma, y) = \mathbb{E}_{PC}[S(\gamma, X) \mid \gamma, y], \tag{20.17}$$

$$I(P) - I(P^C) = \mathbb{E}_{PC}\big[(S - S^C)(S - S^C)^\top\big] \succeq 0. \tag{20.18}$$

Every geometry marginal, including its parameter derivatives, is retained. If $P(\gamma, x) = W_\gamma p_\gamma(x)$ on its occupied fibers, then

$$I(P) - I(P^C) = \sum_{\gamma : W_\gamma > 0} W_\gamma \big[I(p_\gamma) - I(p_\gamma C_\gamma)\big]. \tag{20.19}$$

These are finite specializations of Theorem 10.7 and proposition 18.6. [ESTABLISHED]

Proof. For every positive target, differentiate eq. (20.16). The channel has no parameter derivative, so division by $P^C(\gamma, y)$ gives the conditional expectation in eq. (20.17). Under the augmented law $P(\gamma, x)C_\gamma(x, y)$, write $S = S^C + (S - S^C)$. The residual has zero conditional expectation given the target, so both cross terms in its second moment vanish. This proves eq. (20.18). A directional loss is zero exactly when the corresponding fine score equals its target conditional expectation almost surely.

Summing each channel row gives $\sum_y P^C(\gamma, y) = \sum_x P(\gamma, x) = W_\gamma$ as a function of θ. Hence every available derivative of the label marginal is unchanged. Its score is already measurable from the retained label and cancels from $S - S^C$; equivalently subtract the two complete labeled Fisher decompositions to obtain eq. (20.19). If a target has zero mass, nonnegativity forces every incidence into it from a positive source atom to vanish. It is consequently a structural zero, and no reverse probability or score is assigned to it. A channel varying with θ would contribute a channel derivative and is outside this proof. Positive-semidefinite loss at order zero does not impose a sign on its parameter derivatives. □

Proposition 20.11 (Transport of the complete observation experiment). The complete push preserves E_o and obeys $\mathbf{\Pi}_o^C = \mathbf{\Pi}_o C$. It uses the actual induced likelihood of Theorem 18.3, not an independently chosen likelihood on coarse states. Within a branch, on a reachable reference target, define the following quantities. Sums containing a raw factor or likelihood run over X_γ^+; the branch label is suppressed within these formulas:

$$\nu^C(y) = \sum_x \nu(x) C(x, y), \qquad r^C(y) = \frac{\sum_x \nu(x) r(x) C(x, y)}{\nu^C(y)}, \tag{20.20}$$

$$B_p(x \mid y) = \frac{p(x) C(x, y)}{p^C(y)}, \qquad \ell^C(o \mid y) = \sum_x B_p(x \mid y) \ell(o \mid x). \tag{20.21}$$

The physical partition stays Z. For $r = s\widehat{r}$, let $\widehat{r}^C = (\sum_x \nu(x) \widehat{r}(x) C(x, \cdot))/\nu^C$ on reachable targets. At a reachable anchor y_* set

$$\zeta_C = \widehat{r}^C(y_*), \qquad \widehat{r}_{\mathrm{new}} = \widehat{r}^C/\zeta_C, \qquad s_{\mathrm{new}} = s\zeta_C. \tag{20.22}$$

The physical product is unchanged, as in Proposition 18.4. [ESTABLISHED]

Proof. The reference sums to one, and positivity of the physical raw factor makes its reachable support the same as the prior's. Multiplying the first line by ν^C and summing gives $\sum_y \nu^C(y) r^C(y) = Z$. The prior reverse row B_p sums to one, so its averaged likelihood is normalized. Moreover,

$$p^C(y) \ell^C(o \mid y) = \sum_x p(x) \ell(o \mid x) C(x, y). \tag{20.23}$$

Multiply by the retained W_γ, sum over targets and labels, and use row normalization to obtain the same evidence E_o. Division by $E_o > 0$ proves posterior commutation. This is an identity of C^3 functions, so its existing derivatives agree as well. The reference reverse row proportional to $\nu(x) C(x, y)$ is generally different from B_p and cannot replace it in the likelihood. An anchor division changes the representative only when the compensating scalar is retained; their product recovers the original physical push. On a reference-null target both raw quotient and reverse row remain undefined. Such a target contributes zero to every pushed law, even if a subsequent channel has a nonzero row there. Finite-sum associativity proves composition along any fixed finite word. □

Definition 20.12 (Exact selected-observation free energy). At a fixed parameter and $E_o > 0$, let $\mathbb{Q}(\gamma, x) = q_\gamma Q_\gamma(x)$ be a recognition law on the complete labeled carrier. The conditional Q_γ is defined only when $q_\gamma > 0$. With $0 \log 0 = 0$ and a positive recognition mass over zero selected joint mass assigned infinite cost, define

$$\mathcal{F}_o(\mathbb{Q}) = \sum_{\gamma,x:\mathbb{Q}(\gamma,x)>0} \mathbb{Q}(\gamma, x) \log \frac{\mathbb{Q}(\gamma, x)}{\mathbb{P}(\gamma, x, o)}. \tag{20.24}$$

The exact ELBO is $-\mathcal{F}_o$. This is the same finite selected-joint objective as eqs. (18.4) and (18.5), before any restriction of the recognition family. [DEFINITION]

Proposition 20.13 (Full recognition entropy and the posterior form). On the finite dominated support $\mathbb{Q} \ll \mathbf{\Pi}_o$,

$$\mathcal{F}_o(\mathbb{Q}) = D_{\mathrm{KL}}(q\|W) + \sum_{\gamma:q_\gamma>0} q_\gamma D_{\mathrm{KL}}(Q_\gamma\|p_\gamma) - \sum_{\gamma:q_\gamma>0} q_\gamma \mathbb{E}_{Q_\gamma} \log \ell_\gamma(o \mid X) \tag{20.25}$$

$$= -\log E_o + D_{\mathrm{KL}}(\mathbb{Q}\|\mathbf{\Pi}_o). \tag{20.26}$$

No additional recognition-entropy or correlation-entropy term is required. [ESTABLISHED]

Proof. Insert the factors in eq. (20.3) into the logarithm in eq. (20.24). Split $\log \mathbb{Q} = \log q_\gamma + \log Q_\gamma(x)$ on positive recognition atoms and sum by label. This gives the first expression, including both parts of the single joint recognition entropy. Alternatively, use $\mathbb{P}(\gamma, x, o) = E_o \mathbf{\Pi}_o(\gamma, x)$ inside the same logarithm. Since $\sum \mathbb{Q} = 1$, its common scalar contributes $-\log E_o$ and gives the second expression. Domination makes every logarithm at a recognition-positive atom finite. If domination fails, the second expression is still valid as an extended value $+\infty$; no difference of infinities is used. □

Proposition 20.14 (Exact stochastic conditional-KL loss). Use the same fixed channel C for recognition and the selected posterior, and let $\mathbb{Q}^C = \mathbb{Q}C$. Define the coarse objective from the full induced observation experiment. If $\mathbb{Q} \ll \mathbf{\Pi}_o$, then

$$\mathcal{F}_o(\mathbb{Q}) - \mathcal{F}_o^C(\mathbb{Q}^C) = \sum_{\gamma,y:\mathbb{Q}^C(\gamma,y)>0} \mathbb{Q}^C(\gamma, y) D_{\mathrm{KL}}\big(\mathbb{Q}(X \mid \gamma, y)\|\mathbf{\Pi}_o(X \mid \gamma, y)\big) \geq 0. \tag{20.27}$$

The reverse conditionals are those of the augmented recognition and posterior laws. The loss vanishes exactly when they coincide on every recognition-positive target. This is the finite stochastic chain rule underlying Proposition 18.5. [ESTABLISHED]

Proof. Define the two augmented laws explicitly by

$$\overline{Q}(\gamma, x, y) = \mathbb{Q}(\gamma, x) C_\gamma(x, y), \qquad \overline{\Pi}(\gamma, x, y) = \mathbf{\Pi}_o(\gamma, x) C_\gamma(x, y). \tag{20.28}$$

On every $\overline{Q}$-positive pair, the common channel factor cancels from the likelihood ratio. Summing over y therefore gives $D_{\mathrm{KL}}(\overline{Q}\|\overline{\Pi}) = D_{\mathrm{KL}}(\mathbb{Q}\|\mathbf{\Pi}_o)$. For a positive coarse recognition atom, domination implies a positive coarse posterior atom, and the same ratio also factors as

$$\frac{\overline{Q}(\gamma,x,y)}{\overline{\Pi}(\gamma,x,y)} = \frac{\mathbb{Q}^C(\gamma,y)}{\mathbf{\Pi}_o^C(\gamma,y)}\frac{\mathbb{Q}(x \mid \gamma,y)}{\mathbf{\Pi}_o(x \mid \gamma,y)}. \tag{20.29}$$

Taking logarithms and summing yields the coarse KL plus the displayed conditional KL. The evidence term $-\log E_o$ is identical by Proposition 20.11, so eq. (20.26) gives eq. (20.27). For finite normalized probability vectors, $\log t \le t-1$ implies $D_{\mathrm{KL}}(q\|p) \ge 0$, with equality only for $q = p$; applying this on each occupied reverse fiber proves the sign and equality statements. Applying the same identity successively telescopes the finite-word loss on the same complete experiment. □

If fine recognition is not dominated but its coarse image is dominated, a violating fine atom reaches a positive coarse recognition atom whose posterior reverse fiber is defined and assigns zero probability to that fine atom. Its conditional KL is infinite. The additive identity $\mathcal{F}_o(\mathbb{Q}) = \mathcal{F}_o^C(\mathbb{Q}^C) +$ conditional KL then remains an extended-value identity. If coarse recognition instead charges a posterior-null target, both fine and coarse VFE are infinite, the posterior fiber at that target is undefined, and their subtraction is not a numerical loss. A target with zero coarse recognition weight contributes zero without defining its recognition reverse fiber. These cases are separate from zero evidence, where no selected posterior exists. [ESTABLISHED]

Stochastic loss and deterministic minimization have different types. The preceding identity compares one fine recognition law and its actual stochastic image. It does not assert

$$\mathcal{F}_o^C(q') = \inf_{\mathbb{Q}:\mathbb{Q}C=q'} \mathcal{F}_o(\mathbb{Q}) \tag{20.30}$$

for an arbitrary stochastic channel. For a deterministic retained-label statistic c, a prescribed $q' \ll c_{\#}\mathbf{\Pi}_o$ does admit the fine law $\mathbb{Q}_*(\gamma,x) = q'(\gamma,c_\gamma(x))\mathbf{\Pi}_o(x \mid \gamma,c_\gamma(x))$. Define this expression to be zero when $q'(\gamma,c_\gamma(x)) = 0$, without evaluating a posterior fiber at a null target. Its deterministic image is q' and its occupied reverse fibers equal the posterior fibers, so the chain rule gives equality with the coarse objective and proves the corresponding deterministic infimum statement when all such fine probability laws are allowed. A restricted recognition family need not contain this lift. For a stochastic channel, prescribing the output and choosing posterior reverse fibers need not define a joint with the required conditional channel C given the fine input. [ESTABLISHED]

Finite witness. Take one geometry label, a one-point observation alphabet with likelihood one, $\mathbf{\Pi}_o = (1/2,1/2)$, and

$$C = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad \mathbb{Q} = (1,0), \qquad q' = \mathbb{Q}C = (3/4,1/4). \tag{20.31}$$

The matrix has determinant $1/2$, so $\mathbb{Q}$ is the unique fine probability vector with output q'. Its VFE is $\log 2$, since $E_o = 1$, whereas the coarse VFE is $(3/4)\log(3/2) + (1/4)\log(1/2)$. Their difference

is $2\log 2 - (3/4)\log 3 > 0$, because $2^8 > 3^3$. Thus eq. (20.30) is false for this positive stochastic channel even though the exact conditional-KL loss identity holds. No failure of normalization or discarded geometry label is involved. □ [ESTABLISHED]

20.2 Volume limits and exact renormalization

The arguments in Appendix C apply to actual finite laws. For the native model in Definition 19.1, the source map $\chi_\gamma : \Theta \to \mathbb{R}^2$, with $\chi_\gamma(\theta) = (h_\gamma(\theta), j_\gamma(\theta))$, is C^4 and has image in a compact positive high-temperature box. The native pressure theorem obtains C^4 convergence from local response, and its Fisher consequence is

$$\frac{I(p_{\gamma,L,\theta})}{V} \longrightarrow g_\gamma(\theta) = (D\chi_\gamma)^\top (D^2 \mathsf{A} \circ \chi_\gamma) D\chi_\gamma \quad \text{in } C^2, \qquad V = L^2. \tag{20.32}$$

Here A is the natural log-partition density. The raw-factor pressure differs by the explicit edge and plaquette normalization terms in Theorem C.6. The pullback in Proposition C.7 therefore does not identify Fisher information with a Hessian of the raw pressure after a nonlinear source substitution. The separate free-boundary theorem Theorem C.11 supplies the derivative limit for actual fixed-word pushes. Neither a matching bulk pressure nor a surface bound on log-density ratios supplies that conclusion by itself; Proposition C.15 gives a counterexample to that inference. [ESTABLISHED]

For the retained-label family, write its actual effective log masses as $\lambda_{\gamma,L}$ and use the hypotheses of Equation (C.66): $\lambda_{\gamma,L}/V \to s_\gamma$ in C^3, with C^3 norms of the log masses bounded by a constant times V on each admitted compact neighborhood. The global-factor convention includes $\log Z_{\gamma,L}$ in $\lambda_{\gamma,L}$; the prescribed hierarchy does not. A common label-independent representative cancels from the normalized weights and their scores. These choices affect the bulk functions and therefore which branch dominates. If one fixed label has a uniform positive gap in s_γ over the other labels, and all actual branch densities converge in C^2, then Theorem C.19 gives

$$\frac{I(P_{L,\theta})}{V} \longrightarrow g_{\gamma_*}(\theta) \quad \text{in } C^2. \tag{20.33}$$

This remains a semimetric statement unless the limiting tensor has a positive lower bound on the declared chart. Only on such charts do the inverse metric, Levi-Civita connection, and Fisher curvature inherit the convergence proved in Proposition C.20. A two-source branch pulled back into a higher-dimensional parameter space does not acquire a full-rank metric from this argument. [ESTABLISHED]

At a tie θ_0, actual finite-volume weights are additional data. If they converge to w_γ, let $v_\gamma = \nabla s_\gamma(\theta_0)$. Under the branch $O(V)$ bound, Equation (C.83) gives

$$\frac{I(P_{L,\theta_0})}{V^2} \longrightarrow \mathrm{Cov}_w(v_\gamma). \tag{20.34}$$

A positive covariance direction has order-V^2 information. A zero direction implies only $o(V^2)$; it does not imply an order-V bound. Residual log-weight gradients and rare labels enter before score squaring. The examples in Section C.2 show why bulk pressure data alone cannot determine their contribution or the existence of the tie-weight limit. [ESTABLISHED]

A different experiment uses $u = V(\theta - \theta_0)$. Assume the actual center ratios in Equation (C.97) converge to positive κ_γ on the tied set S, and retain the branch C^2 and effective-log-mass C^3 hypotheses. Then Theorem C.25 identifies

$$\pi_\gamma(u) = \frac{\kappa_\gamma e^{u\cdot v_\gamma}}{\sum_{\eta\in S}\kappa_\eta e^{u\cdot v_\eta}}, \qquad \gamma \in S, \tag{20.35}$$

$$I_L^{(u)}(u) = V^{-2} I(P_{L,\theta_0+u/V}) \longrightarrow \mathrm{Cov}_{\pi(u)}(v_\gamma) \quad \text{in } C^2. \tag{20.36}$$

The appendix constructs the comparison kernels for this experiment and its identifiable quotient. It also treats an actual moving center for a transverse two-label tie. The reciprocal-volume chart is not a smooth extension of the extensive density across every tie, nor a universal physical time parameter. [ESTABLISHED]

Exact renormalization retains one normalized fine experiment, its reference, its observation, and its declared recognition law. For a fixed word of total block factor B, the fine and coarse cell counts are $M = B^2V$. The finite score and conditional-KL identities above therefore give the scale accounting in Section C.3: an extensive density carries B^2, while a density normalized by squared volume carries B^4. The denominator, parameter coordinate, and observed experiment must be the same on both sides of each comparison. Every limiting argument keeps the word fixed before volume tends to infinity. The exact finite VFE difference is a posterior conditional relative entropy; a general stochastic channel does not turn it into an equality of variational infima. [ESTABLISHED]

Prescribed geometry actions change the declared raw weights. Their relative center values, as well as their volume orders and derivative bounds, govern the resulting label experiments. Gauge connection curvature, statistical Fisher curvature, and curvature of the base manifold remain distinct inputs with separate metric and measure conventions. This mapping does not solve a self-consistent curvature action. Unbounded-depth regular inverses, critical or continuum limits, arbitrary local-law gluing, and physical identification require additional constructions outside the present chapter. [ESTABLISHED]

Part IV

The Multivariate-Gaussian Realization

Chapter 21

A Linear-Gaussian Generative Realization

This chapter realizes the fixed directed law of Chapter 4 in two associated bundles induced from the same principal G-bundle, using possibly different representations and separately chosen local frames. The Gaussian is a tractable subclass, not the ambient theory. The specialization assumes that at a root the model-to-belief kernel induces a map from model laws to belief laws that may realize $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$. The sample matrix inside that kernel is not itself the associated-bundle morphism. At a nonroot the kernel is parent-indexed unless its domain is enlarged. The existence of $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ creates no reverse conditional.

21.1 Directed forest and conditional laws

Take

$$k_i \in \mathbb{R}^K, \qquad m_i \in \mathbb{R}^{d_m}, \qquad o_i \in \mathbb{R}^{d_{o,i}}, \tag{21.1}$$

and specialize the directed acyclic graph to a forest, so each nonroot i has one parent $j = \mathrm{par}(i)$. This is a displayed subclass; the general DAG replaces each parent term by a finite sum. [DEFINITION]

Fix one design point c_a. Let $\Theta^b_{ij} := \Theta^b_{a,ij} \in G$ and $\Theta^m_{ij} := \Theta^m_{a,ij} \in G$ abbreviate the design-indexed same-channel graph-link copies of Proposition 4.7. A submodel that constrains one common link across all design points retains the stabilizer of that constraint, namely the frame rechoices for which $h^x_{a,i}\Theta^x_{ij}(h^x_{a,j})^{-1}$ is independent of a. Context-independent rechoices form a sufficient, but not generally maximal, subgroup; any larger law must be declared explicitly. For a nonroot,

$$m_i \mid m_j, X \sim \mathcal{N}\left(F^m_i \rho_m(\Theta^m_{ij}) m_j + \eta^m_i, \Sigma^m_i\right), \tag{21.2}$$

$$k_i \mid k_j, m_i, X \sim \mathcal{N}\left(F^b_i \rho_b(\Theta^b_{ij}) k_j + B_i m_i + \eta^b_i, \Sigma^b_i\right), \tag{21.3}$$

$$o_i \mid k_i, m_i, X \sim \mathcal{N}\left(H^b_i k_i + H^m_i m_i + \eta^o_i, \Sigma^o_i\right). \tag{21.4}$$

The parameter shapes are

$$\rho_b(\Theta^b_{ij}), F^b_i \in \mathbb{R}^{K\times K}, \quad \rho_m(\Theta^m_{ij}), F^m_i \in \mathbb{R}^{d_m\times d_m}, \quad B_i \in \mathbb{R}^{K\times d_m}, \tag{21.5}$$

$$H^b_i \in \mathbb{R}^{d_{o,i}\times K}, \quad H^m_i \in \mathbb{R}^{d_{o,i}\times d_m}, \quad \eta^b_i \in \mathbb{R}^K, \quad \eta^m_i \in \mathbb{R}^{d_m}, \quad \eta^o_i \in \mathbb{R}^{d_{o,i}}. \tag{21.6}$$

The represented links are invertible. The conditional endomorphisms F^b_i, F^m_i need not be: positive definiteness of the receiving covariance, not invertibility of the mean map, normalizes a Gaussian kernel.

The bridge matrix $B_i : \mathbb{R}^{d_m} \to \mathbb{R}^K$ acts on sample coordinates. It is one parameter of a Gaussian kernel, not a map between probability-law fibers. At a nonroot, the kernel also depends on the parent belief coordinate. Only after fixing that parent, or enlarging the domain to include it, does integration against a model law define a belief law. The left-right covariance law below is a parameter transformation that makes this induced law map covariant; it does not identify B_i with $\widetilde{\Phi}$. No reading supplies a belief-to-model factor.

Each root r has the proper laws

$$m_r \mid X \sim \mathcal{N}(\bar{m}_r, \Sigma^m_{r,0}), \tag{21.7}$$

$$k_r \mid m_r, X \sim \mathcal{N}(B_r m_r + \eta^b_{r,0}, \Sigma^b_{r,0}), \tag{21.8}$$

followed by (21.4). Every receiving covariance is positive definite:

$$\Sigma^m_i, \Sigma^m_{r,0} \in \mathbb{S}^{d_m}_{++}, \qquad \Sigma^b_i, \Sigma^b_{r,0} \in \mathbb{S}^K_{++}, \qquad \Sigma^o_i \in \mathbb{S}^{d_{o,i}}_{++}. \tag{21.9}$$

[HYPOTHESIS]

Proposition 21.1 (Normalization and root moments). Under (21.9), every kernel above is a nondegenerate normalized density and the assembled directed law is a probability measure. The root belief coordinate has

$$\bar{k}_r = B_r\bar{m}_r + \eta^b_{r,0}, \qquad \Sigma^{b,\mathrm{marg}}_{r,0} = \Sigma^b_{r,0} + B_r\Sigma^m_{r,0}B_r^\top. \tag{21.10}$$

[ESTABLISHED]

More generally, if a root model law is $s = \mathcal{N}(\mu_m, C_m)$, then the law map induced by the root kernel is

$$\widetilde{\phi}_r(s) = \mathcal{N}\left(B_r\mu_m + \eta^b_{r,0}, \quad \Sigma^b_{r,0} + B_r C_m B_r^\top\right). \tag{21.11}$$

This map, rather than B_r alone, is the candidate local realization of $\widetilde{\Phi}$. [ESTABLISHED]

Proof. A Gaussian density with positive definite covariance integrates to one for every mean, so the normalization proposition of Chapter 4 applies. The tower property gives the first moment and the law of total covariance gives the second identity. Its added term is positive semidefinite, so the marginal covariance dominates the conditional covariance. □

The Gaussian observation kernel may be replaced by any normalized Markov kernel from $\mathbb{R}^K \times \mathbb{R}^{d_m}$ to the observation space. That changes the likelihood but not the exact ELBO identity, which uses only the fixed normalized joint. [ESTABLISHED]

21.2 Independent channel-frame covariance

For the global coordinate formulas in this section, invoke Hypothesis 2.11 on a declared region $\mathcal{C}_0$ containing the contexts under study. Choose $\sigma_0 : \mathcal{C}_0 \to \mathscr{P}_G|_{\mathcal{C}_0}$ and restrict each local frame to $\mathcal{C}_i \cap \mathcal{C}_0$. Then write the independent frame coordinates as

$$U_i^b : \mathcal{C}_i \cap \mathcal{C}_0 \to G, \qquad U_i^m : \mathcal{C}_i \cap \mathcal{C}_0 \to G, \qquad u_i^x = \sigma_0 \cdot (U_i^x)^{-1}. \tag{21.12}$$

Thus σ_0 is common only on the explicitly trivialized region; no global triviality of $\mathscr{P}_G$ is inferred. The corresponding relative principal-frame field is

$$h_i = U_i^b (U_i^m)^{-1}, \qquad u_i^m = u_i^b h_i. \tag{21.13}$$

This G-valued field exists even when ρ_b and ρ_m are inequivalent or have different dimensions. Its represented actions $\rho_b(h_i)$ and $\rho_m(h_i)$ live in different endomorphism spaces and do not define the bridge B_i or either associated-bundle morphism. An independent pair of channel-frame re-expressions is $(g_i^b, g_i^m) \in G \times G$, with represented matrices $R_i^b = \rho_b(g_i^b)$ and $R_i^m = \rho_m(g_i^m)$ as in (4.23). The pair records two choices of local trivialization of bundles associated to the same $\mathscr{P}_G$; it does not replace the principal structure group G by $G \times G$. In the convention of (2.10), the corresponding principal-section rechoices are $u_i^{b\prime} = u_i^b \cdot (g_i^b)^{-1}$ and $u_i^{m\prime} = u_i^m \cdot (g_i^m)^{-1}$. This accounts for the opposite-sided inverse in the geometric local-map law and makes the formulas below identical to (2.15).

The same-channel links obey (4.25). The conditional maps and offsets transform as

$$F_i^{b\prime} = R_i^b F_i^b (R_i^b)^{-1}, \qquad F_i^{m\prime} = R_i^m F_i^m (R_i^m)^{-1}, \tag{21.14}$$

$$B_i' = R_i^b B_i (R_i^m)^{-1}, \qquad \eta_i^{b\prime} = R_i^b \eta_i^b, \qquad \eta_i^{m\prime} = R_i^m \eta_i^m. \tag{21.15}$$

The left-right law for B_i, together with the covariance and offset laws, makes the induced root law map (21.11) obey the local law of $\widetilde{\Phi}$. The parameter B_i alone is not that morphism, and the two frames remain independent. Receiving covariances transform by their own congruences:

$$\Sigma_i^{b\prime} = R_i^b \Sigma_i^b (R_i^b)^\top, \qquad \Sigma_i^{m\prime} = R_i^m \Sigma_i^m (R_i^m)^\top. \tag{21.16}$$

For either channel,

$$\Lambda_i = \Sigma_i^{-1}, \qquad \Sigma_i' = R_i \Sigma_i R_i^\top, \qquad \Lambda_i' = R_i^{-\top} \Lambda_i R_i^{-1}. \tag{21.17}$$

At a root,

$$\bar{m}'_r = R^m_r \bar{m}_r, \qquad \Sigma^{m\prime}_{r,0} = R^m_r \Sigma^m_{r,0} (R^m_r)^\top, \tag{21.18}$$

$$B'_r = R^b_r B_r (R^m_r)^{-1}, \qquad \eta^{b\prime}_{r,0} = R^b_r \eta^b_{r,0}, \qquad \Sigma^{b\prime}_{r,0} = R^b_r \Sigma^b_{r,0} (R^b_r)^\top. \tag{21.19}$$

Gauge-inert observations require

$$H^{b\prime}_i = H^b_i (R^b_i)^{-1}, \qquad H^{m\prime}_i = H^m_i (R^m_i)^{-1}, \qquad \eta^{o\prime}_i = \eta^o_i, \qquad \Sigma^{o\prime}_i = \Sigma^o_i. \tag{21.20}$$

[DEFINITION]

Proposition 21.2 (Gaussian kernel covariance). The transformation laws above satisfy the three kernel pushforward identities (4.28)–(4.30), and hence the normalized joint and its evidence obey (4.32)–(4.33) under all independent channel-frame re-expressions. [ESTABLISHED]

Proof. For the belief kernel, the transformed conditional mean evaluated at transformed inputs is

$$F^{b\prime}_i \rho_b(\Theta^{b\prime}_{ij})(R^b_j k_j) + B'_i (R^m_i m_i) + \eta^{b\prime}_i = R^b_i \left(F^b_i \rho_b(\Theta^b_{ij}) k_j + B_i m_i + \eta^b_i \right). \tag{21.21}$$

The model-channel calculation is the same without a bridge. Congruence (21.16) therefore makes each transformed Gaussian the receiving-frame pushforward of the original. At the observation kernel,

$$H^{b\prime}_i (R^b_i k_i) + H^{m\prime}_i (R^m_i m_i) + \eta^{o\prime}_i = H^b_i k_i + H^m_i m_i + \eta^o_i, \tag{21.22}$$

and its covariance is fixed. The root calculation is identical. Composition now invokes the frame-covariance result of Chapter 4. □

Equation (21.10) transforms by belief-channel congruence, because both of its summands do. Positive definiteness is preserved independently in each channel.

Group-valued frame coordinates remain primary. An expression $U = \exp \phi$ is valid only inside an exponential chart; for example, $\mathrm{diag}(-1, -2) \in \mathrm{GL}^+(2)$ has no real logarithm because each negative eigenvalue occurs in one Jordan block [26]. [ESTABLISHED]

21.3 Common-frame specialization

Under the common-frame specialization of Section 2.11, impose $u^b_i = u^m_i$ and use one frame re-expression $g_i \in G$ with

$$R^b_i = \rho_b(g_i), \qquad R^m_i = \rho_m(g_i). \tag{21.23}$$

If the graph links are also identified, Θ^b_{ij} and Θ^m_{ij} are the same group link seen through the two representations. Equations (21.14)–(21.20) then reduce to the familiar common-frame, two-representation formulas. [ESTABLISHED]

This reduction ties the gauge choices but does not identify the latent dimensions, make B_i invertible, turn Φ and $\widetilde{\Phi}$ into inverses, or add a reverse generative factor. Compatibility of a cross map with the

two connections remains the parallelness condition of Chapter 2; an equal connection is a further hypothesis.

Finally, the assembled Gaussian precision need not have the Laplacian-plus-self-terms form used later. A general directed transition can contribute an off-diagonal block that is neither symmetric nor sign-correct. The interaction family is therefore a separately declared Gaussian operator subfamily, not a consequence of linear-Gaussian directedness. [NOT-CLAIMED]

Chapter 22

The Gaussian Information Form and the Interaction Family

The preceding chapters first allow general belief-law and model-law fibers, with smooth manifolds selected only at the differential tier, and then identify regular exponential families as a tractable dominated subclass. Nothing in that construction selects the multivariate Gaussian. This chapter makes that additional choice. It develops one finite-dimensional Gaussian realization in information form, where the primitive coordinates are a vector and a matrix of natural parameters rather than a mean and a covariance, and it then isolates the still narrower subfamily of Gaussian precisions on which the renormalization half of this document acts. None of the matrix identities below is asserted for an arbitrary belief fiber.

Two things are worth separating at the outset, because conflating them is the standing hazard of the whole subject. Writing a Gaussian in information form is a change of coordinates and costs nothing. Restricting the precision to the Laplacian-plus-self-terms shape of Section 22.3 is a substantive restriction that costs a great deal, and it does not follow from the generative constructions of the preceding part. Section 22.4 proves that it does not, exhibits the exact obstruction, and then states precisely the parameter condition under which the shape is recovered. The restriction is carried forward as a declared hypothesis, and every later result that uses it says so.

Throughout, the development is written for the state sector, with agents $i \in V = \{1, \dots, N\}$, state latents $k_i \in \mathbb{R}^K$, and stacked latent Y of dimension $n := NK$. The model sector, with latents $m_i \in \mathbb{R}^{d_m}$, is identical with K replaced by d_m, and a partition mixing the two sectors is admissible wherever a partition is required. The design $D = \{c_a\}_{a=1}^M$ plays no role in the algebra below; either fix one design point, or read the index i as ranging over agent-design pairs, and every statement stands unchanged.

22.1 The recognition family in information form

Let ν be Lebesgue measure on $\mathbb{R}^n$. The Gaussian family in *information form* is parameterized by a vector $h \in \mathbb{R}^n$ and a symmetric positive definite matrix $J \in \mathbb{S}^n_{++}$ through the density

$$q(y \mid h, J) = \exp\left(h^\top y - \tfrac{1}{2} y^\top J y - \mathsf{A}(h, J)\right), \qquad y \in \mathbb{R}^n. \tag{22.1}$$

The pair (h, J) are the natural parameters, J is the precision, and A is the log normalizer. The symbol A is reserved for the log normalizer and never for a self term; the self terms of Section 22.3 are written A_i with a subscript and are $K \times K$ matrices.

Proposition 22.1 (Log normalizer, moments, and convexity). For every $(h, J) \in \mathbb{R}^n \times \mathbb{S}^n_{++}$ the density in Equation (22.1) is normalizable, with

$$\mathsf{A}(h, J) = \tfrac{1}{2} h^\top J^{-1} h - \tfrac{1}{2} \log\det J + \tfrac{n}{2} \log 2\pi. \tag{22.2}$$

Its differential is

$$d\mathsf{A} = \mu^\top dh - \tfrac{1}{2} \operatorname{Tr}\left[\left(C + \mu\mu^\top\right) dJ\right], \qquad \mu = J^{-1}h, \qquad C = J^{-1}, \tag{22.3}$$

so the mean and covariance of $q(\cdot \mid h, J)$ are μ and C, and the Hessian of A in h at fixed J is C. Regarded as a function of the natural parameter pair (h, Θ) with $\Theta = -\frac{1}{2}J$, the log normalizer is strictly convex on the open convex domain $\mathbb{R}^n \times \{\Theta \prec 0\}$. [ESTABLISHED]

Proof. Complete the square in the exponent,

$$h^\top y - \tfrac{1}{2} y^\top J y = -\tfrac{1}{2} \left(y - J^{-1}h\right)^\top J \left(y - J^{-1}h\right) + \tfrac{1}{2} h^\top J^{-1} h, \tag{22.4}$$

and integrate the Gaussian kernel, whose value is $(2\pi)^{n/2}(\det J)^{-1/2}$ because $J \succ 0$. Exponentiating and taking logarithms gives Equation (22.2). Differentiating Equation (22.2) termwise, $d(\frac{1}{2} h^\top J^{-1} h) = h^\top J^{-1} dh - \frac{1}{2} \operatorname{Tr}(J^{-1} h h^\top J^{-1} dJ)$ and $d(-\frac{1}{2} \log\det J) = -\frac{1}{2} \operatorname{Tr}(J^{-1} dJ)$, which is Equation (22.3). The identification of $\nabla_h \mathsf{A}$ with the mean and $\nabla^2_h \mathsf{A}$ with the covariance is the standard exponential-family cumulant property, valid here because the family is regular: the domain is open and differentiation under the integral is licensed on the interior of the natural domain [20]. Convexity is the convexity of a log-Laplace transform, and strictness follows because the family is minimal, the statistics $(y, yy^\top)$ being affinely independent [20, 87]. Since $J \mapsto -\frac{1}{2}J$ is affine, convexity in (h, Θ) transfers to convexity in (h, J) on $\{J \succ 0\}$. □

Two consequences of Equation (22.3) organize everything that follows. The first is that (h, J) and (μ, C) are two coordinate systems on one family, related by the Legendre correspondence of a minimal exponential family, so no information is lost by working in either. The second is that the Fisher information of the family in the natural parameter h is exactly $C = J^{-1}$, while the Fisher information in the mean parameter μ at fixed covariance is exactly J. The precision is therefore simultaneously a coupling matrix and a metric, and the renormalization chapters use both readings.

22.2 What an off-diagonal precision block is

Let $\mathfrak{B}$ be a finite partition of the n coordinates into blocks, with block dimensions n_b summing to n, and write J_{bc} for the corresponding submatrices. The meaning of J_{bc} for $b \neq c$ is fixed by the

following statement, which is standard but is stated here in full because the rest of the document depends on reading it correctly.

Proposition 22.2 (Conditional reading of the precision). Let $Y \sim \mathcal{N}(\mu, J^{-1})$ with $J \succ 0$. For any block b,

$$\mathbb{E}\left[Y_b \mid Y_{-b} = y_{-b}\right] = \mu_b - J_{bb}^{-1} \sum_{c \neq b} J_{bc}\,(y_c - \mu_c)\,, \qquad \operatorname{Cov}\left(Y_b \mid Y_{-b}\right) = J_{bb}^{-1}. \tag{22.5}$$

For any two distinct blocks b and c, the variables Y_b and Y_c are conditionally independent given all remaining coordinates if and only if $J_{bc} = 0$. [ESTABLISHED]

Proof. Fix y_{-b} and read the exponent of Equation (22.1) as a function of y_b alone. It is $-\frac{1}{2} y_b^{\top} J_{bb} y_b$ plus a linear term, so the conditional law is Gaussian with precision J_{bb}, and completing the square in y_b around μ gives the displayed conditional mean. For the second claim, condition on the coordinates outside $b \cup c$ and repeat the argument: the conditional law of (Y_b, Y_c) is Gaussian with precision $\left(\begin{smallmatrix} J_{bb} & J_{bc} \\ J_{cb} & J_{cc} \end{smallmatrix}\right)$, and a Gaussian density factorizes into a function of y_b times a function of y_c if and only if the quadratic form has no cross term, that is if and only if $J_{bc} = 0$. This is the Gaussian case of the pairwise Markov property [54, Ch. 5]; the same statement organizes the Gaussian Markov random field literature [77, Ch. 2]. □

Equation (22.5) also fixes the sign convention that Section 22.3 will exploit. A negative semidefinite off-diagonal block $J_{bc} \preceq 0$ makes the conditional mean of Y_b move *toward* y_c as y_c departs from its own mean, so negative off-diagonal precision is attractive coupling. The interaction family is the family in which every off-diagonal block is attractive in this sense.

Precision blocks are not covariance values. A zero off-diagonal precision block is a statement about conditional dependence and carries no implication about the marginal covariance. The tridiagonal example

$$J = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad J^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \tag{22.6}$$

has $J_{13} = 0$ and $(J^{-1})_{13} = \frac{1}{4} \neq 0$. In general the marginal covariance of a block is the inverse of a Schur complement,

$$\left(J^{-1}\right)_{bb} = \left(J_{bb} - J_{b,-b} J_{-b,-b}^{-1} J_{-b,b}\right)^{-1}, \tag{22.7}$$

so it records the effect of every path through the remaining coordinates, whereas J_{bb}^{-1} records only the conditional dispersion of Equation (22.5) [92]. The two coincide exactly when $J_{b,-b} = 0$. [ESTABLISHED]

The distinction exhibited by (22.6) is the reason this document works in information form. A restriction imposed on the recognition family is a restriction on conditional dependence, and it is stated cleanly only in coordinates in which conditional dependence is a coordinate. The exact price is computed in Chapter 23.

22.3 The declared interaction subfamily

Definition 22.3 (Interaction family). Let $\Lambda \in \mathbb{S}^{NK}$ be partitioned into $K \times K$ blocks indexed by agents. Say that Λ belongs to the *interaction family* $\mathcal{I}(V)$ when there exist matrices $A_i \in \mathbb{S}^K_+$ and $W_{ij} = W_{ji} \in \mathbb{S}^K_+$ for $i \neq j$ with

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \qquad \Lambda_{ij} = -W_{ij} \quad (i \neq j). \tag{22.8}$$

The A_i are the *self terms*, the W_{ij} the *edge weights*, and $L := \Lambda - \mathrm{blockdiag}(A_1, \ldots, A_N)$ the *Laplacian part*. Nothing is proved by this definition; it declares a type. [DEFINITION]

The edge weights are matrices, so L is a matrix-weighted graph Laplacian rather than a scalar-weighted one. That object, and the ways in which it fails to behave like its scalar counterpart, are the subject of the matrix-weighted consensus literature [81], and one of those failures appears immediately below.

Proposition 22.4 (Energy and kernel). For $\Lambda \in \mathcal{I}(V)$ and any $y = (y_1, \ldots, y_N)$,

$$y^\top \Lambda y = \sum_{i \in V} y_i^\top A_i y_i + \sum_{i<j} (y_i - y_j)^\top W_{ij} (y_i - y_j). \tag{22.9}$$

Consequently $\Lambda \succeq 0$ for every admissible choice of parameters; the consensus subspace $\mathbf{1} \otimes \mathbb{R}^K$ lies in $\ker L$; and $\Lambda \succ 0$ if and only if the only y satisfying $A_i y_i = 0$ for all i together with $W_{ij}(y_i - y_j) = 0$ for all $i < j$ is $y = 0$. A necessary condition for $\Lambda \succ 0$ is $\sum_i A_i \succ 0$. [ESTABLISHED]

Proof. Expand the right-hand side of Equation (22.9). The self sum contributes $\sum_i y_i^\top A_i y_i$ to the diagonal blocks. Each unordered pair contributes $y_i^\top W_{ij} y_i + y_j^\top W_{ij} y_j - 2 y_i^\top W_{ij} y_j$, which places W_{ij} on the ii and jj diagonal blocks and $-W_{ij}$ on the ij and ji off-diagonal blocks. Summing reproduces Equation (22.8). Every term of Equation (22.9) is nonnegative because $A_i \succeq 0$ and $W_{ij} \succeq 0$, so $\Lambda \succeq 0$; and a sum of nonnegative terms vanishes only when each vanishes, which for a positive semidefinite matrix A and vector v means $v^\top A v = 0$ if and only if $Av = 0$. Taking $y = \mathbf{1} \otimes v$ makes every difference vanish, so the consensus subspace lies in $\ker L$, and evaluating Equation (22.9) there gives $v^\top (\sum_i A_i) v$, whence the necessary condition. □

The kernel criterion in Proposition 22.4 is where the matrix-weighted setting departs from the scalar one. In the scalar case a connected interaction graph forces $\ker L$ to be exactly the consensus subspace. With matrix weights, a singular W_{ij} leaves the difference $y_i - y_j$ free along $\ker W_{ij}$, so $\ker L$ can be strictly larger than the consensus subspace even on a connected graph, and connectivity alone is not a definiteness criterion [81]. The criterion stated in the proposition is exact and replaces it.

Proposition 22.5 (The interaction family is thin). For $K \geq 2$ and $N \geq 2$, $\mathcal{I}(V)$ is contained in a proper linear subspace of $\mathbb{S}^{NK}$ of codimension at least $\binom{N}{2}\frac{K(K-1)}{2}$, hence has Lebesgue measure zero in $\mathbb{S}^{NK}$. [ESTABLISHED]

Proof. Symmetry of a matrix Λ already forces $\Lambda_{ji} = \Lambda_{ij}^{\top}$ for every ordered pair, so this identity carries no information about the family. Membership additionally requires each off-diagonal block Λ_{ij} to be symmetric *as a* $K \times K$ *matrix*, since $W_{ij} = W_{ji}$ and $W_{ji} = W_{ij}^{\top}$ together give $W_{ij} = W_{ij}^{\top}$. That is $\frac{K(K-1)}{2}$ independent linear equations per unordered pair, and there are $\binom{N}{2}$ pairs. The remaining requirements are conic inequalities, which can only shrink the set further. □

Proposition 22.5 is the quantitative form of the claim that the interaction shape is a restriction in a fixed product chart rather than a normalization. It is not invariant under arbitrary independent block-diagonal coordinate changes, so membership may be created or destroyed by such a change and no coordinate-orbit exclusion follows from thinness. An exact witness starts from the interaction precision with $N = 2$, $K = 2$, $A_1 = A_2 = I_2$, and $W_{12} = I_2$. Congruence by $G = \mathrm{diag}(I_2, S)$, $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, changes the off-diagonal block from $-I_2$ to $-S$, which is not symmetric and hence is outside the family; applying G^{-1} to that transformed precision creates membership again. The invariant statement is only that a declared trivialization must be fixed before Equation (22.8) is imposed. Section 22.4 asks when the generative construction supplies such a distinguished chart.

22.4 The interaction form is a hypothesis and not a consequence

22.4.1 Trivialized coordinates and flat comparison

In this Gaussian belief-channel subsection, write $\Theta_{ij} := \rho_b(\Theta_{ij}^b) \in \mathrm{GL}^+(K)$ for the represented graph link. This abbreviation does not identify it with a base-manifold parallel transport.

The state latents carry frames. A residual between two agents is not $k_i - k_j$, which compares vectors in different fibers, but $k_i - \Theta_{ij}k_j$, where Θ_{ij} is the represented belief-channel graph link. The interaction shape of Definition 22.3 involves plain differences, so it can only be a statement in coordinates in which the graph links have been removed. When that removal is possible is settled exactly by the following.

Proposition 22.6 (Simultaneous trivialization is exactly coboundary). Let $G = (V, E)$ be a connected graph and let $\{\Theta_{ij}\}_{(i,j)\in E} \subset \mathrm{GL}^+(K)$ satisfy $\Theta_{ji} = \Theta_{ij}^{-1}$. There exist frames $\{U_i\} \subset \mathrm{GL}^+(K)$ with $U_i^{-1}\Theta_{ij}U_j = I$ for every edge if and only if $\Theta_{ij} = U_iU_j^{-1}$ for those frames, and this holds if and only if the ordered product of graph links around every cycle of G is the identity. When it holds, the frames are unique up to a common right factor $U_i \mapsto U_ig$ with $g \in \mathrm{GL}^+(K)$ independent of i. [ESTABLISHED]

Proof. The condition $U_i^{-1}\Theta_{ij}U_j = I$ is literally $\Theta_{ij} = U_iU_j^{-1}$, which gives the first equivalence. For the second, suppose such frames exist and let $i_0, i_1, \ldots, i_r = i_0$ be a cycle. Then $\prod_t \Theta_{i_ti_{t+1}} = \prod_t U_{i_t}U_{i_{t+1}}^{-1} = U_{i_0}U_{i_0}^{-1} = I$ by telescoping. Conversely, assume every cycle product is the identity, fix

a spanning tree and a root r, set $U_r = I$, and define U_i as the ordered product of graph links along the tree path from r to i. The relation $\Theta_{ij} = U_i U_j^{-1}$ then holds on every tree edge by construction, and on a non-tree edge it holds precisely because the cycle formed by that edge and the tree path between its endpoints has trivial product. For uniqueness, if $U_i U_j^{-1} = U_i' U_j'^{-1}$ on every edge then $U_i'^{-1} U_i = U_j'^{-1} U_j$ for adjacent i, j, so this quantity is constant on a connected graph. □

Hypothesis 22.7 (Flat comparison). The declared graph links are a coboundary, $\Theta_{ij} = U_i U_j^{-1}$, so that the trivialization $z_i = U_i^{-1}\mu_i$ of the means, and the same map $z_i = U_i^{-1} k_i$ applied to the latents themselves, turn every residual into a plain difference,

$$\mu_i - \Theta_{ij}\mu_j = U_i\,(z_i - z_j)\,. \tag{22.10}$$

This is a choice, not a derivation. By Proposition 22.6 it excludes exactly the graph-link assignments with nontrivial holonomy around some cycle, which are permitted in Regime II. It is used in two places: it makes the interaction shape of Definition 22.3 a statement in one fixed coordinate system, and it supplies a clean sufficient domain for closure under aggregation. In the flat chart the represented link $\widehat{\Theta}_{ij} := U_i^{-1}\Theta_{ij}U_j$ equals I, so merging two agents by $z_i = z_j = z$ annihilates their internal residual. More generally, in an arbitrary common chart the internal edge energy becomes

$$z^\top (I - \widehat{\Theta}_{ij})^\top W_{ij} (I - \widehat{\Theta}_{ij}) z,$$

which vanishes for every z exactly when $W_{ij}^{1/2}(I - \widehat{\Theta}_{ij}) = 0$. Positive-definite W_{ij}, or faithfulness of the weight on every represented direction, reduces this condition to $\widehat{\Theta}_{ij} = I$; with merely positive-semidefinite weights, a nontrivial represented link can survive in invisible null directions. Thus Proposition 22.6 remains the exact coboundary theorem, while the converse from energy annihilation to a trivial represented link requires the stated nondegeneracy. The coarse-graining chapter carries out the weighted computation and separates the sufficient trivializing domain from degenerate exceptions. [HYPOTHESIS]

Two remarks fix the status of Equation (22.10). First, the change of variables $y = Uz$ with $U = \mathrm{blockdiag}(U_1, \ldots, U_N)$ acts on precisions by congruence, $\Lambda^{(y)} = U^{-\top}\Lambda^{(z)}U^{-1}$, which preserves inertia by Sylvester's law [40, Ch. 4] but does not preserve membership in $\mathcal{I}(V)$, since the transformed off-diagonal block $U_i^{-\top}\Lambda_{ij}^{(z)}U_j^{-1}$ need not be symmetric when $U_i \neq U_j$. Membership in the interaction family is therefore a frame-dependent assertion. Second, that frame dependence is exactly why the invariants of the renormalization chapters are stated as generalized eigenvalues of the pair (L, Λ) rather than as eigenvalues of either matrix alone: the pair transforms by the same congruence and the generalized spectrum is unchanged. [ESTABLISHED]

22.4.2 What a directed linear-Gaussian model actually contributes

Consider the state sector of a directed linear-Gaussian model of the kind used in the preceding part: each non-root agent i has a single parent $j = \mathrm{pa}(i)$ and receiving kernel

$$k_i \mid k_j \sim \mathcal{N}\left(G_i \Theta_{ij} k_j, \Lambda_i^{-1}\right), \qquad \Lambda_i \in \mathbb{S}_{++}^K, \qquad G_i \in \mathbb{R}^{K\times K}, \tag{22.11}$$

and each root r carries a proper prior $\mathcal{N}(0, \Lambda_r^{-1})$, a nonzero prior mean shifting only the information vector h and not the precision. The gain is written G_i rather than A_i because the latter symbol is reserved for the self terms of Definition 22.3. Unlike the represented graph link Θ_{ij}, the conditional mean map G_i need not be invertible: positive definiteness of the innovation covariance normalizes the receiving kernel for every matrix G_i, exactly as Chapter 4 declares. A single subscript on Λ denotes the innovation precision of one receiving kernel; a double subscript denotes a block of the assembled precision. When an invertible-gain submodel is wanted it will be named explicitly.

Proposition 22.8 (Assembled precision blocks). Write $T_i := G_i \Theta_{i\,\mathrm{pa}(i)}$ and $\mathrm{ch}(i) = \{\ell : \mathrm{pa}(\ell) = i\}$. The joint precision of the model above has blocks

$$\Lambda_{i\,\mathrm{pa}(i)} = -\Lambda_i T_i, \qquad \Lambda_{ii} = \Lambda_i + \sum_{\ell \in \mathrm{ch}(i)} T_\ell^\top \Lambda_\ell T_\ell, \tag{22.12}$$

and $\Lambda_{ij} = 0$ whenever i and j are not adjacent in the parent graph. [ESTABLISHED]

Proof. The negative log joint density is $\frac{1}{2}\sum_i (k_i - T_i k_{\mathrm{pa}(i)})^\top \Lambda_i (k_i - T_i k_{\mathrm{pa}(i)})$ plus terms not depending on k, with the parent term absent at a root. Expanding one summand gives $\frac{1}{2} k_i^\top \Lambda_i k_i - k_i^\top \Lambda_i T_i k_{\mathrm{pa}(i)} + \frac{1}{2} k_{\mathrm{pa}(i)}^\top T_i^\top \Lambda_i T_i k_{\mathrm{pa}(i)}$. The precision is the Hessian of this function, and collecting the three contributions of each factor gives Equation (22.12). □

By Equation (22.12), membership in the interaction family requires the matrix

$$W_{i\,\mathrm{pa}(i)} := \Lambda_i T_i = \Lambda_i G_i \Theta_{i\,\mathrm{pa}(i)} \tag{22.13}$$

to be symmetric and positive semidefinite. Neither property is automatic, and both fail on explicit parameter values.

Two sources of failure must be kept apart, and a convention does it. Under Hypothesis 22.7 the trivialized residual is $z_i - \widetilde{T}_i z_j$ with

$$\widetilde{T}_i := U_i^{-1} T_i U_j = U_i^{-1} G_i U_i, \qquad \widetilde{\Lambda}_i := U_i^\top \Lambda_i U_i, \tag{22.14}$$

the second equality in the first display using $\Theta_{ij} = U_i U_j^{-1}$, so that (22.12) holds verbatim in trivialized coordinates with (Λ_i, T_i) replaced by $(\widetilde{\Lambda}_i, \widetilde{T}_i)$. The graph links have then already been removed and any remaining obstruction is attributable to the gain alone. For the rest of this section the tildes are dropped and (Λ_i, T_i) denote the trivialized quantities.

Proposition 22.9 (Two global-condition witnesses). There exist admissible parameters for which no interaction-family representation of the assembled precision exists.

$$\Lambda_i = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right) \succ 0, \qquad T_i = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in \mathrm{GL}^+(2) \qquad \Longrightarrow \qquad -\Lambda_i T_i = \left(\begin{smallmatrix} 1 & -2 \\ 3 & -1 \end{smallmatrix}\right), \tag{22.15}$$

which differs from its own transpose by a matrix of Frobenius norm $5\sqrt{2}$, so the off-diagonal block is not symmetric and no W exists at all. Separately, taking $K = 1$ with $G_i = -1$ and $\Theta_{ij} = \Lambda_i = 1$

gives a symmetric block with $W = -1 < 0$, so the sign requirement fails; the same construction with $G_i = -I$, $\Theta_{ij} = I$ and $\Lambda_i = I$ gives $W = -I \prec 0$ at every K. [ESTABLISHED]

Proof. Direct computation. For the first, $\Lambda_i T_i = \left(\begin{smallmatrix} -1 & 2 \\ -3 & 1 \end{smallmatrix}\right)$, so $-\Lambda_i T_i$ is as displayed and $-\Lambda_i T_i - (-\Lambda_i T_i)^\top = \left(\begin{smallmatrix} 0 & -5 \\ 5 & 0 \end{smallmatrix}\right)$ with Frobenius norm $\sqrt{50}$. Positive definiteness of Λ_i follows from $\det = 5 > 0$ and trace $5 > 0$; invertibility of T_i from $\det T_i = 1 > 0$. For the second, (22.13) evaluates to $\Lambda_i G_i \Theta_{ij} = -I$, and $-I$ is not positive semidefinite for any $K \geq 1$. □

Proposition 22.9 settles the status of the interaction shape. The gain matrix G_i is a free generative parameter in the construction of the preceding part, not required to be symmetric, positive, or even close to the identity, and for the displayed values the assembled precision has no representation of the form of Equation (22.8). Whatever else may be said for the interaction family, it is not a consequence of the general linear-Gaussian directed model.

22.4.3 Exactly when the form does hold

Having shown that the form fails, the obligation is to say when it holds. Two statements are available, one global and one local, and they answer different questions.

Proposition 22.10 (Exact global interaction condition). The assembled precision of Equation (22.12) lies in $\mathcal{I}(V)$ with the edge assignment of Equation (22.13) if and only if, for every non-root i,

$$W_i := \Lambda_i T_i \in \mathbb{S}_+^K, \tag{22.16}$$

and, for every agent i,

$$A_i := \Lambda_i - W_i + \sum_{\ell \in \mathrm{ch}(i)} \left(W_\ell \Lambda_\ell^{-1} W_\ell - W_\ell\right) \succeq 0, \tag{22.17}$$

with the term $-W_i$ omitted at a root. [ESTABLISHED]

Proof. Condition (22.16) is exactly the requirement that the off-diagonal blocks have the form $-W$ with W symmetric positive semidefinite. Given it, $T_i = \Lambda_i^{-1} W_i$ and $T_\ell^\top \Lambda_\ell T_\ell = W_\ell \Lambda_\ell^{-1} \Lambda_\ell \Lambda_\ell^{-1} W_\ell = W_\ell \Lambda_\ell^{-1} W_\ell$, using symmetry of W_ℓ. The self term is then the residual $A_i = \Lambda_{ii} - \sum_{j \sim i} W_{ij}$, where the edges at i are the one to its parent, of weight W_i, and one to each child ℓ, of weight W_ℓ. Substituting Equation (22.12) gives Equation (22.17), and family membership requires exactly that this residual be positive semidefinite. □

Condition (22.17) is a balance at each node and permits cancellation: a parent deficit $\Lambda_i - W_i \prec 0$ may be offset by child surpluses. That freedom disappears if one asks the stronger and more natural question, namely whether each generative factor separately lies in the family, so that the interaction structure can be read off the model factor by factor rather than after a global cancellation. The answer is sharp.

The cancellation is not merely formal. Take one scalar receiving factor $j \to i$ with $\Lambda_i = 1$, $T_i = \frac{1}{2}$, and root precision $\Lambda_j = 1$. In the block order (i, j), Equation (22.12) gives

$$\Lambda_{\text{joint}} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{4} \end{pmatrix} = \begin{pmatrix} A_i + W & -W \\ -W & A_j + W \end{pmatrix}, \qquad W = \tfrac{1}{2}, \quad A_i = \tfrac{1}{2}, \quad A_j = \tfrac{3}{4}. \tag{22.18}$$

Thus the assembled precision belongs to $\mathcal{I}(\{i, j\})$, although $T_i^2 = \frac{1}{4} \neq \frac{1}{2} = T_i$, so the receiving factor does not belong locally: its parent-side residual is $T_i^2 \Lambda_i - W = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$, and the root prior offsets it to $A_j = 1 - \frac{1}{4} = \frac{3}{4}$. The matrix is positive definite because its leading principal minors are 1 and 1. This exact witness proves that factorwise membership is sufficient but not necessary for global membership. [ESTABLISHED]

Proposition 22.11 (Edge-local characterization with singular gains allowed). Consider a single receiving factor $j \to i$ in isolation, contributing Λ_i to the ii block, $T_i^\top \Lambda_i T_i$ to the jj block, and $-\Lambda_i T_i$ to the ij block. This factor lies in $\mathcal{I}(\{i, j\})$ if and only if

$$X := \Lambda_i^{-1/2} \, (\Lambda_i T_i) \, \Lambda_i^{-1/2} \text{ is an orthogonal projection,} \tag{22.19}$$

equivalently if and only if $\Lambda_i T_i$ is symmetric and T_i is idempotent, equivalently if and only if T_i is a Λ_i-orthogonal projection. In that case the self terms are $A_i = \Lambda_i^{1/2}(I - X)\Lambda_i^{1/2}$ and $A_j = 0$. The full-rank case $X = I$ is $T_i = I$, in which the factor is the pure agreement factor whose mean is the parent's own value in the trivialized coordinates and whose edge weight is $W = \Lambda_i$. [ESTABLISHED]

Proof. Membership requires three things: $W := \Lambda_i T_i \succeq 0$ and symmetric; the i-side residual $\Lambda_i - W \succeq 0$; and the j-side residual $T_i^\top \Lambda_i T_i - W \succeq 0$. Assume W symmetric and set $X = \Lambda_i^{-1/2} W \Lambda_i^{-1/2}$, which is symmetric. Then $W \succeq 0$ reads $X \succeq 0$; $\Lambda_i - W \succeq 0$ reads $X \preceq I$; and, using $T_i = \Lambda_i^{-1} W$ so that $T_i^\top \Lambda_i T_i = W \Lambda_i^{-1} W = \Lambda_i^{1/2} X^2 \Lambda_i^{1/2}$, the third reads $X^2 \succeq X$. Now $0 \preceq X \preceq I$ implies $X^2 \preceq X$, since X and X^2 are simultaneously diagonalizable and $\lambda^2 \leq \lambda$ for $\lambda \in [0, 1]$. Combining, $X^2 = X$, so X is a symmetric idempotent, that is an orthogonal projection. Conversely any orthogonal projection satisfies all three conditions, and then $T_i = \Lambda_i^{-1/2} X \Lambda_i^{1/2}$ satisfies $T_i^2 = T_i$ and $\Lambda_i T_i = \Lambda_i^{1/2} X \Lambda_i^{1/2}$ is symmetric positive semidefinite. The self terms follow by substitution, the j-side residual being $\Lambda_i^{1/2}(X^2 - X)\Lambda_i^{1/2} = 0$. □

Corollary 22.12 (Invertible-gain factor). If the receiving gain T_i is required to be invertible, the isolated factor lies in $\mathcal{I}(\{i, j\})$ if and only if $T_i = I$. Indeed, Proposition 22.11 makes T_i idempotent, and multiplying $T_i^2 = T_i$ by T_i^{-1} gives $T_i = I$; the converse is the full-rank case of that proposition. [ESTABLISHED]

Proposition 22.11 describes the stronger factor-local requirement, not global family membership. With singular gains allowed, a proper projection constrains only the components of the residual inside $\text{range}(X)$ and leaves the complement to the receiving-side self term. In the invertible-gain submodel, Corollary 22.12 removes every proper projection and leaves only pure agreement.

The deterministic replacement check `CHK-GAUSS-PROJECTION` tests 4000 generic gains and exact controls of ranks 0 through 3 under a fully specified protocol; its current output corroborates the characterization but does not prove generic measure zero. [NUMERICAL]

Hypothesis 22.13 (Global interaction form). The renormalization half of this document acts on recognition precisions that belong globally to Equation (22.8), in trivialized coordinates, and this is adopted as a declared restriction. For a precision assembled from the directed model of Equation (22.11), the restriction is exactly Proposition 22.10; it does *not* require every receiving factor to satisfy Proposition 22.11, because the balance terms in Equation (22.17) may cancel as Equation (22.18) proves. It is used in the aggregation closure statement, in the definition of the flow, and in the fixed-point analysis; each of those results names it. The document does not claim that it is forced by anything, and Proposition 22.5 records that it is a measure-zero restriction on the space of Gaussian recognition laws. [HYPOTHESIS]

Hypothesis 22.14 (Optional factorwise-agreement realization). When the interaction parameters are required to be readable from the generative model one factor at a time, every receiving factor is separately assumed to satisfy Proposition 22.11. This is a strictly stronger hypothesis than global membership: it permits Λ_i-orthogonal projections in the relaxed singular-gain model and only $T_i = I$ in the invertible-gain model, while it excludes the globally admissible witness (22.18). The operator-level renormalization results require Hypothesis 22.13, not this stronger factorwise realization, unless they explicitly invoke a factor interpretation. [HYPOTHESIS]

22.5 Nonemptiness, and a note on conditioning

A declared subfamily is worthless if it is empty or numerically degenerate, so both should be checked. Nonemptiness is a construction, not an experiment.

Proposition 22.15 (Constructive nonemptiness and exact kernel). Under Hypothesis 22.7, let each edge of a graph $G = (V, E)$ carry an agreement factor with residual $\mu_i - \Theta_{ij}\mu_j$ and link covariance $R_{ij} \in \mathbb{S}^K_{++}$, and let each vertex carry a quadratic self term $A_i \succeq 0$ in the trivialized coordinates. A self term is the precision of a proper full-dimensional Gaussian prior at vertex i only when $A_i \succ 0$. The assembled trivialized energy is

$$E(z) = \tfrac{1}{2}\sum_i z_i^\top A_i z_i + \tfrac{1}{2}\sum_{i<j} (z_i - z_j)^\top W_{ij}\,(z_i - z_j)\,, \qquad W_{ij} = U_i^\top R_{ij}^{-1} U_i, \tag{22.20}$$

where the edge sum is over E. The associated precision Λ lies in $\mathcal{I}(V) \cap \mathbb{S}^{NK}_+$, every represented edge weight is positive definite, and

$$\ker \Lambda = \{z : A_i z_i = 0 \text{ for every } i, \quad z_i = z_j \text{ for every } (i,j) \in E\}\,. \tag{22.21}$$

If V_α are the connected components of G, then equivalently

$$\ker \Lambda = \bigoplus_{\alpha} \left\{ \mathbf{1}_{V_\alpha} \otimes v : v \in \bigcap_{i \in V_\alpha} \ker A_i \right\}. \tag{22.22}$$

Consequently $\Lambda \succ 0$ exactly when $\bigcap_{i \in V_\alpha} \ker A_i = \{0\}$ for every connected component. In particular, one positive-definite self term in each component suffices; taking every $A_i \succ 0$ makes every vertex self term a proper prior and constructs an element of $\mathcal{I}(V) \cap \mathbb{S}^{NK}_{++}$. [ESTABLISHED]

Proof. By (22.10) the residual is $U_i(z_i - z_j)$, so the quadratic form $(\mu_i - \Theta_{ij}\mu_j)^\top R_{ij}^{-1}(\mu_i - \Theta_{ij}\mu_j)$ equals $(z_i - z_j)^\top U_i^\top R_{ij}^{-1} U_i (z_i - z_j)$. The matrix $U_i^\top R_{ij}^{-1} U_i$ is symmetric and positive definite because R_{ij}^{-1} is and U_i is invertible. Comparing with (22.9) identifies the parameters and proves positive semidefiniteness. Equality in the energy holds exactly when $A_i z_i = 0$ at every vertex and, because every $W_{ij} \succ 0$, when $z_i = z_j$ on every edge. The latter conditions make z constant on each connected component, which proves (22.21) and (22.22). A positive-semidefinite matrix is positive definite exactly when its kernel is trivial, yielding the final criterion and the stated proper witnesses. □

Proposition 22.15 also makes the role of Hypothesis 22.7 concrete by contrast. Without trivialization the same agreement factor contributes the off-diagonal block $-R_{ij}^{-1}\Theta_{ij}$, which is symmetric only in special position. The deterministic replacement check `CHK-GAUSS-TRIVIALIZATION` emits its matrices and compares that asymmetric raw block with the symmetric positive-definite trivialized weight. It corroborates the proposition and is not part of its proof. [NUMERICAL]

Conditioning is a different question and admits only protocol-specific computational evidence. The deterministic replacement check `CHK-GAUSS-CONDITIONING` specifies and runs a 200-draw interaction ensemble together with a zero-self consensus control and 200 unstructured positive-definite controls. Its output concerns only that sampler. Positive semidefiniteness and the general null-space criterion are proved by Proposition 22.4; Proposition 22.15 specializes that criterion to the constructed positive-definite edge weights and supplies explicit proper witnesses. Thinness is proved by Proposition 22.5; no universal conditioning theorem is claimed. [NUMERICAL]

22.6 Marginalization belongs to the Gaussian coarse chapter

Aggregation identifies variables, whereas exact Gaussian marginalization eliminates them by a Schur complement. The interaction cone is closed under the former and not under unrestricted matrix-weighted Kron reduction. The exact positive-definite $N = 3, K = 2$ rational counterexample, including the asymmetric manufactured coupling, is stated and proved once in Chapter 25. [ESTABLISHED]

The same chapter gives the positive result in its sharp current form: every fixed congruence-diagonal cone $\mathcal{C}_H = \{HDH^\top : D \succeq 0 \text{ diagonal}\}$ is closed under every defined node-block Schur elimination. This strictly contains the common-orthogonal-eigenbasis case because congruence-diagonal

matrices need not commute. Maximality is proved only for rich, independently parameterized closed convex product cones with an SPD order unit and closure under every defined one-node elimination. Correlated, nonconvex, variable-chart, rank-changing, and nonflat families remain outside that theorem. [ESTABLISHED]

Chapter 23

Gaussian Recognition Restrictions

Fix the general restriction principle of Proposition 5.11. This chapter specializes it to nondegenerate multivariate-Gaussian laws, computes the exact costs of block factorization and mean restriction, and states when comparisons between recognition structures are meaningful.

23.1 The block restriction and its exact optimum

For the rest of this chapter the exact conditional is Gaussian and nondegenerate,

$$P^\star = \mathcal{N}\left(\mu, J^{-1}\right), \qquad J = J^\top \succ 0, \qquad \mu = J^{-1}h, \tag{23.1}$$

on $\mathbb{R}^n$ with $n = NK$ in the state sector of the previous chapter. A partition $\mathfrak{B}$ of the n coordinates into blocks of dimensions n_b is declared, and the restricted family is

$$\mathcal{Q}_{\mathfrak{B}} = \left\{\mathcal{N}\left(\mu_Q, C_Q\right) : \mu_Q \in \mathbb{R}^n, \quad C_Q = \operatorname{blockdiag}_{b\in\mathfrak{B}}\left(C_b\right), \quad C_b \in \mathbb{S}_{++}^{n_b}\right\}. \tag{23.2}$$

The blocks are whatever the declared partition says they are. A block may collect one agent's coordinates, may split one agent's state and model channels, or may collect several agents; nothing below identifies a matrix block with an agent, and every statement is relative to $\mathfrak{B}$.

Proposition 23.1 (Gaussian reverse relative entropy). For $Q = \mathcal{N}(\mu_Q, C_Q)$ with $C_Q \succ 0$ and $P^\star$ as in Equation (23.1),

$$D_{\mathrm{KL}}\left(Q\|P^\star\right) = \frac{1}{2}\left[\operatorname{Tr}\left(JC_Q\right) + \left(\mu_Q - \mu\right)^\top J\left(\mu_Q - \mu\right) - n - \log\det J - \log\det C_Q\right]. \tag{23.3}$$

[ESTABLISHED]

Proof. Write $D_{\mathrm{KL}}(Q\|P^\star) = \mathbb{E}_Q[\log q] - \mathbb{E}_Q[\log p^\star]$. The first term is the negative differential entropy of a Gaussian, $-\frac{1}{2}\log\det(2\pi e C_Q)$. For the second, $-\log p^\star(y) = \frac{1}{2}(y-\mu)^\top J(y-\mu) + \frac{n}{2}\log 2\pi - \frac{1}{2}\log\det J$, and taking the expectation under Q uses $\mathbb{E}_Q[(Y-\mu)^\top J(Y-\mu)] = \operatorname{Tr}(JC_Q) + (\mu_Q - \mu)^\top J(\mu_Q - \mu)$, which is the standard expansion of a quadratic form about a shifted center. Adding the two and canceling the $\frac{n}{2}\log 2\pi$ terms gives Equation (23.3). □

Equation (23.3) separates: the mean enters only through the quadratic term and the covariance only through the trace and the log determinant, with no cross term. That separation is used repeatedly

below and is the reason the two kinds of restriction studied in this chapter have exactly the costs they have.

Theorem 23.2 (Exact block optimum). Over the family of Equation (23.2) the objective of Equation (23.3) has a unique minimizer, given by

$$\mu_Q^\star = \mu, \qquad C_b^\star = J_{bb}^{-1} \quad \text{for every } b \in \mathfrak{B}. \tag{23.4}$$

The optimal recognition mean is exact even though the recognition covariance is restricted. [ESTABLISHED]

Proof. On the block-diagonal set, $\operatorname{Tr}(JC_Q) = \sum_b \operatorname{Tr}(J_{bb}C_b)$, because only the diagonal blocks of J meet a block-diagonal C_Q under the trace, and $\log\det C_Q = \sum_b \log\det C_b$. The objective is therefore

$$D_{\mathrm{KL}} = \frac{1}{2}\left(\mu_Q - \mu\right)^\top J \left(\mu_Q - \mu\right) + \frac{1}{2}\sum_{b\in\mathfrak{B}} \left[\operatorname{Tr}\left(J_{bb}C_b\right) - \log\det C_b\right] + \text{const}, \tag{23.5}$$

a sum of a strictly convex quadratic in μ_Q, since $J \succ 0$, and one term per block. Each block term is strictly convex on $\mathbb{S}^{n_b}_{++}$, since $C_b \mapsto \operatorname{Tr}(J_{bb}C_b)$ is linear and $C_b \mapsto -\log\det C_b$ is strictly convex on the positive definite cone [19, §3.1.5]. The parameter set is convex, being $\mathbb{R}^n$ times a product of positive definite cones, so the minimizer is unique if it exists. Differentiating on the open cone gives

$$\nabla_{\mu_Q} D_{\mathrm{KL}} = J\left(\mu_Q - \mu\right), \qquad \nabla_{C_b} D_{\mathrm{KL}} = \tfrac{1}{2}\left(J_{bb} - C_b^{-1}\right), \tag{23.6}$$

whose simultaneous zero is Equation (23.4), admissible because $J_{bb} \succ 0$ as a principal submatrix of a positive definite matrix. □

The same optimum arrives from coordinate ascent, which is worth recording because the two derivations fail in different ways and agreeing is informative. Holding all factors but b fixed, the exact coordinate maximizer of the bound is Gaussian with covariance J_{bb}^{-1} and mean

$$\mu_b^{\mathrm{new}} = J_{bb}^{-1}\left(h_b - \sum_{c\neq b} J_{bc}\mu_c\right), \tag{23.7}$$

which is Equation (22.5) read as an update rule. Any fixed point of the full sweep satisfies $J\mu_Q = h$, hence $\mu_Q = \mu$ [17, 18]. The covariance is pinned in one step and the mean only at a fixed point, which is the standard asymmetry of Gaussian coordinate ascent.

Theorem 23.3 (Determinant gap). Let $\mathcal{L}^\star_{\mathfrak{B}}$ denote the bound at the optimum of Theorem 23.2. Then

$$\log p_\theta(o \mid X) - \mathcal{L}^\star_{\mathfrak{B}} = \frac{1}{2}\left[\sum_{b\in\mathfrak{B}} \log\det J_{bb} - \log\det J\right] \geq 0, \tag{23.8}$$

with equality if and only if $J_{bc} = 0$ for every pair of distinct blocks of $\mathfrak{B}$. [ESTABLISHED]

Proof. At the optimum, $\operatorname{Tr}(JC_Q^\star) = \sum_b \operatorname{Tr}(J_{bb}J_{bb}^{-1}) = \sum_b n_b = n$ and $\log\det C_Q^\star = -\sum_b \log\det J_{bb}$, while the quadratic mean term vanishes. Substituting into Equation (23.3) leaves

$$\inf_{Q\in\mathcal{Q}_{\mathfrak{B}}} D_{\mathrm{KL}}\left(Q\|P^\star\right) = \frac{1}{2}\left[\sum_{b\in\mathfrak{B}} \log\det J_{bb} - \log\det J\right], \tag{23.9}$$

and Equation (5.23) converts this into Equation (23.8).

For nonnegativity and the equality condition, two lemmas suffice. The first is that $0 \prec S \preceq T$ implies $\det S \le \det T$, with equality only when $S = T$. To see it, set $R := T^{-1/2}ST^{-1/2}$, which is symmetric with eigenvalues in $(0, 1]$ and with $\det R = \det S/\det T$. That ratio is a product of numbers in $(0, 1]$, hence at most one, and it equals one only when every eigenvalue equals one, which for a symmetric matrix forces $R = I$ and so $S = T$. The second lemma is that, for $J \succ 0$ partitioned into a first block and its complement,

$$\det J = \det J_{11}\cdot\det\left(J_{22} - J_{21}J_{11}^{-1}J_{12}\right), \tag{23.10}$$

with the Schur complement positive definite [92]. Since $J_{21}J_{11}^{-1}J_{12} \succeq 0$, that Schur complement is $\preceq J_{22}$, so the first lemma gives $\det J \le \det J_{11}\det J_{22}$, with equality if and only if $J_{21}J_{11}^{-1}J_{12} = 0$, which since $J_{11}^{-1} \succ 0$ holds if and only if $J_{11}^{-1/2}J_{12} = 0$, that is if and only if $J_{12} = 0$. Applying this with the first block of $\mathfrak{B}$ against the union of the rest, and then inducting on the number of blocks, gives Equation (23.8) and the stated equality condition. This is Fischer's determinant inequality for positive definite matrices [40, §7.8]. □

Canonical-correlation form and telescoping identity. For a two-block precision, let $A \in \mathbb{S}_{++}^p$, $D \in \mathbb{S}_{++}^q$, and $C \in \mathbb{R}^{p\times q}$, and write

$$J = \begin{pmatrix} A & C \\ C^\top & D \end{pmatrix}, \qquad A = L_A L_A^\top, \qquad D = L_D L_D^\top, \qquad R := L_A^{-1}CL_D^{-\top}. \tag{23.11}$$

Both diagonal blocks are positive definite. The Schur complement factorizes as

$$D - C^\top A^{-1}C = L_D\left(I_q - R^\top R\right)L_D^\top, \qquad \det J = (\det A)(\det D)\det\left(I_q - R^\top R\right). \tag{23.12}$$

If ρ_a are the singular values of R, positive definiteness of the Schur complement gives $0 \le \rho_a < 1$, and the two-block increment is exactly

$$\Delta(A, D; C) = -\frac{1}{2}\log\det\left(I_q - R^\top R\right) = -\frac{1}{2}\sum_{a=1}^{\min(p,q)} \log\left(1 - \rho_a^2\right). \tag{23.13}$$

For ordered blocks $B_1, \dots, B_m$, define the cumulative coordinate sets $U_k := \bigcup_{j=1}^k B_j$. The left-fold increment at step $k \ge 2$ is

$$\delta_k = \frac{1}{2}\left[\log\det J_{U_{k-1}U_{k-1}} + \log\det J_{B_kB_k} - \log\det J_{U_kU_k}\right]. \tag{23.14}$$

Consequently the intermediate cumulative determinants cancel exactly:

$$\sum_{k=2}^{m} \delta_k = \frac{1}{2}\left[\sum_{k=1}^{m} \log\det J_{B_k B_k} - \log\det J\right]. \tag{23.15}$$

The total is order independent in exact arithmetic; the increments and the finite-precision trajectory can depend on the declared order. [ESTABLISHED]

Operational stable realization. The ordinary numerical path computes L_A and L_D by Cholesky factorization, obtains R by triangular solves, computes its singular values, and evaluates each summand as `-0.5*log1p(-rho**2)`. It forms neither A^{-1} nor D^{-1} and does not subtract large floating-point log determinants. Validation and ordinary linear algebra use exact power-of-two normalization and range-safe residuals. If that scaling loses a nonzero entry, or a binary64 Cholesky, solve, SVD, or condition estimate is unusable, an adaptive-precision exact-dyadic Cholesky calculation certifies positive definiteness and supplies the log-determinant gap. A separate boundary policy handles a squared singular value within an operational $64n\epsilon_{\mathrm{mach}}$ neighborhood of one. There the exact stored binary64 entries are reevaluated at a minimum of 200 decimal digits, including the determinant-gap value; a second binary64 solve backend makes the diagnostic the maximum excursion observed across the two declared binary64 backends. The code rejects an exact-input value outside the Schur domain and rejects a raw binary64 excursion above one when its discrepancy exceeds that operational guard. It also uses the exact-input value path when $\kappa_2(J)\epsilon_{\mathrm{mach}} \geq 10^{-4}$, an explicitly operational conditioning trigger selected for the frozen protocol rather than asserted as a universal error threshold. A nonfinite binary64 condition estimate is also a fallback trigger, not a rejection criterion. An ill-conditioned but in-domain value is reevaluated rather than rejected merely for triggering that path. An early high-precision fallback selected before residual evaluation records the Cholesky residual, solve residual, residual tolerance, and backward-error diagnostics as `null` because they were not evaluated. If the residual-health gate is reached, every finite measurement is retained. A field is `null` only when it has no finite usable numeric value, either because an attempted measurement is nonfinite or because a derived diagnostic requires an unavailable constituent; finite sibling fields remain numeric. The evaluation method distinguishes a finite threshold excursion from nonfinite Cholesky, solve, or tolerance diagnostics and lists multiple nonfinite causes in that fixed order. Thus `null` denotes neither failure nor zero, and the zero-step one-block case retains its exact zero diagnostics. The `GapStep` API retains this fieldwise quartet; the `FactorizationCaseRecord` protocol surface stores only the aggregate backward diagnostic, maximum Cholesky residual, and ordered evaluation-method tuple, not per-step solve or tolerance values. Despite the compatibility name `backward_error_bound`, the numeric quantity is a one-ULP-up summary of already rounded local residual measurements, not a certified perturbation or forward-error bound. These guarded fallbacks are declared finite-precision procedures, not theorems that Cholesky or solve residuals universally control canonical-correlation or gap forward error. [DEFINITION]

Proposition 23.4 (The optimum is dominated by the exact marginal). For every block b,

$$C_b^\star = J_{bb}^{-1} \preceq \left(J^{-1}\right)_{bb}, \tag{23.16}$$

with equality if and only if $J_{b,-b} = 0$. [ESTABLISHED]

Proof. By Equation (22.7), $(J^{-1})_{bb} = S_b^{-1}$ with $S_b = J_{bb} - J_{b,-b}J_{-b,-b}^{-1}J_{-b,b}$, and $S_b \preceq J_{bb}$ because the subtracted matrix is positive semidefinite. Inversion reverses the Loewner order on positive definite matrices: if $0 \prec S \preceq T$ then $T^{-1/2}ST^{-1/2} \preceq I$, so the eigenvalues of $T^{1/2}S^{-1}T^{1/2}$ are at least one, giving $S^{-1} \succeq T^{-1}$ [40, §7.7], [14, Ch. V]. Applying this to $S_b \preceq J_{bb}$ gives Equation (23.16). Equality forces $S_b = J_{bb}$, hence $J_{b,-b}J_{-b,-b}^{-1}J_{-b,b} = 0$, hence $J_{b,-b} = 0$ as in the proof of Theorem 23.3. □

Proposition 23.4 is the exact form of a statement usually made loosely, that a factorized reverse-relative-entropy approximation understates posterior uncertainty [17, 18, 82]. Here it is a Loewner inequality with a proof and an equality condition, and it holds blockwise rather than merely in some scalar summary. It also fixes what the restriction is and is not.

Conditional dependence, not marginal covariance. The family of Equation (23.2) constrains the recognition covariance, hence the recognition precision, to be block diagonal. By Proposition 22.2 that is a statement that the recognition law makes distinct blocks conditionally independent given the rest. It is not, and does not imply, the assertion that the target's marginal covariance blocks $(J^{-1})_{bc}$ vanish; (22.6) shows that those can be nonzero while $J_{bc} = 0$. The reverse-relative-entropy projection changes only the candidate Q: it sets $C_{bc}^\star = 0$ for $b \neq c$ and $C_{bb}^\star = J_{bb}^{-1} \preceq (J^{-1})_{bb}$. It neither leaves nor replaces any covariance block of the fixed target $P = \mathcal{N}(\mu, J^{-1})$. Reading the block restriction as a claim about the target's marginal correlation is a category error, and it is the error that makes the determinant gap look like a correlation-strength measurement rather than what Theorem 23.3 says it is, a coupling-structure measurement. [ESTABLISHED]

23.2 Observation independence, and its exact scope

Theorem 23.3 expresses the cost as a pure log-determinant functional of J. Whether that makes the cost independent of the realized observation depends on where J comes from, and the answer differs for the two kinds of restriction this document uses. Getting the difference right matters because a later chapter compares recognition structures across data and would draw the wrong conclusion from the wrong version.

Proposition 23.5 (Conditional precision is observation free). Suppose the joint law of (o, Y) given X is Gaussian with precision $\mathcal{J}$ and information vector η, both functions of (θ, X) alone. Then the exact conditional law of Y given o has precision and mean

$$J = \mathcal{J}_{YY}, \qquad \mu = J^{-1}\left(\eta_Y - \mathcal{J}_{Yo}o\right), \tag{23.17}$$

so J does not depend on the realized value of o while μ is an affine function of it. [ESTABLISHED]

Proof. Read the exponent of the joint density as a function of y at fixed o. The quadratic part is $-\frac{1}{2}y^\top \mathcal{J}_{YY} y$ and the linear part is $(\eta_Y - \mathcal{J}_{Yo} o)^\top y$, so by Equation (22.1) the conditional law is Gaussian in information form with the displayed natural parameters, and $\mu = J^{-1}h$. □

Corollary 23.6 (Observation independence of the determinant gap). Under the hypothesis of Proposition 23.5, the gap of Theorem 23.3 is a function of (θ, X) and the declared partition alone, and does not depend on the realized observation. [ESTABLISHED]

Proof. The gap depends on Q and o only through J, by Theorem 23.3, and through nothing else, because the mean sector contributes exactly zero at the optimum by Theorem 23.2. Proposition 23.5 makes J observation free. □

Two hypotheses are doing the work in Corollary 23.6 and both are load bearing. The first is joint Gaussianity with an observation-independent joint precision, which excludes, for instance, a model whose observation noise covariance is itself a function of the realized observation; without it Equation (23.17) fails and J carries data. The second, and the one more easily overlooked, is that the restriction is a *factorization*, so that the optimal recognition mean coincides with the target mean and the quadratic term in Equation (23.3) vanishes identically at the optimum. A restriction that constrains the mean does not have that property, and the next proposition computes exactly what survives.

Proposition 23.7 (Cost of a mean restriction). Let $[B\ B_\perp]$ be an orthogonal matrix with $B \in \mathbb{R}^{n\times r}$, and restrict the recognition family to

$$\mathcal{Q}_B = \left\{ \mathcal{N}(\mu_Q, C_Q) : \mu_Q \in \operatorname{range}(B), \quad C_Q \in \mathbb{S}^n_{++} \right\}, \tag{23.18}$$

with the covariance unrestricted. Writing $\mu_\perp := B_\perp^\top \mu$ for the component of the target mean orthogonal to the retained subspace, the exact minimum of the relative entropy over $\mathcal{Q}_B$ is

$$\inf_{Q\in\mathcal{Q}_B} D_{\mathrm{KL}}(Q\|P^\star) = \frac{1}{2}\mu_\perp^\top \left(B_\perp^\top J^{-1} B_\perp \right)^{-1} \mu_\perp, \tag{23.19}$$

attained at $C_Q^\star = J^{-1}$ and $\mu_Q^\star = \mu - J^{-1}B_\perp (B_\perp^\top J^{-1} B_\perp)^{-1}\mu_\perp$. The matrix $(B_\perp^\top J^{-1} B_\perp)^{-1}$ is the *marginal* precision of the restricted directions, that is the inverse of their exact marginal covariance. [ESTABLISHED]

Proof. By the separation noted after Equation (23.3), the covariance sector is unconstrained and its minimum is attained at $C_Q^\star = J^{-1}$, where it contributes zero. What remains is to minimize $\frac{1}{2}(\mu_Q - \mu)^\top J (\mu_Q - \mu)$ subject to $B_\perp^\top \mu_Q = 0$. Put $v = \mu_Q - \mu$, so the constraint reads $B_\perp^\top v = -\mu_\perp$. Stationarity of the Lagrangian gives $Jv = B_\perp \lambda$, hence $v = J^{-1}B_\perp\lambda$; imposing the constraint gives $\lambda = -(B_\perp^\top J^{-1} B_\perp)^{-1}\mu_\perp$, which is well defined because $B_\perp$ has full column rank and $J^{-1} \succ 0$. The

objective at that point is $\frac{1}{2}v^\top Jv = \frac{1}{2}\lambda^\top(B_\perp^\top J^{-1}B_\perp)\lambda$, which is Equation (23.19), and the minimizer is as displayed. The final identification holds because $\mathrm{Cov}(B_\perp^\top Y) = B_\perp^\top J^{-1}B_\perp$ under $P^\star$. □

Corollary 23.8 (A mean restriction carries the data). Under the hypothesis of Proposition 23.5, the cost in Equation (23.19) is a quadratic function of the realized observation, since $\mu_\perp = B_\perp^\top J^{-1}(\eta_Y - \mathcal{J}_{Yo}o)$ is affine in o. It is observation free if and only if $B_\perp^\top J^{-1}\mathcal{J}_{Yo} = 0$. Thus the observation independence of Corollary 23.6 does not extend to restrictions of the mean, and a construction that reports one cost as though it were the other is reporting a data-dependent quantity as a structural one. [ESTABLISHED]

Proposition 23.9 (Marginal precision against restricted precision). With $[B\ B_\perp]$ as above,

$$B_\perp^\top JB_\perp - \left(B_\perp^\top J^{-1}B_\perp\right)^{-1} = B_\perp^\top JB\left(B^\top JB\right)^{-1}B^\top JB_\perp \succeq 0, \tag{23.20}$$

so the marginal precision of the restricted directions is dominated in the Loewner order by the restriction $B_\perp^\top JB_\perp$ of the precision to those directions, the difference has rank at most r, and the two coincide if and only if $B^\top JB_\perp = 0$, that is if and only if the retained and restricted subspaces are J-orthogonal. [ESTABLISHED]

Proof. Since $[B\ B_\perp]$ is orthogonal, conjugating by it gives $\widetilde{J} := [B\ B_\perp]^\top J[B\ B_\perp]$ and also $[B\ B_\perp]^\top J^{-1}[B\ B_\perp] = \widetilde{J}^{-1}$. Hence $B_\perp^\top J^{-1}B_\perp$ is the lower right block of $\widetilde{J}^{-1}$, and by the block inversion formula its inverse is the Schur complement

$$\widetilde{J}_{\perp\perp} - \widetilde{J}_{\perp\parallel}\widetilde{J}_{\parallel\parallel}^{-1}\widetilde{J}_{\parallel\perp}, \tag{23.21}$$

where $\parallel$ and $\perp$ index the two groups of columns [92]. Subtracting Equation (23.21) from $\widetilde{J}_{\perp\perp} = B_\perp^\top JB_\perp$ leaves exactly the displayed product, which is positive semidefinite and of rank at most $\mathrm{rank}(B^\top JB_\perp) \le r$. It vanishes if and only if $(B^\top JB)^{-1/2}B^\top JB_\perp = 0$, that is if and only if $B^\top JB_\perp = 0$. □

Proposition 23.9 is the reason Equation (23.19) must be written with the marginal precision and not with the restriction of the precision. The two are different matrices, ordered but not equal, and substituting one for the other overstates the cost of a mean restriction by the rank-r correction of Equation (23.20) except in the J-orthogonal case. The deterministic replacement check `CHK-RESTRICTION-SCHUR` independently reconstructs the block-inversion and constrained-optimization endpoints and includes the orthogonal equality control. It corroborates the proved identities and is not evidence for them. [NUMERICAL]

Corollary 23.10 (Costs of combined restrictions add). Restricting the mean to $\mathrm{range}(B)$ and the covariance to the block-diagonal set of Equation (23.2) simultaneously has exact cost

$$\frac{1}{2}\left[\sum_{b\in\mathfrak{B}}\log\det J_{bb} - \log\det J\right] + \frac{1}{2}\mu_\perp^\top\left(B_\perp^\top J^{-1}B_\perp\right)^{-1}\mu_\perp, \tag{23.22}$$

the first term structural and the second data dependent. [ESTABLISHED]

Proof. Equation (23.3) has no term coupling μ_Q to C_Q, and the two constraint sets act on disjoint parameter blocks, so the minimizations are independent and their values add. Each value is supplied by Theorem 23.3 and Proposition 23.7 respectively. □

23.3 Comparing recognition structures at fixed model

Equation (23.8) assigns a number to a partition, which invites the question of whether one recognition structure is better than another. The question is well posed only under conditions that are easy to violate, and this section states both the affirmative result and its boundary.

Proposition 23.11 (Refinement monotonicity for every comparable pair). Let $\mathfrak{B}$ and $\mathfrak{B}'$ be any two partitions such that $\mathfrak{B}'$ refines $\mathfrak{B}$, meaning every block of $\mathfrak{B}'$ is contained in a block of $\mathfrak{B}$. Then $\mathcal{Q}_{\mathfrak{B}'} \subseteq \mathcal{Q}_{\mathfrak{B}}$ and

$$\mathcal{L}^{\star}_{\mathfrak{B}} - \mathcal{L}^{\star}_{\mathfrak{B}'} = \sum_{b \in \mathfrak{B}} \frac{1}{2} \left[\sum_{b' \in \mathfrak{B}' \colon b' \subseteq b} \log\det J_{b'b'} - \log\det J_{bb} \right] \geq 0, \tag{23.23}$$

each summand being the determinant gap of the positive definite matrix J_{bb} under the partition induced on b, and vanishing exactly when J_{bb} is block diagonal under that induced partition. Thus the coarser structure dominates the finer one for every comparable pair in the refinement partial order. The mechanism is the Loewner ordering of Theorem 23.3: within each coarse block, a Schur complement is dominated by the corresponding diagonal block. [ESTABLISHED]

The selected hierarchy invokes this proposition only on its declared edges or, by composition, on comparable nodes of its declared chain or rooted tree. It imposes no ordering on nonnested partitions. [DEFINITION]

Proof. A block-diagonal matrix under $\mathfrak{B}'$ is block diagonal under $\mathfrak{B}$ whenever $\mathfrak{B}'$ refines $\mathfrak{B}$, giving the inclusion of families. Subtracting two instances of Equation (23.8) cancels $\log\det J$ and regroups the remaining terms by coarse block, giving Equation (23.23). Each bracket is Theorem 23.3 applied to $J_{bb} \succ 0$ with the induced partition, hence nonnegative with the stated equality condition. □

Proposition 23.12 (Non-nested structures are not ordered). Refinement is a partial order, and outside it the bounds are not ordered by any structural feature such as the number or the sizes of the blocks. For $n = 3$ take

$$J(a,b) = \begin{pmatrix} 1 & a & 0 \\ a & 1 & b \\ 0 & b & 1 \end{pmatrix}, \qquad a^2 + b^2 < 1, \tag{23.24}$$

which is positive definite, and compare the two partitions $\mathfrak{B}_1 = \{\{1,2\},\{3\}\}$ and $\mathfrak{B}_2 = \{\{1\},\{2,3\}\}$, which have identical block-size multisets and neither of which refines the other. Their gaps are $\frac{1}{2}\log\frac{1-a^2}{1-a^2-b^2}$ and $\frac{1}{2}\log\frac{1-b^2}{1-a^2-b^2}$, so the ordering is decided by whether $|a|$ exceeds $|b|$ and reverses when they are exchanged. At $(a,b) = (0.8, 0.2)$ the values are 0.0589 and 0.5493; at $(0.2, 0.8)$ they are exchanged. [ESTABLISHED]

Proof. Positive definiteness follows from the leading minors 1, $1-a^2$ and $1-a^2-b^2$. The gaps follow from Equation (23.8) with $\det J = 1-a^2-b^2$, $\det J_{\{1,2\}\{1,2\}} = 1-a^2$, $\det J_{\{2,3\}\{2,3\}} = 1-b^2$ and the remaining diagonal entries equal to one. The numerical values are $\frac{1}{2}\log(0.36/0.32)$ and $\frac{1}{2}\log(0.96/0.32)$. □

Proposition 23.12 says that a recognition structure is well chosen relative to a given coupling matrix, not well chosen in the abstract, and it is what makes any partition-selection rule an optimization over structures rather than a reading off of a scalar. It also disposes of the temptation to treat block count as a proxy for quality.

Proposition 23.13 (The gap does not select a structure). Along any declared nested refinement chain containing the one-block partition, the gap of Equation (23.8) is nonincreasing under coarsening by Proposition 23.11, and it attains its minimum value zero at that one-block partition for every $J \succ 0$. This monotonicity supplies no ordering of nonnested partitions. Minimizing the gap therefore selects the unrestricted family and returns no information about structure. [ESTABLISHED]

Proof. For the one-block partition, $\sum_b \log\det J_{bb} = \log\det J$ and the gap is zero; nonnegativity is Theorem 23.3, while monotonicity holds along the declared chain by Proposition 23.11. Proposition 23.12 supplies the nonnested boundary. □

Proposition 23.14 (ELBO values do not order unrelated joints). Let P_θ and $P_{\theta'}$ be normalized joints over the same observation but possibly different latent inventories, with recognition families $\mathcal{Q}$ and $\mathcal{Q}'$ on their respective latent spaces. Assume the selected o belongs to the regular-observation set of each joint, or that pointwise versions have been declared for both. If no evidence-preserving channel relating the two joints is given, then

$$\mathcal{L}^\star - \mathcal{L}'^\star = [\log p_\theta(o \mid X) - \log p_{\theta'}(o \mid X)] - \left[\inf_{\mathcal{Q}} D_{\mathrm{KL}} - \inf_{\mathcal{Q}'} D_{\mathrm{KL}}\right], \tag{23.25}$$

which combines an evidence difference, a model-comparison quantity, with a tightness difference, a recognition-quality quantity. Neither bracket is identifiable from the value of the left side, so the sign of that value orders neither the models nor the recognition families. [ESTABLISHED]

Proof. Apply Equation (5.24) to each model and subtract. To show that both failures occur within normalized models, let $\ell \le 0$, $g \ge 0$, take $O \in \{o, \bar{o}\}$, $Y \in \{0,1\}$, and define a normalized joint by

$$P_{\ell,g}(O=o, Y=1) = e^{\ell-g}, \qquad P_{\ell,g}(O=o, Y=0) = e^{\ell}(1-e^{-g}), \qquad P_{\ell,g}(O=\bar{o}, Y=0) = 1-e^{\ell},$$

with the remaining atom assigned mass zero. Then

$$\log P_{\ell,g}(O = o) = \ell, \qquad P_{\ell,g}(Y = 1 \mid O = o) = e^{-g}.$$

For the singleton recognition family $\mathcal{Q} = \{\delta_1\}$, the minimized gap is g, so the optimized bound is $\ell - g$. To make the two latent inventories literally different, append an unused deterministic binary coordinate $Z = 0$ to either second model and replace its recognition law by $\delta_1 \otimes \delta_0$; taking a product with this point mass changes neither normalization, evidence, relative entropy, nor the optimized ELBO. The pairs

$$(\ell_1, g_1) = (-1, 0), \quad (\ell_2, g_2) = (0, 2)$$

give $\mathcal{L}_1^\star = -1 > \mathcal{L}_2^\star = -2$ although the evidence ordering is $\ell_1 < \ell_2$. The pairs

$$(\ell_1, g_1) = (0, 1), \quad (\ell_2, g_2) = (-2, 0)$$

again give $\mathcal{L}_1^\star = -1 > \mathcal{L}_2^\star = -2$, although the first recognition family has the larger gap and hence the less tight bound. Thus a bound ordering determines neither bracket in Equation (23.25). □

Proposition 23.14 is a statement about unrelated joints, not about latent inventory by itself. If a normalized Markov kernel $K : x \rightsquigarrow z$ defines the coarse joint $\bar{P} = (\mathrm{id}_O \otimes K)_\# P$ and the coarse recognition law is $\bar{Q} = QK$, then the observation marginal and evidence are unchanged and data processing gives

$$\bar{\mathcal{L}}(\bar{Q}; o) = \log p(o) - D_{\mathrm{KL}}(QK \| P_o K) \geq \log p(o) - D_{\mathrm{KL}}(Q \| P_o) = \mathcal{L}(Q; o).$$

Thus different latent spaces can carry ordered bounds when both laws are related by the same evidence-preserving channel. Absent such a relation, an ELBO difference mixes the evidence difference with the tightness difference as in (23.25). The general hypotheses and equality conditions are stated in Chapter 10. [ESTABLISHED]

Open Problem 23.15 (An intrinsic criterion). Propositions 23.13 and 23.14 leave a gap that this document does not close. There is no criterion here, internal to a single evidence bound at a fixed model, that selects a nondegenerate recognition structure: the determinant gap is minimized at the trivial partition, the mean-restriction cost of Equation (23.19) is a cost at fixed structure, and an arbitrary replacement joint supplies no common evidence. Evidence-preserving Markov pushforwards are the important exception, but their data-processing order still does not select a nondegenerate partition. What would settle the problem positively is a functional on admissible channels or partitions, expressible relative to one fixed joint and not monotone to an endpoint, whose optimizer is nondegenerate. A negative result would have to prove endpoint degeneracy for a precisely stated class of functionals. Absent either, the choice of structure remains a model-selection or RG-scheme declaration. [OPEN]

Chapter 24

Gaussian Information Geometry

Chapter 6 distinguishes general law fibers from smooth statistical manifolds, regular exponential families, and the multivariate-Gaussian realization. This chapter works only in the last tier. It identifies the dual Gaussian charts, computes their Fisher metrics, and separates frame-invariant generalized spectral data from ordinary chart-dependent eigenvalues. It does not assert that an RG flow contracts any information-geometric quantity.

24.1 Canonical and expectation coordinates

Use the nondegenerate Gaussian family and log normalizer of Proposition 22.1. For the stacked latent $Y \in \mathbb{R}^n$, $n = NK$, write

$$q(y) = \exp\left(h^\top y - \tfrac{1}{2} y^\top J y - \mathsf{A}(h, J)\right), \qquad h \in \mathbb{R}^n, \quad J \in \mathbb{S}^n_{++}, \tag{24.1}$$

where

$$\mathsf{A}(h, J) = \tfrac{1}{2} h^\top J^{-1} h - \tfrac{1}{2} \log\det J + \tfrac{n}{2} \log 2\pi. \tag{24.2}$$

For sufficient statistic $T(y) = (y, yy^\top)$, the canonical natural and expectation coordinates are

$$\eta = \left(h, -\tfrac{1}{2} J\right), \tag{24.3}$$

$$\tau = \nabla_\eta \mathsf{A} = (\mu, M_2), \qquad M_2 = C + \mu\mu^\top = \mathbb{E}_q[YY^\top], \tag{24.4}$$

The information-form chart (h, J) ranges over $\mathbb{R}^n \times \mathbb{S}^n_{++}$, whereas the canonical matrix coordinate $-J/2$ is negative definite. The family is regular and minimal, so the Legendre map is a bijection on its natural domain [20]. [ESTABLISHED]

The Fisher matrix in natural coordinates is the Hessian of the potential and equals the covariance of the sufficient statistic,

$$g_\eta = \nabla^2_\eta \mathsf{A}(\eta) = \mathrm{Cov}_q\left(T(Y)\right), \tag{24.5}$$

which is the standard exponential-family fact [2, 3, 20]. Because $\tau = \nabla_\eta \mathsf{A}$, the Jacobian of the coordinate change is $\partial\tau/\partial\eta = g_\eta$, and the Fisher metric expressed in the expectation coordinates is

$$g_\tau = \left(\frac{\partial \eta}{\partial \tau}\right)^\top g_\eta \left(\frac{\partial \eta}{\partial \tau}\right) = g_\eta^{-1}, \tag{24.6}$$

the Hessian of the convex conjugate $\mathsf{A}^*(\tau) = \sup_\eta (\langle \eta, \tau \rangle - \mathsf{A}(\eta))$, which for this family is the negative differential entropy of q [3, 67]. [ESTABLISHED]

Four coordinate presentations must be kept apart. The canonical natural chart is $\eta = (h, -J/2)$; the affinely equivalent information-form chart is (h, J); the expectation chart is $\tau = (\mu, M_2)$; and the moment chart is (μ, C). The last is useful for sampling and covariance transport but is not Legendre dual to the natural chart, because $C = M_2 - \mu\mu^\top$. [DEFINITION]

Symmetric-matrix coordinates carry one further convention. Derivatives in the matrix argument are taken with respect to the Frobenius pairing $\langle X, Z \rangle = \operatorname{Tr}(X^\top Z)$ on the space of symmetric matrices. Concretely this is the chart whose coordinates are the diagonal entries together with the off-diagonal entries scaled by $\sqrt{2}$, for which the Euclidean inner product of coordinate vectors equals the Frobenius pairing of the matrices they represent. With the unscaled half-vectorization instead, every off-diagonal coordinate derivative acquires a factor of two and the matrix representing (24.6) changes accordingly. All matrix blocks displayed below are in the Frobenius-paired chart. [DEFINITION]

When the recognition precision has the interaction form of Chapter 22, $J = \Lambda$, with $\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}$, $\Lambda_{ij} = -W_{ij}$, and $C = \Lambda^{-1}$. Below, Λ emphasizes the interaction operator and J the natural parameter; they are the same matrix.

24.2 The chart correction

It is often asserted that the mean sector of the Fisher metric of a Gaussian family *is* the precision. The assertion is true for the mean block in the moment chart. In the expectation chart it is false for the restriction to the coordinate mean directions, while the metric induced on the quotient by the second-moment directions is again exactly the precision. Thus “the mean sector” is ambiguous in the expectation chart until restriction or quotient is named. Both readings are computed below.

Proposition 24.1 (Fisher metric in the moment chart). *In the chart (μ, C) the Fisher metric of the Gaussian family is block diagonal, and for tangent vectors (u, A) and (v, B) with $u, v \in \mathbb{R}^n$ and A, B symmetric,*

$$g_{(\mu,C)}\big((u, A), (v, B)\big) = u^\top \Lambda v + \tfrac{1}{2} \operatorname{Tr}(\Lambda A \Lambda B), \qquad \Lambda = C^{-1}. \tag{24.7}$$

[ESTABLISHED]

Proof. Write $e = y - \mu$ and $\log q(y) = -\frac{1}{2} e^\top \Lambda e + \frac{1}{2} \log\det\Lambda - \frac{n}{2} \log 2\pi$. A displacement (u, A) in the moment chart moves μ by u and C by A, hence Λ by $-\Lambda A \Lambda$, and the directional score is

$$\partial_{(u,A)} \log q = u^\top \Lambda e + \tfrac{1}{2} e^\top \Lambda A \Lambda e - \tfrac{1}{2} \operatorname{Tr}(\Lambda A). \tag{24.8}$$

Under $e \sim \mathcal{N}(0, C)$ the first term has second moment $u^\top \Lambda C \Lambda u = u^\top \Lambda u$. The cross term between the linear and quadratic pieces is a third central moment of a Gaussian and vanishes, which is

exactly why the metric is block diagonal in this chart. Setting $S = \Lambda A \Lambda$, the remaining piece is $\frac{1}{4}\operatorname{Var}(e^\top S e) = \frac{1}{2}\operatorname{Tr}(SCSC) = \frac{1}{2}\operatorname{Tr}(\Lambda A \Lambda A)$. Polarization gives (24.7). □

Expression (24.7) is the classical Fisher–Rao metric of the multivariate normal model; it is the metric whose geodesic equations and Rao distance are studied by Skovgaard [79] and by Atkinson and Mitchell [5], and it appears in the survey of Pinele et al. [74] as their equation (7), in the general reparameterized form $g_{ij} = \partial_i \mu^\top \Sigma^{-1} \partial_j \mu + \frac{1}{2}\operatorname{Tr}\left(\Sigma^{-1}\partial_i \Sigma \Sigma^{-1} \partial_j \Sigma\right)$. The printed equation in that reference carries a typographical slip, repeating the index i in place of j in the second factor of the trace; the intended and correct expression is the one displayed here, and it agrees with (24.7) on setting $\theta = (\mu, C)$. That reference also records, as its equation (8), that the affine action $(\mu, \Sigma) \mapsto (Q\mu + c, Q\Sigma Q^\top)$ is an isometry of this metric for every $Q \in \mathrm{GL}_n(\mathbb{R})$ and $c \in \mathbb{R}^n$, a fact used again in Section 24.4.

Proposition 24.2 (Fisher metric in the expectation chart). *In the chart (μ, M_2) with $M_2 = C + \mu\mu^\top$, the Fisher metric has mean block*

$$g_{[\mu\mu]} = \left(1 + \mu^\top \Lambda \mu\right)\Lambda + \Lambda \mu \mu^\top \Lambda, \tag{24.9}$$

cross block

$$g_{[\mu M_2]}(u, B) = -u^\top \Lambda B \Lambda \mu, \tag{24.10}$$

and second block $g_{[M_2 M_2]}(A, B) = \frac{1}{2}\operatorname{Tr}(\Lambda A \Lambda B)$, unchanged from (24.7). Hence the metric is not block diagonal in this chart unless $\mu = 0$. [ESTABLISHED]

Proof. The chart change is $C = M_2 - \mu\mu^\top$, so a displacement (u, B) in the expectation chart induces

$$\delta\mu = u, \qquad \delta C = B - \left(u\mu^\top + \mu u^\top\right), \tag{24.11}$$

and the metric in the new chart is the pullback of (24.7) along (24.11). For the mean block take $B = 0$, so $\delta C = -(u\mu^\top + \mu u^\top)$. Then

$$\begin{aligned} \operatorname{Tr}\left(\Lambda \delta C \Lambda \delta C\right) &= \operatorname{Tr}\left[\left(\Lambda u \mu^\top + \Lambda \mu u^\top\right)\left(\Lambda u \mu^\top + \Lambda \mu u^\top\right)\right] \\ &= 2\left(u^\top \Lambda \mu\right)^2 + 2\left(\mu^\top \Lambda \mu\right)\left(u^\top \Lambda u\right), \end{aligned} \tag{24.12}$$

each of the four traces being evaluated by cycling. Adding the mean contribution $u^\top \Lambda u$ from (24.7) and halving (24.12) gives the quadratic form

$$u^\top \Lambda u + \left(\mu^\top \Lambda \mu\right)\left(u^\top \Lambda u\right) + \left(u^\top \Lambda \mu\right)^2 = u^\top \left[\left(1 + \mu^\top \Lambda \mu\right)\Lambda + \Lambda \mu \mu^\top \Lambda\right] u, \tag{24.13}$$

which is (24.9). For the cross block pair $(u, 0)$ with $(0, B)$: the mean parts do not meet, and the covariance parts contribute $\frac{1}{2}\operatorname{Tr}\left(\Lambda\left[-(u\mu^\top + \mu u^\top)\right]\Lambda B\right) = -u^\top \Lambda B \Lambda \mu$, which is (24.10). For the second block take $u = 0$, so $\delta C = B$ and nothing changes. □

Corollary 24.3 (Restriction and quotient mean sectors). *Decompose the expectation-chart tangent space as $\mathbb{R}^n \oplus \mathbb{S}^n$, with the second summand tangent to M_2. The metric induced on the quotient by the second-moment directions is the Schur complement*

$$g_\mu^{\mathrm{quot}} = g_{[\mu\mu]} - g_{[\mu M_2]} g_{[M_2 M_2]}^{-1} g_{[M_2 \mu]} = \Lambda. \tag{24.14}$$

Equivalently, it is the restriction of g_τ to the g_τ-orthogonal complement of the second-moment directions, expressed through the quotient coordinate μ. [ESTABLISHED]

Proof. The Fisher metric is positive definite on this minimal Gaussian family, so the second block and its Schur complement are invertible. The block-inverse identity gives

$$\left(g_\mu^{\mathrm{quot}}\right)^{-1} = \left(g_\tau^{-1}\right)_{[hh]}.$$

By (24.6), $g_\tau^{-1} = g_\eta$, and its hh block is $\mathrm{Cov}_q(Y) = C = \Lambda^{-1}$ by (24.5). Inverting proves (24.14). The Schur complement is the standard formula for the metric on the orthogonal quotient. □

Corollary 24.4 (Size of the discrepancy and the exact equality case). *Write $v = \Lambda^{1/2}\mu$. Then*

$$g_{[\mu\mu]} = \Lambda^{1/2}\left[\left(1 + \|v\|^2\right) I_n + vv^\top\right]\Lambda^{1/2}, \tag{24.15}$$

so the eigenvalues of $\Lambda^{-1} g_{[\mu\mu]}$ are $1 + \mu^\top\Lambda\mu$ with multiplicity $n-1$ and $1 + 2\mu^\top\Lambda\mu$ with multiplicity one. Consequently $g_{[\mu\mu]} \succeq \Lambda$ always, and $g_{[\mu\mu]} = \Lambda$ if and only if $\mu = 0$. [ESTABLISHED]

Proof. Equation (24.15) is (24.9) rewritten. The bracket is $\left(1 + \|v\|^2\right) I + vv^\top$, whose eigenvalues are $1 + \|v\|^2 + \|v\|^2$ along v and $1 + \|v\|^2$ on $v^\perp$; since $\Lambda^{-1} g_{[\mu\mu]} = \Lambda^{-1/2}\,[\cdot]\,\Lambda^{1/2}$ is similar to the bracket, its spectrum is that list. For the equality case, $g_{[\mu\mu]} - \Lambda = \left(\mu^\top\Lambda\mu\right)\Lambda + \Lambda\mu\mu^\top\Lambda$ is a sum of two positive semidefinite matrices, so it vanishes only if both vanish; the first forces $\mu^\top\Lambda\mu = 0$ and hence $\mu = 0$ because $\Lambda \succ 0$. The converse is immediate. □

The discrepancy between the two *restriction* mean blocks is therefore governed by the squared Mahalanobis norm of the mean, $\mu^\top\Lambda\mu$, and it is unbounded. It is not a normalization constant and it does not disappear in any limit that keeps the mean away from the origin. This statement does not concern the quotient metric, which equals Λ exactly by Corollary 24.3.

The mechanism behind the discrepancy is worth stating plainly, because it is what makes the error easy to commit. Both charts call their first coordinate μ, but they do not have the same coordinate vector fields. The vector $\partial/\partial\mu$ at fixed C moves the mean while holding the covariance fixed, so M_2 changes; the vector $\partial/\partial\mu$ at fixed M_2 moves the mean and compensates in C. These are different tangent vectors at the same point of the manifold, and a metric assigns them different lengths. Writing $g_{[\mu\mu]}$ without saying which of the two is meant does not denote anything.

The same observation resolves the phrase “the mean sector” in the expectation chart. A metric tensor has a well defined restriction to a subspace of the tangent space only once that subspace is declared; what is not chart independent is the *complement*. In the moment chart the covariance

directions are Fisher-orthogonal to the mean directions, by the vanishing third-moment argument in the proof of Proposition 24.1, so "the mean sector" can be taken to mean either the restriction to the mean directions or the orthogonal quotient, and the two agree. In the expectation chart (24.10) is nonzero, the mean and second-moment directions are not Fisher-orthogonal. The restriction is (24.9), whereas the orthogonal quotient is (24.14); the former differs from Λ away from $\mu = 0$, and the latter equals Λ at every point. Absent a declared reading, "the mean sector" in that chart names no unique object.

The chart discipline. A coordinate identity that mixes the mean and covariance sectors of a Gaussian family must name the chart and whether it uses restriction or quotient. Identity (24.7) holds in (μ, C); identities (24.9) and (24.10) give the expectation-chart restriction and cross blocks; and (24.14) gives its quotient mean metric. By Corollary 24.4 the two restriction mean blocks agree only at $\mu = 0$, while the quotient mean metric equals Λ everywhere. [ESTABLISHED]

Verification

The statements are proved above in general dimension; the machine checks reported here test those proofs and are not a substitute for them. Two symbolic routes and two numerical routes were run, all in exact or double precision on the CPU.

The first symbolic route is independent of Proposition 24.1: the Hessian of (24.2) was formed by computer algebra in the natural chart $(h, \mathrm{svec}(-J/2))$ with the Frobenius-paired symmetric coordinates, evaluated at exact rational (μ, Λ), inverted exactly as in (24.6), and its blocks compared with (24.9) and (24.10). At dimensions $n = 1, 2, 3$ the difference was the exact zero matrix in both blocks, and at $n = 2$ the cross block became exactly zero when $\mu = 0$ was substituted and was nonzero otherwise, as Proposition 24.2 requires. The second symbolic route carries general symbols: at dimensions $n = 2$ and $n = 3$, with μ a vector of free symbols and Λ a general symmetric matrix of free symbols, the pullback of (24.7) along (24.11) was simplified to zero difference against all three blocks of Proposition 24.2. Those two runs are proofs at those dimensions, not samples.

The deterministic numerical route `CHK-IG-EXPECTATION-METRIC` uses the closed-form Gaussian relative entropy and its Fisher quadratic limit in six directions at a fully specified base point. It compares the expectation-chart block with Equation (24.9), uses the moment-chart precision as a negative control against chart conflation, and checks the predicted generalized spectrum. The computation is reproducible corroboration, not a proof. [NUMERICAL]

24.3 Sample space and parameter space

The renormalization chapters will import vocabulary from a literature that computes Fisher metrics of likelihoods with respect to *model* parameters and reads their spectra as a scale [13]. The object developed above is not that object, and keeping the two typed apart is a precondition for importing anything at all.

Four spaces are in play. The latent sample space is $\mathbb{R}^n$, where Y takes its values; the interaction energy $y \mapsto \frac{1}{2} y^\top \Lambda y$ is a quadratic form on it. The recognition parameter manifold is the set of laws (24.1), coordinatized by any of the four charts of Section 24.1; the Fisher metric (24.7) is a Riemannian metric on it. The model parameter space is where θ lives and indexes the generative kernel. For a regular conditional model and a declared fixed input X, its conditional Fisher tensor is

$$I_X(\theta)(u,v) = \mathbb{E}_\theta \left[\partial_u \log p_\theta(O \mid X) \partial_v \log p_\theta(O \mid X) \,\middle|\, X \right]. \tag{24.16}$$

If inputs are randomized or a design ensemble is used, a declared probability law $r(dX)$ supplies the averaged tensor

$$I_r(\theta) = \int I_X(\theta) r(dX). \tag{24.17}$$

Different choices of X or r can change both the tensor and its rank. Only where the selected tensor is nondegenerate is it a Riemannian metric on the model parameter space; that space has the dimension of θ, which has no relation to $n = NK$ [1, 3]. Finally, the contextual base $\mathcal{C}$ indexes law-bundle sections. Once a connection and a statistical section are chosen, Definition 8.3 gives a generally semidefinite tensor on $T\mathcal{C}$; it is not any of the preceding three objects. [ESTABLISHED]

What makes the confusion easy is an exact but scoped component identity in this family, and stating it precisely is more useful than warning against it. The recognition family is a location family in its mean, so the mean parameter takes values in the same vector space $\mathbb{R}^n$ as the sample variable, and the mean block (24.7) of the parameter-space metric has the same components as the sample-space form Λ. The equality is of components in matched charts, not of objects: one is a metric on the tangent spaces of a manifold of probability laws, the other is a bilinear form on the latent vector space. For a family that is not a location family the two have different dimensions and no comparison arises. The identity is therefore a feature of the Gaussian mean sector and cannot be used as evidence that a construction defined on one space transfers to the other. [ESTABLISHED]

There is a corresponding exact but scoped relation to the interaction Laplacian L. If the transported endpoint laws on an edge (i,j) have means in $\mathbb{R}^K$ and share one fixed covariance $C_{ij} = J_{ij}^{-1} \in \mathbb{S}_{++}^K$ within the mean submodel, then

$$D_{\mathrm{KL}}\big(\mathcal{N}(\mu_i, C_{ij}) \| \mathcal{N}(\mu_j, C_{ij})\big) = \frac{1}{2}(\mu_i - \mu_j)^\top J_{ij} (\mu_i - \mu_j) = \frac{1}{2} g^{\mathrm{mean}}_{(\mu, C_{ij})}(\mu_i - \mu_j, \mu_i - \mu_j). \tag{24.18}$$

Thus pairwise KL is exactly one half of the associated mean-sector Fisher quadratic. Summing these typed edge quadratics defines the corresponding $NK \times NK$ connection-Laplacian energy; the assembled matrix L is not inserted into a single K-dimensional edge term. For a regular non-Gaussian family, freezing the Fisher tensor at one parallel base law gives the analogous form only as the local second-order term of transported KL [3]. Outside these Gaussian or frozen-local hypotheses, a sample-space L is not a recognition Fisher tensor or a natural-gradient operator. [ESTABLISHED]

For a smooth section of the full-Gaussian law bundle, the Fisher metric proved in Proposition 24.1 is the vertical tensor pulled back by the connection-relative first jet in Definition 8.3. The resulting base tensor depends on the section and chosen connection, and its rank is the rank seen by that first jet. The Gaussian component formulas do not identify the contextual base with the sample or parameter spaces. [ESTABLISHED]

The word "pullback" consequently has three typed uses in this manuscript. First, the connection-relative construction of Chapter 8 pulls a vertical statistical tensor back to the contextual base by $D^{\omega}s$. Second, the restriction below pulls a tensor back along an injective map of parameter manifolds. Third, the pullback attractor terminology in Section 11.4 describes backward asymptotic dependence for a nonautonomous cocycle; it is not a tensor operation. No theorem transfers between these uses merely because the same word appears. [DEFINITION]

The distinction has a sharp consequence for coarse-graining, which the next chapters will need. Restricting the recognition mean to a subspace and marginalizing onto that subspace are different operations and give different metrics.

Proposition 24.5 (Pullback versus pushforward). *Let $\Lambda \succ 0$ on $\mathbb{R}^n$, let $B \in \mathbb{R}^{n\times k}$ have orthonormal columns, and let $B_\perp$ span the orthogonal complement. The Fisher metric of the submodel obtained by constraining the mean to $\mathrm{range}(B)$ is the restriction $B^\top \Lambda B$; the Fisher metric of the mean of the pushforward law under $y \mapsto B^\top y$ is the marginal precision $\left(B^\top \Lambda^{-1} B\right)^{-1}$. They satisfy*

$$B^\top \Lambda B - \left(B^\top \Lambda^{-1} B\right)^{-1} = \left(B^\top \Lambda B_\perp\right)\left(B_\perp^\top \Lambda B_\perp\right)^{-1}\left(B_\perp^\top \Lambda B\right) \succeq 0, \tag{24.19}$$

with equality if and only if $B^\top \Lambda B_\perp = 0$, that is if and only if the retained and discarded subspaces are Λ-orthogonal. [ESTABLISHED]

Proof. The matrix $[B \quad B_\perp]$ is orthogonal, so in that basis Λ has blocks $\Lambda_{11} = B^\top \Lambda B$, $\Lambda_{12} = B^\top \Lambda B_\perp$, $\Lambda_{22} = B_\perp^\top \Lambda B_\perp$. The block inversion formula gives $B^\top \Lambda^{-1} B = \left(\Lambda_{11} - \Lambda_{12}\Lambda_{22}^{-1}\Lambda_{21}\right)^{-1}$ [40], and inverting again yields (24.19). The right side is a congruence of $\Lambda_{22}^{-1} \succ 0$ and is therefore positive semidefinite, and it vanishes exactly when $\Lambda_{12} = 0$. The identification of the two sides as pullback and pushforward metrics is the Gaussian conditional and marginal precision respectively [54]. □

Corollary 24.6 (Full-rank inclusion and averaging). *Let $S \in \mathbb{R}^{n\times k}$ have full column rank, set $M_S = S^\top S$, and let $S^\dagger = M_S^{-1} S^\top$. The mean submodel $\mu = S\bar{\mu}$ has Fisher matrix $S^\top \Lambda S$, while the mean of the pushforward under $y \mapsto S^\dagger y$ has Fisher matrix*

$$\Lambda_{\mathrm{av}} = M_S \left(S^\top \Lambda^{-1} S\right)^{-1} M_S. \tag{24.20}$$

Writing $B = S M_S^{-1/2}$, so that $B^\top B = I_k$, and completing B to an orthogonal matrix $[B\ B_\perp]$ gives

$$S^\top \Lambda S - \Lambda_{\mathrm{av}} = M_S^{1/2}\left[B^\top \Lambda B - \left(B^\top \Lambda^{-1} B\right)^{-1}\right] M_S^{1/2} \succeq 0. \tag{24.21}$$

Equality holds exactly when $B^\top \Lambda B_\perp = 0$, equivalently when $\mathrm{range}(S)$ and its Euclidean orthogonal complement are Λ-orthogonal. [ESTABLISHED]

Proof. Pulling the mean metric back along $\bar{\mu} \mapsto S\bar{\mu}$ gives $S^\top \Lambda S$. The pushforward covariance is

$$S^\dagger \Lambda^{-1} (S^\dagger)^\top = M_S^{-1} S^\top \Lambda^{-1} S M_S^{-1},$$

whose inverse is (24.20). Substituting $S = B M_S^{1/2}$ into both matrices yields (24.21), and Proposition 24.5 supplies its positivity and equality condition. □

Restriction is a pullback along an injective map of parameters, and it makes the metric larger in the Loewner order; marginalization is a pushforward along a map of the sample space, and it makes the metric smaller. The monotonicity theorems of information geometry govern the second operation, not the first, a point taken up in Section 24.5. The deterministic replacement check `CHK-IG-PULLBACK-PUSHFORWARD` tests the Schur identity and Loewner order over a specified ensemble and includes the Λ-orthogonal equality control. Computation is not proof and the proof is above. [NUMERICAL]

Whether the scale-setting construction of the model-space literature transfers to Λ is therefore not settled by observing that both are called Fisher metrics. The transfer requires a declaration that makes the coarse recognition mean a parameter of the generative law fitted against the bound, after which the relevant Fisher metric is the one with respect to that parameter and the import becomes a statement rather than an analogy. This section records the requirement and does not discharge it. What would close it is that declaration together with a check that the resulting parameter-space metric is nondegenerate, since the construction in that literature is written throughout with an inverted metric and does not treat the singular case. [OPEN]

24.3.1 Ambient score tangents and exponential action charts

A third distinction is needed before the renormalization chapters use the word “tangent”. At a fixed Gaussian law three objects are in play, and they are nested rather than equal. The first is the finite-dimensional tangent space of the declared regular exponential family, whose metric is (24.5); its dimension is that of the chart (h, J). The second is the ambient centered score Hilbert space $L^2_0(\gamma)$ of (12.54), which contains every quadratic-mean score of every path through the law, whether or not that path stays in the Gaussian family. The third is the bounded action quotient $L^\infty/\mathbb{R}\mathbf{1}$, which is the domain of the nonlinear action calculus of Theorem 12.4. By Lemma 12.9 every element of the second space is an actual two-sided score, and by Proposition 12.12 the third embeds isometrically into the second with dense but generally proper range. The first is a finite-dimensional subspace of the second, of dimension $n + n(n+1)/2$ by (24.8), and is a proper subspace whenever $L^2_0(\gamma)$ is infinite dimensional, which holds for every nondegenerate Gaussian on $\mathbb{R}^n$. [DEFINITION]

The gap between the second and third objects is not a technicality, and the Hermite directions show exactly where it opens.

Proposition 24.7 (Exact two-sided domain of a Hermite exponential action). *Let $\gamma = N(0,1)$, let He_k be the probabilists' Hermite polynomial of degree $k \geq 1$, and set*

$$N_k(t) = \int e^{-t\mathrm{He}_k(x)}\gamma(dx) \in (0, +\infty]. \tag{24.22}$$

Then $N_1(t) < \infty$ for every real t; $N_2(t) < \infty$ exactly for $t > -\frac{1}{2}$, where $N_2(t) = e^t(1+2t)^{-1/2}$; for odd $k \geq 3$, $N_k(t) = +\infty$ for every $t \neq 0$; and for even $k \geq 4$, $N_k(t) < \infty$ exactly for $t \geq 0$. Consequently every He_k with $k \geq 1$ lies in $L^2_0(\gamma)$ and is realized as a two-sided quadratic-mean score, while for $k \geq 3$ no two-sided exponential action neighborhood of the origin contains it. [ESTABLISHED]

Proof. For $k = 1$ the integral is the Gaussian moment-generating function $e^{t^2/2}$, finite for every t. For $k = 2$, $\mathrm{He}_2(x) = x^2 - 1$, so $N_2(t) = e^t \int e^{-tx^2}\gamma(dx)$, and the Gaussian integral $\int e^{-tx^2}\gamma(dx)$ converges exactly when $t + \frac{1}{2} > 0$, with value $(1+2t)^{-1/2}$. For $k \geq 3$ write $\mathrm{He}_k(x) = x^k + O(x^{k-2})$, so the exponent of the integrand of (24.22) against the Gaussian density is $-t\mathrm{He}_k(x) - x^2/2$, whose leading behavior is $-tx^k$ because $k \geq 3$ dominates degree two. For odd k and any $t \neq 0$, exactly one of the two tails has $-tx^k \to +\infty$, so the integrand grows superpolynomially there and the integral diverges. For even k and $t < 0$, both tails have $-tx^k \to +\infty$ and the integral diverges; for $t > 0$ both tails have $-tx^k \to -\infty$, so the integrand is eventually bounded by e^{-cx^k} for some $c > 0$ and the integral converges; and $N_k(0) = 1$. Membership in $L^2_0(\gamma)$ holds because Hermite polynomials are square integrable against γ and orthogonal to the constants, and the realization is Lemma 12.9. □

The two Gaussian score directions of (24.8) sit inside the benign part of this classification. At $n = 1$, $\mu = 0$, and $\Lambda = 1$, that score reads $ux + \frac{A}{2}(x^2 - 1) = u\mathrm{He}_1 + \frac{A}{2}\mathrm{He}_2$, so the mean and variance directions occupy Hermite degrees one and two, and by Proposition 24.7 both admit two-sided exponential neighborhoods; the constraint $t > -\frac{1}{2}$ in the variance direction is the natural-domain boundary $J \succ 0$ seen through the exponential chart. Every ambient direction of degree three or higher behaves differently. A complete Hermite basis of $L^2_0(\gamma)$ is therefore not a chart of the Gaussian model and not a set of directions of the bounded action space. In particular the spectral statement of Theorem 12.13 is a theorem about the L^2_0 tangent, and it must not be read as a statement about the nonlinear bounded action map, whose domain is the third object above. [ESTABLISHED]

24.4 Gauge invariants of the interaction precision

A local reframing is a change of local trivialization of the fibers, and it changes the components of every object while changing no law. Let

$$T = \mathrm{diag}\,(T_1, \ldots, T_N)\,, \qquad T_i \in \mathrm{GL}^+(K), \tag{24.23}$$

act on the assembled latent coordinate by $y \mapsto Ty$. Every quadratic form in y then transforms by inverse congruence,

$$\Lambda \longmapsto T^{-\top}\Lambda T^{-1}, \qquad L \longmapsto T^{-\top}LT^{-1}, \qquad A \longmapsto T^{-\top}AT^{-1}, \tag{24.24}$$

where L is the Laplacian part of the interaction precision and $A = \mathrm{diag}(A_1, \ldots, A_N)$ collects the self terms, so that $\Lambda = L + A$. Two hypotheses are being used and should be visible. First, the splitting $\Lambda = L + A$ is a declared decomposition of the assembled precision, made once in the frame-dependent chart; (24.24) then says that congruence acts on the two summands separately, which is immediate from linearity. Second, the reframing is block diagonal with respect to the agent partition, which is what a change of local trivialization is; a coordinate change mixing agents would not preserve the block structure that makes A block diagonal. [HYPOTHESIS]

Under a nonconstant reframing the image $T^{-\top}LT^{-1}$ need not itself be a matrix-weighted Laplacian, since the row sums that define that normal form are not preserved when neighboring blocks are reframed differently. This is a statement about normal forms, not about the object: L is a bilinear form on the latent space, and (24.24) is how a bilinear form reads in a new chart. The invariants below compare forms and are therefore unaffected.

Proposition 24.8 (Frame invariance of the generalized spectrum). *Let $\Lambda \succ 0$ and $L = L^{\top}$, and let T be invertible. The generalized eigenvalue problem*

$$Lx = d\Lambda x \tag{24.25}$$

and the reframed problem $\left(T^{-\top}LT^{-1}\right)x' = d'\left(T^{-\top}\Lambda T^{-1}\right)x'$ have the same eigenvalues, with eigenvectors related by $x' = Tx$. [ESTABLISHED]

Proof. Suppose $Lx = d\Lambda x$ and set $x' = Tx$. Then $T^{-\top}LT^{-1}x' = T^{-\top}Lx = dT^{-\top}\Lambda x = d\left(T^{-\top}\Lambda T^{-1}\right)x'$. The map $x \mapsto Tx$ is a bijection, so the correspondence is onto, and the argument run with T^{-1} gives the converse. □

Because $\Lambda \succ 0$, the pair (L, Λ) is a symmetric-definite, hence regular, pencil: it is simultaneously diagonalizable by congruence, all n generalized eigenvalues are real, and they obey a Rayleigh characterization [36, 40]. The positive-definiteness hypothesis is load bearing. Congruence covariance alone also holds for a singular pencil, but it does not imply that the determinant polynomial has degree n or even that it is nonzero. For example, at a common singular ray $(L, \Lambda) = (L, L)$ with $\ker L \neq \{0\}$,

$$\det(L - d\Lambda) = \det\big((1 - d)L\big) \equiv 0,$$

so the full-space determinant does not define an n-element generalized spectrum. A quotient, pin, or mass repair must first be declared and the resulting pencil proved regular before the conclusions below can be invoked. Thus Propositions 24.8 to 24.10 apply to the Gaussian probability interior

$\Lambda \succ 0$ (or to a separately proved regular repair), not automatically to an operator-layer boundary point. That qualification is the pencil-level instance of the hierarchy in Chapter 6.

Proposition 24.9 (Localization and endpoints). *If in addition $L \succeq 0$ and $A = \Lambda - L \succeq 0$, then every generalized eigenvalue of (24.25) lies in $[0, 1]$. The value 0 is attained exactly on $\ker L$, and the value 1 exactly on $\ker A$. If, further, L is the matrix-weighted Laplacian part of (22.8), with $W_{ij} \succeq 0$, then every configuration constant across agents lies in $\ker L$, and hence $\dim \ker L \geq K$.* [ESTABLISHED]

Proof. By the Rayleigh characterization, each generalized eigenvalue is a value of $d(x) = x^\top Lx / x^\top \Lambda x$ at a stationary point, and $x^\top \Lambda x = x^\top Lx + x^\top Ax$ with both terms nonnegative and the sum positive. Hence $0 \leq d(x) \leq 1$ for all $x \neq 0$, with $d(x) = 0$ iff $x^\top Lx = 0$, which for $L \succeq 0$ is $x \in \ker L$, and $d(x) = 1$ iff $x \in \ker A$. Under the additional interaction-Laplacian hypothesis, the difference form

$$x^\top Lx = \sum_{i<j} (x_i - x_j)^\top W_{ij} (x_i - x_j)$$

vanishes on every $x = \mathbf{1} \otimes w$, $w \in \mathbb{R}^K$. These K independent configurations lie in $\ker L$, proving the dimension clause. □

The generalized eigenvalue is therefore readable, and its reading is frame independent: d_k is the fraction of the total precision along the k-th pencil direction that is supplied by interaction rather than by self terms. The consensus directions sit at $d = 0$ because interaction supplies no precision there; a direction that no self term constrains sits at $d = 1$.

Proposition 24.10 (Frame dependence of absolute spectra and subspace determinants). *(i) For $T = cI_n$ with $c > 0$, which is a legitimate reframing since $\det(cI_K) = c^K > 0$, the ordinary spectrum obeys $\operatorname{spec}\left(T^{-\top} \Lambda T^{-1}\right) = c^{-2} \operatorname{spec}(\Lambda)$. (ii) Let $U \subset \mathbb{R}^n$ have orthonormal basis B, let $U' = TU$ have orthonormal basis B', and let $\Lambda' = T^{-\top} \Lambda T^{-1}$. Then*

$$\log\det\left(B'^\top \Lambda' B'\right) = \log\det\left(B^\top \Lambda B\right) - \log\det\left(B^\top T^\top T B\right), \tag{24.26}$$

so a subspace log-determinant is frame independent if and only if T restricted to U has unit determinant in absolute value; every orthogonal T satisfies this condition. [ESTABLISHED]

Proof. Part (i) is immediate. For (ii), write $TB = B'S$ with S invertible, so $B' = TBS^{-1}$; orthonormality of B' gives $S^{-\top}\left(B^\top T^\top T B\right) S^{-1} = I$ and hence $\det(S)^2 = \det\left(B^\top T^\top T B\right)$. Then $B'^\top \Lambda' B' = S^{-\top} B^\top T^\top T^{-\top} \Lambda T^{-1} T B S^{-1} = S^{-\top}\left(B^\top \Lambda B\right) S^{-1}$, and taking log determinants gives (24.26). □

The consequence is a prohibition, and it is the reason this section exists. A reframing changes no law; it changes only the components in which the law is written. A quantity computed from those components that is not invariant under (24.24) is therefore a property of the chart. By Proposition 24.10(i), any threshold criterion applied to the absolute eigenvalues of Λ can be made

to accept or to reject any given direction by choosing c, so no such criterion states anything about the model. Any admissible criterion must be phrased in generalized-eigenvalue form, that is as a statement about a pencil in which both members transform by the same congruence. By Proposition 24.10(ii) the same warning applies to determinant criteria evaluated on a subspace: they are invariant under orthogonal reframings and not otherwise, so a construction that uses one must either declare its frame or carry the Jacobian term in (24.26) explicitly. [ESTABLISHED]

Verification

Proposition 24.8 is proved above and needs no support. The deterministic replacement check `CHK-GENERALIZED-SPECTRUM` measures the generalized spectrum over twelve fully specified block congruences and uses the moving ordinary spectrum as its negative control. It establishes numerical usability only for that protocol; congruence invariance is proved above. [NUMERICAL]

24.5 What is not claimed

The Fisher metric is used here as a declared choice of geometry on the recognition manifold, and the classical uniqueness results are not strong enough to make that choice forced in this setting. On a finite normalized simplex the Fisher metric is, up to a positive scale, the unique Riemannian metric invariant under congruent Markov embeddings; this is Chentsov's theorem [85]. Campbell's extension to the positive cone of nonnormalized measures is a characterization, but not uniqueness up to scale: writing $s = \sum_i x_i$, every metric in the characterized family has coordinate components

$$g_x(\partial_i, \partial_j) = A(s) + \delta_{ij} \frac{sB(s)}{x_i}, \qquad B(s) > 0, \qquad A(s) + B(s) > 0, \tag{24.27}$$

for two otherwise arbitrary smooth functions A and B, and conversely every such pair defines an invariant family [21]. On the normalized simplex, tangent vectors satisfy $\sum_i u_i = \sum_i v_i = 0$, so the A term vanishes and (24.27) restricts to $B(1) \sum_i u_i v_i / x_i$, a constant multiple of Fisher. Thus simplex uniqueness and the larger nonnormalized cone family are separate statements. [ESTABLISHED]

The latent space here is $\mathbb{R}^n$, so neither finite-sample statement applies directly. Extensions to general sample spaces exist and are established, but they carry their own hypotheses: Ay et al. [6] and Lê [55] obtain uniqueness under a strong-continuity assumption on the information metric, and the monograph treatment collects the required regularity [7]. [ESTABLISHED]

The development therefore adopts the Fisher metric because it is the Hessian of the exponential-family potential (24.5) and because that is the structure the calculations use, and it does not claim that no other invariant metric is available in the continuous setting. [HYPOTHESIS]

No Markov/data-processing monotonicity theorem is asserted for the deterministic Galerkin aggregation map. [NOT-CLAIMED]

The established general statement is that the Fisher metric contracts under a Markov kernel: pushing a family forward through a stochastic map cannot increase Fisher information [3, 7]. [ESTABLISHED]

That score-projection theorem yields the contextual-base tensor defect in Theorem 8.21 only when the channel is realized by a bundle morphism satisfying the isotropy criterion (8.56) and the fine and coarse sections obey (8.44); without that criterion the exact signed comparison (8.52) replaces it and positivity holds only in the directions meeting (8.53). [ESTABLISHED]

The theorem does not cover the aggregation operation of the following chapters, because by Proposition 24.5 the coarse operator is a restriction, a pullback along an inclusion of mean submodels, and not a pushforward along a Markov kernel; the two differ by the positive semidefinite Schur term in (24.19), and the restriction is the larger of the two in the Loewner order. [ESTABLISHED]

That same distinction is why the pointwise-Markov hypothesis of Theorem 9.70 is not satisfied by the Galerkin aggregation, so no configuration-level Fisher contraction or duration comparison may be attached to it by that route. [ESTABLISHED]

Whether a monotone functional decreases under the full declared aggregation/RG map is open. Closing it requires either exhibiting a Markov kernel whose pushforward realizes the map, after which the classical theorem applies verbatim, or exhibiting a Lyapunov functional and proving its decrease directly. [OPEN]

The later fixed-covariance Gaussian relaxation in Section 25.5.1 is a different continuous-time dynamics: its quadratic energy has an exact dissipation identity, and that scoped flow is a Fisher natural-gradient flow. [ESTABLISHED]

It does not turn the coarse map itself into a Markov contraction. [NOT-CLAIMED]

Nothing here should be read as importing a uniqueness statement from the quantum setting. There, the metrics monotone under completely positive trace-preserving maps form a family parameterized by operator monotone functions rather than a single metric [72], so the phrase “the monotone metric” does not denote in that context, and an argument that relies on uniqueness cannot be transported from it. [ESTABLISHED]

No Markov-morphism contraction theorem for the deterministic aggregation maps is asserted. [NOT-CLAIMED]

Finally, this section supplies a metric and not a dynamics. No natural-gradient identification should be inferred merely from the appearance of Λ as a preconditioner: an update is a natural gradient only when its metric is the Fisher metric of the family being updated, in a named chart, and with the objective differential typed in the matching chart [1, 58]. The fixed-covariance Gaussian mean flow in Section 25.5.1 satisfies those requirements and is identified there. [ESTABLISHED]

No broader identification for the remaining updates is claimed. [NOT-CLAIMED]

Even for a declared natural-gradient flow, the metric does not turn its curve parameter into a clock. The oriented-history and Fisher-duration constructions of Chapter 9 quotient that parameterization explicitly. [ESTABLISHED]

No physical-time identification is claimed. [NOT-CLAIMED]

Chapter 25

Multivariate-Gaussian Coarse-Graining

This chapter realizes the three operations of Chapter 10 in the multivariate-Gaussian interaction family. It does not repeat the Markov or graph-exponential theory. Its claims concern hard identification, Gaussian marginalization and recognition restrictions, all at finite dimension. Unless a channel index is displayed, the symbols $K, \Theta, \Lambda, L, A, W$ in this chapter denote the belief/recognition channel. A model-channel realization is separate and uses its own $K_m, \Theta^m, \Lambda_m, L_m, A_m, W_m$; no equality of the two channels is implied. [DEFINITION]

Within the connection-Laplacian realization, Proposition 25.26 applies only to invertible represented transports with inverse reverse transports and positive-definite represented internal weights. Under those hypotheses, trivial represented internal holonomy is the criterion for a connected cluster to retain a full K-dimensional, state-independent structural fixed section; nontrivial holonomy may still leave a lower-dimensional fixed sector. [ESTABLISHED]

Marginal similarity is not identified with this structural criterion. [NOT-CLAIMED]

25.1 Hard aggregation of the interaction precision

Assume the declared interaction family of Chapter 22,

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \qquad \Lambda_{ij} = -W_{ij} \quad (i \neq j), \qquad A_i \succeq 0, \quad W_{ij} = W_{ji} \succeq 0. \tag{25.1}$$

In flat trivialized coordinates its energy is

$$\mathcal{E}(z) = \sum_i z_i^\top A_i z_i + \sum_{i<j} (z_i - z_j)^\top W_{ij} (z_i - z_j). \tag{25.2}$$

[HYPOTHESIS]

This form is a Gaussian subfamily, not a consequence of an arbitrary linear-Gaussian directed model. [NOT-CLAIMED]

For a hard partition $\mathcal{P}$, let

$$S = \widehat{S} \otimes I_K, \qquad \widehat{S}_{iI} = \mathbf{1}_{\{i \in I\}}, \qquad \Lambda_{\mathrm{c}} = S^\top \Lambda S.$$

The columns of S are orthogonal but not normalized:

$$S^\top S = \operatorname{diag}(n_I) \otimes I_K. \tag{25.3}$$

[DEFINITION]

Proposition 25.1 (Exact Gaussian aggregation). For every hard partition,

$$(\Lambda_c)_{IJ} = -\sum_{\substack{i\in I\\ j\in J}} W_{ij} \quad (I \neq J), \qquad (\Lambda_c)_{II} = \sum_{i\in I} A_i + \sum_{\substack{i\in I\\ j\notin I}} W_{ij}. \tag{25.4}$$

Thus $A_I = \sum_{i\in I} A_i$, $W_{IJ} = \sum_{i\in I, j\in J} W_{ij}$, and every internal edge is annihilated. [ESTABLISHED]

Proof. Substitute one coordinate z_I for every z_i in a cluster. An internal edge contributes zero. A cut edge contributes its difference energy between z_I and z_J, and self terms add. Finite sums preserve positive semidefiniteness; independently, $\Lambda_c \succeq 0$ follows by congruence. □

This is the Galerkin coarse operator with a piecewise-constant prolongator, not a new numerical construction [56, 84]. The parameter map is exact for hard identification. A separately fitted coupling is outside that exact trace construction, while learned or soft pooling is a different approximate model. [NOT-CLAIMED]

25.2 Kron reduction: coefficient families and graph boundaries

Gaussian marginalization eliminates nodes by a Schur complement. It is not the aggregation above and is not closed on the unrestricted matrix-weighted interaction family.

Proposition 25.2 (Unrestricted Kron reduction leaves the family). Take $N = 3$, $K = 2$, $A_1 = A_2 = A_3 = I_2$, and

$$W_{12} = 0, \qquad W_{13} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad W_{23} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{25.5}$$

Then the assembled Λ is positive definite. Eliminating node 3 manufactures

$$-\left(\Lambda^{\mathrm{Kron}}\right)_{12} = W_{13}\Lambda_{33}^{-1}W_{23} = \frac{1}{19}\begin{pmatrix} 9 & 3 \\ 4 & 14 \end{pmatrix}, \tag{25.6}$$

which is not symmetric and hence is not a positive-semidefinite interaction weight. [ESTABLISHED]

Proof. $\Lambda_{33} = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ has determinant 19, and direct multiplication gives (25.6). The leading principal minors of the full precision are $2, 6, 18, 48, 102, 233$, so Sylvester's criterion gives $\Lambda \succ 0$. The full Schur complement is symmetric; its two mirrored off-diagonal blocks are transposes. Membership in (25.1) additionally requires each individual block to be symmetric, which fails. □

The scalar loopy-Laplacian family is closed under every defined elimination [28]. The matrix analog is closed on a larger subfamily than common orthogonal diagonalization.

Definition 25.3 (Congruence-diagonal cone). For $H \in \mathrm{GL}(K)$, define

$$\mathcal{C}_H = \{HDH^\top : D = \mathrm{diag}(d_1, \dots, d_K),\ d_a \geq 0\}. \tag{25.7}$$

[DEFINITION]

Theorem 25.4 (Congruence-diagonal Kron closure). If every A_i and W_{ij} lies in one fixed $\mathcal{C}_H$, then every defined node-block Schur complement remains in the interaction family with all reduced parameters in $\mathcal{C}_H$. [ESTABLISHED]

Proof. Factor

$$\Lambda = (I_N \otimes H)\Lambda_D(I_N \otimes H^\top), \tag{25.8}$$

where, after grouping coordinates by channel, Λ_D is a direct sum of K scalar loopy Laplacians. For retained nodes R and eliminated nodes E,

$$\mathrm{Sc}_E(\Lambda) = (I_R \otimes H)\Big[(\Lambda_D)_{RR} - (\Lambda_D)_{RE}(\Lambda_D)_{EE}^{-1}(\Lambda_D)_{ER}\Big](I_R \otimes H^\top). \tag{25.9}$$

The middle operator eliminates channel by channel. Each scalar Schur complement is another loopy Laplacian: its off-diagonals remain nonpositive because the eliminated block is a nonsingular M-matrix, and its row sums remain nonnegative. Reassembling the channels gives reduced self terms and weights in $\mathcal{C}_H$. □

This family strictly contains the common-orthogonal-eigenbasis family. For

$$H = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad X = H\,\mathrm{diag}(1,2)H^\top, \quad Y = H\,\mathrm{diag}(3,1)H^\top,$$

one has $XY - YX = \left(\begin{smallmatrix} 0 & -5 \\ 5 & 0 \end{smallmatrix}\right) \neq 0$, so X, Y cannot be simultaneously orthogonally diagonalized, while they belong to the same $\mathcal{C}_H$. [ESTABLISHED]

Theorem 25.5 (Qualified maximality). Let $\mathcal{C} \subseteq \mathbb{S}_+^K$ be a closed convex cone with an SPD order unit $M \in \mathrm{relint}\,\mathcal{C}$. Assume that anchors and incident edge coefficients range independently over $\mathcal{C}$, arbitrary positive edge scaling is allowed, every defined one-node elimination must remain in the family, and every manufactured off-diagonal block must be minus a symmetric member of $\mathcal{C}$. Then $\mathcal{C} \subseteq \mathcal{C}_H$ for some H, and a maximal cone under these assumptions equals $\mathcal{C}_H$. [ESTABLISHED]

Proof. For $X, Y \in \mathcal{C}$, relative interior gives an $\varepsilon > 0$ such that

$$A_e = M - \varepsilon(X + Y) \in \mathcal{C}.$$

Use $\varepsilon X, \varepsilon Y$ as the two incident weights and M as retained anchors. The eliminated diagonal is M, and the manufactured coupling is $\varepsilon^2 X M^{-1} Y$. Its required symmetry forces

$$XM^{-1}Y = YM^{-1}X.$$

After whitening by $M^{-1/2}$, every pair in the whitened cone commutes. Commuting real symmetric matrices are simultaneously orthogonally diagonalizable, so

$$\mathcal{C} \subseteq \{M^{1/2} O D O^\top M^{1/2} : D \succeq 0 \text{ diagonal}\} = \mathcal{C}_H.$$

The preceding theorem proves that $\mathcal{C}_H$ itself is closed, yielding maximality. □

The qualified theorem concerns coefficient cones with independently variable anchors. Three further questions have different quantifiers: which arbitrary coefficient families pass tests at one fixed positive anchor, which fixed graphs close for every independent SPD assignment, and which nonflat graphs close while retaining their link marks. We treat these questions separately. In every case, Schur reduction means integration over coordinates, with the linear source and scalar normalizer retained. [DEFINITION]

25.2.1 The complete Gaussian marginal action

Fix a finite coordinate split $z = (z_R, z_E)$, an SPD precision Λ, a source h, and a real scalar offset $\mathfrak{c}$. Write $n = \dim z$, $n_E = \dim z_E$, and use the declared coordinate Lebesgue measures. Write $\mathbb{P}_\Lambda(dz) = p_\Lambda(z)dz$. The normalized Gaussian prior is

$$\begin{aligned} \Phi(z) &= \frac{1}{2} z^\top \Lambda z - h^\top z + \mathfrak{c}, \\ p_\Lambda(z) &= Z_\Lambda^{-1} e^{-\Phi(z)}, \\ Z_\Lambda &= e^{-\mathfrak{c}} (2\pi)^{n/2} (\det \Lambda)^{-1/2} \exp\left(\frac{1}{2} h^\top \Lambda^{-1} h\right). \end{aligned} \tag{25.10}$$

The first line declares the action; the last line follows by translating z by $\Lambda^{-1}h$, orthogonally diagonalizing Λ, and multiplying the one-dimensional integrals $\int_{\mathbb{R}} e^{-\lambda x^2/2} dx = (2\pi/\lambda)^{1/2}$ for $\lambda > 0$. Thus the prior is the actual probability law $\mathcal{N}(\Lambda^{-1}h, \Lambda^{-1})$. [ESTABLISHED]

Proposition 25.6 (Normalized Schur marginal and reverse kernel). Let $M = \Lambda_{EE}$. The exact marginal action and all-input conditional kernel are

$$\begin{aligned} \Lambda_R^{\text{eff}} &= \Lambda_{RR} - \Lambda_{RE} M^{-1} \Lambda_{ER}, \\ h_R^{\text{eff}} &= h_R - \Lambda_{RE} M^{-1} h_E, \\ \mathfrak{c}_R^{\text{eff}} &= \mathfrak{c} + \frac{1}{2} \log\det M - \frac{n_E}{2} \log(2\pi) - \frac{1}{2} h_E^\top M^{-1} h_E, \\ \mathbb{P}_\Lambda(dz_E \mid z_R = x) &= \mathcal{N}\left(M^{-1}(h_E - \Lambda_{ER} x), M^{-1}\right)(dz_E). \end{aligned} \tag{25.11}$$

Both M and Λ_R^{eff} are SPD. Integration preserves the raw mass Z_Λ, and the normalized marginal is the coordinate pushforward of p_Λ. Successive eliminations agree with direct elimination, including their sources and scalar offsets. [ESTABLISHED]

Proof. The hidden square has center $M^{-1}(h_E - \Lambda_{ER}x)$:

$$\begin{aligned}\Phi(x, z_E) = \frac{1}{2}\big(z_E - M^{-1}(h_E - \Lambda_{ER}x)\big)^\top M\big(z_E - M^{-1}(h_E - \Lambda_{ER}x)\big) \\ + \frac{1}{2}x^\top \Lambda_{RR}x - h_R^\top x + \mathfrak{c} - \frac{1}{2}(h_E - \Lambda_{ER}x)^\top M^{-1}(h_E - \Lambda_{ER}x).\end{aligned} \tag{25.12}$$

Integrating this square gives $(2\pi)^{n_E/2}(\det M)^{-1/2}$; expansion of the last term proves every line of (25.11). A principal block of an SPD matrix is SPD. For any nonzero retained x, minimizing $(x, z_E)^\top \Lambda (x, z_E)$ over z_E gives $x^\top \Lambda_R^{\mathrm{eff}} x > 0$, proving the other positivity assertion. For a nonnegative Borel function on a finite product of Euclidean spaces, the product integral equals its iterated integrals in any order, with infinite values allowed. Here the displayed Gaussian integrals are finite, so this fact proves equality of all elimination orders. It also proves composition of the conditional reconstruction kernels when each is formed from the corresponding marginal. □

A normalized observed experiment can be retained as well. Declare the independent noise model $o = D_R z_R + D_E z_E + \eta$, with $\eta \sim \mathcal{N}(0, \Gamma_o)$ and $\Gamma_o \succ 0$. Integrating this actual sensor against the conditional kernel above gives

$$o \mid z_R = x \sim \mathcal{N}\left((D_R - D_E M^{-1}\Lambda_{ER})x + D_E M^{-1} h_E, \Gamma_o + D_E M^{-1} D_E^\top\right). \tag{25.13}$$

The covariance is SPD, and mixing this likelihood with the retained prior recovers the original positive, finite Gaussian evidence at every record. This follows by writing the hidden conditional as its affine mean plus independent centered Gaussian noise. Hence the same coordinate projection pushes the entire observed joint, not just its prior. The finite-KL chain identity of Section 10.2 then gives the exact VFE difference using the same evidence on both sides. The posterior hidden conditional generally differs from the prior conditional in (25.11); when $D_E = 0$, the sensor depends only on retained coordinates and those conditionals agree. [ESTABLISHED]

25.2.2 Arbitrary independent families at a fixed anchor

Fix $Q_0 = Q_0^\top \succ 0$ and an arbitrary subset $\mathscr{W} \subseteq \mathbb{S}_+^K$. No closedness, convexity, scaling closure, or membership of Q_0 in $\mathscr{W}$ is assumed. For each $A, B \in \mathscr{W}$, require that a two-edge test with some $t, s > 0$ is admitted, with eliminated anchor Q_0, incident weights tA, sB, and manufactured coefficient

$$M = Q_0 + tA + sB, \qquad W_{A,B}(t, s) = tsAM^{-1}B. \tag{25.14}$$

This is independent pair coverage of the test domain. It does not assert that tA, sB, or the output already belong to $\mathscr{W}$. [HYPOTHESIS]

Giving the two retained vertices the anchor Q_0 realizes every test as a proper three-node Gaussian: its energy is at least $\lambda_{\min}(Q_0)\|z\|^2/2$. [ESTABLISHED]

Theorem 25.7 (Fixed-anchor classification of arbitrary families). Under the preceding test hypothesis, symmetry of at least one $W_{A,B}(t,s)$ for every pair $A, B \in \mathscr{W}$ is equivalent to

$$AQ_0^{-1}B = BQ_0^{-1}A \quad \text{for all } A, B \in \mathscr{W}. \tag{25.15}$$

It is also equivalent to the existence of one $O \in O(K)$ such that

$$H = Q_0^{1/2}O, \qquad Q_0 = HH^\top, \qquad \mathscr{W} \subseteq \mathcal{C}_H. \tag{25.16}$$

The same H works for the whole family, even if it is uncountable. Under these equivalent conditions every positive pair of scales gives $W_{A,B}(t,s) \in \mathcal{C}_H$. The inclusion-maximal families satisfying the test property are precisely these full cones $\mathcal{C}_H$. [ESTABLISHED]

Proof. Whiten by Q_0, writing $a = Q_0^{-1/2}AQ_0^{-1/2}$, $b = Q_0^{-1/2}BQ_0^{-1/2}$, and $N = I + ta + sb$. Substituting $b = (N - I - ta)/s$ into the two terms gives

$$\begin{aligned} aN^{-1}b - bN^{-1}a &= s^{-1}(N^{-1}a - aN^{-1}) = N^{-1}(ab - ba)N^{-1}, \\ W_{A,B} - W_{A,B}^\top &= tsQ_0M^{-1}(AQ_0^{-1}B - BQ_0^{-1}A)M^{-1}Q_0. \end{aligned} \tag{25.17}$$

All outside factors are invertible or positive scalars. Thus one positive tested pair already forces (25.15); zero scales would give no information.

The whitened family is real symmetric and pairwise commuting. Simultaneous orthogonal diagonalization for the entire family follows by induction on dimension. If every member is scalar, choose any orthonormal basis. Otherwise select one nonscalar member. Every other member preserves its orthogonal eigenspaces, since commutation preserves each eigenspace. Those spaces have smaller dimensions, so apply the induction to the restricted families and concatenate their bases. Only finitely many dimension reductions are used; no countability premise is needed. For an empty family any basis works. PSD positivity makes all diagonal entries nonnegative, proving (25.16). Conversely, in these coordinates

$$W_{A,B}(t,s) = H\,\mathrm{diag}\left(\frac{tsa_\ell b_\ell}{1 + ta_\ell + sb_\ell}\right)_{\ell=1}^{K} H^\top, \qquad a_\ell, b_\ell \geq 0. \tag{25.18}$$

This proves sufficiency and positivity.

Every tested family therefore lies in a full tested cone. A maximal family equals that cone. Conversely, a PSD matrix compatible with every member of a full cone becomes, after whitening and the same orthogonal change, a matrix commuting with every coordinate rank-one projector. Commutation with the i-th projector forces all off-diagonal entries in its row and column to vanish. The matrix is already nonnegative diagonal, so it belongs to the cone. This proves maximality. □

The fixed anchor and independent pair coverage are material. If only equal pairs $A = B$ are admitted, every $A(I + (t+s)A)^{-1}A$ is symmetric PSD, even for the family $\{I, \mathrm{diag}(1,2), \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)\}$, which has no common congruence-diagonal cone containing all three matrices. Indeed, a cone

containing $I = HD_0H^\top$ gives the common orthogonal basis $HD_0^{1/2}$, whereas the latter two matrices do not commute. A cone accommodating the weights but not the anchor is also insufficient: for $A = \mathrm{diag}(1,0)$, $B = \mathrm{diag}(0,1)$, $Q_0 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, and $t = s = 1$, the manufactured coefficient is $\left(\begin{smallmatrix} 0 & -1/8 \\ 0 & 0 \end{smallmatrix}\right)$. These are exact failures of the omitted premises. [ESTABLISHED]

Proposition 25.8 (Stability inside a prescribed diagonal domain). Fix H and independent coordinate domains $\mathscr{D}_A, \mathscr{D}_W \subseteq [0,\infty)^K$, with $0 \in \mathscr{D}_W$. For $A_i = H\,\mathrm{diag}(a_i)H^\top$ and $W_{ij} = H\,\mathrm{diag}(w_{ij})H^\top$, a defined elimination of e has $d = a_e + \sum_{i\neq e} w_{ie} > 0$ coordinatewise and updates

$$w'_{ij} = w_{ij} + \frac{w_{ie} \odot w_{ej}}{d}, \qquad a'_i = a_i + \frac{w_{ie} \odot a_e}{d}. \tag{25.19}$$

Products and division are coordinatewise. Closure inside the prescribed domains for every finite PSD graph and every defined one-node elimination is equivalent to

$$b_i + \frac{u_i \odot a}{d} \in \mathscr{D}_A, \qquad c_{ij} + \frac{u_i \odot u_j}{d} \in \mathscr{D}_W, \qquad d = a + \sum_i u_i > 0, \tag{25.20}$$

for every finite star inventory $a, b_i \in \mathscr{D}_A$ and $u_i, c_{ij} \in \mathscr{D}_W$. If only full-space SPD graphs are admitted, necessity is restricted to the inventories producing such graphs. [ESTABLISHED]

Proof. The off-diagonal Schur blocks give the first update. In each scalar coordinate, subtracting the newly computed incident-weight sum from the new diagonal leaves

$$a_i + w_{ie} - \frac{w_{ie}(w_{ie} + \sum_{j\neq i,e} w_{ej})}{d} = a_i + \frac{w_{ie}a_e}{d}. \tag{25.21}$$

This proves the anchor update. Applying these formulas at every actual star proves sufficiency of (25.20). Conversely, realize any displayed inventory as a star with the specified retained anchors and pair weights, and eliminate its center. Its pivot is positive by the assumption on d, so the outputs must satisfy exactly the displayed membership conditions. For several eliminated vertices, positivity of the eliminated principal block ensures positive successive pivots by the minimum-energy argument in Proposition 25.6. □

Containing a family in $\mathcal{C}_H$ therefore does not imply closure inside that family. Already in one dimension, the coefficient set $\{1\}$, with anchor one at all three vertices and two unit edges, produces a weight $1/3$ and retained anchors $4/3$. The failure is membership in the chosen scalar domain, despite positivity and diagonalization. [ESTABLISHED]

Corollary 25.9 (The shared-kernel version). If $Q_0 \succeq 0$, $N = \ker Q_0$, and $N \subseteq \ker A$ for every $A \in \mathscr{W}$, all forms descend to $\mathbb{R}^K/N$. The descended anchor is SPD, and Theorem 25.7 applies there using an ordinary inverse. In particular, if $E_0 : \mathbb{R}^r \to N^\perp$ is an orthonormal inclusion and $r = \mathrm{rank}\, Q_0$, the pair-test property is equivalent to

$$H_r = E_0(E_0^\top Q_0 E_0)^{1/2}O, \qquad Q_0 = H_r H_r^\top, \qquad A = H_r\,\mathrm{diag}(a_A)H_r^\top, \quad a_A \geq 0, \tag{25.22}$$

for one $O \in O(r)$ and all members A. This is a quotient statement, not an inverse or a normalized density on the original null directions. [ESTABLISHED]

Proof. Symmetry and $An = 0$ make each form independent of the representative of a quotient class. Positivity gives $\ker(Q_0 + tA + sB) = N$, since a zero of its quadratic must annihilate every nonnegative summand, including Q_0. The quotient pivot is therefore positive, and whitening in the coordinates E_0 gives (25.22). When $r = 0$, all the forms vanish and the quotient is a point. Without the shared-kernel hypothesis, $Q_0 = 0$ and SPD A, B instead satisfy

$$\left[A(tA + sB)^{-1}B\right]^{-1} = tB^{-1} + sA^{-1} \succ 0. \tag{25.23}$$

even when A, B do not commute. The three-matrix family containing I above excludes a common congruence cone, so it is also a counterexample to dropping the support premise. A positive-dimensional null direction has infinite Lebesgue integral, with either a constant or an exponential linear factor along it; no full-space Gaussian normalizer has been supplied. □

25.2.3 Variable contexts and ordered diagonal frames

On a declared smooth context domain $U \subseteq \mathcal{C}$, let $Q_0(c) \succ 0$ vary smoothly and let $\mathscr{W}_c$ be an inclusion-maximal family for the fixed-anchor pair test at each c. Assume that the whitened spans

$$\mathscr{D}_c = \operatorname{span}_{\mathbb{R}}\{Q_0(c)^{-1/2} A Q_0(c)^{-1/2} : A \in \mathscr{W}_c\}. \tag{25.24}$$

form a smooth rank-K vector subbundle of $U \times \mathbb{S}^K$. Pointwise commutation does not supply this regularity assumption, and no pair using coefficients from different contexts is admitted by the present hypothesis. [HYPOTHESIS]

Theorem 25.10 (Local frames and a global ordering obstruction). Every $c_0 \in U$ has a neighborhood U_0 and a smooth $O : U_0 \to O(K)$ for which

$$\begin{aligned} \mathscr{D}_c &= O(c)\operatorname{Diag}(K)O(c)^\top, \\ \mathscr{W}_c &= H(c)\operatorname{Diag}_+(K)H(c)^\top, \\ H(c) &= Q_0(c)^{1/2}O(c). \end{aligned} \tag{25.25}$$

A global continuous ordered diagonal frame need not exist, even for a smooth maximal family over the circle with $Q_0 = I_2$. [ESTABLISHED]

Proof. The finite-dimensional implicit-function theorem says that a smooth equation $F(x, y) = 0$ with invertible derivative in y at a solution has a unique nearby smooth solution $y(x)$. Apply it on symmetric matrices to $S^2 - Q_0 = 0$ at the positive square root. In an orthonormal eigenbasis of S, its derivative $X \mapsto SX + XS$ multiplies entry ij by $s_i + s_j > 0$. Thus $Q_0^{1/2}$, and hence its inverse, are smooth; uniqueness of the positive square root identifies the local solutions.

By Theorem 25.7, $\mathscr{D}_{c_0}$ contains a matrix with K distinct eigenvalues. A smooth local subbundle frame extends it to a smooth section $T(c) \in \mathscr{D}_c$. For $\det(\lambda I - T(c)) = 0$, the derivative in λ at each initial eigenvalue is the product of its nonzero spectral gaps. The same implicit-function theorem therefore supplies K distinct smooth real eigenvalues $\lambda_i(c)$ on a smaller neighborhood. Their spectral projectors are

$$P_i(c) = \prod_{j \neq i} \frac{T(c) - \lambda_j(c) I}{\lambda_i(c) - \lambda_j(c)}. \tag{25.26}$$

In an eigenbasis this polynomial is one on the i-th eigenspace and zero on the others, so these are smooth orthogonal rank-one projectors. Choose a unit $v_i \in \operatorname{im} P_i(c_0)$. After shrinking the neighborhood, $e_i(c) = P_i(c)v_i / \|P_i(c)v_i\|$ is defined and smooth. Its columns form $O(c)$. Every member of $\mathscr{D}_c$ commutes with $T(c)$, hence preserves these one-dimensional eigenspaces. Both algebra spaces have dimension K, so the resulting diagonal containment is equality. Undoing whitening gives (25.25).

For the global obstruction write $\theta \in S^1 = \mathbb{R}/(2\pi\mathbb{Z})$, let $\mathsf{R}(\varphi)$ be the planar rotation by φ, and set

$$\mathscr{W}_\theta = \mathsf{R}(\theta/4)\,\mathrm{Diag}_+(2)\mathsf{R}(\theta/4)^\top, \qquad T_\theta = \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2) \\ \sin(\theta/2) & -\cos(\theta/2) \end{pmatrix}. \tag{25.27}$$

Increasing θ by 2π rotates by another $\pi/2$, which permutes the two coordinate axes and preserves the cone. The algebra is $\operatorname{span}\{I_2, T_\theta\}$. Its trace-inner-product projection is

$$X \longmapsto \frac{1}{2}\operatorname{Tr}(X) I_2 + \frac{1}{2}\operatorname{Tr}(T_\theta X) T_\theta. \tag{25.28}$$

This is smooth and periodic because $T_{\theta+2\pi} = -T_\theta$, and has constant rank two. It therefore defines the required smooth subbundle. The two trace-one extreme projectors are $P_\pm(\theta) = (I_2 \pm T_\theta)/2$, which interchange after a full circuit. Any continuous ordered first projector has $\operatorname{Tr}(T_\theta P(\theta)) \in \{1, -1\}$, a continuous function on $[0, 2\pi]$, so its sign is constant. Its endpoint is therefore the opposite projector from its initial value, contradicting periodicity. An ordered orthogonal diagonal frame would supply such an ordered projector, so none exists. □

Smooth global coefficients still give smooth Schur complements, determinant offsets, and normalized Gaussian laws wherever the actual precision blocks remain SPD: matrix inversion is the adjugate divided by a nonzero determinant, and $\log\det$ is smooth there. No diagonal frame is needed to evaluate those formulas. The obstruction is global ordering of the axes, not probabilistic marginalization. If every cross-context pair were instead tested against one common fixed anchor, Theorem 25.7 would apply to the union and force one common frame. [ESTABLISHED]

25.2.4 A graph criterion for unrestricted matrix weights

Now vary every present edge weight independently over all SPD $K \times K$ matrices and restrict the graph instead. Let $\mathcal{G}$ be a finite undirected graph with parallel edge copies allowed and no self-loops. If anchors occur, add a ground vertex 0, represent each present SPD anchor by an edge to 0, and retain 0 throughout elimination. Fix its coordinate to zero only when forming the ordinary anchored

probability law. In the augmented common frame,

$$\mathcal{E}_{\mathrm{aug}}(z) = \frac{1}{2} \sum_{e \in \mathscr{E}(\mathcal{G})} (z_{i_e} - z_{j_e})^\top W_e (z_{i_e} - z_{j_e}), \qquad W_e \succ 0. \tag{25.29}$$

Let E be the eliminated ordinary vertices and R the remaining vertices, including ground. For each connected component C of the induced graph on E, define its retained terminal boundary

$$B(C) = \{r \in R : \text{some edge joins } r \text{ to a vertex of } C\}. \tag{25.30}$$

Assume $B(C) \neq \varnothing$ for every C. [HYPOTHESIS]

This is precisely the structural condition making the hidden Dirichlet block positive for every SPD assignment: with retained coordinates zero, zero edge energy forces equality throughout a hidden component and then zero at its boundary. A component without boundary instead has a nonzero constant hidden null vector. [ESTABLISHED]

Theorem 25.11 (Universal closure at a fixed retained boundary). For $K \geq 2$, the Schur complement of (25.29) is an augmented matrix-weighted Laplacian for every independent SPD edge assignment if and only if

$$|B(C)| \leq 2 \quad \text{for every connected hidden component } C. \tag{25.31}$$

Ground counts as one terminal. Here membership means symmetric nonpositive off-diagonal blocks and diagonal blocks equal to the negative sum of the off-diagonal blocks. A two-terminal component contributes one SPD edge, and a one-terminal component contributes zero quadratic form. For $K = 1$, every graph satisfying the nonempty-boundary hypothesis closes, so the bound of two is not necessary. [ESTABLISHED]

Proof. The hidden Dirichlet block is block diagonal over the components. Each component's Schur form is the minimum of its energy with boundary values fixed, hence is PSD. Equal boundary values admit the same constant value at every hidden vertex and give zero energy. With one terminal this makes the entire form zero. With two terminals, write its symmetric block matrix as M. The equations $M_{11} + M_{12} = 0$ and $M_{21} + M_{22} = 0$, together with $M_{21} = M_{12}^\top$, force

$$M = \begin{pmatrix} W_C & -W_C \\ -W_C & W_C \end{pmatrix}, \qquad W_C = W_C^\top \succeq 0. \tag{25.32}$$

If a nonzero terminal difference had zero energy, its unique minimizing hidden extension would make every edge difference zero. Connectedness would force the two terminal values equal, a contradiction. Thus $W_C \succ 0$. Summing these component contributions and direct retained edges proves sufficiency, including an anchor when one terminal is ground.

For necessity, a hidden star with three retained terminals a, b, c and weights

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad B = I_2, \qquad C_0 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{25.33}$$

has pivot and manufactured ab coefficient

$$D = A + B + C_0 = \begin{pmatrix} 4 & 1 \\ 1 & 5 \end{pmatrix}, \qquad M_{ab} = AD^{-1}B = \frac{1}{19}\begin{pmatrix} 5 & -1 \\ -2 & 8 \end{pmatrix}. \tag{25.34}$$

The coefficient is nonsymmetric. If one terminal is ground, give it the C_0 leg, leaving the defect between two ordinary terminals. Direct sums with I_{K-2} extend the obstruction to every $K > 2$.

Consider any hidden component with at least three terminal neighbors. Select one incident edge from each of three such terminals, connect their hidden endpoints by a minimal tree inside the component, and extend it to a tree containing every remaining hidden vertex. The three-terminal core has a unique branching vertex, allowing coincident hidden endpoints and legs of length one. Extra hidden branches have no terminal and minimize to their attachment value, so they contribute no quadratic form. On a core leg, minimize $\sum_j d_j^\top W_j d_j/2$ subject to $\sum_j d_j = d$. Stationarity gives $W_j d_j = \lambda$, hence

$$W_{\text{leg}} = \left(\sum_j W_j^{-1} \right)^{-1}. \tag{25.35}$$

The positive Hessian makes this the unique minimum. For a leg of length l, assigning lA to each edge realizes A; do the same for B, C_0. Assign I_K to the extra hidden tree edges. Temporarily set unused edges to zero. The selected tree still joins every hidden vertex to a terminal, so its Dirichlet block is SPD and its reduced core has the defect (25.34). Restore every unused present edge with weight εI_K. The Dirichlet inverse and Schur complement are continuous at $\varepsilon = 0$, where the determinant is nonzero. Their nonzero antisymmetric block therefore persists for sufficiently small positive ε. Every final edge weight is SPD. Other hidden components can use scalar identity weights, and direct retained edges are symmetric; neither can cancel this antisymmetric block. This constructs a failure on every excluded graph.

In a scalar graph, one-node elimination with pivot $d = \sum_i w_i > 0$ adds $w_i w_j/d \geq 0$ between distinct retained neighbors. The constant-vector kernel preserves row sums. Every successive pivot is positive because the eliminated Dirichlet block and its Schur complements are SPD. Repeating the formula proves scalar closure, including the scalar-identity assertion used above. □

The sufficiency implication extends to PSD weights whenever the specified hidden Dirichlet block remains SPD. Add εI_K to every present weight and pass to $\varepsilon \downarrow 0$. Continuity of the inverse at the positive limiting block preserves the Schur matrix, and symmetry, zero row sums, and PSD edge weights are closed conditions. The two-terminal output may then be only semidefinite. A singular limiting Dirichlet block is excluded from this argument. [ESTABLISHED]

For a declared nested collection of eliminated sets, universal closure at every declared level is equivalent to (25.31) at every such set. Direct and successive Schur elimination agree by Proposition 25.6. This does not assert closure at all intermediate one-vertex prefixes: a component with two final terminals can expose three or more temporary terminals during an ordering. Closure at every prefix requires the same criterion at every prefix. An excluded graph has some failing SPD assignments, not necessarily failure for every assignment. Flatness here means plain differences in a common frame; a graph coboundary can first be trivialized, but genuinely nonflat links require another result. [ESTABLISHED]

25.2.5 A proper Gaussian beyond every positive two-node decomposition

The three-terminal star also distinguishes failure of the original difference-weight ansatz from failure of every positive pair representation on the same retained node spaces.

Proposition 25.12 (Rational obstruction to positive pair decomposition). There is a proper four-node Gaussian with SPD edge weights and retained anchors whose marginal on three two-dimensional nodes is not a sum of PSD quadratic terms each supported on at most two of those nodes. Thus arbitrary new invertible link transports, rectangular pair residuals, parallel edges, self-loops, and freely redistributed PSD anchors cannot represent that marginal within the positive two-node class. [ESTABLISHED]

Proof. Let $L_\star$ be the six-dimensional retained Schur Laplacian of the unanchored star (25.33), and put

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad t = \frac{1}{104}, \qquad \varepsilon = \frac{1}{20000}. \tag{25.36}$$

Define a symmetric matrix Y in three two-dimensional blocks by

$$\begin{aligned} Y_{ii} &= (1+t^2)I_2, \\ Y_{ab} &= I_2 + tJ, \qquad Y_{ba} = I_2 - tJ, \\ Y_{ij} &= I_2 \quad \text{for the other distinct pairs.} \end{aligned} \tag{25.37}$$

Every principal submatrix on one or two nodes is PSD. For the special pair, $(I_2+tJ)^\top(I_2+tJ) = (1+t^2)I_2$, so its cross-block norm is $\sqrt{1+t^2} \le 1+t^2$; other cross-block norms equal one. For diagonal blocks αI and a cross block C, the quadratic is bounded below by $\alpha(\|u\|^2+\|v\|^2) - 2\|C\|\|u\|\|v\| \ge 0$ when $\|C\| \le \alpha$. Consequently $\operatorname{Tr}(YP) \ge 0$ for any PSD matrix P supported on at most two nodes: diagonalize its supporting principal block as a sum of nonnegative multiples of vector outer products and pair each with that PSD principal block of Y. Every sum of positive one-node or two-node terms has nonnegative pairing with Y.

Direct substitution into $A - AD^{-1}A$, $B - BD^{-1}B$, and $C_0 - C_0D^{-1}C_0$ gives diagonal traces $36/19, 29/19, 39/19$, respectively, hence $\operatorname{Tr} L_\star = 104/19$. Write $Y = (\mathbf{1}\mathbf{1}^\top) \otimes I_2 + tS + t^2 I_6$, where $S_{ab} = J$, $S_{ba} = -J$, and other blocks are zero. The constant-vector kernel makes the first pairing

vanish. Since $M_{ab} - M_{ab}^{\top} = J/19$, the two skew pairings give

$$\begin{aligned} \operatorname{Tr}(SL_\star) &= \operatorname{Tr}\big(J(M_{ab} - M_{ab}^{\top})\big) = -\frac{2}{19}, \\ \operatorname{Tr}(YL_\star) &= \frac{-2t + 104t^2}{19} = -\frac{1}{1976}. \end{aligned} \tag{25.38}$$

Put the actual anchor εI_2 at each of the three retained vertices in the original star, leaving its hidden anchor zero. Zero total energy forces all anchored terminal coordinates zero and then the connected hidden coordinate zero, so the full precision is SPD. Its retained precision is $\Lambda_R = L_\star + \varepsilon I_6 \succ 0$, but

$$\begin{aligned} \operatorname{Tr}(Y\Lambda_R) &< -\frac{1}{1976} + \frac{7}{20000} < 0, \\ \operatorname{Tr} Y &= 6(1 + t^2) < 7. \end{aligned} \tag{25.39}$$

The final inequality follows from $7 \cdot 1976 = 13832 < 20000$. This contradicts the necessary nonnegative pairing for every positive two-node decomposition. All constants are rational, and (25.10) supplies its finite Gaussian normalizer. □

Block-diagonal invertible changes of retained vertex frames preserve the positive two-node support class by congruence, so they cannot remove this obstruction. Mixing distinct retained nodes, adding latent vertices, using signed pair terms, or allowing positive terms supported on three or more nodes changes the representation problem. The SPD marginal remains a valid complete quadratic Gaussian action. [ESTABLISHED]

25.2.6 Nonflat series and cycle marginalization

We now retain link data instead of requiring a flat difference presentation. Fix a finite multigraph, one orientation for each edge copy $e = (i, j)$, invertible represented links $\Theta_e : \mathbb{R}^K \to \mathbb{R}^K$, SPD weights W_e at the receiving endpoint, PSD vertex anchors A_i, arbitrary sources h_i, and a scalar offset $\mathfrak{c}$. The action is

$$\Phi(z) = \mathfrak{c} + \frac{1}{2} \sum_{e=(i,j)} (z_i - \Theta_e z_j)^{\top} W_e (z_i - \Theta_e z_j) + \frac{1}{2} \sum_i z_i^{\top} A_i z_i - \sum_i h_i^{\top} z_i. \tag{25.40}$$

Choose retained vertices R and hidden vertices E. Every hidden vertex has total degree two, counting a self-loop twice, and has zero quadratic anchor $A_i = 0$. Hidden sources need not vanish. Retained anchors are protected throughout elimination. Assume the assembled precision Λ is SPD. These are hypotheses on a finite Gaussian realization, not on arbitrary statistical fibers. [HYPOTHESIS]

Positive definiteness has an explicit criterion. The kernel of Λ consists exactly of vectors satisfying $z_i = \Theta_e z_j$ on every edge and $A_i z_i = 0$ at every vertex, since its quadratic is a sum of nonnegative residuals and anchor terms. In each connected component choose a root and a spanning tree, with invertible tree transports T_i from that root and $T_r = I$. Every tree-compatible vector has the form

$z_i = T_i u$. Therefore

$$\Lambda \succ 0 \iff \bigcap_{\substack{e=(i,j)\\ \text{non-tree}}} \ker(T_i - \Theta_e T_j) \cap \bigcap_i \ker(A_i T_i) = \{0\} \quad \text{in each component.} \tag{25.41}$$

One retained SPD anchor in every component is sufficient. An all-hidden unanchored cycle with holonomy M is admitted exactly when $I - M$ is invertible; $M \neq I$ alone is insufficient. [ESTABLISHED]

Proposition 25.13 (Exact nonflat series pivot). Orient the two non-loop edges through a hidden vertex e so their residuals are $x - Tz_e$ and $z_e - Sy$, with SPD weights W, V, respectively. The endpoints x, y may be coordinates of the same retained or hidden vertex. Set

$$D = T^\top W T + V, \qquad U = (W^{-1} + T V^{-1} T^\top)^{-1}, \qquad \Theta_{\text{new}} = TS. \tag{25.42}$$

Eliminating z_e replaces the two quadratic edge terms by $(x - TSy)^\top U (x - TSy)/2$. With endpoints i, j, the complete source and scalar updates are

$$\begin{aligned} h_i^{\text{new}} &= h_i + W T D^{-1} h_e, \\ h_j^{\text{new}} &= h_j + S^\top V D^{-1} h_e, \\ \mathfrak{c}^{\text{new}} &= \mathfrak{c} + \frac{1}{2}\log\det D - \frac{K}{2}\log(2\pi) - \frac{1}{2} h_e^\top D^{-1} h_e. \end{aligned} \tag{25.43}$$

If $i = j$, both source increments add at that vertex and the new edge is the marked self-loop (TS, U). No new quadratic vertex anchor is generated. [ESTABLISHED]

Proof. The hidden terms are

$$\frac{1}{2} z_e^\top D z_e - z_e^\top q + \frac{1}{2} x^\top W x + \frac{1}{2} y^\top S^\top V S y, \qquad q = T^\top W x + V S y + h_e. \tag{25.44}$$

Here $D \succ 0$, so completing the square multiplies the remaining density by

$$(2\pi)^{K/2} (\det D)^{-1/2} \exp(q^\top D^{-1} q/2).$$

Let $P = W - W T D^{-1} T^\top W$. From $D = T^\top W T + V$, $PT = W T D^{-1} V$; consequently

$$\begin{aligned} P(W^{-1} + T V^{-1} T^\top) &= I, \\ V - V D^{-1} V &= T^\top P T. \end{aligned} \tag{25.45}$$

The first factor's inverse is SPD, so $P = U$. The second identity and $PT = W T D^{-1} V$ identify the endpoint diagonal and cross terms as exactly the new edge energy. The linear and constant terms in $q^\top D^{-1} q/2$ give (25.43). If the two endpoints coincide, substituting $x = y$ gives the loop and the sum of source increments. Reversing an original edge requires the exact pair $(\Theta, W) \mapsto (\Theta^{-1}, \Theta^\top W \Theta)$,

because the old residual is minus Θ times the reversed residual; this permits the displayed orientation without changing the energy. □

Theorem 25.14 (Arbitrary-order nonflat series closure). Under the hypotheses of (25.40), eliminating any subset of hidden vertices in any order gives the exact normalized Gaussian marginal within the same marked multigraph class. Every surviving hidden vertex still has degree two and zero quadratic anchor, and the remaining precision is SPD. Distinct parallel paths remain distinct edges. Along an oriented path $v_0, \dots, v_m$, define $P_0 = I$ and $P_j = \Theta_1 \cdots \Theta_j$, where Θ_i maps from v_i to v_{i-1}. Its fused data are

$$\Theta_{\mathrm{path}} = P_m, \qquad W_{\mathrm{path}}^{-1} = \sum_{i=1}^{m} P_{i-1} W_i^{-1} P_{i-1}^{\top}. \tag{25.46}$$

A returning path has retained loop holonomy $M = P_m$ and quadratic

$$Q_{\mathrm{loop}} = (I - M)^{\top} W_{\mathrm{path}} (I - M). \tag{25.47}$$

Each cycle meeting the retained set keeps its nonflat holonomy mark. [ESTABLISHED]

Proof. A degree-two hidden vertex without a loop has exactly two incident edges counted with multiplicity, so Proposition 25.13 applies. If its neighbors differ, replacing the two edges by one preserves their degrees. If they coincide, a loop carries the same two incidences. No surviving hidden anchor is introduced. A degree-two vertex already carrying a loop is an isolated component. It occurs initially or as the last vertex of an all-hidden cycle. Its loop block is SPD by the global positivity hypothesis and can be integrated as one Gaussian block, contributing only the scalar in (25.11). Positivity of every remaining Schur complement was proved in Proposition 25.6. These observations prove closure by induction for every elimination order.

Write a path segment as (Θ, C), with $C = W^{-1}$. The binary series operation is

$$(\Theta, C) * (\Psi, F) = (\Theta\Psi,\ C + \Theta F \Theta^{\top}). \tag{25.48}$$

For a third pair (Ξ, G), either parenthesization gives transport $\Theta\Psi\Xi$ and covariance $C + \Theta F \Theta^{\top} + \Theta\Psi G \Psi^{\top}\Theta^{\top}$. This proves associativity with the stated matrix order and yields (25.46) by induction. Each covariance summand is SPD. Substitution of equal endpoints proves (25.47). The complete scalar and source terms agree for all orders by the finite nonnegative Gaussian integration argument in Proposition 25.6; equivalently, the product of successive pivot determinants is the eliminated-block determinant in that proposition. □

A component meeting R decomposes into maximal hidden paths between retained endpoints, which may coincide. A cycle with one retained vertex becomes one loop; with several retained vertices its ordered coarse link product equals its original cycle product. An all-hidden cycle can instead be integrated away entirely only under the invertibility condition in (25.41). Its raw partition factor survives, but its normalized marginal on the empty coordinate set does not identify its holonomy.

More generally, a retained quadratic Q_{loop} need not identify its separate marks M, W_{path}. These marks are declared structural data, and the theorem does not infer them uniquely from the Gaussian law. [ESTABLISHED]

Proposition 25.15 (Gauge and rescaling covariance with scalar factors). For nodewise passive coordinates $z_i' = G_i z_i$, $G_i \in \mathrm{GL}(K)$, the transformed coefficients are

$$\begin{aligned} \Theta_e' &= G_i \Theta_e G_j^{-1}, \qquad W_e' = G_i^{-\top} W_e G_i^{-1}, \\ A_i' &= G_i^{-\top} A_i G_i^{-1}, \qquad h_i' = G_i^{-\top} h_i. \end{aligned} \tag{25.49}$$

The series rule commutes with this change: $D' = G_e^{-\top} D G_e^{-1}$, $U' = G_i^{-\top} U G_i^{-1}$, and $\Theta'_{\mathrm{new}} = G_i T S G_j^{-1}$. With the raw scalar $\mathfrak{c}$ unchanged,

$$\begin{aligned} \mathfrak{c}_R' &= \mathfrak{c}_R - \sum_{e \in E} \log |\det G_e|, \\ Z' &= \left(\prod_{i \in R \cup E} |\det G_i| \right) Z, \\ p_R'(x') &= |\det G_R|^{-1} p_R(G_R^{-1} x'), \qquad G_R = \mathrm{blkdiag}_{i \in R} G_i. \end{aligned} \tag{25.50}$$

Alternatively, transforming the raw measure itself uses $\mathfrak{c}' = \mathfrak{c} + \sum_{i \in R \cup E} \log |\det G_i|$. Then $Z' = Z$ and $\mathfrak{c}_R' = \mathfrak{c}_R + \sum_{i \in R} \log |\det G_i|$. Both conventions give the same normalized pushforward law. [ESTABLISHED]

Proof. Substitution into each residual proves (25.49). The inverse covariance in (25.42) transforms by

$$W'^{-1} + T' V'^{-1} T'^{\top} = G_i (W^{-1} + T V^{-1} T^{\top}) G_i^{\top}. \tag{25.51}$$

The source increments transform as endpoint covectors; $h_e^{\top} D^{-1} h_e$ is invariant, while $\det D' = \det D / |\det G_e|^2$. This proves the hidden scalar shift. Changing variables in the full integral gives its total Jacobian and then the retained Jacobian in (25.50). Adding the total log-Jacobian to the initial action instead gives the raw-measure convention. A retained loop transforms by $M' = G_r M G_r^{-1}$, preserving its conjugacy class and noncommutation with another loop based at r. □

A declared invertible retained-coordinate rescaling $x' = Tx$ has the normalized Gaussian pushforward

$$\mu_R' = T \mu_R, \qquad C_R' = T C_R T^{\top}, \qquad p_R'(x') = |\det T|^{-1} p_R(T^{-1} x'). \tag{25.52}$$

This follows from the same change of variables. Block-diagonal T preserves the marked graph presentation; general T can mix vertices and need not do so. Any restriction of the links to a matrix subgroup requires closure under products and inverses, and gauges in that subgroup preserve the restricted domain. Arbitrary $\mathrm{GL}(K)$ gauges carry it to its transformed domain. No preferred physical scale transformation is selected by this finite pushforward. [ESTABLISHED]

Proposition 25.16 (A noncommuting nonflat proper example). Take one retained vertex a, hidden vertices e, f, and two distinct two-edge cycles $a-e-a$ and $a-f-a$. Their edge copies are distinct interactions, not repeated orientations of one edge. Give a the anchor I_2, both hidden vertices zero quadratic anchors, and choose

$$T_1=\begin{pmatrix}1&1\\0&1\end{pmatrix},\qquad T_2=\begin{pmatrix}1&0\\1&1\end{pmatrix},\qquad W=\begin{pmatrix}1&0\\0&2\end{pmatrix},\qquad V=\begin{pmatrix}2&1\\1&2\end{pmatrix}. \tag{25.53}$$

The first cycle has transports T_1, I_2 and weights W, V; the second has transports T_2, I_2 and weights I_2, I_2. Put $h_e=(1,0)^\top$, all other sources zero, and $\mathfrak{c}=0$. This is a proper Gaussian with noncommuting loop holonomies and coefficients outside every one fixed congruence-diagonal cone containing its anchor. [ESTABLISHED]

Proof. Every edge weight is SPD, and the retained SPD anchor makes (25.41) hold. The two holonomies satisfy $T_1T_2-T_2T_1=\mathrm{diag}(1,-1)\neq 0$. They cannot both become flat or diagonal under one based conjugation. The coefficient triple I_2, W, V cannot lie in a common congruence-diagonal cone: including I_2 would give a common orthogonal basis as above, but $WV-VW=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$. This fixed-chart coefficient statement and the gauge-invariant holonomy statement are distinct.

For the exact marginal, the hidden pivots and loop weights are

$$\begin{aligned} D_e&=\begin{pmatrix}3&2\\2&5\end{pmatrix},\quad U_e=\frac{1}{11}\begin{pmatrix}7&-2\\-2&10\end{pmatrix},\\ D_f&=\begin{pmatrix}3&1\\1&2\end{pmatrix},\quad U_f=\frac{1}{5}\begin{pmatrix}3&-1\\-1&2\end{pmatrix}. \end{aligned} \tag{25.54}$$

Here $\det D_e=11$, $\det D_f=5$, and the two loop quadratics are $\mathrm{diag}(0,7/11)$ and $\mathrm{diag}(2/5,0)$. The formulas above therefore give

$$\begin{aligned} \Lambda_a^{\mathrm{eff}}&=\mathrm{diag}(7/5,18/11), & h_a^{\mathrm{eff}}&=(1,-3/11)^\top,\\ \mathfrak{c}_a^{\mathrm{eff}}&=\frac{1}{2}\log 55-2\log(2\pi)-\frac{5}{22}, & \mu_a&=(5/7,-1/6)^\top,\\ C_a&=\mathrm{diag}(5/7,11/18), & Z&=(2\pi)^3\exp(17/28)/\sqrt{126}. \end{aligned} \tag{25.55}$$

Indeed, $\det\Lambda=11\cdot 5\cdot(126/55)=126$, $D_e^{-1}h_e=(5,-2)^\top/11$, and the total completed-square value is $5/11+5/7+1/22=17/14$. These calculations verify the source, covariance, scalar, and normalization in the same proper law. □

The zero hidden quadratic anchor is essential for the particular series rule. In the scalar chain with unit edges and hidden anchor $\alpha>0$, the actual pivot is $D=2+\alpha$, and the retained quadratic matrix is

$$\begin{pmatrix}1-1/D&-1/D\\-1/D&1-1/D\end{pmatrix}=\frac{1}{D}\begin{pmatrix}1&-1\\-1&1\end{pmatrix}+\frac{\alpha}{D}I_2. \tag{25.56}$$

New anchors arise, so the zero-anchor rule cannot be applied unchanged. This does not refute all larger anchored closure classes. In the flat graph theorem a hidden anchor is an edge to ground, which can create a third terminal; in the independent-family theorem the fixed positive hidden anchor is precisely the premise forcing the commutation constraint. Degree two in the nonflat theorem guarantees every intermediate order, whereas the graph criterion concerns each selected final eliminated set. These quantifiers remain distinct. [ESTABLISHED]

25.2.7 Finite semialgebraic classification

The preceding analytical classifications impose recognizable structural conditions. A broader exact decision theorem is available after specifying how the admissible families are described. Fix finitely many labeled nodes with finite real coordinate spaces, a finite list of oriented edge copies and loops, and a finite list of retained coordinate sets. The reference measures are the declared coordinate Lebesgue measures. A fine parameter $x \in \mathbb{R}^p$ belongs to A when it satisfies a supplied finite formula of polynomial equalities and strict or weak inequalities with real algebraic coefficients. Each coefficient has an exact finite representation, for example a defining integer polynomial and a rational isolating interval. Finite Boolean combinations and existential auxiliary variables are allowed. Matrix entries are rational expressions in the parameters on an explicitly specified nonzero-denominator domain. Every admitted fine precision $\Lambda(x)$ is the actual assembled symmetric positive-definite precision, with source $h(x)$. These finite-description and properness hypotheses are introduced for the decision theorem; they are not assumed for every possible RG family. Singular or rank-changing Gaussian supports are excluded from this full-SPD statement. Section 25.2.8 supplies a separate finite extension with a declared quotient reference and map contract.

A retained presentation has a finite tuple $y \in \mathbb{R}^q$, a supplied semialgebraic admissibility predicate $B(y)$, and rational precision and source matrices $\Lambda_c(y), h_c(y)$. Let $\mathcal{E}(x, y)$ record any further finite semialgebraic compatibility conditions. Correlated weights, nonconvex domains, variable anchors, nonflat transports, and polynomial path identities are allowed. A finite union of permitted graph inventories is allowed as well. The target is the statement that *every* admitted fine parameter has *some* admitted retained presentation of its exact normalized Gaussian marginal, satisfying every declared compatibility condition. It does not ask for an unencoded continuous function, a global naturality law, or an unbounded inventory of individually marked edges. [DEFINITION]

The imported real-closed-field theorem has the following effective form: given a finite first-order formula in polynomial equalities and inequalities over an effective real algebraic coefficient field, an algorithm returns an equivalent quantifier-free formula in its free variables; closed sentences are therefore decidable [24, Sec. 6]. The operations, coefficient representations, and finite variables just specified match these hypotheses. This theorem asserts termination, without a useful complexity bound or a numerical sign tolerance.

Encoding the exact Gaussian marginal. For a nonempty hidden coordinate set E, write $D = \Lambda_{EE}$, $d = \det D > 0$, and $J = \operatorname{adj} D$. Define

$$\begin{aligned} N_\Lambda(x) &= d\Lambda_{RR} - \Lambda_{RE} J \Lambda_{ER}, \\ N_h(x) &= dh_R - \Lambda_{RE} J h_E, \\ d\Lambda_{\mathrm{c}}(y) &= N_\Lambda(x), \qquad dh_{\mathrm{c}}(y) = N_h(x). \end{aligned} \tag{25.57}$$

For an empty hidden set, use $d = 1$, $N_\Lambda = \Lambda$, and $N_h = h$. Thus no empty-block inverse is needed. The adjugate identity $D^{-1} = J/d$, together with Proposition 25.6, proves that the last line is exactly equality of marginal precision and source.

All admissions in this formula have polynomial encodings. For a symmetric matrix, positive leading principal determinants are equivalent to SPD: completion of the first coordinate's square gives a positive first pivot and a Schur block whose leading determinants are the subsequent leading determinants divided by that pivot; induction proves the assertion. PSD is equivalent to an existential Gram equation $M = TT^\top$, since a real symmetric PSD matrix has an orthogonal eigendecomposition with nonnegative eigenvalues, and the converse follows from $v^\top TT^\top v = \|T^\top v\|^2$. Invertibility is $(\det \Theta)^2 > 0$. A rational denominator q can be replaced by an inverse variable u with $uq = 1$. This keeps its nonzero domain and makes every remaining expression polynomial. Multiplying an equation by a nonzero denominator is valid; multiplying an inequality by a denominator of unknown sign is not. The inverse-variable construction, or a finite split into its two sign cases, handles that distinction exactly.

There is no loss of information about the normalized Gaussian law. The Schur precision is SPD because for nonzero retained z,

$$z^\top \Lambda_R^{\mathrm{eff}} z = \min_t (z,t)^\top \Lambda (z,t) > 0, \qquad t_{\min} = -D^{-1}\Lambda_{ER} z. \tag{25.58}$$

The minimum is attained, and its full vector still has nonzero retained part. For nonsingular Gaussian laws, equality of covariances gives equality of precisions; equality of means then gives equality of sources because $h = \Lambda\mu$. The reverse implication follows from their explicit normalized densities. Equation (25.57) therefore encodes exact law equality on every retained coordinate, not equality at sampled points.

Theorem 25.17 (Exact finite semialgebraic Kron classification). Under the preceding finite-description hypotheses, let $M(x,y)$ be the conjunction of $B(y)$, $\mathcal{E}(x,y)$, the retained SPD conditions, all denominator admissions, and (25.57). There is an algorithm producing a quantifier-free formula $G(x)$ for exactly the admissible fine parameters with an exact retained presentation. It also constructs the full failure locus and decides both universal closure and nonemptiness:

$$\begin{aligned} &G(x) \iff A(x) \wedge \exists y\big[M(x,y)\big], \\ &F(x) = A(x) \wedge \neg G(x), \\ &\qquad \forall x\big(A(x) \Rightarrow \exists y\big[M(x,y)\big]\big), \qquad \exists x\big[A(x)\big]. \end{aligned} \tag{25.59}$$

The witnesses include any introduced inverse or Gram variables. The nonemptiness sentence is separate: vacuous closure is not an inhabited construction. [ESTABLISHED]

Proof. Determinants, adjugates, and matrix products use finitely many additions and multiplications. The preceding encodings therefore produce a finite polynomial-sign formula over exactly the stated coefficient field. Effective quantifier elimination produces G and F, and decides the two closed sentences. Conversely, every satisfying retained tuple gives an admitted precision and source equal to those of the exact marginal. Every admitted presentation of that law satisfies those equalities and all specified constraints, so it supplies a witness for the formula. This proves both directions. The corresponding existential sentences distinguish no good inputs, some good inputs, and all good inputs. The theorem decides the declared problem; it does not assert that closure always holds. □

A nonempty failure locus has a point with real algebraic coordinates. Indeed, apply quantifier elimination to its existential sentence over both $\mathbb{R}$ and the real closed field of real algebraic numbers. The resulting signs of algebraic constants agree, so the sentence has the same truth value in the two fields. Enumeration of finite tuples of algebraic numbers and exact sign evaluation of F therefore eventually produces a witness. Density alone would not justify this assertion for a thin algebraic set. The witness need not be rational or small. Formula construction is effective from the declared finite input; evaluation at arbitrary real parameters given only by approximations still lacks an exact equality or sign oracle.

Sources, raw scalars, and observed laws. The source equation in (25.57) is part of the classification. The raw scalar is retained by the exact analytic formula

$$\mathfrak{c}_{\mathrm{c}} = \mathfrak{c} + \frac{1}{2}\log\det D - \frac{\dim z_E}{2}\log(2\pi) - \frac{1}{2}h_E^{\top}D^{-1}h_E. \tag{25.60}$$

The prior reverse conditional is $\mathcal{N}(D^{-1}(h_E - \Lambda_{ER}z_R), D^{-1})$; integration preserves the full raw mass. These statements follow from the completed square (25.12). Formula (25.60) is not an ordered-ring constraint. Additional conditions on logarithms, exponentials, fixed local-factor normalizers, or numerical evidence enter the decision theorem only after a valid polynomial reduction has actually been supplied. Introducing an unconstrained variable named for a logarithm does not encode its graph.

One additive constant independent of every random coordinate and observation cancels from a globally normalized joint and hence from its evidence, although it changes the raw density and its mass. An offset depending only on an externally fixed model parameter likewise cancels for each fixed model. A factor depending on retained random coordinates, an observation, or a random branch is not such a constant and need not cancel. Exact raw marginal equality retains the scalar pointwise. For a linear Gaussian sensor with algebraic matrix data and SPD noise covariance, the full observed joint is again a proper Gaussian with rational natural parameters: its precision is positive because its quadratic is the sum of the positive prior quadratic and $(o - Lz)^{\top}\Gamma_o^{-1}(o - Lz)$. Keeping the observation coordinates in the marginal therefore includes the entire

observed experiment in the same finite classification. Its evidence and the conditional-KL identity of Section 10.2 are preserved under their stated support and finite-KL conventions. A sensor using hidden coordinates generally changes the posterior reverse conditional. An arbitrary nonlinear sensor or singular noise is not admitted by this proper-Gaussian polynomial argument.

Simultaneous finite diagrams. For a fixed finite acyclic diagram of retained coordinate sets, introduce one common tuple y_v at each object. Put all arrow relations, path identities, and specified structural conditions into one formula, and quantify the whole tuple simultaneously:

$$\forall x \left(A(x) \Rightarrow \exists (y_v)_{v \in V_{\mathrm{diag}}} M_{\mathrm{diag}}\big(x, (y_v)_{v \in V_{\mathrm{diag}}}\big) \right). \tag{25.61}$$

The same theorem decides this sentence. Separate witnesses for each arrow would not give a compatible value at a shared object. At the law level, coordinate marginalization composes by the equality of iterated integrals for the positive, integrable Gaussian density; the inherited sources and raw scalars agree as well. Extra marks remain additional constraints, since equal densities do not identify holonomies or choose a structural frame.

For a fixed finite list of rational operations on one declared domain, totality and universal one-step domain invariance imply admission of every finite composition by induction. Undefined denominators must count as failure of totality, rather than being removed to make the implication vacuous. This implication does not require an a priori bound on the number of iterations, but it supplies neither a limit nor a fixed-point theorem and does not encode an arbitrary changing-dimension hierarchy.

Bounding an unmarked positive presentation. An unrestricted number of independently scalable unmarked summands can also be reduced to finite input. Let $\mathbb{R}^N = \bigoplus_{i \in V} \mathbb{R}^{d_i}$, let P_i denote the coordinate projection, and put $D_N = N(N+1)/2$. An anchor contributes $P_i^\top A_i P_i$ with $A_i \succeq 0$. For nodes of equal dimension an allowed square link contributes

$$S_e = (P_i - \Theta_e P_j)^\top W_e (P_i - \Theta_e P_j), \qquad W_e \succ 0, \quad \det \Theta_e \neq 0. \tag{25.62}$$

The target equivalence is equality of the summed matrix in these fixed retained coordinates. Allowed endpoint types form a fixed finite list. Every surviving generator may be independently rescaled by an arbitrary positive number, and unmarked copies may be deleted. These are substantive assumptions; a correlated admissibility predicate need not permit either operation.

Proposition 25.18 (Finite positive pair-presentation bound). If $Q \in \mathrm{Sym}_N$ has any finite sum representation by the anchors and links just declared, it has one with at most D_N nonzero summands. With unrestricted PSD anchors, it can have at most one anchor at each node and no self-loops. Every retained edge weight remains strictly positive definite and every transport remains invertible. Membership in this unmarked presentation class is equivalent to one finite disjunction of existential polynomial systems. [ESTABLISHED]

Proof. A loop is the PSD anchor

$$P_i^\top (I - \Theta_e)^\top W_e (I - \Theta_e) P_i.$$

Absorb unmarked loops into anchors, sum anchors at each node, and delete zero terms. Reversal of a non-loop orientation replaces Θ by Θ^{-1} and W by $\Theta^\top W \Theta$, preserving the residual quadratic; a fixed orientation can therefore be used when the transport domain is the unrestricted general linear group. For a restricted domain, retain its original finite orientation types.

Write the remaining sum as $Q = \sum_{r=1}^{m} c_r T_r$, initially with $c_r = 1$, where each $T_r \neq 0$ is an admitted PSD generator. If $m > D_N = \dim \mathrm{Sym}_N$, choose a nonzero linear dependence $\sum_r \alpha_r T_r = 0$. Every nonzero PSD matrix has positive trace, so the coefficients α_r have both signs. Set

$$\tau = \min_{\alpha_r > 0} \frac{c_r}{\alpha_r}, \qquad c_r' = c_r - \tau \alpha_r. \tag{25.63}$$

All $c_r' \geq 0$, at least one is zero, and $\sum_r c_r' T_r = Q$. Delete the zero coefficients. Repetition terminates because the number of summands strictly decreases. Absorb each surviving positive coefficient into its A_i or W_e. PSD anchors, strict SPD weights, and unchanged invertible transports satisfy the same admission conditions. There is no limiting argument. The zero matrix uses an empty presentation; when $N = 0$, the bound is zero.

Only finitely many endpoint and anchor patterns have at most D_N terms. For each such pattern, use symmetric anchor variables, symmetric weight variables, and unrestricted transport variables, and impose

$$\begin{aligned} Q &= \sum_i P_i^\top A_i P_i + \sum_e (P_i - \Theta_e P_j)^\top W_e (P_i - \Theta_e P_j), \\ A_i &= T_i T_i^\top, \qquad \mathrm{tr}(T_i T_i^\top) > 0, \\ \det W_e[1:k, 1:k] &> 0 \quad (1 \leq k \leq d_e), \qquad (\det \Theta_e)^2 > 0. \end{aligned} \tag{25.64}$$

All factor variables T_i are existential. These are polynomial equalities and inequalities; the Gram and leading-minor arguments above give their exact meanings. Every solution is a permitted presentation, and the support reduction puts every finite presentation into one listed pattern. The finite disjunction is therefore necessary and sufficient. Quantifier elimination makes its projection onto the entries of Q semialgebraic and decides exact algebraic inputs. □

This compression cannot be applied to arbitrary correlated predicates B. For instance, in one scalar dimension, generators restricted to the fixed value 1 represent 2 with two copies, but cannot meet the bound $D_N = 1$ without rescaling. Individually marked edges, their inventories, or their prescribed loop products also cannot be discarded. Such problems retain their declared finite inventory in Theorem 25.17. If $Q \succ 0$, repacking Q while carrying the original $h, \mathfrak{c}$, and reference measure unchanged preserves the complete Gaussian action, its mass, and its normalized law. Independently normalized local factors introduce determinant scalars which must be compensated if exact raw action equality is required.

The positive GL-link class is smaller than arbitrary PSD two-node forms. For two two-dimensional nodes and $u = (1, 0)^\top$, consider

$$Q_0 = \begin{pmatrix} u \\ -u \end{pmatrix} \begin{pmatrix} u \\ -u \end{pmatrix}^\top . \tag{25.65}$$

It has rank one and nonzero cross block. Every non-loop term (25.62) on those nodes has rank two, because $(I, -\Theta)$ has full row rank and $W \succ 0$. If PSD matrices sum to Q_0, each vanishes on $\ker Q_0$: their nonnegative quadratic forms sum to zero there. Their ranges are therefore contained in its one-dimensional range, excluding every non-loop term. Anchors and loops have zero cross blocks, so they cannot represent Q_0 either. Conversely the admissible single links with $\Theta = I$ and $W_\varepsilon = \mathrm{diag}(1, \varepsilon)$, $\varepsilon > 0$, tend to Q_0. Thus the presentation class need not be closed, and replacing strict SPD by PSD would change membership. This singular operator example is not a normalized full-space Gaussian; the proper example in Proposition 25.12 excludes even the larger PSD two-node class.

Proposition 25.19 (Measurable selection and structural obstructions). For the good locus G of Theorem 25.17, there is a semialgebraic, hence Borel, selector $s : G \to \mathbb{R}^q$ with $M(x, s(x))$. Neither a continuous selector nor a gauge equivariant structural selector follows from these hypotheses, even when every accompanying Gaussian law is proper and identical. [ESTABLISHED]

Proof. First select from a nonempty one-dimensional semialgebraic fiber $T \subset \mathbb{R}$. Prefer its nonnegative part when nonempty; otherwise reflect the negative part and reverse the selected sign. Select zero if present. The remaining set is positive and has finite infimum $a \geq 0$. Select a if it belongs. If not, the finitely many real roots of the nonzero defining univariate polynomials partition the line into intervals of constant polynomial signs. Hence some interval (a, b) lies in T. Let b be the supremum of the right endpoints of such intervals. Select $(a + b)/2$ if $b < \infty$, and $a + 1$ if $b = \infty$. The selected point lies in T: every point strictly between a and that supremum belongs to a defining interval.

Infimum, supremum, interval inclusion, and unboundedness are expressible by finite first-order formulas in the fiber parameters. Quantifier elimination therefore makes this selection rule semialgebraic, including each branch. For a fiber in $\mathbb{R}^q$, select from its first-coordinate projection, restrict to that choice, select from the next projection, and continue. Each selected prefix has a nonempty completion, so the resulting tuple is feasible. Projection and composition are semialgebraic by the same theorem. Preimages of rational open boxes are semialgebraic projections, hence Borel; such boxes form a countable base. The empty locus and zero-dimensional target have their unique empty or singleton selections.

Continuity fails in an explicit polynomial family. Let the context be $z = (\cos\theta, \sin\theta) \in S^1$, attach its rotation R_θ, and require $B \in \mathrm{SO}(2)$ with $B^2 = R_\theta$. Each fiber has two elements. Along $[0, 2\pi]$, a continuous selection must be $B(\theta) = \sigma R_{\theta/2}$, where $\sigma \in \{I, -I\}$ is continuous and therefore constant. Then $B(2\pi) = -B(0)$, although the two endpoints are the same context, a contradiction.

All membership conditions are polynomial in z and the entries of B, and the state law may remain $\mathcal{N}(0, I_2)$ throughout.

For the distinct symmetry obstruction, let the input covariance be I_2 and require a factor B from a fixed reference space with $BB^\top = I_2$. The gauge action is $B \mapsto gB$ for $g \in \mathrm{SO}(2)$. The input is fixed by every such g, so an equivariant selector would obey $B = gB$ for every g. Taking $g = -I_2$ forces $B = 0$, contradicting the factor condition. The isotropic Gaussian and its tensor transformation law remain well-defined. □

The selected tuple supplies a measurable presentation of the exact marginal; it does not promise a smooth frame, an equivariant section, or compatibility with an unspecified global composition law. Explicit finite diagram compatibility is already handled by (25.61), with its shared variables.

Proposition 25.20 (Obstruction for computably enumerated domains). There is no terminating algorithm which, from arbitrary code for a total computable sequence of rational parameters, decides whether every sequence member has a positive pairwise coarse presentation. This remains true when the topology and node dimensions are fixed, every fine law is proper, and the range is promised to be a subset of one fixed two-point set. The obstruction concerns the input description, even though every such range is finite and semialgebraic as a set. [ESTABLISHED]

Proof. Use the four-node, two-dimensional star of Proposition 25.12. Let x_{bad} be its rational edge data and retained anchors εI, with $\varepsilon = 1/20000$. Its proper marginal has no positive one-node or two-node presentation on the declared retained nodes. Let x_{good} have the same topology and anchors, with all three star weights equal to I. Both complete precisions are SPD. In the good case the hidden pivot is $3I$, so the retained interaction diagonal blocks are $2I/3$ and off-diagonal blocks are $-I/3$. The complete retained graph with weights $I/3$, together with the original anchors, is an exact positive presentation.

Given a finite program e, define

$$x_n(e) = \begin{cases} x_{\mathrm{bad}}, & e \text{ halts within } n \text{ steps}, \\ x_{\mathrm{good}}, & \text{otherwise}. \end{cases} \tag{25.66}$$

Computing a term requires only n simulation steps, so this is always a total computable rational sequence. Its range is contained in $\{x_{\mathrm{good}}, x_{\mathrm{bad}}\}$, and all its members admit positive pairwise coarse presentations if and only if e never halts. An algorithm deciding the claimed property from the enumerator code would decide halting by reversing its answer.

For completeness, suppose $H(p, q)$ decided whether program p halts on input q. Construct $D(q)$ to loop when $H(q, q)$ predicts halting and to halt otherwise. Applying D to its own code contradicts either prediction. A procedure for empty-input programs would supply H by hardcoding q into a wrapper. Thus the procedure implied by (25.66) cannot exist. □

There is no conflict with the finite decision theorem: an explicit complete semialgebraic formula for the range is different input from code that enumerates it. The proposition does not forbid a

theorem about a particular infinite family, a nonalgorithmic classification, or exact full-Gaussian marginalization.

25.2.8 Finite Gaussian quotients with variable support

The full-SPD assumption in Theorem 25.17 can be replaced by an explicit quotient reference, giving a different finite classification that includes variable ranks. Fix $V = \mathbb{R}^n$, a symmetric PSD quadratic form Q, an SPD reference metric H, a covector $h \in \operatorname{ran} Q$, an affine origin $a \in V$, and a real scalar $\mathfrak{c}$. Represent $V/\ker Q$ by

$$S = (\ker Q)^{\perp_H}, \qquad \Phi(a+s) = \mathfrak{c} + \frac{1}{2} s^\top Q s - h^\top s, \quad s \in S. \tag{25.67}$$

Use the volume induced by H on S, translated to $a+S$; in dimension zero it is unit mass on the point. Because Q is symmetric, $h \in \operatorname{ran} Q$ is equivalent to annihilation of $\ker Q$. Thus both the quadratic and linear energies descend to the quotient. The metric, affine embedding, and reference volume are introduced data. They specify an actual law on a representative slice, not a normalization of singular ambient Lebesgue density. Adding a law on the discarded null coordinates would define a different completion. [DEFINITION]

Proposition 25.21 (Polynomial quotient covariance). For every such Q, H, there is exactly one symmetric C satisfying

$$QCQ = Q, \qquad CQC = C, \qquad HCQ = QCH. \tag{25.68}$$

It is PSD, its range is S, and $P = CQ$ is the H-orthogonal projection onto S with kernel $\ker Q$. In particular these polynomial equations specify the covariance at every rank. [ESTABLISHED]

Proof. The SPD square root of H exists by orthogonal diagonalization. Put

$$M = H^{-1/2} Q H^{-1/2}, \qquad Z = H^{1/2} C H^{1/2}. \tag{25.69}$$

Equation (25.68) becomes $MZM = M$, $ZMZ = Z$, and $ZM = MZ$, with Z symmetric. Decompose into the positive eigenspace of M and its kernel. Commutation makes the off-diagonal blocks between these spaces vanish. On the positive block $MZM = M$ gives $Z = M^{-1}$; on the null block $ZMZ = Z$ gives $Z = 0$. This proves uniqueness and

$$C = H^{-1/2} M^{+} H^{-1/2}, \tag{25.70}$$

where M^+ inverts the positive eigenvalues and vanishes on the kernel. It also proves positivity. In the whitened coordinates ZM is the Euclidean orthogonal projection onto $\operatorname{ran} M$. Since $\ker M = H^{1/2} \ker Q$, undoing the isometry $s \mapsto H^{1/2} s$ gives the asserted projection and range. Equivalently $P^2 = P$, $\ker P = \ker Q$, and $CQC = C$ makes $\operatorname{ran} P = \operatorname{ran} C$. □

The covariance graph uses only polynomial equations in Q, H, C. PSD and SPD admissions have the finite encodings already proved, and $h \in \operatorname{ran} Q$ is the existential equation $Qw = h$. The square root and spectral inverse in the proof establish the equations' semantics; neither is an operation required in the polynomial input.

Theorem 25.22 (Normalized quotient Gaussian law). Let $r = \operatorname{rank} Q$ and let $\det_+(M)$ be the product of the positive eigenvalues of M, with empty product one. Relative to the declared volume on $a + S$, the action (25.67) has positive finite mass and defines the Gaussian probability measure with ambient mean μ, covariance C, and characteristic function

$$\begin{aligned} Z_Q &= e^{-\mathfrak{c}}(2\pi)^{r/2} \det_+(M)^{-1/2} \exp\left(\frac{1}{2} h^\top C h\right), \\ \mu &= a + Ch, \\ \widehat{\mathbb{P}}_Q(t) &= \exp\left(it^\top \mu - \frac{1}{2} t^\top C t\right). \end{aligned} \tag{25.71}$$

For $Q = 0$, these formulas give $h = C = 0$, $Z_Q = e^{-\mathfrak{c}}$, and the point mass at a. [ESTABLISHED]

Proof. In the coordinates $y = H^{1/2}s$, the slice becomes $\operatorname{ran} M$ and its H-volume becomes Euclidean volume. The restriction of M there is positive definite. The source $H^{-1/2}h$ belongs to that range because it annihilates $\ker M$. Diagonalizing the positive restriction and completing its r scalar squares gives the first line of (25.71). Transforming the mean and covariance back gives Ch and C; translation adds a. The same one-dimensional Gaussian integrations give the displayed characteristic function. In dimension zero the integral is over one point. All cases have $0 < Z_Q < \infty$. □

Passive coordinates and the reference metric. Under an invertible change $x' = Gx$, transform all declared tensors:

$$Q' = G^{-\top} Q G^{-1}, \quad H' = G^{-\top} H G^{-1}, \quad h' = G^{-\top} h, \quad a' = Ga. \tag{25.72}$$

Substitution into (25.68) and uniqueness give $C' = GCG^\top$. Hence $\mu' = G\mu$ and $\mathbb{P}' = G_\# \mathbb{P}$. The map is an isometry between the declared representative metrics, so it pushes their volume measures and leaves Z_Q and $\mathfrak{c}$ unchanged. A convention fixing coordinate Lebesgue measures instead must include the relevant support Jacobian.

The metric condition is material. Take $Q = \operatorname{diag}(1, 0)$, $H = I_2$, and $G = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The covariance pushed from the original slice is $\operatorname{diag}(1, 0)$, while

$$Q' = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad (Q')^+_{\text{Euclidean}} = \frac{1}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{25.73}$$

Resetting the transformed metric to I_2 has changed the slice and its law. This does not contradict covariance when H transforms as in (25.72).

Ambient affine maps and intrinsic quotient maps. Every declared affine map $x \mapsto Lx + b$, including rectangular or rank-deficient L, has exact pushforward moments

$$\mu_{\mathrm{push}} = L(a + Ch) + b, \qquad C_{\mathrm{push}} = LCL^{\top}. \tag{25.74}$$

This follows by substituting $L^{\top}t$ in the characteristic function and multiplying by $e^{it^{\top}b}$. Two Gaussian measures on the same finite ambient space, singular or nonsingular, agree if and only if their means and covariances agree. Necessity follows from their finite first and second moments; sufficiency follows from the same orthogonal Gaussian construction with the given PSD covariance, including its zero directions.

A target tuple $(Q_{\mathrm{c}}, H_{\mathrm{c}}, h_{\mathrm{c}}, a_{\mathrm{c}})$ has its own unique covariance C_{c} from (25.68). It represents this ambient affine push exactly when

$$C_{\mathrm{c}} = LCL^{\top}, \qquad a_{\mathrm{c}} + C_{\mathrm{c}}h_{\mathrm{c}} = L(a + Ch) + b. \tag{25.75}$$

These are equalities of probability measures, including their supports, rather than equalities of densities against different support volumes.

An intrinsic map $[x] \mapsto [Lx]$ between the energy quotients has the additional necessary and sufficient condition $L \ker Q \subseteq \ker Q_{\mathrm{c}}$. Because $I - CQ$ projects onto $\ker Q$, this is exactly

$$Q_{\mathrm{c}}L(I - CQ) = 0. \tag{25.76}$$

Necessity follows by replacing x by $x + k$ with $k \in \ker Q$; the same calculation proves sufficiency. Let $P_{\mathrm{c}} = C_{\mathrm{c}}Q_{\mathrm{c}}$. With a declared affine quotient offset b_0 relative to the origins a, a_{c}, the actual target representative map is

$$a + s \longmapsto a_{\mathrm{c}} + P_{\mathrm{c}}(Ls + b_0).$$

Its exact law conditions replace (25.75) by

$$C_{\mathrm{c}} = P_{\mathrm{c}}LCL^{\top}P_{\mathrm{c}}^{\top}, \qquad C_{\mathrm{c}}h_{\mathrm{c}} = P_{\mathrm{c}}(LCh + b_0). \tag{25.77}$$

They follow from the affine push calculation applied to the displayed representative map. Together with descent, they encode an intrinsic quotient push. Choosing a map on one slice alone would not prove descent. A target metric or projection that varies with the model parameter does not by itself define one parameter-independent statistical channel, so Fisher contraction for a family requires that additional admission separately.

A descending unanchored chain. Take three scalar nodes with two unit edges, no anchors, zero source, and $H = I_3$. The fine precision and its quotient covariance are

$$Q_{\mathrm{f}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad C_{\mathrm{f}} = \frac{1}{9}\begin{pmatrix} 5 & -1 & -4 \\ -1 & 2 & -1 \\ -4 & -1 & 5 \end{pmatrix}. \tag{25.78}$$

Multiplication gives $Q_{\mathrm{f}}C_{\mathrm{f}} = I_3 - \mathbf{1}\mathbf{1}^{\top}/3$, which verifies the covariance equations. Let L select the first and third coordinates. At the two coarse nodes use $H_{\mathrm{c}} = I_2$ and

$$Q_{\mathrm{c}} = C_{\mathrm{c}} = P_{\mathrm{c}} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad LC_{\mathrm{f}}L^{\top} = \frac{1}{9}\begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}. \tag{25.79}$$

The fine common-translation kernel maps into the coarse common-translation kernel, establishing descent. Direct multiplication gives $P_{\mathrm{c}}LC_{\mathrm{f}}L^{\top}P_{\mathrm{c}}^{\top} = C_{\mathrm{c}}$. Thus the target positive edge of weight $1/2$ is an exact quotient push with zero sources and offsets. Equivalently, the fine residuals $x_1 - x_2$ and $x_2 - x_3$ are independent standard normals on normalization, and their sum has variance two. Their fixed coordinate Jacobian cancels from that normalized law. The unprojected endpoint covariance in (25.79) differs from C_{c}; omitting P_{c} would confuse the ambient endpoint law with the quotient law.

Theorem 25.23 (Finite variable-support Gaussian classification). Fix finite ambient dimensions, finitely many declared affine or intrinsic quotient maps, and finite semialgebraic input and target predicates over effectively represented real algebraic coefficients. Allow rational matrix expressions on explicit nonzero-denominator domains. At each object require $Q \succeq 0$, $H \succ 0$, and $h \in \operatorname{ran} Q$, with the reference and affine origin declared as above. Ranks and supports may vary with the parameters. Then an algorithm constructs the exact good and failure loci for quotient-Gaussian representability and decides universal closure, separate nonemptiness, and any finite simultaneous diagram with its supplied polynomial structural compatibilities. For ambient affine maps it uses (25.75); for intrinsic quotient maps it uses both (25.76) and (25.77). The asserted equivalence is equality of the normalized measures on their declared representative spaces. [ESTABLISHED]

Proof. Introduce a symmetric covariance variable at each object and impose (25.68), together with $Qw = h$ and the PSD and SPD admission formulas. Uniqueness in Proposition 25.21 ensures that an existential covariance cannot select a different law. Use inverse variables or denominator sign splits exactly as in the full-SPD classification. The selected map equations are then finite polynomial relations. By the quotient Gaussian calculation, they are necessary and sufficient for the specified normalized pushforward, with descent included precisely when an intrinsic map is requested.

Additional support restrictions can be encoded in the same language. At specified rank $r > 0$, all $(r+1)$-minors vanish and the sum of squares of all r-minors is positive; the rank-zero condition is that all entries vanish. The finitely many possible ranks can be treated as a finite disjunction; at full rank there are no larger minors to constrain. For declared column matrices U, V, $\operatorname{ran} U \subseteq \operatorname{ran} V$

is the existential equation $U = VB$; equality uses both inclusions. Affine support relations also impose membership of the displacement in the appropriate column space. Every further support condition still needs its actual finite formula.

Write $\mathcal{A}(x)$ for the admitted fine data and $\mathcal{M}(x, y)$ for all target, covariance, map, support, and structural conditions. Effective quantifier elimination, with the same primary theorem and exact coefficient hypotheses as Theorem 25.17, produces

$$\begin{aligned} G_{\mathrm{quot}}(x) &\Longleftrightarrow \mathcal{A}(x) \wedge \exists y\big[\mathcal{M}(x, y)\big], \\ F_{\mathrm{quot}}(x) &= \mathcal{A}(x) \wedge \neg G_{\mathrm{quot}}(x), \end{aligned} \tag{25.80}$$

and decides both $\forall x(\mathcal{A}(x) \Rightarrow \exists y[\mathcal{M}(x, y)])$ and $\exists x[\mathcal{A}(x)]$. For a finite diagram, use one common target tuple and its unique covariance at each object, with all arrow equations simultaneously. For quotient arrows, include descent and the declared identities of the induced quotient maps. Equality modulo a target null direction must be encoded as such; it is not ambient equality. When the admitted representative maps compose, their moment transformations compose by direct affine substitution. These finite formulas therefore express the entire stated diagram with the correct shared quantifiers. No nonsingular inverse or common support is assumed. □

A preferred positive pair factorization is an extra target condition, expressible using (25.64) when its assumptions hold, or using the supplied fixed marked inventory. The bound of Proposition 25.18 applies only when independent positive scaling, deletion, and its anchor operations remain admitted. Variable support does not relax strict edge-weight positivity, erase meaningful holonomy marks, or replace a prescribed target domain by a larger one.

Complete raw finite measures. The normalized classification retains an exact analytic normalizer at each object. Write

$$\begin{aligned} \mathcal{A}_i^{\mathrm{mass}} &= \frac{r_i}{2}\log(2\pi) - \frac{1}{2}\log\det{}_{+}(M_i) + \frac{1}{2}h_i^{\top} C_i h_i, \\ \mathfrak{c}_{\mathrm{c}} &= \mathfrak{c}_{\mathrm{f}} + \mathcal{A}_{\mathrm{c}}^{\mathrm{mass}} - \mathcal{A}_{\mathrm{f}}^{\mathrm{mass}}. \end{aligned} \tag{25.81}$$

When the target scalar is free, this choice gives $Z_{\mathrm{c}} = Z_{\mathrm{f}}$ by (25.71). Equality of the normalized laws then proves equality of the pushed raw finite measures, expressed using the target quotient volume. A separately prescribed target scalar must satisfy this analytic identity; logarithms, positive pseudodeterminants, and rank-dependent normalizers are not silently added to the polynomial decision theorem. A common positive mass factor cancels from a normalized law, while raw mass equality is a separate requirement.

An observed experiment can be included by retaining its observation coordinates in the finite Gaussian joint and requiring the actual representative map to leave them unchanged. For intrinsic arrows this condition applies to $a_{\mathrm{c}} + P_{\mathrm{c}}(Ls + b_0)$, including its projection and offset. Equality of that complete pushed joint preserves its observation marginal, including a singular observation law. Conditional versions and conditional-KL inference refer to that one experiment with their stated

support and every-record or almost-everywhere convention. Changing observation support does not imply a positive density at every ambient record against a common scalar reference measure.

Proposition 25.24 (Rank change obstructs statistical regularity). Finite semialgebraic feasibility across ranks does not imply total-variation continuity, differentiability in quadratic mean, or a smooth positive Fisher geometry. These failures occur even for a smooth PSD coefficient path whose quotient law is normalized at every parameter. [ESTABLISHED]

Proof. Take

$$Q(t) = \operatorname{diag}(1, t^2), \qquad H = I_2, \qquad h = a = 0. \tag{25.82}$$

For $t \neq 0$, the covariance is $\operatorname{diag}(1, t^{-2})$ and the support is $\mathbb{R}^2$. At zero, the covariance is $\operatorname{diag}(1, 0)$ and the support is the first coordinate axis. That axis has probability zero for every $t \neq 0$ and one at zero. In the convention $\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}} = \sup_B |\mathbb{P}(B) - \mathbb{Q}(B)|$, the distance is therefore exactly one. For every finite R,

$$\mathbb{P}_t(|X_2| \leq R) = \mathbb{P}(|Z| \leq R|t|) \longrightarrow 0, \qquad Z \sim \mathcal{N}(0, 1). \tag{25.83}$$

There is no weak probability limit as $t \to 0$. For example, the bounded continuous positive function $e^{-X_2^2}$ has expectation $|t|/\sqrt{t^2 + 2} \to 0$, whereas its integral against any probability measure on $\mathbb{R}^2$ is strictly positive. This directly contradicts the defining bounded-continuous-function test for a putative weak probability limit.

On $t > 0$, the second-coordinate log density has derivative $1/t - tX_2^2$. Its mean is zero and $\operatorname{Var}(X_2^2) = 2/t^4$, giving Fisher information $2/t^2$. Here two-dimensional Lebesgue measure plus the one-dimensional measure on the axis is a common dominating measure. For densities relative to that measure, differentiability in quadratic mean at zero would require a square-integrable score ℓ_0 with

$$\left\| \sqrt{p_t} - \sqrt{p_0} - \frac{t}{2} \ell_0 \sqrt{p_0} \right\|_2 = o(|t|).$$

The triangle inequality would give $\|\sqrt{p_t} - \sqrt{p_0}\|_2 \to 0$. Cauchy–Schwarz applied to $(\sqrt{p_t} - \sqrt{p_0})(\sqrt{p_t} + \sqrt{p_0})$ then gives $\|\mathbb{P}_t - \mathbb{P}_0\|_{\mathrm{TV}} \leq \|\sqrt{p_t} - \sqrt{p_0}\|_2 \to 0$, contradicting the distance-one calculation. Thus the normalized quotient laws admit no such DQM continuation across this rank change. □

Even the smooth covariance path $C(t) = \operatorname{diag}(1, t^2)$ has total-variation distance one from its line-supported law at zero. Those laws do converge weakly: couple them as (Z_1, tZ_2) and use dominated convergence for every bounded continuous function. Their quotient precisions form the finite semialgebraic graph specified by (25.68). Thus weak convergence alone would not supply the missing regular statistical geometry. The finite classification concerns exact laws at each parameter; it does not assert an automatic smooth passage between support strata.

Remaining classification. The finite algebraically encoded full-SPD problem is decided by Theorem 25.17; Theorem 25.23 extends the decision result to the declared metric quotient laws with variable ranks and supports. Correlated, nonconvex, variable-context, and branching nonflat

domains are included when their actual constraints fit the corresponding finite input contract. These are algorithmic classifications, not a closed-form catalog of every such family. The scalar formulas remain exact analytic outputs; extra transcendental scalar constraints require their own reduction or argument. General smooth, equivariant, or functorial structural selection does not follow, and the explicit selector and rank-change counterexamples rule out the stated universal continuity and equivariance claims. Arbitrary enumerator input admits no universal decision procedure by Proposition 25.20. For a family outside these finite encodings or a specified global regularity target, the remaining obligation is to supply its domain and compatibility conditions and establish the requested invariance, limit, or global construction. Nonalgebraic domains, changing-dimension limits, alternative null-coordinate completions, and global constructions outside the declared hypotheses retain their separate proof obligations. [OPEN]

25.3 Nonflat links and effective support

We now return to hard identification. The exact Gaussian marginal constructed in Theorem 25.14 does not identify this hard energy trace with a coordinate pushforward. [DEFINITION]

For a graph edge $e = (i, j)$, write the twisted energy as

$$(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j). \tag{25.84}$$

An internal edge under hard identification contributes

$$(I - \Theta_e)^\top W_e (I - \Theta_e), \tag{25.85}$$

which vanishes exactly when

$$W_e^{1/2}(I - \Theta_e) = 0. \tag{25.86}$$

[ESTABLISHED]

Positive-definite W_e forces $\Theta_e = I$; a singular weight may hide a nonidentity twist. [ESTABLISHED]

For $N_e = \ker W_e$ and quotient map $q_e : \mathbb{R}^K \to \mathbb{R}^K/N_e$,

$$W_e^{1/2}(I - \Theta_e) = 0 \quad \Longleftrightarrow \quad q_e \Theta_e = q_e. \tag{25.87}$$

[ESTABLISHED]

This is edge-local. A single common quotient $\mathbb{R}^K/N$ on which every induced weight is positive definite exists exactly when

$$\ker W_e = N, \qquad \Theta_e N = N \quad \text{for every relevant edge.} \tag{25.88}$$

On that quotient ordinary group-valued holonomy is defined. When the kernels differ, there is no canonical product of the edge-local quotients and no single “represented support.” [ESTABLISHED]

A triangle makes the boundary exact:

$$\begin{aligned} W_{12} &= \operatorname{diag}(1,0), & \Theta_{12} &= \operatorname{diag}(1,2), \\ W_{23} &= \operatorname{diag}(0,1), & \Theta_{23} &= \operatorname{diag}(2,1), \\ W_{31} &= I, & \Theta_{31} &= I. \end{aligned}$$

Every edge satisfies (25.86), yet the loop product is $2I$, fixing no nonzero vector. The local kernels differ, so no common effective quotient exists. [ESTABLISHED]

25.4 Cut links and rectangular sheaf closure

For positive-definite cut weights between two already trivialized clusters, define

$$C = \sum W_e, \qquad B = \sum W_e \Theta_e, \qquad A = \sum \Theta_e^\top W_e \Theta_e.$$

One group-valued coarse pair reproduces the cut exactly if and only if all cut twists coincide. Indeed, with $\bar{\Theta} = C^{-1}B$,

$$\Delta = A - B^\top C^{-1} B = \sum_e (\Theta_e - \bar{\Theta})^\top W_e (\Theta_e - \bar{\Theta}) \succeq 0, \tag{25.89}$$

and $\Delta = 0$ exactly under common twists. [ESTABLISHED]

The weighted mean has the correct gauge law but can be singular; equal weights with twists I_2 and $-I_2$ give $\bar{\Theta} = 0$. [ESTABLISHED]

The exact closed category is larger than one invertible link.

Theorem 25.25 (Rectangular endpoint-map closure). Choose a minimal factor

$$W_e = C_e^\top C_e, \qquad C_e : \mathbb{R}^K \to E_e.$$

Then

$$(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j) = \| R_{i,e} z_i - R_{j,e} z_j \|_{E_e}^2, \qquad R_{i,e} = C_e, \quad R_{j,e} = C_e \Theta_e. \tag{25.90}$$

For partial injections $S_{iI} : V_I \to \mathbb{R}^K$, the coarse endpoint maps are

$$R^c_{I,e} = R_{i,e} S_{iI}, \qquad R^c_{J,e} = R_{j,e} S_{jJ}. \tag{25.91}$$

They reproduce $S^\top \Lambda S$ exactly, parallel cut edges compress losslessly by the direct sum of their E_e, and nested blocking is associative. [ESTABLISHED]

Proof. Expanding (25.90) gives the original pair form. Precomposition gives (25.91); its cross block is

$$-(R^c_{I,e})^\top R^c_{J,e} = -S_{iI}^\top W_e \Theta_e S_{jJ},$$

and the diagonal blocks agree by the same expansion. Vertical stacking in $\bigoplus_e E_e$ sums the energies without loss. A second injection T gives $(RS)T = R(ST)$, proving repeated closure. □

This is the degree-zero cellular-sheaf Laplacian category [37]. Vertex spaces may have unequal dimensions and automorphism groups $\mathrm{GL}(V_I)$; edge basis changes act on both endpoint maps and the edge metric. It is gauge covariant and strictly larger than the fixed-K group-link family. Open questions are canonical minimal compression, characterization of compression to one invertible transporter and one SPD weight, and compatibility with normalized laws and RG rescaling. [OPEN]

25.5 Partial fixed-section aggregation

For each undirected internal edge, choose one orientation $e = (i, j)$, so Θ_e transports from j to i. Assume $\Theta_e \in \mathrm{GL}(K)$, $\Theta_{\bar{e}} = \Theta_e^{-1}$, and $W_e \succ 0$. Positive definiteness is the nondegeneracy assumption used below. [HYPOTHESIS]

The cluster connection Laplacian L_I is defined by

$$z^\top L_I z = \sum_{e=(i,j)\in E_I} (z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j). \tag{25.92}$$

[DEFINITION]

With semidefinite weights, the exact condition remains the edgewise constraint in (25.86), and the resulting effective-support problem is not replaced by the following proposition. [ESTABLISHED]

Let $(I_\alpha)_\alpha$ be the connected components of the internal graph. Choose a root r_α and a spanning tree in each component. For a tree path from r_α to v, let $P_{v\leftarrow r_\alpha}$ be the ordered product of its represented transports. Let Hol_{r_α} be the group of represented transport products around all closed walks based at r_α, and define

$$\mathrm{Fix}(\mathrm{Hol}_{r_\alpha}) = \{u \in \mathbb{R}^K : Hu = u \text{ for every } H \in \mathrm{Hol}_{r_\alpha}\}. \tag{25.93}$$

[DEFINITION]

Proposition 25.26 (Kernel–holonomy fixed-space isomorphism). On every connected component I_α, root evaluation is a linear isomorphism

$$\mathrm{ev}_{r_\alpha} : \ker L_{I_\alpha} \xrightarrow{\sim} \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}), \qquad z \longmapsto z_{r_\alpha}. \tag{25.94}$$

Consequently,

$$\ker L_I \cong \bigoplus_\alpha \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}), \qquad f_I := \dim \ker L_I = \sum_\alpha \dim \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}). \tag{25.95}$$

[ESTABLISHED]

Proof. By (25.92) and $W_e \succ 0$, a vector is in $\ker L_{I_\alpha}$ exactly when $z_i = \Theta_e z_j$ on every edge. Such a parallel section is determined by its root value, so root evaluation is injective. Transporting its value around any rooted closed walk returns the same value, placing its image in $\mathrm{Fix}(\mathrm{Hol}_{r_\alpha})$.

Conversely, take $u \in \mathrm{Fix}(\mathrm{Hol}_{r_\alpha})$ and define $z_v = P_{v\leftarrow r_\alpha} u$ along the spanning tree. Tree-edge constraints hold by construction. For a non-tree edge $e = (i, j)$, the tree path from r_α to j, followed by e, followed by the reverse tree path from i to r_α, has holonomy

$$H_e = P_{r_\alpha\leftarrow i}\Theta_e P_{j\leftarrow r_\alpha}. \tag{25.96}$$

Since $H_e u = u$ and $P_{r_\alpha\leftarrow i} = P_{i\leftarrow r_\alpha}^{-1}$, multiplication by $P_{i\leftarrow r_\alpha}$ gives $\Theta_e z_j = z_i$. Thus every edge constraint holds, proving surjectivity. The full Laplacian is the direct sum over connected components, which yields (25.95). □

For a connected cluster, $f_I = K$ exactly when every represented holonomy element is the identity. This is the full fixed sector. The case $0 < f_I < K$ is a partial fixed sector, while $f_I = 0$ leaves no retained cluster degree. A nonfaithful representation can make represented holonomy trivial even when principal holonomy is not. [ESTABLISHED]

This rank is a state-independent structural admissibility criterion. It assigns no partition cost, singletons remain admissible, and it does not require the constituent belief means to coincide. [NOT-CLAIMED]

Define the abstract fixed fiber

$$\mathsf{F}_I = \bigoplus_\alpha \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}). \tag{25.97}$$

After choosing bases, let $\iota_I : \mathsf{F}_I \hookrightarrow (\mathbb{R}^K)^I$ be the full-column matrix that extends each root value by the spanning-tree construction in the proof. Its range is independent of those choices and equals $\ker L_I$. For a partition $\mathcal{P}$, put $\iota_\mathcal{P} = \mathrm{blkdiag}_{I\in\mathcal{P}}\, \iota_I$. Then

$$L_I\iota_I = 0, \qquad \Lambda_c = \iota_\mathcal{P}^\top \Lambda \iota_\mathcal{P} \succeq 0, \qquad \Lambda_c \succ 0 \iff \mathrm{range}(\iota_\mathcal{P}) \cap \ker\Lambda = \{0\}. \tag{25.98}$$

[ESTABLISHED]

Internal edges vanish, while cut edges become the rectangular endpoint data of (25.91). An ordinary smooth vector bundle over varying parameters additionally requires locally constant f_I; rank jumps produce strata rather than a smooth bundle. [ESTABLISHED]

Proposition 25.27 (Nested fixed sections compose exactly). Let a larger cluster A be a union of first-level clusters, and let

$$\iota_{A\leftarrow\mathcal{P}} = \mathrm{blkdiag}_{I\subset A}\, \iota_I, \qquad L_A^{(1)} = \iota_{A\leftarrow\mathcal{P}}^\top L_A \iota_{A\leftarrow\mathcal{P}}. \tag{25.99}$$

Then the fine and first-level kernels obey

$$\ker L_A = \iota_{A\leftarrow\mathcal{P}} \ker L_A^{(1)}. \tag{25.100}$$

If $\iota_A^{(2)} : \mathsf{F}_A \hookrightarrow \bigoplus_{I\subset A} \mathsf{F}_I$ has range $\ker L_A^{(1)}$, then the fine-level fixed-section injection is

$$\iota_A^{\mathrm{fine}} = \iota_{A\leftarrow\mathcal{P}}\iota_A^{(2)}. \tag{25.101}$$

[ESTABLISHED]

Proof. Every zero mode of L_A has zero energy in each positive-semidefinite edge summand. Its restriction to every first-level cluster therefore belongs to $\ker L_I$, so it lies in $\mathrm{range}(\iota_{A\leftarrow\mathcal{P}})$. Write it as $\iota_{A\leftarrow\mathcal{P}}y$. Since $L_A \succeq 0$,

$$y \in \ker L_A^{(1)} \iff y^\top L_A^{(1)} y = 0 \iff \iota_{A\leftarrow\mathcal{P}}y \in \ker L_A. \tag{25.102}$$

This proves (25.100); substituting the range of $\iota_A^{(2)}$ gives (25.101). Matrix composition is associative, so the same statement iterates through further levels. □

Thus the vertex fixed spaces and sheaf endpoint maps are exactly functorial under nested partitions. [ESTABLISHED]

Probability normalization, rescaling, and minimal group-link compression remain separate obligations. [OPEN]

25.5.1 Constant-metric Gaussian relaxation

Proposition 25.28 (Relaxation onto the fixed sector). Let $R = R^\top \succ 0$ and $L = L^\top \succeq 0$ be constant matrices on $\mathbb{R}^d$. If the columns of N_0 form a basis of $\ker L$, define the R-orthogonal projector

$$\Pi_0^R = N_0(N_0^\top R N_0)^{-1}N_0^\top R, \tag{25.103}$$

and set $\Pi_0^R = 0$ when $\ker L = \{0\}$. The flow

$$R\dot{z} = -Lz, \qquad z(0) = z_0, \tag{25.104}$$

converges to $\Pi_0^R z_0$. If the positive spectrum is nonempty and

$$\lambda_+ = \lambda_{\min}^+(R^{-1/2}LR^{-1/2}), \tag{25.105}$$

then

$$\|z(t) - \Pi_0^R z_0\|_R \le e^{-\lambda_+ t}\|z_0 - \Pi_0^R z_0\|_R. \tag{25.106}$$

For $\mathcal{E}(z) = \frac{1}{2}z^\top Lz$, the exact dissipation identity is

$$\frac{d}{dt}\mathcal{E}(z(t)) = -\dot{z}^\top R\dot{z} = -z^\top LR^{-1}Lz \le 0. \tag{25.107}$$

[ESTABLISHED]

Proof. Set $y = R^{1/2}z$ and $A = R^{-1/2}LR^{-1/2} \succeq 0$. Then $\dot{y} = -Ay$. The spectral theorem gives convergence to the Euclidean projection of y_0 onto $\ker A = R^{1/2} \ker L$, with transverse norm decaying at the smallest positive eigenvalue of A. Transforming back gives (25.103) and (25.106). Direct differentiation gives (25.107). □

For a constant passive reframing $z' = Tz$, $T \in \mathrm{GL}(d)$, covariance of the equation requires the congruence laws

$$R' = T^{-\top} R T^{-1}, \qquad L' = T^{-\top} L T^{-1}, \qquad \Pi_0^{R'} = T \Pi_0^R T^{-1}. \tag{25.108}$$

[ESTABLISHED]

The choice $R = I_d$ is therefore equivariant only under orthogonal reframings. If $T = T(t)$, the ordinary derivative acquires the term $\dot{T}T^{-1}z'$. Covariance requires a time covariant derivative; when the original temporal connection is zero, it is

$$D'_t z' = \dot{z}' - \dot{T}T^{-1}z' = T\dot{z}, \qquad R' D'_t z' = -L' z'. \tag{25.109}$$

[ESTABLISHED]

The information-geometric identification is exact on the fixed-covariance Gaussian mean family $q_z = \mathcal{N}(z, R^{-1})$. Its mean Fisher matrix is R, and

$$D_{\mathrm{KL}}(q_z \| q_w) = \frac{1}{2}(z-w)^\top R (z-w). \tag{25.110}$$

Thus (25.104) is exactly the Fisher natural-gradient flow of $\mathcal{E}$ on that mean family. For a general regular statistical family, only the local expansion

$$D_{\mathrm{KL}}(q_\theta \| q_{\theta+d\theta}) = \frac{1}{2} d\theta^\top F(\theta) d\theta + o(\|d\theta\|^2), \tag{25.111}$$

is available under the standard regularity hypotheses [3]. [ESTABLISHED]

Freezing $F(\theta_0) = R$ yields a local linearization. The development does not identify a generic, state-dependent Fisher flow with the constant global projector and rate above. [NOT-CLAIMED]

The solution of (25.104) is a parameterized dynamical trajectory. Away from stationary points, its oriented unparameterized orbit is a history in the sense of Definition 9.4, and its Fisher arclength is invariant under orientation-preserving reparameterization by Theorem 9.54. The exponential rate and the dissipation per unit t are properties of this chosen parameterization, not of the unparameterized history. [ESTABLISHED]

Here t is a declared flow parameter; it is neither a coordinate on the timeless contextual base, an RG-depth index, nor physical time. No physical-time identification is claimed. [NOT-CLAIMED]

25.5.2 Gaussian marginal-law specialization

Specialize Theorem 10.13 after transporting every constituent marginal to one root. For this specialization, declare the admissible parent family to be the full nondegenerate Gaussian family. Trivial represented holonomy then makes the holonomy-fixed admissible subfamily equal to that declared parent; it does not itself select the parent family. Write the transported marginals as $q_i^{(r)} = \mathcal{N}(m_i, C_i)$, with $C_i \succ 0$, and choose weights $a_i > 0$ satisfying $\sum_i a_i = 1$. [HYPOTHESIS]

Proposition 25.29 (Full-Gaussian forward-KL barycenter). The unique minimizer of

$$\mathcal{D}_I(m, C) = \sum_i a_i D_{\mathrm{KL}}\big(\mathcal{N}(m_i, C_i) \| \mathcal{N}(m, C)\big), \tag{25.112}$$

matches the first two moments:

$$\bar{m} = \sum_i a_i m_i, \qquad \bar{C} = \sum_i a_i \left[C_i + (m_i - \bar{m})(m_i - \bar{m})^\top\right]. \tag{25.113}$$

Its minimized score is

$$\mathcal{D}_I^\star = \frac{1}{2}\left[\log\det \bar{C} - \sum_i a_i \log\det C_i\right]. \tag{25.114}$$

[ESTABLISHED]

Proof. The Gaussian KL formula gives

$$\begin{aligned} 2\mathcal{D}_I(m, C) &= \mathrm{Tr}(C^{-1} M(m)) - K + \log\det C - \sum_i a_i \log\det C_i, \\ M(m) &= \sum_i a_i \left[C_i + (m_i - m)(m_i - m)^\top\right]. \end{aligned} \tag{25.115}$$

Completing the weighted square gives $M(m) = \bar{C} + (m - \bar{m})(m - \bar{m})^\top$, so $m = \bar{m}$ is the unique minimizer. At that value, $\mathrm{Tr}(C^{-1}\bar{C}) - \log\det(C^{-1}\bar{C}) \geq K$, with equality exactly when $C = \bar{C}$. Substitution proves (25.114). □

Under nontrivial represented holonomy, the moment-matched Gaussian need not belong to the holonomy-fixed parent family. [ESTABLISHED]

The applicable optimization is then the constrained one in Theorem 10.13. The development does not claim that (25.113) removes that constraint. [NOT-CLAIMED]

For an explicit witness, let $K = 2$, $\mathcal{H} = \{I_2, -I_2\} \subset \mathrm{SO}(2)$, and take the one-law ensemble $\mathcal{N}(u, I_2)$ with $u \neq 0$. Its unconstrained moment-matched Gaussian is itself and is not $\mathcal{H}$-invariant. The Haar-constrained optimizer below is instead $\mathcal{N}(0, I_2 + uu^\top)$, with score $\frac{1}{2}\log(1 + \|u\|^2) > 0$. Thus the holonomy constraint can be active even when there is no pairwise disagreement among the transported constituents.

Corollary 25.30 (Compact-closure holonomy Gaussian barycenter). Let $\mathcal{H} \subset \mathrm{GL}(K)$ be a represented holonomy group whose closure $\overline{\mathcal{H}}$ is compact in $\mathrm{GL}(K)$, and let $\mu_{\overline{\mathcal{H}}}$ be normalized Haar probability measure on $\overline{\mathcal{H}}$. A nondegenerate Gaussian is $\mathcal{H}$-invariant exactly when it is $\overline{\mathcal{H}}$-invariant. Define

$$a = \sum_i a_i m_i, \qquad M_2 = \sum_i a_i (C_i + m_i m_i^\top), \tag{25.116}$$

and their Haar projections

$$a_{\overline{\mathcal{H}}} = \int_{\overline{\mathcal{H}}} h a d\mu_{\overline{\mathcal{H}}}(h), \qquad M_{2,\overline{\mathcal{H}}} = \int_{\overline{\mathcal{H}}} h M_2 h^\top d\mu_{\overline{\mathcal{H}}}(h), \qquad C_{\overline{\mathcal{H}}} = M_{2,\overline{\mathcal{H}}} - a_{\overline{\mathcal{H}}} a_{\overline{\mathcal{H}}}^\top. \tag{25.117}$$

Then $C_{\overline{\mathcal{H}}} \succ 0$, and $\mathcal{N}(a_{\overline{\mathcal{H}}}, C_{\overline{\mathcal{H}}})$ is the unique minimizer of (25.112) over all $\mathcal{H}$-invariant nondegenerate Gaussians. Its score retains the log-determinant form

$$\mathcal{D}^\star_{I,\overline{\mathcal{H}}} = \frac{1}{2}\left[\log\det C_{\overline{\mathcal{H}}} - \sum_i a_i \log\det C_i\right]. \tag{25.118}$$

[ESTABLISHED]

Proof. The stabilizer of a Gaussian under the continuous $\mathrm{GL}(K)$ pushforward action is closed. Hence invariance under $\mathcal{H}$ is equivalent to invariance under $\overline{\mathcal{H}}$. For every such invariant candidate q, invariance of KL under a common invertible pushforward gives

$$\sum_i a_i D_{\mathrm{KL}}(q_i^{(r)} \| q) = \int_{\overline{\mathcal{H}}} \sum_i a_i D_{\mathrm{KL}}(h_\# q_i^{(r)} \| q) d\mu_{\overline{\mathcal{H}}}(h). \tag{25.119}$$

The orbit-averaged source ensemble on the right has first moment $a_{\overline{\mathcal{H}}}$ and second moment $M_{2,\overline{\mathcal{H}}}$. Haar invariance makes both moments $\overline{\mathcal{H}}$-invariant, so its moment-matched Gaussian is an admissible candidate. Its covariance is positive definite because it contains the Haar average of the positive-definite matrices $hC_ih^\top$. The calculation in Proposition 25.29 therefore proves uniqueness. Every element of the compact matrix group $\overline{\mathcal{H}}$ has $|\det h| = 1$, so $\log\det(hC_ih^\top) = \log\det C_i$, which yields (25.118). □

Compactness cannot be dropped from this averaging formula. If the represented holonomy contains $2I_K$, an invariant Gaussian would require $m = 2m$ and $C = 4C$; hence $m = 0$ and $C = 0$, contradicting nondegeneracy. The generated expanding holonomy group therefore admits no invariant SPD Gaussian. [ESTABLISHED]

There is a separate exact identity when every transported constituent and the parent are restricted to one fixed covariance $C_0 \succ 0$. Define the symmetrized convention

$$D_{\mathrm{SKL}}(p, q) := \frac{1}{2}\left[D_{\mathrm{KL}}(p\|q) + D_{\mathrm{KL}}(q\|p)\right]. \tag{25.120}$$

For $q_i^{(r)} = \mathcal{N}(m_i, C_0)$ and $\bar{q}_{C_0} = \mathcal{N}(\bar{m}, C_0)$, the fixed-covariance barycenter score satisfies

$$\sum_i a_i D_{\mathrm{KL}}(q_i^{(r)} \| \bar{q}_{C_0}) = \sum_{i<j} a_i a_j D_{\mathrm{SKL}}(q_i^{(r)}, q_j^{(r)}) = \frac{1}{2} \sum_{i,j} a_i a_j D_{\mathrm{SKL}}(q_i^{(r)}, q_j^{(r)}). \tag{25.121}$$

[ESTABLISHED]

This follows from the weighted variance identity in the C_0^{-1} inner product. It is not the free-covariance optimization of Proposition 25.29, whose covariance includes the between-mean term in (25.113). [ESTABLISHED]

Ordinary pairwise KL between untransported local marginals is not a cross-fiber quantity. After choosing a path, a second path can change the transported law by a loop-holonomy action. Stabilization by that action guarantees path independence; without it the two KL values need not agree. For example, using the holonomy H in (25.123) and $q_i = q_j = \mathcal{N}(e_2, I_3)$, the identity path gives KL zero, whereas a path differing by H gives KL equal to 2. Moreover, an unpenalized sum of nonnegative within-cluster pairwise KL values is minimized by the singleton partition, where it vanishes. It therefore supplies no nondegenerate partition selector by itself. It also contains only marginal information and does not test the common recovery-kernel condition for joint recovery in Section 10.2. [ESTABLISHED]

Two elementary examples separate the structural and marginal statements. On a two-vertex tree with identity transport and $W = I_K$, represented holonomy is trivial, yet the marginals $\mathcal{N}(-ae_1, I_K)$ and $\mathcal{N}(ae_1, I_K)$ have transported divergence

$$D_{\mathrm{KL}}\big(\mathcal{N}(-ae_1, I_K) \| \mathcal{N}(ae_1, I_K)\big) = 2a^2, \tag{25.122}$$

which is arbitrarily large. Conversely, take a triangle with $K = 3$, unit edge weights, identity transports on two edges, and closing transport

$$H = \mathrm{diag}(1, -1, -1) \in \mathrm{SO}(3). \tag{25.123}$$

Its represented holonomy is nontrivial and $\mathrm{Fix}(H) = \mathrm{span}(e_1)$, so the structural fixed sector has $f_I = 1 < K$. Nevertheless, H preserves $\mathcal{N}(0, \sigma^2 I_3)$; assigning that marginal at every vertex makes all transported marginal KL values zero. [ESTABLISHED]

Transported KL therefore measures marginal occupation of the available law modes, not the dimension of the state-independent structural fixed section. [ESTABLISHED]

25.6 Quotient volume for hard membership

Let $P = \widehat{S} \in \{0,1\}^{n \times m}$, with nonempty cluster sizes $s_1, \ldots, s_m$, $D = \mathrm{diag}(s_i)$, $s = (s_i)$, and $n = \sum_i s_i$. For a positive-semidefinite matrix, pdet denotes the product of its positive eigenvalues.

On the translation quotient, the metric induced from the fine Euclidean quotient is

$$H = D - \frac{ss^\top}{n}, \qquad \ker H = \operatorname{span}(\mathbf{1}_m), \qquad \operatorname{pdet} H = \frac{m \prod_i s_i}{n}. \tag{25.124}$$

[ESTABLISHED]

Proof. For a coarse class $[x]$,

$$\min_a \|Px + a\mathbf{1}_n\|^2 = x^\top \left(D - \frac{ss^\top}{n} \right) x.$$

Deleting row and column r, the matrix determinant lemma gives the principal cofactor $\prod_i s_i/n$. Since a rank-$(m-1)$ Laplacian with kernel $\mathbf{1}_m$ has pseudodeterminant m times any principal cofactor, (25.124) follows. □

For K channels, standard quotient Lebesgue measure and fine-induced Hausdorff measure are related by

$$\mu_{\text{ind}} = J_{\mathcal{P}} \mu_{\text{std}}, \qquad J_{\mathcal{P}} = \left(\frac{m \prod_i s_i}{n} \right)^{K/2}. \tag{25.125}$$

If

$$\operatorname{range}(S) \cap \ker \Lambda = \mathbf{1}_n \otimes \mathbb{R}^K,$$

the quotient Gaussian is proper and its generalized determinant satisfies

$$\det_{H \otimes I_K} (\Lambda_{\text{c}}) := \frac{\operatorname{pdet} \Lambda_{\text{c}}}{(\operatorname{pdet} H)^K} = \frac{\operatorname{pdet} \Lambda_{\text{c}}}{J_{\mathcal{P}}^2}. \tag{25.126}$$

This is the determinant of the precision relative to the quotient metric $H \otimes I_K$. For the common unnormalized factor $\exp(-x^\top \Lambda_{\text{c}} x/2)$, let $d_q = (m-1)K$. Its normalizers are

$$\begin{aligned} Z_{\text{std}} &= (2\pi)^{d_q/2} (\operatorname{pdet} \Lambda_{\text{c}})^{-1/2}, \\ Z_{\text{ind}} &= (2\pi)^{d_q/2} \det_{H \otimes I_K} (\Lambda_{\text{c}})^{-1/2} = J_{\mathcal{P}} Z_{\text{std}}. \end{aligned} \tag{25.127}$$

The factor cancels from the normalized law because the measure changes by the same amount. [ESTABLISHED]

Raw partition functions differ by $\log J_{\mathcal{P}}$, so a partition score needs a fixed volume convention, an exact correction, or a compensating prior. [ESTABLISHED]

25.7 Recognition-side costs

Let $B \in \mathbb{R}^{NK \times mK}$ have orthonormal columns spanning $\operatorname{range} S$, let $B_\perp \in \mathbb{R}^{NK \times r}$, $r = NK - mK$, complete it so that $[B \; B_\perp]$ is orthogonal, and write $p = \mathcal{N}(\mu, \Lambda^{-1})$. These definitions apply throughout this section.

The admissible mean-only family leaves covariance unrestricted. For $q = \mathcal{N}(\nu, \Sigma)$, $\nu \in \operatorname{range} S$, its exact optimized cost is

$$\mathcal{G}_{\text{tie}} = \inf_{\nu \in \operatorname{range} S,\ \Sigma \succ 0} D_{\mathrm{KL}}\big(\mathcal{N}(\nu, \Sigma) \| p\big) = \frac{1}{2} m_\perp^\top \left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} m_\perp, \qquad m_\perp = B_\perp^\top \mu. \tag{25.128}$$

[ESTABLISHED]

Proof. Optimizing the Gaussian KL over covariance gives Λ^{-1}, leaving $\frac{1}{2}(\nu - \mu)^\top \Lambda (\nu - \mu)$. Minimizing over $\nu = Ba$ gives the Λ-projection of μ. In the orthogonal basis $[B\ B_\perp]$, its remaining block is the Schur complement

$$\Lambda_{22} - \Lambda_{21}\Lambda_{11}^{-1}\Lambda_{12} = \left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1}.$$

This proves (25.128). □

A recognition law supported exactly on $\operatorname{range} S$ is singular against p, so its relative entropy to p is $+\infty$. The following declared full-rank regularization makes the divergence and every finite term explicit. Let $\nu_\star$ be the mean optimizer above and set

$$\Sigma_\varepsilon = B(B^\top \Lambda B)^{-1} B^\top + \varepsilon B_\perp B_\perp^\top, \qquad q_\varepsilon = \mathcal{N}(\nu_\star, \Sigma_\varepsilon), \qquad \varepsilon > 0. \tag{25.129}$$

Then, exactly,

$$\begin{aligned} \mathcal{G}_\varepsilon := D_{\mathrm{KL}}(q_\varepsilon \| p) = \mathcal{G}_{\text{tie}} + \frac{r}{2}\log\frac{1}{\varepsilon} - \frac{r}{2} + \frac{1}{2}\log\det(B_\perp^\top \Lambda^{-1} B_\perp) \\ + \frac{\varepsilon}{2}\operatorname{Tr}(B_\perp^\top \Lambda B_\perp). \end{aligned} \tag{25.130}$$

[ESTABLISHED]

Proof. In the orthogonal basis $[B\ B_\perp]$, the covariance in (25.129) has determinant $\det(B^\top \Lambda B)^{-1}\varepsilon^r$, while

$$\operatorname{Tr}(\Lambda \Sigma_\varepsilon) = mK + \varepsilon \operatorname{Tr}(B_\perp^\top \Lambda B_\perp).$$

The Gaussian KL formula and the optimized mean term give all terms in (25.130) after using

$$\det \Lambda = \det(B^\top \Lambda B) \det\left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1},$$

which is the block determinant identity. □

Thus $\mathcal{G}_\varepsilon \to +\infty$ as $\varepsilon \downarrow 0$: identification is a generative construction, not a finite recognition restriction.

The marginal precision in (25.128) obeys

$$\left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} \preceq B_\perp^\top \Lambda B_\perp, \tag{25.131}$$

with equality exactly when $B^\top \Lambda B_\perp = 0$. Using the restriction precision instead therefore overstates the cost except under Λ-orthogonality. [ESTABLISHED]

If only covariance factorization across a partition is imposed, the exact Gaussian optimum has gap

$$\mathcal{G}_{\text{fact}} = \frac{1}{2}\left[\sum_I \log\det \Lambda_{II} - \log\det \Lambda\right]. \tag{25.132}$$

[ESTABLISHED]

The mean-tie cost is nonnegative and vanishes at the finest partition; the factorization gap decreases under merging by Fischer's determinant inequality and vanishes at the coarsest partition. The quotient volume is convention dependent across partitions. Therefore these terms do not supply a nondegenerate intrinsic scale without an additional coefficient or a different nonmonotone functional. Whether such a selector exists remains open. [OPEN]

The corrected cross-inventory boundary is the general theorem of Section 10.4: a common evidence-preserving Markov pushforward makes ELBOs comparable and ordered. The hard energy trace analyzed here is not such a pushforward unless a separate theorem proves it, so its cross-partition difference can mix evidence and tightness. [ESTABLISHED]

25.8 The frame sector

If a coarse Gaussian (U_I, μ_I, Σ_I) is sent to constituent i through $U_i U_I^{-1}$, then

$$\mathcal{N}(U_i U_I^{-1}\mu_I, U_i U_I^{-1}\Sigma_I U_I^{-\top} U_i^\top) = \mathcal{N}(U_i \bar{m}, U_i \bar{S} U_i^\top), \tag{25.133}$$

with $\bar{m} = U_I^{-1}\mu_I$ and $\bar{S} = U_I^{-1}\Sigma_I U_I^{-\top}$. Thus the coarse frame cancels. With faithful representation the level set is exactly one gauge orbit; without faithfulness only unidentifiability follows. [ESTABLISHED]

For $K \geq 2$, $n \geq 2$, no map $F : \mathrm{GL}^+(K)^n \to \mathrm{GL}^+(K)$ is both left equivariant and permutation symmetric. Choose an order-n rotation $v \in \mathrm{SO}(K)$ and inputs $(I, v, \ldots, v^{n-1})$. Left multiplication by v cyclically permutes the tuple. Symmetry keeps F fixed, while equivariance sends it to vF; hence $(I - v)F = 0$, making F singular. [ESTABLISHED]

Physicist summary. Hard Gaussian aggregation is exact Galerkin restriction: internal springs disappear and cut springs add. Nonflat or rank-changing hard restrictions retain their energy in the larger sheaf category of endpoint maps. With positive-definite internal weights, holonomy determines the parallel modes available to a structural fixed section: trivial represented holonomy retains all K dimensions, while nontrivial holonomy can retain a smaller sector or none. [ESTABLISHED]

Gaussian marginalization is a different operation and keeps the hidden Gaussian normalization. Its closure has distinct exact boundaries. Independent coefficient tests at one fixed positive anchor force common congruence axes. For $K \geq 2$, a defined Schur reduction on a fixed flat graph closes for every independent SPD weight assignment precisely when each hidden component has at most two retained terminals, counting ground. Proper nonflat degree-two hidden paths and cycles with

zero hidden quadratic anchors instead close by composing their marked transports and covariance contributions, with sources, scalar factors, and loop holonomies retained. Branching can produce a proper Gaussian marginal outside every positive two-node representation, so a complete quadratic action may be necessary. [ESTABLISHED]

When attained, transported KL tests whether marginal laws occupy an available holonomy-fixed mode; without attainment, a zero infimum proves only approximability. It is not a structural-holonomy test or a partition selector, and joint recovery remains separate. Quotient volumes, mean tying, and determinant gaps are explicit, but these terms alone supply no nondegenerate intrinsic blocking scale. [ESTABLISHED]

Chapter 26

Multivariate-Gaussian Operator Renormalization

Chapter 11 separates a genuine scale diagram from a sequence of differently sized objects. This chapter instantiates that diagram on the multivariate-Gaussian (MVG) interaction sector of Chapter 25. The flowed object is an operator parameter, not automatically a normalized Gaussian law: a law-level flow also needs a coarse reference measure, an information-vector rule, and a finite normalizer at every scale. Likewise, the hard aggregation below is only one admissible coarse arrow. The effective-support, sheaf, and Kron reductions of Chapter 25 give other arrows on narrower domains. Unless a channel index is displayed, this chapter flows the belief/recognition operator sector; a model-channel flow is a separate typed construction with its own dimension, links, and parameters. [DEFINITION]

26.1 The MVG operator arrow

At level ℓ, let $\Xi_\ell^{\mathrm{MVG}} = ((A_i^{(\ell)}), (W_{ij}^{(\ell)}))$ with $A_i^{(\ell)}, W_{ij}^{(\ell)} \in \mathbb{S}_+^K$, and assemble

$$(\Lambda_\ell)_{ii} = A_i^{(\ell)} + \sum_{j \neq i} W_{ij}^{(\ell)}, \qquad (\Lambda_\ell)_{ij} = -W_{ij}^{(\ell)}. \tag{26.1}$$

For a hard partition, write $S_\ell = \widehat{S}_\ell \otimes I_K$. Its unrescaled aggregation map is

$$\mathcal{A}_{S_\ell}(\Lambda_\ell) = S_\ell^\top \Lambda_\ell S_\ell, \qquad \mathcal{A}_{S_\ell}(\Xi_\ell^{\mathrm{MVG}}) = \left(\left(\sum_{i \in I} A_i^{(\ell)} \right)_I, \left(\sum_{\substack{i \in I \\ j \in J}} W_{ij}^{(\ell)} \right)_{I \neq J} \right). \tag{26.2}$$

[DEFINITION]

The second identity is exact on the hard-aggregation domain proved in Section 25.1; nonflat data must first satisfy the effective-support or rectangular endpoint conditions of Sections 25.3 and 25.4. [ESTABLISHED]

Proposition 26.1 (Aggregation is a positive semigroup action). If S_1 maps fine nodes to intermediate clusters and S_2 maps those clusters to coarse clusters, then

$$\mathcal{A}_{S_2}\mathcal{A}_{S_1} = \mathcal{A}_{S_1 S_2}. \tag{26.3}$$

Each $\mathcal{A}_S$ is linear and positive on the product cone of node and edge weights. It is generally noninjective because every edge internal to a cluster is annihilated. [ESTABLISHED]

Proof. Congruence gives $S_2^\top(S_1^\top \Lambda S_1)S_2 = (S_1S_2)^\top \Lambda (S_1S_2)$. The parameter formula follows by associativity of finite sums. Sums preserve positive semidefiniteness, so the map is positive. Changing an internal weight while holding every cut weight and self term fixed leaves $\mathcal{A}_S(\pi)$ unchanged, proving noninjectivity.

An RG arrow additionally declares a positive rescaling:

$$\Lambda_{\ell+1} = \zeta_\ell^{-1}\mathcal{A}_{S_\ell}(\Lambda_\ell), \qquad \zeta_\ell > 0. \tag{26.4}$$

This is the MVG component of the comparison data required by Section 11.1. The source and target usually have different dimensions. A fixed ray is therefore typed only after comparison maps identify a repeated hierarchy with one fixed parameter space. Without that identification, (26.4) is a cocycle of positive maps, not an autonomous dynamical system.

Equal-block spectral bounds. If every cluster has b nodes, then $S^\top S = bI$. For every symmetric Λ,

$$b\lambda_{\min}(\Lambda) \le \lambda_{\min}(S^\top \Lambda S), \qquad \lambda_{\max}(S^\top \Lambda S) \le b\lambda_{\max}(\Lambda). \tag{26.5}$$

Indeed, congruence preserves the Loewner inequalities $\lambda_{\min}(\Lambda)I \preceq \Lambda \preceq \lambda_{\max}(\Lambda)I$. These are finite-level bounds. Divergence or convergence as $\ell \to \infty$ additionally requires an infinite hierarchy and a declared normalization. [ESTABLISHED]

26.2 The biadditive closed family

Let each node carry $x_i \ge 0$, and fix $A, M \in \mathbb{S}_+^K$. The family

$$A_i = x_i A, \qquad W_{ij} = x_i x_j M \tag{26.6}$$

is exactly closed under every hard partition: with $x_I = \sum_{i\in I} x_i$,

$$A_I = x_I A, \qquad W_{IJ} = x_I x_J M. \tag{26.7}$$

This follows from distributivity and proves closure, not attraction. [ESTABLISHED]

Proposition 26.2 (Positive additive rules are forced to be linear). Suppose $\alpha : \mathbb{R}_+ \to \mathbb{S}_+^K$ is additive and $w : \mathbb{R}_+^2 \to \mathbb{S}_+^K$ is symmetric and separately additive. Then there are $A, M \in \mathbb{S}_+^K$ such

that

$$\alpha(x) = xA, \qquad w(x,y) = xyM. \tag{26.8}$$

Thus (26.6) is the maximal positive family among rules depending only on additive scalar node masses and obeying those additivity hypotheses. [ESTABLISHED]

Proof. If $y \geq x$, additivity gives $\alpha(y) - \alpha(x) = \alpha(y - x) \succeq 0$; hence α is Loewner monotone. For every v, the scalar function $x \mapsto v^\top \alpha(x) v$ is additive, nonnegative, and therefore monotone. A monotone additive function on $\mathbb{R}_+$ is linear, so $v^\top \alpha(x) v = x v^\top \alpha(1) v$. Polarization yields $\alpha(x) = x\alpha(1)$. For fixed y, the same argument gives $w(x,y) = xw(1,y)$; applying it to the second argument gives $w(1,y) = yw(1,1)$. Take $A = \alpha(1)$ and $M = w(1,1)$.

On a homogeneous complete graph with equal blocks of size b, the unrescaled map acts on the common pair (A, W) by

$$A \longmapsto bA, \qquad W \longmapsto b^2 W. \tag{26.9}$$

The dense normalization $\zeta = b^2$ fixes the coupling sector and contracts the self sector by $1/b$. Consequently the candidate limiting ray has $A = 0$. It is a boundary operator: its consensus directions lie in the kernel, so it does not define a full-space normalized Gaussian density. Pinning, quotienting the kernel, and retaining an external positive mass are different repairs and define different probability problems. A pseudoinverse inserted into the original Lebesgue density is not a fourth repair.

The exponent 2 is not topology independent. For a typical block $B(b)$ and coarse cut $\partial B(b)$, define the weighted sums

$$W_\partial(b) = \sum_{e \in \partial B(b)} W_e, \qquad A_B(b) = \sum_{i \in B(b)} A_i.$$

If, in a declared matrix norm, $\|W_\partial(b)\| \asymp b^{s_W}$ and $\|A_B(b)\| \asymp b^{s_A}$, then a scalar normalization that keeps the coupling sector nondegenerate scales as b^{s_W}, while the normalized self sector scales as $b^{s_A - s_W}$. Edge count alone does not determine either exponent when the fine weights vary with b.

The count-based exponents follow only under explicit nonzero-average assumptions. If $n_{\mathrm{cut}}(b) \asymp b^s$ and

$$\frac{W_\partial(b)}{n_{\mathrm{cut}}(b)} \longrightarrow \overline{W} \neq 0, \qquad \frac{A_B(b)}{b} \longrightarrow \overline{A} \neq 0,$$

then $s_W = s$, $s_A = 1$, and the coupling and normalized-self exponents are respectively s and $1 - s$. For the homogeneous complete-graph scheme above, a cut between two $\Theta(b)$-sized blocks has $s = 2$; consecutive blocks in a nearest-neighbor chain have $s = 0$. Relevance of the self term is therefore a property of the declared weighted graph family and normalization scheme, not of (26.4) alone. [ESTABLISHED]

26.3 Cone dynamics: what Birkhoff does and does not prove

Let

$$\mathcal{K}_N = \left(\prod_i \mathbb{S}_+^K\right) \times \left(\prod_{i<j} \mathbb{S}_+^K\right) \tag{26.10}$$

be the full MVG parameter cone. On the interior of a proper cone $\mathcal{K}$, Hilbert's projective metric is

$$d_{\mathrm{H}}(x,y) = \log\left(\frac{\inf\{\beta>0 : x \preceq_{\mathcal{K}} \beta y\}}{\sup\{\alpha>0 : \alpha y \preceq_{\mathcal{K}} x\}}\right). \tag{26.11}$$

A positive endomorphism of finite projective diameter contracts this metric; a primitive finite-dimensional positive matrix is the familiar special case [16, 68]. The hypotheses fail for sum-only MVG aggregation on the full cone.

Proposition 26.3 (The full-cone route is blocked). After a hierarchical identification makes $\mathcal{A}_S$ an endomorphism, sum-only aggregation is not primitive on $\mathcal{K}_N$. Moreover, the dense candidate ray with $A = 0$ lies on $\partial\mathcal{K}_N$, whereas a full-cone Birkhoff contraction has an interior Perron ray. Hence Birkhoff's full-cone theorem cannot prove attraction to that candidate. [ESTABLISHED]

Proof. An absent coarse cut remains a zero edge coordinate, so the map can meet the boundary even from positive data on the represented fine edges. More sharply, define for a subspace $U \subseteq \mathbb{R}^K$

$$\mathcal{F}_U = \{(W_e)_e : \operatorname{ran} W_e \subseteq U \text{ for every } e\}. \tag{26.12}$$

Each $\mathcal{F}_U$ is an invariant face because the range of a sum of positive semidefinite matrices contained in U is still contained in U. Proper invariant faces rule out primitivity. Finally, $A = 0$ is a boundary coordinate of $\prod_i \mathbb{S}_+^K$, while the Perron ray of a strict interior contraction is interior.

The coupling cone alone removes the vanishing self coordinate but not the matrix obstruction. Suppose an identified spatial map $B \geq 0$ has a positive eigenvector $c = (c_e)_e$. Because aggregation only adds weights, for every $M \in \mathbb{S}_+^K$

$$W_e = c_e M \tag{26.13}$$

is an eigenray with the same spatial eigenvalue. Different matrix rays are not mixed; on the raw product cone their projective separation can be infinite. Thus sum-only aggregation cannot select a fiber direction. It can at most contract a spatial cone after the fiber ray has been declared.

Conjecture 26.4 (Fixed-B scalarized attraction). Fix $M_0 \succ 0$ and restrict to

$$\mathcal{K}_{M_0}^W = \{(c_e M_0)_e : c_e \geq 0\}. \tag{26.14}$$

If a hierarchical identification yields one fixed spatial endomorphism B, some power B^q is strictly positive, and its Perron vector factorizes as $c_{ij} = x_i x_j$, then Birkhoff's theorem gives a unique attracting spatial ray $W_{ij} = x_i x_j M_0$ on this scalarized cone. [CONJECTURE]

This statement freezes every matrix direction. It is not a theorem about the full coupling cone, a nonautonomous sequence B_ℓ, or the common-principal RG state of Section 11.3. Alternating positive maps may contract projective distances while approaching a periodic orbit, as Section 11.4 shows. The missing obligations are a fixed hierarchical identification, primitivity of the resulting B, and the factorization of its Perron vector.

Proposition 26.5 (Residual fiber label). For $M \in \mathbb{S}_+^K$, the orbit under residual global reframing $M \mapsto h^\top M h$, $h \in \mathrm{GL}(K)$, is determined exactly by $\operatorname{rank} M$. Hence the biadditive ansatz has $K+1$ internal congruence classes and no continuous internal modulus. [ESTABLISHED]

Proof. Sylvester's law of inertia classifies real symmetric matrices under congruence [40]. A positive semidefinite matrix has no negative eigenvalues, so its inertia is determined by its rank. In particular, every $M \succ 0$ is congruent to I_K, witnessed by $h = M^{-1/2}$.

Rank does not classify the spatial graph, the limiting spectral measure, corrections to scaling, basin, or scheme. Nor does the proposition extend unchanged to a nonflat pair (W, Θ), whose residual action combines congruence on W with conjugation on Θ.

26.4 An exact finite-sector gate

The homogeneous calculation (26.9) is exact. On $\mathbb{S}^K \oplus \mathbb{S}^K$, its eigenvalues are b^2 on the coupling sector and b on the self sector, each with multiplicity $K(K+1)/2$. The spectral ratio outside the dominant eigenspace is therefore

$$\frac{b}{b^2} = \frac{1}{b}, \tag{26.15}$$

equal to 0.5 at $b = 2$. Comparing the two largest entries of the sorted spectrum would incorrectly return one because the dominant eigenvalue is degenerate. [ESTABLISHED]

Equation (26.15) is the complete finite-sector result. Attraction, uniqueness, and scheme independence remain open because they require a declared invariant cone, hierarchy, and limiting protocol. [OPEN]

26.5 Gauge-invariant spectral data

Let L be the interaction Laplacian obtained from (26.1) by setting all $A_i = 0$. Both L and $\Lambda = L + A$ are bilinear forms. Represent a local reframing by the invertible new-to-old coordinate map

$$G : \mathbb{R}^{NK}_{\mathrm{new}} \xrightarrow{\sim} \mathbb{R}^{NK}_{\mathrm{old}}, \qquad G = \operatorname{diag}(g_1, \ldots, g_N).$$

Pullback to the new coordinates sends both forms by congruence, $(L, \Lambda) \mapsto (G^\top L G, G^\top \Lambda G)$. Ordinary eigenvalues of either matrix are therefore frame dependent. The generalized spectrum of the pair is invariant when the pencil is regular.

Proposition 26.6 (Regular-pencil invariance). Suppose $\det(L - d\Lambda)$ is not the zero polynomial. Then (L, Λ) and $(G^\top LG, G^\top \Lambda G)$ have the same generalized eigenvalues with the same multiplicities. If Λ is invertible, there are exactly NK finite roots. [ESTABLISHED]

Proof.

$$\det(G^\top LG - dG^\top \Lambda G) = \det(G)^2 \det(L - d\Lambda). \tag{26.16}$$

The nonzero factor does not change roots or regularity. When Λ is invertible, the coefficient of d^{NK} is $(-1)^{NK} \det \Lambda \neq 0$.

Proposition 26.7 (Regularity criterion). For $L, A \succeq 0$ and $\Lambda = L + A$, the pencil (L, Λ) is regular if and only if

$$\ker L \cap \ker A = \{0\}. \tag{26.17}$$

In that case $\Lambda \succ 0$. [ESTABLISHED]

Proof. Positive semidefiniteness gives $\ker(L + A) = \ker L \cap \ker A$. A nonzero common-kernel vector annihilates $L - d\Lambda$ for every d, so the determinant is identically zero. Conversely, if the intersection is trivial then $L + \Lambda = 2L + A \succ 0$; evaluating at $d = -1$ proves that the determinant polynomial is not identically zero. The same intersection identity gives $\ker \Lambda = \{0\}$.

At the dense candidate ray $A = 0$, one has $\Lambda = L$ and $\det(L - dL) \equiv 0$. Passing to a complement of $\ker L$ produces the constant generalized spectrum $\{1\}$, not a graph-sensitive label. Pinning and quotienting therefore do not preserve the same spectral object.

Proposition 26.8 (Mass-pencil transfer). Let $R \succ 0$, $a > 0$, and $\Lambda = L + aR$. Generalized eigenvalues of (L, R) and (L, Λ) are related by

$$d = \frac{\lambda}{\lambda + a}, \qquad \lambda = \frac{ad}{1 - d}. \tag{26.18}$$

The correspondence preserves eigenvectors, multiplicities, and ordering. Consequently,

$$N_d(t) = N_\lambda\left(\frac{at}{1 - t}\right), \qquad 0 \leq t < 1. \tag{26.19}$$

If $N_\lambda(s) - N_\lambda(0) \sim Cs^\alpha$ near zero, then at fixed $a > 0$, $N_d(t) - N_d(0) \sim Ca^\alpha t^\alpha$. [ESTABLISHED]

Proof. The equation $Lv = d(L + aR)v$ is equivalent, for $d \neq 1$, to $Lv = (ad/(1 - d))Rv$. The case $d = 1$ would require $Rv = 0$, which is impossible. Monotonicity of the fractional map gives the counting identity and its asymptotic consequence.

Mass removal and the infrared limit do not commute:

$$\lim_{a \downarrow 0} \lim_{\lambda \downarrow 0} \frac{\lambda}{\lambda + a} = 0, \qquad \lim_{\lambda \downarrow 0} \lim_{a \downarrow 0} \frac{\lambda}{\lambda + a} = 1. \tag{26.20}$$

Thus a positive mass regularizes each finite problem but does not supply a mass-independent fixed-ray spectrum.

Thermodynamic spectral labels. For a declared sequence (V_n, L_n, R_n), define

$$N_n(t) = \frac{1}{|V_n|K} \#\{k : \lambda_k^{(n)} \le t\}. \tag{26.21}$$

A cumulative exponent $N(t) - N(0) \sim Ct^\alpha$ is meaningful only after proving convergence $N_n \to N$ in a stated mode. A fixed finite graph has an atomic measure, so a density exponent is not defined there. RG depth ℓ and thermodynamic size n form the two-index family of Section 11.6; neither the existence nor the commutation of those limits follows from the finite congruence identity. [OPEN]

Equality of one exponent is not equality of universality classes. The full limiting measure, amplitudes, corrections to scaling, invariant faces, basin, and blocking scheme may still differ. Projective eigenvalue ratios, block-normalized scaling dimensions, and matched heat/IDS exponents are candidate invariant observables; the raw running matrices are scheme-dependent coordinates.

26.6 Finite stochastic refinement on matrix cones

Section 11.7 proves the general typing statement: a stochastic right section samples a fine preimage and is not an inverse of the realized coarse map. The matrix cone sharpens both the obstruction and the available finite construction.

The central Wishart family has Laplace transform $\det(I + \sigma\Theta)^{-p}$. Apart from the degenerate law δ_0, it is infinitely divisible exactly when $\operatorname{rank}\sigma = 1$ [59]. Therefore a full-rank Wishart prior cannot provide convolution roots of every order for $K \ge 2$; the surviving rank-one case is an embedded scalar Gamma law on a proper face. This does not mean full-cone convolution semigroups are absent. Define the trace-one cone

$$\mathbb{S}^K_{+,1} = \{U \succeq 0 : \operatorname{tr} U = 1\}.$$

Let α be a finite positive Borel measure on $\mathbb{S}^K_{+,1}$, and let $\beta : \mathbb{S}^K_{+,1} \to (0,\infty)$ be measurable with

$$\int_{\mathbb{S}^K_{+,1}} \log\left(1 + \beta(U)^{-1}\right) \alpha(dU) < \infty.$$

For $\Theta \succeq 0$, the corresponding matrix-Gamma law on $\mathbb{S}^K_+$ has Laplace transform

$$\mathcal{L}_{\alpha,\beta}(\Theta) = \exp\left[-\int_{\mathbb{S}^K_{+,1}} \log\left(1 + \frac{\operatorname{tr}(U\Theta)}{\beta(U)}\right) \alpha(dU)\right], \tag{26.22}$$

and, for fixed β, replacing α by $t\alpha$, $t \ge 0$, produces a convolution semigroup; $t = 0$ is δ_0 [71]. Generic full-cone draws need not lie in one Kron-closed chart, however.

Theorem 26.9 (Congruence-diagonal Gamma–Dirichlet bridge). Fix the Kron-closed cone $\mathcal{C}_H$ from Section 25.2, channel parameters $a_r, \beta_r > 0$, and intensities $t_e > 0$. Independently draw

$$X_{e,r} \sim \mathrm{Gamma}(t_e a_r,\ \text{rate } \beta_r), \qquad W_e = H\,\mathrm{diag}(X_{e,1}, \dots, X_{e,K})H^\top. \tag{26.23}$$

For a coarse coordinate f formed from constituents E_f, $\sum_{e\in E_f} W_e$ has the same form with intensity $\sum_{e\in E_f} t_e$. Conditioned on

$$W_f = H\,\mathrm{diag}(c_1, \dots, c_K)H^\top,$$

sample independently over channels

$$(P_{e,r})_{e\in E_f} \sim \mathrm{Dirichlet}((t_e a_r)_{e\in E_f}), \qquad X_{e,r} = c_r P_{e,r}. \tag{26.24}$$

Then every $W_e \in \mathcal{C}_H$ and $\sum_{e\in E_f} W_e = W_f$ exactly. Applied independently to coarse self terms and cut edges, this is a stochastic right section of any fixed finite aggregation map on the represented coordinates. [ESTABLISHED]

Proof. Scalar Gamma variables with a common rate add by addition of their shape parameters. Conditional on their sum, their proportions have the Dirichlet law in (26.24). The identities hold channelwise, and multiplication by the fixed congruence $H(\cdot)H^\top$ preserves both $\mathcal{C}_H$ and addition.

Internal fine edges are absent from the coarse coordinates and require a separate prior. The theorem neither recovers the discarded realization nor constructs a two-sided inverse. An infinite graph-level construction still owes compatible internal-edge priors, nested bridge kernels, gauge-covariant chart data, cylinder consistency, and tightness or another projective-limit theorem. [OPEN]

26.7 Relations to Bayesian, Laplacian, network, and learned RG

Bayesian renormalization. For a Gaussian mean family with fixed precision Λ, the Fisher metric in the mean coordinates is Λ, and the inverse Fisher cometric is Λ^{-1}. This local identity makes an MVG comparison with Bayesian renormalization natural. Berman, Klinger, and Stapleton, however, begin with a posterior over model parameters. In a late-data local Gaussian regime, the delta method pushes an inverse Fisher cometric through a forward map to obtain a sample-space diffusion tensor, exactly for an affine map and to leading local order for a nonlinear map [11, 13]. Howard, Klinger, Maiti, and Stapleton subsequently apply Bayesian RG to neural network field theories [41]. [ESTABLISHED]

Hard energy precomposition $\Lambda \mapsto S^\top \Lambda S$ is not that posterior flow. [ESTABLISHED]

No theorem identifies their scale orientation, connection data, or global continuation with (26.4); the comparison therefore supplies no “opposite-flow” theorem between this operator coarsening and Bayesian learning. [NOT-CLAIMED]

Laplacian RG. With a positive reference form R, the generalized heat generator $R^{-1}L$ transforms by similarity, so its spectrum and heat trace are gauge invariant. The scalar Laplacian-RG construction motivates the entropy scale diagnostic and the positive susceptibility

$$C(\tau) = -\frac{dS}{d\log\tau} \tag{26.25}$$

[86]. Raw off-diagonal heat-kernel blocks transform as $K_{ij} \mapsto g_i^{-1} K_{ij} g_j$, however, and are not invariant scalar node affinities. [ESTABLISHED]

A gauge-invariant real-space block selector, zero-mode treatment, and entropy normalization remain open. The heat scale diagnostic does not by itself prove a unique or scheme-independent partition. [OPEN]

Network RG. Exact closure occurs in the independent-edge multiscale family of Garuccio, Lalli, and Garlaschelli, whose hidden variables aggregate additively [34]. It is a characterized family, not a theorem for arbitrary hidden-variable networks. Interaction closure after a separately specified elimination is a different result. Interacting-network RG can generate higher-order interactions outside restricted coordination classes [23]; it does not imply exact closure of arbitrary matrix-weighted node aggregation. Conversely, (26.2) does not integrate out edge variables and should not be presented as that network transformation. The broader network-RG taxonomy is reviewed by Gabrielli et al. [33]. [ESTABLISHED]

Graph learning and variational RG. The Galerkin identity $L_c = S^\top L S$ preserves the restricted quadratic energy. It does not imply spectral approximation, message-passing commutation, or a graph neural network realization; those require the approximation and residual controls separated in Section 10.7 [43, 57]. DiffPool instead learns soft cluster assignments end to end [91]; that modeling choice does not by itself provide an exact RG semigroup or a closure theorem. Oono and Suzuki prove conditional exponential information loss for deep graph convolutional networks under spectral and weight hypotheses [69]; their result is an oversmoothing boundary, not a Galerkin-commutation theorem. Likewise, Mehta and Schwab's exact variational-RG/deep-network correspondence assumes a hidden-variable kernel satisfying its trace condition [60]. A deterministic identification $y = S\widetilde{y}$ does not supply that kernel or trace condition, so their exact mapping is a precedent within its own construction, not a proof that (26.4) is a deep network. [ESTABLISHED]

26.8 Remaining MVG obligations

The operator recursion is exact on its declared domain, but the following extensions remain open. Partial fixed-section fibers and rectangular cellular-sheaf endpoint maps are already structurally closed under nested blocking by Sections 25.4 and 25.5. [ESTABLISHED]

This structural result does not supply a canonical compression to one invertible link and one SPD weight or an RG rescaling across variable-rank fibers. A full MVG law flow must specify

the information vector, reference measure, and normalizer and prove that every iterate stays in the Gaussian natural domain. Attraction needs a typed autonomous endomorphism or a cocycle theorem, a basin, and a matrix mixing mechanism or an explicit scalarized restriction. Universality needs scheme-invariant observables and the two-index convergence required by Section 11.6. A gauge-invariant, nondegenerate block selector remains open. Stochastic inverse RG needs graph-level projective priors, including priors for annihilated internal edges. [OPEN]

Physicist summary. The MVG interaction precision has a clean positive coarse map: cut weights and self terms add, and a declared scalar restores the chosen units. On dense homogeneous graphs the coupling sector scales like b^2 while the self sector scales like b, giving an exact relative factor $1/b$. That algebra produces a boundary operator ray, not a normalized Gaussian fixed law. Birkhoff contraction becomes plausible only after freezing the matrix direction and constructing one primitive spatial endomorphism; sum-only aggregation preserves matrix faces and cannot select that direction. Generalized spectra and Gamma–Dirichlet finite refinements are exact gauge-compatible tools, but neither solves scheme independence, nonflat hierarchy, or the infinite projective-limit problem.

Part V

Boundaries, Evidence, and Interpretation

Chapter 27

Obstructions

The construction developed in this document is deliberately narrow. Its general belief-law and model-law layers fix one normalized joint and one exact bound, with manifolds introduced only on selected finite-dimensional tiers; the concrete obstructions in this chapter concern particular Gaussian realizations and must not be promoted to no-go theorems for arbitrary belief or model fibers. Each statement below names its architecture, and the weaker constructions that survive it are kept explicit.

Every discrete map in this chapter is a graph-link datum. We write Θ_e^b and Θ_e^m for belief- and model-sector links on an oriented edge copy e; the Gaussian examples suppress the sector superscript. Reverse orientation of the same edge copy obeys $\Theta_{\bar{e}} = \Theta_e^{-1}$. The reciprocal-pair example instead uses two *distinct parallel edge copies* $e \neq f$, traversed in opposite directions, so their link values are not required to be inverses. These symbols are distinct from the base-connection transports $\Omega_\gamma, \widetilde{\Omega}_\gamma$ and from the cross-bundle morphisms $\Phi, \widetilde{\Phi}$. No graph link is identified with base parallel transport without a curve assignment and an equality hypothesis.

The material organizes around one temptation. A tower of scales in which every level's prior is supplied by the level above must terminate, and its top level has no level above. It is natural to try to close the tower by a reciprocal Gaussian agreement pair. The exact result is narrower than "cycles fail": the *flat, unanchored reciprocal pair* is singular, and an objective that appends its energy while omitting a required global normalizer is not an evidence bound. Cyclic globally normalized Markov random fields with proper anchors are legitimate probability models; directed auxiliary-latent models and cycles of inference updates are legitimate as well. Sections 27.1 through 27.2 isolate the failed architecture, Section 27.3 separates it from a distinct Bayesian-renormalization construction, and Sections 27.4 and 27.5 prove that an anchored Gaussian star supplies a unique, geometrically attracting coordinate-ascent fixed point. Section 27.6 finally shows that nontrivial graph-link holonomy can remove the kernel of this reciprocal pair, while making no claim about cyclic models in general.

27.1 The flat unanchored reciprocal Gaussian fold is singular

Consider two latents $u, v \in \mathbb{R}^K$ joined by distinct parallel edge copies $e : v \to u$ and $f : u \to v$. This is the minimal reciprocal fold: v supplies a penalty for u, and u supplies one for v. The two residuals

contribute the assembled precision

$$J = E_e^\top R_e^{-1} E_e + E_f^\top R_f^{-1} E_f, \qquad E_e = \begin{pmatrix} I & -\Theta_e \end{pmatrix}, \quad E_f = \begin{pmatrix} -\Theta_f & I \end{pmatrix}, \tag{27.1}$$

with $R_e, R_f \succ 0$ and $\Theta_e, \Theta_f \in \mathrm{GL}^+(K)$. Write $H = \Theta_f \Theta_e$ for the graph holonomy of the parallel-edge two-cycle based at v. The typed inverses $\Theta_{\bar{e}} = \Theta_e^{-1}$ and $\Theta_{\bar{f}} = \Theta_f^{-1}$ exist but do not appear in this loop.

Proposition 27.1 (Kernel of a reciprocal pair). For every pair of positive definite link covariances,

$$\ker J = \{(\Theta_e v, v) : v \in \ker(H - I)\}, \qquad \dim \ker J = \dim \ker(H - I). \tag{27.2}$$

The kernel does not depend on R_e or R_f: its dimension is the multiplicity of the eigenvalue one in the loop holonomy. [ESTABLISHED]

Proof. J is a sum of two positive semidefinite matrices, so $w \in \ker J$ if and only if w lies in the kernel of each summand. For positive definite R, the form $w \mapsto w^\top E^\top R^{-1} E w$ vanishes exactly when $Ew = 0$, so $\ker J = \ker E_e \cap \ker E_f$. Writing $w = (u, v)$, the first condition is $u = \Theta_e v$ and the second is $v = \Theta_f u$; substituting gives $v = \Theta_f \Theta_e v = Hv$, and conversely any v fixed by H produces a solution with $u = \Theta_e v$. The map $v \mapsto (\Theta_e v, v)$ is injective, so the dimensions agree.

Corollary 27.2 (A flat parallel-edge fold is singular). If the two distinct edge copies give the same flat endpoint comparison, equivalently $\Theta_f = \Theta_e^{-1}$, then $H = I$ and

$$\ker J = \{(\Theta_e v, v) : v \in \mathbb{R}^K\}, \qquad \dim \ker J = K. \tag{27.3}$$

The assembled precision is singular for every positive definite pair of link covariances, and its kernel is exactly the set of transport-consistent configurations.[ESTABLISHED]

Boundary of the no-go. Corollary 27.2 is not a theorem against cycles. If $P_u, P_v \succ 0$, then

$$J_{\mathrm{anch}} = J + \mathrm{diag}(P_u, P_v) \succ 0, \tag{27.4}$$

because a positive definite matrix added to a positive semidefinite matrix is positive definite. Hence

$$p(u, v) = Z^{-1} \exp\left[-\tfrac{1}{2}(u, v)^\top J_{\mathrm{anch}}(u, v) + h^\top (u, v)\right]$$

is a legitimate globally normalized Gaussian model built from the reciprocal parallel-edge energy, with finite $Z = (2\pi)^K (\det J_{\mathrm{anch}})^{-1/2} \exp(\tfrac{1}{2} h^\top J_{\mathrm{anch}}^{-1} h)$. The same proof covers a finite cycle with any number of vertices: its Gaussian edge energies sum to a positive semidefinite precision, and an SPD block anchor at every vertex makes the total precision positive definite. Such a Markov random field loses the convenience of a product of locally normalized directed kernels, not existence as a probability law. Likewise, Proposition 27.1 shows that the unanchored parallel-edge pair is definite

when $1 \notin \operatorname{spec}(H)$. The obstruction is therefore exactly the conjunction of the reciprocal-pair energy, no SPD anchor, and flat parallel-edge holonomy $H = I$. [ESTABLISHED]

Within that scope the direction of the degeneracy is important. A Gaussian energy whose precision has a K-dimensional kernel does not define a normalized Lebesgue density, so the exact identity of Chapter 5 has no joint to which it can apply. The flat direction consists precisely of those configurations in which u and v agree under transport. The unanchored flat pair charges exactly nothing there, so it supplies no coercivity along its intended agreement direction.

There are two independent ways to restore definiteness. Equation (27.4) adds declared coercivity and remains valid under flat parallel-edge holonomy. Alternatively, (27.2) makes the unanchored pair positive definite exactly when 1 is not an eigenvalue of H, which requires abandoning the Regime-I coboundary and declaring at least one of the two parallel edge copies independently. That route leaves the fixed-K flat interaction family used by the standard aggregation flow: an internal twisted cycle leaves a self residue, and the cut sector needs the enlarged nonflat rule left open in Chapter 25. This is a limitation of that flow, not a no-go for globally normalized cyclic models. [ESTABLISHED]

An objective-only reading fails for a different reason. If the reciprocal energy is appended outside the declared joint, the evidence $\log p_\theta(o \mid X)$ is determined by the factors that are in the joint and is unaffected, so the appended term has nothing to bound. If instead the energy is included in a globally normalized Markov random field, it is part of a legitimate joint whenever its global partition function is finite, but that partition function must be retained. Thus there are three distinct cases: the flat unanchored energy is improper; the properly anchored or non-flat energy with its global normalizer is a valid undirected model; and the energy appended while its normalizer is omitted is not an evidence bound. Chapter 5 applies to the second case and not to the other two. [ESTABLISHED]

27.2 A proper anchor restores definiteness and makes the global normalizer essential

Adding a proper prior to each folded latent replaces J by $J + \operatorname{diag}(P_0, P_0)$ with $P_0 \succ 0$. This restores definiteness, since the added term is positive definite and the sum of a positive semidefinite and a positive definite matrix is positive definite. What it does not restore is the ability to treat the normalization as a constant.

For a Gaussian in information form the log normalizer is

$$\mathsf{A}(h, J) = \tfrac{1}{2} h^\top J^{-1} h - \tfrac{1}{2} \log\det J + \tfrac{n}{2} \log 2\pi, \tag{27.5}$$

so any dependence of J on the transports passes directly into A. In (27.1) the transports appear inside E_e and E_f, and the question is whether their contribution to $\det J$ cancels. It does not.

Proposition 27.3 (The global normalizer depends on the model graph link). Take $K = 1$, unit link variances, flat parallel-edge values $\Theta_e = a$ and $\Theta_f = a^{-1}$ with $a > 0$, and a coordinate-isotropic anchor precision $p_0 > 0$. Then

$$\det\big(J + p_0 I\big) = p_0^2 + p_0\big(a + a^{-1}\big)^2, \tag{27.6}$$

which is a nonconstant function of a, uniquely minimized on $\mathrm{GL}^+(1) = (0, \infty)$ at $a = 1$, with value $p_0^2 + 4p_0$.[ESTABLISHED]

Proof. With $E_e = (1, -a)$ and $E_f = (-a^{-1}, 1)$, the two rank-one contributions sum to

$$J = \begin{pmatrix} 1 + a^{-2} & -(a + a^{-1}) \\ -(a + a^{-1}) & 1 + a^2 \end{pmatrix},$$

whose determinant is $(1+a^{-2})(1+a^2) - (a+a^{-1})^2 = (2+a^2+a^{-2}) - (a^2+2+a^{-2}) = 0$, confirming Corollary 27.2 in this case. Adding $p_0 I$ gives $\det(J + p_0 I) = p_0^2 + p_0\big[(1 + a^{-2}) + (1 + a^2)\big] = p_0^2 + p_0(2 + a^2 + a^{-2})$, and $2 + a^2 + a^{-2} = (a + a^{-1})^2$. Since $a + a^{-1} \geq 2$ for $a > 0$, with equality only at $a = 1$, the stated minimizer is unique.

The witness settles nonconstancy, because variation in one admissible instance refutes a constant-normalizer claim. For zero information vector, write

$$D(a) = p_0^2 + p_0(a + a^{-1})^2, \qquad \mathsf{A}(a) = -\tfrac{1}{2}\log D(a) + \mathrm{const.}$$

Then

$$\mathsf{A}'(1) = 0, \qquad \mathsf{A}''(1) = -\frac{4}{p_0 + 4} < 0. \tag{27.7}$$

Thus the identity is a strict *maximum* of the isolated log-normalizer contribution, not a minimum. Minimizing the corresponding isolated contribution to the negative log density pushes away from $a = 1$; the quadratic and information-vector terms also vary with a, so no universal direction follows for the complete objective. What is universal in this witness is only that the global term $-\frac{1}{2}\log\det(J + p_0 I)$ cannot be absorbed into per-agent constants or dropped from a normalized joint. [ESTABLISHED]

This variation is a change of model, not a pure coordinate change. It holds the coordinate matrices $p_0 I$, $R_e = R_f = I$, and the reference Lebesgue coordinates fixed while varying a. Under a genuine reframing, all covariances, precisions, information vectors, and the reference-density Jacobian transform together, and Proposition 4.9 leaves the probability measure and evidence invariant. Proposition 27.3 therefore proves transport-parameter dependence of a declared anchored model; it does not prove a force along a gauge orbit. [ESTABLISHED]

27.3 Bayesian posterior flow is a different construction

Bayesian renormalization, as formulated by Berman et al. [13], declares a prior and a likelihood and studies posterior and model-space descriptions indexed by an information-theoretic scale. A posterior is the output of that declared Bayesian model; the construction does not, merely by allowing the posterior to flow, make a generative kernel read its own posterior. The typing prohibition of Chapter 4 therefore does not refute Bayesian renormalization. [ESTABLISHED]

The two frameworks nevertheless cannot be identified without an additional construction. If one takes a posterior from one scale and inserts it as the generative top factor at another, then one has declared either a new model at each scale or an explicit transition rule between models. That can be legitimate, but the evidence and ELBO at different scales then refer to different declared joints unless a common target and an exact comparison map are supplied. Proposition 4.3 gives the precise warning: a family of exact pointwise identities does not itself designate one member or one evidence, and comparing two bounds certifies an improvement in one fixed quantity only on an evidence level set. This is a condition on a proposed composition with the present model, not a criticism of a posterior flow considered on its own. [ESTABLISHED]

The resulting boundary is narrower than a prohibition on comparing different latent inventories. If two normalized joints are declared independently, with no evidence-preserving channel between them, their optimized ELBO difference mixes an evidence difference with a tightness difference and supplies no universal order. By contrast, a parameter-independent probability kernel $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ satisfying $K(x, \mathsf{Y}) = 1$ for every x, applied to both the posterior and recognition law through the joint channel $(o, x) \mapsto \delta_o \otimes K(x, \cdot)$, preserves the observation evidence and orders the bounds by data processing. The normalization equation is the hypothesis that makes the pushforward preserve total mass; a sub-Markov kernel is outside this statement. The arbitrary-model caveat applies only in the absence of that relation. [ESTABLISHED]

The asymptotic covariance $C_T = T^{-1}I^{-1} + o(T^{-1})$ is a $T \to \infty$ statement and supplies no bound, limit, or monotonicity as $T \downarrow 0$. Properness at zero information depends on the declared prior and normalizer; with a proper prior, the no-data posterior is that prior. Hence this expansion supports no improper-endpoint claim about Bayesian renormalization. [NOT-CLAIMED]

Nor does a verbal contrast between “posterior widening” and “precision addition” establish that the two scale directions are opposites. The first depends on the declared Bayesian path, and the second is an unrescaled Gaussian aggregation identity. Relating them would require an explicit map between their scale parameters and an operator comparison in a common theory space; none is supplied here. [NOT-CLAIMED]

27.4 What survives

Two constructions remain, and the second is the interesting one.

The first is a declared top prior on a truncated tower. The tower is finite, nothing is folded, and the top level carries a prior that is part of the declared parameters. This is fully typed, it satisfies the hypotheses of Chapter 5 without amendment, and it costs the closure ambition entirely: the top is given, not constituted.[DEFINITION]

The second replaces the fold by an apex latent inside the joint. Let $b \in \mathbb{R}^K$ carry a declared proper prior $\mathcal{N}(0, P_0^{-1})$ with $P_0 \succ 0$, and let the constituents be conditionally independent given it,

$$b \sim \mathcal{N}(0, P_0^{-1}), \qquad y_i \mid b \sim \mathcal{N}(\Theta_i b, R_i), \qquad i = 1, \dots, n. \tag{27.8}$$

The scale graph is a star, hence a tree, so no cycle can form and Proposition 27.1 has nothing to act on. Three properties make this the right replacement rather than merely an admissible one.

Proposition 27.4 (Definiteness of the anchored star). The assembled precision $J_\star$ of (27.8) is positive definite for every choice of $\Theta_i \in \mathrm{GL}^+(K)$ and $R_i \succ 0$.[ESTABLISHED]

Proof. The quadratic form is $b^\top P_0 b + \sum_i (y_i - \Theta_i b)^\top R_i^{-1} (y_i - \Theta_i b)$, a sum of positive semidefinite terms. It vanishes only if $b^\top P_0 b = 0$ and each residual vanishes; the first forces $b = 0$ because $P_0 \succ 0$, and then $y_i = \Theta_i b = 0$. The kernel is trivial.

Proposition 27.5 (The exact mean-field coordinate is precision addition). Under a mean-field recognition family factorizing over b and the constituents, the optimal factor for b at fixed remaining factors is Gaussian with precision

$$P_b = P_0 + \sum_{i=1}^{n} \Theta_i^\top R_i^{-1} \Theta_i, \tag{27.9}$$

independent of the remaining factors, and with mean $P_b^{-1} \sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$.[ESTABLISHED]

Proof. The coordinate-ascent optimum satisfies $\log q^*(b) = \mathbb{E}_{-b}[\log p(b, y)] + \text{const}$. Only terms quadratic in b contribute to the precision, and taking expectations over the other factors leaves them unchanged, so the coefficient of $-\frac{1}{2} b^\top (\cdot) b$ is the bb block of $J_\star$, which is $P_0 + \sum_i \Theta_i^\top R_i^{-1} \Theta_i$. The linear coefficient is $\sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$.

Equation (27.9) is the transported precision-pooling rule, obtained here as an exact consequence of the joint rather than posited as an aggregation convention. It is also the sufficiency statement of Chapter 10 read in the recognition direction: the statistic that the constituents supply to the apex is $\sum_i \Theta_i^\top R_i^{-1} y_i$, and (27.9) is the precision that statistic carries.

Proposition 27.6 (The declared prior supplies exactly the missing coercivity). Let $w = (v, \Theta_1 v, \dots, \Theta_n v)$ be a transport-consistent configuration of (27.8). Then

$$w^\top J_\star w = v^\top P_0 v \tag{27.10}$$

exactly, for every $v \in \mathbb{R}^K$ and every choice of transports and link covariances.[ESTABLISHED]

Proof. Every constituent residual $y_i - \Theta_i b$ vanishes on such a configuration, so the only surviving term of the quadratic form is $v^\top P_0 v$.

The three propositions together locate precisely what the flat unanchored pair lacks and what the star supplies. By Corollary 27.2 the pair charges nothing on the transport-consistent ray; by (27.10) the star charges exactly $v^\top P_0 v$ there, no more and no less. The deterministic check `CHK-OBS-STAR-FOLD-NEW-PROTOCOL` constructs star and flat-pair instances and verifies the precision-addition identity and coercivity contrast under its declared protocol. The propositions, not the computation, establish the claim. [NUMERICAL]

That a prior must be declared somewhere is not a shortcoming of this particular construction, and it is worth recording as a general fact rather than as a concession.

Proposition 27.7 (A declared root is unavoidable). In any finite directed acyclic generative model there is at least one latent with no parents, whose marginal law is therefore part of the declared parameters. Hence no finite acyclic construction can have every prior constituted by other latents.[ESTABLISHED]

Proof. A finite directed acyclic graph admits a topological order, and its first element has no incoming edges. Its factor in the joint is unconditional, so it is declared.

The ambition behind the fold was to eliminate the declared root. Proposition 27.7 says that within finite acyclic models the ambition is unachievable for structural reasons rather than for want of ingenuity, and Corollary 27.2 says that the particular flat unanchored reciprocal-pair escape is improper. Properly anchored globally normalized cyclic models remain available, but their anchors and partition functions are declarations rather than derivations from the population.

27.5 The participatory content survives in the inference

What the fold was reaching for is nonetheless available, and it is available exactly where Propositions 27.5 and 27.7 jointly place it: in the inference rather than in the model.

At a variational fixed point of (27.8), the apex factor has the mean $P_b^{-1} \sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$ of Proposition 27.5, which is a transported pooling of every constituent's current belief; and each constituent's effective prior, in its own coordinate update, is that apex belief transported back down. Each agent's effective coordinate input is therefore constituted by the apex update, and the apex input is constituted by the agents' updates. This is a loop in an inference algorithm, not a cycle in the factorization of the model. Every step is a coordinate update of a recognition law against a joint that is fixed, acyclic, and normalized, so nothing reads a posterior into a generative kernel and the prohibition of Chapter 4 is respected throughout.[ESTABLISHED]

The distinction being drawn is not verbal. A cycle in the model concerns the factorization or global potential of a joint; Corollary 27.2 rejects only one flat unanchored reciprocal Gaussian potential. A

cycle in the inference is a property of an iterative scheme applied to a fixed joint. What the star gives up is the ambition that the apex prior be derived; what it keeps is a bidirectional dependence of exact mean-field coordinates during inference.

Theorem 27.8 (Exact Gaussian-star fixed point and contraction). Allow arbitrary fixed information vectors $r_b, r_1, \dots, r_n$ by adding $r_b^\top b + \sum_i r_i^\top y_i$ to the log density of (27.8); the displayed centered star is the case $r_b = r_i = 0$. Define

$$B := \sum_{i=1}^{n} \Theta_i^\top R_i^{-1} \Theta_i, \qquad P_b := P_0 + B, \qquad M := P_b^{-1} B. \tag{27.11}$$

Consider the exact block schedule that updates every constituent factor from the current apex factor, in parallel or in any order, and then updates the apex factor. The factor covariances become R_i and P_b^{-1}, respectively, on their first updates, and the means obey

$$m_i^{(t+1)} = \Theta_i m_b^{(t)} + R_i r_i, \tag{27.12}$$

$$m_b^{(t+1)} = P_b^{-1} \left(r_b + \sum_i \Theta_i^\top R_i^{-1} m_i^{(t+1)} \right) = c + M m_b^{(t)}, \qquad c = P_b^{-1} \left(r_b + \sum_i \Theta_i^\top r_i \right). \tag{27.13}$$

There is a unique fixed point,

$$m_b^\star = P_0^{-1} \left(r_b + \sum_i \Theta_i^\top r_i \right), \qquad m_i^\star = \Theta_i m_b^\star + R_i r_i, \tag{27.14}$$

and, in the norm $\|x\|_{P_b} = (x^\top P_b x)^{1/2}$,

$$\|m_b^{(t+1)} - m_b^\star\|_{P_b} \le \rho \|m_b^{(t)} - m_b^\star\|_{P_b}, \qquad \rho = \lambda_{\max}\left(P_b^{-1/2} B P_b^{-1/2} \right) < 1. \tag{27.15}$$

Thus the exact schedule converges geometrically from every initial collection of means. [ESTABLISHED]

Proof. Taking the expectation of the joint log density over all factors except q_i leaves a quadratic in y_i with precision R_i^{-1} and information vector $R_i^{-1}\Theta_i m_b^{(t)} + r_i$, which gives (27.12). Proposition 27.5 with the added information vector gives the first expression in (27.13); substitution gives the affine recursion. Since

$$I - M = P_b^{-1}(P_b - B) = P_b^{-1} P_0,$$

the fixed-point equation $(I - M)m_b^\star = c$ has the unique solution (27.14), because $P_0 \succ 0$.

The matrix M is similar to the symmetric matrix

$$S := P_b^{-1/2} B P_b^{-1/2} = I - P_b^{-1/2} P_0 P_b^{-1/2}.$$

Both $B \succeq 0$ and $P_0 \succ 0$, so every eigenvalue of S lies in $[0, 1)$. If $e_t = m_b^{(t)} - m_b^\star$, then

$$\|e_{t+1}\|_{P_b} = \|S P_b^{1/2} e_t\|_2 \le \lambda_{\max}(S) \|e_t\|_{P_b},$$

which proves (27.15). After the first updates, each factor covariance already equals its fixed-point covariance. Consequently $D_{\mathrm{KL}}(q_b^{(t)} \| q_b^\star) = \frac{1}{2}\|e_t\|_{P_b}^2$ contracts by at least ρ^2 per full sweep, and the sum of constituent divergences after their next update is $\frac{1}{2} e_t^\top B e_t \le \frac{1}{2} e_t^\top P_b e_t$. This also proves convergence of the entire product factor in relative entropy. □

The strict anchor $P_0 \succ 0$ is essential. If it is weakened to $P_0 \succeq 0$, the symmetric matrix S can have eigenvalue one, and neither uniqueness nor geometric contraction follows. When $B = 0$, $\rho = 0$ and the apex reaches its fixed point in one update. Delayed, noisy, asynchronous, or inexact updates are not covered by Theorem 27.8 and remain a separate convergence problem. [OPEN]

If a one-parameter family is genuinely wanted, for a construction that interpolates between an uninformative and a data-driven description, a legitimate one is available and does not repeat the failures of Section 27.3. Declare a tempering path $p_\beta \propto p(Y)p(o \mid Y)^\beta$ for $\beta \in [0,1]$, whose endpoints are the prior and the posterior of one fixed model, both proper, and whose evidence satisfies the exact path-sampling identity

$$\log p(o) = \int_0^1 \mathbb{E}_{p_\beta}\big[\log p(o \mid Y)\big]\,\mathrm{d}\beta \tag{27.16}$$

[35]. The family is indexed by a declared temperature and built from one declared prior–likelihood pair; its normalizing constant is retained at every β, both endpoints are proper when the prior and posterior are proper, and (27.16) is an identity rather than a bound. The identity itself is established and cited; the construction is not developed here, and it is not identified with the information-scale construction of Section 27.3.[ESTABLISHED]

27.6 In the reciprocal-pair architecture, the kernel is a flatness artifact

Proposition 27.1 measures the obstruction of Section 27.1 by one number, $\dim\ker(H - I)$, and Corollary 27.2 evaluates that number under flat parallel-edge holonomy, where it is K. Reading the corollary as a statement about cyclicity is a misreading, and the correct reading inverts the relation between the two halves of this document.

Corollary 27.9 (Nontrivial holonomy shrinks the reciprocal-pair kernel). Under the hypotheses of Proposition 27.1, with $H = \Theta_f \Theta_e$ the holonomy of the parallel-edge two-cycle,

$$\begin{aligned} \dim\ker J = K &\iff H = I, \\ H \neq I &\implies \dim\ker J \le K - 1, \\ \ker J = \{0\} &\iff 1 \notin \operatorname{spec}(H). \end{aligned} \tag{27.17}$$

The set of $H \in \mathrm{GL}^+(K)$ at which the last condition fails is the intersection of $\mathrm{GL}^+(K)$ with the zero set of $\det(H - I)$, a polynomial that is not identically zero since $H = 2I$ gives $\det(H - I) = 1$. It is therefore a proper algebraic subset, relatively closed in $\mathrm{GL}^+(K)$, with dense complement there,

and has Lebesgue measure zero in the ambient matrix space. The flat case $H = I$ is the maximally degenerate point of this reciprocal-pair family, while a generic H gives no kernel.[ESTABLISHED]

Proof. All three statements are read off (27.2), which gives $\dim\ker J = \dim\ker(H - I)$ with no dependence on the link covariances. The kernel of $H - I$ is all of $\mathbb{R}^K$ exactly when $H - I = 0$; if $H \neq I$ then $H - I$ has rank at least one, so its kernel has dimension at most $K - 1$; and the kernel is trivial exactly when 1 is not an eigenvalue of H, that is when H fixes no nonzero vector. For the genericity statement, $\det(H - I)$ is a polynomial in the entries of H, and a nonzero polynomial has a closed, nowhere-dense, Lebesgue-null zero set in the ambient Euclidean matrix space; intersecting with $\mathrm{GL}^+(K)$ gives the stated relative properties.

The same computation controls the normalization, and in a form that makes the transport dependence of Section 27.2 exact rather than exhibited at one witness.

Proposition 27.10 (Graph-link dependence of the determinant factors through holonomy). For the reciprocal pair of (27.1),

$$\det J = \frac{\big(\det(I - H)\big)^2}{\det R_e \det R_f}. \tag{27.18}$$

The graph links enter the determinant only through the holonomy of the loop, and J is singular exactly when $\det(I - H) = 0$, in agreement with Proposition 27.1.[ESTABLISHED]

Proof. Stack the two residual maps as $\widetilde{E} = \left(\begin{smallmatrix} I & -\Theta_e \\ -\Theta_f & I \end{smallmatrix}\right)$ and write $\widetilde{R} = \mathrm{diag}(R_e, R_f)$, so that $J = \widetilde{E}^\top \widetilde{R}^{-1} \widetilde{E}$ by (27.1). Hence $\det J = (\det \widetilde{E})^2/(\det R_e \det R_f)$, and the Schur formula gives $\det \widetilde{E} = \det(I - \Theta_f\Theta_e) = \det(I - H)$.

What holonomy removes. Exactly one defect of this exact architecture: the unanchored pair's kernel. When $1 \notin \mathrm{spec}(H)$ the assembled precision is positive definite, so the reciprocal energy with its global normalizer defines a Gaussian law and the failure recorded in Corollary 27.2 does not occur. By (27.2), the transport-consistent zero modes are $\{(\Theta_e v, v) : v \in \ker(H - I)\}$, which collapse to the origin. Cyclicity did not produce that kernel; the flat loop product did. This statement concerns the two-residual precision (27.1) and is not generalized to other cyclic potentials. [ESTABLISHED]

What holonomy does not remove. First, it does not turn the reciprocal energy into a pair of locally normalized directed kernels. It can instead define an undirected Gaussian potential, and then its *global* partition function is part of the model. Section 27.2 exhibits its parameter dependence in an anchored scalar model; Proposition 27.10 gives the unanchored non-flat determinant exactly. By (27.5) the latter contains

$$-\tfrac{1}{2}\log\det J = -\log\big|\det(I - H)\big| + \tfrac{1}{2}\log\det R_e + \tfrac{1}{2}\log\det R_f, \tag{27.19}$$

a nonconstant model term that cannot be absorbed into per-agent constants. It diverges as H approaches the singular flat locus, exactly where the unanchored law ceases to exist. Retaining this

term gives a legitimate normalized model; omitting it changes the objective. Proposition 27.3 and Equation (27.19) both vary declared model transports while holding other coordinate parameters fixed, so neither is a statement about motion along a pure gauge orbit. [ESTABLISHED]

Second, nontrivial H does not eliminate declarations. A loop with $H \neq I$ requires at least one link declared independently of the pointwise frames, because a Regime-I loop product telescopes to the identity. Replacing a declared prior by a declared edge link relocates the declaration. [ESTABLISHED]

A conditional incompatibility with the standard flat flow. If one insists simultaneously on an *unanchored* reciprocal pair and on the fixed-K flat aggregation flow, the two requirements conflict. The standard flow requires trivial graph-link holonomy on every cluster and, for exact cut closure, on every union of two clusters; Corollary 27.9 says the unanchored pair is definite only when $1 \notin \operatorname{spec}(H)$. In that scoped architecture,

$$\begin{aligned} \text{standard coarsening needs} \quad & \operatorname{Hol}(\gamma) = e_G \quad \text{on the cycles in its declared domain,} \\ \text{the unanchored reciprocal pair needs} \quad & 1 \notin \operatorname{spec}(H). \end{aligned} \tag{27.20}$$

An SPD anchor removes the right-hand demand and therefore avoids this incompatibility, at the cost of explicitly declaring the coercive top term. The star avoids it for the same reason and also has no cycles. Equation (27.20) is not a constraint on all cyclic models or on the general belief-law and model-law theory. [ESTABLISHED]

Placing the reciprocal cycle permanently between clusters does not evade the scoped conflict along a finite flow to the coarsest resolution. Admissibility is inherited under refinement, while each nontrivial blocking strictly decreases the number of clusters. At a level with two nonempty clusters, exact fixed-K cut closure is the condition $\Delta = 0$ in (25.89); after the two clusters have been trivialized, that equality forces their cut twists to coincide and hence makes their union, the whole population, trivializing. With one cluster, whole-population triviality is already the cluster condition. Thus a non-flat reciprocal cycle can remain outside the standard flow only by stopping at a finite maximal resolution or by changing the coarse family. [ESTABLISHED]

Remaining non-flat compression problem. The rectangular endpoint-map or cellular-sheaf category constructed in Chapter 25 is already gauge covariant and closed under repeated linear-quadratic blocking. [ESTABLISHED]

What remains open is compression of that richer cut datum to one invertible fixed-rank graph link, together with a normalized probability/RG lift that retains the global normalizer. A solution must preserve closure under another merge; it is not an open question about anchored cyclic Gaussian models, which (27.4) settles positively. [OPEN]

Both new statements are proved above. The deterministic replacements `CHK-OBS-HOLONOMY-DET-KERNEL` and `CHK-OBS-NORMALIZER-WITNESS` construct their transports and covariances explicitly, check the determinant identity and prescribed fixed-space dimensions, and include a normalized

anchored control. Rank tolerance remains part of the numerical protocol; the proofs settle the statements. [NUMERICAL]

Chapter 2 distinguishes three holonomies, and everything in this section uses only graph-link holonomy: the ordered product of a declared comparison assignment on a population graph. Neither side of (27.20) refers to the base holonomy of a smooth connection along curves of $\mathcal{C}$. No conclusion about bundle topology or base curvature follows from the reciprocal-pair calculation. [ESTABLISHED]

Chapter 28

Empirical Commitments and Interpretation

This chapter proves no mathematical result. It distinguishes declared readings, hypotheses consumed by the mathematics, and extensions that would require new evidence. A reading is declared. [DEFINITION]

The sole interpretive restriction used later is observational closure. [HYPOTHESIS]

Unresolved empirical or interpretive consequences remain open. [OPEN]

Availability is not support, and no preference below changes a theorem proved in the preceding chapters. [NOT-CLAIMED]

28.1 What the base is, and what it is not

The general construction has distinct belief and model associated bundles, $\mathcal{E}_b$ and $\mathcal{E}_m$, over the contextual base $\mathcal{C}$. An agent carries sections such as

$$q_i^b \in \Gamma\left(\mathcal{C}_i, \mathcal{E}_b|_{\mathcal{C}_i}\right), \qquad q_i^m \in \Gamma\left(\mathcal{C}_i, \mathcal{E}_m|_{\mathcal{C}_i}\right), \tag{28.1}$$

whose values are declared probability laws in their respective fibers. The cross maps $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ and $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$ do not identify those bundles, and neither is assumed inverse or connection-parallel. The base indexes the sections; the finite design is a declared subset, not a random sample from a law on $\mathcal{C}$. No expectation over contexts is used. The projective-limit chapter separately identifies what is established for cylinder laws and what remains open for measures supported on smooth section spaces. [DEFINITION]

The informational pullbacks of Chapter 8 can give this fixed base an agent-, section-, and connection-relative positive-semidefinite tensor. Passive gauge invariance of that tensor does not make it independent of the chosen connection, and degeneracy can leave contextual directions unmeasured. [ESTABLISHED]

This construction supplies no Lorentzian signature, causal cones, spacetime measure, gravitational field equation, or identification of $\mathcal{C}$ with physical spacetime. [NOT-CLAIMED]

The word “timeless” has an equally narrow meaning in Chapter 9: an oriented inference history retains order while quotienting away its arbitrary orientation-preserving parameterization, and Fisher length supplies a nonnegative information-geometric duration. [DEFINITION]

It does not assert that physical time is unreal or emergent, nor identify Fisher duration with relativistic proper time, thermal time, or a quantum clock. [NOT-CLAIMED]

The Neo-Kantian reading concerns the agent-relative conditions under which a record becomes inferentially meaningful: local frames, model evaluations, and admissible observation interfaces encoded by access maps. Their role is constitutive within the model but revisable across learning or theory change. This is an interpretation of the formal architecture, not a consequence of its probability theorems. [DEFINITION]

The reading belongs to a historical and interpretive lineage running from Kant’s account of constitutive conditions through Reichenbach’s relativized a priori, Cassirer’s functional account of scientific concepts, and Friedman’s revisable constitutive frameworks [22, 32, 47, 76]. [DEFINITION]

Those citations locate the lineage only. They neither prove the probability construction nor establish this interpretation of its formal objects. [NOT-CLAIMED]

The fixed contextual base $\mathcal{C}$, common principal bundle $\mathscr{P}_G$, finite design D, and declared measurable spaces remain analyst-declared coordination scaffolds. They make the agent-relative interfaces comparable but are not constructed by the local inference laws. The architecture is local-first, not background-free. [DEFINITION]

A noumenal reading remains separately compatible with the typing: the base and bundles may be read as structure approached only through agent-side representations. The absence of a probability law on the base discriminates neither that reading nor the Neo-Kantian one. Noumenal objects are not ordinary uncertain latents, while declared scaffolding is likewise not an inferential target. [DEFINITION]

No equation selects between the two. [NOT-CLAIMED]

28.2 When a noumenal reading earns its keep

This chapter adopts an explicit idle-wheel criterion: a posit with no trace in any declared observable is removed by parsimony. [DEFINITION]

No result in this chapter derives the declared rule. [NOT-CLAIMED]

For attributional scope in this chapter, “constructive empiricism” denotes the position combining a literal reading of scientific theories with the claim that accepting a theory involves belief only in its empirical adequacy [83, 11–12]. This is a declared use of the cited position, not a theorem of the manuscript. [DEFINITION]

The manuscript does not claim that constructive empiricism entails its removal rule. [NOT-CLAIMED]

The criterion keeps three kinds of typed geometric or comparison data separate. Discrete population links are Θ^b_{ij} and Θ^m_{ij}, connection parallel transports along base curves are Ω_γ and $\widetilde{\Omega}_\gamma$, and the cross-bundle maps are Φ and $\widetilde{\Phi}$. An edge-labeled graph loop $\gamma = (e_0, \dots, e_{r-1})$ determines, for example,

$$H^b_\gamma = \Theta^b_{e_0}\Theta^b_{e_1}\cdots\Theta^b_{e_{r-1}}, \qquad [H^b_\gamma]_{\mathrm{conj}}, \tag{28.2}$$

and analogously in the model sector. [DEFINITION]

Under vertex reframing, the loop product is conjugated by the rechoice at its base vertex, so its conjugacy class is invariant. [ESTABLISHED]

The graph-link observable is not claimed to be evidence of base curvature or bundle topology. [NOT-CLAIMED]

In the flat graph-link specialization every loop product telescopes. [ESTABLISHED]

Under the declared idle-wheel criterion, the resulting identity observable gives the noumenal reading no additional work. A declared nontrivial graph-link holonomy supplies a model-internal referent, but only within the interaction complex. [DEFINITION]

Identifying Θ_e with Ω_{γ_e} requires an embedded interaction complex, assigned piecewise-smooth base curves, and equality with the path-ordered exponential of a specified connection. The explicit cover construction supplies one such assignment; a general realization or recovery theorem remains open. Even an exact assignment concerns connection holonomy, not topological nontriviality. [OPEN]

A specified connection can now be associated with a normalized population record through the construction in Theorem 14.7. Using that association as a reading of the formalism is a declared interpretation, not an identification of physical geometry. [DEFINITION]

A positive result requires a named principal bundle and connection, an assigned base loop, a gauge-invariant population-record statistic, a typed map from the connection data to its record law, and two admissible connection data sets for which every declared non-holonomy input is fixed, the represented loop holonomies have distinct conjugacy classes, and the statistic's induced distributions differ. A stronger sufficient formulation requires the controlled record-law map to factor nontrivially through the conjugacy class of the represented loop holonomy. The specified Gaussian record in Theorem 14.7 supplies a mathematical tuple; no empirical identification is supplied. Extending sensitivity to other record classes, or recovering a connection from observations, remains open. A theorem that every statistic in a declared admissible controlled class is insensitive to that holonomy class would close the route negatively. [OPEN]

Proposition 8.5 proves exact change formulas for the covariant Fisher and Amari–Chentsov pull-backs when a chosen connection changes. [ESTABLISHED]

Theorem 8.4 proves their invariance under passive reframing of a fixed connection–section pair. [ESTABLISHED]

The manuscript does not derive a canonical connection from either result. [NOT-CLAIMED]

The existence of a constructed holonomy-sensitive record does not establish a canonical observable or an empirical realization of that record. Neither is claimed here. [NOT-CLAIMED]

28.3 Which realism this is

The desktop analogy belongs to the interface theory of perception, whose evolutionary argument permits an arbitrary measurable relation between representation and world [38, 39]. The present formalism is narrower: agents are sections of associated law bundles, and a global reference frame exists only under the corresponding principal-bundle trivialization hypothesis. Local frames can exist when a global one does not, and a pointwise graph coboundary does not provide a smooth Čech trivialization. [DEFINITION]

What survives a change of frame is structural data, schematically

$$\mathcal{I}_{\text{struct}} = \left\{\text{ELBO and KL values},\ \operatorname{rank} M,\ \operatorname{spec}(L,\Lambda),\ [h^{\omega_b}_{q_i^b}],\ [h^{\omega_m}_{q_i^m}],\ [H^b_\gamma]_{\text{conj}},\ [H^m_\gamma]_{\text{conj}}\right\}, \quad (28.3)$$

on the exact domains established earlier. This supports an epistemic structural-realist reading in Worrall's sense [89]. Here the brackets on the pullback tensors denote their passive-coordinate equivalence classes for the stated sections and fixed connections, not connection-independent canonical geometries. [DEFINITION]

Ontic structural realism in the eliminativist sense, in which structure is all that exists, is available but unsupported [52, 53]. [OPEN]

Supporting it would require an argument from formal unidentifiability to ontological elimination and discriminating evidence for that bridge. [OPEN]

In this chapter, "moderate structural realism" denotes Esfeld and Lam's position on which objects and relations occupy the same ontological footing and objects are characterized only by their relations [30]. This is a declared use of the cited position, not a theorem of the manuscript. [DEFINITION]

The manuscript proposes, but does not establish, that its invariance and unidentifiability results fit that moderate reading. [OPEN]

Establishing the fit would require a typed interpretation relating the formalism's objects and relations to the source's spacetime ontology and discriminating evidence for that interpretation. [OPEN]

The manuscript does not claim that either structural reading follows from its invariance results. [NOT-CLAIMED]

28.4 Two agreements, and the cost of conflating them

For a graph loop and a configuration of Gaussian means, the two relevant conditions are

$$H^b_\gamma = I \qquad \text{and} \qquad \mu_i = \Theta^b_e \mu_j \quad \text{on each edge copy } e : j \to i. \quad (28.4)$$

The first is frame agreement: the graph links admit a common trivialization on the cluster. The second is belief agreement: a particular configuration is link-consistent. [DEFINITION]

They are independent. A graph coboundary constrains links but leaves the means free. Conversely, under nontrivial holonomy, belief agreement can survive on the fixed subspace

$$\mu_{i_0} \in \ker(H_\gamma^b - I). \tag{28.5}$$

[ESTABLISHED]

The manuscript therefore avoids using “consensus” for both. Common graph trivialization is an admissibility condition for the declared fixed-rank coarse recursion, exact under positive-definite represented weights and sufficient with semidefinite weights. Belief agreement is a configuration and a null direction of the interaction energy. Aggregation never consults the realized means, so belief agreement does not select a partition. [DEFINITION]

Semidefinite null-direction exceptions belong to the typed effective-support problem rather than to an interpretive identification of the two agreements. [DEFINITION]

28.5 The participatory reading and its exact cost

Operational observational closure declares that every datum directly available to an interior agent is a clamped state or message of another interior or boundary environment node. [HYPOTHESIS]

The kernel-level equivalence is proved in Theorem 7.8. [ESTABLISHED]

Neither $\mathbb{P}_{\theta,V}$ nor $\mathbb{Q}_{V,o,X}$ is an agent, a mind, a population mind, a super-agent, or a universal observer. They are complete joint laws at the finite-population level: $\mathbb{P}_{\theta,V}$ is the constructed latent–record law and $\mathbb{Q}_{V,o,X}$ is the selected full latent coupling. Evaluating a block conditional does not by itself construct a meta-agent. A meta-agent exists only after its own coarse state, observation interface, evaluated generative mechanism, recognition law, and update rule have been separately constructed. [DEFINITION]

The interpretive hypothesis is the decision to call every such boundary node an agent. [HYPOTHESIS]

It does not eliminate the conditioning record, require the boundary node to be autonomous, exclude an unobserved common cause, prove convergence of an RG orbit, or identify a fixed object with a physical law. [NOT-CLAIMED]

Participation means reciprocal inferential dependence mediated by relational records inside one fixed normalized model. Those records may alter posterior and predictive states, but no live posterior is fed into a same-step generative factor. This is the fixed-joint fence of Requirement 4.2. [DEFINITION]

The obstruction chapter locates the participatory loop in inference rather than in a self-referential generative kernel. In the proper Gaussian-star model, the apex update pools the constituents’ beliefs and each constituent update receives the apex belief; the joint remains fixed, acyclic, and

normalized. Reading that reciprocal inferential dependence as the content of the construction resembles Wheeler's participatory proposal [88], but is neither derived from nor evidence for it. [DEFINITION]

Likewise, a relational inference history tracks a recognition-state orbit against the fixed normalized joint and conditioning record. [DEFINITION]

Quotienting its parameterization does not feed the orbit back into the generative kernel or make the joint self-referential. [NOT-CLAIMED]

Three commitments often hidden in that slogan must be separated. Operational observational closure is the declared model restriction above. [HYPOTHESIS]

Strong agent-only ontology additionally requires a criterion of agency, such as persistent state, action, a Markov blanket, or a local VFE, for every boundary node. The formalism supplies no unique such criterion. [NOT-CLAIMED]

The manuscript claims neither that participation entails consciousness nor that observation physically creates what exists. It also claims no ontological closure, background independence, or identification of variational free energy with a physical action principle. [NOT-CLAIMED]

Identifying a repaired limiting object with a physical law is the independent semantic claim E.1; it lacks a target system, validation criterion, probability-level closure, and discriminating evidence. [OPEN]

The formalism does not exclude an unobserved environment affecting all agents without becoming the argument of any agent's belief. [NOT-CLAIMED]

The present graph-holonomy reading and Gaussian RG also occupy different domains. The former becomes non-idle under nontrivial graph-link holonomy, whereas the standard fixed-rank iterated flow uses its declared trivializing domain. Rectangular endpoint-map closure resolves the linear-quadratic blocking problem, but compression to one invertible link, a normalized probability lift, effective-support typing across rank changes, and external scale selection remain unsettled. [OPEN]

A base-holonomy reading is a different extension and needs the operational bridge stated above. The formalism does not adjudicate between these readings. [NOT-CLAIMED]

28.6 What would make the interpretation testable

The current mathematics permits an internal admissibility check: compute fundamental graph-loop products and test whether a proposed cluster lies in the trivializing domain; with semidefinite weights, also test weighted invisibility on the represented support. [DEFINITION]

This checks declared model data. It is not empirical detection of principal-bundle topology or base curvature. [NOT-CLAIMED]

A cross-scale empirical test would require a preregistered target system, sampling units, noise model, and estimator for gauge-typed parameters with uncertainty. It would also require at least two admissible partitions, declared blocking and rescaling maps, gauge alignment, a held-out statistic, an acceptance tolerance, and Gaussian-graphical and spectral baselines. One would fit only at fine resolution, push the result through each blocking without coarse refitting, and compare the predicted held-out coarse statistic both with data and across admissible blockings. Failure would falsify that added cross-scale hypothesis, not the finite-dimensional theorems. [OPEN]

No such target, estimator, tolerance, baseline margin, attraction theorem, or scheme-robust invariant has been supplied. The present empirical status is therefore inconclusive. A viable interpretation must continue to distinguish graph-link holonomy, base-connection holonomy, bundle topology, and gauge redundancy, and must state which of them its evidence actually bears on.

Appendix A

Proofs for Finite-Depth Adaptive Formation

This appendix proves the estimates used in theorem 16.1. The carrier, normalized experiment, and declared kinetics are those of chapter 16. Throughout, the depth $L \geq 2$ is fixed. Constants denoted by c_L, C_L are positive and may change between displays; they depend on that depth and the fixed model data, but not on ϵ, A, T, or the descending strengths unless this dependence is displayed. All limits have $\epsilon > 0$ and $T > 0$. The results concern the stated family at one base point. [HYPOTHESIS]

The orthogonal decomposition is $\mathcal{N} \oplus \mathcal{H} \oplus \mathcal{K}$. On $M_p = \mathcal{N}_p \oplus \mathcal{H}$, write H_p for the Gaussian Schur precision, $k_p = \dim M_p$, and $d_p = k_0 - k_p$ for the number of eliminated continuous coordinates. The normalized loop vector is h. The entire assignment S has values j, k in the fixed finite set $\mathcal{J}$; its value evolves, whereas its coordinate set is retained at every level. Locally, $\pi_j(\vartheta)$ denotes its generative prior mass, $\pi^p_\vartheta(x, j)$ the restricted optimizer density, and $q^p_t(x, j)$ the actual joint recognition density, with respect to Lebesgue measure times counting measure. The physical posterior $\mathbf{\Pi}^p_\vartheta$ remains a different law. The independent equilibrium root factor is omitted from non-root integrals and appended unchanged. These are local notation conventions. [DEFINITION]

In the proofs, $D(\nu\|\mu) = D_{\mathrm{KL}}(\nu\|\mu)$ denotes relative entropy. The same symbol applied to a density means its law with the stated reference measure. [DEFINITION]

A.1 Exact marginals, rates, and parameter derivatives

Lemma A.1 (Weighted geometry and exact parent score). For the declared triangle-chain experiment, all marked marginal laws and inherited proposal rows are normalized and compose exactly. On the compact parameter interval $1 \leq \vartheta \leq 2.1$, the exact parent potentials and complete model covectors have the forms established below, including the eliminated Gaussian scalar and the source-prior score. Their derivatives obey uniform energy-weighted bounds. These statements are established for every fixed finite depth and all sufficiently small positive ϵ. [ESTABLISHED]

Proof. Let B_s be the residual rows of level s. Independence follows directly from their columns: a linear relation first vanishes on every independent link column, forcing all internal-contrast and bridge coefficients to vanish, and then vanishes on the model columns, forcing all belief–model discrepancy coefficients to vanish. The normalized loop rows are orthogonal to these rows. Choose

fixed orthonormal flag columns Q_s for $\mathcal{N}_{s-1} \cap \mathcal{N}_s^{\perp}$. The matrix $\mathsf{B} = B[\mathsf{Q}_1 \ \cdots \ \mathsf{Q}_L]$ is block lower triangular with invertible diagonal blocks, as is its inverse. If $D_J = \mathrm{diag}(\sqrt{J_s} I)$, every nonzero block (s, r) of $D_J \mathsf{B} D_J^{-1}$ is multiplied by $\sqrt{J_s/J_r} \le 1$. Its Frobenius norm and the corresponding norm of its inverse are therefore bounded by fixed graph-dependent constants. Consequently

$$c_L \sum_{s=1}^{L} J_s |w_s|^2 \le w^T H_{\mathcal{N}} w \le C_L \sum_{s=1}^{L} J_s |w_s|^2, \qquad H|_{M_0} = H_{\mathcal{N}} \oplus J_1 I_{\mathcal{H}}. \tag{A.1}$$

The Schur quadratic is the infimum of this quadratic over the eliminated coordinates. Minimizing both bounds proves their retained versions. In particular $H_p \succeq h_L I$ for a fixed $h_L > 0$, since $J_L = 1$. The normal part of H_p is bounded above by $C_L J_{p+1} I$, with $J_{L+1} = 0$; the loop part still has weight J_1.

For a cumulative hidden/retained split (z, x) of M_0, write $F_p = H_{EE}$ and $G_p = H_{ER}$. Completing the square gives the conditional-mean lift and its two identities

$$L_p x = (-F_p^{-1} G_p x, x), \qquad L_p^T H L_p = H_p, \qquad L_p^T H z = 0. \tag{A.2}$$

Here and below z in the last expression is embedded in the hidden subspace. The centered hidden Gaussian has covariance $(T/\vartheta) F_p^{-1}$, so its mean does not depend on ϑ. Let $\gamma_{p,\vartheta}$ denote this centered law. Exact integration of the original source factor gives

$$V_{p,j}(x, \vartheta) = \frac{\vartheta}{2} x^T H_p x + C_p^G(\vartheta) + T \psi_{p,j}(x, \vartheta), \tag{A.3}$$

$$\psi_{p,j}(x, \vartheta) = -\log \int e^{-\phi_j(L_p x + z)} \gamma_{p,\vartheta}(dz), \qquad 0 \le \psi_{p,j} \le B_0, \tag{A.4}$$

$$C_p^G(\vartheta) = -\frac{T d_p}{2} \log \frac{2\pi T}{\vartheta} + \frac{T}{2} \log \det F_p, \qquad (C_p^G)'(\vartheta) = \frac{T d_p}{2\vartheta}. \tag{A.5}$$

At $p = 0$ the integral is absent and $C_0^G = 0$. These formulas integrate over whole Euclidean fibers. Positivity and Gaussian domination justify iterated integration and differentiation on compact parameter intervals. In particular the positive finite normalizer is retained, and repeated integration agrees with direct integration.

The normalized non-root optimizer can also be written

$$\pi_\vartheta^p(x, j) = Z_p(\vartheta)^{-1} e^{-\psi_{p,j}(x,\vartheta)} G_\vartheta^p(x) \pi_j(\vartheta), \qquad G_\vartheta^p = N\big(0, (T/\vartheta) H_p^{-1}\big), \qquad e^{-B_0} \le Z_p \le 1. \tag{A.6}$$

The microscopic proposal probability for the complete assignment is

$$b_k(x, \vartheta) = \frac{\pi_k(\vartheta) e^{-\phi_k(x)}}{\sum_j \pi_j(\vartheta) e^{-\phi_j(x)}}. \tag{A.7}$$

It is the product of the normalized receiver rows at this fixed configuration. The exact inherited row is

$$R_{jk}^p(x, \vartheta) = \mathbb{E}_{\pi_\vartheta^0}[b_k(X, \vartheta) \mid P_p X = x, S = j]. \tag{A.8}$$

Every row sums to one, including the self entry, and $R^p_{jk} \geq e^{-B_0}\pi_k(\vartheta) > 0$. The conservative generator has diagonal $R^p_{jj} - 1$, rather than R^p_{jj}. Its full proposal clock has rate one and changes no continuous coordinate. The microscopic identity $\pi^0_\vartheta(X, j)b_k = \pi^0_{\vartheta,X}(X)b_j b_k$ is symmetric in j, k. Integrating over the same retained fiber proves

$$\pi^p_\vartheta(x, j)R^p_{jk}(x, \vartheta) = \pi^p_\vartheta(x, k)R^p_{kj}(x, \vartheta). \tag{A.9}$$

The conditional-expectation tower proves composition of these rows under another fixed retention. Separate integration of the physical, optimizing, and actual joint laws through the same channel proves their respective probability composition identities. No identification of those laws is needed. In particular the retained actual source conditional is not replaced by the optimizing conditional.

The fixed selected-source potential is $\phi_j = \delta \sum_i f(\sqrt{J_1/T}r_{i,j_i})$, where $f(t) = 1 - e^{-t^2/2}$. Thus $|f'|, |f''| \leq 1$. Write D_{int} for the matrix with rows $\sqrt{J_1}b_e$ on internal edges. Then $D^T_{\mathrm{int}}D_{\mathrm{int}} \preceq H$. An edge is selected $k_e \in \{0, 1, 2\}$ times and $\sum_e k_e = n$. Expressing the force as $\delta\sqrt{T}D^T_{\mathrm{int}}c$, with $c_e = k_e f'(\sqrt{J_1/T}r_e)$, gives $|c|^2 \leq 2n$. Hence, on M_0,

$$|H^{-1/2}T\nabla\phi_j| \leq B_*\sqrt{T}, \qquad -2\delta H \preceq T\nabla^2\phi_j \preceq 2\delta H, \qquad B_* = \delta\sqrt{2n}. \tag{A.10}$$

For the marginal derivative, keep the centered Gaussian variable in eq. (A.4) fixed. In its normalized tilted law $\nu_{p,x,j}$, differentiation gives

$$T\nabla\psi_{p,j} = \mathbb{E}_{\nu_{p,x,j}}L^T_p(T\nabla\phi_j), \tag{A.11}$$

$$T\nabla^2\psi_{p,j} = \mathbb{E}_{\nu_{p,x,j}}L^T_p(T\nabla^2\phi_j)L_p - T^{-1}\operatorname{Cov}_{\nu_{p,x,j}}(L^T_p T\nabla\phi_j). \tag{A.12}$$

The covariance has not been omitted. The isometry $H^{1/2}L_pH_p^{-1/2}$ and eq. (A.10) bound it by $B^2_*H_p$. With $\theta_* = 2\delta + 2n\delta^2 < 1/2$, this proves

$$|H_p^{-1/2}T\nabla\psi_{p,j}| \leq B_*\sqrt{T}, \qquad (\vartheta - \theta_*)H_p \preceq \nabla^2 V_{p,j} \preceq (\vartheta + 2\delta)H_p. \tag{A.13}$$

Evenness of the original source factor and centered Gaussian integration also imply $\psi_{p,j}(-x, \vartheta) = \psi_{p,j}(x, \vartheta)$. Every non-root sheet therefore has its unique minimum at zero.

The hidden Gaussian scale score is $s_{d_p} = (d_p - |Z|^2)/(2\vartheta)$, with Z standard Gaussian. It has zero mean and second moment $d_p/(2\vartheta^2)$. The tilted density ratio lies in $[e^{-B_0}, e^{B_0}]$, so

$$\partial_\vartheta\psi_{p,j} = -\mathbb{E}_{\nu_{p,x,j}}s_{d_p}, \qquad |\partial_\vartheta\psi_{p,j}| \leq (e^{B_0} - 1)\frac{\sqrt{d_p}}{\sqrt{2}\vartheta}. \tag{A.14}$$

This bound is uniform in the retained input because the Gaussian conditional center is independent of ϑ.

The full restricted objective retains its common scalar C_T. Differentiating it at fixed actual q gives

$$g_p(q,\vartheta) = \vartheta - 2 + \frac{1}{2}\mathbb{E}_q x^T H_p x + \frac{T d_p}{2\vartheta} + T\mathbb{E}_q \partial_\vartheta \psi_{p,S} - \frac{T}{2}\operatorname{Tr}(A_0^{-1}H) - T\mathbb{E}_q \partial_\vartheta \log \pi_S(\vartheta). \tag{A.15}$$

The first term is the calibration derivative, and the trace is the derivative of the proper prior's full Gaussian normalizer. The source term is present even though it has no continuous-coordinate gradient. For the distinguished binary prior its absolute unscaled score is at most four. A time derivative of actual source masses is not a term in this partial derivative.

For completeness, the complete generative Fisher is

$$M(\vartheta) = T^{-1} + \frac{1}{2}\operatorname{Tr}[(A_0^{-1}H)^2] + \operatorname{Var}_{\pi_{i_0}(\vartheta)}(t), \qquad t_+ = 4, \quad t_- = 0, \qquad m(\vartheta) = TM(\vartheta). \tag{A.16}$$

The prior, calibration, and source-prior scores have zero mean and zero cross expectations in the normalized joint. Its other conditional kernels have no explicit parameter derivative at fixed inputs. This proves the formula without substituting a posterior or marginal Fisher. Since R is fixed, annihilates $\mathcal{K}$, and is supported on the first normal layer, $A_0 = \vartheta H + P_{\mathcal{K}} - TR$ is positive for sufficiently small T. Conjugation by the positive quadratic core bounds the trace terms and their derivatives uniformly. Thus

$$1 \le m(\vartheta) \le 1 + C_L T, \qquad |m'(\vartheta)| \le C_L T. \tag{A.17}$$

All normalizers, determinant derivatives, and categorical terms in eq. (A.15) follow from normalized integrals. The physical prior correction $-TR$ has not been inserted into the optimizing Gaussian precision. □

A.2 Prepared actual laws and coupled continuation

Lemma A.2 (Prepared actual-law continuation). For the admitted common-center preparation and every finite sequence of actual marked activations, the coupled diffusion, source proposals, and own-law parameter equation have a unique nonexplosive solution through $H_* = 2/\epsilon$. The actual laws remain positive smooth joint densities and satisfy the moment estimates below. The compact parameter and speed bounds are established from those moments before any conditional-entropy or source-history estimate is used. [ESTABLISHED]

Proof. For a fixed positive instance and a prescribed continuous parameter path in a compact neighborhood of $[1, 2.1]$, each sheet drift has a globally bounded Hessian and linear growth by eq. (A.13). All proposal rows are smooth, have bounded derivatives for this instance, and have a positive finite-depth lower bound. Their denominator in the Gaussian integral representation is at least e^{-B_0}. Interlacing the sheet diffusions on an independent rate-one Poisson grid therefore constructs a unique nonexplosive process with all fixed-order polynomial moments.

To close the parameter feedback, let $u_j^{\vartheta}(s,x) = \mathbb{E}_{\vartheta}[f(X_t,S_t) \mid X_s = x, S_s = j]$ for a smooth terminal function whose derivatives have polynomial growth. Condition on the proposal times and a finite label sequence. The resulting expectation is a composition of Lipschitz sheet flows and products of positive row weights. Flow derivatives are bounded on compact time intervals; a differentiated weight is bounded by a constant times that weight, since the weights have a positive lower bound. The proposal count has every exponential moment. Polynomial diffusion moments therefore justify differentiating this representation and give polynomial bounds for u and ∇u. Applying Itô's formula to $u^{\widetilde{\vartheta}}$ along the ϑ process, first stopped in a bounded ball and then using these moments, proves

$$|\mathbb{E}_{\vartheta} f(X_t,S_t) - \mathbb{E}_{\widetilde{\vartheta}} f(X_t,S_t)| \le C \int_0^t |\vartheta(s) - \widetilde{\vartheta}(s)| ds. \tag{A.18}$$

The second-order coefficients agree, the drift difference is bounded by $C|\vartheta - \widetilde{\vartheta}|(1+|x|)$, and the row difference by $C|\vartheta - \widetilde{\vartheta}|$. The observable terms in eq. (A.15) meet these hypotheses. Their explicit terminal parameter dependence adds a local Lipschitz term. Hence the integral map for $\dot{\vartheta} = -\epsilon m(\vartheta)^{-1} g_p(q,\vartheta)$ is a contraction on a sufficiently short interval. This proves local coupled existence and uniqueness. These local constants may depend on the fixed positive instance; no uniform short-time contraction constant is asserted.

Stop only at the first possible exit of ϑ from $[1, 2.1]$. The Gaussian comparison starts at common non-root center Av_0, zero loop center, and covariance TH_0^{-1}. Its initial coupling to every admitted sheet satisfies $|H_0^{1/2}(X_0 - X_0^G)| < \epsilon\sqrt{T}$. The root is the prescribed independent equilibrium factor. Small compactly supported smooth transports in standardized non-root coordinates satisfy this displacement bound and a fixed bounded log-density-ratio bound, so the preparation neighborhood is nonempty. In particular no order-A spread of sheet centers is admitted.

On an active level let the Gaussian comparison use the same actual deterministic parameter and Brownian increments: $dX^G = -\vartheta(t) H_p X^G dt + \sqrt{2T} dW$. Continue its projected reference at an activation, while initializing the actual parent at the full actual retained sample. For $e = X - X^G$, the noise cancels and eq. (A.13) gives

$$\frac{d}{dt}|e|^2 \le -e^T H_p e + C_L T, \qquad \sup_{t \le H_*} |e_t| \le |e_0| + C_L \sqrt{T H_*}. \tag{A.19}$$

A source proposal leaves e continuous. The estimate is valid for every adapted label history, without conditioning Brownian increments on that history.

For the energy estimate, variation of constants uses the commuting propagator $\exp[-H_p \int_s^t \vartheta(r) dr]$. Spectral maximization gives

$$\int_0^{\infty} \|H_p e^{-H_p u}\| du \le C_L \left[1 + \log \frac{\lambda_{\max}(H_p)}{\lambda_{\min}(H_p)}\right]. \tag{A.20}$$

Indeed split at the inverse largest and smallest eigenvalues. The three bounds are the largest eigenvalue, $1/(eu)$, and the smallest eigenvalue times its decaying exponential. Their integrals give the constant, logarithmic, and tail terms. Euclidean projection contracts eq. (A.19), and the Schur

variational identity contracts its energy counterpart at each activation. With $\ell_\epsilon = 1+\log(1/\epsilon)$, finitely many such stages give

$$\sup_t |H_p^{1/2} e_t| \le C_L \sqrt{T}\ell_\epsilon. \tag{A.21}$$

The Gaussian covariance barrier $\Sigma^G \preceq C_L T H_p^{-1}$ uses only $\vartheta \ge 1$. It follows either from the stochastic integral or by substituting the constant upper barrier into $\dot\Sigma^G = -\vartheta H_p \Sigma^G - \vartheta \Sigma^G H_p + 2TI$. At retention, $P H_p^{-1} P^T = H_s^{-1}$ preserves that barrier. The deterministic mean contracts, has norm $O_L(A)$, and has zero loop part. Combining this with eq. (A.21) yields

$$\mathbb{E}_{q_t^p} x^T H_p x \le C_L[A^2 J_{p+1} + T\ell_\epsilon^2], \tag{A.22}$$

$$\mathbb{E}_{q_t^p}|x|^2 \le C_L[A^2 + T\ell_\epsilon^2], \qquad J_1 \mathbb{E}_{q_t^p}|h|^2 \le C_L T. \tag{A.23}$$

The sharper loop bound follows separately from its scalar core rate ϑJ_1 and force at most $C_L\sqrt{TJ_1}$; its forced error is at most $C_L\sqrt{T/J_1}$. This does not assert that the nonlinear loop itself is an independent Ornstein–Uhlenbeck process.

Set

$$r_\epsilon = A^2 J_1 + T\ell_\epsilon^2. \tag{A.24}$$

Substitution in the complete covector and metric bounds proves $\dot\vartheta = \epsilon(2-\vartheta) + O_L(\epsilon r_\epsilon)$ before assuming any bound on $\dot\vartheta$. The field points inward at both endpoints of $[1, 2.1]$ for small ϵ, excluding the stopped exit. The moment bounds exclude explosion. This extends the solution through H_* and gives $|\dot\vartheta| \le C_L\epsilon$. Positive-temperature elliptic smoothing and positive finite rates preserve the positive smooth densities. The same construction applies after an actual activation because its full retained law, rather than an equilibrium replacement, is the initial law. □

Lemma A.3 (Comparison of independently learned parameters). For every actual process or finite replacement lineage in lemma A.2, the following estimates are established:

$$1 \le \vartheta_p(t) \le 2.1, \qquad |\dot\vartheta_p(t)| \le C_L\epsilon, \qquad \sup_{t\le H_*} |\vartheta_p(t) - (2-e^{-\epsilon t})| \le C_L r_\epsilon. \tag{A.25}$$

Every independently evolving descendant uses its own actual law and copies only the actual starting parameter. [ESTABLISHED]

Proof. The first two bounds were proved without entropy or averaging. Subtract $\dot\vartheta_0 = \epsilon(2-\vartheta_0)$, $\vartheta_0(0) = 1$, from the actual equation and integrate with factor $e^{\epsilon t}$. The forcing contribution is at most $C_L r_\epsilon$ through the full horizon. Copying the actual parameter at each of finitely many activations adds only a finite multiple of that error. Under the declared regime

$$r_\epsilon = \epsilon^{2Q_1+20} + \epsilon^{3Q_1+26}\ell_\epsilon^2 \longrightarrow 0. \tag{A.26}$$

In particular each coefficient exceeds $7/4$ on the last ten units of the horizon, for sufficiently small ϵ, since $2 - e^{-2} > 7/4$. The comparison trajectory is a consequence of the own-law equation, not an imposed coefficient schedule. □

A.3 Entropy estimates for actual joint laws

Lemma A.4 (Continuous entropy and score tools). Let $\mu(dx) \propto e^{-W(x)/T}dx$ on a finite-dimensional Euclidean space, where W is smooth, has a globally bounded Hessian, and $\nabla^2 W \succeq \lambda I$ with $\lambda > 0$. For a probability density f relative to μ, the following inequalities hold whenever their right sides are finite:

$$\mathrm{Ent}_\mu(f) \le \frac{T}{2\lambda}\int \frac{|\nabla f|^2}{f}d\mu, \tag{A.27}$$

$$\log \mathbb{E}_\mu e^{u(F-\mathbb{E}_\mu F)} \le \frac{TK^2u^2}{2\lambda}, \qquad \mathrm{Lip}(F) \le K. \tag{A.28}$$

Here $\mathrm{Ent}_\mu(f) = \int f \log f d\mu$ when $\int f d\mu = 1$; the unnormalized convention used below is $\mathrm{Ent}_\mu(f) = \int f \log(f/\int f d\mu)d\mu$. These statements apply to every conditional sheet potential in this construction. [ESTABLISHED]

Proof. For the semigroup P_t generated by $T\Delta - \nabla W \cdot \nabla$, the derivative of the synchronous stochastic flow has operator norm at most $e^{-\lambda t}$. Weighted Cauchy–Schwarz therefore gives $|\nabla P_t f|^2/P_t f \le e^{-2\lambda t}P_t(|\nabla f|^2/f)$. Invariance of μ implies $I_\mu(P_t f) \le e^{-2\lambda t}I_\mu(f)$. Integration by parts gives $d\,\mathrm{Ent}_\mu(P_t f)/dt = -TI_\mu(P_t f)$. For bounded positive Lipschitz f, contraction against a stationary initial sample yields $P_t f(x) \to \int f d\mu$. Bounded convergence then makes its entropy tend to zero. Integrating the Fisher bound proves eq. (A.27) with its stated constant. Cutoff, positive truncation, and short semigroup smoothing extend it by lower semicontinuity. Gaussian tails from strong convexity justify the integrations. This is the curvature entropy argument in Bakry and Émery [9, pp. 187, 194–200], with diffusion coefficient T made explicit.

Apply eq. (A.27) to e^{uF}. For $K_F(u) = \log \mathbb{E}_\mu e^{uF}$ it gives $uK_F'(u) - K_F(u) \le TK^2u^2/(2\lambda)$. Integration of the derivative of $K_F(u)/u$ from zero proves eq. (A.28), also for negative u. The entropy variational inequality

$$u\mathbb{E}_\nu F \le D(\nu\|\mu) + \log \mathbb{E}_\mu e^{uF} \tag{A.29}$$

follows from nonnegativity of relative entropy to the normalized exponential tilt of μ. Consequently $|\mathbb{E}_\nu F - \mathbb{E}_\mu F| \le K\sqrt{2TD(\nu\|\mu)/\lambda}$. If instead a centered observable ξ obeys

$$\log \mathbb{E}_\mu e^{u\xi} \le \frac{vu^2}{2(1-b|u|)}, \qquad |u| < b^{-1}, \tag{A.30}$$

optimization in eq. (A.29), using $|u| = \sqrt{2D/v}/(1 + b\sqrt{2D/v})$, gives

$$|\mathbb{E}_\nu \xi| \le \sqrt{2vD(\nu\|\mu)} + bD(\nu\|\mu). \tag{A.31}$$

The $D = 0$ case follows by continuity. Finally $\|\nu - \mu\|_{\mathrm{TV}} \le \sqrt{D(\nu\|\mu)/2}$: push both laws to a set realizing their total variation, and use the binary relative entropy's second derivative $[t(1-t)]^{-1} \ge 4$ and its zero value and derivative at equality. □

All normalized hidden Gaussian conditionals have a center independent of ϑ. Their logarithmic parameter scores, after including the bounded marginal tilt, are centered versions of $-|Z|^2/(2\vartheta) - \partial_\vartheta \psi_{p,j}$. The full joint score adds $\partial_\vartheta \log \pi_j(\vartheta)$, whose absolute value is at most four. These scores satisfy eq. (A.30) with constants v_L, b_L independent of the retained input and of T, J_s. To check this uniformity directly, a bounded tilt of a standard Gaussian satisfies

$$\mathbb{E}|Z|^{2m} \le e^{B_0} 2^{k_0/2} 4^m m!, \qquad m \ge 0. \tag{A.32}$$

Indeed $r^m \le m! 4^m e^{r/4}$ and the Gaussian exponential moment is $2^{k_0/2}$. The other score terms are uniformly bounded by eq. (A.14). Centering increases a suitable geometric moment constant by at most a factor two. Expanding the exponential series, removing the zero linear term, and using $\log(1+x) \le x$ proves eq. (A.30). Conditional expectations of such scores retain the same bound by Jensen's inequality. [ESTABLISHED]

Lemma A.5 (Entropy at actual activations). For the actual active law at level p, let $E_p(t) = TD(q_t^p \| \pi^p_{\vartheta_p(t)})$, with the independent root factor canceled. At an activation time τ_p,

$$E_p(\tau_p) \le C_L[A^2 J_{p+1} + T\ell_\epsilon^2]. \tag{A.33}$$

On each ensuing active interval,

$$T^2 \int_{\tau_p}^{t} I_p(u)du \le C_L[E_p(\tau_p) + T], \tag{A.34}$$

where I_p is the full continuous relative Fisher information of the joint actual law. Neither estimate assumes a positive lower bound on its time-dependent source masses. [ESTABLISHED]

Proof. We first control actual differential/counting entropy $h(q) = -\sum_j \int q_j \log q_j$ across changes of dimension. Write $q_j = \lambda_j f_j$, with terms of zero mass omitted. On a d-dimensional sheet, diffusion gives

$$h'(q)_{\text{diff}} = T\sum_j \lambda_j I(f_j) - \mathbb{E}_q \Delta V_{p,S} \ge T\sum_j \lambda_j I(f_j) - d\Lambda_p, \qquad \Lambda_p \le C_L J_1. \tag{A.35}$$

The entropy–Fisher bound $I(f) \ge 2\pi e d e^{-2h(f)/d}$ follows from the Gaussian case of eq. (A.27): expand both sides relative to $N(0, \sigma^2 I)$, integrate the cross term $\int x \cdot \nabla f = -d$, cancel the second moments, and minimize at $\sigma^2 = d/I(f)$. Jensen's inequality and $h(q) = \sum_j \lambda_j h(f_j) + H(\lambda)$, with $H(\lambda) \ge 0$, then give $\sum_j \lambda_j I(f_j) \ge 2\pi e d e^{-2h(q)/d}$.

At fixed x, a proposal moves the conditional label row α in the direction $\alpha R - \alpha$. Concavity of categorical entropy yields its directional derivative at least $H(\alpha R) - H(\alpha) \ge -\log|\mathcal{J}|$. The continuous marginal is unchanged by the proposal. Hence

$$h'(q) \ge 2\pi e d T e^{-2h(q)/d} - d\Lambda_p - \log|\mathcal{J}|. \tag{A.36}$$

Scalar comparison bounds $h(q)$ below by the lesser of its initial value and $\frac{d}{2}\log[2\pi edT/(d\Lambda_p + \log|\mathcal{J}|)]$. At a projection dropping k coordinates z, the chain rule and the maximum-entropy Gaussian bound imply

$$h(Y,S) \geq h(X,S) - \frac{k}{2}\log\frac{2\pi e\mathbb{E}|z-z_*|^2}{k}. \tag{A.37}$$

Here z_* can be the projected comparison mean. The bound follows conditionally on (Y,S) and then from Jensen; it also follows from nonnegativity of relative entropy to the common isotropic Gaussian. The moment proof of lemma A.2 bounds this centered covariance by $C_L T\ell_\epsilon^2 H_p^{-1}$. Initial entropies and all logarithms in these estimates are $O_L(\ell_\epsilon^2)$: the covariance eigenvalues, A, and T are fixed powers of ϵ, and the admitted initial density ratios are bounded. There are finitely many projections. Expanding $TD(q\|G_\vartheta^p\pi(\vartheta))$ and using eq. (A.22) proves eq. (A.33). The bounded optimizer tilt adds at most TB_0. The source prior has a positive minimum on $[1,2.1]$, so its log contribution is bounded independently of the actual λ_j.

For the time-integrated estimate put $V = x^T H_p x/2$. The normalized reference satisfies

$$\mathbb{E}_{\pi_\vartheta^p}\exp\frac{\vartheta V}{2T} \leq R_L, \qquad R_L = e^{B_0}2^{k_0/2}. \tag{A.38}$$

Entropy duality gives $\vartheta\mathbb{E}_q V \leq 2E_p + 2T\log R_L$; Jensen gives $\vartheta\mathbb{E}_\pi V \leq 2T\log R_L$. In the normalized full score, the remaining centered terms have a uniform bound $2M_L$. Therefore

$$|T\mathbb{E}_q\partial_\vartheta\log\pi_\vartheta^p| \leq \frac{2E_p + 4T\log R_L}{\vartheta} + 2TM_L. \tag{A.39}$$

Diffusion dissipates $T^2 I_p$. Detailed balance eq. (A.9) makes the source contribution nonpositive. Thus, for a fixed $C_{0,L}$,

$$E_p' \leq -T^2 I_p + \frac{2|\dot\vartheta|}{\vartheta}(E_p + C_{0,L}T). \tag{A.40}$$

The integrating factor is $\exp[-2\int_{\tau_p}^t|\dot\vartheta|/\vartheta]$. Its exponent is bounded over H_* by the already proved speed bound. Integration proves eq. (A.34). At an exact activation both actual and reference laws are pushed by the same channel at the same current parameter. The relative-entropy chain rule makes E_p decrease, so the estimates continue through the declared finite lineage. All entropy differentiations follow first for positive truncated densities on bounded domains; the moment and entropy bounds above justify removing these cutoffs. □

Lemma A.6 (Conditional entropy and complete score error). For every retention $p = s-1 \to s$, the actual conditional relative entropy with respect to the optimizing hidden conditional satisfies

$$\sup_{t\in I_s \text{ and subsequent active times}} \Delta_s(t) \leq \begin{cases} C_L\epsilon^{14}, & s < L, \\ C_L\epsilon^2, & s = L. \end{cases} \tag{A.41}$$

The early bound also applies until the next activation to the corresponding un-replaced comparison process. At every such time the complete fine/parent covector difference is at most $C_L T(\sqrt{\Delta_s}+\Delta_s)$. The terminal assertion is proved below using joint relaxation. [ESTABLISHED]

Proof of the early bound and the covector estimate. For an early retention use the auxiliary continuous split

$$Z = (\mathcal{N}_p \cap \mathcal{N}_s^{\perp}) \oplus \mathcal{H}, \qquad Y = \mathcal{N}_s, \qquad S \text{ retained.} \tag{A.42}$$

The true parent retains the loops. The entropy chain rule therefore bounds its hidden conditional defect Δ_s by the auxiliary defect $\widehat{\Delta}_s$ obtained by also integrating h. The independent root cancels. Conditional strong convexity in Z is at least $\lambda_s = c_{\mathrm{early},L} J_s$ after reducing the fixed positive constant if necessary. The mixed Hessian satisfies

$$\|\nabla_Z \nabla_Y V_{p,j}\| \le \Gamma_s = C_L J_{s+1}, \qquad \Gamma_L = 0. \tag{A.43}$$

Here is the scale check, which a bound by J_1 would not replace. The only nonlinear residuals lie in the first normal layer and the loop space. At $p = 0$ the original nonlinear potential has no future normal dependence, so its mixed future derivatives vanish directly. For $p \ge 1$, use coordinates $r_E = B_E w_E$ on eliminated normal rows. The Gaussian conditional quadratic has precision $A_E = D_E + C^T D_f C$. Its perturbation $\mathcal{R}_E = C^T D_f C$ has norm at most $C_L J_{p+1}$, while A_E^{-1} has norm at most C_L/J_p. The identity

$$A_E^{-1} = D_E^{-1} - D_E^{-1} \mathcal{R}_E A_E^{-1} \tag{A.44}$$

bounds the first block row by C_L/J_1. A future-layer r force column has size $C_L J_r$, so the first conditional mean responds by $C_L J_r/J_1$. Its conditional covariance is comparable to T/J_1.

Differentiate the normalized Gaussian integral defining $\psi_{p,j}$. A first-layer or loop derivative of ϕ_j has size $C_L\sqrt{J_1/T}$, and a second derivative has size $C_L J_1/T$. Each derivative in a future direction brings the conditional-mean response just obtained. The covariance term in the second derivative is bounded by the product of the same first derivative bounds. Consequently

$$T|\partial_h \partial_{w_r} \psi_{p,j}| \le C_L J_r, \qquad T|\partial_{w_i} \partial_{w_r} \psi_{p,j}| \le C_L \frac{J_i J_r}{J_1}. \tag{A.45}$$

The Gaussian Schur cross block has size at most $C_L J_{s+1}$, by the lower triangular residual form and its variational bound. Summing finitely many blocks proves eq. (A.43). This calculation includes the generated conditional covariance.

Write the joint density as $q = r(y,j)a(z \mid y,j)$ and its optimizing reference as $\pi = \pi_Y(y,j)k(z \mid y,j)$. Let $w_Y = \nabla_Y \log(q/\pi)$, $v_Y = \nabla_Y \log(r/\pi_Y)$, and $e_Y = \mathbb{E}_a \nabla_Y \log k$. Normalization gives $\mathbb{E}_k \nabla_Y \log k = 0$ and $\mathbb{E}_a w_Y = v_Y - e_Y$. The actual marginal flux is $Tr(v_Y - e_Y)$. Subtracting its entropy derivative from the full derivative yields the exact continuous identity

$$\begin{aligned} \widehat{\Delta}_s'|_{\mathrm{cont}} = & -TI_Z - T\mathbb{E}_r \operatorname{Var}_a(w_Y) + T\mathbb{E}_r[e_Y \cdot \mathbb{E}_a w_Y] \\ & - \dot{\vartheta}\mathbb{E}_q \partial_\vartheta \log k. \end{aligned} \tag{A.46}$$

The conditional logarithmic Sobolev inequality gives $TI_Z \ge 2\lambda_s \widehat{\Delta}_s$. Apply eq. (A.28) to each unit-vector component of $\nabla_Y \log k$, whose Z Lipschitz constant is at most Γ_s/T. This gives

$|e_Y| \le \Gamma_s\sqrt{2D(a\|k)/(T\lambda_s)}$. Cauchy–Schwarz and Young's inequality bound the mixed term in eq. (A.46) by $\lambda_s\widehat{\Delta}_s/2 + T\Gamma_s^2 I_Y/\lambda_s^2$; I_Y is the actual full relative Fisher component. The normalized conditional parameter score has the uniform exponential-moment bound above. Its contribution is at most $\lambda_s\widehat{\Delta}_s/2 + v_L|\dot{\vartheta}|^2/\lambda_s + b_L|\dot{\vartheta}|\widehat{\Delta}_s$.

A finite-rate source update is not a conditional-entropy contraction, so its contribution must be retained. At fixed $Y = y$, compare k_j with the common label-independent hidden Gaussian γ. Each $\log(\gamma/k_j)$ lies in $[-B_0, B_0]$. The conditional entropy relative to γ splits as $D(q_Z\|\gamma) + I(Z;S)$. A proposal leaves q_Z unchanged. Mutual information is convex as a function of the conditional label channel at fixed q_Z, by joint convexity of relative entropy. Its directional derivative towards the new channel is at most $\log|\mathcal{J}|$. The difference from the actual target is a linear expectation of $\log(\gamma/k_j)$ and changes at rate at most $2B_0$. Thus

$$\widehat{\Delta}_s'|_{\text{jump}} \le C_J := \log|\mathcal{J}| + 2B_0. \tag{A.47}$$

This argument remains valid when a source mass vanishes, by convex approximation, and divides by no such mass.

For sufficiently small ϵ, $b_L|\dot{\vartheta}| \le \lambda_s/2$. Dropping the negative variance term gives

$$\widehat{\Delta}_s' \le -\frac{\lambda_s}{2}\widehat{\Delta}_s + \frac{T\Gamma_s^2}{\lambda_s^2}I_Y + \frac{v_L|\dot{\vartheta}|^2}{\lambda_s} + C_J. \tag{A.48}$$

The initial conditional entropy is at most $E_p(\tau_p)/T$. Integrate the inequality and apply eqs. (A.33) and (A.34). Since $A^2/T = \epsilon^{-6}$, the result is

$$\begin{aligned}\widehat{\Delta}_s(t) \le C_L\Bigg[&e^{-c_{\text{early},L}J_s(t-\tau_p)}(\epsilon^{-6}J_s + \ell_\epsilon^2) \\ &+ \epsilon^{-6}\frac{J_{s+1}^2}{J_s} + \ell_\epsilon^2\frac{J_{s+1}^2}{J_s^2} + \frac{\epsilon^2}{J_s^2} + \frac{C_J}{J_s}\Bigg],\end{aligned} \tag{A.49}$$

where the constant named $c_{\text{early},L}$ is chosen to include the preceding factor $1/2$. For $s < L$, $J_{s+1}^2/J_s = \epsilon^{20}$ and $J_s \ge \epsilon^{-20}$. The declared entrance time and $c_{\text{early},L}k_L/2 > Q_1 + 20$ make the first term no larger than $C_L\epsilon^{14}$. The other terms meet the same bound. This proves the early assertion. The last term would be C_J at $s = L$; it is not discarded there.

For the true fixed retention the exact objective chain rule is

$$F_T^p(q,\vartheta) - F_T^s((C_{s\leftarrow p})_\# q,\vartheta) = T\Delta_s(q,\vartheta). \tag{A.50}$$

Differentiating at fixed actual q and fixed channel gives $g_p - g_s = -T\mathbb{E}_q\partial_\vartheta \log k_{\text{true}}$. The normalized conditional score has the same uniform exponential-moment bound. Apply eq. (A.31) conditionally and then Jensen to obtain

$$|g_p - g_s| \le C_L T(\sqrt{\Delta_s} + \Delta_s). \tag{A.51}$$

The common scalar and source-prior derivative cancel through this exact chain rule; they were not removed from either objective. □

Lemma A.7 (Joint relaxation at the fixed weakest strength). There is a constant $c_{\mathrm{joint},L} > 0$, independent of ϵ, such that every actual lineage satisfies

$$D(q_t^p \| \pi^p_{\vartheta_p(t)}) \le C_L e^{-c_{\mathrm{joint},L} t}[\epsilon^{-(Q_1+6)} + \ell_\epsilon^2] + C_L \epsilon^2. \tag{A.52}$$

Consequently the terminal part of eq. (A.41) holds once $t \ge t_L = k_L \log(1/\epsilon)$. [ESTABLISHED]

Proof. Fix a level and parameter, write $p = \pi^p_\vartheta$, $\rho = G^p_\vartheta \pi(\vartheta)$, and $u_j = q_j/p_j$. The bounded tilt gives $e^{-B_0} \le p/\rho \le e^{B_0}$, and eq. (A.8) gives

$$p_j(x) R^p_{jk}(x) \ge e^{-2B_0} G^p_\vartheta(x) \pi_j(\vartheta) \pi_k(\vartheta). \tag{A.53}$$

For a probability measure μ, $\mathrm{Ent}_\mu(u) = \inf_{r>0} \int [u \log(u/r) - u + r] d\mu$. The integrand is nonnegative, so $\mathrm{Ent}_p(u) \le e^{B_0} \mathrm{Ent}_\rho(u)$. The product chain rule and convexity imply

$$\mathrm{Ent}_\rho(u) \le \sum_j \pi_j \mathrm{Ent}_G(u_j) + \mathbb{E}_G \mathrm{Ent}_\pi(u(x)). \tag{A.54}$$

The Gaussian entropy inequality bounds the first term by $T/(2\vartheta h_L)$ times its continuous Fisher information. For any positive finite vector v,

$$\begin{aligned} \frac{1}{2} \sum_{j,k} \pi_j \pi_k (v_j - v_k)(\log v_j - \log v_k) &= \sum_j \pi_j v_j \log v_j - \bar{v} \sum_j \pi_j \log v_j \\ &\ge \mathrm{Ent}_\pi(v), \qquad \bar{v} = \sum_j \pi_j v_j. \end{aligned} \tag{A.55}$$

The inequality is Jensen's inequality for $\log$. Compare the Fisher measures and the conductances using their positive pointwise bounds. For

$$\mathcal{D}_{\mathrm{jump}} = \frac{1}{2} \int \sum_{j,k} p_j R^p_{jk} (u_j - u_k)(\log u_j - \log u_k) dx, \tag{A.56}$$

one obtains

$$D(q\|p) \le e^{2B_0} \frac{T}{2\vartheta h_L} I_p + e^{3B_0} \mathcal{D}_{\mathrm{jump}}. \tag{A.57}$$

No lower bound on the actual label masses, nor a vanishing source time scale, occurs in this inequality. Let $c_{0,L} = \min\{2h_L e^{-2B_0}, e^{-3B_0}\} > 0$. Then $TI_p + \mathcal{D}_{\mathrm{jump}} \ge c_{0,L} D$. The time-dependent normalized full score satisfies eq. (A.30). Hence

$$D' \le -c_{0,L} D + C_L \epsilon(\sqrt{D} + D) \le -\frac{c_{0,L}}{2} D + C_L \epsilon^2 \tag{A.58}$$

for sufficiently small ϵ. Take $c_{\mathrm{joint},L} = c_{0,L}/2$. Actual activations decrease D at the unchanged parameter; each descendant then satisfies the same inequality with its own parameter speed. The

initial entropy estimate is $D_0 \le C_L[\epsilon^{-(Q_1+6)} + \ell_\epsilon^2]$. Integrating through finitely many such decreases proves eq. (A.52). The condition $c_{\mathrm{joint},L} k_L > Q_1 + 8$ makes its decaying term $O_L(\epsilon^2)$ at t_L. Both the auxiliary and true hidden conditional entropies are bounded by full joint relative entropy. This proves the terminal assertion without treating the nonzero conditional jump remainder as small. □

A.4 Exact fast marginals and complete source histories

Lemma A.8 (Common exact fast marginal and environmental error). There is a normalized fast law, conditional on each assignment, whose averaged proposal matrix is common to every exact optimizing parent at a fixed ϑ. Replacing the actual environment by this auxiliary fast process incurs the explicit errors below; the common matrix is an exact conditional integral, not an independently specified source dynamics. [ESTABLISHED]

Proof. Use the fixed orthonormal normal flag from eq. (A.1). In these coordinates

$$w^T H_{\mathcal{N}} w = J_1 |\mathsf{B}_{11} w_1|^2 + \sum_{s=2}^{L} J_s \left| \sum_{r \le s} \mathsf{B}_{sr} w_r \right|^2 . \tag{A.59}$$

The nonlinear factor ϕ_j depends only on (w_1, h). For fixed w_1, h, change all later w_s to residuals $r_s = \sum_{r \le s} \mathsf{B}_{sr} w_r$. The domain remains the whole Euclidean product and the Jacobian is the constant $\prod_{s=2}^{L} |\det \mathsf{B}_{ss}|$. Integrating those Gaussian residuals contributes a scalar independent of w_1, h, j. Set $A_* = J_1 \mathsf{B}_{11}^T \mathsf{B}_{11}$. The exact fast optimizing law conditional on $S = j$ is therefore

$$\mu^*_{\vartheta,j}(dw_1, dh) = \frac{1}{Z^*_{\vartheta,j}} \exp\left\{ -\frac{\vartheta}{2T}(w_1^T A_* w_1 + J_1 |h|^2) - \phi_j(w_1, h) \right\} dw_1 dh. \tag{A.60}$$

Write $\mu^{*,h}_{\vartheta,j}$ for its loop marginal and $\kappa^*_{\vartheta,j}(dw_1 \mid h)$ for its first-normal conditional. Define

$$R^{*,h}_{jk}(h, \vartheta) = \int b_k(w_1, h, \vartheta) \kappa^*_{\vartheta,j}(dw_1 \mid h), \tag{A.61}$$

$$\mathcal{B}_{jk}(\vartheta) = \int b_k(w_1, h, \vartheta) \mu^*_{\vartheta,j}(dw_1, dh) = \int R^{*,h}_{jk}(h, \vartheta) \mu^{*,h}_{\vartheta,j}(dh). \tag{A.62}$$

The prior factor $\pi_j(\vartheta)$ cancels on conditioning on j. The Gaussian change of variables and the tower property show that averaging the inherited row R^p_{jk} against the exact optimizing law conditional on j gives this same $\mathcal{B}_{jk}$ for every p. In particular $\mathcal{B}$ need not have invariant assignment law equal to the original prior.

For an intermediate $p \ge 1$, keep the future normal coordinates $y = (w_{p+1}, \dots, w_L)$ fixed and integrate $w_2, \dots, w_p$. The unperturbed conditional first-normal law is

$$g_{p,\vartheta,y} = N\left(m_p(y), \frac{T}{\vartheta} \mathsf{C}_p^{-1} \right), \qquad A_* \preceq \mathsf{C}_p \preceq A_* + C_L J_2 I, \qquad |m_p(y)| \le C_L \frac{J_2}{J_1} |y|. \tag{A.63}$$

Here C_p is a precision matrix, distinct from the Gaussian integration scalar C_p^G. For the lower precision bound, minimize the sum of nonnegative later residual squares with $y = 0$. For the upper bound, set all eliminated later coordinates to zero and use $J_s \le J_2$ for $s \ge 2$. At $p = L$, integration of every later residual gives $\mathsf{C}_L = A_*$ and $m_L = 0$. The mean bound is the first-block-row estimate eq. (A.44); each future coordinate of level s contributes at most $C_L J_s/J_1$.

Put

$$\rho_\epsilon = \frac{J_2}{J_1}, \qquad \gamma_\epsilon = \frac{J_2}{\sqrt{J_1 T}}, \qquad d_\epsilon(y) = \rho_\epsilon + \gamma_\epsilon |y|. \tag{A.64}$$

The Gaussian divergence from $g_{p,\vartheta,y}$ to $g_\vartheta^* = N(0, (T/\vartheta)A_*^{-1})$ is

$$\frac{1}{2}\left[\frac{\vartheta}{T} m_p^T A_* m_p + \mathrm{Tr}(A_* \mathsf{C}_p^{-1}) - \dim w_1 - \log\det(A_* \mathsf{C}_p^{-1})\right] \le C_L[\rho_\epsilon^2 + \gamma_\epsilon^2 |y|^2]. \tag{A.65}$$

Indeed the eigenvalues of the whitened covariance lie between $(1+C_L\rho_\epsilon)^{-1}$ and 1, and $t-1-\log t = O((t-1)^2)$ uniformly there for small ϵ. Pinsker's inequality gives a total-variation error at most $C_L d_\epsilon(y)$.

Tilting two probability measures by the same weight $w \in [e^{-B_0}, 1]$ increases their total variation by at most $2e^{B_0}$. To see this, subtract $w\mu/\mu(w)$ and $w\nu/\nu(w)$, split numerator and normalization differences, and use $\mu(w), \nu(w) \ge e^{-B_0}$. Apply this to $w = e^{-\phi_j(w_1,h)}$, uniformly for every h. The exact parent and fast-reference rows and loop forces therefore satisfy

$$\|R_{j\cdot}^p(y,h,\vartheta) - R_{j\cdot}^{*,h}(h,\vartheta)\|_{\mathrm{TV}} \le C_L d_\epsilon(y), \tag{A.66}$$

$$|\nabla_h V_{p,j}(y,h,\vartheta) - \nabla_h V_j^{*,h}(h,\vartheta)| \le C_L \sqrt{T J_1}\, d_\epsilon(y). \tag{A.67}$$

The second formula uses the common Gaussian force $\vartheta J_1 h$ and $T\mathbb{E}_\kappa \nabla_h \phi_j$. Furthermore

$$\mathrm{Lip}_{(w_1,h)}\, b_{j\cdot} + \mathrm{Lip}_h\, R_{j\cdot}^{*,h} \le C_L \sqrt{J_1/T}. \tag{A.68}$$

The derivative of $R^{*,h}$ includes the covariance of b_k with $\nabla_h \phi_j$; the conditional Gaussian g_ϑ^* is independent of h. Its bounded source derivatives and this covariance prove the displayed estimate. The exact marginal covariance has therefore been included in both the force and row calculations. □

Lemma A.9 (Fast mixing with a learned coefficient). Let the auxiliary process have, on a proposal-free interval, fixed assignment j and either the full fast potential of eq. (A.60) or its exact loop marginal. Its actual deterministic coefficient satisfies $1 \le \vartheta \le 2.1$ and $|\dot\vartheta| \le C_L\epsilon$. For a point start z at time s and $t - s \ge (c_{\mathrm{fast},L} J_1)^{-1}$,

$$\|P_{s,t}^*(z,\cdot) - \mu_{\vartheta(t),j}^*\|_{\mathrm{TV}} \le C_L(1 + \sqrt{J_1/T}|z|)e^{-c_{\mathrm{fast},L} J_1 (t-s)} + C_L\epsilon/J_1. \tag{A.69}$$

The notation μ^* in the loop case means its loop marginal. Also $\mathcal{B}_{jk}(\vartheta)$ is uniformly Lipschitz in ϑ. [ESTABLISHED]

Proof. The relative Hessian bounds and the normalized marginal covariance calculation give $c_L J_1 I \preceq \nabla^2 V^* \preceq C_L J_1 I$ for both potentials. Both are even, minimize at zero, and their equilibrium second moments are at most $C_L T/J_1$. For two point starts, reflection coupling before meeting gives their distance a one-dimensional noise of size $2\sqrt{2T}$ and a drift at most $-\lambda D$, where $\lambda = c_L J_1$. Multiplying the comparison equation by $e^{\lambda(t-s)}$ gives a time-changed Brownian motion of variance $4T(e^{2\lambda(t-s)} - 1)/\lambda$. The reflection principle therefore bounds the probability of nonmeeting by

$$C|z - z'|\sqrt{\lambda/T}(e^{2\lambda(t-s)} - 1)^{-1/2}. \tag{A.70}$$

This construction uses an orthogonal predictable reflection of one Brownian motion, so both marginals have the required Brownian law. The drifts are globally Lipschitz for each fixed instance; construct away from the diagonal, then join the copies at meeting. The one-dimensional comparison and the displayed hitting bound follow by stopping before the diagonal and taking the stopping radius to zero. Averaging z' against the equilibrium at time s gives the first term in eq. (A.69).

It remains to compare that evolving equilibrium start with the instantaneous equilibrium at time t. After standardizing (w_1, h) by $\sqrt{J_1/T}$, its normalized parameter score is a centered quadratic with uniform exponential moments. The loop-marginal score is its conditional expectation and has the same bound by Jensen. The moving-reference entropy calculation, using the continuous entropy inequality with curvature $c_L J_1$, gives

$$D' \le -c_L J_1 D + C_L \epsilon^2/J_1, \qquad D(s) = 0. \tag{A.71}$$

The linear-in-D parameter term is absorbed for small ϵ. It follows that $D \le C_L \epsilon^2/J_1^2$, and Pinsker gives the second term in eq. (A.69). Finally, differentiating $\mathcal{B}_{jk} = \mathbb{E}_{\mu^*_{\vartheta,j}} b_k$ gives the explicit row derivative and a covariance with this centered scale score. The sum of absolute explicit row derivatives is bounded by eight, and the standardized score has bounded first moment. Thus $\sum_k |\partial_\vartheta \mathcal{B}_{jk}| \le C_L$. □

Lemma A.10 (Coupling of complete source histories). The actual source-assignment histories of a process and of an activated exact parent can be coupled, together with their common actual initial marked sample, so that the probability of any disagreement through H_* is at most $C_L \epsilon^6$. The result applies through the finite activation lineage. It concerns the full finite-state path, including arbitrary actual label correlations, and uses the declared rate-one proposal clock. [ESTABLISHED]

Proof. We first bound an actual proposal row on one proposal-free interval. For a microscopic process let $\xi = (w_1, h)$. Couple its auxiliary fast diffusion to the same initial fast point, Brownian increments, assignment, and actual parameter path. The omitted Gaussian environmental force is $\vartheta[(H_{11} - A_*)w_1 + H_{1,>1} w_{>1}]$ and has norm at most $C_L J_2 |w|$. Strong contraction gives the pathwise bound

$$|\xi_t - \xi^*_t| \le C_L J_2 \int_\sigma^t e^{-c_L J_1 (t-v)} |w_v| dv. \tag{A.72}$$

For a parent at any $p \geq 1$, couple its retained loop to the auxiliary loop process. By eq. (A.67),

$$|h_t - h_t^*| \leq C_L \sqrt{T J_1} \int_\sigma^t e^{-c_L J_1 (t-v)} d_\epsilon(y_v) dv. \tag{A.73}$$

The environment in these two inequalities remains the actual adapted environment. It has not been frozen or differentiated. Multiplication by the row Lipschitz constant, with the direct parent row error in eq. (A.66), converts them to proposal-row errors.

The bound needed at random proposal times is stronger than a fixed-time unconditional moment. Condition on the whole independent future Poisson grid, but not its marks or any future continuous path. Brownian law is unchanged under this conditioning. Evenness and the uniform lower curvature on M_p give, on every sheet and level,

$$(T\Delta - \nabla V_{p,j} \cdot \nabla)|x|^2 \leq 2Tk_0 - 2c_L|x|^2. \tag{A.74}$$

Proposals preserve x, and projections contract its norm. The common actual preparation has second moment at most $C_L(A^2 + T)$. Thus

$$\sup_{t \leq H_*} \mathbb{E}[|X_{M_p,t}|^2 \mid \text{Poisson grid}] \leq C_L(A^2 + T). \tag{A.75}$$

If the common actual initial sample is also fixed, the right side first contains its squared norm; integrating that sample returns the displayed bound. The future grid after a deterministic activation is independent of its actual past, so the same argument applies at each activation. In particular it applies at grid-selected times.

Average the row errors from eqs. (A.72) and (A.73) using eq. (A.75). The exponential convolution has integral at most C_L/J_1. Their common expected per-proposal bound is

$$\eta_\epsilon = C_L \left[\frac{J_2}{J_1} + \frac{J_2}{\sqrt{J_1}} \left(1 + \frac{A}{\sqrt{T}} \right) \right] = O_L(\epsilon^7). \tag{A.76}$$

Indeed $Q_1 = 2Q_2 + 20$ and $A/\sqrt{T} = \epsilon^{-3}$. This averaging uses the actual distribution of every past prefix. There is no division by a probability of a rare source prefix.

Define one comparison source process on the same proposal grid, starting at the actual initial assignment and using row $\mathcal{B}_{j\cdot}(\vartheta_0(t))$, where $\vartheta_0(t) = 2 - e^{-\epsilon t}$. This is a single comparison law common to both actual processes. The actual dynamics continue to use their own laws and coefficients. At a proposal time t, let σ be the most recent preceding proposal or activation. Condition on the common actual starting sample, the full future grid, and all previous proposal marks, including self marks. The assignment and active level are fixed on (σ, t); no future continuous trajectory has been conditioned on. Starting the auxiliary diffusion from the actual fast point at σ, use the environmental coupling, then eq. (A.69), then the Lipschitz comparison of $\vartheta(t)$ with $\vartheta_0(t)$. For gaps at least

$$h_* = \frac{k_L}{J_1} \log(1/\epsilon), \tag{A.77}$$

the averaged conditional row error is at most

$$C_L\left[(1+A\sqrt{J_1/T})e^{-c_{\mathrm{fast},L}J_1h_*}+\epsilon/J_1+\eta_\epsilon+r_\epsilon\right]. \tag{A.78}$$

The moment of the point-start mixing factor includes $\sqrt{J_1/T}(A+\sqrt{T})$; its $\sqrt{J_1}$ term is absorbed by $A\sqrt{J_1/T}$ because $A/\sqrt{T}\geq 1$. After averaging over the actual past marks, their conditional expectations return the grid-conditioned moments exactly.

Short gaps use the trivial total-variation bound one. For a rate-one Poisson grid, the expected number of ordered pairs of proposal times less than h_* apart is at most H_*h_*, by direct integration of the two-time intensity on $\{0<s<t<H_*, t-s<h_*\}$. This bounds the count of short consecutive gaps. Proposals within h_* after the initial time or one of the finitely many activations add at most $(L+1)h_*$. The expected total proposal count is H_*.

For clarity, the passage from conditional row errors to the entire history uses a finite-chain identity, rather than pointwise marginal agreement. Conditional on a grid with N proposals, write the two joint mark laws as products of their conditional kernels. Telescope by replacing kernels in order while keeping the earlier prefix distributed according to the actual law and the later tail according to the comparison law. Integration of the tail, whose mass is one, gives

$$\|\mathcal{L}(J_1,\dots,J_N)-\mathcal{L}(\overline{J}_1,\dots,\overline{J}_N)\|_{\mathrm{TV}}\leq\sum_{m=1}^{N}\mathbb{E}_{\text{actual prefix}}\|K_m(\cdot\mid J_{<m})-\overline{K}_m(\cdot\mid J_{<m})\|_{\mathrm{TV}}. \tag{A.79}$$

The J_m in this display are proposal marks, not energy strengths. Integrating the grid and using the preceding bounds gives

$$\begin{aligned}\mathcal{E}_{\mathrm{hist}}\leq C_L\big[&(H_*+L+1)h_*+H_*(1+A\sqrt{J_1/T})e^{-c_{\mathrm{fast},L}J_1h_*}\\&+H_*\epsilon/J_1+H_*\eta_\epsilon+H_*r_\epsilon\big].\end{aligned} \tag{A.80}$$

With $c_{\mathrm{fast},L}k_L>Q_1/2+12$, the five contributions are, respectively, bounded by constants times $\epsilon^{19}\ell_\epsilon$, ϵ^8, ϵ^{Q_1}, ϵ^6, and $\epsilon^{2Q_1+19}+\epsilon^{3Q_1+25}\ell_\epsilon^2$. Thus $\mathcal{E}_{\mathrm{hist}}\leq C_L\epsilon^6$.

For each common actual initial sample and grid, maximally couple the two finite mark sequences to the comparison sequence. Glue the two couplings through that common comparison law. This preserves the actual source initial sample and its exact parent projection. Fill in each continuous trajectory from its original regular conditional distribution given its initial sample, grid, and marks. Such distributions exist on finite-dimensional continuous path spaces and finite-state right-continuous path spaces, which are standard Borel spaces. Every actual full-process marginal is therefore preserved. Forgetting self proposals makes the source path a function of this grid and mark sequence and cannot increase the disagreement probability. Summing over finitely many activations changes only C_L.

This coupling need not be causal. It makes no total-variation claim for the continuous trajectories. Continuous-path estimates can be combined with it by retaining each original marginal path

event; they cannot be justified by treating Brownian motion conditioned on future marks as a new independent Brownian motion. □

A.5 Actual coherence, source movement, and continuous paths

Lemma A.11 (Successive coherence with the original conditional marks). Under the preparation and regime of eqs. (16.30) to (16.32), there are fixed positive constants η_L, c_L, C_L and a finite $k_{\mathrm{geom},L}$ such that the windows $I_s = [t_s, 2t_s]$, $t_s = k_L \log(1/\epsilon)/J_s$, with $k_L > k_{\mathrm{geom},L}$, have the following property. Except on an event of probability at most $C_L(1 + H_*)\exp(-c_L/\epsilon^6)$, the fine process and its actual successive parent descriptions have coherent components $3\cdot 2^{L-1} \to 2^{L-1} \to \cdots \to 1$. On I_s, residuals of levels at most s and the local discrepancies d_i are below the fixed tolerance $\eta_L A$, whereas future bridges remain separated by a fixed positive multiple of A. For eliminated coordinates this statement uses the conditional root-mean-square original residuals in eq. (16.19), separately for the physical posterior and restricted optimizer. Final coherence persists through H_*. All loop coordinates remain retained. [ESTABLISHED]

Proof. Write the non-root Gaussian comparison of lemma A.2 on active level $p < s$ as a current normal component $u \in \mathcal{N}_p \cap \mathcal{N}_s^{\perp}$, a future component $y \in \mathcal{N}_s$, and the separate loop component. The weighted flag gives, for the current/future blocks of H_p,

$$F \succeq c_L J_s I, \qquad \|G\| \le C_L J_{s+1}, \qquad \|D\| \le C_L J_{s+1}, \qquad J_{L+1} = 0. \tag{A.81}$$

Indeed rows of levels at most s annihilate $\mathcal{N}_s$. Consequently the original cross and future blocks contain only strengths at most J_{s+1}. In an earlier elimination through p, the eliminated-current block is $O_L(J_{p+1})$, its inverse hidden block is $O_L(J_p^{-1})$, and the eliminated-future block is $O_L(J_{s+1})$. Their Schur correction is therefore still $O_L(J_{s+1})$, since $J_{p+1} \le J_p$. For $p = 0$ there is no such correction. Loop/normal cross blocks vanish in the Gaussian core.

Let μ^G be the deterministic reference mean. Its norm is at most $C_L A$, and its loop component is zero. Integrating the current equation $\dot{\mu}_u^G = -\vartheta(t)(F\mu_u^G + G\mu_y^G)$, using $1 \le \vartheta \le 21/10$, gives

$$|\mu_u^G(t)| \le C_L A \left[e^{-c_L J_s (t - t_0)} + \frac{J_{s+1}}{J_s} \right]. \tag{A.82}$$

The full future row has norm $O_L(J_{s+1})$. Integrating that row from the original common center and noting that earlier projections preserve $P_s \mu^G$ gives, for a sequential process begun at τ_{s-1},

$$|\mu^G(t) - A P_s v_0| \le C_L A \left[e^{-c_L J_s (t - \tau_{s-1})} + \frac{J_{s+1}}{J_s} + J_{s+1} t \right]. \tag{A.83}$$

The future terms are absent when $s = L$. For a fine process use $t_0 = 0$ in eq. (A.82).

Here is the required uniform-in-time fluctuation bound. On each active non-root space, $Z = X^G - \mu^G$ obeys $dZ = -\vartheta(t) H_p Z dt + \sqrt{2T} dW$, with $\vartheta H_p \succeq h_L I$ and initial covariance at most $C_L T H_p^{-1}$. Choose $a_* > 0$ below h_L and small enough for the initial Gaussian exponential moment. For

$V(z) = \exp(a_*|z|^2/(4T))$, put $v = a_*|z|^2/(4T)$. Direct differentiation gives

$$\mathcal{L}_t V = \left[\frac{a_* d}{2} + \frac{a_*^2|z|^2}{4T} - \frac{a_*\vartheta z^T H_p z}{2T}\right] V \le a_*(d/2 - v)e^v \le C_L. \tag{A.84}$$

The initial expectation of V is bounded. Orthogonal retention decreases V, so the same estimate survives the finite sequence of projections. Apply Itô's formula up to the first exit from a ball, localize the coefficients, and use the nonnegative stopped value of V. Removing the localization yields, for every $\rho > 0$,

$$\Pr\left\{\sup_{t\le H_*} |X_t^G - \mu^G(t)| > \rho A\right\} \le C_L(1 + H_*)\exp\left[-c_L\rho^2 A^2/T\right]. \tag{A.85}$$

Higher stiffnesses only strengthen the negative term in eq. (A.84); no upper stiffness enters this probability estimate.

Let M_L bound the norms of the finitely many original residual and d_i covectors. Choose η_L smaller than a fixed fraction of m_L and then a fixed fluctuation radius ρA so that $M_L\rho A$ is smaller than the reserved tolerance margin. Take, for example, $k_{\mathrm{geom},L} = 2/c_L$ for the restoring constant in eq. (A.82); increasing this finite constant is allowed. For sufficiently small ϵ, $\tau_{s-1} < t_s/2$, so the exponential term in eq. (A.83) tends to zero uniformly on I_s. Also

$$\frac{J_{s+1}}{J_s} \longrightarrow 0, \qquad J_{s+1}t_s \longrightarrow 0 \quad (s < L), \qquad \frac{\tau_L}{H_*} \longrightarrow 0. \tag{A.86}$$

The nonlinear correction from eq. (A.19) is at most $C_L A\epsilon^{5/2}$, including the allowed initial transport and finitely many inherited errors. This bound holds for every adapted assignment path. Since $A^2/T = \epsilon^{-6}$, the strict prepared residual margins and eq. (A.85) now give the asserted probability and separation on every window. The stable current-block equation continues to bound previously formed residuals as their future forcing decreases. After the last normal block disappears there is no future forcing. Thus the final coherent regime persists through H_* on the same event.

It remains to check what an eliminated edge means. For a cumulative elimination through $p \ge 1$, write a fine non-root state as $z + y$, where $z \in W_p^\perp$ and $y \in M_p$. The optimizing Gaussian precision is ϑH; the physical one is $\vartheta H - TR$. Every row of b_d belongs to the first normal layer, so $R = b_d^T b_d/(1 - c)$ annihilates $\mathcal{N}_1$, $\mathcal{H}$, and $\mathcal{K}$. The correction therefore has no hidden/future or hidden/loop block. The admitted inequality $TR \preceq H/2$ preserves a hidden precision lower bound $c_L J_p I$ for both laws. Their hidden/future block is $O_L(J_{p+1})$, while the hidden/loop block is zero.

Consequently the corresponding Gaussian conditional mean has norm at most $C_L(J_{p+1}/J_p)|y_{\mathcal{N}_p}|$ and its covariance trace is at most $C_L T/J_p$. Conditional on the full assignment, each exact hidden density is a normalized tilt of its own Gaussian conditional by a factor with log oscillation at most B_0. Its normalized density ratio is at most e^{B_0}. For every retained state with $|y_{\mathcal{N}_p}| \le C_L A$,

$$\mathbb{E}_{\mu_\vartheta}[|z|^2 \mid P_p X = y, S = j] \le C_L e^{B_0}\left[A^2\left(\frac{J_{p+1}}{J_p}\right)^2 + \frac{T}{J_p}\right], \quad \mu_\vartheta \in \{\mathbf{\Pi}_{\vartheta,\mathrm{o}}, \pi_\vartheta\}. \tag{A.87}$$

The normal-state bound holds on the event already constructed. For every original residual covector r_e, the triangle inequality and its reverse in conditional L^2 give

$$\left|\sqrt{\mathbb{E}_{\mu_\vartheta}[r_e(X)^2 \mid P_pX = y, S = j]} - |r_e(y)|\right| \le \|r_e\|\sqrt{\mathbb{E}_{\mu_\vartheta}[|z|^2 \mid P_pX = y, S = j]}. \tag{A.88}$$

The right side is $o(A)$ under the common regime, uniformly in the full assignment and separately for the two laws. The same argument applies to d_i. It transfers both the small formed residuals and the separated future residuals to eq. (16.19). No erased mark is replaced by its value at P_px, and no complete mixture KL is identified with a scalar residual. The bridge levels join successive pairs of already formed components, giving the stated counts. □

Lemma A.12 (Actual source envelope and changing hidden kernels). Every actual process or finite replacement lineage in lemma A.2 satisfies eq. (16.39) for the distinguished binary source, measured from any of its actual activation times. Between $\vartheta = 1$ and its final learned parameter, the first hidden physical conditional and the first hidden optimizing conditional each change by KL greater than $1/200$, uniformly at the same retained input and full assignment. [ESTABLISHED]

Proof. At the distinguished receiver the posterior log odds equal

$$\log \frac{\beta^P_{i_0,+}(x,\vartheta)}{\beta^P_{i_0,-}(x,\vartheta)} = 4(\vartheta - 3/2) - \delta\left[f\big(\sqrt{J_1/T}r_{i_0,+}(x)\big) - f\big(\sqrt{J_1/T}r_{i_0,-}(x)\big)\right]. \tag{A.89}$$

Since $0 \le f \le 1$, this row lies between $\sigma(4(\vartheta - 3/2) - \delta)$ and $\sigma(4(\vartheta - 3/2) + \delta)$, where $\sigma(u) = (1 + e^{-u})^{-1}$. Summing the exact inherited proposal row over all assignments with distinguished label $+$ and taking the actual expectation preserves these bounds. Apply the generator to the indicator of that label. With $\lambda(t) = \Pr\{S_{i_0}(t) = +\}$, the result is

$$\lambda' = b_\lambda - \lambda, \qquad \sigma(4(\vartheta - 3/2) - \delta) \le b_\lambda(t) \le \sigma(4(\vartheta - 3/2) + \delta). \tag{A.90}$$

Neither the diffusion nor a marked projection changes this indicator. Thus variation of constants remains valid across every activation, even when the inherited row depends on the current assignment and continuous state.

By lemma A.3, uniformly before $\tau_L = 2k_L \log(1/\epsilon)$, $\vartheta \le 1 + C_L\epsilon\log(1/\epsilon) + C_Lr_\epsilon$. A solution of eq. (A.90) cannot exceed the larger of its starting value and the supremum forcing. The preparation tolerance and $\|\sigma'\|_\infty \le 1/4$ give

$$\lambda(\tau_s) \le \sigma(-2) + 10^{-2} + \delta/4 + C_L[\epsilon\log(1/\epsilon) + r_\epsilon], \qquad 0 \le s \le L. \tag{A.91}$$

There is no estimate on the number of proposals in this argument. For small ϵ, every activation precedes $H_* - 10$ and $\vartheta \ge 7/4$ throughout $[H_* - 10, H_*]$. Nonnegativity of the earlier contribution in variation of constants yields

$$\lambda(H_*) \ge (1 - e^{-10})\sigma(1 - \delta). \tag{A.92}$$

For $\delta < 10^{-4}$, the elementary estimates $e^{1-\delta} > 249/100$, $e^2 > 7$, and $e^{10} > 2^{10}$ imply

$$(1-e^{-10})\sigma(1-\delta) - \sigma(-2) - 10^{-2} - \delta/4 > \frac{1023}{1024}\frac{249}{349} - \frac{1}{8} - \frac{1}{100} - \frac{1}{40000} > \frac{57}{100}. \tag{A.93}$$

The remaining error in eq. (A.91) tends to zero. Hence $\lambda(H_*) - \lambda(\tau_s) > 1/2$ for every admitted activation. For a binary row this is its total-variation change. The calculation uses the actual recognition marginal throughout.

For the hidden-kernel statement, fix the retained input and full assignment, and first omit the bounded source factor. On the first hidden normal space, let D_1, D_f be the initial and final Gaussian precisions, with covariances TD_1^{-1}, TD_f^{-1}. For the optimizer the precision scales with ϑ. For the physical law, on the non-root space,

$$D_1 = H - TR \succ 0, \qquad D_f = D_1 + (\vartheta_f - 1)H \succeq \vartheta_f D_1, \qquad \vartheta_f \geq 7/4. \tag{A.94}$$

Restriction to a conditional block preserves the order. If an earlier Gaussian marginal is taken, its Schur quadratic is the minimum over the discarded coordinates, which also preserves this order. In either case the weaker common bound $D_f \succeq (5/4)D_1$ holds. The conditional Gaussian means may differ; equality of means is not needed.

For Gaussian laws G_1, G_f on a hidden space of dimension $k \geq 1$, whiten D_1 and diagonalize the conjugated final precision, whose eigenvalues satisfy $\rho_i \geq 5/4$. Completing the Gaussian square in their Hellinger affinity gives

$$\mathcal{A}(G_1, G_f) = \prod_{i=1}^{k} \left(\frac{2\sqrt{\rho_i}}{1+\rho_i}\right)^{1/2} e^{-M}, \qquad M \geq 0. \tag{A.95}$$

The factor decreases for $\rho_i \geq 1$, so this is at most $[2\sqrt{5/4}/(1+5/4)]^{k/2}$. For each of the two exact conditionals k_1, k_f, its normalized source tilt relative to its own Gaussian conditional is bounded above by e^{B_0}, including after exact hidden integration. Therefore $\mathcal{A}(k_1, k_f) \leq e^{B_0}\mathcal{A}(G_1, G_f)$. Jensen's inequality under k_1 gives $-D_{\mathrm{KL}}(k_1 \| k_f)/2 \leq \log \mathcal{A}(k_1, k_f)$ and consequently

$$D_{\mathrm{KL}}(k_1 \| k_f) \geq \frac{k}{2}\log\frac{81}{80} - 2B_0 \geq \frac{1}{162} - \frac{1}{5000} > \frac{1}{200}. \tag{A.96}$$

Here $\log(1 + 1/80) \geq 1/81$. The means and bounded tilts make the likelihood-ratio use integrable. This argument applies separately to the physical and optimizing conditionals, uniformly in retained input and assignment. It does not reset an actual hidden law. □

Lemma A.13 (Continuous displacement compatible with source histories). For an actual source and its exact parent started from a common actual retained sample and parameter at τ_s, the coupling of lemma A.10 can be extended without changing either full process marginal so that

$$\Pr\left\{\sup_{\tau_s \leq t \leq H_*} |P_s X_t - \overline{X}_t| > C_L A \epsilon^2\right\} \leq C_L(1 + H_*)e^{-c_L/\epsilon^2}. \tag{A.97}$$

Its common actual activation information and complete assignment history bound are retained. A finite number of successive replacements changes only the constants. [ESTABLISHED]

Proof. First compare Gaussian means at the same deterministic parameter path. In the current/future split of eq. (A.81), the inherited normal precision is $H_s = D - G^T F^{-1} G$. Put $v = \mu_u^G + F^{-1} G \mu_y^G$. The source future mean satisfies

$$\dot{\mu}_y^G = -\vartheta H_s \mu_y^G - \vartheta G^T v, \qquad |v(t)| \le C_L A \left[e^{-c_L J_s (t-\tau_s)} + \frac{J_{s+1}}{J_s} \right]. \tag{A.98}$$

The second bound follows from eq. (A.82) and $\|F^{-1}G\| \le C_L J_{s+1}/J_s$. Integrating the contractive future propagator from a matching retained mean gives

$$\sup_{\tau_s \le t \le H_*} |\mu_y^G(t) - \overline{\mu}^G(t)| \le C_L A \left[\frac{J_{s+1}}{J_s} + H_* \frac{J_{s+1}^2}{J_s} \right]. \tag{A.99}$$

There is no future normal contribution when $s = L$. The loop Gaussian core is unchanged by the reduction and has zero mean.

Now allow the independently learned parameters to differ. Write $\chi = \sup |\vartheta - \overline{\vartheta}| \le C_L r_\epsilon$ by lemma A.3. For homogeneous parent means $m, \overline{m}$ and their difference d, Young's inequality in the H_s quadratic form gives

$$\frac{d}{dt}|d|^2 \le -\vartheta d^T H_s d + C_L \chi^2 \overline{\vartheta} \overline{m}^T H_s \overline{m}. \tag{A.100}$$

Since $\frac{d}{dt}|\overline{m}|^2 = -2\overline{\vartheta}\overline{m}^T H_s \overline{m}$, the time integral of the final energy is at most $C_L A^2$. The extra displacement is therefore at most $C_L A \chi$; no largest-stiffness factor times the horizon is introduced.

Attach to each correct actual marginal its own Gaussian comparison from lemma A.2. At activation its reference starts from the projected preceding Gaussian reference, while the actual parent starts from the full actual retained sample. Their inherited discrepancy is already included in eq. (A.19). Apply eq. (A.85) to each centered reference with $\rho = \epsilon^2$, and then use the two nonlinear force bounds. The triangle inequality gives, outside a set of probability $C_L(1 + H_*)e^{-c_L/\epsilon^2}$,

$$\sup_{\tau_s \le t \le H_*} |P_s X_t - \overline{X}_t| \le C_L A \left[\epsilon^2 + \epsilon^{5/2} + r_\epsilon + \frac{J_{s+1}}{J_s} + H_* \frac{J_{s+1}^2}{J_s} \right]. \tag{A.101}$$

For $s < L$, the last term equals $2\epsilon^{19}$ and $J_{s+1}/J_s = \epsilon^{Q_{s+1}+20}$. The quantity $r_\epsilon = A^2 J_1 + T[1 + \log(1/\epsilon)]^2$ is $o(\epsilon^2)$. Both future terms vanish at $s = L$. This proves eq. (A.97) for the non-root coordinates. The autonomous root law and dynamics are identical and independent of the source history, so their root coordinates can be coupled identically.

Finally take the coupling of complete source histories and common activation information from lemma A.10. Extend it by the conditional law of each actual path together with its own reference, given that process's history and activation information. These conditional probability kernels exist on the finite-dimensional continuous-path and finite-state path spaces, which are standard Borel. Each process-reference pair retains its correct marginal joint law. The force comparison holds

for every adapted assignment path, and each reference concentration event retains its marginal probability under this extension. Their union bound therefore gives the same displacement estimate on the history coupling. This argument does not declare Brownian motions to remain shared after conditioning on source histories. Common activation samples are preserved, and no claim of total variation convergence of continuous path laws is required. A finite sequence of replacements uses a finite triangle inequality and union bound. □

A.6 Joining the estimates in one regime

The proofs use the single regime $Q_L = 0$, $Q_s = 2Q_{s+1} + 20$, $J_s = \epsilon^{-Q_s}$, $A^2 = \epsilon^{3Q_1+20}$, and $T = A^2\epsilon^6$. Choose the fixed entrance constant to satisfy

$$k_L > \max\left\{k_{\mathrm{geom},L}, \frac{2(Q_1+20)}{c_{\mathrm{early},L}}, \frac{Q_1+8}{c_{\mathrm{joint},L}}, \frac{Q_1/2+12}{c_{\mathrm{fast},L}}\right\}. \tag{A.102}$$

Each denominator is positive for this finite graph and fixed nonlinear coefficient. The first condition supplies the geometric windows, the second the early conditional-entropy entrance, the third terminal joint relaxation, and the fourth the complete source-history estimate. For sufficiently small ϵ these windows are ordered, their activations precede H_*, and the same admitted preparation satisfies every estimate. Thus the separate bounds prove the conjunction in theorem 16.1; no estimate uses a different temperature, source rate, or initialization. [ESTABLISHED]

Normalization, marked pushforward composition, inherited proposal rows, and the objective chain rule are exact identities at each positive parameter instance. Formation probabilities, conditional entropy, complete covector error, parameter comparison, and path couplings are asymptotic bounds in the displayed regime. The positive nonlinear coefficient and source proposal rate remain fixed, as does $J_L = 1$. All constants and neighborhoods may depend on the finite depth. Extension to mixed-level source support, uniform depth, or base-manifold transport is not proved here. [NOT-CLAIMED]

Appendix B

Proofs for the constructive spatial RG family

This appendix proves the constructions in chapter 17. The geometric inventories, reference measures, and parameter meanings are those of that chapter. The symbol μ denotes a lattice probability law here, not a Gaussian mean. [DEFINITION]

B.1 The source measure and boundary comparison

Proof of theorem 17.2. For fixed matter signs the coordinate change $u \mapsto z$ is a bijection, preserving normalized counting. Its inverse is $u_e = \phi_x z_e \phi_{x+e_a}$. Thus the finite raw law is independent uniform matter times its dressed law in these coordinates. The exterior η is dressed data. Varying an exterior matter frame while holding η fixed requires changing its reconstructed raw links as well.

For $\Delta \subset \Lambda$, conditioning on $z_{\Lambda\setminus\Delta}$ cancels the edge factors outside Δ and the plaquettes not meeting Δ. All remaining factors are exactly W_Δ with the completed exterior. Hence the finite specification is pointwise consistent. Its one-edge conditional plus probability is

$$q_e = \frac{1+\tanh H_e}{2}, \qquad H_e = h + j\sum_{p\ni e}\prod_{f\in p\setminus\{e\}} z_f, \qquad |H_e| \le h + 2j. \tag{B.1}$$

Every conditional sign probability is at least $\epsilon_f = (1-\rho)/2 > 0$. Averaging over any unrevealed variables preserves this bound, so a specified assignment on A has probability at least $\epsilon_f^{|A|}$.

Let $n(e,f)$ count plaquettes containing both distinct edges. Each edge has two plaquettes and three other edges in each, whence $\sum_f n(e,f) = 6$. Changing one plaquette contribution gives

$$\frac{|\tanh(s+j) - \tanh(s-j)|}{2} = \frac{\sinh(2j)}{\cosh(2s)+\cosh(2j)} \le t. \tag{B.2}$$

Telescoping changes gives influence matrix $D_{ef} = tn(e,f)$, with row sums at most $\alpha = 6t < 1$.

A random-coordinate heat bath on the finite free set preserves its Gibbs law by conditional factorization. For $n = |\Lambda|$, updating each coordinate in a prescribed order to any target has probability at least $(\epsilon_f/n)^n$. Therefore every entry of the n-step transition matrix has that positive lower bound. Removing its common row mass gives a strict total-variation contraction, proving uniqueness of the

stationary law. Couple two such chains by the same chosen edge and the same uniform number in their conditional Bernoulli rows. Cesaro averages of the finite pair chain have a convergent subsequence; the limit is stationary because one application changes the average by a difference divided by its length. Its marginals are the two identified Gibbs laws. If d_e is its probability of disagreement, stationarity cancels the unchanged update probability and gives

$$d_e \le \sum_{f\in\Lambda} D_{ef} d_f + \beta_e, \qquad \beta_e = \sum_{f\notin\Lambda} D_{ef}\mathbf{1}_{\{\eta_f\neq\zeta_f\}}. \tag{B.3}$$

Iteration removes the remainder $D_\Lambda^n d$, bounded by α^n. The surviving sum consists of paths making their first exit from Λ at a differing exterior edge. Dividing each transition weight by α, and adding a cemetery state for missing row mass, makes their weights disjoint first-exit probabilities whose sum is at most one. Every exit path from e has at least $D(e,\Lambda^c)$ steps, so $d_e \le \alpha^{D(e,\Lambda^c)}$. A union bound on target disagreements proves eq. (17.2).

A larger finite-volume restriction is a mixture of the smaller specifications, by finite consistency. Integrating the same bound over any two boundary mixtures proves comparison across volumes containing a common interior. On growing boxes each fixed cylinder is consequently Cauchy, independently of all exterior choices. Its limit is normalized, consistent, and has every atom at least $\epsilon_f^{|A|}$. To construct a measure, enumerate the countable edges. Split $[0,1)$ into half-open intervals with the probabilities of the first edge, then split each interval into the joint probabilities of the next edge. Consistency makes every subdivision exact. Each point specifies a unique infinite sign sequence; its cylinder preimages are finite unions of intervals. The pushforward of Lebesgue measure is a countably additive law with these cylinders. Cylinders form a pi-system, and the pi-lambda argument makes the extension unique.

For a cylinder function f, the finite update $\gamma_\Lambda f$ remains a cylinder function, because the interaction boundary is finite. Passing finite compatibility to the cylinder limit gives $\mu f = \mu\gamma_\Lambda f$. Cylinder determination extends the identity to bounded measurable functions. For exterior-measurable g, $\gamma_\Lambda(gf) = g\gamma_\Lambda f$, so this is the conditional DLR identity. Any two DLR laws have target marginals that are mixtures of the same finite specifications. The boundary bound tending to zero makes all their cylinders equal, hence their measures equal. Translation of the unique law is another law for the same specification and therefore agrees. The conditional law in an ordinary box embedded in a full torus has the same factors and a random dressed exterior. The same comparison proves the torus cylinder limit while retaining all winding sectors.

For the raw lift, append independent uniform matter signs and apply the inverse coordinate map. A finite raw window with free matter vertices and all their incident links converts to independent free frames and the correct dressed conditional. Such windows are cofinal; conditioning further proves the raw DLR equations for every finite window. Conversely, a local gauge flip of one matter vertex and its incident links preserves each raw conditional density and reference. Any raw Gibbs law is therefore invariant under every finite product of these flips. In dressed coordinates this flips arbitrary finite sets of matter signs without changing z. Averaging those flips proves that every finite matter assignment has probability $2^{-|V|}$ independently of every dressed cylinder. Its dressed marginal has

the already identified DLR specification, hence equals $\mu_{b,t}$. This proves uniqueness of the raw lift. □

B.2 Constrained parity laws uniformly in blocking depth

Proof of theorem 17.6. Partition the fine edges into the retained B-paths and unused singleton edges. No plaquette contains two edges of one retained path. A coarse anchor cell contains two retained paths and $2B(B-1)$ unused singleton blocks, hence $2B^2$ fine edges. A pattern prescribes each retained block to be unconstrained or to have one specified parity; singleton blocks remain unconstrained. Use the product of uniform probability references on these supports, multiply all edge and touching plaquette factors, and normalize. These are positive finite specifications on their actual supports. A block row with at most B edges has every allowed atom at least

$$\xi_B = 2^{-B}\left(\frac{1-b}{1+b}\right)^B\left(\frac{1-t}{1+t}\right)^{2B} > 0. \tag{B.4}$$

Two product sign laws obtained from one positive product law by conditioning on opposite total parities admit a coupling at Hamming distance exactly one. For one bit this is immediate. For n bits, couple the last-bit marginals by any coupling. If those bits differ, the required prefix parities agree; couple the prefixes identically. If they agree, apply the induction coupling to the opposite prefix parities. In either branch the prefix marginal on each side is its correct product law conditioned on the required parity, independently of the other branch. This proves the induction including both marginals.

Changing a single coordinate field from $+j$ to $-j$ in a product row or fixed-parity product row changes that coordinate's marginal by at most t, by eq. (B.2) with the other-coordinate masses absorbed into s. Couple it maximally. If it differs in a fixed-parity row, the remaining prefix parities are opposite, so the preceding coupling adds exactly one further discrepancy. The expected Hamming cost is at most $2t$; a one-bit fixed fiber has zero cost. Successive tilts can be coupled by composing their conditional coupling kernels. The endpoint Hamming distance is bounded by the sum of successive distances, proving the required triangle inequality directly.

For target block i and exterior fine edge e, let $n(i,e)$ be the number of plaquettes meeting both. An exterior flip changes at most $n(i,e)$ single-coordinate fields. The preceding coupling gives a row cost at most $2t\sum_e n(i,e)\mathbf{1}_{\{z_e\neq z'_e\}}$. The column identity, not a block-state row bound, is $\sum_i n(i,e) = 6$: each of the two plaquettes of e meets three other blocks. Incident anchors differ by at most one.

Finite block heat baths have unique stationary laws by the common-mass argument using ξ_B. A stationary pair coupling exists by the same Cesaro construction as above. Let d_e be fine-edge disagreements and $D_i = \sum_{e\in i} d_e$. Canceling the update probability separately in each block gives

$$D_i \le 2t\sum_{e\ \text{free}} n(i,e)d_e + 2t\sum_{e\ \text{exterior}} n(i,e)\mathbf{1}_{\{\eta_e\neq\zeta_e\}}. \tag{B.5}$$

For target anchor set A, multiply by $w_i = e^{-\theta \operatorname{dist}_\infty(\operatorname{anchor}(i),A)}$ and sum. The column identity and $w_i \le e^{\theta} w_{\operatorname{block}(e)}$ when $n(i,e) > 0$ yield

$$S \le qS + q \sum_{e \in \partial_f \Lambda} w_{\operatorname{block}(e)} \mathbf{1}_{\{\eta_e \neq \zeta_e\}}, \qquad S = \sum_{i \text{ free}} w_i D_i. \tag{B.6}$$

Use all blocks anchored in the radius R neighborhood of A. Every contributing exterior edge is anchored in its next shell, whose cardinality is at most $8|A|(R+1)$, because a square shell around one cell has exactly $8(R+1)$ cells. There are $2B^2$ fine edges per anchor. Thus the exterior sum is at most $16|A|B^2(R+1)e^{-\theta(R+1)}$. Dividing by $1-q$ and taking a union bound over target disagreements proves eq. (17.6) for these finite constrained laws.

For every fixed full pattern, the bound makes all finite-cylinder limits Cauchy on expanding whole-block volumes. Interval subdivision constructs their measure. Finite compatibility passes to the limit, giving its DLR law. Any two such laws have cylinders within the same vanishing boundary bound and hence agree. The argument also compares patterns agreeing in a large interior; different supports outside it are only exterior forcing. No conditioning on an infinite event was used.

Finite conditioning on previously free path parities has positive probability by eq. (B.4). Restricting its finite conditional reference is precisely the updated support pattern; uniqueness identifies the resulting infinite law. In the all-unconstrained pattern the law is the original fine Gibbs law, because its finite DLR identities also hold on unions of blocks. Therefore its finite parity conditionals are these constrained laws. To define a target coarse kernel, leave its paths unconstrained, prescribe every exterior retained parity, and take the target marginal of the constructed constrained law.

Conditional on all other block states, the signs along a free path are independent, with conditional means bounded by ρ. Its parity mean has magnitude at most ρ^B, also after conditioning on any already revealed disjoint parities. Revealing the two target parities successively gives the asserted one-cell lower and upper probabilities. The boundary comparison makes every kernel continuous and measurable. Conditioning on a finite set of additionally revealed parities identifies the corresponding updated pattern; finite Bayes identities therefore give pointwise specification consistency at every exterior.

For an exterior cylinder event, the defining finite conditional integral holds under μ_B. Increase the revealed exterior set; uniform convergence permits passing the integral to the just constructed kernel. Cylinder determination extends this conditional identity to the full exterior sigma algebra. Thus μ_B is DLR for this everywhere-defined specification. Any two coarse DLR laws can be mixed over increasingly remote exteriors; the same uniform comparison makes their target cylinders equal. This proves uniqueness. Translation covariance follows by transporting factors and patterns and invoking constrained uniqueness. At $t = 0$, independent product-fiber laws give the separate exact control with coarse edge mean b^B; no formula dividing by q is needed. □

B.3 The complete canonical interaction

Proof of theorem 17.8. The one-cell bounds give log-odds amplitude $4\,\mathrm{atanh}(\rho^B) \le A_0\rho^B$. Atom differences are bounded by total variation, and $|\log u - \log v| \le |u-v|/m$ for $u, v \ge m$, so reference-state log odds have spatial variation at most $D_0B^2(R+1)e^{-\theta(R+1)}$. The canonical supports have diameter at most $2r$, and at most $(2r+1)^2$ translates contain the origin. Their maximal lexicographic cell uniquely specifies the anchor, so no assignments collide. Consequently

$$\|\Phi^{(B)}_{A_{x,r}}\|_\infty \le \min\{2A_0\rho^B, D_0B^2re^{-\theta r}\}, \qquad r \ge 1. \tag{B.7}$$

The inequality $\min(u,v) \le \sqrt{uv}$, the bound $\sqrt{r} \le r$, and $\lambda_v = \theta/8$ reduce the weighted shell sum to $\sqrt{2A_0D_0}B\rho^{B/2}\sum_{r\ge1} r(2r+1)^2z_v^r$. Differentiating the geometric series three times gives exactly $z_v(9+14z_v+z_v^2)/(1-z_v)^4$. This proves the norm bound; its explicit tail after N shells replaces that series with $\sum_{r>N} r(2r+1)^2z_v^r$. Since $u \le e^{\lambda_v u}/\lambda_v$, it also proves the first moment. Both terms in $\mathcal{E}_B$ tend to zero; for an exponential bound, $B\rho^{B/4} \le \rho^{1/4}/(1-\rho^{1/4})^2$.

For a configuration differing from $*$ on finitely many sites, change those sites in increasing lexicographic order. Positive finite specification consistency expresses its log probability ratio against the vacuum as the sum of the corresponding full-past log odds. The absolutely convergent canonical shells telescope to exactly those odds. Terms anchored outside the changed set vanish, since their anchor stays at $*$. Thus the interaction and specification give identical vacuum-exterior ratios. Compare two finite-support configurations and cancel their common exterior to obtain equality for every finite-support exterior. Approximate an arbitrary exterior by its values in larger boxes and vacuum outside. Quasilocality gives uniform convergence on the specification side; absolute summability of all terms touching the finite changed set gives uniform convergence of interaction energy differences. Positive normalization then gives equality for every exterior. Infinite total Hamiltonians have not been used as unnormalized densities.

A continuous specification of a full-support law is unique everywhere: a difference at one exterior would persist on a cylinder of positive probability, contradicting almost-sure equality of conditional versions. The canonical interaction therefore depends on the law. Path composition and this uniqueness give the stated action on its orbit, even if two indices happen to describe the same interaction. At the fair product law all odds are zero, so the interaction is zero and fixed. At $t = 0$ the singleton interaction is

$$\Phi^{(B)}_x(s_1, s_2) = -\sum_{a=1}^{2} \log \frac{1 + b^B s_a}{1 + b^B}, \qquad p_B = -2\log(1 + b^B), \tag{B.8}$$

where p_B is its normalized-counting pressure. This finite scalar is not zero despite vacuum normalization. □

B.4 The open-ball coefficient map and its law

Proof of theorem 17.10. The coefficient space is complete: a norm-Cauchy sequence converges in every anchored weighted ℓ^1 space, its coordinate limits agree on overlaps, and the uniform Cauchy bound passes to the supremum over anchors. Translation invariance is a closed coordinate condition. Finite translation-orbit families of arbitrarily large supports give linearly independent members, so the positive-radius ball is infinite dimensional.

Condition the fair fine reference on all retained a-path parities. Its blocks are independent uniform parity fibers and unused fair singletons. A character has conditional expectation zero unless its support is a union of whole paths; a partially occupied path is killed by flipping one included and one excluded sign, and an occupied unused singleton is killed by its own flip. A surviving support $\mathrm{lift}_a(T)$ has expectation χ_T. For an ordered list of source occurrences $S_1, \dots, S_n$, repetitions allowed, every product of moments in the joint cumulant is therefore either zero or the same character corresponding to $S_1 \triangle \cdots \triangle S_n$. Denote its constant coefficient by $c_a(S_1, \dots, S_n)$. The degree-n map is

$$P_{a,n}(J_1, \dots, J_n)_T = \frac{1}{n!} \sum_{S_1 \triangle \cdots \triangle S_n = \mathrm{lift}_a(T)} c_a(S_1, \dots, S_n) \prod_{i=1}^{n} (J_i)_{S_i}, \qquad T \neq \varnothing. \tag{B.9}$$

For finite source inventories this is differentiation of the finite conditional log partition, including its $1/n!$ and repeated occurrences.

Join two occurrence vertices when their supports meet a common reference block. Disconnected groups then depend on independent reference variables. The bounded dependency-graph cumulant inequality is $|c_a| \leq 2^{n-1}\,\mathrm{ST}(G)$, where ST counts spanning trees of this occurrence graph [31]. Its hypotheses hold uniformly in every parity assignment: the variables are sign characters of magnitude one, and the independence is at the block level. This is the finite inequality used here; convergence and the infinite-law identification are proved below.

Write $s_i = |S_i|$ and $d_i = \operatorname{diam} S_i$. For nonempty output T, disjoint paths give $a|T| \leq \sum_i s_i$. Along an occurrence tree, travel within a source at cost at most d_i, and between two source edges in one reference block at cost at most $a - 1$. The first fine edges of two output paths realizing its diameter then give

$$a \operatorname{diam} T \leq \sum_i d_i + (a-1)(n-1). \tag{B.10}$$

Fix an output edge and its first fine edge. At least one occurrence contains that fine edge, so sum over n possible roots. Pay $|T|$ by $a^{-1}\sum_i s_i$, summing over n marked occurrences. A child support touching a parent reference block contains an edge in the union of blocks met by the parent, at most as_i edges. Expanding this witness condition and summing from leaves costs a per tree edge and one parent support-size factor per child.

For an occurrence with k_i children, the weighted anchored sum is bounded by its input norm times $k_i!\delta_a^{-k_i}$. Indeed the unmarked residual factor is $s_i^{k_i-1}e^{-\delta_a s_i}$, and the marked one is $s_i^{k_i}e^{-\delta_a s_i}$; both are at most $k_i!\delta_a^{-k_i}$, using $(\delta_a s_i)^{k_i}/k_i! \leq e^{\delta_a s_i}$ and $s_i \geq 1$. The unused spatial weight only decreases

the bound. Summing $\prod_i k_i!$ over labeled trees with fixed root gives $(n-1)!\,\mathrm{Cat}_{n-1}$: ordering the children counts plane rooted trees, and the other $n-1$ labels are free. Balanced-parenthesis encoding gives $\mathrm{Cat}_{n-1} \le 4^{n-1}$. Combining the cumulant tree inequality, roots, mark, $1/n!$, and eq. (B.10) proves

$$\|P_{a,n}(J_1,\dots,J_n)\| \le \frac{n}{a} K_a^{n-1} \prod_{i=1}^{n} \|J_i\|. \tag{B.11}$$

All rearrangements were of positive majorants, so they are justified also for infinite source inventories.

Define $\mathcal{R}_a J = \sum_{n\ge 1} P_{a,n}(J,\dots,J)$ for $K_a\|J\| < 1$. The majorant gives locally uniform convergence in the Banach norm and locally uniform convergence of $D\mathcal{R}_a(J)H = \sum_{n\ge 1} nP_{a,n}(H,J,\dots,J)$ in operator norm. The multilinear increment expansion leaves a uniformly summable remainder with at least two increments; hence this is the actual Fréchet derivative, and its locally uniform series is continuous. The first-order survival rule gives L_a. The lift is injective, has size $a|T|$, contains the first edge of every output path, and has diameter at least $a\,\mathrm{diam}\,T$, so comparing anchored weights gives $\|L_a\| \le \ell_a$. For $x \le 1/2$, $\sum_{n\ge 2} nx^{n-2} \le 6$ and $\sum_{n\ge 2} n^2 x^{n-2} \le 22$, by differentiating the geometric series. These prove the quadratic bound with $6K_a/a$ and the derivative bound with $22K_a/a = 2M_a$, hence the stated bounds. The chosen radius makes the radial contraction at most $(1+\ell_a)/2$.

It remains to identify an actual normalized law. For any real input in the ball, its one-edge half-log-odds are $\sum_{S\ni e} J_S \chi_{S\setminus\{e\}}$, absolutely and uniformly convergent. Flipping f changes the Bernoulli row by at most $\sum_{S\ni e,f} |J_S|$. Its influence row sums are at most $\|J\| < 1$. Finite heat-bath coupling gives the resolvent comparison $d \le \sum_{m\ge 0} D^m \beta$. For a fixed target, the contribution from a remote boundary tends to zero: each fixed power has summable rows, and the remaining powers have a common geometric tail. Finite-cylinder subsequences exist by finite-state compactness; uniform convergence of finite local conditional energies passes their DLR equations to the limit. The comparison then makes the limit unique and makes the entire exhaustion converge.

Take finite edge boxes E_m and source truncations $J_S^{[m]} = J_S \mathbf{1}_{\{S\subset E_m\}}$. Each is a finite inventory and has no larger norm. No global norm convergence of these inhomogeneous truncations is asserted. Every fixed source tuple is eventually present; the rooted majorant therefore gives, for every fixed output edge, weighted incident convergence of $\mathcal{R}_a J^{[m]}$ to $\mathcal{R}_a J$. In the finite inventory the convergent scalar and nonempty Taylor series exponentiate to the exact conditional moment generating function near zero. The identity of the convergent analytic series extends over its real convergence interval, and that function is strictly positive there. Its real logarithm therefore has precisely these coefficients throughout the admitted ball. The scalar cancels only when the finite density is normalized.

The truncated fine law is the free-boundary Gibbs law on E_m, with fair signs outside. A path crossing its boundary retains those outside fair signs; incomplete characters are canceled rather than discarded by an altered channel. Its path push has exactly the finite coefficient expansion above. Incident convergence makes every finite coarse energy difference, hence every normalized conditional row, converge uniformly in the exterior. The fine laws converge locally to their unique

law and the path map preserves convergence of cylinder observables. Passing the finite coarse DLR integral to this limit proves that the path-pushed law is Gibbs for $\mathcal{R}_a J$. Its norm lies in the ball, so it is the unique coarse Gibbs law by the same comparison.

For completeness, every prescribed parity exterior has a continuous conditional version. In the whole-reference-block representation, changing one outside block gives an influence row sum bounded by $2a\|J\| < 1$, since a block has at most a fine edges. This estimate holds on every fixed-parity or unconstrained support, with all crossing sources present. Its summable resolvent constructs unique laws for every pattern and gives local continuity when remote patterns change. Free retained parity probabilities are at least $e^{-2a\|J\|}/2 > 0$. Finite parity conditioning corresponds exactly to these product-fiber supports. Testing the uniform limit of its finite conditional rows against exterior cylinders identifies it as a conditional version of the path-pushed law. The coefficient Gibbs kernel is also continuous. Since the law has full cylinder support, the two versions agree everywhere.

Finally, integrating the conditional half-log-odds at e against the fair exterior character $\chi_{S\setminus\{e\}}$ recovers J_S by orthogonality and uniform convergence. This proves uniqueness of every nonempty coefficient. Path composition, unique laws, and this extraction give $\mathcal{R}_2\mathcal{R}_3 = \mathcal{R}_3\mathcal{R}_2 = \mathcal{R}_6$. The same reasoning applies to any finite word with its directly defined path channel. Zero is the fair product interaction and is fixed. The original field and plaquette source has norm $e^{\kappa}h + 8e^{4\kappa+\lambda}j$, so the ball includes a nonempty region with both positive couplings. □

B.5 Complete finite Taylor calculations and their errors

Proof of theorem 17.12. Represent each nonempty support once modulo integer translation, by moving its lexicographically first edge basepoint to zero and retaining its axis. If its shape is S, size s, diameter d, and it contains $n_i(S)$ axis-i edges, exactly $n_i(S)$ distinct translates contain a specified axis-i edge. A nonempty finite set has no nonzero translation period. Therefore a finite library's exact norm is

$$\|J\| = \max_{i=1,2} \sum_S n_i(S)|S|e^{\kappa|S|+\lambda \operatorname{diam} S}|J_S|. \tag{B.12}$$

Duplicates are consolidated by exact addition before this formula is used. Only translations are identified; rotations and reflections are not.

At order n, let $d_{\max}$ and $s_{\max}$ be the largest input shape diameter and size. A connected source-occurrence list contributing an output with first edge at the origin stays within fine basepoint radius $R_n = nd_{\max} + (n-1)(a-1)$, by the occurrence-tree path argument. Thus every contributing literal instance belongs to the finite catalog of all input shapes translated by $[-R_n, R_n]^2 \cap \mathbb{Z}^2$. Reference-block ownership uses Euclidean division: a horizontal edge is retained exactly when its transverse coordinate is divisible by a, and its owner uses the floor of its parallel coordinate divided by a; vertical edges have the rotated rule. Negative coordinates follow the same floor rule. Unused singleton owners have a distinct type. A support is a union of complete paths exactly when every owner is retained and has all a distinct members.

For a set I of occurrence labels, let $m(I)$ be one if their XOR support is a union of complete paths, including the empty support, and zero otherwise. At all positive parities this is precisely its conditional moment. With r the first occurrence label in a nonempty I, the cumulant obeys

$$c(I) = m(I) - \sum_{\substack{D \subsetneq I \\ r \in D}} c(D) m(I \setminus D). \tag{B.13}$$

The partition block containing r uniquely decomposes the moment–cumulant formula; summing all other blocks gives the second moment factor. This proves the recurrence and its singleton base. Occurrence labels remain distinct even when their supports repeat. Computing all subsets takes finitely many exact integer operations; fewer than 3^n subset-pair contributions suffice.

For each order and each output first axis, enumerate every size-n multiset of literal catalog instances once. Keep connected occurrence graphs with nonempty complete-path XOR support whose output is already canonically anchored on that axis. A multiset with multiplicities m_τ contributes exactly $c(I) \prod_\tau J_\tau^{m_\tau} / \prod_\tau m_\tau!$: its $n!/\prod m_\tau!$ ordered occurrences cancel the $n!$ in eq. (B.9). Root choices and spanning trees are bounds, not extra multiplicities. Disconnected occurrences have zero cumulant by block independence. The radius argument loses no connected term. Output canonicalization is a selection of the already canonical output, not accumulation of all translated copies. Consolidation over all accepted terms gives the complete coefficient at every support through the specified order. Surviving supports satisfy $a|T| \le n s_{\max}$ and $a \operatorname{diam} T \le n d_{\max} + (a-1)(n-1)$, so the output is finite. Every finite catalog, multiset list, recurrence, and rational operation terminates at fixed order.

Summing eq. (B.11) over orders above N gives eq. (17.13) by differentiating the geometric tail $x^{N+1}/(1-x)$. Its positive-series representation makes it monotone in the norm and K_a, so certified upper bounds for both are valid despite the minus sign in the closed numerator. It tends to zero for $x < 1$. The zero input gives the zero table and zero tail.

All transcendental constants admit convergent rational enclosures. For a nonnegative rational x, put $t_0 = 1$, $t_{k+1} = t_k x/(k+1)$, $s_m = \sum_{k=0}^{m} t_k$. Once $v = x/(m+2) < 1$, positivity and decreasing successive term ratios give

$$s_m \le e^x \le s_m + \frac{t_{m+1}}{1-v}. \tag{B.14}$$

The width tends to zero. Reciprocal endpoints enclose e^{-x}. Applying these termwise to eq. (B.12) and eq. (17.11) gives convergent upper norm and lower radius bounds. Every strict admission or positive tolerance gap is therefore decided at some finite precision. The strict image margin and vanishing Taylor tail eventually admit the approximate image as well.

For one certified $r < \varepsilon$, the derivative estimate along a segment gives Lipschitz constant $L_a(r) < 1$, while the value estimate gives radial factor $q_a(r) = \ell_a + M_a r < 1$. The triangle inequality separates one step's Taylor error from its input error and gives the stated recurrence. Choosing every tail below both a prescribed small fraction of the total tolerance and $(1 - q_a(r))r/2$ keeps all approximate states in the ball and achieves the requested error for a finite word. Two differently truncated direct and staged tables need not be equal; their difference is bounded by the sum of their errors relative

to the same exact composed operator. The empty character was excluded only from the coefficient table. For example, an unused fine field has a nontrivial $\cosh$ normalizer although its retained interaction is zero; no evidence value can be inferred from that table. □

B.6 A constructive basin-entry theorem in the prescribed norm

Proof of theorem 17.13. The proof uses the specified compact box and retains the prescribed κ, λ. For $B \geq 3$, conditionally integrate all but the first and last edges of each retained path. No plaquette meets two eliminated edges, and the other three edges in its plaquette are outside every retained path. These interiors are therefore conditionally independent. For a horizontal path at the origin its conditional parity mean is

$$g_p = z_{(0,0,1)} z_{(B-1,0,1)} \prod_{s=1}^{B-2} f(u_s, v_s), \qquad f(u,v) = \tanh(h + j(u+v)), \tag{B.15}$$

where u_s, v_s are the three-edge characters in the adjacent plaquettes. Its strip consists of its endpoints, the horizontal edges at $(s, \pm 1)$, $1 \leq s \leq B-2$, and the vertical edges at $(s, 0), (s, -1)$, $1 \leq s \leq B-1$. It has $4B-4$ edges and diameter at most B. These strips and their rotations are pairwise disjoint. For horizontal edges, horizontal-strip interiors have first coordinate residues $1, \ldots, B-2$, whereas vertical-strip interiors have residues $0, B-1$; the transverse residues identify the unique adjacent grid line for $B \geq 3$. Retained endpoints belong only to their own path. Rotation proves the other orientation. This also proves conditional independence of all eliminated interiors.

Set $v = 2t/(1+t^2)$. The four Walsh coefficients of f are a_0, a_1, a_1, a_{12}, with

$$a_0 - a_{12} = b, \qquad 2a_1 = \frac{v(1-b^2)}{1-b^2v^2}, \qquad a_{12} = -\frac{bv^2(1-b^2)}{2(1-b^2v^2)}. \tag{B.16}$$

Here $a_0, a_1 \geq 0$, $a_{12} \leq 0$, so the unweighted coefficient sum is $R(b,v) = b + v(1-b^2)/(1-b^2v^2)$. It increases in v, and $\partial_b R \geq 1 - 2v/(1-v^2) > 0$ for $v \leq 24/145$, the value at $t = 1/12$. Thus its upper-endpoint value R_* bounds the compact box, with the explicit slack $1 - R_* \geq (13489/20449)(1-b_+) > 0$. Put

$$\Delta = 1 - R_*, \qquad \alpha = \Delta/24, \qquad \sigma = \frac{R_*}{1-\Delta/4} < 1, \qquad G_B = \frac{\sigma^{B-2}}{1-2\alpha}. \tag{B.17}$$

The fine Walsh norm with weight $e^{\alpha|S|}$, including its scalar, is submultiplicative under symmetric difference. Since u, v have three edges each, $\|f\|_\alpha \leq e^{6\alpha} R_* \leq \sigma$ and $\|g_p\|_\alpha \leq G_B$, using $e^x \leq 1/(1-x)$ for $0 \leq x < 1$, including the two endpoints.

For a finite path set P, the actual parity density relative to fair counting is

$$f_P(y) = \mathbb{E}_{\mu_{b,t}} \prod_{p \in P} (1 + y_p g_p). \tag{B.18}$$

Conditional independence of the eliminated interiors proves this identity; averaging over y gives one. The auxiliary fine law tilted by this product has the correct remaining-sign marginal for these observations. It is not the hard-parity posterior on all original fine variables. This typing matters for its normalized conditional rows below.

For a root path p_0 with fine anchor x_0, let $D_{p_0}(S,T)$ be the maximum fine-coordinate distance from x_0 of any fine-edge anchor in S, or of B times any coarse-edge anchor in T. An empty maximum is zero. Set $\beta = \kappa + 1$ and $\zeta = 2\lambda/B$, and complete the joint fine/coarse Walsh algebra in

$$\|F\|_{p_0} = \sum_{S,T} |F_{S,T}| e^{\alpha|S| + \beta|T| + \zeta D_{p_0}(S,T)}. \tag{B.19}$$

Fine signs and coarse labels are distinct variables. Cardinalities and rooted radii are subadditive under multiplication, proving the Banach algebra inequality. For a local update at a fine edge use the same norm centered there. Removing that edge lowers the fine weight by $e^{-\alpha}$; replacing it by a local polynomial increases the rooted radius by at most that polynomial's local radius.

Put $p = (1+\sigma)/2$, and increase B until $2\lambda/B < 1$ and $\sigma/(1 - 2\lambda/B) \le p < 1$. The original conditional mean at an edge has local norm at most p, because it reaches distance at most one. Choose rational upper bounds $A_\beta \ge e^\beta$, $A_{2\lambda} \ge e^{2\lambda}$, and $A_\kappa \ge e^\kappa$, and set $H_B = A_\beta A_{2\lambda} G_B$. Each observed strip has radius at most B from any of its members, hence local activity at most H_B.

A fine edge belongs to at most one observed strip. If it belongs to that strip, write $g_p = g_0 + z_e g_1$ and let f_e be its original conditional mean. Direct normalization of the two probabilities after multiplying by the complete observed factor gives

$$F_e = \frac{f_e + y_p(f_e g_0 + g_1)}{1 + y_p(g_0 + f_e g_1)} = f_e + \frac{y_p(1 - f_e^2) g_1}{1 + y_p(g_0 + f_e g_1)}. \tag{B.20}$$

If no observed strip touches the edge, use $F_e = f_e$; omit the target strip when computing its own conditional mean. Both $y_p g_0$ and $y_p g_1$ have local norm at most H_B. Increase B until

$$H_B \le \min\left\{ \frac{1}{2(1+p)}, \frac{1-p}{4(1+p^2)} \right\}. \tag{B.21}$$

The denominator perturbation then has norm at most $1/2$, so its geometric inverse has norm at most two. The correction has norm at most $2(1+p^2)H_B \le (1-p)/2$. Thus every normalized conditional mean has local norm at most $r = (1+p)/2 < 1$. The denominator and its even contribution have both been retained.

Let Π project onto terms with empty fine support. For nonempty S, define

$$\mathcal{A}(\chi_S y_T) = \frac{1}{|S|} \sum_{e \in S} \chi_{S \setminus \{e\}} y_T F_e, \qquad N = (1 - \Pi)\mathcal{A}, \tag{B.22}$$

and let $\mathcal{A}$ vanish on fine-constant terms. Removing the fine edge, using its local row norm, and averaging give $\|\mathcal{A}(1-\Pi)\| \le e^{-\alpha} r \le r$. The average is necessary to avoid a support-size factor.

Conditional expectation under the actual normalized finite tilted law satisfies $\mathbb{E}\chi_S y_T = \mathbb{E}\mathcal{A}(\chi_S y_T)$. Absorbing the fine-constant terms once and iterating proves

$$\mathbb{E}F = \Pi F + \sum_{k\geq 1} \Pi \mathcal{A} N^{k-1}(1-\Pi)F. \tag{B.23}$$

The unabsorbed remainder is bounded geometrically in the same norm. At any fixed substitution order only finitely many fine variables and coarse labels are reached; the coefficients stabilize once the finite inventories contain that region. Fixing exterior signs decreases norms. Finite weights include every crossing original and observed factor. The uniform geometric remainder therefore proves boundary-independent continuous limits for arbitrary full observed patterns.

For a finite target set A, observing its complement gives the conditional density $\mathbb{E}_{\mu^{y,A^c}} \prod_{e\in A}(1 + x_e g_e)$, relative to fair counting on A. It is normalized, and lies between $(1-G_B)^{|A|}$ and $(1+G_B)^{|A|}$. Increasing finite observed subsets gives this formula uniformly for every exterior by the preceding stabilization and tail. Finite Bayes consistency passes to these limits. Testing their conditional integrals against exterior cylinders identifies them as the actual pushed law's DLR kernels. This supplies the full finite kernels, not only one mean or an almost-sure version.

For a single target, start eq. (B.23) at g_{p_0}, whose rooted norm is at most $A_{2\lambda}G_B$. Its conditional mean m_{p_0} therefore has norm at most $M_B = A_{2\lambda}G_B/(1-r)$. Increase B until $M_B \leq 1/2$. The norm-convergent series $u_{p_0} = \operatorname{atanh} m_{p_0}$ gives its complete half-log-odds and bound $\|u_{p_0}\| \leq U_B = M_B/(1-M_B^2)$. Positive two-edge conditional kernels satisfy the conditional square identity

$$u_e(y^{f,+}) - u_e(y^{f,-}) = u_f(y^{e,+}) - u_f(y^{e,-}). \tag{B.24}$$

Both sides are half the same log cross ratio. Taking Walsh coefficients, justified by absolute convergence, shows that the coefficient of $\chi_{T\setminus\{e\}}$ in u_e is independent of the marked edge $e \in T$. Define it to be $J_T^{(B)}$. It reproduces every single-edge ratio; connectedness of the finite sign cube and normalization then reproduce all the conditional kernels. Translation covariance gives one canonical orbit representative.

For a size-n orbit its translated incidence and norm prefactor are at most n^2. The inequality $n^2 e^{\kappa n} \leq 2e^{\kappa} e^{(\kappa+1)(n-1)}$ follows from $\sup_{x\geq 1} x^2 e^{-(x-1)} = 4/e < 2$. Because the root belongs to T, its diameter is at most twice its rooted radius. Selecting only supports whose literal first edge is the root and summing the two root orientations therefore gives

$$\|J^{(B)}\|_{\kappa,\lambda} \leq \mathcal{B}_B := 4A_\kappa \frac{M_B}{1-M_B^2}. \tag{B.25}$$

This estimate counts all supports and translated incidences in the original prescribed weights. The fine weight α was auxiliary, and λ has not been replaced by a smaller output weight.

Every displayed gate is eventually true and remains true for larger B: the base gate improves with B, and G_B decreases exponentially. Rational exponential enclosures as in eq. (B.14) give a positive lower bound $\varepsilon_0 < \varepsilon$. A finite rational search for B_0 ensuring the gates and $\mathcal{B}_B < \varepsilon_0/16$

terminates, uniformly over the compact box. The local RG theorem now supplies uniqueness in its ball. Extraction of conditional coefficients and exact path composition identify $\mathcal{R}_a J^{(B)} = J^{(aB)}$.

To obtain a finite rational interaction, keep absorbed substitution orders through K, obtaining m_K, and keep the odd powers of $\operatorname{atanh} m_K$ through $2L-1$. Every local row in eq. (B.20) is a rational function of finitely many signs when b, t are rational. Its denominator is positive. Evaluate it on the complete cube and take its finite Walsh transform, retaining every coefficient. One may reduce the cube exactly to a basis for the $\mathbb{F}_2$-span of its character supports: the basis-character map is surjective by row rank; the function is constant on its fibers; every outside-span coefficient vanishes by pairing with a kernel element that flips that character. Thus the complete transform on the basis cube, lifted to literal supports, is the entire raw-coordinate function. The weights still use literal cardinalities and distances, not the number of basis generators. The local strip row's character rank is at most $2B$, from its $2(B-2)$ three-edge characters, endpoint character, possibly removed edge, and two original row characters.

All finite substitution orders and powers have finite inventories. The conservative mean remainder is $d_K = M_B r^K$. Both exact and truncated absorbed sums have norm at most M_B. Telescoping the odd powers gives an atanh Lipschitz bound $(1-M_B^2)^{-1}$, and its omitted powers give the complete error

$$\|\widehat{J} - J^{(B)}\|_{\kappa,\lambda} \le 4A_\kappa \left[\frac{d_K}{1-M_B^2} + \frac{M_B^{2L+1}}{(2L+1)(1-M_B^2)} \right]. \tag{B.26}$$

The incidence argument applies also to omitted coefficients and to approximate rows that need not obey exact square identities. Canonical selection gives one valid approximate interaction. After K substitutions its labels lie within coarse radius $K+1$; the full atanh of that finite-variable mean introduces no new labels. Hence supports of diameter greater than $2(K+1)$ have total norm at most $4A_\kappa d_K/(1-M_B^2)$. The finite odd polynomial pays its separate power tail in eq. (B.26).

Increasing K, L makes the error arbitrarily small, and each complete rational transform, substitution, and product terminates with sufficient finite resources. A bounded incomplete table does not have this conclusion. For a finite RG word, choose the input error small enough that both the true interaction and its finite approximation lie in one certified radius. Comparing their exact trajectories by the Lipschitz bounds gives $\delta \prod_k L_{a_k}(r)$; adding the separate Taylor word error proves the stated total. The argument retains normalized fine tilts throughout but computes no discarded original scalar partition. □

B.7 Higher response and the actual Fisher-density geometry

Proof of theorem 17.14. Enlarge the chosen natural-parameter compact rectangle to $\widehat{K} = [h_-, h_+] \times [0, j_+]$, with $h_- > 0$, $0 < j_+ < \operatorname{atanh}(1/12)$. The positive-$j$ interior contains the original compact set. The boundary $j = 0$ is used only for the direct one-sided continuation below. Write $t_+ = \tanh j_+$, choose $\theta > 0$ with $q = 12 t_+ e^\theta < 1$, and put $C = 16q/(1-q)$, $\rho = \tanh(h_+ + 2j_+)$, $d = 1-\rho > 0$. These constants work for every volume, depth, support pattern, and fixed fine exterior. Finite supports and references are parameter independent. The finite density is an exponential

family in (h, j); its field statistic sums free edge signs and its plaquette statistic includes every touching plaquette with its exterior completion.

B.7.1 Localized source cumulants and actual derivatives

Let U be a finite vector of block observables supported on an anchor set A, and V another outside its radius R neighborhood. Condition on the fine blocks outside that neighborhood. The weighted Hamming proof compares any two resulting interior rows. If the original finite whole-block volume is irregular, its inherited exterior is identical on both sides and contributes zero disagreement. Only the newly revealed shell contributes. Thus

$$\mathrm{TV}(\mathcal{L}(U,V), \mathcal{L}(U)\otimes\mathcal{L}(V)) \le \min\{1, C|A|B^2(R+1)e^{-\theta(R+1)}\}. \tag{B.27}$$

Each conditional law of U is within this bound of their mixture; integrating with the exterior and marginalizing proves the displayed group comparison. It passes to infinite constrained laws by finite-cylinder convergence, also at $j = 0$ by the exact product-fiber law.

For m bounded observables define

$$\mathrm{cum}(F_1, \dots, F_m) = \sum_{\pi\in\mathcal{P}_m} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{D\in\pi} \mathbb{E} \prod_{i\in D} F_i. \tag{B.28}$$

Its individual absolute bound is $k_m \prod_i \|F_i\|_\infty$, and a change of joint law by total variation v changes it by at most $q_m v \prod_i \|F_i\|_\infty$, where $k_m = \sum_\pi (|\pi|-1)!$ and $q_m = 2\sum_\pi |\pi|!$. For $m = 2, 3, 4$, these pairs are $(2, 6), (6, 26), (26, 150)$. The second estimate telescopes each product of moments, each moment changing by at most twice total variation. A cumulant of two independent nonempty groups vanishes: its generating function factorizes and its logarithm has no mixed coefficient. Expanding that finite logarithm also proves eq. (B.28).

Let a local f have anchor support A of size $a_0 \ge 1$, and let $r = 1, 2, 3$ labeled edge or plaquette sources have base anchors $x_1, \dots, x_r$. Put $M = \max_s \mathrm{dist}_\infty(x_s, A)$. On the complete graph of these labeled anchors and a root representing A, a minimum-weight spanning tree has an edge of length at least M/r. Removing a longest edge splits the graph into groups with all cross distances at least that length; otherwise exchanging a shorter cross edge would reduce the tree. Plaquette supports enlarge anchor sets by at most one on either side. For $M \ge 4r$, use $R = \lfloor M/r \rfloor - 3$ in eq. (B.27); the root group has at most $a_0 + 4r$ anchors. For smaller M, use the individual cumulant bound. With

$$L_r(a_0) = e^{4\theta}[k_{r+1} + q_{r+1}C(a_0 + 4r)], \qquad S_r(a_0) = \frac{(8a_0)^r L_r(a_0)(2r+1)!}{(1-e^{-\theta/r})^{2r+2}}, \tag{B.29}$$

this proves a bound $L_r(a_0)B^2(M+1)e^{-\theta M/r}\|f\|_\infty$ for each cumulant. There are at most $2B^2$ choices of a source of one derivative type at each anchor, and at most $a_0(2M+1)^2$ anchors within

distance M. Counting ordered tuples, including repeats, gives

$$\begin{aligned}\sum_{s_1,\dots,s_r} |\operatorname{cum}(f,T_{s_1},\dots,T_{s_r})| &\le S_r(a_0)B^{2r+2}\|f\|_\infty,\\ \sum_{\max \operatorname{dist}(x_s,A)>L} |\operatorname{cum}(f,T_{s_1},\dots,T_{s_r})| &\le S_r(a_0)B^{2r+2}(L+2)^{2r+1}e^{-\theta(L+1)/r}\|f\|_\infty.\end{aligned} \tag{B.30}$$

Here T_s is the appropriate edge or plaquette source for its labeled derivative direction. To sum the bound, use $(M+1)^k \le k!\binom{M+k}{k}$ in the geometric series. For the tail, write $M = L+1+n$ and bound $(M+1)^k \le (L+2)^k(n+1)^k$. Exterior constant sources have zero cumulant and may safely be overcounted.

Finite differentiation of the normalized exponential family gives these cumulant sums for each derivative of a parameter-independent observable. For a varying observable, the actual formula is

$$\partial^\alpha \mathbb{E}_g f_g = \sum_{\beta\le\alpha}\binom{\alpha}{\beta}\sum_{s_1,\dots,s_{|\alpha-\beta|}} \operatorname{cum}_g(\partial^\beta f_g, T_{s_1},\dots,T_{s_{|\alpha-\beta|}}), \tag{B.31}$$

where an empty source list means ordinary expectation. Differentiating the finite generating function or its normalized finite sum proves all multiplicities, including repeated directions. For each fixed source tuple, finite cylinder convergence gives convergence of its moments, uniformly on $\widehat{K}$. Near tuples form a finite set and far tuples have the uniform tails in eq. (B.30). First sending the source cutoff to infinity and then the volume proves uniform convergence of finite derivatives through order three. The sums are continuous by the same finite-near, uniform-tail argument. Passing finite coordinate fundamental-theorem-of-calculus identities to the limit proves that these are actual continuous derivatives, successively through order three, with agreeing mixed derivatives. At $j = 0$ the argument is on right-sided coordinate segments and gives actual one-sided C^3 continuation. No infinite normalizer or null-event probability has been differentiated.

B.7.2 Full local jets and interaction derivatives

For a free path its conditional mean given other fine blocks is the product G_g of B hyperbolic tangents with field $h + ja_e$, $|a_e| \le 2$, on at most nine anchor cells. For $n \le 3$, differentiating the product in labeled directions gives

$$\|\partial^\beta G_g\|_\infty \le g_n B^n \rho^{B-n}, \qquad g_n = 2^n n!, \quad |\beta| = n. \tag{B.32}$$

The first three hyperbolic-tangent derivatives have bounds $1, 2, 6$; each j differentiation adds a factor at most two. There are at most B^n assignments to product factors and the product of their factorials is at most $n!$. Negative exponents when $B < n$ only enlarge the upper bound. Apply eqs. (B.30) and (B.31) to G_g. With

$$D_n = g_n + \sum_{b'=0}^{n-1}\binom{n}{b'} g_{b'} S_{n-b'}(9), \qquad M_n(B) = D_n B^{2n+2}\rho^{B-n}, \quad M_0(B) = \rho^B, \tag{B.33}$$

every actual order-n free-path mean derivative is bounded by $M_n(B)$. These bounds tend to zero with B.

For patterns agreeing through Q_R, choose $L = \lfloor R/2 \rfloor - 1$ when $R \geq 2$. A near source tuple has joint support within Q_{L+1}; apply the group comparison on its remaining buffer and count $[2B^2(2L+1)^2]^n$ ordered tuples. Use eq. (B.30) for each of the two far sums. Writing $N_n = 2^n q_{n+1} C(1+4n)$, the resulting derivative modulus is

$$V_n(B,R) = \min\{2M_n(B), B^{2n+2}\big[N_n(2L+1)^{2n}(R-L)e^{-\theta(R-L)} + 2S_n(1)(L+2)^{2n+1}e^{-\theta(L+1)/n}\big]\}. \tag{B.34}$$

For $R = 0, 1$ take $V_n = 2M_n$. For values use $V_0 = \min\{2\rho^B, 2CB^2(R+1)e^{-\theta(R+1)}\}$. The same proof compares a finite whole-block conditional calculation against its infinite kernel: near sources are interior, and finite crossing sources and infinite far sources satisfy the two separate tail bounds. For $R \geq 2$, the bracket in eq. (B.34) is at most $Z_n(R+1)^{2n+1}e^{-\theta R/(2n)}$, with $Z_n = N_n + 2e^{\theta/(2n)}S_n(1)$.

Every Bernoulli factor is $(1+sm)/2$, with $1 + sm \geq d$. Revealing both target parities sequentially retains all their dependence. Their positive four-state row is actual C^3. For one Bernoulli log factor its derivative amplitudes are

$$A_1 = \frac{M_1}{d}, \qquad A_2 = \frac{M_2}{d} + \frac{M_1^2}{d^2}, \qquad A_3 = \frac{M_3}{d} + \frac{3M_2M_1}{d^2} + \frac{2M_1^3}{d^3}. \tag{B.35}$$

The multiplicities follow by differentiating $\log(1+sm)$ three times. Its derivative moduli are

$$\begin{aligned} T_1 &= V_1/d + M_1V_0/d^2, \\ T_2 &= V_2/d + (M_2V_0 + 2M_1V_1)/d^2 + 2M_1^2V_0/d^3, \\ T_3 &= V_3/d + (M_3V_0 + 3M_2V_1 + 3M_1V_2)/d^2 \\ &\quad + (6M_2M_1V_0 + 6M_1^2V_1)/d^3 + 6M_1^3V_0/d^4. \end{aligned} \tag{B.36}$$

Subtract the actual derivative ratios and use $|u^{-k} - v^{-k}| \leq k|u-v|/d^{k+1}$ to obtain these formulas. One-cell log probabilities use $2A_n, 2T_n$; reference-state log odds use $4A_n, 4T_n$. Each modulus can also be capped by twice its amplitude. The same already revealed first parity is fixed when comparing the second factor, so the support patterns still agree in their common interior.

Here is an explicit summability check for the complete interaction jets. Set $E_1 = D_1/d$, $E_2 = D_2/d + D_1^2/d^2$, and $E_3 = D_3/d + 3D_2D_1/d^2 + 2D_1^3/d^3$. Then $F_n(B) = 4A_n(B) \leq 4E_nB^{4n}\rho^{B-n}$. Each M_j is bounded over all B by $U_j = D_j\rho^{1-j}(2j+2)!/(1-\rho)^{2j+3}$. Put $Z_0 = 2C$ and obtain P_n from the right side of eq. (B.36) by replacing every M_j with U_j and every V_j with Z_j. Thus these are explicit positive constants; the powers of d are unchanged. For $R \geq 2$, $T_n \leq P_nB^{2n+2}(R+1)^{2n+1}e^{-\theta R/(2n)}$. The constant

$$H_n = 4P_n + 8E_n\rho^{-n}\frac{(2n-2)!}{(1-\rho)^{2n-1}}e^{\theta/(2n)} \tag{B.37}$$

also covers shell radii one and two using the amplitude cap. Therefore its order-n canonical shell is bounded by both $2F_n(B)$ and $H_nB^{2n+2}r^{2n+1}e^{-\theta(r-1)/(2n)}$. Choose one common $\lambda_f = \theta/48$.

Geometric interpolation, $\operatorname{diam} A_{x,r} \le 2r$, and the translate count give

$$\|\partial^\alpha \Phi^{(B)}\|_{\lambda_f} \le F_n(B) + \sqrt{2F_n(B)H_n}B^{n+1}e^{\theta/(4n)}\sum_{r\ge 1}(2r+1)^2 r^{n+1}e^{-[\theta/(4n)-2\lambda_f]r}, \quad |\alpha| = n \le 3. \tag{B.38}$$

The bracketed exponent is positive. The positive series is bounded by $9(n+3)!/(1-e^{-[\theta/(4n)-2\lambda_f]})^{n+4}$, and its omitted tail is an explicit convergent bound. The right side decays as a constant times $B^{3n+1}\rho^{(B-n)/2}+B^{4n}\rho^{B-n}$. Values use the preceding canonical-interaction proof in the same weaker λ_f norm. Finite-shell maps are C^3; their derivatives through order three converge uniformly in this Banach norm. Successive coordinate FTC identities prove actual C^3 of the entire interaction. No shell or derivative has been replaced by a finite truncation in the statement.

B.7.3 Moving expectations and actual marginal information

For any finite or infinite subset S of the lexicographic past, let $p_S(g,s;y)$ be the four-state predictive row with exactly those past coordinates prescribed and the other paths, including the future, free. Let $l = \log p_S$, with subscripts denoting natural derivatives. Define $\mathsf{I}_{S,ij} = \sum_s p_S l_i l_j$, an observable of the prescribed past. Its actual first and second parameter derivatives, keeping y fixed, are

$$\begin{aligned}
\mathsf{I}_{S,ij,k} &= \sum_s p_S(l_k l_i l_j + l_{ik} l_j + l_i l_{jk}),\\
\mathsf{I}_{S,ij,kv} &= \sum_s p_S\big[(l_{kv}+l_k l_v)l_i l_j + l_k(l_{iv}l_j + l_i l_{jv}) + l_v(l_{ik}l_j + l_i l_{jk})\\
&\quad + l_{ikv}l_j + l_{ik}l_{jv} + l_{iv}l_{jk} + l_i l_{jkv}\big].
\end{aligned} \tag{B.39}$$

These formulas differentiate the probabilities as well as the scores. Put $L_i = 2A_i$,

$$E_B^F = \max\{L_1^2, L_1^3 + 2L_1L_2, L_1^4 + 5L_1^2L_2 + 2L_1L_3 + 2L_2^2\}, \qquad H_B^F = \max(1, L_1, L_2, L_3). \tag{B.40}$$

Every matrix entry and its first two derivatives are bounded by E_B^F. For patterns agreeing in Q_R, every row probability changes by at most V_0, and each log jet by at most $2T_n$. Telescoping the at most ten expanded monomials in eq. (B.39), at most five factors per term and four states, gives the common modulus

$$\omega_B(R) = \min\{2E_B^F, 200(H_B^F)^4[V_0(B,R) + 2T_1(B,R) + 2T_2(B,R) + 2T_3(B,R)]\}. \tag{B.41}$$

At fixed B it is bounded by an explicit constant times $(R+1)^7 e^{-\theta R/6}$. These are differences of actual jets, not derivatives of an earlier scalar continuity bound.

Expectations now use the original unconstrained fine law, with coarse observables pulled back by the fixed path map. For a local C^2 observable on Q_N, put $a_N = (2N+1)^2$ and $W_B(N) = 1 + 2S_1(a_N)B^4 + S_2(a_N)B^6$. The moving response formula gives

$$\max_{|\alpha|\le 2}\sup_{g\in\widehat{K}} |\partial^\alpha \mathbb{E}_{\mu_g} f_N(g)| \le W_B(N) \max_{|\alpha|\le 2}\sup_{g,y} |\partial^\alpha f_N(g,y)|. \tag{B.42}$$

In particular, the second derivative includes $\mathbb{E}f_{ij}$, both source covariance sums involving f_i, f_j, and the complete three-variable cumulant sum involving f. Formula eq. (B.30) justifies each passage from finite volumes; neither the observable nor the probability law is held fixed incorrectly. The function $W_B(N)$ is polynomial in N. For a shell series $f = \sum_N d_N$ with $\sum_N W_B(N)\|d_N\|_{C^2} < \infty$, finite partial sums and their expectation derivatives are uniformly Cauchy. Repeated FTC gives actual C^2 of its expectation and termwise differentiation.

Apply this to $f_N^S = \mathsf{I}_{S\cap Q_N}$. Its initial shell is bounded by E_B^F, and its later shell differences through order two by $\omega_B(N-1)$. Thus

$$Z_B^F = W_B(0)E_B^F + \sum_{N\geq 1} W_B(N)\omega_B(N-1) < \infty, \qquad \epsilon_B(R) = 2\sum_{N>R} W_B(N)\omega_B(N-1) \longrightarrow 0. \tag{B.43}$$

If two past sets agree inside Q_R, their finite prefixes agree exactly through that radius, so their full-law Fisher-observable expectations differ in entrywise C^2 norm by at most $\epsilon_B(R)$. This holds uniformly over the remote shape of the past. In particular the full-past expectation $\mathcal{I}_B = \mathbb{E}\mathsf{I}_{\mathrm{past}}$ is actual C^2.

The finite positive marginal on Λ_L factors into its actual predictive rows in lexicographic order. Each score increment is centered conditionally on its earlier revealed cells. Two increments are orthogonal because the earlier one is measurable for the later past. Consequently the exact finite Fisher matrix is the sum of expected row-Fisher matrices with their actual finite pasts. Each is a C^2 function by the preceding varying-expectation argument. Translation invariance removes the shifts. For $L \geq 2R$, all but at most $4RL$ cells have radius R inside the square; their finite pasts and full past agree on that radius. Each remaining difference is bounded by $2Z_B^F$. Therefore

$$\|L^{-2}I_{B,\Lambda_L} - \mathcal{I}_B\|_{C^2,\mathrm{entry}} \leq \epsilon_B(R) + 8Z_B^F R/L. \tag{B.44}$$

First send L to infinity at fixed R, then R to infinity. This proves the asserted actual marginal C^2 limit and includes all boundary-score derivatives. The canonical row with future fixed to vacuum was never substituted for the predictive row.

B.7.4 An exhibited positive chart and its curvature

At $j = 0$, with any fixed complete past, the blocks are independent product parity fibers and the two target path parities are independent with mean b^B. For one fine edge of a target path set

$$a(s) = \mathbb{E}[z_e \mid Y = s] - b = \frac{sb^{B-1}(1-b^2)}{1+sb^B}, \qquad v_B = \mathbb{E}a^2 = \frac{b^{2B-2}(1-b^2)^2}{1-b^{2B}} > 0. \tag{B.45}$$

Multiplying the product law by its parity indicator and summing proves this formula; $\mathbb{E}a = 0$, and conditional-variance decomposition gives $v_B \leq 1-b^2 \leq 1$. The actual field score is $s_h = B(a_1+a_2)$. The actual plaquette score is $s_j = b^2a_1a_2 + \alpha_1a_1 + \alpha_2a_2$, with $|\alpha_i| \leq 2B$. To see the mixed term, the plaquette at the common starting vertex is the unique one meeting both outgoing target paths. Its other two edges are future paths at $B = 1$, or unused singleton edges at $B > 1$, and have mean

b. Its centered contribution is $b^2a_1a_2 + b^3(a_1 + a_2)$. Every other incident plaquette meets only one target path, giving a multiple of a_i of magnitude at most one; there are $2B - 1$ such incidences. Nonincident plaquettes contribute zero by product-block independence. These are derivatives of the full family through its source formula and one-sided limit, not derivatives of a formula restricted to $j = 0$.

Independence and centering make a_1a_2 orthogonal to both linear contrasts. The conditional Fisher entries are therefore $I_{hh} = 2B^2v_B$, $I_{hj} = Bv_B(\alpha_1 + \alpha_2)$, and $I_{jj} = v_B(\alpha_1^2 + \alpha_2^2) + b^4v_B^2$. Its determinant is $B^2v_B^2(\alpha_1 - \alpha_2)^2 + 2B^2b^4v_B^3$, while its trace is at most $11B^2v_B$. Dividing the determinant by the trace bounds its smaller eigenvalue below by $2b^4v_B^2/11$, uniformly in the past. On a compact positive-field interval with $b \in [b_-, b_+]$, this is at least

$$\eta_{0,B} = \frac{2}{11}b_-^{4B}(1 - b_+^2)^4 > 0. \tag{B.46}$$

Use the bounds A_1, A_2 on the enlarged rectangle and put $Q_B = 16(A_1^3 + A_1A_2) > 0$. The score norm is at most $2\sqrt{2}A_1$, and its j derivative at most $2\sqrt{2}A_2$. Differentiating the four-state Fisher sum gives operator-norm derivative at most Q_B, including the probability derivative term. Actual one-sided C^3 and FTC imply

$$\mathcal{I}_B(h, j) \succeq \frac{\eta_{0,B}}{2}I_2 \quad \text{for} \quad 0 < j < \delta_B := \min\{j_+/2, \eta_{0,B}/(2Q_B)\}. \tag{B.47}$$

The same floor holds rowwise and hence after the full-law expectation. An enlarged field interval supplies an open neighborhood of each original compact interval. For a prescribed finite list of depths, take the finite minima of positive widths and floors. No positive minimum over every integer depth is asserted.

On a compact subchart of this region write $g_{ij} = \mathcal{I}_{B,ij}$. The inverse is C^2, by differentiating $gg^{-1} = I$ twice with its uniform positive floor. Define

$$\Gamma^l_{jk} = \frac{1}{2}g^{lm}(\partial_j g_{mk} + \partial_k g_{mj} - \partial_m g_{jk}), \qquad R^l{}_{kij} = \partial_i\Gamma^l_{jk} - \partial_j\Gamma^l_{ik} + \Gamma^l_{im}\Gamma^m_{jk} - \Gamma^l_{jm}\Gamma^m_{ik}. \tag{B.48}$$

Direct substitution proves torsion freedom and metric compatibility. Adding and subtracting their cyclic index equations recovers the displayed Γ, proving uniqueness. The covariant-derivative commutator gives the displayed tensor, so it is coordinate covariant. The connection is C^1, and curvature, Ricci contraction, and scalar contraction are continuous because they use metric derivatives through order two. Equation eq. (B.44) and the positive floor make $L^{-2}I_{B,\Lambda_L}$ invertible for every sufficiently large L. Continuous inversion and the displayed formulas give compact convergence of its connections and curvature, at this fixed B.

These are also actual local law charts. At a chart point choose a sufficiently large finite marginal with positive Fisher matrix. Its probability Jacobian has rank two because $I = (Dp)^\top \operatorname{diag}(1/p)Dp$; hence two atom probabilities have nonsingular Jacobian. After multiplication by its inverse at the point, shrink a convex parameter ball until that map's derivative is within $1/2$ of the identity. Segment integration proves a lower Lipschitz bound. The iteration $x \mapsto x + y - F(x)$ is a contraction on a

smaller ball for nearby y, giving a local inverse. Differentiating $F(F^{-1}(y)) = y$ gives its derivatives through order three. This exhibits a chart of the actual law without assuming a global quotient manifold. □

A vanishing sequence of Fisher tensors does not establish vanishing Fisher curvature. If a regular metric is multiplied by a positive constant a, its connection and its Riemann tensor with one raised index are unchanged, whereas its scalar curvature is divided by a. Thus metric collapse can coexist with divergent scalar curvature. This elementary scaling obstruction is separate from the positive fixed-depth charts proved above. [ESTABLISHED]

Appendix C

Proofs for Thermodynamic Fisher Geometry

This appendix proves the conditional thermodynamic Fisher statements for the actual finite-free laws and their exact fixed-word pushes. The source, boundary, support, and regular-chart hypotheses are part of each result; no unrestricted simultaneous depth and volume limit is asserted.

The earlier spatial appendix supplies the previously stated constrained specification and local-response setting. The arguments here make the additional finite-free boundary transfer explicit, then state the exact inputs used by the retained-label and action calculations. Every regular inverse is restricted to its declared chart, and every volume limit retains its boundary and fixed-word convention. [ESTABLISHED]

C.1 Native response and the finite-free boundary bridge

The response construction in sections B.7.1 and B.7.3 uses finite all-touching specifications and actual infinite-law marginals. Here the finite outer boundary is free: plaquettes leaving the original square are omitted. We prove the additional response comparison needed for these finite laws and their exact fixed-word pushes. Their common limiting Fisher density does not identify their finite probability laws. In particular, a free law of a canonical coarse interaction is a fourth family and does not enter the conclusion below. [ESTABLISHED]

C.1.1 Source box, complete carriers, and fixed words

Hypothesis C.1 (Compact native source box). Let $g = (h, j)$, $b = \tanh h$, and $t = \tanh j$. Fix a compact rectangle K inside $h > 0$, $0 < j < \operatorname{atanh}(1/12)$, and choose an enlarged rectangle $\widehat{K} = [h_-, h_+] \times [0, j_+]$, with $h_- > 0$ and $0 < j_+ < \operatorname{atanh}(1/12)$, whose positive-$j$ interior contains K. Choose $\tau > 0$ and define

$$q = 12 \tanh(j_+) e^{\tau} < 1, \qquad C = \frac{16q}{1-q}, \qquad \rho_* = \tanh(h_+ + 2j_+), \qquad d = 1 - \rho_* > 0. \tag{C.1}$$

The decay constant τ is the constant denoted by θ in section B.7.1; it is unrelated to the parameter coordinate θ used for a source pullback below. All constants in this proof are fixed on $\widehat{K}$. Its edge $j = 0$ is used only for a separately labeled one-sided product control. These restrictions are

adopted hypotheses for the development that follows, with the exclusions and scope stated above. [HYPOTHESIS]

Definition C.2 (Native free law and its exact pushes). For $\Lambda_L = \{0, \dots, L-1\}^2$, let $V = L^2$ and retain all $2V$ signs $z_{x,a}$, $a = 1, 2$. Put

$$P_x = z_{x,1} z_{x+e_1,2} z_{x+e_2,1} z_{x,2}, \qquad r_L(g, z) = \prod_{x \in \Lambda_L} \prod_{a=1}^{2} (1 + b z_{x,a}) \prod_{x \in \{0,\dots,L-2\}^2} (1 + t P_x). \tag{C.2}$$

Let $\nu^{\mathrm{f}}_{L,g}$ be this raw factor normalized against counting probability 4^{-V}, and let Z_L be its reference partition. The unique infinite native Gibbs law is μ_g, constructed by its normalized local specifications. For a fixed integer $B \geq 1$, retain the paths $P_B(x, a) = \{(Bx + u e_a, a) : 0 \leq u < B\}$ and complete them to a partition of fine edges with unused singleton blocks. The parameter-independent map C_B records their parities. Define

$$P^{\mathrm{f}}_{B,L,g} = (C_B)_{\#} \nu^{\mathrm{f}}_{BL,g}, \qquad \mu_{B,g} = (C_B)_{\#} \mu_g, \qquad \mu_{B,L,g} = \mu_{B,g}|_{\Lambda_L}. \tag{C.3}$$

The finite free law, its exact push, and the actual infinite-law marginal use their own normalized references. A coarse translation by x is a fine translation by Bx. Local limits use centered squares, with fine translation $-B\lfloor L/2 \rfloor$ and coarse translation $-\lfloor L/2 \rfloor$, or equivalent bulk neighborhoods.

A fixed finite word of integer blocking factors has total linear factor B. Successive parity products concatenate disjoint straight paths and equal C_B on the same finite fine square. Fine area is $M = B^2 V$ and final coarse area is V. The word, hence B, is fixed throughout every volume limit. Restricted parity conditionals use the product of uniform probability references on their prescribed fibers; unrevealed paths and unused singleton blocks remain unconstrained. These are declared definitions and conventions; no proof is asserted by their adoption. [DEFINITION]

C.1.2 Constrained contraction and finite-free local response

Lemma C.3 (Weighted comparison with omitted outer plaquettes). The constrained weighted comparison and local source-cumulant estimates of sections B.2 and B.7.1 hold with the constants in eq. (C.1) when plaquettes at the original free outer boundary are omitted. They are uniform in the finite whole-block inventory, prescribed-or-absent path pattern, and fixed exterior completion. [ESTABLISHED]

Proof. A plaquette meets four distinct blocks and at most one edge of any retained path. Conditional on the other fine blocks, a path row is therefore a positive product sign law, possibly restricted to one parity fiber. To recall why the row transport constant is independent of path length, consider one product law conditioned on opposite total parities. It admits a coupling at Hamming distance exactly one. Induct on its number of bits: couple the last bits, use identical prefixes when their required parities agree, and use the induction coupling when those parities differ. In each branch

the conditional prefix marginal is the required product law with its specified parity, so both complete marginals are correct.

Changing one incident plaquette tilts one coordinate by opposite fields. If u collects the positive baseline masses of its two signs, the coordinate-marginal total variation is

$$\frac{\sinh(2j)}{\cosh(2u)+\cosh(2j)} \le \tanh j \le \tanh j_+. \tag{C.4}$$

Couple that bit maximally. A disagreement in a fixed parity fiber needs one additional discrepancy in the remaining bits, by the preceding coupling; the mean whole-path Hamming cost is at most $2\tanh j_+$. The unconstrained row costs at most $\tanh j_+$, and a prescribed one-bit fiber costs zero. Intermediate individual tilts remain positive product laws on the same support. Composing their coupling kernels and using the triangle inequality for Hamming distance permits telescoping all changed incident plaquettes.

For a target block i and exterior fine edge e, let $n(i,e)$ count incident plaquettes. Each fine edge belongs to two plaquettes and each of those meets three other blocks, so $\sum_i n(i,e) \le 6$. Omitting outer plaquettes only removes nonnegative incidences. Finite positive heat baths are irreducible on the product support; their pair kernel has a stationary law by finite-simplex Cesaro averaging. If D_i is its expected block Hamming discrepancy, canceling the block-update probability gives the inequality in eq. (B.5) with this restricted incidence list. There is no extra factor B: D_i already sums all fine-bit discrepancies in the updated block.

For target anchor set A, use $w_i = e^{-\tau\,\mathrm{dist}_\infty(\mathrm{anchor}(i),A)}$. Incident anchors differ by at most one, and consequently

$$S \le qS + q\sum_{e\in\partial_f D} w_{\mathrm{block}(e)}\delta_e, \qquad S=\sum_{i\ \text{free}} w_i D_i, \quad \delta_e = \mathbf{1}_{\{\eta_e\neq\zeta_e\}}. \tag{C.5}$$

This is eq. (B.6) for the actual retained plaquettes. A shell at anchor distance $R+1$ from A has at most $8|A|(R+1)$ cells and $2B^2$ fine edges per cell. Solving eq. (C.5) and bounding the probability of a target discrepancy gives the joint-target bound $\min\{1, C|A|B^2(R+1)e^{-\tau(R+1)}\}$.

To obtain group decoupling, condition on all existing free blocks outside the radius-R neighborhood of A. Any boundary inherited from the original inventory, including its omitted outer interactions, is identical in the two interior conditionals. It produces no near discrepancy. Only the newly conditioned shell contributes. Each conditional target law is within the preceding bound of their mixture. Integrating with the exterior and marginalizing therefore gives, for a vector U supported on A and a vector W outside its radius-R neighborhood,

$$\mathrm{TV}(\mathcal{L}(U,W), \mathcal{L}(U)\otimes\mathcal{L}(W)) \le \min\{1, C|A|B^2(R+1)e^{-\tau(R+1)}\}. \tag{C.6}$$

The argument permits irregular whole-block inventories. It compares joint groups, not only pairwise covariances. Infinite constrained laws follow from consistent finite-cylinder limits as in section B.2;

the same bounds pass through finite-alphabet total variation. No infinite prescribed event is used as a probability denominator. □

Lemma C.4 (Ordered source tails and the finite-free local modulus). Let a bounded parameter-independent observable f have anchor support A, with $a_0 = |A| \geq 1$. Its finite-free expectations have actual compact-uniform derivatives through degree three. If the finite inventory contains $A_R = \{x : \mathrm{dist}_\infty(x, A) \leq R\}$ and the finite and infinite constrained patterns agree there, then

$$\max_{|\alpha|=r} \sup_{g \in \widehat{K}} \left| \partial^\alpha \mathbb{E}_{\nu_g^D} f - \partial^\alpha \mathbb{E}_{\mu_g} f \right| \leq \Delta_r(B, a_0, R) \|f\|_\infty, \qquad 0 \leq r \leq 3, \tag{C.7}$$

where μ_g denotes the matching constrained law in this statement and the explicit moduli below tend to zero at fixed B, a_0. [ESTABLISHED]

Proof. Use cum for the cumulant of bounded observables. The moment-partition formula and its constants from eq. (B.28) are

$$\begin{gathered} \mathrm{cum}(F_0, \ldots, F_r) = \sum_\pi (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{D \in \pi} \mathbb{E} \prod_{i \in D} F_i, \\ (k_2, q_2) = (2, 6), \qquad (k_3, q_3) = (6, 26), \qquad (k_4, q_4) = (26, 150). \end{gathered} \tag{C.8}$$

Here $k_m = \sum_\pi (|\pi| - 1)!$ bounds the individual cumulant, and $q_m = 2 \sum_\pi |\pi|!$ multiplies the total-variation change of the joint argument law. The latter estimate telescopes block moments, each changing by at most twice total variation. Independent nonempty groups have zero mixed cumulant because their moment-generating function factorizes. Expanding its finite logarithm gives the displayed partition formula, so neither cancellation nor its constants require an infinite analyticity theorem.

For $r \in \{1, 2, 3\}$, let $T_{i_1, s_1}, \ldots, T_{i_r, s_r}$ be individual edge or plaquette sources for the prescribed ordered natural derivative directions $i_1, \ldots, i_r$. Sources may repeat. Let M be their maximum base-anchor distance from A. Connect their labeled anchors and a root representing A by a minimum spanning tree. A path to a source at distance M has at most r edges and length at least M, hence a longest tree edge has length at least M/r. Removing it separates groups with no shorter cross edge. Plaquette supports enlarge anchor sets by at most one on either side. For $M \geq 4r$, the groups are therefore separated by a neighborhood of radius $\lfloor M/r \rfloor - 3$. The root group has at most $a_0 + 4r$ anchors. Apply eq. (C.6) and cumulant cancellation; for $M < 4r$, use the individual cumulant bound. With

$$\begin{aligned} L_r(a_0) &= e^{4\tau} [k_{r+1} + q_{r+1} C(a_0 + 4r)], \\ S_r(a_0) &= \frac{(8a_0)^r L_r(a_0)(2r+1)!}{(1 - e^{-\tau/r})^{2r+2}}, \qquad U_r(B, a_0) = S_r(a_0) B^{2r+2}, \quad U_0(B, a_0) = 1, \end{aligned} \tag{C.9}$$

each cumulant is bounded by $L_r(a_0) B^2 (M+1) e^{-\tau M/r} \|f\|_\infty$. There are at most $2B^2$ choices of each source type per anchor and at most $a_0(2M+1)^2$ anchors within distance M. Thus the number

of ordered tuples of radius at most M is no greater than $(8a_0)^r B^{2r}(M+1)^{2r}$. Using this cumulative count also for the exact-radius count only enlarges the sum. The result is

$$\begin{aligned}\sum_{s_1,\dots,s_r} |\operatorname{cum}(f, T_{i_1,s_1}, \dots, T_{i_r,s_r})| &\le U_r(B, a_0)\|f\|_\infty, \\ \sum_{\max \operatorname{dist}(x_s,A)>N} |\operatorname{cum}(f, T_{i_1,s_1}, \dots, T_{i_r,s_r})| &\le U_r(B, a_0)(N+2)^{2r+1} e^{-\tau(N+1)/r}\|f\|_\infty.\end{aligned} \tag{C.10}$$

For completeness, $(M+1)^k \le k!\binom{M+k}{k}$ sums the geometric power series. For its tail write $M = N+1+n$ and use $(M+1)^k \le (N+2)^k(n+1)^k$. These are the counts in eq. (B.30), now valid for the actual free-boundary source list. Constant exterior-completed sources have zero higher cumulant; missing outer sources only remove terms.

We now compare local derivatives, without differentiating exterior mixing weights. For values, condition both laws on a whole-block neighborhood one interaction layer inside A_R. Every plaquette touching that smaller region is present in both conditionals. Its exterior assignments may have different parameter-dependent distributions, but at each fixed g the comparison is uniform over both assignments. Integrating it proves the conservative modulus

$$\Delta_0(B, a_0, R) = \min\{2, 2Ca_0B^2(R+1)e^{-\tau(R-1)}\}, \qquad R \ge 4. \tag{C.11}$$

For derivatives, first differentiate each finite normalized law with its actual source list. Normalized finite exponential differentiation gives the complete ordered cumulant sum. Put $N_R = \lfloor R/2 \rfloor - 2$. A near source tuple and f occupy at most $a_0 + 4r$ anchors inside A_{N_R+1}. Their neighborhood of radius $R - N_R - 2$ has a full extra interaction layer inside A_R. The joint near-argument total variation is at most $C(a_0+4r)B^2(R-N_R+1)e^{-\tau(R-N_R-2)}$. There are at most $[2B^2a_0(2N_R+1)^2]^r$ near tuples. Cumulant telescoping bounds their differences; the two far sums are separately bounded by eq. (C.10). Therefore

$$\begin{aligned}\widetilde{\Delta}_r(B, a_0, R) &= q_{r+1}[2B^2a_0(2N_R+1)^2]^r C(a_0+4r)B^2(R-N_R+1)e^{-\tau(R-N_R-2)} \\ &\quad + 2U_r(B, a_0)(N_R+2)^{2r+1}e^{-\tau(N_R+1)/r}, \\ \Delta_r(B, a_0, R) &= \min\{2U_r(B, a_0), \widetilde{\Delta}_r(B, a_0, R)\}, \qquad 1 \le r \le 3.\end{aligned} \tag{C.12}$$

For $R < 4$, set $\Delta_r = 2U_r$, including $r = 0$. Each modulus tends to zero at fixed B, a_0. An omitted original outer source cannot occur in the common near inventory, so it is covered by the far comparison rather than erased from a common interior formula.

To justify the infinite derivatives used in this comparison, first exhaust the matching all-touching fixed-exterior specifications. Each fixed near tuple has a uniformly convergent finite-alphabet moment law, while eq. (C.10) controls both far sums uniformly. The derivatives through degree three are uniformly Cauchy on compact rectangles. Their limits are continuous. Pass the finite coordinate fundamental-theorem-of-calculus identities to the limit, first for the expectation, then for its first and second derivatives. This identifies ordinary continuous derivatives through order three, with agreeing mixed derivatives. The same argument on coordinate segments starting at $j = 0$

proves continuous one-sided derivatives. The finite-free comparison just proved converges to those actual derivatives. Neither an infinite partition nor an infinite-event probability is differentiated. □

Lemma C.5 (Local moving observables). For a local C^2 observable f_g, finite and infinite expectations satisfy

$$\partial^\alpha \mathbb{E}_g f_g = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} \sum_{s_1,\dots,s_{|\alpha-\beta|}} \operatorname{cum}_g(\partial^\beta f_g, T_{i_1,s_1}, \dots, T_{i_{|\alpha-\beta|},s_{|\alpha-\beta|}}), \qquad |\alpha| \le 2, \tag{C.13}$$

where the source directions are those of $\alpha - \beta$ and an empty source list means expectation. If all pointwise C^2 jets of f on A are bounded by E_f, then the finite/infinite comparison of lemma C.4 gives

$$\|\mathbb{E}_{\nu^D} f - \mathbb{E}_\mu f\|_{C^2(K)} \le E_f[\Delta_0(B, a_0, R) + 2\Delta_1(B, a_0, R) + \Delta_2(B, a_0, R)]. \tag{C.14}$$

[ESTABLISHED]

Proof. Differentiate the finite normalized sum. At each step, a derivative either acts on f or inserts the corresponding centered source; successive insertions give the cumulant. The choices of derivatives on f give the multi-index binomial coefficient. Thus the second derivative includes $\mathbb{E} f_{ik}$, both source covariance sums involving f_i, f_k, and the complete three-argument cumulant sum involving f. This is the finite identity eq. (B.31) on the actual source list. The source-tail bound permits taking the infinite limit termwise after a finite cutoff; coordinate FTC identifies the actual derivatives there. Apply eq. (C.7) separately to the observable derivatives in the formula. At degree two their total coefficients are $1, 2, 1$, proving the displayed comparison. Both the observable and its averaging law vary. □

C.1.3 Fourth pressure derivatives and the native information tensor

Theorem C.6 (Native free pressure in C^4). Define the natural statistics and pressure by

$$\begin{aligned} T_{h,L} &= \sum_{x \in \Lambda_L} \sum_{a=1}^{2} z_{x,a}, \qquad T_{j,L} = \sum_{x \in \{0,\dots,L-2\}^2} P_x, \\ \mathsf{A}_L(h,j) &= V^{-1} \log \mathbb{E}_{\mathrm{ref}} e^{hT_{h,L} + jT_{j,L}}, \qquad \mathfrak{p}_L = V^{-1} \log Z_L. \end{aligned} \tag{C.15}$$

There are actual C^4 functions $\mathsf{A}, \mathfrak{p}$ on the positive high-temperature domain such that $\mathsf{A}_L \to \mathsf{A}$ and $\mathfrak{p}_L \to \mathfrak{p}$ in $C^4(K)$ for every compact source box in hypothesis C.1. Their exact finite and limiting conversions are

$$\begin{aligned} \mathfrak{p}_L &= \mathsf{A}_L - 2\log\cosh h - q_L \log\cosh j, \qquad q_L = \frac{(L-1)^2}{L^2}, \\ \mathfrak{p} &= \mathsf{A} - 2\log\cosh h - \log\cosh j. \end{aligned} \tag{C.16}$$

[ESTABLISHED]

Proof. For a sign z, $1+\tanh(h)z = e^{hz}/\cosh h$. Apply this identity to all $2V$ edges and to the $(L-1)^2$ included plaquettes. Both partitions use normalized counting, so no further cardinality factor enters. This gives the first line of eq. (C.16).

The value limit in theorem 19.2 applies to this raw native interaction. Its singleton diameter is zero and each cell has three plaquette incidences, giving first diameter moment at most $\overline{M} = 3t_+/(1-t_+)$. To recall the uniformity, tile a large square by n-squares and a leftover set. Terms touching the leftover set cost its size times a common finite incident norm. Crossing a tile boundary costs at most $4\overline{M}n$ per tile. The exponential sandwich of finite partition sums therefore traps limiting upper and lower pressures within $4\overline{M}/n$ of $\mathfrak{p}_n$. Their separation is at most $8\overline{M}/n$ for every n, so the limit exists and $\|\mathfrak{p}_L - \mathfrak{p}\|_{C^0(\widehat{K})} \le 4\overline{M}/L$. The finite conversion gives a uniform value limit for A_L. This step supplies no pressure derivative.

Put $f_h = z_{0,1} + z_{0,2}$ and $f_j = P_0$, whose anchor supports have sizes one and three, and let $m_i(g) = \mathbb{E}_{\mu_g} f_i$. These are actual C^3 functions by lemma C.4. Finite differentiation gives

$$\partial_h \mathsf{A}_L = \frac{1}{V}\sum_{x\in\Lambda_L} \mathbb{E}_{\nu^{\mathrm{f}}_{L,g}} f_h \circ \vartheta_x, \qquad \partial_j \mathsf{A}_L = \frac{1}{V}\sum_{x\in\{0,\dots,L-2\}^2} \mathbb{E}_{\nu^{\mathrm{f}}_{L,g}} f_j \circ \vartheta_x, \tag{C.17}$$

where ϑ_x translates the local observable to x. For $0 \le r \le 3$, define

$$M_r = \max\{2U_r(1,1), U_r(1,3)\}, \qquad D_r(R) = \max\{2\Delta_r(1,1,R), \Delta_r(1,3,R)\}. \tag{C.18}$$

For $L \ge 2(R+2)$, at most $4(R+2)L$ base cells lack a complete target support and its radius-R neighborhood. The bulk differences of local derivatives are bounded by $D_r(R)$. Each finite and infinite boundary derivative is bounded by M_r; an omitted plaquette is an identically zero finite term. Therefore

$$\max_{i\in\{h,j\},|\alpha|=r} \|\partial^\alpha \partial_i \mathsf{A}_L - \partial^\alpha m_i\|_{C^0(K)} \le D_r(R) + 8M_r\frac{R+2}{L}, \qquad 0 \le r \le 3. \tag{C.19}$$

At fixed R, take L large; then take R large. Equivalently, for a prescribed tolerance first choose R for its local modulus and then L for its boundary fraction. The gradients converge in C^3.

On any coordinate segment in the enlarged rectangle,

$$\mathsf{A}_L(g+ue_i) - \mathsf{A}_L(g) = \int_0^u \partial_i \mathsf{A}_L(g+ve_i)dv. \tag{C.20}$$

Uniform value and gradient convergence pass through the integral, giving $\partial_i \mathsf{A} = m_i$. Apply the same argument to the first, second and third gradient derivatives in eq. (C.19). It proves actual C^4 regularity, convergence of every derivative through order four, and equality of mixed derivatives. Since $|q_L - 1| \le 2/L$ and q_L is parameter independent, the conversion proves the same result for raw pressure. In particular, its fourth derivative requires only three local-response derivatives: its first derivative is already a local source density. The argument never differentiates a surface log-ratio estimate. □

Proposition C.7 (Native Fisher density and a source pullback). The actual native free family satisfies

$$\frac{1}{V} I(\nu^{\mathrm{f}}_{L,g}) = \nabla_g^2 \mathsf{A}_L(g) \longrightarrow H(g) := \nabla_g^2 \mathsf{A}(g) \quad \text{in } C^2(K), \qquad H = \nabla_g^2 \mathfrak{p} + \mathrm{diag}(2\,\mathrm{sech}^2 h, \mathrm{sech}^2 j). \tag{C.21}$$

If $\chi : \Theta \subset \mathbb{R}^d \to \mathbb{R}^2$, $\chi(\theta) = (h(\theta), j(\theta))$, is C^4 and sends a neighborhood of a compact parameter set into an admitted source box, then

$$g_L(\theta) = \frac{1}{V} I(\nu^{\mathrm{f}}_{L,\chi(\theta)}) = D\chi(\theta)^\top \nabla_g^2 \mathsf{A}_L(\chi(\theta)) D\chi(\theta) \longrightarrow g(\theta) = D\chi(\theta)^\top H(\chi(\theta)) D\chi(\theta) \quad \text{in } C^2. \tag{C.22}$$

[ESTABLISHED]

Proof. The natural normalized log score is $(T_{h,L}, T_{j,L}) - \mathbb{E}(T_{h,L}, T_{j,L})$. Its covariance is exactly $V \nabla_g^2 \mathsf{A}_L$. The pressure theorem and its scalar conversion yield eq. (C.21). The ordinary chain rule for the normalized log probability multiplies the source score by $D\chi^\top$, proving the finite pullback. Its C^2 convergence follows from the C^2 source-Hessian limit and the product and composition rules. In contrast,

$$\nabla_\theta^2 (\mathsf{A}_L \circ \chi) = D\chi^\top (\nabla_g^2 \mathsf{A}_L) D\chi + \sum_{i=1}^{2} (\partial_i \mathsf{A}_L) \nabla_\theta^2 \chi_i. \tag{C.23}$$

The last term is absent from the score covariance. Thus neither the raw pressure Hessian nor a Hessian after nonlinear source substitution can be used without its stated correction. □

Proposition C.8 (A native finite-energy information floor). On an admitted compact source box put $H_* = h_+ + 2j_+$, $m_* = (1 + e^{2H_*})^{-1}$, and $v_* = \mathrm{sech}^2 H_*$. For $L \geq 12$,

$$\frac{1}{V} I(\nu^{\mathrm{f}}_{L,g}) \succeq \frac{v_* m_*^6}{8} I_2, \qquad H(g) \succeq \frac{v_* m_*^6}{8} I_2. \tag{C.24}$$

Consequently the pullback in eq. (C.22) has rank $\mathrm{rank}\, D\chi$; positivity on an arbitrary d-parameter chart is not presumed. [ESTABLISHED]

Proof. Every edge conditioned on all other signs has field magnitude at most H_*, so either sign has conditional probability at least m_* and conditional variance at least v_*. Choose horizontal edges based at $(1 + 3u, 1 + 3v)$, where $u, v \geq 0$ and both coordinates are at most $L - 2$. There are $\lfloor L/3 \rfloor^2 \geq V/16$ such edges for $L \geq 12$, and no two share a plaquette. Condition on all unselected signs. The selected signs are then independent. For $F = w_h T_{h,L} + w_j T_{j,L}$, a selected edge e has coefficient $w_h + w_j R_e$, where R_e is the sum of the two neighboring three-sign products. Thus

$$\mathrm{Var}(F) \geq \mathbb{E}\,\mathrm{Var}(F \mid \text{unselected signs}) \geq v_* \sum_{e \text{ selected}} \mathbb{E}(w_h + w_j R_e)^2. \tag{C.25}$$

Each selected edge has six distinct neighboring signs, all unselected and inside the square. Setting all six positive forces $R_e = 2$. Setting the horizontal edge immediately above and the horizontal

edge immediately below negative, and the four vertical neighbors positive, forces $R_e = -2$. Each of these disjoint assignments has probability at least m_*^6. To see this under partial conditioning, reveal the six signs successively: averaging the full one-edge conditional lower bound over all unspecified signs preserves m_* at every step. Therefore

$$\frac{\mathrm{Var}(F)}{V} \geq \frac{v_* m_*^6}{16}[(w_h + 2w_j)^2 + (w_h - 2w_j)^2] = \frac{v_* m_*^6}{8}(w_h^2 + 4w_j^2) \geq \frac{v_* m_*^6}{8}|w|^2. \tag{C.26}$$

This is the finite matrix bound, and the proved source-Hessian limit inherits it. A positive definite middle matrix in $D\chi^\top H D\chi$ has kernel exactly $\ker D\chi$, proving the rank statement. On a compact full-column-rank chart the smallest singular value of $D\chi$ has a positive minimum. No positive metric on more than two independent source coordinates, or global quotient of a singular pullback, follows. □

C.1.4 Predictive rows under the finite free law

Lemma C.9 (Finite-free predictive rows through third order). Let D be a finite anchor inventory, $x \in D$, and $S \subset D$ a set of cells preceding x in the fixed lexicographic order. Write $p_S^D(g, s; y)$ for the four-state row of the exact finite free push, conditional on $Y_S = y_S$. It is positive and C^3, uniformly in S, y, D, x with the bounds below. The corresponding infinite-law row p_S prescribes exactly this past and leaves every future path free. [ESTABLISHED]

Proof. The finite conditioning restricts precisely the observed path fibers and retains every original internal plaquette. It deletes no additional interaction, and an absent future is an unconstrained future. A currently free target path conditioned on all other fine blocks has independent signs with fields $h + ja_e$, $|a_e| \leq 2$. Its parity mean is $G_g = \prod_{e \in P} \tanh(h + ja_e)$, of magnitude at most ρ_*^B, with support in at most nine anchors. Successively revealing the two target parities gives

$$p_S^D(g, s; y) \geq \frac{(1 - \rho_*^B)^2}{4} \geq \frac{d^2}{4}. \tag{C.27}$$

Finite constrained response applies at the original free boundary by lemma C.3.

For $n \leq 3$, assigning n labeled derivatives to the B hyperbolic-tangent factors gives $\|\partial^\beta G_g\|_\infty \leq g_n B^n \rho_*^{B-n}$, where $g_n = 2^n n!$ and $|\beta| = n$. The first three tangent derivatives are bounded by $1, 2, 6$, each j derivative contributes at most two, and the product of the factorials of assigned derivative orders is at most $n!$. At $B < n$ the negative exponent only enlarges the bound. Apply the moving-observable formula to $\mathbb{E}G_g$, keeping both derivatives of the amplitude and source insertions. Exactly as in eq. (B.33), this yields

$$\begin{aligned} D_n &= g_n + \sum_{a=0}^{n-1} \binom{n}{a} g_a S_{n-a}(9), \\ M_0(B) &= \rho_*^B, \qquad M_n(B) = D_n B^{2n+2} \rho_*^{B-n}, \qquad 1 \leq n \leq 3. \end{aligned} \tag{C.28}$$

These bound every order-n target parity-mean derivative. The actual infinite derivatives and their one-sided continuation follow from lemma C.4.

For one Bernoulli factor $(1+sm)/2$, the denominator $1+sm$ is at least d. Direct logarithmic differentiation gives

$$A_1=\frac{M_1}{d},\qquad A_2=\frac{M_2}{d}+\frac{M_1^2}{d^2},\qquad A_3=\frac{M_3}{d}+\frac{3M_2M_1}{d^2}+\frac{2M_1^3}{d^3}. \tag{C.29}$$

A whole-cell log row $l=\log p_S^D$ has order-n derivatives bounded by $\ell_n=2A_n$. Its second target-path factor conditions on the same already revealed first parity, retaining their dependence. These are the finite-free counterparts of eq. (B.35).

Define the row-Fisher observable $\mathsf{I}_{S,ik}^D=\sum_s p_S^D l_i l_k$, whose configuration argument depends only on y_S. Its derivatives at fixed y are

$$\begin{aligned}\partial_u \mathsf{I}_{S,ik}^D&=\sum_s p_S^D(l_u l_i l_k+l_{iu}l_k+l_i l_{ku}),\\ \partial_v\partial_u \mathsf{I}_{S,ik}^D&=\sum_s p_S^D\big[(l_{uv}+l_u l_v)l_i l_k+l_u(l_{iv}l_k+l_i l_{kv})+l_v(l_{iu}l_k+l_i l_{ku})\\ &\quad+l_{iuv}l_k+l_{iu}l_{kv}+l_{iv}l_{ku}+l_i l_{kuv}\big].\end{aligned} \tag{C.30}$$

In particular $\partial_u p=pl_u$ and $\partial_v\partial_u p=p(l_{uv}+l_u l_v)$ have been included. The ten expanded second-derivative monomials give the common entrywise bound

$$E_B^F=\max\{\ell_1^2,\ell_1^3+2\ell_1\ell_2,\ell_1^4+5\ell_1^2\ell_2+2\ell_1\ell_3+2\ell_2^2\},\qquad H_B^F=\max(1,\ell_1,\ell_2,\ell_3), \tag{C.31}$$

for the observable and its first two derivatives.

Suppose two target parity means have jet differences bounded by v_0,v_1,v_2,v_3. Subtract the differentiated logarithms and use $|a^{-k}-b^{-k}|\le k|a-b|/d^{k+1}$. The one-factor log-jet differences are bounded by

$$\begin{aligned}\mathcal{T}_1(v)&=v_1/d+M_1v_0/d^2,\\ \mathcal{T}_2(v)&=v_2/d+(M_2v_0+2M_1v_1)/d^2+2M_1^2v_0/d^3,\\ \mathcal{T}_3(v)&=v_3/d+(M_3v_0+3M_2v_1+3M_1v_2)/d^2\\ &\quad+(6M_2M_1v_0+6M_1^2v_1)/d^3+6M_1^3v_0/d^4.\end{aligned} \tag{C.32}$$

These reconstruct eq. (B.36). A complete row uses $2\mathcal{T}_n$; its probability difference is at most v_0, because each of its two Bernoulli factors changes by at most $v_0/2$. Telescope the at most five factors in each monomial of eq. (C.30), over four states, to obtain

$$\mathcal{W}_B(v)=\min\{2E_B^F,200(H_B^F)^4[v_0+2\mathcal{T}_1(v)+2\mathcal{T}_2(v)+2\mathcal{T}_3(v)]\}. \tag{C.33}$$

It bounds every entrywise row-Fisher derivative difference through degree two, with a separately valid cap $2E_B^F$.

For two past patterns in the same finite D agreeing through Q_R, let $a_R = \lfloor R/2 \rfloor - 1$ for $R \geq 2$, and put $N_n = 2^n q_{n+1} C(1+4n)$. Their parity-jet differences obey

$$\begin{aligned} V_0(B,R) &= \min\{2\rho_*^B, 2CB^2(R+1)e^{-\tau(R+1)}\}, \\ V_n(B,R) &= \min\big\{2M_n(B), B^{2n+2}\big[N_n(2a_R+1)^{2n}(R-a_R)e^{-\tau(R-a_R)} \\ &\qquad + 2S_n(1)(a_R+2)^{2n+1}e^{-\tau(a_R+1)/n}\big]\big\}, \qquad 1 \leq n \leq 3. \end{aligned} \tag{C.34}$$

For $R = 0, 1$, take $V_n = 2M_n$. To justify the same finite-domain modulus near an original free boundary, split source tuples at a_R. Near joint supports occupy at most $1+4n$ anchors in Q_{a_R+1}; there are at most $[2B^2(2a_R+1)^2]^n$ tuples. Group comparison on the remaining buffer gives the first term. Original omitted interactions are identical in the two laws and produce no near discrepancy. Each far sum obeys eq. (C.10). This is the argument of eq. (B.34) on the retained finite source list, including irregular intersections with the original inventory. Consequently

$$\|\mathsf{I}_S^D - \mathsf{I}_{S'}^D\|_{C^2(K),\infty,\mathrm{entry}} \leq \omega_B(R) := \mathcal{W}_B(V_0, V_1, V_2, V_3). \tag{C.35}$$

At fixed B, $\omega_B(R)$ is bounded by a finite constant times $(R+1)^7 e^{-\tau R/6}$. The order-three response supplies the exponent $\tau/6$; a first-order rate has not been substituted.

If the target has $Q_R \subset D$, compare a finite row with the infinite row of the same local past. Apply eq. (C.7) with $a_0 = 1$ separately to each target parity. When comparing the second factor, the first parity is prescribed identically on both sides. Thus

$$\|\mathsf{I}_S^D - \mathsf{I}_{S'}\|_{C^2(K),\infty,\mathrm{entry}} \leq \Omega_B(R) := \mathcal{W}_B\big(\Delta_0(B,1,R), \Delta_1(B,1,R), \Delta_2(B,1,R), \Delta_3(B,1,R)\big) \longrightarrow 0. \tag{C.36}$$

This bound is pointwise in the common past coordinates and controls actual row derivatives. It has not yet differentiated their expectations under a moving law. □

Lemma C.10 (Weighted shells under the moving unconditioned law). For a local C^2 observable supported in Q_N, put $a_N = (2N+1)^2$ and

$$\|f\|_{2,K,\infty} = \max_{|\alpha|\leq 2} \sup_{g\in K, y} |\partial^\alpha f(g,y)|, \qquad W_B(N) = 1 + 2S_1(a_N)B^4 + S_2(a_N)B^6. \tag{C.37}$$

The original finite free and infinite unconstrained fine laws satisfy

$$\|\mathbb{E}_g f_g\|_{C^2(K)} \leq W_B(N)\|f\|_{2,K,\infty}. \tag{C.38}$$

Every finite-past row expectation, including targets at the finite free boundary, has entrywise C^2 bound

$$Z_B^F = W_B(0)E_B^F + \sum_{n\geq 1} W_B(n)\omega_B(n-1) < \infty, \qquad \mathcal{T}_B(N) = \sum_{n>N} W_B(n)\omega_B(n-1) \longrightarrow 0. \tag{C.39}$$

Here $\mathcal{T}_B(N)$ bounds the C^2 error of truncating the past at Q_N. The same statements hold for the infinite full-past row and its actual expectation under μ_g. [ESTABLISHED]

Proof. Pull the coarse observable back along the fixed parity map. Its fine support uses at most a_N anchors. In the first derivative of its expectation, the moving-response formula has an expectation of f_i and a source covariance sum. In the second derivative it has an expectation of f_{ik}, the two source covariance sums of f_i, f_k, and the two-source cumulant sum of f. Applying eq. (C.10) to these terms proves eq. (C.38). The infinite version follows by finite-near and uniform-tail passage before coordinate FTC, as in lemma C.5. This supplies the finite-free counterpart of eq. (B.42).

For a finite past S, let $S_n = S \cap Q_n$, $f_n^D = \mathsf{I}_{S_n}^D$, $d_0^D = f_0^D$, and $d_n^D = f_n^D - f_{n-1}^D$. The two prefixes agree through Q_{n-1}, so

$$\|d_0^D\|_{2,K,\infty} \le E_B^F, \qquad \|d_n^D\|_{2,K,\infty} \le \omega_B(n-1), \qquad n \ge 1. \tag{C.40}$$

Each difference is a function only of y_{S_n}, although its parameter-dependent coefficients were computed from the entire finite law. Its pointwise derivatives include that coefficient dependence. Thus configuration locality is exactly what the moving-response bound needs; no derivative of the finite normalizer is discarded.

The degree of $S_1(a)$ in a is at most two, and that of $S_2(a)$ at most three. Hence $W_B(n)$ grows at most as a degree-six polynomial in n, whereas $\omega_B(n-1)$ is bounded by a degree-seven polynomial times $e^{-\tau n/6}$ at fixed B. Their product is summable. The finite series terminates exactly once Q_n contains S. The common weighted sum bounds every finite expected row, independently of its target, domain and remote past shape, and bounds its truncation remainder by $\mathcal{T}_B(N)$.

For an infinite past the pointwise row jets are uniformly Cauchy by eq. (C.35). Their shell expectations and derivatives through degree two are uniformly Cauchy under the same majorant. Passing finite-sum coordinate FTC identities identifies an actual C^2 expectation and allows termwise differentiation. Absolute convergence permits reordering the source and shell sums. Throughout, the averaging law is the original unconditioned fine law. The constrained law defines the row; it is not substituted for this outer expectation. □

C.1.5 Exact finite-free pushes and the three-cutoff limit

Theorem C.11 (Exact pushed-free Fisher density). For every fixed integer $B \ge 1$ and compact positive source box,

$$\frac{1}{L^2} I(P_{B,L,g}^{\mathrm{f}}) \longrightarrow \mathcal{I}_B(g) := \mathbb{E}_{\mu_g} \mathsf{I}_{\mathcal{P}}(g, Y) \quad \text{in } C^2(K), \tag{C.41}$$

where $\mathcal{P}$ is the complete lexicographic past and every future path is unconstrained. This is the same density as the actual infinite-law marginal limit in eq. (B.44). The finite laws in the two limits need not agree. [ESTABLISHED]

Proof. Order the cells of Λ_L lexicographically and use the true finite predictive rows of $P^{\mathrm{f}}_{B,L,g}$. Their log scores add to the complete score. Each row score has conditional mean zero given its finite past, since differentiating its finite normalization gives zero. An earlier row score is measurable with respect to any later past; conditioning on that past therefore kills every cross expectation. With $S_{x,L} = \Lambda_L \cap \{y : y < x\}$, the exact finite identity is

$$I(P^{\mathrm{f}}_{B,L,g}) = \sum_{x\in\Lambda_L} \mathbb{E}_{\nu^{\mathrm{f}}_{BL,g}} \mathsf{I}^{D_L}_{S_{x,L}}(g, Y). \tag{C.42}$$

It is an equality of finite C^2 functions, so its first two derivatives are still exact finite identities.

Fix an inner past cutoff $N \geq 0$, then an outer radius $R \geq N + 4$. Consider a bulk target x with $x + Q_R \subset \Lambda_L$ and translate it to zero. Its finite past and the complete past have exactly the same intersection with Q_N. Truncating each of their expected row observables to that common prefix costs at most $2\mathcal{T}_B(N)$ in entrywise C^2, by lemma C.10.

The finite and infinite truncated rows are functions of the same finite past coordinates. Their pointwise C^2 difference is at most $\Omega_B(R)$, by eq. (C.36). Its expectation under the moving finite law is therefore bounded by $W_B(N)\Omega_B(R)$. This step differentiates the finite averaging law; it does not multiply a pointwise error by an uncontrolled extensive score.

It remains to compare expectations of the same infinite truncated row under $\nu^{\mathrm{f}}_{BL,g}$ and μ_g. Its pointwise jets are bounded by E^F_B, its support is Q_N, and this support has a radius-$(R - N)$ neighborhood in D_L. Thus eq. (C.14) bounds the remaining cost by $E^F_B[\Delta_0+2\Delta_1+\Delta_2](B, a_N, R-N)$. Combining these three comparisons gives the bulk bound

$$\begin{aligned}\mathcal{B}_B(N, R) = {} & 2\mathcal{T}_B(N) + W_B(N)\Omega_B(R) \\ & + E^F_B[\Delta_0(B, a_N, R - N) + 2\Delta_1(B, a_N, R - N) + \Delta_2(B, a_N, R - N)].\end{aligned} \tag{C.43}$$

At most $4RL$ cells are outside the bulk for $L \geq 2R$. Each finite and infinite expected-row jet is bounded by Z^F_B, including near the free outer boundary. Stationarity identifies the infinite target expectation at each bulk translation. Dividing eq. (C.42) by L^2 therefore yields

$$\left\| L^{-2} I(P^{\mathrm{f}}_{B,L,g}) - \mathcal{I}_B(g) \right\|_{C^2(K),\mathrm{entry}} \leq \mathcal{B}_B(N, R) + 8Z^F_B \frac{R}{L}. \tag{C.44}$$

Given a tolerance, first choose N for the weighted shell tail. Next choose R for the two local comparisons at that fixed N, and finally choose L for the boundary fraction. This order retains the growth of the support-dependent constants in a_N; no bound uniform in N has been presumed. All terms vanish in that order, with B fixed. This proves the asserted C^2 limit.

For actual infinite-law marginals the same finite score-row identity uses μ_g as its outer expectation at every volume. Their finite pasts and full past agree through Q_R at all but $4RL$ sites. The weighted shell argument then gives $2\mathcal{T}_B(R) + 8Z^F_B R/L$, exactly the proof of eq. (B.44). Hence its limit is the same $\mathcal{I}_B$. At $B = 1$, C_1 is the identity, and eqs. (C.21) and (C.41) are limits of the same finite

matrices. Consequently

$$\mathcal{I}_1 = H = \nabla_g^2 \mathsf{A}. \tag{C.45}$$

For $B > 1$, an exact push need not have a two-statistic exponential representation, so it has not been identified with a Hessian of native or canonical coarse pressure. □

Proposition C.12 (Area normalization of the fixed-word comparison). For fixed positive integers a, d_0,

$$\mathcal{I}_{ad_0} \preceq d_0^2 \mathcal{I}_a, \qquad \frac{\mathcal{I}_{ad_0}}{(ad_0)^2} \preceq \frac{\mathcal{I}_a}{a^2}, \qquad \mathcal{I}_B \preceq B^2 H. \tag{C.46}$$

[ESTABLISHED]

Proof. The laws $P^{\mathrm{f}}_{a,d_0 L}$ and $P^{\mathrm{f}}_{ad_0,L}$ are obtained from the same fine free square of side $ad_0 L$, and the latter is the exact d_0-path push of the former. Differentiating a finite parameter-independent channel gives an output score equal to the conditional expectation of the input score. Conditional covariance decomposition proves $I(P^{\mathrm{f}}_{ad_0,L}) \preceq I(P^{\mathrm{f}}_{a,d_0 L})$. Divide by the final coarse area L^2 and take the two fixed-depth limits to obtain the first inequality; the second is its rearrangement. Direct comparison with the fine native law gives the last inequality. These are tensor inequalities at each source point. Differentiating a positive-semidefinite inequality does not order its derivatives. □

C.1.6 Product controls and nonempty regular charts

Proposition C.13 (Product-boundary scores and fixed-depth positive charts). For every prescribed finite list of depths, the tensors $\mathcal{I}_B$ have a common nonempty open positive-j source region on which each is positive definite. Each compact subchart has a positive minimum eigenvalue at every listed depth. The minimum need not persist over all integer depths. [ESTABLISHED]

Proof. First consider the native product control $j = 0$. Write $b = \tanh h$ and $v = 1 - b^2$. Independent fine signs give

$$H(h, 0) = \begin{pmatrix} 2v & 4b^3 v \\ 4b^3 v & 1 - b^8 + 4b^6 v \end{pmatrix}, \qquad \det H(h, 0) = 2v^3(1 + 2b^2 + 3b^4) > 0. \tag{C.47}$$

The field covariance counts two signs per cell. The mixed covariance counts four edges per plaquette, each contributing $b^3 v$. A plaquette has variance $1 - b^8$ and four neighbors sharing one edge, each with covariance $b^6 v$; all other plaquette covariances vanish. Expanding the determinant gives $2v(1 - b^8) - 8b^6 v^2 = 2v^3(1 + 2b^2 + 3b^4)$. These are derivatives of the full source family at the boundary, including its transverse plaquette response. They are not obtained by holding j identically zero while differentiating. The one-sided response construction justifies the values and their continuation. This product argument is separate from the positive-coupling compact finite-energy bound in proposition C.8.

For a general fixed B, prescribe any complete lexicographic past, leave future paths free, and work at $j = 0$. The constrained fine law is a product of its block-fiber laws. A free target path has parity

mean b^B. For its parity value s, the contrast of any one path edge is

$$a(s) = \mathbb{E}[z_e \mid Y = s] - b = \frac{sb^{B-1}(1-b^2)}{1+sb^B}, \qquad \mathbb{E}a = 0, \qquad v_B = \mathbb{E}a^2 = \frac{b^{2B-2}(1-b^2)^2}{1-b^{2B}} > 0. \tag{C.48}$$

Multiply the independent sign law by its parity indicator to obtain this conditional mean, and sum its two parity outcomes to obtain the variance. Conditional variance decomposition also gives $v_B \le 1 - b^2 \le 1$. The two target contrasts a_1, a_2 are independent and centered. Finite source differentiation gives the actual field row score $s_h = B(a_1 + a_2)$.

The plaquette at the common starting vertex is the unique plaquette meeting both outgoing target paths. Its other two edges are future edges at $B = 1$ and unused singleton edges at $B > 1$, hence both have mean b. Its centered contribution is $b^2 a_1 a_2 + b^3(a_1 + a_2)$. Each other incident plaquette meets one target path, and contributes a multiple of that path's contrast with coefficient of magnitude at most one. There are $2B - 1$ such incidences per target path. Nonincident plaquettes cancel by block independence. Thus the actual plaquette score is

$$s_j = b^2 a_1 a_2 + \alpha_1 a_1 + \alpha_2 a_2, \qquad |\alpha_i| \le 2B. \tag{C.49}$$

These finite source identities pass to the one-sided limiting derivatives by lemma C.4, agreeing with the explicit construction in section B.7.4.

The bilinear contrast is orthogonal to both linear contrasts. Therefore

$$\begin{aligned} I^{\mathrm{row}}_{hh} &= 2B^2 v_B, \qquad I^{\mathrm{row}}_{hj} = B v_B(\alpha_1 + \alpha_2), \\ I^{\mathrm{row}}_{jj} &= v_B(\alpha_1^2 + \alpha_2^2) + b^4 v_B^2, \\ \det I^{\mathrm{row}} &= B^2 v_B^2(\alpha_1 - \alpha_2)^2 + 2B^2 b^4 v_B^3, \qquad \operatorname{Tr} I^{\mathrm{row}} \le 11 B^2 v_B. \end{aligned} \tag{C.50}$$

The smaller eigenvalue is at least the determinant divided by the trace, hence at least $2b^4 v_B^2/11$, uniformly in the past. On a compact positive field interval, $b \in [b_-, b_+] \subset (0,1)$, define

$$\eta_{0,B} = \frac{2}{11} b_-^{4B}(1 - b_+^2)^4 > 0, \qquad Q_B = 16(A_1^3 + A_1 A_2) > 0, \qquad \delta_B = \min\{j_+/2, \eta_{0,B}/(2Q_B)\} > 0. \tag{C.51}$$

The floor follows from $v_B \ge b_-^{2B-2}(1 - b_+^2)^2$. The row-score vector has norm at most $2\sqrt{2}A_1$, and its j derivative has norm at most $2\sqrt{2}A_2$. Differentiating its four-state Fisher sum gives matrix norm at most $16A_1^3$ for the probability derivative and at most $16A_1A_2$ for the two score derivatives. Its total derivative is therefore at most Q_B. The actual one-sided derivatives and FTC give

$$I^{\mathrm{row}}(h, j \mid \text{past}) \succeq \frac{\eta_{0,B}}{2} I_2, \qquad \mathcal{I}_B(h, j) \succeq \frac{\eta_{0,B}}{2} I_2, \qquad 0 < j < \delta_B. \tag{C.52}$$

The second inequality is the expectation of the first. Enlarge the field interval before fixing the constants to obtain an open neighborhood of the desired compact interval. For a prescribed finite depth list, take finite minima of widths and floors. There is no positive all-depth minimum in this argument. □

Proposition C.14 (Geometry on the admitted regular charts). On the compact regular charts just exhibited, the normalized finite-free and exact-push Fisher metrics converge with their inverses in C^2, their Levi-Civita coefficients in C^1, and their Riemann, Ricci and scalar curvatures in C^0. The actual-marginal metrics obey the same conclusion. A general source pullback needs an admitted full-column-rank chart and its positive compact singular-value minimum. [ESTABLISHED]

Proof. Write G for a limiting positive metric and G_L for its indicated normalized finite metric. The proved C^2 convergence makes G_L positive for every sufficiently large L, with a common inverse bound. Differentiating $GG^{-1} = I$ gives

$$\partial_i G^{-1} = -G^{-1}(\partial_i G)G^{-1}, \tag{C.53}$$

and a second differentiation is a finite sum of products of inverse and metric derivatives through order two. These formulas prove C^2 inverse convergence. In local coordinates the formulas

$$\begin{aligned} \Gamma^l_{ik} &= \frac{1}{2}G^{lm}(\partial_i G_{mk} + \partial_k G_{mi} - \partial_m G_{ik}), \\ R^l{}_{kiv} &= \partial_i \Gamma^l_{vk} - \partial_v \Gamma^l_{ik} + \Gamma^l_{im}\Gamma^m_{vk} - \Gamma^l_{vm}\Gamma^m_{ik} \end{aligned} \tag{C.54}$$

then give the asserted derivative orders. Their torsion-free, metric-compatible and tensorial interpretation is the one proved in eq. (B.48); only metric derivatives through degree two occur. Ricci and scalar contractions use the same convergent inverse. These statements concern the normalized metrics, whose constant volume rescaling cannot be suppressed when comparing scalar curvature.

These source coordinates also give local charts of the actual law. At a point, choose a sufficiently large actual finite marginal with positive Fisher matrix. Its positive probability vector satisfies $I = (Dp)^\top \operatorname{diag}(1/p)Dp$, so two atom probabilities have a nonsingular two-by-two Jacobian. Multiply that two-coordinate map by the inverse Jacobian at the point and shrink a convex ball until its derivative differs from the identity by less than $1/2$. Segment integration gives a lower Lipschitz bound. For nearby target values the iteration $x \mapsto x + y - F(x)$ preserves a smaller ball and contracts distances by $1/2$; its successive differences form a convergent geometric series, yielding a local inverse. Differentiating its defining identity gives the inverse derivatives. Thus distinct nearby parameters give distinct actual laws. This does not assert a global quotient of a singular tensor. Fisher curvature, gauge-connection curvature and base curvature retain their different underlying spaces. □

C.1.7 Why scalar boundary control is insufficient

Proposition C.15 (A bounded boundary ratio with divergent information density). Uniform subvolume control of log-density ratios and agreement of bulk cylinder limits do not imply convergence of volume-normalized Fisher tensors, even for positive smooth finite families on one fixed carrier at each volume. [ESTABLISHED]

Proof. Use the complete square carrier with $2V$ signs, $V = L^2$. Let P_L be the fair product law, independent of a real parameter θ. Choose one sign on the boundary of the recentered square and let $Q_{L,\theta}$ change only its mean to $m_L(\theta) = a\sin(V\theta)$, with fixed $0 < a < 1/2$. All probabilities are positive and smooth on the same open parameter interval, and for every configuration

$$\left|\log \frac{Q_{L,\theta}}{P_L}\right| \le \max\{-\log(1-a), \log(1+a)\}. \tag{C.55}$$

The varying sign leaves every fixed recentered cylinder for sufficiently large L, so the two sequences have the same bulk-cylinder limit. For a sign of mean m, differentiating its two probabilities gives Fisher information $(m')^2/(1-m^2)$. Independence of all other signs therefore gives

$$I(Q_{L,\theta}) = \frac{a^2V^2\cos^2(V\theta)}{1-a^2\sin^2(V\theta)}, \qquad I(Q_{L,0}) = a^2V^2, \qquad V^{-1}I(Q_{L,0}) = a^2V \longrightarrow \infty, \tag{C.56}$$

whereas $I(P_L) = 0$. The witness violates the native uniform local jet bounds: its boundary response grows with V. It refutes the scalar boundary inference, not the native theorems above. □

At the separately labeled native product control $j = 0$, the raw partition is one for every h, so $\partial_h^2\mathfrak{p}(h,0) = 0$, while $H_{hh}(h,0) = 2\,\mathrm{sech}^2 h > 0$. Thus even in the native family the raw-pressure Hessian needs the conversion in eq. (C.21). More generally, $f_L(\theta) = L^{-1}\sin(L^2\theta)$ converges uniformly to zero but has derivative L at zero, showing directly why the pressure proof must use local response rather than differentiation of its value error. [ESTABLISHED]

An exact finite shifted-source identity permits the preceding native and fixed-word proofs to be applied after its admitted C^4 source map, with scalar evidence accounted for separately in the raw geometry weights. A general positive likelihood or a free canonical coarse-interaction family requires its own response comparison or an exact finite family identity. The common infinite law and a surface log-ratio alone do not supply that derivative bridge. No arbitrary-observation, critical, continuum, simultaneous depth-volume, or all-depth inverse conclusion is included in this proof fragment. [ESTABLISHED]

C.2 Proofs for retained labels, ties, and reciprocal-volume experiments

This section uses the actual normalized finite experiments of Section 20.1. The sample space retains the geometry label $\gamma \in \mathfrak{G}$, even when conditional branches coincide. All supports are fixed on the parameter neighborhood in use; structural zero atoms are omitted from score sums and have no conditional score. The limit theorems below state their branch response inputs explicitly. For the admitted native families these are supplied by Section C.1; other branch families require their own matching input. The label conclusions also require the stated representative growth, gaps, or actual center ratios and do not follow from pressure convergence alone. Every volume limit is taken at fixed catalog and fixed block word. [ESTABLISHED]

C.2.1 Native inputs and effective representatives

Definition C.16 (Volume families and ordinary derivative norms). Let $\Theta \subset \mathbb{R}^d$ be open, let $V = L^2$, and let $p_{\gamma,L,\theta}$ be a positive normalized C^3 mass family on its fixed active finite carrier. Carriers may depend on L. Define

$$P_{L,\theta}(\gamma, x) = W_{\gamma,L}(\theta) p_{\gamma,L,\theta}(x), \qquad W_{\gamma,L} = \frac{e^{\lambda_{\gamma,L}}}{\sum_\eta e^{\lambda_{\eta,L}}}, \qquad \xi_{\gamma,L} = \nabla \lambda_{\gamma,L}. \tag{C.57}$$

For a compact K inside an open neighborhood, $\|f\|_{C^k(K)}$ denotes the maximum of the suprema of its ordinary coordinate derivatives through degree k, with a fixed finite-dimensional norm for vector and matrix entries. Derivatives of order zero are included. Put $g_{\gamma,L} = V^{-1} I(p_{\gamma,L})$ and $g_L = V^{-1} I(P_L)$. Unless stated otherwise, every convergence in this section is uniform in the indicated C^k norm on each such compact set. The bulk function s_γ below is an effective log-mass density; it is distinct from the physical scalar in a raw-factor representation. [DEFINITION]

Proposition C.17 (Conditional map from native response to branch inputs). Consider the complete native free square with normalized counting reference 4^{-V}, all $2V$ edge signs, and $(L-1)^2$ internal plaquettes. Write

$$T_L = \sum_e z_e, \qquad U_L = \sum_{p \text{ internal}} P_p, \qquad \mathsf{A}_L(h,j) = \frac{1}{V} \log \sum_x 4^{-V} e^{hT_L(x) + jU_L(x)}. \tag{C.58}$$

The actual C^4 convergence $\mathsf{A}_L \to \mathsf{A}$ is supplied by Theorem C.6 on the compact source boxes of Hypothesis C.1, inside $h > 0$, $0 < j < \operatorname{arctanh}(1/12)$. This uses the finite free law of Definition C.2, with the same complete carrier and normalized reference as above. If $\chi_\gamma : \Theta \subset \mathbb{R}^d \to \mathbb{R}^2$, $\chi_\gamma(\theta) = (h_\gamma(\theta), j_\gamma(\theta))$, is C^4 on a neighborhood of K and sends that neighborhood into a common compact admitted source box, then Proposition C.7 gives

$$\mathfrak{p}_{\gamma,L} := \frac{1}{V} \log Z_{\gamma,L} = \mathsf{A}_L \circ \chi_\gamma - 2 \log\cosh h_\gamma - \frac{(L-1)^2}{V} \log\cosh j_\gamma, \tag{C.59}$$

$$g_{\gamma,L} = (D\chi_\gamma)^\top (D^2 \mathsf{A}_L \circ \chi_\gamma) D\chi_\gamma \longrightarrow g_\gamma = (D\chi_\gamma)^\top (D^2 \mathsf{A} \circ \chi_\gamma) D\chi_\gamma \quad \text{in } C^2(K). \tag{C.60}$$

In particular $\mathfrak{p}_{\gamma,L}$ has an actual C^4 limit $\mathfrak{p}_\gamma$ and the branch densities have bounded C^2 norms. This records the application of the native derivative theorem; its local-response and finite-free boundary proof is Theorem C.6. [ESTABLISHED]

Proof. For a sign z, $1 + (\tanh h) z = e^{hz} / \cosh h$. Apply this identity to every edge and plaquette, with $b = \tanh h$ and $t = \tanh j$, and count the factors. This gives eq. (C.59) exactly. The normalized branch mass is the exponential family with sufficient statistic (T_L, U_L): its scalar hyperbolic-cosine factors cancel in its own normalization. Differentiating the finite sum in eq. (C.58) gives $D^2 \mathsf{A}_L = V^{-1} \operatorname{Cov}(T_L, U_L)$ in natural source coordinates. Multiplying the centered natural score by $(D\chi_\gamma)^\top$ gives the displayed tensor pullback. Ordinary chain and product rules through degree two use derivatives of A_L through degree four and of χ_γ through degree three. The declared C^4 source

maps more than supply these derivatives. The raw-pressure convergence follows by composition and by $(L-1)^2/V \to 1$. No derivative of a C^0 pressure error is taken. The Hessian of $\mathsf{A}_L \circ \chi_\gamma$ would additionally contain $\sum_a (\partial_a \mathsf{A}_L \circ \chi_\gamma) D^2 \chi_{\gamma,a}$; the Hessian of raw pressure would also retain the scalar correction. Neither is the Fisher pullback in general. □

For an exact fixed-word push, use the distinct family in Definition C.2. Let C_B be its fixed parameter-independent straight-path parity channel, including the unrevealed paths and unused singleton blocks in the fine carrier. A fixed finite word of the admitted integer path maps concatenates to this C_B on the same fine square, as in Definition C.2 and proposition C.12. Write $p^{\mathrm{free}}_{\gamma,BL} C_B$ for the actual push from $B^2 V$ fine cells to $V = L^2$ coarse cells. The source-coordinate theorem Theorem C.11, followed by the same C^4 source map, supplies

$$\frac{1}{V} I_\theta(p^{\mathrm{free}}_{\gamma,BL} C_B) \longrightarrow (D\chi_\gamma)^\top (\mathcal{I}_B \circ \chi_\gamma) D\chi_\gamma \quad \text{in } C^2(K). \tag{C.61}$$

Here $\mathcal{I}_B$ is the natural-source Fisher density per coarse cell of actual infinite-law marginals. To make the parameter dimension explicit, at each finite volume the normalized score chain rule gives

$$I_\theta(p^{\mathrm{free}}_{\gamma,BL} C_B) = (D\chi_\gamma)^\top I_{(h,j)}(P^{\mathrm{f}}_{B,L,\chi_\gamma(\theta)}) D\chi_\gamma. \tag{C.62}$$

The source theorem is a C^2 tensor limit in two natural coordinates; composition with χ_γ and the ordinary product rule through degree two give eq. (C.61) in the declared d coordinates. No rank assumption is needed for this semidefinite pullback. The theorem's three-cutoff boundary estimate eq. (C.44) identifies its actual finite-free limit with the actual-marginal density while keeping these finite families distinct. The actual-marginal theorem alone, a scalar pressure estimate, an arbitrary channel, or a separately normalized free canonical coarse family does not supply this input. The word and B remain fixed before the volume limit. [ESTABLISHED]

Proposition C.18 (Effective growth in both normalization modes). Suppose the positive raw weights have actual C^3 logarithms and satisfy

$$\|\log a_{\gamma,L}\|_{C^3(K)} \le C_K V, \qquad V^{-1} \log a_{\gamma,L} \longrightarrow \alpha_\gamma \quad \text{in } C^3(K). \tag{C.63}$$

Using Theorem C.6 through Proposition C.17, valid effective representatives and bulk limits for the latent experiment are

$$\lambda^{\mathrm{f}}_{\gamma,L} = \log a_{\gamma,L} + \log Z_{\gamma,L}, \qquad s^{\mathrm{f}}_\gamma = \alpha_\gamma + \mathfrak{p}_\gamma, \tag{C.64}$$

$$\lambda^{\mathrm{h}}_{\gamma,L} = \log a_{\gamma,L}, \qquad s^{\mathrm{h}}_\gamma = \alpha_\gamma. \tag{C.65}$$

Each satisfies

$$\|\lambda_{\gamma,L}\|_{C^3(K)} \le C_K V, \qquad \lambda_{\gamma,L}/V \longrightarrow s_\gamma \quad \text{in } C^3(K). \tag{C.66}$$

The same assertion holds for the admitted all-edge selected posterior, with the effective representatives given below. Common normalizers must be removed algebraically before imposing this growth condition. [ESTABLISHED]

Proof. The finite formulas in eqs. (20.4) and (20.7) give eqs. (C.64) and (C.65). Replacing every $\lambda_{\gamma,L}$ by $\lambda_{\gamma,L} + c_L(\theta)$ leaves every $W_{\gamma,L}$ unchanged and adds the same ∇c_L to every $\xi_{\gamma,L}$. Their covariance is unchanged as a function, including all its available derivatives. Thus a hierarchy generated from raw a uses $\log a$ for eq. (C.66). One need not, and cannot in general, impose the same bound on $\log W_\gamma = \log a_\gamma - \log \sum_\eta a_\eta$ at a tie. For example, with $a_0 = 1$, $a_1 = e^{V\theta}$, the second derivative of $\log(a_0 + a_1)$ at zero is $V^2/4$. Its exact common cancellation preserves the stated raw-weight domain. A fixed positive C^3 prescribed hierarchy w_γ is a separate admissible case with $\lambda_{\gamma,L} = \log w_\gamma$ and bulk limit zero.

For the all-edge binary sensor of eq. (20.15), let its mean parameter be $\eta_\gamma(\theta)$ and set

$$A_\gamma^{\mathrm{sens}} = \frac{1 + b_\gamma \eta_\gamma}{2}, \qquad b'_\gamma = \frac{b_\gamma + \eta_\gamma}{1 + b_\gamma \eta_\gamma}. \tag{C.67}$$

Assume these data are C^4 on a neighborhood of K, the sensor is strictly positive, and both original (h_γ, j_γ) and shifted $(\operatorname{arctanh} b'_\gamma, j_\gamma)$ sources lie in common compact admitted boxes. Multiplication over exactly $2V$ signs gives $Z^o_\gamma = (A^{\mathrm{sens}}_\gamma)^{2V} Z_{\gamma,L}(b'_\gamma, t_\gamma)$. The selected conditional branch is exactly the shifted native free law. Up to a common log mass, the posterior representatives are

$$\lambda^{o,\mathrm{f}}_{\gamma,L} = \log a_{\gamma,L} + 2V \log A^{\mathrm{sens}}_\gamma + \log Z_{\gamma,L}(b'_\gamma, t_\gamma), \tag{C.68}$$

$$\lambda^{o,\mathrm{h}}_{\gamma,L} = \log a_{\gamma,L} + 2V \log A^{\mathrm{sens}}_\gamma + \log Z_{\gamma,L}(b'_\gamma, t_\gamma) - \log Z_{\gamma,L}(b_\gamma, t_\gamma). \tag{C.69}$$

Divide by V and apply Theorem C.6 to both the original and shifted sources in their admitted boxes. The bulk limits are respectively $\alpha_\gamma + 2\log A^{\mathrm{sens}}_\gamma + \mathfrak{p}'_\gamma$ and $\alpha_\gamma + 2\log A^{\mathrm{sens}}_\gamma + \mathfrak{p}'_\gamma - \mathfrak{p}_\gamma$ in C^3. Proposition C.7 applied to the shifted source supplies the conditional branch input; its exact fixed-word push uses Theorem C.11 and eq. (C.62). All sensor derivatives remain in these expressions. For a prescribed hierarchy replace $\log a$ by $\log w$ and keep the same physical partition difference. Strict positivity makes the finite selected event well defined. This argument is confined to this exact source-shift identity; it asserts no derivative theorem for an arbitrary positive likelihood or a null infinite observation event. □

When the volume family instead denotes a fixed-word push, its branches in eq. (C.57) use coarse volume V, whereas its label weights are inherited from the fine volume $M = B^2 V$. Thus $\lambda_{\gamma,BL}/V \to B^2 s_\gamma$ in C^3 and the same strict winner is selected. This conversion is required before applying a volume-normalized statement. It neither changes B with L nor turns an arbitrary stochastic channel into spatial blocking.

C.2.2 Strict phases and regular charts

Theorem C.19 (Strict-phase convergence with both metric derivatives). Assume eq. (C.66) on a neighborhood of K and actual branch convergence $g_{\gamma,L} \to g_\gamma$ in $C^2(K)$ for every label. Suppose a fixed label γ_* obeys $s_{\gamma_*} - s_\gamma \geq \delta_K > 0$ on K for every $\gamma \neq \gamma_*$. Then

$$\|g_L - g_{\gamma_*}\|_{C^2(K)} \longrightarrow 0. \tag{C.70}$$

More precisely, for $j = 0, 1, 2$ the losing label term obeys

$$\left\| D^j \left[V^{-1} \operatorname{Cov}_{W_L}(\xi_{\gamma,L}) \right] \right\|_K \le C_K V^{j+1} e^{-\delta_K V/2}, \tag{C.71}$$

and the weighted branch-density error relative to its winning branch has $C^2(K)$ norm at most $C_K V^2 e^{-\delta_K V/2}$ for all sufficiently large L. Only three derivatives of the effective log masses are used. [ESTABLISHED]

Proof. Uniform convergence of λ/V implies $\lambda_{\gamma,L} - \lambda_{\gamma_*,L} \le -\delta_K V/2$ for each loser when L is large. The softmax denominator includes the winning mass, so $W_{\gamma,L} \le e^{-\delta_K V/2}$. Write $\overline{\xi}_i = \sum_\eta W_\eta \xi_{\eta,i}$, suppressing L. Finite differentiation gives

$$\partial_i W_\gamma = W_\gamma(\xi_{\gamma,i} - \overline{\xi}_i), \tag{C.72}$$

$$\partial_j \overline{\xi}_i = \sum_\eta W_\eta \partial_j \xi_{\eta,i} + \operatorname{Cov}_W(\xi_i, \xi_j), \tag{C.73}$$

$$\partial_{ij} W_\gamma = W_\gamma \big[(\xi_{\gamma,i} - \overline{\xi}_i)(\xi_{\gamma,j} - \overline{\xi}_j) + \partial_j \xi_{\gamma,i} - \partial_j \overline{\xi}_i \big]. \tag{C.74}$$

Every ξ and its first two derivatives is $O(V)$. The covariance in eq. (C.73) is $O(V^2)$, so for every ordinary multi-index β of degree at most two,

$$\|\partial^\beta W_{\gamma,L}\|_K \le C_K V^{|\beta|} e^{-\delta_K V/2} \quad (\gamma \ne \gamma_*). \tag{C.75}$$

The same bound applies to $W_{\gamma_*,L} - 1$ and its derivatives, since it is minus the sum of losing weights.

Put $d_\gamma = \xi_\gamma - \xi_{\gamma_*}$ and $m_d = \sum_{\gamma \ne \gamma_*} W_\gamma d_\gamma$. Exact centering gives

$$\operatorname{Cov}_W(\xi) = \sum_{\gamma \ne \gamma_*} W_\gamma d_\gamma d_\gamma^\top - m_d m_d^\top. \tag{C.76}$$

Each derivative of d_γ through degree two is $O(V)$. A term with j total derivatives in the first sum has bound $C V^{j+2} e^{-\delta_K V/2}$: at most j derivatives land on a weight, and the two score factors each contribute at most V. The product $m_d m_d^\top$ has the same polynomial bound and two exponential factors, hence also this bound. Dividing by V proves eq. (C.71).

For the branch term use the exact identity

$$\sum_\gamma W_\gamma g_{\gamma,L} - g_{\gamma_*,L} = \sum_{\gamma \ne \gamma_*} W_\gamma (g_{\gamma,L} - g_{\gamma_*,L}). \tag{C.77}$$

The branch differences have uniformly bounded C^2 norms. The ordinary Leibniz sum, including all derivatives of W, and eq. (C.75) yield its stated C^2 bound. Add the winning branch's C^2 limit error and the covariance error, using eq. (20.7). Exponentials dominate all the displayed powers of V, proving eq. (C.70). □

The native positivity input is Proposition C.8. On its compact source box use $H_* = h_+ + 2j_+$, $m_* = (1+e^{2H_*})^{-1}$, and $v_* = \operatorname{sech}^2 H_*$. Its exact finite and limiting bounds are

$$D^2\mathsf{A}_L \succeq \frac{v_* m_*^6}{8}\,\mathrm{Id}_2 \quad (L \geq 12), \qquad D^2\mathsf{A} \succeq \frac{v_* m_*^6}{8}\,\mathrm{Id}_2\,. \tag{C.78}$$

The separated-edge conditional-variance proof is given there. The pullback in eq. (C.60) therefore has kernel $\ker D\chi_\gamma$ and rank $\operatorname{rank} D\chi_\gamma$, since a positive matrix has zero quadratic form only at the zero vector. [ESTABLISHED]

Proposition C.20 (Inverse, connection, and curvature on regular charts). Let symmetric tensor fields $G_L \to G$ in C^2 on a neighborhood of a compact parameter chart, and suppose G is positive definite there. Then $G_L^{-1} \to G^{-1}$ in C^2, the Levi-Civita coefficients converge in C^1, and the Riemann, Ricci, and scalar curvatures converge in C^0 on the compact chart. This applies to the strict-phase density of theorem C.19 precisely on such a chart. [ESTABLISHED]

Proof. For the fixed Euclidean coordinate norm, minimize $z^\top G(\theta) z$ on the compact product of the chart with the unit sphere. Positivity gives a minimum $\epsilon > 0$. Uniform convergence implies $G_L \succeq (\epsilon/2)\,\mathrm{Id}$ for large L. The identity $G_L^{-1} - G^{-1} = G_L^{-1}(G - G_L)G^{-1}$ proves uniform inverse convergence. Direct differentiation of $G_L G_L^{-1} = \mathrm{Id}$ gives, writing $H_L = G_L^{-1}$,

$$\partial_i H_L = -H_L(\partial_i G_L)H_L, \tag{C.79}$$

$$\partial_{ij} H_L = H_L(\partial_j G_L)H_L(\partial_i G_L)H_L + H_L(\partial_i G_L)H_L(\partial_j G_L)H_L - H_L(\partial_{ij} G_L)H_L. \tag{C.80}$$

Every factor converges uniformly, proving C^2 inverse convergence. Use the explicit coefficients and curvature convention

$$\Gamma^k_{ij}(G_L) = \frac{1}{2}(G_L)^{k\ell}(\partial_i (G_L)_{j\ell} + \partial_j (G_L)_{i\ell} - \partial_\ell (G_L)_{ij}), \tag{C.81}$$

$$R^\ell{}_{kij}(G_L) = \partial_i \Gamma^\ell_{jk} - \partial_j \Gamma^\ell_{ik} + \Gamma^\ell_{im}\Gamma^m_{jk} - \Gamma^\ell_{jm}\Gamma^m_{ik}. \tag{C.82}$$

The first line and its first derivatives converge; the second line and its contractions therefore converge continuously. Lowering the first index multiplies by the convergent metric, and scalar contraction uses the convergent inverse. These arguments prove exactly the stated orders, without third derivatives of the metric. □

For a parameter-independent positive constant V, replacing G_L by $I_L = VG_L$ leaves the connection, type-$(1,3)$ Riemann tensor, and covariant Ricci tensor unchanged. The fully covariant Riemann tensor is multiplied by V, and scalar curvature is divided by V, as follows by substitution in eq. (C.82) and its contractions. A finite positive Fisher tensor implies that the Jacobian of the finite probability map has full column rank: its Gram form is the Fisher quadratic form with positive diagonal weights. A full-rank probability minor supplies a locally identifiable chart. It does not give global identifiability.

The native floor makes the natural two-source chart nonempty. A general d-parameter native pullback has rank at most two; directions acting only on losing labels can disappear in the strict-

phase limit. For the exact push input eq. (C.61), Proposition C.13 supplies the required source floor on its specific positive-j region. On its compact positive field interval, the constants $\eta_{0,B} > 0$ and $\delta_B > 0$ in eq. (C.51) give $\mathcal{I}_B(h, j) \succeq (\eta_{0,B}/2)\,\mathrm{Id}_2$ for $0 < j < \delta_B$, by eq. (C.52). For a prescribed finite list $\mathcal{B}$ choose a parameter compact whose source image lies in this field interval and $0 < j < \min_{B\in\mathcal{B}} \delta_B$. The source derivative on the chosen chart must have full column rank, with compact minimum singular value $\sigma_K > 0$. Its pulled-back limiting metrics then have the common floor $(\sigma_K^2/2) \min_{B\in\mathcal{B}} \eta_{0,B} > 0$. Together with the strict-phase convergence above this supplies the hypotheses of Proposition C.20, consistently with the branch geometry in Proposition C.14. An arbitrary positive high-temperature source image is not thereby shown to lie in these small-j charts; that inclusion and full column rank remain explicit conditions for this inverse statement. No positive floor uniform in unbounded depth, inverse on a singular stratum, or quotient inferred solely from constant rank is asserted.

C.2.3 Fixed ties and lower information scales

Proposition C.21 (Squared-volume Fisher tensor at an actual finite-weight tie). Fix θ_0 and assume eq. (C.66) on a neighborhood. Let S be the nonempty maximizing set of $s_\gamma(\theta_0)$, suppose the actual finite weights satisfy $W_{\gamma,L}(\theta_0) \to w_\gamma$, and assume $I(p_{\gamma,L})(\theta_0) = O(V)$ for every label. Write $v_\gamma = \nabla s_\gamma(\theta_0)$ and $\overline{v} = \sum_\gamma w_\gamma v_\gamma$. Then w is supported on S and

$$\frac{I(P_L)(\theta_0)}{V^2} \longrightarrow C_w(v) := \sum_\gamma w_\gamma (v_\gamma - \overline{v})(v_\gamma - \overline{v})^\top. \tag{C.83}$$

The rank is the dimension of the span of $v_\gamma - v_\eta$ over pairs with positive limiting weights, at most $\min(d, |\operatorname{supp} w| - 1)$. A direction with positive variance has Fisher information asymptotic to a positive multiple of V^2. Zero variance implies only $o(V^2)$ information in that direction. [ESTABLISHED]

Proof. There is a positive bulk gap from S to every nonmaximizing label, since the catalog is finite. Comparing to any one maximizing label and using convergence of λ/V shows that every nonmaximizing weight is exponentially small at θ_0. Thus its limit is zero. Also $\xi_{\gamma,L}/V \to v_\gamma$ by C^1 bulk convergence. Dividing the complete finite decomposition by V^2 makes the branch contribution $O(1/V)$ and leaves $\mathrm{Cov}_{W_L}(\xi_{\gamma,L}/V)$, which converges by finite summation to eq. (C.83). For any vector z,

$$z^\top C_w(v) z = \sum_\gamma w_\gamma \big[z \cdot (v_\gamma - \overline{v})\big]^2. \tag{C.84}$$

This vanishes exactly when $z \cdot v_\gamma$ is constant on the positive-weight labels. The kernel is the orthogonal complement of the stated span, proving the rank. A positive limiting quadratic form gives the asserted asymptotic by multiplication by V^2; a zero one gives only the stated little-oh conclusion. □

Proposition C.22 (Residual gradients and rare labels before score squaring). At every finite volume the label contribution has the exact form

$$\frac{1}{V}\operatorname{Cov}_{W_L}(\xi_{\gamma,L}) = \frac{1}{2V}\sum_{\gamma,\eta} W_{\gamma,L}W_{\eta,L}(\xi_{\gamma,L}-\xi_{\eta,L})(\xi_{\gamma,L}-\xi_{\eta,L})^{\top}. \tag{C.85}$$

Consequently, if the weighted branch-density term converges, convergence of the full density is equivalent to convergence of the matrix on the right. Its weights cannot be replaced by their limits before its score differences are squared.

Write $\lambda_{\gamma,L} = Vs_\gamma + r_{\gamma,L}$ with $\|r_{\gamma,L}\|_{C^3(K)}/V \to 0$. In the setting of proposition C.21, suppose every label of S has positive limiting weight. Fix a direction z for which $z \cdot v_\gamma$ is the same for all $\gamma \in S$, choose $\gamma_0 \in S$, and assume for some $0 \le \kappa < 1$ that

$$V^{-\kappa}\partial_z(r_{\gamma,L} - r_{\gamma_0,L})(\theta_0) \longrightarrow d_\gamma \quad (\gamma \in S), \qquad d_{\gamma_0} = 0. \tag{C.86}$$

Then the directional label information divided by $V^{2\kappa}$ tends to $\operatorname{Var}_w(d_\gamma)$. Positive variance is below the extensive scale if $\kappa < 1/2$, contributes a finite nonzero density if $\kappa = 1/2$, and makes the density diverge if $\kappa > 1/2$. If all the vector slopes on S agree and the corresponding relative residual gradients divided by $\sqrt{V}$ tend to vectors d_γ, then, provided $g_{\gamma,L}(\theta_0) \to g_\gamma(\theta_0)$,

$$g_L(\theta_0) \longrightarrow \sum_{\gamma\in S} w_\gamma g_\gamma(\theta_0) + \operatorname{Cov}_w(d_\gamma). \tag{C.87}$$

These are fixed-parameter limits, with no conclusion about derivatives of the limiting tensor. If a tied weight vanishes, the exact pairwise formula still applies, but this positive-weight residual conclusion requires separate control of that rare label. [ESTABLISHED]

Proof. Expand the right side of eq. (C.85). The two squared terms together give

$$V^{-1}\sum_{\gamma} W_\gamma \xi_\gamma \xi_\gamma^{\top},$$

and the two cross terms together give

$$-V^{-1}\left(\sum_{\gamma} W_\gamma \xi_\gamma\right)\left(\sum_{\gamma} W_\gamma \xi_\gamma\right)^{\top}.$$

This proves the identity and the convergence equivalence after adding the branch term.

On S, equal bulk directional slopes cancel exactly from directional score differences at θ_0. Divide those differences by V^κ and use eq. (C.86) in the finite pairwise sum. The limiting sum is $\frac{1}{2}\sum_{\gamma,\eta\in S} w_\gamma w_\eta (d_\gamma - d_\eta)^2 = \operatorname{Var}_w(d_\gamma)$. Pairs involving a nonmaximizing label contribute at most $CV^{2-2\kappa}e^{-cV}$ after this normalization, since all original score differences are $O(V)$; they vanish even when $\kappa = 0$. Comparison of 2κ with one proves the threshold claims. Repeat the same calculation

with outer products and $\kappa = 1/2$ for eq. (C.87); finite summation of the branch-density limits gives its remaining term. No derivative of any fixed-point limit has been taken. □

C.2.4 Explicit obstructions inside the admitted label-growth domain

Proposition C.23 (Native realizations with divergent or nonconvergent ties). The growth condition eq. (C.63), even with identical positive high-temperature native branches, does not imply a finite extensive Fisher limit at a tie, convergence of the label law, or C^2 convergence of a uniformly convergent Fisher density. The following families give explicit counterexamples to these assertions. [ESTABLISHED]

Proof. Use two labels $0, 1$ and identical complete native free laws at fixed $0 < b_0 < 1$ and $0 < t_0 < 1/12$. The positive coupling t_0 is independent of the statistical parameter θ in a fixed open interval about zero. Every branch mass is positive on its complete carrier; its conditional Fisher tensor in θ is zero. Its physical partition is common to the two labels, so it cancels from their relative effective mass in either normalization mode. Set $a_{0,L} = 1$ and $a_{1,L} = e^{f_L(\theta)}$. The binary weight and its complete Fisher information are therefore exactly

$$W_{1,L} = \frac{e^{f_L}}{1 + e^{f_L}}, \qquad I_L = q(f_L)(f_L')^2, \qquad q(x) = \frac{e^x}{(1+e^x)^2} = \frac{1}{4\cosh^2(x/2)}. \tag{C.88}$$

Indeed $W_1' = W_1(1-W_1)f_L'$ and the two centered label scores are $(1-W_1)f_L'$ and $-W_1 f_L'$. Squaring and averaging gives this formula. Thus no pressure derivative theorem is needed to evaluate these examples. All assertions about ties concern the effective bulk scores, whose common native pressure has no θ dependence.

First choose $f_L = V\theta$. Its C^3 norm on any compact interval is at most $C_K V$, and $f_L/V = \theta$ exactly. At the tie $\theta = 0$,

$$I_L(\theta) = \frac{V^2}{4\cosh^2(V\theta/2)}, \qquad \frac{I_L(0)}{V} = \frac{V}{4}. \tag{C.89}$$

The extensive density diverges. The pointwise $V^{-2}I_L$ limit equals $1/4$ at zero and zero at every nonzero θ, so even that pointwise rescaling is discontinuous across the fixed-parameter tie.

Next take $f_L = 0$ for even L and $f_L = \log 2$ for odd L. Its derivatives vanish, its C^3 norm divided by V tends to zero, and $I_L = 0$ identically. But $W_{1,L}$ alternates between $1/2$ and $2/3$. The retained event $\{\gamma = 1\}$ already fails to converge, regardless of the changing microscopic carriers. Thus convergence of Fisher information does not force convergence of the labeled law.

For $0 < \kappa < 1$ take $f_L = V^\kappa\theta$. Its C^3 norm divided by V is at most $C_K V^{\kappa-1} \to 0$, so the two bulk scores are identical throughout the interval. At zero both weights equal $1/2$, and

$$\frac{I_L(0)}{V} = \frac{1}{4}V^{2\kappa-1}. \tag{C.90}$$

This realizes all three thresholds in proposition C.22. With $f_L = c_L + \sqrt{V}\theta$, where c_L alternates between 0 and $\log 2$, the same growth bound holds while the density at zero alternates between $q(0) = 1/4$ and $q(\log 2) = 2/9$. Hence even bounded extensive density need not converge.

A rare tied label is obtained with $f_L = V\theta - a \log V$, for any fixed $a > 0$. Here $f_L/V \to \theta$ in C^3 and its C^3 norm is at most $C_K V$ for large V. At zero $W_{1,L} = (1 + V^a)^{-1} \to 0$, but

$$\frac{I_L(0)}{V} = \frac{V^{1+a}}{(1+V^a)^2} \sim V^{1-a}. \tag{C.91}$$

Thus the limiting V^2 coefficient is zero while the density diverges for $a < 1$, tends to one for $a = 1$, and vanishes for $a > 1$. Discarding the rare label before squaring its score would miss all of these cases.

Finally put

$$f_L(\theta) = A_L \sin(k_L\theta), \qquad A_L = V^{-1/6}, \qquad k_L = V^{1/3}. \tag{C.92}$$

The largest bound among its derivatives through degree three is $A_L k_L^3 = V^{5/6}$, so $\|f_L\|_{C^3(K)}/V = O(V^{-1/6}) \to 0$. Positivity of the raw weights is automatic, and $|f_L| \le A_L$ makes both label weights converge uniformly to $1/2$. The density $F_L = V^{-1} q(f_L)(f_L')^2$ obeys $0 \le F_L \le V^{-2/3}/4$, so it converges uniformly to zero. Differentiate this expression twice before passing to a limit:

$$F_L'' = \frac{1}{V}\left[q''(f_L)(f_L')^4 + 5q'(f_L)(f_L')^2 f_L'' + 2q(f_L)(f_L'')^2 + 2q(f_L) f_L' f_L'''\right]. \tag{C.93}$$

At zero, $f_L = 0$, $f_L' = A_L k_L$, $f_L'' = 0$, and $f_L''' = -A_L k_L^3$. Also $q(0) = 1/4$, $q'(0) = 0$, and $q''(0) = -1/8$, obtained by differentiating $[4\cosh^2(x/2)]^{-1}$. Consequently

$$F_L''(0) = -\frac{A_L^2 k_L^4}{2V} - \frac{A_L^4 k_L^4}{8V} = -\frac{1}{2} - \frac{1}{8}V^{-1/3} \longrightarrow -\frac{1}{2}. \tag{C.94}$$

The uniform limit has second derivative zero. This proves failure of C^2 convergence under the original C^3 bulk growth and convergent finite weights. Every example has smooth positive finite data and fixed strictly positive native coupling; none is a product-boundary exception. □

Proposition C.24 (A scalar boundary estimate does not control information). Uniform sub-volume bounds on the log ratio of two finite laws, even with a common bulk-cylinder limit, do not imply convergence of Fisher densities or their parameter derivatives. [ESTABLISHED]

Proof. Let P_L be the fair product law on $V = L^2$ signs, independent of θ, and choose one sign at the outer boundary. Let $Q_{L,\theta}$ change only that sign to mean $a\sin(V\theta)$, with fixed $0 < a < 1/2$. Every finite law is positive and smooth on the same open parameter interval. Pointwise $Q_{L,\theta}/P_L = 1 + a\sin(V\theta) z_{\text{boundary}}$, whence

$$\left|\log \frac{Q_{L,\theta}}{P_L}\right| \le \max\{-\log(1-a), \log(1+a)\}. \tag{C.95}$$

Every fixed bulk cylinder eventually avoiding that boundary sign has exactly the same law under P_L and $Q_{L,\theta}$. A binary sign with mean $m(\theta)$ has Fisher information $(m')^2/(1-m^2)$, obtained directly by summing the squared scores of $(1 \pm m)/2$. Thus

$$I(P_L) = 0, \qquad I(Q_{L,\theta}) = \frac{a^2 V^2 \cos^2(V\theta)}{1 - a^2 \sin^2(V\theta)}, \qquad \frac{I(Q_{L,0})}{V} = a^2 V. \tag{C.96}$$

The scalar log-ratio bound cannot transfer the density limit. This example deliberately violates the native uniform local-response bounds, so it refutes that inference and does not refute the conditional native inputs. The still simpler functions $L^{-1}\sin(L^2\theta)$ converge uniformly to zero while their first derivatives at zero equal L; a scalar convergence assertion has no implicit permission to differentiate it. □

C.2.5 The reciprocal-volume statistical experiment

Theorem C.25 (Actual center ratios, smooth weights, and the full Fisher limit). Assume eq. (C.66) and actual branch limits $g_{\gamma,L} \to g_\gamma$ in C^2 on a neighborhood of θ_0. Let S be the maximizing set of $s_\gamma(\theta_0)$, choose $\gamma_0 \in S$, and impose the additional actual center-ratio condition

$$\exp[\lambda_{\gamma,L}(\theta_0) - \lambda_{\gamma_0,L}(\theta_0)] \longrightarrow \kappa_\gamma \in (0,\infty) \quad (\gamma \in S), \qquad \kappa_{\gamma_0} = 1. \tag{C.97}$$

Set $v_\gamma = \nabla s_\gamma(\theta_0)$ and define the actual finite experiment on its original labeled carrier by

$$\theta_L(u) = \theta_0 + u/V, \qquad \mathcal{E}_L(u) = P_{L,\theta_L(u)}. \tag{C.98}$$

On every fixed compact u-set, its label law converges in C^3 to

$$\pi_\gamma(u) = \frac{\kappa_\gamma e^{u\cdot v_\gamma}}{\sum_{\eta\in S}\kappa_\eta e^{u\cdot v_\eta}} \quad (\gamma \in S), \qquad \pi_\gamma(u) = 0 \quad (\gamma \notin S). \tag{C.99}$$

Its full Fisher tensor in the coordinate u satisfies

$$I_L^{(u)}(u) = V^{-2} I(P_{L,\theta_L(u)}) \longrightarrow G(u) = \mathrm{Cov}_{\pi(u)}(v_\gamma) = D_u^2 \log \sum_{\gamma\in S} \kappa_\gamma e^{u\cdot v_\gamma} \quad \text{in } C^2. \tag{C.100}$$

The center-ratio condition is additional finite-volume data. Equality of bulk pressure or effective bulk scores supplies no such condition. [ESTABLISHED]

Proof. A fixed compact u-set has bounded radius, so its image and the line segments to θ_0 lie in the chosen parameter neighborhood for large L. Define $\Delta_{\gamma,L}(u) = \lambda_{\gamma,L}(\theta_0 + u/V) - \lambda_{\gamma,L}(\theta_0)$. The fundamental theorem of calculus on each segment gives

$$\Delta_{\gamma,L}(u) = \int_0^1 V^{-1} \nabla\lambda_{\gamma,L}(\theta_0 + \tau u/V)\cdot u\, d\tau. \tag{C.101}$$

The integrand tends uniformly to $v_\gamma \cdot u$, by the compact C^1 convergence in eq. (C.66) and continuity of ∇s_γ. Direct differentiation in u yields

$$D_u \Delta_{\gamma,L} \to v_\gamma, \qquad \|D_u^2 \Delta_{\gamma,L}\| = O(V^{-1}), \qquad \|D_u^3 \Delta_{\gamma,L}\| = O(V^{-2}). \tag{C.102}$$

Thus $\Delta_{\gamma,L} \to u \cdot v_\gamma$ in C^3. For tied labels, subtract the reference increment and add the log of the actual center ratio. The resulting relative log masses converge in C^3 to $\log \kappa_\gamma + u \cdot (v_\gamma - v_{\gamma_0})$. Exponentiation and the finite softmax are smooth on the resulting compact ranges; their denominator is bounded below by the reference mass, which equals one in relative coordinates.

For a nonmaximizing label the relative center log mass is at most $-cV$ for some $c > 0$ and large L. Its window increment is $O(1)$ and its u-derivatives through degree three are bounded. Differentiating its relative exponential through degree three therefore bounds each term by Ce^{-cV}. These masses and all those derivatives vanish. Dividing the complete finite vector of relative masses by its sum proves eq. (C.99) in C^3, including the zero limiting coordinates outside S.

Define the normalized effective score $\zeta_{\gamma,L}(u) = V^{-1}\xi_{\gamma,L}(\theta_L(u))$. Equation eq. (C.102) implies $\zeta_{\gamma,L} \to v_\gamma$ in C^2, for all labels. The affine parameter change in eq. (C.98) has derivative $V^{-1}\,\mathrm{Id}$, so the full finite Fisher identity becomes

$$I_L^{(u)}(u) = \mathrm{Cov}_{W_L(\theta_L(u))}(\zeta_{\gamma,L}(u)) + \frac{1}{V}\sum_\gamma W_{\gamma,L}(\theta_L(u)) g_{\gamma,L}(\theta_L(u)). \tag{C.103}$$

The covariance converges in C^2 by finite sums and products. This form never divides by a weight tending to zero. The branch densities and their first two u-derivatives are bounded, since a u derivative is V^{-1} times a θ derivative. The weights have bounded C^2 norms by the preceding argument. Every term in the ordinary second-order Leibniz sum for the last term in eq. (C.103) is therefore $O(V^{-1})$. The conditional-state information vanishes in the limit but is not identically zero in general. Finally, differentiating the finite log-sum-exponential gives gradient $\sum \pi_\gamma v_\gamma$ and Hessian $\mathrm{Cov}_\pi(v)$, completing the proof. □

Proposition C.26 (Two parameter-independent comparison kernels). Under theorem C.25, let $\Pi(u)$ denote the finite-label experiment on the full catalog with masses $\pi_\gamma(u)$, including its structural zeros. There are kernels in both directions between $\mathcal{E}_L(u)$ and $\Pi(u)$ that are independent of u and have uniformly vanishing total-variation errors on every compact u-set. The replacement of a conditional branch by its actual center law has error $O(V^{-1/2})$; the additional label error has only the convergence rate furnished by eq. (C.97). [ESTABLISHED]

Proof. Let K_L extract the label and let R_L sample the corresponding actual finite center branch:

$$K_L(\delta \mid \gamma, x) = \mathbf{1}_{\{\delta=\gamma\}}, \qquad R_L(\delta, x \mid \gamma) = \mathbf{1}_{\{\delta=\gamma\}} p_{\gamma,L,\theta_0}(x). \tag{C.104}$$

Both are normalized, preserve labels, and may depend on L but not on u. The center branch is defined for every finite positive label, even when its limiting label mass is zero. Put $\varepsilon_L =$

$\sup_u \frac{1}{2}\sum_\gamma |W_{\gamma,L}(\theta_L(u)) - \pi_\gamma(u)|$ over the chosen compact. The first comparison has error at most $\varepsilon_L \to 0$.

For the reverse comparison, each branch has $I(p_{\gamma,L,\theta}) \preceq CV\,\mathrm{Id}$ along the segments in question, by its bounded density. On the fixed active support, direct differentiation gives

$$\left\| \frac{d}{d\tau} \sqrt{p_{\gamma,L,\theta_0+\tau u/V}} \right\|_2^2 = \frac{1}{4V^2} u^\top I(p_{\gamma,L,\theta_0+\tau u/V}) u \le \frac{C|u|^2}{4V}. \tag{C.105}$$

The square root vector is differentiable because the support is fixed and positive; zero coordinates can simply be left identically zero. Integrating its derivative and using the triangle inequality gives square-root Euclidean distance at most $C'|u|/\sqrt{V}$ between the branch at $\theta_L(u)$ and its center. For any probability vectors p, q, Cauchy-Schwarz gives

$$\|p - q\|_{\mathrm{TV}} = \frac{1}{2}\sum_x |\sqrt{p(x)} - \sqrt{q(x)}|(\sqrt{p(x)} + \sqrt{q(x)}) \le \|\sqrt{p} - \sqrt{q}\|_2, \tag{C.106}$$

since $\|\sqrt{p} + \sqrt{q}\|_2 \le 2$. On the disjoint labeled carrier, split $W_\gamma p_{\gamma,\theta_L} - \pi_\gamma p_{\gamma,\theta_0}$ into $W_\gamma(p_{\gamma,\theta_L} - p_{\gamma,\theta_0})$ and $(W_\gamma - \pi_\gamma)p_{\gamma,\theta_0}$, then sum absolute values. This proves the explicit bound

$$\sup_u \|\mathcal{E}_L(u) - \Pi(u)R_L\|_{\mathrm{TV}} \le C_K V^{-1/2} + \varepsilon_L. \tag{C.107}$$

This constructs comparison of actual experiments without an infinite microscopic partition sum or a common limiting state carrier. □

Proposition C.27 (The actual Euclidean null-slope quotient). Let

$$E = \mathrm{span}\{v_\gamma - v_{\gamma_0} : \gamma \in S\}, \qquad r = \dim E. \tag{C.108}$$

For every u, $\ker G(u) = E^\perp$, and $\pi(u) = \pi(u')$ holds exactly when $u - u' \in E^\perp$. The limiting family therefore descends to the Euclidean quotient $\mathbb{R}^d/E^\perp$, identified with E. On E its probability map is injective and, if $r > 0$, its Fisher tensor is positive definite with a positive minimum on every compact chart. The restrictions of the finite u-metrics to this fixed subspace converge in C^2 and have the inverse, connection, and curvature limits of proposition C.20. [ESTABLISHED]

Proof. All limiting weights on S are positive at every finite u. The directional quadratic form of G is their variance of $z \cdot v_\gamma$, so it vanishes exactly when z is orthogonal to all slope differences. This proves the kernel statement. The probability ratios satisfy the exact formula

$$\log \frac{\pi_\gamma(u)}{\pi_{\gamma_0}(u)} = \log \kappa_\gamma + u \cdot (v_\gamma - v_{\gamma_0}). \tag{C.109}$$

Equal laws therefore imply orthogonality of $u - u'$ to every generating slope difference, and that orthogonality conversely makes every ratio, hence every probability, equal. Orthogonal projection onto E gives the claimed quotient identification. Choosing r independent slope differences gives

a nonsingular linear map on E; their log ratios recover the coordinates on E, which also proves injectivity and a smooth inverse onto the image of the probability map.

Choose orthonormal coordinates on E. The restriction of G is positive definite, since its only null vector in E is zero. Its continuous quadratic form attains a positive minimum on each compact chart times the unit sphere in E. Constant linear restriction preserves the C^2 convergence in eq. (C.100), and proposition C.20 applies. When $r = 0$ the limiting experiment is constant and the quotient is a point; no positive-dimensional inverse is asserted. The finite metrics need not descend along $E^\perp$: this statement restricts them to a chosen fixed copy of E. Slower information scales in the other directions are not excluded. □

Proposition C.28 (A nonempty native realization of the limiting experiment). Every finite list of slopes $v_\gamma \in \mathbb{R}^d$ and positive constants κ_γ, with $\kappa_{\gamma_0} = 1$, has a realization on identical native branches with strictly positive high-temperature plaquette coupling for which the crossover weights, Fisher tensor, and both comparison kernels above are exact at every finite volume. [ESTABLISHED]

Proof. Choose common fixed $0 < b_0 < 1$, $0 < t_0 < 1/12$ and the complete native free branch for every label. Set $a_{\gamma,L}(\theta) = \kappa_\gamma e^{V v_\gamma \cdot \theta}$. The weights are positive and smooth, their log C^3 norms on compact sets are $O(V)$, and their bulk logarithms converge in C^3 to $v_\gamma \cdot \theta$. The identical physical partitions cancel in all label ratios, both globally and hierarchically. At $\theta_0 = 0$ the actual center ratios are κ_γ exactly, and at u/V the normalized label weights are precisely eq. (C.99). Conditional branches are independent of θ, so their Fisher term is zero and the reconstruction kernel uses exactly their unchanged law. Both comparison errors are zero. With two labels in one coordinate, $v_0 = 0$, $v_1 = 1$, and $\kappa_0 = \kappa_1 = 1$, this gives

$$I_L^{(u)} = G(u) = \frac{1}{4\cosh^2(u/2)} > 0. \tag{C.110}$$

Its one-dimensional intrinsic curvature is zero, since the two antisymmetrized derivative indices in the Riemann formula must coincide. The branch coupling is strictly positive; only its dependence on the crossover coordinate is absent. □

Theorem C.29 (A transverse two-label moving center). Assume the effective growth and actual branch C^2 inputs of theorem C.25. Suppose exactly two labels $0, 1$ maximize at θ_0, all others have a strict gap there, and for a fixed unit vector n, $a := \partial_n(s_1 - s_0)(\theta_0) \neq 0$. Reverse n if necessary so $a > 0$. On a sufficiently short fixed segment through θ_0 there is, for all large L, exactly one center $\theta_{c,L} = \theta_0 + t_L n$ satisfying $\lambda_{1,L}(\theta_{c,L}) = \lambda_{0,L}(\theta_{c,L})$. It obeys

$$|t_L| \le \frac{4|r_{1,L}(\theta_0) - r_{0,L}(\theta_0)|}{Va} \longrightarrow 0, \qquad r_{\gamma,L} = \lambda_{\gamma,L} - V s_\gamma. \tag{C.111}$$

The experiment $P_{L,\theta_{c,L}+u/V}$ has the C^3 weight and C^2 full Fisher limits of theorem C.25, with $S = \{0, 1\}$, $\kappa_0 = \kappa_1 = 1$, and slopes $v_\gamma = \nabla s_\gamma(\theta_0)$. Its comparison kernel samples $p_{\gamma,L,\theta_{c,L}}$, independently of u. [ESTABLISHED]

Proof. Let $f = s_1 - s_0$. Choose $\rho > 0$ such that the closed segment $\theta_0 + [-\rho, \rho]n$ is inside the parameter neighborhood, $\partial_n f \geq a/2$ on it, and both labels $0, 1$ remain uniformly above every other bulk score there. Such a choice exists by continuity, the nonzero transverse derivative, and the finite set of strict gaps at θ_0. Since $f(\theta_0) = 0$, integrating its derivative gives $f(\theta_0 + \rho n) \geq a\rho/2$ and $f(\theta_0 - \rho n) \leq -a\rho/2$.

For $F_L = \lambda_{1,L} - \lambda_{0,L}$, C^1 bulk convergence gives $\partial_n F_L \geq Va/4$ throughout this segment for large L. Uniform bulk convergence also gives opposite signs of F_L at its endpoints. Continuity supplies a zero and strict monotonicity makes it unique. Since $F_L(\theta_0) = r_{1,L}(\theta_0) - r_{0,L}(\theta_0) = o(V)$, integration of the derivative lower bound between zero and t_L gives eq. (C.111). In particular the centers converge to θ_0 and eventually lie in its interior neighborhood.

At the center the relative mass of label one to zero is exactly one. Other center relative masses are exponentially small by the uniform bulk gaps. For compact u-sets, apply the segment identity eq. (C.101) with base point $\theta_{c,L}$ instead of θ_0. Compact C^3 growth and $\theta_{c,L} \to \theta_0$ give the same first-derivative limits and second/third derivative bounds in eq. (C.102). The actual centered relative masses therefore converge in C^3 to $\exp[u \cdot (v_1 - v_0)]$, with exponentially vanishing complementary masses. The covariance proof and the $O(V^{-1})$ branch bound in eq. (C.103) give C^2 full Fisher convergence. Finally use the two kernels in eq. (C.104) with the actual moving center branch. The square-root argument is uniform on the neighborhood and again gives the bound in eq. (C.107). □

The displacement bound permits $|t_L|$ much larger than $V^{-1/2}$. Sampling the fixed θ_0 branch instead would require control of that additional displacement: the same information estimate bounds its total-variation cost only by $C\sqrt{V}|t_L|$, which need not vanish. No such replacement is used. More than two tied labels would require solving several relative-mass equations simultaneously; the single transverse argument does not assert that solvability. The reciprocal coordinate u defines a statistical experiment at a declared volume scale. It is neither physical time nor a universal smooth continuation of the extensive tensor across arbitrary ties.

C.3 Fixed-word scales, finite jets, and prescribed actions

The exact retained experiment fixes the meaning of every scale comparison. Throughout this section, $\Theta \subset \mathbb{R}^d$ is open and the geometry catalog $\mathfrak{G}$ is finite and nonempty. At each finite volume, $P_\theta(\gamma, x) = W_\gamma(\theta)p_{\gamma,\theta}(x)$ is a normalized law on the disjoint union of finite branch carriers. Its positive support is fixed on the parameter neighborhood under discussion, and its masses are C^3. Conditional objects are defined only on occupied fibers. A channel is normalized, independent of θ, and preserves γ. The source may be a prior or a positive selected posterior. For a random observation joint, the observation is retained as part of both endpoint variables. These conventions are those of Definition 20.9; they exclude replacing the induced observation model by a new local sensor. These restrictions are adopted hypotheses for the development that follows, with the exclusions and scope stated above. [HYPOTHESIS]

Proposition C.30 (Finite cell-area accounting). Fix a finite block word with total linear factor $B \geq 1$. Its fine and final coarse cell counts are $M = B^2V$ and V, respectively. Let I_f and I_c be the full Fisher tensors of one finite experiment and its exact image, evaluated in the same parameter coordinates, and put $D = I_f - I_c$. Then

$$B^2 \frac{I_f}{M} = \frac{I_c}{V} + \frac{D}{V}, \tag{C.112}$$

$$B^4 \frac{I_f}{M^2} = \frac{I_c}{V^2} + \frac{D}{V^2}. \tag{C.113}$$

The loss is positive semidefinite, and more precisely

$$\begin{aligned} D &= \sum_\gamma W_\gamma \big[I(p_\gamma) - I(p_\gamma C_\gamma) \big], \\ 0 &\preceq D \preceq \sum_\gamma W_\gamma I(p_\gamma). \end{aligned} \tag{C.114}$$

All identities have their ordinary parameter derivatives through order two; the sign assertions concern the tensors themselves. [ESTABLISHED]

Proof. The conditional-score identity of Proposition 20.10 gives $D \succeq 0$ and its weighted branch representation. Each pushed branch Fisher tensor is positive semidefinite because it is a score Gram matrix. Subtracting these tensors from the weighted fine branch sum proves the upper bound in eq. (C.114). Division of the exact identity $I_f = I_c + D$ by V and by V^2, followed by $M = B^2V$, gives eqs. (C.112) and (C.113). Volumes and B are fixed integers when parameter derivatives are taken, so the same equalities hold after each available derivative. No sign for those derivative matrices follows by differentiating an order inequality. Two edge signs per cell change the native statistic counts, not the relation between the two cell areas. □

If $I_f/M \to g_f$ and $I_c/V \to g_c$ in $C^2(K)$ on a compact parameter set K, then the first identity gives $D/V \to B^2 g_f - g_c$ in $C^2(K)$, and its limiting value is positive semidefinite. This conclusion requires both density limits as inputs. The fixed word precedes every volume limit in this section.

For the native family, these branch inputs are supplied by Proposition C.7 and theorem C.11. Precisely, let each $\chi_\gamma : \Theta \to \mathbb{R}^2$ be C^4 on a neighborhood of K, with image in a common compact source box of Hypothesis C.1, and take $p^f_{\gamma,M} = \nu^{\mathrm{f}}_{BL,\chi_\gamma}$ and $p^c_{\gamma,L} = (C_B)_\# \nu^{\mathrm{f}}_{BL,\chi_\gamma}$. Here $M = (BL)^2$ and the word is the exact parity word of Definition C.2. The source-score chain rule and those two results give

$$\begin{aligned} \frac{I(p^f_{\gamma,M})}{M} &\longrightarrow D\chi_\gamma^\top (H \circ \chi_\gamma) D\chi_\gamma, \\ \frac{I(p^c_{\gamma,L})}{V} &\longrightarrow D\chi_\gamma^\top (\mathcal{I}_B \circ \chi_\gamma) D\chi_\gamma \end{aligned} \quad \text{in } C^2(K). \tag{C.115}$$

The coarse expression is a pulled-back exact-push density. It is not identified with the Hessian of a native coarse pressure. Compact C^2 convergence bounds the fine densities uniformly; finitely many

remaining small sizes are bounded by finite smoothness and compactness. Taking the maximum over the finite catalog supplies eq. (C.116).

For the full labeled tensors, also impose the effective C^3 growth and strict-gap hypotheses of Theorem C.19, with fine area M in place of its volume V. The same representative $\lambda^f_{\gamma,M}$ can be used after the label-preserving push. Consequently $\lambda^c_{\gamma,L}/V = B^2\lambda^f_{\gamma,M}/M$ converges in C^3 to $B^2 s_\gamma$, and its C^3 norm is $O(V)$ for fixed B. The winner is unchanged and the limiting gap is multiplied by B^2. Applying the strict-phase result to each endpoint therefore supplies g_f and g_c above as the corresponding winning branch limits in eq. (C.115). Proposition C.18 supplies the effective growth from the admitted raw weights in each normalization mode; it does not supply a strict gap. For other channels or branch families, the two density limits remain explicit inputs: a fine limit alone does not produce a coarse one or a canonical free coarse ensemble. [ESTABLISHED]

Proposition C.31 (Preservation of a proved superextensive scale). Suppose that on a compact $K \subset \Theta$ there is a constant C_K independent of M and γ such that

$$I(p_{\gamma,M,\theta}) \preceq C_K M \,\mathrm{Id} \quad \text{for all } \theta \in K. \tag{C.116}$$

Let $a_M > 0$ be a parameter-independent information scale with $M/a_M \to 0$. If $I_f/a_M \to H$ pointwise on K, or uniformly there, then $I_c/a_M \to H$ in the same sense. In particular, for a fixed exponent $p > 1$, a proved limit $I_f/M^p \to H$ implies

$$\frac{I_c}{V^p} \longrightarrow B^{2p}H, \qquad \frac{D}{V^p} \longrightarrow 0. \tag{C.117}$$

No convergence of parameter derivatives is asserted from eq. (C.116) alone. [ESTABLISHED]

Proof. The branch bound and $\sum_\gamma W_\gamma = 1$ imply $0 \preceq D \preceq C_K M \,\mathrm{Id}$. Consequently $\|I_f/a_M - I_c/a_M\| \le C_K M/a_M \to 0$ in operator norm, uniformly on K. This proves the first assertion with the common fine-volume denominator a_M. For $a_M = M^p$, use $M^p = B^{2p}V^p$ to obtain eq. (C.117). No relation between a_M and a_V is implied for an arbitrary sequence a; the power-law multiplier follows from the specified power itself. □

For example, the finite score argument in Proposition C.21 applies with fine area M. Suppose at a fixed parameter the normalized label weights converge to w, the effective label log masses have $M^{-1}\nabla\lambda_{\gamma,M} \to v_\gamma$, and eq. (C.116) holds. The complete score decomposition then gives $I_f/M^2 \to \mathrm{Cov}_w(v_\gamma)$. The coarse tensor per squared coarse area tends to $B^4\,\mathrm{Cov}_w(v_\gamma)$. Likewise any separately proved scale $M^{2\kappa}$ with $\kappa > 1/2$ is preserved, with multiplier $B^{4\kappa}$ when measured per $V^{2\kappa}$. The label marginal is already preserved exactly at every finite volume; the branch bound shows that the discarded state information cannot contribute at these larger scales. Equality of bulk log-mass densities alone supplies neither the normalized weight limit nor any such information limit.

Proposition C.32 (Finite-word score and VFE telescopes). Let $C_1, \dots, C_m$ be a fixed finite sequence of channels of the stated kind. Put $P_j = P_0 C_1 \cdots C_j$, let S_j be its normalized score, and

write $I_j = I(P_j)$. Then

$$I_0 - I_m = \sum_{j=1}^{m} D_j, \qquad D_j = I_{j-1} - I_j \succeq 0. \tag{C.118}$$

For a fixed selected observation with evidence $E_o > 0$, let Π_0 be the selected posterior and let $Q_0 \ll \Pi_0$ be a recognition law. Push both by this same word and set $\mathcal{F}_j = -\log E_o + D_{\mathrm{KL}}(Q_j \| \Pi_j)$. If K_j^{KL} is the expected recognition-to-posterior conditional KL on the reverse fiber of step j, then

$$\mathcal{F}_0 - \mathcal{F}_m = \sum_{j=1}^{m} K_j^{\mathrm{KL}}, \qquad K_j^{\mathrm{KL}} \geq 0. \tag{C.119}$$

For total area factor B^2, this also gives

$$B^2 \frac{\mathcal{F}_0}{M} = \frac{\mathcal{F}_m}{V} + \frac{1}{V} \sum_{j=1}^{m} K_j^{\mathrm{KL}}. \tag{C.120}$$

[ESTABLISHED]

Proof. Use the joint path law $P_0(dx_0)C_1(x_0, dx_1) \cdots C_m(x_{m-1}, dx_m)$, including all retained labels and observations in each x_j. Finite differentiation shows $S_j(X_j) = \mathbb{E}[S_0(X_0) \mid X_j]$ and $\mathbb{E}[S_{j-1}(X_{j-1}) \mid X_j] = S_j(X_j)$. Put $\varepsilon_j = S_{j-1}(X_{j-1}) - S_j(X_j)$. The Markov property gives zero conditional mean of ε_j given $X_j, \ldots, X_m$. For $j < k$, ε_k is measurable with respect to these later variables, so $\mathbb{E}[\varepsilon_j \varepsilon_k^\top] = 0$. Thus the telescoping score residual $S_0 - S_m = \sum_j \varepsilon_j$ has second moment $\sum_j \mathbb{E}[\varepsilon_j \varepsilon_j^\top]$, and each summand is D_j by the one-step conditional-score identity. This proves eq. (C.118) on one augmented path space; unrelated output spaces need not have nested sigma-algebras.

For VFE, finite domination persists under each channel, so all displayed KL values are finite. Apply Proposition 20.14 at each step to obtain $\mathcal{F}_{j-1} - \mathcal{F}_j = K_j^{\mathrm{KL}}$. The evidence is unchanged because every step pushes the complete observation experiment, as in Proposition 20.11. Summing cancels the intermediate objectives and proves eq. (C.119); division by V gives eq. (C.120). A new likelihood at an intermediate stage would destroy this common-evidence argument. The formula compares the given recognition law with its actual images; it asserts neither a stochastic variational-infimum identity nor a thermodynamic VFE limit. □

Proposition C.33 (Fine and coarse crossover coordinates). Let $\theta_{c,L}$ be a declared center, independent of the local coordinate, and compare

$$u = M(\theta - \theta_{c,L}), \qquad v = V(\theta - \theta_{c,L}), \qquad u = B^2 v. \tag{C.121}$$

In the common u coordinate, the fine and coarse tensors are $M^{-2} I_f(\theta_{c,L} + u/M)$ and $M^{-2} I_c(\theta_{c,L} + u/M)$. Under the uniform branch bound on a compact neighborhood of these windows, their difference tends to zero uniformly on compact u sets. If their common limit is $H(u)$, the coarse

tensor in its own reciprocal-coarse-volume coordinate is

$$I_c^{(v)}(v) = B^4 I_c^{(u)}(B^2 v) \longrightarrow B^4 H(B^2 v). \tag{C.122}$$

[ESTABLISHED]

Proof. The differential of $\theta_{c,L} + u/M$ with respect to u is $M^{-1}\,\mathrm{Id}$, so the two factors in the Fisher pullback give M^{-2}. Their loss is bounded by C_K/M using eq. (C.114). This proves uniform value convergence of the difference wherever the windows stay in the compact parameter neighborhood. The differential of $u = B^2 v$ is $B^2\,\mathrm{Id}$; pulling back once more gives both the factor B^4 and the argument $B^2 v$ in eq. (C.122). The retained label law at the same u is identical before and after the channel at every finite volume, not merely in its limit. □

A derivative version requires derivative inputs. Write $g^f_{\gamma,M} = I(p^f_{\gamma,M})/M$ and $g^c_{\gamma,M} = I(p^c_{\gamma,L})/M$, using the fine denominator for both. Suppose these tensors have uniformly bounded C^2 norms in the original θ chart on a compact neighborhood of the windows, and the reparameterized label weights have uniformly bounded C^2 norms on compact u sets. The within-branch contribution to the loss in the u chart is then

$$\frac{1}{M}\sum_{\gamma} W_{\gamma,M}(\theta_{c,L} + u/M)(g^f_{\gamma,M} - g^c_{\gamma,M})(\theta_{c,L} + u/M). \tag{C.123}$$

Twice applying the product and chain rules bounds its C^2 norm by $O(M^{-1})$: every derivative of a composed branch density contributes an additional M^{-1}, and the label-weight derivatives are bounded. Under these inputs, C^2 convergence of the fine window tensor transfers to the coarse one, and eq. (C.122) holds in C^2. The original value bound alone does not justify this upgrade.

For the native branches of eq. (C.115), the needed branch bounds follow from those actual C^2 limits. The coarse density there is divided by V; the density here is its B^{-2} multiple. For a fixed center θ_0, assume the effective growth and the actual positive center ratios in Theorem C.25, using the fine area M. That theorem gives C^3 convergence of the reparameterized weights and the fine C^2 window limit $H(u) = \mathrm{Cov}_{\pi(u)}(v_\gamma)$, where $v_\gamma = \nabla s_\gamma(\theta_0)$ and s_γ is normalized per fine cell. The preceding loss bound therefore gives this same $H(u)$ for the exact push in the common u chart. In the coarse v chart the limit is $B^4 H(B^2 v)$; equivalently the coarse bulk slopes are $B^2 v_\gamma$. Neither the factor nor the argument may be removed by identifying the two reciprocal-volume coordinates.

For a moving center, Theorem C.29 supplies the weight and fine-tensor inputs when exactly two labels maximize at θ_0, the other labels have a strict gap there, and their bulk log-mass density difference has a nonzero derivative along the declared transverse direction. Apply it on the fine-square subsequence of side BL, and use its actual equal-log-mass center as $\theta_{c,L}$. The weights are unchanged by the push, so this is the same center at both endpoints. The theorem gives unit relative center ratios and the required C^3 weight and C^2 Fisher limits; the native bounds again transfer the limit. An arbitrary declared center, a nontransverse tie, or a tie of more than two labels does not acquire these center properties from this argument. For fixed centers the positive ratio

condition remains an additional premise, since the bulk pressure limit alone does not imply it. [ESTABLISHED]

C.3.1 Ordinary finite probability jets

For a positive integer d, write $\mathcal{O}_m = \{\alpha \in \mathbb{N}^d : |\alpha| \le m\}$, $\alpha! = \prod_i \alpha_i!$, and $\binom{\alpha}{\beta} = \prod_i \binom{\alpha_i}{\beta_i}$ for $\beta \le \alpha$ componentwise. Every jet below stores ordinary derivatives at one point θ_0, not Taylor coefficients. Thus a coefficient in a Taylor polynomial is divided by $\alpha!$. These are declared definitions and conventions; no proof is asserted by their adoption. [DEFINITION]

Proposition C.34 (Local realization on fixed support). Let X be a finite nonempty carrier and prescribe real rows $\pi_\alpha(x)$ for $\alpha \in \mathcal{O}_3$. Assume

$$\begin{aligned} \pi_0(x) \ge 0, \qquad \sum_x \pi_0(x) = 1, \\ \sum_x \pi_\alpha(x) = 0 \quad (0 < |\alpha| \le 3), \end{aligned} \tag{C.124}$$

and require $\pi_\alpha(x) = 0$ for every supplied α whenever $\pi_0(x) = 0$. These rows are the ordinary third-order jet of a smooth normalized family with exactly the same positive support on a neighborhood of θ_0. [ESTABLISHED]

Proof. Set

$$\pi_{\theta_0+z}(x) = \sum_{\alpha \in \mathcal{O}_3} \frac{\pi_\alpha(x)}{\alpha!} z^\alpha. \tag{C.125}$$

Summing the finite polynomial over x and using eq. (C.124) gives one identically. Each atom with $\pi_0(x) > 0$ remains positive on some neighborhood of $z = 0$ by continuity. There are finitely many such atoms, so the intersection of these neighborhoods is a common open neighborhood. Every other column is the zero polynomial. The family is therefore a probability law with the required fixed support there, and differentiation at zero recovers each supplied ordinary row. This construction gives a local realization; it does not identify the polynomial with a particular native law that might have produced the same derivatives. □

The support condition is substantive. A nonnegative family may have $\pi_{\theta_0}(x) = 0$ and a nonzero second derivative at that atom while creating positive mass nearby. Such a family is outside this fixed-support realization and score interface. A zero atom retained only as part of the declared carrier has no score at θ_0 and contributes zero to all score sums here.

Lemma C.35 (Product, reciprocal, and normalized mass recurrences). For scalar jets a, b through degree m, their product has rows

$$(ab)_\alpha = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} a_\beta b_{\alpha-\beta}. \tag{C.126}$$

If $a_0 > 0$, the reciprocal rows are uniquely determined by

$$r_0 = a_0^{-1}, \qquad r_\alpha = -a_0^{-1} \sum_{0<\beta\le\alpha} \binom{\alpha}{\beta} a_\beta r_{\alpha-\beta} \quad (0 < |\alpha| \le m). \tag{C.127}$$

For unnormalized mass rows $w_\alpha(x)$ on a fixed finite support, let $Z_\alpha = \sum_x w_\alpha(x)$ and assume $Z_0 > 0$ and positive base masses on that support. If r is the reciprocal jet of Z, then

$$\pi_\alpha(x) = \sum_{\beta\le\alpha} \binom{\alpha}{\beta} w_\beta(x) r_{\alpha-\beta} \tag{C.128}$$

is the normalized probability jet, provided structural zero columns have all their supplied derivatives zero. [ESTABLISHED]

Proof. The multivariate Leibniz rule gives eq. (C.126). Differentiate $ar = 1$. At order $\alpha \neq 0$, the term with $\beta = 0$ is $a_0 r_\alpha$; all other terms have reciprocal order strictly less than $|\alpha|$. Solving for r_α gives eq. (C.127), and induction on total order proves uniqueness. A local polynomial realization of a stays positive near its positive base value, so these are the actual derivatives of its reciprocal. The same observation applies to the finite sum Z. Each active raw mass stays positive in a common neighborhood, and its normalized mass is $w(x)/Z$. Applying the product rule to that quotient gives eq. (C.128). Summing over x yields the derivatives of $Z/Z = 1$, giving exactly the normalization constraints; zero columns remain identically zero. The argument uses only scalar reciprocals of positive masses or partitions, never a matrix inverse. □

Proposition C.36 (Second-order Fisher jets from third-order masses). For a probability jet satisfying eq. (C.124), let $r_{\delta,x}$ be the reciprocal jet of $\pi(x)$ for each positive atom. For $\alpha \in \mathcal{O}_2$ and coordinate unit multi-indices e_i, e_j,

$$\partial^\alpha I_{ij}(\theta_0) = \sum_{x:\pi_0(x)>0} \sum_{\beta+\gamma+\delta=\alpha} \frac{\alpha!}{\beta!\gamma!\delta!} \pi_{\beta+e_i}(x) \pi_{\gamma+e_j}(x) r_{\delta,x}. \tag{C.129}$$

Thus third-order mass data determine the full second-order Fisher jet, including singular zeroth-order tensors. [ESTABLISHED]

Proof. On the local positive support supplied by Proposition C.34, the Fisher tensor is $I_{ij} = \sum_x (\partial_i \pi(x))(\partial_j \pi(x))/\pi(x)$. Apply the three-factor Leibniz rule to each summand. A derivative of order at most two of $\partial_i \pi$ needs a mass derivative of order at most three, giving precisely eq. (C.129). All sums are finite and the reciprocals are smooth on the common positive neighborhood. For any $v \in \mathbb{R}^d$,

$$v^\top I(\theta_0) v = \sum_{x:\pi_0(x)>0} \pi_0(x) \left(\sum_i v_i \frac{\pi_{e_i}(x)}{\pi_0(x)} \right)^2 \ge 0. \tag{C.130}$$

This establishes the zeroth-order sign without requiring rank. The higher derivative matrices need not be positive semidefinite, and a singular tensor supplies no Riemannian inverse. □

Proposition C.37 (Differentiated label, conditional, and channel identities). Partition X by geometry labels. Form $W_{\gamma,\alpha} = \sum_{x\in \mathsf{X}_\gamma} \pi_\alpha(x)$. On each fiber with $W_{\gamma,0} > 0$, use scalar normalization to form $p_\gamma = \pi|_{\mathsf{X}_\gamma}/W_\gamma$. Then, for $|\alpha| \le 2$,

$$\begin{aligned}\partial^\alpha I(\pi) &= \partial^\alpha I(W) \\ &\quad + \sum_{\gamma: W_{\gamma,0}>0} \sum_{\beta\le\alpha} \binom{\alpha}{\beta} (\partial^\beta W_\gamma)(\partial^{\alpha-\beta} I(p_\gamma)).\end{aligned} \tag{C.131}$$

Every derivative is evaluated at θ_0. An unoccupied fiber has zero weighted contribution and no defined conditional probability or conditional jet. A fixed finite channel pushes each mass row linearly; if it preserves labels, the categorical law and all its available mass and Fisher derivatives are unchanged. [ESTABLISHED]

Proof. The local realization gives positive W_γ near θ_0 on occupied fibers, so its quotient defines the actual conditional family. The full score is $\nabla \log W_\gamma + \nabla \log p_\gamma(x)$, and the second term has conditional mean zero by differentiation of its normalization. Expanding the score Gram gives $I(\pi) = I(W) + \sum_\gamma W_\gamma I(p_\gamma)$ as a function identity. Its ordinary product derivatives give eq. (C.131), including derivatives of W_γ. If $W_{\gamma,0} = 0$, every atom in that fiber has zero base mass and hence an identically zero realization; dividing by the fiber mass is neither needed nor defined.

For a fixed channel C, finite summation gives $(\pi C)_\alpha(y) = \sum_x \pi_\alpha(x) C(x,y)$. A target with zero base mass receives no channel incidence from any positive source atom, because every term is nonnegative. Source atoms of zero base mass have zero derivative columns. Thus all target derivative rows at such a target are zero, and the fixed-support condition persists. If C has zero entries between different labels, summing its normalized rows over each target label recovers $W_{\gamma,\alpha}$ for every α. The full Fisher formulas and the fixed-area divisions therefore apply without dropping any derivative term. □

All these finite algebraic statements hold over real input derivatives. When every supplied mass, factor derivative, and channel entry is rational, all displayed products, finite sums, positive scalar reciprocals, and integer-area divisions remain rational. They then define an exact finite arithmetic interface in every finite parameter dimension. This does not assert that arbitrary real smooth generative data have rational values or rational derivatives. A two-coordinate specialization is an ordinary restriction of the multi-index formulas, not a restriction of the mathematical theorem to two parameters. [ESTABLISHED]

C.3.2 Physical native raw-factor and partition jets

Use the complete native free square of Definition 19.1. Its edge set E_L contains the two outgoing signs per cell, and its internal plaquette set $\mathcal{P}_L$ has $(L-1)^2$ elements. For $z \in \{-1,1\}^{E_L}$, put

$P_q(z) = \prod_{e\in\partial q} z_e$. With $|b| < 1$, $|t| < 1$, and normalized counting reference $\nu_L(z) = 2^{-2L^2}$, define

$$\begin{aligned} R_L(b,t;z) &= \prod_{e\in E_L}(1+bz_e)\prod_{q\in\mathcal{P}_L}(1+tP_q(z)),\\ Z_L(b,t) &= 2^{-2L^2}\sum_z R_L(b,t;z). \end{aligned} \tag{C.132}$$

The high-temperature setting additionally restricts $b > 0$ and $0 < t < 1/12$; no volume-limit assertion is used in the finite formulas below. These are declared definitions and conventions; no proof is asserted by their adoption. [DEFINITION]

Proposition C.38 (Complete native ordinary derivatives). For nonnegative integers r, s, direct differentiation gives

$$\begin{aligned} \partial_b^r\partial_t^s R_L = r!s!\sum_{\substack{A\subset E_L,\ |A|=r\\ D\subset\mathcal{P}_L,\ |D|=s}}\Bigg[&\left(\prod_{e\in A} z_e\right)\left(\prod_{q\in D}P_q(z)\right)\\ &\cdot\prod_{e\in E_L\setminus A}(1+bz_e)\prod_{q\in\mathcal{P}_L\setminus D}(1+tP_q(z))\Bigg], \end{aligned} \tag{C.133}$$

$$\partial_b^r\partial_t^s Z_L = 2^{-2L^2}\sum_z \partial_b^r\partial_t^s R_L. \tag{C.134}$$

An empty subset sum is zero and an empty product is one. The normalized mass jet is the quotient of $\nu_L R_L$ by this same physical Z_L. [ESTABLISHED]

Proof. Each edge factor is affine in b and each plaquette factor is affine in t. A nonzero rth b derivative must differentiate r distinct edge factors; every such subset has $r!$ orders of differentiation. The independent choice of s distinct plaquette factors contributes $s!$. This gives eq. (C.133), including zero when there are too few factors. The finite reference is fixed in b, t, so differentiation commutes with its finite sum and gives eq. (C.134). Every factor is positive on the stated domain, hence $Z_L > 0$ and the scalar quotient recurrence produces the normalized mass derivatives. There are no outer crossing plaquettes and no fixed exterior in this inventory. □

For $L = 1$, there are two independent signs and no plaquette, so the normalized counting expectation of each edge factor is one and $Z_1 = 1$. The b score of a sign is $z/(1+bz)$ and has Fisher information $1/(1-b^2)$; independence gives $I_{bb} = 2/(1-b^2)$, while the t direction is null. For $L = 2$, the edge-factor product with reference 2^{-8} is already a normalized independent-sign law of mean b. Its one internal plaquette has mean b^4, so

$$\begin{aligned} Z_2 &= 1 + tb^4,\\ \partial_b \log p_2 &= \partial_b \log R_2 - \frac{4tb^3}{1+tb^4},\\ \partial_t \log p_2 &= \partial_t \log R_2 - \frac{b^4}{1+tb^4}. \end{aligned} \tag{C.135}$$

These examples exhibit the physical normalization terms explicitly. A finite arithmetic restriction to at most eight independently enumerated signs permits exactly $L = 1, 2$ because $2L^2 \le 8$; it places no restriction on the general finite algebraic theorem and supplies no thermodynamic evidence.

At rational (b, t), the raw and partition derivatives in eqs. (C.133) and (C.134) and the resulting normalized Fisher jet are rational. For a different smooth chart, ordinary composition differentiates every varying source factor. In particular, the selected all-positive sensor identity of Proposition 19.7 multiplies the raw partition by $(A^{\mathrm{sens}})^{2L^2}$ and evaluates the native factor at its shifted field b'. A hierarchical posterior retains the original $Z_L(b, t)$ divisor. Derivatives of a varying sensor strength therefore enter both the scalar and b', as well as the original branch normalization. For the thermodynamic posterior, the matching interface is Proposition C.18: impose its strictly positive C^4 sensor data and its C^4 original and shifted source maps, both with images in common compact admitted high-temperature boxes. Theorem C.6 then supplies the actual free-boundary pressure limits at both maps. Their composition, the $2L^2 \log A^{\mathrm{sens}}$ term, and the original-partition subtraction give the stated C^3 effective growth. The selected branch is the shifted native free law, so eq. (C.115) applies to that shifted map and its exact fixed-word image. The finite source-shift identity alone supplies none of the source regularity, compact-box, strict-gap, or center-ratio conditions. An arbitrary positive likelihood is not covered by this all-edge sensor specialization. [ESTABLISHED]

C.3.3 Prescribed geometry actions and relative label weights

An action constant within a branch is distinct from an action common to all random variables. Fix real geometry-only functions $\mathscr{A}_{\gamma,L}(\theta)$, independent of the branch state, and replace the positive raw geometry weight by $\widetilde{a}_{\gamma,L} = a_{\gamma,L} e^{-\mathscr{A}_{\gamma,L}}$. For a prescribed normalized hierarchy this means explicitly reweighting and renormalizing its label masses by the same factor; it is a new specified ensemble. Conditional branch laws and their physical partitions remain unchanged. A factor depending on the state would instead change the branch and needs a separate admission argument. [ESTABLISHED]

Proposition C.39 (Action growth and the exact relative multiplier). Suppose $\log a_{\gamma,L}$ is C^3 on a neighborhood of each compact $K \subset \Theta$, with

$$\|\log a_{\gamma,L}\|_{C^3(K)} \le C_K V, \qquad V^{-1} \log a_{\gamma,L} \longrightarrow \alpha_\gamma \quad \text{in } C^3(K). \tag{C.136}$$

If the prescribed actions satisfy

$$\|\mathscr{A}_{\gamma,L}\|_{C^3(K)} \le C'_K V, \qquad V^{-1} \mathscr{A}_{\gamma,L} \longrightarrow \mathscr{A}_\gamma^{\mathrm{bulk}} \quad \text{in } C^3(K), \tag{C.137}$$

then the reweighted raw data have the same growth class, with bulk log weight $\alpha_\gamma - \mathscr{A}_\gamma^{\mathrm{bulk}}$. For either normalization mode, the exact relative label law is

$$\frac{\widetilde{W}_{\gamma,L}}{\widetilde{W}_{\gamma_0,L}} = \frac{W_{\gamma,L}}{W_{\gamma_0,L}} \exp\bigl[-\mathscr{A}_{\gamma,L} + \mathscr{A}_{\gamma_0,L}\bigr]. \tag{C.138}$$

[ESTABLISHED]

Proof. The logarithm of the new raw weight is $\log a_{\gamma,L} - \mathscr{A}_{\gamma,L}$. Subtraction preserves the C^3 bound of order V and gives the claimed bulk limit. In the global-factor ensemble the unnormalized label mass is $a_{\gamma,L}Z_{\gamma,L}$, and in the hierarchy generated from raw weights it is $a_{\gamma,L}$. Since $Z_{\gamma,L}$ is unchanged, either log-mass representative is shifted by $-\mathscr{A}_{\gamma,L}$. Taking the ratio to γ_0 cancels the new common normalization and gives eq. (C.138). The same cancellation applies when a prescribed hierarchy is explicitly reweighted. All quantities are finite and positive at fixed volume, so the resulting finite catalog has a finite positive normalizer. □

An extensive action $V\mathscr{A}_\gamma(\theta)$ changes the bulk log-mass density by $-\mathscr{A}_\gamma$. A fixed smooth order-one action has zero bulk rate. An action $V^\kappa\mathscr{A}_\gamma$ with $0 < \kappa < 1$ is also subextensive in the full C^3 sense, but its finite value and derivatives remain in eq. (C.138). For two identical fixed native branches with positive b, t in the high-temperature domain, initial equal weights and actions $\mathscr{A}_{0,L} = 0$, $\mathscr{A}_{1,L} = V^\kappa\theta$ give $W_{1,L}/W_{0,L} = e^{-V^\kappa\theta}$. At $\theta = 0$ the categorical Fisher information is $V^{2\kappa}/4$, so its extensive density is $V^{2\kappa-1}/4$. The common native law is independent of θ and contributes no score in this example. Thus a vanishing bulk action does not justify deleting its derivatives: it can vanish in the extensive density, add a finite correction, or dominate that density according to whether κ is below, equal to, or above $1/2$. These real geometry actions define positive finite weights even where their values are negative; nonnegativity is an additional property of particular energy constructions, not a requirement for this finite normalization argument.

Proposition C.40 (Conditional limits and a bounded-action obstruction). Let $\theta_{c,L}$ be declared centers and suppose the original positive relative weights there converge to $k_\gamma \in (0, \infty)$. If

$$\mathscr{A}_{\gamma,L}(\theta_{c,L}) - \mathscr{A}_{\gamma_0,L}(\theta_{c,L}) \longrightarrow a_\gamma, \tag{C.139}$$

then the new relative weights converge to $k_\gamma e^{-a_\gamma}$. When these are all labels in the catalog, their normalized weights converge to the normalization of those relative masses. Boundedness of the actions, even with the bulk convergence in eq. (C.137), does not imply eq. (C.139) or a limiting ratio. [ESTABLISHED]

Proof. Evaluate eq. (C.138) at the centers. Continuity of the exponential and multiplication proves the stated relative limit. Normalization is continuous on a finite positive vector, giving the full-catalog assertion. For fixed C^3 actions $\mathscr{A}_\gamma$ and centers tending to θ_0, continuity supplies the particular limit $a_\gamma = \mathscr{A}_\gamma(\theta_0) - \mathscr{A}_{\gamma_0}(\theta_0)$. A common action has $a_\gamma = 0$ and cancels exactly at every volume, as the relative formula requires.

For the obstruction, choose two identical fixed interacting native branches with $0 < b < 1$ and $0 < t < 1/12$, and initial unit raw weights. Their finite physical partitions coincide, so both normalization modes have equal labels initially. Set

$$\mathscr{A}_{0,L} = 0, \qquad \mathscr{A}_{1,L} = 1 + (-1)^L. \tag{C.140}$$

These are nonnegative, bounded by two, and independent of θ. Their parameter derivatives vanish; their C^3 norms divided by $V = L^2$ tend to zero. They satisfy eq. (C.137) with zero bulk actions.

Nevertheless, the ratio of label one to label zero is e^{-2} for even L and one for odd L, so the corresponding label-one weights are $e^{-2}/(1+e^{-2})$ and $1/2$. They do not converge. No phase or regularity boundary of the native branches has been crossed. The failure is the missing relative center limit, not a failure of positive finite normalization. □

C.3.4 Which curvature supplies a prescribed action

The three curvature objects retain the types fixed in Definition 18.7. Gauge curvature belongs to a declared principal connection ω; base curvature belongs to a declared nondegenerate base metric; Fisher curvature belongs to a declared regular normalized statistical family and its Fisher metric. Naming one of these objects does not specify the pairing, contraction, integration measure, coupling, or normalization needed for an action. Once those data give a finite geometry-only function with eq. (C.137), the preceding propositions apply to it without identifying the three curvatures. [ESTABLISHED]

For example, use a unitary representation of dimension n, and write $F_{\mu\nu}$ for its represented, anti-Hermitian gauge curvature. Let the base carry a prescribed Riemannian metric g^{base}, raise both base indices with that metric, and use its volume measure. With $\kappa_{\mathrm{g}} \geq 0$, the sign and normalization of eq. (18.14) are

$$\mathscr{S}_{\mathrm{g}} = -\frac{\kappa_{\mathrm{g}}}{4n} \int_{\mathcal{C}} \mathrm{Re}\,\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\, d\,\mathrm{vol}_{g^{\mathrm{base}}}\,. \tag{C.141}$$

The integral must be finite on the prescribed geometry. If this construction is used for each finite geometry label, its parameter regularity and extensive growth must still be checked through eq. (C.137); the sign of the integrand does not establish those bounds. [ESTABLISHED]

Proof: Pairing and sign. For anti-Hermitian represented matrices, the real bilinear pairing $\langle X, Y\rangle = -\mathrm{Re}\,\mathrm{Tr}(XY)/n$ satisfies $\langle X, X\rangle = \mathrm{Tr}(X^*X)/n \geq 0$, with equality only for $X = 0$. In an orthonormal Riemannian base frame, the contraction in eq. (C.141) is a sum of these matrix squares over μ, ν, proving its nonnegative sign. The corresponding unitary loop expression also has a direct square form,

$$\frac{\mathrm{Re}\,\mathrm{Tr}(2I - U - U^*)}{2n} = \frac{\mathrm{Tr}((I-U)^*(I-U))}{2n} \geq 0, \qquad U^*U = I. \tag{C.142}$$

These signs use both the represented matrix class and the Riemannian base contraction. They do not assert a positive contraction for an indefinite base metric. Nor does a nonnegative action alone normalize a probability measure on all connections; that would require a declared reference measure and a finite nonzero partition function. □ [ESTABLISHED]

For real $\mathrm{GL}(n)$, $n \geq 2$, no positive norm on the entire matrix Lie algebra is invariant under every conjugation. Indeed, conjugation of E_{12} by $\mathrm{diag}(s, 1, \ldots, 1)$, $s > 0$, gives sE_{12}. Invariance and norm homogeneity would imply $\|E_{12}\| = s\|E_{12}\|$ for every s, contradicting positivity on $E_{12} \neq 0$. In particular, an unqualified trace-square expression cannot be interpreted as that positive invariant norm. [ESTABLISHED]

A separately prescribed SPD fiber metric H supplies a different, explicitly metric-dependent construction:

$$\mathrm{Tr}(H^{-1}F^{\top}HF) = \left\|H^{1/2}FH^{-1/2}\right\|_{\mathrm{F}}^{2} \geq 0. \tag{C.143}$$

For $G \in \mathrm{GL}(n)$, take the simultaneous changes $F' = GFG^{-1}$ and $H' = G^{-\top}HG^{-1}$. Substitution into the left-hand side of eq. (C.143) gives $\mathrm{Tr}(GH^{-1}F^{\top}HFG^{-1})$, equal to the original trace by cyclicity. Positivity therefore comes with the additional covariant metric data; it is not a norm invariant under arbitrary conjugation of F while holding all metric data fixed. A base contraction and measure would still have to be specified before integrating this quantity. [ESTABLISHED]

A base-curvature functional or a functional of the curvature of a separately prescribed statistical family can likewise be evaluated as geometry data, if its metric is regular and the resulting action meets the finite and C^3 growth assumptions above. If instead the action is required to be the Fisher curvature of the same unknown reweighted law, the weight equation becomes a new self-consistency problem. The present normalization argument proves no existence, uniqueness, or regularity for that problem. It also does not introduce a measure on random base metrics or connections, an all-depth inverse, a critical or continuum limit, or an identification with a physical clock or physical law. [ESTABLISHED]

Appendix D

Typed Notation Contract

This appendix is a type checker, not a second development of the theory. Superscripts b and m always denote the belief and model channels. They may have different representations, fiber dimensions, connections, and coarse maps, but the ambient theory induces both channels from one principal G-bundle.

Symbol	Type	Meaning and prohibited identification
$\mathcal{C}$	smooth base manifold	Context space. It is neither the population graph nor a probability sample space.
$\mathcal{C}_i;\ U_A = \bigcap_{i\in I} \mathcal{C}_i$	agent support and common overlap patch	A finite child block I has parent label A, and the pointwise program fixes $c_* \in U_A$. Ordinary R is not an overlap region.
$q_i^{b;o,X}, q_i^{m;o,X}$	measurable belief-law and model-law response sections	The map $(o, X) \mapsto (q_i^{b;o,X}, q_i^{m;o,X})$ is measurable into declared section spaces, with measurable evaluation at every admitted context. These are context-indexed occupants of the generic section slots, not bare aliases.
$\mathbb{Q}_{i,o,X}$	agent-indexed normalized recognition law on $\mathsf{Y}_{i,D}$	The finite recognition law is evaluated from the declared access map Acc_i; it is not a population coupling or a continuum section value.
$\mathrm{Acc}_i : \mathsf{O}_D \times \mathsf{X} \to \mathsf{I}_i$	measurable agent access map	It specifies the information interface admitted to agent i. An identity access map is the minimal interface; decentralized claims require a stricter declared map.
$\mathsf{M}_{i,a};\ m_i \in \mathsf{M}_{i,a};\ \mathrm{ev}_{i,a}(m_i) = K_{i,m_i}^{X,a}$	incidence model-presentation space, sample, and normalized evaluated kernel	The sample m_i is never a law. Each incidence evaluation is measurable, keeps its declared X and parent dependencies, and reads no recognition or posterior. The bundle $\mathrm{ev}_i = (\mathrm{ev}_{i,a})_{a\in D_i}$ is a typed finite family, not one common-codomain map.
$G_{\theta,i,a}; G_{\theta,i,D} = \bigotimes_{a\in D_i} G_{\theta,i,a}$	agent component kernel and finite-design mechanism	$G_{\theta,i,a} : (\mathsf{X} \times \mathsf{Y}_{\mathrm{pa}(i),a}) \rightsquigarrow \mathsf{Y}_{i,a}$ is normalized in its target coordinate. The bundled mechanism is distinct from an evaluated model kernel and from a recognition law.
$K_a : (\mathsf{X} \times \mathsf{Y}_{\partial a,D}) \rightsquigarrow \mathsf{O}_a$	normalized interaction-record kernel	It is normalized in the record coordinate and owns one record once. It is neither the evaluated model mechanism $K_{i,m_i}^{X,a}$, directed transition $T_{i\leftarrow j}$, nor unnormalized potential ψ_{ij}.

$\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$	class of population couplings with marginals $\mathbb{Q}_{i,o,X}$	It contains all normalized joint laws on $\prod_{i\in V}\mathsf{Y}_{i,D}$ with the declared agent-indexed marginals; local marginals do not determine a member.
$\mathsf{S}_{o,X}$	declared measurable population-coupling selector	It maps $(\mathbb{Q}_{i,o,X})_{i\in V}$ into $\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$. It is additional probabilistic data. Gauge re-expression transports its already selected output; equivariance of a rerun selector is a separate, unclaimed hypothesis.
$\mathbb{P}_{\theta,V};\mathbb{Q}_{V,o,X};\ \mathbf{\Pi}_{\theta,V,o,X}$	constructed normalized population joint, selected recognition coupling, and posterior slice	The joint is constructed from local mechanisms and record kernels, the coupling is selected from the declared coupling class, and the posterior is derived from the joint. None is an agent.
$X;\ X_A=\chi_A(X);\ \xi_A\in\mathbf{\Xi}_A$	fixed fine/parent structure and retained random interface	X, X_A remain outside the random channel. The random interface ξ_A is not structural X_A.
$\mathsf{Z}_A=\mathsf{B}_A\times\mathsf{M}_A\times\mathbf{\Xi}_A\times\mathsf{H}_A$	parent random state space	Contains random retained state only, not structural X_A or the unchanged observation coordinate.
$\mathbb{P}_I(Do,DY\mid X);\ \mathbb{Q}_{I,o,X};\ \mathbf{\Pi}_{I,o,X}$	fine generative joint, correlated recognition law, and selected posterior	The generative joint is fixed before recognition; the posterior is derived from it at an admitted observation.
$\mathbb{P}_A(Do,Dz\mid X);\ \mathbb{Q}_{A,o,X};\ \mathbf{\Pi}_{A,o,X}$	parent generative joint, recognition law, and posterior	All three use the same C_A, with the observation unchanged. Bare global P_A, Q_A are forbidden.
$q_A^b, q_A^m;\ p_A^b, p_A^m;\ \mathbf{\Pi}_{A,o,X}^b, \mathbf{\Pi}_{A,o,X}^m$	derived recognition, generative-conditional, and posterior marginals	These are projections of their typed full laws. No marginal pair replaces a correlated joint. Historical Q_q, Q_m are frozen local aliases only.
$\mathrm{ev}_A : m_A\mapsto K_{A,m_A}^{X_A}$	measurable parent model-evaluation family	Its codomain is $\mathrm{Kern}(\mathbf{\Xi}_A,\mathsf{B}_A\times\mathsf{O}\times\mathsf{H}_A)$. The observation space is unchanged. An induced evaluator exists by disintegration; a predeclared evaluator needs a separate almost-sure compatibility condition on the parent generative conditional.
$C_A:\mathsf{Y}_I\rightsquigarrow\mathsf{Z}_A$	normalized recognition-independent Markov kernel	One channel acts on generation, posterior, and recognition. It is neither an aggregation matrix nor a deterministic moving map.
$\Delta_A(o,X)$	conditional-information defect in the common-channel KL chain	The extended KL identity is additive in $[0,+\infty]$. Zero defect is equivalent to equality of the discarded conditional laws $\mathbb{Q}_{A,o,X}$-almost surely, without a finite-KL premise. Ordinary VFE subtraction and the two-way pairwise common-recovery equivalence require finite fine KL; family-wide recovery requires simultaneous hypotheses.
c_t	differentiable moving coarse map in a declared smooth dynamics	This special-case deterministic moving map is not C_A. The auxiliary t is not physical time, base position, or RG depth by implication.
$R\to E\to O$	retained input or parameter, intervened target, and retained output	This is the typed intervention comparison chain, not an overlap notation or an ontic-action claim.
$\mathscr{S}$	variational action functional	Lowercase behavioral action is unavailable until separately declared.

$\mathscr{P}_G \xrightarrow{\pi} \mathcal{C}$	principal G-bundle	The common source of belief and model associated bundles. It is not either statistical bundle.
ρ_b, ρ_m	actions on sample fibers	Both are representations of the same G; they need not be equivalent, faithful, or of equal dimension. $\rho_x(g)$ acts on a sample coordinate in channel x.
$\widehat{\rho}_b, \widehat{\rho}_m$	actions on laws	$\widehat{\rho}_x(g)q = (\rho_x(g))_{\#}q$. A hatted action is not an action on a sample vector.
$\mathcal{E}_b = \mathscr{P}_G \times_{\widehat{\rho}_b} \mathcal{B}_b, \quad \mathcal{E}_m = \mathscr{P}_G \times_{\widehat{\rho}_m} \mathcal{B}_m$	associated law bundles over $\mathcal{C}$	Their fibers contain belief and model laws. They are not principal bundles.
$u_i^b, u_i^m;\ h_i$ with $u_i^m = u_i^b h_i$	two local sections of $\mathscr{P}_G$; unique map $h_i : \mathcal{C}_i \to G$	The relative principal frame exists independently of ρ_b, ρ_m. It does not define a map between their representation spaces.
$u_i^{x\prime} = u_i^x a_i^x; \quad R_i^x = \rho_x(a_i^x)^{-1} = \rho_x(g_i^x)$, with $g_i^x = (a_i^x)^{-1}$	passive section rechoice and old-to-new sample-coordinate map in channel x	If $v_x' = R_i^x v_x$, then $q_i^{x\prime} = (R_i^x)_{\#} q_i^x$ and a same-channel transport obeys $T_{r\leftarrow i}^{x\prime} = R_r^x \circ T_{r\leftarrow i}^x \circ (R_i^x)^{-1}$. A linear cross-channel map transforms by $F_i' = R_i^m F_i (R_i^b)^{-1}$, covariance by $\Sigma_i' = R_i^x \Sigma_i (R_i^x)^\top$, and precision by inverse congruence. This R_i^x is neither a belief–model map nor an RG restriction.
$\Phi : \mathcal{E}_b \to \mathcal{E}_m$	belief-to-model associated-bundle morphism	Covers $\mathrm{id}_{\mathcal{C}}$. It is not a gauge automorphism or a same-channel transport.
$\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$	model-to-belief associated-bundle morphism	Covers $\mathrm{id}_{\mathcal{C}}$. It is independent of Φ and is not presumed to be its inverse.
ω_b, ω_m, or H^b, H^m	two principal connections on $\mathscr{P}_G$, or induced horizontal distributions	They are separate geometric data on the same principal bundle. Vector-bundle covariant derivatives are a linear specialization.
$\Omega_\gamma, \widetilde{\Omega}_\gamma$	$(\mathcal{E}_b)_{\gamma(0)} \to (\mathcal{E}_b)_{\gamma(1)}$ and $(\mathcal{E}_m)_{\gamma(0)} \to (\mathcal{E}_m)_{\gamma(1)}$	Same-channel parallel transports induced by the two connections along a piecewise-smooth base curve $\gamma \subset \mathcal{C}$.
$\varpi_x : \mathcal{E}_x \to \mathcal{C}, \quad V\mathcal{E}_x$	associated law-bundle projection and its vertical tangent bundle	The vertical space is tangent to a statistical fiber. It is neither a sample space nor the tangent space of the contextual base.
$g_x^F, \mathcal{T}_x$	vertical symmetric tensors of orders two and three on $\mathcal{E}_x$	The descended Fisher and Amari–Chentsov tensors exist under the declared regular-model hypotheses. They are not tensors on $\mathcal{C}$ until a section and connection are supplied.
$D^{\omega_x} s = \mathrm{ver}^{\omega_x} \circ Ts$	map $T\mathcal{U} \to s^* V\mathcal{E}_x$ for a smooth section $s : \mathcal{U} \to \mathcal{E}_x$	Connection-relative vertical first jet. For nonlinear statistical fibers it is not a linear covariant derivative on a vector space of sections.
$h_s^{\omega_x}, c_s^{\omega_x}$	sections of $\mathrm{Sym}^2 T^*\mathcal{U}$ and $\mathrm{Sym}^3 T^*\mathcal{U}$	Pullbacks of g_x^F and $\mathcal{T}_x$ by $D^{\omega_x} s$. They are passive-gauge covariant but connection-relative, and they are not fixed by the active gauge group at fixed connection; $h_s^{\omega_x}$ is only a semimetric until nondegeneracy is proved.
$N;\ N_\star;\ q = N_\star\|_{\mathcal{B}}$	sample kernel $\mathsf{K} \times \bar{\mathscr{K}} \to [0,1]$; affine law map $\mathcal{P}(\mathsf{K}) \to \mathcal{P}(\bar{\mathsf{K}})$; law-fiber map $\mathcal{B} \to \bar{\mathcal{B}}$	Three distinct objects. N determines $N_\star$, and $N_\star$ restricts to q only under the declared family closure $N_\star(\mathcal{B}) \subseteq \bar{\mathcal{B}}$; neither arrow reverses. Sample-level equivariance of N implies the law-level intertwining, and not conversely. None of the three is the associated-bundle morphism.

$\Psi;\ T^V\Psi$	associated-bundle morphism $E \to \bar{E}$ over f; its vertical differential $VE \to V\bar{E}$	Ψ is induced by $(\mathcal{P}, \kappa, q)$ exactly under (11.14). For a fixed principal map $\mathcal{P}$ and associated-bundle morphism Ψ, the representative $q = \iota_{\mathcal{P}(u)}^{-1}\Psi\iota_u$ is unique. Replacing the target frame changes that representative by the full represented target gauge action, not merely by image isotropy; Ψ never determines N. $T^V\Psi$ equals Tq only in a frame pair $(u, \mathcal{P}(u))$; in any other frame pair the identification uses coarse-side invariance of $\bar{g}^F$.
$A_\Psi = \mathcal{D}\Psi;\ \mathfrak{A}_\mathcal{P}$	section of $\varpi^*T^*\mathcal{C} \otimes \Psi^*V\bar{E}$; element of $\Omega^1(\mathcal{C}; f^* \operatorname{Ad}\bar{P})$	Horizontal defect and scale-connection defect form, related by (8.55). The sign convention is transported fine horizontal minus coarse horizontal lift. They compose by the ordered law (8.57); an unweighted sum of two stage defects is a type error.
$\Delta_F^\Psi;\ \delta_\Psi;\ \mathcal{X}_\Psi;\ \mathcal{Q}_\Psi$	vertical form on $\operatorname{Sym}^2 V^*E$; its section pullback and the two anomaly tensors on $\operatorname{Sym}^2 T^*\mathcal{U}$	Δ_F^Ψ is the fiber Fisher defect and δ_Ψ its base pullback; they are different objects on different bundles. The base comparison retains *two* tensors, $-\mathcal{X}_\Psi$ with $\mathcal{X}_\Psi$ signed and $-\mathcal{Q}_\Psi$ with $\mathcal{Q}_\Psi \succeq 0$; neither may be summarized as one mismatch term.
$\varpi_i : \mathcal{E}_i \to \mathcal{C}_i;\ \boldsymbol{\omega}_i$	$\mathcal{E}_b\|_{\mathcal{C}_i} \times_{\mathcal{C}_i} \mathcal{E}_m\|_{\mathcal{C}_i} \to \mathcal{C}_i$; product connection	Agent i's paired law bundle and the pair $\boldsymbol{\omega}_i = (\omega_b, \omega_m)$. The common projection does not select relative channel weights or a joint Fisher metric.
$v^\omega(\Gamma), L^\omega(\Gamma)$	vertical velocity and vertical Fisher length of a curve $\Gamma : J \to E$	The parameter on J is arbitrary. This length selects neither an inference orbit nor a physical time.
$\mathfrak{S}_i$	declared regular configuration space of belief–model sections for agent i	Its tangent vectors are section variations evaluated vertically at each fixed context. It is not the contextual base or an undeclared infinite-dimensional probability space.
$\mathscr{H}_i$	oriented unparameterized curve in $\mathfrak{S}_i$	An equivalence class under orientation-preserving reparameterization. Its orientation is selected by the VFE descent ray only after a regular metric or mobility is declared.
$\Sigma_i : \mathscr{H}_i \times \mathcal{C}_i \to \mathcal{E}_i$	adjoint evaluation map of an inference history	$\varpi_i \circ \Sigma_i(r, c) = c$: history changes the section value while the base context remains fixed.
$\pi_i^{\mathrm{conf}}, \iota_i$	configuration extraction from an admissible conditional/full-law family and its smooth right-inverse	$\pi_i^{\mathrm{conf}} \circ \iota_i = \mathrm{id}$ is the commuting condition that makes ι_i a genuine joint-law lift of the displayed agent configuration, rather than an unrelated parameterized family.
$\overline{\mathcal{F}}_{B,o}$	outside-averaged conditional block VFE at fixed Q_{B^c}	Under the local–global support, finiteness, integrability, and differentiation hypotheses, it differs from the restricted collective VFE only by a configuration-independent term.
$\mathsf{G}_i^F = \iota_i^* G^F_{\mathfrak{R}_B}$	exact Fisher metric induced from the lifted conditional/full recognition law	A weighted marginal Fisher sum has this status only under separately proved block orthogonality or fixed dependence; degeneracy requires a justified quotient.

$\mathcal{Q}_\ell;\ \mathsf{G}_\ell$	declared configuration manifold at scale ℓ; its *strong* configuration metric	Determined by the declared base measure, channel weights, basis, and topology, not by the bundle. G_ℓ is a section of $\mathrm{Sym}^2 T^*\mathcal{Q}_\ell$ and is neither the fiber metric g_x^F nor the base semimetric $h_s^{\omega_x}$. It carries a scale or agent subscript always, and is not the unsubscripted M-step preconditioner G of Chapter 5. A labeled weighted product equals a joint-law pullback only under the criterion (9.19).
$\mathsf{R}_\ell : \mathcal{Q}_\ell \to \mathcal{Q}_{\ell+1}$	smooth coarse configuration map	Separately declared data. It is not the sample kernel K_ℓ, the law map $N_\star$, the bundle morphism Ψ or $C_{\ell,s}$, the nonlinear action map $\mathcal{R}^H$, the block measure-pair map $\mathcal{R}_b$, the derivative D_ℓ, the reference-space endomorphism $\widehat{\mathcal{R}}_\ell$, the conjugated retained projection $\widehat{R}_\ell$, the root-vertex set $\mathcal{R}$, or the descent ray $\mathcal{R}^-_{\mathcal{F}_i}$. A pointwise bundle morphism induces it only on the projectable set, which can have empty interior.
$a_\ell;\ Q^{(\ell)}(r);\ \tau^{(\ell)}(r)$	positive semiconjugacy factor; configuration at depth ℓ and orbit position r; accumulated Fisher duration at that depth	Distinct typed quantities plus a parameter-conversion datum. Depth ℓ is a discrete scale index; r is a chosen oriented orbit parameter; and $\tau^{(\ell)}$ is metric-relative arc length from a chosen origin. After choosing metric, origin, orientation, and path, τ can be a function of r, but it is not canonically determined by an unparameterized orbit or by scale depth. The factor $a_\ell = \mathrm{d}\sigma/\mathrm{d}r$ is the coarse-flow-parameter rate relative to the fine parameter, not a physical-time or Fisher-duration rate; ordinary arc length uses $\lvert a_\ell\rvert$, and positivity selects the oriented semiconjugacy.
$\nu_F, L_F, \tau_{Q,\lambda_0}$	Fisher speed, realized path length, and cumulative clock from a chosen origin	They are defined on a selected regular history. Length and cumulative duration are invariant under orientation-preserving reparameterization; neither is physical time.
$T^b_{ij}(c), T^m_{ij}(c)$	elements of G at one $c \in \mathcal{C}_i \cap \mathcal{C}_j$	Transition functions computed from two frame atlases of the same $\mathscr{P}_G$. Their cocycles are cohomologous descriptions of one principal-bundle class. They use no nonconstant curve and are not graph links.
Θ^b_e, Θ^m_e	oriented-edge-copy elements of G on $\mathfrak{I} = (V, E, {}^-)$	Channel-specific graph-link data, transforming under discrete vertex frame changes h^x_i, with $\Theta^x_{\bar{e}} = (\Theta^x_e)^{-1}$. Parallel copies retain distinct labels. Equality with represented base transport requires vertex contexts, assigned curves, $h^x_i = g^x_i(c_i)$, and the explicit link–transport hypothesis. The superscripted contextual re-expression h^x_i is not the relative principal-frame field h_i.
$\mathcal{H}, \overline{\mathcal{H}}$	represented graph holonomy group and its closure in the represented matrix group	A finitely generated $\mathcal{H}$ need not be closed. Haar averaging in the compact-closure theorem is over $\overline{\mathcal{H}}$, after explicitly assuming that this closure is compact.

$\nu_{\mathrm{mix}} = \lambda + \sum_n 2^{-n}\delta_{a_n}$	fixed sigma-finite reference for a mixed family	Exists in this form exactly when the union of all positive atomic supports is countable, assuming no singular-continuous components. It is fixed for the family, not selected separately at each parameter.
$M_o;\ z = M_o(\mathsf{Y}_D);\ \Pi_o = M_o/z$	finite evidence-slice measure; positive-finite mass; posterior probability	The slice is primary at a selected regular observation. M_o is not itself a probability unless $z = 1$, and normalization requires $0 < z < \infty$.
$\mathcal{L}^{\mathrm{ext}}, \mathcal{F}^{\mathrm{ext}}$	maps on every latent probability law with values in $[-\infty, \log z]$ and $[-\log z, +\infty]$	$\mathcal{L}^{\mathrm{ext}} = \log z - D_{\mathrm{KL}}(Q\|\Pi_o)$ and $\mathcal{F}^{\mathrm{ext}} = -\mathcal{L}^{\mathrm{ext}}$. The safe gap identity is $\log z - \mathcal{L}^{\mathrm{ext}} = D_{\mathrm{KL}}$; the sum $\mathcal{L}^{\mathrm{ext}} + D_{\mathrm{KL}}$ is not formed in the singular branch.
$\mathcal{L}, \mathcal{F}$	finite classical split of the extended functionals	The expected-log-joint minus expected-log-recognition formula is used only under absolute continuity and separate log-integrability. It is not the definition on arbitrary Q.
$\mathrm{TC}(Q) = D_{\mathrm{KL}}(Q\|\bigotimes_b Q_b)$	extended nonnegative total correlation	May equal $+\infty$. Its product-reference chain identity holds in $[0, +\infty]$ without subtracting entropies; marginal-entropy formulas are finite-domain corollaries.
$K : \mathsf{X} \rightsquigarrow \mathsf{Y}$	normalized, parameter-independent Markov kernel	It satisfies $K(x, \mathsf{Y}) = 1$ for every x, pushes laws forward, and obeys data processing. For right-acting scale kernels, $K_{\ell r} := K_{\ell k}K_{kr}$ means $(\mu K_{\ell k})K_{kr} = \mu K_{\ell r}$, representing the abstract composite $K_{kr} \circ K_{\ell k}$. A kernel is not an energy restriction or a Galerkin matrix.
$\mathscr{K}_{\mathrm{op}}$	declared operator category	Its objects are topological vector spaces and its morphisms are continuous linear maps; the analytic tier uses Banach spaces and bounded linear maps. Abstract arrows compose with $\circ$, not with right-acting kernel juxtaposition.
$I_{\mathbb{R}^b};\ I_b$	the b-dimensional identity covariance; declared rescaling/identification kernel	The covariance $I_{\mathbb{R}^b}$ occurs in replicated Gaussian laws. The plain I_b is a normalized kernel composed with C_b in the RG measure-pair map; it is neither that covariance nor the score lift $\mathscr{I}_b$.
$\iota_{\mathcal{P}}$	coarse-to-fine deterministic embedding	Imposes a hard block constraint on an energy family. Normalization and an induced probability channel require separate proofs.
C, P	compression and prolongation on a declared linear feature tier	Here $\mathsf{CP} = I$. A message propagator may be compressed as $\mathsf{C}A\mathsf{P}$, but a hard-tied quadratic energy uses $\mathsf{P}^*H\mathsf{P}$, an exact Gaussian marginal uses a Schur complement, and a general law uses Markov pushforward.
$K_a, E_{a,o}$	normalized interaction-record kernel and its fixed-record negative log density	One actual record is one global factor. It may occur in several incident coordinate-local VFEs without being duplicated in the joint.
$\alpha_i^x;\ \eta_{ij}^b = \alpha_i^b\beta_{ij};\ \eta_{ij}^m = \alpha_i^m\gamma_{ij}$	receiver occupancy and channel-typed joint edge-event laws	Exact meta-attention pushes η^x and then disintegrates. A row β alone does not determine receiver weights.
$c_*;\ U \ni c_*;\ \ell;\ f_\ell : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$	fixed base point, base neighborhood, RG depth, and actual base coarse map	The fixed-fiber theory is vertical over c_*. It neither constructs sections over U nor implies a base map f_ℓ. RG depth is not a base coordinate or physical time.

$m_\ell(I \mid i, R_\ell);\ M_\ell(I, J \mid i, j, R_\ell)$	normalized one-endpoint and correlated two-endpoint membership kernels	Their target sums equal one. A hard partition gives a deterministic kernel; a product formula for M_ℓ requires conditional independence. Literal unit-weight multiple membership is replication of a finite measure, not a normalized pushforward. The fully argumented $M_\ell(\cdot)$ is distinct from the scalar evidence mass M_ℓ below.
η^b_ℓ, η^m_ℓ	retained recursive marked receiver–source event laws	Receiver occupancies are their first marginals, and the conditional attention rows are disintegrations on positive occupancy. Chapman–Kolmogorov composition of membership kernels also retains every mark needed by the next step.
$\widehat{\mathsf{Y}}_{\ell+1\leftarrow\ell}$	augmented fixed-fiber transition state	It is $(\mathsf{Y}_{\ell,c_*}, R_\ell, \mathsf{Y}_{\ell+1,c_*})$. The current Y_{ℓ,c_*} includes η^b_ℓ, η^m_ℓ, rooted-holonomy, and dressed-boundary marks. Interaction-record or memory marks must additionally be retained whenever the claimed exact closure requires them.
$C^{\mathrm{struct},c_*}_{\ell+1\leftarrow\ell}$	common structural channel	Its type is $\mathsf{Y}_{\ell,c_*} \rightsquigarrow \mathsf{R}_\ell \times \mathsf{Y}_{\ell+1,c_*}$. It samples and retains R_ℓ; its transition joint is the augmented state above. When declared, the normalized recognition-independent channel pushes $\mathbb{P}_\ell$, the selected $\mathbf{\Pi}_\ell$, and $\mathbb{Q}_\ell$.
$\tau^x_{I\leftarrow i};\ H^x_I;\ V^x_e = \tau^x_{I\leftarrow i}\Theta^x_e(\tau^x_{J\leftarrow j})^{-1}$	root transport, based holonomy representation, and dressed boundary edge	For $e : j \to i$ from J to I, $V^{x\prime}_e = (a^x_{r_I})^{-1} V^x_e a^x_{r_J}$. These are the retained parent-frame objects; no undeclared coarse dressing legs are used.
$\mathscr{P}_\ell \xrightarrow{\varpi_\ell} \mathcal{C}_\ell$	scale-ℓ principal bundle and its principal projection	$\mathscr{P}_\ell$ is distinct from the root-level notation $\mathscr{P}_G$, from the normalized law π_ℓ, and from every interaction-coordinate operator. The projection is ϖ_ℓ, not π_ℓ.
$\mathbf{\Xi}^{\mathrm{MVG}}_\ell = ((A^{(\ell)}_i), (W^{(\ell)}_{ij}))$	scale-ℓ MVG operator-parameter tuple	It determines the assembled interaction precision in Chapter 26. It is neither the normalized law π_ℓ nor a bundle projection.
$(\rho_\ell, m_\ell);\ M_\ell, \pi_\ell$	scale-ℓ probability reference, finite evidence measure, its mass, and normalized law	Here $M_\ell = m_\ell(\mathsf{X}_\ell) \in (0, \infty)$ and $\pi_\ell = m_\ell / M_\ell$. The measure pair is the primary law-level RG state; normalization does not discard the separately tracked evidence mass. The bare m_ℓ here is a finite measure and is distinct from the fully argumented membership kernel $m_\ell(I \mid i, R_\ell)$.
$c_{\mathrm{ray}}(b);\ n_{\mathrm{cut}}(b)$	additive fixed-action-ray shift; cut-edge count for the declared block ratio b	It is present only on the projective action tier of the b-step map $\mathcal{R}^H_b$. A fixed tracked measure pair restores mass preservation and forces $c_{\mathrm{ray}}(b) = 0$. The count $n_{\mathrm{cut}}(b)$ occurs only in the operator-scale counting law.
$B(b);\ \partial B(b)$	typical block at ratio b; its coarse cut	Local notation for the weighted scaling comparison. Neither object is fixed by the block ratio alone; both belong to the declared weighted graph family.
$W_\partial(b), A_B(b);\ s_W, s_A$	cut-coupling and block-self sums; their declared norm-growth exponents	$W_\partial(b) = \sum_{e\in\partial B(b)} W_e$ and $A_B(b) = \sum_{i\in B(b)} A_i$. The exponents satisfy $\|W_\partial(b)\| \asymp b^{s_W}$ and $\|A_B(b)\| \asymp b^{s_A}$ in the declared norm; edge count alone does not determine them.

$b > 1;\ C_b,\ I_b,\ K_b = C_b I_b$	block-RG structural coarse kernel, rescaling/identification kernel, and composite kernel	C_b is the structural coarse Markov kernel and I_b the declared rescaling/identification kernel returning its target to the common measurable state space; their right-acting composite is $K_b = C_b I_b$.
$\mathcal{R}_b, \mathcal{R}_b^\rho;\ \mathcal{R}_b^H, \mathfrak{B}_b^H$	block-RG measure-pair and reference maps; finite-action map and action beta	$\mathcal{R}_b(\rho, m) = (\rho K_b, m K_b)$ is the primary measure-pair map and $\mathcal{R}_b^\rho \rho = \rho K_b$ its reference component. On the dominated, finite-action tier, $\mathcal{R}_b^H[H;\rho]$ is the reference-dependent action map and $\mathfrak{B}_b^H[H;\rho] = (\mathcal{R}_b^H[H;\rho] - H)/\log b$. Neither the action nor its beta is defined outside its stated finite-action domain.
$K_\ell;\ \Pi_\ell(z, dx)$	normalized forward scale kernel; reverse conditional for $\pi_\ell(dx)K_\ell(x, dz)$	K_ℓ pushes ρ_ℓ and m_ℓ forward together. Π_ℓ is a fixed version defined only $\pi_{\ell+1}$-almost everywhere; it is not an inverse of K_ℓ.
$Q_\ell,\ U_\ell = DQ_\ell(0)$	bounded conditional log-Laplace action increment and its linearization	$Q_\ell(\varphi) = -\log U_\ell(e^{-\varphi})$ is finite on bounded actions and locally analytic at every bounded center by the recentering identity, whereas $U_\ell\varphi = \mathbb{E}[\varphi(X_\ell) \mid X_{\ell+1}]$ extends as an L^p contraction. The larger linear domain is not a nonlinear action domain.
$\mathfrak{B}_\ell, \overline{\mathfrak{B}}_\ell$	$L^\infty(\pi_\ell; \mathbb{R})$ and its quotient by $\mathbb{R}\mathbf{1}$	The full space tracks additive evidence-mass changes. The quotient is used only on the explicitly projective tier and carries the oscillation norm.
$\nu_\ell = \bigotimes_i \nu_{\ell i};\ C_{\ell,A}, P_{\ell,A}$	product interaction reference; coordinate expectation and Boolean-lattice Hoeffding projector	The interaction tier separately assumes $\nu_\ell \sim \pi_\ell$ at every admitted scale. A normalized Markov kernel need not preserve this premise; the diagonal-cloning channel is the exact obstruction.
$\mathcal{G}_\ell, \phi_g;\ E_\ell, \mathsf{H}_\ell$	full nonempty-subset interaction Banach space and canonical representative; quotient assembly and Hoeffding extraction	Here $\phi_g = \sum_{A \neq \varnothing} g_A$, $E_\ell g = [\phi_g]$, and $\mathsf{H}_\ell E_\ell = E_\ell \mathsf{H}_\ell = I$ on their typed spaces. The extraction bound $3^{\lvert V_\ell \rvert} - 1$ is finite for every finite network but is not dimension-free.
$T_\ell^{\mathcal{G}};\ U_\ell^{\phi_g}$	exact nonlinear interaction RG and its tilted conditional-expectation factor	$T_\ell^{\mathcal{G}} = \mathsf{H}_{\ell+1}\overline{Q}_\ell E_\ell$, while $DT_\ell^{\mathcal{G}}(g) = \mathsf{H}_{\ell+1}\overline{U_\ell^{\phi_g}} E_\ell$. The untilted U_ℓ occurs only at $g = 0$.
$R_\ell; r_{\ell+1}^{\mathcal{G}}, \overline{r}_{\ell+1}^{Q}$	bounded idempotent retained projection; coordinate and action-quotient residuals	The retained map is projected unless the exact-image invariance condition holds. This indexed projection is unrelated to the stochastic refinement kernel R below; the residual norm is always reported when nonzero.
$\delta_\ell,\ B_{n\leftarrow\ell}, \overline{U}_{n\leftarrow\ell}$	essential Dobrushin coefficient, accumulated block factor, and ordered quotient cocycle	The accumulated log-δ inequality is a sufficient irrelevance certificate, with $\log 0 = -\infty$; its failure is not a relevance or marginality theorem.
$r_{\mathrm{ess}}(U) = r(U + \mathcal{K}(X))$	essential spectral radius in the Calkin algebra $\mathcal{L}(X)/\mathcal{K}(X)$	This convention quotients by all compact operators. It is not a finite-rank quotient, and the qualified positive-unital theorem additionally requires its stated Banach-lattice, quasi-interior-unit, pole, and positive dual-eigenfunctional hypotheses.

$L^2_0(\mu)$; $\mathscr{S}_\pi[\varphi] = -\varphi + \pi(\varphi)$	centered square-integrable score tangent; score of the normalized exponential action path	$\mathscr{S}_\pi$ is an isometry from the action quotient with the Fisher norm $\|[\varphi]\|_F$ onto the bounded centered subspace, whose Fisher completion is $L^2_0(\pi)$. The Fisher norm is not the oscillation norm, and the completion is not a domain of the nonlinear action map Q_ℓ.
$\mathscr{I}_b$; U_b; $\mathscr{L}_b = U_b \mathscr{I}_b$	iid score replication lift; block reverse conditional expectation; score renormalization operator	$\mathscr{I}_b h = \sum_{i=1}^b h(x_i)$ with $\|\mathscr{I}_b\| = \sqrt{b}$. The replication $\mathscr{I}_b$ is not the Hoeffding assembly E_ℓ; $\mathscr{L}_b$ is not the ELBO $\mathcal{L}$. $\mathscr{L}_b$ is an endomorphism only when the block statistic is normalized so that its law equals the input law.
He_k; $e_k = \mathrm{He}_k/\sqrt{k!}$	probabilists' Hermite polynomials and their normalized orthonormal basis	Complete orthonormal in $L^2(\gamma)$; deleting e_0 gives a basis of $L^2_0(\gamma)$. Every e_k is unbounded, so a Hermite basis is not a basis of $L^\infty/\mathbb{R}\mathbf{1}$ and, for $k \geq 3$, admits no two-sided exponential action neighborhood.
$\Delta s_\ell = \log b_\ell$; $s_{n\leftarrow\ell} = \sum_{k=\ell}^{n-1} \Delta s_k$	adjacent and cumulative log block scale	$s_{n\leftarrow\ell} = \log B_{n\leftarrow\ell}$. It is a scale index, not a duration; every reported growth rate is divided by it.
$D_\ell = DT_\ell(x_\ell)$; $D_{n\leftarrow\ell} = D_{n-1}\cdots D_\ell$; $\widehat{D}_{n\leftarrow\ell}$	derivative along an exact orbit and its ordered composition	The rightmost factor acts first. The hatted cocycle is the one transported by the declared comparison trivializations. These are maps between different spaces; an ordinary spectrum requires a declared identification.
$v_{\ell,a}, \lambda_{\ell,a}$; $\lambda_{n\leftarrow\ell,a}$	compatible cross-scale mode line and its coefficients	Defined by $D_\ell v_{\ell,a} = \lambda_{\ell,a} v_{\ell+1,a}$ and the ordered product law. The equation $D_\ell v = \lambda v$ is ill typed and is not used.
$\chi_{n\leftarrow\ell}(u)$; $\upsilon_{n\leftarrow\ell,a}$	finite-time norm growth rate and scalar mode exponent	Both are relative to the declared norms, the declared extensive normalization, the block ratios, and the comparison maps. A Lyapunov exponent is asserted only when the upper and lower asymptotic rates agree; no Oseledets splitting is claimed. The exponent υ_a is distinct from the generative latent variable y_a.
$J_\ell : \mathcal{G}_\star \to \mathcal{G}_\ell$; $\widehat{T}^{\mathcal{G}}_\ell = J^{-1}_{\ell+1} T^{\mathcal{G}}_\ell J_\ell$	comparison trivialization and the pulled exact interaction map	This is the inverse orientation of I_ℓ, so $J_\ell = I_\ell^{-1}$ when both are declared. Exponent invariance requires tempered $\|J_n\|$ and $\|J_n^{-1}\|$, not only tempered mode normalization.
$\beta^{\mathrm{ex}}_\ell, \beta^{\mathrm{ret}}_\ell, \delta\beta_\ell$	exact, retained, and omitted discrete interaction beta	All three live in $\mathcal{G}_\star$ and divide by Δs_ℓ. $\delta\beta_\ell$ is the transported coordinate residual; it vanishes on the retained sector exactly under exact-image invariance. Neither is the action beta $\mathfrak{B}^H_b$, whose reference change is inhomogeneous.
$\nabla^{\mathrm{scale}}, \beta^{\mathrm{scale}}$; $\mathsf{V}(s,t)$	scale connection on a declared C^1 Banach bundle over a scale manifold, its covariant scale derivative, and a scale evolution family	The scale connection has base the scale manifold and is not either contextual principal connection ω_b, ω_m, nor the inference-orbit parameter, nor Fisher duration. Discrete endpoint data do not determine it.
$\widehat{\mathcal{M}}^{(\tau)}_\ell$	base-ℓ reference monodromy of a p-periodic identified scale sequence	It is an endomorphism of the declared reference space $Y^{(\tau)}_\star$. A reference monodromy fixed object and all points of its p-cycle lie in that space and need not be fixed by any one-step reference map.

$\Pi^{\mathrm{res}} = \mathsf{PC};\ \mathsf{Q} = I - \Pi^{\mathrm{res}}$	resolved projection and its complement on the linear observable tier	Idempotent by $\mathsf{CP} = I$. This projection is not the reverse kernel Π_ℓ, not the posterior Π_o, and not the retained interaction projection R_ℓ.
$R_\rho(z, dy)$	reverse conditional of the probability reference ρ under the coarse channel	Used by the conditional-partition formula and by the reference-change law for the action beta. It is neither the retained projection R_ℓ nor the stochastic refinement kernel R below.
$C_{\ell k} : \mathfrak{X}_\ell \to \mathfrak{X}_k$	arrow in the scale diagram	Composes abstractly as $C_{kr} \circ C_{\ell k} = C_{\ell r}$. Right-acting kernel juxtaposition is the separate convention stated above. A renormalization dynamics also needs comparison and rescaling maps.
$f_\ell, C_\ell^b, C_\ell^m$	base and channelwise adjacent-scale maps	C_ℓ^x covers $f_\ell : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$. Their failures to commute with $\Phi_\ell, \widetilde{\Phi}_\ell$ or parallel transport are covariant defect diagrams. This base-changing tier is distinct from a fixed-c_* vertical coarse channel.
Q, R	coarse map and stochastic refinement kernel	For noninjective Q, $Q_\# R(c, \cdot) = \delta_c$ is only a stochastic right section; it does not recover the discarded realized microstate.
$P_b, P_m;\ G_b \times G_m$	optional product-gauge extension	Used only when independent principal topology or independent physical gauge symmetries are intended. It is not required for separate frames, representations, or connections in the ambient theory.
$U_A;\ V_A$	fixed-child intersection; availability domain	A union-domain parent requires its declared weights and compatible ambient law; it is not a fixed-child parent on an empty intersection.
$b = \tanh h,\ t = \tanh j$	chapter-local lattice parameters	Field and plaquette strengths on $0 < b < 1$, $0 < t < 1/12$. The lattice t is not a time or contextual coordinate.
$J_S;\ \Psi_B$	complete character coefficient; vacuum interaction	Distinct action representatives. A scalar-free coefficient list does not specify the raw partition or its geometry weight.
$\gamma \in \mathfrak{G}$	finite geometry catalog label	Local to the normalized-geometry and pressure chapters. It is not their base-curve coordinate; the catalog is generative data, not another agent.
$\nu_\gamma, r_\gamma, Z_\gamma, a_\gamma$	reference, raw factor, branch partition, label factor	All enter normalization. Hierarchical weights and global factor weights are different unless their specified branch partitions agree.
$\zeta_C;\ \sigma_a$	finite anchor scalar; scalar density	Representative normalization, not a principal gauge transformation. Density composition carries the intermediate area factor.
$\mathfrak{p}(\Psi);\ f;\ h$	pressure; specific VFE; specific relative entropy	Limits per lattice cell of finite quantities. They are not an infinite partition or ordinary KL between infinite laws.

Thermodynamic Fisher draft notation (OPEN; pending adjudication). The following symbols are local to Chapters C and 20. Their proof drafts retain their stated hypotheses; these rows assign no established status.

$L;\ V = L^2$ $M = B^2V$	side length; coarse and fine cell counts	For a fixed finite word with total integer linear factor B, the fine square has side BL. Two signs per cell do not replace cell area by edge count. Every volume limit fixes the word first.
$g = (h, j);\ \chi_\gamma$	natural source pair; map $\Theta \to \mathbb{R}^2$	Here g is a source coordinate, not a metric. Native derivative limits use C^4 maps into compact boxes with $h > 0$ and $0 < j < \operatorname{atanh}(1/12)$. The product edge $j = 0$ is a separate control.
$\mathsf{A}_L;\ \mathfrak{p}_L$	natural and raw-factor log-partition densities	Both are per cell. Their difference contains the edge and internal-plaquette $\log\cosh$ terms in eq. (C.16); they are not interchangeable under differentiation. Neither is an infinite partition function.
$H(g)$	native two-source Fisher density	Here $H = \nabla_g^2\mathsf{A}$. Its source-map pullback is $D\chi^\top(H \circ \chi)D\chi$. This is not generally the Hessian of $\mathsf{A} \circ \chi$, nor the Hessian of raw pressure. Its derivative limits are those of Proposition C.7.
$\nu^{\mathrm{f}}_{L,g};\ \mu_g$	finite native free law; infinite native Gibbs law	The finite law retains all outgoing signs and only internal plaquettes, with its physical reference partition. μ_g is the infinite law; its restriction need not equal a finite free-boundary law.
$C_B;\ P^{\mathrm{f}}_{B,L,g}$	fixed parity map; exact pushed finite free law	$P^{\mathrm{f}}_{B,L,g} = (C_B)_\# \nu^{\mathrm{f}}_{BL,g}$. The map concatenates the declared straight paths; it is neither a parameter-dependent channel nor a canonical free coarse Gibbs prescription.
$\mu_{B,g};\ \mu_{B,L,g}$ $\mathcal{I}_B$	pushed infinite law; actual marginal; Fisher density	$\mu_{B,L,g} = \mu_{B,g}\vert_{\Lambda_L}$. The exact finite-free push and this actual marginal have separately stated C^2 density limits in Theorem C.11. Equality of their limits neither identifies their finite laws nor identifies $\mathcal{I}_B$ with a coarse pressure Hessian.
$P_{L,\theta};\ W_{\gamma,L}$ $p_{\gamma,L,\theta}$	full labeled law; categorical mass; conditional law	$P_{L,\theta}(\gamma, x) = W_{\gamma,L}p_{\gamma,L,\theta}(x)$. The conditional exists only on an occupied label. Global masses are proportional to $a_{\gamma,L}Z_{\gamma,L}$; a raw-weight hierarchy has masses proportional to $a_{\gamma,L}$. A prescribed hierarchy must be declared separately.
$\lambda_{\gamma,L};\ s_\gamma$	effective log-mass representative; bulk log-mass density	W is the softmax of λ, and $\lambda_{\gamma,L}/V \to s_\gamma$ in the admitted C^3 sense. The representatives are $\log a + \log Z$ globally and $\log a$ for the raw-weight hierarchy. A common scalar cancels exactly; the $O(V)$ derivative bound need not hold for $\log W$.
$\xi_{\gamma,L};\ v_\gamma$	log-mass gradient; limiting gradient at a declared center	$\xi_{\gamma,L} = \nabla\lambda_{\gamma,L}$ and $v_\gamma = \nabla s_\gamma(\theta_0)$. The normalized categorical score is $\xi_\gamma - \mathbb{E}_W\xi$. Equality of bulk log-mass densities at a tie does not imply equality of these gradients.
$I(P_L);\ g_{\gamma,L}$	full Fisher tensor; branch Fisher density	$g_{\gamma,L} = I(p_{\gamma,L})/V$, while the full tensor also contains $\mathrm{Cov}_W(\xi_\gamma)$. $I(P_L)/V$ is the extensive density; $I(P_L)/V^2$ is a different normalization used at fixed ties. A singular limit remains a semimetric.

$S;\ w_\gamma;\ \kappa_\gamma$	tied maximizing labels; limiting masses; relative center limits	S maximizes $s_\gamma(\theta_0)$. The fixed-tie result assumes actual masses tend to w; the crossover result instead assumes positive relative center limits κ, with one reference value equal to one. Bulk pressure alone supplies neither limit.
$u;\ v;\ \theta_{c,L}$	fine and coarse window coordinates; actual declared center	For the fixed-word comparison, $u = M(\theta - \theta_{c,L})$, $v = V(\theta - \theta_{c,L})$, and $u = B^2 v$. The single-volume label theorem writes $u = V(\theta - \theta_0)$; its volume must be replaced by M at the fine endpoint. Neither coordinate is a physical clock.
$\pi_\gamma(u);\ G(u)$	limiting crossover law and its Fisher tensor	The admitted family has weights proportional to $\kappa_\gamma e^{u\cdot v_\gamma}$, with $G(u) = \mathrm{Cov}_{\pi(u)}(v_\gamma)$. This C^2 window limit is not an unrestricted extension of the extensive density through a tie. The coarse coordinate changes both the tensor multiplier and its argument: $B^4 G(B^2 v)$.
$E;\ E^\perp;\ \mathbb{R}^d/E^\perp$	span of slope differences; null space; identifiable quotient	Here $E = \mathrm{span}\{v_\gamma - v_{\gamma_0} : \gamma \in S\}$. The limiting crossover family descends to the quotient identified with E. The finite metrics are restricted to a fixed copy of E; they need not descend along $E^\perp$. An arbitrary constant-rank semimetric does not acquire a global quotient by this example.
$G_L;\ G$	metric candidates on a declared parameter chart	For $G_L \to G$ in C^2 with a compact positive floor for G, the regular-chart draft gives inverse convergence in C^2, Levi-Civita convergence in C^1, and curvature convergence in C^0. The floor is not uniform over unbounded depth and is not supplied by notation or by rank-changing data.
$D;\ D_j$	finite Fisher loss; one-step loss in a fixed word	In the same parameter coordinates, $D = I_f - I_c \succeq 0$. It is the weighted loss of conditional branch information for a label-preserving channel. With $M = B^2 V$, the extensive and squared-volume identities carry B^2 and B^4, respectively. Their derivatives are identities, not ordered derivative tensors.
$Z_\gamma^o;\ e_\gamma(o)$ E_o	likelihood-weighted raw partition; branch evidence; full evidence	$e_\gamma(o) = Z_\gamma^o / Z_\gamma$ and $E_o = \sum_\gamma W_\gamma e_\gamma(o)$. The selected posterior requires $E_o > 0$; a zero posterior label has no conditional law. A hierarchy retains the original physical Z_γ divisor.
$A_\gamma^{\mathrm{sens}};\ b'_\gamma$	all-edge sensor scalar; shifted field	The strictly positive binary sensor supplies the finite factor $(A_\gamma^{\mathrm{sens}})^{2V}$ and shifted native field b'_γ. Thermodynamic derivatives require both original and shifted source maps in admitted C^4 compact boxes. This is not an arbitrary-likelihood theorem or a geometry action.
$\mathbb{Q};\ \mathbf{\Pi}_o$ K_j^{KL}	full recognition law; selected posterior; conditional KL loss	One fixed channel pushes both laws and the complete observation experiment. Under finite domination the VFE loss is the expected conditional KL of their augmented reverse laws. Its word sum compares the given recognition law with its images; it is not an equality of stochastic variational infima.

$\pi_\alpha;\ \alpha!;\ \mathcal{O}_m$	ordinary finite probability jet; multi-index factorial; index set	$\pi_\alpha = \partial^\alpha \pi(\theta_0)$ and $\mathcal{O}_m = \{\alpha \in \mathbb{N}^d : \lvert\alpha\rvert \le m\}$. Taylor coefficients divide by $\alpha!$. Structural-zero atoms have zero derivative columns; C^3 masses determine the C^2 Fisher jet without an inverse metric. Exact rational arithmetic assumes rational input jets and channel entries.
$\mathscr{A}_{\gamma,L}$	prescribed real geometry-only action	It replaces $a_{\gamma,L}$ by $a_{\gamma,L}e^{-\mathscr{A}_{\gamma,L}}$, with separate finite normalization and C^3 growth admission. Relative label multipliers are $e^{-\mathscr{A}_{\gamma,L}+\mathscr{A}_{\gamma_0,L}}$. A bounded action need not have convergent relative center values.
$F_\omega;\ g^{\mathrm{base}}$	principal-connection curvature; prescribed base metric	Gauge curvature needs a declared represented pairing, base contraction, measure, and coupling to define an action. Curvature of the base metric and Levi-Civita curvature of a regular Fisher metric are distinct objects. No one of these selects either of the others.
H in the fiber-pairing paragraph	separately prescribed SPD fiber metric	This H is not the native source Hessian $H(g)$, the base metric, or a Fisher inverse. The positive pairing $\mathrm{Tr}(H^{-1}F^\top HF)$ uses its simultaneous covariant transformation. An arbitrary $\mathrm{GL}(n)$ trace square is not thereby a positive conjugation-invariant norm.
$\vartheta;\ M(\vartheta);\ m = TM$	local learned scalar, model Fisher metric, and scaled metric	In Chapter 16, ϑ is not the design index a. Every actual parent uses its own full-law model score and the inherited metric.
$x = (u, v, \zeta);\ S \in \mathcal{J}$	continuous configuration and complete source assignment	These are local coordinates of the finite-depth realization. Its stochastic X_t is not structural X; S retains every source label through each parent activation.
$\mathfrak{h}_q;\ h_q$	raw triangle holonomy and normalized loop coordinate	$\mathfrak{h}_q = \sum_e \zeta_e$ around the oriented triangle and $h_q = \mathfrak{h}_q/\sqrt{3}$. These are network marks at fixed c_*, not holonomy along base paths.
$J_s;\ Q_s;\ A;\ T;\ H_*$	local energy strength, exponent, amplitude, scale and horizon	Here $J_s = \epsilon^{-Q_s}$ is a scalar energy strength, not Gaussian precision. Q_s is not a recognition law. T is not physical time; H_* is the chosen inference horizon.
$P_p;\ C_{s\leftarrow p}$	orthogonal projection and its induced normalized Markov kernel	The projection acts on continuous coordinates; the channel also retains the complete assignment S. Exact pushed laws do not imply exact autonomous dynamical closure.
$\mathbf{\Pi}^p_\vartheta;\ \pi^p_\vartheta;\ q^p_t$	physical posterior, restricted optimizer, actual recognition density	These chapter-local probability objects have distinct definitions. Optimizer-conditioned inherited rates are not actual recognition conditionals or regenerated parent heat baths.

Three transports that must remain distinct. Ω_γ and $\widetilde{\Omega}_\gamma$ move laws between fibers over different base points in one channel. $T^x_{ij}(c)$ compares two coordinates in the same principal fiber. Θ^x_e is a declared edge-copy variable on the interaction complex. Only an explicit hypothesis may identify the represented Θ^x_e with parallel transport along an assigned base curve.

Three uses of "pullback." The informational tensor pullback of Chapter 8 uses the connection-relative vertical jet $D^{\omega}s$ to obtain tensors on the contextual base. The parameter restriction of Proposition 24.5 is an ordinary tensor pullback along an injective map of statistical parameter manifolds. A pullback attractor in Section 11.4 is instead a backward-asymptotic object for a nonautonomous cocycle. These are three different typed constructions; their shared word transfers no theorem.

Four notions of order and path. A base curve $\gamma \subset \mathcal{C}$ orders contexts and carries no verticality or horizontality predicate at all. A bundle curve Γ has stationary, vertical, horizontal, or mixed velocity at each parameter value relative to a connection, and the interval labels obtained by quantifying those pointwise types are not exhaustive. An inference history $\mathscr{H}_i$ is an oriented unparameterized orbit in a section configuration space; its evaluation at each fixed context is vertical, and verticality predicates apply to that evaluation rather than to the configuration curve. An ordered interaction-record path retained by exact coarse-graining is a sample-space coordinate, while RG depth is a scale index. None is identified with physical time.

Finite-design link copies. The full context-dependent product of frame-coordinate changes uses $\Theta^x_{a,e}$ with discrete vertex action $h^x_{a,i} = g^x_i(c_a)$, abbreviated as $\Theta^x_{a,ij}$ when the edge copy is unique. A submodel that constrains one shared link across all a retains the stabilizer for which $h^x_{a,i}\Theta^x_{ij}(h^x_{a,j})^{-1}$ is independent of a. Context-independent rechoices are sufficient but need not exhaust that stabilizer; any larger law requires a declared compensation.

Two uses of "coarse." A Markov channel discards sample information while preserving normalization. A hard restriction substitutes a lower-dimensional configuration into an energy. A Galerkin map compresses an operator. These operations can agree in a special realization, but no shared word changes their types or transfers a theorem from one to another.

Appendix E

Central Claim-Status and Open-Obligation Ledger

This appendix collects the manuscript's unresolved obligations together with the resolved, refuted, and conditional boundaries needed to state the surviving scope. It does not upgrade any claim beyond its local theorem statement or certificate. A *conjecture* is a favored, falsifiable statement that has not been proved; an *open problem* has no presumed answer. The local theorem statements, certificates, and hypotheses remain authoritative.

E.1 Mathematical and probabilistic closure

Continuum law theory (open). Construct a refining projective family of normalized finite laws, prove tightness in a declared topology on sections, control gauges and reference measures, and pass the ELBO, connections, and observables to the limit. Finite-dimensional consistency alone does not do this. [OPEN]

Pointwise-dominated and nondominated likelihood families (open). The finite interaction theory uses one family-level dominating measure for each record kernel and a finite jointly measurable density version. If domination exists only separately at each latent state, pointwise Radon–Nikodym versions need not supply a common measurable likelihood $L_o(y)$. Extend the construction directly at the joint-measure and regular conditional-law level, or prove hypotheses giving jointly measurable versions, and specify how record-null version choices propagate through local disintegrations and coarse maps. [OPEN]

Regular frame-coordinate quotient (open). Determine generic stabilizers of the passive $G \times G$ action generated by independent belief and model trivializations, and identify strata on which the orbit space is a regular quotient. This coordinate action is not the principal structure group. A dimension subtraction is invalid without freeness and properness. [OPEN]

Optimization, scheduling, and projection (open). The finite conditional-gap and local–global potential identities establish objective compatibility, not convergence. For each proposed algorithm, declare the sequential, asynchronous, or parallel schedule and give sufficient compact-sublevel, lower-semicontinuity, differentiability, and exactness or line-search/damping hypotheses for convergence to a stationary point, or give a counterexample in the declared family. Parallel replacement of correlated full conditionals additionally requires proof that the replacements define one joint law. Separately, prove when Bregman projection is gauge compatible and commutes with a coarse parameter map, and determine whether a positive mass pin is relevant, marginal, or irrelevant under that map. [OPEN]

Joint-law lift and selector boundary (open/refuted split). An exact VFE history requires the commuting conditional/full-law lift of Hypothesis 9.9, including the dependence coordinates omitted by marginal

belief–model sections. Existence of such a smooth right inverse, nondegeneracy of its Fisher pullback, and preservation under coarse graining must be proved for each declared recognition family; they do not follow from the two associated bundles alone. The governing finite correction in Theorem 9.11 closes one smooth positive categorical instance: one parity direction inside the 57-dimensional Frechet fiber for three paired binary agents.

The governing selector correction of 2026-08-14 supersedes the unqualified claim that canonical selection is merely open. On finite typed lists, naturality under coordinatewise finite Markov kernels whose randomness is independently tensored uniquely forces the product section; this quantifier does not cover all of FinStoch. A wide marginal-compatible enlargement containing that local class admits a natural section exactly when every added morphism preserves products. Separately, the frozen contract explicitly admits both split kernels $R_{1/3}$ and $R_{1/2}$. From one fair source they have the same fair/fair target marginals but respective atom multisets $\{1/3, 1/3, 1/6, 1/6\}$ and $\{3/8, 3/8, 1/8, 1/8\}$. Naturality under both declared splits would force one single-valued target section to return both unequal laws, even modulo relabeling. The admission of both splits and single-valuedness are load-bearing. The absolute selector with those quantifiers is *REFUTED*.

Under the declared full naturality and canonicity requirements, selecting nonproduct dependence requires either restricting the admitted morphism class or adding declared relational, constraint, or reference data. Relative to a finite reference p, statistic T, and target m, a finite-KL minimizer of $D_{\mathrm{KL}}(q\|p)$ subject to $\mathbb{E}_q T = m$ exists exactly when $m \in \operatorname{conv} T(\operatorname{supp} p)$ and is unique as a law. For deterministic $f : X \to Y$, conditional-KL completion is defined for $r \ll f_{\#}p$ and composes along nested coarse maps when each stage uses the pushed reference. These relative constructions do not close arbitrary-family existence, coarse preservation, odd independent relabelings, $\mathrm{GL}(K)$ equivariance, a continuum lift, or renormalization. [OPEN]

Operational intervention quotient and nonidentifiability (established). Fix a monoid A and a response Φ. Contextual equivalence is the largest congruence contained in the kernel relation of Φ. For a compatible quotient q, write

$$\pi : A \twoheadrightarrow \mathrm{Syn}(\Phi), \qquad q : A \twoheadrightarrow B,$$
$$\Phi = \psi q.$$

There is one unique surjective unital homomorphism satisfying

$$h : B \twoheadrightarrow \mathrm{Syn}(\Phi),$$
$$\pi = hq, \qquad \bar{\Phi} h = \psi.$$

For finite A, this gives minimum protocol-class cardinality and uniqueness over A, not a minimum raw DAG or latent realization. Under compact-metrizable monoid, continuous response, and metrizable-Hausdorff target hypotheses, a countable dense context signature realizes a compact metrizable quotient with continuous multiplication and response; see Proposition 9.29 and theorem 9.30.

The declared groupoid $\mathsf{FinTIP}^{\mathrm{iso}}_{R,O}$ contains finite normalized typed DAG presentations and all partial hard assignments. Total right override forms the intervention monoid, and equality of every two-sided contextual retained response is a congruence. Thus $\mathrm{Red} : \mathsf{FinTIP}^{\mathrm{iso}}_{R,O} \to \mathsf{FinRIE}^{\mathrm{iso}}_{R,O}$ and $\overline{U}_{\mathrm{pass}} : \mathsf{FinRIE}^{\mathrm{iso}}_{R,O} \to \mathsf{FinObs}^{\mathrm{iso}}_{R,O}$ are well-defined.

Within the declared BSC subcategory, $L(1/4, 1/3)$ and $L(1/3, 1/4)$ have the same passive retained law but nonisomorphic fifteen-class hard experiments. The independent null extension collapses and is a control. Consequently no universal $R\overline{U}_{\mathrm{pass}} \cong \mathrm{id}$ recovers every reduced object, while $\overline{U}_{\mathrm{pass}} R \cong \mathrm{id}$ may still select one representative.

The same pair has unequal exact marked-soft mediator-face diameters, $(1-2\epsilon)/3$ and $(1-2\epsilon)/2$, for $0<\epsilon<1/2$, including strict-interior witnesses. For independent affine randomization, a fifteen-coordinate contextual minor has determinant $(2b-1)^6(2\delta-1)^3/32$, so any admitted randomized isomorphism would restrict on simplex vertices to the forbidden hard isomorphism; see Theorems 9.37 and 9.39. The randomized proof does not reuse the hard unmatched-response invariant, which convexification destroys.

For a finite DAG of standard-Borel node spaces and palettes, declared normalized pointwise kernel families with jointly measurable evaluations give a Borel retained response. The construction supplies an algebraic contextual quotient but does not by itself establish a standard-Borel quotient; that requires an exhibited smooth classifier or stronger topology. Compact-Polish node spaces, compact palettes, isolated baseline symbols, and jointly Feller kernels give a compact metrizable quotient with weakly continuous response. The circle heat pair has equal passive retained law but a strict one-way Blackwell order and strict soft response-set inclusion under the retained mediator, heat geometry, ordered boundary, and one global response map; see Proposition 9.40, corollary 9.41, and theorem 9.42. [ESTABLISHED]

Category-independent intervention semantics (open). The released extension supersedes blanket open wording only for its fixed hard, normalized marked-soft, independently randomized, declared standard-Borel, compact-Feller, and circle categories. Target-erasing soft maps, correlated or adaptive randomization, and identification of null-version point interventions from almost-sure passive observational conditionals remain open, as do noncompact quotients, category canonicity, arbitrary latent dilation, and minimal raw realization. No result establishes fixed-observation ELBO or VFE equality, autonomous agency, continuum base-manifold/gauge/RG dynamics, or any identification with physical geometry, time, units, constants, or ontology; see Open Problem 9.43. [OPEN]

Global relational information clock (open). The finite-dimensional Fisher length of one regular oriented history is defined and reparameterization invariant. A scalar clock shared across agents or patches additionally requires the exactness and period conditions isolated in Theorem 9.59, compatibility across critical and null strata, and, for a continuum of sections, the missing base measure and function-space theory. None of those follow from local dissipation alone. [OPEN]

E.2 General coarse maps and renormalization

Full pointwise probabilistic datum (established). Fix a nonempty finite child block I, a parent label A, one $c_* \in U_A = \bigcap_{i\in I} \mathcal{C}_i$, and structural data X with $X_A = \chi_A(X)$ outside the random channel. Under the hypotheses of Theorem 12.2, one normalized recognition-independent vertical channel $C_{A,c_*} : \mathsf{Y}_{I,c_*} \rightsquigarrow \mathsf{Z}_{A,c_*}$, covering id_{c_*}, pushes the fixed fine generative joint, selected posterior-version family, and correlated recognition law to a normalized parent triple while leaving the observation in O unchanged. An induced parent evaluator exists by standard-Borel disintegration; a predeclared evaluator requires separate almost-sure compatibility. All named recognition, prior, and posterior marginals are projections of their typed full laws and do not reconstruct the corresponding joints.

The KL chain has the nonnegative conditional-information defect $\Delta_A(o,X)$ as an additive identity in $[0,+\infty]$. Adding the same finite real $-\log p_X(o)$ to both KL terms gives an extended-real VFE identity; a finite VFE may be negative. Without a finiteness premise, $\Delta_A = 0$ exactly when the discarded conditional recognition and posterior laws agree $\mathbb{Q}_{A,o,X}$-almost surely. Finite fine KL is required for ordinary subtraction $\mathcal{F}_I - \mathcal{F}_A = \Delta_A$ and for the stated two-way pairwise common-recovery equivalence. Holonomy blindness requires typed actions, full fine-law covariance, compatible

selected posterior versions, C_{A,c_*} equivariance, and evaluator covariance; same-slice invariance requires fixed-(o, X) isotropy. Raw holonomy retention uses component roots r_I, tree transports $\tau^x_{I\leftarrow i}$, based representations H^x_I, and dressed boundary marks $V^x_e = \tau^x_{I\leftarrow i}\Theta^x_e(\tau^x_{J\leftarrow j})^{-1}$. It is the alternative and makes no blindness or membership-selection claim. The mathematical implication remains `ESTABLISHED`. The earlier bound release recorded ledger state `EVIDENCE_VERIFIED` and terminal package status `COMPLETE_AFFIRMATIVE` for its pinned pre-amendment artifact; those formal states do not bind this amended Part II revision. [ESTABLISHED]

Membership pushforward and selection (established/open split). Gauge, path, and root invariance of the marginal-law distortion follows from Theorem 10.13. Its attained zero characterizes a holonomy-stabilized parallel marginal section. The normalized evidence-preserving full pointwise joint-law pushforward and finite pairwise recovery criterion are supplied by Theorem 12.2; what remains open is one common recovery kernel for an entire statistical experiment without separately imposing simultaneous family hypotheses.

Given a declared membership law, normalized assignment kernels $m_\ell(I \mid i, R_\ell)$ include hard partitions as deterministic special cases. Correlated endpoint membership uses the normalized joint kernel $M_\ell(I, J \mid i, j, R_\ell)$, not an unjustified product. Literal unit-weight multiple membership replicates a finite measure and is not a probability pushforward. Exact recursion retains η^b_ℓ, η^m_ℓ, recovers α and the conditional rows by marginalization and disintegration, and requires Chapman–Kolmogorov composition together with all marks used by the next assignment. When adaptive membership is random, one normalized recognition-independent $C^{\mathrm{struct},c_*}_{\ell+1\leftarrow\ell} : \mathsf{Y}_{\ell,c_*} \rightsquigarrow \mathsf{R}_\ell \times \mathsf{Y}_{\ell+1,c_*}$ samples and retains R_ℓ. The current state already contains η^b_ℓ, η^m_ℓ and the declared rooted-holonomy and dressed-boundary marks. Interaction-record or memory marks must additionally be retained whenever the claimed exact closure requires them. Its induced transition joint on $\widehat{\mathsf{Y}}_{\ell+1\leftarrow\ell} := (\mathsf{Y}_{\ell,c_*}, R_\ell, \mathsf{Y}_{\ell+1,c_*})$ is the augmented law. The same declared channel pushes the generative, selected-posterior, and recognition laws. Conditional on that declaration, normalization and the typed pushforward are established; a relabeling-natural, gauge-compatible, nondegenerate selector remains open. [OPEN]

Neighborhood parents and descent (established/open split). Chapter 14 constructs a cover with pairwise and triple intersections and empty total intersection, compatible ambient-law restrictions, availability parents, and joint attention-event fields with explicit zero-occupancy conditional domains. Its nonflat connection, quotient, and base-descent results retain their stated compatibility and projectability hypotheses. These are established realizations of geometric parents. [ESTABLISHED]

Arbitrary independently specified local-law gluing, a canonical choice of ambient dependence, unrestricted rank-changing quotient geometry, and canonical channel or membership selection remain open. A compatible ambient law supplies one extension; it does not reconstruct every marginal family. [OPEN]

Parent mechanisms, memory, and agency (established/open split). Chapter 15 gives normalized transitions that depend on parents, retained hidden-state path laws, a finite-memory KL projection, and the stated quantitative memory and duration bounds. Projected dynamics need not be Markov, and the persistence comparison has no universal advantage over its waiting control. These are established mechanism and approximation statements. [ESTABLISHED]

The prepared finite-depth family in Theorem 16.1 supplies successive actual-law activations and controlled approximate dynamics; its theorem and proof remain in Chapters A and 16. Persistent nested lineage from arbitrary descent, a separated slow manifold, nondegenerate intrinsic hierarchy selection, autonomous agency, and physical time remain open. General bidirectional optimization still requires parent dependence transmitted through the chosen VFE parameterization. The operational

comparison category and its retained intervention boundary remain separate from an observational model. [OPEN]

Constructive lattice RG (established/open split). Chapter 17 supplies a unique lattice law and its conditional specification at every exterior in the high-temperature regime on $0 < b < 1$, $0 < t < 1/12$, exact path blocking, complete generated interactions, a local RG operator on a positive open interaction ball, and a constructive weighted-basin bridge. Its law, interaction-norm, and regular fixed-depth Fisher results retain their separate hypotheses. Chapter 19 supplies pressure and specific-VFE limits with a finite retained geometry catalog and the complete induced observation experiment. [ESTABLISHED]

These realizations do not establish arbitrary standard-Borel globally equivariant conditional versions, a universal rescaling family, noncompact/continuum law closure, or convergence of every geometric component. The failed earlier rescaling construction remains failed; a different admitted path family does not repair it. These broader construction obligations remain open. [OPEN]

Two-index limits and universality (established/open split). The specified spatial cylinder limits and their admitted volume/block comparisons are established in Chapter 17. The geometry-pressure identities in Chapter 19 fix every finite blocking word before the volume limit. These are different quantifiers. [ESTABLISHED]

Unrestricted simultaneous volume/depth limits for the full labeled experiment, critical or continuum universality, and blocking-scheme robustness beyond the admitted path semigroup remain open. A finite congruence identity, one matched exponent, or a norm contraction cannot replace those constructions. [OPEN]

Geometry weights and statistical curvature (established/open split). Chapters 18 and 19 establish exact branch normalization, scalar transport, strict-gap selection and the necessary tied-weight criterion. Geometry erasure can destroy Gibbs regularity; a vanishing label can make KL to a limiting joint infinite while the finite-ensemble density exists. These are established boundaries, not unresolved warnings. [ESTABLISHED]

The regular spatial Fisher theorem supplies fixed-depth parameter regions and finite-hierarchy charts, not one uniform inverse at all depths. The scoped constructions below are established with proofs in Chapters C and 20. Each derivative, limiting, and regularity conclusion retains the hypotheses listed with it. The strict-phase, fixed-tie, and reciprocal-volume results are distinct statements and are not interchangeable.

Complete finite experiment and native derivative inputs (established under the stated hypotheses). Section 20.1 gives written proofs of the full retained-label Fisher decomposition and stochastic conditional-KL VFE loss, with one fixed channel, the complete induced observation experiment, and the stated support and finite-domination conditions. Theorems C.6 and C.11 and proposition C.7 provide the written native C^4 pressure and C^2 Fisher arguments for actual free laws and their exact fixed-word pushes on compact positive high-temperature source boxes. The source maps are C^4, all physical normalizations remain present, and fixed word depth precedes the volume limit. Equality of limiting densities does not identify a finite free law, an actual infinite-law marginal, and an exact push. The all-edge selected posterior additionally requires the admitted original and shifted source maps of Proposition C.18; it is not an arbitrary positive-likelihood extension. [ESTABLISHED]

Strict-phase extensive geometry (established under the stated hypotheses). Theorem C.19 proves convergence of the full retained-label tensor divided by V in C^2 on compact parameter sets. It requires actual branch C^2 density limits, effective log-mass C^3 bounds of order V and C^3 bulk limits, and one uniformly separated maximizing bulk log-mass density. The common log-mass scalar cancels before those growth conditions are imposed; global and hierarchical representatives retain their distinct

physical-partition terms. This statement does not extend the strict-gap hypothesis through a tie or imply a nondegenerate limiting metric. [ESTABLISHED]

Fixed-tie squared-volume information (established under the stated hypotheses). Proposition C.21 proves the fixed-parameter statement for $I(P_L)/V^2 \to \mathrm{Cov}_w(v_\gamma)$ under its actual finite-weight limit, effective growth, and branch $O(V)$ hypotheses, where $v_\gamma = \nabla s_\gamma(\theta_0)$. Equal bulk log-mass densities do not imply equal gradients or convergent finite weights. A null direction of this covariance gives $o(V^2)$, not an extensive bound; residual gradients and rare-label mass can produce intermediate scales. This is a value statement at the fixed tie, not a C^2 extension of the extensive metric across it. The accounting proof in Proposition C.31 preserves any separately proved superextensive scale under its branch $O(M)$ bound and exact label-preserving channel. [ESTABLISHED]

Reciprocal-volume experiments (established under the stated hypotheses). Theorem C.25 proves the C^3 label-law and C^2 Fisher-limit arguments in the coordinate $u = V(\theta - \theta_0)$ under actual positive relative center limits, effective C^3 growth, and actual branch C^2 limits. Proposition C.26 supplies its written comparison-kernel argument. The moving-center theorem in Theorem C.29 instead treats the actual equal-log-mass center at a transverse two-label tie with a gap to all other labels. It does not supply arbitrary declared centers, nontransverse centers, or a general multi-label centering construction. For a fixed block word, Proposition C.33 uses fine area $M = B^2V$ and retains both $u = M(\theta - \theta_{c,L})$ and $v = V(\theta - \theta_{c,L})$; its coarse limit is $B^4G(B^2v)$. This window geometry is a distinct parameter scaling, not unrestricted smoothness of the extensive density through ties. [ESTABLISHED]

Regular charts and the explicit quotient (established under the stated hypotheses). Proposition C.20 derives inverse convergence in C^2, Levi-Civita convergence in C^1, and curvature convergence in C^0 only on charts with an actual positive limiting metric and therefore a compact positive floor. The native fixed-depth chart theorem Proposition C.13 supplies its stated source regions; a general source pullback still needs the required rank and source-image conditions. Proposition C.27 constructs the limiting crossover quotient by the orthogonal complement of the span of slope differences. Its finite metric statement is restriction to a fixed representative subspace, not descent of every finite metric to that quotient. Neither this example nor constant rank alone constructs an unrestricted global regular quotient or an inverse through rank changes. [ESTABLISHED]

Prescribed curvature actions (established under the stated hypotheses). Propositions C.39 and C.40 contain proofs for separately prescribed geometry-only actions with the stated finite normalization, C^3 growth, and relative-center conditions. The exact relative multiplier is $\exp(-\mathscr{A}_{\gamma,L} + \mathscr{A}_{\gamma_0,L})$; boundedness or zero bulk action alone does not ensure a relative limit. The typed action discussion in Section C.3.4 retains the declared pairing, base metric, integration measure, coupling, and sign. Gauge connection curvature, Fisher curvature of a regular statistical family, and base-metric curvature remain distinct. A positive pairing that uses an additional SPD fiber metric does not provide a positive conjugation-invariant norm for an unrestricted $\mathrm{GL}(n)$ trace square. These are prescribed-action arguments, not a self-consistency construction for curvature of the same unknown reweighted law. [ESTABLISHED]

Full-program limits and identification (OPEN). The preceding scoped theorems do not assert smooth extensive geometry through arbitrary label ties, an everywhere nonsingular limiting Fisher metric, uniform regular inverses at unbounded depth, or unrestricted simultaneous volume/depth limits. Critical or continuum universality, robustness beyond the admitted path semigroup, general regular rank-changing quotients, and a self-consistent unknown-metric action remain separate construction obligations. Nor do these theorems supply a probability measure on all connections or base metrics, arbitrary local-law gluing, canonical geometric selection, or physical-time or physical-law identification.

The earlier established finite and fixed-depth results remain in force within their own hypotheses. [OPEN]

Extensive relevance beyond the scalar Gaussian realization (open). Theorem 12.13 inhabits all three relevance classes for independent scalar Gaussian coordinates under the declared replication sum, block statistic, and fixed Fisher norm. Proposition 12.15 shows that correlated coordinates, unrestricted multivariate covariance, and general $\mathrm{GL}(d)$ actions each leave that theorem. Obtaining the corresponding spectrum for a correlated, anisotropic, or gauge-mixed block requires a declared transported family of laws and tangent norms together with a separate diagonalization. [OPEN]

Nonlinear attraction at the Gaussian tangent (open). The Hermite spectrum controls the derivative of the score map at the Gaussian law. A basin of attraction, a remainder estimate for the nonlinear map, and the fate of the marginal degree-two direction do not follow from a linear spectral statement and are not supplied here. [OPEN]

Multiplicative ergodic structure of the interaction cocycle (open). Definition 11.10 and Theorem 11.14 define compatible cross-scale modes and give exponent invariance under tempered comparison data. An Oseledets splitting would additionally require a measure-preserving base flow, measurability of the cocycle in the base variable, integrability of $\log^+\|D\|$ and of $\log^+\|D^{-1}\|$, and either finite dimension or a compactness hypothesis. A deterministic blocking sequence supplies none of these, so the growth rates remain definitions rather than an existence theorem. [OPEN]

Smooth scale tier and continuous beta (open). Definition 12.30 declares the scale manifold, Banach bundle, and scale connection that a continuous beta requires, and Proposition 12.31 shows that a discrete blocking sequence does not determine them. Selecting a canonical interpolation, or proving that a reported continuous beta is independent of the choice inside a declared class, remains open. [OPEN]

Nontrivial invariant retained subspace (open). (12.154) settles exactly when a retained beta function is exact. Whether any retained subspace other than the zero and full ones is invariant under $T_\ell^{\mathcal{G}}$ for an admitted network and blocking channel is not settled here, so every reported projected flow carries its residual $\delta\beta_\ell$. [OPEN]

Bayesian-RG bridge (open). To compare this flow with Bayesian renormalization, place both on one state space, transport the relevant Fisher tensors, declare a common scale map, and prove a monotone. Similar Gaussian formulas do not prove opposite flow directions. [OPEN]

Fine–coarse history semiconjugacy for the declared RG maps (open). Theorem 9.60 compares Fisher duration along a single path pushed through a parameter-independent record channel, and Theorems 9.66, 9.68 and 9.69 and proposition 9.67 settle what oriented semiconjugacy does and does not give once it holds. What is not proved is that this manuscript's own independently recomputed coarse natural-gradient flow satisfies it. One structured sufficient route is the one isolated at Proposition 9.72: exhibit χ_ℓ with $\chi_\ell' > 0$ and an open set containing the fine orbit on which Hypothesis 9.71 holds, prove for the exact-contraction coarse objective that the orbit lies in the interior of the attaining set, and then verify horizontal conformality of the declared R_ℓ. This route is not asserted necessary or minimal. Behavior at critical and singular strata remains part of the open work. The route is satisfiable, since the exhibited tier of Definition 9.6 meets it with $a_\ell = \frac{1}{2}$. [OPEN]

Configuration tier for the declared recognition family (open). Definition 9.6 and Theorem 9.7 exhibit a nonempty finite-dimensional configuration manifold with a strong constant Gram metric, a rank test for nondegeneracy, and a closed gauge-orbit tangent, and the infinite-dimensional L^2 and Sobolev tiers are separated with the failure witness attached. What is not settled is whether the manuscript's

own declared recognition family can be brought to one of those tiers at every scale at which a natural-gradient field or a Fisher duration is asserted. Until that is done, the history theorems hold on the exhibited tier and are conditional elsewhere. [OPEN]

Smooth structure of the projectable set (open). Theorem 8.13 characterizes which fine sections descend, and Proposition 8.14 shows the descendable set is proper and can have empty interior. Whether that set is a smooth submanifold of a declared configuration manifold carrying a smooth induced arrow needs a transversality or elliptic-regularity hypothesis that is not supplied. This manuscript avoids the question by declaring R_ℓ separately; the pointwise-induced route remains open for anyone who prefers it. [OPEN]

Strong Hilbert quotients and the weak-Fisher boundary. Theorem 9.56 proves the quotient-speed formula for a free proper smooth isometric Banach-Lie action on a strong Riemannian Hilbert manifold, including the smooth principal quotient and orthogonal splitting. Proposition 9.57 gives an actual normalized normal-mean experiment on a Hilbert-model section space with a weak Fisher metric. Its gauge action is free, proper, smooth and isometric, but the metric-orthogonal lift fails. The trace obstruction in Proposition 9.58 also prevents interpreting that quotient as the indicated fixed Markov statistical channel. The distinction is between strong and weak metrics, not finite and infinite dimensions. [ESTABLISHED]

E.3 Multivariate-Gaussian realization

Scalarized attraction (conjecture). Conjecture 26.4 predicts a unique attracting spatial ray only after the matrix direction M_0 and one primitive spatial map B are fixed and the Perron vector factorizes. Primitivity, factorization, and the fixed hierarchy still require proof in any application. The claim does not cover the full matrix cone or a nonautonomous cocycle. [CONJECTURE]

Kron families and finite semialgebraic classification. Theorem 25.7 removes convexity and closedness from the independently pair-tested coefficient family with a fixed positive hidden anchor. Its maximal families are precisely the full anchor-congruence-diagonal cones. Proposition 25.8 retains the exact scalar-domain stability test, and Corollary 25.9 treats the common-support quotient. Theorem 25.10 supplies local frames for a smooth maximal-algebra bundle and an obstruction to global ordered frames. For independently arbitrary positive matrix weights, Theorem 25.11 classifies universal closure by the number of retained terminals per hidden component. Theorem 25.14 supplies an exact marked nonflat series sector, with sources, Gaussian mass and coordinate Jacobians. Proposition 25.12 gives a proper Gaussian marginal outside every positive one-node/two-node energy presentation on the same retained node spaces. These results are established. [ESTABLISHED]

For a supplied finite semialgebraic domain with effective real algebraic coefficients, Theorem 25.17 constructs the complete exact-representability locus and decides universal closure, including finite shared-variable diagrams. It allows correlated and nonconvex predicates and variable anchors. The support bound in Proposition 25.18 requires independently scalable unmarked summands. Exact Gaussian source and scalar data are retained, while additional transcendental constraints require their own reduction. Proposition 25.19 distinguishes measurable selection from smooth or equivariant selection, and Proposition 25.20 excludes a universal decision procedure from arbitrary enumerator code. These are established at their declared quantifiers. [ESTABLISHED]

Proposition 25.21 and theorem 25.22 construct the actual normalized energy-quotient Gaussian for PSD complete precision with a declared representative metric and volume. Theorem 25.23 extends the finite semialgebraic decision theorem to parameter-dependent kernels and specified descending

quotient maps, with the allowed retained pair presentation encoded explicitly. Exact covariance and source conditions, metric covariance, and a raw mass-matching scalar are retained. Proposition 25.24 refutes automatic DQM continuation through rank changes, even for smooth PSD energy data. This settles the finite encoded variable-support tier, without an implicit full-space normalization or a global smoothness conclusion. [ESTABLISHED]

Nonsemialgebraic admissibility families need a specified description and an actual argument beyond that finite algorithm. A global smooth, gauge-compatible or functorial presentation is an additional target; pointwise feasibility does not supply it. The enumerator obstruction is algorithmic and does not refute every possible nonalgorithmic classification. No universal classification of unspecified constraints is asserted. [OPEN]

Nonflat-link compression (open). Characterize when the rectangular endpoint data of a blocked cut compress to one invertible transporter and one SPD weight, remains closed under another merge, and admits a normalized probability and rescaling lift. [OPEN]

Intrinsic scale selection (open). Find a nondegenerate criterion internal to one fixed evidence problem, or prove endpoint degeneracy for a stated class. A successful alternative may be a normalized Markov pushforward or a proved Lyapunov functional; the determinant gap and quotient-volume convention do not supply one. [OPEN]

Information-geometric transfer (open). Identify the coarse recognition parameter with a parameter of the fitted generative law, prove nondegeneracy of the relevant Fisher metric, and then test the proposed scale construction. Recognition-space and model-space Fisher tensors are not interchangeable by name. [OPEN]

Stochastic inverse RG (open). Choose compatible priors for discarded internal edges, nested refinement kernels, chart data, and a projective-limit theorem. Infinite divisibility can redraw compatible microstates in distribution; it cannot invert a many-to-one realized blocking. [OPEN]

Update robustness (open). Extend the anchored Gaussian convergence theorem, or find its boundary, for delayed, noisy, asynchronous, or inexact updates. The present contraction proof covers only its stated synchronous exact regime. [OPEN]

E.4 Geometric and physical interpretation

Operational base holonomy (established/open split). Theorem 14.7 supplies a specified base loop, connection family, normalized Gaussian record, and a pair with differing record distributions while the non-holonomy model inputs are held fixed. This establishes existence of an operational tuple in that declared family. [ESTABLISHED]

Canonical connection recovery, sensitivity for arbitrary record classes, and empirical identification remain open. A controlled example does not infer a connection from unrestricted population data or identify bundle topology. [OPEN]

Graph-to-base identification (established/open split). The cover construction with nonzero curvature assigns base curves and connection transports in Chapter 14. This supplies one compatible graph/base realization. [ESTABLISHED]

For arbitrary graph data, an embedding, oriented curves, and proof that represented links equal the specified base transports remain additional obligations. Even with that identification, connection holonomy does not imply nontrivial bundle topology. [OPEN]

Canonical nondegenerate pullback geometry (open). Theorem 8.4 proves passive gauge covariance for a fixed section and chosen connection, while Theorem 8.8 identifies the conditional quotient geometry. A canonical or empirically specified connection, relative channel weights, constant rank, integrability of the radical, a regular Hausdorff leaf space, basicness of both induced tensors, and a physical interpretation remain separate obligations. No canonical connection is selected anywhere in this manuscript. Every pullback tensor, and every mixed or section-induced length that uses a horizontal–vertical split, stays connection relative; every Fisher duration stays relative to its declared metric. By contrast, the vertical Fisher length of a curve confined to one fixed statistical fiber uses only the vertical Fisher metric and is connection independent, although it remains metric relative. [OPEN]

Retained Fisher quotient and declared blocks (conditional boundary). For a smooth retained-law map $\rho : \Theta \to N$ and a positive-semidefinite target tensor g, the exact pullback radical is

$$\operatorname{rad}(\rho^* g) = d\rho^{-1}(\operatorname{rad} g).$$

It equals $\ker d\rho$ exactly when $\operatorname{im} d\rho \cap \operatorname{rad} g = \{0\}$. Under this transversality and constant rank, the visible tangent is a smooth positive-definite quotient vector bundle. A global quotient manifold still requires a simple Hausdorff null-leaf quotient. Under the equality $\operatorname{rad}(\rho^* g) = \ker d\rho$ and a regular connected-leaf quotient, the specific tensor $\rho^* g$ is basic because ρ factors through that quotient. This does not make an arbitrary tensor, a declared block projector, or the connection-relative covariant jet of Hypothesis 8.1 automatically basic.

For a declared tangent splitting, direct quotient blocks require blockwise kernel splitting; additive Fisher energy additionally requires pairwise orthogonality; and block descent requires projectable/basic image data, including the global disconnected-fiber condition when applicable. In the promoted six-bit parity family the full-joint map has rank seven everywhere, whereas singleton retention has rank six with kernel exactly $\operatorname{span}\{\partial_\kappa\}$. This rank statement neither supplies node typing nor agentizes κ. The vector-bundle identities are established under their stated hypotheses; a global regular quotient, canonical block data, autonomous agency, and physical interpretation remain open. [OPEN]

Intrinsic Fisher positivity and cocycle solutions. The exact criteria of Theorems 8.17 and 8.24 have complete finite-dimensional parameterizations in the actual fundamental-action image. Theorem 8.25 gives every positive solution by a contraction from a positive-form quotient. Theorem 8.26 gives the positive-semidefinite cocycle solutions by isometric embeddings, and Theorem 8.27 retains the radical cross-pairing, inertia conditions and totally isotropic freedom required by an indefinite defect. All kernel lifts are included. Proposition 8.28 gives the stated constant-rank smoothness, while Theorem 8.29 characterizes realization by a changed target connection with the source fixed. Pointwise admissibility does not remove the all-fiber descent condition or imply smooth extension through rank changes. These classification statements are established at their stated quantifiers. [ESTABLISHED]

Physical-time identification (open). Fisher duration on an oriented inference history is unchanged by an orientation-preserving change of parameter. It is a statistical length, not a physical clock. A physical-time claim would require a named target clock, a calibration map and uncertainty model, causal and compositional consistency, and discriminating evidence against alternative parameterizations. None is supplied here. [OPEN]

Open Problem E.1 (Physical-law identification). Identifying a repaired limiting object with a physical law requires a named target system, an observable and estimator, a frozen validation and uncertainty protocol, a baseline margin, probability-level closure, and held-out evidence robust across admissible

blocking schemes. None is supplied here. Accordingly, the present mathematical theory neither proves a physical law nor supports ontological closure. [OPEN]

Appendix F

Numerical Provenance

The reproducibility package is part of the manuscript record:

`verification/claims.json`	claim inventory, seeds, sample counts, tolerances, mappings, and open obligations
`verification/run_checks.py`	single deterministic CPU runner
`verification/current-results.json`	current environment, source and protocol hashes, individual outcomes, residuals, and overall result
`verification/VERIFICATION.md`	human-readable protocol and interpretation

From the repository root, the declared run is

```
& "C:\Python314\python.exe" manuscripts/gauge_vfe_rg/verification/run_checks.py
```

The generated JSON record is the authority for exact counts and numerical values; this appendix intentionally does not duplicate seed lists or result tables that can drift.

Frozen factorization-gap stress protocol. The check `CHK-CG-FACTOR-GAP-STRESS-3138` is a new deterministic protocol, not a reconstruction of an undocumented historical run. Seed 20260803 fixes exactly $3{,}138$ case identities: $2{,}400$ general, 200 exactly block diagonal, 200 nearly decoupled, 120 scale, 120 permutation, 80 nested-refinement, and 18 100-decimal-digit control cases. Every stratum covers dimensions 2 through 16 and condition-number tiers $1, 10^2, \ldots, 10^{14}$ as nominal pre-transform generator labels. The legacy field `condition_number` stores that nominal label, while every digest-bound record also stores `achieved_condition_number`, defined as `numpy.linalg.cond` of the regenerated matrix. The observed global range is $[1, 1.015265553314175 \times 10^{14}]$, but this is only a global coverage claim. The achieved ranges are:

Stratum	Minimum	Maximum
Global	1	$1.015265553314175 \times 10^{14}$
General	1	$1.015265553314175 \times 10^{14}$
Exactly block diagonal	1	$1.0075938532153164 \times 10^{14}$
Nearly decoupled	1.0000000000000002	$9.273791153286738 \times 10^{8}$
Scale	1	$1.0036120669923527 \times 10^{14}$
Permutation	1	$1.0071935024197442 \times 10^{14}$
Nested refinement	1	$1.0077056571347422 \times 10^{14}$
100-digit controls	1.0000000000000004	$1.0048132000124055 \times 10^{14}$

The two scalar blocks in the dimension-two exactly-block-diagonal cases cannot realize their nominal targets, and the nearly-decoupled transformation does not attain its high nominal tiers. Each record

binds the regenerated binary64 matrix and partition by SHA-256, and repeated executions must agree exactly. The named runtime check separately records all $3{,}138$ attempted schedule cases, successfully evaluated 100-decimal-digit exact-binary64 references, and completed production-reference comparisons. PASS requires $3{,}138$ in each category and zero failures; the 18 designated controls remain a named stratum rather than the sole accuracy evidence. [NUMERICAL]

The ordinary value path uses Cholesky triangular solves, canonical-correlation singular values, and `log1p(-rho**2)`. Near the floating-point boundary, including a raw excursion above one, the runner compares two binary64 solve backends, applies a $64n\epsilon_{\mathrm{mach}}$ operational rejection guard, and uses a 200-decimal-digit evaluation of the exact stored binary64 input for the reported gap. The value path also falls back when $\kappa_2(J)\epsilon_{\mathrm{mach}} \geq 10^{-4}$; this is a frozen-protocol conditioning trigger, not a proved perturbation radius. A nonfinite binary64 condition estimate also triggers exact-input evaluation, rather than rejection, after high-precision Cholesky certification of positive definiteness. The public diagnostic name `residual_derived_clip_bound` is retained for protocol compatibility, but on this boundary path its value is the outward-rounded discrepancy between the maximum raw binary64 ρ^2 and the high-precision exact-input ρ^2. It is not presented as a general residual perturbation bound. The positive excursion witness is a separate focused regression fixture, not an outcome of the ordinary $3{,}138$-case schedule. Its matrix digest is `45310a74550d3759fed0f83f71a6cf3b0f45942499361d723a2248bbe243e2e3`, its 200-digit exact-input gap rounds to 22.777105858844084, its raw excursion is $4.440892098500626 \times 10^{-16}$, and its outward allowance is $9.432369402326953 \times 10^{-16}$. The check mechanically recomputes all four quantities and requires PASS. Its value comes from the exact-input fallback rather than from clipping ρ^2 below one. The compatibility fields `clipping_applied` and `clipping_amount` record that a positive cross-backend excursion was admitted and its maximum excess above one; they do not say that this amount was substituted into the returned value. Exact-input domain violations and raw excursions whose discrepancies exceed the guard are rejected. Cholesky and solve residuals remain local backward diagnostics; the protocol does not promote them to canonical-correlation or gap, relative-entropy, monotonicity, scale, or oracle-comparison forward-error bounds, and no forward tolerance includes them. Forward comparisons instead use independent exact-binary64 high-precision references plus separately declared endpoint-roundoff or protocol-acceptance policies. Their finite-health gate is $64\gamma_n$, with $\gamma_n = nu/(1-nu)$ and binary64 unit roundoff $u = \epsilon_{\mathrm{mach}}/2$. Early high-precision fallbacks selected before residual evaluation serialize the residual, tolerance, and backward-error diagnostics as `null` because they were not evaluated. A residual-health fallback instead retains each finite binary64 measurement, including finite values that exceed the health gate. A field is `null` only when it has no finite usable numeric value, either because an attempted measurement is nonfinite or because a derived diagnostic requires an unavailable constituent; finite sibling fields remain numeric. The evaluation method distinguishes the finite threshold case from nonfinite Cholesky, solve, or tolerance causes and lists multiple causes in that fixed order. These retained measurements remain local backward diagnostics and do not enter a forward-error tolerance. In every branch, `null` denotes neither failure nor zero. The exact zero diagnostics for a one-block, zero-step result remain numeric zero. The `GapStep` API retains the fieldwise residual quartet. A `FactorizationCaseRecord` serializes only the aggregate backward diagnostic, maximum

Cholesky residual, and ordered evaluation-method tuple, not per-step solve or tolerance values. The compatibility field `backward_error_bound` is a sum of one-ULP-up summaries of already rounded local residual measurements; it is not a certified perturbation or forward-error bound. [NUMERICAL]

What a passing run means. It means that the current deterministic, double-precision CPU implementations satisfy their declared finite checks and controls within their recorded tolerances. Those checks corroborate algebraic identities, exercise finite witnesses, and detect regressions in the encoded endpoints. They do not prove a theorem, genericity, an asymptotic limit, universality, or a physical interpretation. Analytic claims rest on derivations or eligible cited results, not on numerical agreement.

Freshness and scope. The result artifact records interpreter and dependency versions and binds the discovered manuscript sources and verification protocol by SHA-256. Any source, protocol, input, dependency, or environment change requires a new run before the numerical evidence is current. The suite uses no GPU and makes no GPU claim. A missing protocol for an empirical statement remains inconclusive even when every implemented check passes.

The checked-in `verification/current-results.json` is the older 29-check artifact. It is not evidence for this 30-check package. A governed 30-check regeneration remains required before that evidence is current. This appendix does not claim that deferred artifact is current.

F.1 Finite thermodynamic Fisher controls

The following finite computational comparisons concern the retained-label experiments of Section 20.1. They used exact rational arithmetic, without a floating-point acceptance tolerance or random sampling; no random seed was used. Affine-simplex controls in parameter dimensions one, two, and three matched their independent rational Fisher formulas through derivative degree two, including a null third coordinate and reordered input data. Two-label logistic controls matched the corresponding rational Fisher derivatives at declared tie, subextensive-score, and decreasing-weight configurations. Their volume parameters specify finite algebraic examples rather than enumerations of larger spatial lattices. These are numerical observations, not proofs of arbitrary-dimensional or thermodynamic claims. [NUMERICAL]

The varying-label and varying-branch comparison matched the full Fisher decomposition and its differentiated weight-times-branch terms. A fixed stochastic-channel control matched the zeroth residual-score covariance and the degree-two derivative accounting for supplied areas $(V_{\mathrm{fine}}, V_{\mathrm{coarse}}) = (4, 1)$ and $B = 2$, including both multipliers B^2 and B^4. The supplied areas do not identify that channel as a spatial block map. Coherent geometry and carrier relabelings preserved the results. An empty conditional fiber remained undefined and contributed zero after weighting. These numerical comparisons retain their fixed-support and fixed-channel scope. [NUMERICAL]

For the native side-two square at $(b, t) = (1/4, 1/20)$, exhaustive summation over all 256 configurations of eight signs matched every raw-factor and physical-partition derivative through total degree

three. There is one internal plaquette. The independent normalized-score Gram comparison was at Fisher derivative degree zero. At side one with $(b, t) = (1/3, 1/20)$, the comparison used two signs, native partition one, a null plaquette coordinate, and the stated field Fisher derivatives. These finite numerical observations do not supply derivative-convergence estimates as the square grows. [NUMERICAL]

A separate selected-posterior comparison used two side-one branches with global factor normalization, a fixed reference, varying raw masses, geometry weight and physical scalar, and a binary sensor on one edge. Its mean was $\eta_\gamma = k_\gamma b$, with $k_\gamma \in \{1/2, 1/3\}$. The posterior derivatives matched an independent two-coordinate Taylor-quotient calculation through total degree three; the induced pushed posterior and evidence derivatives also agreed. The physical scalar varied, but the underlying native partition was identically one. This numerical comparison therefore does not exercise a hierarchical posterior with a nonconstant native partition divisor. [NUMERICAL]

These retained finite comparisons contain no independently evaluated all-edge source-shift derivative, multistage information telescope, prescribed curvature-action derivative, or recognition-to-posterior conditional-KL VFE comparison. Their scientific scope is the finite comparisons stated above; they supply no computational proof of native thermodynamic response, limiting label weights, regular-chart inverses, curvature limits, or physical identification. Those conclusions require their stated mathematical arguments and outstanding adjudication. No new calculation is asserted by recording these numerical observations here. [NUMERICAL]

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